2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan

AREA-WIDE ELEMENTS

VOLUME 1

HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

TETRA TECH

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan

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EXECUTIVE SUMMARY

Hazard Mitigation Overview

Hazard mitigation is the use of long-term and short-term policies, programs, projects, and other activities to minimize the loss of life, injury, and property damage that can result from a disaster. Communities, residents, and businesses across the United States have been faced with continually increasing costs associated with natural and human-caused hazards. Hazard mitigation is the first step in reducing risk and is the most effective way to reduce costs associated with hazards. This hazard mitigation plan examines Harris County's vulnerability to natural and technological hazards and identifies actions that Harris County and participating agencies can take to mitigate those hazards.

Harris County first prepared a hazard mitigation plan in 2005, which included 26 jurisdictions participating as partners in the plan. Since then, Harris County has updated the plan every five years, in compliance with State of Texas and federal guidelines. This 2025 update to the plan was coordinated with 42 planning partners in the County. This coordination in planning facilitates a more efficient planning process than each jurisdiction developing their own plan and allows these planning partners to pool resources and eliminate redundant activities.

The plan complies with federal and state hazard mitigation planning requirements to establish the Planning Partners' eligibility for funding under Federal Emergency Management Agency (FEMA) grant programs. FEMA has issued guidelines for the development of multi-jurisdictional hazard mitigation plans. The federal Disaster Mitigation Act of 2000 requires state and local entities to implement pre-disaster mitigation planning and develop hazard mitigation plans. The Texas Department of Emergency Management (TDEM) supports plan development for jurisdictions in Texas, ensuring alignment with federal standards and enhancing community resilience through coordinated, proactive risk-reduction strategies.

The Planning Process

The effort to update the Harris County Multi-Jurisdictional Hazard Mitigation Plan (MJ-HMP) was led by a core planning team of staff from the Harris County Office of Homeland Security and Emergency Management (HCOHSEM) and consultant Tetra Tech, Inc. In addition, working as subconsultants to Tetra Tech, Resalire Infrastructure Solutions, Incorporated, and Bent Ear Solutions, LLC, also played an active role in development of the plan. In initiating the planning project, Harris County sought input and involvement from the Harris County community through the following activities:

- Developed a Steering Committee consisting of key stakeholders.
- Reviewed the 2020 Harris County Multi-Hazard Mitigation Action Plan (HC MH-MAP)
- Identified hazards of concern to the County to be included in the update
- Profiled the hazards of concern

- Estimated the inventory at risk and potential losses associated with these hazards
- Reviewed and updated the mitigation goals and objectives
- Reviewed mitigation strategy and actions outlined in the 2020 HC MH-MAP to indicate progress
- Developed new mitigation actions to reduce the vulnerability of assets from hazards of concern
- Involved a wide range of stakeholders and the public in the plan update process
- Developed mitigation plan maintenance procedures to be executed after obtaining approval of the plan from TDEM and FEMA

Involvement by Stakeholders and the Public

The Planning Partners kept stakeholders and the public informed throughout the planning process and provided opportunities for public comment and input. In addition, numerous agencies and stakeholders participated as core or support members of the Steering Committee or Planning Partnership, providing input and expertise throughout the planning process.

Participating Jurisdictions Involved in the Mitigation Planning Effort

The following are the local governments/agencies in Harris County that participated as Planning Partners in the 2025 MJ-HMP update:

- Harris County
- Harris County Flood Control District
- Harris County Office of County Administration
- City of Bellaire
- City of Bunker Hill Village
- City of Deer Park
- City of El Lago
- City of Friendswood
- City of Galena Park
- City of Hedwig Village
- City of Humble
- City of Hunters Creek Village
- City of Jacinto City
- City of Jersey Village
- City of Katy
- City of La Porte
- City of Missouri City
- City of Morgan's Point
- City of Nassau Bay
- City of Pasadena
- City of Piney Point Village
- City of Seabrook
- City of Shoreacres
- City of Southside Place
- City of Spring Valley Village
- City of Stafford
- City of Taylor Lake Village
- City of Tomball
- City of Waller
- City of Webster
- City of West University Place
- Cypress Creek Drainage District
- Cypress Fairbanks Independent School District
- Goose Creek Independent School District
- Gulf Coast Authority
- Harris Health System
- Houston Community College
- Lee College
- San Jacinto College
- Spring Independent School District
- University of Houston
- Varnett Public School

The participating jurisdictions provided significant input into the preparation of the plan, in particular the preparation of jurisdiction-specific annexes included in Volume II.

Multiple Agency Support for Hazard Mitigation

Primary responsibility for the development and implementation of mitigation strategies and policies lies with local governments. However, local governments are not alone; various partners and resources at the regional, state, and federal levels are available to assist communities in the development and implementation of mitigation strategies. In Texas, TDEM is the lead agency providing hazard mitigation planning assistance to local jurisdictions. In addition, FEMA provides grants, tools, guidance, and training to support mitigation planning.

In updating the MJ-HMP, the participating jurisdictions fully coordinated with and solicited participation from county and local governments, relevant organizations and groups, state and federal agencies, and the public. This coordination ensured that stakeholders had established communication channels and relationships to support mitigation planning and mitigation actions included in the plan.

Additional input and support for this planning effort was obtained from a wide range of agencies as well as through public involvement. Under the project management of HCOHSEM, the Harris County Hazard Mitigation Steering Committee provided oversight for the preparation of this plan. The Steering Committee includes representatives from the following agencies:

- HCOHSEM
- Harris County Office of County Administration
- Harris County Flood Control District
- City of Bellaire
- City of Deer Park
- City of El Lago
- City of Friendswood
- City of Jersey Village
- City of La Porte
- City of Missouri City
- City of Pasadena
- City of Seabrook
- City of Shoreacres
- City of Taylor Lake Village
- North Houston Association
- Houston Community College
- San Jacinto College
- Houston-Galveston Area Council (H-GAC)
- Cypress Creek Drainage Improvement District
- Coastal Prairie Conservancy
- Greens Bayou Coalition
- Cy-Fair Independent School District
- Partnership Lake Houston
- SouthEast Texas Regional Advisory Council
- Houston Arts Alliance
- Houston Society for the Prevention of Cruelty to Animals
- Harris County Department of Education

Risk Assessment for Local Hazards of Concern

The Planning Partners evaluated each jurisdiction's risk and vulnerability due to each of the hazards of concern, based on past events, past and predicted future losses, and the expected probability of future occurrence. From these evaluations, hazards were ranked as high, medium, or low risk to each jurisdiction. The hazard rankings were used to focus and prioritize individual jurisdictional mitigation strategies. Summary overall hazard rankings for all of Harris County are presented in Table ES-1. Information on each of the hazards of concern can be found in Part 2: Risk Assessment of Volume I of this plan.

Table ES-1. Countywide Ranking for Harris County Hazards of Concern

Hazard of Concern	Hazard Ranking
Coastal Erosion	Medium
Dam/Levee Failure	Low
Drought	Low
Earthquake	Low
Energy Pipeline Failure	Medium
Extreme Cold	Medium
Extreme Heat	Medium
Flood	High
Hurricane/Coastal Storms	High
Mass Movement	Low
Public Health Emergency	Medium
Severe Weather	High
Technology Failure	High
Toxic Release/Hazardous Materials Incident	High
Tsunami	Low
Utility Failure	Medium
Wildfire	Medium

Capability Assessment and Plan Integration into Other Local Mechanisms

Effective mitigation is achieved when hazard awareness and risk management approaches and strategies become an integral part of public activities and decision-making. Within the County, there are many existing plans and programs that support hazard risk management. It is critical that this MJ-HMP integrate, complement, and reference those plans and programs to the extent practical in order for it to be a comprehensive resource for hazard mitigation.

The MJ-HMP includes a capability assessment to review relevant local mechanisms for each participating jurisdiction. This assessment identifies where each jurisdiction is currently able to implement hazard mitigation measures and where each would benefit from improved capabilities

for such measures. The capability assessment also provides a summary and description of the existing plans, programs, and regulatory mechanisms at all levels of government (federal, state, county and local) that support hazard mitigation in the County. In the jurisdictional annexes, each participating jurisdiction identifies how it has integrated hazard risk management into its existing planning, regulatory and operational/administrative framework, and how it intends to continue to promote this integration.

Mitigation Strategy

Hazard Mitigation Plan Goals and Objectives

It is a federal requirement for hazard mitigation plans to include a description of mitigation goals to reduce or avoid long-term vulnerabilities to the identified hazards of concern.

The Harris County MJ-HMP planning process included a review and update of mitigation goals and objectives that were previously established to guide the selection of mitigation actions addressing all hazards of concern. Mitigation goals were updated based on the updated risk assessment, discussions, research, and input from plan participants and stakeholders. The goal development process also considered the goals expressed in the 2023 Texas State Hazard Mitigation Plan, as well as other relevant county and local planning documents. The goals and objectives for the 2025 MJ-HMP are stated in Table ES-2.

Implementation of the 2020 Plan

The status of the mitigation projects identified in the 2020 HC MH-MAP was reviewed for this MJ-HMP. Numerous projects and programs have been implemented that have reduced hazard vulnerability to assets in the planning area. Uncompleted projects have been revaluated, modified as necessary, and incorporated into this plan. The Planning Partners' annexes describe these mitigation activities in more detail, and plan maintenance procedures have been developed to encourage thorough integration with local decisions and processes and regular review of implementation progress.

Table ES-2. 2025 Hazard Mitigation Goals and Objectives

2025 Goals	Description of Goal	2025 Objectives	Description of Objective
1	Improve and coordinate data collection efforts to fully maximize mitigation capabilities	1a	Implement wildfire mitigation and watershed protection strategies through local, state, tribal, federal, and private partnerships
		1b	Improve understanding of the locations, potential impacts, and linkages among threats, hazards, vulnerability, and measures needed to protect life safety, health, and property
2	Enhance public outreach strategies to improve information sharing regarding hazards, including risk reduction strategies	2a	Improve and expand systems that provide ongoing preparedness education, warning, and emergency communications to the whole community
3	Reduce the impacts of hazards on people and property and protect natural and cultural resources	3a	Reduce the impacts of hazards on individuals with disabilities and others with access and functional needs
		3b	Coordinate state and local efforts to reduce greenhouse gas emissions and implement climate adaptation strategies through hazard mitigation plans and actions
		3c	Control access and provide buffers to maximize resource protection where possible
		3d	Encourage all state, regional, and local hazard mitigation projects and planning programs to protect the environment and promote implementation of sustainable mitigation and climate actions
		3e	Encourage hazard mitigation measures that promote and enhance natural processes and minimize adverse impacts on the ecosystem
		3f	Encourage the building of redundant systems needed to support public health and quality of life
4	Work to improve and coordinate existing local plans, policies, codes, and regulations to reduce hazard impacts	4a	Promote enforcement of relevant state regulations and development of local ordinances that significantly reduce life loss and injuries
		4b	Encourage all cities, counties, special districts, Councils of Governments and tribal organizations to develop, adopt, and implement local hazard mitigation principles to be integrated with local general plan safety elements, local coastal plans, facilities master plans, and other local plan initiatives
		4c	Implement wildfire mitigation and watershed protection strategies through local, state, tribal, federal and private partnerships
		4d	Advance community resilience through preparation, adoption, and implementation of state, regional and local multi-hazard mitigation plans and projects
5	Implement property protection measures to reduce or eliminate repetitive loss occurrences.	5a	Retrofit, elevate, purchase, or relocate structures in high-hazard areas, especially those known to be repetitively damaged
6	Investigate and implement a range of structural projects that will reduce the effects of hazards on public and	6a	Encourage projects that simultaneously reduce risk while increasing resilience and sustainability
		6b	Support the protection of vital and essential records, and strengthening or replacement of buildings, infrastructure, and lifelines to minimize post-disaster disruption and facilitate short-term and long-term recovery

2025 Goals	Description of Goal	2025 Objectives	Description of Objective
	private property throughout the County	6c	Encourage life and property protection measures for all communities and structures located in the greater Harris County region
		6d	Manage development in geologically hazardous areas and floodplains to protect life and property
7	Investigate and implement a range of nature-based solutions that utilize, enhance, and protect natural resources and their ability to reduce hazard impacts.	7a	Support the protection of vital and essential records, and strengthening or replacement of buildings, infrastructure, and lifelines to minimize post-disaster disruption and facilitate short-term and long-term recovery
		7b	Encourage life and property protection measures for all communities and structures located in the greater Harris County region
		7c	Promote disaster resistant development
		7d	Manage development in geologically hazardous areas and floodplains to protect life and property
8	Advance hazard analysis and mitigation strategies to reflect interdependencies across infrastructure and community lifelines	8a	Encourage private and public sector agencies and organizations to work together in the development of plans, strategies, and projects to mitigate identified hazards
9	Prioritize mitigation investments to inform and support efforts to seek funding	9a	Identify funding sources and increase awareness of funding sources to obtain funding for mitigation projects from a variety of federal, state, regional, and local entities.

2025 Mitigation Strategy

There are 811 hazard mitigation actions identified in the plan by the 42 planning partners. The general categories of these hazard mitigation actions include the following:

- Drainage improvements
- Generators
- Building codes/code enforcement
- Flood proofing structures (elevation, retrofitting, or relocation)
- Harden critical infrastructure
- Public threat warning systems
- Surge protection
- Public education
- NFIP compliance
- Cooling and/or warming centers
- Tree management
- Chemical release warning systems/air monitoring systems
- Cybersecurity measures
- Saferooms

More detail on the hazard mitigation actions identified by each of the planning partners can be found in the jurisdictional annexes found within Volume II of this plan.

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Part 1: The Planning Process and Planning Area

1. INTRODUCTION

Harris County, Texas, has developed a multi-jurisdictional hazard mitigation plan (MJ-HMP) to reduce risks from disasters to the people, property, economy, and environment within the County. Harris County and 45 local communities worked together to update the 2020 Hazard Mitigation Plan. This new Multi-Jurisdictional Hazard Mitigation Plan (MJ-HMP) helps the County better prepare for and reduce the risks from natural disasters. The updated 2025 MJ-HMP includes countywide analyses and assessment of hazards, risk, and capabilities. Jurisdiction-specific analyses are included in Volume II.

1.1 OVERVIEW TO HAZARD MITIGATION PLANNING

1.1.1 What Is Hazard Mitigation?

Hazard mitigation is any sustained action taken to reduce or eliminate the long-term risk and effects that can result from hazards. The Federal Emergency Management Agency (FEMA) defines a hazard mitigation plan as the documentation of a state or local government's evaluation of natural hazards and strategies to mitigate them.

Effective mitigation planning helps people, organizations, and government agencies better prepare for and respond to disasters. It also allows local governments to remain eligible for FEMA grant funding for mitigation projects that will reduce the impact of future disaster events. The long-term benefits of mitigation planning and implementation include the following:

- An increased understanding of hazards faced by local communities
- A more sustainable and disaster-resistant community
- Financial savings through partnerships that support planning and mitigation efforts
- Focused use of limited resources on hazards that have the biggest impact on the community
- Reduced long-term impacts and damage to human health and structures
- Reduced costs associated with response and recovery efforts, including repairs

FEMA estimates that for every $1 spent on damage prevention (mitigation), $6 is saved by not having to perform post-disaster repairs. (FEMA 2018)

1.1.2 History of Hazard Mitigation Planning in Harris County

Harris County has been included in 144 federal hazard-related declarations (major disaster, fire management, and emergency) since 1954. The County adopted its first hazard mitigation plan in 2015. The plan has been updated since then, with the last update adopted in 2020. Table 1-1 summarizes information contained in the 2025 update and previous versions of the MJ-HMP.

Table 1-1. Harris County MJ-HMP Comparison Matrix

	2015 (Initial MJ-HMP)	2020 (Previous Update)	2025 (Current Update)
Participating Municipalities	28	29	30
Participating Special Districts	4	10	16
Number of Mitigation Actions	672	837	811
Hazards of Concern	1. Coastal Erosion 2. Dam/Levee Failure 3. Drought 4. Earthquakes 5. Energy Pipeline Failure 6. Extreme Heat 7. Flooding 8. Hail 9. Hurricanes & Coastal Storms 10. Landslides 11. Severe Thunderstorms 12. Sinkholes 13. Subsidence 14. Tornadoes 15. Toxic Release 16. Tsunami 17. Wildfire 18. Winter Storm and Freezes	1. Coastal Erosion 2. Dam/Levee Failure 3. Drought 4. Earthquakes 5. Energy Pipeline Failure 6. Flooding 7. Hurricanes & Coastal Storms 8. Mass Movement (Landslides, Sinkholes, and Subsidence) 9. Severe Weather (Thunderstorms, Hail, Extreme Heat, Tornadoes and Damaging Winds, Winter Storms and Freezes) 10. Toxic Release/ Hazardous Materials 11. Tsunami 12. Wildfire	1. Coastal Erosion 2. Dam/Levee Failure 3. Drought 4. Earthquakes 5. Extreme Cold 6. Extreme Heat 7. Flood 8. Hurricanes & Coastal Storms 9. Mass Movement 10. Severe Weather (Thunderstorms, Hail, Tornadoes, Damaging Winds) 11. Tsunami 12. Wildfire 13. Energy Pipeline Failure 14. Public Health Emergency 15. Technology Failure 16. Toxic Release/ Hazardous Materials 17. Utility Failure

1.1.3 Key Changes in the Current Update

The key changes in the 2025 MJ-HMP update reflect some of the changes in Harris County's priorities compared to previous versions. The County's current priorities include robust and expanded community engagement as well as a risk assessment that is tailored to the changing landscape of hazard impacts in the County. Significant changes made between the 2020 Harris County Multi-Hazard Mitigation Action Plan (HC MH-MAP) and the 2025 MJ-HMP are as follows:

- The Steering Committee and Planning Partnership for the 2025 MJ-HMP update identified extreme heat and extreme cold as individual hazards separate from the severe weather hazard.
- The Steering Committee and Planning Partnership identified three new hazards to be included in the 2025 MJ-HMP update: public health emergency, technology failure, and utility failure.
- The following jurisdictions that were involved in the 2020 HC MH-MAP were not involved in the 2025 MJ-HMP update:
  - City of Baytown
  - Houston Independent School District
  - Huffman Independent School District
  - Lone Star College

  - Harris County Emergency Services District 12
  - Texas Medical Center
- The following jurisdictions participated in the MJ-HMP for the first time for 2025:
  - City of Jacinto City
  - City of Southside Place
  - City of Waller
  - A+ Unlimited Potential Charter School
  - Cypress Fairbanks Independent School District
  - Cypress Creek Drainage Improvement District
  - Goose Creek Consolidated Independent School District
  - Harris County Office of County Administration (OCA)
  - Lee College
  - San Jacinto College
  - Southwest Public Schools
  - Spring Independent School District
  - Gulf Coast Authority
  - Varnett Public Schools

1.2 GENERAL MITIGATION PLANNING APPROACH

FEMA provides hazard mitigation planning support to local communities through guidance, resources, and plan reviews. Preparation of this MJ-HMP accorded with the following regulations and guidance:

- Mitigation Ideas: A Resource for Reducing Risk to Natural Hazards (FEMA, January 2013)
- Integrating Hazard Mitigation into Local Planning (FEMA, March 1, 2013)
- Plan Integration: Linking Local Planning Efforts (FEMA, July 2015)
- Local Mitigation Planning Policy Guide (FEMA, April 2025)
- Local Mitigation Planning Handbook (FEMA, May 2023)
- Disaster Mitigation Act (DMA) 2000 (Public Law 106-390, October 30, 2000)
- 44 *Code of Federal Regulations* (CFR) 201 and 206 (including Feb. 26, 2002; Oct. 1, 2002; Oct. 28, 2003; and Sept. 13, 2004, Interim Final Rules)

In addition to the above-listed regulations and guidance, the MJ-HMP update was informed by the 2023 Texas State Hazard Mitigation Action Plan and the 2020 HC MH-MAP.

1.2.1 Regulatory Framework

The Disaster Mitigation Act of 2000 (DMA 2000) amended the Robert T. Stafford Disaster Relief and Emergency Assistance Act to encourage hazard mitigation planning by states, tribes, and local governments. Under the DMA 2000, communities seeking certain hazard-related federal funding must have a plan that identifies actions to mitigate hazards, risks, and vulnerabilities and establish a strategy to implement those actions. Regulations implementing the DMA 2000 are included in Title 44 of the Code of Federal Regulations, Section 201 (44 CFR 201).

To be eligible for hazard mitigation assistance from the federal government, communities must prepare, maintain, and update a hazard mitigation plan. The hazard mitigation plan must identify potential natural hazards that could affect the health, safety, and well-being of residents and identify and prioritize actions by the community to mitigate those hazards before disaster strikes. The federal regulations require that local governments update their hazard mitigation plans on a five-year basis. Table 1-2 summarizes the 44 CFR 201 requirements and where each is addressed in this hazard mitigation plan.

Table 1-2. FEMA Local Mitigation Plan Review Crosswalk

Plan Criteria	Primary Location in Plan
Prerequisites	
Adoption by the Local Governing Body: §201.6(c)(5)	Chapter 2; Appendix A
Planning Process	
Documentation of the Planning Process: §201.6(b) and §201.6(c)(1)	Chapter 2
Risk Assessment	
Identifying Hazards: §201.6(c)(2)(i)	Chapter 5
Profiling Hazards: §201.6(c)(2)(i)	Chapters 6 – 22
Assessing Vulnerability: Overview: §201.6(c)(2)(ii)	Chapters 6 – 22
Assessing Vulnerability: Identifying Structures: §201.6(c)(2)(ii)(A)	Chapters 6 – 22
Assessing Vulnerability: Estimating Potential Losses: §201.6(c)(2)(ii)(B)	Chapters 6 – 22
Assessing Vulnerability: Analyzing Development Trends: §201.6(c)(2)(ii)(C)	Chapters 6 – 22, Volume II
Mitigation Strategy	
Local Hazard Mitigation Goals: §201.6(c)(3)(i)	Chapter 25; Volume II
Identification and Analysis of Mitigation Actions: §201.6(c)(3)(ii)	Chapter 25; Volume II
Implementation of Mitigation Actions: §201.6(c)(3)(iii)	Chapter 25; Volume II
Multi-Jurisdictional Mitigation Actions: §201.6(c)(3)(iv)	Chapter 25; Volume II
Plan Maintenance Process	
Monitoring, Evaluating, and Updating the Plan: §201.6(c)(4)(i)	Chapter 26
Incorporation into Existing Planning Mechanisms: §201.6(c)(4)(ii)	Chapter 26; Volume II Annexes
Continued Public Involvement: §201.6(c)(4)(iii)	Chapter 26

One goal of the federal regulations is to facilitate cooperation between state and local authorities, prompting them to work together. This enhanced planning process enables state and local

governments to better articulate accurate needs for mitigation, resulting in faster allocation of funding and more effective risk reduction projects.

1.3 PLAN ORGANIZATION

The Harris County MJ-HMP provides a detailed review and analysis of each hazard of concern, resources, and relevant statistical information for the Planning Partners and the community. The plan is organized into two volumes: Volume I includes all information that applies to the entire planning area (Harris County), and Volume II includes specific information for each participating jurisdiction.

Volume I is a resource for ongoing mitigation analysis. It includes a description of the County and its jurisdictions as well as information on mitigation planning and how the risk assessment and capability assessment were performed. Volume I of the plan includes the following chapters:

- Part 1: The Planning Process and Planning Area
  - Chapter 1: Introduction
  - Chapter 2: Planning Process—A description of the plan methodology and development process, committee and stakeholder roles and activities, and how the plan will be incorporated into existing programs. It also includes information regarding the adoption of the plan by each participating jurisdiction.
  - Chapter 3: County Profile—An overview of Harris County, including: general information and physical conditions, land use patterns and trends, population and demographics, economy, general building stock inventory, community lifelines, and natural, historic, and cultural resources.
- Part 2: Risk Assessment
  - Chapter 4: Risk Assessment Methodology and Tools—Description of the methodology used to assess hazard risk and the status of local data.
  - Chapter 5: Identified Hazards of Concern—Documentation of the process of identifying the natural hazards of concern for further profiling and evaluation.
  - Chapters 6–22—Hazard profiles and findings of the risk assessment (estimates of the impact of hazard events on life, safety, and health; general building stock; critical facilities; the economy, and natural, historic, and cultural resources).
  - Chapter 23: Hazard Ranking—Description and summary of the hazard ranking process.
- Part 3: Capability Assessment
  - Chapter 24: Capability Assessment—A summary and description of the existing plans, programs, and regulatory mechanisms at all levels of government (federal, state, county, local) that support hazard mitigation within the County.
- Part 4: Mitigation Strategy
  - Chapter 25: Mitigation Strategy—Information regarding the mitigation goals and objectives identified by the Steering Committee in response to priority hazards of concern, and the process by which County and local mitigation strategies have been developed or updated.

- Part 5: Plan Maintenance
  - Chapter 26: Plan Maintenance Procedures—A system to continue to monitor, evaluate, maintain, and update the plan.

Volume II consists of annexes for each participating jurisdiction. Each annex summarizes the jurisdiction's planning, regulatory, and fiscal capabilities; evaluates vulnerabilities to hazards; describes the status of past mitigation actions; and provides a specific mitigation strategy. The annexes provide each jurisdiction with a resource for implementing mitigation projects and maximizing future grant opportunities.

The MJ-HMP includes the following appendices:

- Appendix A: Sample Resolution of Plan Adoption—Documentation that supports the plan approval signatures included in Chapter 2 of this plan.
- Appendix B: Meeting Documentation—Agendas, attendance sheets, minutes, and other documentation (as available and applicable) of planning meetings convened during the development of the plan.
- Appendix C: Public and Stakeholder Outreach Documentation—Documentation of the public and stakeholder outreach effort, including webpages, informational materials, public and stakeholder meetings and presentations, surveys, and other methods used to receive and incorporate public and stakeholder comment and input to the plan update process.
- Appendix D: Participation Matrix
- Appendix E: Action Worksheet Template and Instructions
- Appendix F: Plan Review Tools—Examples of plan review templates available to support annual plan review and example FEMA Guidance Worksheets (FEMA 386-4).
- Appendix G: Risk Assessment Supplementary Data—Details regarding past hazard events since those documented in the 2020 HC MH-MAP.
- Appendix H: Linkage Procedures—Procedures for linking to the MJ-HMP.
- Appendix I: Public Survey Results.
- Appendix J: NFIP Guidance Survey Results.

1.4 SPECIALIZED TERMS AND CONCEPTS

Like any technical field, the hazard mitigation field has its own set of terms and concepts with particular meanings. The list below provides a quick reference for specialized terms used in this MJ-HMP (EPA 2023, FEMA 2023, National Academies 2022, FEMA 2022):

- Adaptive capacity: The ability of a human or natural system to adjust to climate change by moderating potential damage, taking advantage of opportunities, or coping with the consequences.
- Asset: Anything that is important to the character and function of a community (e.g., people, structures, community lifelines, the economy, and natural, or historic resources).

- Capability assessment: An evaluation of which authorities, policies, programs, funding, and resources a participant has on hand to carry out the actions described in the plan and increase resilience.
- Cascading hazards: A primary event, such as heavy rainfall, seismic activity, or rapid snowmelt, followed by a chain of consequences that may range from modest (lesser than the original event) to substantial.
- Community lifelines: The most fundamental services in a community that, when stabilized, enable all other aspects of society to function.
- Extent: The range of anticipated intensities of the identified hazards within a community, expressed using various scientific scales.
- Hazard profile: A description of a hazard's location, extent, previous occurrences, and probability of future events within a community.
- Hazard ranking: The process of identifying the hazards that pose the greatest risk to a community based on how likely the hazard is to occur, the potential consequences if the hazard does occur, and other relevant local factors.
- Impact: The consequences or effects of a hazard on a community's assets identified in the vulnerability assessment.
- Integration: The inclusion of hazard mitigation principles, vulnerability information, and mitigation actions into other existing community planning to leverage activities that have co-benefits, reduce risk, and increase resilience.
- Mitigation action: Measures, projects, plans, or activities proposed to reduce the current and future vulnerabilities identified in the risk assessment.
- Mitigation strategy: The long-term blueprint for reducing the potential hazard-related losses identified in the risk assessment; the strategy consists of mitigation goals, mitigation actions, and a plan for implementing the actions.
- Natural hazard: A source of harm or difficulty created by a meteorological, environmental, or geological event.
- Plan maintenance: Monitoring and updating a hazard mitigation plan as warranted by changing conditions, availability of new information, and progress on the proposed mitigation actions.
- Planning process: The procedures used to develop a hazard mitigation plan with broad acceptance across the community
- Risk: The potential for damage or loss when natural hazards interact with people or assets.
- Risk assessment: A data-driven analysis to find out where a local jurisdiction is vulnerable to hazards.
- Social vulnerability: The potential for loss within an individual or social group, as affected by traits that influence an individual's or group's resilience, which is their ability to prepare, respond, cope, or recover from an incident.
- Stakeholder: Individuals or groups that a mitigation action or policy affects, including businesses, private organizations, and residents.

- Vulnerability: A description of which assets within locations identified to be hazard-prone are at risk from the effects of the hazard.

2. THE PLANNING PROCESS

This chapter describes the planning process used to update the Harris County MJ-HMP, including how it was prepared, who was involved in the process, and how the public was involved. The planning approach aimed to achieve the following results:

- Serve as a multi-jurisdictional plan, to include the County as well as most municipalities that lie within the County boundary. The County invited each jurisdiction within the county boundaries to join in the planning process. At the time of the plan's development, all but four participated. Collectively known as the Planning Partnership, the entities listed below are those who participated during the 2025 plan update process:
  - Harris County
  - Harris County Flood Control District
  - Harris County Office of County Administration (OCA)
  - City of Bellaire
  - City of Bunker Hill Village
  - City of Deer Park
  - City of El Lago
  - City of Friendswood
  - City of Galena Park
  - City of Hedwig Village
  - City of Hilshire Village
  - City of Humble
  - City of Hunters Creek Village
  - City of Jacinto City
  - City of Jersey Village
  - City of Katy
  - City La Porte
  - City of Missouri City
  - City of Morgan's Point
  - City of Nassau Bay
  - City of Pasadena
  - City of Piney Point Village
  - City of Seabrook
  - City of Shoreacres
  - City of Southside Place
  - City of Spring Valley Village
  - City of Stafford
  - City of Taylor Lake Village
  - City of Tomball
  - City of Waller
  - City of Webster
  - City of West University Place
  - Cypress Fairbanks Independent School District (ISD)
  - Goose Creek CISD
  - Spring ISD
  - Varnett Public Schools
  - Houston Community College
  - Lee College
  - San Jacinto College
  - University of Houston
  - Cypress Creek Drainage Improvement District
  - Harris Health
  - Gulf Coast Authority
- Provide opportunities for jurisdictions to adopt the plan at any point during the five-year implementation period if they are unable to participate in the update process. The plan is structured in a manner that allows additional entities to seamlessly join later during subsequent updates to the five-year regulatory plan.
- Assess all natural hazards that pose a risk to the area, as required by 44 CFR 201. Non-natural hazards that pose significant risk were considered as well.
- Comply with current FEMA regulations and state guidance. This ensures that all the requirements are met and supports plan review. In addition, this plan will meet criteria for the National Flood Insurance Program (NFIP) Community Rating System (CRS) and the Hazard Mitigation Assistance (HMA) grant programs.

Harris County applied for and was awarded CDBG-MIT funds, which supported the development of this MJ-HMP. Grant administration is the responsibility of Harris County Office of Homeland Security and Emergency Management (HCOHSEM).

2.1 ORGANIZATION OF PLANNING PROCESS

2.1.1 Planning Process Participants

Project Management and Planning Consultant

Project management was the responsibility of HCOHSEM, and their office was tasked with the following:

- Organization of the Steering Committee and the Planning Partnership
- Development and implementation of a public and stakeholder outreach program
- Identification of the 2025 hazards of concern
- Data collection
- Facilitation of Steering Committee meetings, public outreach workshops, and social media campaigns using www.readyharris.gov and social media platforms

HCOHSEM contracted with the planning consultant Tetra Tech, Inc. The Tetra Tech planning team, consisting of planners and subject matter experts from Tetra Tech and partner firms Bent Ear Solutions and Resalire Infrastructure Solutions, was tasked with the following:

- Assisting in the organization and facilitation of Steering Committee meetings as well as attendance
- Supplementing geospatial data collection efforts
- Reviewing and updating the hazards of concern, hazard profiles, and risk assessment
- Assisting with the review and update of mitigation planning goals and objectives
- Assisting in the review of past mitigation action progress
- Assisting with the screening of mitigation actions and the identification of appropriate actions
- Assisting with the prioritization of mitigation actions
- Authoring of the draft and final plan documents (Volume I and II)
- Digitizing of the 2025 HC MJ-HMP once formally approved by FEMA Region 6
- Developing an interactive web-viewer of the 2025 HC MJ-HMP
- Assisting in the development and authoring of jurisdictional annexes included in Volume II

Planning Partnership

In March 2024, the County invited all municipalities and other partners to participate in the HC MJ-HMP update. Jurisdictions were asked to formally notify the County of their intent to participate, via a letter of intent, and to identify planning points of contact to facilitate their participation and represent the interests of their communities. All participating jurisdictions, including the County, are recognized as Planning Partners and belong to the Planning Partnership for this MJ-HMP. The Planning Partnership was charged with the following:

- Representing their jurisdiction throughout the planning process
- Ensuring participation of all departments and functions within their jurisdiction that have a stake in mitigation (e.g., planning, engineering, code enforcement, police and emergency services, public works)
- Assisting in gathering information for inclusion in the MJ-HMP update, including the use of previously developed reports and data
- Supporting and promoting the public involvement process
- Reporting on progress of mitigation actions identified in prior or existing hazard mitigation plans, as applicable
- Identifying, developing, and prioritizing appropriate mitigation actions
- Reporting on progress of integration of prior or existing hazard mitigation plans into other planning processes and municipal operations
- Supporting and developing a jurisdictional annex
- Reviewing, amending, and approving all sections of the plan update
- Adopting, implementing, and maintaining the plan update

The jurisdictions in Harris County have differing levels of capabilities and resources available to apply to the plan update process, as well as differing levels of vulnerability to and impacts from the natural hazards being considered in this plan. Out of the 46 jurisdictions that had initial interest in joining, including Harris County, 42 ultimately were able to participate in the plan update. It was Harris County's intent to encourage participation by all jurisdictions and to accommodate their specific needs and limitations while still meeting the intent and purpose of plan update participation. Such accommodations have included establishing a Steering Committee, engaging a contract consultant to assume certain elements of the plan update process on behalf of the jurisdictions, and providing alternative mechanisms for planning participation.

Ultimately, jurisdictional participation is evidenced by a completed annex of the MJ-HMP, wherein jurisdictions individually identify their planning points of contact, evaluate their risk from the hazards of concern, identify their capabilities to effect mitigation in their community, identify and prioritize a suite of actions to mitigate their hazard risk, and adopt the updated plan via resolution. Annexes are included in Volume II of this MJ-HMP.

Table 2-1 shows the current members of the Planning Partnership as of the time of publication of this plan update. Appendix D (Participation Matrix) identifies how individuals who represented the jurisdictions during this planning effort contributed to the planning process.

Table 2-1. Harris County Hazard Mitigation Planning Partnership Members

Jurisdiction	Primary Point of Contact		Alternative Point of Contact		Sector/ Community Lifeline
	Name	Title	Name	Title	
Harris County	Brittany Ray	Assistant Planning Section Chief	Brian Murray	PIO and Deputy Emergency Management Coordinator	• Emergency Management • Safety & Security
Harris County Flood Control District	Jeremy Phillips	Director, Asset Management Division	Todd Ward	Risk Mitigation Dept. Manager	• Land Use & Development (Floodplain Management) • Safety and Security • Natural & Cultural Resources
Harris County Office of County Administration (OCA)	Leah Barton	Deputy County Administrator, Director of Harris County Public Health	Jesse Dickerman	Interim County Administrator	• Economic Development • Transportation • Communication • Health & Medical
City of Bellaire	Deacon Tittel	Fire Chief / EMC	Travis Tanner	Dev. Services Director	• Emergency Management • Safety & Security
City of Bunker Hill Village	Gerardo Barrera	City Administrator	Howard Miller	Fire Chief / EMC	• Emergency Management • Safety & Security
City of Deer Park	Jamie Galloway	Director of Emergency Services	Buddy Rice	Asst. Director Emerg. Services	• Emergency Management • Safety & Security • Hazardous Materials
City of El Lago	Carl Nunn	EMC / Chief of Police	Bob Kosar	Floodplain Administrator	• Emergency Management • Safety & Security
City of Friendswood	Brian Mansfield	Director of Emergency Services	Steven Simmons	Deputy Fire Marshal	
City of Galena Park	Thomas Ehlers	Fire Chief / EMC	Mayra Gonzales	City Secretary	• Emergency Management • Safety & Security
City of Hedwig Village	Wendy Baimbridge	City Administrator	Howard Miller	Fire Chief / EMC	• Emergency Management • Safety & Security
City of Hilshire Village	Cassie Stephens	City Administrator	Howard Miller	Fire Chief / EMC	• Emergency Management • Safety & Security
City of Humble	James Nykaza	EMC	Steve Forbes	Asst. EMC	• Emergency Management • Safety & Security
City of Hunters Creek Village	Tom Fullen	City Administrator	Howard Miller	Fire Chief / EMC	• Emergency Management • Safety & Security
City of Jacinto City	Lon Squyers	City Manager	Christal Rodriguez	City Secretary	• Emergency Management • Economic Development
City of Jersey Village	Mark Bitz	Fire Chief / EMC	Jacob Daily	Asst. Fire Chief / EMC	• Emergency Management • Safety & Security • Health & Medical • Hazardous Materials
City of Katy	Greg Goedecker	EMC	Jason Rivera	Public Works Director	• Emergency Management • Safety & Security

Jurisdiction	Primary Point of Contact		Alternative Point of Contact		Sector/ Community Lifeline
	Name	Title	Name	Title	
City of La Porte	Johnny Morales	EMC	Amber Castro-Beltran	Deputy EMC	• Emergency Management • Safety & Security • Hazardous Materials
City of Missouri City	Trameka Jewett	EMC	Marcus Snell	City Engineer	• Emergency Management • Safety & Security
City of Morgan's Point	Kevin Lagana	Police Chief	Brian Schneider	City Administrator	• Emergency Management • Safety & Security
City of Nassau Bay	Tom George	Fire Chief / EMC	Sherri Ditrich	Contract Specialist	• Emergency Management • Safety & Security • Health & Medical • Hazardous Materials
City of Pasadena	Azell Carter	EMC	Andrew Aleman	Deputy EMC	• Emergency Management • Safety & Security
City of Piney Point Village	Bobby Pennington	City Administrator	Howard Miller	Fire Chief / EMC	• Emergency Management • Safety & Security • Health & Medical • Hazardous Materials
City of Seabrook	Kevin Rodgers	Director of Emergency Management	Nick Kondejewski	Building Official	• Emergency Management • Safety & Security
City of Shoreacres	Troy Harrison	City Manager	N/A	N/A	• Emergency Management • Infrastructure
City of Southside Place	Don McCalll	Chief of Police	N/A	N/A	• Emergency Management • Safety & Security
City of Spring Valley Village	John McDonald	City Administrator	Howard Miller	Fire Chief / EMC	• Emergency Management • Safety & Security
City of Stafford	Larry Di Camillo	Fire Chief / EMC	Ashley Merchant	Grant Manager	• Emergency Management • Safety & Security
City of Taylor Lake Village	Carl Nunn	EMC / Chief of Police	Bob Kosar	Floodplain Consultant	• Emergency Management • Safety & Security
City of Tomball	Joe Sykora	Fire Chief	Chris Melton	Firefighter	• Emergency Management • Safety & Security
City of Waller	Charles Mistric	EMC	N/A	N/A	• Emergency Management • Safety & Security
City of Webster	Dena Mahan	EM Specialist	Dean Spencer	EMC / Fire Chief	• Emergency Management • Safety & Security
City of West University Place	James Novak	Fire Chief / EMC	Adam Smith	Captain	• Emergency Management • Safety & Security
A+ Unlimited Potential Charter Schools	Kap McWhorter	Superintendent	N/A	N/A	• Health, Education and Human Services • Infrastructure

Jurisdiction	Primary Point of Contact		Alternative Point of Contact		Sector/ Community Lifeline
	Name	Title	Name	Title	
Cypress Fairbanks ISD	Scott Hudson	Director of Emergency Management	Grant Burnett	EMC	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Goose Creek ISD	Dr. Tricia Times	Director, Safe and Secure Schools	Dr. Anthony Price	Assistant Superintendent	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Pasadena ISD	Kevin Fornof		N/A	N/A	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Southwest Public Schools	Dr. Michael Love	Chief of Ops, Strategy & Innovation	John Bishop	EMC	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Spring ISD	Artilia Haughton	EMC	David Borchardt	Police Lt.	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Varnett Public Schools	Dr. Anthony Pineda	Executive Director Safety & Ops	Dr. Nita White	Superintendent	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Houston Community College	Robert McCracken	Associate Vice Chancellor, Risk Management & Resilience	David Potts	N/A	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Lee College	Tom Quinn	Manager, Emerg & Safety Ops	Eric Williams	Security Chief	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
San Jacinto College	Ali Shah	Director of Emergency Management	Khenley Harvell	EMC	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure

Jurisdiction	Primary Point of Contact		Alternative Point of Contact		Sector/ Community Lifeline
	Name	Title	Name	Title	
University of Houston	Ginger Walker	Director, EM	Maia Solomon	EM Specialist	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Cypress Creek Drainage District	Calvin Cobb	President	N/A	N/A	• Emergency Management • Safety & Security • Water Systems • Infrastructure • Natural and Cultural Resources
Gulf Coast Authority	Scott Harris	Chief Operating Officer	Anna Brown	Facility Manager, Blackhawk Facility	• Infrastructure • Water Systems
Harris Health	Aaron Velasquez	Emergency Management Director	N/A	N/A	• Emergency Management • Safety & Security • Health, Education and Human Services

Note: EMC = Emergency Management Coordinator; ISD = Independent School District; N/A = Not Available

After completion of the plan, implementation and ongoing maintenance will become a function of the Planning Partnership as described in Chapter 26 (Plan Maintenance). The Planning Partnership will be responsible for reviewing the draft plan and soliciting public comment as part of an annual review and as part of the five-year mitigation plan updates.

Steering Committee

Harris County developed a Steering Committee to provide guidance and direction to the MJ-HMP update effort, and to ensure that the resulting document will be embraced by local government leaders as well as all who live and work within the planning area. Steering Committee members were charged with the following:

- Providing guidance and oversight of the planning process on behalf of the general planning partnership
- Attending and participating in Steering Committee meetings
- Assisting with the development and completion of certain planning elements, including:
  - Reviewing and updating the hazards of concern
  - Developing a public and stakeholder outreach program
  - Ensuring that the data and information used in the plan update process is the best available
  - Reviewing and updating the hazard mitigation goals and objectives
  - Identifying and screening appropriate mitigation strategies and activities
- Reviewing and commenting on plan documents prior to submission to the TDEM and FEMA.

The Steering Committee provided guidance, leadership, and oversight of the planning process and acted as the point of contact for all participating jurisdictions and various interest groups in the planning area. Table 2-2 lists the members of the Steering Committee.

Table 2-2. Harris County Hazard Mitigation Steering Committee Members

Affiliation	Primary Point of Contact		Alternative Point of Contact		Sector/Community Lifeline
	Name	Title	Name	Title	
City of Bellaire	Deacon Tittle	Fire Chief / Emergency Management Coordinator	Travis Tanner	Development Services Director	• Emergency Management • Safety & Security
City of Deer Park	Jamie Galloway	Emergency Services Director	Buddy Rice	Asst. Director Emergency Services	• Emergency Management
City of Friendswood	Steven Simmons	Deputy Fire Marshal	Brian Mansfield	Director of Emergency Services	• Safety & Security
City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal	Jacob Daily	Asst. Fire Chief / EMC	• Emergency Management
City of La Porte	Johnny Morales	Emergency Management Coordinator	Amber Castro-Beltran	Deputy EMC	• Safety & Security
City of Missouri City	Trameka Jewett	Emergency Management Coordinator	Marcus Snell	City Engineer	• Emergency Management
City of Pasadena	Azell Carter	Emergency Management Coordinator	Andrew Aleman	Deputy EMC	• Safety & Security
City of Seabrook	Kevin Rodgers	Office of Emergency Management Director	Nick Kondejewski	Building Official	• Emergency Management
City of Shoreacres	Troy Harrison	City Manager	-	-	• Safety & Security
City of Taylor Lake Village / City of El Lago	Carl Nunn	Chief of Police / Emergency Management Coordinator	Bob Kosar	Floodplain Consultant	• Emergency Management
Coastal Prairie Conservancy	Mary Anne Piacentini	President and CEO	Elisa Donovan	-	• Natural & Cultural Resources
Cy-Fair ISD	Scott Hudson	Director of Emergency Management	Grant Burnett	EMC	• Emergency Management • Safety & Security
Cypress Creek Drainage Improvement District	Calvin Cobb	President	-	-	• Emergency Management • Safety & Security • Water Systems • Infrastructure • Natural and Cultural Resources

Affiliation	Primary Point of Contact		Alternative Point of Contact		Sector/Community Lifeline
	Name	Title	Name	Title	
Greens Bayou Coalition	Colleen Gilbert	Executive Director	Robert Fiederlein	-	• Natural and Cultural Resources
Harris County OCA	Leah Barton	Deputy County Administrator	Jesse Dickerman	Interim County Administrator	• Economic Development • Housing • Health, Education, and Human Services
Harris County Dept. of Education	Leslie Etheridge	Director, Center for Safe and Secure Schools	-	-	• Health, Education, and Human Services
Harris County Flood Control District	Jeremy Phillips	Director, Asset Management Division	Todd Ward	Risk Mitigation Dept. Manager	• Land Use & Development (Floodplain Management) • Safety and Security • Natural & Cultural Resources
HCOHSEM	Brian Murray	Deputy EMC / Public Information Officer	-	-	• Emergency Management • Safety & Security • Communications
HCOHSEM	Brittany Ray	Assistant Planning Section Chief	-	-	• Emergency Management • Safety & Security
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience	-	-	• Emergency Management • Safety & Security • Natural & Cultural Resources
Houston Community College	Robert McCracken	Associate Vice Chancellor, Risk Management & Resilience	-	-	• Emergency Management • Safety & Security • Health, Education and Human Services • Infrastructure
Houston Society for the Prevention of Cruelty to Animals	Dawn Blackmar	Director, Emergency Management	-	-	• Safety & Security • Natural & Cultural Resources
Houston-Galveston Area Council (H-GAC)	Amanda Ashcroft	Planner	Sarai Osorio	-	• Housing • Economic Development
North Houston Association	Marlisa Briggs	President	-	-	• Elected Official • Housing • Economic Development

Affiliation	Primary Point of Contact		Alternative Point of Contact		Sector/Community Lifeline
	Name	Title	Name	Title	
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Officer	-	-	• Economic Development • Natural & Cultural Resources
San Jacinto College	Ali Shah	Director, Emergency Management	Khenley Harvell	-	• Emergency Management • Safety & Security
SouthEast Texas Regional Advisory Council	James Meaux	Regional Training & Exercise Coordinator	Lisa Spivey	-	• Emergency Management • Safety & Security

2.1.2 Planning Activities

Members of the Planning Partnership (individually and as a whole), as well as key stakeholders, met and communicated as needed to share information. This included workshops to identify hazards, assess risks, update inventories of critical facilities, and assist in updating mitigation goals and strategies. All members of the Planning Partnership had the opportunity to review the draft plan, support interaction with other stakeholders, and assist with public involvement efforts. These activities provided continuity through the process to ensure that natural hazard vulnerability information and appropriate mitigation strategies were incorporated.

Table 2-3 summarizes meetings and other planning activities conducted during the development of the plan. It also identifies the activities designed to meet 44 CFR 201 requirements. Documentation of meetings (agendas, sign-in sheets, minutes, etc.) may be found in Appendix C (Public and Stakeholder Outreach). In addition to the formal meetings and milestone events in the plan update process listed in the table, there was significant communication between Planning Partnership members and the consultant team through individual local meetings, phone calls, and email.

Table 2-3. Summary of Mitigation Planning Activities / Efforts

Date	44 CFR 201 Requirement	Description of Activity	Participants
February 2024	-	Meetings with TDEM to discuss planning process timeline and Texas State and FEMA requirements	HCOHSEM, TDEM
March 20, 2024	2	RENS messaging sent to announce kickoff meeting information and to invite entities to participate in the planning process.	HCOHSEM, Fire EMS, HC Commissioner Precincts 1-4, HC Department Heads, HC Department Liaisons, HC EMCs, HC Mayors, Houston-Galveston Area Council (H-GAC) EMCs, of HC elected officials, Public Information Officer, State Agency Liaison
April 12, 2024	2	Project Kickoff Meeting: Discuss proposed planning process and scope of work including documenting participation, schedule, and public and stakeholder outreach and involvement.	HCOHSEM, Eligible jurisdictions

Date	44 CFR 201 Requirement	Description of Activity	Participants
April -June to June 20, 2024	1c, 2	Interested jurisdictions submit Letters of Intent to Participate in this planning process, acknowledging municipal participation requirements and identifying planning point(s) of contact.	See Volume II
July 24, 2024	1b, 2, 3a, 3b, 3c, 4a, 5c	Steering Committee Workshop #1: Defined the planning area as well as critical facilities and infrastructure; reviewed Hazards of Concern and proposed new ones; discussed public involvement strategy	HCOHSEM, TDEM, Steering Committee Members, Public, Participating Jurisdictions
September 11, 2024	2, 3	Steering Committee Workshop #2: Discussion of CDBG-MIT timeline update; public involvement strategy; stakeholder and partner engagement; development of mission statement; goal setting	HCOHSEM, TDEM, Steering Committee Members, Public, Participating Jurisdictions
October 1, 2024	2	Online Public Hazard Preparedness and Mitigation survey developed and deployed	HCOHSEM
October 2, 2024	1b	Steering Committee Workshop #3: Discussion of CDBG-MIT/timeline update; public survey; public workshops/ education campaign (Oct-Nov)	HCOHSEM, TDEM, Steering Committee Members, Public, Participating Jurisdictions
October 16, 2024		RENS message sent to solicit public engagement strategy	HCOHSEM, Fire EMS, HC Commissioner Precincts 1-4, HC Department Heads, HC Department Liaisons, HC EMCs, HC Mayors, H-GAC EMCs, of HC elected officials, Public Information Officer, State Agency Liaison
October 19, 2024	1b, 3a	Community Workshop #1: Hazard mitigation planning overview, purpose and benefits of local mitigation planning; review of 2020 HC MJ-HMP; presentation from HCOHSEM hazard mitigation planning overview and 2020 plan review; HCFCD and HCED project updates; HCOCA Climate Justice Action Plan; SBA post-disaster resources; and invitation to join 2025 HC MJ-HMP planning process	HCOHSEM, HCFCD, Harris County Public Health (HCPH), Harris County Office of the County Engineer (HCOCE), Harris County Office of County Administration (HCOCA), United States Small Business Administration (U.S. SBA), public
October 22, 2024	1b, 3a	Community Workshop #2: Hazard mitigation planning overview, purpose and benefits of local mitigation planning; review of 2020 HC MJ-HMP; presentation from HCOHSEM hazard mitigation planning overview and 2020 plan review; HCFCD and HCED project updates; HCOCA Climate Justice Action Plan; SBA post-disaster resources; and invitation to join 2025 HC MJ-HMP planning process	HCOHSEM, HCFCD, HCOCE, HCOCA, HCPH, U.S. SBA, public

Date	44 CFR 201 Requirement	Description of Activity	Participants
October 30, 2024	1b, 3a	Online Public Hazard Preparedness and Mitigation survey closes Community Workshop #3: Hazard mitigation planning overview, purpose and benefits of local mitigation planning; review of 2020 HC MJ-HMP; presentation from HCOHSEM hazard mitigation planning overview and 2020 plan review; HCFCD and HCED project updates; HCOCA Climate Justice Action Plan; SBA post-disaster resources; and invitation to join 2025 HC MJ-HMP planning process	HCOHSEM, HCFCD, HCOCE, HCOCA, U.S. SBA, public
November 9, 2024	1b, 3a	Community Workshop #4: Hazard mitigation planning overview, purpose and benefits of local mitigation planning; review of 2020 HC MJ-HMP; presentation from HCOHSEM hazard mitigation planning overview and 2020 plan review; HCFCD and HCED project updates; HCOCA Climate Justice Action Plan; SBA post-disaster resources; and invitation to join 2025 HC MJ-HMP planning process	HCOHSEM, HCFCD, HCOCE HCOCA, HCPH, U.S. SBA, public
March 12, 2025	3, 4	Steering Committee Workshop #4: Discussion of new planning partners; project status/timeline; review of public survey results; discussion on 2025 Hazards of Concern; Q&A with steering committee members on GIS vulnerability assessment question Worksheet A: Hazard Event History Worksheet B: Capability Assessment Worksheet C: Past Mitigation Actions Worksheet D: NFIP Compliance Worksheet E: Building Permits	HCOHSEM, TDEM, Steering Committee Members, Tetra Tech, public, Participating Jurisdictions
March 28, 2025	1b	Workshop held at Houston TranStar to assist partners with completing Worksheet B: Capability Assessments.	HCOHSEM, Participating Jurisdictions
April 9, 2025	3b, 3c, 4a, 4b, 4c	Steering Committee Workshop #5—project status/timing; hazard risk ranking overview; discussion and approval of 2025 Goals and Objectives; assessment of strengths, weaknesses, obstacles and opportunities; and development of new mitigation actions. Worksheet F: Hazard Risk Evaluation Worksheet G: New Mitigation Actions	HCOHSEM, TDEM, Steering Committee Members, Tetra Tech, public, Participating Jurisdictions
April 29, 2025	3a, 4a, 4c	Webinar tailored for Harris County Department Heads—basic Q&A about mitigation benefits and potential mitigation project they can pursue	HCOHSEM, Tetra Tech

Date	44 CFR 201 Requirement	Description of Activity	Participants
June 4, 2025	1b, 2, 3, 4, 5	Steering Committee Workshop #6—Plan Maintenance; review of 2025 MJ-HMP Update DRAFT; overview of the risk assessment and vulnerability assessment GIS results; demonstration of the interactive HMP Viewer (currently under construction)	HCOHSEM, TDEM, Steering Committee Members, Tetra Tech, Public, Participating Jurisdictions
June 25, 2025	2	Draft Plan posted to public project website	Public and Stakeholders
June 25-July 9, 2025	1b, 2	Public and stakeholder comments to Draft Plan received and incorporated into Final Plan.	Public and Stakeholders
June 26, 2025	1b	Public comment workshop held at George H.W. Bush Community Center for Draft Plan	Public and Stakeholders
June 28, 2025	1b	Public comment workshop held at Richard and Meg Weekley Community Center for Draft Plan	Public and Stakeholders
July 1, 2025	1b	Public comment workshop held at Leonel Castillo Community Center for Draft Plan	Public and Stakeholders
July 2, 2025	1b	Public comment workshop held at Julia C. Hester House for Draft Plan	Public and Stakeholders
July 16, 2025	All requirements	Final plan submitted to TDEM and FEMA Region 6	HCOHSEM, TDEM, FEMA Region 6
Upon plan approval by FEMA	1a	Plan adoption by resolution by the governing bodies of all participating jurisdictions	All plan participants

Note: HC = Harris County; TBD = to be determined; RENS = Regional Emergency Notification System
Numbers in column 2 identify specific federal requirements, as follows:
1a—Prerequisite—Adoption by the Local Governing Body
1b—Public Participation
2—Planning Process—Documentation of the Planning Process
3a—Risk Assessment—Identifying Hazards
3b—Risk Assessment—Profiling Hazard Events
3c—Risk Assessment—Assessing Vulnerability: Identifying Assets
3d—Risk Assessment—Assessing Vulnerability: Estimating Potential Losses
3e—Risk Assessment—Assessing Vulnerability: Analyzing Development Trends
4a—Mitigation Strategy—Local Hazard Mitigation Goals
4b—Mitigation Strategy—Identification and Analysis of Mitigation Measures
4c—Mitigation Strategy—Implementation of Mitigation Measures
5a—Plan Maintenance Procedures—Monitoring, Evaluating, and Updating the Plan
5b—Plan Maintenance Procedures—Implementation through Existing Programs
5c—Plan Maintenance Procedures—Continued Public Involvement

2.2 STAKEHOLDER OUTREACH AND INVOLVEMENT

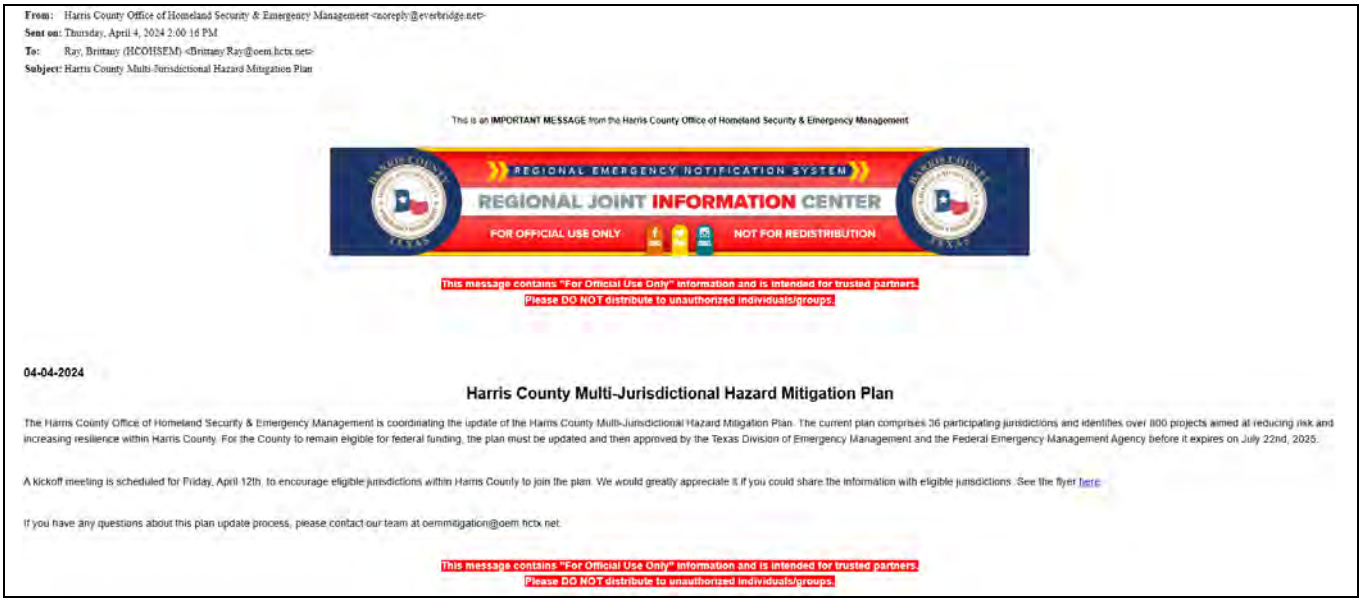
The Harris County MJ-HMP update was written using the best available information obtained from a wide variety of sources. Throughout the MJ-HMP update process, a concerted effort was made to gather information from municipal and regional agencies and staff as well as stakeholders, federal and state agencies, and the residents of the County. The Steering Committee solicited information

from local agencies and individuals with specific knowledge of certain natural hazards and past historical events. In addition, the Steering Committee and Planning Partnership took into consideration planning and zoning codes, ordinances, and recent land use planning decisions.

Diligent efforts were made to ensure broad regional, county, and local representation in this planning process. A comprehensive list of stakeholders was developed with the support of the Steering Committee and Planning Partnership. Stakeholder outreach was performed early and throughout the planning process, including mass media notification efforts. Identified stakeholders were invited to attend the Planning Partnership risk assessment meeting, and key stakeholders were requested to participate on the Steering Committee and/or Planning Partnership. Information and input provided by these stakeholders has been included throughout this plan.

The Harris County Office of Homeland Security and Emergency Management utilized the Regional Emergency Notification System (RENS) to message partners countywide in March 2024, as shown in Figure 2-1.

Figure 2-1. RENS Messaging from March 20, 2024



From: Harris County Office of Homeland Security & Emergency Management <noreply@everbridge.net>
Sent on: Thursday, April 4, 2024 2:00:16 PM
To: Ray, Brittany (HCOHSEM) <Brittany.Ray@oem.hctx.net>
Subject: Harris County Multi-Jurisdictional Hazard Mitigation Plan

This is an **IMPORTANT MESSAGE** from the Harris County Office of Homeland Security & Emergency Management

REGIONAL EMERGENCY NOTIFICATION SYSTEM
REGIONAL JOINT INFORMATION CENTER
FOR OFFICIAL USE ONLY NOT FOR REDISTRIBUTION

This message contains "For Official Use Only" information and is intended for trusted partners.
Please DO NOT distribute to unauthorized individuals/groups.

04-04-2024

Harris County Multi-Jurisdictional Hazard Mitigation Plan

The Harris County Office of Homeland Security & Emergency Management is coordinating the update of the Harris County Multi-Jurisdictional Hazard Mitigation Plan. The current plan comprises 36 participating jurisdictions and identifies over 800 projects aimed at reducing risk and increasing resilience within Harris County. For the County to remain eligible for federal funding, the plan must be updated and then approved by the Texas Division of Emergency Management and the Federal Emergency Management Agency before it expires on July 22nd, 2025.

A kickoff meeting is scheduled for Friday, April 12th, to encourage eligible jurisdictions within Harris County to join the plan. We would greatly appreciate it if you could share the information with eligible jurisdictions. See the flyer here.

If you have any questions about this plan update process, please contact our team at oemmitigation@oem.hctx.net.

This message contains "For Official Use Only" information and is intended for trusted partners.
Please DO NOT distribute to unauthorized individuals/groups.

Recipients of RENS messages include Fire EMS, Harris County (HC) Commissioner Precincts 1, 2, 3, and 4, HC Department Heads, HC Department Liaisons, HC Emergency Management Coordinators (EMCs), HC Mayors, Houston-Galveston Area Council (H-GAC) EMCs, other HC elected officials, the Public Information Officer, as well as the State Agency Liaison.

The following sections detail the outreach to, and involvement of, the agencies, departments, organizations, non-profits, districts, authorities, and other entities that have a stake in managing hazard risk and mitigation, commonly referred to as stakeholders. They list the stakeholders who were invited to participate in the development of this plan and describe how they contributed to the plan. This summary information demonstrates the scope and breadth of the stakeholder outreach efforts during the planning process. Beyond those described here, many stakeholders were aware of and/or contributed to this plan through formal and informal outreach efforts by the Planning Partners involved in the plan update.

2.2.1 Federal and State Agencies

The federal and state agencies listed in Table 2-4 were contacted during the planning process. The table describes how each participant participated.

Table 2-4. Participation of Federal and State Agencies

Agency	Participation
FEMA Region 6	Provided updated planning guidance; provided summary and detailed NFIP data for planning area; presented preliminary regulatory flood products to municipalities and the public; attended meetings; conducted plan review.
Texas Department of Emergency Management (TDEM)	Administered planning grant and facilitated FEMA review; provided updated planning guidance; attended meetings; participated in the Mitigation Strategy Workshop, provided review of Draft and Final Plan.

2.2.2 County and Regional Agencies

The County and regional agencies listed in Table 2-5 were those who were invited to participate during the planning process, and their level of participation is described below.

Table 2-5. County and Regional Agencies

Agency	Participation
• Coastal Prairie Conservancy • Cy-Fair ISD • Cypress Creek Drainage Improvement District • Greens Bayou Coalition • Harris County Office of County Administration (OCA) • Harris County Office of Homeland Security and Emergency Management • Harris County Flood Control District (HCFCD) • Harris County Public Health • Houston Community College • Houston Society for the Prevention of Cruelty to Animals • Houston-Galveston Area Council (H-GAC) • North Houston Association • Partnership Lake Houston • San Jacinto College • SouthEast Texas Regional Advisory Council	Served on steering committee; attended meetings; completed hazard of concern exercise; goals and objectives exercise; provided input and reviewed draft plan.

2.2.3 Stakeholders by Community Lifeline Category

FEMA defines community lifelines as fundamental services in a community that, when stabilized, enable all other aspects of society. Following a disaster, intervention is required to stabilize community lifelines. Many jurisdictions also directly involved representatives of these agencies in the planning process, as identified in Table 2-1.

This section describes outreach to and participation by stakeholders in the planning process associated with FEMA's eight designated community lifeline categories. More detailed information

about community lifelines in the planning area is provided in Section 3.10. The community lifeline represented by each Steering Committee member is listed in Table 2-2.

- Safety and Security—29 municipalities directly involved police, fire, and/or other law enforcement representatives in the planning process. Other representatives came from hazardous materials teams and rescue teams. To address High Hazard Potential Dams, outreach was conducted alongside the Harris County Flood Control District to gain information on inundation threat areas, as well as to understand the scope of potential flooding which may/may not impact the planning area from the following:
  - Addicks Dam
  - Barker Dam
  - Bayport Reservoir Dam
  - Highlands Reservoir Dam
  - Lake Houston Dam
  - Lynchburg Reservoir Dam
  - Sheldon Reservoir Dam
  - Warren Lake Dam
- Food, Hydration, and Shelter—Department heads and liaisons from this lifeline were invited to participate in the planning process, including the review of the draft plan. See Appendix C.
- Health and Medical—Harris County Public Health, Southeast Texas Regional Advisory Committee (SETRAC), and Harris Health served as contributors to the plans development and served on the Steering Committee board. Department heads and liaisons from this lifeline were invited to participate in the planning process, including the review of the draft plan. See Appendix C.
- Energy—Department heads and liaisons from this lifeline were invited to participate in the planning process, and review of the draft plan. See Appendix C.
- Communications—Each jurisdiction was asked to provide information on emergency communication and warning systems. Department heads and liaisons from this lifeline were invited to participate in the planning process, including the review of the draft plan. See Appendix C.
- Transportation—Department heads and liaisons from this lifeline were invited to participate in the planning process, including the review of the draft plan. See Appendix C.
- Hazardous Materials—Department heads and liaisons from this lifeline were invited to participate in the planning process, including the review of the draft plan. See Appendix C.
- Water Systems—Department heads and liaisons from this lifeline were invited to participate in the planning process, including the review of the draft plan. See Appendix C.

2.2.4 Additional Stakeholder Groups

Additional stakeholder outreach was made to academia and organizations that support socially vulnerable populations and underserved populations, as listed in the sections below.

School Districts and Other Academic Institutions

Many jurisdictions directly involved school district representatives in the planning process, as identified in Table 2-1. Additionally, all Independent School Districts within the County were contacted and invited to participate in the planning process. The following school districts, colleges, and academic organizations accepted the invitation and contributed to the development and review of the draft plan:

- Cypress Fairbanks ISD
- Goose Creek CISD
- Houston Community College
- Lee College
- San Jacinto College
- Spring ISD
- University of Houston
- Varnett Public Schools

Groups Supporting Socially Vulnerable Populations and Underserved Communities

Department heads and liaisons from this lifeline were invited to participate in the planning process, including the review of the draft plan. See Appendix C.

2.2.5 Adjacent Jurisdictions

The County kept surrounding jurisdictions informed of the project, invited them to participate in the planning process, and requested their review of the draft plan. Although invited, the City of Houston was unable to be a part of the Steering Committee. However, they participated in the plan update process by attending meetings and reviewing the draft Plan.

2.3 PUBLIC OUTREACH

To facilitate better coordination and communication between the Planning Partnership and all community members and to involve the public in the planning process, draft documents were made available to the public through a variety of venues, including printed and online format. The initial online platform was an in-house HMP Viewer. Prior to the review of the 2025 MJ-HMP draft, the Steering and Planning Partnership made a concerted effort to engage the public through a series of Community Workshops held throughout the months of October and November of 2024. In the workshops, the following information was presented:

- An overview of hazard mitigation planning, along with the purpose and benefits of local mitigation planning
- An overview of the 2020 Harris County Multi-Jurisdictional Hazard Mitigation Plan, including the hazards of concern, the participating entities, and Steering Committee members,
- Presentation and overview of the Harris County Flood Control District and what kinds of mitigation projects are tied to this department
- Presentation and overview from the Harris County Office of the County Engineer on mitigation projects the department is working on and/or recently completed.

- Presentation and overview from the Harris County Office of County Administration on the Harris County Climate Justice Plan
- Presentation of the U.S. Small Business Administration on available resources after the federally declared disasters.

Examples of virtual outreach via websites and social media completed by the Planning Partners are provided in Figure 2-2 through Figure 2-4. These, as well as material for all public and stakeholder outreach efforts, are compiled in Appendix C.

All Steering Committee meetings were open to the public, with agendas posted one week in advance and meeting notes uploaded no later than one week after each meeting. To encourage community engagement, meeting information was also shared through social media platforms. As part of the broader outreach effort, four public review workshops were conducted between June 28 and July 1, 2025 in different areas of the County. These workshops were scheduled during evenings and weekends to ensure accessibility for a wide range of participants. In addition to the public review workshops, announcements were made on Harris County's Facebook page as well as multiple jurisdictional Facebook pages. Each of the four public review workshops had feedback worksheets printed out for any attendees as well as QR codes and URL links for the public to offer their feedback/input on the plan. A hard copy of the plan was positioned at Houston TranStar.

Public Survey

The public survey was developed to assess the level of knowledge of tools and techniques to assist in reducing risk and loss associated with hazards. It asked quantifiable questions about the public's perception of risk, knowledge of mitigation, and support of community programs. The County launched the survey on October 1, 2024 and advertised the survey on its website and social media accounts. By the time the survey closed on November 22, 2024, there were 1,227 responses. A summary of public survey results can be found in Appendix C.

Harris County HMP Data Viewer

As part of the hazard mitigation planning process, the planning team collaborated with Bent Ear Solutions to develop a web-based data viewer aimed at increasing public accessibility and engagement. This interactive platform provides access to data collected throughout the planning effort, including information on hazard risks specific to individual communities, proposed mitigation actions, and all maps generated for the plan. The viewer includes interactive features that allow users to explore and engage with the content in a dynamic and informative way.

Draft Harris County MJ-HMP Public Comment Period

Between June 25 and July 9, 2025, the County made the draft hazard mitigation plan available for public review and feedback. The plan was accessible online at ReadyHarris.org, with printed copies provided at Houston TranStar. To further encourage community engagement, four public comment workshops were held at various locations throughout the County. Each session included a presentation summarizing the plan, followed by a Q&A segment to address questions and gather input from attendees.

Figure 2-2. Harris County Community Workshop Flyer

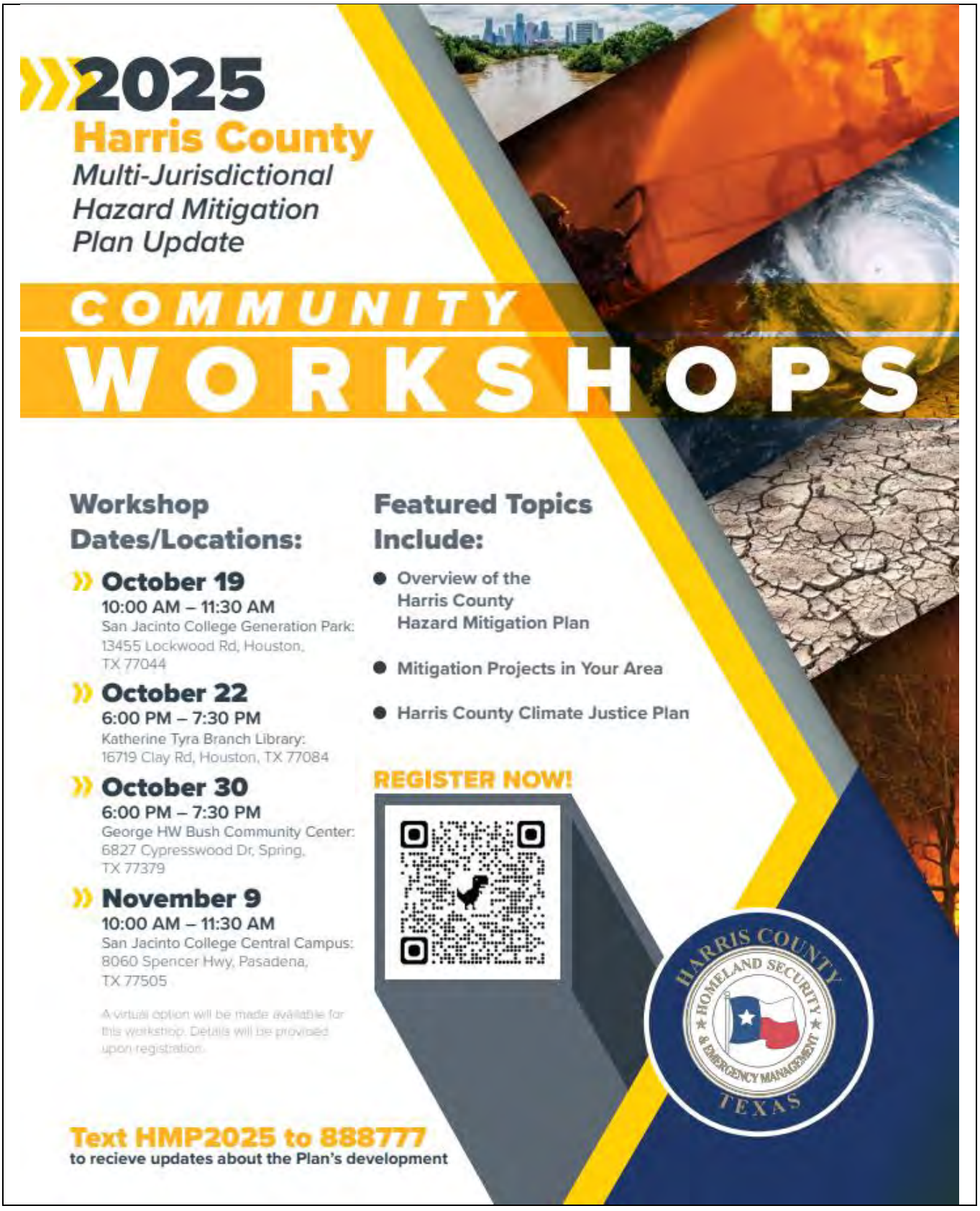


Figure 2-3. Harris County Public Survey Outreach Flyers and Materials

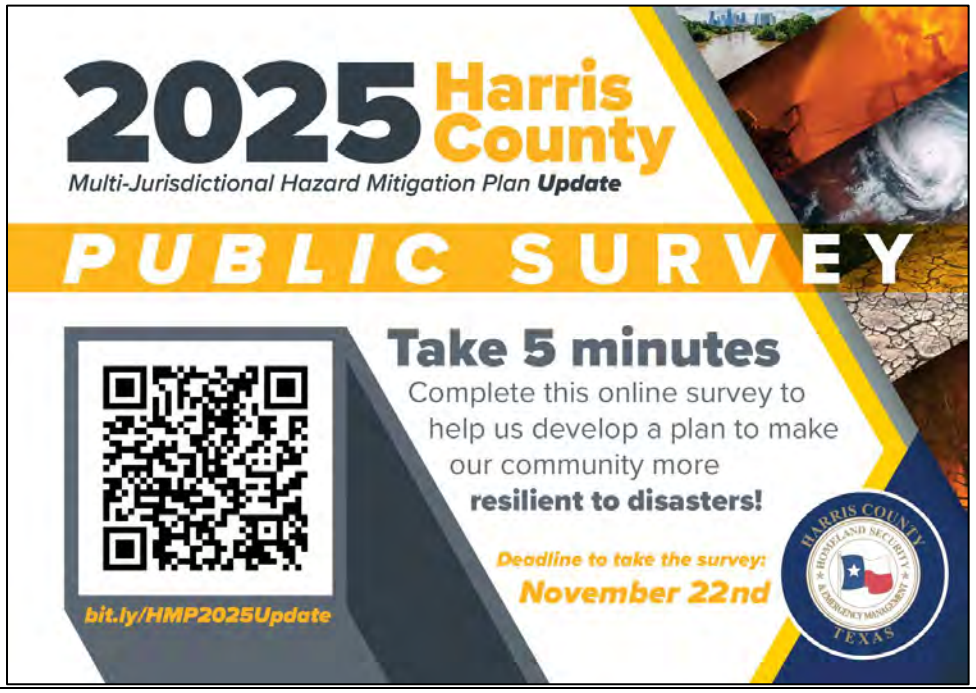


Figure 2-4. Public Review Workshop Flyer

2.4 INCORPORATION OF EXISTING PLANS, STUDIES, REPORTS AND TECHNICAL INFORMATION

The Harris County MJ-HMP uses the best available information to support hazard profiling, risk assessment, review and evaluation of mitigation capabilities, and the development and prioritization of County and local mitigation strategies. Plans, reports, and other technical information were identified and accessed online through independent research by the planning consultant or provided directly by the County, participating jurisdictions, and stakeholders involved in the planning effort. Detailed sources of technical data and information used are listed in the References section.

The asset inventory data used for the risk assessment is presented in the County Profile (Chapter 3). Details of the source of this data, along with technical information on how the data was used to develop the risk assessment, are presented in Chapter 4, as well as throughout the hazard profiles in this MJ-HMP. The County and participating jurisdictions provided relevant jurisdiction-specific planning and regulatory documents, which were reviewed to identify the following:

- Existing jurisdictional capabilities
- Needs and opportunities to develop or enhance capabilities, which may be identified in the County or local mitigation strategies
- Mitigation-related goals or objectives, considered in the review and update of the overall Goals and Objectives
- Proposed, in-progress, or potential mitigation actions to be incorporated into the updated County and local mitigation strategies

The following regulations, codes, ordinances, and plans were reviewed to develop mitigation planning goals and objectives and mitigation strategies that are consistent across local and regional planning and regulatory mechanisms:

- Comprehensive/master plans
- Building codes
- Zoning and subdivision ordinances
- Flood insurance studies
- Flood insurance rate maps
- NFIP flood damage prevention ordinances
- Site plan requirements
- Stormwater management plans
- Emergency management and response plans
- Land use and open space plans
- Community rating system

The County and participating jurisdictions were tasked with updating the assessment of their planning and regulatory capabilities (see capability assessment section of each jurisdictional annex in Volume II). They reviewed relevant plans contributing to the capability of the County and each jurisdiction to integrate effective mitigation efforts into their daily activities. This review is reflected in the capability assessment table in each of the municipal annexes. These tables list plan types, names, and dates, as well as a summary of how each plan supports mitigation and resilience.

2.5 INTEGRATION WITH EXISTING PLANNING MECHANISMS AND PROGRAMS

Effective mitigation is achieved when hazard awareness and risk management approaches and strategies become an integral part of public activities and decision-making. Many existing plans and programs support hazard mitigation in the County. It is critical that this MJ-HMP integrate, coordinate with, and complement, those existing plans and programs.

The capability assessment presented in Chapter 24 provides a summary and description of the existing plans, programs, and regulatory mechanisms at all levels of government (federal, state, county and local) that support hazard mitigation in the County. In the jurisdictional annexes in Volume II, each participating jurisdiction identifies how it has already integrated hazard mitigation into its planning, regulatory and administrative framework ("integration capabilities") and how it intends to promote this integration ("integration actions").

A description of continued efforts toward a holistic approach to hazard mitigation is presented in Chapter 26.

2.6 PLAN ADOPTION

Adoption by the local governing bodies of each participating jurisdiction demonstrates the commitment of the Planning Partners to fulfill the mitigation goals and strategies outlined in this MJ-HMP. Adoption via a municipal resolution legitimizes the MJ-HMP and authorizes responsible agencies to execute their responsibilities.

All participating jurisdictions proceed with formal adoption. Each jurisdiction must submit a copy of its formal adoption resolution or other legal instrument to the HCOHSEM.

In addition to being required by DMA 2000, adoption of the MJ-HMP is necessary because:

- It lends authority to the plan to serve as a guiding document for all local and state government officials.
- It gives legal status to the plan in the event it is challenged in court.
- It certifies to program and grant administrators that the plan's recommendations have been properly considered and approved by the jurisdictions' governing authority and citizens.
- It helps to ensure the continuity of mitigation programs and policies over time because elected officials, staff, and other community decision-makers can refer to the official document when making decisions about the community's future.

Source: FEMA. 2003. How to Series: Bringing the Plan to Life (FEMA 386-4).

FEMA transmitted acknowledgement of verification of formal plan adoption and the official approval of the plan to HCOHSEM. The plan approval date began the five-year approval period and set the expiration date for the plan. All participating jurisdictions have the same expiration date regardless of their own jurisdiction's adoption date. The date indicated on FEMA's approval letter is the official approval date. The resolutions issued by each jurisdiction to support adoption of this MJ-HMP are included in Appendix A.

2.7 CONTINUED PUBLIC INVOLVEMENT

The Planning Partners are committed to the continued involvement of the public in the hazard mitigation process. This Plan update will be posted on-line (currently at https://www.readyharris.org/), and jurisdictions will be encouraged to maintain links to the plan website. Further, the County will make hard copies of the Plan available for review at public locations as identified on the public plan website. A notice regarding annual updates of the plan and the location of plan copies will be publicized annually after the Planning Partnership's annual evaluation and posted on the County's website shown above.

The public will have an opportunity to comment on the plan as part of the annual mitigation planning evaluation process and the next five-year mitigation plan update. The Resilience Planner at HCOHSEM will serve as the MJ-HMP Coordinator, and this person is responsible for coordinating the plan evaluation portion of the meeting, soliciting feedback, collecting, and reviewing the comments, and ensuring their incorporation in the 5-year plan update as appropriate. Members of the Planning Partnership will assist the MJ-HMP Coordinator. Additional meetings may be held as deemed necessary by the Planning Partnership to provide the public with an opportunity to express concerns, opinions, and ideas about the plan.

After completion of this plan, implementation and ongoing maintenance will continue to be a function of the Planning Partnership. Each jurisdiction's governing body will be responsible for receiving, tracking, and filing public comments regarding this plan. The Planning Partnership will review the plan and accept public comments as part of an annual review and as part of the five-year mitigation plan updates. A notice regarding annual updates of the plan and the location of plan copies will be publicized annually after the MJ-HMP Committee's annual evaluation and posted on the County's homepage.

Mrs. Brittany Ray of HCOHSEM has been identified as the ongoing County MJ-HMP Coordinator (see Chapter 26), and is responsible for receiving, tracking, and filing public comments regarding this Plan Update. Contact information is:

Mailing Address:	Harris County Office of Homeland Security and Emergency Management 6922 Old Katy Rd. Houston, TX 77024
Contact Name:	Brittany Ray
Email Address:	Brittany.Ray@harriscountytx.gov
Telephone:	713-881-3100

3. COUNTY PROFILE

Understanding the history and physical setting of a community helps to understand the potential for risk. Risk occurs when hazards impact a community's assets, such as people, buildings, the environment, and the economy. The following sections provide an overview of Harris County's history, government, physical setting, community composition, land use and development, housing, and physical assets. The County Profile informs the Risk Assessment (Chapter 4) to assess the economic, structural, and population assets at risk and the concerns that may be present related to the hazards analyzed. Vulnerability, and therefore risk, may be increased or decreased based on land use, governance, and allocation of resources.

3.1 LOCATION

Harris County is located on the upper Gulf Coast in Southeast Texas and is bounded by Waller County on the north and west, Montgomery County on the north, Liberty and Chambers counties on the east, Galveston County and Brazoria County on the south, and Fort Bend County on the west (see Figure 3-1). The County comprises 1,778 square miles and is the largest Texas county east of the Nueces River (Henson 2020). The planning area for this MJ-HMP is all of Harris County. This MJ-HMP includes additional information for the jurisdictions within the County, found in Volume II.

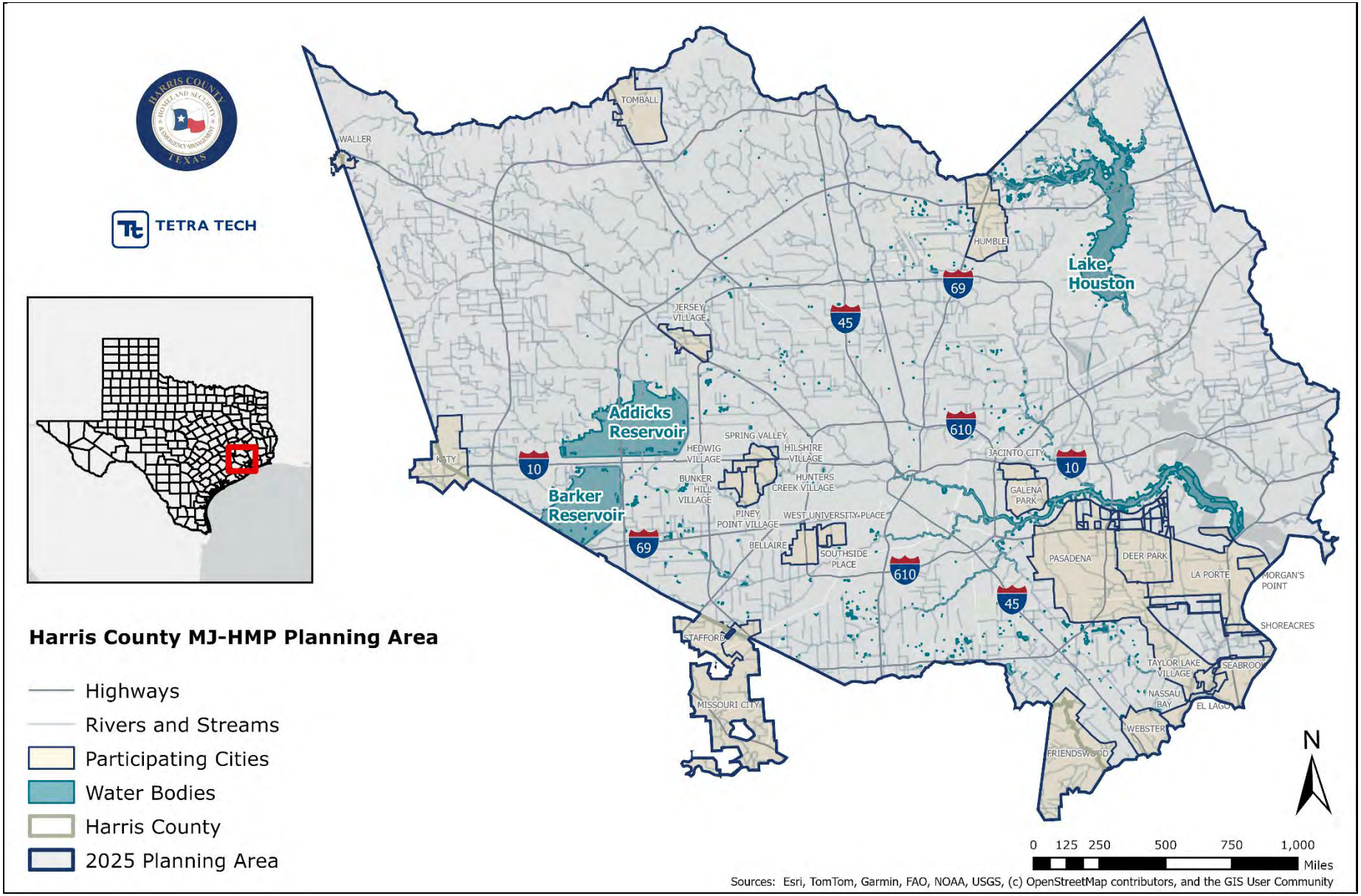
3.2 HISTORY

The early development of Harris County is closely tied to the establishment of Harrisburg, one of the region's earliest Anglo-American settlements. Founded by John Richardson Harris in 1826, Harrisburg was situated near the confluence of Buffalo and Brays Bayous. Harris selected this location to build his home and a steam-powered sawmill, capitalizing on the abundant timber resources and navigable waterways to supply lumber to markets, including the Mexican port city of Tampico (Harris County Historical Commission 2010).

By the early 1830s, Harrisburg had evolved into a modest community featuring two dry goods stores, a steam sawmill, and approximately 20 homes. Despite its growth, the town lacked certain institutions such as churches, schools, and a courthouse (Harris County Historical Commission 2010). The surrounding primeval pine forests facilitated the rafting of logs to the mills, supporting the town's lumber industry. Structures like the bakery operated by German immigrant Schilling exemplify the craftsmanship of the era, with buildings constructed using hand-hewn cedar beams secured with wooden pegs (Harris County Historical Commission 2010).

Harrisburg County was established by the First Congress on December 22, 1836 (Henson 2020). Andrew Briscoe was appointed chief justice, and Houston, then a young city, was designated as the county seat and national capital. Initially, the County included the area of the old municipality along with Galveston Island, while the mainland was part of Brazoria County. In May 1838, the County's current boundaries were defined. In December 1839, Harrisburg County was renamed Harris County by Congress in honor of John R. Harris (Henson 2020).

Figure 3-1. Harris County Hazard Mitigation Plan Area



The strategic location of Harrisburg made it a focal point during the Texas War for Independence. In 1836, the town served as a temporary capital of the Republic of Texas and played a significant role in the conflict's logistics and communications (Harris County Historical Commission 2010). Following the war, Harrisburg continued to develop, and in 1853, it became the eastern terminus of the Buffalo Bayou, Brazos, and Colorado Railway, the first operating railroad in Texas. This advancement bolstered Harrisburg's position as a transportation and trade hub (Harris County Historical Commission, Heritage Tourism Chair, Will Howard 2015).

As industrialization expanded throughout the 20th century, particularly in the petrochemical and shipping sectors, land use patterns began to reflect both economic opportunity and environmental vulnerability. Industrial and residential development sprawled into areas with natural flood risk, especially in eastern Harris County near the Houston Ship Channel.

The rise in development and higher rainfall amounts have resulted in more runoff into the reservoirs. Over time, development has grown, exposing more people and property to a greater risk of flooding, particularly during major events like Hurricane Harvey, when inflows exceed the reservoirs' capacity to release water (U.S. Army Corps of Engineers 2020).

Like many urban counties across the United States, Harris County also bears the legacy of racial segregation and housing discrimination. During the mid-20th century, redlining by the Homeowners' Loan Corporation and exclusionary zoning policies led to the systemic disinvestment in neighborhoods of color (National Community Reinvestment Coalition, Bruce Mitchell, Juan Franco 2018). Black and Hispanic communities were often relegated to low-lying or industrial-adjacent areas, amplifying their exposure to flooding, air pollution, and other environmental hazards (National Community Reinvestment Coalition, Bruce Mitchell, Juan Franco 2018).

The effects of historic redlining and other discriminatory practices are still seen today. Communities that received "D" or "C" ratings in the 1930s generally have lower homeownership rates compared to those with "A" grades. For instance, only 31.2 percent of residents in the Fifth Ward (historically redlined) own homes, whereas 48.1 percent of residents in the Museum District and 65.6 percent of residents in the Heights (rated "A") are homeowners. These differences highlight the ongoing challenges that families in Houston's redlined neighborhoods encounter in achieving homeownership (Greater Houston Community Foundation 2021).

Since 1990, Harris County's population has grown significantly, becoming the third most populous county in the U.S. (Harris County Intergovernmental & Global Affairs n.d.). Harris County's economy has continued to diversify over time, with energy, health care, and aerospace playing major roles. Residents and businesses flourish due to a cost of living that is 3 percent below the national average, a strategic geographic location, and health and transportation costs that are 13 percent lower than the national average (Harris County Economic Development 2024). The County is well-connected globally through a major seaport, two international airports, an excellent highway system, and three major railroad companies, linking Harris County and the Gulf of Mexico to the U.S. interior and the world.

Despite economic growth, vulnerability in Harris County has been compounded by sprawling development, rapid population growth, and aging infrastructure. The impacts of historic land use

decisions, such as wetland loss, have altered the County's hydrology, often increasing downstream flood risk (Brody 2011). As the climate changes, the consequences of these historical decisions become more pronounced, especially for low-income and historically marginalized populations.

3.3 GOVERNMENT

Harris County operates under the authority of the Texas Constitution and is governed by a Commissioners Court, a structure that differs from municipal governments. The Commissioners Court serves as both the executive and legislative body for the County and consists of a County Judge elected countywide and four Commissioners elected by precinct. The County Judge serves as the presiding officer and has responsibilities like a chief executive, while each Commissioner represents one of four precincts and manages infrastructure and services within their areas (Harris County Intergovernmental & Global Affairs n.d.).

The Commissioner's Court is responsible for setting the County tax rate, adopting the budget, and overseeing County services such as flood control, criminal justice, emergency management, and transportation infrastructure. The Court enacts policy by adopting orders and resolutions rather than ordinances, and it appoints or oversees the heads of various departments and agencies ((Harris County Precint 2 n.d.). Day-to-day operations are managed by various elected officials and department heads. Elected officials include the Sheriff, District Attorney, Tax Assessor-Collector, County Clerk, and District Clerk, each of whom operates independently within their statutory responsibilities (Harris County Precint 4 n.d.).

Emergency management and disaster preparedness fall under the responsibility of the Harris County Office of Homeland Security and Emergency Management (HCOHSEM). HCOHSEM coordinates countywide planning for emergency response, recovery, and mitigation efforts and ensures compliance with state and federal emergency requirements. It operates in collaboration with city governments within Harris County, including the City of Houston, and is supported by departments such as Public Health, Engineering, and Flood Control (Texas Gulf Coast Regional Voluntary Organizations Active in Disaster n.d.).

3.4 JURISDICTIONS WITHIN THE COUNTY

As listed in Table 3-1, 34 jurisdictions (26 cities and 8 villages) and 15 special purpose districts or academic institutions participated. Special purpose districts are typically board-governed political subdivisions that are independent from general purpose local governments. They include independent school districts, hospital districts, river authorities, and water districts. Several additional special-purpose jurisdictions and institutions were also involved in the planning process; those listed below are the ones that actively participated in the Plan.

Table 3-1. Harris County Municipalities

Cities		Special Purpose Districts and Institutions	
• Baytown • Bellaire • Deer Park • El Lago • Friendswood • Galena Park • Houston • Humble • Jacinto City • Katy • La Porte • League City • Missouri City • Morgan's Point • Nassau Bay • Pasadena • Pearland	• Seabrook • Shoreacres • South Houston • Southside Place • Stafford • Tomball • Waller • Webster • West University Place • Bunker Hill Village • Hedwig Village • Hilshire Village • Hunters Creek Village • Jersey Village • Piney Point Village • Spring Valley Village • Taylor Lake Village	• A+ Unlimited Potential Charter Schools • Cypress Creek Drainage District • Cypress Fairbanks Independent School District (ISD) • Goose Creek ISD • Gulf Coast Authority • Harris County Flood Control District • Harris Health • Houston Community College	• Lee College • Pasadena ISD • San Jacinto College • Southwest Public Schools • Spring ISD • University of Houston • Varnett Public Schools

3.5 PHYSICAL SETTING

3.5.1 Topography and Geology

The landscape in Harris County is level to gently undulating with slow surface drainage (Texas State Historical Association, 2019b). The elevation slowly rises from sea level at the County's southeast edge where the land meets Galveston Bay, to approximately 50 feet above sea level at mid-County, to as high as 300 feet at the far northwest corner of the County. Buffalo Bayou runs through the middle of the County from the City of Katy at the west end of the County to the Houston Ship Channel on the southeast corner of the County. The Houston Ship Channel empties into Galveston Bay and leads to the Gulf of Mexico.

Several fault lines lie within the County. The largest of these, the Long Point-Eureka Heights fault system, runs from the center of the Houston metropolitan area out to the west side of the County. While there have been no earthquakes reported in the County, the potential risk cannot be discounted.

3.5.2 Hydrography and Hydrology

Major waterways in Harris County include Buffalo Bayou, Brays Bayou, Greens Bayou, White Oak Bayou, and the San Jacinto River. In addition to many creeks and ponds, Galveston Bay provides the southeastern border of the County. The San Jacinto River is fed by numerous tributaries, including Spring Creek and Cypress Creek, and flows into Galveston Bay.

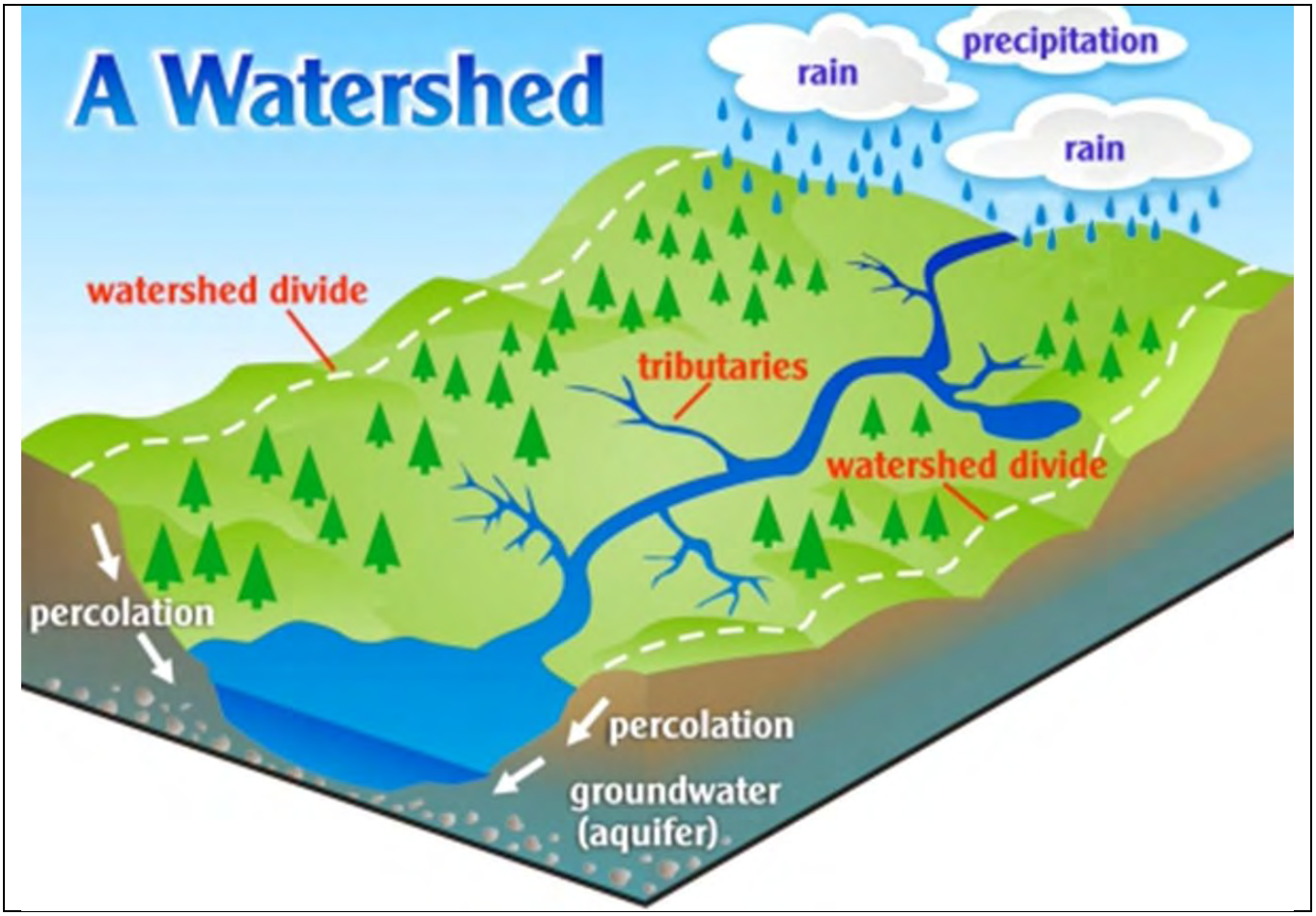
Galveston Bay, an estuary along the Texas Gulf Coast, is a predominant feature in Harris County, as it forms the southeastern border of the City of Houston and the Towns of Pasadena, Baytown, and La Porte and is an important aesthetic, economic, environmental, and cultural resource for the

County. The Buffalo Bayou is also significant, as it runs through the heart of Houston and drains into the San Jacinto River in the eastern part of the County. Topography ranges from flat coastal plains in the southeastern part of the County to gently rolling hills in the northwestern sections.

3.5.3 Watersheds

A watershed is a land area that ultimately drains rainfall runoff (or stormwater) to a common body of water. As shown in Figure 3-2, the watershed not only includes the waterway itself but also the entire land area that drains to it. For example, the watershed of a lake includes the streams entering the lake as well as the land area that drains into those streams. Drainage basins refer to the large areas that encompass the watersheds of many smaller rivers and streams.

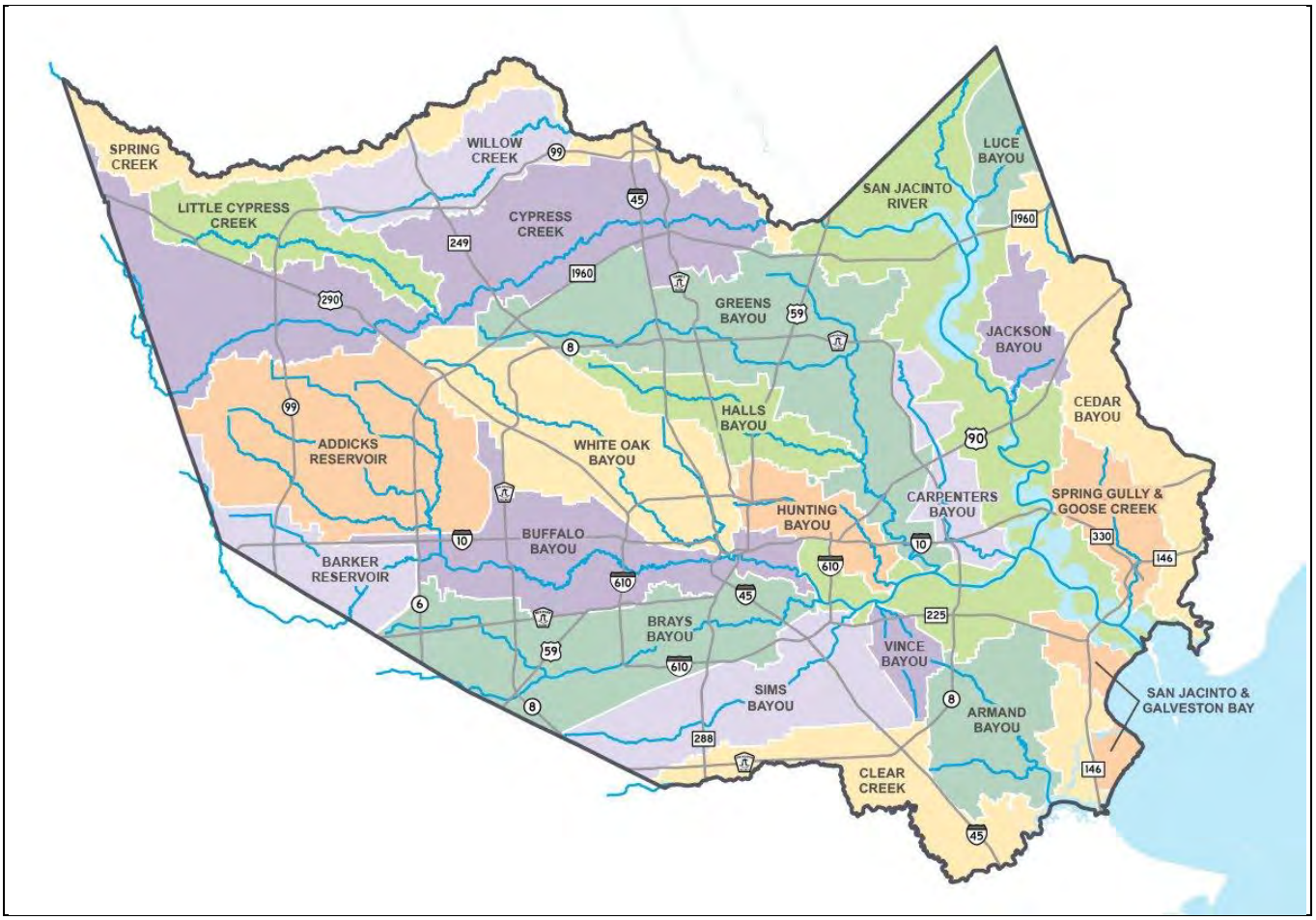
Figure 3-2. Watershed



Source: Riverside-Corona Resource Conservation District 2022

Harris County has classified 23 major watersheds, each collecting and transporting stormwater downstream to Galveston Bay. Each watershed varies in size includes creeks, bayous and other water bodies. Each watershed also varies in primary land use, density of development and flood vulnerability. Figure 3-3 shows the 23 watersheds across Harris County.

Figure 3-3. Watersheds in Harris County



Source: (HCFCD 2025)

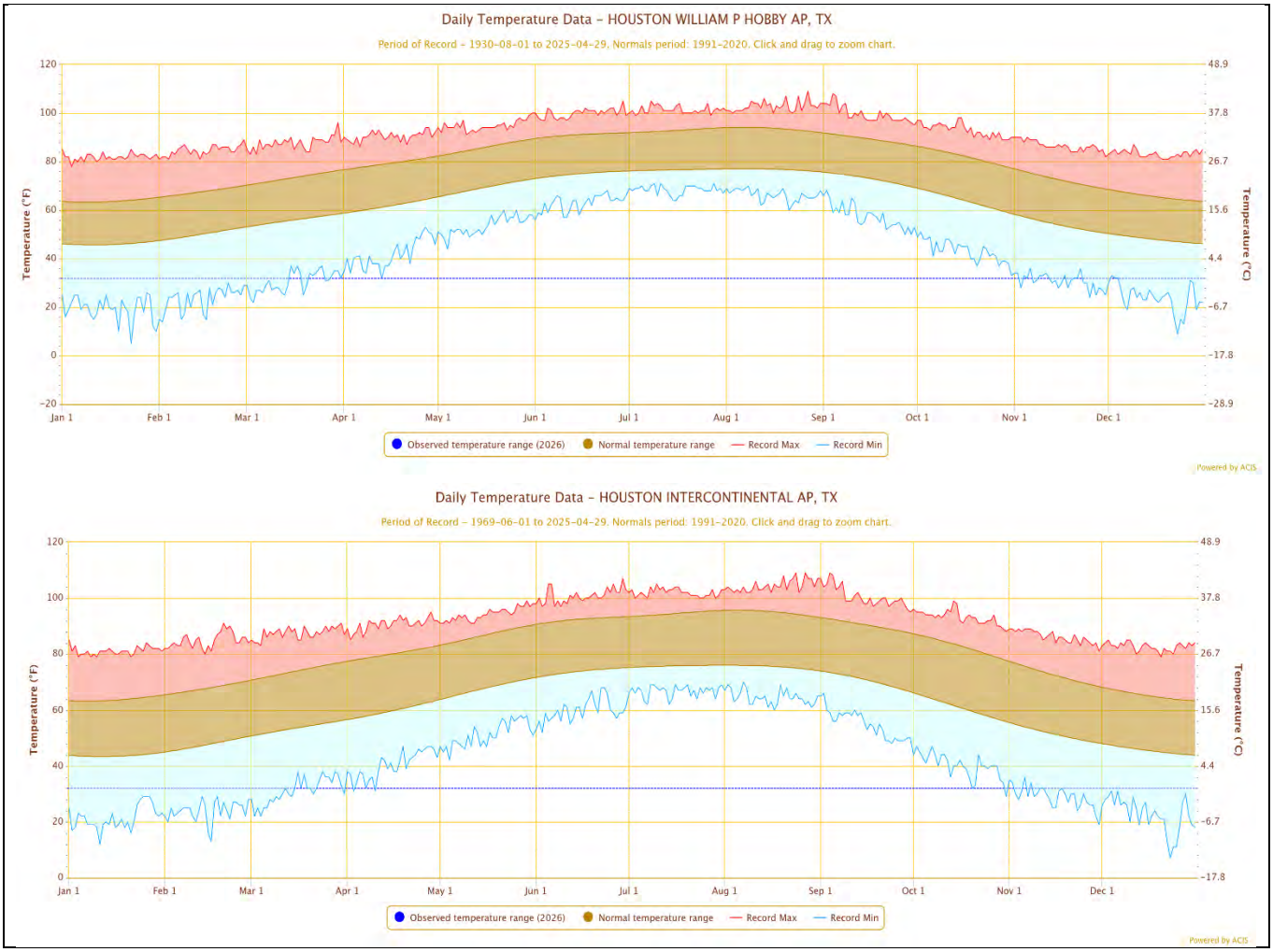
A dam below the East and West Forks of the San Jacinto River created Lake Houston in 1954. The lake is now a popular recreational destination. This reservoir for the City of Houston lessened dependence on subsurface water, the use of which has caused up to 9 feet of subsidence around the confluence of Buffalo Bayou and the San Jacinto River (Texas State Historical Association, 2019a). In western Harris County, Addicks and Barker Dams provide substantial flood control.

3.5.4 Climate

Harris County lies within Climate Zone 2, a humid subtropical climate zone characterized by hot, humid summers and mild to cool winters (U.S. Energy Information Administration 2020). Climate zones are based on weather factors like winter and summer temperatures, along with humidity and rainfall (to define the "Dry" and "Marine" sub-climates) (Built Egineers 2024). This climate zone results in seasonal variability with distinct summer and winter patterns, though extreme weather events such as heat waves, hurricanes, and intense rainfall are increasingly common due to climate change.

Annual temperature data, shown in Figure 3-4, from Houston's Intercontinental and Hobby airports, show that summer temperatures in Harris County regularly exceed 90°F, with record highs reaching above 100°F, particularly from June through September. Winter temperatures, while relatively mild, can occasionally fall below freezing. In 2024, Houston's observed daily temperatures demonstrated typical seasonal patterns, with average highs peaking near 95°F in August and lows dipping into the 30s during January and December. The frequency of extremely hot days and nights each year is rising, along with the temperature of the hottest day each year. The duration of the longest heat wave has almost doubled, and future heat waves are anticipated to be even hotter and longer as the number of days per year with temperatures reaching or exceeding 100°F is expected to increase (HAYHOE 2020).

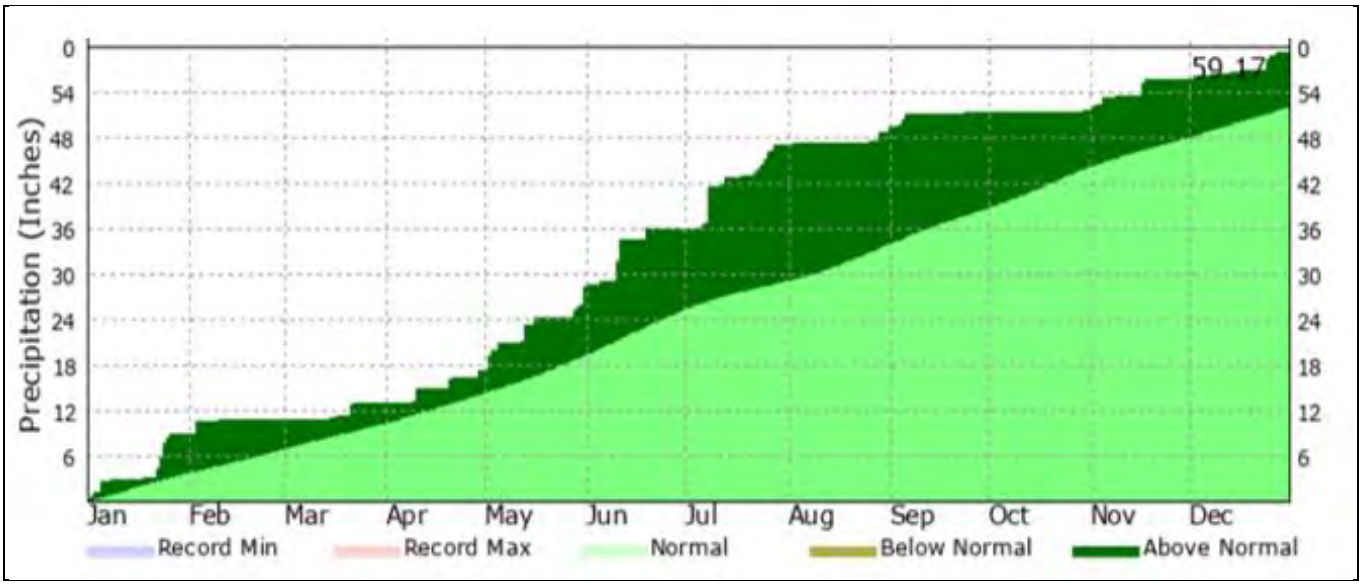
Figure 3-4. Daily Temperature Data for Houston Hobby and Intercontinental Airport



Source: NOAA Regional Climate Center

Average annual precipitation in Harris County is approximately 50 inches, though precipitation is not evenly distributed throughout the year. Data for Houston from 2024 show a cumulative total of 59.17 inches of precipitation, with peaks in spring and fall, as shown in Figure 3-5.

Figure 3-5. Annual Precipitation Data for Houston, 2024



Source: NWS 2024

The heaviest rainfall typically occurs from May through September, often associated with severe thunderstorms and tropical systems (NOAA, 2024). Houston's rainfall totals regularly exceed the national average and contribute to its chronic flooding issues, especially when combined with the County's flat topography and expansive impervious surfaces.

Climate projections for the region indicate that the beginning of summer is occurring earlier in the year and the end of summer is occurring later in the year, resulting in longer summers (HAYHOE 2020). This will result in hotter temperatures, more variable rainfall, and higher energy consumption for air conditioning for both residents and businesses. These shifts are expected to increase the frequency and intensity of weather-related hazards such as extreme heat, inland flooding, and drought.

Harris County's exposure to climate-related hazards underscores the importance of climate adaptation and mitigation. The County has begun incorporating resilience planning into infrastructure, emergency management, and land use policies, though ongoing investments are necessary to address projected risks. Mitigation efforts, such as expanding green infrastructure and reducing greenhouse gas emissions, remain critical for reducing long-term vulnerability.

3.5.5 Land Cover

The original primeval forest in Harris County was a mix of several different forest communities. In general, oak and pine dominated on upland sites while cypress was most prevalent in wetter lowland areas. Other common species included pecan trees, elm, and sycamore trees. Current vegetation consists of urban development, agriculture, and mixed forests such as loblolly pine, live oak, water oak, and sweetgum (Houston-Galveston Area Council, 2022).

3.6 LAND USE

3.6.1 Current Land Use

According to the 2024 Houston-Galveston Area Council, the greatest share of land use in Harris County is residential, with 37.3 percent of all land cover categorized as one of many residential land use categories (in terms of acreage). The next largest shares are vacant land with 19.7 percent, followed by multiple-use land and parks/open spaces, with 11.1 percent and 8.4 percent, respectively. Table 3-2 summarizes the land use categories, and Figure 3-6 shows land use distribution across the county. Ranked by number of properties, the top three land uses are Residential, Vacant Land, and Industrial (Houston-Galveston Area Council, 2024).

Table 3-2. Harris County 2024 Land Use Classification

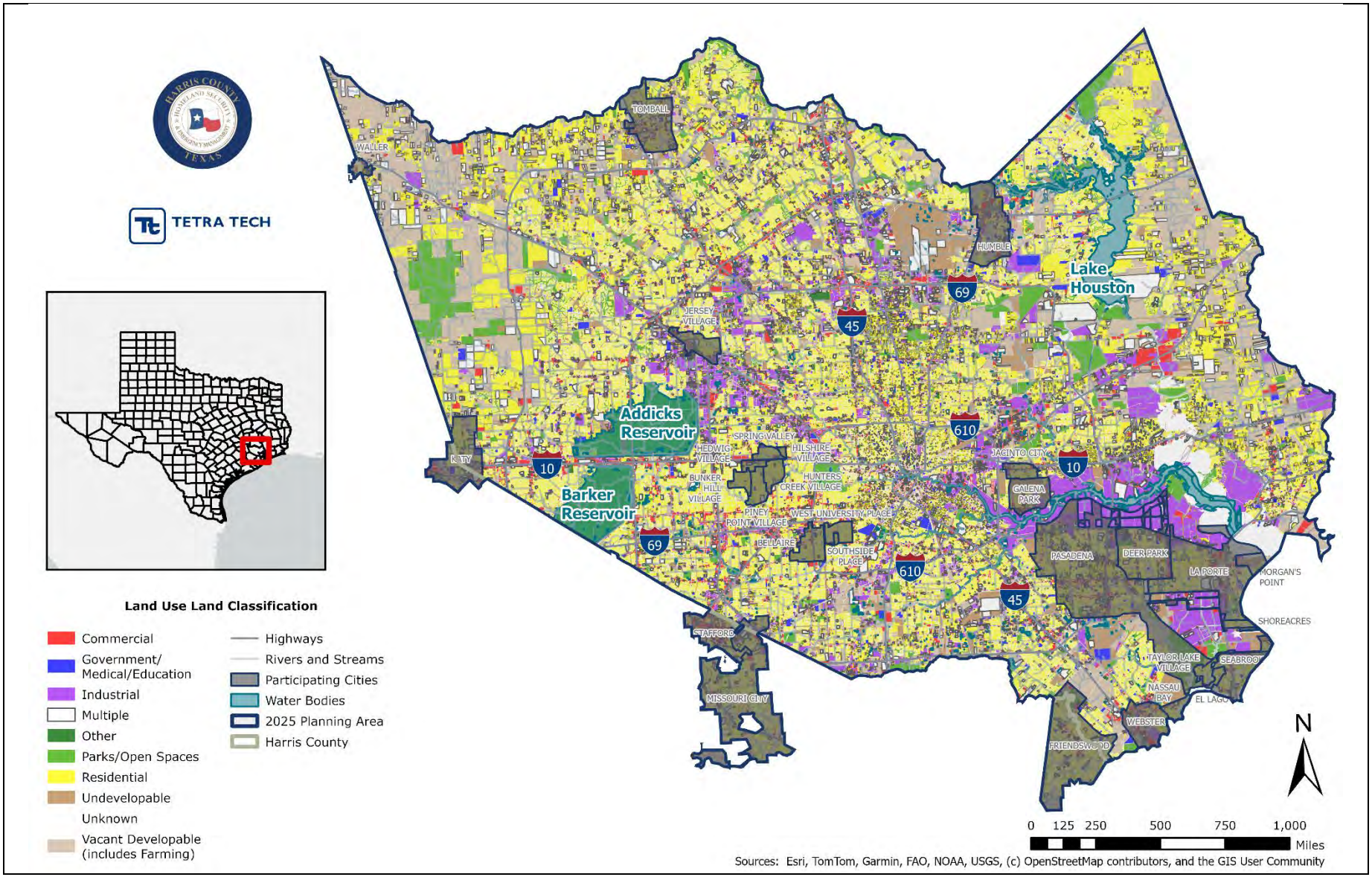
	2024 Data	
Land Use Category	Total Acreage	% of County
Commercial	53,138	4.8%
Government/Medical/Educational	17,368	1.6%
Industrial	87,476	7.9%
Multiple	123,504	11.1%
Other	5,393	0.5%
Parks/Open Spaces	93,087	8.4%
Residential	414,355	37.3%
Undevelopable	85,325	7.7%
Unknown	12,819	1.2%
Vacant	219,172	19.7%
Total	1,111,637	100.0%

Source: Houston-Galveston Area Council 2024

3.6.2 Land Use Trends

In Texas, land use regulatory authority is vested in counties, cities, and municipalities. However, many development and preservation issues transcend local political boundaries. 44 CFR 201 requires that communities consider land use trends, which can impact the need for, and priority of, mitigation options over time. Land use trends can also significantly affect vulnerability to various hazards and the potential impacts from hazard events. For example, significant development in a hazard area increases the building stock and population exposed to that hazard.

Figure 3-6. Harris County Land Use



Residential

Harris County's housing landscape is expansive, with 1,262,384 housing units serving just over 4,893,207 residents. Homeownership is increasingly expensive, with the median home sales price reaching $315,000 in Harris County and $335,000 in Houston in 2023 (Kinder Institute for Urban Research, 2023). While home prices have increased, buying power has not. Since 2018, the affordability gap has grown 275 percent in Harris County and 54 percent in Houston (Kinder Institute for Urban Research, 2023). To meet the needs of the population through 2042, it is estimated that an average of 20,000 new housing units will be required each year (Harris County Redevelopment Authority, n.d.).

The lack of proximate and affordable housing options forces many residents employed in large employment areas to endure long commutes. This situation exacerbates greenhouse gas emissions, increases vehicle counts, contributes to roadway congestion, and negatively impacts residents' work-life balance.

Agriculture

Agriculture in Harris County, Texas, has experienced notable shifts in recent years due to urban expansion and changing market dynamics. As non-farm development continues to encroach on agricultural land, farmers face increasing pressure to convert farmland for other uses. Despite these challenges, Harris County maintains a robust agricultural sector, particularly in the production of vegetables, melons, and sweet potatoes (Harris Central Appraisal District). Additionally, agriculture in the planning area encompasses 50,000 irrigated acres planted with rice, soybeans, grains, hay, corn, and vegetables. Cattle, hogs, horses, and poultry are raised on agricultural lands, as well (Texas State Historical Association, 2019a).

According to the Harris Central Appraisal District, the County's agricultural land is subject to productivity valuation, which lowers the taxable value based on the land's capacity to produce crops, forage, or timber rather than its market value. This valuation method helps reduce property taxes for farmers and ranchers, making agricultural operations more financially viable despite urban pressures (Harris Central Appraisal District).

Harris County also benefits from Texas Wildlife Management Qualifications, which allows landowners to receive tax savings for land used for wildlife conservation. This initiative supports the preservation of natural habitats while providing economic incentives for landowners to maintain agricultural and timber land. The County's agricultural advisory board plays a crucial role in determining eligibility and ensuring that land use standards are met (Harris Central Appraisal District).

3.7 POPULATION AND DEMOGRAPHICS

Population trends can provide a basis for making decisions on the type of mitigation approaches to consider, and the locations in which these approaches should be applied. Information about population is a critical part of planning because it directly relates to land needs such as housing,

industry, stores, public facilities and services, and transportation. This information can also be used to support planning decisions regarding future development in vulnerable areas. Evaluating development trends and population and demographic changes provides insight into how vulnerability may evolve over time.

3.7.1 Current Population

According to the 2020 U.S. Census, Harris County has a population of 4,731,145. The planning area for this MJ-HMP includes the portions of planning partner cities that extend beyond the county boundaries; therefore, the planning area 2020 population was slightly higher than the county total, at 4,859,444. Table 3-3 shows the planning area population breakdown by jurisdiction based on two datasets: the 2020 decennial census and the 2023 5-year American Community Survey (ACS) estimates.

Population density is a critical factor when assessing hazard risk and planning mitigation strategies. Harris County's population density is 2,771 people per square mile, considerably higher than the national average (U.S. Census Bureau 2020). Higher density can exacerbate the impact of natural hazards by increasing the number of people and structures exposed to risk. In such areas, evacuations, emergency response coordination, and service restoration are more complex and resource intensive. Figure 3-7 shows the distribution of population density across the County.

3.7.2 Population Trends

Population changes are useful indicators: a growing population can indicate a growing economy, and a decreasing population may signify economic decline. Harris County continues to experience significant population growth, primarily driven by international migration and natural increase.

Recent Changes

According to the U.S. Census Bureau's Vintage 2023 estimates, Harris County added approximately 32,694 housing units between July 2021 and July 2022, ranking among the top counties in the nation for numeric housing growth (U.S. Census Bureau, 2023). This growth reflects the County's continued appeal to both domestic and international migrants. International migration has become an increasingly important component of population growth in Harris County. Recent analyses show that immigration is helping offset domestic outmigration in major metro areas, including Harris County, contributing to the nation's highest annual population growth rate in over two decades (AP News, 2023).

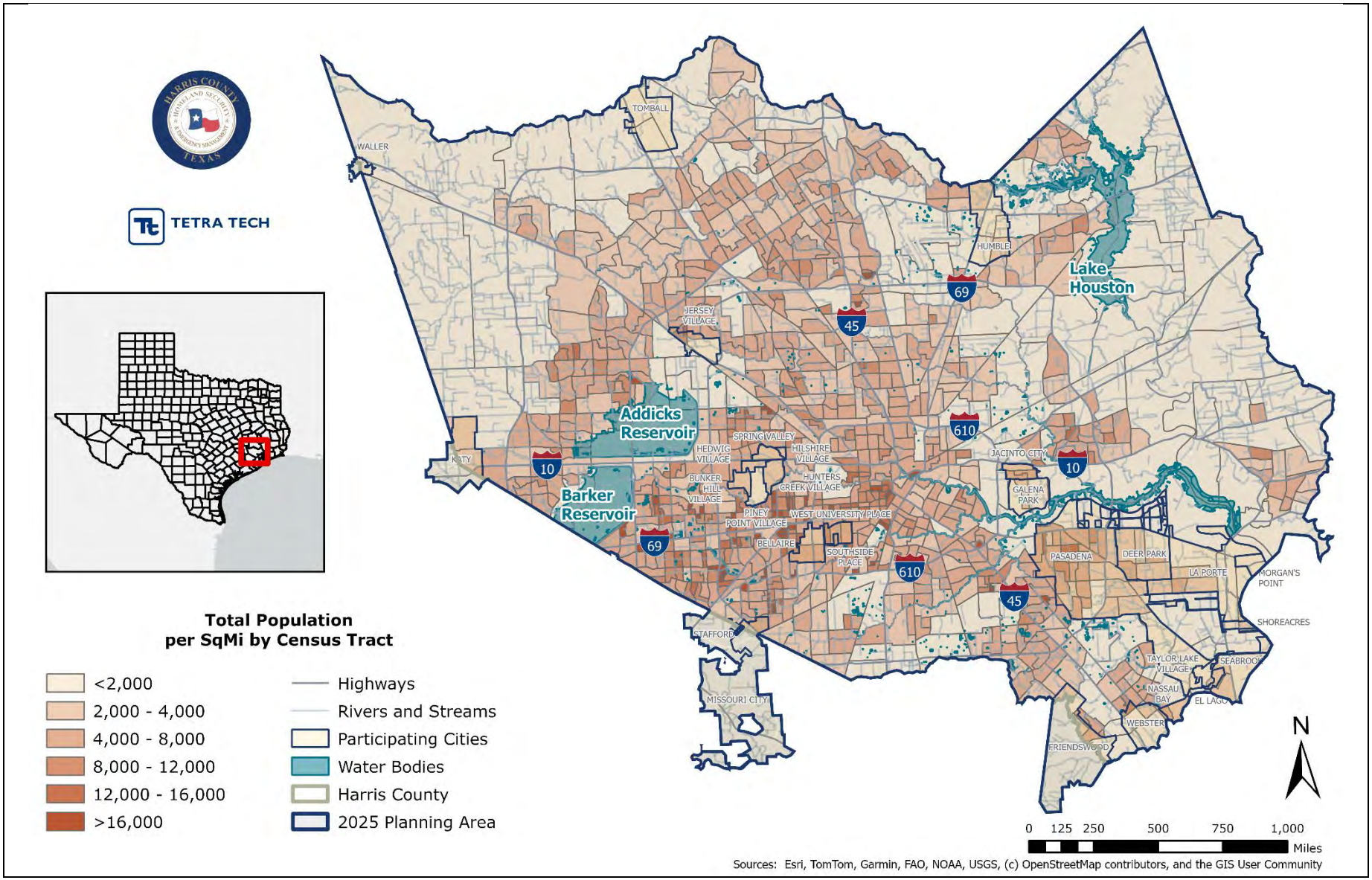
Table 3-3. Planning Area Population Breakdown by Jurisdiction

	2020 Decennial	2023 ACS 5-Year Estimates
Baytown[b]	78,840	79,635
Bellaire	17,202	17,116
Bunker Hill Village	3,822	3,801
Deer Park	34,495	33,823
El Lago	3,090	3,016
Friendswood[a]	41,213	40,827
Galena Park	10,740	10,527
Hedwig Village	2,370	2,029
Hilshire Village	816	1,091
Houston[b]	2,255,442	2,252,110
Humble	16,795	16,489
Hunters Creek Village	4,385	4,358
Jacinto City	9,613	9,458
Jersey Village	7,921	7,817
Katy[a]	21,894	23,900
La Porte	35,124	36,077
League City[b]	2,527	2,527
Missouri City[a]	74,259	75,234
Morgan's Point	273	255
Nassau Bay	5,347	5,211
Pasadena	151,950	149,345
Pearland[b]	8,128	8,189
Piney Point Village	3,128	3,090
Seabrook	13,618	13,617
Shoreacres	1,566	1,327
South Houston[b]	16,153	16,017
Southside Place	1,835	1,631
Spring Valley Village	4,229	4,226
Stafford[a]	17,666	17,279
Taylor Lake Village	3,704	3,678
Tomball	12,341	13,032
Waller[a]	2,682	3,062
Webster	12,499	12,283
West University Place	14,955	14,907
Unincorporated County	1,968,822	2,001,319
Total	4,859,444	4,888,304

[a]. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

[b]. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

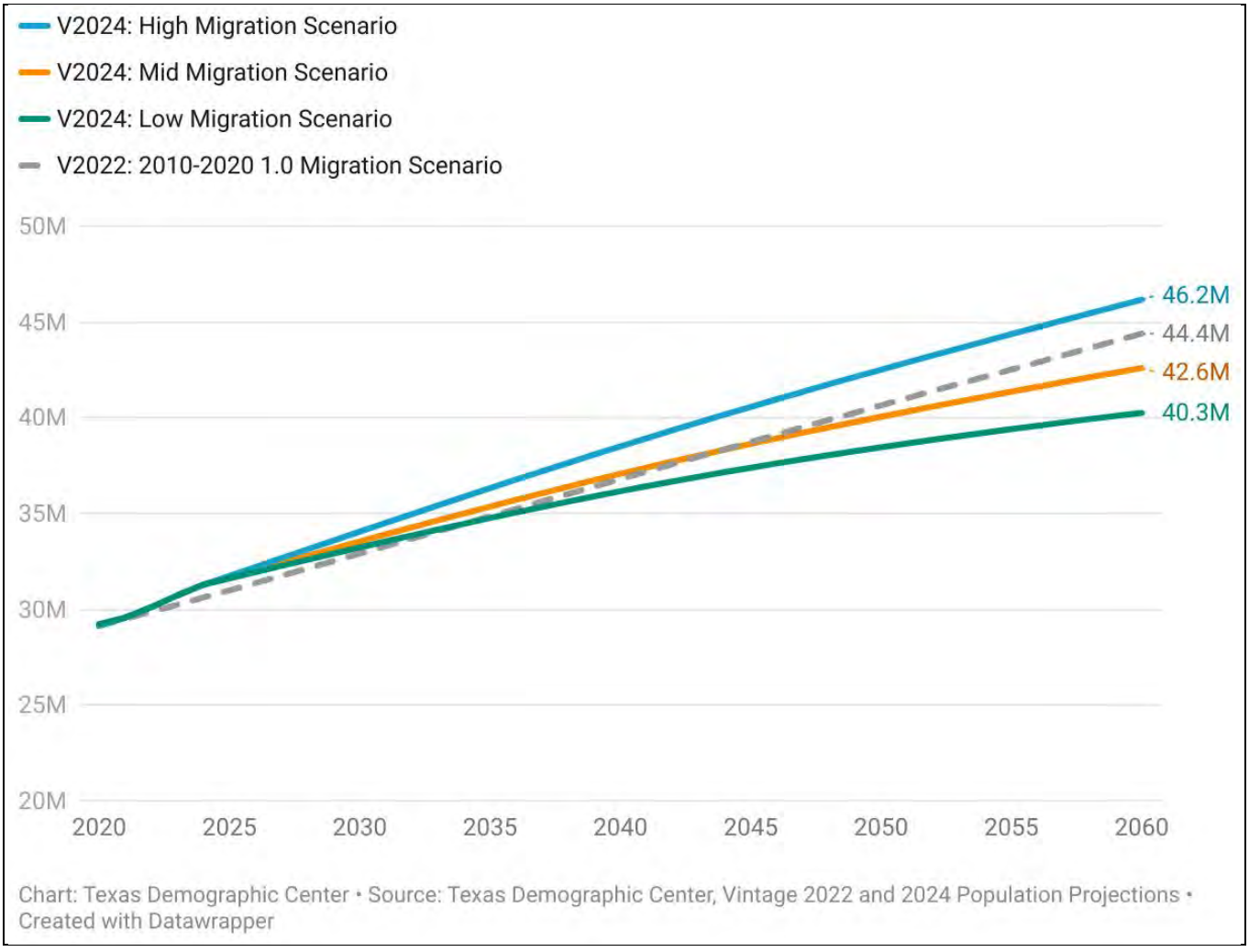
Figure 3-7. Harris County Population Density



Projected Future Population

Texas as a whole is expected to experience significant population increases through 2060 under all migration scenarios. According to the Texas Demographic Center's 2024 projections (Figure 3-8), the state's population could grow to between 40.3 million (low migration) and 46.2 million (high migration) by 2060 (Texas Demographic Center, 2024). Harris County, as a core urban area, is anticipated to receive a large share of this growth, increasing the demand for housing, transportation, public services, and emergency management, especially in hazard-prone areas like floodplains and low-income communities lacking adequate infrastructure.

Figure 3-8. Projected Population Growth in Texas with Various Migration Scenarios



The projected increase in population presents both challenges and opportunities. A higher concentration of people means more lives, and assets are potentially at risk from climate-exacerbated hazards such as hurricanes, flooding, and extreme heat. As hazards continue to increase in frequency and intensity due to climate change, areas with projected growth will continue to become more vulnerable to natural hazards, making it crucial for these areas to have

updated evacuation plans, alert systems, backup power for critical facilities, and the capacity to protect the increasing population from these hazards.

3.7.3 Socially Vulnerable Populations

The ability of an individual or community to withstand and quickly recover from hazards and threats is critical to building community-wide resilience. The same disaster or emergency can impact different populations in different ways. For example, differences in age, income, disabilities, and English proficiency affect people's ability to cope with the effects of disasters. Individuals may also face compounding barriers because they may fall within multiple categories of vulnerability.

Identifying concentrations of socially vulnerable and underserved populations can assist communities in targeting preparedness and mitigation actions. Often, populations and communities are categorized based on shared characteristics that create additional barriers to accessing resources, leading to increased vulnerability.

These populations can be more susceptible to hazard events based on a number of factors, including their physical and financial ability to react or respond during a hazard and the location and construction quality of their housing. This MJ-HMP considers several socially vulnerable population groups: older adults (persons over the age of 65), children (persons under the age of 5), individuals with limited English proficiency, individuals with disabilities, and those living below the poverty level (as defined by the U.S. Census Bureau). Refer to Table 3-4 for population statistics for these socially vulnerable populations, for each municipality in the County based on the 2023 American Community Survey data.

In Harris County, 11.5 percent of the population is over the age of 65, and 6.9 percent of the population is under the age of 5. The 2023 American Community Survey data indicate a total of 15.2 percent of all persons living in households fall below the poverty level, and 3.9 percent of the County's residents live in non-English speaking households.

The distributions of the general population density (persons per square mile) for five metrics of social vulnerability are shown in Figure 3-9 through Figure 3-13.

Table 3-4. Harris County Population and Demographic Statistics, 2023

Jurisdiction	American Community Survey 5-Year Population Estimates (2023)									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Baytown[b]	9,203	11.6%	5,903	7.4%	2,504	3.1%	10,201	12.8%	1,390	1.7%
Bellaire	3,283	19.2%	846	4.9%	279	1.6%	1,114	6.5%	444	2.6%
Bunker Hill Village	523	13.8%	115	3.0%	11	0.3%	164	4.3%	131	3.4%
Deer Park	5,555	16.4%	2,052	6.1%	371	1.1%	4,329	12.8%	2,764	8.2%
El Lago	621	20.6%	106	3.5%	31	1.0%	248	8.2%	121	4.0%
Friendswood[a]	6,432	15.8%	2,465	6.0%	94	0.2%	3,543	8.7%	1,134	2.8%
Galena Park	1,037	9.9%	827	7.9%	577	5.5%	1,326	12.6%	2,460	23.4%
Hedwig Village	393	19.4%	112	5.5%	15	0.7%	190	9.4%	104	5.1%
Hilshire Village	184	16.9%	44	4.0%	2	0.2%	39	3.6%	7	0.6%
Houston[b]	271,304	12.0%	151,876	6.7%	119,249	5.3%	245,929	10.9%	436,448	19.4%
Humble	2,031	12.3%	841	5.1%	512	3.1%	1,931	11.7%	2,946	17.9%
Hunters Creek Village	721	16.5%	181	4.2%	6	0.1%	235	5.4%	176	4.0%
Jacinto City	1,307	13.8%	736	7.8%	448	4.7%	1,496	15.8%	1,779	18.8%
Jersey Village	1,684	21.5%	495	6.3%	174	2.2%	810	10.4%	395	5.1%
Katy[a]	2,860	12.0%	1,795	7.5%	300	1.3%	2,204	9.2%	1,284	5.4%
La Porte	4,816	13.3%	2,486	6.9%	390	1.1%	4,487	12.4%	4,415	12.2%
League City[b]	313	12.4%	154	6.1%	22	0.9%	244	9.6%	113	4.5%
Missouri City[a]	12,289	16.3%	4,572	6.1%	1,163	1.5%	8,141	10.8%	4,593	6.1%
Morgan's Point	54	21.2%	11	4.3%	0	0.0%	36	14.1%	32	12.5%
Nassau Bay	1,115	21.4%	127	2.4%	225	4.3%	707	13.6%	693	13.3%
Pasadena	17,050	11.4%	11,207	7.5%	6,379	4.3%	17,970	12.0%	27,175	18.2%

Jurisdiction	American Community Survey 5-Year Population Estimates (2023)									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Pearland[b]	984	12.0%	582	7.1%	135	1.6%	601	7.3%	344	4.2%
Piney Point Village	674	21.8%	27	0.9%	6	0.2%	159	5.1%	24	0.8%
Seabrook	1,962	14.4%	685	5.0%	111	0.8%	1,394	10.2%	904	6.6%
Shoreacres	236	17.8%	56	4.2%	0	0.0%	285	21.5%	43	3.2%
South Houston[b]	1,387	8.7%	1,107	6.9%	794	5.0%	1,591	9.9%	3,764	23.5%
Southside Place	303	18.6%	74	4.5%	3	0.2%	107	6.6%	50	3.1%
Spring Valley Village	704	16.7%	205	4.9%	20	0.5%	253	6.0%	55	1.3%
Stafford[a]	1,942	11.2%	1,223	7.1%	653	3.8%	1,610	9.3%	1,620	9.4%
Taylor Lake Village	1,124	30.6%	99	2.7%	21	0.6%	513	13.9%	10	0.3%
Tomball	2,794	21.4%	996	7.6%	151	1.2%	2,211	17.0%	1,238	9.5%
Waller[a]	201	6.6%	124	4.0%	37	1.2%	302	9.9%	732	23.9%
Webster	1,742	14.2%	761	6.2%	508	4.1%	1,304	10.6%	1,107	9.0%
West University Place	2,481	16.6%	1,145	7.7%	9	0.1%	684	4.6%	291	2.0%
Unincorporated County	203,620	10.2%	144,118	7.2%	57,378	2.9%	192,353	9.6%	245,872	12.3%
Harris County Total	562,929	11.5%	338,154	6.9%	192,578	3.9%	508,711	10.4%	744,658	15.2%

[a]. *City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties*

[b]. *City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.*

Figure 3-9. Distribution of Population with Disability by Census Block for Harris County

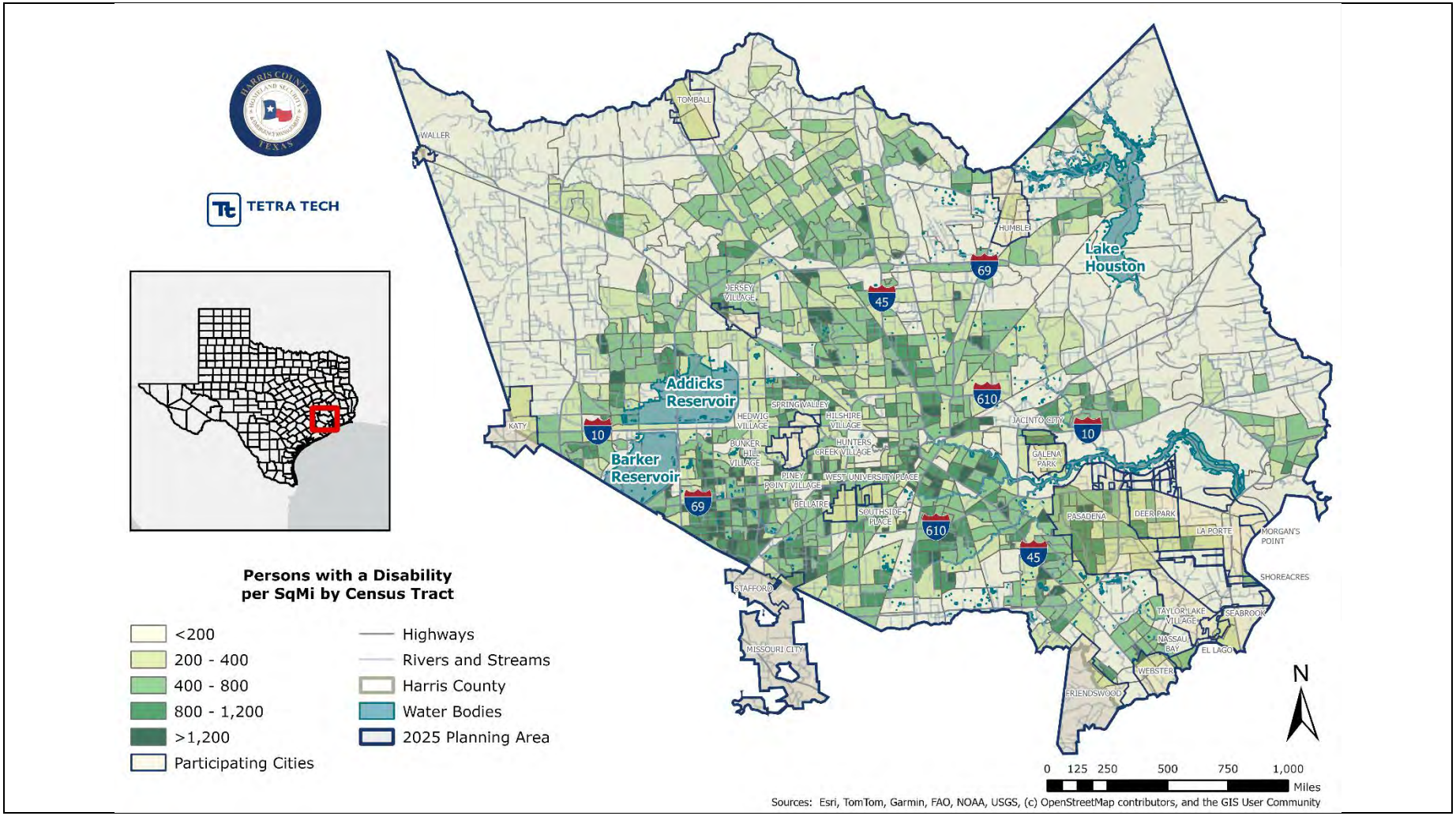


Figure 3-10. Distribution of Low-Income Population by Census Block for Harris County

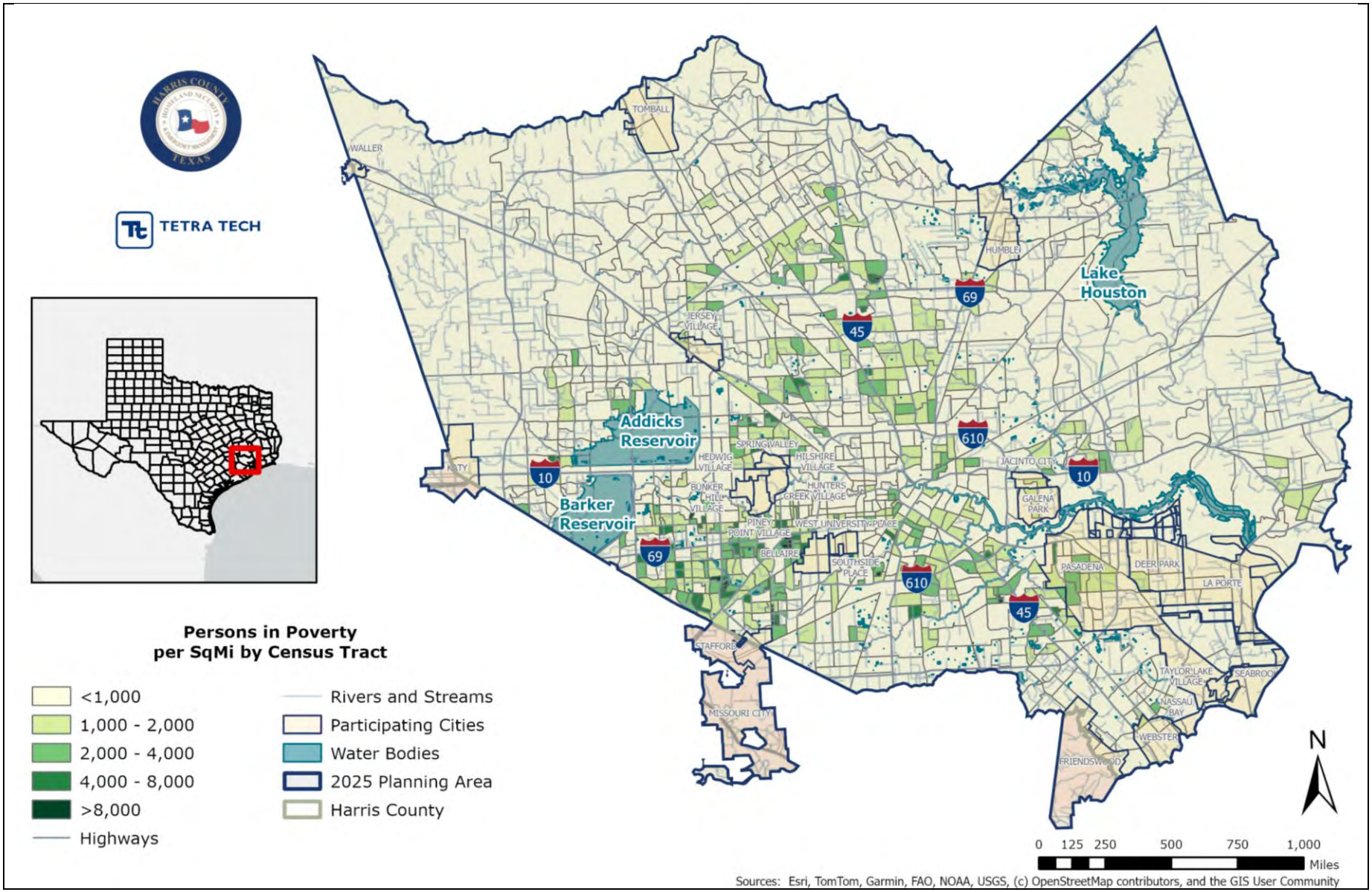


Figure 3-11. Distribution of Population Under 5 Years Old for Harris County

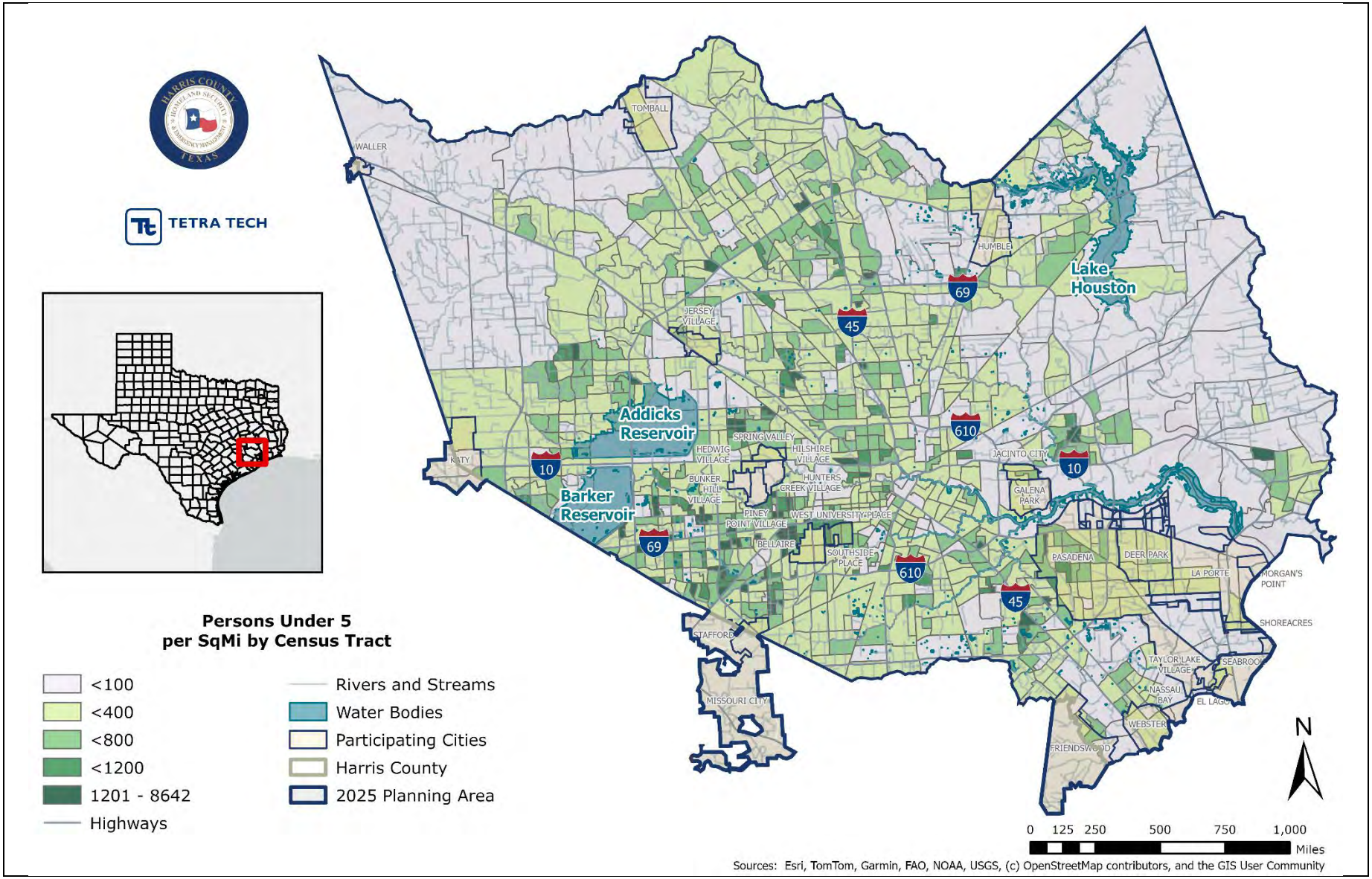


Figure 3-12. Distribution of Population Over 65 Years Old for Harris County

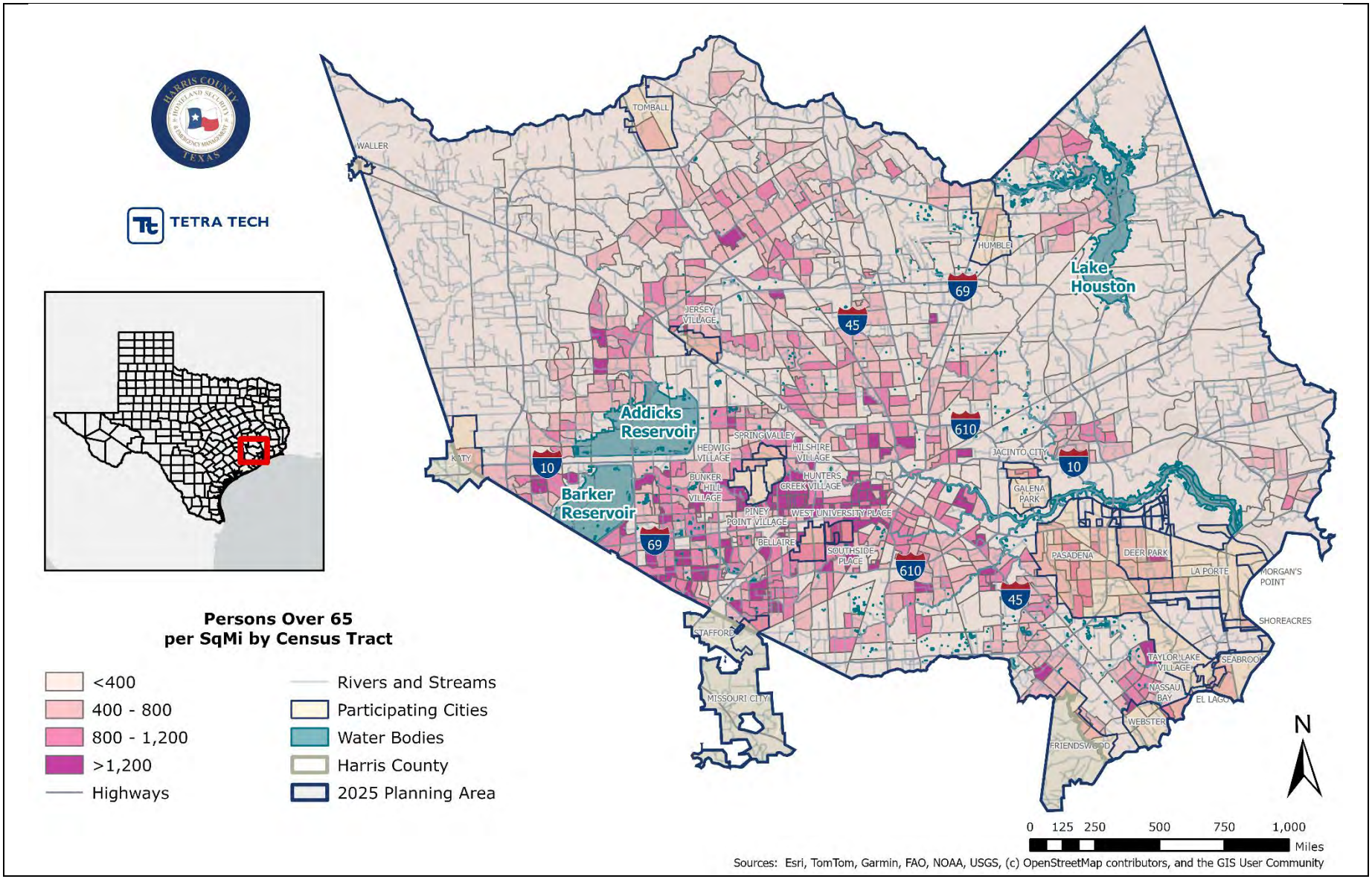
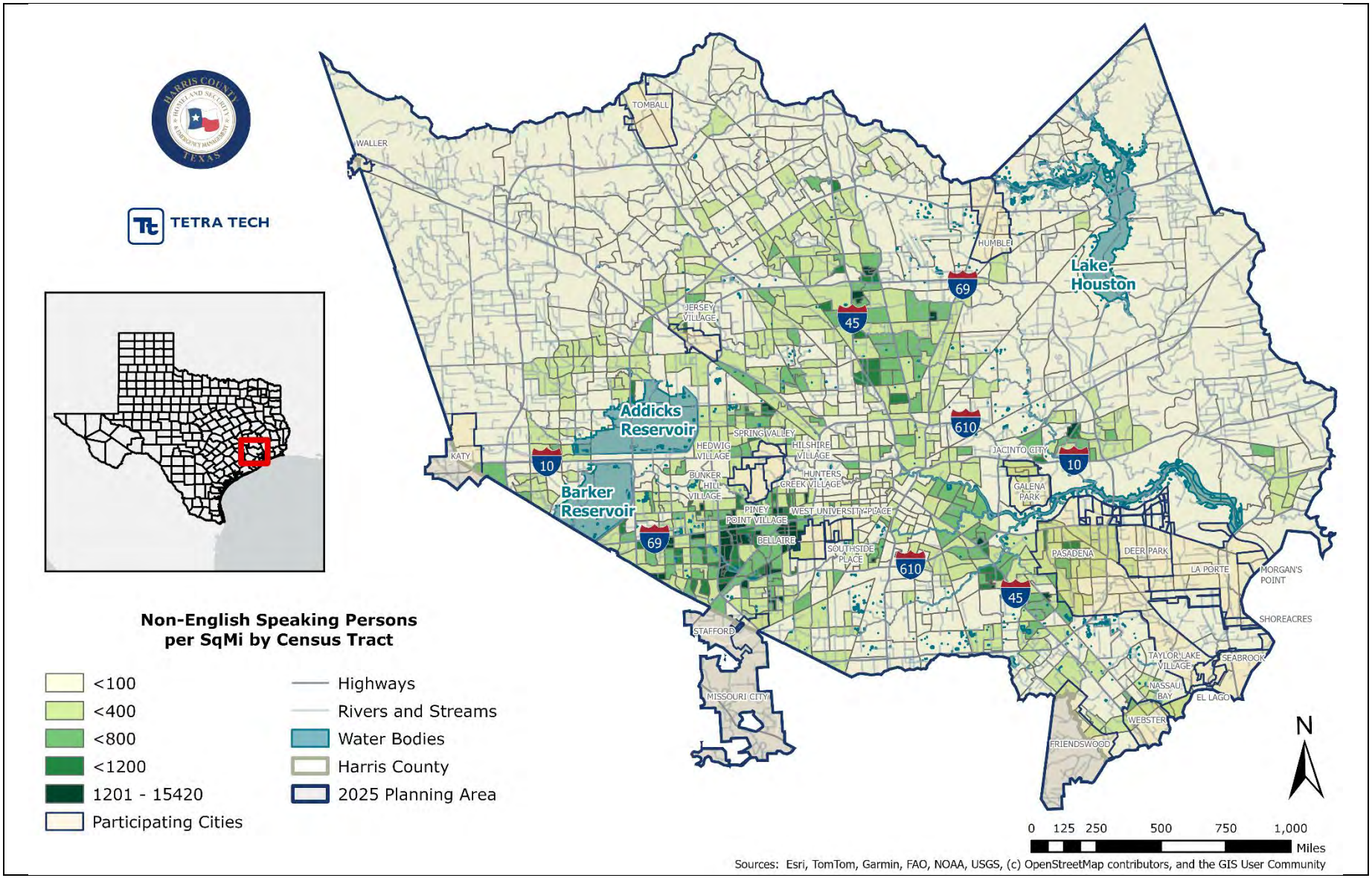


Figure 3-13. Distribution of Non-English-Speaking Population by Census Block for Harris County

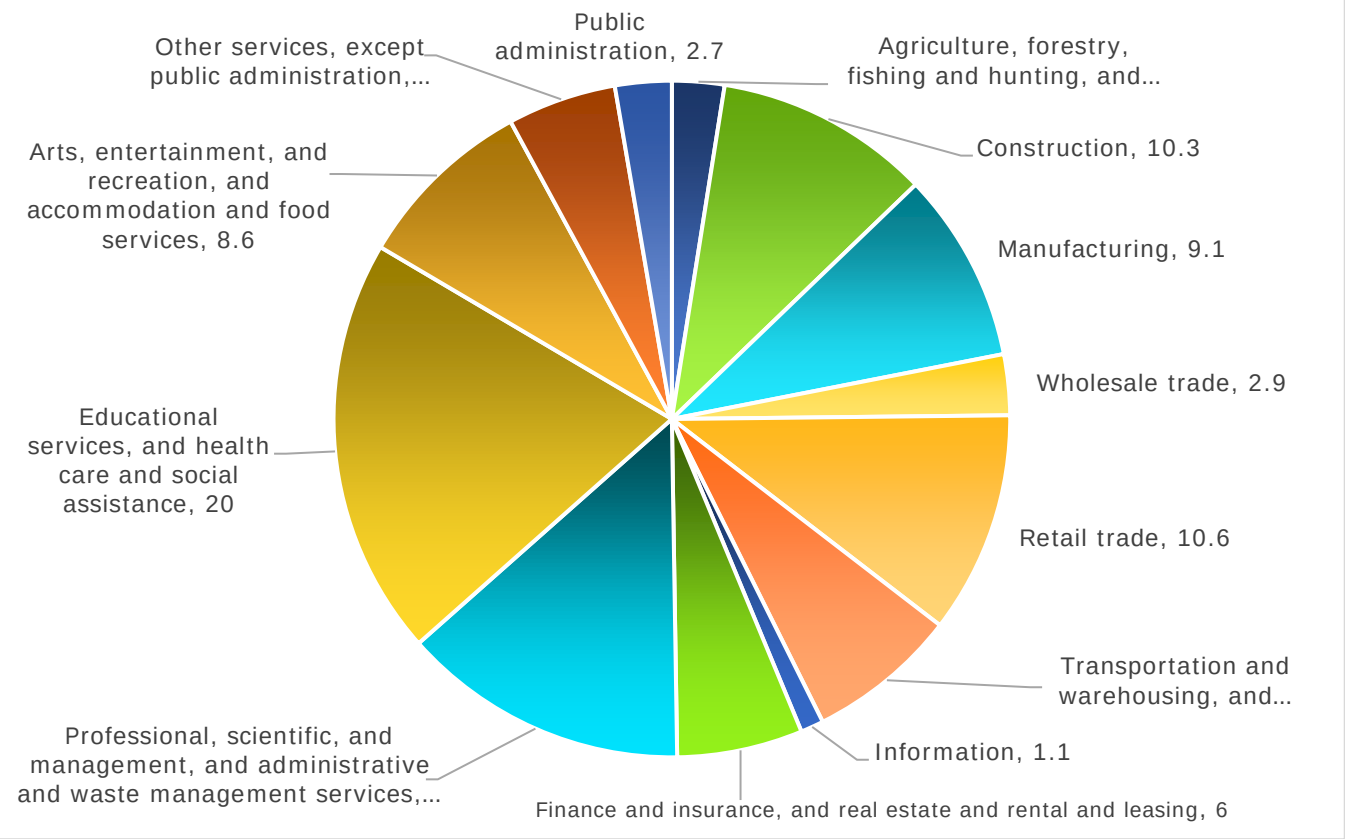


3.8 ECONOMY

3.8.1 Employment

Harris County's industry is diverse, with significant employment in several key sectors. The largest industry is educational services, and health care and social assistance account for 20.0 percent of the County's industry. See Figure 3-14 for a breakdown of industries within the County.

Figure 3-14. Industry for Civilian Employed Population 16 Years and Over



Source: (U.S. Census n.d.)

Harris County has experienced fluctuations in unemployment rates over the years. In 2020, the unemployment rate peaked at 8.9 percent due to the economic impacts of the COVID-19 pandemic (U.S. Bureau of Labor Statistics, 2021). However, the rate gradually decreased, reaching 4.4 percent by the end of 2024, and historically Harris County's unemployment rate has mirrored national trends, with notable increases during economic downturns and decreases during periods of economic recovery (U.S. Bureau of Labor Statistics, 2025).

3.8.2 Major Institutions

Harris County, Texas, is a hub of major employers, educational institutions, human services providers, and recreational attractions. The County hosts several Fortune 500 companies, underscoring its significance in the energy and technology sectors (Feser 2024). Also located in Harris County are many health care institutions, some of which are among the largest employers, reflecting the County's robust health care industry (Sixel 2018). The County's diversity also extends into the educational landscape, which is enriched by institutions and universities that offer diverse academic programs, all of which contribute to workforce development in the planning area (Development n.d.). Table 3-5 summarizes the roster of major institutions and employers in Harris County.

Harris County provides extensive human services and recreational opportunities, all of which offer vital support services to residents (Harris County Resources for Children and Adults n.d.); (The Harris Center for Mental Health and IDD n.d.). The County's commitment to public health is evident through initiatives by Harris County Public Health, which delivers a range of clinical and preventive services (Harris County Public Health, n.d.).

Recreationally, the County boasts numerous parks and attractions, including George Bush Park, Terry Hershey Park, and the Houston Arboretum & Nature Center, providing residents and visitors with ample opportunities for outdoor activities and nature exploration (Harris County Precint 4 n.d., Komoot.com n.d.)

3.8.3 Income

In the United States, individuals and households are generally expected to rely on private resources to prepare for, respond to, and recover from disasters. This places an increased burden on individuals experiencing economic hardship, who may face systemic barriers to accessing the resources needed for adequate preparedness and recovery. Individuals living below the poverty line may reside in housing that is older, poorly maintained, or structurally vulnerable, such as mobile or modular homes. These types of housing are more susceptible to damage from hazards like floods, high winds, or earthquakes.

In urban areas, residents facing economic challenges may live in aging housing stock, including older apartment complexes or buildings constructed with unreinforced masonry, which is known to perform poorly during seismic events. Additionally, these individuals are less likely to have financial safety nets, such as insurance, to recover from disaster-related losses. As a result, those experiencing economic insecurity are at greater risk of losing essential belongings, shelter, and stability in the aftermath of a hazard event. These conditions highlight the need for equitable risk reduction strategies and targeted support for economically vulnerable populations to ensure all residents have the resources necessary to prepare for and recover from disasters.

Table 3-5. Major Institutions and Employers Located in Harris County

Name	Description
Fortune 500 Companies	
ExxonMobil	One of the world's largest publicly traded oil and gas companies; HQ in Spring (Harris County)
Chevron	Global headquarters are located in Harris County as well as numerous gas stations and convenience stores
ConocoPhillips	Energy corporation with a major presence in Houston
Hewlett-Packard Enterprise	Global headquarters are located in the new facility, which opened in 2022
Sysco Corporation	Global leader in food service distribution; HQ in Houston
Halliburton	Multinational oilfield services company; HQ in Houston
Phillips 66	Energy manufacturing and logistics; HQ in Houston
Health care Institutions	
Texas Medical Center	The largest medical complex in the world includes hospitals, research institutions, and universities
Houston Methodist	Major hospital system with several campuses in Harris County
Memorial Hermann Health System	One of the largest not-for-profit health systems in Texas
Harris Health System	Public health care provider serving Harris County residents
MD Anderson Cancer Center	Leading cancer treatment and research center; part of UT Health System
Educational Institutions	
University of Houston System	Major public university system headquartered in Houston
Texas Southern University	A historically black university offering undergraduate and graduate degrees
Rice University	A private research university located near the Texas Medical Center
Houston Community College System	Provides community college education across Harris County
Lone Star College System	One of the largest community college systems in Texas
Government & Human Services	
Harris County Government	One of the largest local employers; includes departments such as Public Health, Flood Control, etc.
City of Houston Government	Employs thousands in public works, emergency management, police, fire, and other services
METRO (Metropolitan Transit Authority)	Provides public transportation services throughout Harris County
Houston Food Bank	Largest food bank in the U.S. by distribution volume
Recreational & Cultural	
Houston Zoo	Popular zoological park located in Hermann Park
Houston Museum District	Includes museums such as the Museum of Fine Arts, Houston Museum of Natural Science, and more
NRG Park	Includes NRG Stadium (home of the Houston Texans), convention facilities, and event venues
Houston Livestock Show & Rodeo	Largest livestock exhibition and rodeo in the world
Harris County Precinct Parks	Large public parks supporting recreation, conservation, and emergency sheltering

Source: (Feser 2024, Sixel 2018, Development n.d., Harris County Resources for Children and Adults n.d., Komoot.com n.d.), (Harris County Public Health, n.d.), (Harris County Precinct 4, n.d.)

According to the U.S. Census Bureau's 2019–2023 estimates, the median household income in Harris County was $73,104, and the per capita income was $39,978, both adjusted to 2023 dollars (U.S. Census Bureau 2020). These figures represent significant increases from 2017, when the median household income was $57,791 and the per capita income was $30,856. In comparison, the State of Texas reported a median household income of $73,035 and a per capita income of $36,245 during the same period (U.S. Census Bureau 2020). Nationally, the United States had a median household income of $77,719 and a per capita income of $41,132 (U.S. Census Bureau 2020). These figures indicate that Harris County's median household income slightly surpasses the state average but remains below the national median.

Income plays a crucial role in determining the ability of individuals at risk to recover from natural disasters and extreme weather events. Higher income levels can provide greater access to resources such as insurance, emergency funds, and resilient housing, which are essential for effective recovery. Conversely, lower-income households may struggle to rebuild and recover, exacerbating the long-term impacts of such events.

3.8.4 Economic Trends

Harris County's economy has greatly diversified since 2020, shifting from a traditional dependence on energy and manufacturing toward a more varied structure that includes health care, education, construction, and financial services. The health care sector has become a key economic driver, anchored by the Texas Medical Center, the largest medical complex in the world, and supported by investments in health-related startups and biomedical research. The education sector has also expanded rapidly, with the number of establishments increasing by 61 percent from 2010 to 2023 (Understanding Houston, 2023). Meanwhile, the construction industry added 17,800 jobs over the year ending in October 2024, representing a 7.7 percent increase, while financial activities contributed 5,700 new jobs during the same period (U.S. Bureau of Labor Statistics [BLS], 2024).

Although manufacturing added 3,800 jobs in the same timeframe, the information sector experienced a decline, losing 500 jobs and highlighting a broader shift toward service and tech-oriented industries (BLS, 2024). Wage growth in Harris County has also outpaced national trends, with average weekly wages rising by 5.4 percent from Q3 2023 to Q3 2024, compared to the U.S. average of 4.5 percent. This brought the County's average weekly wage to $1,608, placing it among the top 50 nationwide (BLS, 2024). Additionally, economic development efforts have spurred major investments in renewable energy, electric vehicles, and advanced technology, with over $260 million funneled into clean energy manufacturing and innovation hubs, positioning Harris County as a growing center for sustainable industry (Financial Times, 2024).

3.9 GENERAL BUILDING STOCK

3.9.1 Existing Development

For the purposes of this plan, approximately 1,356,713 structures were identified by the tax data and spatial data available. These structures account for a replacement cost value of approximately $692,795,818,339. Estimated content value was calculated by using 50 percent of the residential

replacement cost value and 100 percent of the nonresidential replacement values. Using this methodology, approximately $538,633,327 in contents exist within these properties. Approximately 93.2 percent of the total buildings in the County are residential, which make up 73.5 percent of the total building stock value. Table 3-6 presents building stock statistics by occupancy class for Harris County.

According to 2020 Census data, 1,607,498 households are located in Harris County. A household includes all the people who occupy a housing unit as their usual residence. The Census data identified 1,780,203 housing units in the County. A housing unit is a house, apartment, mobile home or trailer, a group of rooms, or a single room occupied as separate living quarters (or if vacant, intended for occupancy as separate living quarters). According to the 2020 Census, there are 172,705 vacant housing units in the County (U.S. Census 2020).

Figure 3-15 through Figure 3-17 show the distribution and value density of residential, commercial, and industrial buildings in Harris County based on the Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018. Value density is the dollar value of structures per unit area, including building content value. The densities are shown in units of $1,000 ($K) per square mile. Viewing value distribution maps can assist communities in visualizing areas of high loss potential and in evaluating aspects of the study area in relation to the specific hazard risks.

3.9.2 New Development

In 2024, Harris County municipalities issued 12,909 new residential permits compared to 12,773 in 2021. There were two major multi-family projects proposed throughout the County in 2020, which provided about 470 housing units. Individual development projects are detailed in Volume II under each jurisdictional annex.

Table 3-6. Building Stock Count and Replacement Cost Value (RCV) by Occupancy Class

	Residential		Commercial		Industrial		Other[c]	
	Building Count	Total Replacement Cost Value	Building Count	Total Replacement Cost Value	Building Count	Total Replacement Cost Value	Building Count	Total Replacement Cost Value
Baytown[b]	18,985	$7,918,018,424	1,594	$3,983,197,317	12	$37,232,704	79	$538,654,610
Bellaire	6,307	$5,050,121,866	181	$1,683,162,118	0	$0	15	$188,097,253
Bunker Hill Village	1,300	$1,716,615,430	1	$1,785,121	0	$0	0	$0
Deer Park	10,575	$4,425,111,583	693	$2,081,805,450	77	$558,775,577	25	$154,674,131
El Lago	1,078	$544,347,226	18	$20,522,354	0	$0	7	$14,687,250
Friendswood[a]	12,934	$5,661,553,053	441	$1,738,672,012	3	$6,904,978	10	$73,720,793
Galena Park	3,032	$702,838,507	208	$411,151,025	20	$192,809,923	28	$177,343,254
Hedwig Village	818	$720,805,582	88	$735,350,875	0	$0	3	$7,609,559
Hilshire Village	311	$258,885,107	8	$6,508,684	0	$0	2	$2,040,523
Houston[b]	540,261	$328,062,000,000	44,154	$262,653,000,000	1,702	$14,980,530,605	2,027	$14,322,469,395
Humble	3,918	$1,827,039,643	831	$3,177,175,294	47	$281,025,185	33	$195,240,773
Hunters Creek Village	1,552	$2,161,337,984	17	$92,858,316	0	$0	3	$65,803,002
Jacinto City	2,745	$777,934,026	216	$699,632,609	4	$75,145,135	21	$66,997,660
Jersey Village	2,260	$1,893,847,425	169	$918,313,188	4	$20,618,824	6	$30,732,038
Katy[a]	5,611	$1,824,884,190	581	$1,717,514,393	40	$21,944,620	90	$225,442,787
La Porte	11,465	$4,138,437,606	877	$3,455,546,808	43	$133,078,203	39	$409,523,092
League City[b]	386	$324,574,120	7	$13,723,839	0	$0	1	$5,460,840
Missouri City[a]	23,411	$10,910,463,445	850	$3,413,361,754	22	$536,742,751	42	$84,474,104
Morgan's Point	150	$78,252,206	7	$137,654,342	3	$108,600,868	0	$0
Nassau Bay	1,504	$1,220,766,693	70	$883,692,435	0	$0	0	$0
Pasadena	37,128	$16,669,778,744	3,266	$9,604,829,312	82	$667,549,608	130	$990,051,186
Pearland[b]	766	$481,650,798	30	$194,202,676	13	$202,481,686	0	$0
Piney Point Village	1,188	$2,108,406,327	3	$59,467,270	0	$0	4	$215,839,084
Seabrook	4,102	$2,536,569,234	264	$542,652,046	2	$20,487,890	6	$16,240,412

	Residential		Commercial		Industrial		Other[c]	
	Building Count	Total Replacement Cost Value	Building Count	Total Replacement Cost Value	Building Count	Total Replacement Cost Value	Building Count	Total Replacement Cost Value
Shoreacres	656	$257,563,840	1	$10,478,745	0	$0	0	$0
South Houston[b]	4,050	$1,219,015,779	885	$1,286,572,286	26	$84,290,752	40	$98,266,369
Southside Place	491	$533,888,803	16	$67,508,484	0	$0	1	$3,348,050
Spring Valley Village	1,466	$1,131,096,866	38	$399,171,612	0	$0	5	$38,201,117
Stafford[a]	3,967	$1,915,414,324	1,047	$4,813,388,927	28	$447,971,497	13	$56,079,114
Taylor Lake Village	1,385	$850,001,766	5	$4,882,526	0	$0	0	$0
Tomball	2,916	$1,588,407,159	653	$2,309,312,201	26	$178,582,466	37	$215,886,899
Waller[a]	867	$164,702,559	191	$103,331,321	18	$31,574	21	$2,913,757
Webster	1,774	$2,254,329,733	583	$4,545,478,656	2	$5,990,111	20	$203,856,005
West University Place	5,468	$4,628,007,780	52	$137,529,918	0	$0	5	$16,883,964
Unincorporated County	546,176	$267,069,000,000	27,412	$97,915,344,602	2,647	$23,593,982,739	1,241	$13,573,672,659
Harris County Total	1,261,003	$683,625,667,828	85,457	$409,818,778,517	4,821	$42,154,777,698	3,954	$31,994,209,678

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018, RS Means 2024

[a]. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

[b]. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

[c]. Other = Government, Religion, Agricultural, and Education

Figure 3-15. Distribution of Residential Building Stock and Value Density in Harris County

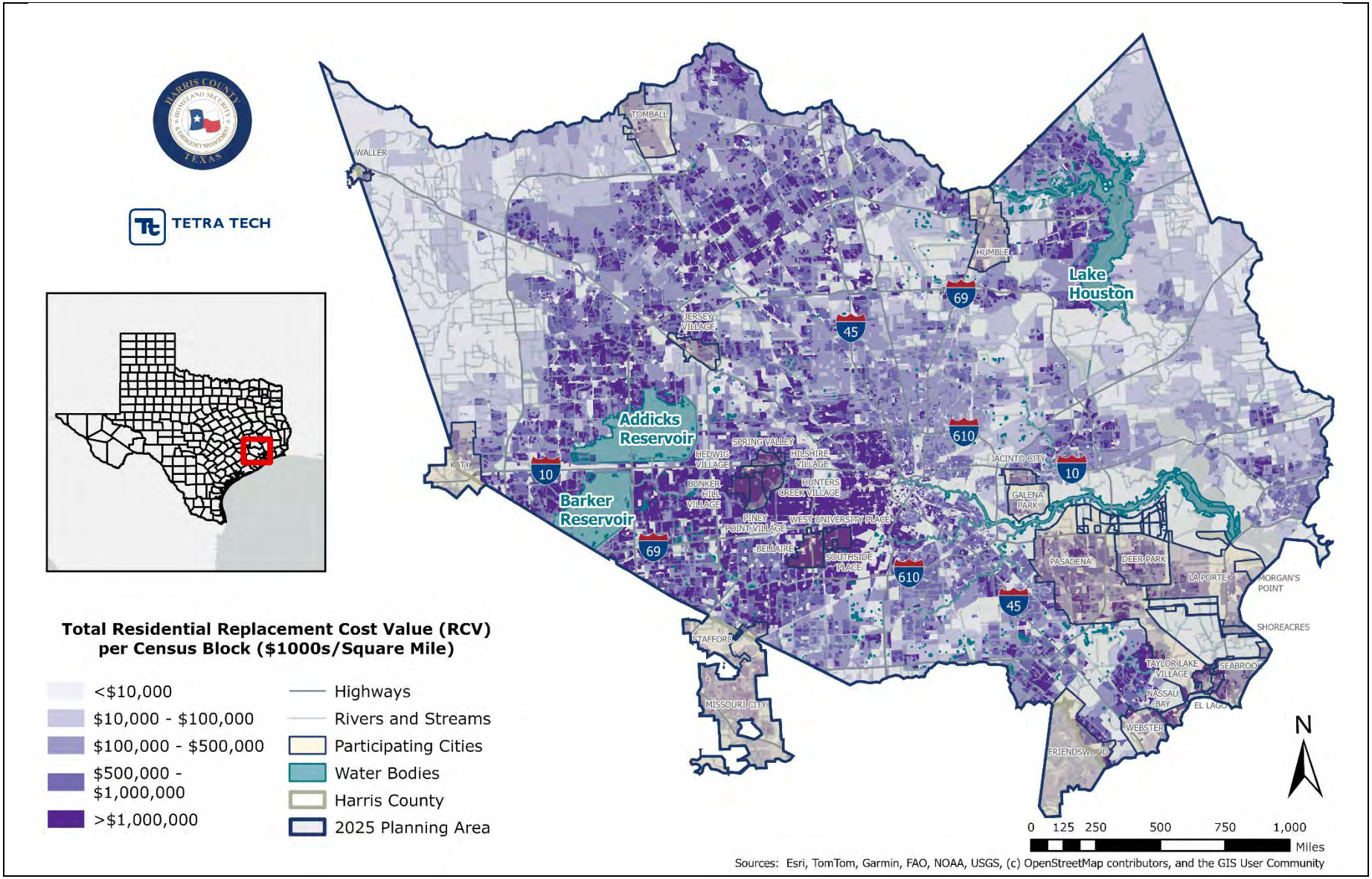


Figure 3-16. Distribution of Commercial Building Stock and Value Density in Harris County

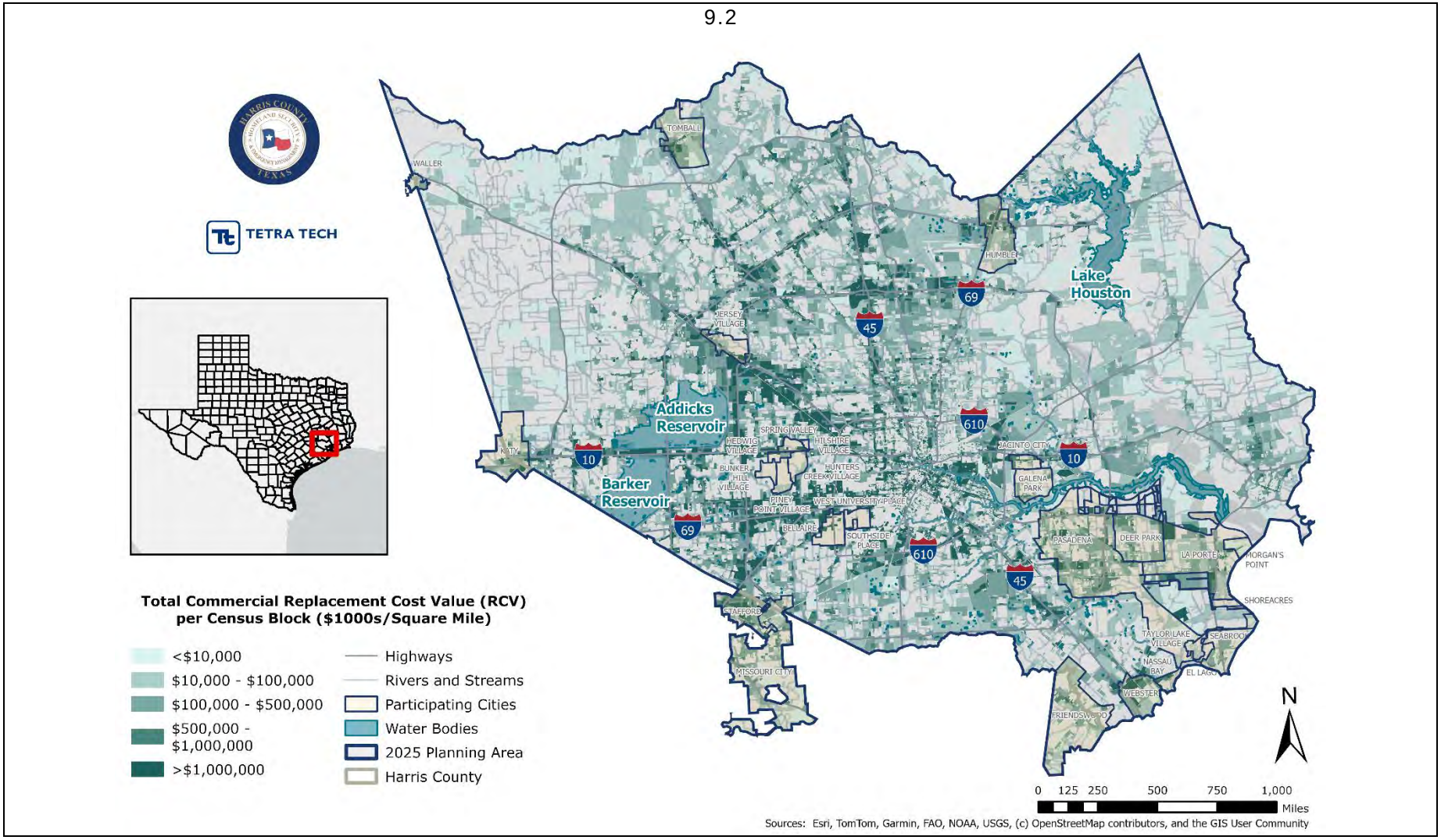
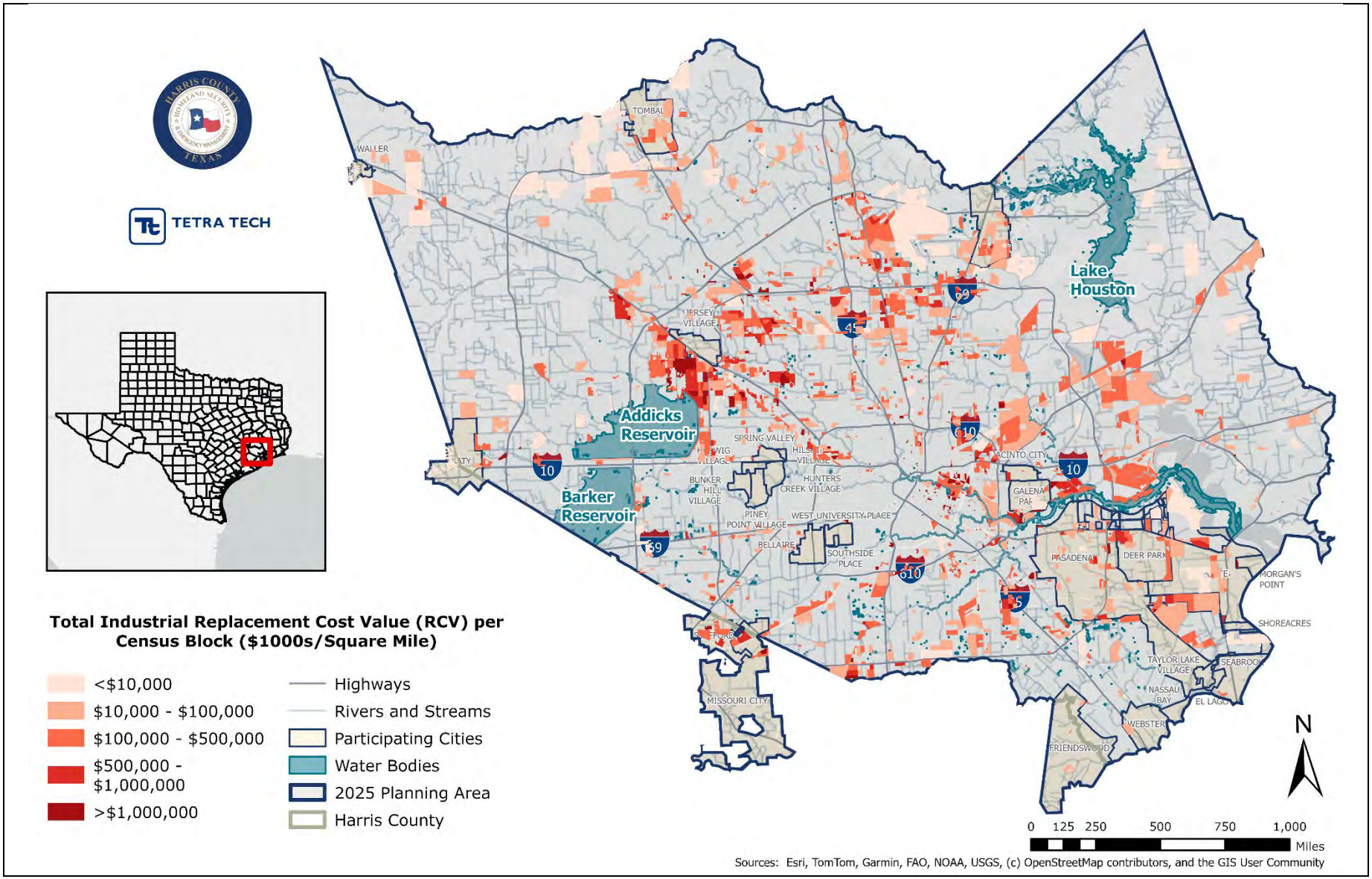


Figure 3-17. Distribution of Industrial Building Stock and Value Density in Harris County



3.10 COMMUNITY LIFELINES

Facilities that are essential to the health and welfare of the population and maintain essential and emergency functions are designated as critical facilities. These typically include police and fire stations, schools, emergency operations centers, and infrastructure such as roads, bridges, and utilities that provide water, electricity, and communications. Facilities that use or store hazardous materials are designated as critical facilities as well. All of these facilities are especially important after any hazard event (FEMA 1997).

FEMA defines some types of critical facilities, as well as public services or activities, as "community lifelines." Community lifelines provide the fundamental services in a community that, when stabilized, enable all other aspects of society. Following a disaster event, intervention is required to stabilize community lifelines. FEMA defines eight categories of community lifelines as summarized in Table 3-7.

A comprehensive inventory of community lifelines in Harris County was developed from various sources, including input from the Steering Committee and Planning Partnership. The following sections describe the inventory of community lifelines that was used for the risk assessment in this MJ-HMP. Although many lifeline facilities could fall within numerous categories, each lifeline facility identified for this planning effort was categorized according to its primary function.

3.10.1 Safety and Security

Figure 3-18 shows the location of safety and security facilities included in the lifelines inventory. Key facilities and services considered for the inventory under this category are as follows:

- HCOHSEM works along with the Texas Division of Emergency Management (TDEM) to ensure the County and state are prepared for and can respond to various emergencies and disasters. HCOHSEM is led by the County Judge, who also serves as the County's emergency manager.
  - Aids communities in emergency planning and response.
  - Provides training and equipment for first responders and volunteers.
  - Develops, maintains, and executes Harris County's Comprehensive Emergency Management Plan for disaster relief before, during, and after any type of natural or human-caused disaster (or a war-time situation).
- There are 115 fire department facilities in Harris County.

3.10.2 Food, Hydration, Shelter

Figure 3-19 shows the location of food, hydration, and shelter facilities included in the critical lifelines inventory.

Table 3-7. FEMA-Defined Categories of Community Lifelines

Community Lifeline Category		Types of Facilities and Services Included
	Safety and security	Law enforcement/security, fire service, search and rescue, government service, community safety
	Food, hydration, shelter	Food, hydration, shelter, agriculture
	Health and medical	Medical care, public health, patient movement, medical supply chain, fatality management
	Energy	Power grid, fuel
	Communications	Infrastructure, responder communications, alerts warnings and messages, finance, 911 and dispatch
	Transportation	Highway/roadway/motor vehicle, mass transit, railway, aviation, maritime
	Hazardous materials	Facilities, hazmat, pollutants, contaminants
	Water systems	Potable water infrastructure, wastewater management

Source: FEMA 2023

Figure 3-18. Safety and Security Facilities in Harris County

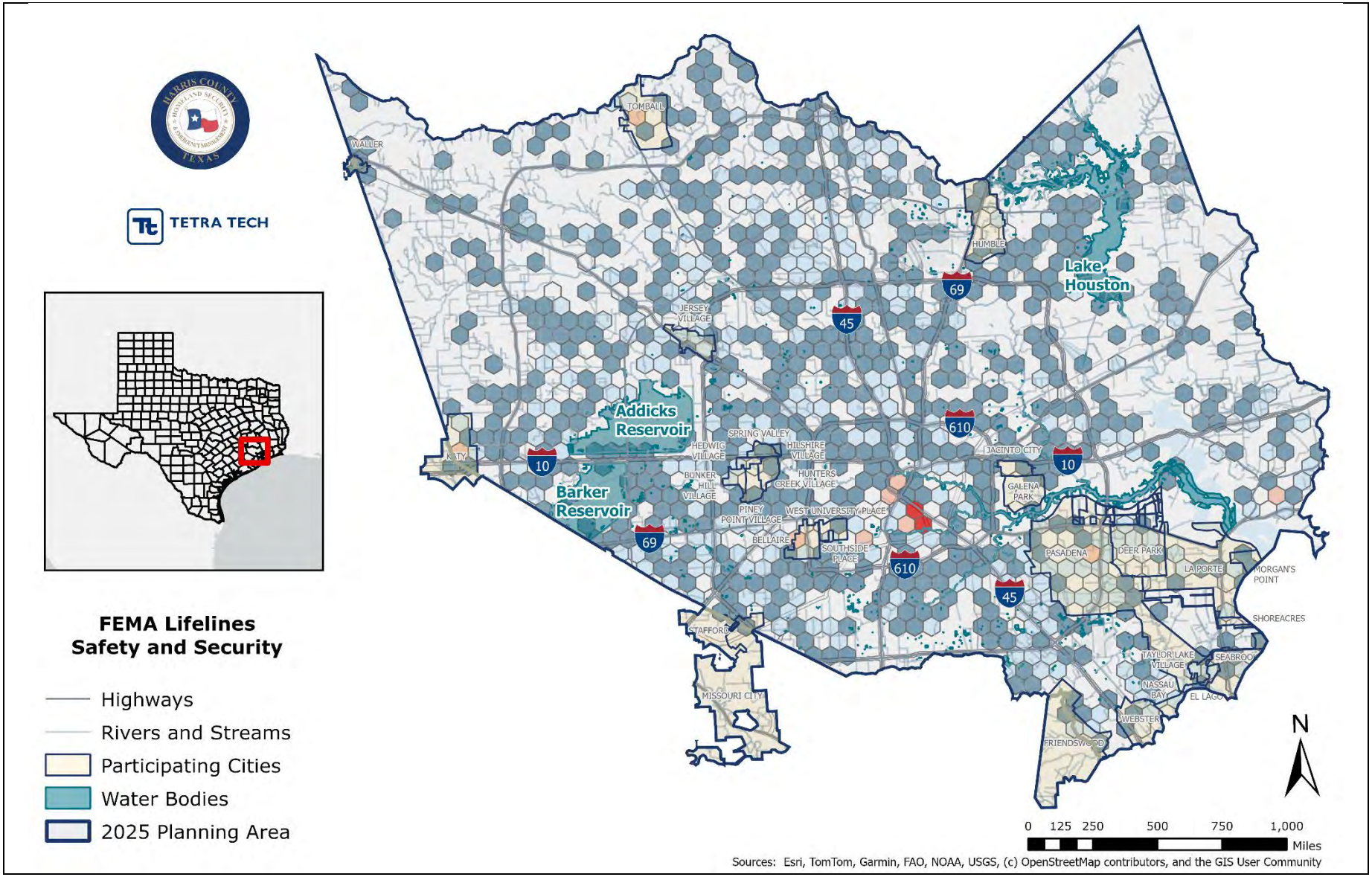
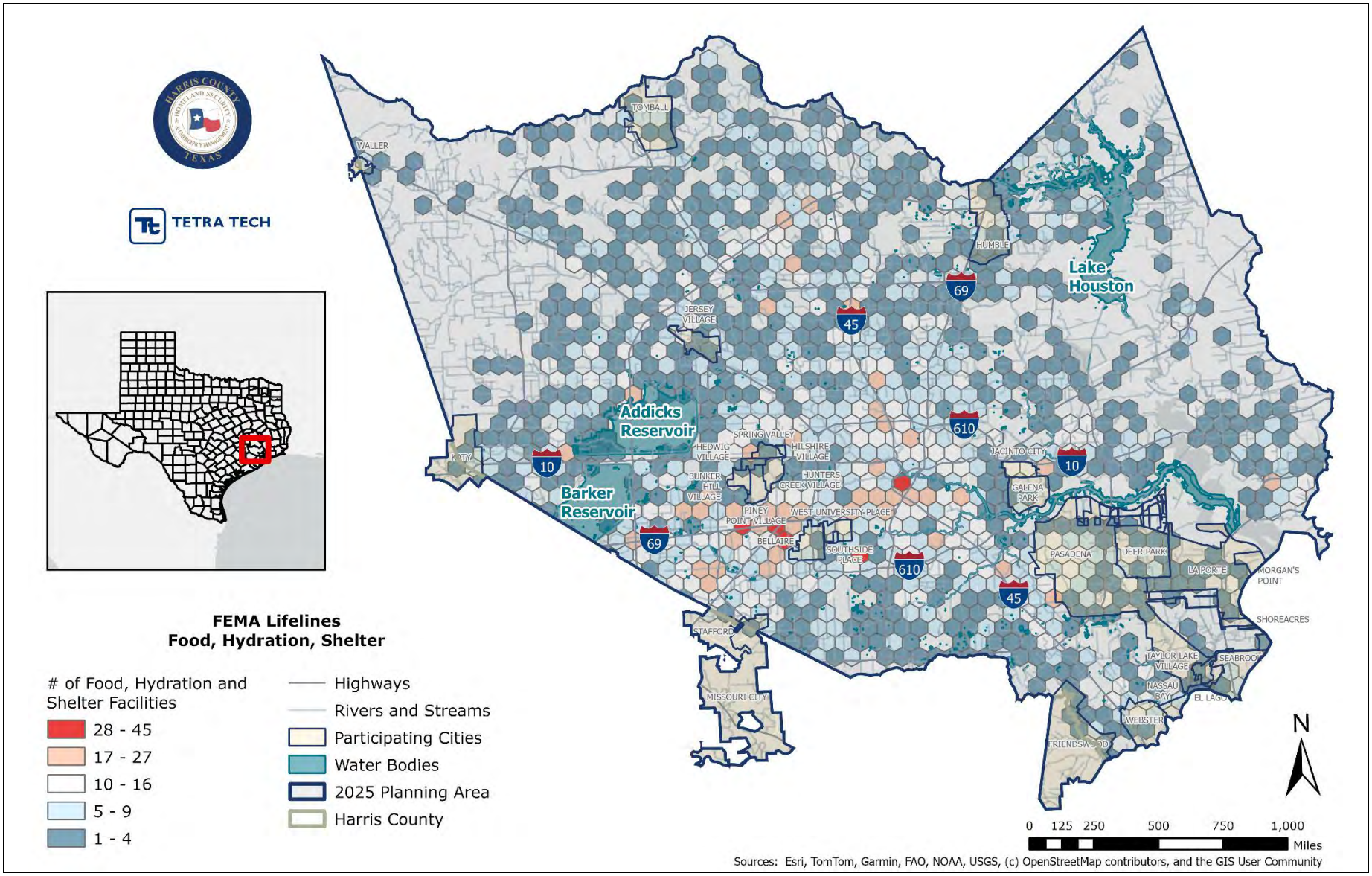


Figure 3-19. Food, Hydration, and Shelter Facilities in Harris County



3.10.3 Health and Medical

Figure 3-20 shows the location of health and medical facilities in the critical lifelines inventory. This list of facilities may not be completely comprehensive. Key facilities and services considered for the inventory under this category are as follows:

- Hospitals in Harris County consist of nursing homes, health centers or clinics, and hospital-sponsored medical practices:
  - HCA Houston Health care, Houston Methodist Hospital, Memorial Hermann Hospitals
  - Pam Rehabilitation Hospital
  - Avalon Memory, BROOKDALE NURSING HOMES, CYPRESS ASSISTED LIVING
- Harris County is served by a network of about 71 federally qualified Community Health Centers (FQHCs)

3.10.4 Energy

Figure 3-21 shows the location of energy facilities included in the lifelines inventory. Key facilities and services considered for the inventory under this category are as follows:

- Gas and electric power in Harris County are transmitted and distributed by these companies: CenterPoint Energy, Reliant Energy, TXU Energy, Direct Energy, Green Mountain Energy, Cirro Energy, Just Energy, Constellation, and Pulse Power.
- Homes in the County are heated by many different sources, with a majority using utility gas or electricity. A few residential areas are dependent on fuel oil or kerosene.

3.10.5 Communications

Figure 3-22 shows the location of communications facilities included in the lifelines inventory. Key facilities and services considered for the inventory under this category are as follows:

- Harris County HCOHSEM operates an emergency operations center in Houston, which serves as the coordination hub for emergency response, disaster management, and public safety efforts across the County during severe weather, disaster events, or other public emergencies.
- Communications systems in Harris County include traditional land lines, fiber optic, and cellular service provided by multiple companies. Each carrier has individual plans for emergency situations during hazard events and post-disaster recovery efforts.
- Harris County has an extensive radio communications network that is utilized by emergency services agencies, hospitals, law enforcement, public works, transportation, and other supporting organizations.
- There are 874 communication facilities in Harris County identified as critical facilities.

Figure 3-20. Health and Medical Facilities in Harris County

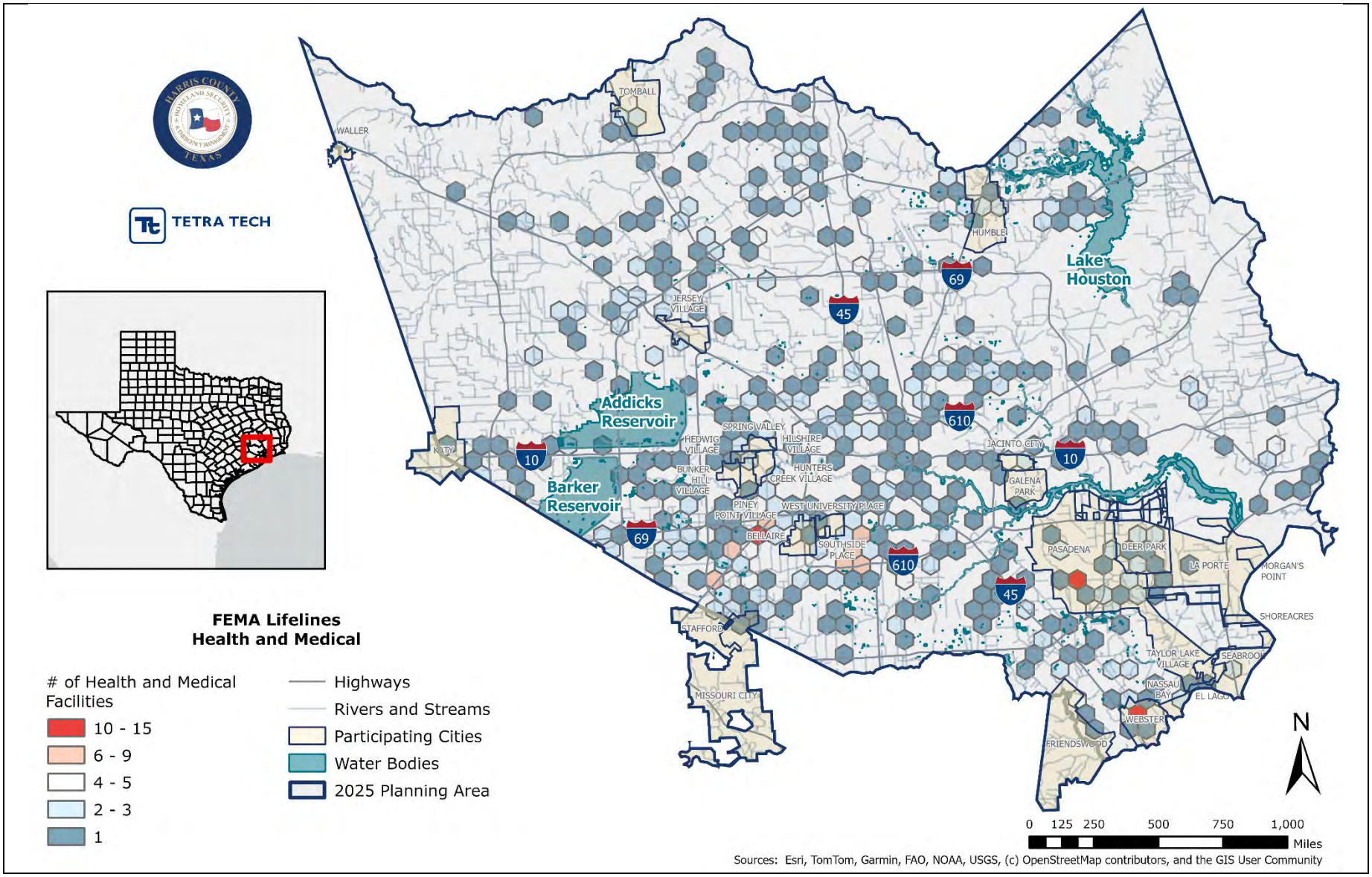


Figure 3-21. Energy Facilities in Harris County

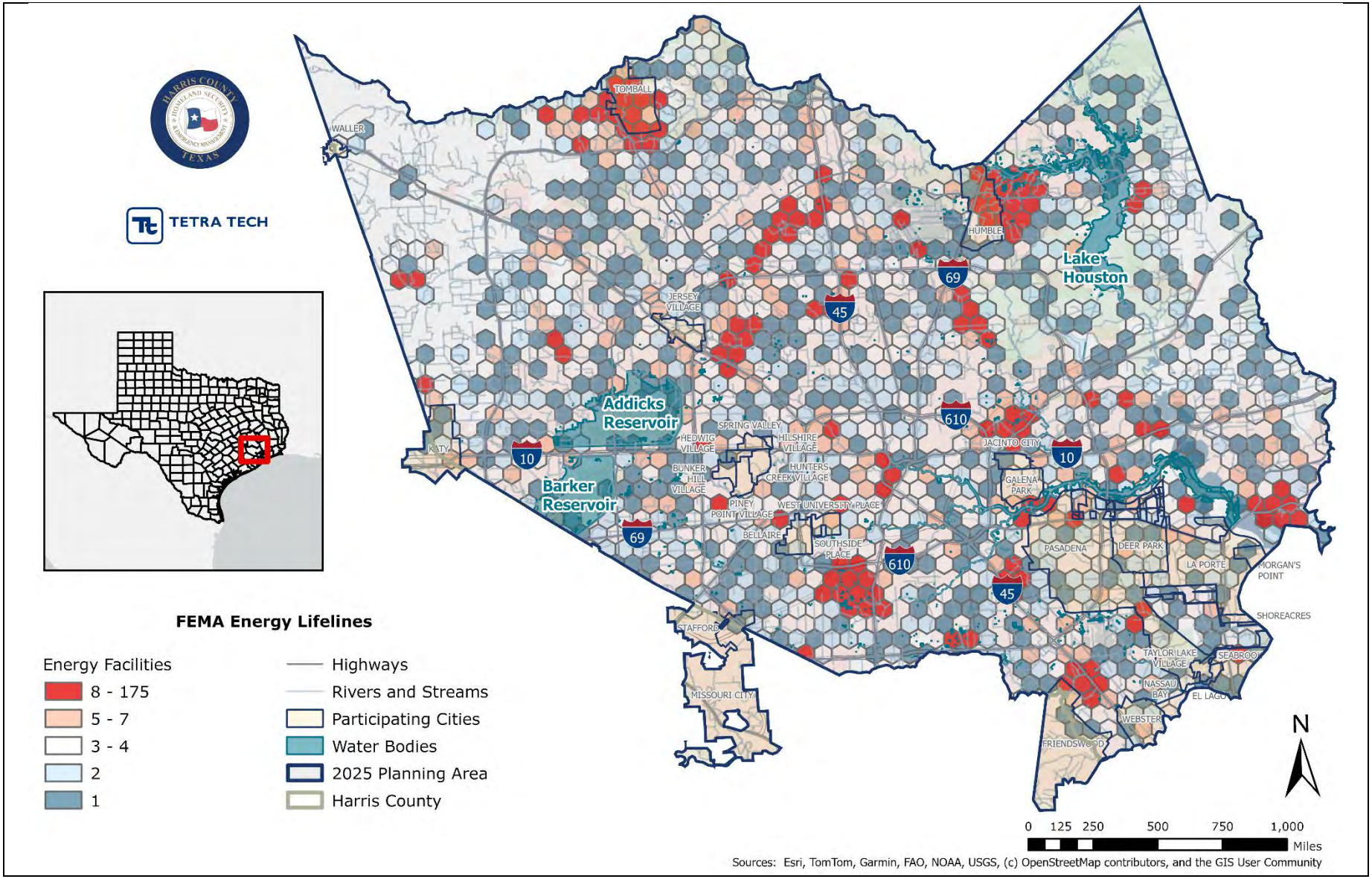
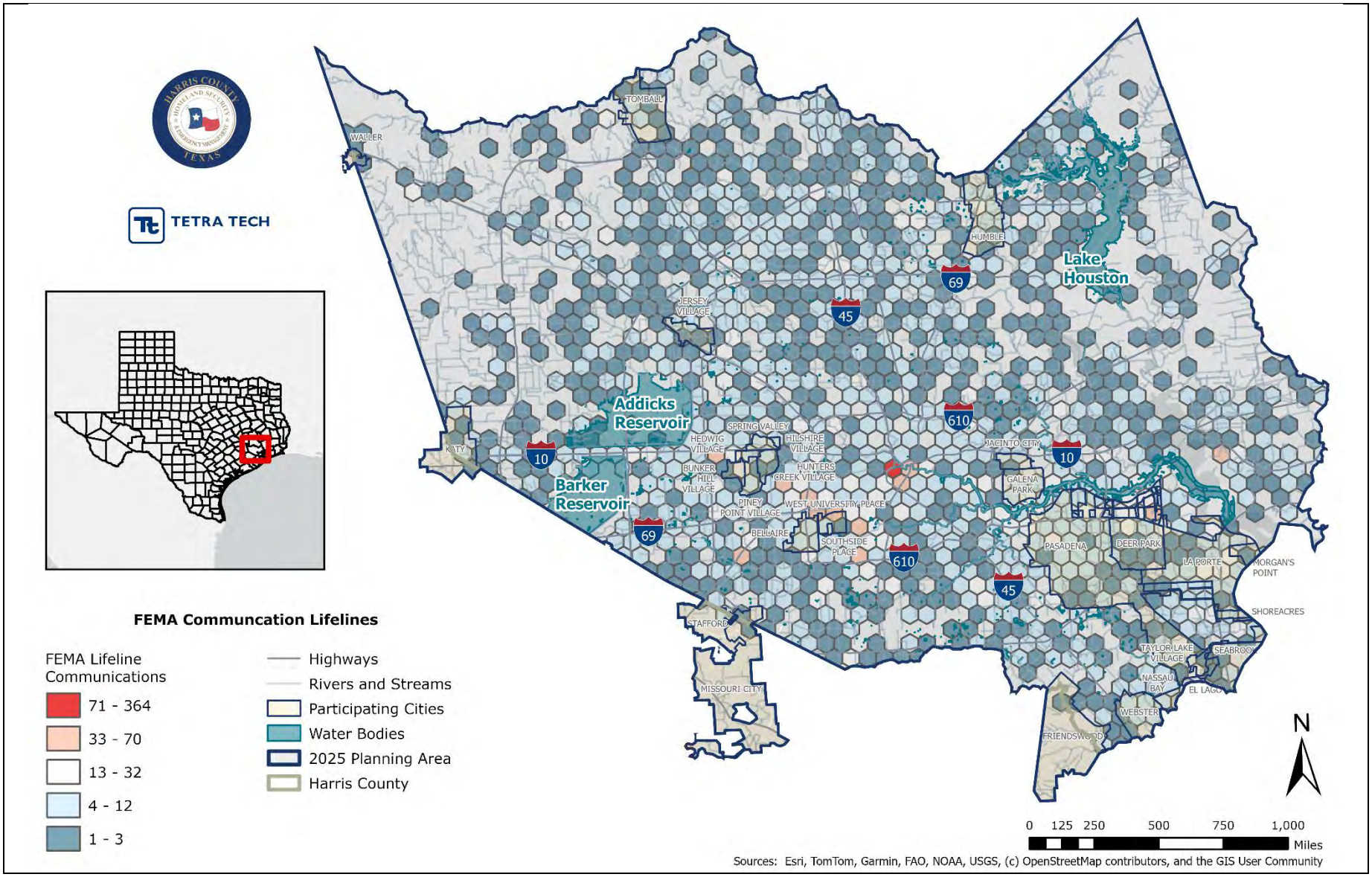


Figure 3-22. Communications Facilities in Harris County

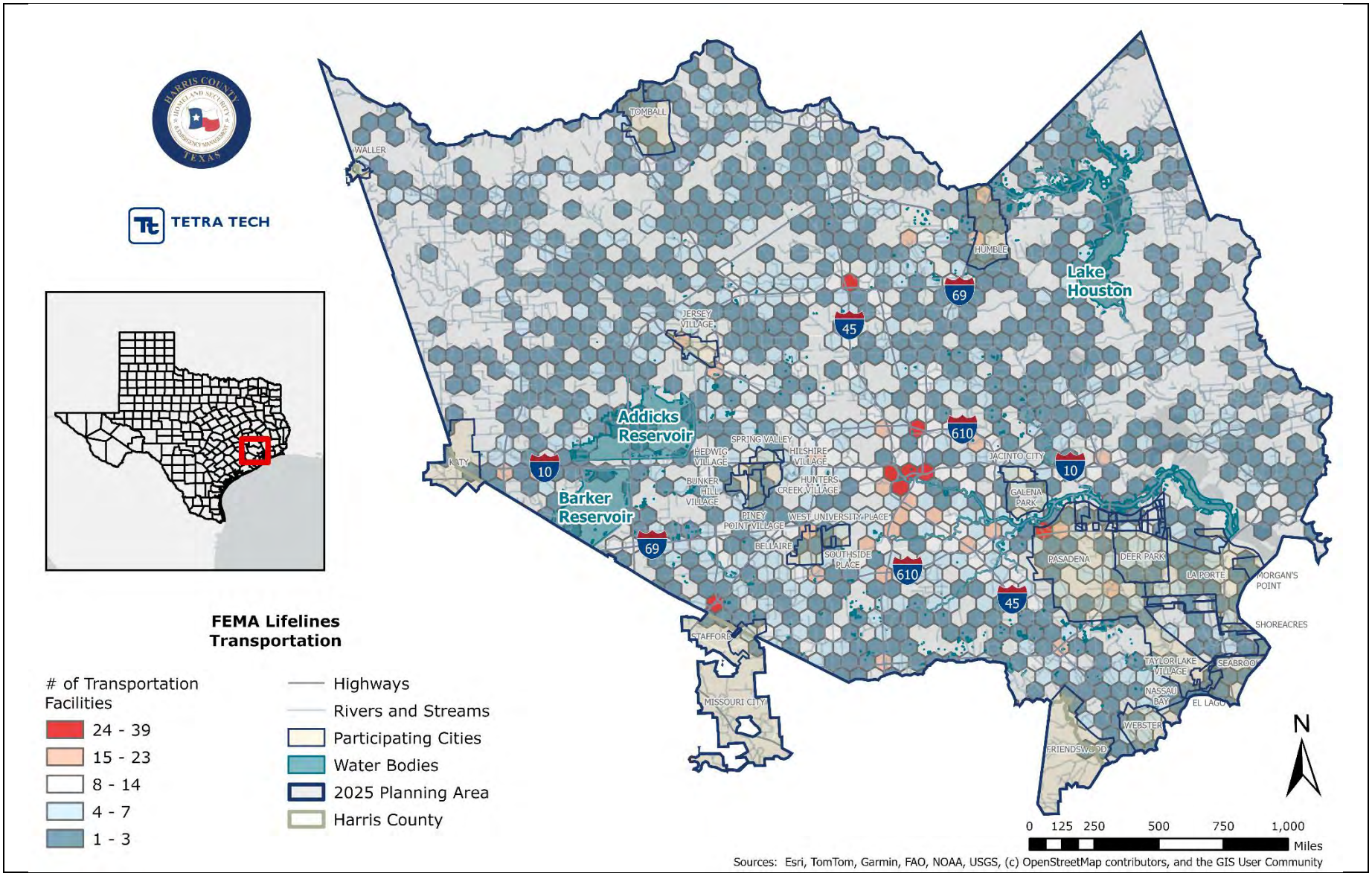


3.10.6 Transportation

Figure 3-23 shows the location of transportation facilities included in the lifelines inventory. Key facilities and services considered for the inventory under this category are as follows:

- I-10, also known as the Katy Freeway, was built in Harris County in 1972. I-10 enters the County from the west in Katy and passes through Houston, then continues east toward Beaumont.
- I-45, also known as the Gulf Freeway or North Freeway, was opened in 1948 and connects Houston to Galveston in the south and Houston to Dallas in the north.
- I-69 and I-59 enter Harris County from the southwest near Sugar Land, traverses downtown Houston, and continues northeast toward Cleveland.
- U.S. Highway 290 was originally designated in 1927 and has undergone several reconstruction projects, including the 2011 reconstruction project that expanded capacity between I-610 and the Grand Parkway with significant portions completed in 2018. The highway begins in northwest Houston and extends toward Austin.
- Public transportation by bus and light rail across most of the County is provided by the Metro Transit Authority. Amtrak provides passenger rail services.
- Harris County is a major freight rail hub served by Union Pacific Railroad, BNSF Railway, and Kansas City Southern. Key rail yards include Englewood Yard, Settegast Yard, and Hardy Street Yard, all located in or near Houston. These rail lines support intermodal and bulk freight movement for the petrochemical, construction, agriculture, and manufacturing sectors.
- Amtrak's Sunset Limited line runs through Harris County, with service at Houston Station. This long-distance line connects Los Angeles to New Orleans, with stops in San Antonio and other Texas cities. The station is located at 902 Washington Avenue and serves regional passengers three times a week in each direction (Amtrak 2023).
- Harris County is home to two major commercial airports: George Bush Intercontinental Airport (IAH) and William P. Hobby Airport (HOU). IAH is one of the nation's top 15 busiest airports, handling over 40 million passengers annually, and serves as a major hub for United Airlines. Hobby Airport supports both domestic and limited international flights and is a base for Southwest Airlines (Houston Airport System 2024).
- The Houston Ship Channel is a 52-mile-long waterway supporting deep-draft vessel access to the Port of Houston. It accommodates more than 200 private and public terminals, facilitating transport of petroleum, chemicals, steel, and containerized cargo. The port supports over 1.35 million jobs statewide and generates $802 billion in annual economic impact.

Figure 3-23. Transportation Facilities in Harris County



3.10.7 Hazardous Materials

Figure 3-24 shows the location of hazardous materials facilities included in the lifelines inventory. Due to security concerns, local hazardous materials' lifeline data was only partially obtained. Key facilities and services considered for the inventory under this category are as follows:

- Harris County contains several active and former Superfund sites, including the San Jacinto River Waste Pits, Brio Refining, French Ltd., and Highlands Acid Pit. These sites are associated with contamination from industrial activities involving dioxins, volatile organic compounds (VOCs), and heavy metals.
- Under the Emergency Planning and Community Right-to-Know Act, facilities in Harris County storing hazardous chemicals above certain thresholds are required to submit Tier II reports. These reports provide critical information about chemical types, quantities, and storage locations, aiding emergency responders in risk assessment and preparedness.
- The Harris County Fire Marshal's Office mandates operational permits for facilities that store, handle, or use hazardous materials beyond specified quantities. These permits ensure compliance with safety regulations and facilitate proper emergency response planning.
- The Harris County Household Hazardous Waste Collection Facility provides residents with a designated site to safely dispose of household hazardous materials. The facility operates by appointment, ensuring controlled handling of items like paints, solvents, and batteries.
- The Harris County Hazardous Materials Response Team, established post-September 11, 2001, operates under the Harris County Fire Marshal's Office. The team is equipped to respond to chemical spills, leaks, and other hazardous material incidents, enhancing the County's emergency preparedness and public safety infrastructure.

3.10.8 Water Systems

Figure 3-25 shows the location of water system facilities included in the lifelines inventory. Due to security concerns, water system lifeline data was only partially obtained. Key facilities and services considered for the inventory under this category are as follows:

- Harris County's public water supply is managed by a combination of municipal utilities, regional water authorities, and the City of Houston. Additionally, regional water authorities like the West Harris County Regional Water Authority and the North Harris County Regional Water Authority coordinate with municipal utility districts to deliver water services across the County.
- In the unincorporated and rural areas of Harris County, some residents rely on private wells for their water supply. These wells are largely unregulated, and homeowners are responsible for testing and maintaining water quality. The Texas Groundwater Protection Committee provides resources to assist private well owners in ensuring safe drinking water.
- Harris County hosts a network of wastewater treatment infrastructure. There are about 450 municipal sewage wastewater treatment plants and 212 active industrial wastewater treatment facilities permitted to discharge treated wastewater throughout the County.

Figure 3-24.Hazardous Material Lifelines in Harris County

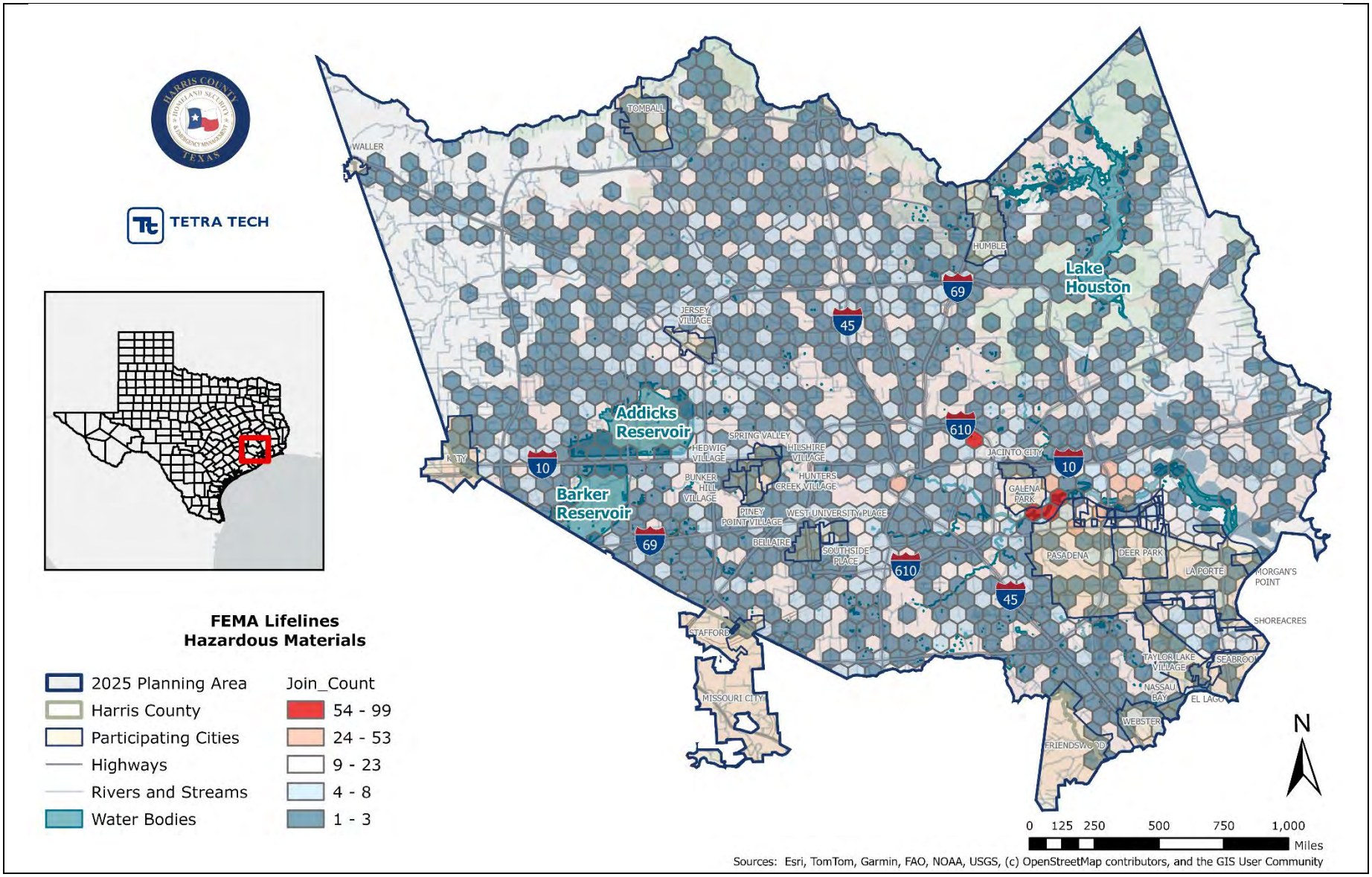
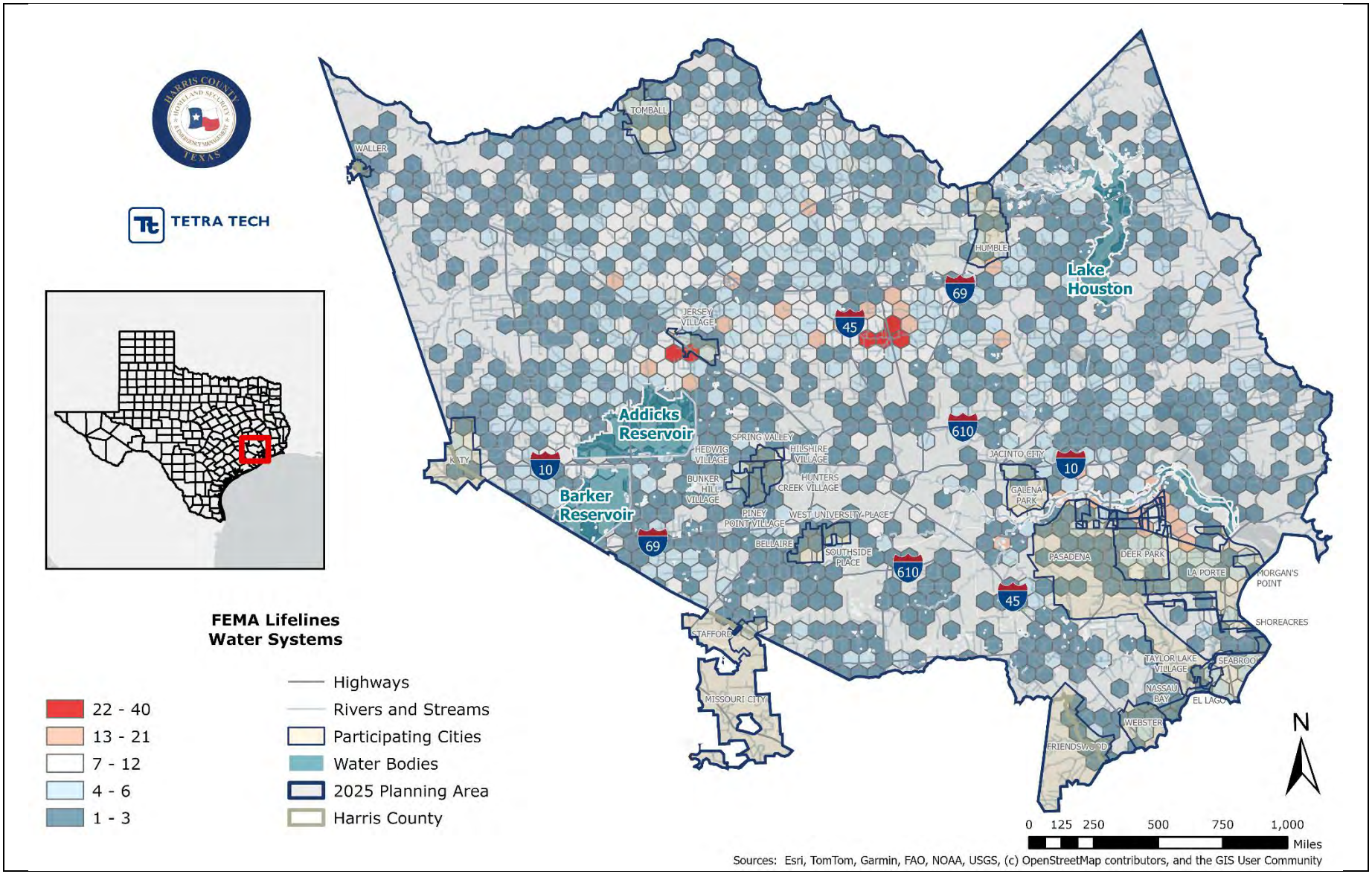


Figure 3-25. Water System Lifelines in Harris County



- In areas lacking adequate fire hydrant coverage, Harris County fire departments use alternative water sources for firefighting. These include tanker trucks capable of transporting over 3,000 gallons of water, as used by departments like the Spring Fire Department. Additionally, some departments access ponds or other surface water sources equipped with suction points to supply water during firefighting operations.
- The Texas Commission on Environmental Quality (TCEQ) oversees public drinking water systems in Harris County, ensuring compliance with state and federal standards. Harris County Public Health also plays a role in assessing water quality and addressing potential health risks. Regular monitoring and reporting help maintain the safety and reliability of the County's water supply.

3.11 OTHER CRITICAL FACILITIES

Some facilities that are identified as critical for hazard mitigation in Harris County do not fit in any of FEMA's community lifeline categories.

3.12 NATURAL, HISTORIC, AND CULTURAL RESOURCES

Harris County is rich in natural, historic, and cultural resources that shape its identity and contribute to the region's economic vitality, quality of life, and resilience. As the home of Houston—the fourth-largest city in the United States—Harris County offers a diverse array of heritage landmarks, parks, cultural institutions, and ecological features that attract millions of visitors each year (Visit Houston n.d.).

Harris County's proximity to the Gulf Coast and its extensive bayou network define its natural landscape. The County contains over 2,500 miles of natural and engineered waterways, including Buffalo Bayou, Brays Bayou (see Figure 3-26), and the San Jacinto River. These natural features are integral to flood management and provide opportunities for recreation and ecological preservation (Harris County Flood Control District n.d.).

Figure 3-26. Mason Park, Brays Bayou

The County is home to over 400 parks and green spaces, including the expansive George Bush Park and Hermann Park, which serve as vital community gathering places and nature preserves (Harris County Precinct 4 n.d.). Trails in Texas cater to three of the state's most favored recreational activities: walking, bicycling, and running. They accommodate a larger number of people at a lower cost compared to most other recreational facilities. According to a study by Texas Parks and Wildlife, nearly 70 percent of Texans enjoy walking for leisure, making it the most popular recreational activity in the state (Harris County Flood Control District n.d.).

Harris County is home to several protected habitats and wildlife areas, including the Armand Bayou Nature Center and Sheldon Lake State Park. These areas support biodiversity, offer environmental education programs, and act as buffers against climate impacts like flooding and extreme heat. Sheldon Lake State Park & Environmental Learning Center, a 2,800-acre park in northeast Harris County, was purchased by Texas Parks and Wildlife Department in the 1950s to provide a refuge for migratory waterfowl, a public fishing lake, and a fish hatchery (Texas Parks and Wildlife n.d.). It opened in 1955 as the Sheldon Wildlife Management Area and became a state park in 1984. Urban development has transformed the surrounding area, making Sheldon Lake a green and blue oasis for wildlife and people. The Sheldon Reservoir, constructed in 1942, now encompasses over 1,200 acres, including 800 acres of water and 400 acres of marsh and swampland (Texas Parks and Wildlife n.d.).

Armand Bayou Nature Center was founded in 1974 to preserve 2,500 acres of land (Armand Bayou Nature Center n.d.). In 2024, the nature center expanded its preserve by adding 1,147 acres through collaboration with Harris County Precinct 2 and ExxonMobil, bringing the total acreage to 3,947, making it the largest urban wilderness preserve in Texas (Armand Bayou Nature Center n.d.).

Culturally, Harris County is one of the most diverse regions in the country and features institutions that reflect its multicultural heritage. Houston's Museum District includes the Houston Museum of Natural Science, the Museum of Fine Arts, and the Contemporary Arts Museum, which collectively attract millions of visitors annually. Other notable sites include the Holocaust Museum Houston and the Buffalo Soldiers National Museum, which highlight important facets of American and African American history (Houston Museum District, n.d.).

The County is also home to numerous historic sites listed on the National Register of Historic Places, including the San Jacinto Battleground State Historic Site, where Texas won independence from Mexico in 1836. The San Jacinto Monument and Museum of History provide educational opportunities and commemorate the state's origins (Texas Historical Commission, n.d.). Additionally, many neighborhoods in Houston, such as the East End and Freedmen's Town, are designated historic districts that celebrate architectural heritage and cultural resilience (City of Houston, n.d.).

These natural, historic, and cultural resources contribute to the vibrancy and livability of Harris County. However, they are also vulnerable to natural hazards, such as hurricanes, flooding, and extreme heat. Integrating these resources into resilience and mitigation planning efforts is critical to preserving their benefits for future generations. Doing so not only enhances community identity and cohesion but also supports sustainable recovery.

Part 2: Risk Assessment

4. RISK ASSESSMENT METHODOLOGY

A risk assessment is the process of evaluating the potential loss of life, personal injury, and economic and property damage that could result from the identified hazards. Identifying potential hazards and vulnerable assets allows planning personnel to address and reduce hazard impacts and allows emergency management personnel to establish early response priorities. Results of the risk assessment are used in subsequent mitigation planning processes, including determining and prioritizing mitigation actions that reduce each jurisdiction's risk from each hazard. Past, present, and future conditions must be evaluated to assess risk most accurately for the county and participating jurisdictions. The process focuses on the following elements:

- Identify Hazards of Concern—Use all available information to determine what types of hazards may affect a jurisdiction.
- Profile Each Hazard—Understand each hazard in terms of:
  - Extent—The potential severity of each hazard
  - Location—Geographic area most likely to be affected by the hazard
  - Previous occurrences and losses
  - Impacts of Climate Change
  - Probability of future hazard events
- Assess Risk—Use all available information to estimate to what extent populations and assets may be adversely affected by a hazard:
  - Determine vulnerability—Estimate the total number of assets in the jurisdiction that are likely to experience a hazard event if it occurs by overlaying hazard maps with the asset inventories.
  - Estimate potential impacts—Assess the impact of hazard events on the people, property, economy, and lands of the region, including estimates of the cost of potential damage or cost that can be avoided by mitigation.
  - Evaluate future changes that may affect vulnerability and impacts—Analyze how demographic changes, projected development and climate change impacts can alter current vulnerability and potential impacts.

The Harris County risk assessment was updated using the following best-available information:

- An updated building stock inventory was generated using 2018 building footprints, tax assessor and parcel data provided by Harris County Appraisal District; 2016 building footprints, 2018 tax assessor and parcel data provided by Fort Bend Appraisal District; 2018 tax assessor and parcel data provided by Galveston County Appraisal District; and 2024 RS Means cost adjustment values.
- 2020 Decennial Census Population data, and 2019-2023 American Community Survey 5-year Population Estimates.
- Critical facilities data were provided by Harris County. The critical facilities have been derived from various sources. The critical facilities analyzed are only within the boundaries of Harris County, as critical facilities for the entire planning area were not made available.

- Lifelines were identified in the critical facility inventory to align with FEMA's community lifeline definition.
- FEMA's Hazus program was used to estimate potential impacts from the flood, wind, and seismic hazards.
- Best-available hazard data were used, as described in this section.

4.1 ASSET INVENTORIES

Harris County assets were identified to assess potential vulnerability and impacts associated with the hazards of concern. For the HMP update, Harris County assessed vulnerability and potential hazard impacts for the following types of assets: population, buildings, critical facilities, community lifelines, and the environment Some assets may be more susceptible to impacts because of their physical characteristics or socio-economic uses. Each asset type is described below. To protect individual privacy and the security of critical facilities, information on properties assessed is presented in aggregate, without details about specific individual personal or public properties.

4.1.1 Population

The risk assessment included the collection and use of an expanded and enhanced asset inventory to estimate hazard vulnerability and impacts.

Statistics from the 2020 Decennial Census Population estimate and 2019-2023 American Community Survey (ACS) 5-year estimate were used to estimate the vulnerability of and potential impacts on the County's population. To determine population statistics for the cities that go beyond the Harris County limits, StatsAmerica was used to determine the percentage of the population that lives within each respective jurisdiction by county. Population counts at the jurisdictional level were averaged among the residential structures in the county to estimate the population at the structure level. This estimate provides a more precise distribution of population across the county compared to only using the Census block or Census tract boundaries. Limitations of these analyses are recognized, and thus the results are used only to provide a general estimate for planning purposes.

FEMA's Hazus program was used to model estimated potential losses to flood, seismic and wind hazards, as discussed further later in this section. Hazus still contains 2020 U.S. Census data and was used to estimate sheltering and injuries as part of the hazard analysis.

As discussed in Section 4, County Profile, research has shown that some populations are at greater risk from hazard events because of decreased resources or physical abilities. Vulnerable populations in Harris County included in the risk assessment are children, older adults, population below the poverty level, individuals with limited English proficiency, and persons institutionalized with a disability.

4.1.2 Buildings

Replacement cost values and structure information derived from parcel and tax assessor data provided by Harris County (2018) were loaded into Hazus. Replacement cost is the cost to replace the entire structure with one of equal quality and utility. Replacement cost is based on industry-

standard cost-estimation models published in *RS Means Square Foot Costs* (RS Means, 2024). It is calculated using the RS Means square foot cost for a structure, which is based on the Hazus occupancy class (i.e., multi-family residential or commercial retail trade), multiplied by the square footage of the structure from the tax assessor data. The construction class and number of stories for single-family residential structures also factor into determining the square foot costs.

As several of the planning partner jurisdictions extend into neighboring counties (Fort Bend and Galveston County), parcel and tax assessor data (2018) for these areas were also acquired. The Hazus default inventory was used for Waller County.

A regional location factor for the project area was applied based on the zip code locations, as listed in Table 4-1. Location factors are designated based upon a city or a larger rural area. The following location factors were applied to the custom-built inventories (Galveston, Fort Bend, Harris).

Table 4-1. RCV Location Factor Based on Zip Code

First Three Digits of the ZIP Code	RCV Regional Location Factor	
	Residential	Non-Residential
770xx – 772xx	0.85	0.87
773xx	0.76	0.81
774xx	0.77	0.82
775xx	0.81	0.84
777xx	0.83	0.85

4.1.3 Critical Facilities and Community Lifelines

A lifeline enables the continuous operation of critical government and business functions and is essential to human health and safety or economic security

A critical facility inventory, which includes essential facilities, utilities, transportation features and user-defined facilities, were sent by HCOHSEM. The critical facility inventory is derived from various sources such as: ATLAS; BTS/USDOT; CIMA; City of La Porte; City of Seabrook; FleetCor; Harris County; HCOHSEM; Harris County Public Library; HCUS; HCPCS; HIFLD; METRO; NPMS; SETRAC; TCEQ; Regional Catastrophic Preparedness Grant Program; DSHS; TEA; THC; USACE. The critical facilities analyzed are only within the boundaries of Harris County. To protect individual privacy and the security of assets, information is presented in aggregate, without details about specific individual properties or facilities.

4.1.4 Environment and Land Use

The Land Use data was created by the Houston-Galveston Area Council (2022). It was used to assess land use characteristics of the county. The land cover datasets were produced by the Data Analytics and Research department at the Houston-Galveston Area Council for the purpose of developing land cover data product with a level of accuracy suitable for planning and research applications at the regional scale.

The dataset was developed based on Landsat 8 Operational Land Imager and the Thermal Infrared Sensor scenes of 30m resolution acquired from the USGS. Twelve scenes with 0-5 percent cloud cover were used to cover the entire area for diverse seasonal coverage. Using pixel classification with Deep Learning techniques available in ESRI ArcGIS Pro product with Image Analyst extension, the imageries were classified into 15 major land cover classes. Classification classes were determined based on classification schemes from the National Land Cover Database. The output cell size resolution is 30 meters (98 feet).

H-GAC provided the following disclaimer regarding the accuracy, timeliness, and completeness of the data.:

> "The data which you are being provided is the property of the Houston-Galveston Area Council (H-GAC). H-GAC has developed this data from a variety of datasets and inputs, and the data represents H-GAC's best judgment as to the values provided within the datasets provided. H-GAC does not warrant the accuracy, timeliness, or completeness of the data being provided. Moreover, H-GAC makes no guarantees regarding the positional accuracy and/or value accuracy of the attribute data being provided. H-GAC is not required to notify the user of any changes, revisions, additions, and/or deletions to the data being provided. H-GAC cannot be held responsible for any changes, revisions, additions, deletions, and/or misrepresentations that the user makes regarding this data. H-GAC also does not warrant the accuracy, timeliness, and/or completeness of such changes, revisions, additional, deletions, and/or representations."

4.1.5 New Development

Integrating new development into the risk assessment provides information for communities to consider when developing a mitigation strategy to reduce vulnerabilities in the future. This MJ-HMP considers recent development over the last five years and anticipated new development in the next five years. In lieu of an analysis conducted in the geographic information system (GIS), a qualitative analysis was employed using information provided by the county or obtained from online research.

4.2 METHODOLOGY

Harris County used standardized tools, combined with local, state, and federal data and expertise to assess potential vulnerability and losses associated with hazards of concern. Three levels of analysis were used, depending upon the data available for each hazard:

- Historical Occurrences and Qualitative Analysis – This analysis includes an examination of historical impacts to understand potential impacts of future events of similar size. Potential impacts and losses are discussed qualitatively using best-available data and professional judgment.
- Vulnerability Analysis – This analysis involves overlaying available spatial hazard layers, for hazards with defined extent and locations, on asset mapping in GIS to determine which assets are located in the impact area of the hazard.

- Loss Estimation — The FEMA Hazus modeling software was used to estimate potential losses for the following hazards: flood, earthquake, and hurricane.

Table 4-2 summarizes the type of analysis conducted by the hazard of concern.

Table 4-2. Summary of Risk Assessment Analyses

Hazard	Population	General Building Stock	Critical Facilities	New Development
Coastal Erosion	Q	Q	Q	Q
Dam/Levee Failure	V	V	V	Q
Drought	Q	Q	Q	Q
Earthquake	H	H	H	Q
Extreme Cold	Q	Q	Q	Q
Extreme Heat	Q	Q	Q	Q
Flooding	V, H	V, H	V, H	Q
Hurricane, Coastal Storm	H	H	H	Q
Mass Movement	Q	Q	Q	Q
Severe Weather	Q	Q	Q	Q
Tsunami	Q	Q	Q	Q
Wildfire	V	V	V	Q

Note: V = Vulnerability analysis; H = Hazus analysis; Q = Qualitative analysis

4.2.1 Hazus

Hazus is a GIS-based software tool developed by FEMA that applies engineering and scientific risk calculations, which have been developed by hazard and information technology experts, to provide defensible damage and loss estimates. These methodologies are accepted by FEMA and provide a consistent framework for assessing risk across a variety of hazards. The GIS framework also supports the evaluation of hazards and assessment of inventory and loss estimates for these hazards.

Hazus uses GIS technology to produce detailed maps and analytical reports that estimate a community's direct physical damage to building stock, critical facilities, transportation systems and utility systems. To generate this information, Hazus uses default data for inventory, vulnerability, and hazards; this default data can be supplemented with local data to provide a more refined analysis. Damage reports can include induced damage (inundation, fire, threats posed by hazardous materials and debris) and direct economic and social losses (casualties, shelter requirements, and economic impact) depending on the hazard and available local data. Hazus' open data architecture can be used to manage community GIS data in a central location. The use of this software also promotes consistency of data output now and in the future and standardization of data collection and storage. More information on Hazus is available at http://www.fema.gov/hazus.

In general, modeled losses were estimated in the program using depth grids for the flood analysis and probabilistic analyses were performed to develop expected or estimated distribution of losses (mean return period losses) for hurricane wind and seismic hazards. The probabilistic model generates estimated damages and losses for specified return periods (e.g., 100- and 500-year). Table 4-3 displays the levels of analysis that can be conducted using the Hazus software.

Table 4-3. Summary of Hazus Analysis Levels

Level 1	Hazus provides hazard and inventory data with minimal outside data collection or mapping.
Level 2	Analysis involves augmenting the Hazus provided hazard and inventory data with more recent or detailed data for the study region, referred to as "local data"
Level 3	Analysis involves adjusting the built-in loss estimation models used for the hazard loss analyses. This Level is typical done in conjunction with the use of local data.

4.2.2 Hazard-Specific Methodologies

Coastal Erosion

A detailed hazard exposure and vulnerability assessment was not performed for coastal erosion. It can be assumed that the risk of coastal erosion in the Harris County planning area and all participating jurisdictions, as well as potential losses due to coastal erosion impact, is negligible.

Dam/Levee Failure

Exposure to dam failure hazard was assessed by use of spatial analysis. The U.S. Army Corps of Engineers (USACE) has inundation areas for Addicks Dam and Barker Dam, both dams were analyzed based upon an MH-Breach. "MH Breach Dam" likely refers to a dam breach scenario modeled under the "Maximum Height" (MH) assumption, which is used in dam safety assessments and emergency planning. This involves simulating the maximum possible breach size and downstream flooding consequences, often used in "worst-case" scenarios for risk analysis. The inundation area for the Lake Houston Dam was developed using DSS-WISE. DSS-WISE is a FEMA product that was designed for water infrastructural safety studies. DSS-WISE can simulate dam/levee breach floods, fluvial floods, storm/tidal surges, landslide waves, and their consequences. The inundation areas for Addicks Dam, Barker Dam and Lake Houston Dam, for which inundation mapping was available, were the sole basis for this analysis.

The assets (population, buildings, critical facilities) were examined to determine if they are located in any of the dam failure hazard areas. Assets with their centroid located in the hazard area were totaled to estimate the totals and values at risk.

Using qualitative approaches, the following communities within the planning area are estimated to have some degree of dam failure exposure, due to their location downstream to the dams.

Drought

All of Harris County is at risk from the impacts of drought events. A qualitative analysis was conducted to assess the county's vulnerability to this hazard of concern.

Earthquake

A probabilistic assessment was conducted for Harris County for the 750-year mean-return-period events through a Level 2 analysis in Hazus to analyze the earthquake hazard and provide a range of loss estimates. The 750-year mean-return period means that there is a 1/750 (or 13 percent) chance of this event occurring in any given year. Events with lower return periods, such as a 500-year mean-return-period (0.2 percent), returned zero results, meaning that there were no estimated impacts.

The probabilistic method uses information from historical earthquakes and inferred faults, locations, and magnitudes, and computes the probable ground shaking levels that may be experienced during a recurrence period by Census tract.

The default assumption is a magnitude 7.0 earthquake for all return periods. Although damage is estimated at the census tract level, results were presented at the municipal level. Because there are multiple Census tracts that contain more than one jurisdiction, an area analysis was used to extract the percent of each tract that falls within individual jurisdictions. The percentage was multiplied against the results calculated for each tract and summed for each jurisdiction.

Damage estimates are calculated for losses to buildings (structural and non-structural) and contents; structural losses include load carrying components of the structure, and non-structural losses include those to architectural, mechanical, and electrical components of the structure, such as nonbearing walls, veneer and finishes, HVAC systems, boils, etc. All critical facilities are assumed to have been built to high code standards. This may not be the case, especially for older facilities.

Extreme Cold

All of Harris County is at risk from the impacts of extreme temperature events. A qualitative analysis was conducted to assess the county's vulnerability to this hazard of concern. All of Harris County is exposed and vulnerable to the winter storm hazard. In general, structural impacts include damage to roofs and building frames, rather than building content. Current modeling tools are not available to estimate specific losses for this hazard. A qualitative analysis was conducted to assess the county's vulnerability to this hazard of concern.

Extreme Heat

All of Harris County is at risk from the impacts of extreme temperature events. A qualitative analysis was conducted to assess the county's vulnerability to this hazard of concern.

Flooding

The 1-percent annual chance flood (riverine & coastal) and Hurricane Harvey flood events were examined to evaluate the county's risk from the flood hazard. These flood events are generally those considered by planners and evaluated under federal programs such as NFIP.

The following data were used to evaluate vulnerability and determine potential future losses for this plan update:

- The effective Digital Flood Insurance Rate Map (DFIRM) for the planning area was used to delineate flood hazard areas and estimate potential losses from the 1-percent-annual-chance flood event. The Harris County DFIRM is effective as of May 2, 2019, with the latest incorporated letter of map revision dated May 13, 2019. A 1-percent-annual-chance flood depth grid provided by the Harris County Flood Control District was used as the base and updated to reflect the effective DFIRM and include coastal flood zones and the pending DFIRM (to be effective November 15, 2019). Updates were generated using the floodplain boundaries and the U.S. Geological Survey's (USGS) 2-meter digital elevation model. For areas where planning partner jurisdictions extended into neighboring counties, the flood depth grid was extended using the DFIRM data for those counties.
  - Fort Bend County: December 21, 2017; latest letter of map revision June 12, 2018
  - Galveston County: February 28, 2018
  - Montgomery County: December 21, 2017; latest letter of map revision June 12, 2018
  - Waller County: February 18, 2019; latest letter of map revision September 30, 2016
- For the Hurricane Harvey flooding analysis, a depth grid of calculated inundation areas generated by FEMA Region 6 in November 2017 was integrated into the Hazus model. This depth grid was created using high water mark data from Harris County Flood Control District, USGS, and Individual Assistance data.

A quantitative assessment of exposure to the flood hazard was conducted using the asset inventory developed for this plan and three mapped flood hazard areas:

- The area that was flooded during Hurricane Harvey
- The 1-percent-annual-chance riverine flood hazard area
- The 1-percent-annual-chance coastal flood hazard area

To estimate vulnerability of the flood events, flood boundaries were overlaid on the maps of updated assets (population, building stock, and critical facilities). Centroids that intersected the flood boundaries were totaled to estimate the building RCV and population vulnerable to the flood inundation areas.

A Level 2 Hazus riverine flood analysis was performed. Hazus calculated the estimated potential losses to the population (default 2020 U.S. Census data across dasymetric blocks), potential damage to the general building stock, and potential damage to critical facility inventories based on the depth grids generated and the default Hazus damage functions in the flood model.

Both the critical facility and building inventories were formatted to be compatible with Hazus and its Comprehensive Data Management System. Once updated with the inventories, the Hazus riverine flood model was run to estimate potential losses in Harris County and the surrounding counties for the 1-percent annual chance flood events. Buildings located within the floodplain were imported as user-defined facilities to estimate potential losses to the building stock at the structural level.

Hurricane and Coastal Storm

A Hazus probabilistic and historical analysis was performed to analyze the wind hazard losses for Harris County for the:

- Probabilistic 20-year mean return period event
- Probabilistic 100-year mean return period event
- Hurricane Alicia (1983) historic event

The probabilistic Hazus hurricane model activates a database of thousands of potential storms that have tracks and intensities reflecting the full spectrum of Atlantic hurricanes observed since 1886 and identifies those with tracks associated with Harris County. Hazus contains data on historical hurricane events and wind speeds. It also includes surface roughness and vegetation (tree coverage) maps for the area. Surface roughness and vegetation data support the modeling of wind force across various types of land surfaces. Default demographic and updated building and critical facility inventories in Hazus were used for the analysis. Although damages are estimated at the census tract level, results were presented at the municipal level. Because there are multiple census tracts that contain more than one jurisdiction, a density analysis was used to extract the percent of building structures that fall within each tract and jurisdiction. The percentage was multiplied against the results calculated for each tract and summed for each jurisdiction.

Mass Movement

The exposure and vulnerability for the people, property, critical facilities/infrastructure, and environment are low to none for the planning area. This is based on the lack of historical occurrence of the types of hazards and the planning area's proactive approach to mitigating subsidence issues starting in 1999. This does not mean that these types of events could not happen in the future, it just means that there is currently no empirical data to assess impacts as of this planning effort.

Severe Weather

All the people, property (including critical facilities and infrastructure), and environment of the planning area are exposed to some degree to the severe weather hazard.

Tsunami

The exposure and vulnerability for the people, property, critical facilities/infrastructure, and environment are low to none for the planning area. This is based on the lack of historical

occurrence of the tsunami hazard and/or the lack of extent and location data for the planning area to support risk assessment of the tsunami hazard. This does not mean that these types of events could not happen in the future, it just means that there is currently no empirical data to assess impacts as of this planning effort. Therefore, no exposure and vulnerability analysis were performed for this hazard.

Wildfire

A quantitative assessment of exposure to the wildfire hazard was conducted using the Wildfire Threat and Ignition Density obtained through the Texas A&M Forest Service.

The Wildfire Threat data was derived at a 30-meter resolution. This scale of data was chosen to be consistent with the accuracy of the primary surface fuels dataset used in the assessment. While not appropriate for site specific analysis, it is ideal for state and regional scale analysis. To determine what assets are exposed to wildfire, available and appropriate GIS data were overlaid with the hazard area. Assets with their centroid located in the hazard area were totaled to estimate the totals and values at risk from the impacts of a wildfire event.

The Ignition Density classifications analyzed were: Low, Moderate, High, and Very High.

4.3 RATING PROBABILITY OF OCCURRENCE

Based on records of previous hazard events and consideration of potential future changes that could affect the frequency of future events, the risk assessment for each hazard assigns a rating for the probability of occurrence of that hazard in the future. These ratings were assigned as follows:

- Unlikely—A hazard event is not likely to occur or is unlikely to occur with less than a 1% annual chance probability
- Rare—Between 1% and 10% annual probability of a hazard event occurring
- Occasional—Between 10% and 100% annual probability of hazard event occurring
- Frequent—100% annual probability; a hazard event may occur multiple times per year

4.4 DATA SOURCE SUMMARY

Table 4-4 summarizes the data sources used for the risk assessment for this plan.

Table 4-4. Risk Assessment Data Documentation

Data	Source	Date	Format
Population data	• U.S. Census Bureau • American Community Survey 5-Year Estimates	• 2020 • 2023	Spreadsheet Spreadsheet
Building Inventory	• Harris County Appraisal District • Harris County Flood Control District • Fort Bend Central Appraisal District • Fort Bend County • Galveston Central Appraisal District • RS Means	• 2018 • 2018 • 2018 • 2016 • 2018 • 2024	Digital (GIS & Tabular)
Wildfire Hazard Data	• Texas A&M Forest Service	• 2025	Digital (GIS)
Critical Facilities and Lifelines	• ATLAS • BTS/USDOT • CIMA • City of La Porte • City of Seabrook • FleetCor • Harris County • HCOHSEM • Harris County Public Library • HCUS • HCPCS • HIFLD • METRO • NPMS • SETRAC • TCEQ • Regional Catastrophic Preparedness Grant Program • DSHS • TEA • THC • USACE	• 2022 • 2023 • 2024 • 2025 • 2025 • 2024 • 2020, 2024 • 2022-2025 • 2022 • 2020 • 2024 • 2021-2024 • 2024 • 2024 • 2024 • 2023-2024 • 2023 • 2024 • 2023 • 2023 • 2024	Digital (GIS)
Effective DFIRM maps (Harris, Fort Bend, Montgomery, and Waller Counties) Preliminary DFIRM maps (Brazoria, Galveston Counties)) Harris County Pending DFIRM maps	• FEMA • FEMA • FEMA	• 2016, 2018 & 2019 • 2018 • 2019	Digital (GIS)
2-Meter Digital Elevation Model	U.S. Geological Survey	• 2008	Digital (GIS)
Depth Grid of Calculated Inundation Areas for Disaster Declared Counties, Texas, USA, DR-4332 (Hurricane Harvey)	FEMA	• 2017	Digital (GIS)
Addicks Dam Breach Analysis Inundation Area (MH Breach)	U.S. Army Corps of Engineers	• 2025	Digital (GIS)
Barker Dam Breach Analysis Inundation Area (MH Breach)	U.S. Army Corps of Engineers	• 2025	Digital (GIS)

Data	Source	Date	Format
Lake Houston Dam Breach Analysis Inundation Area (Sudden and Complete Breach)	FEMA, DSS-WISE	• 2025	Digital (GIS)
Texas Wildfire Risk Assessment Portal Wildfire Threat & Ignition Density	Texas A&M Forest Service	Unknown	Digital (GIS)

Note: BTS = Bureau of Transportation Statistics; CIMA = Channel Industries Mutual Aid; DOT = Department of Transportation; DSHS = Texas Department of State Health Services; DSS-WISE = Decision Support System for Water Infrastructure Security; FEMA = Federal Emergency Management Agency; HCOHSEM = Harris County Office of Homeland Security and Emergency Management; HCPCS = Healthcare Common Procedure Coding System; HCUS = Harris County Universal Services; HIFLD = Homeland Infrastructure Foundation-Level Data; METRO = Metropolitan Transit Authority; NPMS = National Pipeline Mapping System; SETRAC = SouthEast Texas Regional Advisory Council; TCEQ = Texas Commission on Environmental Quality; TEA = Texas Education Agency; Texas A&M = Texas Agricultural and Mechanical College; THC = Texas Historical Commission; USACE = U.S. Army Corps of Engineers; USGS = U.S. Geological Survey

4.5 LIMITATIONS

Loss estimates, vulnerability analyses, and hazard-specific impact evaluations rely on the best-available data and methodologies. Uncertainties are inherent in any loss estimation methodology and arise in part from incomplete scientific knowledge concerning natural hazards and their effects on the built environment. Uncertainties also result from the following:

- Approximations and simplifications necessary to conduct such a study
- Incomplete or dated inventory, demographic, or economic parameter data
- The unique nature, geographic extent, and severity of each hazard
- Mitigation measures already employed by the participating jurisdictions
- The amount of advance notice residents has to prepare for a specific hazard event
- Uncertainty of climate change projections

These factors can result in a range of uncertainty in loss estimates, possibly by a factor of two or more. Therefore, potential vulnerability and loss estimates are approximate. These results do not predict precise results and should be used to understand relative risk. Over the long term, Harris County will collect additional data and update and refine existing inventories to assist in estimating potential losses.

Potential economic loss is based on the present value of the general building stock using best-available data. The county acknowledges that significant impacts may occur to critical facilities and infrastructure as a result of these hazard events, causing great economic loss. However, monetized damage estimates to critical facilities and infrastructure, and economic impacts were not quantified and require more detailed loss analyses. In addition, economic impacts on industry, such as tourism and the real estate market, were not analyzed.

4.6 CONSIDERATIONS FOR MITIGATION AND NEXT STEPS

The following items are to be discussed for consideration for the next plan update to enhance the risk assessment:

- All Hazards
  - Create an updated user-defined general building stock dataset using up-to-date parcels, footprints, and RSMeans values.
  - Utilize updated and current demographic data.
  - Utilizing assessor data, include updated occupancy class attributes in general building stock.
- Earthquake
  - Identify unreinforced masonry in critical facilities and privately-owned buildings (i.e., residences) by accessing local knowledge, tax assessor information, and/or pictometry/orthophotos. These buildings may not withstand earthquakes of certain magnitudes and plans to provide emergency response or recovery efforts at these properties can be developed.
- Extreme Cold and Extreme Heat
  - Track extreme temperature data for injuries, deaths, shelter needs, pipe freezing, agricultural losses, and other impacts to determine distributions of the most at-risk areas.
- Flooding
  - The general building stock inventory can be updated to include attributes regarding first floor elevation and foundation type (basement, slab on grade, etc.) to enhance loss estimates.
  - As more current FEMA floodplain data become available (i.e., DFIRMs), update the exposure analysis and generate a more detailed flood depth grid that can be integrated into the current Hazus version.
  - Conduct a Hazus loss analysis for more frequent flood events (e.g., 10- and 50-year flood events).
  - Conduct a repetitive loss area analysis.
  - Continue to expand and update urban flood areas to further inform mitigation.
- Severe Storm
  - The general building stock inventory can be updated to include attributes regarding protection against strong winds, such as hurricane straps, to enhance loss estimates.
- Wildfire
  - General building stock inventory can be updated to include attributes such as roofing material, fire detection equipment, or distance to fuels as another measure of vulnerability.

5. IDENTIFICATION OF HAZARDS OF CONCERN

To provide a strong foundation for mitigation actions in this plan, Harris County considered a full range of hazards that could impact the area and then identified and ranked those that present the greatest concern. These hazards of concern were identified based on the following:

- Input from all Steering Committee Members
- Review of the 2023 Texas State Hazard Mitigation Plan
- Review of the 2020 Harris County Multi-Jurisdictional Hazard Mitigation Plan
- Research on the frequency, magnitude, and costs associated with hazards that have previously or could feasibly impact the region
- Qualitative information regarding natural (not human-caused) hazards and the perceived vulnerability of the study area's assets to them.

Hazards of Concern are those hazards that are considered most likely to impact a community. These are identified using available data and local knowledge.

Natural Hazards are those hazards that are a source of harm or difficultly created by a meteorological, environmental, or geological event.

5.1 REVIEW OF HAZARDS CONSIDERED

Table 5-1 documents the process of reviewing hazards to identify those that are of concern for this MJ-HMP update. Based on the review of potential hazards of concern, 17 hazards of concern were identified as significant hazards affecting the entire County, to be addressed at the County level in this plan (shown here in alphabetical order):

- Coastal erosion
- Dam/levee failure
- Drought
- Earthquake
- Extreme cold
- Extreme heat
- Flooding
- Hurricanes and coastal storms
- Mass movement
- Severe weather
- Tsunami
- Wildfire
- Energy pipeline failure
- Public health emergency
- Technology failure
- Toxic release/hazardous materials
- Utility failure

Other natural and human-caused hazards of concern have occurred in Harris County but have a low potential to occur, are addressed by other planning mechanisms, and/or do not result in significant impacts within the County. Therefore, these hazards are not addressed in this update. If deemed necessary by the County, these hazards may be considered in future plan updates.

Table 5-1. Identification of Hazards of Concern for Harris County

Hazard	May Occur in the County?	Significant Threat to the County?	Why was this determination made?	Sources
Avalanche	No	No	The FEMA National Risk Index (NRI) indicates that avalanche hazards are not applicable to the planning area, and thus, avalanche hazards are not considered for the 2025 Harris County MJ-HMP Update.	FEMA NRI, 2023 State of Texas Hazard Mitigation Action Plan (HMAP)
Coastal Erosion	Yes	No	While the 2023 State of Texas HMAP identifies severe coastal erosion as a hazard of concern, the HMAP identifies this hazard as being possible along all of Texas's coastal counties. Due to its location on the western side of Galveston Bay, the coastal erosion hazard threat to Harris County is fairly low, as the barrier island to the east provides a level of protection. However, the Steering Committee and Planning Partnership agreed with HCOHSEM that coastal erosion should be included in the 2025 Harris County MJ-HMP Update.	2023 State of Texas HMAP, HCOHSEM, and input from the Steering Committee and Planning Partnership
Coastal Flooding	Yes	No	The NRI indicates that the County has a relatively low risk for coastal flooding. The effects of Coastal Flooding are discussed in the Hurricane and Coastal Storm hazard profile in the 2025 Harris County MJ-HMP Update.	NOAA/NCEI, FEMA NRI
Dam/Levee Failure	Yes	No	The 2023 State of Texas HMAP does not specify dam/levee failure as a primary hazard of concern; however, after consideration was given to the dam/levee locations around Harris County, the Steering Committee decided to include this hazard.	2023 State of Texas HMAP, HCOHSEM, and input from the Steering Committee and Planning Partnership
Drought	Yes	Yes	Drought hazards continue to present a significant concern for the County, and it was determined drought should be included in the 2025 Harris County MJ-HMP Update. Additionally, drought was identified as a hazard of concern in previous MJ-HMPs and was profiled in the 2023 State of Texas HMAP.	U.S. Drought Monitor, 2023 State of Texas HMAP, HCOHSEM, and input from the Steering Committee and Planning Partnership
Earthquake	Yes	No	The NRI indicates that the County has a relatively low risk of earthquake. The 2023 State of Texas HMAP indicates that the USGS has no recorded significant earthquakes within the state, including Harris County. However, the Steering Committee and Planning Partnership agreed that earthquakes should continue to be assessed in the 2025 Harris County MJ-HMP.	FEMA NRI, HCOHSEM, input from the Steering Committee and Planning Partnership, U.S. Geological Survey (USGS)—Earthquake Hazards Program, Review of USGS Seismic Maps
Energy Pipeline Failure	Yes	Yes	Since 2015, Harris County has identified energy pipeline failure as a primary hazard of concern based on the high concentration and volume of energy pipelines within the planning area.	HCOHSEM, and input from the Steering Committee and Planning Partnership

Hazard	May Occur in the County?	Significant Threat to the County?	Why was this determination made?	Sources
Extreme Cold	Yes	Yes	The 2025 Harris County MJ-HMP includes extreme cold as a hazard of concern, following input from the Steering Committee and Planning Partnership, who agreed that this hazard should be addressed separately from severe weather-related threats following Winter Storm Uri in 2021. Historical data indicate that extreme cold events have caused substantial damage in the past, and FEMA's NRI designates this hazard as a very high risk for Harris County. Between 1956 and April 2025, Harris County was included in two FEMA declarations related to extreme cold.	FEMA NRI, 2023 State of Texas HMAP, HCOHSEM, and input from Steering Committee and Planning Partnership
Extreme Heat	Yes	Yes	The NRI indicates that the County has a relatively high risk for heat waves. Based on input from HCOHSEM, the Steering Committee, and the Planning Partnership, it was determined that extreme heat should be its own hazard of concern in the 2025 County MJ-HMP update.	FEMA NRI, HCOHSEM, and input from Steering Committee and Planning Partnership
Flooding	Yes	Yes	Flooding remains a significant concern throughout the entire planning area, including heavy rain events, tropical systems, riverine flooding, and urban flooding. According to FEMA's NRI, Harris County continues to face a very high risk of riverine flooding. Between 1956 and April 2025, Harris County was included in 12 FEMA flood-related declarations, including FEMA. Based on historical flooding data, its impacts on Harris County, and input from the Steering Committee and Planning Partnership, flooding has been identified as a critical hazard for the County.	NOAA/NCEI, NWS, FEMA NRI, 2023 State of Texas HMAP,
Hail	Yes	Yes	See Severe Weather.	NOAA/NCEI, FEMA NRI, 2023 State of Texas HMAP, HCOHSEM, and input from the Steering Committee and Planning Partnership
Heat Wave	Yes	Yes	See Extreme Heat.	NOAA/NWS, FEMA NRI, 2023 State of Texas HMAP, HCOHSEM, and input from the Steering Committee and Planning Partnership
Hurricane, Coastal Storm	Yes	Yes	The NRI indicates that the County is at very high risk for hurricanes. Historical occurrences and impacts to the planning area are well-documented. Between January 1956 and April 2025, Harris County was included in 17 hurricane or coastal storm disaster declarations. Based on the history of hurricanes and coastal storms and their impacts to Harris County and input from the Steering Committee and Planning Partnership, the hurricane and coastal storm hazard of concern remains a priority for the 2025 Harris County MJ-HMP update.	National Hurricane Center, FEMA NRI, 2023 State of Texas HMAP, HCOHSEM, and input from the Steering Committee and Planning Partnership

Hazard	May Occur in the County?	Significant Threat to the County?	Why was this determination made?	Sources
Ice Storm	Yes	Yes	See Extreme Cold.	NOAA/NCEI and NWS, 2023 State of Texas HMAP, and input from the Steering Committee and Planning Partnership
Landslide	Yes	No	The NRI indicates that the County has a relatively moderate risk for landslides. After consultation with the Steering Committee and Planning Partnership, it was decided to include landslide hazards in the Mass Movement hazard profile in the 2025 Harris County MJ-HMP.	FEMA NRI
Lightning	Yes	Yes	See Severe Weather.	2023 State of Texas HMAP
Mass Movement	Yes	No	For the 2025 Harris County MJ-HMP Update, the mass movement hazard of concern includes three types of hazards: landslide, subsidence, and sinkholes. While no FEMA disasters related to mass movement have been declared for Harris County since 1965, input from HCOHSEM, Steering Committee members, and the Planning Partnership confirmed that mass movement should be examined as a hazard of concern for the 2025 County MJ-HMP Update	HCOHSEM and input from the Steering Committee and Planning Partnership
Public Health Emergency	Yes	Yes	New in 2025, HCOHSEM consulted with Harris County Public Health to structure a new hazard profile for this year's update. Between January 1965 and April 2025, Harris County has been included in two public health emergency-related disaster declarations. Based on the history of health-related emergencies and their impact on Harris County and input from the Steering Committee and Planning Partnership, public health emergencies have been identified as a hazard of concern for the County.	FEMA, HCOHSEM, and input from the Steering Committee and Planning Partnership
Riverine Flooding	Yes	Yes	See Flooding.	FEMA NRI, 2023 State of Texas HMAP
Severe Weather	Yes	Yes	The Severe Weather hazard profile from the 2020 Harris County Multi-Jurisdictional Hazard Mitigation Action Plan (MJ-HMAP) has been updated for 2025 to encompass additional hazards associated with severe thunderstorms, including hail, lightning, tornadoes, and high winds. Unlike the 2020 plan, which categorized hazards such as extreme heat, winter storms, and freezes under severe weather, these sub-groups have been reorganized for the 2025 MJ-HMP Update. Specifically, extreme heat is now recognized as a separate hazard, while winter storms and freezes will be incorporated under the extreme cold hazard profile. From January 1956 to April 2025, Harris County experienced 10 disaster declarations related to Severe Weather. Based on historical events and input from the Steering Committee and Planning Partnership, maintaining a comprehensive severe weather hazard profile remains a key focus for the 2025 Harris County MJ-HMP Update.	NOAA/NCEI, NWS, 2023 State of Texas HMAP, HCOHSEM, and input from the Steering Committee and Planning Partnership

Hazard	May Occur in the County?	Significant Threat to the County?	Why was this determination made?	Sources
Strong Wind	Yes	Yes	See Severe Weather and Hurricane/Coastal Storm.	
Technology Failure	Yes	Yes	In response to recent technological failures during emergency situations, the Steering Committee and Planning Partners have decided to designate technology failure as an official hazard.	HCOHSEM and input from the Steering Committee and Planning Partnership
Tornado	Yes	Yes	See Severe Weather.	
Toxic Release / Hazardous Materials	Yes	Yes	Due to its inclusion in the 2020 Harris County Multi-Jurisdictional Hazard Mitigation Action Plan (MJ-HMAP), the Steering Committee and Planning Partnership, in collaboration with HCOHSEM, have agreed to include this hazard in the 2025 MJ-HMP Update. The county's proximity to and the high concentration of petrochemical facilities and industrial plants in the eastern region were influential factors in choosing to revisit this hazard in 2025.	HCOHSEM and input from the Steering Committee and Planning Partnership
Tsunami	Yes	No	FEMA's NRI explains that there is insufficient data available for tsunami hazards and how it impacts Harris County; however, after consulting with HCOHSEM and with the direction of the Steering Committee members, the tsunami hazard was again chosen to be included in the 2025 MJ-HMP Update.	HCOHSEM and input from the Steering Committee and Planning Partnership
Utility Failure	Yes	Yes	Harris County experiences utility failures (generally power outages) several times each year. These failures are usually due to severe storms or severe winter storms that affect the county. The Steering Committee and Planning Partnership consider utility failure a cascading impact of severe storms, severe winter storms, and flooding events, and as such, the Steering Committee and Planning Partnership agreed that utility failure should be designated as one of the hazards of concern in the 2025 Harris County MJ-HMP.	HCOHSEM and input from the Steering Committee and Planning Partnership
Wildfire	Yes	Yes	Between January 1956 and April 2025, Harris County was designated in three wildfire-related disaster declarations. Considering the historical wildfire hazards, their impacts on Harris County, and feedback from the Steering Committee and Planning Partnership, the wildfire hazard profile continues to be a priority for the 2025 Harris County MJ-HMP update.	FEMA NRI, HCOHSEM, and input from the Steering Committee and Planning Partnership
Windstorm	Yes	Yes	See Severe Storm.	

Note: DR = Presidential Disaster Declaration Number; EM = Presidential Disaster Emergency Number; FEMA = Federal Emergency Management Agency; NCEI = National Centers for Environmental Information; NOAA = National Oceanic and Atmospheric Administration; NWS = National Weather Service; USGS = United States Geologic Survey

5.2 HAZARD GROUPINGS

HCOHSEM and members of the Steering Committee approved the use of the following hazard event groupings:

Natural Hazards

- Coastal Erosion: The gradual breakdown and movement of shoreline due to physical and chemical processes of water, wind, and meteorological conditions. The extent of coastal erosion is measured in terms of loss of shoreline.
- Dam/Levee Failure: The uncontrolled release of impounded water, which can result in rapid and potentially catastrophic flooding.
- Drought: A period characterized by long durations of below-normal precipitation that can affect agriculture, water supply, aquatic ecology, wildlife, and plant life. Types of droughts include meteorological, hydrological, agricultural, and socioeconomic.
- Earthquake: The sudden movement of the Earth's surface caused by the release of stress accumulated within or along the edge of the Earth's tectonic plates, a volcanic eruption, or man-made activities that cause geologic stresses.
- Extreme Cold: When temperatures drop well below normal in an area. In regions relatively unaccustomed to winter weather, near-freezing temperatures are considered "extreme cold." In Harris County, extreme cold temperatures are characterized by the ambient air temperature dropping to 0°F or below (NWS n.d.). For the 2025 MJ-HMP, the extreme cold hazard profile includes discussions on cold temperatures as well as other winter-related events such as snowstorms and ice storms.
- Extreme Heat: A prolonged period of excessively high temperatures and exceptionally humid conditions. The National Weather Service criteria for an excessive heat warning is a heat index of 105°F or greater that will last for 2 hours or more. The 2025 MJ-HMP incorporates a discussion of the urban heat island effect in the extreme heat hazard profile
- Flooding: Flooding occurs when water overflows onto land that is normally dry; flooding may be caused by heavy rainfall events, slow-moving thunderstorms, tropical storm systems, and/or coastal storms. Other types of flooding arise from dam/levee failure. Types of flooding most likely to impact Harris County are riverine (inland) flooding, flash flooding, and stormwater/urban flooding.
- Hurricanes and Coastal Storms: Hurricanes are classified as tropical cyclones, which include any closed circulation developing around a low-pressure center in which the winds rotate counterclockwise in the northern hemisphere and whose diameter averages 10 to 30 miles across. Coastal storms refer to any type of tropical or non-tropical low-pressure circulation that develops and tracks along the coastal shores, bringing high winds, high surf, and heavy precipitation in the process.
- Mass Movement: The transport of large volumes of earth material primarily by gravity. Mass movements can be divided into four main classes based on how quickly the rock and sediment move: falls, slides, creeps, and flows. Three types of mass movements have been

assessed for the 2025 MJ-HMP, and those hazard subtypes are landslides, subsidence, and sinkholes.

- Severe Weather: Conditions produced by thunderstorms, including damaging winds, tornadoes, large hail, freezing rain, sleet, snow, and strong winds (NOAA n.d.).
- Tsunami: A series of high-energy waves that radiate outward like pond ripples from an area where a generating event occurs, arriving at shorelines over an extended period. Tsunamis can be induced by earthquakes, landslides, and submarine volcanic explosions.
- Wildfire: Any non-structural fire that occurs in the wildland. Three distinct types of wildland fires have been defined and include naturally occurring wildfire, human-caused wildfire, and prescribed fire.

Other Hazards of Concern

- Energy Pipeline Failure: The disruption, damage, or failure of pipelines that transport natural gas, crude oil, refined petroleum products, and other energy resources in Harris County.
- Public Health Emergency: An event that has direct impacts and concerns for human health across the County. Types of public health emergencies encompass pandemics, such as COVID-19, as well as other human health-related events, including concerns related to ozone and air pollution and mosquito-borne viruses.
- Technology Failure: The disruption, malfunction, or compromise of digital, mechanical, or communication systems essential to daily operations, public safety, and critical infrastructure.
- Toxic Release/Hazardous Materials: The discharge or emission of harmful substances into the environment. Harmful substances may include materials and wastes as defined by the U.S. Environmental Protection Agency (EPA) Comprehensive Environmental Response, Compensation, and Liability Act, also known as Superfund.
- Utility Failure: When essential services such as electricity, natural gas, water, wastewater, or telecommunications systems are interrupted or become unavailable, whether temporarily or for an extended period. These failures may result from various causes, including severe weather events, equipment or infrastructure failure, cyberattacks, human error, or cascading impacts from other hazards (e.g., flooding damaging substations or water mains).

6. COASTAL EROSION

6.1 HAZARD PROFILE

6.1.1 Hazard Description

Coastal erosion is the gradual breakdown and movement of shoreline due to physical and chemical processes of water, wind, and meteorological conditions. Natural erosion has occurred since the Earth's formation and continues at a slow and uniform rate. In the U.S., coastal erosion causes approximately $500 million in property loss annually (U.S. Climate Resilience Toolkit n.d.). While coastal erosion affects all shorelines, the impacts and rates of erosion are highly localized.

Coastal erosion is measured as the rate of change in the position or horizontal displacement of a shoreline and can be chronic or episodic. It is a natural process caused by storm surges, coastal storms/hurricanes, and high waves (such as king tides). A healthy shoreline will naturally recover from erosion events over time. However, it may take months or years for a coastline to recover from a significant episodic erosion event. When shorelines do not recover quickly enough, they retreat inland, potentially damaging buildings, infrastructure, and natural areas. The extent and severity of naturally occurring coastal erosion are accelerated by global sea level rise.

6.1.2 Causes

The U.S. Geologic Survey (USGS) states that the primary causes of coastal land loss include both natural processes and human activities (see Figure 6-1). Coastal land loss encompasses not only beach erosion but also land losses along bluffs and wetlands around interior bays and estuaries. The primary causes include erosion, inadequate sediment supply, and coastal submergence, with land loss often resulting from a combination of these factors.

Figure 6-1. Primary Causes of Coastal Land Loss.

Natural Processes	
Agent	**Examples**
Erosion	waves and currents storms landslides
Sediment Reduction	climate change stream avulsion source depletion
Submergence	land subsidence sea-level rise
Wetland Deterioration	herbivory freezes fires saltwater intrusion

Human Activities	
Agent	**Examples**
Transportation	boat wakes, altered water circulation
Coastal Construction	sediment deprivation (bluff retention) coastal structures (jetties, groins, seawalls)
River Modification	control and diversion (dams, levees)
Fluid Extraction	water, oil, gas, sulfur
Climate Alteration	global warming and ocean expansion increased frequency and intensity of storms
Excavation	dredging (canal, pipelines, drainage) mineral extraction (sand, shell, heavy mins.)
Wetland Destruction	pollutant discharge traffic failed reclamation burning

The potential for erosion is determined by the following factors:

- Soils—Soils composed of a large percentage of silt and fine sand are most susceptible to erosion. As the clay and organic material content of soils increases, the potential for erosion decreases. Well-drained and well-graded gravels and gravel-sand mixtures are the least likely to erode. Coarse gravel soils are highly permeable and have a good capacity for absorption, which can prevent or delay the amount of surface runoff.
- Vegetative Cover—Vegetative cover can help control erosion by shielding the soil surface from falling rain, absorbing water from the soil, and slowing the velocity of runoff.
- Topography—Runoff is affected by the topography of the area, including size, shape and slope. The greater the slope length and gradient, the more potential an area has for erosion.
- Climate—Climate can affect the amount of runoff, especially the frequency, intensity and duration of rainfall and storms. When rainstorms are frequent, intense, or of long duration, erosion risks are increased, all other factors held constant. Seasonal changes in temperature and rainfall amounts define the period of highest erosion risk

Major storms such as hurricanes may cause significant erosion by combining high winds with heavy surf and storm surge to significantly impact the rate of shoreline erosion.

6.1.3 Location

The Texas Natural Resource Code, Section 33.601 defines coastal erosion "loss of land, marshes, wetlands, beaches, or other coastal features within the coastal zone because of the actions of wind, waves, tides, storm surges, subsidence, or other forces." According to the Texas General Land Office's 2023 CERPA Report, Texas has 367 miles of gulf-facing shoreline, approximately 3,300 miles of bay shoreline, and some of the highest erosion rates in the country. The report also states that, on average, four feet of land are lost annually across the coast, with certain areas experiencing severe erosion losing over 62 feet each year. The geomorphic features that span the Texas Gulf coast include deltaic headlands, peninsulas, barrier islands, and spits, and their depositional origins provide clues on sediment erodibility. Variations in the wave climate impacting those features influence the amounts of sediment that are supplied to the littoral drift (the amount of sediment carried along the shoreline) as well as the direction the sediment travels. Elevations of beaches and dunes can control whether episodic high wave and tidal events allow over wash and landward barrier migration or merely erosion of the beach-face that can be repaired by quieter waves. If left unmitigated or managed, the severity of the coastal erosion hazard in Texas could significantly impact critical assets in identified coastal erosion zones.

The only participating jurisdictions at-risk to Coastal Erosion hazard events are the following: La Porte, Morgan's Point, Shoreacres, Seabrook, El Lago, Nassau Bay, and portions of unincorporated Harris County.

6.1.4 Extent

Coastal erosion is primarily measured as a rate of shoreline recession in linear feet per year or as volumetric sediment loss per linear foot of shoreline. Shorelines along the eastern edge of Harris

County—particularly in areas bordering Galveston Bay, Trinity Bay, and the Houston Ship Channel—are experiencing erosion at varying rates, with some sections retreating by more than 4 feet per year depending on local geomorphology, wave energy, and subsidence factors (Texas General Land Office 2023). While some areas of the shoreline remain relatively stable or are artificially protected, others—especially low-lying marshes and bayfronts—continue to erode, placing adjacent infrastructure, habitats, and communities at risk.

Harris County lost almost 30 percent of its freshwater wetlands, including most prominently the iconic prairie pothole-pimple mound complexes (NOAA 2017). These wetlands are vital to the region's ecological health and resilience, performing essential functions such as stormwater detention, erosion control, water purification and storage, and providing habitat for wildlife as well as recreational spaces for people. Among their most critical roles, wetlands act as natural buffers that help protect inland areas from storm surge and coastal flooding. When these ecosystems are degraded or destroyed, the protective services they offer are also diminished, leaving communities more vulnerable to extreme weather events. This vulnerability was starkly illustrated after Hurricane Harvey, when the Texas coast experienced an average shoreline retreat of just over 16 meters. In some areas near Rockport, where the storm made landfall, erosion reached nearly 60 meters—far exceeding the region's typical annual retreat rate of 1.27 meters per year (Luhn 2024). The loss of wetlands, combined with accelerated coastal erosion, underscores the urgent need for conservation and restoration efforts to safeguard both natural systems and human communities.

According to data from the Texas General Land Office's Coastal Erosion Planning & Response Act, several locations along the San Jacinto Battleground, Morgan's Point, and the Baytown Nature Center are experiencing active shoreline retreat, driven by both persistent wave action and long-term land subsidence. Coastal storms—particularly hurricanes and tropical systems—intensify these processes, often accelerating erosion in areas lacking natural vegetation or engineered defenses.

"Two critical parameters that influence the erosion potential of a tropical cyclone are surge height and surge duration: the longer sea level is elevated above normal during storm passage, the greater the potential for redistribution of sediment eroded from the beach (Bureau of Economic Geology 2020)." Harris County's vulnerability to both chronic and storm-induced erosion highlights the critical need for ongoing shoreline monitoring and investment in a combination of structural and nature-based mitigation strategies. Among the most effective natural defenses are barrier islands, which play a crucial role in shielding the mainland coast from wave energy and storm surge. These islands act as buffers, absorbing the brunt of coastal forces and helping to reduce erosion along more populated and ecologically sensitive shorelines. Preserving and restoring barrier islands, alongside wetland systems and other protective features, is essential to enhancing the region's resilience against future coastal hazards.

6.1.5 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2023, Harris County was not included in any major disaster (DR) or emergency (EM) declarations for coastal erosion-related events (FEMA 2023)

State Emergency Proclamations

The Governor of Texas can issue emergency proclamations to access powers authorized by the Emergency Services Act. Harris County has not been included in any coastal erosion-related state emergency proclamations.

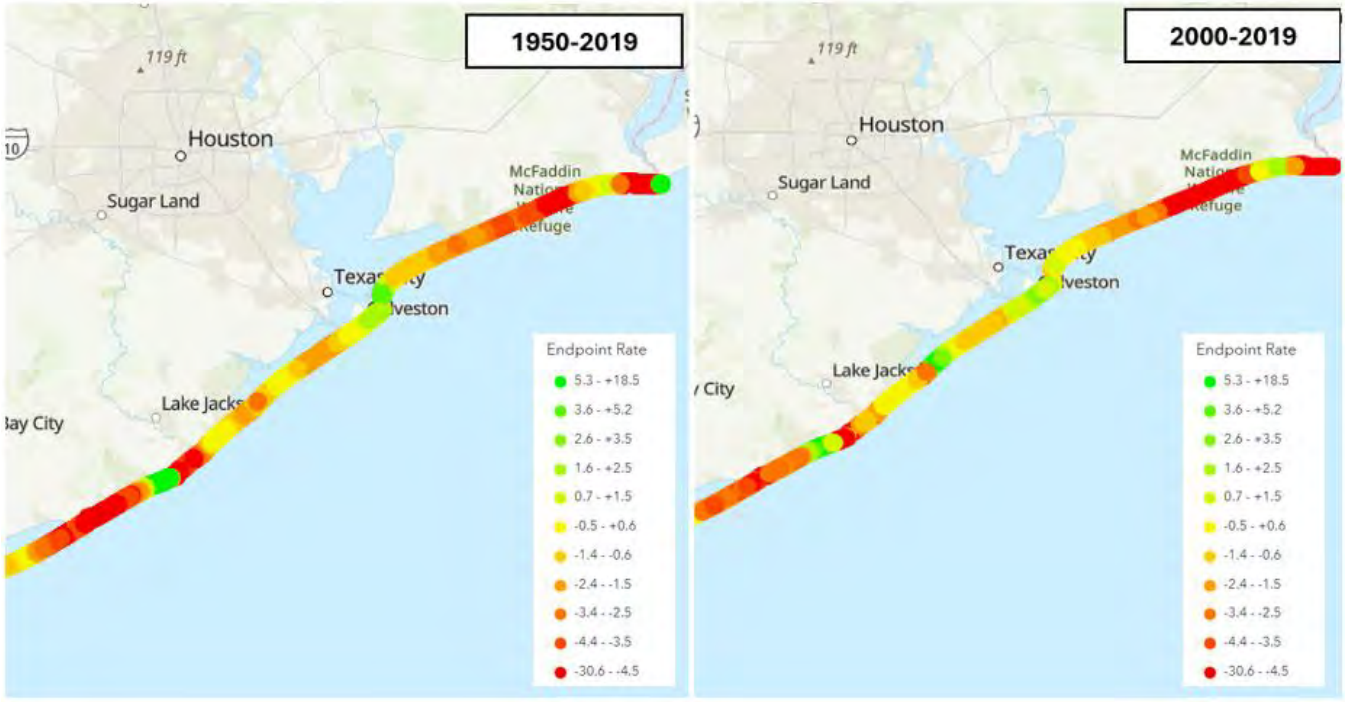
USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2024, Harris County was not included in any USDA coastal erosion-related agricultural disaster declarations.

Previous Events

No historical damage values could be identified for losses due to erosion within the planning area. Figure 6-2 shows the variability of shoreline change rates along the upper-Texas coast.

Figure 6-2. Shoreline Change Rates at Galveston Island



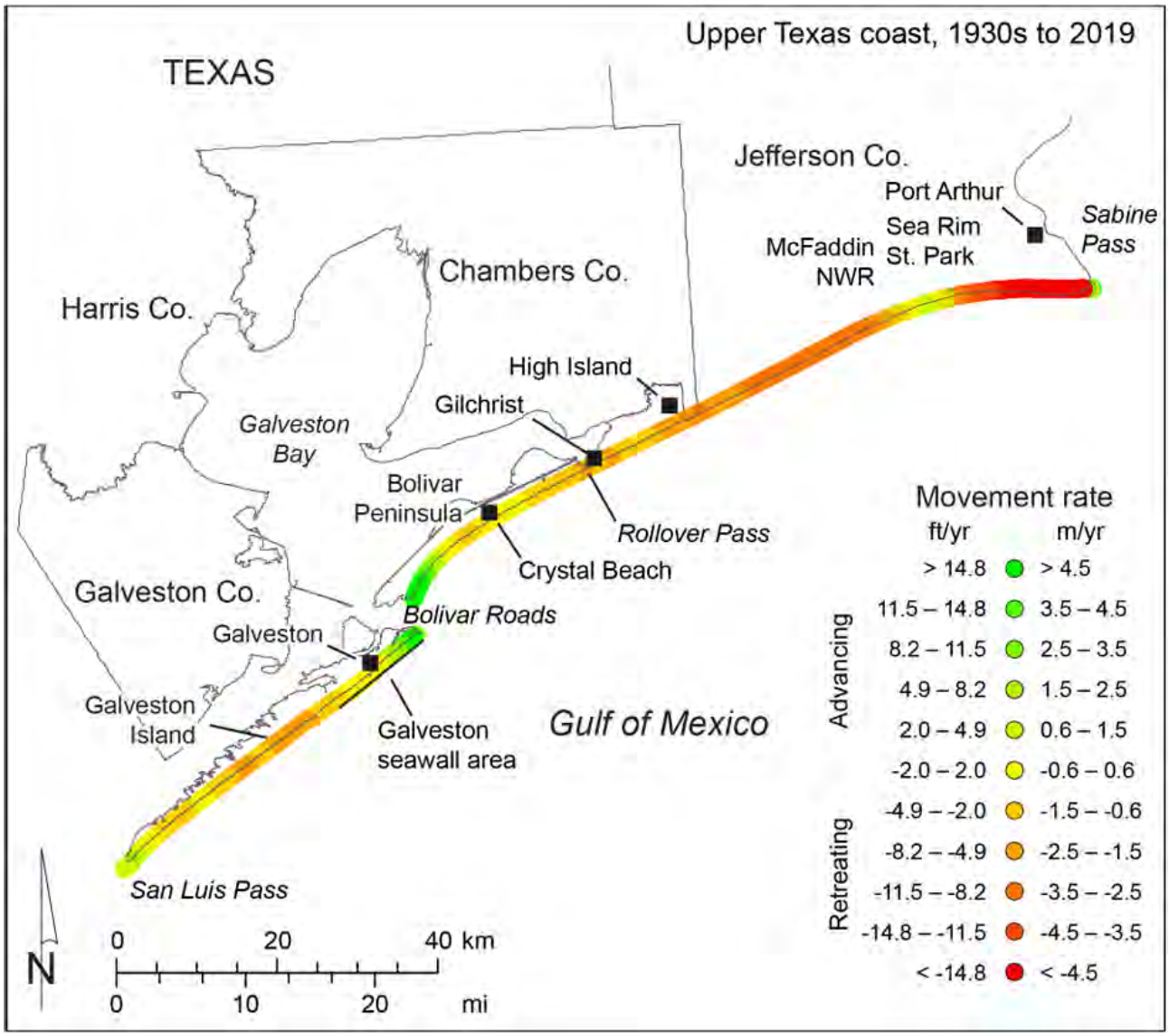
Rates greater than 2 feet per year are shown in orange and red, and stable or accreting (gaining land) areas are shown in green.

Note: Data for shoreline changes across coastal Harris County are not included in the 2019 report

Source: Shoreline Change Rates at Galveston Island (Jeffrey Paine T. 2019)

Identifying coastal erosion events is challenging since this hazard occurs over a much longer timescale than some of the other hazards. Published in 2020, the "Shoreline Movement Along the Texas Gulf Coast, 1930 to 2019" details areas where there have been notable movements in the shoreline location. Using lidar, the report shows how shoreline movement varies along the coast. Figure 6-3 shows the rate of movement observed along the shoreline over recent decades.

Figure 6-3. Net Rates of Long-Term Movement for the Upper Texas Gulf Shoreline



Source: (Bureau of Economic Geology 2020)

A new study released from the University of Houston shows just how impactful a tropical system can be, especially when it comes to coastal erosion. In the months before Hurricane Beryl made landfall, the study collected data and drone photography at several locations along the Gulf. One of these locations was on Galveston Island. Aside from observing substantial coastal erosion, the findings mirror what was also observed a few years earlier when Hurricane Harvey struck the area.

At the coast, the study found that after Harvey, there was an average shoreline retreat of just over 16 meters − values that far exceed the average short and long-term retreat of 1.27 meters per year for the Texas Coast (UH 2024). While this annual retreat rate may not be representative of coastal communities in Harris County, it does confirm that coastal erosion occurs every year, particularly when tropical systems make landfall.

6.1.6 Probability of Future Occurrences

Information on previous coastal erosion occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 6-1. Historical observations as well as the findings in many scientific papers demonstrate that coastal erosion events occur annually. In order to calculate probabilities for future coastal erosion events, this assessment will adopt the widely accepted notion that on average, the Texas coastline retreats about four feet every year (UH 2024). When hurricanes or tropical systems impact the planning area, these erosion events can be greatly exacerbated. The probability of occurrence for coastal erosion in the County is considered "frequent."

Table 6-1. Probability of Future Hazard Events in Harris County

Hazard Type	Number of Occurrences Between 1999 and 2023	Percent Chance of Occurring in Any Given Year
Coastal Erosion	24	2.67%

Source: (Bureau of Economic Geology 2020); (UH 2024); (Salt Bayou Marsh Workgroup 2013)
Note: Due to limitations in data, not all coastal erosion events occurring between 1999 and 2023 are accounted for in the tally of occurrences. As a result, the number of hazard occurrences is calculated using the number of occurrences between 1999 and 2023.

Climate Change Projections

Climate change is projected to significantly intensify coastal erosion in Harris County due to a combination of sea-level rise, increased storm intensity, and ongoing land subsidence. The Texas coast is experiencing relative sea level rise at one of the highest rates in the country, with projections indicating an increase of 1.2 to 1.5 feet by 2050 in the Galveston Bay region (NOAA, 2022). This rise is exacerbated by subsidence caused by groundwater withdrawal and natural geologic processes, which further lower land elevations and expose coastal areas to inundation and wave action. As the shoreline recedes, critical ecosystems such as wetlands and barrier islands—natural defenses against erosion—are expected to deteriorate, leaving inland areas more vulnerable to flooding and storm surge (Texas General Land Office [GLO], 2021).

Additionally, the frequency and intensity of hurricanes and heavy rainfall events are expected to increase as climate change progresses. Warmer Gulf waters provide more energy for tropical storms, increasing their potential for rapid intensification and coastal damage (Intergovernmental Panel on Climate Change [IPCC], 2021). Storm surge, when combined with higher base sea levels, leads to more severe and persistent erosion during storm events. Furthermore, projections from the Union of Concerned Scientists suggest that coastal Harris County could face over 130 high-tide

flooding days annually by 2050, up from fewer than 10 in the early 2000s (Union of Concerned Scientists, 2020). These trends emphasize the urgency of integrating climate adaptation strategies, such as managed retreat, shoreline stabilization, and green infrastructure, into local planning and development.

6.1.7 Cascading Impacts on Other Hazards

Coastal erosion can have a significant impact on the Texas Gulf Coast, where the loss of shoreline can exacerbate the effects of other natural hazards. As coastal land erodes, natural barriers such as beaches, wetlands, and barrier islands like the Galveston Island outside of Harris County diminish. The loss of these natural defenses reduces their ability to absorb storm surges and buffer inland areas from severe weather events. Additionally, future hurricane and tropical storm events will have a larger impact on vulnerable communities because the removal of natural defenses allows storm surges to reach further inland leading to an increase in flooding events and property damage or loss.

Coastal erosion contributes to increased flooding risks by destabilizing shorelines and removing protective wetlands that naturally absorb excess rainwater. In Harris County, where urban development is dense, stormwater drainage systems rely heavily on these natural buffers. When erosion depletes these areas, floodwaters have fewer places to disperse, leading to higher flood levels and prolonged inundation, especially during heavy rain events or hurricanes. The result is a greater likelihood of both flash flooding and chronic flooding, further stressing drainage infrastructure and increasing costs for flood mitigation and recovery.

Beyond immediate storm and flood impacts, coastal erosion can also intensify drought conditions. Wetlands and coastal ecosystems play a crucial role in maintaining regional water cycles, helping to regulate humidity and precipitation patterns. As these landscapes shrink due to erosion, disruptions in evapotranspiration and groundwater recharge can reduce available water supplies. This can compound drought conditions, particularly in times of low rainfall, as freshwater sources become more limited. With Harris County's growing population and water demand, such cascading effects can have long-term consequences on water availability, agriculture, and overall resilience to climate variability.

6.2 VULNERABILITY AND IMPACT ASSESSMENT

A detailed hazard exposure and vulnerability assessment was not conducted for coastal erosion due to the localized and infrequent nature of its impacts in this region. Unlike other hazards with broader geographic reach or more consistent historical data, coastal erosion in Harris County tends to affect limited areas along the shoreline and does not pose a widespread threat to the county's overall building stock, critical facilities, or economy. As such, a traditional vulnerability assessment—typically used to quantify potential losses and exposure—was deemed not applicable for this hazard. Coastal Erosion hazards have unique and varied risks associated with the Harris County planning partners, and these risks are summarized below.

Unique and Varied Risks

While Harris County is predominantly urbanized and inland, its southeastern boundary along Galveston Bay, Clear Lake, and the Houston Ship Channel is directly exposed to coastal erosion processes. Coastal erosion in Harris County is largely a chronic, slow-moving hazard—driven by storm surge, wave action, relative sea-level rise, land subsidence, and human modifications to natural shoreline systems.

Although only a portion of Harris County's jurisdictions are situated directly on the coast, the countywide impacts of coastal erosion extend inland through shoreline retreat, loss of wetlands and habitat, increased storm surge penetration, and degradation of protective natural buffers. These changes reduce the effectiveness of bayfront infrastructure and increase vulnerability to hurricanes, flooding, and chemical releases from nearby industrial facilities

Southern and Coastal Jurisdictions

The Cities of Seabrook, El Lago, Taylor Lake Village, Shoreacres, La Porte, and Morgan's Point experience the highest exposure to coastal erosion in Harris County. Seabrook and La Porte have documented shoreline retreat along Galveston Bay and Clear Lake, where bulkheads and revetments have been installed to slow erosion along residential and marina areas. Morgan's Point, located near the mouth of the Houston Ship Channel, is experiencing both natural shoreline retreat and anthropogenic erosion caused by vessel wake action, channel dredging, and high-energy storm events. El Lago and Taylor Lake Village, located along Taylor Lake and Clear Lake, face gradual shoreline loss that threatens recreational areas, residential properties, and local roadways. Repetitive exposure to tropical storms and hurricanes accelerates erosion, particularly when storm surge removes shoreline vegetation and deposits sediment into navigation channels.

In these communities, shoreline stabilization, habitat restoration, and green-infrastructure measures—such as living shorelines and oyster-reef restoration—are increasingly prioritized as long-term mitigation solutions.

Eastern Jurisdictions – Industrial, and Port Areas

In Baytown, Pasadena, La Porte, and Deer Park, coastal erosion intersects with industrial and port operations along the Houston Ship Channel and San Jacinto River. The San Jacinto Battleground State Historic Site and surrounding wetlands are actively retreating, threatening cultural resources and historical assets. Industrial facilities use levees and bulkheads to protect against erosion, but subsidence and sea-level rise continue to undermine these defenses. Goose Creek CISD and Lee College in Baytown are located inland but could experience indirect impacts through increased surge penetration and degradation of floodplain storage caused by shoreline loss. The Harris Health System's east-county clinics, located near coastal industrial corridors, face access and infrastructure risks from erosion-related flooding or containment failures of protective berms.

Here, coastal erosion poses not only environmental concerns but also technological and economic risks due to potential impacts on refinery operations, port logistics, and hazardous-material containment.

Northern and Western Jurisdictions

Jurisdictions such as Tomball, Jersey Village, Katy, and Waller, and school districts such as Cy-Fair ISD and Spring ISD, are located well inland and do not experience direct coastal erosion. However, they are indirectly affected by the regional consequences of erosion and land subsidence downstream. Accelerated shoreline retreat and wetland loss reduce coastal surge buffering capacity, increasing downstream flood volumes within bayous and reservoirs that drain toward the coast. Inland jurisdictions thus experience heightened flooding frequency and stormwater management challenges as sediment transport dynamics shift throughout the watershed.

Central and Urban Jurisdictions

While Houston, Bellaire, and West University Place are not exposed to active shoreline erosion, they are indirectly affected by subsidence and channelization within the bayou system. The University of Houston is located inland, but the county's overall loss of wetland storage capacity from coastal erosion increases downstream flood peaks in watersheds such as Brays and Buffalo Bayous. Harris Health System facilities, especially Ben Taub Hospital and LBJ Hospital, rely on urban drainage systems whose long-term effectiveness can be compromised by reduced estuarine outflow capacity along the coastal fringe.

Educational, Health Facilities and Institutions

San Jacinto College South Campus lies near Clear Lake and is one of the educational institutions most directly exposed to shoreline retreat and wind-wave erosion during tropical events. Coastal erosion here can undermine campus grounds, limit emergency-vehicle access, and heighten flood risk. Lee College and Goose Creek CISD facilities in Baytown may not be directly adjacent to eroding shorelines but rely on access routes vulnerable to surge and erosion-related flooding. Cy-Fair ISD and Spring ISD facilities, though inland, may face cascading effects from the broader regional degradation of coastal and bayou ecosystems, which diminishes the county's natural flood mitigation capacity.

6.2.1 Life, Health, and Safety

Coastal erosion in Harris County poses direct threats to life, health, and safety, especially for residents in low-lying coastal communities such as those near Galveston Bay and along the Bolivar Peninsula. Erosion can undermine foundations, contribute to sudden land loss, and increase the risk of injury or death during storm surge events. The loss of natural protective features like dunes and wetlands also exposes inland areas to greater impact from hurricanes and flooding, compounding health and safety risks during extreme weather events (National Oceanic and Atmospheric Administration [NOAA], 2020).

Overall Population

While the overall population of Harris County is largely urban and concentrated inland, the growth of coastal development and recreational communities has increased the number of residents exposed to erosion-related hazards. Population expansion into high-risk coastal zones, often for economic or

lifestyle reasons, raises long-term vulnerability. Residents in these areas may face displacement, repeated evacuation orders, and reduced access to emergency services during hazard events linked to erosion (Texas General Land Office [GLO], 2021).

Socially Vulnerable Population

Socially vulnerable populations, including low-income residents, older adults, and those without reliable transportation, are disproportionately impacted by coastal erosion in Harris County. These groups are more likely to live in at-risk areas due to lower property costs and may lack the resources to relocate or implement erosion-resistant improvements. Furthermore, they often face barriers to accessing emergency assistance, recovery funding, and insurance coverage, exacerbating their risk during and after coastal hazard events (Federal Emergency Management Agency [FEMA], 2020).

6.2.2 General Building Stock

Coastal erosion can have significant impacts on the general building stock, particularly in unincorporated or less regulated parts of Harris County where shoreline protection measures may be limited. Homes, businesses, and public facilities built too close to receding shorelines are increasingly at risk of damage or total loss. Saltwater intrusion and foundation destabilization from soil movement can also degrade building materials over time, increasing maintenance costs and reducing property values (NOAA, 2020).

6.2.3 Community Lifelines and Other Critical Facilities

Critical infrastructure such as roads, bridges, power lines, and water systems along the Harris County coastline are vulnerable to coastal erosion. Shoreline retreat can undermine transportation corridors, isolate communities, and disrupt access to emergency response routes. Water and wastewater treatment facilities located near the coast are at increased risk of saltwater intrusion, structural damage, and service interruptions, potentially leading to public health emergencies (U.S. Army Corps of Engineers [USACE], 2019).

6.2.4 Economy

According to the Texas General Land Office's CERPA Report (2023), Texas, which leads the nation in water-based commerce with an annual economic value exceeding $82.8 billion (TCS, 2019), is significantly concerned about erosion's impact on economic resources like the Gulf Intracoastal Waterway, ship channels, ports, petrochemical facilities, road infrastructure, and commercial businesses (Texas General Land Office, 2019). The tourism, fishing, and shipping industries are particularly vulnerable, as shoreline retreat affects beaches, piers, port access, and coastal ecosystems. Property loss and decreased land values in erosion-prone areas also reduce the local tax base, potentially affecting funding for schools, services, and infrastructure. Additionally, erosion-related recovery and protection measures create long-term financial burdens on local and state governments (GLO, 2021).

6.2.5 Natural, Historic and Cultural Resources

Harris County's historic and cultural assets located near the coast, such as historic districts, cemeteries, and cultural landmarks, are at risk of permanent loss due to coastal erosion. Erosion not only damages physical structures but can also displace or destroy archaeological resources buried within vulnerable shorelines. Once lost, these resources cannot be recovered, representing a significant cultural and historical loss for the region (National Park Service, 2020).

6.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

6.3.1 Potential or Planned Development

Future development in erosion-prone areas of Harris County may be limited or require enhanced protective measures, such as setback regulations, shoreline stabilization projects, or managed retreat. Without comprehensive planning and enforcement, development in high-risk coastal zones could exacerbate long-term erosion risks and increase the exposure of people and property to future loss (Houston-Galveston Area Council [H-GAC], 2021).

6.3.2 Projected Changes in Population

Population projections for coastal Harris County suggest continued growth, which may place more residents in areas at risk of coastal erosion. Without policy interventions or incentive-based relocation programs, this trend could increase the burden on emergency services, insurance systems, and disaster recovery funds. Planned development must consider erosion trends and direct population growth toward safer inland zones (Texas Demographic Center, 2021).

6.3.3 Climate Change

The Texas Coastwide Erosion Response Plan (2020) highlights Texas's vulnerability to damages caused by hurricanes and tropical systems. Both the Gulf and bay shorelines are susceptible to wave attacks, and the offshore seabed can undergo substantial changes as waves suspend bottom sediments, causing erosion ranging from centimeters (0.3 inches) to meters (3.2 feet) (Xu et al., 2016). Hurricane Ike significantly altered the shoreface profile offshore Bolivar Peninsula, creating large scarps that took over five years to stabilize. Erosion from storm surges and excessive rainfall, which can lead to severe flooding, poses major threats to coastal zones. Sedimentation studies at McFaddin National Wildlife Refuge compared Hurricanes Ike and Harvey, revealing that both storms deposited substantial sediment on the marsh surface. Notably, Hurricane Harvey's flood sedimentation was equivalent to seven years of typical sedimentation, significantly contributing to marsh growth (Xu et al., 2016).

Climate change is expected to accelerate coastal erosion in Harris County through sea level rise, stronger storm surges, and increased wave action. As rising seas encroach on the shoreline, previously stable areas may become erosion hotspots, endangering homes, infrastructure, and ecosystems. The loss of wetlands and barrier islands that act as natural buffers will further amplify the impact of erosion and reduce the county's resilience to future climate-related hazards (Intergovernmental Panel on Climate Change [IPCC], 2021).

6.3.4 Other Identified Conditions

Coastal erosion interacts with other hazards such as flooding, storm surge, and infrastructure degradation. For example, erosion can compromise the structural integrity of flood protection systems, including levees and seawalls, reducing their effectiveness during hurricanes. It also contributes to wetland loss, which diminishes the landscape's natural capacity to absorb water and mitigate storms—thereby creating a feedback loop that increases regional vulnerability to multiple hazards (USACE, 2019).

7. DAM/LEVEE FAILURE

7.1 HAZARD PROFILE

7.1.1 Hazard Description

A dam failure is any malfunction that adversely affects the primary function of the dam, releasing water or other liquid stored behind the dam (UNDRR 2023). When dams fail or overtop, they can lead to major flooding and cause catastrophic impacts in downstream areas. Throughout history, hundreds of dams have failed in the United States, causing property and environmental damage, injuries, and fatalities.

Definitions

Dams

A dam is an artificial barrier that has the ability to store water, wastewater, or liquid-borne materials for many reasons—flood control, human water supply, irrigation, livestock water supply, energy generation, containment of mine tailings, recreation, or pollution control. Many dams fulfill a combination of these functions.

Dams can be classified according to their purpose, the construction material or methods used, their slope or cross-section, the way they resist the force of the water pressure, or the means used for controlling seepage. Materials used to construct dams include earth, rock, tailings from mining or milling, concrete, masonry, steel, timber, plastic, rubber, and combinations of these.

Levees

A levee is a man-made earthen structure designed and constructed for the primary purpose to provide protection related to seasonal high flood water, storm surges, hurricanes, significant precipitation, and other weather events. They are normally subjected to temporary high water which can last from a few hours up to weeks during a year. A levee system usually consists of one or more levees and associated structures, such as floodwalls, closures, and drainage devices. These associated structures are constructed and operated in accordance with standard engineering practices. Privately owned levees sometimes have multiple owners, which can extend for miles (ASCE Texas Section 2025).

Levees are not feasible in most cases in Harris County for a variety of reasons. Levees typically remove valuable floodplain storage and block the ability of the channel to move water. There are also concerns with rainfall that falls on the levee itself. Most important is the possibility for catastrophic and sudden failure under extreme flood events, potentially resulting in loss of life and total destruction of property (National Geographic 2022).

Causes of Failure

Dams

A dam failure is any malfunction that adversely affects the primary function of the dam, releasing water or other liquid stored behind the dam (FEMA 2018). When dams fail or overtop, they can lead to major flooding and cause catastrophic impacts in downstream areas. Throughout history, hundreds of dams have failed in the United States, causing property and environmental damage, injuries, and fatalities.

Dam failures typically occur when spillway capacity is inadequate and excess flow overtops the dam, or when internal erosion (piping) through the dam or foundation occurs. Complete failure occurs if internal erosion or overtopping results in a complete structural breach, releasing a high-velocity wall of debris-filled water that rushes downstream, damaging anything in its path. Dam failures in the United States typically occur in one of four ways:

- Overtopping of the primary dam structure, which accounts for 34 percent of all dam failures, can occur due to inadequate spillway design, settlement of the dam crest, blockage of spillways, and other factors.
- Foundation defects due to differential settlement, slides, slope instability, uplift pressures, and foundation seepage can also cause dam failure. These account for 30 percent of all dam failures.
- Failure due to piping and seepage accounts for 20 percent of all failures. These are caused by internal erosion due to piping and seepage, erosion along hydraulic structures such as spillways, erosion due to animal burrows, and cracks in the dam structure.
- Failure due to problems with conduits and valves, typically caused by the piping of embankment material into conduits through joints or cracks, constitutes 10 percent of all failures.

The remaining 6 percent of U.S. dam failures are due to miscellaneous causes. Many dam failures in the United States are secondary results of other disasters. The prominent causes are earthquakes, landslides, extreme storms, massive snowmelt, equipment malfunction, structural damage, foundation failures, and sabotage (Association of State Dam Safety Officials, 2016).

Levees

A levee breach occurs when part of a levee gives way, creating an opening through which floodwaters may pass. A breach may occur gradually or suddenly. The most dangerous breaches happen quickly during periods of high water. Earthen levees can be damaged in several ways. Strong river currents and waves can erode the surface. Trees growing on a levee can blow over, leaving a hole where the root wad and soil used to be. Burrowing animals can create holes that enable water to pass through a levee. If severe enough, any of these situations can lead to a zone of weakness that could cause a levee breach. In seismically active areas, earthquakes and ground shaking can cause a loss of soil strength, weakening a levee and possibly resulting in failure. Seismic activity can also cause levees to slide or slump, both of which can lead to failure.

Regulatory Oversight

Dams

National Dam Safety Act

Potential for catastrophic flooding due to dam failures led to passage of the National Dam Safety Act (Public Law 92-367). The National Dam Safety Program requires a periodic engineering analysis of the majority of dams in the country; exceptions include the following:

- Dams under jurisdiction of the Bureau of Reclamation, Tennessee Valley Authority, or International Boundary and Water Commission
- Dams constructed pursuant to licenses issued under the Federal Power Act
- Dams that the Secretary of the Army determines do not pose any threat to human life or property.

The goal of this FEMA-monitored effort is to identify and mitigate the risk of dam failure so as to protect lives and property. The National Dam Safety Program is a partnership among states, federal agencies, and other stakeholders that encourages individual and community responsibility for dam safety. Under FEMA's leadership, state assistance funds have allowed all participating states to improve their programs through increased inspections, emergency action planning, and purchases of needed equipment. FEMA has also expanded existing and initiated new training programs. Grant assistance from FEMA provides support for improvement of dam safety programs that regulate most of the dams in the United States.

Texas State Dam Safety Program

The Texas Commission on Environmental Quality Dam Safety Program monitors and regulates private and public dams in Texas. The program periodically inspects dams that pose a high or significant hazard and makes recommendations and reports to dam owners to help them maintain safe facilities. This program is governed by Texas Administrative Code Title 30, Part 1, Chapter 299: Dams and Reservoirs.

Effective September 1, 2013, dams are exempt from safety requirements if they meet all of the following criteria:

- Are located on private property
- Have a maximum impoundment capacity of less than 500 acre-feet
- Are classified as low or significant hazard
- Are located in a county with a population of less than 350,000 (per the 2010 census)
- Are not located within the corporate limits of a municipality.

Owners of exempt dams still have to comply with maintenance and operation requirements. There is no exemption expiration date. Harris County is not an exempt county (TCEQ 2025). Table 7-1 summarizes how the Texas Commission on Environmental Quality defines dam hazard potential.

Table 7-1. USACE and TCEQ Dam Classification Criteria

Category	TCEQ Definition	USACE Definition
Low	A dam in the low-hazard potential category has: no loss of human life expected (no permanent habitable structures in the breach inundation area downstream of the dam); and minimal economic loss (located primarily in rural areas where failure may damage occasional farm buildings, limited agricultural improvements, and minor highways.	Dams assigned the low hazard potential classification are those where failure or mis-operation results in no probably loss of human life and low economic and/or environmental losses. Losses are principally limited to the owner's property.
Significant	A dam in the significant-hazard potential category has: loss of human life possible (one to six lives or one or two habitable structures in the breach inundation area downstream of the dam); or appreciable economic loss, located primarily in rural areas where failure may cause: damage to isolated homes damage to secondary highways as defined in §299.2(58);damage to minor railroads; or interruption of service or use of public utilities, including the design purpose of the utility.	Dams assigned the significant hazard potential classification are those dams where failure or mis-operation results in no probable loss of human life but can cause economic loss, environmental damage, disruption of lifeline facilities, or impact other concerns. Significant hazard potential classification dams are often located in predominantly rural or agricultural areas but could be in areas with population and significant infrastructure.
High	A dam in the high-hazard potential category has: loss of life expected (seven or more lives or three or more habitable structures in the breach inundation area downstream of the dam); or excessive economic loss, located primarily in or near urban areas where failure would be expected to cause extensive damage to: public facilities; agricultural, industrial, or commercial facilities; public utilities, including the design purpose of the utility; main highways and/or railroads used as a major transportation system.	Dams assigned the high hazard potential classification are those where failure or mis-operation will probably cause loss of human life.

Source: (HCOHSEM 2020)

U.S. Army Corps of Engineers Dam Safety Program

The U.S. Army Corps of Engineers operates and maintains approximately 740 dams nationwide. It is also responsible for safety inspections of some federal and non-federal dams in the United States that meet the size and storage limitations specified in the National Dam Safety Act. The Corps has inventoried dams; surveyed each state and federal agency's capabilities, practices and regulations regarding design, construction, operation and maintenance of the dams; and developed guidelines for inspection and evaluation of dam safety. The Corps maintains the National Inventory of Dams, which contains information about a dam's location, size, purpose, type, last inspection and regulatory status (U.S. Army Corps of Engineers 2021).

Federal Energy Regulatory Commission Dam Safety Program

The Federal Energy Regulatory Commission (FERC) cooperates with a large number of federal and state agencies to ensure and promote dam safety. Dam safety is a critical part of the Commission's hydropower program and receives top priority. Before projects are constructed, the Commission staff reviews and approves the designs, plans, and specifications of dams, powerhouses, and other structures. During construction, Commission staff engineers frequently inspect a project, and once construction is complete, Commission engineers continue to inspect it on a regular basis. More than 3,000 dams are part of regulated hydroelectric projects in the FERC program.

FERC inspects hydroelectric projects on an unscheduled basis to investigate the following:

- Potential dam safety problems
- Complaints about constructing and operating a project
- Safety concerns related to natural disasters
- Issues concerning compliance with the terms and conditions of a license.

Every five years, an independent engineer approved by the FERC must inspect and evaluate projects with dams higher than 32.8 feet (10 meters), or with a total storage capacity of more than 2,000 acre-feet.

FERC monitors seismic research and applies it in performing structural analyses of hydroelectric projects. FERC also evaluates the effects of potential and actual large floods on the safety of dams. During and following floods, FERC visits dams and licensed projects, determines the extent of damage, if any, and directs any necessary studies or remedial measures the licensee must undertake. The FERC publication Engineering Guidelines for the Evaluation of Hydropower Projects guides the FERC engineering staff and licensees in evaluating dam safety. The publication is frequently revised to reflect current information and methodologies.

FERC requires licensees to prepare emergency action plans and conducts training sessions on how to develop and test these plans. The plans outline an early warning system if there is an actual or potential sudden release of water from a dam due to failure. The plans include operational procedures that may be used, such as reducing reservoir levels and reducing downstream flows, as well as procedures for notifying affected residents and agencies responsible for emergency management. These plans are frequently updated and tested to ensure that everyone knows what to do in emergency situations.

Levees

Corps of Engineers and FEMA Levee Oversight

The Corps and FEMA have differing roles and responsibilities related to levees. The Corps addresses operation and maintenance, risk communication, risk management, and risk reduction issues as part of its responsibilities under the Levee Safety Program. FEMA addresses mapping and floodplain management issues related to levees, and it accredits levees as meeting requirements set forth by the National Flood Insurance Program.

The Corps and FEMA may be involved with a levee sponsor and community independently or jointly. The two agencies' goals are similar: reduce risk and lessen the devastating consequences of flooding. Corps and FEMA partnering activities related to levees include the following:

- Joint meetings with levee sponsors and other stakeholders
- Integration of levee information into the National Levee Database
- State Silver Jackets teams
- Sharing of levee information
- Targeted task forces to improve program alignment.

Coordination between the Corps and FEMA on levees is now standard within many of each agency's practices. Over the past several years, both agencies coordinated policies where appropriate; jointly participated in meetings with stakeholders; and participated in many multiagency efforts, such as the National Committee on Levee Safety, the Federal Interagency Floodplain Management Task Force, and the Silver Jackets Program, which brings together state, federal, tribal, and local agencies to learn from each other and apply their knowledge to reduce risk from hazards.

National Committee on Levee Safety

Congress created the National Committee on Levee Safety—made up of representatives from state, regional, and local agencies; the private sector; the Corps; and FEMA—to develop a national levee safety program. The Committee has been working toward this goal since October 2008.

Maintenance Requirements

No levee provides protection from events for which it was not designed, and levees require maintenance to continue to provide the level of protection they were designed and built to offer. Maintenance responsibility belongs to a variety of entities including local, state, and federal government and private landowners. Well-maintained levees may obtain certification through independent inspections.

Levees may not be certified for maintaining flood protection when the levee owner does not maintain the levee or pay for an independent inspection. The impacts of an un-certified levee include higher risk of levee failure. In addition, insurance rates may increase because FEMA identifies on Flood Insurance Rate Maps that the structures are not certified to protect from a 1-percent annual chance flood event.

7.1.2 Location

The "inundation zone" is the area downstream of the dam/levee that would be flooded in the event of a failure (breach) or uncontrolled release of water. These inundation zones are generally much larger than the area for the normal river or stream flood event. It is assumed that dam breaks happen at the time of maximum capacity and that the released water would inundate a downstream area proportional to the maximum capacity of the dam (TDEM, Texas State Hazard Mitigation Plan 2023).

Dams

The National Inventory of Dams lists 33 dams in Harris County (USACE/NID 2025), and information on each is summarized in Table 7-2. Figure 7-1 shows the locations of these dams within the planning area. Dam failure inundation mapping to the scope and scale needed for risk and vulnerability assessment of the dam failure hazard was not available for all 33 dams, however, inundation files were provided for Barker Dam, Addicks Dam, and Lake Houston Dam. These inundation scenarios were utilized to support this plan update. Inundation areas for these three, as shown in Figure 7-2, were the only three dam structures for which inundation mapping was available during the planning process.

Table 7-2 Harris County Dam and Levee Information

Dam/ Levee Name	Federal ID	Primary Purpose *a*	River or Stream Name	Owner Names	Dam/ Levee Type	Dam/ Levee Height (Ft)	Dam/ Levee Length (Ft)	Year Completed	Max Storage (Acre-Ft)	Hazard Potential Classification

DAM HAZARD DATA NOT AVAILABLE FOR PUBLIC REVIEW
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Dam/ Levee Name	Federal ID	Primary Purpose *a*	River or Stream Name	Owner Names	Dam/ Levee Type	Dam/ Levee Height (Ft)	Dam/ Levee Length (Ft)	Year Completed	Max Storage (Acre-Ft)	Hazard Potential Classification

DAM HAZARD DATA NOT AVAILABLE FOR PUBLIC REVIEW
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Source: (USACE/NID 2025); (USACE/NLD 2025)
Notes: N/A=Not Applicable or Not Available
a. I= Irrigation, WS= Water Supply, FP=Fire Protection, R= Recreation, O= Other, FC= Flood Control

Figure 7-1. Dam and Levee Locations across Harris County

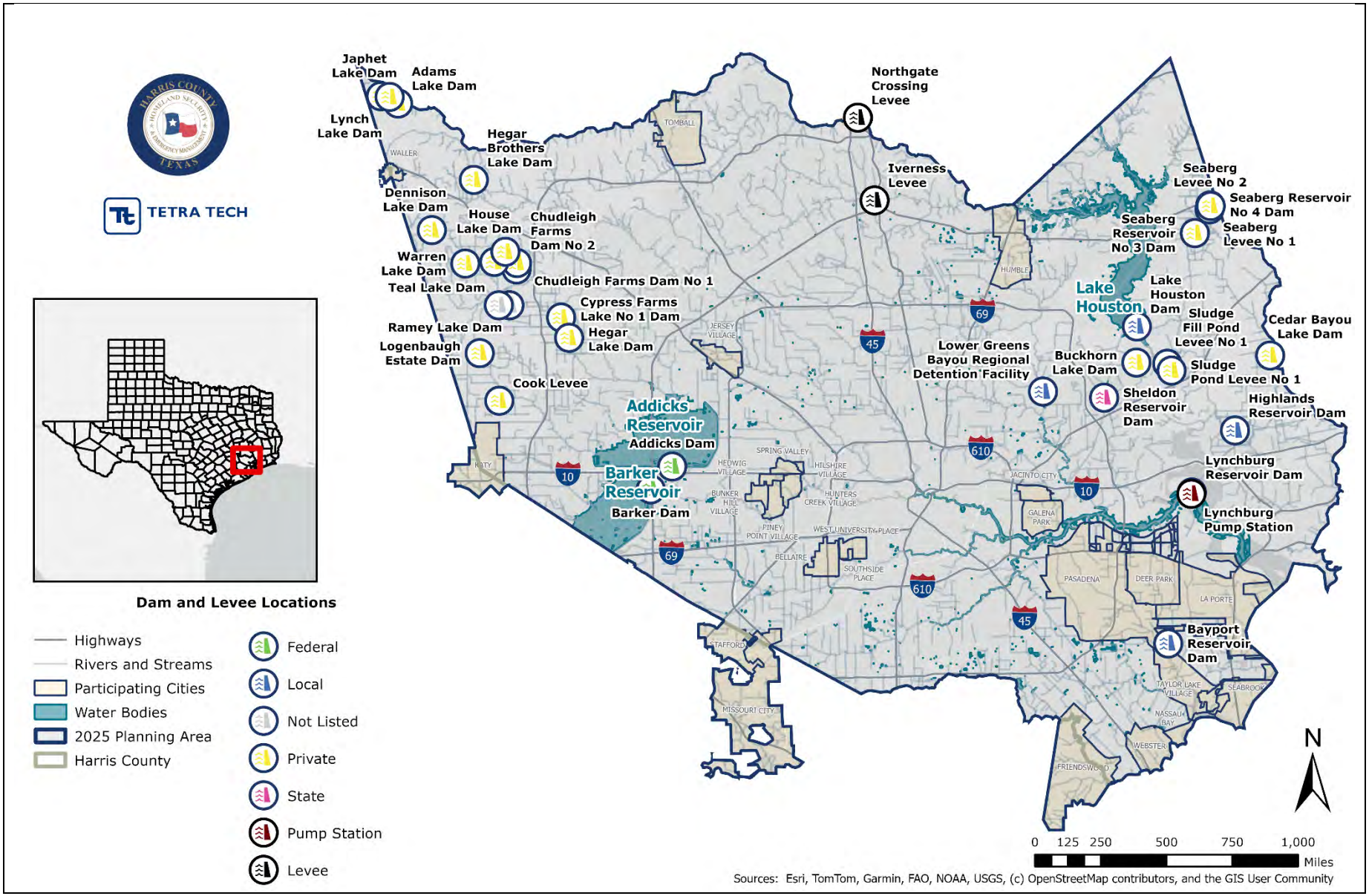

Figure 7-2. Dam Failure Inundation Area for Harris County High Hazard Dams

DAM INUNDATION MAPPING NOT AVAILABLE FOR PUBLIC REVIEW

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Levees

The USACE National Levee Database identifies three levee systems located in the County, and each is summarized in Table 7-2 and mapped in Figure 7-1. Brief summaries of these levee systems are further discussed below.

Cypress Creek System

Also referred to as "Iverness Levee", the Cypress Creek System is located along Cypress Creek in the community of Spring, north of Houston. The length of embankment for this system is 0.92 miles. The system protects a population of 545 and 179 structures with property values an estimated to be $64 million (USACE/NLD 2025). This system is accredited by FEMA and is mapped as such on the Flood Insurance Rate Map (FIRM) for its area.

Lynchburg Pump Station

The Lynchburg Pump Station system was federally constructed in 2002. The system is 4,012 feet long, consisting of 2,851 feet of earthen levee, and 1,161 feet of floodwalls. Both ends of the system tie into the existing reservoir levee on the north side of the project. A southern portion of the levee system was integrated into the Independence Parkway roadway. To facilitate crossing for the system, there is a single swing gate closure structure. A gravity drainage structure allows passage of rainwater from within the system (USACE/NLD 2025).

The Lynchburg Pump Station system reduces damage associated with hurricane storm surge to the pump station that provides drinking water for the City of Houston. No permanent residents are located inside the levee area. Total protected property value is estimated to be $1 million. The consequences associated with inundation to the protected area also include potentially leaving Houston without enough drinking water. The system is currently operated and maintained by Coastal Water Authority (USACE/NLD 2025).

The risk associated with this levee system is considered low. For the Lynchburg Pump Station the record surge loading is 75% of the levee height which occurred in 2008 during Hurricane Ike. The likelihood of failure of the system prior to surge water elevation reaching the top of the levees and floodwalls is low. The primary risk driver in this scenario is seepage of water through the foundation of the levees and floodwalls causing soil to be eroded away until they collapsed. After Hurricane Ike there was no evidence of this seepage thus increasing the certainty of the system to perform well in the next flood event. The likelihood of failure of the system when surge water elevation is greater than the heights of the levees and floodwalls is high. This overtopping of the structure by water and can quickly erode the backsides of the earthen levees, and the foundations soil of floodwalls until they collapse. If the system were to fail the flood depths could be over 22 feet for some areas (USACE/NLD 2025).

Spring Creek System

This system is located along Spring Creek in unincorporated Harris County. The length of embankment for this system is 0.98 miles. The system protects a population of 1,271 and 446

structures with an estimated value of $218 million. This system is accredited by FEMA and is mapped as such on the FIRM for its area.

7.1.3 Extent

Dams

Dam failure can be catastrophic to all life and property downstream. The U.S. Army Corps of Engineers developed a classification system for the hazard potential of dam failures. There are currently 7,402 dams in the State of Texas, of which 1,607 are classified as high hazard potential, 558 are classified as significant hazard potential, and 5,237 are classified as low hazard potential (USACE/NID 2025). In addition to USACE classifications, the Texas Commission on Environmental Quality (TCEQ) classifies hazards of dams based on potential loss of human life or property damage in the event of failure or malfunction. Table 7-1 provides the hazard classification definitions for TCEQ and USACE.

The Texas Commission on Environmental Quality lists six High-Hazard Potential Dams (HHPD) dams in Harris County as high-risk, listed below:

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The high-risk dam information is deemed sensitive by the Commission; therefore, detailed information such as location is not released to the general public. This information is maintained by HCOHSEM and is available upon request by other governmental entities with the appropriate need to know. Appendix C contains a one-page summary sheet discussing the benefits and purpose of the HHPD Rehabilitation Program, and this material was distributed to HHPD dam owners/agencies during the planning process.

Levees

In the event of a levee failure, floodwaters may ultimately inundate the protected area landward of the levee. The extent of inundation is dependent on the flooding intensity. Failure of a levee during a 1-percent annual chance flood will inundate the floodplain previously protected by the levee. Residential and commercial buildings near the levee overtopping or breach location will suffer the most damage from the initial embankment failure flood wave. Landward buildings will be damaged by inundation.

7.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2024, Harris County was included in zero major disaster (DR) or emergency (EM) declarations for Dam/Levee Failure-related events (FEMA 2023).

State Emergency Proclamations

Similarly, Harris County has not been included in Dam/Levee Failure-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2017 and 2024, Harris County was included in no Dam/Levee Failure-related agricultural disaster declarations.

Previous Events

Known hazard events that impacted Harris County between January 2017 and June 2025 are discussed in Table 7-3. For events prior to 2020, refer to the 2020 Harris County Multi-Jurisdictional Hazard Mitigation Action Plan (MJ-HMAP).

Table 7-3. Dam/Levee Failure Events in Harris County (2017 to 2024)

Event Date	Declaration or Proclamation Number	Location Impacted	Description
August 19, 2017	DR-4332-TX	Harris County	Addicks Reservoir Spillway Overtopping – during Hurricane Harvey, the Addicks Reservoir exceeded its capacity, resulting in uncontrolled releases and significant downstream flooding. To prevent dam failure during Harvey, controlled releases from Barker Reservoir led to flooding in adjacent neighborhoods.

Source: (CBS 2017)

Note: DR-4332-TX was declared due to Harvey, not specifically for dam/levee failures, however, these two events were cascading hazards which resulted from Hurricane Harvey.

7.1.5 Probability of Future Occurrences

Texas has recorded relatively few historical dam failures but the Texas Commission on Environmental Quality (TCEQ) and local emergency managers closely monitor the risk. Based solely on past events, the likelihood of a major future dam failure in Texas is low. However, the growing population and urban development in high-risk areas are increasing the overall risk. Future increases in rainfall intensity and duration will add more stress to dam systems during floods. Aging dams also heighten the risk of failure and catastrophic flooding in dam inundation zones. The probability of a dam failure event in Texas is considered "Occasional," meaning it could occur within the next five years.

Information on previous dam/levee failure occurrences in the County was used to calculate the probability of future occurrences of such events, as summarized in Table 7-4. Based on historical

records and input from the Steering Committee, the probability of occurrence for dam/levee failure events in the County is considered "rare."

Table 7-4. Probability of Future Dam/Levee Failure Hazard Events in Harris County

Hazard Type	Number of Occurrences Between 1950 and 2024	Percent Chance of Occurring in Any Given Year
Dam/Levee Failure	2	2.6%

Source: (CBS 2017)

Climate Change Projections

Texas is experiencing changes in its climate, with most areas warming by 0.5 to 1 degree Fahrenheit over the past century. The eastern two-thirds of the state are seeing more intense rainstorms and increasingly severe floods. In the future, Texas is likely to face more severe storms. Extreme precipitation events heighten the risk of dam and levee failures (EPA 2016) (Centers for Climate and Energy Solutions n.d.). These changes pose significant challenges for infrastructure and water management systems. Additionally, the increased frequency and intensity of storms could lead to greater economic and environmental impacts, affecting communities and ecosystems across the state.

Assumptions about river flow patterns, depicted through hydrographs, play a crucial role in the design of dams and levees. Alterations in weather patterns can significantly impact the hydrographs used for these designs. If the hydrograph changes, the dam or levee may lose some or all of its intended safety margin, known as freeboard. This loss of safety margin increases the risk of floodwaters overtopping the dam or levee or creating unexpected loads, potentially leading to structural failure. Ensuring the accuracy of hydrographs is vital for maintaining the integrity of these structures. As weather patterns continue to evolve, regular updates and assessments of hydrographs are essential to safeguard against potential failures.

7.1.6 Cascading Impacts on Other Hazards

Dam and levee failures in Harris County can trigger severe cascading hazards that significantly amplify flood risk and disrupt community systems. A sudden structural failure can result in catastrophic downstream flooding, particularly in low-lying areas of the county such as Brays Bayou, Buffalo Bayou, and along the Addicks and Barker reservoirs. These failures may overwhelm stormwater systems, submerge critical roadways, and disrupt evacuation routes, thereby increasing the potential for loss of life and injury (FEMA 2020). Additionally, rapid flooding can destabilize bridges, culverts, and transportation corridors, delaying emergency response efforts and exposing populations to prolonged hazard conditions (NIAC 2018).

The cascading impacts extend to public health, the environment, and infrastructure resilience. A failure event can damage wastewater treatment plants, leading to contamination of drinking water sources and localized environmental hazards, such as exposure to untreated sewage or hazardous materials stored in flooded industrial sites. Utility services—including electricity, telecommunications,

and potable water—may be interrupted for days or weeks depending on the magnitude of the breach, particularly in socially vulnerable neighborhoods with limited adaptive capacity. These cascading failures can also increase the likelihood of mold outbreaks, disease transmission, and long-term displacement, compounding the initial disaster into a multi-dimensional crisis.

7.2 VULNERABILITY AND IMPACT ASSESSMENT

Although large-scale dam or levee failures are rare in Harris County, the potential consequences of such an event could be catastrophic for downstream communities. The county's flood control network includes two major federal flood-retention structures—Addicks and Barker Reservoirs—managed by the U.S. Army Corps of Engineers (USACE), and numerous smaller local detention basins and levee systems managed by the Harris County Flood Control District (HCFCD), municipal drainage authorities, and private entities.

Potential failure modes include overtopping or breach due to extreme rainfall, structural failure of earthen embankments, or localized failure of smaller levees protecting subdivisions, schools, or industrial facilities. While most jurisdictions are indirectly affected through flood propagation, a few lie directly within reservoir or levee-protected areas and face significant residual risk. Section 4.2.2 provides a detailed summary of the methodology and process used to assess Harris County's risk to dam and levee failure.

Unique and Varied Risks

Western Jurisdictions − Addicks and Barker Reservoirs

The greatest concentration of dam-related risk occurs in western Harris County, where the Addicks and Barker Reservoirs store runoff from the upper Buffalo Bayou watershed. The Cities of Katy, Jersey Village, and portions of unincorporated Harris County lie within or immediately downstream of these reservoirs. A significant number of residential subdivisions, schools, and public facilities were built within reservoir pools that can retain stormwater for extended periods during major rain events. Cy-Fair ISD has multiple campuses situated downstream of Addicks Reservoir; a catastrophic breach or controlled release could inundate portions of these areas. The City of Katy and nearby MUDs have invested in floodproofing and elevation studies, but the residual risk remains for neighborhoods near Mason Creek and Bear Creek tributaries.

If a breach or overtopping event occurred, the resulting flooding would flow eastward along Buffalo Bayou and its tributaries, impacting both western suburbs and downstream urban centers

Central and Urban Jurisdictions

Jurisdictions located downstream of Addicks and Barker, including Houston, Bellaire, West University Place, and Southside Place, face secondary risk from the rapid propagation of reservoir floodwaters. During Hurricane Harvey (2017), controlled releases from both reservoirs inundated neighborhoods along Buffalo Bayou, damaging homes, businesses, and critical transportation corridors such as I-10. The University of Houston and Harris Health System facilities, while outside the immediate reservoir zone, are vulnerable to downstream flooding and loss of access due to

transportation and utility disruptions. Critical infrastructure such as wastewater treatment plants and medical supply routes could be indirectly impacted by sustained high water levels following a dam release.

Northern Jurisdictions – Levee and Detention Systems along Cypress and Spring Creeks

The Cities of Tomball, Waller, and portions of unincorporated northern Harris County contain numerous smaller stormwater detention basins, earthen embankments, and privately maintained levees designed to protect subdivisions and roadways. These systems are typically designed for rainfall storage and local drainage control rather than large flood containment. Failure or overtopping of these structures, particularly during multi-day rainfall events, could flood schools, neighborhoods, and roadways along Cypress Creek, Little Cypress Creek, and Spring Creek. Spring ISD campuses near Cypresswood Drive and the Hardy Toll Road corridor lie within downstream drainage basins that would experience flood surge from any local detention or levee failure.

Eastern and Industrial Jurisdictions – San Jacinto and Ship Channel Systems

Communities along the San Jacinto River and Houston Ship Channel, including Baytown, Deer Park, Morgan's Point, and La Porte, are at risk from levee or containment structure failures associated with industrial facilities. Many petrochemical and port complexes maintain on-site levees or berms designed to contain floodwater and hazardous materials. Failure of such systems during a storm surge or extreme flood could lead to both inundation and secondary technological hazards such as hazardous material releases. Goose Creek CISD and Lee College campuses in Baytown could be affected by flooding from the San Jacinto River or industrial drainage channels. The Harris Health System clinics operating in Pasadena and eastern Harris County would face access challenges and potential service interruptions if levee failures or chemical contamination occurred.

Areawide Critical Facilities and Institutional Exposure

Across Harris County, schools, colleges, and health facilities often serve dual purposes as community shelters and response centers, magnifying the consequence of any dam or levee failure. School Districts (Cy-Fair ISD, Spring ISD, Goose Creek CISD) as they rely on bus networks and emergency shelter capabilities that could be disrupted by flood surge from structural failures. Higher Education Institutions (University of Houston, San Jacinto College, Lee College) also maintain research and operational facilities vulnerable to prolonged power loss and access restrictions following major inundation events. Harris Health System facilities must maintain continuity of care through backup power, elevated critical systems, and redundant access routes, particularly for Ben Taub and LBJ Hospitals, which may be isolated by flooded roadways even if not directly inundated.

7.2.1 Life, Health, and Safety

The impact of dam failure on life, health, and safety is dependent on several factors, such as the class of dam, the area that the dam is protecting, the location of the dam, and the proximity of structures, infrastructure, and critical facilities to the dam structure.

The estimation for population exposure to the dam inundation hazard area is shown in Table 7-5 for the Addicks Reservoir Dam, Lake Houston Dam, and Barker Dam. Based on the spatial analysis, there are an estimated 1,320,689 residents living in the Addicks Reservoir Dam Inundation Area, or 27 percent of the County's total population. There are an estimated 1,341 residents living in the Lake Houston Dam Inundation Area, or less than 0.1 percent of the County's total population. There are an estimated 1,077,214 residents living in the Barker Dam Inundation Area, or 22 percent of the County's total population. The City of Bellaire has the greatest number of residents living in the Addicks Reservoir Dam Inundation Area with approximately 17,013 residents and 16,954 residents in the Barker Dam Inundation Area.

Table 7-5. Estimated Number of Persons in Harris County Living in Dam Inundation Areas

	Total Population	Population Living in Dam Failure Inundation Area					
		Addicks Reservoir Dam		Barker Dam		Lake Houston Dam	
Jurisdiction	(American Community Survey 2023)	Number of People	Percent of Total	Number of People	Percent of Total	Number of People	Percent of Total
Baytown[b]	79,635	0	0.0%	0	0.0%	251	0.3%
Bellaire	17,116	17,013	99.4%	16,954	99.1%	0	0.0%
Bunker Hill Village	3,801	3,387	89.1%	2,414	63.5%	0	0.0%
Deer Park	33,823	0	0.0%	0	0.0%	0	0.0%
El Lago	3,016	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	40,827	0	0.0%	0	0.0%	0	0.0%
Galena Park	10,527	4,784	45.4%	3,209	30.5%	0	0.0%
Hedwig Village	2,029	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	1,091	629	57.7%	46	4.2%	0	0.0%
Houston[b]	2,252,110	1,160,942	51.5%	958,471	42.6%	33	<0.1%
Humble	16,489	0	0.0%	0	0.0%	0	0.0%
Hunters Creek Village	4,358	2,879	66.1%	1,969	45.2%	0	0.0%
Jacinto City	9,458	24	0.3%	0	0.0%	0	0.0%
Jersey Village	7,817	3,673	47.0%	557	7.1%	0	0.0%
Katy[a]	23,900	0	0.0%	0	0.0%	0	0.0%
La Porte	36,077	0	0.0%	0	0.0%	0	0.0%
League City[b]	2,527	0	0.0%	0	0.0%	0	0.0%
Missouri City[a]	75,234	6	0.0%	0	0.0%	0	0.0%
Morgan's Point	255	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	5,211	0	0.0%	0	0.0%	0	0.0%
Pasadena	149,345	6,488	4.3%	2,016	1.3%	12	<0.1%
Pearland[b]	8,189	0	0.0%	0	0.0%	0	0.0%
Piney Point Village	3,090	2,114	68.4%	1,768	57.2%	0	0.0%
Seabrook	13,617	0	0.0%	0	0.0%	0	0.0%
Shoreacres	1,327	0	0.0%	0	0.0%	0	0.0%
South Houston[b]	16,017	984	6.1%	0	0.0%	0	0.0%
Southside Place	1,631	1,627	99.8%	1,627	99.8%	0	0.0%

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Jurisdiction	Total Population (American Community Survey 2023)	Population Living in Dam Failure Inundation Area					
		Addicks Reservoir Dam		Barker Dam		Lake Houston Dam	
		Number of People	Percent of Total	Number of People	Percent of Total	Number of People	Percent of Total
Spring Valley Village	4,226	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	17,279	0	0.0%	0	0.0%	0	0.0%
Taylor Lake Village	3,678	0	0.0%	0	0.0%	0	0.0%
Tomball	13,032	0	0.0%	0	0.0%	0	0.0%
Waller[a]	3,062	0	0.0%	0	0.0%	0	0.0%
Webster	12,283	0	0.0%	0	0.0%	0	0.0%
West University Place	14,907	14,602	98.0%	14,236	95.5%	0	0.0%
Unincorporated County	2,001,319	101,537	5.1%	90,901	4.5%	1,045	0.1%
Total	4,888,304	1,320,689	27.0%	1,077,214	22.0%	1,341	<0.1%

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; USACE 2025

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Overall Population

Dam and levee failures pose a significant risk to the overall population of Harris County. According to 2023 American Community Survey data, approximately 27% of the county's population—over 1.3 million people—are located within the Addicks Reservoir dam inundation area, while 22% are within the Barker Dam area. Though fewer individuals (about 1,341) are at risk from Lake Houston Dam, the potential for cascading impacts remains present. These populations face the threat of sudden, high-velocity floodwaters in the event of a dam breach, with limited time for evacuation. This level of exposure demonstrates how concentrated flood risk can be in downstream urbanized areas (FEMA 2022).

The risks are not limited to physical flooding alone; dam and levee failures can overwhelm emergency response systems and create long-term displacement, particularly in areas like unincorporated Harris County, which alone has more than 190,000 people located in dam inundation zones. Populations within these areas may experience cascading infrastructure failures—such as damaged transportation corridors and utility outages—that further delay emergency services, disrupt access to medical care, and hinder post-disaster recovery (USACE n.d.). These risks are especially pronounced in high-density areas near Buffalo Bayou and other downstream channels, where structural vulnerabilities are compounded by existing drainage limitations.

Moreover, the socioeconomic consequences for affected populations can be severe. Flooding due to dam failure can result in housing loss, long-term relocation, and decreased property values, which disproportionately affect low-income communities with limited resources for recovery. As these communities face repeated hazards, the cumulative impacts can deepen cycles of poverty and reduce community resilience (NIAC 2018). Given the scale of population exposure and the multifaceted

nature of risk, mitigation efforts must integrate structural improvements, early warning systems, and population-based planning to reduce the threat posed by dam and levee failure in Harris County.

Socially Vulnerable Population

Social vulnerability refers to the susceptibility of social groups to the adverse impacts of natural hazards, such as death, injury, loss, or disruption of livelihood. It considers the social, economic, demographic, and housing characteristics that affect a community's ability to prepare for, respond to, cope with, recover from, and adapt to environmental hazards.

Vulnerable populations downstream from dam or levee failures include those unable to escape in time, such as older adults individuals, children, and individuals with disabilities or access and functional needs who may be unable to get themselves out of the inundation area. It also includes people who may not receive adequate emergency warnings, like residents and visitors, and those relying on the dam for potable water.

Table 7-6, Table 7-7, and Table 7-8 summarize the vulnerable populations that are vulnerable to inundation waters during a maximum high breach scenario at the Addicks Reservoir Dam, Barker Dam, and Lake Houston Dam.

Of the population exposed to dam and levee failure in Harris County, the economically disadvantaged and individuals over the age of 65 are among the most vulnerable. Low-income residents may face barriers to evacuation due to limited access to transportation, an inability to afford temporary housing, or the absence of safe relocation options. These constraints can leave them trapped in high-risk areas during a flood event. Similarly, older adults, who make up approximately 11.5% of Harris County's population, are at greater risk due to mobility limitations, chronic health conditions, and increased reliance on medical services that may be inaccessible during emergencies. The combination of financial, physical, and health vulnerabilities makes these populations especially susceptible to the cascading impacts of dam failure, including prolonged displacement and reduced access to recovery resources. The best preparation for these effects includes awareness that they can occur, education of the public on prevention, and planning to deal with them during responses to dam failure events.

Table 7-6. Estimated Vulnerable Persons Located in Addicks Dam Inundation Area

Jurisdiction	Persons over 65		Persons Under 5		Non-English Speaking		Persons with a Disability		Persons Living in Poverty	
	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total
Baytown[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bellaire	3,263	99.4%	840	99.3%	277	99.3%	1,107	99.4%	441	99.3%
Bunker Hill Village	464	88.7%	105	91.3%	10	90.9%	150	91.5%	128	97.7%
Deer Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
El Lago	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Galena Park	472	45.5%	375	45.3%	262	45.4%	601	45.3%	1,116	45.4%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	106	57.6%	25	56.8%	1	50.0%	22	56.4%	4	57.1%
Houston[b]	139,880	51.6%	78,286	51.5%	61,445	51.5%	126,742	51.5%	224,893	51.5%
Humble	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hunters Creek Village	476	66.0%	119	65.7%	4	66.7%	155	66.0%	116	65.9%
Jacinto City	3	0.2%	1	0.1%	1	0.2%	3	0.2%	4	0.2%
Jersey Village	791	47.0%	232	46.9%	81	46.6%	380	46.9%	185	46.8%
Katy[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
La Porte	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
League City[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Missouri City[a]	1	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Morgan's Point	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Pasadena	740	4.3%	486	4.3%	277	4.3%	780	4.3%	1,180	4.3%
Pearland[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Piney Point Village	461	68.4%	18	66.7%	4	66.7%	108	67.9%	16	66.7%
Seabrook	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Shoreacres	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%

Jurisdiction	Persons over 65		Persons Under 5		Non-English Speaking		Persons with a Disability		Persons Living in Poverty	
	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total
South Houston[b]	85	6.1%	68	6.1%	48	6.0%	97	6.1%	231	6.1%
Southside Place	302	99.7%	73	98.6%	2	66.7%	106	99.1%	49	98.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Taylor Lake Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Tomball	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Waller[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Webster	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
West University Place	2,430	97.9%	1,121	97.9%	8	88.9%	670	98.0%	285	97.9%
Unincorporated County	10,335	5.1%	7,311	5.1%	2,915	5.1%	9,761	5.1%	12,485	5.1%
Total	159,809	28.4%	89,060	26.3%	65,335	33.9%	140,682	27.7%	241,133	32.4%

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; USACE 2025

Note: Max High breach scenario

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b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Table 7-7 Estimated Vulnerable Persons Located in Barker Dam Inundation Area

Jurisdiction	Persons over 65		Persons Under 5		Non-English Speaking		Persons with a Disability		Persons Living in Poverty	
	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total
Baytown[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bellaire	3,251	99.0%	837	98.9%	276	98.9%	1,103	99.0%	439	98.9%
Bunker Hill Village	331	63.3%	75	65.2%	10	90.9%	108	65.9%	93	71.0%
Deer Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
El Lago	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Galena Park	316	30.5%	251	30.4%	175	30.3%	403	30.4%	749	30.4%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	7	3.8%	1	2.3%	0	0.0%	1	2.6%	0	0.0%
Houston[b]	115,487	42.6%	64,632	42.6%	50,726	42.5%	104,634	42.5%	185,661	42.5%
Humble	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hunters Creek Village	325	45.1%	81	44.8%	2	33.3%	106	45.1%	80	45.5%
Jacinto City	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jersey Village	120	7.1%	35	7.1%	12	6.9%	57	7.0%	28	7.1%
Katy[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
La Porte	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
League City[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Missouri City[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Morgan's Point	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Pasadena	230	1.3%	151	1.3%	86	1.3%	242	1.3%	366	1.3%
Pearland[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Piney Point Village	385	57.1%	15	55.6%	3	50.0%	90	56.6%	13	54.2%
Seabrook	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%

Jurisdiction	Persons over 65		Persons Under 5		Non-English Speaking		Persons with a Disability		Persons Living in Poverty	
	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total
Shoreacres	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
South Houston[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Southside Place	302	99.7%	73	98.6%	2	66.7%	106	99.1%	49	98.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Taylor Lake Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Tomball	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Waller[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Webster	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
West University Place	2,369	95.5%	1,093	95.5%	8	88.9%	653	95.5%	277	95.2%
Unincorporated County	9,249	4.5%	6,545	4.5%	2,606	4.5%	8,737	4.5%	11,169	4.5%
Total	129,121	22.9%	72,952	21.6%	53,630	27.8%	115,137	22.6%	198,485	26.7%

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; USACE 2025

Note: Max High breach scenario

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b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Table 7-8. Estimated Vulnerable Persons Located in Lake Houston Dam Inundation Area

Jurisdiction	Persons over 65		Persons Under 5		Non-English Speaking		Persons with a Disability		Persons Living in Poverty	
	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total
Baytown[b]	29	0.3%	18	0.3%	7	0.3%	32	0.3%	4	0.3%
Bellaire	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bunker Hill Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Deer Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
El Lago	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Galena Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Houston[b]	4	<0.1%	2	<0.1%	1	<0.1%	3	<0.1%	6	<0.1%
Humble	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hunters Creek Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jacinto City	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jersey Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Katy[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
La Porte	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
League City[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Missouri City[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Morgan's Point	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Pasadena	1	<0.1%	0	0.0%	0	0.0%	1	<0.1%	2	<0.1%
Pearland[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Piney Point Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Seabrook	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Shoreacres	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%

Jurisdiction	Persons over 65		Persons Under 5		Non-English Speaking		Persons with a Disability		Persons Living in Poverty	
	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total	Count	% of Total
South Houston[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Southside Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Taylor Lake Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Tomball	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Waller[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Webster	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
West University Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Unincorporated County	106	0.1%	75	0.1%	29	0.1%	100	0.1%	128	0.1%
Total	140	<0.1%	95	<0.1%	37	<0.1%	136	<0.1%	140	<0.1%

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; USACE 2025

Note: Max High breach scenario

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

7.2.2 General Building Stock

Properties located within the dam or levee failure inundation zones are at significant risk, as they would be subjected to the most intense and destructive water surges. This surge can lead to severe damage or complete destruction of these properties. Additionally, transportation infrastructure such as roads, railways, and bridges within these zones are highly vulnerable. If these routes are compromised, it could result in isolation of affected areas, making it difficult for residents to evacuate or receive emergency services. The most at-risk transportation routes are those already in poor condition, as they are less likely to withstand the force of a large water surge.

Moreover, essential utilities like overhead power lines, cable, and telephone lines are also susceptible to damage in the event of a dam or levee failure. The loss of these utilities can exacerbate isolation issues, cutting off communication and power to the inundation areas, further complicating rescue and recovery efforts. Ensuring the resilience of these infrastructures is crucial to mitigate the impact of such disasters.

Table 7-9 summarizes the number of structures located in the dam failure inundation area for Addicks Reservoir Dam, Lake Houston Dam, and Barker Dam. In summary, there are 358,396 buildings located in the Addicks Reservoir Dam Inundation Area with an estimated $452 billion of replacement cost value (i.e., building and content replacement costs). In total, this represents approximately 26.4 percent of the County's total general building stock inventory. In addition, there are 472 buildings located in the Lake Houston Dam Inundation Area with an estimated $406 billion of building stock and contents exposed, representing less than 0.1 percent of the County's total general building stock inventory. Lastly, there are 290,620 buildings located in the Barker Dam Inundation Area with an estimated $386 billion of building stock and contents exposed, which represents approximately 21.4 percent of the County's total general building stock inventory.

7.2.3 Community Lifelines and Other Critical Facilities

Transportation routes are at risk of being severely impacted by dam or levee flooding, potentially leading to isolation and major travel disruptions. This includes all roads, railways, and bridges in the affected areas. Routes in poor condition are particularly susceptible and may not withstand a significant water surge. Utilities such as overhead power lines, cable, and phone lines within the flood zone are also at risk. The loss of phone lines could cause major communication issues, especially in areas with limited cell phone coverage. Additionally, emergency response efforts would be hampered due to the loss of transportation routes and some critical facilities within the flood zone. The recovery period to restore essential services, including wastewater, potable water, and other community facilities, could be prolonged. In conclusion, the vulnerability of transportation routes and utilities to dam or levee flooding underscores the need for robust infrastructure and emergency preparedness plans. Ensuring that critical routes and services are resilient can mitigate the impact of such events and facilitate quicker recovery.

Table 7-9. Estimated General Building Stock Located in the Dam Inundation Area

Jurisdiction	Addicks Reservoir Dam Inundation Area				Barker Dam Inundation Area				Lake Houston Dam Inundation Area			
	Count	% of Total	Total RCV of Buildings	% of Total	Count	% of Total	Total RCV of Buildings	% of Total	Count	% of Total	Total RCV of Buildings	% of Total
Baytown[b]	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	65	0.3%	$23,608,751	0.2%
Bellaire	6,502	100.0%	$6,920,624,681	100.0%	6,477	99.6%	$6,844,463,475	98.9%	0	0.0%	$0	0.0%
Bunker Hill Village	1,170	89.9%	$1,474,513,784	85.8%	832	64.0%	$945,981,760	55.1%	0	0.0%	$0	0.0%
Deer Park	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
El Lago	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Friendswood[a]	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Galena Park	1,473	44.8%	$450,748,898	30.4%	948	28.8%	$221,257,143	14.9%	0	0.0%	$0	0.0%
Hedwig Village	563	61.9%	$782,506,866	53.5%	202	22.2%	$199,340,833	13.6%	0	0.0%	$0	0.0%
Hilshire Village	188	58.6%	$140,010,070	52.4%	14	4.4%	$13,161,430	4.9%	0	0.0%	$0	0.0%
Houston[b]	305,524	51.9%	$398,306,000,000	64.2%	252,564	42.9%	$347,928,000,000	56.1%	33	<0.1%	$29,319,944	<0.1%
Humble	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Hunters Creek Village	1,036	65.9%	$1,554,811,104	67.0%	710	45.2%	$1,108,647,865	47.8%	0	0.0%	$0	0.0%
Jacinto City	7	0.2%	$1,466,124	0.1%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Jersey Village	1,163	47.7%	$1,145,507,613	40.0%	196	8.0%	$540,149,759	18.9%	0	0.0%	$0	0.0%
Katy[a]	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
La Porte	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
League City[b]	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Missouri City[a]	2	<0.1%	$687,623	<0.1%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Morgan's Point	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Nassau Bay	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Pasadena	1,868	4.6%	$1,065,165,726	3.8%	600	1.5%	$522,855,952	1.9%	8	<0.1%	$6,891,237	<0.1%
Pearland[b]	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Piney Point Village	825	69.0%	$1,744,290,351	73.2%	691	57.8%	$1,536,547,267	64.5%	0	0.0%	$0	0.0%
Seabrook	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%

Jurisdiction	Addicks Reservoir Dam Inundation Area				Barker Dam Inundation Area				Lake Houston Dam Inundation Area			
	Count	% of Total	Total RCV of Buildings	% of Total	Count	% of Total	Total RCV of Buildings	% of Total	Count	% of Total	Total RCV of Buildings	% of Total
Shoreacres	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
South Houston[b]	262	5.2%	$66,864,138	2.5%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Southside Place	508	100.0%	$604,745,337	100.0%	508	100.0%	$604,745,337	100.0 %	0	0.0%	$0	0.0%
Spring Valley Village	1,471	97.5%	$1,500,042,828	95.6%	1,148	76.1%	$1,215,038,199	77.5%	0	0.0%	$0	0.0%
Stafford[a]	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Taylor Lake Village	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Tomball	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Waller[a]	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Webster	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
West University Place	5,520	99.9%	$4,774,165,795	99.8%	5,376	97.3%	$4,614,392,442	96.5%	0	0.0%	$0	0.0%
Unincorporated County	30,314	5.2%	$31,489,306,528	7.8%	26,831	4.6%	$27,019,011,352	6.7%	366	0.1%	$346,271,602	0.1%
Harris County Total	358,396	26.4%	$452,021,457,467	38.7%	290,620	21.4%	$386,469,129,340	33.1%	472	<0.1%	$406,091,533	<0.1%

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018, RS Means 2024; DSS-WISE FEMA 2025

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Community lifelines are essential structures that ensure the safety of Harris County residents during and after hazard events. These lifelines include transportation systems like highways, ports, and railways, as well as communication networks such as phone towers, satellites, radios, and mass notification systems. When these lifelines are disrupted, the risk associated with the hazard increases. Dam failures can potentially damage or interrupt these lifelines, heightening Harris County's overall risk during and after such events.

Table 7-10 details the community lifelines within the dam failure inundation areas, with transportation being the most prevalent, followed by health and medical services, communication systems, and safety and security measures.

Table 7-10. Lifelines in the Addicks, Barker, and Lake Houston Dam Inundation Areas

FEMA Lifeline Category	Number of Lifelines	Number of Lifelines Located in the Addicks Reservoir Dam Inundation Area	Number of Lifelines Located in the Barker Dam Inundation Area	Number of Lifelines Located in the Lake Houston Dam Inundation Area
Communications	7,017	2,445	2,089	17
Energy	6,149	789	618	4
Food, Water, Shelter	6,462	2,161	1,819	0
Hazardous Materials	3,947	1,026	847	57
Health and Medical	806	273	220	0
Safety and Security	2,485	716	586	0
Transportation	4,649	1,499	1,331	69
Water Systems	4,801	807	641	121
Total	36,316	9,716	8,151	268

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; DSS-WISE FEMA 2025

7.2.4 Economy

Severe flooding resulting from a dam or levee failure can lead to extensive structural damage and disrupt essential services. The recovery costs from such flood damages will vary based on the hazard risk associated with each dam. Flooding can cause significant damage to public utilities and interrupt service delivery, potentially leading to power outages and communication failures. Drinking water and wastewater treatment facilities may be temporarily shut down, and debris from nearby buildings can accumulate, especially if the dam failure mimics major flood events like the 1-percent annual chance flood event mentioned in Section 12 (Flood).

The economic impact of dam or levee failure events can be substantial, affecting both local and regional economies. Similar to other flooding events, the losses can include damage to buildings and infrastructure, agricultural losses, business interruptions, and reductions in the tax base. The disruption of essential services such as power and communications, along with the temporary shutdown of water treatment facilities, can exacerbate the situation, making recovery efforts more challenging and costly. Additionally, the accumulation of debris can hinder rescue and recovery operations, further complicating the response to such disasters.

7.2.5 Natural, Historic and Cultural Resources

The environmental impacts of a dam failure in Harris County could include severe water quality degradation, widespread hazardous material distribution, and long-lasting ecosystem disruption. Floodwaters resulting from a breach at the Addicks, Barker, or Lake Houston dams could overwhelm multiple sanitary sewer systems and inundate nearby wastewater treatment facilities, leading to the discharge of untreated sewage into neighborhoods and bayous. Industrial and residential areas at risk of flooding may also contain unsecured containers of oil, fertilizers, pesticides, or other hazardous chemicals that could be swept into the floodplain, further contaminating the water and soil. These materials may settle in flood-damaged buildings and natural habitats, requiring extensive remediation (FEMA 2020). Water supply and wastewater infrastructure could be inoperable for weeks, affecting thousands of residents and businesses.

Beyond environmental degradation, the cultural impacts of dam failure could significantly affect Harris County's historic assets. Many historic districts and landmarks within the Buffalo Bayou corridor and central Houston—such as Freedmen's Town, the San Felipe Courts, and portions of the Houston Heights—fall within potential dam inundation zones. These areas are vulnerable to the same types of structural damage, mold, and contamination seen in other flood-prone regions. As with typical flooding in mapped Special Flood Hazard Areas (SFHAs), post-flood recovery would involve removing contaminated sediment, restoring damaged infrastructure, and addressing the loss of irreplaceable cultural resources.

7.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

7.3.1 Potential or Planned Development

Ongoing and future development in Harris County, particularly in western and central areas downstream of the Addicks and Barker reservoirs, faces elevated risk if dam or levee failures occur. As population growth drives suburban expansion into previously undeveloped flood-prone zones, residential subdivisions, commercial centers, and infrastructure projects may be built within dam inundation areas. Without adequate mitigation measures, these developments could face catastrophic losses in the event of structural failure. Updates to building codes, zoning regulations, and land-use planning that incorporate dam failure risk are critical for reducing exposure. Incorporating elevation requirements, floodproofing standards, and setbacks from downstream flow paths can help limit damage and support faster recovery.

In addition to suburban growth, urban redevelopment and high-rise construction in central Houston, especially in neighborhoods near Buffalo Bayou, also fall within dam inundation zones. These developments often include dense residential and mixed-use structures, making evacuation and emergency response more challenging during flood emergencies. While high-rises may be structurally resilient, the loss of access roads, power, and critical services can still render these

buildings uninhabitable. Planned infrastructure improvements, including green stormwater systems and emergency communication upgrades, must account for the dual risks of routine flooding and dam failure. By aligning urban growth strategies with hazard mitigation efforts, Harris County can minimize long-term vulnerabilities while accommodating continued development.

7.3.2 Projected Changes in Population

According to the U.S. Census Bureau, Harris County had an estimated population of 4,888,304 in 2023, making it the most populous county in Texas. The Texas Demographic Center projects continued population growth through 2040, particularly in suburban and unincorporated areas (Texas Demographic Center 2023). This increase in population, especially in areas downstream of the Addicks, Barker, and Lake Houston dams, raises both economic activity and the overall risk exposure to dam failure events. Many dam inundation zones in Harris County overlap with mapped flood hazard areas, meaning that population growth in these regions directly increases the number of individuals and properties at risk. Without proactive planning, more residents may unknowingly move into high-risk zones. However, targeted mitigation efforts, such as enhanced early warning systems, evacuation planning, and public outreach, can help reduce the vulnerability of a growing population and increase countywide resilience to dam-related disasters.

7.3.3 Climate Change

An increasing average annual temperature in Harris County is expected to directly influence atmospheric moisture potential, leading to more frequent and intense rainfall during storm events. As the atmosphere warms, it can hold and release greater volumes of water vapor, intensifying the scale of precipitation that occurs during hurricanes and frontal storms (Intergovernmental Panel on Climate Change [IPCC], 2021). This heightened rainfall increases the volume of water entering the region's reservoirs and waterways, placing greater strain on the existing dam and levee systems during future riverine flood events. Overall, Harris County continues to be vulnerable to dam failures. With continued population growth, in western and unincorporated areas, more people and structures are being placed within or near downstream inundation zones, amplifying the exposure and risk associated with potential dam or levee failure events.

8. DROUGHT

8.1 HAZARD PROFILE

8.1.1 Hazard Description

The 2017 Texas Water Plan explains that drought can be defined in various ways and is commonly categorized into four main types: meteorological, agricultural, hydrological, and socioeconomic.

- Meteorological drought begins with a period of abnormally dry weather resulting in less than the long-term average rainfall for that period. It does not necessarily impact water supply.
- Agricultural drought often follows or coincides with meteorological drought and can appear suddenly and cause rapid impacts to agriculture. It reduces soil moisture, which decreases crop or range production, and increases irrigation demands. It often leads to drought disaster declarations and in many cases is an indicator of an impending hydrological drought.
- Hydrological drought is a period of below-average stream flows and water volume in aquifers and reservoirs, resulting in reduced water supplies. It is the focus of regional water planning since it impacts water supplies.
- Socioeconomic drought occurs when physical water needs affect the health, safety, and quality of life of the general public or when the drought affects the supply and demand of an economic product.

Drought is a natural and recurring phenomenon that can occur in nearly all climate zones, regardless of typical rainfall levels. It is generally caused by a prolonged reduction in precipitation—often lasting for a season or longer—and can be intensified by high temperatures, strong winds, and low humidity. Human activities, such as increased water consumption and land use changes, can further amplify the impacts of drought. These conditions reduce natural streamflow and deplete soil moisture, leading to a wide range of social, environmental, cultural, and economic consequences. The term "drought" is typically reserved for moisture deficits that are both widespread and prolonged.

Within Harris County, drought poses a unique challenge due to the region's growing population and high-water demand. While the area is more commonly associated with flooding and tropical storms, periods of drought can strain municipal water supplies, stress local ecosystems, and increase the risk of wildfires in undeveloped areas. Understanding drought as a complex hazard, rather than a rare anomaly, is essential for long-term water planning and resilience in Harris County, especially as climate variability continues to influence regional weather patterns.

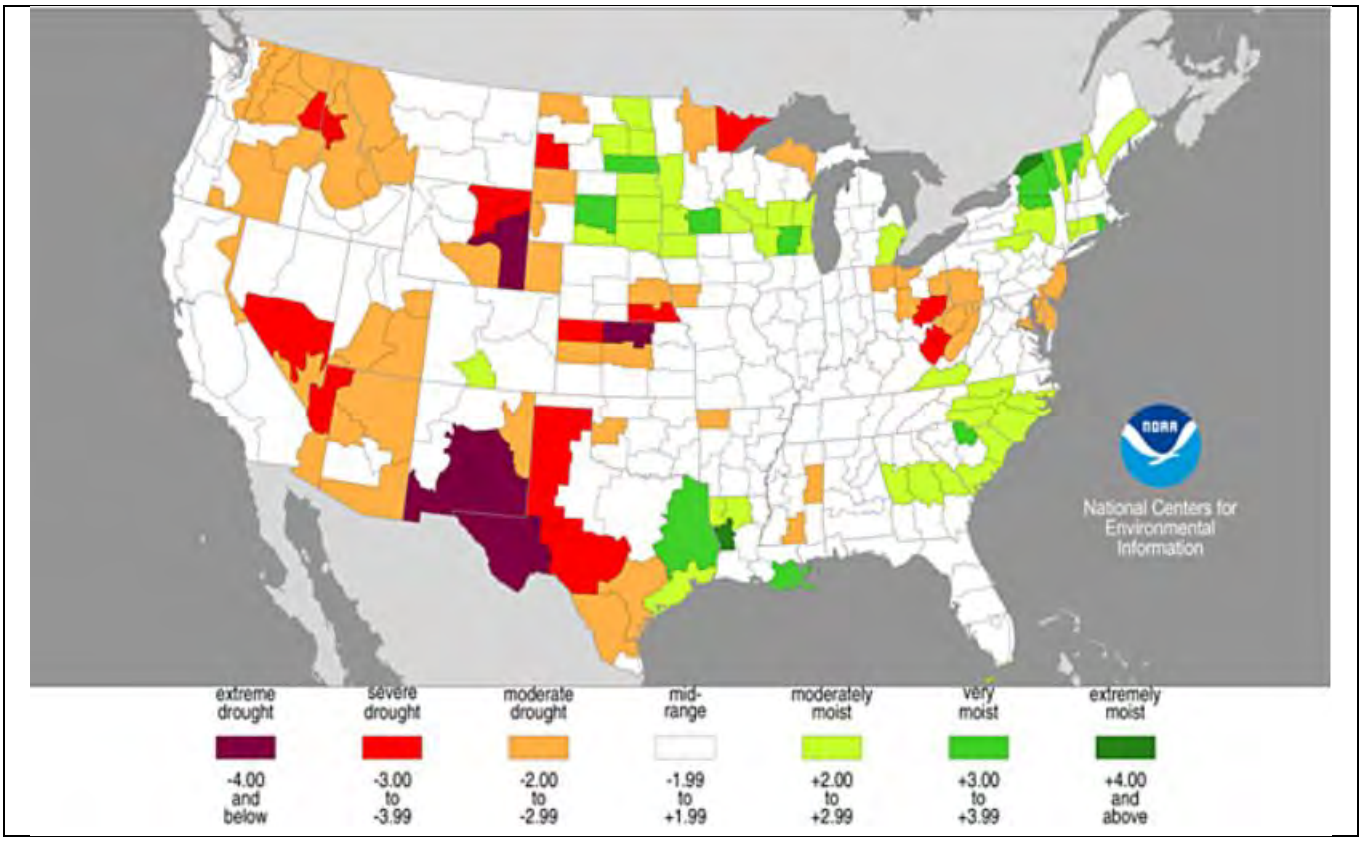
Monitoring and Categorizing Drought

The National Oceanic and Atmospheric Administration (NOAA) has developed several indices to measure drought impacts and severity and to map their extent and locations:

- The Palmer Z Index measures short-term drought on a monthly scale.
- The Palmer Drought Severity Index (PDSI) measures the duration and intensity of long-term weather patterns. The intensity of drought in a given month is dependent on current weather plus the cumulative patterns of previous months. Weather patterns can change quickly, and the PDSI can respond fairly rapidly.
- The Palmer Hydrological Drought Index quantifies hydrological effects (reservoir levels, groundwater levels, etc.), which take longer to develop and last longer. This index responds more slowly to changing conditions than the Palmer Drought Index.
- The Standardized Precipitation Index considers only precipitation. In the Standardized Precipitation Index, an index of zero indicates the median precipitation amount; the index is negative for drought and positive for wet conditions. The Standardized Precipitation Index is computed for time scales ranging from one month to 24 months.
- The Palmer Crop Moisture Index measures short-term drought on a weekly scale to quantify impacts on agriculture.

Maps of these indices show drought conditions nationwide at a given point in time. They are not necessarily indicators of any given area's long-term susceptibility to drought. The most current versions of the map at the time of this plan's preparation is shown on Figure 8-1.

Figure 8-1. Palmer Z Index Short-Term Drought Conditions (September 2024)



Source: NOAA National Center for Environmental Information, 2024

8.1.2 Location

The Harris County planning area is susceptible to all ranges of drought as defined by the indices described above; therefore, the entire planning area is at risk for drought. However, since the Harris County planning area is not primarily an agricultural region, its vulnerability to drought is relatively low.

Drought is a regional phenomenon. A drought affects all aspects of the environment and the community simultaneously and has the potential to directly or indirectly impact every person in the county as well as adversely affect the local economy.

The 2023 *State of Texas Hazard Mitigation Plan* identifies all of Texas as vulnerable to droughts. However, the area's most vulnerable to drought are west Texas areas around the cities of Amarillo, Lubbock, Midland, Odessa, Fort Stockton, San Angelo, Laredo, and El Paso to be the most at risk in the state to drought. The Harris County planning area is not located within that region and is not identified as a high-risk area.

The Harris County planning area is located in the Texas Climate Division 8—Upper Coast. According to the South-Central Climate Science Center, this is an area with recurring drought conditions

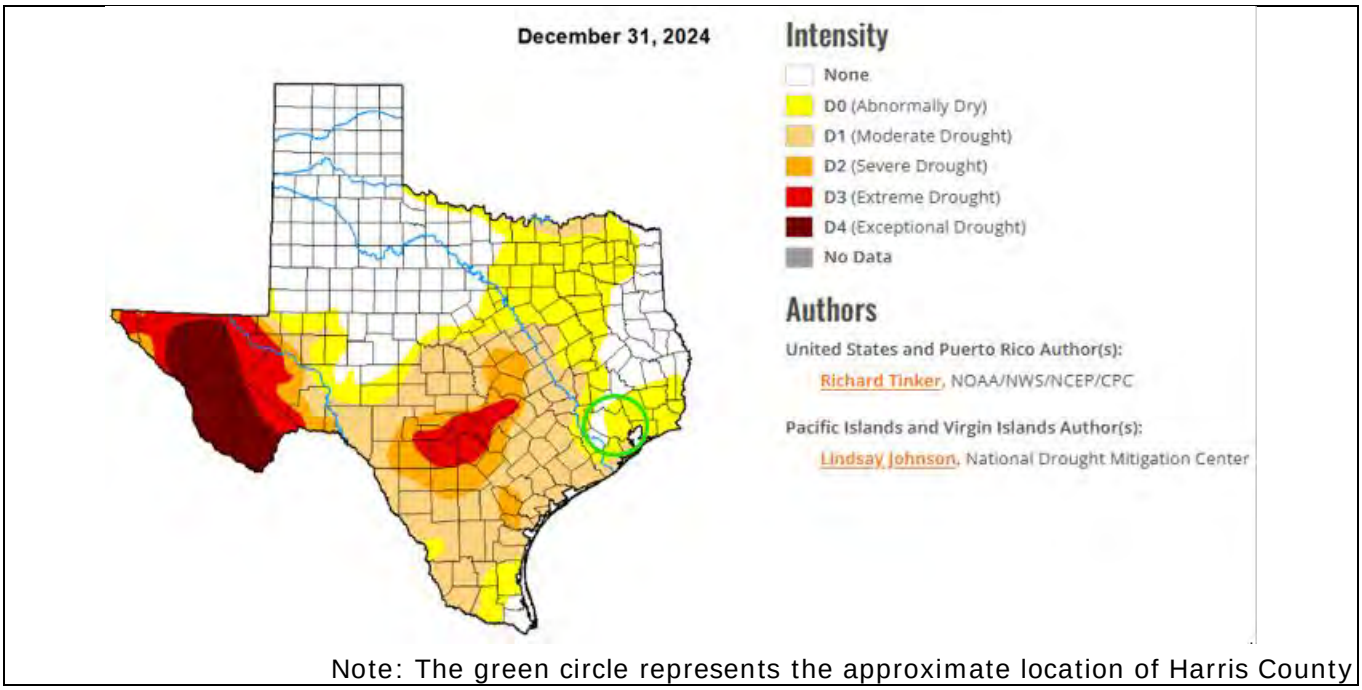
8.1.3 Extent

U.S. Drought Monitor

The U.S. Drought Monitor (USDM) categorizes droughts by impact type and intensity. Impact type indicates whether a drought in a given area is short-term or long-term. Short-term is generally less than six months and impacts are expected on agriculture and grasslands. Long-term drought is typically longer than 6 months and impacts are seen on hydrology and ecology in the area impacted. The USDM uses a five-category system: Abnormally Dry (D0) a precursor to drought but not actually a drought, Moderate Drought (D1), Severe Drought (D2), Extreme Drought (D3), and Exceptional Drought (D4).

Drought categories show experts' assessments of conditions related to dryness and drought, including observations of how much water is available in streams, lakes, and soils compared to usual for the same time of year. Figure 8-2 shows that Harris County has abnormally dry drought conditions for the week of December 31, 2024.

Figure 8-2. U.S. Drought Monitor for Texas, December 2024



Note: The green circle represents the approximate location of Harris County

Source: U.S. Drought Monitor 2024

Palmer Drought Severity Index

The PDSI is primarily based on soil conditions. Soil with decreased moisture content is the first indicator of an overall moisture deficit. Table 8-1 lists the PDSI classifications. At the one end of the spectrum, 0 is used as normal, and drought is indicated by negative numbers. For example, -2 is moderate drought, -3 is severe drought, and -4 is extreme drought. The PDSI can reflect excess precipitation using positive numbers; however, this is not shown in Table 8-1. The PDSI is commonly converted to the Palmer Drought Category (U.S. Drought Monitor n.d.).

Keetch-Byram Drought Index

The Keetch-Byram Drought Index (KBDI) is an index used to determine forest fire potential (refer to Table 8-2). The drought index is based on a daily water balance, where a drought factor is balanced with precipitation and soil moisture (assumed to have a maximum storage capacity of eight inches) and is expressed in hundredths of an inch of soil moisture depletion. The index ranges from 0 to 800, where a drought index of 0 represents no moisture depletion, while an index of 800 represents absolutely dry conditions (USFS n.d.).

This index is currently derived from ground-based estimates of temperature and precipitation resulting from weather stations and interpolated manually by experts at the Texas Forest Service for counties across the state. Figure 8-3 shows the KBDI for the State of Texas for May 1, 2025. The figure shows KBDI value of 0-500 for the County.

Table 8-1. Palmer Drought Category and Palmer Drought Index Descriptions

Category	Description	Possible Impacts (for Texas)	Palmer Drought Index
D0	Abnormally Dry	• Producers begin supplemental feeding for livestock • Planting is postponed; forage germination is stunted; hay cutting is reduced • Grass fires increase • Surface water levels decline	-1.0 to -1.9
D1	Moderate Drought	• Dryland crops are stunted • Early cattle sales begin • Wildfire frequency increases • Stock tanks, creeks, streams are low; voluntary water restrictions are requested	-2.0 to -2.9
D2	Severe Drought	• Pasture conditions are very poor • Soil is hard, hindering planting; crop yields decrease • Wildfire danger is severe; burn bans are implemented • Wildlife moves into populated areas • Hydroelectric power is compromised; well water use increases; mandatory water restrictions are implemented	-3.0 to -3.9
D3	Extreme Drought	• Soil has large cracks; soil moisture is very low; dust and sandstorms occur • Row and forage crops fail to germinate; decreased yields for irrigated crops and very large yield reduction for dryland crops are reported • Need for supplemental feed, nutrients, protein, and water for livestock increases; herds are sold • Increased risk of large wildfires is noted • Many sectors experience financial burden • Severe fish, plant, and wildlife loss reported • Water sanitation is a concern; reservoir levels drop significantly; surface water is nearly dry; river flow is very low; salinity increases in bays and estuaries	-4.0 to -4.9
D4	Exceptional Drought	• Exceptional and widespread crop loss is reported; rangeland is dead; producers are not planting fields • Culling continues; producers wean calves early and liquidate herds due to importation of hay and water expenses • Seafood, forestry, tourism, and agriculture sectors report significant financial loss • Extreme sensitivity to fire danger; firework restrictions are implemented • Widespread tree mortality is reported; most wildlife species' health and population are suffering • Devastating algae blooms occur; water quality is very poor • Exceptional water shortages are noted across surface water sources; water table is declining • Boat ramps are closed; obstacles are exposed in water bodies; water levels are at or near historic lows	-5.0 or less

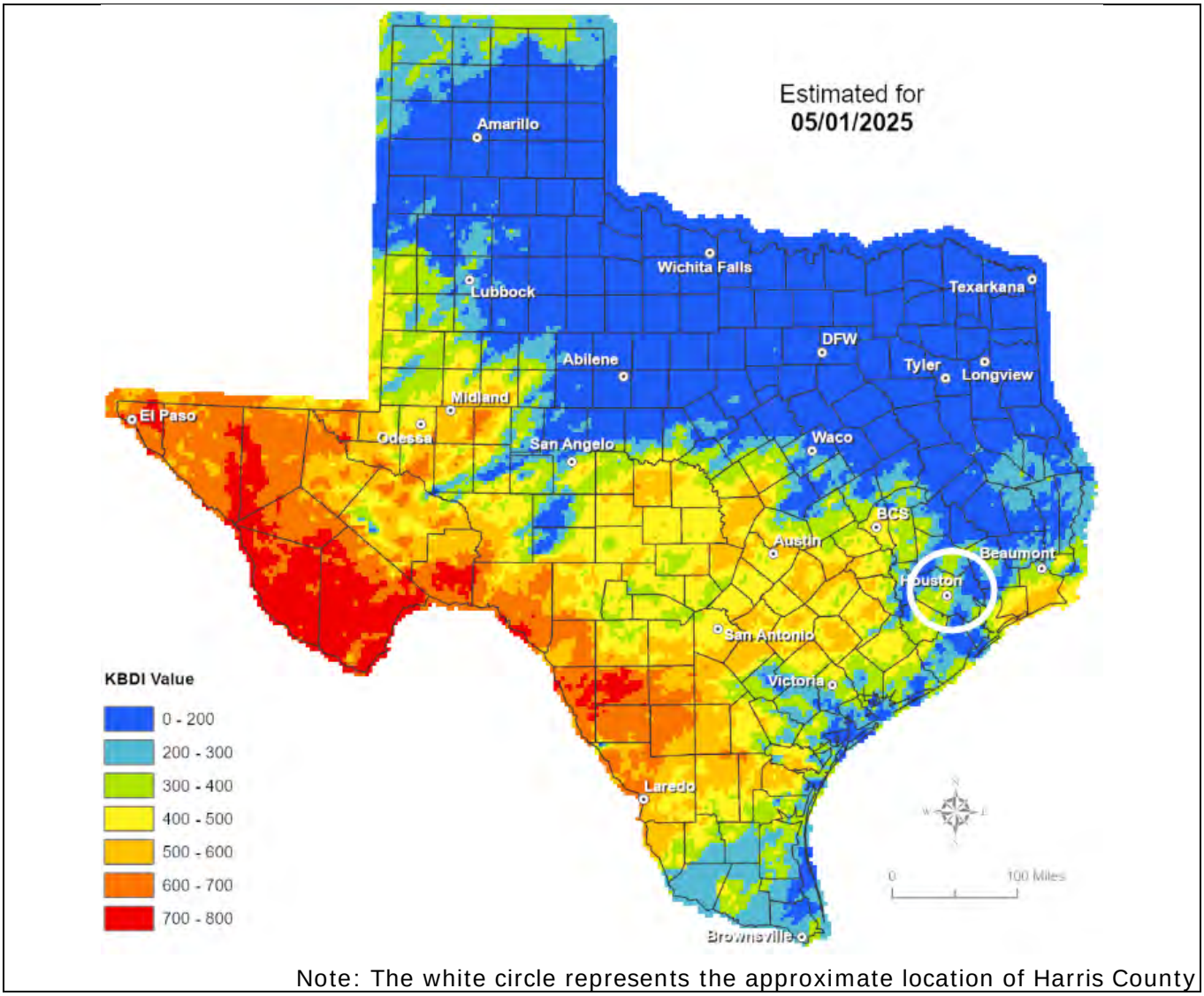
Source: U.S. Drought Monitor 2021

Table 8-2. KBDI Index

KBDI Value	Description
0 to 200	Soil moisture and large-class fuel moistures are high and do not contribute much to fire intensity. Typical of spring dormant season following winter precipitation
200 to 400	Typical of late spring, early growing season. Lower litter and duff layers are drying and beginning to contribute to fire intensity
400 to 600	Typical of late summer, early fall. Lower litter and duff layers actively contribute to fire intensity and will burn actively.
600 to 800	Often associated with more severe drought with increased wildfire occurrence. Intense, deep burning fires with significant downwind spotting can be expected. Live fuels can also be expected to burn actively at these levels.

Source: TAMU n.d.

Figure 8-3. KBDI for the State of Texas, May 1, 2025



Note: The white circle represents the approximate location of Harris County

Source: TAMU 2025

A prolonged drought classified as D4 on the PDSI represents the worst-case scenario for Harris County. The occurrence of a multi-year drought, within the southeastern region of Texas, could significantly strain the county's water resources. If another severe drought occurs before the water systems have had sufficient time to recover, it could further exacerbate the existing stress on the Planning Area's water supply. Additionally, severe droughts can lead to substantial crop and livestock losses, which would negatively impact the food supply and the local economy.

8.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 2020 and 2024, Harris County was not included in any major disaster (DR) or emergency (EM) declarations for drought-related events (FEMA 2023).

State Emergency Proclamations

The Governor of Texas can issue emergency proclamations to access powers authorized by the Emergency Services Act. Harris County has not been included in any drought-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2025, Harris County was included in 11 USDA drought-related agricultural disaster declarations. For declarations that occurred between 2020 and 2025, refer to Table 8-3.

Table 8-3. USDA Declarations for Drought Events in Harris County (2020 to 2025)

Event Date	USDA Declaration Number	Description
2/18/2020	S4669	Drought Fast Track
3/15/2022	S5188	Drought Fast Track
5/10/2022	S5197	Drought Fast Track
4/5/2022	S5209	Drought Fast Track
4/12/2022	S5214	Drought Fast Track
6/14/2022	S5221	Drought Fast Track
7/19/2022	S5240	Drought Fast Track
8/2/2022	S5252	Drought Fast Track
8/15/2023	S5499	Drought Fast Track
8/22/2023	S5511	Drought Fast Track
11/01/2023	S5682	Drought Fast Track

Source: USDA Farm Service Agency 2020-2025

Previous Events

Known hazard events that impacted Harris County between January 2000 and January 2024 are listed in Table 8-4.

Table 8-4. Drought Events in Harris County (2000 to 2024)

Event Date	Declaration or Proclamation Number	Location Impacted	Description
August – November 2023	S5499, S5511, S5682	Countywide	Severe Drought (D2) conditions set in across the planning area, with conditions deteriorating to Exceptional drought (D4) levels between August and October 2023.
March – August 2022	S5188, S5209, S5197, S5209, S5214, S5221, S5240, S5252	Countywide	Beginning in late March, severe drought (D2) conditions set in across Harris County. By late June, parts of the county worsened to extreme drought levels, and by August, 100% of the county saw extreme drought (D3) conditions. Drought
2/18/2020	S4669	Countywide	A dry Spring allowed for severe drought (D2) conditions to develop in portions of Harris County, but rainfall in April helped end the month-long severe drought
1/1/2012	S3288	Countywide	Drought
1/1/2011	S3122	Countywide	Drought
1/1/2009	S2863	Countywide	Drought
1/1/2008	S2772a	Countywide	Drought
1/1/2008	S2845a	Countywide	Drought
1/1/2006	S2500	Countywide	Drought
4/1/2005	S2281	Countywide	Severe drought, above normal temperatures, & extremely high number of wildfires
4/1/2005	S2191	Countywide	Drought
1/1/2000	S1501	Countywide	Drought & Excessive Heat

Source: (USDM 2025); U.S. Department of Agriculture, Farm Service Agency—Harris, Montgomery, and Waller County.

a. Contiguous designation—designated due to being located adjacent to a primary designation county. "S" in front of disaster designation number indicates a Secretarial Designation by the U.S. Secretary of Agriculture.

8.1.5 Probability of Future Occurrences

Information on previous drought occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 8-5. Based on historical records and input from the Steering Committee, the probability of occurrence for drought in the County is considered "occasional."

Table 8-5. Probability of Future Drought Events in Harris County

Hazard Type	Number of Occurrences Between 2000 and 2025	Percent Chance of Occurring in Any Given Year
Drought	24	92%

Source: U.S. Drought Monitor
Note: Due to limitations in data, not all drought events occurring between 2000 and 2025 are accounted for in the tally of occurrences. As a result, the number of hazard occurrences is calculated using the number of occurrences between 2000 and 2025

Climate Change Projections

Climate is defined not simply as average temperature and precipitation but also by the type, frequency, and intensity of weather events. Both globally and at the local scale, climate change has the potential to alter the prevalence and severity of extremes such as droughts. While predicting changes in drought events under a changing climate is difficult, understanding vulnerabilities to potential changes is a critical part of estimating future climate change impacts on human health, society, and the environment (EPA 2016).

With a warmer climate, droughts can become more frequent, more severe, and longer lasting. According to the National Climate Assessment, variable precipitation and rising temperatures are intensifying droughts, increasing heavy downpours, reducing snowpack, and causing declines in water survey quality. Future warming will add to the stress on water supplies and impact the availability of water supply (USGCRP 2018).

8.1.6 Cascading Impacts on Other Hazards

Droughts in Harris County can trigger cascading effects that intensify other natural and human-made hazards, compounding the region's overall vulnerability. For instance, extended drought conditions often lead to an increase in extreme heat events, as reduced soil moisture limits the cooling effect of evaporation, contributing to higher ambient temperatures. This combination intensifies the urban heat island effect in densely developed areas like Houston, increasing the likelihood of heat-related illnesses and straining energy grids due to heightened cooling demand (NIDIS 2022). Prolonged drought also reduces vegetation moisture, raising the risk of wildfires, particularly in green spaces, parks, and peri-urban areas of Harris County that border undeveloped land (Texas A&M Forest Service 2021).

Droughts can also compromise flood resilience by degrading natural systems that mitigate stormwater runoff. When soil becomes hardened and compacted due to a lack of moisture, it loses its ability to absorb rainfall effectively. This leads to higher surface runoff and an elevated risk of flash flooding when storms eventually return (FEMA 2020). Additionally, water stress may delay maintenance or upgrades to critical water infrastructure, increasing the risk of utility failure during concurrent events, such as hurricanes or heavy storms. These cascading hazards highlight the need for integrated hazard mitigation planning that accounts for drought's downstream effects on both chronic and acute risks in Harris County.

8.2 VULNERABILITY AND IMPACT ASSESSMENT

A detailed hazard exposure and vulnerability assessment was not conducted for drought due to the localized and infrequent nature of its impacts in this region. Unlike other hazards with broader geographic reach or more consistent historical data, drought events in Harris County tends to affect limited areas along the shoreline and does not pose a widespread threat to the county's overall building stock, critical facilities, or economy. As such, a traditional vulnerability assessment—typically used to quantify potential losses and exposure—was deemed not applicable for this hazard. Instead, the risk of drought events across the planning area and participating jurisdictions is considered minimal, with potential losses expected to be rare and highly site-specific.

8.2.1 Life, Health, and Safety

Drought conditions in Harris County pose several indirect but significant risks to life, health, and safety. Prolonged droughts can exacerbate existing health conditions, especially among vulnerable populations such as older adults, young children, individuals with chronic illnesses, and low-income communities with limited access to cooling and hydration resources. Elevated temperatures associated with drought often lead to increased incidences of heat exhaustion, heatstroke, and respiratory problems, particularly in densely populated urban areas with limited tree canopy and green infrastructure. Furthermore, reduced water availability can compromise hygiene and sanitation, increasing the potential for illness in both residential and institutional settings ((FEMA 2020).

Additionally, the risk of wildfire increases during extended periods of drought, especially in Harris County's urban-wildland interface areas. Dry vegetation becomes highly flammable, and while wildfires are less common in this part of Texas than in other regions, the presence of drought conditions significantly raises the hazard potential. This, in turn, poses safety concerns for residents, emergency responders, and infrastructure. Compounding effects from power outages due to fire or heat-related system overloads can further endanger medically vulnerable individuals who rely on life-sustaining equipment. These cascading risks emphasize the need for drought mitigation planning that prioritizes public health and emergency preparedness (Texas A&M Forest Service 2021).

Overall Population

The entire population of Harris County is vulnerable to drought events. Drought can affect people's health and safety, including health problems related to low water flows, poor water quality, or dust. Droughts can also lead to loss of human life (National Drought Mitigation Center, 2017). Other possible impacts include recreational risks; effects on air quality; diminished living conditions related to energy, air quality, and hygiene; compromised food and nutrition; and increased incidence of illness and disease (Centers for Disease Control and Prevention, 2012).

Socially Vulnerable Population

Social vulnerability is defined as the susceptibility of social groups to the adverse impacts of natural hazards, including disproportionate death, injury, loss, or disruption of livelihood. Social vulnerability

considers the social, economic, demographic, and housing characteristics of a community that influence its ability to prepare for, respond to, cope with, recover from, and adapt to environmental hazards.

Drought conditions often coincide with periods of extreme heat. Older adults (particularly those over 65), young children, pregnant women, outdoor workers, and economically disadvantaged individuals are especially vulnerable to these conditions. In periods of extreme heat, these vulnerable populations may have limited access to air conditioning, over-exhaust more easily, or may be unaware of the toll the heat is affecting their bodily functions.

8.2.2 General Building Stock

Drought conditions in Harris County typically do not result in direct physical damage to the general building stock. However, prolonged drought may cause soil subsidence or shrinking, especially in clay-heavy soils, which can lead to foundation issues in older or poorly constructed buildings. While these effects are gradual, they may increase long-term maintenance costs and reduce structural integrity if not addressed. Buildings with landscaping reliant on irrigation systems may also experience property degradation due to restricted water usage or elevated costs (TWDB 2021).

8.2.3 Community Lifelines and Other Critical Facilities

Community lifelines such as water, energy, and healthcare services can be strained during drought events. In Harris County, increased water demand coupled with potential limitations in water supply from surface sources may affect water pressure and availability, especially in peripheral areas or those reliant on groundwater. Energy facilities that depend on water for cooling, such as thermoelectric power plants, may also experience operational constraints. Hospitals and other critical care facilities may face increased demand due to heat-related illnesses exacerbated by drought (FEMA 2020).

8.2.4 Economy

The economic impacts of drought in Harris County are most apparent in sectors such as landscaping, agriculture (in surrounding rural edges), and water-intensive manufacturing. Extended drought periods may lead to water-use restrictions, increasing operational costs and reducing productivity. Businesses that depend on outdoor maintenance or water-related activities, including golf courses and nurseries, may suffer economic losses due to drought-induced limitations. Additionally, the cost of water may rise during extended dry periods, affecting households and commercial operations alike (NIDIS 2022).

According to recent data, Harris County has approximately 117,469 acres of farmland (AcreValue n.d.). The market value of agricultural products sold in Harris County is substantial, contributing significantly to the local economy. Drought can severely impact the agricultural productivity and economic stability of Harris County. Prolonged periods of drought may lead to reduced crop yields, increased costs for irrigation, and long-term damage to farmland.

8.2.5 Natural, Historic and Cultural Resources

Harris County's historic and cultural resources, particularly outdoor monuments, historic gardens, and heritage landscapes are susceptible to drought-related deterioration. Reduced soil moisture and water availability can negatively impact historic vegetation, while shifting or compacting soil may threaten the structural stability of older buildings and sites with historical significance. In some cases, drought may also accelerate erosion around archaeological sites, exposing them to additional environmental risks (National Park Service, 2020).

8.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

8.3.1 Potential or Planned Development

Drought hazards may influence future development in Harris County, particularly in areas dependent on non-renewable groundwater or those already experiencing water stress. While urban development continues to expand, areas without sustainable long-term water supply planning may be deprioritized. Building codes and land use planning may increasingly integrate drought resilience measures, such as xeriscaping, use of recycled water, and mandatory efficiency standards in new construction (Houston-Galveston Area Council [H-GAC], 2021).

8.3.2 Projected Changes in Population

Harris County's population is projected to continue growing over the next few decades, increasing demand on already strained water resources. As more residents move into suburban and peri-urban areas, drought resilience planning becomes critical to ensure equitable and reliable access to water. Population increases also elevate competition for water among residential, industrial, and ecological needs, making regional drought planning and investment in infrastructure vital to meet long-term needs (Texas Demographic Center, 2021).

8.3.3 Climate Change

Climate change is expected to increase the frequency and severity of drought events in Harris County. Rising temperatures and shifts in precipitation patterns may reduce the reliability of surface water sources such as reservoirs and increase evaporation rates from both soil and water bodies. These climatic shifts may lead to longer dry spells, placing further pressure on the county's water systems and increasing the likelihood of water-use restrictions, heat stress, and economic disruption (Intergovernmental Panel on Climate Change [IPCC], 2021).

8.3.4 Other Identified Conditions

Drought in Harris County often intersects with other hazard conditions, including extreme heat and wildfire risk. Prolonged dry conditions combined with high temperatures can exacerbate the urban heat island effect, especially in areas with limited tree canopy. Moreover, dried vegetation in peri-urban zones may increase the likelihood of grass or brush fires. These cascading effects can strain emergency services and complicate hazard response planning (Texas A&M Forest Service, 2021).

9. EARTHQUAKE

9.1 HAZARD PROFILE

9.1.1 Hazard Description

An earthquake is the vibration of the earth's surface following a release of energy in the earth's crust. This energy can be generated by a sudden dislocation of the crust or by a volcanic eruption. Most destructive quakes are caused by dislocations of the crust. The crust may first bend and then, when the stress exceeds the strength of the rocks, break and snap to a new position. In the process of breaking, vibrations called "seismic waves" are generated. These waves travel outward from the source of the earthquake at varying speeds.

Geologists have found that earthquakes tend to reoccur along faults, which are zones of weakness in the earth's crust. Even if a fault zone has recently experienced an earthquake, there is no guarantee that all the stress has been relieved. Another earthquake could still occur. In fact, relieving stress along one part of a fault may increase it in another part.

Faults are more likely to have future earthquakes on them if they have more rapid rates of movement, have had recent earthquakes along them, experience greater total displacements, and are aligned so that movement can relieve the accumulating tectonic stresses. Geologists classify faults by their relative hazards. "Active" faults, which represent the highest hazard, are those that have ruptured to the ground surface during the Holocene period (about the last 11,000 years). "Potentially active" faults are those that displaced layers of rock from the Quaternary period (the last 1,800,000 years).

Determining if a fault is "active" or "potentially active" depends on geologic evidence, which may not be available for every fault. Nearly all the movement between the two plates, and therefore the majority of the seismic hazards, are on the well-known active faults. However, inactive faults, where no displacements have been recorded, also have the potential to reactivate or experience displacement along a branch sometime in the future.

Earthquakes can be classified as tectonic, volcanic, collapse, or explosion (USGS n.d.).

- Tectonic Earthquakes: These are the most common type, caused by the movement of tectonic plates or fault lines.
- Volcanic Earthquakes: These are associated with volcanic activity, like magma rising and moving beneath the surface.
- Collapse Earthquakes: These are caused by the collapse of underground caverns or mines.
- Explosion Earthquakes: These are caused by explosions, like nuclear tests or quarry blasts.

Earthquakes are recorded by a seismographic network that is part of the USGS. Each seismic station in the network measures the movement of the ground at that site. The slip of one block of rock over another in an earthquake releases energy that makes the ground vibrate. That vibration

pushes the adjoining piece of ground and causes it to vibrate, and thus the energy travels out from the earthquake hypocenter in a wave.

Earthquake Rating

Earthquakes are typically rated in one of two ways: By the amount of energy released, measured as magnitude; or by the impact on people and structures, measured as intensity.

Magnitude

An earthquake's magnitude is a measure of the energy released at the source of the earthquake. Magnitude is commonly expressed by ratings on the moment magnitude scale (M_W), the most common scale used today (USGS, 2017a). This scale is based on the total moment release of the earthquake (the product of the distance a fault moved, and the force required to move it). The scale is as follows:

- Great—Mw > 8
- Major—Mw = 7.0 − 7.9
- Strong—Mw = 6.0 − 6.9
- Moderate—Mw = 5.0 − 5.9
- Light—Mw = 4.0 − 4.9
- Minor—Mw = 3.0 − 3.9
- Micro—Mw < 3

Intensity

The most commonly used intensity scale is the modified Mercalli intensity scale. Ratings of the scale as well as the perceived shaking and damage potential for structures are shown in Table 9-1. The modified Mercalli intensity scale is generally represented visually using shake maps, which show the expected ground shaking at any given location produced by an earthquake with a specified magnitude and epicenter. An earthquake has only one magnitude and one epicenter, but it produces a range of ground shaking at sites throughout the region, depending on the distance from the earthquake, the rock and soil conditions at sites, and variations in the propagation of seismic waves from the earthquake due to complexities in the structure of the earth's crust. A shake map shows the variation of ground shaking in a region immediately following significant earthquakes (for technical information about shake maps see USGS, 2018).

Table 9-1. Mercalli Scale and Peak Ground Acceleration Comparison

Modified Mercalli Scale	Perceived Shaking	Potential Structure Damage		Estimated PGA[a] (%g)
		Resistant Buildings	Vulnerable Buildings	
I	Not Felt	None	None	<0.17%
II-III	Weak	None	None	0.17% – 1.4%
IV	Light	None	None	1.4% – 3.9%
V	Moderate	Very Light	Light	3.9% – 9.2%
VI	Strong	Light	Moderate	9.2% – 18%
VII	Very Strong	Moderate	Moderate/Heavy	18% – 34%
VIII	Severe	Moderate/Heavy	Heavy	34% – 65%
IX	Violent	Heavy	Very Heavy	65% – 124%
X – XII	Extreme	Very Heavy	Very Heavy	>124%

Source: USGS, 2008; USGS, 2010
a. PGA = peak ground acceleration. Measured in percent of g, where g is the acceleration of gravity

Ground Shaking

The ground experiences acceleration as it shakes during an earthquake. The peak ground acceleration (PGA) is the largest acceleration recorded by a monitoring station during an earthquake. PGA is a measure of how hard the earth shakes in a given geographic area. It is expressed as a percentage of the acceleration due to gravity (%g). Horizontal and vertical PGA varies with soil or rock type. Earthquake hazard assessment involves estimating the annual probability that certain ground accelerations will be exceeded and then summing the annual probabilities over a time period of interest.

National maps of earthquake shaking hazards provide information for creating and updating seismic design requirements for building codes, insurance rate structures, earthquake loss studies, retrofit priorities and land use planning. After thorough review of the studies, professional organizations of engineers update the seismic-risk maps and seismic design requirements contained in building codes (Brown et al., 2001). The USGS updated the National Seismic Hazard Maps in 2014. New seismic, geologic, and geodetic information on earthquake rates and associated ground shaking were incorporated into these revised maps.

Liquefaction and Soil Types

Soil liquefaction occurs when water-saturated sands, silts or gravelly soils are shaken so violently that the individual grains lose contact with one another and float freely in the water, turning the ground into a pudding-like liquid. Building and road foundations lose load-bearing strength and may sink into what was previously solid ground. Unless properly secured, hazardous materials can be released, causing significant damage to the environment and people. A program called the National Earthquake Hazard Reduction Program (NEHRP) creates maps based on soil characteristics to help identify locations subject to liquefaction.

Table 9-2 summarizes NEHRP soil classifications. The areas that are commonly most affected by ground shaking have NEHRP Soils D, E and F. In general, these areas are also most susceptible to liquefaction.

Table 9-2. NEHRP Soil Classification System

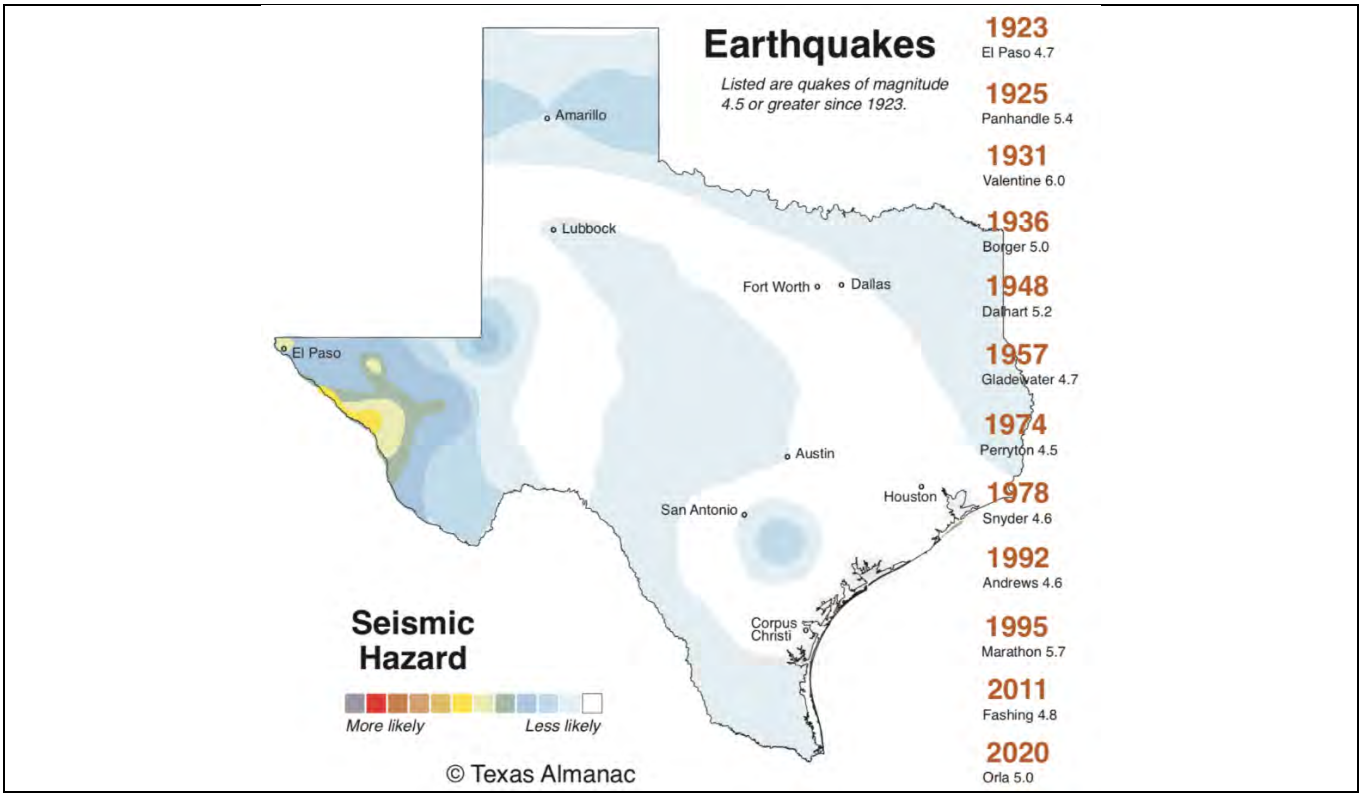
NEHRP Soil Type	Description
A	Hard Rock
B	Firm to Hard Rock
C	Dense Soil/Soft Rock
D	Stiff Soil
E	Soft Clays
F	Special Study Soils (liquefiable soils, sensitive clays, organic soils, soft clays >36 m thick)

Source: (FEMA/NEHRP n.d.)

9.1.2 Location

The entire planning area is located in a low-risk zone for earthquakes. The Harris County planning area is located in the Gulf Coast Region and is not identified as a hazard area in the *State of Texas Hazard Mitigation Plan* (see Figure 9-1). According to the U.S. Geological Survey, the Harris County planning area is located within the lowest hazard risk area across the nation for earthquakes.

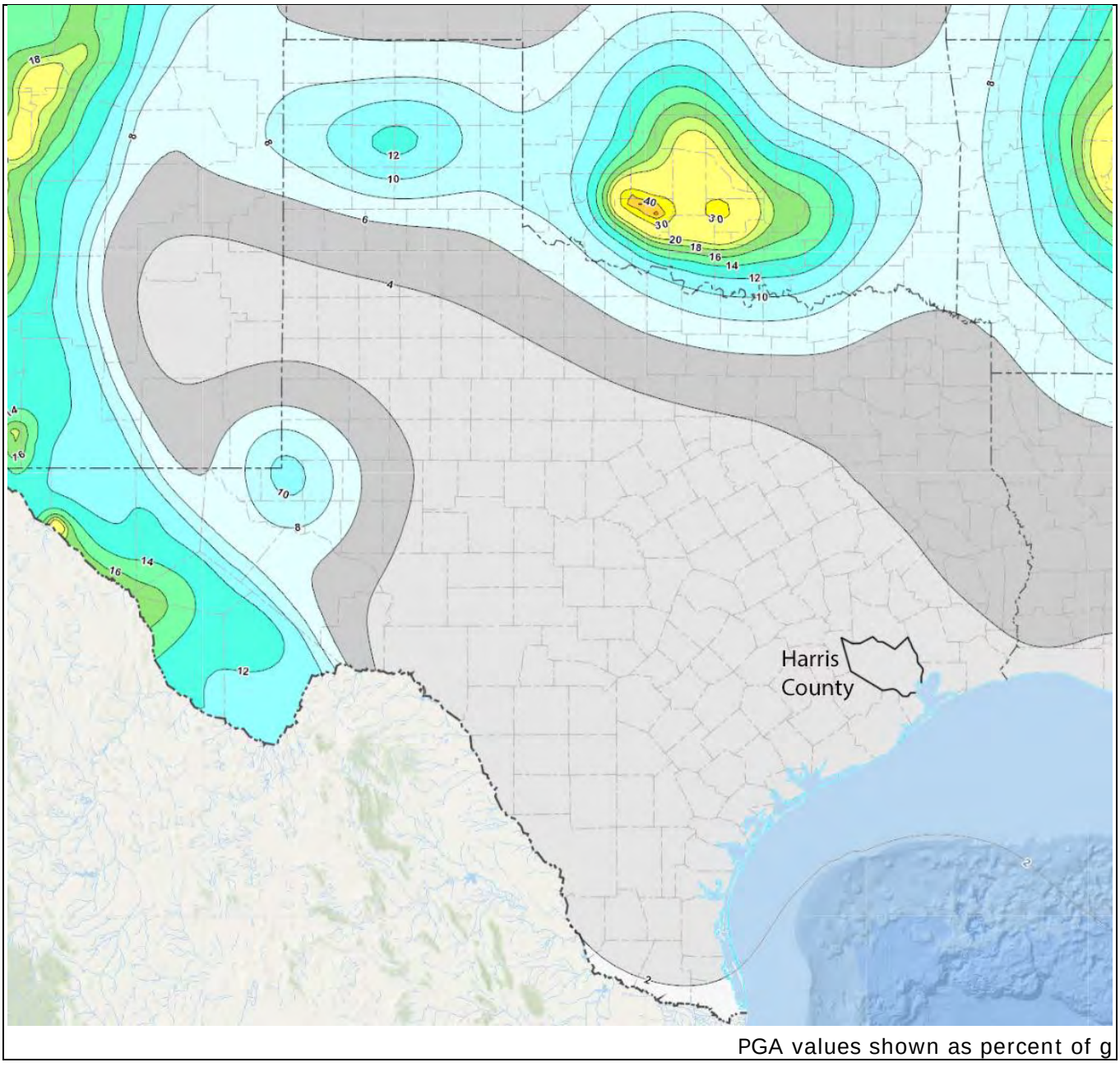
Figure 9-1. Probability of Experiencing an Earthquake in Texas



9.1.3 Extent

Figure 9-2 shows the earthquake-induced PGA with a 2-percent chance of being exceeded in 50 years in Texas. The map was compiled by the U.S. Geological Survey (USGS), which conducts global investigations of earthquake, geomagnetic, and landslide hazards. For Harris County, the PGA with that probability is between 2 and 4 percent of g, representing light perceived shaking on the Modified Mercalli scale (see Table 9-1).

Figure 9-2. PGA with a 2-Percent Chance of Being Exceeded in 50 Years in Texas



Source: (USGS n.d.)

9.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2024, Harris County was not included in any major disaster (DR) or emergency (EM) declarations for earthquake-related events.

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Once an emergency proclamation is declared, The Texas Division of Emergency Management (TDEM) coordinates the state emergency management program, which is intended to ensure the state, and its local governments respond to and recover from emergencies and disasters and implement plans and programs to help prevent or lessen the impact of emergencies and disasters. Harris County has not been included in any earthquake-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2023, Harris County was not included in any USDA earthquake-related agricultural disaster declarations.

Previous Events

There is insufficient data on earthquake events in the planning area, and there are no recorded previous occurrences in the FEMA, NCEI, USGS, or USDA databases.

9.1.5 Probability of Future Occurrences

Information on previous earthquake occurrences in the County was used to calculate the probability of future occurrences of such events, as summarized in Table 9-3. Based on historical records and input from the Steering Committee, the probability of occurrence for Hazard in the County is considered "rare."

Table 9-3. Probability of Future Earthquake Events in Harris County

Hazard Type	Number of Occurrences Between 1950 and 2024	Percent Chance of Occurring in Any Given Year
Earthquake	0	<1%

Source: (USGS 2025)

Note: Due to limitations in data, not all earthquake events occurring between 1954 and 1996 are accounted for in the tally of occurrences. As a result, the number of hazard occurrences is calculated using the number of occurrences between 1996 and 2023

Climate Change Projections

Although Harris County is not traditionally considered an area of high seismic activity, climate change may exacerbate certain geologic hazards such as soil instability, erosion, and land subsidence, factors that can influence or intensify the impacts of low-probability seismic events. These hazards are particularly relevant given Harris County's widespread use of expansive clay soils and its history of groundwater extraction-related subsidence (USGS, 2022).

The climate of Texas is changing. Over the past century, most of the state has warmed between 0.5°F and 1°F, with rainfall events becoming increasingly intense in the eastern part of the state, including Harris County (EPA 2016). As climate change drives more frequent and intense storms, the county faces higher risks of flash flooding and erosion. Heavy rainfall can saturate soils, reduce their cohesion, and trigger shallow landslides or slope failures, particularly in areas with artificial embankments or heavily modified landscapes (C2ES n.d.). These conditions may undermine foundations and exacerbate ground movement during seismic shaking—even minor tremors.

Climate change also affects the behavior of expansive soils. These clay-rich soils swell when wet and shrink when dry, a process that is highly sensitive to variations in temperature and precipitation. With climate projections pointing to more intense drought–flood cycles in the Gulf Coast region, the shrinking and swelling of soil layers may increase. Moreover, climate-related stressors, such as prolonged droughts followed by heavy rains, can accelerate erosion in natural and built environments. Streambank instability, gully formation, and terrain degradation can weaken support for bridges, roads, and underground utilities, further compounding vulnerability to geologic disturbances.

Though earthquakes remain a low-probability hazard in Harris County, climate change is likely to intensify secondary geologic impacts that increase long-term exposure and damage potential. Integrating soil behavior monitoring, resilient construction practices, and climate-informed geotechnical assessments into hazard mitigation planning will be essential for reducing future risk.

9.1.6 Cascading Impacts on Other Hazards

Harris County is not subject to traditional earthquakes, human activity like oil and gas extraction or wastewater injection can have cascading impacts on other hazards. One of the most immediate risks is the destruction or infrastructure such as dams, levees, and drainage systems. If these structures were damaged by an earthquake, the region would become more vulnerable to flooding during heavy rainfall or tropical storms. Since Harris County is susceptible to extreme weather events, a compromised flood control system could lead to widespread inundation, further intensifying the effects of hurricanes and severe storms.

Not only does earthquakes have the potential to intensify flood events, but their occurrence could also alter groundwater systems by shifting aquifers or creating new fractures in the subsurface. These disruptions can reduce the availability of groundwater further intensifying drought conditions.

9.2 VULNERABILITY AND IMPACT ASSESSMENT

A detailed hazard exposure and vulnerability assessment was not conducted for earthquakes due to the localized and infrequent nature of its impacts in this region. Unlike other hazards with broader geographic reach or more consistent historical data, earthquakes in Harris County tends to affect limited areas along the shoreline and does not pose a widespread threat to the county's overall building stock, critical facilities, or economy. As such, a traditional vulnerability assessment — typically used to quantify potential losses and exposure—was deemed not applicable for this hazard. Instead, the risk of earthquakes across the planning area and participating jurisdictions is considered minimal, with potential losses expected to be rare and highly site-specific.

9.2.1 Life, Health, and Safety

Harris County is considered a low-risk area for seismic activity, but earthquakes still pose potential life, health, and safety threats, especially due to unanticipated structural damage, cascading infrastructure failures, and limited local preparedness for such events. Even small-to-moderate magnitude earthquakes can have serious consequences if critical systems or densely populated areas are affected.

A key concern is the county's aging and vulnerable building stock, including older residential homes, schools, and public buildings not constructed to modern seismic standards. Many of these structures, particularly unreinforced masonry buildings or slab-on-grade homes, may sustain damage during an earthquake, leading to potential injuries from falling debris, interior collapse, or structural instability (FEMA, 2021). Public facilities such as hospitals, emergency shelters, and fire stations—if damaged—could limit access to lifesaving care and response services at the exact moment they are most needed.

Secondary hazards such as gas line ruptures, electrical fires, or hazardous material spills pose additional threats to public safety. Earthquakes, even minor ones, can compromise underground infrastructure including pipelines, water mains, and sewer systems. Disruptions to these lifelines can create hazardous conditions, including localized flooding, contamination, and fire risk (USGS, 2022).

Furthermore, Harris County's large and diverse population includes individuals who may be particularly vulnerable in a disaster. People with disabilities, older adults, non-English-speaking residents, and individuals with limited mobility or access to transportation may face significant barriers to safe access to emergency services. In a post-earthquake scenario, lack of familiarity with seismic hazards and limited regional preparedness could delay emergency response or cause confusion among the public, increasing injury risk (CDC, 2018).

While fatalities and injuries from earthquakes are rare in Harris County's historical record, the risk should not be dismissed entirely. Because the region is not typically associated with seismic activity, many residents and institutions lack experience or protocols for earthquake preparedness. A low-frequency but high-consequence event could result in life-threatening situations if proactive measures are not taken to educate the public, retrofit vulnerable structures, and integrate earthquake scenarios into emergency planning.

Overall Population

While Harris County is not located in a seismically active region, the potential for low-to-moderate magnitude earthquakes still presents risks to its population—particularly due to the county's large size, dense urban development, and generally low level of seismic preparedness. As of the 2020 U.S. Census, Harris County is home to 4,888,304 residents, making it the most populous county in Texas and one of the most populous in the United States (U.S. Census Bureau, 2021). The scale and concentration of this population increases the potential for disruption and damage should an earthquake occur, especially in older, vulnerable, or underserved communities.

Most of the county's population has not experienced an earthquake and may be unfamiliar with proper response actions. This knowledge gap, combined with the absence of routine seismic drills or retrofitting programs, could result in higher injury rates or delayed response in the event of even a minor earthquake. Furthermore, many residents live or work in buildings that predate modern seismic design codes, particularly in historic districts and low-income neighborhoods. These areas may be at greater risk of structural damage and injury, despite the low frequency of seismic events (FEMA, 2021).

Certain population groups, such as older adults, persons with disabilities, non-English speakers, and households lacking access to transportation, may face greater challenges in responding to earthquake emergencies. These individuals may require additional assistance during and after an event, especially if power outages, structural damage, or communication barriers arise.

Although earthquakes in Harris County are rare, population vulnerability remains a concern due to the region's aging infrastructure, socioeconomic disparities, and lack of seismic preparedness. As the population continues to grow—particularly in areas with aging building stock or expanding development—it is important to incorporate seismic risk awareness into public outreach, emergency planning, and building safety assessments. Doing so ensures that the entire population, especially those most at risk, is better protected in the face of a low-probability but potentially high-impact seismic event.

Hazus modeling performed for this risk assessment found that the planning area would not have any displaced households or persons seeking short-term sheltering for the 750-year mean return period (MRP) earthquake event for Harris County.

Socially Vulnerable Population

Three population groups are particularly vulnerable to earthquake hazards:

- Linguistically Isolated Populations—–Problems arise when there is an urgent need to inform non-English speaking residents of an earthquake event. They are vulnerable because of difficulties in understanding hazard-related information from predominantly English-speaking media and government agencies. According to the U.S. Census American Community Survey, the linguistically isolated population represented 11.9 percent of the total population in 2018.
- Population Below Poverty Level—These households may lack the financial resources to improve their homes to prevent or mitigate earthquake damage. Poorer residents are also

less likely to have insurance to compensate for losses in earthquakes. According to the American Community Survey, the population below poverty represented 16.8 percent of the total population in 2018.

- Population Over 65 Years Old—This population group is vulnerable because they are more likely to need special medical attention, which may not be available due to isolation caused by earthquakes. Older residents also have more difficulty leaving their homes during earthquake events and could be stranded in dangerous situations. According to the American Community Survey, the population over 65 years old represented 9.6 percent of the total population in 2018.

9.2.2 General Building Stock

There are 1,355,228 buildings in the planning area, with a total replacement cost value of over $1.1 billion. Since all structures in the planning area are susceptible to earthquake impacts to varying degrees, this represents the property exposure to seismic events. Table 9-4 shows the potential damage to the building stock from a 750-year MRP earthquake event. Table 9-5 summarizes Hazus estimates of earthquake damage in the planning area by jurisdiction.

Table 9-4. Planning Area Damage Potential by Occupancy Type

Occupancy Class	Total Number of Buildings in Occupancy	Severity of Expected Damage	750-Year MRP Earthquake	
			Building Count	Percent Buildings in Occupancy Class
Residential Exposure (Single and Multi-Family Dwellings)	1,260,996	None	1,251,728	99.3%
		Minor	7,010	0.6%
		Moderate	2,060	0.2%
		Severe	189	<0.1%
		Destruction	9	<0.1%
Commercial Buildings	85,457	None	84,718	99.1%
		Minor	537	0.6%
		Moderate	181	0.2%
		Severe	21	<0.1%
		Destruction	1	<0.1%
Industrial Buildings	4,821	None	4,747	98.5%
		Minor	53	1.1%
		Moderate	19	0.4%
		Severe	2	<0.1%
		Destruction	0	0.0%
Government, Religion, Agricultural, and Education Buildings	3,954	None	3,919	99.1%
		Minor	25	0.6%
		Moderate	8	0.2%
		Severe	1	<0.1%
		Destruction	0	0.0%

Table 9-5. Estimated Building Damage Due to 750-Year MRP Earthquake Event in the Planning Area

Jurisdiction	Total Replacement Cost Value (RCV)	Estimated Damage Due to Earthquake				
		Total Damage		Damage Breakdown by Occupancy Class		
		Estimated Damage	% of Total RCV	Residential	Commercial	All Other Occupancies
Baytown[b]	$12,477,103,055	$3,939,752	<0.1%	$2,206,315	$1,472,736	$260,701
Bellaire	$6,921,381,237	$0	0.0%	$0	$0	$0
Bunker Hill Village	$1,718,400,551	$0	0.0%	$0	$0	$0
Deer Park	$7,220,366,741	$6,459	<0.1%	$13	$5,146	$1,300
El Lago	$579,556,830	$0	0.0%	$0	$0	$0
Friendswood[a]	$7,480,850,836	$0	0.0%	$0	$0	$0
Galena Park	$1,484,142,708	$0	0.0%	$0	$0	$0
Hedwig Village	$1,463,766,016	$0	0.0%	$0	$0	$0
Hilshire Village	$267,434,315	$0	0.0%	$0	$0	$0
Houston[b]	$620,018,000,000	$16,738,307	<0.1%	$9,824,792	$6,091,369	$822,146
Humble	$5,480,480,895	$1,203,695	<0.1%	$509,214	$582,029	$112,451
Hunters Creek Village	$2,319,999,302	$0	0.0%	$0	$0	$0
Jacinto City	$1,619,709,430	$0	0.0%	$0	$0	$0
Jersey Village	$2,863,511,476	$0	0.0%	$0	$0	$0
Katy[a]	$3,789,785,990	$0	0.0%	$0	$0	$0
La Porte	$8,136,585,709	$36,601	<0.1%	$73	$29,162	$7,365
League City[b]	$343,758,798	$0	0.0%	$0	$0	$0
Missouri City[a]	$14,945,042,054	$0	0.0%	$0	$0	$0
Morgan's Point	$324,507,417	$10,765	<0.1%	$21	$8,577	$2,166
Nassau Bay	$2,104,459,129	$0	0.0%	$0	$0	$0
Pasadena	$27,932,208,850	$0	0.0%	$0	$0	$0
Pearland[b]	$878,335,161	$0	0.0%	$0	$0	$0
Piney Point Village	$2,383,712,681	$0	0.0%	$0	$0	$0
Seabrook	$3,115,949,582	$0	0.0%	$0	$0	$0

Jurisdiction	Total Replacement Cost Value (RCV)	Estimated Damage Due to Earthquake				
		Total Damage		Damage Breakdown by Occupancy Class		
		Estimated Damage	% of Total RCV	Residential	Commercial	All Other Occupancies
Shoreacres	$268,042,585	$0	0.0%	$0	$0	$0
South Houston[b]	$2,688,145,186	$0	0.0%	$0	$0	$0
Southside Place	$604,745,337	$0	0.0%	$0	$0	$0
Spring Valley Village	$1,568,469,594	$0	0.0%	$0	$0	$0
Stafford[a]	$7,232,853,862	$0	0.0%	$0	$0	$0
Taylor Lake Village	$854,884,292	$0	0.0%	$0	$0	$0
Tomball	$4,292,188,724	$1,116,499	<0.1%	$510,442	$532,282	$73,775
Waller[a]	$270,979,210	$0	0.0%	$0	$0	$0
Webster	$7,009,654,505	$0	0.0%	$0	$0	$0
West University Place	$4,782,421,662	$0	0.0%	$0	$0	$0
Unincorporated County	$402,152,000,000	$60,630,048	<0.1%	$40,049,255	$16,572,761	$4,008,031
Total	$1,167,594,514,462	$83,682,125	<0.1%	$53,100,126	$25,294,064	$5,287,935

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018, RS Means 2024; Hazus v6.1

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

9.2.3 Community Lifelines and Other Critical Facilities

Hazus classifies the vulnerability of critical facilities to earthquake damage in five categories: no damage, slight damage, moderate damage, extensive damage, or complete damage. The model was used to assign a probability of each damage state to every critical facility in the planning area. Table 9-6 shows the potential damage and post-event functionality for critical facilities or infrastructure in the planning area due to the 750-year MRP event.

Table 9-6. Average Percent Probability of Sustaining Damage 750-Year MRP

	Average % Probability of Sustaining Damage					Average Percent Functionality			
FEMA Lifeline	None	Slight	Moderate	Extensive	Complete	Day 1	Day 7	Day 30	Day 90
Communications	96.7%	2.4%	0.8%	0.1%	0.0%	99.4%	99.9%	99.9%	99.9%
Energy	96.8%	2.1%	1.0%	0.1%	0.0%	98.2%	99.4%	99.9%	99.9%
Food, Hydration, Shelter	98.2%	1.4%	0.4%	<0.1%	0.0%	98.2%	99.5%	99.9%	99.9%
Hazardous Materials	98.4%	1.3%	0.3%	<0.1%	0.0%	98.4%	99.6%	99.9%	99.9%
Health and Medical	99.2%	0.8%	0.0%	<0.1%	0.0%	99.1%	99.9%	99.9%	99.9%
Safety and Security	98.3%	1.3%	0.4%	<0.1%	0.0%	98.2%	99.5%	99.9%	99.9%
Transportation	99.9%	0.1%	<0.1%	<0.1%	0.0%	99.9%	99.9%	99.9%	99.9%
Water Systems	96.5%	2.2%	1.1%	0.2%	0.0%	98.2%	99.8%	99.9%	99.9%

Source: ATLAS 2022; BTS/USDOT 2023; CIMA 2024; City of La Porte 2025; City of Seabrook 2025; FleetCor 2024; Harris County 2020, 2024; Harris County Office of Emergency Management 2022-2025; Harris County Public Library, HCUS 2020, HCPCS 2024, HIFLD 2021-2024; METRO 2024; NPMS 2024; SETRAC 2024; TCEQ 2023-2024; Regional Catastrophic Preparedness Grant Program 2023; DSHS 2024; TEA 2023; THC 2023; USACE 2024; Hazus v6.1

9.2.4 Economy

Table 9-7 summarizes Hazus estimates of debris that would be generated in the planning area for the 750-year MRP earthquake scenario. The debris estimate includes only structural debris; it does not include additional debris that may accumulate, such as from trees. In addition, these estimates do not include losses that would occur from any local tsunamis or fires stemming from an earthquake.

Table 9-7. Estimated Debris Generated Due to 750-Year MRP Earthquake Event

Jurisdiction	Debris Generated by 750-Year MRP Earthquake	
	Brick/Wood (tons)	Concrete/Steel (tons)
Baytown[b]	3,120.5	754.5
Bellaire	0.0	0.0
Bunker Hill Village	0.0	0.0
Deer Park	3.3	2.5
El Lago	0.0	0.0
Friendswood[a]	0.0	0.0
Galena Park	0.0	0.0
Hedwig Village	0.0	0.0
Hilshire Village	0.0	0.0
Houston[b]	13,802.6	4,045.8
Humble	958.2	305.8
Hunters Creek Village	0.0	0.0
Jacinto City	0.0	0.0
Jersey Village	0.0	0.0
Katy[a]	0.0	0.0
La Porte	18.6	14.2
League City[b]	0.0	0.0
Missouri City[a0]	0.0	0.0
Morgan's Point	5.5	4.2
Nassau Bay	0.0	0.0
Pasadena	0.0	0.0
Pearland[b]	0.0	0.0
Piney Point Village	0.0	0.0
Seabrook	0.0	0.0
Shoreacres	0.0	0.0
South Houston[b]	0.0	0.0
Southside Place	0.0	0.0
Spring Valley Village	0.0	0.0
Stafford[a]	0.0	0.0
Taylor Lake Village	0.0	0.0
Tomball	847.6	277.4
Waller[a]	0.0	0.0
Webster	0.0	0.0
West University Place	0.0	0.0
Unincorporated County	52,590.4	12,602.5
Total	71,346.6	18,006.9

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

9.2.5 Natural, Historic and Cultural Resources

While Harris County is not typically associated with high earthquake risk, it is not entirely immune to seismic activity. Low probability but potentially damaging intraplate earthquakes have occurred in Texas, and minor fault lines do exist in and around the Houston area. Although the threat is minimal compared to flood or wind hazards, understanding the potential impact of seismic events—however rare—is essential for the protection of the county's rich natural, historic, and cultural resources.

Harris County's natural resources, such as bayous, wetlands, and conservation areas, are generally resilient to minor seismic events. However, soft or unconsolidated soils near waterways could amplify ground shaking, which may indirectly affect surrounding infrastructure and buildings. Areas with substantial human-made alterations to the natural landscape, including levees or retention systems, could also be subject to settlement or cracking from seismic motion, which may compromise flood control capacity (U.S. Geological Survey [USGS], 2021).

Historic resources, particularly unreinforced masonry buildings and aging structures in designated historic districts like Freedmen's Town, the East End, and Old Harrisburg, are more susceptible to seismic damage due to their construction type and age. Many of these buildings were constructed before the development of modern seismic building standards and may not be retrofitted to withstand even light tremors. The loss or damage of these culturally significant sites could result in irreversible cultural and historical value loss (City of Houston n.d.).

Cultural institutions such as the San Jacinto Battleground State Historic Site, the Houston Museum District, and historic cemeteries like Glendale and Olivewood may also be vulnerable to damage from ground movement, especially in older or less structurally supported facilities. Collections housed in museums and libraries may be displaced or damaged if storage systems are not designed with seismic considerations in mind.

While the probability of a damaging earthquake in Harris County remains low, the consequences to irreplaceable natural, historic, and cultural assets could be significant. As such, emergency preparedness plans for these resources should include basic seismic risk assessments, protective measures for collections and archives, and integration of cultural asset mapping into broader hazard mitigation strategies.

9.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

9.3.1 Potential or Planned Development

With the low degree of risk to the earthquake hazard within the planning area, any future development should be able to adequately manage that risk through the application of codes and standards such as the International Building Code, which have regionally relevant seismic

standards contained in its base regulations. The core capability assessments by the municipal planning partners to this effort identified a strong commitment to adoption of uniform codes and standards, which will position the planning partnership well for the mange earthquake risk for future development.

9.3.2 Projected Changes in Population

Harris County's population is projected to continue growing significantly in the coming decades, with estimates suggesting the county could surpass 6.6 million residents by 2050 under high-growth scenarios (Texas Demographic Center 2024).This rapid expansion—particularly in suburban and unincorporated areas—may increase exposure to a range of hazards, including those considered low-probability, such as earthquakes. As more people reside in newly developed or historically underserved communities, the strain on infrastructure and emergency response systems could complicate recovery even from minor seismic events. Growth in areas with expansive soils, older construction, or limited hazard awareness may further elevate vulnerability, underscoring the need to incorporate population trends into regional risk assessments and mitigation strategies.

9.3.3 Climate Change

While climate change does not directly cause earthquakes, it can intensify secondary geologic hazards that influence earthquake vulnerability in Harris County. The region's expansive clay soils are particularly sensitive to climate-driven cycles of drought and heavy rainfall. These fluctuations can cause soils to shrink and swell, compromising building foundations, roads, and underground infrastructure. Structures already stressed by shifting ground conditions may be more susceptible to damage from even low-magnitude seismic activity.

Additionally, increased precipitation intensity and flooding—both projected to rise with climate change—can accelerate erosion and destabilize slopes, embankments, and manmade fill. This not only poses hazards on its own but can also amplify ground instability during seismic events. As Harris County continues to grow and urbanize, understanding the compounded effects of climate and geologic processes will be critical for protecting infrastructure, especially in vulnerable areas with older development or limited soil stabilization (EPA, 2016; USGS, 2022).

10. EXTREME COLD

10.1 HAZARD PROFILE

10.1.1 Hazard Description

Extreme cold events are when temperatures drop well below normal in an area. In regions relatively unaccustomed to winter weather, near-freezing temperatures are considered "extreme cold." Extreme cold temperatures are characterized by the ambient air temperature dropping to 0°F or below (NWS n.d.).

Exposure to cold temperatures, whether indoors or outdoors, can lead to serious or life-threatening health problems such as hypothermia, cold stress, frostbite, or freezing of the exposed extremities, such as fingers, toes, nose, and ear lobes. Hypothermia occurs when the core body temperature is lower than 95ºF. Extreme cold can also cause emergencies in vulnerable populations, such as those without shelter, those who are stranded, or those who live in a home that is poorly insulated or without heat (such as mobile homes). Although staying indoors as much as possible can help reduce the risk of car crashes and falls on the ice, individuals may also face indoor hazards. Many homes will be too cold, either due to a power failure or because the heating system is not adequate for the weather. The use of space heaters and fireplaces to keep warm increases the risk of household fires and carbon monoxide poisoning.

Extremely cold temperatures often accompany a winter storm. Severe winter weather includes heavy snow, blizzards, sleet, ice storms or freezing rain, frost/freeze, extreme cold temperatures, or a mix of these. Periods of extreme cold, independent of a storm event, can also be dangerous and damaging. Severe winter weather can down trees, cause widespread power outages, damage property, and cause fatalities and injuries.

- Heavy Snow: According to the National Weather Service (NWS), heavy snow is generally considered snowfall accumulating to a depth of 4 inches or more within 12 hours or less or snowfall accumulating to a depth of 6 inches or more within 24 hours or less. A snow squall is an intense but limited-duration period of moderate to heavy snowfall, also known as a snowstorm, accompanied by strong, gusty surface winds and possibly lightning (generally moderate to heavy snow showers) (NWS, National Weather Service Glossary 2009).
- Blizzard: Blizzards are characterized by low temperatures, wind gusts of 35 miles per hour (mph) or more, and falling and/or blowing snow that reduces visibility to 0.25 miles or less for an extended period (three or more hours). A severe blizzard is defined as having a wind velocity of 45 mph, temperatures of 10°F or lower, and a high density of blowing snow with visibility frequently measured in feet over an extended period. Blizzards take a terrible toll on livestock and people without adequate shelter. Although blizzards in Texas are most likely to occur in the Panhandle and South Plains regions, Harris County is still affected by these events (TDEM, Texas State Hazard Mitigation Plan 2023).
- Sleet or Freezing Rain: Sleet is defined as pellets of ice composed of frozen or mostly frozen raindrops or refrozen, partially melted snowflakes. These pellets of ice usually bounce

after hitting the ground or other hard surfaces. Freezing rain is rain that falls as a liquid but freezes into a glaze upon contact with the ground. Both types of precipitation, even in small accumulations, can cause significant hazards to a community.

- Ice Storm: An ice storm is described as an occasion when damaging volumes of ice are expected to accumulate during freezing rain situations. Significant accumulations of ice pull down trees and utility lines, resulting in loss of power and means of communication. These accumulations of ice render walking and driving extremely dangerous and can cause extreme hazards to motorists and pedestrians.

10.1.2 Location

Extreme cold and winter weather can happen anywhere in Harris County, although its levels can range extensively. Harris County is susceptible to extremely cold temperatures, especially severe winter storms. The record lows for Texas typically occur during October through March. The average first freeze in the County usually occurs from late November to early December, and the last freeze usually occurs from late February to early March, according to data recorded by the National Weather Service.

10.1.3 Extent

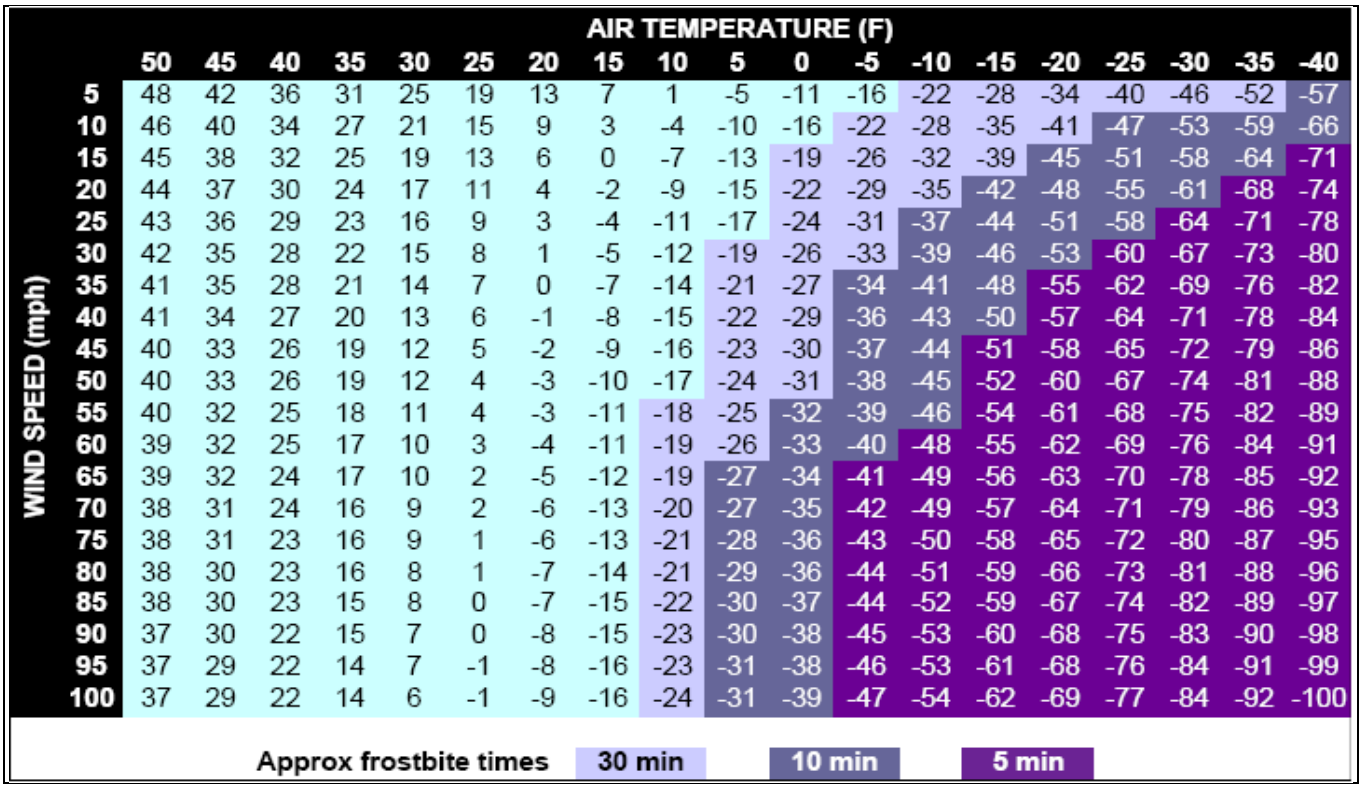
The extent or magnitude of a severe winter storm is measured in intensity based on the temperature and level of accumulation. Table 10-1 provides the description and range of intensities established by NWS for severe winter storms. The Wind Chill Index in Figure 10-1 is a formula developed by the NWS to estimate how cold the air feels on human skin when factoring in wind speed. It combines air temperature and wind speed to give a "feels-like" temperature, and it is helpful when determining the risk of frostbite and hypothermia. The Wind Chill Index is only applicable when temperatures are at or below 50°F and wind speeds are above 3 mph. Wind amplifies the cooling effect by blowing away the thin layer of warm air near a person's skin, making it feel colder than the actual air temperature. Table 10-1 should be read in conjunction with the wind-chill factor described in Figure 10-1 to determine the intensity of a winter storm.

Table 10-1. Magnitude of Severe Winter Storms

Intensity	Temperature	Extent Description
Mild	40°F–50°F	Winds less than 10 mph and freezing rain or light snow falling for short durations with little or no accumulations.
Moderate	30°F–40°F	Winds between 10 and 15 mph and sleet and/or snow up to 4 inches.
Significant	25°F–30°F	Intense snow showers accompanied by strong gusty winds between 15 and 20 mph with significant accumulation.
Extreme	25°F–20°F	Wind-driven snow that reduces visibility, heavy winds (between 20 and 30 mph), and sleet or ice up to 5 millimeters in diameter.
Severe	Below 20°F	Winds of 35 mph or more and snow and sleet greater than 4 inches

Source: NWS

Figure 10-1. NWS Wind Chill Index



Source: NOAA/NWS 2018

The NWS Houston/Galveston office has established certain thresholds for different winter-related hazards. As of October 1, 2024, a Wind Chill Advisory is now called a Cold Weather Advisory, and the criteria for this product changes across southeastern Texas. For Harris County and the entire planning area, Cold Weather Advisories are issued when wind chill values are less than or equal to 25°F. Table 10-2 summarizes these criteria and threshold for the planning area.

Table 10-2. Winter-Related Forecast Products from NWS Houston/Galveston

Product Name	Criteria
Cold Weather Advisory (formerly "wind chill advisory")	The criteria for this product changes across the region, but for Harris County, a Cold Weather Advisory is issued when wind chill values are ≤ 25 °F
Extreme Cold Watch or Warning (formerly "wind chill watch" or "wind chill warning")	The criteria for this product changes across the region, but for Harris County, an Extreme Cold Watch is issued when wind chill values are ≤ 15 °F.

Source: (NWS Houston/Galveston 2024)

The average number of cold days is similar for the entire planning area. Therefore, the intensity or extent of a winter storm event to be mitigated for the area ranges from significant to severe, according to the definitions established by NWS. A winter storm watch, advisory, or warning is issued in advance of an event in order to give people enough time to prepare for an event. Harris County could be under any of these warning types in advance of a winter storm event. Table 10-3 describes when each warning type would be issued.

Table 10-3. Winter Storm Watch, Advisory, Warning Descriptions

Type of Winter Weather	Description
Winter Weather Advisory	This alert may be issued for a variety of severe conditions. Weather advisories may be announced for snow, blowing or drifting snow, freezing drizzle, freezing rain, or a combination of weather events.
Winter Storm Watch	Severe winter weather conditions may affect your area (freezing rain, sleet, or heavy snow may occur separately or in combination).
Winter Storm Warning	Severe winter weather conditions are imminent.
Freezing Rain or Freezing Drizzle	Rain or drizzle is likely to freeze upon impact, resulting in a coating of ice glaze on roads and all other exposed objects.
Sleet	Small particles of ice, usually mixed with rain. If enough sleet accumulates on the ground, it makes travel hazardous.
Blizzard	Sustained wind speeds of at least 35 mph are accompanied by considerable falling or blowing snow. This alert is the most perilous winter storm, with visibility dangerously restricted.
Frost/Freeze	Below-freezing temperatures are expected and may cause significant damage to plants, crops, and fruit trees.
Wind Chill	A strong wind combined with a temperature slightly below freezing can have the same chilling effect as a temperature nearly 50 degrees lower in a calm atmosphere. The combined cooling power of the wind and temperature on exposed flesh is called the wind-chill factor.

Source: NWS

10.1.4 Previous Occurrences

Significant Recent Events

Table 10-4 lists significant cold and winter weather events that impacted Harris County between 2020 and 2024. For events prior to 2020, refer to the 2020 Harris County Multi-Hazard Mitigation Action Plan (MH-MAP).

FEMA Major Disaster and Emergency Declarations

Harris County has been included in five major disaster (DR) or emergency (EM) declarations for any extreme cold and winter weather-related events, as listed in Table 10-5 (FEMA 2024).

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Harris County has been included in one extreme cold and winter weather-related state emergency proclamation, as listed in Table 10-6.

Table 10-4. Extreme Cold and Winter Weather Events in Harris County (2020 to 2024)

Event Date	Declaration Number	Location Impacted	Description
February 12, 2021	EM-3554 DR-4586 Texas Governor's Proclamation (unnumbered)	Inland Harris County and Coastal Harris County	Cold/Wind Chill event. Record cold and windy conditions gripped Southeast Texas behind the passage of an Arctic front. The extreme cold and heightened demand caused widespread power outages, which resulted in many people losing heat. Additionally, the cold weather caused some pipes to burst, leaving many homes without water for several days. Numerous fatalities resulted from hypothermia, carbon monoxide poisoning, and other effects. Winter Storm event. An Arctic front brought extreme cold to Southeast Texas and a few bouts of winter precipitation. The first storm brought snow, sleet, and/or freezing rain to different parts of the area.
February 3, 2022	N/A	Inland Harris County	Winter Weather event. A period of freezing rain fell over areas mainly north and west of Houston, producing an icy glaze that caused numerous car accidents. Portions of 290 in NW Harris County were closed due to ice.
January 18, 2025	N/A	Entire State	Extreme cold covered the State, with snow and icy conditions developing across large portions of the state
February 2025	N/A	Entire State	Texas experienced multiple days of icy conditions across large portions of the state. The entire area saw freezing or below-freezing temperatures for extended periods of time.

Source: NCEI Storm Events Database 2025

Table 10-5. FEMA Declarations for Extreme Cold and Winter Storms in Harris County

Event Date	Declaration Date	Declaration Number	Description
February 11, 2021	February 14, 2021	EM-3554	Severe Winter Storm/Ice Storm
February 11, 2021	February 19, 2021	DR-4586	Severe Winter Storm/Ice Storm
December 12, 2000–January 15, 2001	January 8, 2001	DR-1356	Severe Winter Storm/Ice Storm
December 21, 1989–December 24, 1989	January 9, 1990	DR-850	Texas Severe Freeze
December 22, 1983–December 26, 1983	January 7, 1984	DR-696	Texas Severe Freezing Weather

Table 10-6. State Emergency Proclamations for Extreme Cold and Winter Weather Events in Harris County

Event Date	Proclamation Date	Description
February 11–21, 2021	February 12, 2021	Winter Storm Uri dumped record amounts of snow on Texas, with the frigid temperatures and severe weather impacting all 254 counties in the state in February 2021. Millions of Texans lost power. Snow and ice, paired with ultra-low temperatures, caused widespread road closures and dangerous travel conditions. State emergency management leaders activated warming centers in communities across Texas, and numerous personnel were deployed to assist stranded motorists and conduct welfare checks. Gov. Greg Abbott issued a disaster declaration for all 254 counties in the state.

Source: (Texas 2025)

USDA Declarations

The U.S. Secretary of Agriculture is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2024, Harris County was not included in any USDA extreme cold and winter weather-related agricultural disaster declarations.

10.1.5 Probability of Future Occurrences

Probability Based on Previous Occurrences

Information on previous extreme cold and winter weather occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 10-7. Based on historical records and input from the Steering Committee, the probability of occurrence for extreme cold and winter weather in the County is considered "occasional."

Table 10-7. Probability of Future Extreme Cold and Winter Weather Events in Harris County

Hazard Type	Number of Occurrences Between 1950 and 2024	Percent Chance of Occurring in Any Given Year
Extreme Cold	16	36%

Source: (NOAA/NCEI 2024)
Note: Due to limitations in data, not all extreme cold and winter weather events occurring between 1954 and 1996 are accounted for in the tally of occurrences.

Effect of Climate Change on Future Probability

Climate change is expected to lead to warmer and potentially drier winters in Texas, with fewer cold spells and frost days. However, it is important to note that extreme cold events are still possible, and La Niña patterns can introduce brief periods of arctic temperatures. Winter rainfall is likely to decrease, and the frequency of extreme heat events, including days above 100°F, is projected to increase significantly (NOAA 2024).

10.1.6 Cascading Impacts on Other Hazards

Extreme cold events can escalate severe weather and severe winter weather events for the County. Extreme variation in temperatures could also create ideal atmospheric conditions for severe storms or worsen the outcome of severe winter weather during freezing and thawing periods.

10.2 VULNERABILITY AND IMPACT ASSESSMENT

10.2.1 Life, Health, and Safety

Overall Population

For the purposes of this MJ-HMP, the entire population of Harris County is considered exposed to extreme cold and winter storm events and death from falls, overexertion, and/or hypothermia from exposure while attempting to clear snow and ice. The following health hazards are related to extreme cold temperatures (CDC 2023):

- Frostbite is damage to body tissue caused by extreme cold. A wind chill of -20°F will cause frostbite in 30 minutes. Frostbite can cause a loss of feeling and a white or pale appearance in the extremities.
- Hypothermia is a condition brought on when the body temperature drops to less than 95°F, and it can be deadly. Warning signs of hypothermia include uncontrollable shivering, memory loss, disorientation, incoherence, slurred speech, drowsiness, and apparent exhaustion.

According to the NOAA National Severe Storms Laboratory (NSSL), winter weather indirectly and deceptively kills hundreds of people in the United States every year, primarily from automobile accidents, overexertion, and exposure. Winter storms are often accompanied by strong winds creating blizzard conditions with blinding wind-driven snow, drifting snow, extreme cold temperatures, and dangerous wind chill. Winter storms are considered deceptive killers because most deaths and other impacts or losses are indirectly related to the storm. People can die in traffic accidents on icy roads, of heart attacks while shoveling snow, or of hypothermia from prolonged exposure to cold.

Heavy snow can immobilize a region and paralyze a city, shutting down air and rail transportation, stopping the flow of supplies, and disrupting medical and emergency services. First responders will have to take on additional responsibilities during winter storm events, such as controlling traffic, debris removal from roads, answering a higher-than-normal call volume and demand, and responding to weather-related traffic accidents. First responders' safety may be at risk during on-scene operations, and limited access to roads due to damaged infrastructure and debris may hinder their ability to respond to accidents promptly. Accumulations of snow can collapse buildings and knock down trees and power lines. In rural areas, homes and farms may be isolated for days, and unprotected livestock may be lost. In the mountains, heavy snow can lead to avalanches (NWS 2023).

Heavy accumulations of ice can bring down trees, electrical wires, telephone poles and lines, and communication towers. Communications and power can be disrupted for days while utility companies work to repair the extensive damage. Even small accumulations of ice may cause extreme hazards to motorists and pedestrians. Bridges and overpasses are particularly dangerous because they freeze before other surfaces (NWS 2023).

Surveillance data collected after extreme cold events show that the leading causes of hospitalization are slips, falls, and automobile accidents (PostMashhadi, Maghsoodi and Lincoln 2025). These incidents are typically the result of accumulated ice on roads and walkways, and they affect individuals of all ages. The next most common cause of hospitalization and death is carbon monoxide poisoning. This can occur from the use of faulty heating devices, improper use of gas stoves or fireplaces, or running vehicles inside enclosed garages. Like physical injuries, carbon monoxide exposure poses a serious risk to people of all ages and highlights the importance of public awareness and safety measures during extreme cold conditions.

Socially Vulnerable Population

Unhoused and unsheltered individuals, older adults, and children are considered most susceptible to this hazard. Individuals who are unhoused or unsheltered are directly exposed to the extreme cold hazard. Older adults and children have an increased risk of injuries and loss of life resulting from falls, overexertion, or hypothermia. Winter weather events can reduce the ability of these populations to access emergency services.

Residents with low incomes may not have access to housing, or their housing may be less able to withstand cold temperatures (e.g., homes with poor insulation and heating supply). Chapter 3, County Profile, provides population statistics regarding each participating municipality and a summary of the more vulnerable populations.

10.2.2 General Building Stock

The entire general building stock inventory in Harris County is exposed and vulnerable to the extreme cold hazard. Extreme cold temperature events can damage buildings through freezing/bursting pipes and freeze/thaw cycles as well as increase vulnerability to home fires. Additionally, manufactured homes and older or poorly constructed buildings often lack adequate capabilities to withstand extreme temperatures. These deficiencies can include insufficient insulation and less efficient heating systems.

In general, structural impacts include damage to roofs and building frames rather than building content. Current modeling tools are not available to estimate specific losses from this hazard. An area especially vulnerable to the winter storm hazard is the floodplain. At-risk building stock and infrastructure in floodplains are presented in the flood hazard profile in Section 4.3.6. Generally, losses from flooding associated with winter storms should be less than those associated with a 1 percent or 0.2 percent flood. Snow and ice melt can cause both riverine and urban flooding. Estimated losses caused by riverine flooding in the county are discussed in Section 4.3.6.

10.2.3 Community Lifelines and Other Critical Facilities

Community lifelines and critical facilities will experience similar issues as described for the general building stock. Heavy snowfall and ice storms associated with extreme cold temperatures can interrupt power, resulting in disruption to community lifelines and critical facilities. Infrastructure at risk for this hazard includes roadways that could be damaged due to the application of salt and

intermittent freezing and warming conditions that can damage roads over time. Severe snowfall requires the clearing of roadways and alerting citizens to dangerous conditions; following the winter season, resources for road maintenance and repair are required (NSSL 2022). Additionally, designating and developing emergency heating facilities can enhance the resilience and safety of communities.

10.2.4 Economy

Infrastructure at risk from the winter storm hazard includes roadways that could be damaged by the application of salt and intermittent freezing and warming conditions that can damage roads over time. Snow and ice removal as well as repairs of roads undergoing freeze/thaw cycles can drain local financial resources. Potential secondary impacts from winter storms also impact the local economy, including loss of utilities, interruption of transportation corridors, and loss of business function.

10.2.5 Natural, Historic, and Cultural Resources

Prolonged cold spells can freeze wetland soils, reducing water seepage and causing snowmelt runoff to bypass wetlands (Tiwari 2018). This makes water unavailable during the crucial spring and summer months. Heavy snow and ice accumulation during extreme cold weather can damage trees and crops by breaking vegetation and tree limbs. Prolonged periods of extreme cold can also harm vegetation and crops, negatively impacting the agricultural industry in the region (Tiwari 2018).

Older structures built to less stringent building codes may suffer greater damage as they are typically more vulnerable to the impacts of winter storm events. Similarly, historic buildings and sites are placed at a higher risk of impact due to materials used and the inability to change properties due to their historic status.

10.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure the establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

10.3.1 Potential or Planned Development

Areas targeted for potential future growth and development within the next 5 to 10 years have been identified across the County at the municipal level and are further discussed in Chapter 3 of this MJ-HMP. Because Harris County in its entirety has been identified as a hazard area vulnerable to the winter storm hazard, any new development will be exposed to associated risks. However, because of increased standards and codes, new development may be less vulnerable to severe winter weather hazards compared with the older building stock in the county.

10.3.2 Projected Changes in Population

As discussed in Chapter 3, County Profile, the Harris County population has been increasing and is projected to continue to increase in the coming decades. In addition, the population is aging. As the aging population grows, so too will the number of persons vulnerable to severe winter weather and extreme cold temperatures. Furthermore, continued population growth may contribute to an increase in the unhoused population and economically disadvantaged communities. These groups are particularly susceptible to the impacts of extreme cold and should be considered communities of concern in future mitigation and response planning.

10.3.3 Climate Change

Climate change is expected to lead to milder, shorter winters in Texas with less precipitation. While extreme winter weather events are still possible, they are likely to be less frequent. Overall, Texas is projected to experience a warmer climate, with average annual temperatures increasing by 3.0°F by 2036 compared to the 1950–1999 average (Douglas 2021).

10.3.4 Other Identified Conditions

Extreme cold and winter weather can place significant stress on infrastructure, including roads, bridges, and buildings. There is often an increased demand for energy during extreme temperatures, particularly for heating. This surge in energy use can lead to power outages and increased stress on energy grids.

11. EXTREME HEAT

11.1 HAZARD PROFILE

11.1.1 Hazard Description

Extreme heat is a prolonged period of excessively high temperatures and exceptionally humid conditions. The NWS criteria for an excessive heat warning is a heat index of 105°F or greater that will last for 2 hours or more. While the average high temperature in Harris County is 98°F, the County is susceptible to extreme heat.

In extreme heat, the human body works extra hard to maintain a normal temperature, which can lead to death. Extreme heat is responsible for the highest number of annual deaths among all weather-related hazards (FEMA 2025).

Although heat can damage buildings and facilities, it presents a more significant threat to the safety and welfare of the public. The major human risks associated with extreme heat include heat cramps, sunburn, dehydration, fatigue, heat exhaustion, and even heat stroke. The populations most vulnerable to heat casualties are children, older adults, and individuals with chronic medical conditions.

Depending on the severity, duration, and location, extreme heat events can create secondary hazards, including dust storms, droughts, wildfires, water shortages, and power outages. These secondary hazards could result in a broad and far-reaching set of impacts throughout a local area or an entire region.

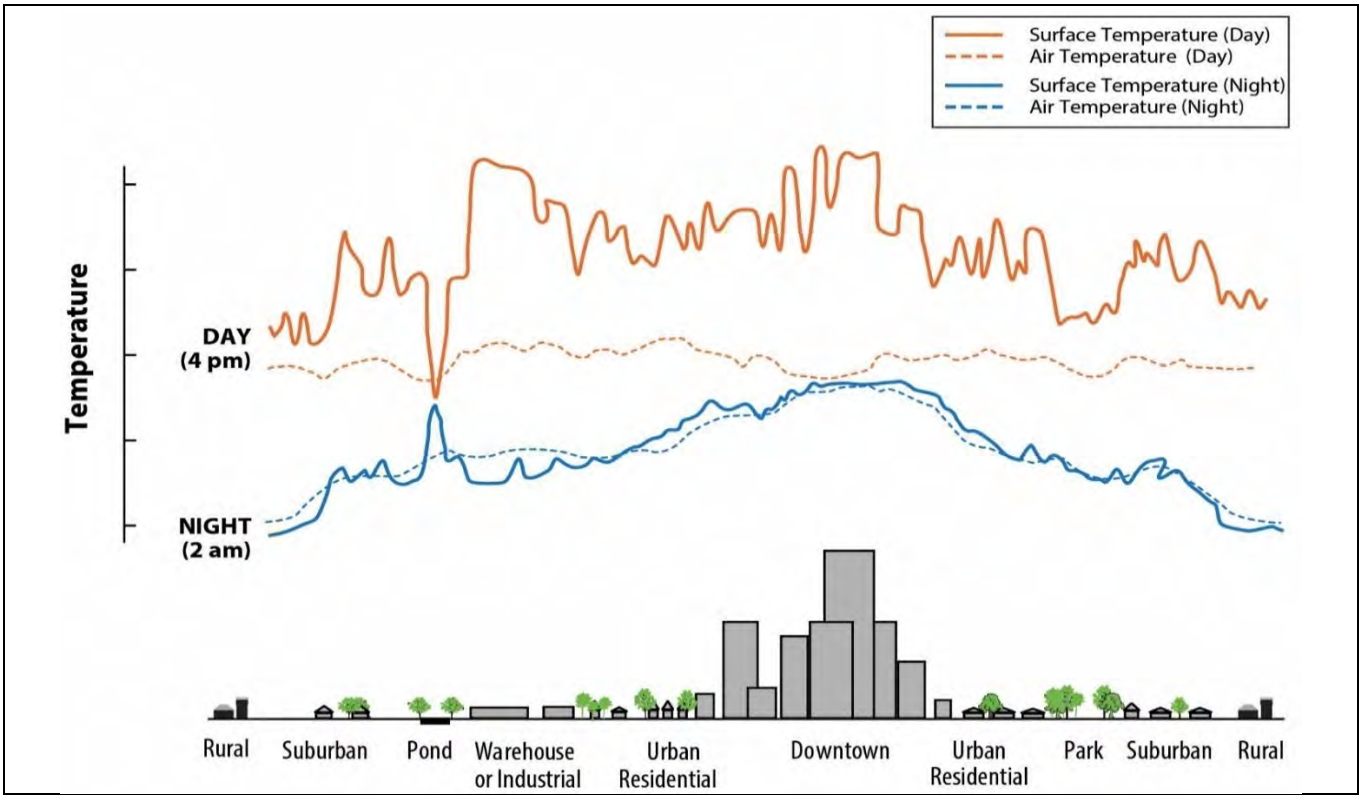
11.1.2 Location

Extreme heat events may occur across the entire Harris County planning area, as there are no defined geographic boundaries for this hazard. Such events have the potential to impact all jurisdictions within the County. Harris County includes urban centers, rural communities, and extensive agricultural land, all of which are susceptible to temperature extremes. However, urban areas are generally at a higher risk of the adverse effects associated with extreme heat.

Urbanization exacerbates risk during an extreme heat event. As urban areas develop, buildings, roads, and other infrastructure replace open land and vegetation. Surfaces that were once permeable and moist become impermeable and dry. These changes cause urban areas to become warmer than the surrounding areas, forming an "island" of higher temperatures. A "heat island" is a built-up area that is hotter than nearby rural areas. The annual mean air temperature of a city with more than 1 million people can be 2°F to 5°F warmer than its surroundings. In the evening, the difference in air temperatures can be as high as 22°F.

As shown in Figure 11-1, surface temperatures vary more than atmospheric air temperatures during the day, but they are generally similar at night. On a hot, sunny day, the sun can heat dry, exposed urban surfaces to temperatures 50°F to 90°F hotter than the air. The dips and spikes in surface temperatures over the pond area show how water maintains a nearly constant temperature day and night because it does not absorb the sun's energy the same way as buildings and paved surfaces. Parks, open land, and bodies of water can create cooler areas within a city. Temperatures are typically lower at suburban-rural borders than in downtown areas.

Figure 11-1. Heat Island Effect



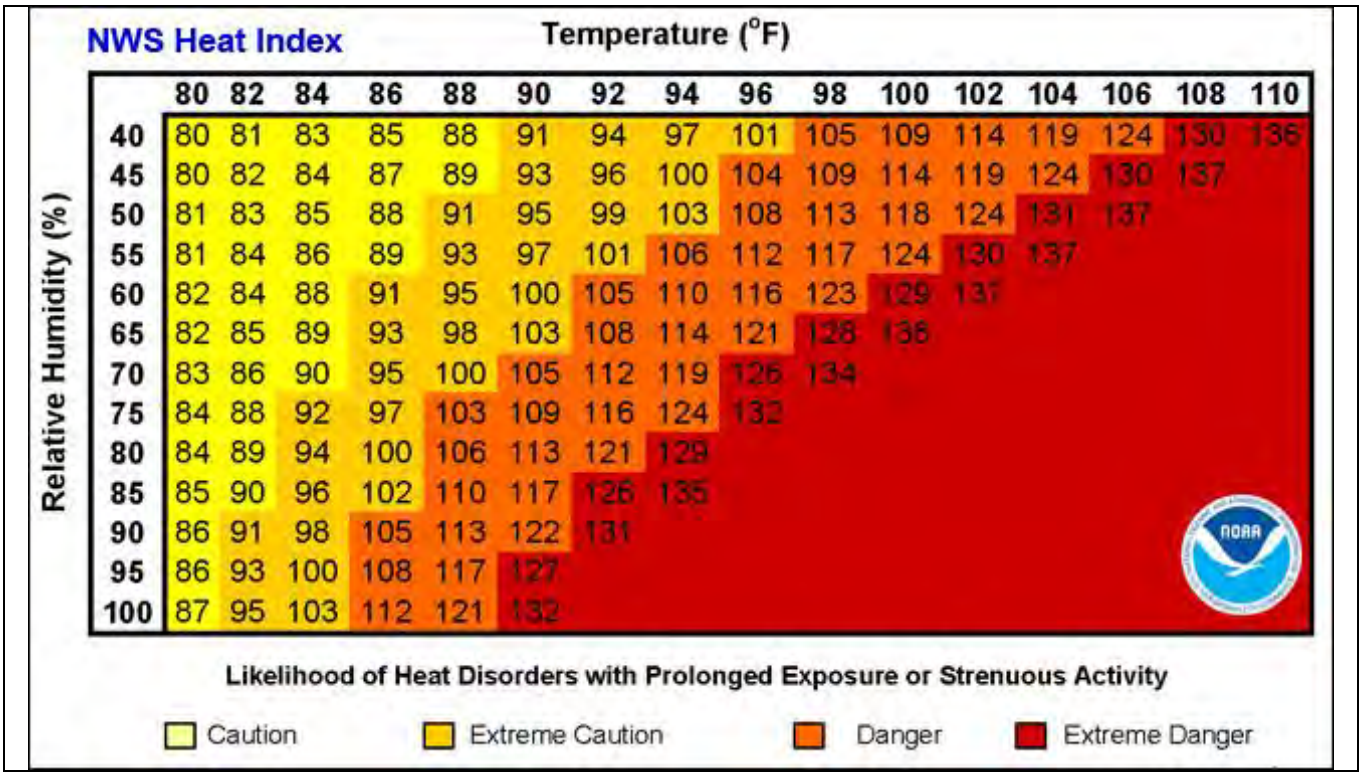
Source: U.S. EPA 2023

Heat islands can affect communities by increasing peak energy demand during the summer, air conditioning costs, air pollution and greenhouse gas emissions, heat-related illness and death, and water quality degradation (EPA 2024).

11.1.3 Extent

NWS has created a heat index chart that takes into account the temperature and relative humidity to estimate the likelihood of heat disorders with prolonged exposure. The heat index is given in degrees Fahrenheit and corresponds with the temperature that the body feels (NWS 2023). Figure 11-2 shows the heat index value for shady areas; full sun exposure can increase the index by up to 15°F. Table 11-1 describes the adverse effects of prolonged exposure to direct sunlight on an individual. NWS offices across the nation initiate heat alerts at different thresholds, with three of those products described in Table 11-2.

Figure 11-2. NWS Heat Index Chart



Source: NOAA/NWS 2023

Table 11-1. Adverse Effects of Prolonged Exposure to Direct Sunlight

Category	Heat Index	Effects on the Body
Caution	80°F – 90°F	Fatigue is possible with prolonged exposure and/or physical activity
Extreme Caution	90°F – 103°F	Heat stroke, heat cramps, or heat exhaustion is possible with prolonged exposure and/or physical activity
Danger	103°F – 124°F	Heat cramps or heat exhaustion is likely, and heat stroke is possible with prolonged exposure and/or physical activity
Extreme Danger	125°F or higher	Heat stroke is highly likely

Source: NWS N.D.

Table 11-2. Heat-Related Forecast Products of the NWS Houston/Galveston

	Criteria
Heat Advisory	Heat Index ≥ 108°F or Temperature ≥ 103°F possible; threshold may be lowered during early season events or other special circumstances (e.g., power outages after tropical events).
Extreme Heat Watch	Heat Index ≥ 113°F or Temperature ≥ 105°F possible; threshold may be lowered during early season events or other special circumstances (e.g., power outages after tropical events).
Extreme Heat Warning	Heat Index ≥ 113°F or Temperature ≥ 105°F expected; threshold may be lowered during early season events or other special circumstances (e.g., power outages after tropical events).

Source: (NWS Houston/Galveston 2024)

In Harris County, NWS Heat Advisory procedures initiate when the Heat Index is greater than or equal 108°F or the air temperature is greater than or equal to 103°F. The NWS issues excessive heat outlooks when the potential exists for an excessive heat event in the next three to seven days. Watches are issued when conditions are favorable for an excessive heat event in the next 24 to 72 hours. Excessive heat warning/advisories are issued when an excessive heat event is expected in the next 36 hours (NWS 2021).

11.1.4 Previous Occurrences

Significant Recent Events

The National Centers for Environmental Information (NCEI) Storm Events database is a national data source organized under the National Oceanic and Atmospheric Administration (NOAA). The NCEI is the largest archive available for historic storm events data. Previous occurrences for extreme heat are derived from the NCEI database, which identifies extreme heat events at the county level for each event. There have been 72 extreme heat events recorded in the NCEI for the Harris County planning between 1998 and 2024. Historical extreme heat information, as provided by the NCEI, shows extreme heat activity across a multicounty forecast area for each event; the appropriate percentage of the total property and crop damage reported for the entire forecast area has been allocated to each county impacted by the event. According to the Texas State Hazard Mitigation Plan and based on historical events from 2000–2021, Harris County had the highest number of heat-related fatalities of any county in Texas.

Table 11-3 lists significant hazard events that impacted Harris County between 2020 and 2024. For events prior to 2020, refer to the 2020 Harris County Multi-Jurisdictional Hazard Mitigation Action Plan (MJ-HMAP).

FEMA Major Disaster and Emergency Declarations

Harris County has not been included in any major disaster (DR) or emergency (EM) declarations for extreme heat-related events (FEMA 2025).

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Harris County has not been included in any extreme heat-related state emergency proclamations.

USDA Declarations

The U.S. Secretary of Agriculture is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2024, Harris County was included in one USDA extreme heat-related agricultural disaster declarations. For declarations that occurred between 2020 and 2024, refer to Table 11-4.

Table 11-3. Extreme Heat Events in (2020 to 2024)

Event Date	Declaration Number	Location Impacted	Description
June 12, 2023	N/A	Inland Harris County	Excessive Heat event. According to the medical examiner, a 40-year-old male died of hyperthermia brought on by excessively hot conditions that occurred across the area throughout the month of June.
June 14, 2023	N/A	Inland Harris County	Excessive Heat event. According to the medical examiner, an 88-year-old woman died of hyperthermia brought on by excessively hot conditions that occurred across the area throughout the month of June.
June 16, 2023	N/A	Inland Harris County Coastal Harris County	Period of extremely hot weather. A 67-year-old man died of hyperthermia in a Houston hospital during the period. He was in a home that was without air conditioning. Long period of at least advisory levels. Periods of warning levels are recorded.
June 22, 2023	N/A	Inland Harris County	According to the medical examiner, a 70-year-old male died of hyperthermia brought on by excessively hot conditions that occurred across the area throughout the month of June.
June 24, 2023	N/A	Inland Harris County	Another stretch of warning-level heat overspread the area with heat indices near 113°F. There were two heat-related deaths in Houston during this period.
June 25, 2023	N/A	Coastal Harris County	Another stretch of warning-level heat overspread the area with heat indices near 113°F.
July 12, 2023	N/A	Coastal Harris County	Period of extremely hot weather, within a long period of at least Heat Advisory conditions. Periods of conditions reaching Excessive Heat Warning levels are recorded.
July 12, 2023	N/A	Inland Harris County	Period of extremely hot weather, within a long period of at least Heat Advisory conditions. Periods of conditions reaching Excessive Heat Warning levels are recorded.
July 14, 2023	N/A	Inland Harris County	A heat event caused an 89-year-old man from Houston to die due to hyperthermia.
August 5, 2023	N/A	Coastal and Inland Harris County	The extremely hot and humid summer continued with excessive heat warnings for most days between August 5 and August 26, and heat advisories for many others.
August 8, 2023	N/A	Inland Harris County	An infant died due to heat-related illness at approximately 3:00 p.m. after being left inside a hot car in southeast Houston. At the time of the incident, the temperature at KHOU was 100°F, and the heat index was 108.
August 23, 2023	N/A	Coastal Harris County	The extremely hot and humid summer continued with excessive heat warnings for most days between August 5 and August 26, and heat advisories for many others. This episode includes the first of two stretches of warning-level conditions, mainly between August 23 and August 26.
August 23, 2023	N/A	Inland Harris County	The extremely hot and humid summer continued with excessive heat warnings for most days between 8/5 and 8/26, and heat advisories for many others. This episode includes the first of two stretches of warning-level conditions, mainly between August 23 and August 26.
September 5, 2023	N/A	Inland and Coastal Harris County	A heat wave gripped much of Southeast Texas in early September, with at least advisory-level conditions experienced for several days between September 5 and September 9.

Source: (NOAA, NOAA Storm Events Database 2025)

Table 11-4. USDA Declarations for Extreme Heat Events in Harris County (2020 to 2024)

Event Date	USDA Declaration Number	Description
September 9, 2022	S5350	Excessive Heat

Source: (USDA 2025)

11.1.5 Probability of Future Occurrences

Probability Based on Previous Occurrences

Information on previous extreme heat occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 11-5. Based on historical records and input from the Steering Committee, the probability of occurrence for extreme heat in the County is considered "frequent."

Table 11-5. Probability of Future Extreme Heat Events in Harris County

Hazard Type	Number of Occurrences Between 1998 and 2024	Percent Chance of Occurring in Any Given Year
Heat, Excessive Heat	83	100%

Source: (NOAA, NOAA Storm Events Database 2025)

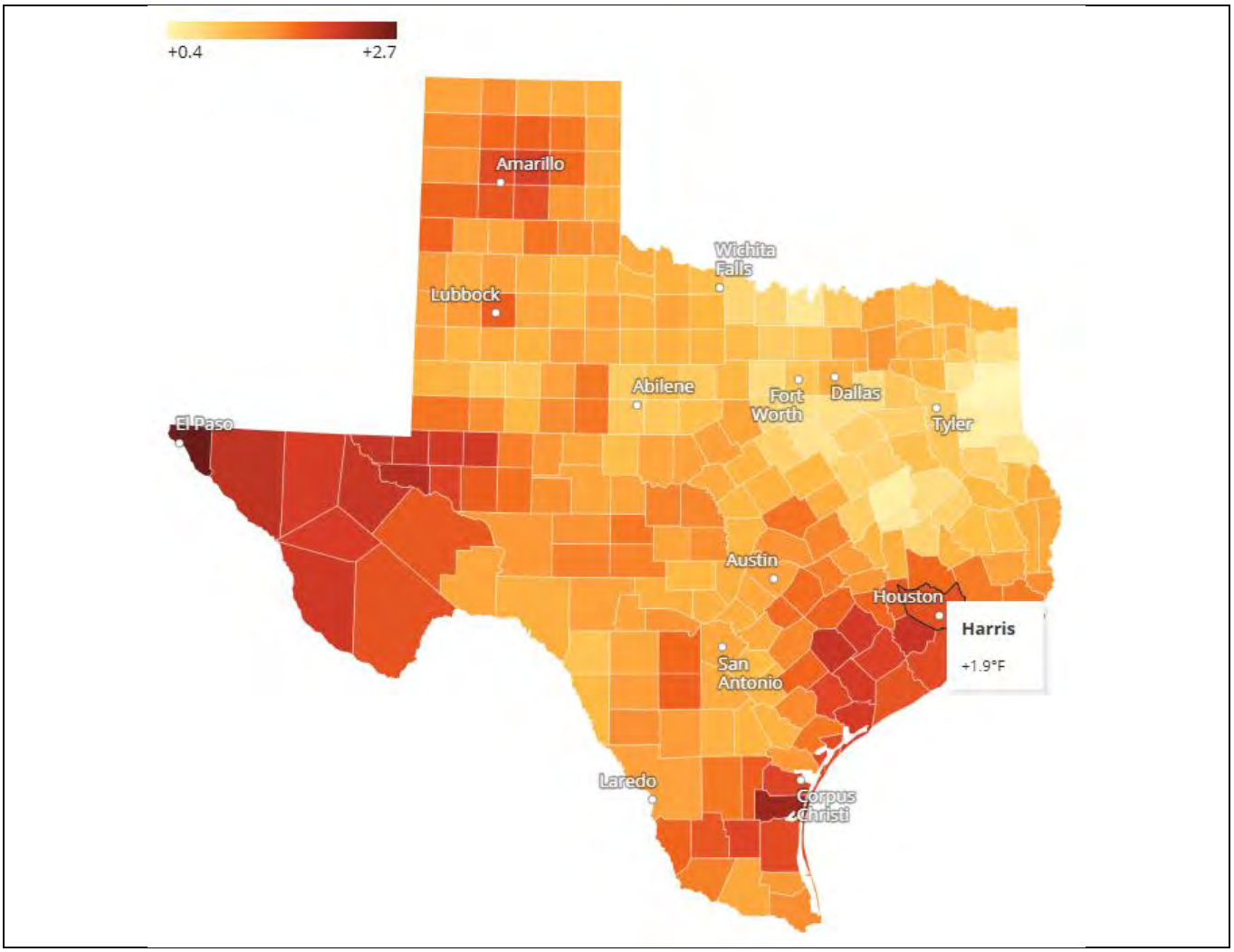
Effect of Climate Change on Future Probability

Climate change can affect the intensity and frequency of extreme heat events as warmer air temperatures and warmer sea surface temperatures mean higher humidity for areas like Harris County. Above-average temperatures, combined with increasing humidity, will mean communities across the planning area can expect to encounter more intense heat waves over the next several decades. Extreme heat under a changing climate regime will also not just occur during daytime hours. Since 2020, there have been increasing numbers of overnight record warm minimum temperatures, and a continued trend is expected moving forward. As overnight temperatures fail to fall to "average" levels, heat waves are perpetuated into the following day.

Climate change is beginning to affect both people and resources in Texas, and these impacts are projected to continue growing. Impacts related to increasing temperatures and heavier precipitation are already being felt in the state. Texans have come face-to-face with the realities of a warming climate, sweating through record-breaking heat waves and lengthy droughts that have taken a toll on agricultural and water resources in many parts of the state (Henkhaus 2024).

While average temperatures have risen across the state, increases have not been felt uniformly. Figure 11-3 shows the differences in average temperatures for each county in the state between two periods: 1895–2012 and 2013–2022.

Figure 11-3. Texas Average Temperatures Between 1895–2012 and 2013–2022



Source: (Schumacher, Ford and Douglas 2023)

11.1.6 Cascading Impacts on Other Hazards

A secondary effect of extreme heat is poor air quality, which can occur during summer when stagnant atmospheric conditions trap humid air and pollutants near the ground, closer to residents. In Harris County, this is especially true when wildfire smoke coincides with periods of prolonged extreme heat.

In addition, extreme heat can cause power outages primarily due to increased electricity demand for cooling, which can strain or overload the electrical grid. Extreme heat can reduce the efficiency of power generation sources like solar and natural gas turbines, further impacting the system (Wendel 2024).

11.2 VULNERABILITY AND IMPACT ASSESSMENT

To understand risk, a community must evaluate what assets are exposed or vulnerable in the identified hazard area. For extreme heat events, the entire County has been identified as the hazard area. Therefore, all assets in the County (population, structures, critical facilities, and lifelines), as described in the County Profile (Section 3), are at risk. The following text evaluates and estimates the potential impact of extreme heat on Harris County, including:

- Overview of vulnerability
- Impact on (1) life, health, and safety of residents; (2) general building stock; (3) critical facilities; (4) economy; (5) the environment; and (6) future growth and development
- Effect of expected future conditions on vulnerability
- Further data collections that will assist understanding of this hazard over time

11.2.1 Life, Health, and Safety

Overall Population

The entire population of Harris County is exposed to extreme heat events. Potential health impacts include injury and death. Health hazards related to extreme high temperatures include heat exhaustion and heat stroke. Adhering to extreme temperature warnings and conducting appropriate mitigation and preparation measures can significantly reduce the risk of temperature-related deaths.

Workers who are exposed to extreme heat, such as outdoor and agricultural workers, may be at risk of heat stroke, heat exhaustion, heat cramps, or heat rashes. Heat can also increase the risk of injuries in workers as it may result in sweaty palms, fogged-up safety glasses, and dizziness. Burns may occur as a result of accidental contact with hot surfaces. Working outdoors increases the chances of getting sunburned (CDC 2020, CDC 2018).

The following health hazards are related to extreme hot temperatures (FEMA 2001):

- Heat exhaustion is the body's response to an excessive loss of water and salt, usually through excessive sweating. Symptoms can include headache, cramping, dizziness, and weakness.
- Heat stroke is the most serious heat-related illness. It occurs when the body can no longer control its temperature: the body's temperature rises rapidly, the sweating mechanism fails, and the body is unable to cool down. When heat stroke occurs, the body temperature can rise to 106°F or higher within 10 to 15 minutes. Heat stroke can cause permanent disability or death if the person does not receive emergency treatment.

Socially Vulnerable Population

Socially vulnerable populations most at risk to extreme heat events include the elderly, who are less able to withstand temperatures extremes due to their age, health conditions, and limited

mobility to access shelters; infants and children up to age 4; individuals with chronic medical conditions (e.g., heart disease, high blood pressure); and low-income persons who cannot afford proper heating and cooling (CDC 2022, CDC 2005).

Extreme heat can lead to heat-related illnesses such as heatstroke and dehydration, with vulnerable groups often having limited access to air conditioning and cooling centers, increasing their risk (NIHHIS n.d.). Many socially vulnerable individuals work in outdoor or non-climate-controlled environments, such as construction and agriculture, making them more susceptible to heat-related health issues and reduced productivity.

Meteorologists can accurately forecast extreme heat events and the severity of the associated conditions with several days of lead time. These forecasts provide an opportunity for public health and other officials to notify vulnerable populations, implement short-term emergency response actions, and focus on surveillance and relief efforts on those at greatest risk.

11.2.2 General Building Stock

All the building stock in the County is exposed to the extreme heat hazard. This hazard generally does not impact buildings; however, elevated summer temperatures increase the energy demand for cooling. Losses can be associated with the overheating of heating, ventilation, and air conditioning (HVAC) systems.

11.2.3 Community Lifelines and Other Critical Facilities

Direct impacts on structures are expected to be minimal. However, it is essential that critical facilities remain operational during natural hazard events. Extreme heat events can sometimes cause short periods of utility failures, commonly referred to as brownouts, due to increased usage from air conditioners, appliances, and similar equipment. To mitigate this risk, backup power is recommended for critical facilities and infrastructure. Additionally, designating and developing emergency cooling facilities can enhance the resilience and safety of communities.

The health and medical sector is especially vulnerable during these events. Brownouts can disrupt the provision of medical services by affecting critical systems such as refrigeration for medications, air conditioning, and life-sustaining equipment. Furthermore, extreme heat often leads to a surge in patients suffering from heat-related illnesses, which can overwhelm healthcare facilities. This combination of increased demand and potential power limitations can significantly strain the healthcare system′s ability to deliver timely and effective care, highlighting the importance of resilience planning for medical infrastructure.

11.2.4 Economy

Extreme temperatures impact crop growth and development, which affects the overall price of goods that are produced using those crops. Farmers who rely on these crops as a primary source of income are also greatly affected due to reduced or unsellable crops.

Extreme heat events can have impacts on the economy such as loss of business function and damage and loss of inventory. Business owners may be faced with increased financial burdens due to higher than normal utility bills.

11.2.5 Natural, Historic, and Cultural Resources

Natural

Extreme heat significantly impacts natural resources, particularly water, agriculture, forests, and wildlife. Elevated temperatures increase evaporation rates, reducing water availability in lakes, rivers, and reservoirs, which exacerbates drought conditions and strains water supplies for various uses. In agriculture, high temperatures stress crops, leading to reduced yields and heightened irrigation demands, further taxing water resources. Forests face increased wildfire risks due to extreme heat, which can devastate ecosystems and wildlife habitats while also making trees more susceptible to diseases and pests. Wildlife struggles to find adequate water and food during heat waves, disrupting breeding cycles and migration patterns, thereby impacting biodiversity.

Historic

Proper strategies help safeguard buildings and their contents. Sudden and dramatic fluctuations in heating should be minimized. Slower heating gives building materials and stored contents time to acclimate to new temperatures in the building and corresponding new humidity levels (CCAHA 2019).

Extreme heat can increase the risk of ignition of fires and their propagation. Fire causes material loss and deformation of cultural heritage assets and may also increase the probability of cracking or splitting in built structures. Under extreme heat, stones can face both macro (e.g., cracking of stones, soot accumulation, color change in stone containing iron) and micro degradation (e.g., mineralogical and textural changes), leading to potential structural instability. The long-term impacts include weakened stones and increased susceptibility to deterioration processes such as salt weathering and temperature cycling (Sesana, et al. 2021).

Cultural

Cultural heritage sites, particularly those exposed to the elements, are subject to weathering. Temperature is a potential threat to these sites as it exacerbates the expected rates of decay and contributes to the appearance of new decay. Extreme heat may aggravate the physical, chemical, and biological mechanisms, causing degradation by affecting the structure or composition of building materials (Sesana, et al. 2021).

11.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure the establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

11.3.1 Potential or Planned Development

New development will change the landscape where buildings, roads, and other infrastructure replace open land and vegetation. Transformation of pervious surfaces (including vegetation) to impervious surfaces causes an island of higher temperatures. Green space preservation will need to continue to be a priority to mitigate increased heat islands. The ability of new development to withstand extreme temperature impacts can be enhanced through land use practices and consistent enforcement of codes and regulations for new construction.

Understanding future changes that impact vulnerability in the County can assist in planning for future development and ensuring that appropriate mitigation, planning, and preparedness measures are in place. Areas targeted for potential future growth and development could be potentially impacted by extreme heat because the entire County is exposed. However, due to increased standards and codes, new development can be less vulnerable to extreme heat in comparison with the aging building stock currently in Harris County.

11.3.2 Projected Changes in Population

Harris County experienced an increase in population between the 2010 Census (4,110,771) and the 2020 Census (4,780,913). The population of the County is expected to continue to increase over the next few years. The increase in population will expose more people to extreme heat (U.S. Census Bureau 2022).

11.3.3 Climate Change

Climate change is expected to increase the intensity and frequency of extreme heat events due to warmer air and sea surface temperatures, leading to higher humidity. Communities can anticipate more intense heat waves over the coming decades, not only during the day but also at night. There has been a rise in overnight record warm minimum temperatures, a trend likely to continue, perpetuating heat waves into the following day.

11.3.4 Other Identified Conditions

Extreme heat can place significant stress on infrastructure, including roads, bridges, and buildings. High temperatures cause materials to expand and contract, leading to cracks and other forms of damage. Additionally, there is often an increased demand for energy during extreme heat events, particularly for cooling. This surge in energy use can lead to power outages and increased stress on energy grids. Transportation systems are also affected by extreme heat, with rail tracks potentially buckling and asphalt softening.

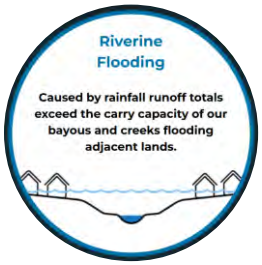
12. FLOODING

12.1 HAZARD PROFILE

12.1.1 Hazard Description

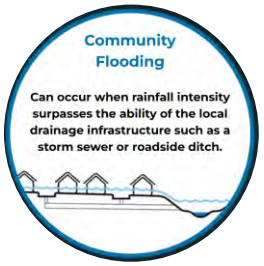
Flooding occurs when water overflows onto land that is normally dry. Flooding can happen during heavy rains, rapid snow melt, or when dams or levees break. Floods are one of the most frequent and costly natural disasters in the United States and in the State of Texas. Flood-related hazards most likely to impact Harris County are riverine (inland) flooding, flash flooding, and stormwater/urban flooding. Another source of flooding in Harris County may come from dam/levee failures, and this is further discussed in the Dam/Levee Failure hazard profile.

Floods are generally the result of excessive precipitation. In Harris County, flooding generally takes one of the following forms:



- Riverine Flooding—Riverine flooding occurs when rivers overflow their banks in response to excessive precipitation levels and water runoff volumes within the watershed. Riverine floodplains may be broad, as when a river crosses an extensive flat landscape, or narrow, as when a river is confined in a canyon.

- Coastal Flooding—Coastal flooding is primarily caused by storm surge, a cascading effect of hurricanes and coastal storms that pushes water toward the shore. The result can be waves that extend further inland, causing damage to development that would not normally be subject to wave action. Storm surge heights, and associated waves, are dependent upon the local width of the continental shelf and the depth of the ocean bottom. A narrow shelf, or one that drops steeply from the shoreline and subsequently produces deep water close to the shoreline, tends to produce a lower surge but higher and more powerful storm waves. Due to the high risk and vulnerability to this flood specific hazard, it was analyzed independently in this chapter rather than as a cascading effect of hurricanes.



- Urban Flooding—Urban flooding occurs when development has obstructed the natural flow of water and decreased the ability of natural groundcover to absorb and retain surface water runoff.
- Flash Flooding—Most flash flooding is caused by slow-moving thunderstorms in a local area or by heavy rains associated with hurricanes and tropical storms. However, flash flooding events can also occur from accelerated snow melt due to heavy rains, or a dam or levee failure within minutes or hours of heavy amounts of rainfall. Flash flooding is common in urbanized areas where much of the ground is covered by impervious surfaces. Flash flood waters move at very high speeds, uprooting trees, destroying buildings, and obliterating bridges and roads.

Measuring Floods and Floodplains

The frequency and severity of flooding are measured using a discharge probability for river systems and storm surge levels for coastal systems. The discharge probability is the probability that a certain river discharge (flow) level will be equaled or exceeded in a given year. Storm surge levels are determined by modeling water depth, wind speed, vegetative cover and other factors to determine the "wave runup" (how far inland waves will reach) and "wave setup" (the height, speed, and slope of waves relative to the still-water elevation).

Flood studies use historical records to determine the probability of occurrence for different discharge levels and storm surge levels. These measurements reflect statistical averages only; it is possible for multiple floods with a low probability of occurrence (such as a 1-percent-annual-chance flood) to occur in a short time period. For riverine flooding, the same flood event can have flows at different points on a river that correspond to different probabilities of occurrence.

The extent of coastal flooding (storm surge) is determined by a variety of factors such as the range of water depth, the width and slope of the gulf/bay, the path of a hurricane, wind speeds, shape of the coastline, and the forward speed. (NOAA). The forward motion of a storm complements the counterclockwise rotation of the wind field, which usually results in the highest coastal flooding and surges from hurricanes occurring on the right-front (northeast) quadrant of the storm's track. A combination of tools/methods can be used to measure coastal flooding/storm surge, such as tidal stations, high water marks, pressure sensors and the NOAA SLOSH model.

The extent of flooding associated with a 1-percent annual probability of occurrence (also called the base flood) is used as the regulatory boundary by many agencies. Also referred to as the special flood hazard area (SFHA), this boundary is a convenient tool for assessing vulnerability and risk in flood-prone communities. Many communities have maps that show the extent and likely depth of flooding for the base flood. Corresponding water-surface elevations describe the elevation of water that will result from a given discharge level, which is one of the most important factors used in estimating flood damage.

FEMA Regulatory Flood Zones

FEMA defines flood hazard areas as areas expected to be inundated by a flood of a given magnitude. These areas are determined via statistical analyses of records of river flow, storm tides, and rainfall; information obtained through consultation with the community; floodplain topographic surveys; and hydrologic and hydraulic analyses. Flood hazard areas are delineated on Flood Insurance Rate Maps (FIRMs) or digital Flood Insurance Rate Maps (DFIRMs), which provide the following information:

- Locations of specific properties in relation to the SFHA
- Base flood elevations (1-percent-annual-chance) at specific sites
- Magnitudes of flood in specific areas
- Undeveloped coastal barriers where flood insurance is not available
- Regulatory floodways and floodplain boundaries (1-percent and 0.2-percent-annual-chance floodplain boundaries).

Land area covered by floodwaters of the base flood is the SFHA on a DFIRM—an area where NFIP floodplain management regulations must be enforced, and where mandatory purchase of flood insurance applies. This regulatory boundary is a convenient tool for assessing vulnerability and risk in flood-prone communities because many communities have maps showing the extent of the base flood and likely depths that will occur.

The base flood elevation (the water elevation of a flood that has a 1-percent chance of occurring in any given year) is one of the most important factors in estimating potential damage from flooding. A structure within a 1-percent-annual-chance floodplain has a 26-percent chance of undergoing flood damage during the term of a 30-year mortgage. The 1-percent-annual-chance flood is used by the NFIP as the basis for insurance requirements nationwide. DFIRMs also depict 0.2-percent-annual-chance flood designations.

FIRMs and DFIRMs are the principle tool used to identify the extent and location of the flood hazard. FEMA and the floodplain management community acknowledge that these maps are not a total depiction of an area's flood risk. They represent the best data source available, but the level of risk they indicate may be understated or overstated compared to current conditions. The following limitations to the accuracy of these maps need to be recognized:

- FIRMs are based on hydrologic conditions at the time they are prepared. FIRMs are not set up to account for changes in hydrology that can occur over time. FIRMs can be up to 25 years old and therefore fail to reflect the conditions of the watershed as they exist today.
- FIRMs do not account for the flood protection benefits of levees unless the levees are certified as providing 100-year flood protection (according to criteria specified in Section 65.10 of 44 CFR). The national levee policy is in a state of flux. Some levees are recognized as 100year levees on FIRMs, others are not. The age of the older maps can draw into question the level of protection provided by levees today. The potential for levees to be certifiable in their current condition requires costly, detailed risk-based analyses.

Floodplain Ecosystems and Beneficial Functions

Connections between a river and its floodplain form a complex physical and biological system that not only supports a variety of natural resources but also provides natural flood and erosion control. When floodwaters recede after a river flood event, they leave behind layers of rock and mud. These gradually build up to create a new floor of the floodplain. Floodplains generally contain unconsolidated sediments (accumulations of sand, gravel, loam, silt, and/or clay), often extending below the bed of the stream. These sediments provide a natural filtering system, with water percolating back into the ground and replenishing groundwater. These are often important aquifers, the water drawn from them being filtered compared to the water in the stream. Fertile, flat reclaimed floodplain lands are commonly used for agriculture, commerce and residential development. When a river is separated from its floodplain with levees and other flood control facilities, natural, built-in benefits can be lost, altered, or significantly reduced.

A floodplain can contain 100 or even 1,000 times as many species as a river. Wetting of the floodplain soil releases an immediate surge of nutrients: those left over from the last flood, and those that result from the rapid decomposition of organic matter that has accumulated since then. Microscopic organisms thrive, and larger species enter a rapid breeding cycle. Opportunistic feeders

(particularly birds) move in to take advantage. The production of nutrients peaks and falls away quickly, but the surge of new growth endures for some time. This makes floodplains valuable for agriculture. Species growing in floodplains are markedly different from those that grow outside floodplains. For instance, riparian trees (trees that grow in floodplains) tend to be very tolerant of root disturbance and very quick-growing compared to non-riparian trees.

Floodplains have many natural and beneficial functions, and disruption of natural systems can have long-term consequences for entire regions. Some well-known, water-related functions of floodplains (noted by FEMA) include:

- Natural flood and erosion control
- Provide flood storage and conveyance
- Reduce flood velocities
- Reduce flood peaks
- Reduce sedimentation
- Surface water quality maintenance
- Filter nutrients and impurities from runoff
- Process organic wastes
- Moderate temperatures of water
- Groundwater recharge
- Promote infiltration and aquifer recharge
- Reduce frequency and duration of low surface flows.

Areas in the floodplain that typically provide these natural functions are wetlands, riparian areas, sensitive areas, and habitats for rare and endangered species.

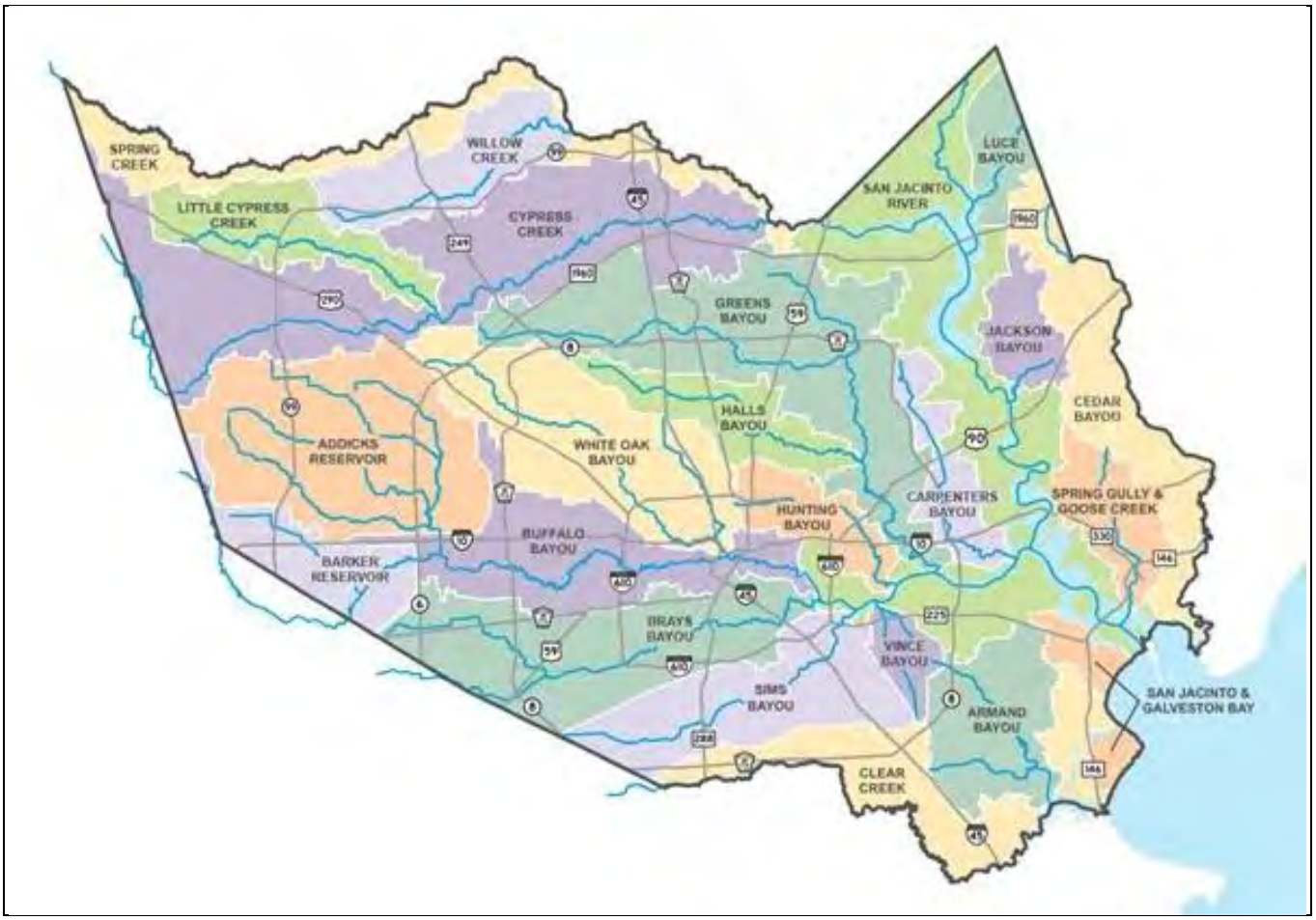
Effects of Human Activities

Because they border water bodies, floodplains have historically been popular sites to establish settlements. Human activities tend to concentrate in floodplains for a number of reasons: water is readily available; riverine floodplain land is fertile and suitable for farming; transportation by water is easily accessible; land is flatter and easier to develop; and there is value placed in ocean views. But human activity in floodplains frequently interferes with the natural function of floodplains. It can affect the distribution and timing of drainage, thereby increasing flood problems. Human development can create local flooding problems by altering or confining drainage channels or causing erosion of natural flood protection systems such as dunes. Flood potential can be increased in several ways: reducing a stream's capacity to contain flows; increasing flow rates or velocities downstream; and allowing waves to extend further inland. Human activities can interface effectively with a floodplain as long as steps are taken to mitigate the activities' adverse impacts on floodplain functions.

12.1.2 Harris County Watersheds and Waterways

Harris County consists of portions of two primary watershed systems—the San Jacinto River and Buffalo Bayou—along with a number of smaller watershed systems. Within the broader region, most watersheds in Harris County flow into one of nearly two dozen major creeks and bayous. For instance, if your home is in the Brays Bayou watershed, any rainfall there will eventually make its way into Brays Bayou (HCFCD n.d.). The planning area is drained by 23 major waterways, as shown in Figure 12-1 and described in the following sections.

Figure 12-1. Harris County Major Watersheds



Source: (HCFCD 2025)

Waterways in the San Jacinto River System

San Jacinto River

The San Jacinto River drains a 3,600 square-mile watershed (including 487 square miles in Harris County) that originates well outside and upstream of Harris County. It drains all or part of Harris, Montgomery, Waller, Walker, Grimes, Liberty, and San Jacinto Counties. The river is formed by the junction of the West and East Forks, which each enter northern Harris County. Lake Houston is a water supply reservoir in northeast Harris County along the San Jacinto River, which includes the confluence of the East and West Forks. The San Jacinto River extends southward through the eastern portion of Harris County from the Lake Houston Dam to its confluence with the Houston Ship Channel continuing on to its mouth at Galveston Bay. The Port of Houston Authority operates the Houston Ship Channel, which originates at the Turning Basin, follows the original alignment of Buffalo Bayou to the San Jacinto River, and continues through the San Jacinto River and San Jacinto Bay to Galveston Bay. Although it is part of the original alignment of Buffalo Bayou, for organizational purposes the ship channel below the Turning Basin is considered a part of the San Jacinto Watershed. Notable tributaries include the East and West Forks. Spring Creek, Cypress

Creek, Little Cypress Creek, Willow Creek, Luce Bayou, Goose Creek, and Jackson Bayou also part of the San Jacinto River system but are described separately. Most of the watershed is rural and undeveloped, although some more moderate levels of development have occurred within Harris County. Communities in the watershed include the Cities of Houston, Baytown, Deer Park, Galena Park, Humble, La Porte, Morgan's Point, Pasadena, and unincorporated Harris County.

Spring Creek

Spring Creek forms the northern boundary of Harris County, bordering Montgomery and Waller Counties. The watershed also includes a portion of Grimes County. The watershed drains 761 square miles in an easterly direction to its confluence with the West Fork of the San Jacinto River upstream of Lake Houston. Notable tributaries and sub-tributaries to Spring Creek from Harris County include Cypress Creek, Little Cypress Creek, and Willow Creek, all of which are described separately in this report. Consequently, this description of the Spring Creek watershed only considers the 59.5 square miles of the watershed that are not in these other watersheds. Notable tributaries from Montgomery County include Lake Creek and Panther Branch. Spring Creek is 68 miles in length. The watershed is lightly developed, with some residential development. Communities in the watershed include the Cities of Houston, Humble, Tomball, and unincorporated Harris County.

Cypress Creek

Cypress Creek is a tributary of Spring Creek and is a part of the San Jacinto River Watershed system. It drains eastward over a length of 50 miles from its origin at the junction of Mound Creek and Snake Creek in Waller County to its confluence with Spring Creek near the West Fork of the San Jacinto River. The Cypress Creek Watershed is 320 square miles, but excluding the Little Cypress Creek Watershed (which is described separately), the watershed is 268 square miles in size. The middle and lower portions of the watershed have a moderate level of residential development, while the upstream portion of the watershed is predominantly rural and agricultural. Notable tributaries include Mound Creek, Snake Creek, Rock Hollow, Dry Creek, Little Cypress Creek, Faulkey Gully, Theiss Gully, Seals Gully, and Turkey Creek. Communities within the watershed include the Cities of Houston and Waller and unincorporated Harris County.

Little Cypress Creek

Little Cypress Creek is a tributary of Cypress Creek. It drains southeastward for a length of 22 miles from its headwaters in far northwest Harris County to its confluence with Cypress Creek. The Little Cypress Creek Watershed consists of 52 square miles and has a light amount of predominantly residential development. The remainder of the watershed is either open land or agricultural. Communities within the watershed include unincorporated Harris County.

Willow Creek

Willow Creek is a tributary of Spring Creek. It drains northeastward for a length of 20 miles from its headwaters near Tomball to its confluence with Spring Creek. The watershed drains an area of 56 square miles. The watershed has a light amount of residential development. The remainder of

the watershed is either open or agricultural. Communities within the watershed include the City of Tomball and unincorporated Harris County.

Luce Bayou

Luce Bayou is a tributary of the East Fork of the San Jacinto River. It drains southward for 35 miles from its headwaters in the Sam Houston National Forest in San Jacinto County to its confluence with the East Fork of the San Jacinto River in the upper portion of Lake Houston. The watershed covers 227 square miles and includes portions of San Jacinto, Liberty, and Harris Counties. The Harris County portion includes only the lower 17 square miles of the watershed. There is minimal development in the watershed; most of the land is forest. Communities within the watershed include the City of Houston and unincorporated Harris County.

Jackson Bayou

Jackson Bayou is a tributary of the San Jacinto River. It drains southward over a length of 7 miles from its headwaters in east Harris County to its confluence with the San Jacinto River. The Jackson Bayou watershed is 50 square miles in size. It is lightly developed, with some rural subdivisions in the lower portion of the watershed. Notable tributaries include Gum Gully. Communities within the watershed include the City of Houston and unincorporated Harris County.

Goose Creek

Goose Creek is a tributary of the San Jacinto River. It drains southward over a length of 15 miles from its headwaters in east Harris County to its confluence with the San Jacinto River just downstream of its confluence with the Houston Ship Channel. The Goose Creek Watershed is 33 square miles in size. It is moderately developed, but the lower half of the watershed has mostly dense residential along with some concentrations of commercial and industrial development. For descriptive purposes, the Goose Creek watershed includes Spring Gully, which drains directly into the San Jacinto River and does not have a common confluence with Goose Creek. Communities within the watershed include the City of Baytown and unincorporated Harris County.

Waterways in the Buffalo Bayou System

Buffalo Bayou

The Buffalo Bayou Watershed is described as the area downstream of Addicks and Barker Reservoirs that drains to Buffalo Bayou and is not part of another designated watershed tributary to Buffalo Bayou. This area totals 102 square miles and drains into Buffalo Bayou as it extends eastward for 50 miles from Barker Reservoir to the Houston Ship Channel Turning Basin just east of downtown Houston. The Buffalo Bayou Watershed is highly urbanized, with a mix of residential and commercial land uses. Features within the watershed include Memorial Park and downtown Houston. Communities within the watershed include the Cities of Houston, Bunker Hill Village, Hedwig Village, Hilshire Village, Hunter's Creek Village, Piney Point Village, Spring Valley Village, and unincorporated Harris County.

Sims Bayou

Sims Bayou is a tributary to Buffalo Bayou. It drains eastward over a length of 25 miles from its headwaters in Fort Bend County to its confluence with the Houston Ship Channel. The Sims Bayou watershed is 93 square miles in size. It is moderately developed, consisting mostly of a relatively high-density (for Harris County) residential, with most of the development concentrated in the downstream and upstream portions of the watershed. The largest tributary to Sims Bayou is Berry Bayou. Communities within the watershed include the Cities of Houston, Missouri City, South Houston, Stafford, and Pasadena, and unincorporated Harris County.

Brays Bayou

Brays Bayou is a tributary of Buffalo Bayou. It drains eastward over a length of 32 miles from its headwaters in Fort Bend County and southwest Harris County to its confluence with the Houston Ship Channel. The Brays Bayou Watershed is 128 square miles in size. It is highly developed, with land use ranging from residential to dense commercial. The watershed includes the Texas Medical Center, Herman Park, and the NRG Park. Notable tributaries include Keegans Bayou and Willow Waterhole Bayou. Communities include the Cities of Houston, Bellaire, Southside Place, Stafford, and West University Place, and unincorporated Harris County.

White Oak Bayou

White Oak Bayou is a tributary of Buffalo Bayou. It drains southeastward over a length of 30 miles from its headwaters in northwest Harris County to its confluence with Buffalo Bayou near downtown Houston. The White Oak Bayou Watershed is 111 square miles in size. It is highly developed, with primarily residential land use. Notable tributaries include Little White Oak Bayou, Brickhouse Gully, Cole Creek, and Vogel Creek. Communities in the watershed include the Cities of Houston and Jersey Village, along with unincorporated Harris County.

Hunting Bayou

Hunting Bayou is a tributary of Buffalo Bayou. It drains eastward and then southward over a length of 15 miles from its headwaters in northeast Houston to its confluence with the Houston Ship Channel near the Washburn Tunnel. The Hunting Bayou Watershed is 30 square miles in size. It is moderately to highly developed, with mostly dense residential development in the upper portion of the watershed and industrial and commercial development in the middle and lower portions of the watershed. Communities within the Hunting Bayou Watershed include the Cities of Houston, Galena Park, and Jacinto City, along with unincorporated Harris County.

Vince Bayou

Vince Bayou is a tributary of Buffalo Bayou. It drains northward for a length of 6 miles from its headwaters in Pasadena to its confluence with the Houston Ship Channel. The watershed is densely developed, with a mixture of residential, commercial, and industrial uses. Communities within the watershed include the Cities of Houston, South Houston, and Pasadena.

Carpenters Bayou

Carpenters Bayou is a tributary of Buffalo Bayou. It drains southward over a length of 13 miles from its headwaters in northeast Harris County to its confluence with the Houston Ship Channel. The Carpenters Bayou Watershed is 31 square miles in size. It has a low to moderate amount of development, consisting mostly of small lot residential and commercial. Sheldon Reservoir is located in the upper basin. This shallow lake and adjoining lands are owned by the State of Texas and were formally used by the Texas Parks and Wildlife Department for fish research and hatchery. Communities within the watershed include the City of Houston and unincorporated Harris County.

Greens Bayou

Greens Bayou is a tributary of Buffalo Bayou. It drains eastward and then southward for a distance of 42 miles from its headwaters in northwest Harris County to its confluence with the Houston Ship Channel. The Greens Bayou Watershed is 211 square miles in size. It is moderately developed; most of the land use is residential and light commercial. The watershed includes George Bush Intercontinental Airport. Notable tributaries include Halls Bayou and Garners Bayou. Communities within the watershed include the Cities of Houston and Humble and unincorporated Harris County.

Barker Reservoir

Barker Reservoir was constructed with Addicks Reservoir to protect downtown Houston and the Houston Ship Channel by impounding flood flows in the upper portion of Buffalo Bayou. The Barker Reservoir Watershed includes all those areas that contribute drainage into the reservoir. This watershed encompasses 129 square miles much of which is within Fort Bend County. A moderate amount of development has occurred in the watershed, consisting predominately of residential development. The primary streams that feed the reservoir are Upper Buffalo Bayou, Cane Island Branch, and Mason Creek. Communities within the watershed include the Cities of Houston and Katy, and unincorporated Harris County.

Addicks Reservoir

Addicks Reservoir was constructed with Barker Reservoir to protect downtown Houston and the Houston Ship Channel by impounding flood flows in the upper tributaries of Buffalo Bayou. The Addicks Reservoir Watershed includes all those areas that contribute drainage into the reservoir. This watershed encompasses 136 square miles of area, all of which is in northwest Harris County. A moderate amount of predominately residential development has occurred in the watershed. The primary streams that feed the reservoir are Langham Creek, South Mayde Creek, Bear Creek, Horsepen Creek, and Dinner Creek. Communities within the watershed include the Cities of Houston and Katy and unincorporated Harris County.

Waterways and Tributaries of Galveston Bay

Clear Creek

Clear Creek forms the southern boundary of Harris County, bordering Galveston and Brazoria Counties and then extending upstream to its headwaters in Fort Bend County. The watershed

drains 198 square miles in an easterly direction into Clear Lake, a natural estuary lake, and then into Galveston Bay. A light to moderate level of development has occurred in the watershed, with most of it concentrated in the downstream and middle reaches. Although the development is mostly residential and commercial, National Aeronautics and Space Administration's Johnson Space Center and associated industries make up a considerable portion of the development in the areas near Clear Lake in the downstream portion of the watershed. Significant tributaries to Clear Creek include Mary's Creek, Cowart Creek, and Hickory Slough, all of which drain from adjacent counties to the south; and Armand Bayou, which is described separately. Communities in the watershed include the Cities of Houston, El Lago, Friendswood, La Porte, League City, Missouri City, Nassau Bay, Pasadena, Pearland, Seabrook, Shoreacres, Taylor Lake Village, Webster, and unincorporated Harris County.

Armand Bayou

Armand Bayou is a tributary of Clear Creek but is treated as a separate watershed. It drains an area of 59 square miles southward into Clear Lake near Galveston Bay. The watershed is moderately developed, with a mix of residential and dense industrial. This development is evenly distributed across the watershed. Communities in the watershed include the Cities of Houston, Deer Park, La Porte, Nassau Bay, Pasadena, Taylor Lake Village, Webster, and unincorporated Harris County.

Cedar Bayou

Cedar Bayou forms the eastern boundary of Harris County, bordering Chambers County. It drains southward for a distance of 51 miles from its headwaters in Liberty County to its confluence with Galveston Bay. The Cedar Bayou Watershed is 199 square miles in size and is lightly developed. Notable tributaries include Pine Gully. Communities within the watershed include the City of Baytown and unincorporated Harris County.

Galveston Bay Tributaries

The Galveston Bay Tributaries are small tributaries along Galveston Bay between Clear Lake and the Houston Ship Channel that drain directly into Galveston Bay and are not included in other watersheds. This area is moderately developed and includes both residential areas and dense commercial and industrial. Notable tributaries include Little Cedar Bayou and Pine Gully. Communities included in this watershed include the Cities of La Porte, Morgan's Point, Pasadena, Seabrook, Shoreacres, and unincorporated Harris County.

12.1.3 Location

Mapping of flood hazard areas was developed for three flood scenarios:

- The 1 percent annual chance riverine flood inundation area (see Figure 12-2)
- The 1 percent annual chance coastal flood inundation area (see Figure 12-3)
- The area that was inundated by Hurricane Harvey, representing a worst-case scenario (see Figure 12-4)

Figure 12-2. Area Inundated by the 1 Percent Annual Chance Riverine Food

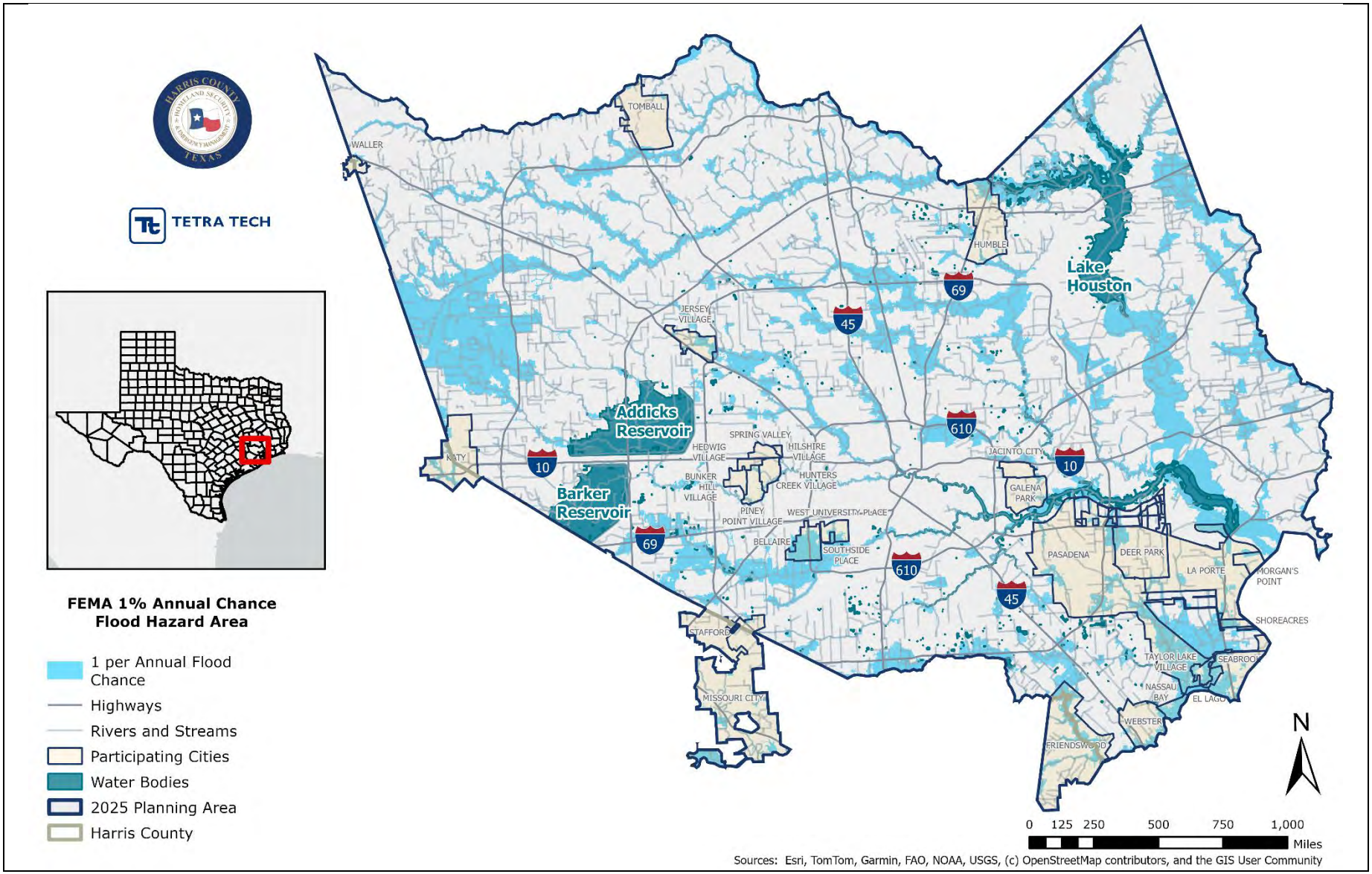


Figure 12-3. Area Inundated by the 1 Percent Annual Chance Coastal Food

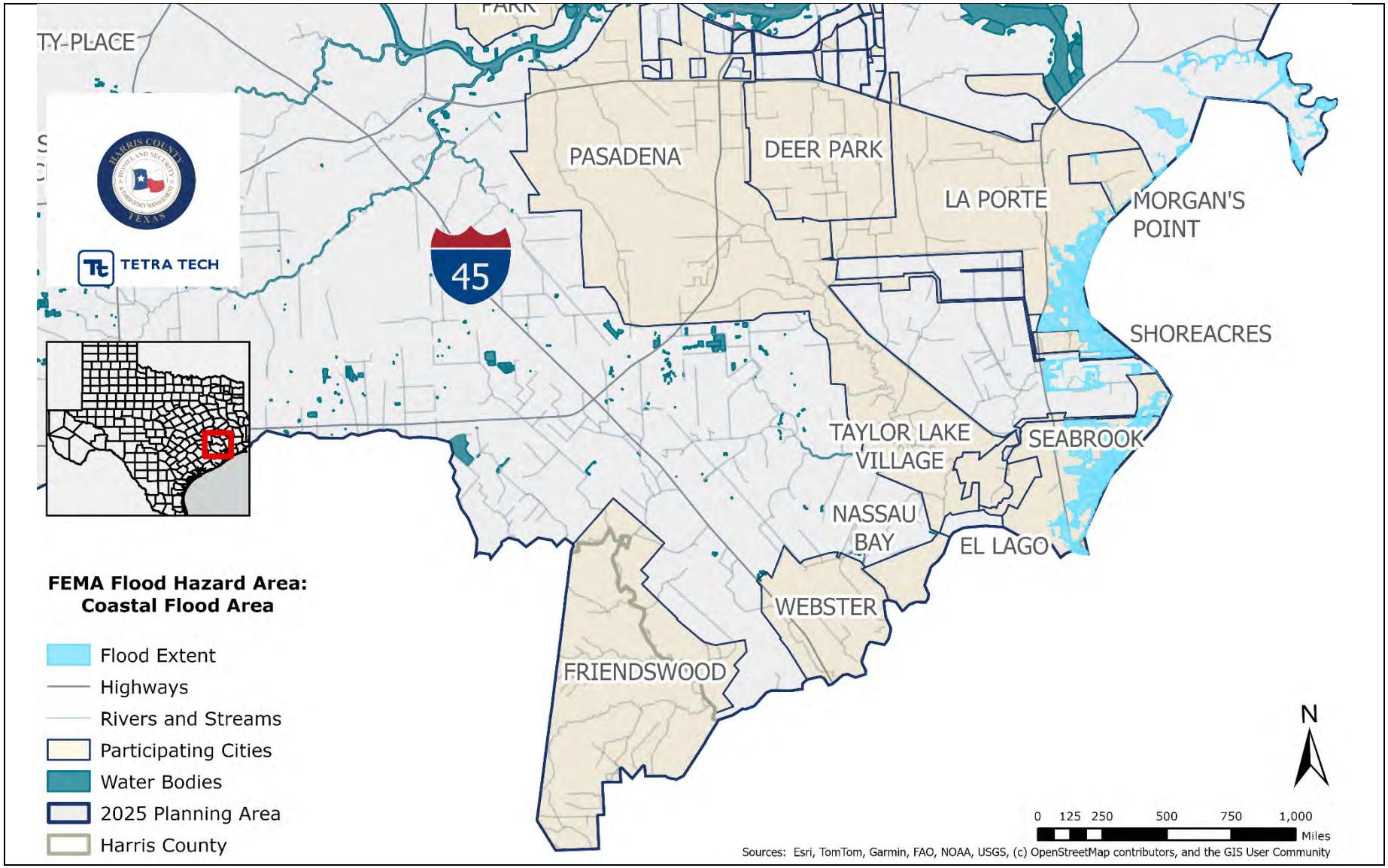
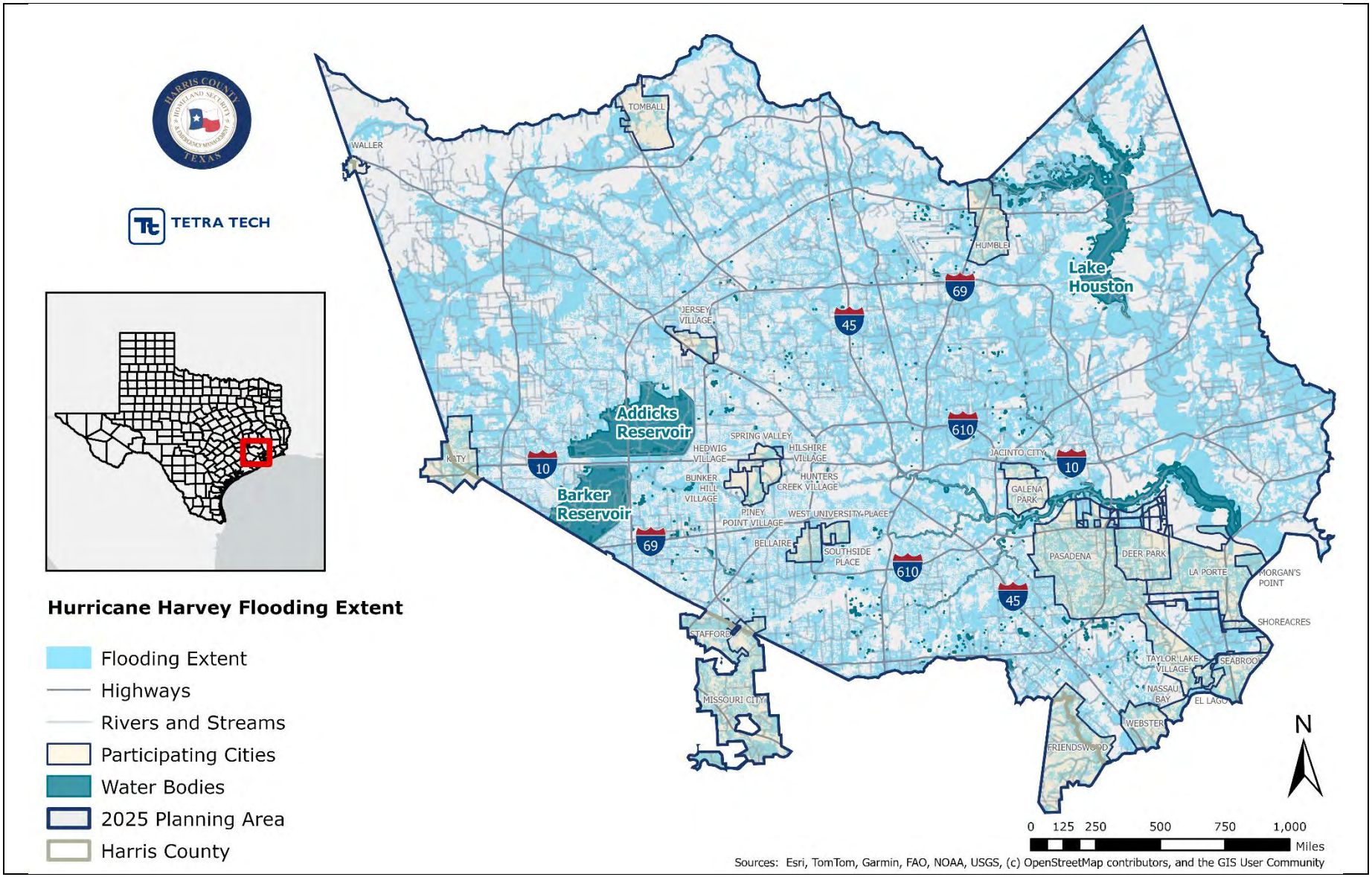


Figure 12-4. Area Inundated by Hurricane Harvey Flooding



While municipalities are identified in the preceding flood maps, this information was used to identify which school districts, colleges, universities, and all remaining non-municipal plan participants are identified in Appendix G:Risk Assessment Supplementary Data.

Table 12-1 shows the number of acres per each land use category exposed to the 1-percent annual chance (riverine and coastal) and those located within the Harvey Inundation Area. These figures are crucial for hazard mitigation planning, as they help identify which types of land use, such as residential, commercial, or critical infrastructure, are most vulnerable to flooding. This information enables the county to prioritize mitigation strategies, allocate resources effectively, and guide future development away from high-risk areas to reduce potential damage and enhance community resilience.

Table 12-1. Existing Land Use Within the Scenario Floodplains

	1-Percent Annual Chance, Riverine		1-Percent Annual Chance, Coastal		Harvey Inundation Area	
Land Use Category	Area (Acres)	% of Total	Area (Acres)	% of Total	Area (Acres)	% of Total
Commercial	8,691	16.4%	205	0.4%	25,731	48.4%
Government/Medical/Educational	2,765	15.9%	87	0.5%	8,810	50.7%
Industrial	15,333	17.5%	238	0.3%	42,292	48.3%
Multiple	21,050	17.0%	235	0.2%	58,834	47.6%
Other	726	13.5%	25	0.5%	2,629	48.7%
Parks/Open Spaces	55,801	59.9%	373	0.4%	66,981	72.0%
Residential	57,688	13.9%	1,489	0.4%	190,990	46.1%
Undevelopable	23,557	27.6%	470	0.6%	49,613	58.1%
Unknown	3,961	30.9%	13	0.1%	8,480	66.1%
Vacant	63,042	28.8%	799	0.4%	113,457	51.8%
Total	252,614	22.7%	3,934	0.4%	567,817	51.1%

12.1.4 Extent

The strength or magnitude of a flood varies based on meteorological, environmental, and geological factors, including latitude, altitude, topography, and atmospheric conditions. Flood is also affected by seasonal variation, storm characteristics, warning time, speed of onset, and duration. Most floods are preceded by a warning period that allows emergency managers to communicate the need to prepare for the event. A flood may last from minutes to days (O'Connor, Grant and Costa 2002).

Warnings issued through official sources, such as the National Weather Service (NWS) and the Storm Prediction Center, provide the most reliable and timely preparedness information, but the exact flood location and depth depends on the amount, duration, and location of rainfall. Many floods, especially flash floods, occur outside of FEMA-designated flood zones.

In the case of riverine flood hazard, once a river reaches flood stage, the flood extent or severity categories used by the NWS include minor flooding, moderate flooding, and major flooding. Each category has a definition based on property damage and public threat:

- Minor Flooding—minimal or no property damage, but possibly some public threat or inconvenience.
- Moderate Flooding—some inundation of structures and roads near streams. Some evacuations of people and/or transfer of property to higher elevations are necessary.
- Major Flooding—extensive inundation of structures and roads. Significant evacuations of people and/or transfer of property to higher elevations (NOAA 2021).

The severity of a flood depends not only on the amount of water that accumulates in a period of time but also on the land's ability to manage this water. The size of rivers and streams in an area and infiltration rates are significant factors. When it rains, soil acts as a sponge. When the land is saturated or frozen, infiltration rates decrease and any more water that accumulates must flow as runoff (Harris 2001).

The frequency and severity of flooding are measured using a discharge probability, which is the probability that a certain river discharge (flow) level will be equaled or exceeded in a given year. Flood studies use historical records to determine the probability of occurrence for the different discharge levels. The flood frequency equals 100 divided by the discharge probability. For example, the 100-year discharge has a 1 percent chance of being equaled or exceeded in any given year. The "annual flood" is the greatest flood event expected to occur in a typical year. These measurements reflect statistical averages only; it is possible for two or more floods with a 100-year or higher recurrence interval to occur in a short time period. The same flood can have different recurrence intervals at different points on a river.

The extent of flooding associated with a 1 percent annual probability of occurrence (the base flood or 100-year flood) is used by the National Flood Insurance Program (NFIP) as the standard for floodplain management and to determine the need for flood insurance, as well as the regulatory flood boundary by many agencies. Also referred to as the Special Flood Hazard Area (SFHA), this boundary is a convenient tool for assessing vulnerability and risk in flood-prone communities. Many communities have maps that show the extent and likely depth of flooding for the base flood. Corresponding water-surface elevations describe the water elevation resulting from a given discharge level, which is one of the most important factors used in estimating flood damage. A structure located within an SFHA shown on an NFIP map has a 26 percent chance of suffering flood damage during the term of a 30-year mortgage.

The term "500-year flood" is the flood that has a 0.2 percent chance of being equaled or exceeded each year. The 500-year flood could occur more than once in a relatively short period of time. Statistically, the 0.2 percent (500-year) flood has a 6 percent chance of occurring during a 30-year period of time, the length of many mortgages. The 500-year floodplain is referred to as Zone X500 for insurance purposes on Flood Insurance Rate Maps (FIRM). Base flood elevations or depths are not shown within this zone, and insurance purchase is not required in this zone (FEMA 2022).

Harris County and all participating jurisdictions consider a rainfall of 2 to 3 inches per hour or a creek rise that stays within the creek bank or drainage channel to be a minor flooding severity. In

addition, street flooding is considered minor flooding. A major severity for the Harris County planning area is identified as a rainfall of 4 inches per hour and greater, or more than 6 inches in 3 hours on saturated ground, or a creek rise that overflows the banks of the creek or drainage channel. Flash flooding can develop quickly in the region, closing roads/bridges, and endangering life and property. The entire planning area can expect minor, moderate and major flooding in the future (HCOHSEM 2020).

Flood extent is also visualized through Figure 12-5 and Figure 12-6 that depict flood water depths across the planning area during a 1% Annual chance flood event as well as during Hurricane Harvey magnitude events. These two graphics appeared in the 2020 Harris County HMAP, and since no changes have occurred to these levels, both figures apply in 2025.

12.1.5 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Since 1950, Harris County has been included in 24 major disasters (DR) and two emergency (EM) declarations for Flood-related events (FEMA 2025).Table 12-2 lists these declarations.

State Emergency Proclamations

The Governor of Texas has the authority to issue emergency proclamations under the Texas Disaster Act of 1975, enabling access to state resources and emergency powers. Once a state of disaster is declared, the Governor may direct the Texas Division of Emergency Management (TDEM) to coordinate response efforts, facilitate access to state and federal assistance, and support local jurisdictions with recovery and mitigation activities. A gubernatorial disaster declaration is also a necessary step when requesting federal disaster assistance under the Robert T. Stafford Disaster Relief and Emergency Assistance Act. Harris County has been included in flood-related state emergency proclamations as listed in Table 12-3.

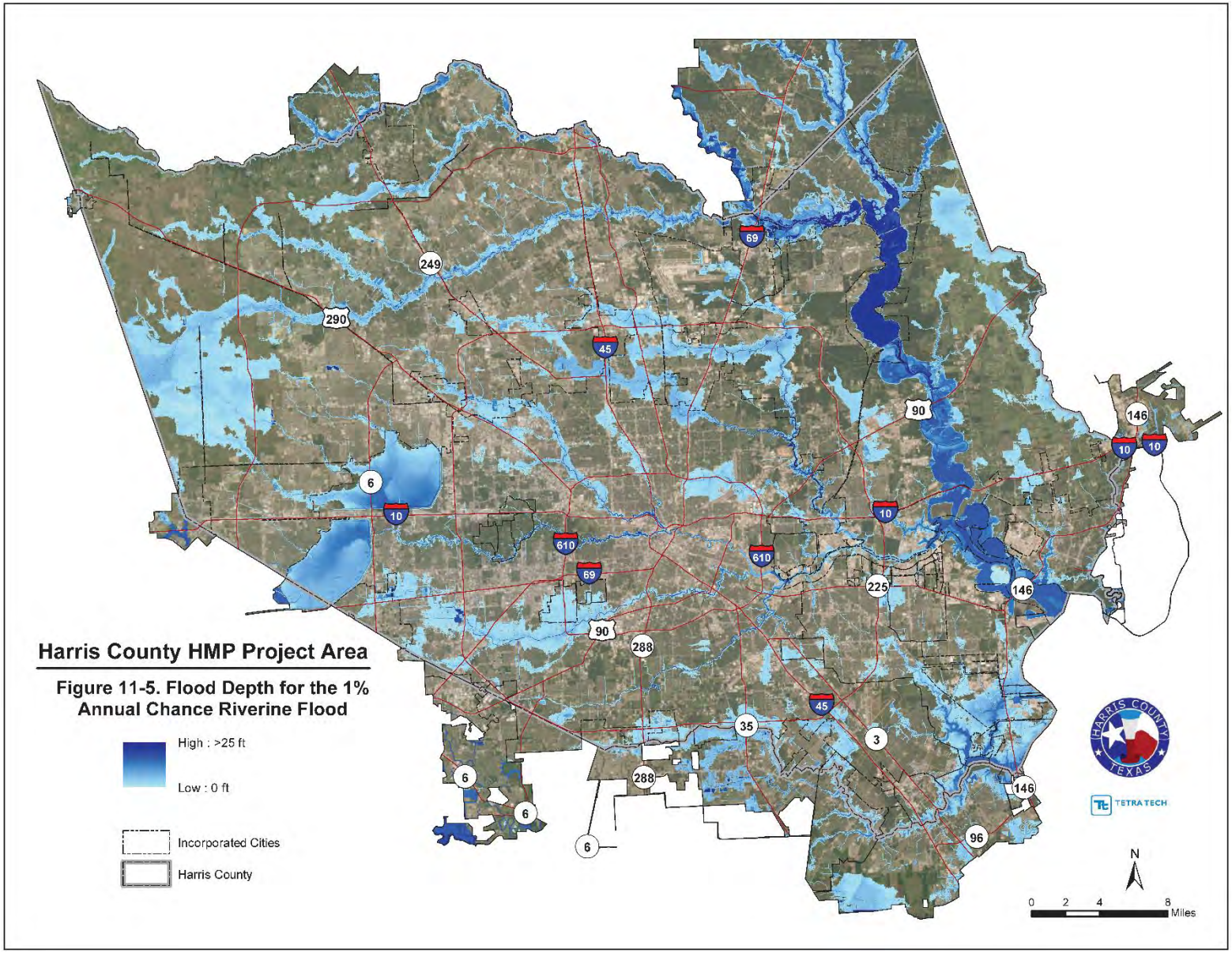
USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2017 and 2023, Harris County was included in 1 USDA flood-related agricultural disaster declarations. For declarations that occurred between 2020 and 2025, refer to Table 12-4.

Previous Events

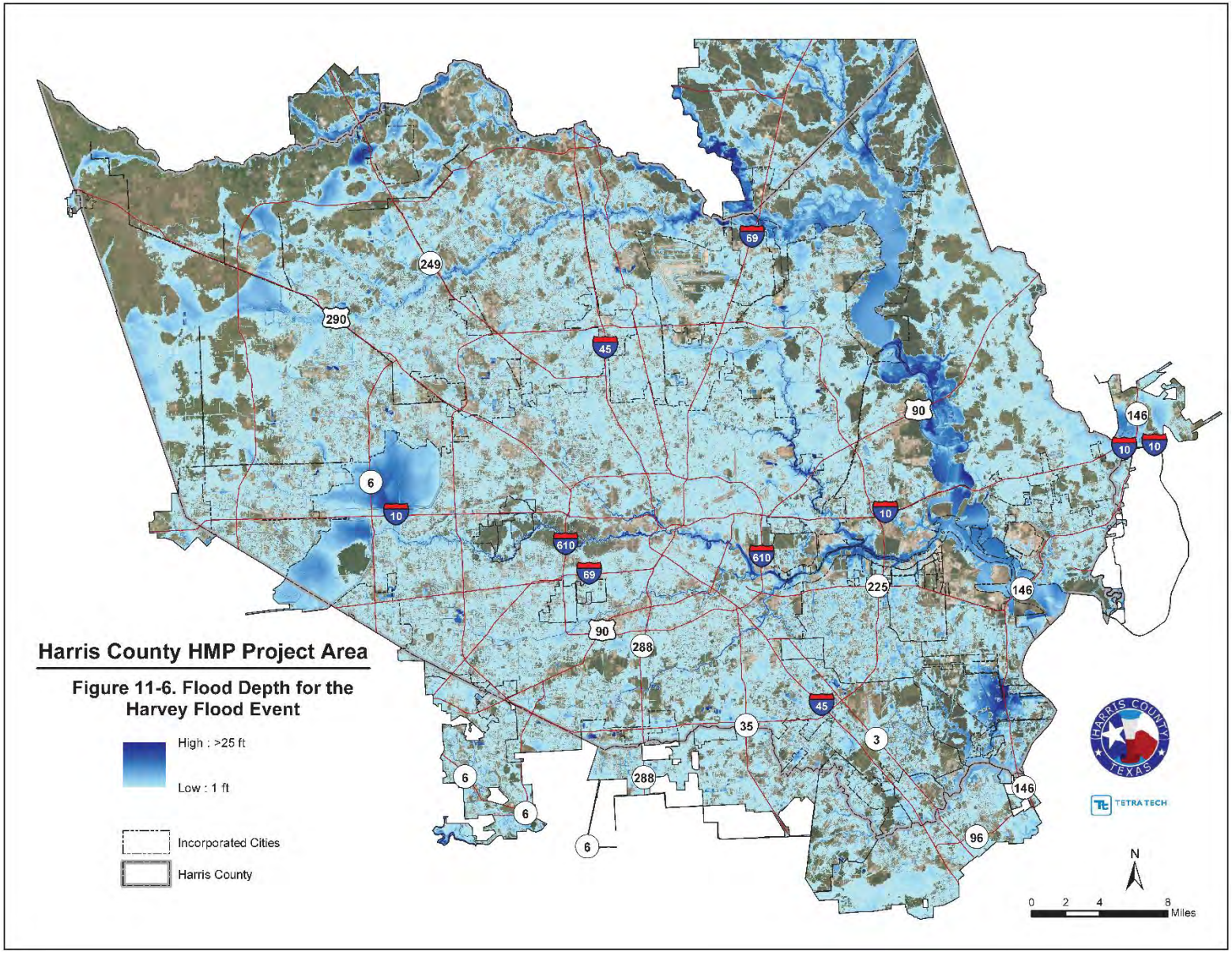
Known hazard events that impacted Harris County between January 2020 and May 2025 are discussed in Table 12-5. For events prior to 2020, refer to the 2019 Harris County HMP.

Figure 12-5. Flood Extent During a 1% Annual Chance Flood Event



Source: (HCOHSEM 2020)

Figure 12-6. Flood Extent During Hurricane Harvey



Source: (HCOHSEM 2020)

Table 12-2. FEMA Declarations for Flood Events in Harris County (1950 to 2025)

Event Date	Declaration Date	Declaration Number	Description
March 26-28, 2025	May 21, 2025	DR-4871-TX	Texas Severe Storms and Flooding
July 5-9, 2024	July 9, 2024	DR-4798-TX	Texas Hurricane Beryl
April 26, 2024 – Jun 5, 2024	May 17, 2024	DR-4781-TX	Severe Storms, Straight-Line Winds, Tornadoes, and Flooding
August 23-27, 2020	August 24, 2020	EM-3540-TX	Texas Tropical Storms Marco and Laura
August 23-27, 2020	December 9, 2020	DR-4572-TX	Texas Hurricane Laura
Jul 25-31, 2020	Jul 26, 2020	EM-3530-TX	Texas Hurricane Hanna
September 17-23, 2019	October 4, 2019	DR-4466-TX	Tropical Storm Imelda
June 24-25, 2019	July 17, 2019	DR-4454-TX	Texas Severe Storms and Flooding
September 10 – November 2, 2018	Feb 25, 2019	DR-4416-TX	Texas Severe Storms and Flooding
June 19, 2018 – July 13, 2018	July 6, 2018	DR-4377-TX	Texas Severe Storms and Flooding
May 22, 2016 – June 24, 2016	June 11, 2016	DR-4272-TX	Texas Severe Storms and Flooding
April 17-30, 2016	April 25, 2016	DR-4269-TX	Severe Storms & Flooding
June 29, 2002 – July 31, 2002	July 4, 2002	DR-1425-TX	Severe Storms and Flooding in Texas
October 22-31, 2015	November 25, 2015	DR-4245-TX	Severe Storms, Tornadoes, Straight-Line Winds, and Flooding
May 4 – June 22, 2015	May 29, 2015	DR-4223-TX	Severe Storms, Tornadoes, Straight-Line Winds, and Flooding
October 24 – November 15, 2002	November 5, 2002	DR-1439-TX	Severe Storms, Tornadoes, and Flooding
October 17 – November 15, 1998	October 21, 1998	DR-1257-TX	TX-Flooding 10/18/98
October 14 – November 8, 1994	October 18, 1994	DR-1041-TX	Severe Thunderstorms & Flooding
March 4-30, 1992	March 20, 1992	DR-937-TX	Severe Storms & Flooding
June 25 – July 7, 1989	July 18, 1989	DR-836-TX	Tropical Storm Allison
May 4 – June 7, 1989	May 19, 1989	DR-828-TX	Severe Storms, Tornadoes, & Flooding
October 30, 1984	October 30, 1984	DR-727-TX	Severe Storms & Flooding
April 26, 1979	April 26, 1979	DR-580-TX	Severe Storms, Tornadoes, & Flooding
September 25, 1979	September 25, 1979	DR-603-TX	Severe Storms & Flooding
June 19, 1976	June 19, 1976	DR-510-TX	Severe Storms & Flooding
July 11, 1973	July 11, 1973	DR-398-TX	Severe Storms & Flooding

Source: FEMA 2025

Table 12-3. State Emergency Proclamations for Flood Events in Harris County

Event Date	State Proclamation Number	Description
August 23-27, 2020	DR-4572-TX	Texas Hurricane Laura
September 14, 2021	EM-3574	Hurricane Nicholas
June 2024	N/A	Tropical Storm Alberto
September 11, 2024	N/A	Hurricane Francine
July 2024	N/A	Hurricane Beryl
April 26-June 5, 2024	DR-4781	Spring Storms and Floods

Source: TDEM 2025

Table 12-4. USDA Declarations for Flood Events in Harris County (2020 to 2025)

Event Date	USDA Declaration Number	Description
April 26, 2024	N/A	Severe Storms, Straight-line Winds, Tornadoes, and Flooding beginning April 26, 2024, and continuing.

Source: USDA 2023

Table 12-5. Flood Events in Harris County (2020 to 2025)

Event Date	Declaration or Proclamation Number	Location Impacted	Description
March 26-28, 2025	DR-4871-TX	Texas	Texas Severe Storms and Flooding
Sept 11, 2024	N/A	Texas coast	Hurricane Francine
Jul 5-Jul 9, 2024	DR-4798-TX	Texas	Texas Hurricane Beryl
Apr 26, 2024 – Jun 5, 2024	DR-4781-TX	Texas	Texas Severe Storms and Flooding
June 2024	N/A	Aransas, Atascosa, Bandera, Bee, Bexar, Brazoria, Brooks, Calhoun, Cameron, Chambers, DeWitt, Dimmit, Duval, Edwards, Fort Bend, Frio, Galveston, Goliad, Gonzales, Harris, Hidalgo, Jackson, Jefferson, Jim Hogg, Jim Wells, Karnes, Kenedy, Kinney, Kleberg, LaSalle, Lavaca, Live Oak, Matagorda, Maverick, McMullen, Medina, Nueces, Orange, Real, Refugio, San Patricio, Starr, Uvalde, Val Verde, Victoria, Webb, Wharton, Willacy, Wilson, Zapata, and Zavala counties	Tropical Storm Alberto
July 2024	N/A	Texas	Hurricane Beryl
September 14, 2021	EM-3574	Aransas, Brazoria, Calhoun, Chambers, Galveston, Harris, Jackson, Jasper, Jefferson, Matagorda, Montgomery, Newton, Nueces, Orange, Refugio, San Patricio, and Victoria	Hurricane Nicholas
Aug 23-27, 2020	DR-4572-TX	Texas	Texas Hurricane Laura
Aug 23- 27, 2020	EM-3540-TX	Texas	Texas Tropical Storms Marco and Laura
Jul 25-31, 2020	EM-3530-TX	Texas	Texas Hurricane Hanna

Source: FEMA 2025, TDEM 2025

12.1.6 Probability of Future Occurrences

Information on previous flood occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 12-6. Since 1950, NCEI Storm Events Database shows there have been 232 flood or flash flood events in Harris County (NOAA/NCEI 2025). Based on historical records and input from the Steering Committee, the probability of occurrence for flooding in the County is considered "occasional."

Table 12-6. Probability of Future Flood Events in Harris County

Hazard Type	Number of Occurrences Between 1950 and 2025	Percent Chance of Occurring in Any Given Year
Flash Flood	63	82.9%
Flood	9	11.84%
Total	72	94.74%

Source: (NOAA/NCEI 2025)

Climate Change Projections

The climate of Texas—and Harris County in particular—is changing. Over the past century, most of the state has warmed by approximately 0.5°F to 1°F, with projections indicating continued warming in the decades to come (Environmental Protection Agency 2016). In eastern Texas, including Harris County, rainstorms have grown more intense, increasing the frequency and severity of flooding. Future storms are expected to become even more powerful due to rising atmospheric moisture and ocean temperatures (Center for Climate and Energy Solutions [C2ES] n.d.). High-frequency flood events, such as those previously categorized as "10-year floods," are now expected to occur more often. This intensification of storm activity leads to more direct runoff and flash flooding, while changing soil moisture and vegetation conditions alter the county's hydrologic behavior. As streamflow velocity and volume increase, erosion patterns will shift, potentially resulting in greater sedimentation behind local dams such as Addicks and Barker, reducing their capacity and impacting water quality and habitat.

The implications of these changes are significant for Harris County's water infrastructure and emergency planning. Increased wildfire potential due to drought and extreme heat could lead to post-fire flooding events, further raising sediment loads and degrading water quality in reservoirs and bayous. As the hydrologic cycle becomes less predictable, events once considered 1-percent annual chance (i.e., 100-year floods) may occur far more frequently, heightening flood risks for thousands of residents. These shifts require planners to rethink the design, regulation, and operation of critical flood protection infrastructure—dams, levees, floodways, and stormwater systems—based on future climate conditions rather than past performance (IPCC 2021). The longstanding reliance on historical hydrologic records for forecasting and design must be replaced with climate-responsive models, flexible decision-making tools, and a revised standard of practice. Harris County water and emergency managers are increasingly recognizing that precipitation and runoff patterns are shifting, demanding enhanced flood protection, drought resilience, and real-time response systems.

12.1.7 Cascading Impacts on Other Hazards

Flooding in Harris County has wide-reaching cascading impacts that extend far beyond the initial inundation. As one of the most flood-prone counties in the United States, Harris County regularly experiences overbank flooding from its complex bayou system, including Buffalo, Brays, and White Oak Bayous. When heavy rainfall overwhelms drainage systems, floodwaters can submerge major roadways such as I-45, I-10, and Beltway 8, disrupting regional mobility and emergency response routes (Harris County Flood Control District 2023). These disruptions impede access to hospitals, shelters, and evacuation centers, especially in lower-income communities, compounding the risk to life and health during emergencies.

Flooding also poses significant threats to critical infrastructure. Power outages are common during flood events, as substations, transformers, and underground utility vaults may be inundated. Extended outages affect not only residential areas but also water and wastewater treatment plants, which are essential for maintaining public health. The interruption of drinking water systems and the overflow of sewage into floodwaters can lead to waterborne disease outbreaks and long-term contamination of local waterways (FEMA 2020). Additionally, many of the region's industrial facilities are located near the Houston Ship Channel, an area vulnerable to both flooding and chemical spills, heightening the risk of hazardous material releases during flood events.

Housing and displacement issues often follow flood events, especially in neighborhoods with repetitive flood losses. Prolonged or repeated flooding can lead to significant property damage and force residents to abandon their homes, exacerbating housing insecurity and straining social services. Low-income renters, many of whom lack flood insurance or relocation resources, are disproportionately affected (National Low Income Housing Coalition 2021). The cumulative effect of these displacements contributes to school interruptions, job losses, and mental health crises. Over time, repeated flooding can erode community cohesion and lead to long-term depopulation of at-risk neighborhoods.

Flooding also intensifies environmental degradation and public health threats across Harris County. Urban runoff carries pollutants, debris, and sediment into the bayous and Galveston Bay, disrupting aquatic ecosystems and threatening fisheries. In heavily industrialized areas, floodwaters may pick up petroleum byproducts, heavy metals, and hazardous waste, spreading contamination over residential areas and natural landscapes. As climate change increases the frequency of extreme rainfall events, these cascading hazards will intensify unless comprehensive mitigation measures—such as green infrastructure, land-use reforms, and updated stormwater regulations—are implemented (Intergovernmental Panel on Climate Change 2021). Understanding these interconnected risks is essential for developing a resilient flood management strategy in Harris County.

12.2 VULNERABILITY AND IMPACT ASSESSMENT

To understand risk, a community must evaluate assets exposed to and vulnerable to the identified hazard. To quantitatively assess Harris County's risk to the flood hazard, a spatial analysis was conducted using the May 2, 2019, FEMA effective Digital Flood Insurance Rate Map (DFIRM). A 1 percent-annual-chance flood depth grid provided by the Harris County Flood Control District was

used as the base and updated to reflect the effective DFIRM and include coastal flood zones and the pending DFIRM (to be effective November 15, 2019). Updates were generated using the floodplain boundaries and the U.S. Geological Survey's (USGS) 2-meter digital elevation model. For areas where planning partner jurisdictions extended into neighboring counties, the flood depth grid was extended using the DFIRM data for those counties. These results are summarized below.

Unique and Varied Risks

Flooding represents the most frequent and damaging natural hazard in Harris County, affecting virtually every jurisdiction to some degree. However, flood risk varies widely depending on topography, watershed location, drainage infrastructure, and degree of urbanization. Flood hazards stem from riverine, coastal, and localized (pluvial) sources, and these types often overlap across the planning area.

Harris County contains an intricate network of bayous, tributaries, and man-made channels that drain into Galveston Bay. The County's low-lying coastal plain, combined with rapid land development, high impervious cover, and frequent tropical rainfall, contributes to recurring flood issues. The Harris County Flood Control District (HCFCD) oversees major drainage infrastructure, but localized street and structure flooding remain common in older or rapidly growing areas.

Northern and Western Jurisdictions

The Cities of Tomball, Waller, and Katy, as well as unincorporated northwest Harris County, experience riverine and urban flooding from tributaries of Cypress Creek, Spring Creek, and the Addicks and Barker Reservoir watersheds. City of Tomball has recurring flood issues along Willow and Spring Creeks where increased impervious cover and aging drainage systems limit stormwater capacity. City of Katy and nearby Municipal Utility Districts (MUDs) experience overflow from the Buffalo Bayou watershed and limited conveyance near the Grand Parkway corridor. The Addicks and Barker Reservoirs, while providing regional flood storage, can exacerbate flooding in surrounding neighborhoods when releases are required during major storm events.

Eastern Jurisdictions

The Cities of Baytown, La Porte, Morgan's Point, and Shoreacres face both coastal and rainfall-induced flooding. Low elevations and tidal influence along the Houston Ship Channel and Galveston Bay amplify storm surge and backwater effects. Baytown experiences chronic flooding in low-lying residential zones near Goose Creek and Cedar Bayou. La Porte and Morgan's Point are vulnerable to surge-related flooding and heavy rainfall from tropical systems, particularly where industrial drainage systems outfall to tidally influenced channels.

Southern and Coastal Jurisdictions

Communities such as El Lago, Taylor Lake Village, and Seabrook are highly exposed to storm surge and heavy rainfall flooding due to their proximity to Clear Lake and Taylor Lake. El Lago and Taylor Lake Village experience periodic street and yard flooding during heavy rain events, with the greatest risk during hurricanes and tropical storms when drainage is impeded by high lake levels. Flood elevations can also be influenced by wind-driven tide and backflow from Galveston Bay,

creating compound flooding scenarios. Infrastructure and evacuation routes such as NASA Road 1 and Old Kirby Road are vulnerable to ponding and overland flow during high-intensity rainfall.

Central and Urban Jurisdictions

Highly urbanized cities including Bellaire, West University Place, Southside Place, and Jacinto City experience pluvial flooding from intense rainfall events overwhelming storm sewers and roadside ditches. Bellaire and West University Place are traversed by Brays Bayou, where high rainfall rates can exceed channel and detention capacity, flooding streets and homes. Jacinto City, adjacent to the industrial Ship Channel area, faces flash flooding from limited drainage infrastructure and flat terrain.

Other Areas

Large portions of unincorporated Harris County remain within mapped Special Flood Hazard Areas (SFHAs) along the county's 22 major watersheds. Rapid growth in the north and west portions of the County has increased runoff volumes, placing additional stress on downstream communities. Coordinated watershed-scale mitigation with HCFCD and the U.S. Army Corps of Engineers is critical to reduce risk across jurisdictional boundaries.

Some County-owned facilities located in adjacent counties—such as communication towers or pumping infrastructure—lie within mapped floodplains and rely on regional coordination with Fort Bend, Montgomery, and Liberty Counties. Maintaining redundancy and elevation protection for these assets ensures continuity of operations for Harris County emergency services.

12.2.1 Life, Health, and Safety

The effects of flooding on people's lives, health, and safety depend on various factors, such as how severe the flood is and whether residents receive timely warnings. "Exposure" refers to the number of people living in or near flood-prone areas who could be affected if a flood occurs. However, exposure is not limited to just those living within designated hazard zones—it also includes anyone who might be impacted by the flood's consequences, such as travelers in flooded areas or individuals who lose access to emergency services. The extent of these impacts can differ widely and cannot be precisely measured.

Understanding exposure in the context of flooding requires a broader perspective that goes beyond geographic boundaries. Vulnerability can extend to individuals who work, commute, or attend school in flood-prone areas, even if they do not live there. Infrastructure such as roads, hospitals, and utilities can also be compromised, amplifying the risk to public safety and disrupting essential services. Moreover, social and economic factors—such as age, mobility, income level, and access to information—can influence how severely different groups are affected. Therefore, assessing flood risk must include a comprehensive view of both direct and indirect exposure, recognizing that the ripple effects of a flood can reach far beyond the immediate area of inundation.

Overall Population

Harris County is one of the most flood-prone regions in the United States, with significant portions of its population living in areas vulnerable to riverine, bayou, and urban flooding. As the county continues to experience rapid population growth—surpassing 4.8 million residents as of the 2020 Census (U.S. Census Bureau, 2021)—flood risk expands alongside development, particularly in low-lying areas, floodplains, and communities with limited infrastructure. The combination of increasing population density, aging infrastructure, and climate change exacerbates exposure and complicates emergency response and recovery.

Flooding not only displaces families but also disrupts access to critical services such as schools, hospitals, and emergency facilities. Neighborhoods in eastern and northeastern Harris County, as well as areas near the Buffalo and Brays Bayous, are particularly susceptible to repetitive loss and high-water events. As population growth accelerates across the region, particularly under high migration scenarios for Texas, the number of people exposed to flood hazards will likely increase unless strategic land use and infrastructure planning are implemented.

Understanding where vulnerable populations live, and how they may be disproportionately affected by flood hazards, is essential for equitable and effective hazard mitigation planning. As such, flood risk must be continuously reassessed in tandem with population trends, housing patterns, and climate projections.

Based on the spatial analysis, there are an estimated 429,404 residents living in the 1 percent annual chance floodplain, or 8.8 percent of the County's total population. The City of Houston has the greatest number of residents living in the 1 percent annual chance flood event hazard area with approximately 259,958 residents, followed by the Unincorporated Areas of Harris County (109,361). Table 12-7 summarizes the population exposed to the flood hazard by jurisdiction.

As a result of flooding, residents can be displaced or require temporary to long-term sheltering. Table 12-8 shows estimated numbers of people displaced or in need of short-term shelter for the evaluated flood scenarios.

Table 12-7 Population Living in the 1 Percent Annual Chance Flood Hazard Areas

Jurisdiction	Total Population (2023 ACS)	Population Living in the Flood Hazard Area					
		1-Percent Annual Chance Riverine Flood		1-Percent Annual Chance Coastal Flood		Hurricane Harvey Flooding	
		Number of Persons	% of Total	Number of Persons	% of Total	Number of Persons	% of Total
Baytown[b]	79,635	2,393	3.0%	201	0.3%	26,938	33.8%
Bellaire	17,116	14,026	81.9%	0	0.0%	7,856	45.9%
Bunker Hill Village	3,801	0	0.0%	0	0.0%	20	0.5%
Deer Park	33,823	683	2.0%	0	0.0%	12,379	36.6%
El Lago	3,016	1,894	62.8%	0	0.0%	1,539	51.0%
Friendswood[a]	40,827	3,717	9.1%	0	0.0%	14,253	34.9%
Galena Park	10,527	370	3.5%	0	0.0%	4,072	38.7%
Hedwig Village	2,029	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	1,091	0	0.0%	0	0.0%	778	71.3%
Houston[b]	2,252,110	255,958	11.4%	0	0.0%	881,997	39.2%
Humble	16,489	727	4.4%	0	0.0%	6,984	42.4%
Hunters Creek Village	4,358	126	2.9%	0	0.0%	857	19.7%
Jacinto City	9,458	0	0.0%	0	0.0%	3,587	37.9%
Jersey Village	7,817	2,238	28.6%	0	0.0%	2,570	32.9%
Katy[a]	23,900	1,547	6.5%	0	0.0%	6,915	28.9%
La Porte	36,077	1,527	4.2%	4,291	11.9%	16,424	45.5%
League City[b]	2,527	204	8.1%	0	0.0%	354	14.0%
Missouri City[a]	75,234	323	0.4%	0	0.0%	18,127	24.1%
Morgan's Point	255	0	0.0%	63	24.7%	116	45.5%
Nassau Bay	5,211	4,135	79.4%	0	0.0%	1,943	37.3%
Pasadena	149,345	9,568	6.4%	504	0.3%	76,344	51.1%
Pearland[b]	8,189	156	1.9%	0	0.0%	1,578	19.3%
Piney Point Village	3,090	162	5.2%	0	0.0%	926	30.0%
Seabrook	13,617	6,110	44.9%	4,066	29.9%	6,419	47.1%
Shoreacres	1,327	546	41.1%	768	57.9%	813	61.3%
South Houston[b]	16,017	3,333	20.8%	0	0.0%	12,047	75.2%
Southside Place	1,631	1,180	72.3%	0	0.0%	523	32.1%
Spring Valley Village	4,226	0	0.0%	0	0.0%	4	0.1%
Stafford[a]	17,279	0	0.0%	0	0.0%	7,850	45.4%
Taylor Lake Village	3,678	1,658	45.1%	0	0.0%	1,410	38.3%
Tomball	13,032	24	0.2%	0	0.0%	2,103	16.1%
Waller[a]	3,062	517	16.9%	0	0.0%	828	27.0%
Webster	12,283	4,774	38.9%	0	0.0%	4,360	35.5%
West University Place	14,907	2,147	14.4%	0	0.0%	4,231	28.4%
Unincorporated County	2,001,319	109,361	5.5%	14	<0.1%	660,032	33.0%
Harris County Total	4,888,304	429,404	8.8%	9,907	0.2%	1,787,177	36.6%

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Table 12-8. Population Displaced or Seeking Shelter Due to Flooding

Jurisdiction	1-Percent Annual Chance Riverine Flood		1-Percent Annual Chance Coastal Flood		Hurricane Harvey Flooding	
	Displaced Population	Persons Seeking Shelter	Displaced Population	Persons Seeking Shelter	Displaced Population	Persons Seeking Shelter
Baytown[b]	2,751	719	269	38	33,364	4,211
Bellaire	13,010	139	0	0	11,048	144
Bunker Hill Village	0	0	0	0	150	19
Deer Park	1,048	241	0	0	15,546	785
El Lago	2,194	85	0	0	1,550	173
Friendswood[a]	3,869	268	0	0	18,626	955
Galena Park	134	49	0	0	4,839	642
Hedwig Village	0	0	0	0	472	94
Hilshire Village	6	0	0	0	700	1
Houston[b]	238,710	37,231	0	0	1,078,139	156,736
Humble	665	112	0	0	8,109	1,271
Hunters Creek Village	85	9	0	0	1,181	31
Jacinto City	25	22	0	0	4,749	668
Jersey Village	1,732	35	0	0	2,913	178
Katy[a]	900	68	0	0	7,909	380
La Porte	1,480	180	4,108	207	17,853	1,093
League City[b]	123	12	0	0	884	25
Missouri City[a]	761	186	0	0	32,598	1,725
Morgan's Point	0	0	29	7	134	17
Nassau Bay	4,086	313	0	0	2,510	587
Pasadena	8,207	1,201	272	13	82,290	9,082
Pearland[b]	144	23	0	0	1,573	81
Piney Point Village	75	2	0	0	1,372	13
Seabrook	6,418	292	5,431	201	6,869	777
Shoreacres	719	1	1,562	4	959	4
South Houston[b]	3,330	457	0	0	12,305	1,028
Southside Place	828	12	0	0	693	12
Spring Valley Village	51	13	0	0	1,730	29
Stafford[a]	26	20	0	0	10,789	734
Taylor Lake Village	1,878	53	0	0	1,474	114
Tomball	41	41	0	0	2,698	439
Waller[a]	223	40	0	0	690	59
Webster	1,960	331	0	0	5,112	1,549
West University Place	1,778	15	0	0	7,229	46
Unincorporated County	106,617	17,639	2	0	924,554	82,343
Harris County Total	403,874	59,809	11,673	470	2,303,611	266,045

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b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Socially Vulnerable Population

Harris County participates in the National Flood Insurance Program (NFIP) and has adopted flood damage prevention regulations that govern how development occurs within mapped SFHAs. Future development in flood-prone zones will require compliance with floodplain management standards, including elevation and floodproofing requirements. However, as growth continues throughout the county—particularly in suburban and unincorporated areas—development pressure on natural floodplains and wetlands may increase. If new construction encroaches upon areas that currently provide natural and beneficial floodplain functions, such as flood detention basins, bayou-adjacent green spaces, and riparian buffers, downstream flooding could be exacerbated. Urban expansion in low-lying areas and along major bayous, including Brays, White Oak, and Greens Bayous, has already worsened flooding by increasing impervious surface, limiting the land's capacity to absorb runoff.

In densely developed areas of Harris County, including Houston's urban core, paved surfaces such as roads, parking lots, and rooftops contribute heavily to urban and stormwater flooding. This is particularly problematic in neighborhoods with high water tables or poor drainage infrastructure, where even moderate rainfall events can result in localized flooding. To manage the hydrologic impacts of ongoing development, communities throughout Harris County can adopt nature-based solutions like rain gardens, bioswales, green roofs, and permeable pavement to enhance stormwater infiltration and reduce runoff. As the county's population continues to grow, the number of residents exposed to flood hazards may rise unless thoughtful planning measures are taken. Strategies such as expanding regulatory floodplain boundaries, increasing freeboard requirements, conducting voluntary buyouts of frequently flooded properties, and steering growth toward safer upland areas can help reduce future risk and promote long-term resilience to flooding.

Estimates of the number of socially vulnerable people living in the mapped flood hazard areas are provided in Table 12-9 through Table 12-11.

Table 12-9. Vulnerable Persons Living in the 1-Percent Annual Chance Riverine Flood Hazard Area

Jurisdiction	Population Within the 1-Percent Annual Chance Riverine Flood Hazard Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Baytown[b]	276	3.0%	177	3.0%	75	3.0%	306	3.0%	419	30.1%
Bellaire	2,690	81.9%	693	81.9%	228	81.7%	912	81.9%	363	81.8%
Bunker Hill Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Deer Park	112	2.0%	41	2.0%	7	1.9%	87	2.0%	55	2.0%
El Lago	387	62.3%	67	63.2%	19	61.3%	157	63.3%	77	63.6%
Friendswood[a]	583	9.1%	224	9.1%	8	8.5%	322	9.1%	103	9.1%
Galena Park	36	3.5%	29	3.5%	20	3.5%	46	3.5%	86	3.5%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Houston[b]	30,841	11.4%	17,259	11.4%	13,549	11.4%	27,946	11.4%	49,586	11.4%
Humble	89	4.4%	37	4.4%	22	4.3%	85	4.4%	130	4.4%
Hunters Creek Village	20	2.8%	5	2.8%	0	0.0%	6	2.6%	5	2.8%
Jacinto City	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jersey Village	482	28.6%	141	28.5%	49	28.2%	231	28.5%	113	28.6%
Katy[a]	185	6.5%	116	6.5%	19	6.2%	142	6.5%	82	6.4%
La Porte	203	4.2%	105	4.2%	16	4.1%	189	4.2%	186	4.2%
League City[b]	25	8.0%	12	7.8%	1	4.5%	19	7.8%	9	7.9%
Missouri City[a]	52	0.4%	19	0.4%	4	0.3%	34	0.4%	19	0.4%
Morgan's Point	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	884	79.3%	100	78.7%	178	79.1%	561	79.3%	549	79.2%
Pasadena	1,110	6.5%	713	6.4%	403	6.3%	1,153	6.4%	1,720	6.3%

Jurisdiction	Population Within the 1-Percent Annual Chance Riverine Flood Hazard Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Pearland[b]	18	1.8%	11	1.9%	2	1.5%	11	1.8%	6	1.7%
Piney Point Village	35	5.2%	1	3.7%	0	0.0%	8	5.0%	1	4.2%
Seabrook	882	45.0%	306	44.7%	49	44.1%	624	44.8%	404	44.7%
Shoreacres	97	41.1%	23	41.1%	0	0.0%	117	41.1%	17	39.5%
South Houston[b]	288	20.8%	230	20.8%	165	20.8%	331	20.8%	783	20.8%
Southside Place	219	72.3%	53	71.6%	2	66.7%	77	72.0%	36	72.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Taylor Lake Village	490	43.6%	48	48.5%	12	57.1%	229	44.6%	4	40.0%
Tomball	5	0.2%	1	0.1%	0	0.0%	4	0.2%	2	0.2%
Waller[a]	34	16.8%	21	16.7%	6	15.8%	50	16.6%	124	16.9%
Webster	677	38.9%	295	38.8%	197	38.8%	507	38.9%	430	38.8%
West University Place	357	14.4%	164	14.3%	1	11.1%	98	14.3%	41	14.1%
Unincorporated County	11,150	5.5%	7,871	5.5%	3,151	5.5%	10,517	5.5%	13,470	5.5%
Harris County Total	52,227	9.3%	28,762	8.5%	18,183	9.4%	44,769	8.8%	68,819	9.2%

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Table 12-10. Vulnerable Persons Living in the 1-Percent Annual Chance Coastal Flood Hazard Area

Jurisdiction	Population Within the 1-Percent Annual Chance Coastal Flood Hazard Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Baytown[b]	23	0.2%	14	0.2%	6	0.2%	25	0.2%	35	2.5%
Bellaire	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bunker Hill Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Deer Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
El Lago	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Galena Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Houston[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Humble	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hunters Creek Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jacinto City	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jersey Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Katy[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
La Porte	574	11.9%	294	11.8%	46	11.8%	536	11.9%	522	11.8%
League City[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Missouri City[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Morgan's Point	12	22.2%	2	18.2%	0	0.0%	8	22.2%	7	21.9%
Nassau Bay	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Pasadena	58	0.3%	36	0.3%	20	0.3%	60	0.3%	87	0.3%
Pearland[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%

Jurisdiction	Population Within the 1-Percent Annual Chance Coastal Flood Hazard Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Piney Point Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Seabrook	585	29.8%	204	29.8%	33	29.7%	416	29.8%	269	29.8%
Shoreacres	135	57.2%	32	57.1%	0	0.0%	163	57.2%	26	60.5%
South Houston[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Southside Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Taylor Lake Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Tomball	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Waller[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Webster	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
West University Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Unincorporated County	1	<0.1%	1	<0.1%	0	0.0%	1	<0.1%	1	<0.1%
Harris County Total	1,388	0.2%	583	0.2%	105	0.1%	1,209	0.2%	947	0.1%

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Table 12-11. Vulnerable Persons Living in the Hurricane Harvey Flood Inundation Area

Jurisdiction	Population Within the Hurricane Harvey Flood Inundation Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Baytown[b]	3,113	33.8%	1,996	33.8%	847	33.8%	3,450	33.8%	470	33.8%
Bellaire	1,506	45.9%	388	45.9%	128	45.9%	511	45.9%	203	45.7%
Bunker Hill Village	2	0.4%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Deer Park	2,029	36.5%	752	36.6%	138	37.2%	1,583	36.6%	1,019	36.9%
El Lago	314	50.6%	54	50.9%	15	48.4%	127	51.2%	62	51.2%
Friendswood[a]	2,240	34.8%	860	34.9%	33	35.1%	1,237	34.9%	397	35.0%
Galena Park	401	38.7%	319	38.6%	223	38.6%	512	38.6%	951	38.7%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	131	71.2%	31	70.5%	1	50.0%	27	69.2%	4	57.1%
Houston[b]	106,268	39.2%	59,479	39.2%	46,685	39.1%	96,299	39.2%	170,862	39.1%
Humble	860	42.3%	356	42.3%	217	42.4%	817	42.3%	1,247	42.3%
Hunters Creek Village	141	19.6%	35	19.3%	1	16.7%	46	19.6%	35	19.9%
Jacinto City	495	37.9%	279	37.9%	169	37.7%	567	37.9%	674	37.9%
Jersey Village	553	32.8%	162	32.7%	57	32.8%	266	32.8%	129	32.7%
Katy[a]	827	28.9%	519	28.9%	86	28.6%	637	28.9%	370	28.8%
La Porte	2,192	45.5%	1,131	45.5%	177	45.4%	2,043	45.5%	2,009	45.5%
League City[b]	44	14.1%	21	13.6%	3	13.6%	34	13.9%	15	13.2%
Missouri City[a]	2,960	24.1%	1,101	24.1%	280	24.1%	1,960	24.1%	1,108	24.1%
Morgan's Point	24	44.4%	5	45.5%	0	0.0%	16	44.4%	14	43.8%
Nassau Bay	415	37.2%	47	37.0%	83	36.9%	263	37.2%	258	37.2%
Pasadena	8,736	51.2%	5,721	51.0%	3,258	51.1%	9,184	51.1%	13,873	51.1%
Pearland[b]	189	19.2%	112	19.3%	26	19.3%	115	19.1%	66	19.2%

Jurisdiction	Population Within the Hurricane Harvey Flood Inundation Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Piney Point Village	202	30.0%	8	29.6%	1	16.7%	47	29.6%	7	29.2%
Seabrook	927	47.2%	322	47.0%	52	46.8%	656	47.1%	424	46.9%
Shoreacres	144	61.0%	34	60.7%	0	0.0%	173	60.7%	27	62.8%
South Houston[b]	1,043	75.2%	832	75.2%	597	75.2%	1,196	75.2%	2,831	75.2%
Southside Place	97	32.0%	23	31.1%	0	0.0%	34	31.8%	16	32.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	881	45.4%	554	45.3%	295	45.2%	731	45.4%	735	45.4%
Taylor Lake Village	417	37.1%	41	41.4%	10	47.6%	195	38.0%	4	40.0%
Tomball	451	16.1%	160	16.1%	24	15.9%	356	16.1%	199	16.1%
Waller[a]	54	27.0%	33	26.9%	10	27.2%	82	27.0%	198	27.0%
Webster	617	35.4%	270	35.5%	181	35.6%	463	35.5%	400	36.1%
West University Place	704	28.4%	325	28.4%	2	22.2%	194	28.4%	82	28.2%
Unincorporated County	67,316	33.1%	47,525	33.0%	18,932	33.0%	63,504	33.0%	81,096	33.0%
Harris County Total	206,293	36.6%	123,495	36.5%	72,531	37.7%	187,325	36.8%	279,785	37.6%

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12.2.2 General Building Stock

Flooding poses a significant threat to Harris County's general building stock, with both residential and commercial structures regularly impacted by floodwaters. The county's flat topography, dense development patterns, aging stormwater infrastructure, and proximity to bayous and waterways increase structural vulnerability to both fluvial (riverine) and pluvial (urban) flooding events.

Table 12-12 through Table 12-14 summarize the number of structures located in the evaluated flood hazard areas by jurisdiction. In summary, there are 119,410 buildings located in the 1 percent annual chance riverine flood hazard area, with an estimated $104 billion of replacement cost value (building and content replacement costs). This represents 8.8 percent of the County's total general building stock inventory.

Residential buildings, particularly slab-on-grade, single-story homes, are susceptible to interior damage from even shallow flooding. Floodwaters can compromise electrical systems, flooring, drywall, insulation, and HVAC units. In multi-family and commercial structures, basements and ground-level retail or parking facilities are highly vulnerable. Prolonged exposure to standing water can also lead to long-term structural issues, such as foundation shifting or mold contamination.

Table 12-15 through Table 12-17 show the number of structures by occupancy class. In summary, there are 110,227 residential buildings, 8,290 commercial buildings, and 506 industrial buildings, located in the 1 percent annual chance riverine flood hazard area.

The Hazus analysis for each flood scenario estimated the potential damage to buildings for each occupancy class. Results are shown in Table 12-18 through Table 12-20.

Table 12-12. General Building Stock Located in the 1 Percent Annual Chance Riverine Flood Hazard Area

	Total Buildings in Jurisdiction		Buildings Within the 1-Percent Annual Chance Riverine Flood Hazard Area			
			Number of Buildings		Replacement Value	
Jurisdiction	Count	Replacement Value	Count	% of Jurisdiction Total	Value	% of Jurisdiction Total
Baytown[b]	20,670	$12,477,103,055	666	3.2%	$437,494,959	3.5%
Bellaire	6,503	$6,921,381,237	5,359	82.4%	$5,648,701,927	81.6%
Bunker Hill Village	1,301	$1,718,400,551	0	0.0%	$0	0.0%
Deer Park	11,370	$7,220,366,741	384	3.4%	$988,161,217	13.7%
El Lago	1,103	$579,556,830	703	63.7%	$380,447,590	65.6%
Friendswood[a]	13,388	$7,480,850,836	1,176	8.8%	$604,799,153	8.1%
Galena Park	3,288	$1,484,142,708	108	3.3%	$24,022,438	1.6%
Hedwig Village	909	$1,463,766,016	0	0.0%	$0	0.0%
Hilshire Village	321	$267,434,315	0	0.0%	$0	0.0%
Houston[b]	588,144	$620,018,000,000	65,929	11.2%	$61,042,384,255	9.8%
Humble	4,829	$5,480,480,895	209	4.3%	$323,469,213	5.9%
Hunters Creek Village	1,572	$2,319,999,302	45	2.9%	$59,748,457	2.6%
Jacinto City	2,986	$1,619,709,430	1	<0.1%	$4,006,611	0.2%
Jersey Village	2,439	$2,863,511,476	668	27.4%	$430,699,735	15.0%
Katy[a]	6,322	$3,789,785,990	450	7.1%	$199,342,685	5.3%
La Porte	12,424	$8,136,585,709	557	4.5%	$368,980,586	4.5%
League City[b]	394	$343,758,798	45	11.4%	$33,526,417	9.8%
Missouri City[a]	24,325	$14,945,042,054	135	0.6%	$184,811,847	1.2%
Morgan's Point	160	$324,507,417	0	0.0%	$0	0.0%
Nassau Bay	1,574	$2,104,459,129	1,217	77.3%	$1,219,383,275	57.9%
Pasadena	40,606	$27,932,208,850	2,695	6.6%	$3,134,733,643	11.2%
Pearland[b]	809	$878,335,161	68	8.4%	$200,702,488	22.9%
Piney Point Village	1,195	$2,383,712,681	65	5.4%	$157,328,878	6.6%

Jurisdiction	Total Buildings in Jurisdiction		Buildings Within the 1-Percent Annual Chance Riverine Flood Hazard Area			
			Number of Buildings		Replacement Value	
	Count	Replacement Value	Count	% of Jurisdiction Total	Value	% of Jurisdiction Total
Seabrook	4,374	$3,115,949,582	2,012	46.0%	$1,579,086,639	50.7%
Shoreacres	657	$268,042,585	274	41.7%	$121,939,439	45.5%
South Houston[b]	5,001	$2,688,145,186	1,245	24.9%	$929,877,300	34.6%
Southside Place	508	$604,745,337	362	71.3%	$436,205,886	72.1%
Spring Valley Village	1,509	$1,568,469,594	7	0.5%	$5,289,300	0.3%
Stafford[a]	5,055	$7,232,853,862	0	0.0%	$0	0.0%
Taylor Lake Village	1,390	$854,884,292	634	45.6%	$431,877,899	50.5%
Tomball	3,632	$4,292,188,724	11	0.3%	$78,820,403	1.8%
Waller[a]	1,097	$270,979,210	191	17.4%	$10,827,620	4.0%
Webster	2,379	$7,009,654,505	720	30.3%	$501,194,149	7.2%
West University Place	5,525	$4,782,421,662	808	14.6%	$749,158,964	15.7%
Unincorporated County	577,476	$402,152,000,000	32,666	5.7%	$24,546,316,160	6.1%
Harris County Total	1,355,235	$1,167,593,433,721	119,410	8.8%	$104,833,339,132	9.0%

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Table 12-13. General Building Stock Located in the 1 Percent Annual Chance Coastal Flood Hazard Area

Jurisdiction	Total Buildings in Jurisdiction		Buildings Within the 1-Percent Annual Chance Coastal Flood Hazard Area			
			Number of Buildings		Replacement Value	
	Count	Replacement Value	Count	% of Jurisdiction Total	Value	% of Jurisdiction Total
Baytown[b]	20,670	$12,477,103,055	54	0.3%	$11,221,969	0.1%
Bellaire	6,503	$6,921,381,237	0	0.0%	$0	0.0%
Bunker Hill Village	1,301	$1,718,400,551	0	0.0%	$0	0.0%
Deer Park	11,370	$7,220,366,741	0	0.0%	$0	0.0%
El Lago	1,103	$579,556,830	0	0.0%	$0	0.0%
Friendswood[a]	13,388	$7,480,850,836	0	0.0%	$0	0.0%
Galena Park	3,288	$1,484,142,708	0	0.0%	$0	0.0%
Hedwig Village	909	$1,463,766,016	0	0.0%	$0	0.0%
Hilshire Village	321	$267,434,315	0	0.0%	$0	0.0%
Houston[b]	588,144	$620,018,000,000	0	0.0%	$0	0.0%
Humble	4,829	$5,480,480,895	0	0.0%	$0	0.0%
Hunters Creek Village	1,572	$2,319,999,302	0	0.0%	$0	0.0%
Jacinto City	2,986	$1,619,709,430	0	0.0%	$0	0.0%
Jersey Village	2,439	$2,863,511,476	0	0.0%	$0	0.0%
Katy[a]	6,322	$3,789,785,990	0	0.0%	$0	0.0%
La Porte	12,424	$8,136,585,709	1,409	11.3%	$780,982,276	9.6%
League City[b]	394	$343,758,798	0	0.0%	$0	0.0%
Missouri City[a]	24,325	$14,945,042,054	0	0.0%	$0	0.0%
Morgan's Point	160	$324,507,417	35	21.9%	$29,141,796	9.0%
Nassau Bay	1,574	$2,104,459,129	0	0.0%	$0	0.0%
Pasadena	40,606	$27,932,208,850	129	0.3%	$41,485,079	0.1%
Pearland[b]	809	$878,335,161	0	0.0%	$0	0.0%
Piney Point Village	1,195	$2,383,712,681	0	0.0%	$0	0.0%

Jurisdiction	Total Buildings in Jurisdiction		Buildings Within the 1-Percent Annual Chance Coastal Flood Hazard Area			
			Number of Buildings		Replacement Value	
	Count	Replacement Value	Count	% of Jurisdiction Total	Value	% of Jurisdiction Total
Seabrook	4,374	$3,115,949,582	1,264	28.9%	$578,205,911	18.6%
Shoreacres	657	$268,042,585	383	58.3%	$146,103,146	54.5%
South Houston[b]	5,001	$2,688,145,186	0	0.0%	$0	0.0%
Southside Place	508	$604,745,337	0	0.0%	$0	0.0%
Spring Valley Village	1,509	$1,568,469,594	0	0.0%	$0	0.0%
Stafford[a]	5,055	$7,232,853,862	0	0.0%	$0	0.0%
Taylor Lake Village	1,390	$854,884,292	0	0.0%	$0	0.0%
Tomball	3,632	$4,292,188,724	0	0.0%	$0	0.0%
Waller[a]	1,097	$270,979,210	0	0.0%	$0	0.0%
Webster	2,379	$7,009,654,505	0	0.0%	$0	0.0%
West University Place	5,525	$4,782,421,662	0	0.0%	$0	0.0%
Unincorporated County	577,476	$402,152,000,000	6	<0.1%	$3,810,610	<0.1%
Harris County Total	1,355,235	$1,167,593,433,721	3,280	0.2%	$1,590,950,786	0.1%

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Table 12-14. General Building Stock Located in the Hurricane Harvey Flood Inundation Area

Jurisdiction	Total Buildings in Jurisdiction		Buildings Within the Hurricane Harvey Flood Inundation Area			
			Number of Buildings		Replacement Value	
	Count	Replacement Value	Count	% of Jurisdiction Total	Value	% of Jurisdiction Total
Baytown[b]	20,670	$12,477,103,055	7,650	37.0%	$3,654,196,898	29.3%
Bellaire	6,503	$6,921,381,237	2,965	45.6%	$3,011,565,979	43.5%
Bunker Hill Village	1,301	$1,718,400,551	7	0.5%	$7,670,690	0.4%
Deer Park	11,370	$7,220,366,741	4,362	38.4%	$3,488,813,869	48.3%
El Lago	1,103	$579,556,830	558	50.6%	$307,948,078	53.1%
Friendswood[a]	13,388	$7,480,850,836	4,536	33.9%	$2,399,294,104	32.1%
Galena Park	3,288	$1,484,142,708	1,296	39.4%	$613,809,453	41.4%
Hedwig Village	909	$1,463,766,016	52	5.7%	$127,901,150	8.7%
Hilshire Village	321	$267,434,315	233	72.6%	$196,298,871	73.4%
Houston[b]	588,144	$620,018,000,000	231,464	39.4%	$227,934,000,000	36.8%
Humble	4,829	$5,480,480,895	2,006	41.5%	$2,092,160,260	38.2%
Hunters Creek Village	1,572	$2,319,999,302	310	19.7%	$461,349,585	19.9%
Jacinto City	2,986	$1,619,709,430	1,127	37.7%	$585,734,625	36.2%
Jersey Village	2,439	$2,863,511,476	799	32.8%	$903,505,005	31.6%
Katy[a]	6,322	$3,789,785,990	1,869	29.6%	$837,513,324	22.1%
La Porte	12,424	$8,136,585,709	5,872	47.3%	$3,948,908,642	48.5%
League City[b]	394	$343,758,798	83	21.1%	$139,662,323	40.6%
Missouri City[a]	24,325	$14,945,042,054	5,872	24.1%	$3,491,577,381	23.4%
Morgan's Point	160	$324,507,417	75	46.9%	$285,604,774	88.0%
Nassau Bay	1,574	$2,104,459,129	598	38.0%	$904,245,265	43.0%
Pasadena	40,606	$27,932,208,850	21,190	52.2%	$15,447,859,233	55.3%
Pearland[b]	809	$878,335,161	539	66.6%	$411,770,459	46.9%
Piney Point Village	1,195	$2,383,712,681	361	30.2%	$655,423,755	27.5%

Jurisdiction	Total Buildings in Jurisdiction		Buildings Within the Hurricane Harvey Flood Inundation Area			
			Number of Buildings		Replacement Value	
	Count	Replacement Value	Count	% of Jurisdiction Total	Value	% of Jurisdiction Total
Seabrook	4,374	$3,115,949,582	2,101	48.0%	$1,436,412,265	46.1%
Shoreacres	657	$268,042,585	405	61.6%	$161,597,902	60.3%
South Houston[b]	5,001	$2,688,145,186	3,777	75.5%	$2,088,150,441	77.7%
Southside Place	508	$604,745,337	159	31.3%	$162,140,165	26.8%
Spring Valley Village	1,509	$1,568,469,594	516	34.2%	$543,476,868	34.7%
Stafford[a]	5,055	$7,232,853,862	2,434	48.2%	$3,577,136,619	49.5%
Taylor Lake Village	1,390	$854,884,292	540	38.8%	$371,383,209	43.4%
Tomball	3,632	$4,292,188,724	705	19.4%	$819,695,569	19.1%
Waller[a]	1,097	$270,979,210	351	32.0%	$233,307	0.1%
Webster	2,379	$7,009,654,505	1,032	43.4%	$4,221,700,450	60.2%
West University Place	5,525	$4,782,421,662	1,610	29.1%	$1,392,370,834	29.1%
Unincorporated County	577,476	$402,152,000,000	189,460	32.8%	$129,780,000,000	32.3%
Harris County Total	1,355,235	$1,167,593,433,721	496,914	36.7%	$416,461,111,350	35.7%

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Table 12-15. General Building Stock Located in the 1 Percent Annual Chance Riverine Flood Hazard Area by Occupancy

Jurisdiction	Buildings Within the 1-Percent Annual Chance Riverine Flood Hazard Area			
	Residential	Commercial	Industrial	Other[c]
Baytown[b]	618	44	2	2
Bellaire	5,199	148	0	12
Bunker Hill Village	0	0	0	0
Deer Park	214	125	44	1
El Lago	680	18	0	5
Friendswood[a]	1,148	27	0	1
Galena Park	106	2	0	0
Hedwig Village	0	0	0	0
Hilshire Village	0	0	0	0
Houston[b]	61,486	4,121	111	211
Humble	173	32	3	1
Hunters Creek Village	45	0	0	0
Jacinto City	0	1	0	0
Jersey Village	666	1	0	1
Katy[a]	372	69	5	4
La Porte	485	72	0	0
League City[b]	43	2	0	0
Missouri City[a]	101	24	0	10
Morgan's Point	0	0	0	0
Nassau Bay	1,196	21	0	0
Pasadena	2,388	291	10	6
Pearland[b]	52	9	7	0
Piney Point Village	63	2	0	0
Seabrook	1,844	164	2	2
Shoreacres	274	0	0	0

Jurisdiction	Buildings Within the 1-Percent Annual Chance Riverine Flood Hazard Area			
	Residential	Commercial	Industrial	Other[c]
South Houston[b]	843	388	10	4
Southside Place	356	5	0	1
Spring Valley Village	7	0	0	0
Stafford[a]	0	0	0	0
Taylor Lake Village	632	2	0	0
Tomball	6	5	0	0
Waller[a]	144	41	4	2
Webster	695	22	2	1
West University Place	804	4	0	0
Unincorporated County	29,587	2,650	306	123
Harris County Total	110,227	8,290	506	387

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Table 12-16. General Building Stock Located in the 1 Percent Annual Chance Coastal Flood Hazard Area by Occupancy

Jurisdiction	Buildings Within the 1-Percent Annual Chance Coastal Flood Hazard Area			
	Residential	Commercial	Industrial	Other[c]
Baytown[b]	52	2	0	0
Bellaire	0	0	0	0
Bunker Hill Village	0	0	0	0
Deer Park	0	0	0	0
El Lago	0	0	0	0
Friendswood[a]	0	0	0	0
Galena Park	0	0	0	0
Hedwig Village	0	0	0	0
Hilshire Village	0	0	0	0
Houston[b]	0	0	0	0
Humble	0	0	0	0
Hunters Creek Village	0	0	0	0
Jacinto City	0	0	0	0
Jersey Village	0	0	0	0
Katy[a]	0	0	0	0
La Porte	1,368	39	0	2
League City[b]	0	0	0	0
Missouri City[a]	0	0	0	0
Morgan's Point	35	0	0	0
Nassau Bay	0	0	0	0
Pasadena	127	2	0	0
Pearland[b]	0	0	0	0
Piney Point Village	0	0	0	0
Seabrook	1,226	34	0	4
Shoreacres	382	1	0	0

Jurisdiction	Buildings Within the 1-Percent Annual Chance Coastal Flood Hazard Area			
	Residential	Commercial	Industrial	Other[c]
South Houston[b]	0	0	0	0
Southside Place	0	0	0	0
Spring Valley Village	0	0	0	0
Stafford[a]	0	0	0	0
Taylor Lake Village	0	0	0	0
Tomball	0	0	0	0
Waller[a]	0	0	0	0
Webster	0	0	0	0
West University Place	0	0	0	0
Unincorporated County	4	1	1	0
Harris County Total	3,194	79	1	6

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Table 12-17. General Building Stock Located in the Hurricane Harvey Flood Inundation Area by Occupancy

Jurisdiction	Buildings Within the Hurricane Harvey Flood Inundation Area			
	Residential	Commercial	Industrial	Other[c]
Baytown[b]	6,956	660	7	27
Bellaire	2,912	46	0	7
Bunker Hill Village	7	0	0	0
Deer Park	3,872	423	48	19
El Lago	552	3	0	3
Friendswood[a]	4,417	49	0	70
Galena Park	1,168	102	4	22
Hedwig Village	34	17	0	1
Hilshire Village	223	8	0	2
Houston[b]	211,854	17,940	784	886
Humble	1,660	307	29	10
Hunters Creek Village	305	2	0	3
Jacinto City	1,042	80	0	5
Jersey Village	765	33	0	1
Katy[a]	1,559	279	8	23
La Porte	5,218	621	13	20
League City[b]	75	7	0	1
Missouri City[a]	5,637	204	7	24
Morgan's Point	66	6	3	0
Nassau Bay	562	36	0	0
Pasadena	18,944	2,106	64	76
Pearland[b]	523	13	3	0
Piney Point Village	359	1	0	1
Seabrook	1,938	156	2	5
Shoreacres	405	0	0	0

Jurisdiction	Buildings Within the Hurricane Harvey Flood Inundation Area			
	Residential	Commercial	Industrial	Other[c]
South Houston[b]	3,047	681	23	26
Southside Place	158	1	0	0
Spring Valley Village	501	15	0	0
Stafford[a]	1,797	614	16	7
Taylor Lake Village	538	2	0	0
Tomball	513	177	4	11
Waller[a]	270	67	10	4
Webster	641	374	2	15
West University Place	1,584	23	0	3
Unincorporated County	178,641	9,466	908	445
Harris County Total	458,743	34,519	1,935	1,717

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Table 12-18. Estimated Damage to Buildings from the 1 Percent Annual Chance Riverine Flood

Jurisdiction	Building Losses Due to the 1-Percent Annual Chance Riverine Flood				
				Total, All Occupancy Classes	
	Residential	Commercial	Other[c]	Loss	% of Jurisdiction Total Building Value
Baytown[b]	$55,837,129	$8,592,720	$1,477,024	$65,906,873	0.5%
Bellaire	$464,139,113	$29,024,624	$4,016,616	$497,180,353	7.2%
Bunker Hill Village	$0	$0	$0	$0	0.0%
Deer Park	$5,052,571	$15,731,694	$2,562,511	$23,346,776	0.3%
El Lago	$59,492,928	$1,303,761	$211,894	$61,008,583	10.5%
Friendswood[a]	$61,955,283	$22,676,080	$59,258	$84,690,621	1.1%
Galena Park	$151,456	$0	$0	$151,456	<0.1%
Hedwig Village	$0	$0	$0	$0	0.0%
Hilshire Village	$0	$0	$0	$0	0.0%
Houston[b]	$4,371,988,046	$1,017,315,672	$101,939,438	$5,491,243,156	0.9%
Humble	$7,842,338	$39,173,359	$294,291	$47,309,988	0.9%
Hunters Creek Village	$2,646,939	$0	$0	$2,646,939	0.1%
Jacinto City	$0	$76,372	$0	$76,372	<0.1%
Jersey Village	$65,398,815	$455,532	$8,370,457	$74,224,804	2.6%
Katy[a]	$6,808,147	$4,094,945	$40,402	$10,943,494	0.3%
La Porte	$16,230,877	$3,893,079	$0	$20,123,956	0.2%
League City[b]	$3,150,833	$0	$0	$3,150,833	0.9%
Missouri City[a]	$21,847	$1,079,458	$1,022,552	$2,123,856	<0.1%
Morgan's Point	$0	$0	$0	$0	0.0%
Nassau Bay	$213,086,461	$8,253,615	$0	$221,340,076	10.5%
Pasadena	$168,183,592	$22,378,593	$2,633,463	$193,195,648	0.7%
Pearland[b]	$4,531,722	$1,249,760	$485,699	$6,267,181	0.7%
Piney Point Village	$7,672,761	$17,259,850	$0	$24,932,611	1.0%
Seabrook	$183,950,988	$34,093,575	$3,127,264	$221,171,827	7.1%

Jurisdiction	Building Losses Due to the 1-Percent Annual Chance Riverine Flood				
				Total, All Occupancy Classes	
	Residential	Commercial	Other[c]	Loss	% of Jurisdiction Total Building Value
Shoreacres	$26,661,555	$0	$0	$26,661,555	9.9%
South Houston[b]	$32,709,363	$26,599,005	$1,167,869	$60,476,237	2.2%
Southside Place	$22,761,492	$125,874	$0	$22,887,366	3.8%
Spring Valley Village	$556,932	$0	$0	$556,932	<0.1%
Stafford[a]	$0	$0	$0	$0	0.0%
Taylor Lake Village	$78,088,908	$393,042	$0	$78,481,951	9.2%
Tomball	$291,354	$245,654	$0	$537,008	<0.1%
Waller[a]	$426,569	$261,233	$275	$688,077	0.3%
Webster	$56,299,639	$6,860,391	$0	$63,160,030	0.9%
West University Place	$25,627,757	$46,688	$0	$25,674,445	0.5%
Unincorporated County	$1,390,176,975	$312,901,974	$74,311,138	$1,777,390,087	0.4%
Harris County Total	$7,331,742,390	$1,574,086,550	$201,720,151	$9,107,549,091	0.8%

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Table 12-19. Estimated Damage to Buildings from the 1 Percent Annual Chance Coastal Flood

Jurisdiction	Building Losses Due to the 1-Percent Annual Chance Coastal Flood				
				Total, All Occupancy Classes	
	Residential	Commercial	Other[c]	Loss	% of Jurisdiction Total Building Value
Baytown[b]	$12,166,481	$0	$267,346	$12,433,827	0.1%
Bellaire	$0	$0	$0	$0	0.0%
Bunker Hill Village	$0	$0	$0	$0	0.0%
Deer Park	$0	$0	$0	$0	0.0%
El Lago	$0	$0	$0	$0	0.0%
Friendswood[a]	$0	$0	$0	$0	0.0%
Galena Park	$0	$0	$0	$0	0.0%
Hedwig Village	$0	$0	$0	$0	0.0%
Hilshire Village	$0	$0	$0	$0	0.0%
Houston[b]	$0	$0	$0	$0	0.0%
Humble	$0	$0	$0	$0	0.0%
Hunters Creek Village	$0	$0	$0	$0	0.0%
Jacinto City	$0	$0	$0	$0	0.0%
Jersey Village	$0	$0	$0	$0	0.0%
Katy[a]	$0	$0	$0	$0	0.0%
La Porte	$170,385,830	$0	$5,034,627	$175,420,458	2.2%
League City[b]	$0	$0	$0	$0	0.0%
Missouri City[a]	$0	$0	$0	$0	0.0%
Morgan's Point	$9,877,902	$0	$0	$9,877,902	3.0%
Nassau Bay	$0	$0	$0	$0	0.0%
Pasadena	$6,427,490	$0	$14,466	$6,441,956	<0.1%
Pearland[b]	$0	$0	$0	$0	0.0%
Piney Point Village	$0	$0	$0	$0	0.0%
Seabrook	$273,976,719	$1,543,522	$35,358,709	$310,878,950	10.0%

Jurisdiction	Building Losses Due to the 1-Percent Annual Chance Coastal Flood				
				Total, All Occupancy Classes	
	Residential	Commercial	Other[c]	Loss	% of Jurisdiction Total Building Value
Shoreacres	$116,052,079	$0	$3,608,869	$119,660,947	44.6%
South Houston[b]	$0	$0	$0	$0	0.0%
Southside Place	$0	$0	$0	$0	0.0%
Spring Valley Village	$0	$0	$0	$0	0.0%
Stafford[a]	$0	$0	$0	$0	0.0%
Taylor Lake Village	$0	$0	$0	$0	0.0%
Tomball	$0	$0	$0	$0	0.0%
Waller[a]	$0	$0	$0	$0	0.0%
Webster	$0	$0	$0	$0	0.0%
West University Place	$0	$0	$0	$0	0.0%
Unincorporated County	$63,725	$0	$20,223	$83,948	<0.1%
Harris County Total	$588,950,227	$1,543,522	$44,304,240	$634,797,989	0.1%

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Table 12-20. Estimated Damage to Buildings from the Hurricane Harvey Flood Scenario

Jurisdiction	Building Losses Due to the Hurricane Harvey Flood Scenario				
				Total, All Occupancy Classes	
	Residential	Commercial	Other[c]	Loss	% of Jurisdiction Total Building Value
Baytown[b]	$48,893,097	$14,490,176	$1,286,901	$64,670,174	0.5%
Bellaire	$361,146,177	$34,236,063	$3,547,823	$398,930,064	5.8%
Bunker Hill Village	$0	$0	$0	$0	0.0%
Deer Park	$9,784,094	$95,781,326	$56,689,705	$162,255,126	2.2%
El Lago	$65,495,006	$474,544	$2,448,300	$68,417,850	11.8%
Friendswood[a]	$137,345,545	$42,762,443	$0	$180,107,987	2.4%
Galena Park	$3,708,390	$283,344	$0	$3,991,734	0.3%
Hedwig Village	$0	$0	$0	$0	0.0%
Hilshire Village	$0	$0	$0	$0	0.0%
Houston[b]	$3,618,577,588	$2,282,898,624	$168,808,195	$6,070,284,407	1.0%
Humble	$19,093,013	$108,831,948	$1,996,941	$129,921,902	2.4%
Hunters Creek Village	$17,337,911	$0	$0	$17,337,911	0.7%
Jacinto City	$0	$1,433,702	$0	$1,433,702	0.1%
Jersey Village	$26,051,545	$0	$9,387,167	$35,438,712	1.2%
Katy[a]	$5,630,710	$21,718,215	$222,876	$27,571,801	0.7%
La Porte	$105,438,756	$86,341,535	$0	$191,780,291	2.4%
League City[b]	$3,841,380	$2,217,607	$0	$6,058,987	1.8%
Missouri City[a]	$14,628,269	$2,793,574	$2,302,238	$19,724,080	0.1%
Morgan's Point	$8,136,725	$0	$0	$8,136,725	2.5%
Nassau Bay	$67,402,123	$21,044,799	$0	$88,446,922	4.2%
Pasadena	$285,169,726	$119,231,729	$25,989,817	$430,391,272	1.5%
Pearland[b]	$7,917,958	$2,267,876	$90,220	$10,276,054	1.2%
Piney Point Village	$26,102,045	$10,638,149	$0	$36,740,194	1.5%
Seabrook	$262,681,973	$60,089,414	$4,226,319	$326,997,706	10.5%

Jurisdiction	Building Losses Due to the Hurricane Harvey Flood Scenario				
				Total, All Occupancy Classes	
	Residential	Commercial	Other[c]	Loss	% of Jurisdiction Total Building Value
Shoreacres	$73,360,601	$0	$0	$73,360,601	27.4%
South Houston[b]	$47,653,562	$81,122,916	$6,023,718	$134,800,196	5.0%
Southside Place	$21,740,151	$96,367	$0	$21,836,518	3.6%
Spring Valley Village	$0	$0	$0	$0	0.0%
Stafford[a]	$0	$0	$0	$0	0.0%
Taylor Lake Village	$110,724,463	$934,741	$0	$111,659,205	13.1%
Tomball	$420,225	$779,996	$0	$1,200,221	<0.1%
Waller[a]	$17,887	$36,704	$10,398	$64,988	<0.1%
Webster	$11,715,944	$27,699,864	$1,890,617	$41,306,424	0.6%
West University Place	$63,975,131	$1,340,674	$0	$65,315,805	1.4%
Unincorporated County	$1,667,867,758	$706,235,688	$325,934,923	$2,700,038,369	0.7%
Harris County Total	$7,091,857,753	$3,725,782,017	$610,856,158	$11,428,495,927	1.0%

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NFIP Statistics

Participating in the NFIP is voluntary. Communities that choose to participate agree to adopt and implement local floodplain management regulations that protect lives and reduce risk from future flooding. In return, the federal government makes flood insurance available to property owners throughout the community. To join, a community must complete an application; adopt a resolution of intent to participate and cooperate with FEMA; and adopt and submit a floodplain management ordinance that meets or exceeds the minimum NFIP criteria. The ordinance must adopt any FEMA-approved flood mapping for the community (FEMA 2020, FEMA 2022).

To inform this risk assessment, FEMA provided a list of residential properties with NFIP policies, past claims, and multiple claims (repetitive loss properties). Properties with multiple claims are defined as repetitive loss (RL) properties or severe repetitive loss (SRL) properties, based on definitions from two FEMA programs as follows:

- The NFIP defines RL properties as structures that meet either of the following qualifiers:
  - Two or more claims of more than $1,000 paid by NFIP within any rolling 10-year period since 1978
  - Two or more claims (building payments only) that, on average, equal or exceed 25 percent of the current value of the property
- FEMA's Flood Mitigation Assistance (FMA) grant program defines RL properties as structures covered by a contract for flood insurance made available under the NFIP that meet both of the following qualifiers:
  - Has incurred flood-related damage on two occasions, in which the cost of the repair, on average, equaled or exceeded 25 percent of the market value of the structure at the time of each flood event
  - At the time of the second incidence of flood-related damage, the contract for flood insurance contained increased cost of compliance coverage.
- The NFIP defines SRL properties as residential properties covered under an NFIP flood insurance policy that satisfy the third condition below and either of the first two:
  - Four or more separate claim payments for the property (including building and contents) over $5,000 each have occurred, and the cumulative amount of such claims payments exceeded $20,000.
  - At least two separate claims payments for the property (building payments only) have occurred, and the cumulative amount of the building portion of such claims exceeded the current value of the property.
  - For either of the above, at least two of the referenced claims must have occurred within any 10-year- period and must have occurred more than 10 days apart.
- The FMA defines SRL properties as structures covered by a contract for flood insurance that meet one of two qualifiers:
  - Four or separate claims payments (includes building and contents) have been made under flood insurance coverage with the amount of each such claim exceeding $5,000, and with the cumulative amount of such claims payments exceeding $20,000.

- At least two separate claim payments (includes only building) have been made under such coverage, with the cumulative amount of such claims exceeding the market value of the insured structured.

Table 12-21 summarizes NFIP claims, payments, and repetitive loss statistics for Harris County. Locations of the properties with policies, claims, and RL and SRL flooding were geocoded by FEMA with the understanding that differences (and variations in those differences) are possible between listed coordinates of properties and actual locations of property addresses—i.e., the indicators of some locations are more accurate than others. The general areas where RL and SRL properties have been identified are shown on Figure 12-7 and Figure 12-8.

Table 12-21. Harris County NFIP Claims

# Policies [a]	201,533
# Claims (Losses) [b]	206,044
Total Loss Payments [b]	$13,071,610,300.02
# Repetitive Loss Properties (NFIP definition) [c]	7,552
# Repetitive Loss Properties (FMA definition) [c]	3,686
# Severe Repetitive Loss Properties [c]	1,760
# Policies Within the 1% Annual Chance Flood Boundary	N/A

Source: Harris County Flood Control District 2025

a. *Policy totals for Harris County include policies associated with the following locations: Atascocita, Baytown, Channelview, Clear Lake City, Crosby, Cypress, Cypress, Dripping Spring, Highland, Highlands, Hockley, Houston, Houston, Houston Harris, Huffman, Huston, Kingwood, Klein, Lane City, League City, Meadows Place, Mont Belvieu, New Caney, Park Row, Pearland, Port Alto, Porter, S, South Houston, Spring, The Woodlands*

b. *Statistics for Harris County include totals from the following NFIP city addresses: Alief, Angleton, Atascocita, Austin, Austin Co, Barzoria Cty, Batyon, Bay, Bayton, Baytonw, Baytowm, Baytown, Baytown Harris, Bayview, Bldg 14 Houston, Bldg 19 Houston, Bldg 2 Houston, Bldg 4 Houston, Bldg A Houston, Brazoria, Brazoria Cty, Camilla Texas, Caplan, Channelview, Channelview T, Chinquapin, Clear Lake, Clear Lake City, Clear Lk City, Clear Lk Cy, Cold Spring, Coldspring, Crosby, Crosley, Crystal Beach, Cypress, Dayton, Daytown, Dewitt, Dripping Spring, Fivekingwood, Forest Houston, Freddiesville, Freeport, Freeway, Galveston, Galveston City, Gilcrest, Harris, Harris County, Harris Cty, Hgihlands, Highland, Highlands, Hockley, Holiday Shores, Housotn, Houstn, Houston, Houston Goettee, Houston Harris, Houston, Houston, Tx., Huffman, Huffman Harris, Huston, Jamaica Bch Gal, Jones Creek Village, Kemah, Kings, Kingwood, Klein, La Margue, La Marque, Laguna Heights, Lake Maxia, Lane City, League City, Leakey, Lomax, Mallard, Manvel, Matagorda, Matagorda Cty, Meadowgreenhous, Meadows Place, Mont Belvieu, Montgomery City, Montgomery Co, Morningside Dr, N Houston, Near Friendswoo, Near Houston, New Braunsfels, New Caney, Nouston, Oak Island, Palacios, Patton Village, Pearland, Pearleans, Port Isabel, Porter, Pt Bolivar, Rye, Sabrook, San Bernard, San Leon, Sattler, Seabreah, Seadrift, Sequin, Shaver Houston, Sheldon, Shepherd, So Houston, South Houston, Spirng, Splendora, Spring, Spring Harris, Sugar Land, Surfside, Sutherland Springs, Terr Houston, Texas, The Woodlands, Treasure Island, Vlg Houston, W Columbia*

c. *NFIP data was organized by municipality, and County total statistics provided above include statistics from the following communities within Harris County: Baytown, Channelview, Crosby, Cypress, Highland, Highlands, Hockley, Houston, Huffman, Kingwood, New Caney, Pearland, Porter, South Houston, and Spring.*

Figure 12-7. Areas Where Multiple RL Properties Have Been Identified

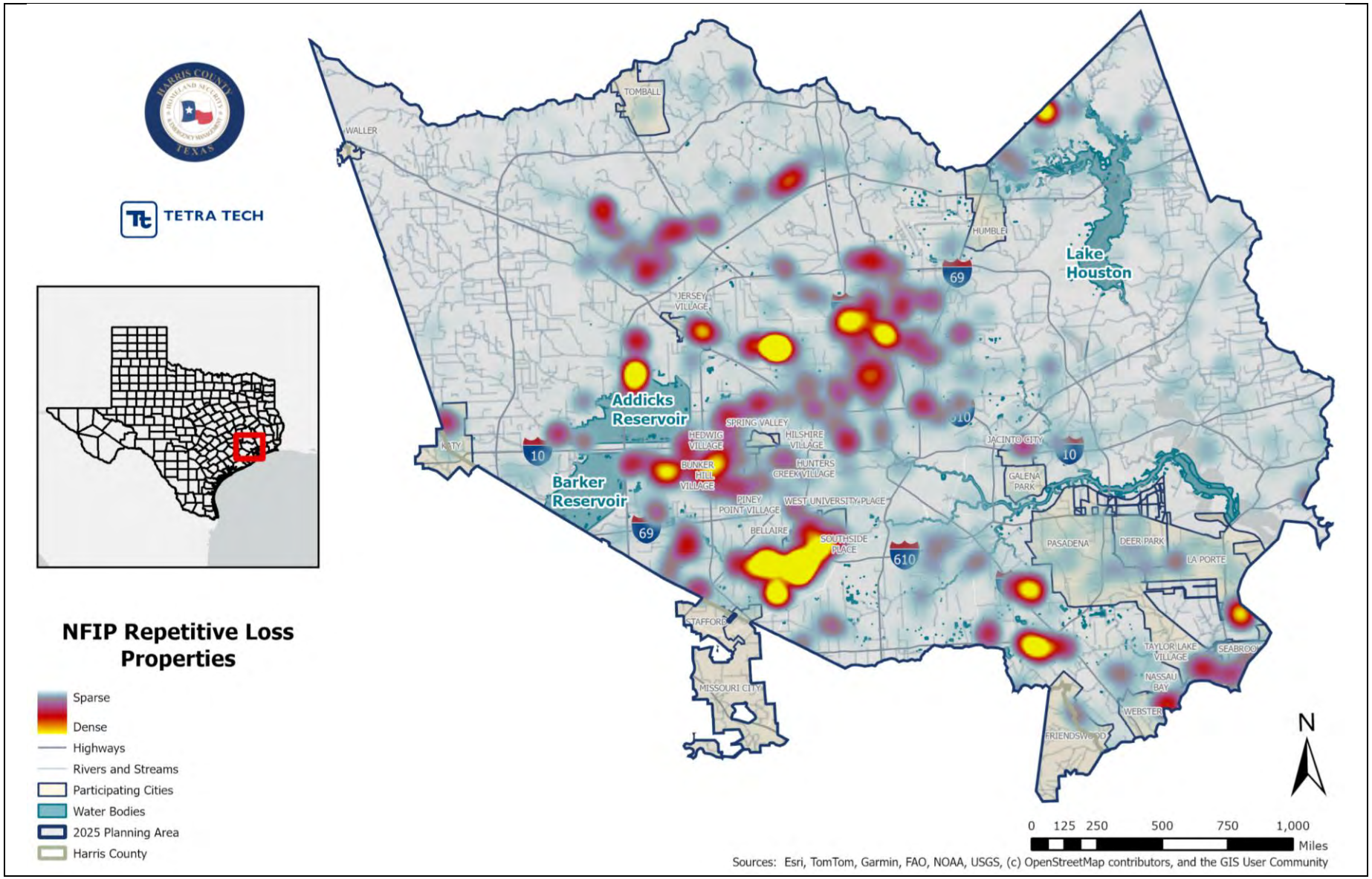
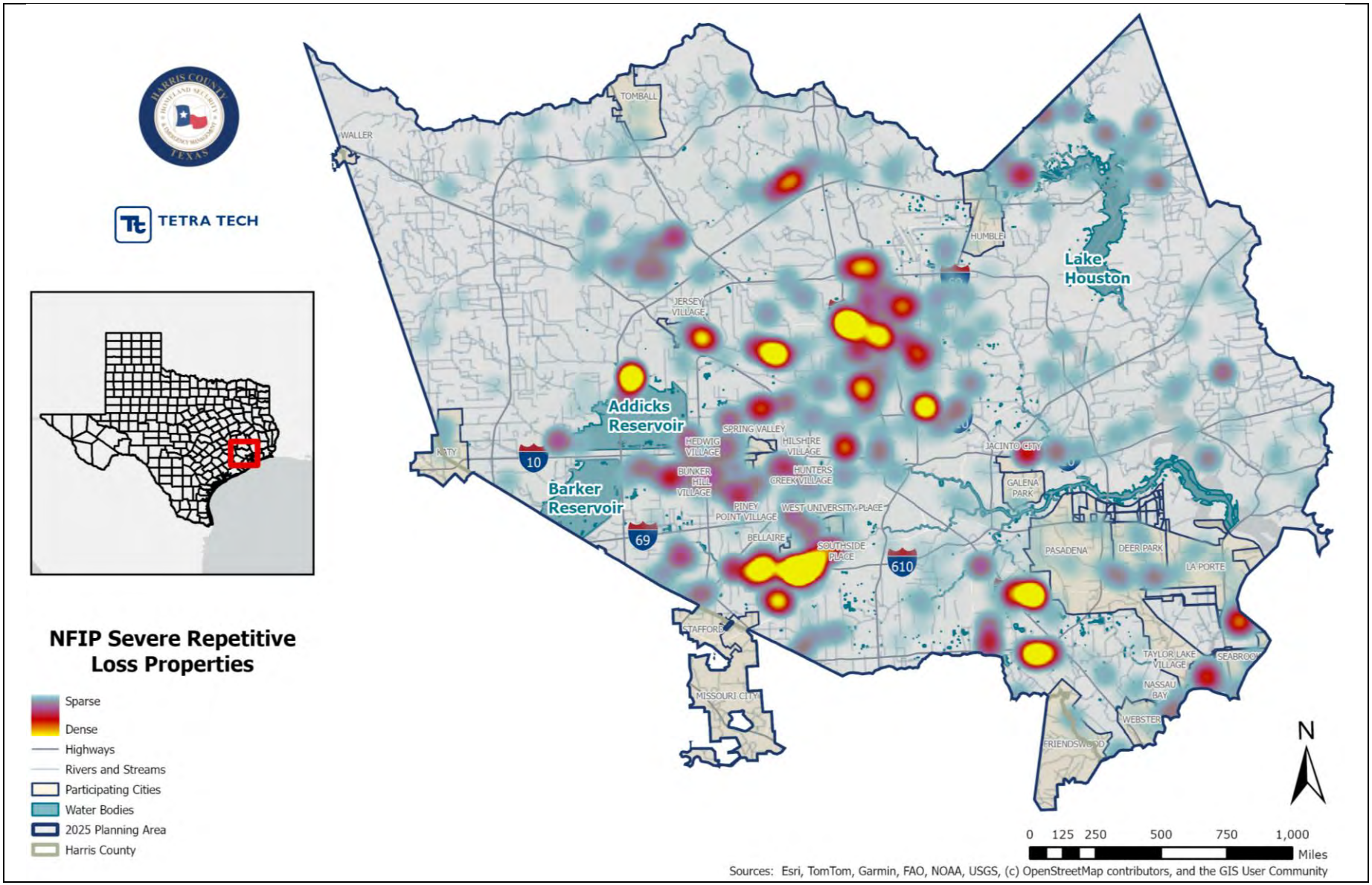


Figure 12-8. Areas Where Multiple SRL Properties Have Been Identified



12.2.3 Community Lifelines and Other Critical Facilities

Flooding in Harris County poses a serious threat to community lifelines and critical facilities, especially during high-intensity rainfall and storm events. Power substations, water treatment plants, emergency shelters, and hospitals located in or near flood-prone areas are vulnerable to inundation, which can disrupt services countywide. Roads and major freeways, such as I-10 and State Highway 288, often become impassable during flooding, cutting off access to emergency care and complicating evacuations. Wastewater systems are also at risk, with floodwaters sometimes causing raw sewage to back up into homes or overflow into bayous, creating serious public health concerns. Protecting lifelines through elevation, redundancy, and relocation is critical to maintaining operational continuity during flood events.

Identifying which critical infrastructure and facilities are vulnerable to flooding is essential, as their damage can have far-reaching consequences. If key services, such as hospitals, emergency shelters, or utility hubs, are directly affected or become inaccessible due to flooded or damaged roads, their ability to function during and after a flood may be severely compromised. Disrupted transportation routes can isolate communities, making it difficult for emergency responders, repair crews, and service providers to reach those in need, especially vulnerable populations. This highlights the importance of ensuring both the resilience of critical facilities and the accessibility of routes leading to them.

Critical facility exposure to the flood hazard was examined. Table 12-22 and Table 12-23 list the number of lifelines and other critical facilities within the 1 percent annual chance riverine and coastal flood hazard areas.

12.2.4 Economy

Flooding can have a major economic impact at both the local and regional levels. These effects may include damage to buildings and the resulting loss in property tax revenue, disruptions to utilities and infrastructure, interruptions to business operations, declines in tourism, and a reduced tax base for Harris County. In flood-affected areas, commercial and industrial properties may require significant repairs, which can interrupt the services they provide. For more details on how buildings in Harris County are directly affected, see the subsection on General Building Stock Impacts.

Flooding can cause extensive damage to public utilities and disruptions to delivery of services. Loss of power and communications may occur, and drinking water and wastewater treatment facilities may be temporarily out of operation. Other economic components, such as loss of facility use, functional downtime, and socio-economic factors, are less measurable with a high degree of certainty.

An important economic impact of flooding is the public cost of removing debris generated from buildings damaged by a flood event. Hazus estimates the amount of debris that may be generated by a given flood event, as listed in Table 12-24.

Table 12-22. Lifelines and Other Critical Facilities in the 1 Percent Annual Chance Riverine Flood Hazard Area

Jurisdiction	Number of Facilities in 1-Percent Annual Chance Flood (Riverine) Hazard Area, by Lifeline Category								Other Critical Facilities	Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems		Count	% of Jurisdiction Total
Baytown	7	150	5	5	0	4	39	5	3	218	19.9%
Bellaire	16	3	11	3	5	29	7	9	22	105	79.5%
Bunker Hill Village	0	0	0	0	0	0	0	0	0	0	0.0%
Deer Park	6	0	0	9	0	1	10	6	1	33	9.8%
El Lago	2	0	1	1	1	5	1	0	1	12	75.0%
Friendswood	3	2	1	1	0	1	5	6	4	23	38.3%
Galena Park	6	3	0	4	0	2	7	7	0	29	18.7%
Hedwig Village	0	0	0	0	0	0	0	0	0	0	0.0%
Hilshire Village	0	0	0	0	0	0	0	0	0	0	0.0%
Houston	443	547	434	373	46	94	1,040	298	371	3,646	19.7%
Humble	3	23	3	5	0	1	21	2	0	58	11.8%
Hunters Creek Village	0	0	0	0	0	0	2	0	0	2	14.3%
Jacinto City	1	0	0	0	0	0	3	1	0	5	7.4%
Jersey Village	1	2	0	1	0	4	15	10	3	36	25.2%
Katy	0	1	0	0	0	0	2	2	2	7	5.2%
La Porte	12	0	3	3	0	2	38	11	3	72	16.8%
League City	2	2	0	0	0	0	0	2	2	8	53.3%
Missouri City	1	0	0	0	0	0	3	0	1	5	10.6%
Morgan's Point	0	0	0	1	0	0	0	0	0	1	3.4%
Nassau Bay	6	1	1	2	1	2	1	2	6	22	59.5%
Pasadena	47	33	18	30	7	8	104	11	15	273	21.9%
Pearland	2	5	0	1	0	0	8	3	1	20	43.5%
Piney Point Village	0	0	0	1	0	0	0	1	0	2	8.7%

Jurisdiction	Number of Facilities in 1-Percent Annual Chance Flood (Riverine) Hazard Area, by Lifeline Category								Other Critical Facilities	Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems		Count	% of Jurisdiction Total
Seabrook	1	5	14	1	0	0	4	2	3	30	34.9%
Shoreacres	4	0	0	0	0	4	3	8	5	24	75.0%
South Houston	3	39	9	8	0	4	13	2	3	81	40.3%
Southside Place	2	0	0	0	0	2	1	4	2	11	55.0%
Spring Valley Village	0	0	0	0	0	0	7	0	1	8	17.0%
Stafford	0	0	0	0	0	0	0	0	0	0	0.0%
Taylor Lake Village	0	0	0	0	0	0	1	2	0	3	21.4%
Tomball	1	19	3	1	0	0	7	4	0	35	8.9%
Waller	0	0	0	0	0	0	1	1	0	2	7.1%
Webster	1	0	1	2	2	0	9	2	2	19	8.5%
West University Place	1	0	0	0	0	0	3	0	11	15	26.3%
Unincorporated County	378	494	132	240	6	63	687	832	165	2,997	18.5%
Harris Total	949	1,329	636	692	68	226	2,042	1,233	627	7,802	19.3%

Note: The critical facilities analyzed are only within the boundaries of Harris County.

Table 12-23. Lifelines and Other Critical Facilities in the 1 Percent Annual Chance Coastal Flood Hazard Area

Jurisdiction	Number of Facilities in 1-Percent Annual Chance Flood (Coastal) Hazard Area, by Lifeline Category								Other Critical Facilities	Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems		Count	% of Jurisdiction Total
Baytown	0	57	0	0	0	0	2	0	0	59	5.4%
Bellaire	0	0	0	0	0	0	0	0	0	0	0.0%
Bunker Hill Village	0	0	0	0	0	0	0	0	0	0	0.0%
Deer Park	0	0	0	0	0	0	0	0	0	0	0.0%
El Lago	0	0	0	0	0	0	0	0	0	0	0.0%
Friendswood	0	0	0	0	0	0	0	0	0	0	0.0%
Galena Park	0	0	0	0	0	0	0	0	0	0	0.0%
Hedwig Village	0	0	0	0	0	0	0	0	0	0	0.0%
Hilshire Village	0	0	0	0	0	0	0	0	0	0	0.0%
Houston	0	0	0	0	0	0	0	0	0	0	0.0%
Humble	0	0	0	0	0	0	0	0	0	0	0.0%
Hunters Creek Village	0	0	0	0	0	0	0	0	0	0	0.0%
Jacinto City	0	0	0	0	0	0	0	0	0	0	0.0%
Jersey Village	0	0	0	0	0	0	0	0	0	0	0.0%
Katy	0	0	0	0	0	0	0	0	0	0	0.0%
La Porte	5	1	5	0	0	3	5	8	11	38	8.9%
League City	0	0	0	0	0	0	0	0	0	0	0.0%
Missouri City	0	0	0	0	0	0	0	0	0	0	0.0%
Morgan’s Point	0	0	0	0	0	0	0	2	0	2	6.9%
Nassau Bay	0	0	0	0	0	0	0	0	0	0	0.0%
Pasadena	2	0	0	0	0	0	0	0	0	2	0.2%
Pearland	0	0	0	0	0	0	0	0	0	0	0.0%
Piney Point Village	0	0	0	0	0	0	0	0	0	0	0.0%
Seabrook	13	2	7	4	0	4	5	6	3	44	51.2%

Jurisdiction	Number of Facilities in 1-Percent Annual Chance Flood (Coastal) Hazard Area, by Lifeline Category								Other Critical Facilities	Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems		Count	% of Jurisdiction Total
Shoreacres	0	0	0	0	0	3	0	3	2	8	25.0%
South Houston	0	0	0	0	0	0	0	0	0	0	0.0%
Southside Place	0	0	0	0	0	0	0	0	0	0	0.0%
Spring Valley Village	0	0	0	0	0	0	0	0	0	0	0.0%
Stafford	0	0	0	0	0	0	0	0	0	0	0.0%
Taylor Lake Village	0	0	0	0	0	0	0	0	0	0	0.0%
Tomball	0	0	0	0	0	0	0	0	0	0	0.0%
Waller	0	0	0	0	0	0	0	0	0	0	0.0%
Webster	0	0	0	0	0	0	0	0	0	0	0.0%
West University Place	0	0	0	0	0	0	0	0	0	0	0.0%
Unincorporated County	7	3	0	2	0	6	0	21	10	49	0.3%
Harris Total	27	63	12	6	0	16	12	40	26	202	0.5%

Note: The critical facilities analyzed are only within the boundaries of Harris County.

Table 12-24. Estimates of Flood-Caused Debris

Jurisdiction	Debris Generated from Flood-Caused Building Damage (tons)		
	1-Percent Annual Chance Riverine Flood	1-Percent Annual Chance Coastal Flood	Hurricane Harvey Flooding
Baytown[b]	13,270.5	2,063.1	32,955.5
Bellaire	9,696.4	0.0	10,828.7
Bunker Hill Village	<0.1	0.0	278.8
Deer Park	318.0	0.0	8,582.5
El Lago	4,940.5	0.1	5,853.4
Friendswood[a]	1,031.0	0.0	3,341.8
Galena Park	242.1	0.0	6,321.9
Hedwig Village	0.0	0.0	787.3
Hilshire Village	34.8	0.0	1,164.1
Houston[b]	172,892.5	0.0	777,778.4
Humble	429.8	0.0	5,875.9
Hunters Creek Village	993.3	0.0	4,082.5
Jacinto City	26.8	0.0	4,299.1
Jersey Village	1,298.4	0.0	972.7
Katy[a]	14.1	0.0	1,576.9
La Porte	1,379.3	11,099.6	16,620.4
League City[b]	5.6	0.0	26.5
Missouri City[a]	48.9	0.0	3,949.8
Morgan's Point	0.0	972.6	1,196.9
Nassau Bay	30,521.9	0.0	17,167.3
Pasadena	9,010.7	452.8	91,731.3
Pearland[b]	0.7	0.0	5.3
Piney Point Village	1,114.6	0.0	4,455.5
Seabrook	20,847.9	30,930.0	33,993.8
Shoreacres	3,713.5	11,435.1	10,653.9
South Houston[b]	2,257.6	0.0	11,938.4
Southside Place	160.4	0.0	284.4
Spring Valley Village	117.1	0.0	3,379.9
Stafford[a]	6.8	0.0	8,457.6
Taylor Lake Village	8,145.5	0.0	13,407.5
Tomball	24.8	0.0	2,897.0
Waller[a]	90.4	0.0	923.6

Jurisdiction	Debris Generated from Flood-Caused Building Damage (tons)		
	1-Percent Annual Chance Riverine Flood	1-Percent Annual Chance Coastal Flood	Hurricane Harvey Flooding
Webster	356.6	0.0	2,322.7
West University Place	362.3	0.0	5,695.9
Unincorporated County	64,605.6	37.5	173,481.8
Harris County Total	347,958.2	56,990.7	1,267,288.8

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

12.2.5 Natural, Historic and Cultural Resources

Flooding significantly impacts Harris County's natural ecosystems, historic landmarks, and cultural institutions. Wetlands and riparian zones that provide flood storage and habitat are often degraded by urban runoff and inundation, especially when these natural areas are fragmented by development. Prolonged flooding can also alter stream channels, increase sedimentation, and degrade water quality in Galveston Bay and local bayous (Environmental Protection Agency \[EPA], 2016). Historic districts such as Freedmen's Town, Houston Heights, and East End—many of which are located near bayous—are at risk of structural damage from repeated flooding. Museums, libraries, and performing arts centers located downtown or near the Museum District may also be affected by rising waters. Without adequate protection, these cultural assets face both physical deterioration and loss of heritage. Preservation planning must integrate flood risk assessments, elevation improvements, and recovery funding strategies to protect Harris County's rich historical and environmental legacy.

12.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

12.3.1 Potential or Planned Development

Future development in Harris County, particularly in western and unincorporated areas, continues to place additional pressure on flood-prone landscapes. As the county experiences sustained growth in residential subdivisions, commercial corridors, and transportation infrastructure, much of this development occurs within or adjacent to mapped floodplains and low-lying bayou systems such as Brays, Greens, and Cypress Creek. Without strategic land-use planning and flood mitigation measures, such as stormwater detention and elevating structures above base flood elevation, new development can displace floodwaters into surrounding neighborhoods, increasing downstream flood risk (Harris County Flood Control District [HCFCD], 2023). Urban expansion also

frequently results in increased impervious cover—like roadways, driveways, and rooftops—reducing natural infiltration and contributing to more frequent and severe localized flooding. To limit future impacts, Harris County is incorporating green infrastructure, buyouts of at-risk properties, and stricter development standards in high-risk areas.

12.3.2 Projected Changes in Population

Harris County is projected to remain one of the fastest-growing counties in Texas, with substantial population increases anticipated through 2045 (Texas Demographic Center, 2023). Much of this population growth is occurring in suburban fringe areas and unincorporated regions that often intersect with flood-prone watersheds. As more people move into these vulnerable areas, the number of residents and critical assets exposed to flooding will rise, straining emergency services and increasing recovery costs after major flood events. The impacts will be particularly pronounced in communities that have historically experienced repetitive loss, including parts of northeast Houston and neighborhoods along the Greens and Halls Bayous. This trend underscores the importance of steering future growth toward areas outside mapped floodplains and enhancing public awareness and regulatory enforcement in high-risk areas.

12.3.3 Climate Change

Climate change is expected to intensify the flooding risks already facing Harris County. Rising temperatures contribute to a more moisture-laden atmosphere, resulting in heavier and more frequent rainfall events, as demonstrated during recent storms such as Hurricane Harvey and Tropical Storm Imelda (IPCC 2021). These events produced rainfall far beyond historical averages, overwhelming bayous, reservoirs, and local stormwater systems. Future conditions are projected to include higher intensity rainfalls over shorter durations, placing greater pressure on aging drainage infrastructure. Additionally, increased temperatures and changing vegetation patterns may alter the hydrologic cycle, reducing soil absorption and increasing surface runoff. These trends suggest that historical data can no longer serve as a sole basis for floodplain mapping or infrastructure design. Climate-responsive planning and engineering practices, including adaptive stormwater systems and updated flood modeling, are essential to protect Harris County’s residents and resources from future climate-related flood hazards.

13. HURRICANES AND COASTAL STORMS

13.1 HAZARD PROFILE

13.1.1 Hazard Description

Hurricanes are classified as tropical cyclones, which include any closed circulation developing around a low-pressure center in which the winds rotate counterclockwise in the northern hemisphere and whose diameter averages 10 to 30 miles across. Coastal storms refer to any type of tropical or non-tropical low pressure circulation that develops and tracks along the coastal shores, bringing high winds, high surf, and heavy precipitation in the process. Tropical cyclones act as a safety valve, limiting the continued build-up of heat and energy in tropical regions by maintaining the atmospheric heat and moisture balance between the tropics and the pole-ward latitudes. The primary damaging forces associated with these storms are high-level sustained winds, heavy precipitation, and tornadoes. Coastal areas are also vulnerable to the additional forces of storm surge, wind-driven waves, and tidal flooding.

The key energy source for these storms is the release of latent heat from the condensation of warm water. Their formation requires a low-pressure disturbance, warm sea surface temperature, rotational force from the spinning of the earth, and the absence of wind shear in the lowest 50,000 feet of the atmosphere. The majority of hurricanes and tropical storms form in the Atlantic Ocean, Caribbean Sea, and the Gulf of Mexico during the Atlantic hurricane season, from June through November. The peak of the Atlantic hurricane season is in early to mid-September. On average, six storms reach hurricane intensity per year (Gray, 2009).

As a hurricane develops, barometric pressure (measured in millibars or inches) at its center falls and winds increase. If the atmospheric and oceanic conditions are favorable, it can intensify into a tropical depression. When maximum sustained winds reach or exceed 39 miles per hour, the system is designated a tropical storm, is given a name, and is closely monitored by the National Hurricane Center in Miami, Florida. When sustained winds reach or exceed 74 miles per hour, the storm is deemed a hurricane. Hurricane intensity is classified using the Saffir-Simpson Hurricane Wind Scale, which rates hurricane intensity on a scale of 1 to 5, with 5 being the most intense.

13.1.2 Location

Hurricane risk in the United States is located along the entire Gulf of Mexico. Thus the entire Texas coast is located within a hurricane susceptible region; including the Harris County planning area. The Harris County planning area coastline is located on the Galveston Bay and Houston Ship Channel, which is directly exposed to the Gulf of Mexico. Thus, hurricane, high-wind and tropical storms are likely to make landfall located within and affect the entire planning area. Due to Harris County's geographic location, this area of the state, and the entire planning area is vulnerable to damage from hurricane winds and to the inland impacts associated with coastal storms.

In evaluating the potential for hazard events of a given magnitude, a mean return period (MRP) is often used. The MRP provides an estimate of the magnitude of an event that may occur within any given year based on past recorded events. MRP is the average period of time, in years, between occurrences of a particular hazard event, equal to the inverse of the annual frequency of exceedance.

Figure 13-1 through Figure 13-3 show the estimated maximum 3-second gust wind speeds that can be anticipated in the study area associated with the 20- and 100-year MRP events as well as a deterministic scenario event looking at observed conditions from Hurricane Alicia. These peak wind speed projections were generated using Hazus. The maximum 3-second hurricane-speed gust wind speeds for Harris County range from 85 mph to 92 mph hurricane speeds for the 20-year MRP event. and 118 mph to 12 mph for the 100-year MRP event. The maximum 3-second gust wind speeds during Hurricane Alicia ranged from 105 mph to 110 mph. The associated impacts and losses from these 20/100-year and Hurricane Alicia model runs are reported in the exposure and vulnerability assessments later in this section.

13.1.3 Extent

The extent of hurricanes and coastal storms in Harris County is characterized by the intensity and physical characteristics of these storm systems as they impact the region. This is defined by a combination of wind speeds, storm surge heights, rainfall totals, storm sizes and duration, and any associated tornado activity. Given the County's geographic location and exposure to the Gulf of Mexico, Harris County remains susceptible to a wide range of storm magnitudes and associated hazards, even from storms that do not make direct landfall within the county boundaries

Storm Surge

Defined as the abnormal rise in seawater generated by a storm, storm surge is particularly relevant for Harris County's low-lying coastal areas and complex bayous, rivers, and channel systems. Storm surge potential is influenced by the storm's intensity, forward speed, angle of approach, among other things. The National Hurricane Center uses the Sea, Lake, and Overland Surges from Hurricanes (SLOSH) model to estimate surge depths; in Harris County, modeled surge heights can range from three to over 20 feet depending on the storm characteristics and landfall location. The National Hurricane Center Storm Surge Risk Maps for the planning area are provided in Figure 13-4.

Wind

Hurricanes are classified based on the Saffir-Simpson Hurricane Wind Scale, which measures sustained wind speeds and categorizes storms from Category 1 (minimal) to Category 5 (catastrophic) as shown in Table 13-1. Due to the planning area's proximity to the Gulf of Mexico, the region is vulnerable to the full spectrum of hurricane categories, with historical events demonstrating the potential for high-intensity storms to reach inland areas. In addition to wind speeds, the extent of hurricanes and coastal storms includes storm surge, central pressure, rainfall rates, and duration.

Figure 13-1. 20-Year Hurricane MRP Peak Wind Gusts

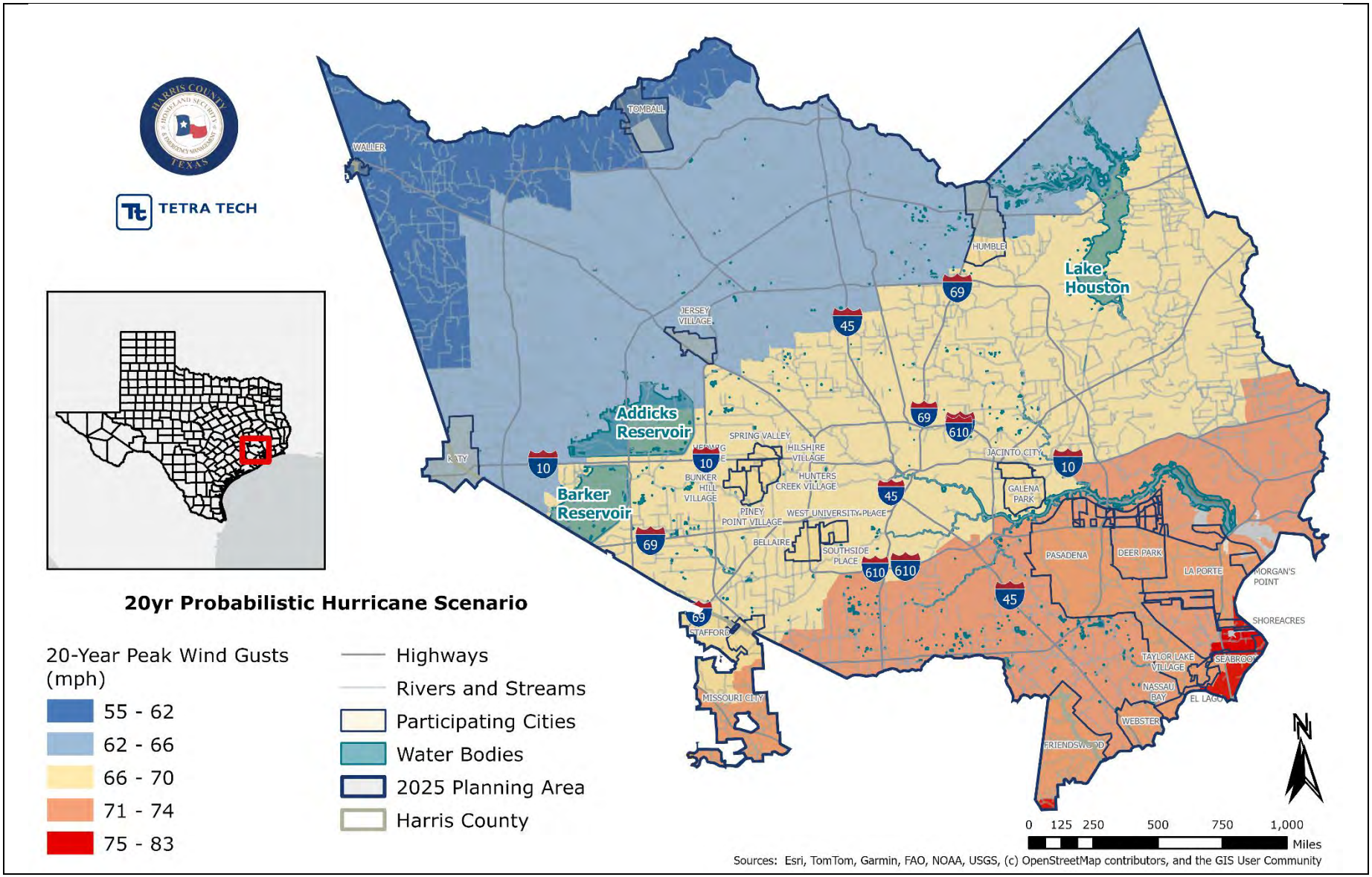


Figure 13-2. 100-Year MRP Hurricane Scenario – Peak Wind Gusts

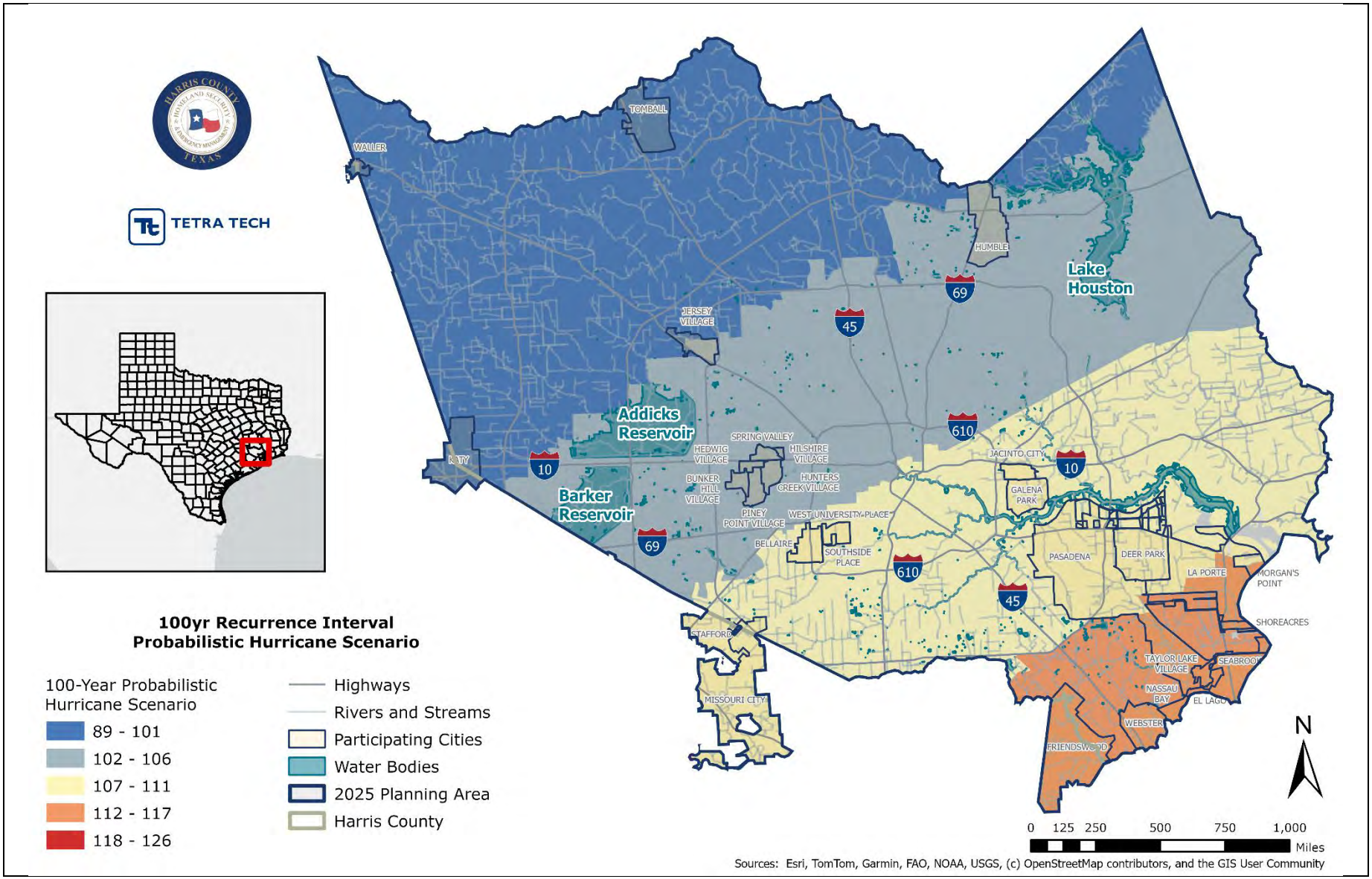


Figure 13-3. Hurricane Alicia Peak Wind Gusts

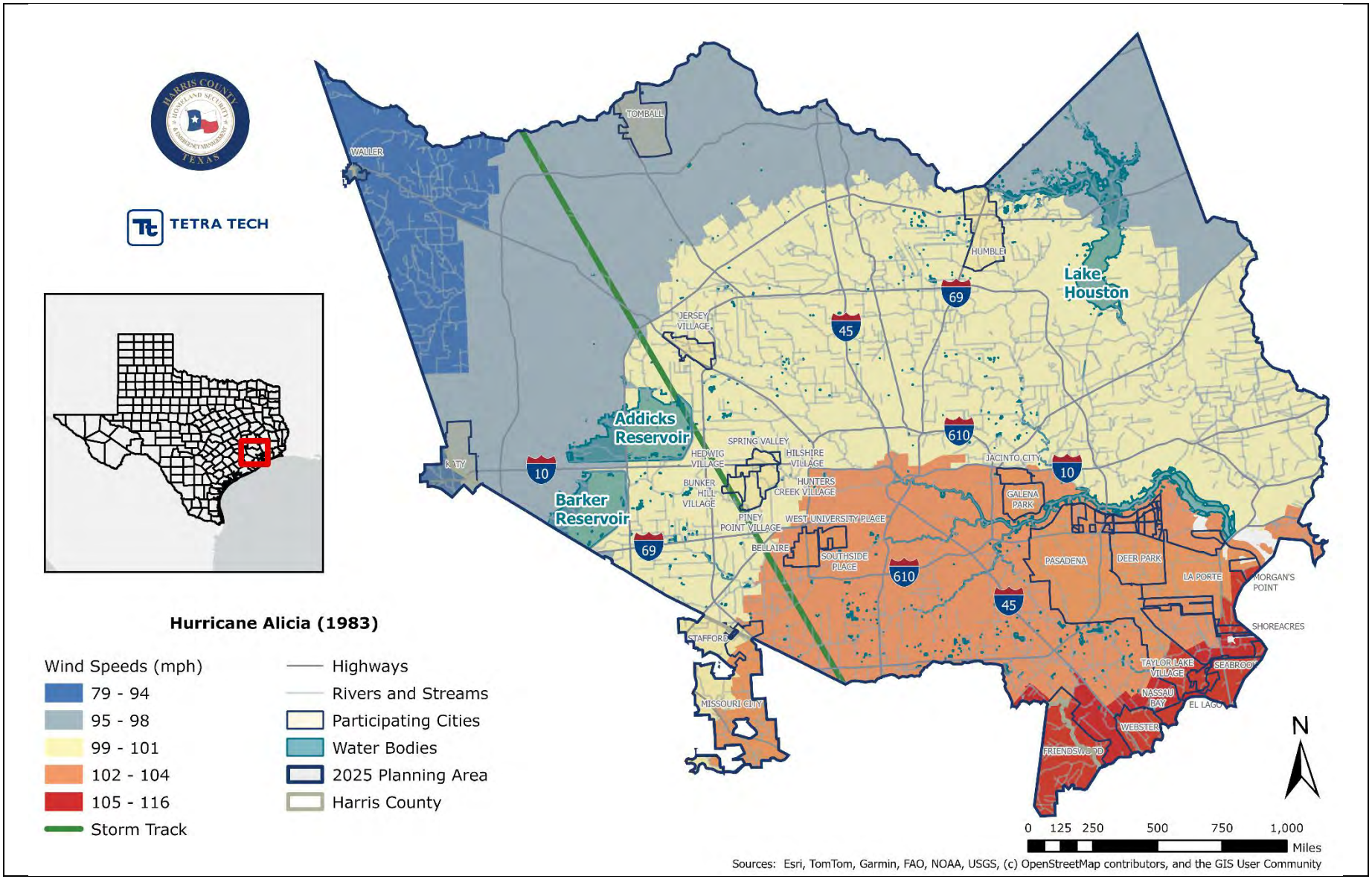
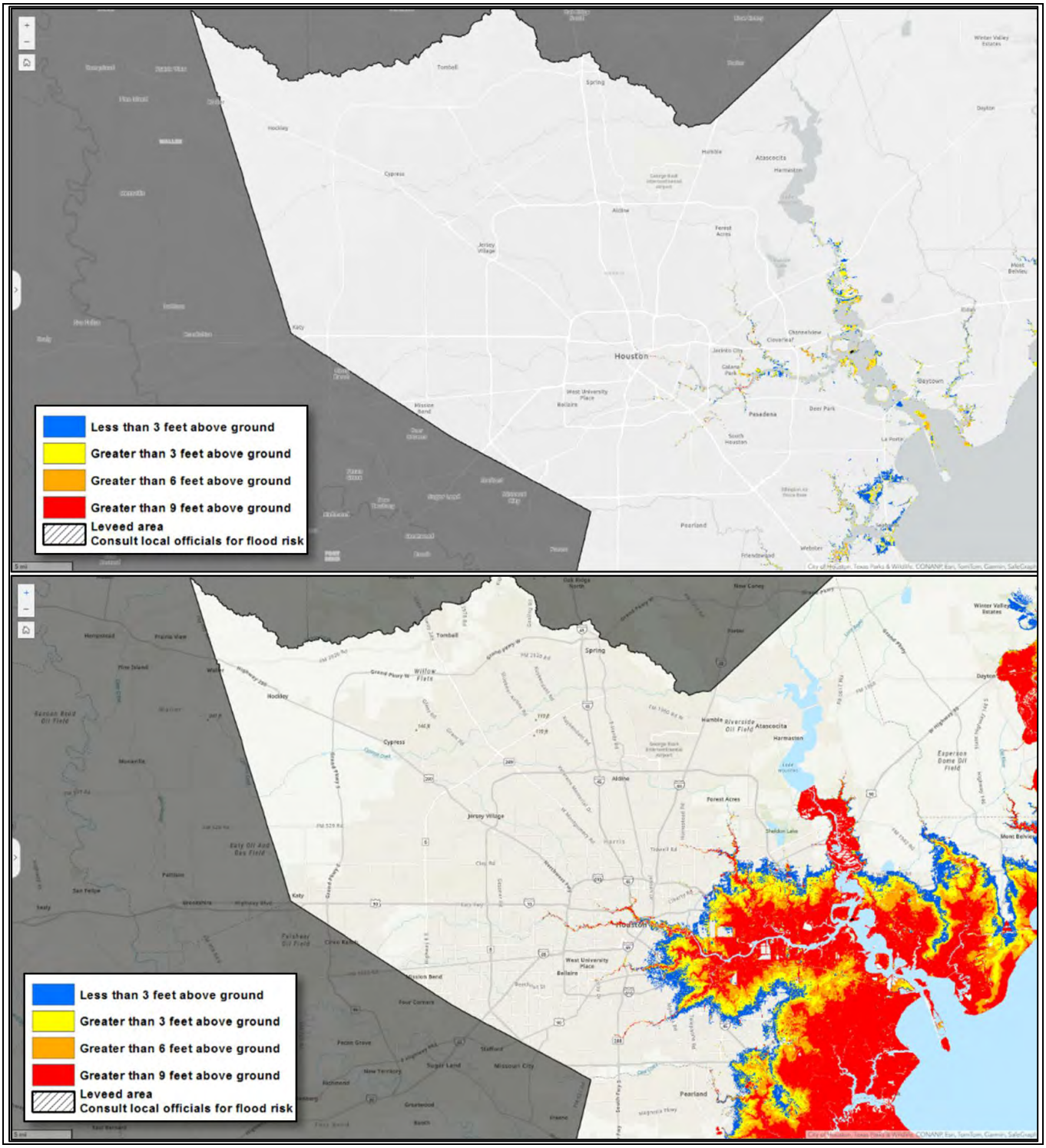


Figure 13-4. SLOSH Storm Surge Inundation Scenarios



Source: (NOAA/NHC 2025)

Table 13-1. Saffir-Simpson Hurricane Wind Speed Scale

Category	Sustained Winds (mph)	Types of Damage Due to Hurricane Winds
Tropical Depression	0 – 38	An area of low pressure, counter-clockwise rotation of clouds, and winds to 38 mph
Tropical Storm	39 – 73	A low pressure area with wind speeds increasing to 39-73 mph. A storm is given a name at his point.
1	74 – 95	Very dangerous winds will produce some damage: Well-constructed frame homes could have damage to roof, shingles, vinyl siding and gutters. Large branches of trees will snap and shallowly rooted trees may be toppled. Extensive damage to power lines and poles likely will result in power outages that could last a few to several days
2	96 – 110	Extremely dangerous winds will cause extensive damage: Well-constructed frame homes could sustain major roof and siding damage. Many shallowly rooted trees will be snapped or uprooted and block numerous roads. Near-total power loss is expected with outages that could last from several days to weeks
3	111 – 129	Devastating damage will occur: Well-built framed homes may incur major damage or removal of roof decking and gable ends. Many trees will be snapped or uprooted, blocking numerous roads. Electricity and water will be unavailable for several days to weeks after the storm passes
4	130 – 156	Catastrophic damage will occur: Well-built framed homes can sustain severe damage with loss of most of the roof structure and/or some exterior walls. Most trees will be snapped or uprooted and power poles downed. Fallen trees and power poles will isolate residential areas. Power outages will last weeks to possibly months. Most of the area will be uninhabitable for weeks or months.
5	157+	Catastrophic damage will occur: A high percentage of framed homes will be destroyed, with total roof failure and wall collapse. Fallen trees and power poles will isolate residential areas. Power outages will last for weeks to possibly months. Most of the area will be uninhabitable for weeks or months.

Source: (NOAA/NHC 2012); (CNRSE n.d.)

Rainfall

The rainfall associated with tropical cyclones can be extreme and long-lasting. Hurricanes and tropical storms may bring sustained rainfall over multiple days, contributing to widespread inland and urban flooding. For example, Hurricane Harvey, in 2017, stalled over southeast Texas and produced over 30 to 50 inches of rainfall in parts of Harris County, demonstrating the potential for catastrophic flooding from slow-moving systems, regardless of wind intensity at landfall. Floodwaters are mapped in Chapter 12.

Size

Other characteristics of storm extent include storm size—defined by the radius of maximum winds and the reach of tropical-storm-force. Large storms, such as Hurricane Ike (2008), impacted a wide swath of the Texas coastline and inland counties, including Harris County, even though the strongest winds did not make direct landfall in the county. Hurricane Alicia, on the other hand, was a much more prominent wind-event for the planning area. See Figure 13-3.

Tornadoes

Tornadoes are also commonly spawned by hurricanes and tropical storms, particularly within the outer rainbands. These tornadoes are typically short-lived but can cause localized damage and further exacerbate storm impacts.

13.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2025, Harris County was included in 16 major disaster (DR) or emergency (EM) declarations for Hurricane, Coastal Storm-related events (FEMA 2025). Table 13-2 lists these declarations.

Table 13-2. FEMA Declarations for Hurricane, Coastal Storm Events in Harris County (1954 to 2025)

Event Date	Declaration Date	Declaration Number	Description
April 29, 1957	April 29, 1957	DR-72-TX	Hurricane, Rain, Wind, Hail & Floods
September 16, 1961	September 16, 1961	DR-118-TX	Hurricane Carla
September 24, 1963	September 24, 1963	DR-159-TX	Hurricane Cindy
August 18 – 20, 1983	August 19, 1983	DR-689-TX	Hurricane Alicia
June 5 – 20, 2001	June 9, 2001	DR-1379-TX	Tropical Storm Allison
September 23 – October 14, 2005	September 24, 2005	DR-1606-TX	Hurricane Rita
August 29 – October 1, 2005	September 2, 2005	EM-3216-TX	Hurricane Katrina Evacuation
September 20 – October 14, 2005	September 21, 2005	EM-3261-TX	Hurricane Rita
August 17 – September 5, 2007	August 18, 2007	EM-3277-TX	Hurricane Dean
September 7 – October 2, 2008	September 13, 2008	DR-1791-TX	Hurricane Ike
August 27 – September 7, 2008	August 29, 2008	EM-3290-TX	Hurricane Gustav
September 7 – 26, 2008	September 10, 2008	EM-3294-TX	Hurricane Ike
August 23 – September 15, 2017	August 25, 2017	DR-4332-TX	Hurricane Harvey
July 25 – 31, 2020	July 26, 2020	EM-3530-TX	Hurricane Hanna
August 23 – 27, 2020	August 24, 2020	EM-3540-TX	Tropical Storms Marco & Laura
July 5 – 9, 2024	July 9, 2024	DR-4798-TX	Hurricane Beryl

Sources: (FEMA 2025)

State Emergency Proclamations

Pursuant to Chapter 433, Texas Government Code, the chief elected official or the governing body of a city or county may request the Governor declare a state of emergency for a jurisdiction or a portion thereof. Furthermore, the Governor may proclaim a state of emergency and issue directives

to control and terminate the emergency and protect life and property. The emergency declaration process is generally *not* used for natural or man-made disasters because:

- It requires action by the Governor to resolve local problems rather than facilitating action by local officials
- The Governor's directives require advance notice before they become effective
- Directives are of very limited duration – 72 hours. Hence, a disaster declaration may be more appropriate for responding to a natural or technological emergencies.

USDA Declarations

The Secretary of Agriculture is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2017 and 2024, was included in zero USDA hurricanes, coastal storms-related agricultural disaster declarations.

Previous Events

NOAA records indicate there have been 86 storms that have either directly impacted Harris County or tracked within 75 miles of the county (see Table 13-3). Of these, eight were classified as "major" hurricanes, with the last one being Hurricane Alicia in 1983. Figure 13-5 summarizes the historical tracks of hurricanes and coastal storms since 1842. Tracks shown are for storms that either crossed through Harris County or are ones that tracked within 75 miles of the county.

Table 13-3. Historical Hurricanes and Coastal Storms in Harris County

Years	Total Storms within 75 miles of Harris County
1842 -- 2024	86
1950 -- 2024	47
2000 -- 2024	14
2020 – 2024	3

Source: (NOAA/NHC 2025)

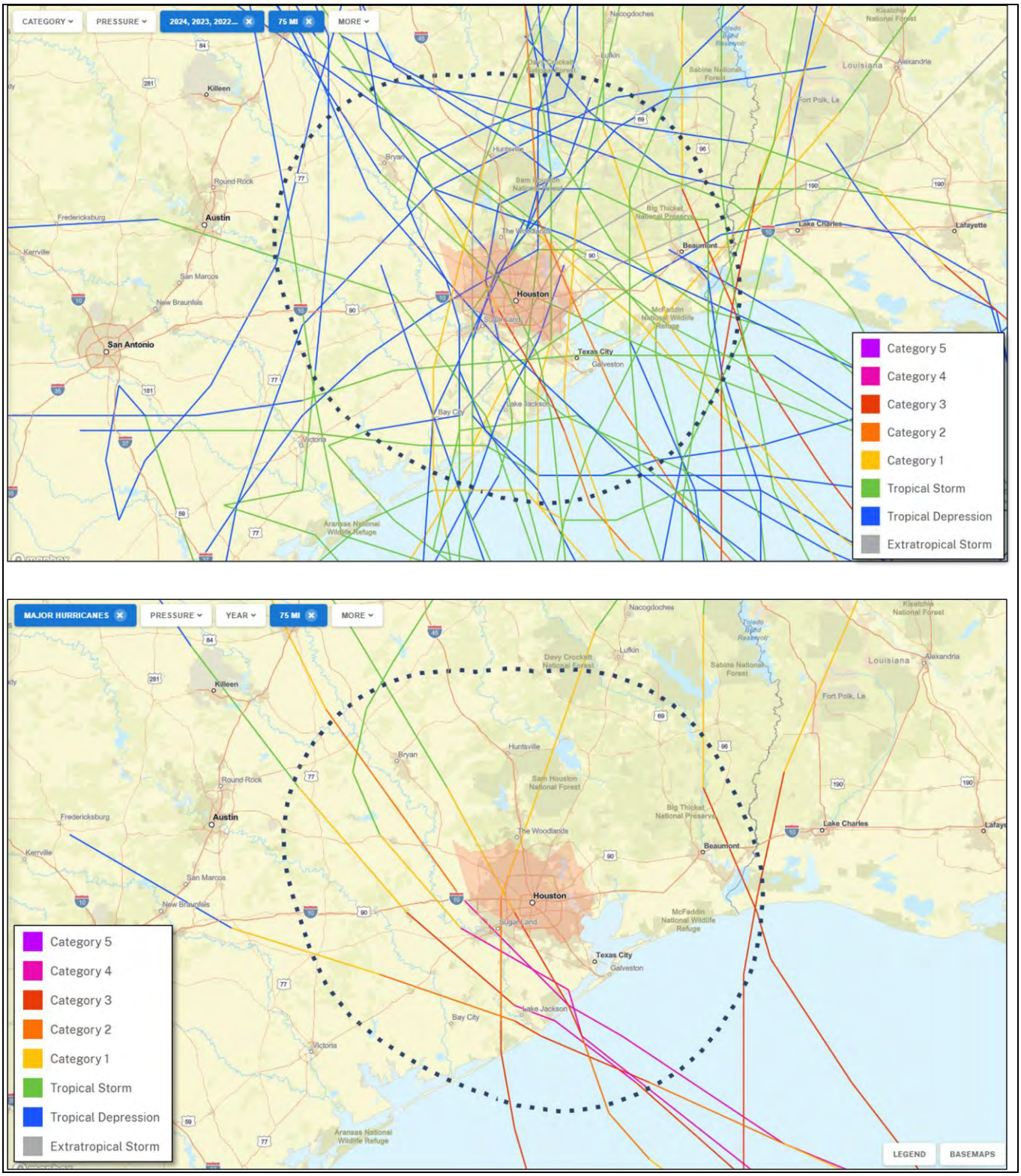
Known hazard events that impacted Harris County between January 2020 and March 2025 are listed in Table 13-4. For events prior to 2020, refer to the 2020 HC MH-MAP.

Table 13-4. Hurricanes, Coastal Storm Events in Harris County (2020 to 2025)

Event Date	Declaration or Proclamation Number	Location Impacted	Description
June 28, 2024 to July 11, 2024	DR-4798-TX	Countywide	Hurricane Beryl
September 12-17, 2021	N/A	Countywide	Hurricane Nicholas
September 17-25, 2020	N/A	Countywide	Tropical Storm Beta

Sources: (NOAA/NHC 2025)

Figure 13-5. Historical Hurricane Tracks Since 1842



Source: (NOAA/NHC 2025)

13.1.5 Probability of Future Occurrences

Information on previous Hurricanes/Coastal Storm occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 13-5. Based on historical records and input from the Steering Committee, the probability of occurrence for Hazard in the County is considered "occasional."

Table 13-5. Probability of Future Hurricane, Coastal Storm Events in Harris County

Hazard Type	Number of Occurrences Between 1950 and 2024	Percent Chance of Occurring in Any Given Year
Hurricane, Coastal Storm	47	62.3%

Sources: (NOAA/NHC 2025)

Climate Change Projections

Climate change is expected to produce warner ocean sea surface temperatures over the coming years, and as water temperatures increase, so will the potential for more frequent and more intense hurricanes and coastal storms.

13.1.6 Cascading Impacts on Other Hazards

Cascading impacts to other hazards are typical with hurricanes and coastal storms, and these are attributed to the impacts brought by high winds and copious rain. The immense size of hurricane wind fields mean that power outages are usually widespread after a hurricane makes landfall. Outages can last for days or event weeks, as was the case with Hurricane Nicholas or Beryl. Widespread utility failures are complicated by extreme heat in many local instances, and the lack of power and air conditioning can lead to serious health risks.

13.2 VULNERABILITY AND IMPACT ASSESSMENT

This risk assessment for the hurricane and coastal storm hazard considered potential impacts of three hurricane scenarios: the 20-year MRP hurricane, the 100-year MRP hurricane, and the Hurricane Alicia scenario.

Unique and Varied Risks

Hurricanes and coastal storms are among the most significant natural hazards affecting Harris County. Although the entire county can experience high winds, heavy rainfall, and tornadoes from tropical systems, the magnitude and type of risk differ substantially across jurisdictions depending on distance from Galveston Bay, local topography, and concentration of vulnerable assets. These events can generate compound hazards, including storm surge, riverine flooding, wind damage, and extended power outages, which impact critical facilities, educational institutions, and health infrastructure.

Harris County's location on the upper Texas Gulf Coast places all jurisdictions within the potential impact area of tropical storms and hurricanes. While coastal and bay-adjacent communities face the most direct exposure to storm surge and high winds, inland jurisdictions remain highly susceptible to rain-induced flooding, tornadoes, and prolonged disruptions to utilities and transportation systems. Historical events such as Hurricanes Ike (2008), Harvey (2017), and Nicholas (2021) demonstrate how widespread hurricane impacts can extend well beyond coastal zones.

Southern and Coastal Jurisdictions

The Cities of Seabrook, El Lago, Taylor Lake Village, La Porte, Morgan's Point, and Shoreacres face the greatest exposure to storm surge and coastal flooding. These communities border Galveston Bay, Clear Lake, and Taylor Lake, where even a moderate hurricane can push significant surge inland. Critical evacuation routes such as NASA Road 1 and SH 146 are prone to inundation, isolating neighborhoods. The City of La Porte and Morgan's Point also face risks to industrial infrastructure, including chemical and petrochemical facilities, where wind damage or floodwaters could create secondary hazardous material releases. San Jacinto College South Campus in the Clear Lake area is susceptible to both surge and wind damage, emphasizing the need for facility hardening and continuity-of-operations planning.

Eastern Jurisdictions

Communities including Baytown, Deer Park, and Pasadena, as well as Goose Creek CISD and Lee College, experience both inland and coastal storm impacts. Baytown lies adjacent to the Houston Ship Channel and is exposed to surge penetration through Cedar Bayou and the San Jacinto River. Industrial complexes and residential areas near the bay are particularly vulnerable. Goose Creek CISD campuses have sustained wind and flood damage in past hurricanes, prompting ongoing retrofits and emergency-shelter coordination. Lee College faces risks from surge and windborne debris due to its proximity to the Baytown industrial corridor.

Central and Urban Jurisdictions

Highly developed areas such as Houston, Bellaire, West University Place, and Southside Place face limited surge risk but remain vulnerable to intense rainfall, flash flooding, and high winds. The University of Houston's main campus sits within the Brays Bayou watershed and is prone to flooding from prolonged rainfall. Hurricane Harvey caused severe impacts to multiple academic buildings and laboratories. Harris Health System facilities—including Ben Taub Hospital, LBJ Hospital, and numerous outpatient clinics—are essential response assets that must maintain operations despite flooding, wind damage, and power loss. Urban tree canopy and older building stock increase exposure to windborne debris hazards and downed power lines.

Northern and Western Jurisdictions

Jurisdictions such as Katy, Tomball, Jersey Village, and Waller, along with Cy-Fair ISD and Spring ISD, are geographically distant from the coast but still subject to hurricane-related rainfall, tornadoes, and prolonged outages. Cy-Fair ISD operates dozens of campuses within the Addicks and Cypress Creek watersheds, where tropical rainfall can inundate roads and schools, disrupting

access and causing closures even when wind speeds have subsided. Spring ISD faces similar rainfall-driven flooding issues in neighborhoods along Cypress and Spring Creeks, compounded by aging drainage infrastructure. These areas also experience occasional wind-generated tornadoes embedded within hurricane rainbands.

Unincorporated and Industrial Areas

Unincorporated portions of east and southeast Harris County, including the Ship Channel industrial corridor, support refineries, pipelines, and energy facilities critical to regional and national supply chains. These facilities face compound hurricane risks—storm surge, wind, and flood impacts that can disrupt fuel production, release pollutants, or cause cascading infrastructure failures. Coordination with HCFCD, HCOHSEM, and the Texas Division of Emergency Management (TDEM) remains essential to ensure continuity and environmental protection.

Areawide Educational Institutions and Health Systems

School Districts (Cy-Fair ISD, Spring ISD, Goose Creek CISD) rely on widespread bus and transportation networks that are disrupted by flood and debris blockages. Many campuses also serve as emergency shelters during hurricanes, requiring backup power and hardened communications. Higher Education Campuses (University of Houston, San Jacinto College, Lee College) must maintain continuity of instruction and protect research facilities from flood and wind damage. Harris Health System operates hospitals and clinics across the county, many of which serve socially vulnerable populations. Maintaining redundant power and flood protection at these facilities is essential for post-storm medical surge capacity.

13.2.1 Life, Health, and Safety

Hurricanes pose a direct threat to the health and safety of residents, particularly in low-lying and coastal areas vulnerable to storm surge. Populations at greatest risk include residents in flood-prone zones, including those along Clear Lake, Buffalo Bayou, and Galveston Bay. It also includes elderly individuals, medically fragile populations, and persons with disabilities, who may have limited mobility or access to evacuation resources. Furthermore, low-income and socially vulnerable populations, who may face challenges in securing temporary housing or transportation.

Primary health and safety impacts include loss of life and injuries from storm surge, debris, falling trees, structural collapse, and traffic accidents during evacuations. Hurricanes also pose health issues related to prolonged power outages (e.g., heat exposure), contaminated floodwaters, and post-storm mold and mildew. Mental health impacts, including stress, anxiety, and trauma stemming from displacement, loss of property, and disaster recovery.

This risk assessment used Hazus to estimate the potential number of people who would be displaced or require short-term shelter following three hurricane scenarios evaluated. Harris County does not open shelters before the landfall of a hurricane or tropical storm. If a shelter is open, for example, after SAR operations, the County defers to the existing agreements in the facilities they have with the American Red Cross, which may or may not be a school facility. Countywide results are shown in Table 13-6.

Table 13-6. Aggregate Estimated Hurricane Impact on Population

Hurricane Scenario	Populations Exposed to the Hazard Area	
	Displaced Population	Persons Seeking Short-Term Sheltering
20-Year MRP	688	887
100-Year MRP	24,733	27,866
Alicia Scenario	4,185	4,943

Source: U.S. Census Bureau 2020; U.S. Census Bureau, American Community Survey 2023; STATS America; Hazus v6.1

13.2.2 General Building Stock

The general building stock in Harris County is vulnerable to damage from high winds, windborne debris, heavy rainfall, and floodwaters associated with hurricanes. Structures built before modern wind and flood-resistant building codes, particularly in older neighborhoods, are more susceptible to wind and water damage. Manufactured homes and mobile housing units, which are especially vulnerable to wind forces. Repetitive loss and severe repetitive loss properties, especially in the floodplains, are at increased risk for repeated hurricane-induced flooding.

Table 13-7 summarizes the estimated damage to buildings county wide from the three evaluated hurricane scenarios. Table 13-8 shows the level of damage for the 20-year and 100-year MRP hurricane scenarios for various occupancy classes. The breakdown by jurisdiction for the 20-year MRP hurricane scenario is presented in Table 13-9.

Table 13-7. Aggregate Estimated Loss for Scenario Hurricane Events

Hurricane Scenario	Estimated Building Loss				
	All Buildings	Residential	Commercial	Industrial	All Other Occupancies
20-Year MRP	$1,298,406,558	$1,179,667,534	$96,152,082	$9,634,197	$12,952,745
100-Year MRP	$21,893,327,157	$17,786,694,277	$3,533,422,545	$275,910,182	$297,300,153
Alicia Scenario	$7,176,816,720	$6,095,688,495	$935,524,985	$69,086,392	$76,516,849

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018, RS Means 2024; Hazus v6.1

Table 13-8. Building Occupancy Impacts

Occupancy Class	Total Number of Buildings in Occupancy	Severity of Expected Damage	20-Year MRP Hurricane		100-Year MRP Hurricane	
			Building Count	% Buildings in Occupancy Class	Building Count	% Buildings in Occupancy Class
Residential Exposure (Single and Multi-Family Dwellings)	1,261,003	NONE	1,215,139	60.9%	768,212	60.9%
		MINOR	37,694	29.4%	370,684	29.4%
		MODERATE	6,140	7.6%	95,436	7.6%
		SEVERE	536	0.6%	7,932	0.6%
		DESTRUCTION	1,494	1.5%	18,739	1.5%
Commercial Buildings	85,457	NONE	82,958	97.1%	56,438	66.0%
		MINOR	1,792	2.1%	18,851	22.1%
		MODERATE	532	0.6%	8,482	9.9%
		SEVERE	169	0.2%	1,673	2.0%
		DESTRUCTION	5	<0.1%	13	<0.1%
Industrial Buildings	4,821	NONE	4,521	93.8%	3,312	68.7%
		MINOR	154	3.2%	746	15.5%
		MODERATE	101	2.1%	560	11.6%
		SEVERE	44	0.9%	197	4.1%
		DESTRUCTION	1	<0.1%	6	0.1%
Government, Religion, Agricultural, and Education Buildings	3,961	NONE	3,587	90.6%	2,783	70.4%
		MINOR	115	2.9%	541	13.7%
		MODERATE	108	2.7%	385	9.7%
		SEVERE	116	2.9%	209	5.3%
		DESTRUCTION	34	0.9%	35	0.9%

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018; Hazus v6.1

Table 13-9. Building Losses Due to a 20-Year MRP Hurricane

Jurisdiction	Estimated Building Losses (All Occupancies)	Estimated Building Losses (Residential)	Estimated Building Losses (Commercial)	Estimated Building Losses (Industrial)	Estimated Building Losses (All Other Occupancies)
Baytown[b]	$210,865,596	$178,783,964	$27,826,897	$315,665	$3,939,069
Bellaire	$1,105,032	$1,063,149	$32,673	$0	$9,210
Bunker Hill Village	$348,234	$348,233	$2	$0	$0
Deer Park	$21,473,721	$19,517,524	$1,632,387	$218,720	$105,091
El Lago	$3,411,507	$3,343,396	$57,379	$1,252	$9,479
Friendswood[a]	$5,674,693	$5,323,264	$342,612	$269	$8,548
Galena Park	$1,897,974	$1,714,667	$84,259	$71,636	$27,412
Hedwig Village	$166,297	$164,489	$1,494	$0	$314
Hilshire Village	$81,510	$80,330	$835	$0	$345
Houston[b]	$286,036,166	$265,170,045	$17,726,032	$1,624,635	$1,515,453
Humble	$10,225,821	$8,756,799	$1,262,946	$85,314	$120,762
Hunters Creek Village	$579,116	$575,830	$0	$0	$3,286
Jacinto City	$1,929,393	$1,712,855	$194,577	$9,014	$12,947
Jersey Village	$418,228	$381,978	$34,006	$833	$1,411
Katy[a]	$0	$0	$0	$0	$0
La Porte	$38,015,062	$28,859,765	$7,711,607	$712,162	$731,528
League City[b]	$644,475	$616,853	$27,417	$0	$205
Missouri City[a]	$241,375	$241,375	$0	$0	$0
Morgan's Point	$783,701	$572,751	$141,645	$12,553	$56,752
Nassau Bay	$3,006,759	$2,652,311	$354,448	$0	$0
Pasadena	$41,583,024	$37,851,184	$3,373,757	$150,328	$207,755
Pearland[b]	$1,461,354	$1,438,389	$10,377	$1,147	$11,441
Piney Point Village	$578,686	$566,348	$1,560	$0	$10,778
Seabrook	$13,701,291	$12,625,475	$975,189	$16,076	$84,550

Jurisdiction	Estimated Building Losses (All Occupancies)	Estimated Building Losses (Residential)	Estimated Building Losses (Commercial)	Estimated Building Losses (Industrial)	Estimated Building Losses (All Other Occupancies)
Shoreacres	$2,571,399	$2,520,103	$50,308	$0	$988
South Houston[b]	$1,892,604	$1,715,972	$157,200	$6,743	$12,689
Southside Place	$181,324	$179,721	$1,322	$147	$134
Spring Valley Village	$393,151	$387,458	$4,027	$0	$1,666
Stafford[a]	$54,436	$54,436	$0	$0	$0
Taylor Lake Village	$4,366,002	$4,293,091	$59,431	$9,620	$3,860
Tomball	$940,484	$834,986	$91,902	$7,074	$6,523
Waller[a]	$6,127	$6,127	$0	$0	$0
Webster	$4,029,724	$2,725,434	$1,270,804	$25	$33,461
West University Place	$1,711,071	$1,708,024	$2,202	$0	$844
Unincorporated County	$638,031,224	$592,881,208	$32,722,787	$6,390,985	$6,036,245
Harris County Total	$1,298,406,558	$1,179,667,534	$96,152,082	$9,634,197	$12,952,745

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018, RS Means 2024; Hazus v6.1

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Potential impacts include structural damage or destruction from wind and debris impacts. Table 13-10 summarizes Hazus estimates of debris that could be generated from the three evaluated hurricane scenarios. Hurricanes and coastal storms can also produce flood damage to foundations, mechanical systems, and interiors, particularly for slab-on-grade structures. Long-term economic losses due to building closures, reduced property values, and rising insurance premiums.

Table 13-10. Aggregate Estimated Hurricane-Caused Debris

Hurricane Scenario	Estimated Debris Created			
	Brick and Wood (tons)	Concrete and Steel (tons)	Tree (tons)	Eligible Tree Volume (cubic yards)
20-Year MRP	143,825.0	352.0	273,640.0	1,117,973.3
100-Year MRP	2,740,362.7	14,878.4	789,433.3	3,997,407.7
Alicia Scenario	916,902.1	2,970.7	560,446.3	2,735,339.1

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018; Hazus v6.1

13.2.3 Community Lifelines and Other Critical Facilities

Hurricanes and coastal storms can severely disrupt community lifelines and critical facilities essential to public safety and recovery operations. Key impacts include the following:

- Transportation Systems—Flooded roadways, damaged bridges, and downed traffic signals restrict evacuation, emergency response, and supply chain continuity. The Port of Houston and nearby airports (e.g., Hobby and Bush Intercontinental) may suspend operations due to storm surge or high winds.
- Energy and Communication Infrastructure—Hurricanes can bring power outages from downed transmission lines and damaged substations that can last days or weeks. They can also create a loss of cell towers and fiber networks, which can hinder emergency communication and public warning systems.
- Water and Wastewater Utilities—Pump stations and wastewater treatment plants located in flood zones are vulnerable to inundation and mechanical failure. Additionally, water supply contamination and boil-water notices are common in the aftermath of major hurricanes.
- Public Safety and Emergency Operations Centers—Police, fire, and EMS facilities may be affected by flooding or wind damage, which can limit emergency response capabilities. Shelters and staging areas must be operational, accessible, and adequately equipped to serve displaced residents.

The potential level of damage to critical facilities from the 20-year and 100-year MRP scenarios is summarized in Table 13-11 and Table 13-12.

Table 13-11. Average Percent Probability of Sustaining Damage (20-Year MRP)

Lifeline Category	Loss of Days	Minor	Moderate	Severe	Complete
Communications	0	2.7%	1.6%	0.5%	<0.1%
Energy	0	4.5%	3.9%	1.2%	<0.1%
Food, Hydration, Shelter	0	1.8%	0.4%	0.1%	<0.1%
Hazardous Materials	0	3.4%	2.0%	0.8%	<0.1%
Health and Medical	0	1.3%	0.6%	<0.1%	<0.1%
Safety and Security	328	2.1%	1.5%	0.1%	<0.1%
Transportation	0	2.9%	1.3%	0.2%	<0.1%
Water Systems	0	3.0%	1.7%	0.4%	<0.1%

Source: ATLAS 2022; BTS/USDOT 2023; CIMA 2024; City of La Porte 2025; City of Seabrook 2025; FleetCor 2024; Harris County 2020, 2024; Harris County Office of Emergency Management 2022-2025; Harris County Public Library, HCUS 2020, HCPCS 2024, HIFLD 2021-2024; METRO 2024; NPMS 2024; SETRAC 2024; TCEQ 2023-2024; Regional Catastrophic Preparedness Grant Program 2023; DSHS 2024; TEA 2023; THC 2023; USACE 2024; Hazus v6.1

Note: The critical facilities analyzed are only within the boundaries of Harris County.

Table 13-12. Average Percent Probability of Sustaining Damage (100-Year MRP)

Lifeline Category	Loss of Days	Minor	Moderate	Severe	Complete
Communications	0	17.2%	14.2%	6.4%	<0.1%
Energy	0	14.7%	17.9%	7.6%	<0.1%
Food, Hydration, Shelter	0	24.2%	8.6%	2.2%	<0.1%
Hazardous Materials	0	15.3%	15.3%	9.6%	<0.1%
Health and Medical	332	13.3%	14.9%	0.8%	<0.1%
Safety and Security	5,423	12.5%	19.5%	1.4%	<0.1%
Transportation	0	20.1%	12.8%	3.2%	<0.1%
Water Systems	0	14.9%	13.7%	5.1%	<0.1%

Source: ATLAS 2022; BTS/USDOT 2023; CIMA 2024; City of La Porte 2025; City of Seabrook 2025; FleetCor 2024; Harris County 2020, 2024; Harris County Office of Emergency Management 2022-2025; Harris County Public Library, HCUS 2020, HCPCS 2024, HIFLD 2021-2024; METRO 2024; NPMS 2024; SETRAC 2024; TCEQ 2023-2024; Regional Catastrophic Preparedness Grant Program 2023; DSHS 2024; TEA 2023; THC 2023; USACE 2024; Hazus v6.1

Note: The critical facilities analyzed are only within the boundaries of Harris County.

13.2.4 Economy

Economic impacts from hurricanes and coastal storms include temporary or longer-term closures, including businesses, local economies across the planning area. Whether planned or unplanned, hurricane hazards can disrupt the Harris County economy as the nation’s fourth largest city.

13.2.5 Natural, Historic and Cultural Resources

Natural resources within Harris County include extensive bayous, wetlands, prairies, and coastal estuaries. These natural provide critical ecosystem services such as floodwater absorption, storm surge protection, and water filtration. However, hurricanes can severely damage these systems through saltwater intrusion, erosion, sediment displacement, and destruction of vegetation. For example, Hurricane Ike (2008) led to large-scale erosion and habitat loss in coastal marshes and wetlands across southeastern Harris County (FEMA 2008).

13.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

These natural features provide critical ecosystem services such as floodwater absorption, storm surge protection, and water filtration. However, hurricanes can severely damage these systems through saltwater intrusion, erosion, sediment displacement, and destruction of vegetation. For example, Hurricane Ike (2008) led to large-scale erosion and habitat loss in coastal marshes and wetlands across southeastern Harris County.

Historic and cultural resources in Harris County include a wide array of sites, such as NASA's Johnson Space Center, the San Jacinto Battleground State Historic Site (including the Battleship Texas), and numerous historic districts, museums, and landmarks throughout Houston and its surrounding municipalities. These assets are often located near waterways or in older, less elevated areas that may not have been designed with modern flood and wind standards in mind. Coastal storms threaten these sites through flooding, wind damage, and mold or deterioration caused by water intrusion. Notably, Hurricane Harvey (2017) caused significant flood damage to historic structures in downtown Houston and along Buffalo Bayou.

Preserving these resources requires proactive measures such as updating building elevations, implementing green infrastructure, and leveraging mitigation funding to retrofit or relocate at-risk assets. The National Register of Historic Places and local heritage preservation groups play an important role in documenting and protecting these resources before and after storm events.

13.3.1 Potential or Planned Development

Harris County remains one of the fastest-growing regions in the United States. According to the U.S. Census and local projections, the population of Harris County is expected to exceed 5 million by 2035, driven by economic growth, immigration, and suburban expansion. Much of this growth is occurring in areas that are increasingly vulnerable to hurricanes and coastal storms, particularly in eastern and southeastern parts of the county. (KHOU 2025).

13.3.2 Projected Changes in Population

Potential and planned development includes new residential subdivisions, commercial centers, industrial zones, and critical infrastructure such as energy facilities and transportation corridors. Many of these developments are taking place in or near the 100- and 500-year floodplains, as mapped by FEMA and local flood control agencies. With intensifying rainfall patterns and sea level rise, areas that were once considered low-risk are now experiencing more frequent flooding. Storm surge modeling also shows that future development along Clear Lake, Galveston Bay, and the Houston Ship Channel corridor faces heightened exposure to wind and water hazards. (KTRK 2025)

Furthermore, the Houston-Galveston Area Council's 2045 Regional Transportation Plan and various city and county comprehensive plans indicate ongoing land use changes in rapidly urbanizing areas such as Cypress, Katy, Baytown, Pasadena, and the unincorporated precincts. If not adequately mitigated, this growth could exacerbate stormwater runoff, strain evacuation routes, and increase the population at risk from hurricanes (H-GAC 2025).

13.3.3 Climate Change

Climate change is expected to produce warner ocean sea surface temperatures over the coming years, and as water temperatures increase, so will the potential for more frequent and more intense hurricanes and coastal storms.

14. MASS MOVEMENT

14.1 HAZARD PROFILE

14.1.1 Hazard Description

Mass movement is the transport of large volumes of earth material primarily by gravity. Mass movements can be divided into four main classes: falls, slides, creeps and flows. The classes are based on how quickly the rock and sediment moves and how much water there is. Steep and unstable slopes are more likely to have a mass movement than gentle and stable slopes. For this assessment, three types of mass movements have been assessed:

Note: The risk from this hazard is considered to be low to negligible, mainly because of the lack of historical occurrence in the planning area. However, this hazard profile is provided in order to address the possibility of a future event in the planning area. This positions the planning partnership for benefit eligibility should an event of this type occur during the performance period of this plan.

- Landslides
- Sinkholes
- Subsidence.

Landslide

Landslides are commonly categorized by the type of initial ground failure. Common types of slides are shown on Figure 14-1 through Figure 14-4. The most common is the shallow colluvial slide, occurring particularly in response to intense, short-duration storms. The largest and most destructive are deep-seated slides, which are less common than other types.

Figure 14-1. Deep Seated Slide

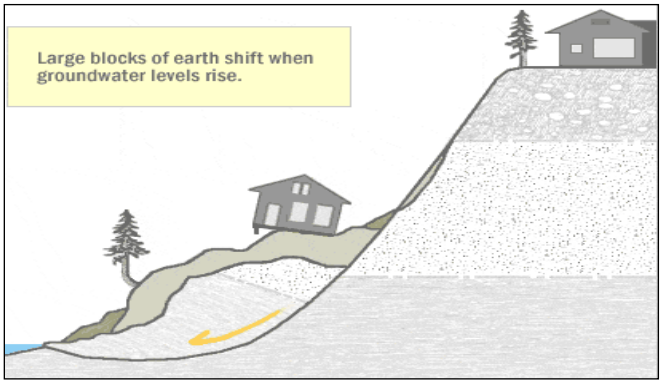


Figure 14-2. Bench Slide

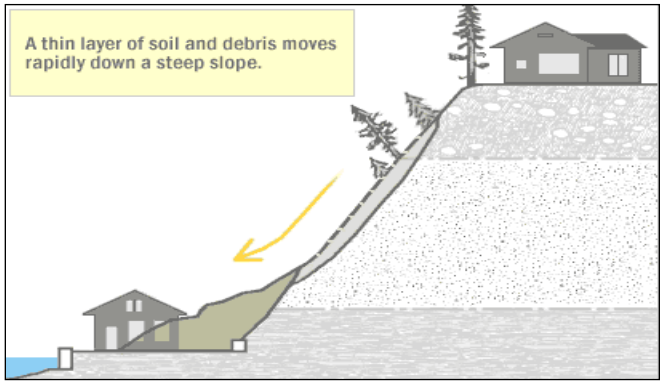


Figure 14-3. Shallow Colluvial Slide

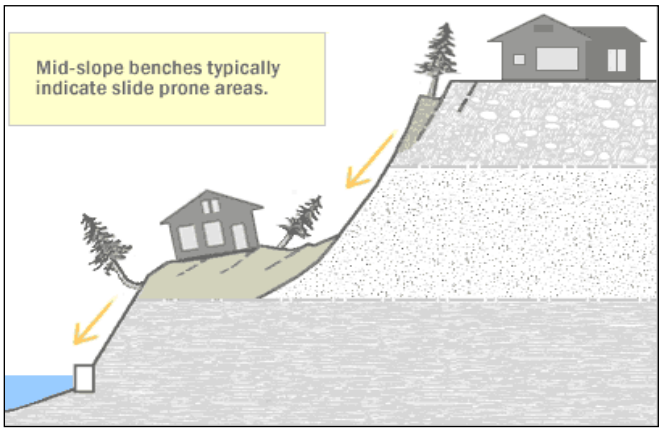
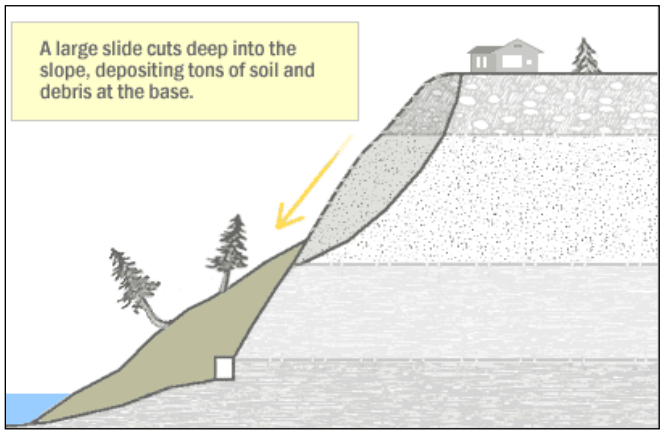


Figure 14-4. Large Slide



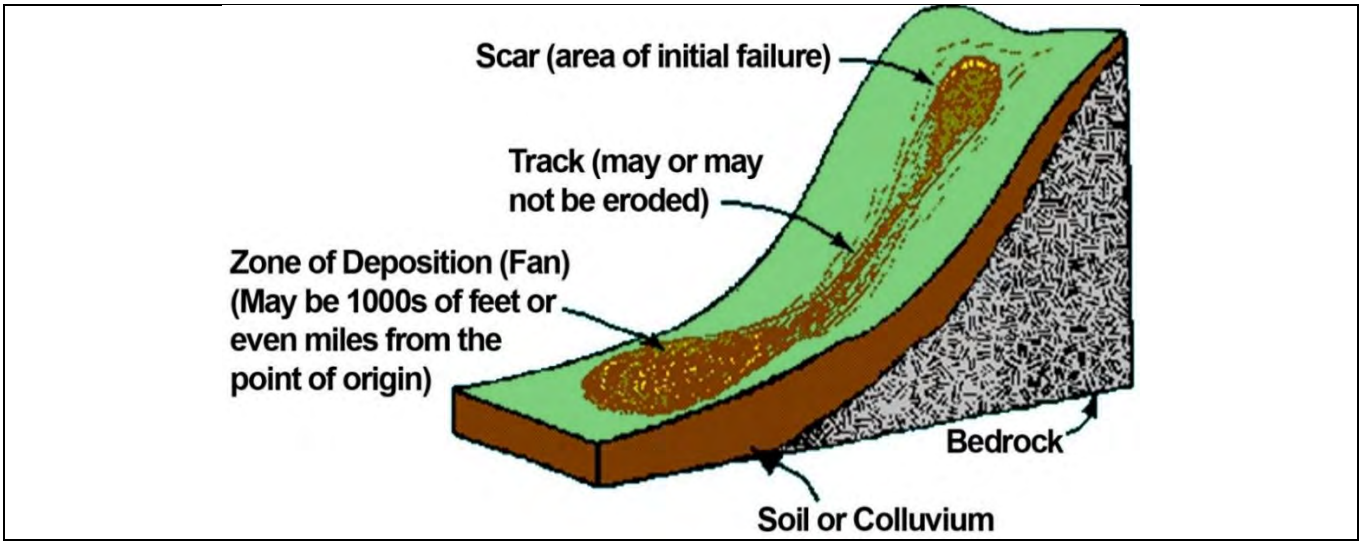
Source: California Department of Conservation, 2017c

Other landslide types include the following:

- Block slides—Blocks of rock that slide along a slip plane as a unit down a slope.
- Creep—A slow-moving landslide often only noticed through crooked trees and disturbed structures.
- Debris avalanche—A debris flow that travels faster than about 10 miles per hour (mph). Speeds in excess of 20 mph are not uncommon, and speeds in excess of 100 mph, although rare, can occur. The slurry can travel miles from its source, growing as it descends, picking up trees, boulders, cars, and anything else in its path Figure 14-5.
- Earth flows—Fine-grained sediments that flow downhill and typically form a fan structure.
- Mudslides or Debris Flows—Rivers of rock, earth, organic matter and other soil materials saturated with water. They develop in the soil overlying bedrock on sloping surfaces when water rapidly accumulates in the ground, such as during heavy rainfall or rapid snowmelt.
- Rock falls—Blocks of rock that fall away from a bedrock unit without a rotational component.

- Rock topples—Blocks of rock that fall away from a bedrock unit with a rotational component.
- Rotational slumps—Blocks of fine-grained sediment that rotate and move down slope.
- Transitional slides—Sediments that move along a flat surface without a rotational component.

Figure 14-5. Typical Debris Avalanche Scar and Track



Landslides are caused by a combination of geological and climate conditions, as well as encroaching urbanization. Vulnerable areas are affected by residential, agricultural, commercial, and industrial development and the infrastructure that supports it. The following human activities have particular influence on the landslide hazard:

- Construction Earthwork—Excavation, grading and fill during construction of buildings or roads on sloping terrain can steepen the terrain and increase weight loads on slopes, potentially increasing the landslide hazard.
- Drainage and Groundwater Alterations—Activities that increase the amount of water flowing into landslide-prone slopes can increase the landslide hazard. This can include broken or leaking water or sewer lines, water retention facilities that direct water onto slopes, lawn irrigation, minor alterations to small streams, and ineffective stormwater management measures. Development that increases impervious surface may redirect surface water to other areas. Road and driveway drains, gutters, downspouts, and other constructed drainage facilities can concentrate and accelerate flow.
- Changes in Vegetation—Removal of vegetation from very steep slopes, by wildfire or land clearing, can increase landslide hazards. In addition, woody debris in stream channels (both natural and man-made) may cause the impacts from debris flows to be more severe.

Other factors that can contribute to landslide include the following:

- Change in slope of the terrain
- Increased load on the land, shocks and vibrations

- Change in water content
- Groundwater movement
- Frost action
- Weathering of rocks
- Removing or changing the type of vegetation covering slopes.

Sinkholes

Sinkholes are a natural and common geologic feature in areas with underlying limestone and other rock types that are soluble in water. Most limestone is porous, allowing the acidic water of rain to percolate through their strata, dissolving the limestone and carrying it away in solution. Over time, this persistent erosion process can create extensive underground voids and drainage systems in the carbonate-based rocks. The collapse of overlying sediments into the underground cavities produces sinkholes.

The three general types of sinkholes are subsidence, solution, and collapse. Collapse sinkholes are most common in areas where the overburden (the sediments and water contained in the unsaturated zone above an aquifer) is thick, but the confining layer is breached or absent. Collapse sinkholes can form with little warning and leave behind a deep, steep-sided hole. Subsidence sinkholes form gradually where the overburden is thin and only a veneer of sediments is overlying the limestone. Solution sinkholes form where no overburden is present, and the limestone is exposed at the surface.

Sinkholes occur in many shapes, from steep-walled holes to bowl or cone shaped depressions. If there is not enough support for the land above the underground spaces, then a sudden collapse of the land surface can occur. Under natural conditions, sinkholes form slowly and expand gradually. However, human activities such as dredging, constructing reservoirs, diverting surface water, and pumping groundwater can accelerate the rate of sinkhole expansions, resulting in the abrupt formation of collapse sinkholes. Development increases water usage, alters drainage pathways, overloads the ground surface, and redistributes soil. According to FEMA, the number of human-induced sinkholes has doubled since 1930, and insurance claims for sinkhole-related damages increased 1,200 percent between 1987 and 1991.

Although a sinkhole can form without warning, specific signs can signal potential development:

- Slumping or falling fence posts, trees, or foundations
- Sudden formation of small ponds
- Wilting vegetation
- Discolored well water
- Structural cracks in walls and floors.

Subsidence

Land Subsidence is the slow, continuous decline of land-surface elevation caused by subsurface agitation of sediment and other earth materials. The level of land subsidence can range from a

gradual lowering of surface elevation to a total collapse of the land surface. An example of extreme land subsidence is referred to as a sinkhole (TDEM, Texas State Hazard Mitigation Plan 2023).

The majority of subsidence is caused by human behavior such as mining, oil and gas extraction, and groundwater pumping from major and minor aquifer systems. The excessive removal of groundwater from aquifers that possess soluble or compressible layers may undergo an increase in erosion, compaction, and subsurface collapse. Even if the groundwater is returned to their average levels, most of land subsidence consequences are irreversible (TDEM, Texas State Hazard Mitigation Plan 2023).

Land surfaces above shallow aquifer systems or areas adjacent to particularly porous earth have an increased risk of experiencing subsidence. Buildings and other infrastructure, such as roads, building foundations, and highways, are extremely susceptible to damage in the event of a land surface collapse. Other effects of land subsidence can include a reduction in an aquifer's storage capability, disruptions in irrigation and collection drains, and flow alterations in waterways (USGS, Houston Subsidence n.d.).

14.1.2 Location

Landslides

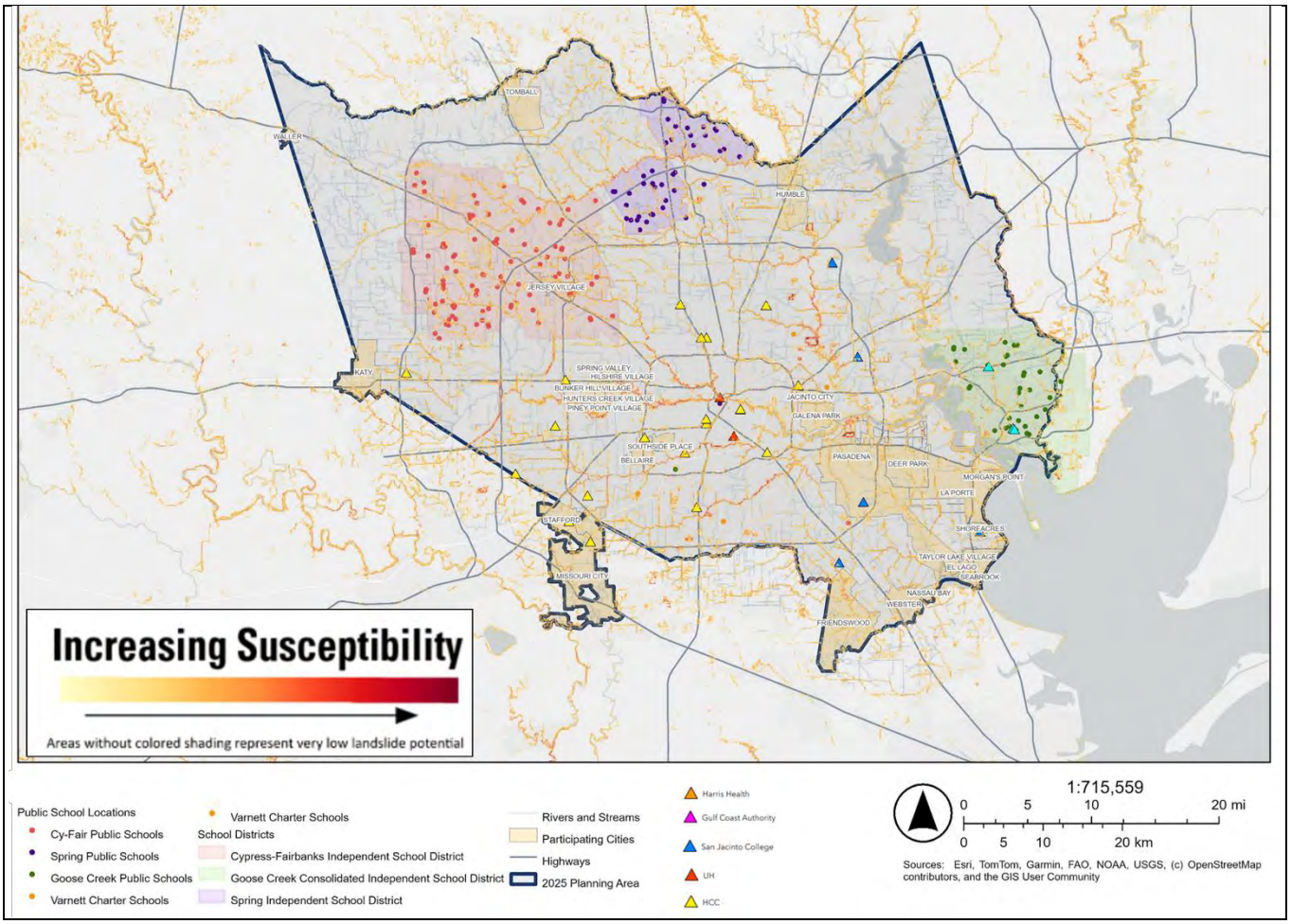
Although most of Harris County is not characterized as highly susceptible to landslides, the landslide hazard remains an issue at some locations across the County, as shown in Figure 14-6.

Sinkholes and Subsidence

No location mapping has been provided for sinkholes or subsidence due to the lack of exposure within the planning area. Future subsidence is not likely in any of the planning area due to existing groundwater reduction measures. Subsidence is being abated by regional limits on groundwater pumping. Many of the cities in Harris and Galveston counties, industries, and others in the coastal region converted to surface water years ago at a considerable cost.

The Harris-Galveston Coastal Subsidence District's 1999 Regulatory Plan requires some areas to continue with groundwater reduction requirements (HGCSD, 1999). The 1999 Regulatory Plan, focused mostly on the north and west part of Harris County, required groundwater reduction plans to show how permittees would reduce dependence on groundwater pumping by 30 percent in 2010, 70 percent in 2020 and 80 percent in 2030.

Figure 14-6. Harris County Landslide Susceptibility Map



Source: (USGS, Landslide Inventory and Susceptibility Map 2025); 2025 Harris County MJ-HMP Web Viewer

14.1.3 Extent

Landslide

Landslides are typically measured by the size of the area affected and how far the ground has moved, this can include the width, length, and depth of the landslide (USGS 2008). Scientists use tools like satellite radar and laser mapping (LiDAR) to track ground changes and figure out how much land has shifted over time. Landslides are typically triggered by localized conditions such as over-saturated soils, manmade slope alterations, or erosional instability along bayous and manmade embankments. In the Houston region, it is classified as having very low landslide susceptibility based on current data, indicating that local conditions such as terrain, geology, and rainfall patterns make landslides rare (ThinkHazard 2025). While these events are usually small in scale—ranging from 10 to 50 feet in width—they can obstruct roads, damage utilities, or undermine the foundations of nearby structures. Slumps and shallow slides involving less than 3 feet of depth are the most common forms observed in Harris County, especially during or following heavy rainfall. Although large-scale landslides with significant depth are not anticipated, rapid

urbanization, poor drainage, and extreme precipitation due to climate change may increase localized slope instability, particularly in areas with artificial elevations or poor soil compaction.

Sinkhole

Sinkholes are measured by the diameter and depth of the collapse, some are just a few feet wide, while others can span hundreds of feet and go deep underground. Experts assess sinkholes using ground surveys, sonar, or radar tools to map the underground voids and determine how much land has caved in or is at risk of collapsing. Sinkholes can occur suddenly and cause significant localized damage when they do. Unlike central Texas regions with karst geology, sinkholes in Harris County are typically caused by collapsed or unstable soils related to decaying underground infrastructure, improperly compacted fill, or localized erosion, rather than natural limestone dissolution. Documented sinkholes in the region have varied widely in size, from small depressions just 2 to 4 feet wide to sudden collapses exceeding 15 feet in diameter and 10 feet deep, capable of damaging roadways, utility corridors, and building foundations (Harris County Engineering Department, 2021). In most cases, these events are highly localized and occur in urbanized areas where water main failures or stormwater infiltration erode the subsurface.

Subsidence

In some rare cases, extreme land subsidence can lead to sinkholes when the subsurface land cannot support the surface land any longer. A sinkhole is described as a portion of the ground that possesses no natural external surface drainage, typically draining water into the subsurface (USGS, Sinkholes n.d.).The development of sinkholes can range from slow to a dramatic collapse which may result in property damage or injury and loss of life. These events can differ in size from a few feet to many acres with equal variety in depth.

In Texas, the forms of land subsidence of greatest concern are related to collapsing soils and sink holes in karst heavy areas. Factors such as soil composition, groundwater withdrawal, drought, erosion, oil or gas extraction, or mining can impact the extent of subsidence. Areas may historically appear free of subsidence threats or occurrences but can suddenly be vulnerable to dramatic events. The extent of land subsidence varies throughout Texas with the most severe cases occurring along the coast and in the Rio Grande Aquifer system. Land subsidence extent is measured by the number of feet of land loss or sunk. The Houston area, with land subsidence issues extending into Galveston and Fort Bend counties, has the greatest known potential of land subsidence in the state.

14.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2025, Harris County was not included in any major disaster (DR) or emergency (EM) declarations for mass movement-related events (FEMA 2023).

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Once an emergency proclamation is declared, The Texas Division of Emergency Management (TDEM) coordinates the state emergency management program, which is intended to ensure the state, and its local governments respond to and recover from emergencies and disasters and implement plans and programs to help prevent or lessen the impact of emergencies and disasters. Harris County has not been included in any mass movement-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2023, Harris County was not included in any USDA mass movement-related agricultural disaster declarations.

Previous Events

The FEMA, NCEI, USGS, or USDA databases have no data on previous occurrences of landslides, sinkholes, and land subsidence in the Planning Area.

14.1.5 Probability of Future Occurrences

Information on previous mass movement occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 14-1. Based on historical records and input from the Steering Committee, the probability of occurrence for mass movements in the County is considered "rare."

Table 14-1. Probability of Future Mass Movement Events in Harris County

Hazard Type	Number of Occurrences Between 1996 and 2023	Percent Chance of Occurring in Any Given Year
Mass Movements	0	1%

Climate Change Projections

Climate change can significantly impact the frequency and severity of landslides, particularly in regions like Harris County. Increased precipitation from more intense and frequent rainfall events can lead to soil saturation, raising the likelihood of landslides. Warmer winters with more freeze-thaw cycles can weaken soil and rock structures, making them more susceptible to landslides. Additionally, changes in temperature and precipitation can affect vegetation cover; loss of vegetation due to drought or other climate stressors can reduce root strength, which helps stabilize slopes. More frequent and intense storms can also trigger landslides by rapidly increasing water infiltration into the soil, leading to slope failure.

Changes to the water balance of an area (including over-withdrawal of groundwater, diverting surface water from a large area, and concentrating it in a single point, artificially creating ponds of surface water, and drilling new water wells) will cause sinkholes. These actions can also serve to accelerate the natural processes of bedrock degradation, which can have a direct impact on sinkhole creation. The potential effects of climate change on Harris County's vulnerability to subsidence/sinkhole events will need to be considered as more information develops regarding regional climate change impacts.

14.1.6 Cascading Impacts on Other Hazards

Mass movement events that damage developed areas can result in explosions from natural gas lines, loss of utilities, such as water and sewer due to shifting infrastructure, and potential failures of reservoir dams. Water and sewer damage can lead to groundwater contamination risking environmental health and safety. Over time land subsidence may cause changes in elevation and slope of waterways reducing or hindering drainage capacity of an area causing excessive flooding; damage public infrastructure such as roadways, bridges, and railways hindering emergency operations; and damage public and private buildings causing foundation issues or lowering finished floor elevations, resulting in higher flood hazards (Devin L. Galloway and Leake 2016).

14.2 VULNERABILITY AND IMPACT ASSESSMENT

To understand risk, a qualitative assessment was performed to evaluate what assets are exposed or vulnerable to the mass movements. The entire County has been identified as exposed to hazard. Therefore, all assets in the County (population, structures, critical facilities, and lifelines), as described in Chapter 3 (County Profile), are exposed and potentially vulnerable.

14.2.1 Life, Health, and Safety

Overall Population

It can be assumed that the entire planning area is exposed to some extent to landslides, sinkholes, and land subsidence. Certain areas are more exposed due to geographic location and human activity. Current growth trends could cause more planning area residents to be exposed to land subsidence. Increased population will increase demands on structure development, as well as sub-surface water use which may lead to higher land subsidence rates.

Socially Vulnerable Population

Mass movement events are rare in Harris County, but the region is highly susceptible to land subsidence and localized sinkhole formation, particularly in areas with expansive clay soils or histories of groundwater withdrawal. Socially vulnerable populations, such as low-income households, older adults, renters, and residents in historically underserved communities, are often the most at risk from the cascading impacts of mass movement hazards. These individuals frequently live in areas with aging infrastructure, limited access to home insurance or mitigation

funding, and fewer resources to repair damage or relocate when hazards occur. In Harris County, subsidence has been most severe in eastern and southeastern areas, including parts of Pasadena, Baytown, and the unincorporated communities near the Houston Ship Channel, locations that also overlap with vulnerable demographic groups (USGS 2022).

Subsidence and sinkholes may not cause immediate loss of life, but they can lead to long-term housing instability, utility disruptions, and infrastructure failure—disproportionately impacting populations with fewer economic or social safety nets. For renters or families in manufactured housing, even minor land shifting or foundation cracking can render a home uninhabitable. Vulnerable communities may also lack awareness of these hazards, especially in areas where land subsidence is gradual and poorly mapped. Additionally, mass movement events may not always be classified as disasters eligible for federal assistance, leaving residents to bear the full financial burden of repairs or displacement. Ensuring that hazard mitigation strategies account for social vulnerability—by prioritizing soil stability assessments, housing reinforcement, and outreach in high-risk areas—is essential to reduce long-term risks for Harris County's most at-risk populations.

14.2.2 General Building Stock

Mass movement hazards pose a moderate threat to Harris County's built environment. The region's widespread use of slab-on-grade foundations, combined with expansive clay soils and historical groundwater withdrawal, makes residential and commercial structures particularly vulnerable to cracking, differential settlement, and eventual structural degradation. Older homes built before geotechnical design improvements or modern building codes are especially susceptible to foundation damage and plumbing failures related to subsurface movement. Unreinforced or lightly reinforced buildings, especially in eastern and southeastern parts of the county are at elevated risk due to past subsidence trends and localized soil instability (USGS 2022).

Commercial properties, industrial sites, and public infrastructure such as schools, fire stations, and underground utilities can also be affected by land deformation. Roadways, pipelines, and drainage systems may buckle, rupture, or shift alignment over time, increasing the cost and complexity of repairs. Since these failures often occur slowly or are not easily visible until significant damage has occurred, they are less likely to be captured by traditional emergency declarations. As Harris County continues to urbanize and infill development increases, structures built on historically unstable or un-remediated land will require targeted mitigation strategies—such as soil stabilization, foundation reinforcement, and geotechnical monitoring—to reduce long-term exposure.

14.2.3 Community Lifelines and Other Critical Facilities

All critical facilities are considered exposed to land subsidence. Several major roads are available to move people and supplies throughout the region. Damage to these roads and other infrastructure could hinder emergency services and affect public health and safety.

14.2.4 Economy

Mass movement hazards could be a significant threat to Harris County's economy, particularly due to their impact on the built environment, public infrastructure, and long-term property value. While often slow and difficult to detect in early stages, these hazards can result in significant cumulative damage over time, including foundation failure, structural deformation, road damage, and utility line disruption. The financial burden of these impacts is amplified when they occur in areas with high development density and valuable property assets.

According to current building inventory data, Harris County contains approximately 1.26 million residential buildings with a total replacement cost value exceeding $683 billion, along with over 85,000 commercial buildings valued at $409.8 billion, and 4,821 industrial structures with a combined value of $42.15 billion. Much of this building stock sits on clay-rich or historically subsiding soils, particularly in southeastern portions of the county. Even marginal ground movement can generate substantial repair costs and business interruptions—especially for small businesses, rental properties, and industrial facilities with heavy equipment or underground infrastructure. Mass movement events may not trigger emergency declarations, but the accumulated economic toll from widespread foundation and infrastructure damage can strain both private and public resources. Proactive soil monitoring, investment in geotechnical assessments, and public education on property-level mitigation can reduce long-term losses and protect the economic health of Harris County.

14.2.5 Natural, Historic and Cultural Resources

Mass movement hazards in Harris County may not be as visibly dramatic as in mountainous regions, but they still pose serious risks to the county's natural, historic, and cultural assets. Land subsidence can contribute to the long-term degradation of wetland ecosystems throughout the county, increasing the risk of flooding and the loss of critical habitats in sensitive areas such as Armand Bayou Nature Center and Sheldon Lake State Park. These natural spaces not only provide ecological services but also function as community landmarks and environmental education sites that could be impacted by subtle but ongoing ground changes.

Cultural and historic sites, such as the San Jacinto Battleground, Harrisburg's early settlement areas, and historic cemeteries may also be vulnerable to ground instability. Shifting soils can compromise building integrity, cause surface cracking, or disrupt underground drainage and burial sites. Many of these landmarks predate modern engineering standards and are not easily retrofitted for subsidence-related impacts. Moreover, cultural institutions such as museums, libraries, and performing arts centers located in older urban cores may also face risks to building integrity and artifact preservation if soil movement is left unmanaged. Protecting these resources requires proactive integration of geotechnical assessments into capital planning, hazard mitigation, and cultural preservation efforts.

14.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

14.3.1 Potential or Planned Development

Due to the lack of exposure and impacts from these hazards, future development in the planning area is not likely to be impacted by these hazards in the short term. However, as a changing climate continues to influence the frequency, severity, and magnitude of hazard events, there could be impacts on future development. Future updates to this plan will have to measure those possibilities as it assesses the mass movement hazards of concern.

14.3.2 Projected Changes in Population

As Harris County's population continues to grow, projected to reach over 6.6 million residents by 2050 under high-growth scenarios, more people and structures may be exposed to areas susceptible to land subsidence, sinkholes, and other mass movement hazards (Texas Demographic Center 2024). Urban expansion often pushes new development into geologically sensitive areas, including regions with expansive clay soils, historical groundwater extraction, or poorly understood subsurface conditions. Infill development and densification in vulnerable neighborhoods may also compound risks, particularly for residents in low-lying or flood-prone zones where the soil is already unstable. Without proactive land use planning, increased population density may intensify damage to roads, homes, and underground utilities when ground movement occurs, and make response and recovery more difficult in densely populated or underserved communities.

14.3.3 Climate Change

While climate change does not directly cause mass movement events, it significantly alters the environmental conditions that trigger or accelerate land subsidence, sinkhole formation, and localized ground failure. Projected increases in the frequency of extreme precipitation, followed by prolonged dry spells, are particularly relevant for Harris County, where expansive soils are common. These soils swell when saturated and contract during drought, putting stress on foundations and underground infrastructure (USDA NRCS 2023). More intense rainfall can also increase infiltration into subsurface cavities, especially in areas with limestone or poorly compacted fill, potentially accelerating sinkhole development. At the same time, the increasing demand for water during hotter, drier periods may further deplete groundwater reserves, contributing to long-term subsidence—especially in parts of the county that have already experienced measurable ground sinking over the past several decades (USGS 2022). These climate-driven dynamics underscore the importance of integrating future climate scenarios into geotechnical risk assessments and urban planning strategies.

15. SEVERE WEATHER

15.1 HAZARD PROFILE

15.1.1 Hazard Description

Severe weather refers to any dangerous meteorological phenomena with the potential to cause damage, serious social disruption, or loss of human life. Types of severe weather phenomena vary, depending on the latitude, altitude, topography, and atmospheric conditions. This assessment profiles five types of severe weather events: thunderstorms; hail; tornadoes; and damaging wind.

Thunderstorms

Thunderstorms create extreme wind events which includes straight line winds. Wind is the horizontal motion of the air past a given point, beginning with differences in air pressures. Pressure that is higher at one place than another sets up a force pushing from high toward low pressure: the greater the difference in pressures, the stronger the force. The distance between the area of high pressure and the area of low pressure also determines how fast the moving air accelerates. Thunderstorms are created when heat and moisture near the Earth's surface are transported to the upper levels of the atmosphere. By-products of this process are the clouds, precipitation, and wind that become the thunderstorm. According to the National Weather Service (NWS), a thunderstorm occurs when thunder accompanies rainfall. Radar observers use the intensity of radar echoes to distinguish between rain showers and thunderstorms.

Lightning occurs as a rapid discharge of electrical energy in the atmosphere between clouds, the air, or the ground. A bolt of lightning can reach temperatures approaching 50,000 degrees Fahrenheit, a temperature five times hotter than the sun's surface. Lightning rapidly heats the sky as it flashes, but the surrounding air quickly cools following the bolt. This rapid heating and cooling of the surrounding air causes thunder which often accompanies lightning strikes. While most often affiliated with severe thunderstorms, lightning often strikes outside of heavy rain and might occur as far as 10 miles away from any rainfall. According to the National Weather Service (NWS), the 10-year (2012–2021) average for fatalities is 23 people with an average of 300 injuries in the United States each year by lightning. Lightning can occur as cloud to ground flashes or as intra-cloud lightning flashes. Direct lightning strikes can cause significant damage to buildings, critical facilities, infrastructure, and communication equipment affecting emergency response.

Hail

Hailstones are formed when raindrops are carried upward by thunderstorm updrafts into extremely cold areas of the atmosphere and freeze. Hailstones then grow by colliding with liquid water drops that freeze onto the hailstone's surface. The hail falls when the thunderstorm's updraft can no longer support the weight of the hailstone, which can occur if the stone becomes large enough or the updraft weakens. The fall speed of hail primarily depends on the size of the hailstone, the friction between the hailstone and surrounding air, the local wind conditions (both horizontal and

vertical), and the degree of melting of the hailstone. For small hailstones smaller than 1-inch in diameter, the expected fall speed is between 9 and 25 mph. Hailstones 1-inch to 1.75 inches in size typically associated with a severe thunderstorm can have an expected fall speed between 25 and 40 mph. In the strongest supercells 2 to 4-inch hail can be produced with an anticipated fall speed between 44 and 72 mph. However, fall speeds fluctuate due to variations in the hailstone's shape, degree of melting, fall orientation, and the environmental conditions. It is possible for very large hailstones, exceeding 4-inches in diameter, to fall at over 100 mph.

Tornadoes

A tornado appears as a rotating, funnel-shaped cloud that extends from a thunderstorm to the ground with whirling winds that can reach 250 miles per hour (mph). Damage paths can be greater than 1 mile wide and 50 miles long. Tornadoes typically develop from either a severe thunderstorm or hurricane as cool air rapidly overrides a layer of warm air. Tornadoes typically move at speeds between 30 and 125 mph and can generate combined wind speeds (forward motion and speed of the whirling winds) exceeding 300 mph. The lifespan of a tornado rarely is longer than 30 minutes (FEMA 1997).

Waterspouts are weak tornadoes that form over warm water and are most common along the Gulf Coast and southeastern states. Waterspouts occasionally move inland, becoming tornadoes that can cause damage and injury. However, most waterspouts dissipate over the open water, threatening marine and boating interests. Typically, a waterspout is weak and short-lived, and because they are so common, most go unreported unless they cause damage.

Damaging Wind

Wind is air moving from high to low pressure. It is the rough horizontal movement of air (as opposed to an air current) caused by uneven heating of the Earth's surface. It occurs at all scales, from local breezes generated by heating of land surfaces and lasting tens of minutes to global winds resulting from solar heating of the Earth. Types of damaging winds include straight-line winds, downdrafts, downbursts, microbursts, macrobursts, gust fronts, derechos, and bow echoes, described as the following (NOAA 2024):

- Straight-line Wind is any thunderstorm wind not associated with rotation (e.g., tornadic winds). Straight-line winds are movements of air from areas of higher pressure to areas of lower pressure—the greater the difference in pressure, the stronger the winds.
- A Downdraft is a small-scale column of air that rapidly sinks toward the ground and usually results in a downburst.
- A Downburst is a strong downdraft with horizontal dimensions larger than 2.5 miles, resulting in an outward burst or damaging winds on or near the ground. It is usually associated with thunderstorms but can occur with rainstorms too weak to produce thunder.
- A Microburst is a small, concentrated downburst that produces an outward burst of damaging winds near the surface. It is typically short-lived, lasting only five to ten minutes, with maximum wind speeds of up to 168 miles per hour (mph).

- A Macroburst is an outward burst of strong winds at or near the surface wider than 2.5 miles across that can occur when a downdraft reaches the surface. While usually associated with thunderstorms, microburst can also occur in storms when no thunder is present.
- A Gust Front is the leading edge of rain-cooled air that clashes with warmer thunderstorm inflow. It is characterized by a wind shift, temperature drop, and gusty winds ahead of a thunderstorm.
- A Derecho is a widespread and long-lived windstorm associated with thunderstorms that are often curved. A derecho generally consists of several microbursts, downbursts, and downburst clusters, and these types of wind events are unique in that they travel great distances.
- A Bow Echo is a formation of showers or thunderstorms that exhibits a curved or bowed appearance. These phenomena typically develop from a cluster of storms; however, they can also originate from a single supercell thunderstorm. When a bow echo travels more than 250 miles and produces widespread wind gusts of 58 mph or greater, it is classified as a derecho (NOAA 2023).

Extreme wind can cause injuries and fatalities in a number of ways. Downed trees may fall on homes or cars, killing or injuring those inside them. Objects that are not secured can be picked up in wind events and become projectiles. Structures that collapse or blow over during damaging wind events, especially tornadoes, may kill or injure those seeking shelter inside.

15.1.2 Location

Thunderstorms

Thunderstorm wind events can develop in any geographic location and are considered a common occurrence in Texas. Therefore, a thunderstorm wind event could occur at any location within Harris County. These storms develop randomly and are not confined to any geographic area within the County. It is assumed that the entire County, including all participating jurisdictions, is uniformly exposed to the threat of thunderstorm winds.

Hail

Hailstorms are an extension of severe thunderstorms that could potentially cause severe damage. As a result, they are not confined to any specific geographic location and can vary greatly in size, location, intensity, and duration. Therefore, the entire County, including all participating jurisdictions, is equally at risk to the hazard of hail.

Tornadoes

Tornadoes can occur at any time of the year, with peak season for Texas from May into early June ((NOAA n.d.). An average of 1,141 tornadoes occurs in the United States each year, based on tornadoes recorded between 1985 and 2014. The State of Texas averages 137 tornadoes each year

(NWS n.d.). In). Since tornadoes can strike anywhere, the entire County is vulnerable to tornadoes and its impacts.

Damaging Wind

Severe summer weathers have the power to produce powerful winds; therefore, strong and powerful winds have a higher chance of occurring in locations that are more likely to experience these storms (NOAA n.d.). In addition, high wind events may occur without a thunderstorm, tornado, or hurricane present and can be just as dangerous and destructive as those hazards.

15.1.3 Extent

Figure 15-1 displays the severe weather event types specific to Harris County, including lightning, tornado, hail and strong wind risk.

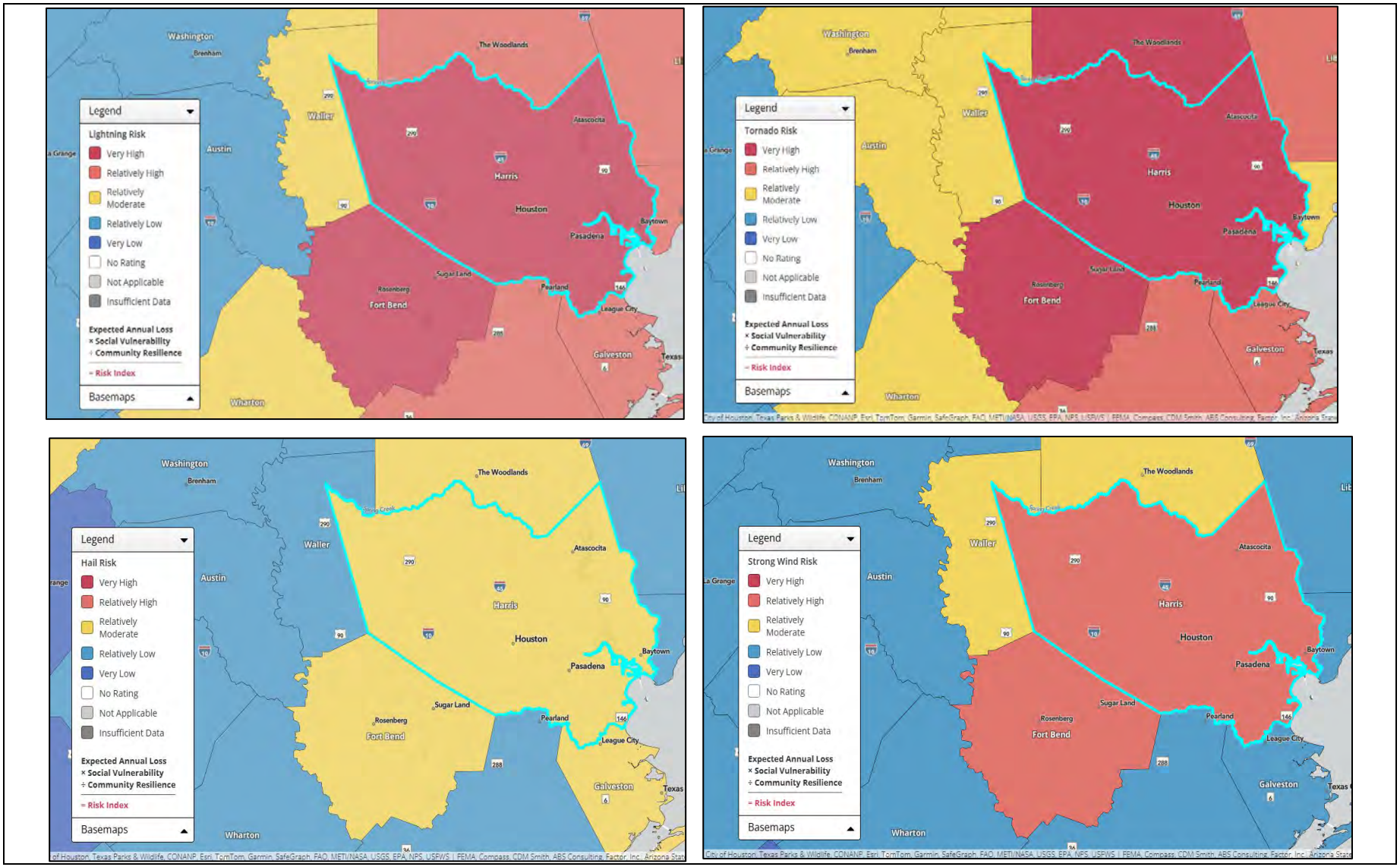
Thunderstorms

Severe thunderstorm statements, watches, and warnings are issued by the local NWS office and the Storm Prediction Center (SPC). The NWS and SPC will update the watches and warnings and notify the public when they are no longer in effect. NWS issues statements, watches, and warnings for thunderstorms:

- Special Weather Statement: Issued for strong storms that are below severe levels but may have impacts. Usually reserved for the threat of wind gust of 40−57 mph or hail of 0.5 inches to 0.99 inches in diameter (NWS n.d.).
- Severe Thunderstorm Watches: A severe thunderstorm watch is issued when severe thunderstorms are possible in and near watch areas (NWS n.d.)
- Severe Thunderstorm Warning: A severe thunderstorm is imminent or occurring; it is either detected by weather radar or reported by storm spotters. A severe thunderstorm is one that produces winds 58 mph or stronger and/or hail 1 inch in diameter or larger. A warning means to take shelter (NWS n.d.).

The NWS has five risk categories for severe weather: marginal, slight, enhanced, moderate, and high. The probabilistic forecast directly expresses the best estimate of a severe weather event occurring within 25 miles of a point (NWS n.d.). Table 15-1 details the thunderstorm risk categories.

Figure 15-1. FEMA National Risk Indices for Severe Weather Hazards in Harris County



Source: (FEMA 2025)

Table 15-1. Convective Outlooks – Thunderstorm Risk Categories

Thunderstorm Risk Label	Thunderstorm Risk Category	Thunderstorm Risk Description
TSTM (light green)	General or Non-Severe Thunderstorms	Delineates, to the right of a line, where a 10% or greater probability of thunderstorms is forecast during the valid period. No severe thunderstorms expected. Lightning/flooding threats exist with all thunderstorms. Winds to 40 mph and small hail.
1-MRGL (dark green)	Marginal Risk	An area of severe storms of either limited organization and longevity, or very low coverage and marginal intensity. Isolated severe thunderstorms possible. Limited in duration and/or coverage and/or intensity. Winds 40-60 mph, hail up to 1 inches and low tornado risk.
2-SLGT (yellow)	Slight Risk	An area of organized severe storms, which is not widespread in coverage with varying levels of intensity. Scattered severe thunderstorms possible. Short-lived and/or not widespread, isolated intense storms possible. One or two tornados, reports of strong winds/wind damage, and hail approximately 1 inch and isolated 2 inches.
3-ENH (orange)	Enhanced Risk	An area of greater (relative to Slight risk) severe storm coverage with varying levels of intensity. Numerous severe storms possible. More persistent and/or widespread, a few intense. A few tornados, several reports of wind damage and damaging hail, 1- 2 inches.
4-MDT (red)	Moderate Risk	An area where widespread severe weather with several tornados and/or numerous severe thunderstorms is likely, some of which should be intense. This risk is usually reserved for days with several supercells producing intense tornados and/or very large hail, or an intense squall line with widespread damaging winds. Widespread severe storms likely. Long-lived, widespread and intense. Strong tornados, widespread wind damage and destructive hail 2 inches (+).
5-HIGH (magenta)	High Risk	An area where a severe weather outbreak is expected from either numerous intense and long-tracked tornados or a long-lived derecho-producing thunderstorm complex that produces hurricane-force wind gusts and widespread damage. This risk is reserved for when high confidence exists in widespread coverage of severe weather with embedded instances of extreme severe (i.e., violent tornados or very damaging convective wind events). Widespread severe storms expected. Long-lived, very widespread and particularly intense. Tornado outbreak and derecho.

Source: NOAA, NWS, Storm Prediction Center
Note: All thunderstorm categories imply lightning and the potential for flooding. Categories are also tied to the probability of a severe weather event within 25 miles.

Lightning is most often associated with moderate to severe thunderstorms. The severity of lightning refers to the frequency of lightning strikes during a storm. The Lightning Activity Level (LAL) is a scale that describes lightning activity (NWCG 2025). The scale, part of the National Fire Danger Rating System, is a range of numbers, from one to six, which reflects the frequency and character of cloud-to-ground lightning, as listed in Table 15-2. Harris County can experience multiple thunderstorms each year, of any severity. Therefore, the entire County can see lightning of any LAL.

Table 15-2. Lightning Activity Level Categories

Lightning Activity Level (LAL)	Condition
1	No thunderstorms.
2	Isolated thunderstorms. Light rain will occasionally reach the ground. Lightning is very infrequent, 1 to 5 cloud-to-ground strikes in a 5-minute period.
3	Widely scattered thunderstorms. Light to moderate rain will reach the ground. Lightning is infrequent, with 6 to 10 cloud-to-ground strikes in a 5-minute period.
4	Scattered thunderstorms. Moderate rain is commonly produced. Lightning is frequent, with 11 to 15 cloud-to-ground strikes in a 5-minute period.
5	Numerous thunderstorms. Rainfall is moderate to heavy. Lightning is frequent and intense, with greater than 15 cloud-to-ground strikes in a 5-minute period.
6	Dry lightning (same as LAL 3 but without rain). This type of lightning has the potential for extreme fire activity and is normally highlighted in fire weather forecasts with a Red Flag Warning.

Source: (NOAA/NWS n.d.)

The National Lightning Detection Network consists of over 100 remote, ground-based sensing stations located across the United States that instantaneously detect the electromagnetic signals given off when lightning strikes the earth's surface. These remote sensors send the raw data via a satellite-based communications network to the Network Control Center operated by Vaisala Inc. in Tucson, Arizona. Within seconds of a lightning strike, the Center's central analyzers process information on the location, time, polarity, and communicate it to users across the country (NASA n.d.).

Hail

The largest hailstone measured in Harris County was a 3.00-inch stone that fell in 2006. The extent of hail in the planning area can range anywhere from 0.75 to 3.00 inches. NOAA utilizes the Modified NOAA/TORRO Hailstorm Intensity Scale to assist in measuring hailstorm intensity (see Table 15-3). The scale extends from H0 to H10, with increments of intensity or damage potential related to hail size (distribution and maximum), texture, numbers, fall speed, speed of storm translation, and strength of the accompanying wind.

Tornadoes

Damage from tornadoes can range from minor damage, such as broken tree limbs, to massive destruction that demolishes homes in its path. The type of damage depends on the intensity, size, and duration of the tornado. The magnitude or severity of a tornado is categorized using the Enhanced Fujita Tornado Intensity Scale (EF Scale). This is the scale now used exclusively for determining tornado ratings by comparing wind speed and actual damage. Table 15-4 illustrates the relationship between EF ratings, wind speed, and the expected damage from tornadoes. The extent of tornado events in the planning area ranges from EF0 to EF5. In 1992, the county was struck by an F4 tornado, which had winds that ranged from 207-260 mph (NOAA/NCEI 2025).

Table 15-3. NOAA/TORRO Scale

Size Code	Intensity Category	Hail Diameter (inches)	Potential Damage
H0	Hard Hail	0.25	No Damage
H1	Potentially Damaging	0.25 – 0.50	Slight general damage to plants, crops
H2	Significant	0.50 – 0.75	Significant damage to fruit, crops, vegetation
H3	Severe	0.75 – 1.25	Severe damage to fruit and crops, damage to glass and plastic structures, paint and wood scored
H4	Severe	1.0 – 1.50	Widespread glass damage, vehicle bodywork damage
H5	Destructive	1.25 – 1.75	Wholesale destruction of glass, damage to tiled roofs, significant risk of injuries
H6	Destructive	1.50 – 2.25	Bodywork of grounded aircraft dented brick walls pitted
H7	Destructive	2.0 – 3.0	Severe roof damage, risk of serious injuries
H8	Destructive	2.50 – 3.50	Severe damage to aircraft bodywork
H9	Super Hailstorms	3.0 – 4.0	Extensive structural damage. Risk of severe or even fatal injuries to persons caught in the open
H10	Super Hailstorms	4.0 +	Extensive structural damage. Risk of severe or even fatal injuries to persons caught in the open

Source: TORRO, 2019

Table 15-4. Enhanced Fujita Scale for Tornadoes

Scale	Wind Speed (mph)	Potential Damage
EF0	65–85	Light damage. Peels surface off some roofs; some damage to gutters or siding; branches broken off trees; shallow-rooted trees pushed over. Confirmed tornados with no reported damage (i.e., those that remain in open fields) are always rated EF0.
EF1	86–110	Moderate damage. Roofs severely stripped; mobile homes overturned or badly damaged; loss of exterior doors; windows and other glass broken.
EF2	111–135	Considerable damage. Roofs torn off well-constructed houses; foundations of frame homes shifted; mobile homes completely destroyed; large trees snapped or uprooted; light-object missiles generated; cars lifted off the ground.
EF3	136–165	Severe damage. Entire stories of well-constructed houses destroyed; severe damage to large buildings such as shopping malls; trains overturned; trees debarked; heavy cars lifted off the ground and thrown; structures with weak foundations blown away some distance.
EF4	166–200	Devastating damage. Well-constructed houses and whole frame houses are completely leveled; cars are thrown, and small missiles are generated.
EF5	>200	Explosive damage. Strong frame houses leveled off foundations and swept away; automobile-sized missiles fly through the air in excess of 100 m (300 ft); steel reinforced concrete structure badly damaged; high-rise buildings have significant structural deformation; incredible phenomena will occur.

Source: (NOAA/NWS Huntsville 2015)

Damaging Wind

Wind is often measured in terms of wind shear. Wind shear is a difference in wind speed and direction over a set distance in the atmosphere. Wind shear is measured both horizontally and vertically. Wind shear is measured in meters per second times kilometers of height. Under normal conditions, the winds move much faster higher in the atmosphere, creating high wind shear in high altitudes. The amount of force that wind is generating is measured according to the Beaufort scale, which is described in Table 15-5. The extent of damaging wind in Harris County can range from 0 to 12 on the Beaufort wind scale.

Table 15-5. Beaufort Scale for Wind Speeds

Scale	Wind (mph)	Description	Specifications for use on land
0	0-1	Calm	Calm; smoke rises vertically
1	1-3	Light Air	Direction of wind shown by smoke drift, but not by wind vanes
2	4-7	Light Breeze	Wind felt on face; leaves rustle; ordinary vanes moved by wind.
3	8-12	Gentle Breeze	Leaves and small twigs in constant motion; wind extends light flag.
4	13-18	Moderate Breeze	Raises dust and loose paper; small branches are moved
5	19-24	Fresh Breeze	Small trees in leaf begin to sway; crested wavelets form on inland waters
6	25-31	Strong Breeze	Large branches in motion; whistling heard in telegraph wires; umbrellas used with difficulty
7	32-38	Near Gale	Whole trees in motion; inconvenience felt when walking against the wind
8	39-46	Gale	Breaks twigs off trees; generally, impedes progress
9	47-54	Severe Gale	Slight structural damage occurs (chimneypots and slates removed)
10	55-63	Storm	Seldom experienced inland; trees uprooted; considerable structural damage occurs
11	64-72	Violent Storm	Very rarely experienced; accompanied by widespread damage.
12	72-83	Hurricane	Hurricane force winds.

Source: (NOAA/NWS 2016)

15.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2024, Harris County was included in 18 major disaster (DR) or emergency (EM) declarations for severe weather-related events (FEMA 2023). Table 15-6 lists these declarations as well as how FEMA designated each disaster.

Table 15-6. FEMA Declarations for Severe Weather Events in Harris County (1954 to 2024)

Event Date	Declaration Date	Declaration Number	Incident Type	Declaration Title
July 11, 1973	July 11, 1973	DR-398-TX	Flood	Severe Storms & Flooding
June 19, 1976	June 19, 1976	DR-510-TX	Flood	Severe Storms & Flooding
April 26, 1979	April 26, 1979	DR-580-TX	Severe Storm	Severe Storms, Tornadoes & Flooding
September 25, 1979	September 25, 1979	DR-603-TX	Flood	Severe Storms & Flooding
October 30, 1984	October 30, 1984	DR-727-TX	Flood	Severe Storms & Flooding
May 4-June 7, 1989	May 19, 1989	DR-828-TX	Severe Storm	Severe Storms, Tornadoes & Flooding
March 4-30, 1992	March 20, 1992	DR-937-TX	Flood	Severe Storms & Flooding
November 21, 1992	December 4, 1992	DR-970-TX	Tornado	Severe Thunderstorms & Tornadoes
October 14 – November 8, 1994	October 18, 1994	DR-1041-TX	Flood	Severe Thunderstorms & Flooding
August 22-31, 1998	August 26, 1998	DR-1239-TX	Severe Storm	Tropical Storm Charley
September 9-October 5, 1998	September 23, 1998	DR-1245-TX	Severe Storm	Hurricane Georges—Texas
October 24-November 15, 2002	November 5, 2002	DR-1439-TX	Severe Storm	Severe Storms, Tornadoes & Flooding
August 14-20, 2007	October 2, 2007	DR-1730-TX	Severe Storm	Tropical Storm Erin
May 4-June22, 2015	May 29, 2015	DR-4223-TX	Severe Storm	Severe Storms, Tornadoes, Straight-Line Winds & Flooding
October 22-31, 2015	November 25, 2015	DR-4245-TX	Severe Storm	Severe Storms, Tornadoes, Straight-Line Winds, & Flooding
April 17-30, 2016	April 25, 2016	DR-4269-TX	Flood	Severe Storms & Flooding
May 22-June24, 2016	June 11, 2016	DR-4272-TX	Flood	Severe Storms & Flooding
April 26-June 5, 2024	May 17, 2024	DR-4781-TX	Flood	Severe Storms, Straight-Line Winds, Tornadoes, & Flooding

Source: (FEMA 2025)

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Once an emergency proclamation is declared, The Texas Division of Emergency Management (TDEM) coordinates the state emergency management program, which is intended to ensure the state, and its local governments respond to and recover from emergencies and disasters and implement plans and programs to help prevent or lessen the impact of emergencies and disasters. Harris County has not been included in severe weather-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2024, Harris County was not included in any USDA severe weather-related agricultural disaster declarations.

Previous Events

Known hazard events that impacted Harris County between January 2020 and December 2024 are discussed in Table 15-7. For events prior to 2020, refer to the 2020 Harris County HMP. Between January 1, 2020, and February 2025, NOAA/NCEI Storm Events Database recorded 92 hail events, 13 lightning events, 110 Thunderstorm wind events, and 12 tornadoes (NOAA/NCEI 2025). Tornado locations are shown in Figure 15-2.

Table 15-7. Severe Weather Events in Harris County (2020 to 2024)

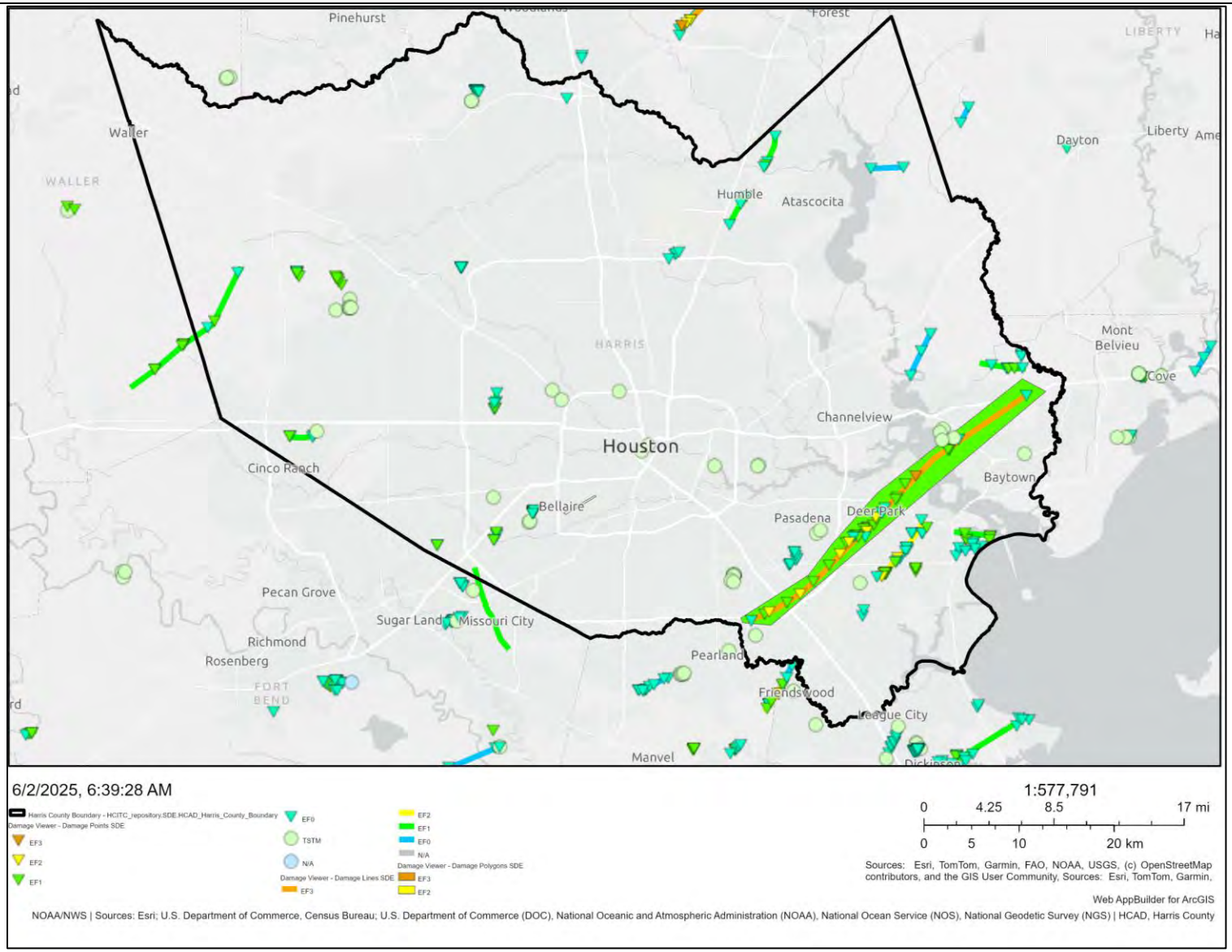
Event Date	Declaration or Proclamation Number	Location Impacted	Description
January 10, 2020	No	Katy, Jacinto City	A strong storm system produced a squall line that moved eastward across the area and produced a lot of wind damage and one tornado
May 27, 2020	No	Pasadena	Severe thunderstorms developed and produced wind damage, hail and few tornadoes. One was rated an EF0, and it resulted in $30,000 in damages as it tracked through Pasadena.
May 18, 2021	No	Crosby, Webster, and western portions of Houston and Unincorporated Harris County	Showers and thunderstorms moved across the area in the evening through late night hours and produced wind damage, lightning damage and some flooding.
June 15, 2021	No	Deer Park, Pasadena, Cypress	A number of severe thunderstorms developed across the area on the evening of June 15, resulting in many reports of tree damage, 1-inch hail, and building damage due to both winds and lightning strikes.
August 12, 2021	No	Cypress	Law enforcement and multiple spotters reported five power poles were downed by a brief EF0 landspout tornado. Total property damages totaled $5,000.
January 8-9, 2022	No	West Houston	Scattered thunderstorms associated with the approach of an offshore low-pressure system produced several tornadoes on the evening of January 8 and the early morning of January 9. These tornadoes resulted in several instances of property and tree damage across the Greater Houston Area. Some non-tornadic storms also produced strong, damaging wind gusts. Later, heavy rainfall associated with continuing thunderstorm activity resulted in flash flooding across the Houston Metro.
May 22, 2022	No	Westpark area	A lightning strike started a fire at an apartment complex.

Event Date	Declaration or Proclamation Number	Location Impacted	Description
January 24, 2023	No	Southeast Houston, Pasadena, Deer Park, Pearland, Baytown	An EF3 tornado produced $6.6 million in property damage. This EF3 tornado touched down near El Franco Lee Park east of Brookside Village. The path of damage extended east northeast, then northeast from there, across portions of Southeast Houston, Pasadena, Deer Park and Baytown. Most of the damage was rated EF0 and EF1 with damage to trees, fences and minor roof damage, but there were some spots with more severe damage, rated EF2. These included the Beamer apartments in Southeast Houston, a few homes in Pasadena, a church in Deer Park near Center Street and East P Street, and a single-family home with roof and some walls removed near the intersection of East X Street and Luella. From an aerial survey, there were three high tension electrical towers flattened northeast of Deer Park near the Houston Ship Channel with EF3 damage. From there, the damage path entered Baytown, north of the Ship Channel. In Baytown there was a mobile home, and a single-family home destroyed along Weaver Street near Bayway. Damage became more spotty and less intense northeast from there with spotty damage as far north as TX 330 Spur. Aerial surveys indicated some minor roof damage to the Exxon building south of West Baker Street. Damage continued up to Interstate 10 where down power lines were observed. No damage was indicated north of Interstate 10.
June 8, 2023	No	Humble, Lake Houston, Unincorporated Harris County	Afternoon through evening severe thunderstorms produced numerous wind damage and hail reports. A lightning strike and some flash flooding occurred too, resulting in $100,000 in damages.
June 21, 2023	No	Northern Harris County	Evening storms produced numerous strong wind reports and associated damage. Property damages totaled $2,515,000
June 23, 2023	No	Jersey Village	Early to mid-evening thunderstorms produced numerous reports of large hail.
September 25, 26, 2023	No	Humble	Late afternoon severe thunderstorms developed and produced some wind damage. Trees were downed at the San Jacinto College Generation Park campus. In all, $3,000 worth of damages were noted from these storms
January 8, 2024	No	Eastern Harris County	Severe thunderstorms developed ahead of a cold front and produced some hail and damage.
March 15, 2024	No	Unincorporated Harris County	Several rounds of severe thunderstorms produced numerous reports of large hail and wind damage.
March 21, 2024	No	Katy, Jersey Village, Unincorporated Harris County	Storms brought numerous evening reports of hail and some wind damage.

Event Date	Declaration or Proclamation Number	Location Impacted	Description
April 10, 2024	No	Western Harris County	A tornado initially touched down near or over Mason Road and immediately produced winds of approximately 90 mph as it destroyed a Firestone commercial building and significantly damaged the commercial strip mall in the vicinity (EF1). The building room and southern outer wall appeared to fail when tornadic winds blew out the west facing windows in the lobby. A large commercial air conditioning unit was thrown from south to north into the parking lot, with the Firestone building and majority of debris pushed more east to west. After leaving the touchdown point, the circulation then broadened and weakened to EF0 after it moved another 1.5 miles or so through the adjacent neighborhood to the east. Many trees were damaged (large limbs and some uprooting). There was only minor structural damage outside of that caused by downed trees and branches. Winds were generally 65 to 75 mph across this track.
May 2, 2024	No	Unincorporated Harris County	A series of disturbances moved across the area and produced flash flooding and severe thunderstorms.
May 16, 2024	DR-4781-TX	Katy, Cypress, Jersey Village, Deer Park, Houston, La Porte, and Unincorporated Harris County	Derecho winds brought widespread damage to many communities in Harris County. Wind speeds ranged between 54 and 90 mph in the thunderstorm while two EF1 tornadoes struck Cypress. In all, property damages exceeded $72.5 million
May 28, 2024	DR-4781-TX	Unincorporated Harris County, Clear Lake, Webster, Lake Houston	Another round of severe thunderstorms moved across the area, resulting in numerous reports of wind damage as well as hail that ranged in size from 1.0″ to 1.75″ in diameter.
December 28, 2024	No	Katy	EF1 tornado developed north of I-10 in the Katy/Brookshire area before tracking northeast towards western Harris County

Source: (NOAA/NCEI 2025) (NOAA/NWS 2025)

Figure 15-2. Historical Tornado and Damaging Wind Events in Harris County (2011-2025)



Note: the Damage Assessment Toolkit provided by the NWS only has archived storm data beginning on January 1, 2011.

Source: (NOAA/NWS 2025)

15.1.5 Probability of Future Occurrences

Information on previous Severe Weather occurrences in the County was used to calculate the probability of future occurrences of such events, as summarized in Table 15-8. Based on historical records and input from the Steering Committee, the probability of occurrence for Hazard in the County is considered "frequent."

Table 15-8. Probability of Future Severe Weather Events in Harris County

Hazard Type	Number of Occurrences Between 1950 and 2024	Percent Chance of Occurring in Any Given Year
Lightning	72	96%
Hail	474	100%
Tornadoes	252	100%
Thunderstorm Wind	828	100%
Total Severe Weather	1,626	100%

Source: (NOAA/NCEI 2025)

Climate Change Projections

Under a changing climate regime, the planning area can expect to encounter more frequent and more intense severe weather events each year. Should more intense and more frequent severe events occur in the County, the population can expect a greater impact.

The climate of Texas is changing. Most of the state has warmed by between 0.5 °F and 1 °F in the past century. In the eastern two-thirds of the state, rainstorms are more intense, and floods are becoming more severe. In the coming decades, storms are likely to become more severe in Texas (EPA 2016). Periods of extreme precipitation increase the risk of hail and lightning (Centers for Climate and Energy Solutions n.d.). Major clusters of summertime thunderstorms in North America will grow larger, more intense, and more frequent later this century in a changing climate, leading to increased rainfall and posing a greater threat of flooding across wide areas (UCAR 2017).

15.1.6 Cascading Impacts on Other Hazards

Severe weather events can cause sudden and widespread disruption to many sectors across Harris County. Cascading impacts can occur because of the county's dense population, extensive infrastructure, petrochemical facilities, and flood-prone landscape. Strong, damaging winds frequently accompany severe weather, knocking down tree branches that can destroy above-ground power lines and utility poles. Cascading impacts include power outages (utility failure), transportation disruption, flooding, structural fires, and hazardous materials incidents to name a few.

15.2 VULNERABILITY AND IMPACT ASSESSMENT

15.2.1 Life, Health, and Safety

Overall Population

The impact of severe weather on life, health, and safety depends on several factors, including the severity of the event and whether adequate warning time was provided to residents. As a result of severe weather events, residents can be displaced or require temporary to long-term sheltering. The most common problems associated with severe storms are immobility and loss of utilities. Residents impacted by severe weather may be displaced or require temporary to long-term sheltering. Additionally, downed trees, damaged buildings, and debris carried by winds associated with severe weather can result in injury or loss of life. In Harris County, the entire population shares a uniform risk of severe weather events.

Socially Vulnerable Population

Social vulnerability is defined as the susceptibility of social groups to the adverse impacts of natural hazards, including disproportionate death, injury, loss, or disruption of livelihood. Social vulnerability refers to the social, economic, demographic, and housing characteristics of a community that influence its capacity to prepare for, respond to, cope with, recover from, and adapt to environmental hazards. Although the entire population is exposed to severe weather, some populations are more vulnerable. Vulnerable populations include people who are elderly, people with low income, linguistically isolated populations, people with life-threatening illnesses, and residents living in areas that are isolated from major roads. Power outages can be life-threatening to those dependent on electricity for life support. In general, populations who lack adequate shelter during a severe weather event, those who are reliant on sustained sources of power in order to survive, and those who live in isolated areas with limited ingress and egress options are the most vulnerable.

15.2.2 General Building Stock

The county's building stock is exposed to severe weather hazards. Damage to buildings depends on several factors, including wind speed, storm duration, and path of the storm track. Building construction also plays a major role in the extent of damage resulting from a storm. Due to differences in construction, residential structures are generally more susceptible to storm damage than commercial and industrial structures. Wood and masonry buildings, in general, regardless of their occupancy class, tend to experience more damage than concrete or steel buildings. Lightning can spark wildfires or building fires, especially if structures are not protected by surge protectors on critical electronic, lighting, or information technology systems.

Manufactured housing (i.e., mobile homes) is particularly vulnerable to high winds. The U.S. Census Bureau defines manufactured homes as "movable dwellings, 8 feet or wider and 40 feet or more long, design to be towed on its own chassis, with transportation gear integral to the unit

when it leaves the factory, and without need of a permanent foundation (U. S. Census 2020)." They can include multi-wide and expandable manufactured homes but exclude travel trailers, motor homes, and modular housing. Due to their lightweight and often unanchored design, manufactured housing is extremely vulnerable to high winds and will generally sustain the most damage. Hailstorms can crack, break, and dent building materials, so building construction plays a major role in the extent of damage resulting from hails. Due to differences in construction, residential structures are generally more susceptible to hail than commercial and industrial structures. Wood and masonry buildings, in general, regardless of their occupancy class, tend to experience more damage than concrete or steel buildings.

15.2.3 Community Lifelines and Other Critical Facilities

Overall, all critical facilities in Harris County are vulnerable to being affected by severe weather. Utility infrastructure could suffer damage from lightning, hail, and high winds, resulting in the loss of power or other utility service. Loss of service can impact residents, critical facilities, and business operations alike. Interruptions in heating or cooling utilities can affect populations, such the young and elderly, who are particularly vulnerable to temperature-related health impacts. Loss of power can impact other public utilities, including potable water, wastewater treatment, and communications. In addition to public water services, property owners with private wells might not have access to potable water until power is restored. Lack of power to emergency facilities, including police, fire, EMS, and hospitals, will inhibit a community's ability to effectively respond to an event and maintain the safety of its citizens.

15.2.4 Economy

Impacts include loss of business function, damage to inventory, relocation costs, wage loss, and rental loss due to the repair or replacement of buildings. Business interruption losses include losses associated with the inability to operate a business because of the damage sustained during a storm or the temporary living expenses for those displaced from their home because of an event. Impacts to transportation lifelines affect both short-term (e.g., evacuation activities) and long-term (e.g., day-to-day commuting and goods transport) transportation needs. Utility infrastructure (power lines, gas lines, electrical systems) could suffer damage and impacts can result in the loss of power, which can impact business operations and can impact heating or cooling provision to the populations.

The wind's power to erode the land can be detrimental to agriculture. Loess, a sediment that can develop into one of the richest soils for farming, is easily swept up by wind. Even when farmers take precautions to protect it, the wind can erode up to 2.5 kilograms of loess per square meter. (1.6 pound per square foot) every year (Turgeon and Morse 2022).

Wind is a renewable resource that does not directly cause pollution. Wind energy is harnessed through powerful turbines. Wind turbines have a tall tubular tower with two or three propeller-like blades rotating at the top. When the wind turns the blades, the blades turn a generator and create electricity. The economic drawback to wind farms, however, is the wind itself. If it is not blowing, there is no electricity generated. (Turgeon and Morse 2022).

According to National Oceanic and Atmospheric Administration (NOAA) Technical Paper on Lightning Fatalities, Injuries, and Damage Reports in the United States from 1959−1994, monetary losses for lightning events range from less than $50 to greater than $5 million (larger losses associated with forest fires with homes destroyed and crop loss) (NOAA 1997).

Hail-producing severe storms impact the economy; impacts include loss of business function, damage to inventory, relocation costs, wage loss, and rental loss due to the repair or replacement of buildings. Additionally, vehicles parked outdoors are vulnerable to hail damage and could increase the economic impacts of a storm.

15.2.5 Natural, Historic and Cultural Resources

The impact of severe weather events on the environment varies, but researchers are finding that the long-term impacts of more severe weather can be destructive to the natural and local environment. National organizations such as United States Geological Survey (USGS) and NOAA have been studying and monitoring the impacts of extreme weather phenomena as it impacts long-term climate change, streamflow, river levels, reservoir elevations, rainfall, floods, landslides, erosion, etc. (USGS 2020). For example, severe weather that creates longer periods of rainfall can erode natural banks along waterways and degrade soil stability for terrestrial species. Severe wind events can tear apart habitats, causing fragmentation across ecosystems. Researchers also believe that a greater number of diseases will spread across ecosystems because of impacts that severe weather and climate change will have on water supplies (NOAA 2019). Overall, as the physical environment becomes more altered, species will begin to contract or migrate in response, which may cause additional stressors to the entire ecosystem within Harris County.

15.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

15.3.1 Potential or Planned Development

Understanding future changes that impact vulnerability in the County can assist in planning for future development and ensuring that appropriate mitigation, planning, and preparedness measures are in place. Areas targeted for potential future growth and development could be potentially impacted by hailstorms since the entire County is exposed. However, due to increased standards and codes, new development can be less vulnerable to extreme temperatures in comparison with the aging building stock currently in Harris County.

15.3.2 Projected Changes in Population

The County experienced an increase in population between the 2010 Census (585,375) and the estimated 2016−2020 American Community Survey estimated population of 790,892. The

population of the County is expected to increase over the next few years. The increase in population will expose more people to the severe weather hazard.

15.3.3 Climate Change

Scientists must attempt to predict how climate change might affect the individual weather "ingredients" that produce storms that produce strong wind events. These weather ingredients are (Geographic n.d.):

- Warm, moist air
- An unstable atmosphere
- Wind at different levels moving in different directions at different speeds, a phenomenon known as wind shear.

As global temperatures rise, the hotter atmosphere can hold more moisture. This increases atmospheric instability, a vital supercell ingredient. On the other hand, as the planet warms, wind shear (another vital ingredient) is likely to decrease. These two forces work against each other, and it is difficult to anticipate which might have a greater impact on tornado formation (Geographic n.d.).

The entire State of Texas is projected to experience an increase in the frequency and severity of extreme storms and rainfall. Climate change may lead to an increase in the number of lightning strikes and lightning-producing storms. Major clusters of summertime thunderstorms in North America will grow larger, more intense, and more frequent later this century in a changing climate, unleashing far more rain and posing a greater threat of flooding across wide areas (UCAR 2017). The changing climate may also increase the frequency of lightning flashes, which could rise by an estimated 50 percent across the continental United States over the next century. A warmer atmosphere can hold more moisture, and moisture is one of the key ingredients for triggering a lightning strike (Sanders 2014).

16. TSUNAMI

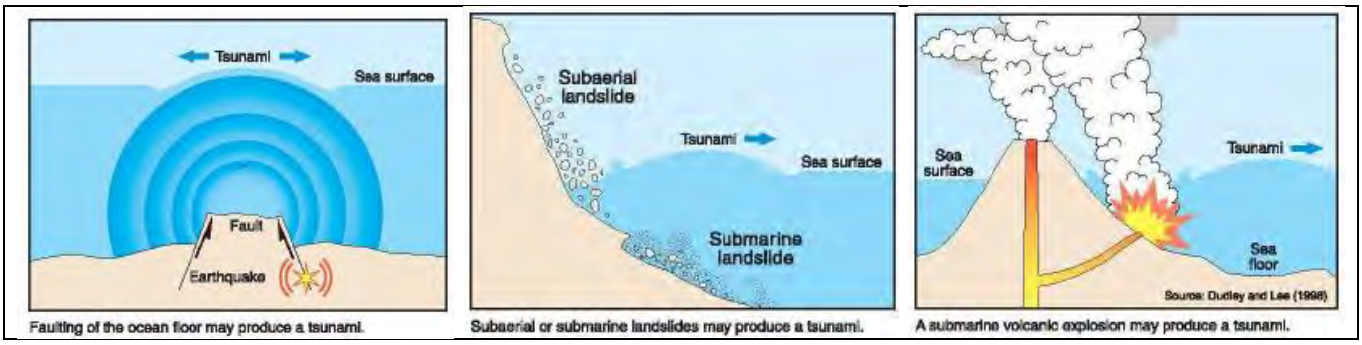
16.1 HAZARD PROFILE

16.1.1 Hazard Description

A tsunami is a series of high-energy waves that radiate outward like pond ripples from an area where a generating event occurs, arriving at shorelines over an extended period. Tsunamis can be induced by earthquakes, landslides and submarine volcanic explosions (see Figure 16-1).

Note: The risk from this hazard is considered to be low to negligible, mainly because of the lack of historical occurrence in the planning area. However, this hazard profile is provided in order to address the possibility of a future event in the planning area. This positions the planning partnership for benefit eligibility should an event of this type occur during the performance period of this plan.

Figure 16-1. Common Sources of Tsunamis



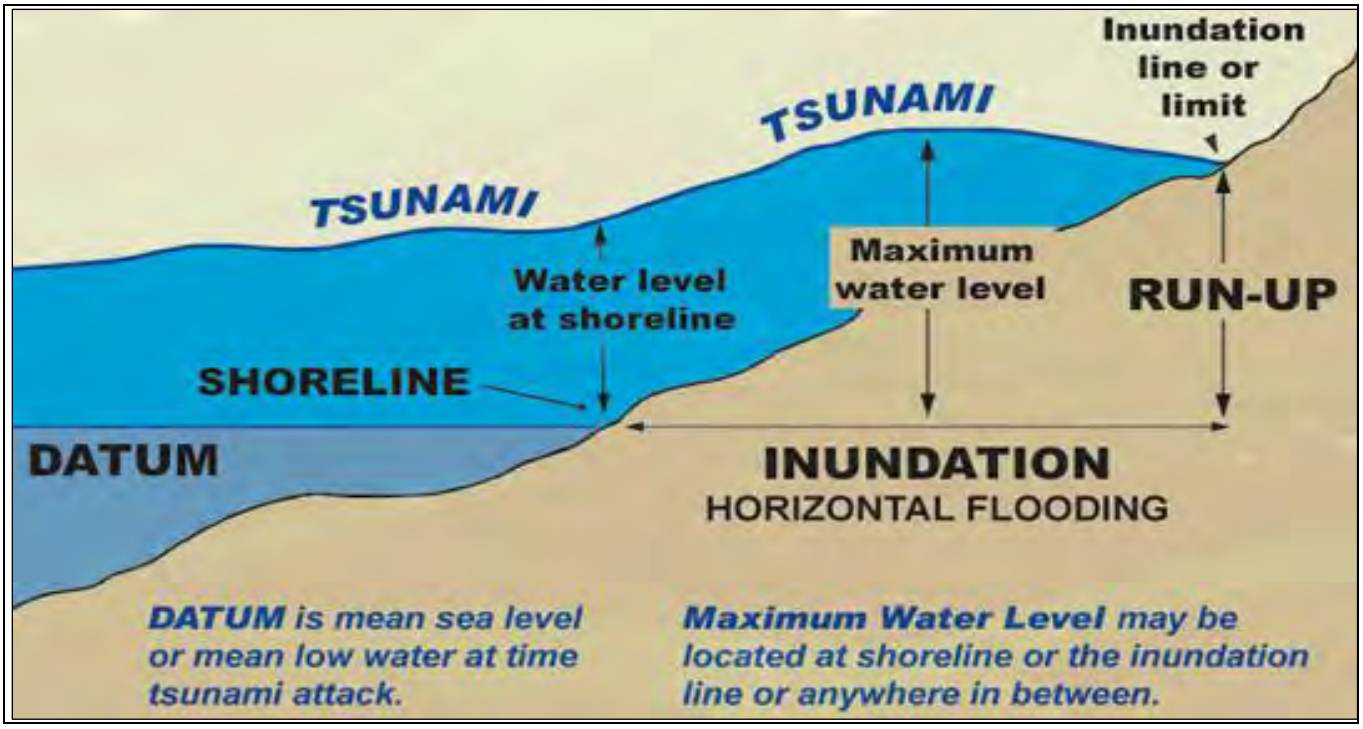
Tsunamis are typically classified as local or distant, depending on the location of their source in comparison to where waves occur:

- The waves nearest to the generating source represent a local tsunami. Such events have minimal warning time, leaving few options except to run to high ground. The damage from the tsunami itself may be accompanied by additional damage from the triggering earthquake due to ground shaking, surface faulting, liquefaction or landslides.
- The waves far from the generating source represent a distant tsunami. Distant tsunamis may travel for hours before striking a coastline, giving a community a chance to implement evacuation plans.

In the open ocean, a tsunami may be only a few inches or feet high, but it can travel with speeds approaching 600 miles per hour. As a tsunami enters the shoaling waters near a coastline, its speed diminishes, its wavelength decreases, and its height increases greatly. At the shoreline, tsunamis may take the form of a fast-rising tide, a cresting wave, or a bore (a large, turbulent wall-like wave). The bore phenomenon resembles a step-like change in the water level that advances rapidly (from 10 to 60 miles per hour). The first wave is usually followed by several larger and more destructive waves.

The configuration of the coastline, the shape of the ocean floor, and the characteristics of advancing waves play important roles in the destructiveness of the waves. Bays, sounds, inlets, rivers, streams, offshore canyons, islands, and flood control channels may cause various effects that alter the level of damage. Offshore canyons can focus tsunami wave energy, and islands can filter the energy. It has been estimated that a tsunami wave entering a flood control channel could reach a mile or more inland, especially if it enters at high tide. The orientation of the coastline determines whether the waves strike head-on or are refracted from other parts of the coastline. A wave may be small at one point on a coast and much larger at other points. The inundation area for a tsunami event is often described as runup as illustrated in Figure 16-2..

Figure 16-2. Runup Distance and Height in Relation to the Datum and Shoreline



Source: UNESCO, Retrieved from Different Directions: Tsunami, n.d.

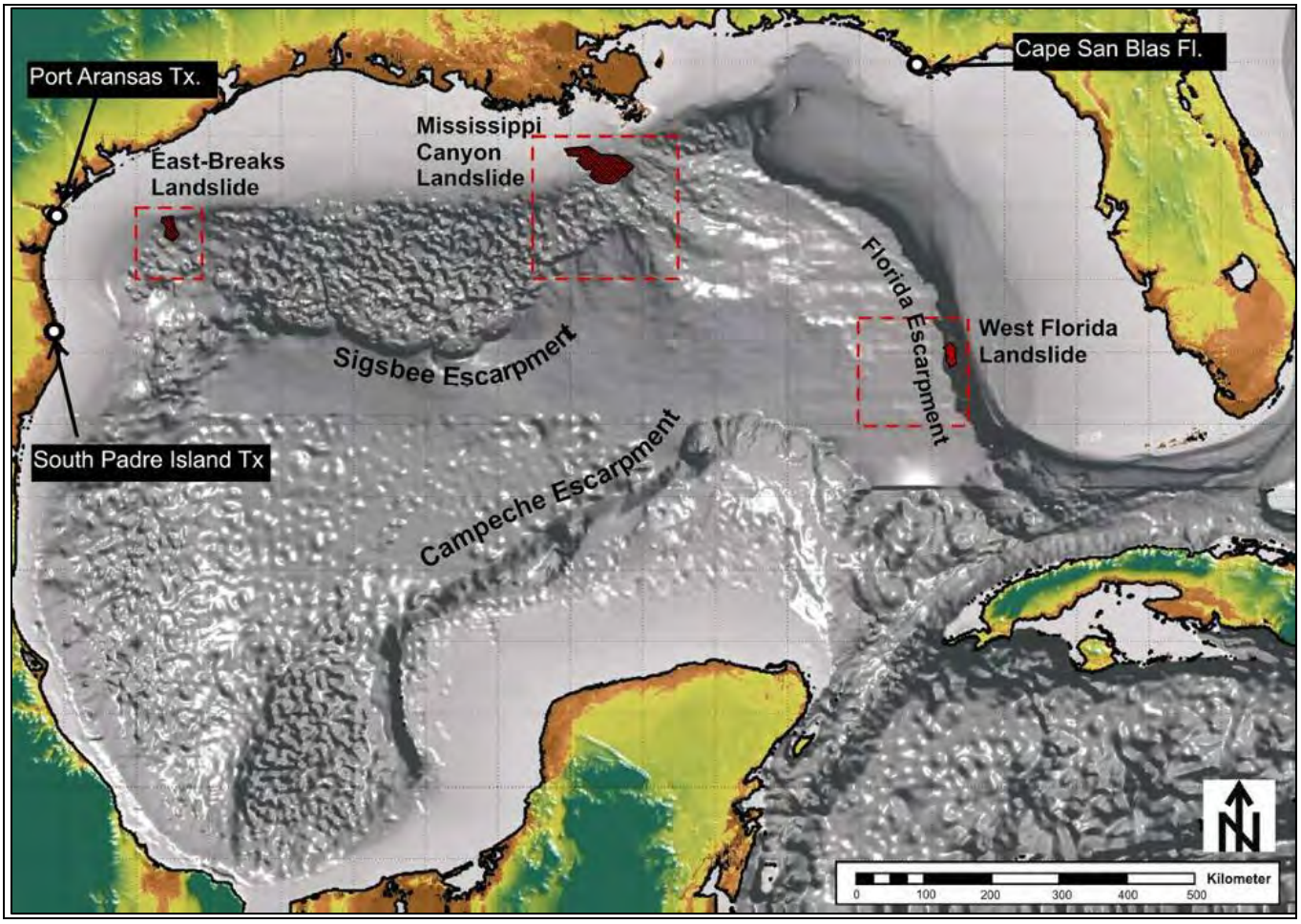
The Gulf of Mexico states have been included to the U.S. Tsunami Warning System since January 2005. The main purpose of the warning system is to enable local emergency management to act in response to warnings. To plan for the warning response, emergency managers must understand what specific areas within their jurisdictions are threatened by tsunamis. Potential tsunami sources for the Gulf of Mexico are local submarine landslides and earthquakes (induced co-seismic tsunami source) along the Caribbean plate boundary faults. However, preliminary modeling of potential tsunami sources outside the Gulf of Mexico has indicated a very low threat and may not pose a tsunami hazard to the Gulf of Mexico coastal communities or infrastructure (TAMU 2025).

Nevertheless, recent assessments of tsunami hazards along the Gulf of Mexico carried out by the U.S. Geological Survey (USGS) and the National Tsunami Hazard Mitigation Program (NTHMP) have identified underwater landslides as the primary potential source of tsunami generation.

16.1.2 Location

The Texas A&M University at Galveston Tsunami Research Group has developed a 3D numerical model for what could happen after a submarine landslide. The 3D model is combined with a 2D model to depict how a tsunami could affect coastal areas, and, in the Gulf, there are three areas that may be the source of future tsunamis along the upper-Texas coastline (see Figure 16-3). There is currently no extent and location mapping of the tsunami hazard available to support tsunami hazard risk assessment for this plan.

Figure 16-3. Gulf of Mexico Bathymetry and Location of Submarine Landslides



Source: (NTHMP 2010)

While no formal tsunami inundation shapefile or data source was available, the jurisdictions that would be at risk to Tsunami include Morgan's Point, La Porte, Shoreacres, Seabrook, El Lago, Nassau Bay, Webster, and portions of unincorporated Harris County. Until tsunami inundation scenarios are made available, it remains unclear if jurisdictions along the Houston Ship Channel (Pasadena, Deer Park, and Galena Park) are at risk of tsunami waves.

16.1.3 Extent

Preliminary modeling of potential tsunami sources outside the Gulf of Mexico by USGS has indicated a very low threat and may not pose a tsunami hazard to the Gulf coastal communities or infrastructure. Nevertheless, recent assessments of tsunami hazards along the Gulf of Mexico carried out by the U.S. Geological Survey (USGS) and the National Tsunami Hazard Mitigation Program have identified underwater landslides as the primary potential source of tsunami generation. Although a massive underwater landslide in the Gulf is considered a potential hazard, the probability of such an event is quite low. The probability of occurrence is related to large ancient landslides which were probably active prior to 7,000 years ago when large quantities of sediments were emptied into the Gulf. However, nowadays sediments continue to empty into the Gulf mainly from the Mississippi river. The sediment supply contributes to slope steepening and also to the increasing of the excess pore water pressure in the underlying soils, which may lead to further landslide activities. Recent evidence from seismic records of small-scale energetic seismic waves in the Gulf have confirmed that there is a probability of recurrence. For the risk assessment for this plan update, the probability of occurrence for this hazard is considered "low."

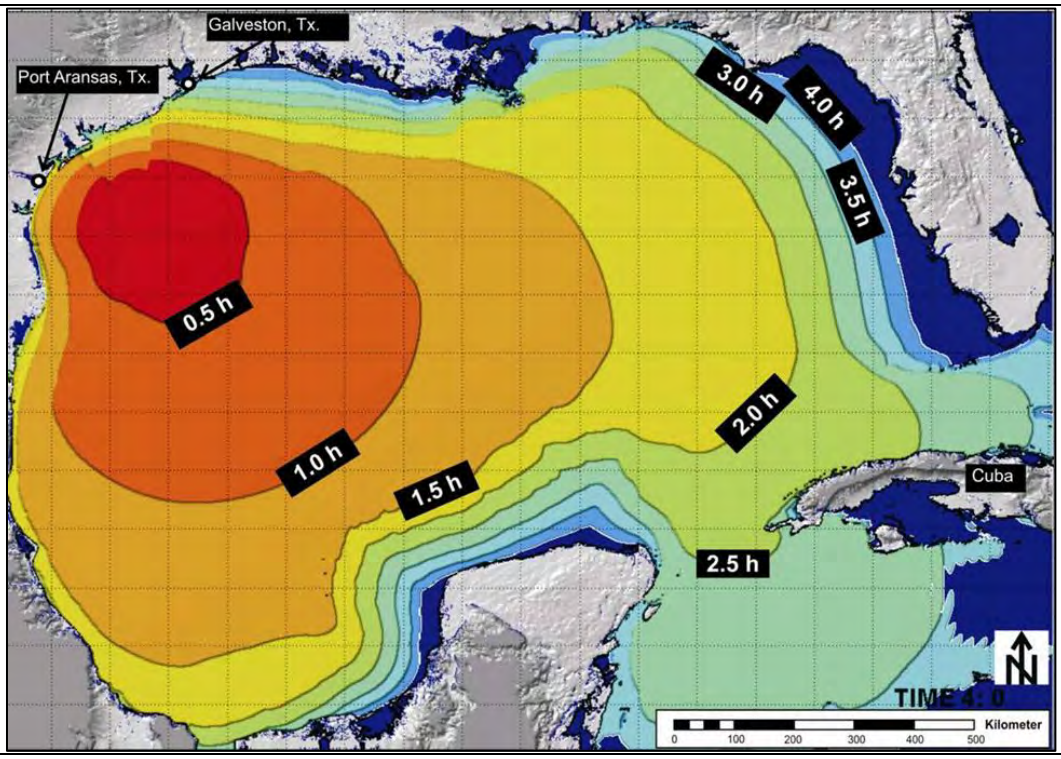
Severity

According to the National Tsunami Hazard Mitigation Program, tsunami events with runups of more than 1 meter are the most likely to be dangerous to people and property. The tsunami's size and speed, as well as the coastal area's form and depth, affect the impact of a tsunami. At some locations, the advancing turbulent wave front will be the most destructive part of the tsunami wave. In other situations, the greatest damage will be caused by the outflow of water back to the sea between crests, sweeping away items on the surface and undermining roads, buildings, bulkheads, and other structures. This outflow action can carry enormous amounts of highly damaging debris, resulting in further destruction. Ships and boats, unless moved away from shore, may be dashed against breakwaters, wharves, and other craft, or be washed ashore and left grounded after the withdrawal of the seawater (National Tsunami Hazard Mitigation Program, 2001). A local tsunami resulting from an underwater landslide in the Gulf of Mexico presents the most severe risk to the planning area.

Warning Time

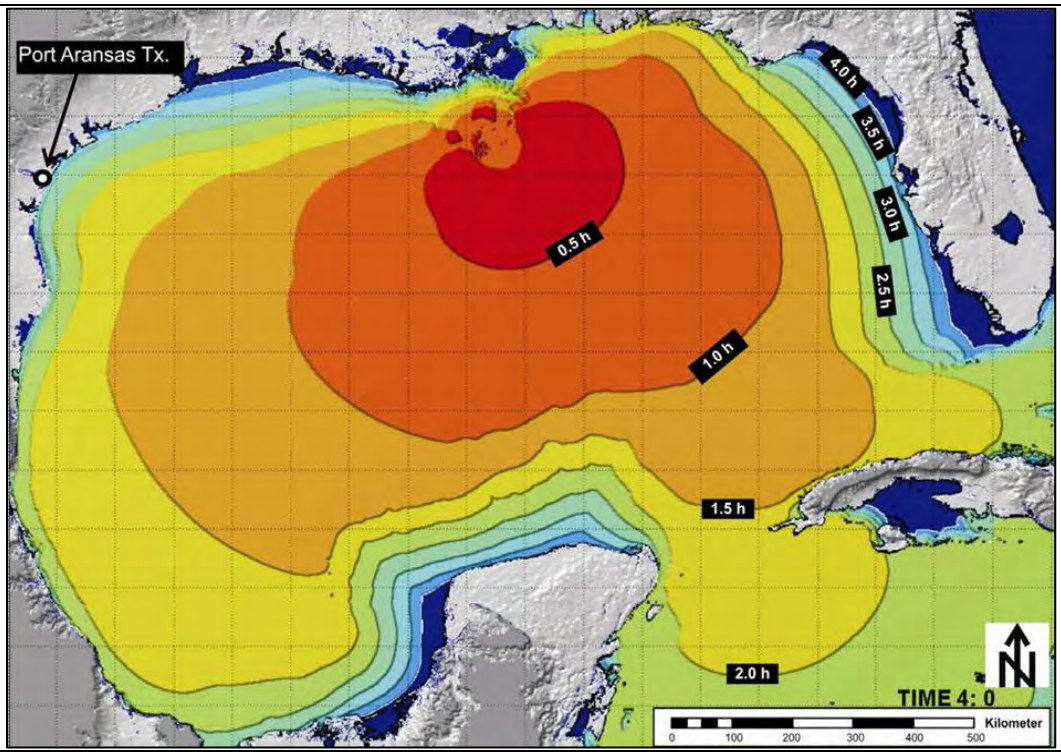
While tsunamis in the Gulf of Mexico are rare, the Gulf states were added to the U.S. Tsunami Warning System in 2005. The main purpose of the warning system is to enable local emergency management to act in response to warnings. To plan for the warning response, emergency managers must understand what specific areas within their jurisdictions are threatened by tsunamis. Potential tsunami sources for the Gulf of Mexico are local submarine landslides and earthquakes (induced co-seismic tsunami source) along the Caribbean plate boundary faults. Figure 16-4 and Figure 16-5 show the estimated travel time for tsunamis generated by the East Break and Mississippi Canyon landslides. However, preliminary modeling of potential tsunami sources outside the Gulf has indicated a very low threat and may not pose a tsunami hazard to the Gulf coastal communities or infrastructure.

Figure 16-4. Tsunami Arrival Time for the East Break Landslide



Source: (NTHMP 2010)

Figure 16-5. Tsunami Arrival Time for the Mississippi Canyon Landslide



Source: (NTHMP 2010)

Visible Indications

Tsunamis are difficult to detect in the open ocean; with waves generally less than 3 feet high. The first visible indication of an approaching tsunami may be either a rise or drop in water surface levels (National Tsunami Hazard Mitigation Program, 2001):

- A drop in water level (draw down) can be caused by the trough preceding the advancing, large inbound wave crest. Rapid draw down can create strong currents in harbor inlets and channels that can severely damage coastal structures due to erosive scour around piers and pilings. As the water's surface drops, piers can be damaged by boats or ships straining at or breaking their mooring lines. The vessels can overturn or sink due to strong currents, collisions with other objects, or impact with the harbor bottom.
- The advancing tsunami may initially arrive as a strong surge increasing the sea level. This can be similar to the rising tide, but the tsunami surge rises faster and does not stop at the shoreline. Even if the wave height appears to be small, 3 to 6 feet for example, the strength of the accompanying surge can be deadly. Waist-high surges can cause strong currents that float cars, small structures, and other debris. Boats and debris are often carried inland by the surge and left stranded when the water recedes.

National Tsunami Warning System

The Tsunami Warning Centers depend on an observation system that includes seismic and water-level networks from around the world to help them determine when and where to issue tsunami messages. These networks are critical to the warning centers' ability to provide timely and accurate messages:

- Seismic Networks—When an earthquake occurs, seismic networks provide information about its location, depth, magnitude, and other source characteristics. The warning centers analyze this information to determine if the earthquake could have generated a tsunami and if a tsunami message is necessary.
- Water-Level Networks—If an earthquake meets certain criteria, the warning centers turn to water-level information, looking for changes in water-level height that could indicate the existence and size of a tsunami. The primary sources of information about water-level change are a network of Deep-ocean Assessment and Reporting of Tsunami systems and an extensive array of coastal water-level stations.

16.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2023, Harris County was not included in any major disaster (DR) or emergency (EM) declarations for tsunami-related events (FEMA 2023).

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Once an emergency proclamation is declared, The Texas Division of Emergency Management (TDEM) coordinates the state emergency management program, which is intended to ensure the state, and its local governments respond to and recover from emergencies and disasters and implement plans and programs to help prevent or lessen the impact of emergencies and disasters. Harris County has not been included in any tsunami-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2023, Harris County was not included in any USDA tsunami-related agricultural disaster declarations

Previous Events

Between 2020 and 2024, Harris County was not included in any tsunami-related disaster events. Though tsunamis are more likely to affect west coast states, historical evidence does show that tsunamis have affected the eastern United States and Gulf of Mexico, including Texas. The following Texas tsunami evidence was gathered from the National Geophysical Data Center (Tsunami Event Database):

- October 24, 1918—A small tsunami, reported to have originated with an earthquake near Puerto Rico, ran ashore in Galveston. Wave height was unknown. No damage, deaths or injuries were recorded.
- May 2, 1922—Three tsunami waves ran ashore in Galveston within a 45-minute period, followed 8 hours later by a similar but smaller train of waves. The maximum water height of the highest wave was 0.6 meters above sea level. Researchers maintain that there is some question regarding the origin and nature of these tsunamis, noting that the waves were probably erroneously ascribed to a small earthquake in Puerto Rico (NOAA Center for Tsunami Research).

Research of historical data shows no recorded property damage or loss of life due to tsunami in the Harris County planning area. In addition, the State of Texas Mitigation Plan does not identify tsunamis as a hazard for any location within the State of Texas.

16.1.5 Probability of Future Occurrences

Information on previous tsunami occurrences in the County was used to calculate the probability of future occurrences of such events, as summarized in Table 16-1. Based on historical records and input from the Steering Committee, the probability of occurrence for tsunamis in the County is considered "rare."

Table 16-1. Probability of Future Tsunami Events in Harris County

Hazard Type	Number of Occurrences Between 1950 and 2025	Percent Chance of Occurring in Any Given Year
Tsunami	0	<1%

Source: (NCEI (formerly NGDC) 2025)
Note: While no historic tsunamis have occurred along the shoreline of Harris County, one event in 1922 produced tidal run-ups, and for that reason, probabilities are not zero.

Climate Change Projections

The earthquakes and landslides that create tsunamis could be impacted by climate change. Some scientists say that melting glaciers could induce tectonic activity. Heavy rainfall could cause soil instability that may increase the likelihood of landslides into water bodies, which can generate tsunamis. Increases in severe storms may result in an increased probability of seiches. Rising seas could result in an increase in wave runup when tsunamis occur. Even modest rises in sea level will dramatically increase the frequency and intensity of flooding when a tsunami occurs, as the tsunami can travel further inland. Future smaller tsunamis could have the same impact as larger tsunamis today. A warming climate can increase the risk of underwater and above ground landslides, thereby increasing the risk of local tsunamis.

16.1.6 Cascading Impacts on Other Hazards

Aside from the tremendous hydraulic force of the tsunami waves themselves, floating debris carried by a tsunami can endanger human lives and batter inland structures. Flooding can cause contamination of drinking water and can result in the spread of disease.

Tsunami inundation can result in erosion and scouring, debris movement and impact, water contamination, and spread of disease due to standing water. Loss of wetlands from erosion and wetland migration due to tsunami inundation can reduce the natural filtration provided by wetland plants, increasing the likelihood of water quality issues. Healthy coastal ecosystems, fisheries, tourism, human health, and public safety can be transformed, degraded, or lost due in part to tsunamis. Indirect economic costs (such as lost business) and adverse socio-psychological impacts on residents and tourists have the potential to negatively affect people and their communities.

16.2 VULNERABILITY AND IMPACT ASSESSMENT

A detailed hazard exposure and vulnerability assessment was not conducted for tsunami due to the localized and infrequent nature of its impacts in this region. Unlike other hazards with broader geographic reach or more consistent historical data, tsunamis in Harris County tends to affect limited areas along the shoreline and does not pose a widespread threat to the county's overall building stock, critical facilities, or economy. As such, a traditional vulnerability assessment—typically used to quantify potential losses and exposure—was deemed not applicable for this hazard. Instead, the risk of tsunami across the planning area and participating jurisdictions is considered minimal, with potential losses expected to be rare and highly site-specific.

Unique and Varied Risks

Harris County lies more than 40 miles inland from the Gulf of Mexico and is bordered to the southeast by Galveston Bay and the Houston Ship Channel, both of which are semi-enclosed, shallow bodies of water. As such, the likelihood of a damaging tsunami directly affecting Harris County is extremely low. Nevertheless, FEMA guidance and the Texas Division of Emergency Management (TDEM) encourage all coastal counties to acknowledge tsunami risk as part of a comprehensive, all-hazards planning framework, especially given that even a distant or localized offshore event could create minor surge effects or seiche-type wave oscillations within Galveston Bay.

While a large Pacific-style tsunami is not expected to reach Harris County, low-probability, high-consequence scenarios—such as a submarine landslide or meteorological tsunami ("meteotsunami") generated by intense squall lines or hurricanes—could induce limited wave amplification along the bay shoreline, causing localized flooding and navigation hazards. These impacts would be confined primarily to bayfront and coastal jurisdictions, with negligible effects inland.

Southern and Coastal Jurisdictions

The Cities of Seabrook, El Lago, Taylor Lake Village, Shoreacres, La Porte, and Morgan's Point are the only jurisdictions in Harris County that could experience direct tsunami or seiche effects due to their proximity to Galveston Bay and Clear Lake. Morgan's Point and La Porte, located near the mouth of the Houston Ship Channel, are most directly exposed to potential water-level oscillations that could accompany distant seismic activity or under-bay slumping. Seabrook and El Lago face limited exposure through wave run-up along marina infrastructure and lakefront residential areas; most impacts would resemble minor storm-surge or boat-wake flooding rather than a catastrophic tsunami. Taylor Lake Village, being more sheltered, could experience brief water-level fluctuations within Taylor Lake, possibly inundating low-lying parklands or docks. Shoreacres and Seabrook maintain small-boat harbors and shoreline facilities that could be affected by amplified oscillations within Galveston Bay's shallow basin.

Although damage potential is minor, these communities would likely experience the earliest observable effects from any Gulf-generated tsunami, emphasizing the importance of regional early-warning systems and public education on spontaneous water-level changes.

Eastern Jurisdictions – Bayfront and Industrial Zones

Communities such as Baytown, Pasadena, and Deer Park are positioned along the upper San Jacinto River estuary and the Houston Ship Channel, which could transmit water-level changes inland. Industrial facilities lining the Ship Channel could face navigation hazards, barge displacement, or overtopping of containment berms in the unlikely event of a rapid oscillatory wave surge. Goose Creek CISD and Lee College in Baytown are inland from the shoreline but depend on access routes (e.g., Bayway Drive, Spur 330) that cross low-lying ground subject to brief inundation in extreme cases. Harris Health System clinics in east Harris County would be indirectly affected if port facilities or transportation corridors were temporarily disrupted. Although the physical threat to life and property is minimal, critical industrial operations and hazardous-

materials containment along the Ship Channel could amplify secondary risks if wave action coincided with ongoing industrial activity or storm events.

Central and Urban Jurisdictions

Houston, Bellaire, West University Place, and Southside Place, as well as the University of Houston and major Harris Health System hospitals (Ben Taub, LBJ), are located far inland and face no direct tsunami risk. However, cascading disruptions could occur if transportation routes, refineries, or port operations along the Ship Channel were affected. Or, if power or fuel supply interruptions were to occur due to coastal facility shutdowns. Another impact would be toward emergency-response assets were redirected toward coastal evacuations or port recovery. Thus, while direct exposure is absent, these inland jurisdictions must still consider indirect economic and logistical effects tied to coastal infrastructure vulnerability.

Northern and Western Jurisdictions

Jurisdictions such as Katy, Tomball, Jersey Village, and Waller, as well as Cy-Fair ISD and Spring ISD, are located at higher elevations far from tidal influence. These areas have no physical tsunami exposure. However, as part of the broader regional emergency-management network, they could serve as host shelters or staging areas if a Gulf-coast tsunami affected Galveston, Chambers, or coastal Harris County communities.

Areawide Educational Institutions

San Jacinto College South Campus (Clear Lake area) is among the few higher-education facilities closest to tidal waters and could experience minimal shoreline inundation under extreme, low-probability tsunami scenarios. Lee College and Goose Creek CISD campuses in Baytown may experience minor backwater flooding or port-related disruptions in transportation and utility service. Cy-Fair ISD, Spring ISD, and the University of Houston are located too far inland for any direct effects but may provide logistical and sheltering support to coastal residents. Harris Health System would play a countywide coordination role in any tsunami-related evacuation or medical-surge response, particularly through its east-county clinics and trauma facilities.

16.2.1 Life, Health, and Safety

Tsunamis pose significant vulnerabilities to coastal communities, impacting life, health, and safety. These natural disasters not only threaten immediate physical harm but also lead to long-term public health challenges and community well-being issues.

Overall Population

The public health impacts of tsunamis closely resemble those associated with coastal flooding. Beyond the immediate threats to life and property, tsunamis can lead to the contamination of drinking water through saltwater intrusion, inundation of hazardous materials facilities, or wastewater treatment plants. Floodwaters may carry hazardous substances, rendering them perilous to navigate. Additionally, flooded structures can rapidly develop mold or wood rot, posing

significant health risks to inhabitants if not properly addressed. These cascading effects highlights the extensive and severe impact tsunamis can have on the overall population, affecting physical safety, long-term public health, and community well-being.

Socially Vulnerable Population

Socially vulnerable populations, such as the economically disadvantaged, elderly, and those with disabilities, face heightened risks during tsunami events due to several factors. Limited financial resources can prevent economically disadvantaged individuals from adequately preparing for or recovering from a tsunami, and they may live in housing that is more susceptible to damage. Elderly and disabled individuals often have mobility issues that hinder quick evacuation and may require assistance, which can be scarce during emergencies.

Health concerns are also significant, as the elderly and those with pre-existing conditions may need medical attention during and after a tsunami, but medical facilities may be overwhelmed or inaccessible. Additionally, tsunamis can disrupt transportation and communication networks, isolating communities and making it difficult for emergency services to reach those in need.

16.2.2 General Building Stock

All current and future buildings and infrastructure, including critical facilities, located within low-lying coastal areas are considered at risk to tsunami.

16.2.3 Community Lifelines and Other Critical Facilities

Tsunamis pose a significant threat to community lifelines and critical facilities, disrupting essential services and hindering emergency response and recovery efforts. Water systems, including treatment plants, pipelines, and distribution networks, can sustain severe damage, leading to contamination and loss of potable water, thereby compromising public health and sanitation. Transportation infrastructure, such as roads, bridges, and airports, is highly susceptible to tsunami damage, impeding evacuation efforts and delaying the arrival of emergency services. This destruction can isolate communities and hinder the delivery of essential supplies.

Energy infrastructure, including power plants, substations, and transmission lines, is also vulnerable, resulting in widespread power outages that affect other critical services like healthcare and telecommunications. Communication networks, including cell towers and fiber optic cables, can be disrupted, making it challenging to coordinate emergency response efforts and communicate with the public, leading to delays in disseminating vital information. Healthcare facilities, such as hospitals and clinics, may be damaged or flooded, reducing their capacity to provide medical care to those injured during the tsunami, and exacerbating the impact on vulnerable populations. Emergency services, including fire stations and police stations, can be impacted, limiting their ability to respond effectively to the disaster and resulting in delayed rescue operations and increased risk to public safety.

The interdependency between these lifelines means that damage to one sector can have cascading effects on others, compounding the disruption of services and delaying recovery efforts. Addressing

these vulnerabilities through proactive hazard mitigation planning is essential to enhance community resilience and ensure a swift and effective response to tsunami events.

16.2.4 Economy

Tsunamis can have wide-reaching economic impacts on Harris County. Direct economic consequences may arise from the destruction of buildings and infrastructure, including power utility poles and lines, potable water and wastewater facilities, and roadways. The force of tsunami waves can devastate piers, breakwaters, and seawalls. Ships, marinas, and fishing fleets may be battered, sunk, moored, or stranded, severely affecting the fishing and tourism industries.

Business interruptions are likely, and damages or alterations to impacted beaches can affect tourism for extended periods after the tsunami. Additionally, the county may experience a decline in tourist numbers due to perceived risks associated with visiting post-tsunami. This reduction in tourism can lead to significant revenue losses for local businesses and the overall economy.

16.2.5 Natural, Historic and Cultural Resources

Natural

Tsunamis can significantly alter the landscape of Harris County, both above and below the water. In some areas, the land may rise, while in others, it may fall. Flooding can extend to unexpected areas if the coast subsides, and tsunami barriers may fail or be overtopped. Associated landslides and sediment shifts can render waterways unnavigable. Other potential consequences include permanent changes to beaches and other coastal features, loss of or changes to wildlife habitats, and alterations in the quality and availability of fresh water. Inundation by salt water, whether short or long term, can render agricultural land useless.

Conversely, subsidence events may lead to the formation of salt marshes or inundated shellfish beds. The force of tsunami waves can alter coastal features, affecting both terrestrial and aquatic ecosystems. Flooding can extend to unexpected areas, and associated landslides and sediment shifts can render waterways unnavigable. These changes can disrupt wildlife habitats, impacting the diverse bird, animal, and plant life that thrive in the park. Additionally, the inundation of salt water can harm agricultural land and natural resources, affecting the services they provide, such as biological, ecological, protective, and recreational functions.

The impact on fisheries can be profound, as the destruction of breeding grounds and changes in water quality can reduce fish populations. This, in turn, affects the livelihoods of local communities that depend on fishing. The disruption of ecosystems can have cascading effects, leading to long-term ecological imbalances. Overall, the environmental damage caused by tsunamis can have lasting effects on natural resources, requiring extensive restoration efforts and long-term monitoring to mitigate the impacts.

Historic

Tsunamis can have devastating effects on the historic resources of Harris County, causing both immediate and long-term damage. Historic buildings, monuments, and landmarks located near the coast are at high risk of being damaged or destroyed by the powerful force of tsunami waves. This can result in the loss of irreplaceable cultural heritage.

The force of the water can erode foundations and walls, while sediment deposition can bury or obscure historic sites, making restoration efforts challenging and sometimes impossible. The inundation of saltwater can cause significant damage to the materials and structural integrity of historic buildings, as wood, metal, and other materials can deteriorate rapidly when exposed to saltwater, leading to long-term preservation issues.

Tsunamis can also wash away or bury artifacts, making them difficult to recover or permanently destroying valuable historical and archaeological information. Additionally, tsunamis can change the topography of coastal areas, altering the context in which historic sites are situated, which can affect their historical significance and interpretation.

Cultural

Tsunamis can have devastating impacts on cultural resources in Harris County. They can destroy sacred sites and disrupt traditional fishing practices and other cultural activities, profoundly affecting the community's way of life. The loss of these cultural resources can diminish the community's connection to its history and traditions, impacting cultural identity and continuity.

16.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

16.3.1 Potential or Planned Development

It is assumed that development and redevelopment trends in Harris County are not such that there is major concern about development in identified tsunami risk areas. Future development in the County will be regulated by their respective general plans.

16.3.2 Projected Changes in Population

Harris County experienced an increase in population between the 2010 Census (4,110,771) and the 2020 Census (4,780,913). The population of the County is expected to increase over the next few years. The increase in population will expose more people to the potential tsunami risk (U.S. Census Bureau 2022).

16.3.3 Climate Change

Climate change may increase tsunami risk by affecting the factors that cause them. Melting glaciers and heavy rainfall can trigger tectonic activity and landslides, respectively, leading to tsunamis. Rising sea levels could amplify the impact of tsunamis, causing more severe flooding. Even smaller future tsunamis could have significant effects due to higher sea levels and increased storm activity.

16.3.4 Other Identified Conditions

Understanding local bathymetry and topography is crucial for tsunami mitigation and planning, as different areas will experience varying levels of impact based on their specific geographical features. Limiting new development in tsunami hazard zones and adopting building codes that address tsunami risks can reduce potential damage, making thoughtful land-use planning essential for protecting vulnerable areas.

17. WILDFIRE

17.1 HAZARD PROFILE

17.1.1 Hazard Description

Wildfires are uncontrolled fires that spread rapidly through vegetated areas, typically fueled by grasses, brush, or forested landscapes. In Harris County, wildfires are typically grass or brush fires, though forested and wooded areas can also support more intense burns. Wildfires can be triggered by natural causes such as lightning but are most frequently the result of human activity, including debris burning, arson, vehicle exhaust, or utility malfunctions. Wildfire behavior is influenced by vegetation type and density (fuel), weather conditions (wind, humidity, temperature), topography (slope and terrain), drought conditions or dry spells, as well as human development patterns in the wildland-urban interface (WUI).

The term WUI refers to the area where human development, such as houses and businesses, directly meets or intermingles with undeveloped, wildland vegetation. This area is a transition zone where built environments and natural ecosystems intersect, making it a focus for conflicts and risks like wildfires (USFA 2022).

17.1.2 Location

While Harris County is mostly urban or suburban, it also contains rural and semi-rural areas prone to wildfire. Key risk locations include:

- Northern and western portions of the county (e.g., Cypress, Tomball, Hockley)
- Wooded parks and nature preserves, including areas within Katy Prairie, Spring Creek, and the Addicks and Barker Reservoirs
- WUI zones where residential development meets forested or brush-covered land
- Utility easements, rights-of-way, and undeveloped land parcels—often near highways or in transition zones

According to the Texas A&M Forest Service, Harris County contains areas with moderate to high wildfire susceptibility, particularly where vegetation is dense, and access is limited for firefighting equipment. Much of the planning area can expect to experience wildfires in the future, especially jurisdictions within the Wildfire Ignition Density and Wildfire Threat areas as mapped in Figure 17-1 and Figure 17-2. In addition to the participating municipalities shown in the figures below, Table 17-1 lists additional planning partners who are also at risk to wildfires.

Figure 17-1. Wildfire Hazard Risk Area in Harris County

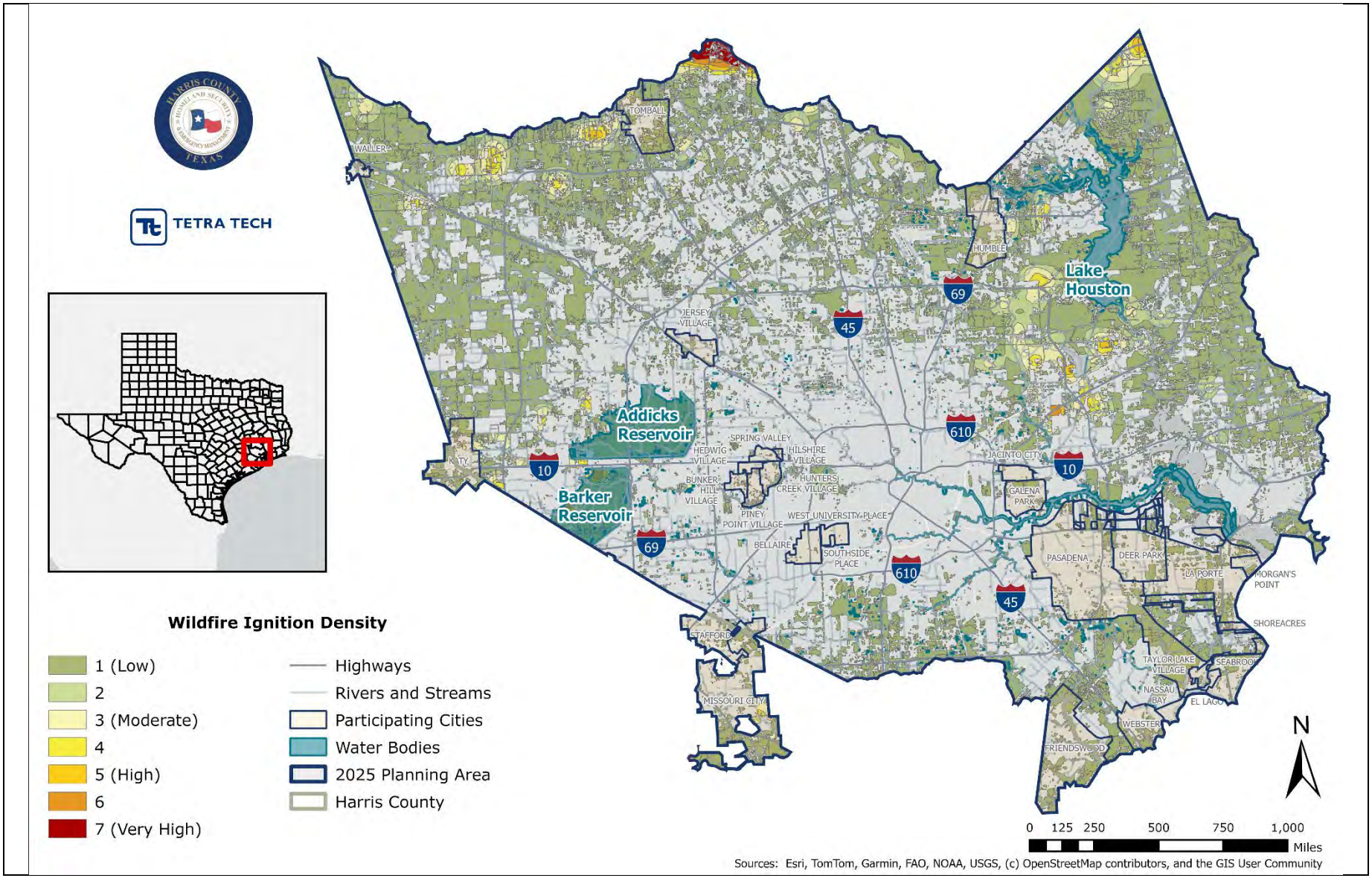


Figure 17-2. Wildfire Threat Areas in Harris County

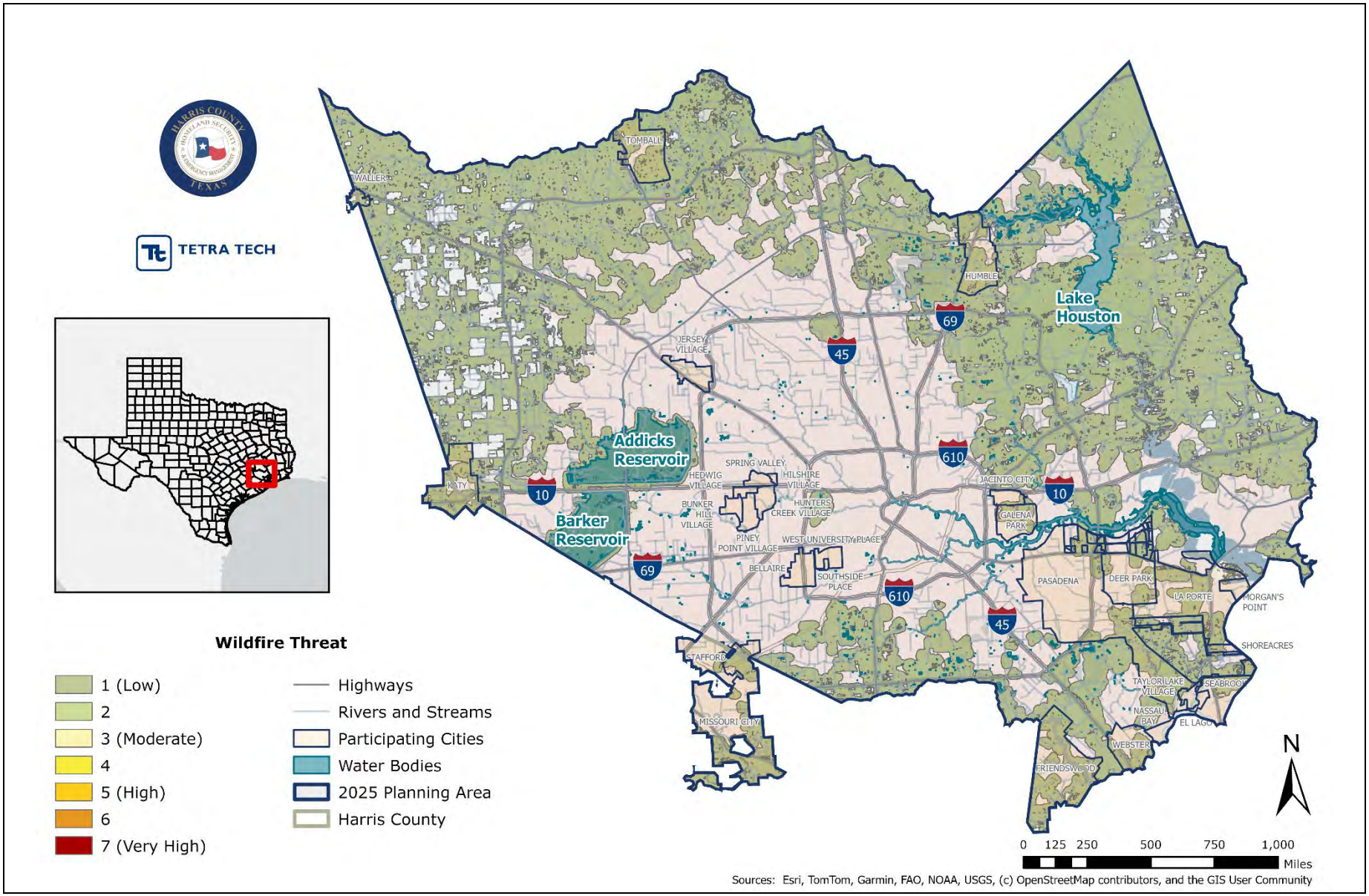


Table 17-1. Participating Jurisdictions at Risk from Wildfires in Harris County

• Deer Park • Friendswood • Galena Park • Humble	• Jacinto City • Katy • La Porte • Missouri City • Morgan's Point	• Nassau Bay • Pasadena • Seabrook • Shoreacres	• Stafford • Taylor Lake Village • Tomball • Waller • Webster
• Cy-Fair ISD • Houston Community College System • Haris Health System	• Goose Creek CISD • Lee College • Gulf Coast Authority	• Spring ISD • San Jacinto Community College	• Varnett Public Schools • Cypress Creek Drainage Improvement District

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; Texas A&M Forest Service;

17.1.3 Extent

The extent of wildfire events in Harris County is generally limited in size compared to more rural counties, but small fires can still threaten lives, property, and infrastructure. Wildfires can range from small grass fires (< 10 acres) along roadways or undeveloped land to multi-hundred-acre events in areas such as national wildlife refuges, flood control reservoirs, and forested parks. Fire intensity is typically lower than in forested areas of Central or West Texas, but wind-driven fires and dry fuel conditions can increase hazard levels.

17.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2025, Harris County was included in three major disaster (DR), emergency (EM), or fire management (FM) declarations for wildfire-related events (FEMA 2025). Table 17-2 lists these declarations.

Table 17-2. FEMA Declarations for Wildfire Events in Harris County (1954 to 2025)

Event Date	Declaration Date	Declaration Number	Description
August 1 – December 10, 1999	September 1, 1999	EM-3142-TX	Extreme Fire Hazards
November 27, 2005, to May 14, 2006	January 11, 2006	FM-2924-TX	Forest Brook Fire
June 17 – 20, 2011	June 17, 2011	DR-1624-TX	Extreme Wildfire Threat

Source: (FEMA 2025)

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Harris County has not been included in any wildfire-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2013 and 2025, Harris County was included in zero wildfire-related agricultural disaster declarations (USDA 2025).

Previous Events

Known hazard events that impacted Harris County between January 2020 and May 2025 is listed in Table 17-3. For events prior to 2020, refer to the 2020 Harris County Multi-Jurisdictional Hazard Mitigation Action Plan (MJ-HMAP).

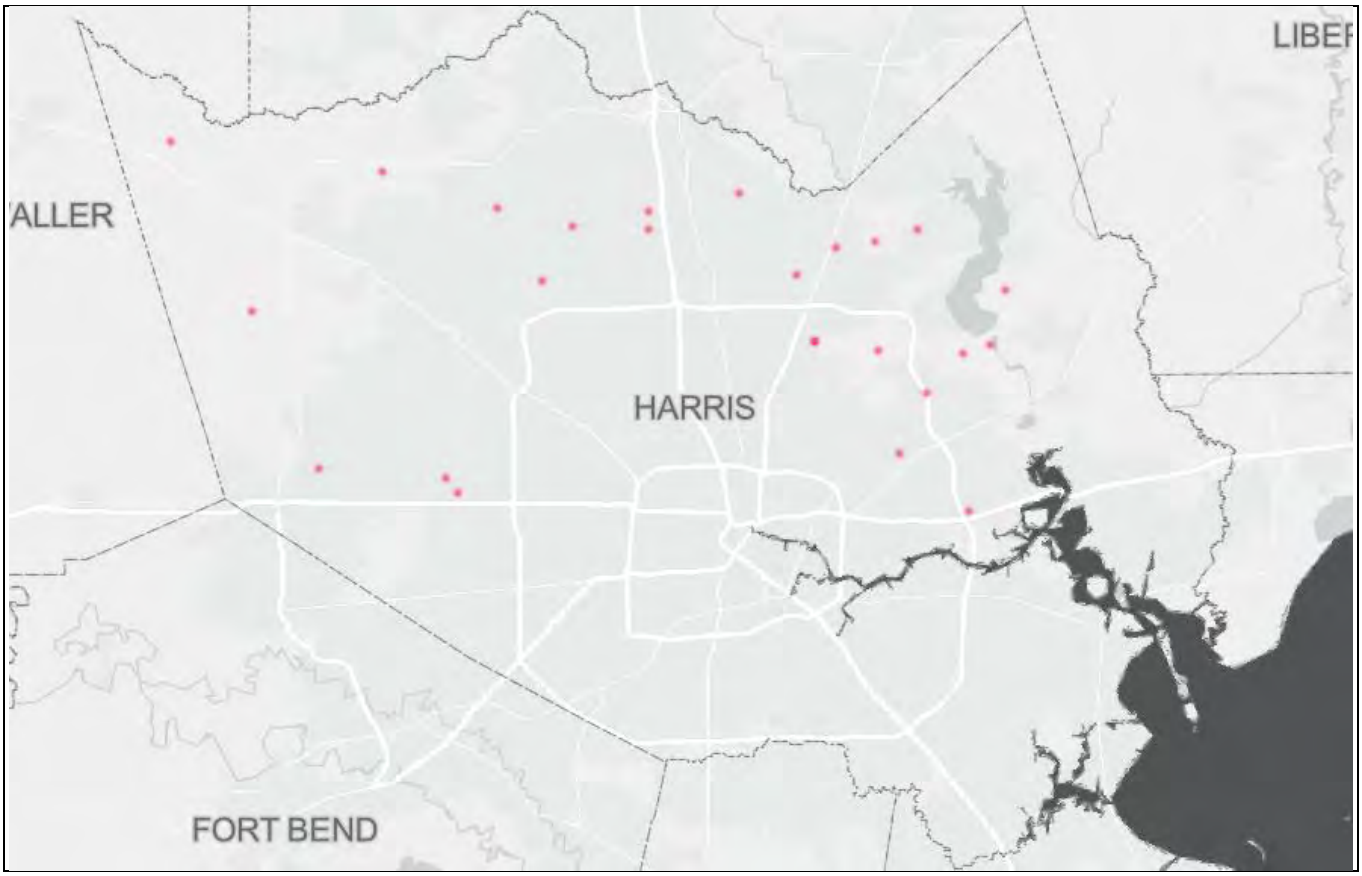
Table 17-3. Wildfire Events in Harris County (2020 to 2025)

Event Date	Declaration or Proclamation Number	Location Impacted	Acres Burned
April 23, 2020	N/A	Harris County	3.80
March 22, 2021	N/A	Harris County	13.60
January 6, 2022	N/A	Harris County	28.00
June 15, 2022	N/A	Harris County	4.30
June 20, 2022	N/A	Harris County	1.00
July 4, 2022	N/A	Harris County	10.00
August 4, 2022	N/A	Harris County	27.00
September 30, 2022	N/A	Harris County	0.50
July 20, 2023	N/A	Harris County	9.00
August 5, 2023	N/A	Harris County	2.00
August 23, 2023	N/A	Harris County	3.00
August 23, 2023	N/A	Harris County	7.00
August 23-24, 2023	N/A	Harris County	33.00
September 3, 2023	N/A	Harris County	1.80
September 5, 2023	N/A	Harris County	4.00
September 9-11, 2023	N/A	Harris County	137
September 20, 2023	N/A	Harris County	29.00
October 6, 2024	N/A	Harris County	44.60
October 17, 2024	N/A	Harris County	1.00
March 15, 2025	N/A	Harris County	11.00

Source: (ABC13 2023) (Texas A&M Forest Servce 2022)

Since 1950, the Texas A&M Forest Service has identified 25 fires that have occurred in Harris County, burning a total of 380.2 acres of land (Texas A&M Forest Servce 2022). The locations of these events are shown in Figure 17-3.

Figure 17-3. Historical Wildfires in Harris County (1950-2025)



Source: (Texas A&M Forest Servce 2022)

17.1.5 Probability of Future Occurrences

Information on previous wildfire occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 17-4. Based on historical records and input from the Steering Committee, the probability of occurrence for wildfires in the County is considered "occasional."

Table 17-4. Probability of Future Wildfire Events in Harris County

Hazard Type	Number of Occurrences Between 1950 and 2025	Percent Chance of Occurring in Any Given Year
Wildfire	25	32.89%

Source: (Texas A&M Forest Servce 2022)

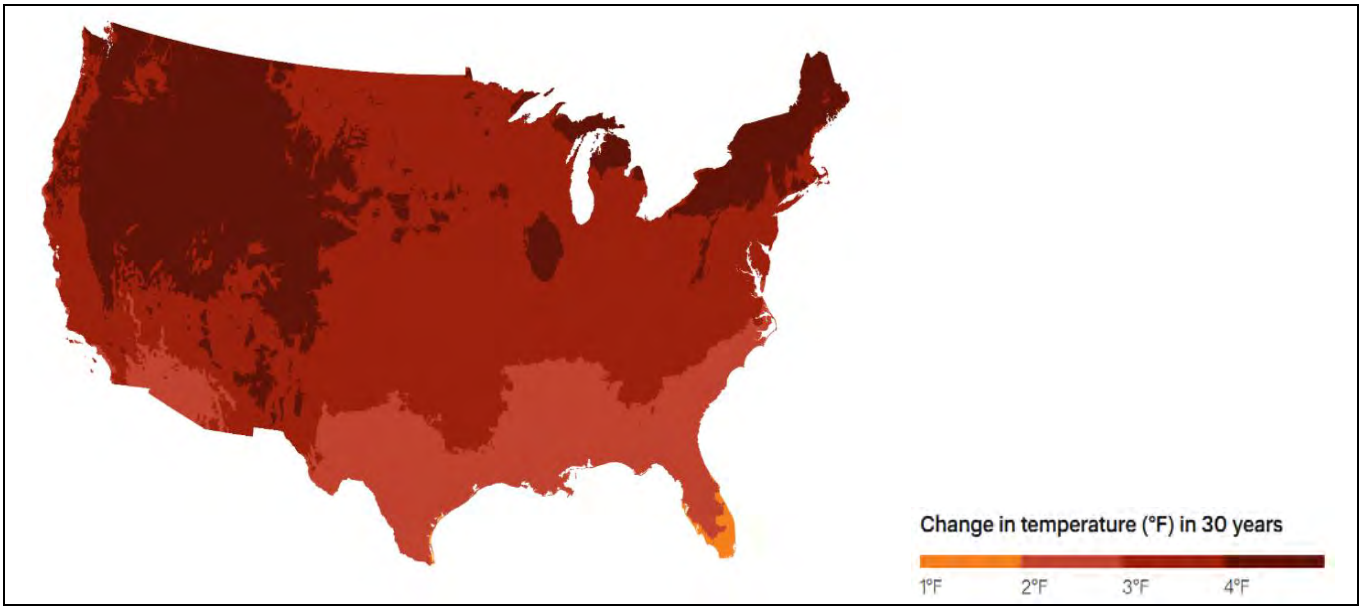
Climate Change Projections

Wildfire risks are changing because of the environment. Under a changing climate regime, the planning area can expect to experience more intense periods of droughts as well as more frequent and more intense periods of extreme heat. Together, these expected future conditions mean wildfire risks in the planning area will be increasing as well (First Street 2024).

Increasing Temperatures

Over the last 30 years, the average temperature in the contiguous United States has increased by 1.2°F compared to the previous half-century, with the largest increases seen in the western United States, where the average increase has been over 1.5°F (see Figure 17-4). As the atmosphere warms, surface temperatures increase in kind. This increases the rate of evaporation in dense wilderness areas, causing soil and vegetation to dry more quickly and more completely than in past seasons.

Figure 17-4. Change in Average Temperature Over the Last 30 years



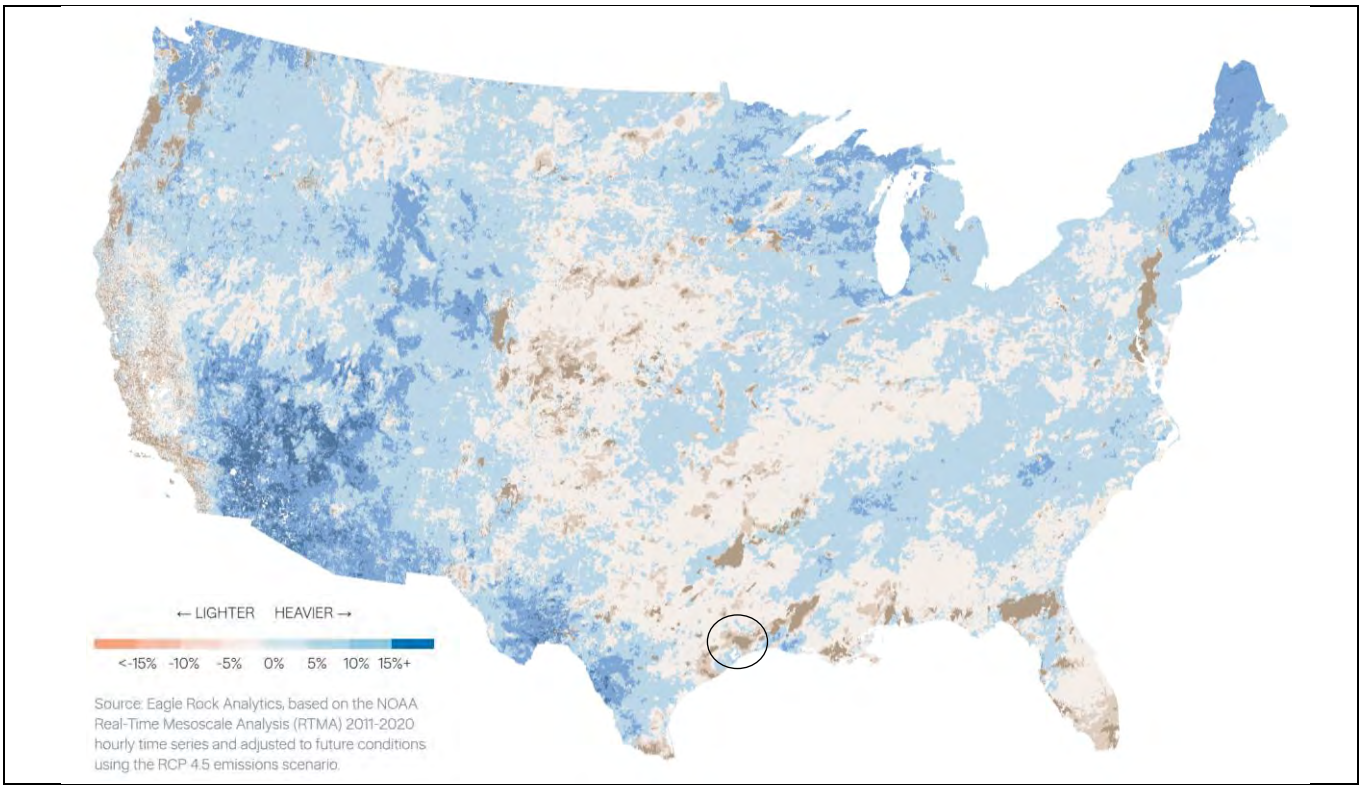
Source: (First Street 2024)

In the next few decades, this warming trend will continue, with temperatures expected to rise over 2.5°F compared to current averages. Rising average temperatures increase the rate of evaporation in dense wilderness areas, causing soil and vegetation to dry more quickly and become flammable. Dried ground and vegetation acts as fuels for wildfires, causing fires that start by natural sources such as lightning to ignite more easily, and causing wildfires with human sources such as campfires and poorly disposed charcoal to get out of control more rapidly. This leads to fires that spread faster and threaten a greater number of homes, especially those in areas designated as part of the Wilderness Urban Interface (First Street 2024).

Changing Precipitation Patterns

As a result of changes in weather patterns across the United States, many areas already prone to wildfires will see changes in precipitation that will result in more severe storms with longer dry spells between them during hot summer months. These weather pattern shifts (see Figure 17-5) can also result in changes in wind direction, blowing wildfires into areas, or allowing them to spread to areas, where they may not have done before. Changing precipitation patterns are exacerbating dry seasons in areas prone to wildfires, causing burns to become more frequent and severe. Lower humidity in the air causes plants to release their moisture to balance the environment, resulting in dried vegetation that is more susceptible to wildfire. Changing precipitation patterns are exacerbating dry seasons in areas prone to wildfires, causing burns to become more frequent and severe. Because the time between rainstorms will increase as climate change intensifies, long dry spells will contribute to the drying of fuel layers brought on by excessive heat, leading to easier wildfire ignitions and harder to control wildfire spread. The growing time between rainfall events also removes a natural check on the spread of wildfires, resulting in longer burns that will require more resources and human intervention in order to stop. (First Street 2024).

Figure 17-5. Changing Precipitation Patterns



Source: (First Street 2024)

Changing Humidity Patterns

Another consequence of changing precipitation patterns is lower air humidity. As the humidity decreases in already arid and vulnerable areas, the resultant dry air creates what is known as a vapor pressure deficit between the ambient atmosphere and the local plant life, causing plants to

release their stored moisture into the air to balance the environment and causing vegetation to dry out in the process. Lower humidity in the air causes plants to release their moisture to balance the environment, resulting in dried vegetation that is more susceptible to wildfire. Dried plants act as highly combustible fuels for wildfires, causing fires to start more easily, spread more readily, and burn hotter than they might if plants were able to retain more of their moisture. This makes more homes, especially those in the WUI, vulnerable to wildfire as the environment continues to change. (First Street 2024)

17.1.6 Cascading Impacts on Other Hazards

Wildfires can trigger or worsen other hazard conditions in several ways. Structure fires can produce embers that can ignite nearby buildings, fences, or vehicles. Wildfires can damage overhead power lines and disrupt gas or electrical utilities to the planning area. They can also cause transportation disruptions (smoke and emergency closures affect roadways, especially near evacuation routes), air quality deterioration (smoke and particulate matter can worsen respiratory conditions, particularly for children and seniors), as well as water contamination (ash and firefighting runoff may impact surface water systems and stormwater infrastructure).

17.2 VULNERABILITY AND IMPACT ASSESSMENT

Wildfire risk varies considerably across Harris County due to differences in vegetation type, land use, population density, and proximity to the wildland–urban interface (WUI). While wildfire is not a widespread hazard throughout the entire planning area, localized conditions in northern and western jurisdictions, and along the periphery of large parks and undeveloped tracts, create elevated exposure for certain communities. It should be noted that impacts from wildfires remain possible in jurisdictions shown to not have any Wildfire Threat or Wildfire Ignition Density zones within their jurisdictional boundaries. Degraded air quality due to wildfire smoke can directly irritate the lungs and, indirectly, lead to health issues during cleanup, from breathing ash and sustaining injuries due to unstable surfaces that remain long after the flames have been put out. Unique and varied wildfire risks are further discussed below as it pertains to each participating jurisdiction.

Unique and Varied Risks

The majority of Harris County is highly urbanized and developed, limiting large-scale wildfire potential. However, open space preserves, right-of-way corridors, and undeveloped tracts in outlying areas still pose localized risk. Periods of drought and high temperatures—common during summer months and prolonged dry spells—significantly increase wildfire ignition and spread potential, particularly in vegetated areas near residential or critical infrastructure zones.

Municipal Planning Partners

Northern and Western Jurisdictions

Jurisdictions such as the Cities of Tomball, Waller, and Katy, and unincorporated areas near Cypress Creek and the Addicks Reservoir have greater exposure due to the presence of mixed

woodland, grassland, and undeveloped acreage. These areas contain numerous residential subdivisions that border natural vegetation, creating defined WUI conditions.

The City of Tomball, for example, has several tracts of pine forest and grassland within city limits and in its extraterritorial jurisdiction. High-voltage transmission corridors and rail lines crossing vegetated areas present ignition sources. The City of Katy experiences moderate risk from grass and brush fires during drought conditions, especially along Interstate 10 and the Grand Parkway corridor where vegetation accumulates

Eastern Municipalities:

Communities such as Morgan's Point and La Porte contain refineries, chemical plants, and port infrastructure adjacent to managed greenbelts and marshland. While the natural vegetation type (marsh and coastal prairie) supports lower fuel loads, these areas are susceptible to ignition from industrial activity. The primary concern here is not large wildland fire spread but secondary fires affecting industrial assets and transportation corridors.

Southern and Coastal Municipalities

In El Lago, Taylor Lake Village, and Seabrook, dense residential development and limited open space reduce wildfire potential. However, extreme drought or prolonged heat events could create conditions where ornamental vegetation and buffer strips ignite, posing localized structure-to-structure fire spread risks. Fire suppression access may be constrained in lake-adjacent neighborhoods.

Central and Urban Municipalities

Within Bellaire, West University Place, and Jacinto City, wildfire risk is minimal due to high structural density, limited vegetation, and robust fire response coverage. Occasional small brush fires along transportation rights-of-way and vacant parcels represent the most likely wildfire incidents

Non-Municipal Planning Partners (Educational Institutions)

Spring ISD

Spring ISD serves areas of north-central Harris County that include forested and suburban zones near Cypress Creek, Spring Creek, and Hardy Toll Road. Several campuses border wooded greenbelts or utility easements containing unmanaged vegetation. During drought or high-wind events, grass and brush fires along roadways or drainage channels could threaten athletic fields, portable classrooms, or bus facilities. The district's key vulnerability lies in smoke exposure and evacuation challenges, as nearby wildfires can impair air quality and temporarily close access routes such as FM 1960 or Cypresswood Drive.

Cy-Fair ISD

Cy-Fair ISD encompasses a broad mix of suburban and semi-rural areas in northwest Harris County, where undeveloped land and open detention basins form an extensive wildland–urban

interface (WUI). Campuses near Huffmeister Road, Skinner Road, and U.S. 290 are bordered by pipeline corridors, power-line ROWs, and drainage ditches that accumulate flammable vegetation. Several campuses were impacted by smoke advisories during the 2011 drought, underscoring air-quality risks. Bus barns and maintenance facilities house fuel and fleet assets that would require pre-fire defensible-space clearing

Goose Creek CISD

Located in Baytown and eastern Harris County, Goose Creek CISD is generally low-lying and urbanized, with limited wildfire fuel. However, periodic grass fires occur along industrial buffer zones and pipeline easements, often ignited by mechanical sparks or roadside debris. The district's concern is less direct flame spread and more exposure to toxic smoke from industrial or grass fires adjacent to refineries and tank farms. Evacuation coordination with the City of Baytown's Emergency Management Office and petrochemical partners remains critical.

Varnett Charter Schools

Varnett Charter campuses are located within urban Houston, where wildfire potential is minimal but urban grass and debris fires can occur near vacant lots or highway corridors. Risks include building HVAC intake of smoke and localized fire spread through ornamental landscaping during droughts. Coordination with the Houston Fire Department for campus-specific fire-prevention education remains the primary mitigation measure.

Houston Community College

HCC operates multiple urban campuses across Houston. Most facilities are not near undeveloped wildlands, but campuses on the city's western and northern edges (e.g., Katy, Alief, and Northwest locations) border vegetated rights-of-way and are susceptible to grass fires ignited along highways. Primary risks involve air-quality degradation, campus closures, and power disruption due to smoke or protective utility shut-offs during drought conditions

University of Houston

UH's main campus lies in central Houston, where wildfire risk is negligible. However, the UH Coastal Center in La Marque (Galveston County) and the UH-Katy campus near Addicks Reservoir are closer to grassland and brush areas where wildfire exposure is moderate. The Katy campus, in particular, faces periodic grass-fire risk due to dry prairie vegetation and strong wind conditions along the I-10 corridor. UH's vulnerability centers on research facility continuity, smoke-related air filtration, and limited evacuation routes during nearby fire events.

San Jacinto Community College

San Jacinto College maintains North, Central, and South campuses across eastern Harris County. The Central Campus in Pasadena includes forested parcels and natural buffers adjacent to industrial zones that can burn under drought conditions. The South Campus near Clear Lake lies within the coastal prairie zone where grass fires may ignite during high-wind, low-humidity periods.

Critical concerns include wildfire-smoke infiltration, transportation disruption on I-45 and Beltway 8, and protection of lab buildings with sensitive ventilation systems.

Lee College

Located in Baytown, Lee College's wildfire exposure is low, but nearby industrial buffer lands, rights-of-way, and drainage corridors can support small brush fires. The campus's main vulnerabilities relate to evacuation coordination and smoke intrusion, particularly if fires originate near the San Jacinto River or industrial sites to the east.

Non-Municipal Planning Partners (Other)

Harris Health System

The Harris Health System operates hospitals, clinics, and administrative facilities throughout Harris County. Ben Taub Hospital (Houston Medical Center) and LBJ Hospital (northeast Houston) are not near wildlands but must maintain HVAC filtration systems capable of managing wildfire-related smoke events, which can affect air quality regionally. Community health clinics in northern and western suburban areas may experience access disruption or smoke-sensitive patient impacts during regional grass-fire events. Harris Health's continuity of operations (COOP) plans incorporate air-quality monitoring and generator testing during red-flag warnings.

Gulf Coast Authority (GCA)

GCA operates multiple wastewater treatment and industrial-effluent facilities, including the Bayport, Washburn Tunnel, and Odessa plants. Facilities near vegetated rights-of-way or coastal marshlands face grass-fire ignition potential from maintenance or electrical equipment. Combustion risk of on-site chemical storage during wildfires is low but would have high consequence if flammable vapors are released. GCA's critical vulnerability lies in maintaining emergency power and fire-suppression capacity to prevent cascading environmental releases.

Cypress Creek Drainage Improvement District

CCDID manages drainage infrastructure, detention basins, and greenway corridors in the Cypress Creek watershed, where wildland–urban interface conditions are most pronounced in Harris County. Vegetated berms, detention slopes, and unmowed channels can accumulate dry fuel during droughts. Maintenance crews may face fire ignition risk during mowing or mechanical work, and smoke from nearby fires can limit visibility for inspection operations. The district's mitigation priorities include vegetation-management programs, fuel-break establishment, and interagency coordination with HCFCD and local fire departments to monitor high-risk corridors.

Harris County Flood Control District

The Harris County Flood Control District (HCFCD) manages an extensive countywide network of flood-risk-reduction infrastructure, including detention basins, earthen levees, drainage channels, natural stream corridors, and associated right-of-way lands. While HCFCD does not maintain permanent residential populations or traditional building stock within its service areas, many of its

assets are located within vegetated open spaces that overlap with Wildland–Urban Interface (WUI) conditions, particularly in northern and western Harris County.

HCFCD infrastructure is vulnerable to wildfire primarily through vegetation-driven exposure rather than direct structural loss. Detention basin slopes, channel banks, and unimproved rights-of-way can accumulate dry grasses and brush during periods of drought or extreme heat, increasing ignition potential. Wildfires within these areas could damage erosion-control features, compromise channel stability, degrade water-quality protection measures, and temporarily limit access for inspection, maintenance, and emergency response operations. In addition, smoke and reduced visibility from nearby grass or brush fires may disrupt field activities and interagency coordination during flood events.

Although mapped wildfire threat data indicate limited direct exposure to people or structures, HCFCD remains vulnerable to operational and cascading impacts, including increased post-fire erosion, sedimentation within drainage channels, and potential obstruction of flood-control capacity following burn events. As climate conditions continue to favor longer dry periods and higher temperatures, wildfire impacts to flood-management infrastructure represent an increasing concern for the District's long-term asset resilience and continuity of operations.

Harris County Office of County Administration (OCA)

The Harris County Office of County Administration (HCOCA) provides centralized administrative oversight and operational coordination for county departments and facilities distributed throughout Harris County. HCOCA does not own or operate facilities with concentrated wildfire fuel loads, nor does it maintain permanent populations within mapped wildfire threat or ignition density zones. As a result, direct wildfire risk to structures and people under HCOCA's purview is limited.

However, HCOCA remains vulnerable to indirect and system-wide wildfire impacts that affect county government continuity. Wildfire events occurring elsewhere in the county—particularly in northern and western areas—may disrupt transportation corridors, degrade regional air quality, and require activation of emergency coordination functions overseen or supported by HCOCA. Smoke impacts can affect employee health, limit access to administrative facilities, and necessitate temporary operational adjustments, including telework or relocation of essential services.

Additionally, wildfire-related power disruptions, telecommunications outages, or emergency resource reallocations may affect countywide administrative functions coordinated through HCOCA. While the Office does not face direct structural wildfire exposure, its vulnerability lies in maintaining continuity of government operations, interdepartmental coordination, and emergency management support during wildfire events that impact other planning partners or unincorporated areas of Harris County.

17.2.1 Life, Health, and Safety

Overall Population

Residents in areas adjacent to forested parks, drainage corridors, or undeveloped brush lands are at elevated risk. Evacuations, power outages, and air quality issues may affect thousands during active fires, especially in rapidly spreading grass fires. Table 17-5 lists the distribution of planning area population by fire risk area.

Socially Vulnerable Population

Low-income residents and those living in mobile homes or manufactured housing are more likely to reside in wildfire-prone fringe areas, and these subgroups of the population often lack the financial or physical resources needed for rapid evacuation or temporary relocation. Similarly, elderly residents, individuals with disabilities, and limited-English-speaking populations may experience communication barriers or medical vulnerabilities related to fire or smoke exposure. Table 17-6, Table 17-7, and Table 17-8 break down the vulnerable population exposure across the planning area.

Table 17-5. Population at Risk to Wildfires Across Harris County

Jurisdiction	Total Population (2023 ACS)	Low Wildfire Threat Hazard Area		Moderate Wildfire Ignition Density Hazard Area		High Wildfire Ignition Density Hazard Area	
		Number of Persons	% of Total	Number of Persons	% of Total	Number of Persons	% of Total
Baytown[b]	79,635	15,072	18.9%	0	0.0%	0	0.0%
Bellaire	17,116	0	0.0%	0	0.0%	0	0.0%
Bunker Hill Village	3,801	0	0.0%	0	0.0%	0	0.0%
Deer Park	33,823	8,250	24.4%	0	0.0%	0	0.0%
El Lago	3,016	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	40,827	24,781	60.7%	0	0.0%	0	0.0%
Galena Park	10,527	3,030	28.8%	0	0.0%	0	0.0%
Hedwig Village	2,029	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	1,091	0	0.0%	0	0.0%	0	0.0%
Houston[b]	2,252,110	230,858	10.3%	965	<0.1%	137	<0.1%
Humble	16,489	12,096	73.4%	0	0.0%	0	0.0%
Hunters Creek Village	4,358	0	0.0%	0	0.0%	0	0.0%
Jacinto City	9,458	2,847	30.1%	0	0.0%	0	0.0%
Jersey Village	7,817	0	0.0%	0	0.0%	0	0.0%
Katy[a]	23,900	17,672	73.9%	85	0.4%	0	0.0%
La Porte	36,077	9,867	27.3%	0	0.0%	0	0.0%
League City[b]	2,527	1,749	69.2%	0	0.0%	0	0.0%
Missouri City[a]	75,234	24,388	32.4%	337	0.4%	0	0.0%
Morgan's Point	255	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	5,211	179	3.4%	0	0.0%	0	0.0%
Pasadena	149,345	16,455	11.0%	0	0.0%	0	0.0%
Pearland[b]	8,189	1,732	21.2%	0	0.0%	0	0.0%
Piney Point Village	3,090	0	0.0%	0	0.0%	0	0.0%
Seabrook	13,617	2,746	20.2%	0	0.0%	0	0.0%

Jurisdiction	Total Population (2023 ACS)	Low Wildfire Threat Hazard Area		Moderate Wildfire Ignition Density Hazard Area		High Wildfire Ignition Density Hazard Area	
		Number of Persons	% of Total	Number of Persons	% of Total	Number of Persons	% of Total
Shoreacres	1,327	546	41.1%	0	0.0%	0	0.0%
South Houston[b]	16,017	0	0.0%	0	0.0%	0	0.0%
Southside Place	1,631	0	0.0%	0	0.0%	0	0.0%
Spring Valley Village	4,226	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	17,279	1,092	6.3%	0	0.0%	0	0.0%
Taylor Lake Village	3,678	922	25.1%	0	0.0%	0	0.0%
Tomball	13,032	9,247	71.0%	0	0.0%	0	0.0%
Waller[a]	3,062	2,299	75.1%	0	0.0%	0	0.0%
Webster	12,283	4,557	37.1%	0	0.0%	0	0.0%
West University Place	14,907	0	0.0%	0	0.0%	0	0.0%
Unincorporated County	2,001,319	800,563	40.0%	14,678	0.7%	3,208	0.2%
Harris County Total	4,888,304	1,190,948	24.4%	16,065	0.3%	3,345	0.1%

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; Texas A&M Forest Service

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Note: Census data was not available for non-municipal partners, and as such, this table does not capture the full scope of risks to persons located in the designated wildfire hazard areas.

Table 17-6. Estimated Number of Vulnerable Persons Located in the Low Wildfire Threat Hazard Area

Jurisdiction	Population in the Low Wildfire Threat Hazard Area									
	Population Over 65		Population Under 5		Non-English-Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Baytown[b]	1,741	18.9%	1,117	18.9%	473	18.9%	1,930	18.9%	263	18.9%
Bellaire	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bunker Hill Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Deer Park	1,355	24.4%	500	24.4%	90	24.3%	1,055	24.4%	674	24.4%
El Lago	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	3,879	60.3%	1,496	60.7%	60	63.8%	2,156	60.9%	699	61.6%
Galena Park	299	28.8%	237	28.7%	166	28.8%	381	28.7%	706	28.7%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Houston[b]	27,816	10.3%	15,567	10.2%	12,219	10.2%	25,209	10.3%	44,705	10.2%
Humble	1,489	73.3%	617	73.4%	375	73.2%	1,416	73.3%	2,161	73.4%
Hunters Creek Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jacinto City	393	30.1%	221	30.0%	134	29.9%	450	30.1%	535	30.1%
Jersey Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Katy[a]	2,114	73.9%	1,326	73.9%	221	73.7%	1,629	73.9%	949	73.9%
La Porte	1,317	27.3%	679	27.3%	106	27.2%	1,227	27.3%	1,207	27.3%
League City[b]	216	69.1%	106	68.6%	15	68.0%	168	68.9%	78	68.8%
Missouri City[a]	3,983	32.4%	1,481	32.4%	376	32.3%	2,638	32.4%	1,488	32.4%
Morgan's Point	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	38	3.4%	4	3.1%	7	3.1%	24	3.4%	23	3.3%
Pasadena	1,890	11.1%	1,229	11.0%	700	11.0%	1,978	11.0%	2,979	11.0%
Pearland[b]	208	21.1%	123	21.1%	28	20.8%	127	21.1%	72	20.9%

Jurisdiction	Population in the Low Wildfire Threat Hazard Area									
	Population Over 65		Population Under 5		Non-English-Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Piney Point Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Seabrook	395	20.1%	138	20.1%	22	19.8%	281	20.2%	182	20.1%
Shoreacres	97	41.1%	23	41.1%	0	0.0%	117	41.1%	17	39.5%
South Houston[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Southside Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	122	6.3%	77	6.3%	41	6.3%	101	6.3%	102	6.3%
Taylor Lake Village	273	24.3%	26	26.3%	6	28.6%	127	24.8%	3	30.0%
Tomball	1,982	70.9%	706	70.9%	107	70.9%	1,568	70.9%	878	70.9%
Waller[a]	151	75.1%	93	75.0%	27	73.7%	226	74.9%	549	75.1%
Webster	646	37.1%	282	37.1%	188	37.0%	483	37.0%	410	37.0%
West University Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Unincorporated County	81,734	40.1%	57,652	40.0%	22,897	39.9%	77,032	40.0%	98,148	39.9%
Harris County Total	132,138	23.5%	83,700	24.8%	38,258	19.9%	120,323	23.7%	156,829	21.1%

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; Texas A&M Forest Service

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Note: Census data was not available for non-municipal partners, and as such, this table does not capture the full scope of risks to vulnerable populations located in the Low Wildfire Threat Hazard Area.

Table 17-7. Estimated Number of Vulnerable Persons in the Moderate Wildfire Ignition Density Hazard Area

Jurisdiction	Population in the Moderate Wildfire Ignition Density Hazard Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Baytown[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bellaire	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bunker Hill Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Deer Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
El Lago	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Galena Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Houston[b]	116	<0.1%	65	<0.1%	51	<0.1%	105	<0.1%	187	<0.1%
Humble	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hunters Creek Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jacinto City	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jersey Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Katy[a]	10	0.3%	6	0.3%	1	0.3%	7	0.3%	4	0.3%
La Porte	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
League City[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Missouri City[a]	55	0.4%	20	0.4%	5	0.4%	36	0.4%	20	0.4%
Morgan's Point	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Pasadena	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Pearland[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%

Jurisdiction	Population in the Moderate Wildfire Ignition Density Hazard Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Piney Point Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Seabrook	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Shoreacres	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
South Houston[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Southside Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Taylor Lake Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Tomball	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Waller[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Webster	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
West University Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Unincorporated County	1,502	0.7%	1,057	0.7%	419	0.7%	1,416	0.7%	1,801	0.7%
Harris County Total	1,683	0.3%	1,148	0.3%	476	0.2%	1,564	0.3%	2,012	0.3%

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; Texas A&M Forest Service

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b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Note: Census data was not available for non-municipal partners, and as such, this table does not capture the full scope of risks to vulnerable populations located in the Moderate Wildfire Ignition Density Hazard Area.

Table 17-8. Estimated Vulnerable Populations Located in the High Wildfire Ignition Density Hazard Area

Jurisdiction	Population in the High Wildfire Ignition Density Hazard Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Baytown[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bellaire	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Bunker Hill Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Deer Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
El Lago	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Friendswood[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Galena Park	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hedwig Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hilshire Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Houston[b]	16	<0.1%	9	<0.1%	7	<0.1%	15	<0.1%	26	<0.1%
Humble	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Hunters Creek Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jacinto City	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Jersey Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Katy[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
La Porte	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
League City[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Missouri City[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Morgan's Point	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Nassau Bay	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Pasadena	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Pearland[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%

Jurisdiction	Population in the High Wildfire Ignition Density Hazard Area									
	Population Over 65		Population Under 5		Non-English Speaking Population		Population with Disability		Population Below Poverty Level	
	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total	Total	% of Jurisdiction Total
Piney Point Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Seabrook	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Shoreacres	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
South Houston[b]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Southside Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Spring Valley Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Stafford[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Taylor Lake Village	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Tomball	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Waller[a]	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Webster	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
West University Place	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Unincorporated County	327	0.2%	231	0.2%	91	0.2%	308	0.2%	393	0.2%
Harris County Total	343	0.1%	240	0.1%	98	0.1%	323	0.1%	419	0.1%

Source: U.S. Census Bureau 2023 American Community Survey; STATS America; Texas A&M Forest Service

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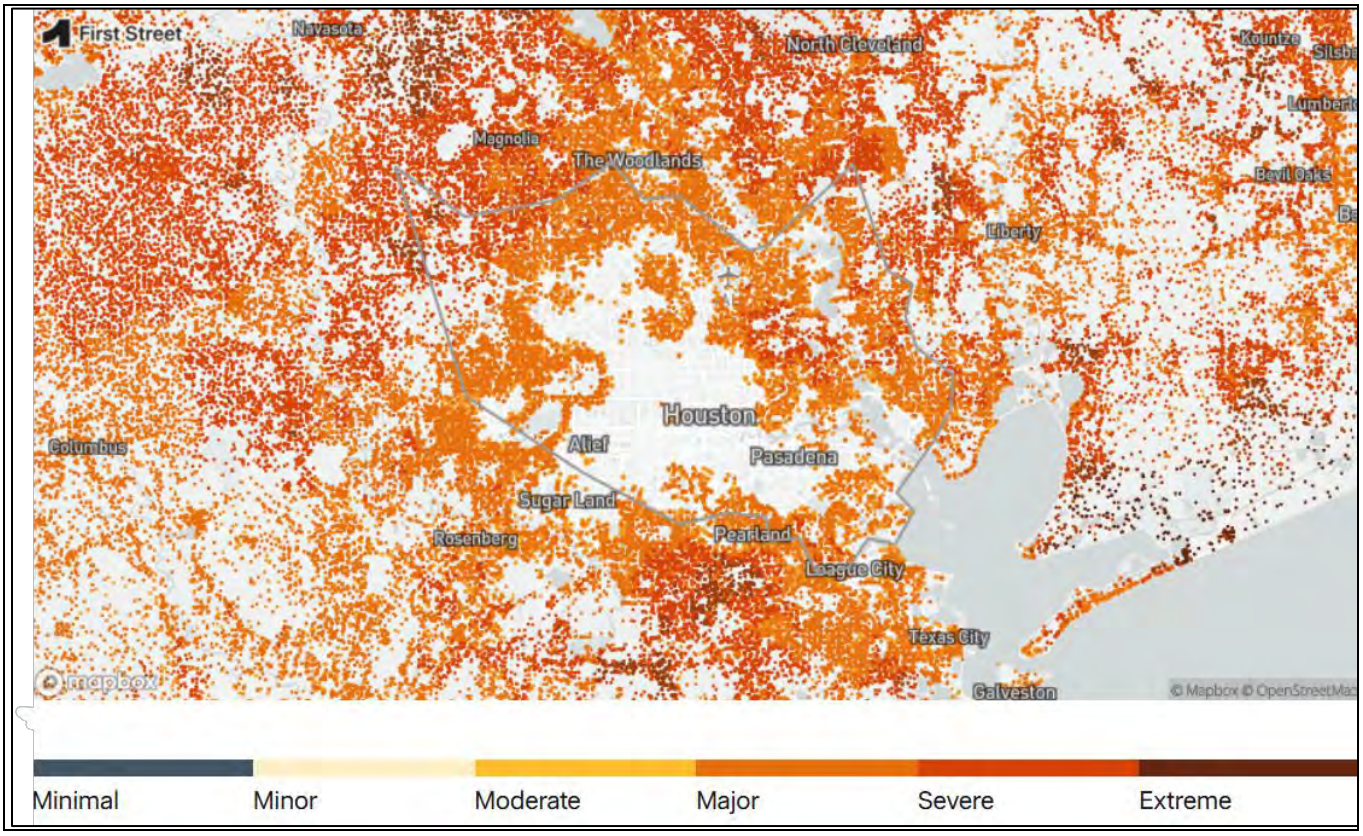
b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Note: Census data was not available for non-municipal partners, and as such, this table does not capture the full scope of risks to vulnerable populations located in the High Wildfire Ignition Density Hazard Area.

17.2.2 General Building Stock

Structures in the WUI are particularly at risk if not built with fire-resistant materials. Mobile homes, barns, sheds, and older frame structures are especially vulnerable to heat and flame. Buildings with flammable landscaping, wood fencing, or limited firebreaks are more exposed. Figure 17-6 summarizes some of the areas where properties are more at risk to wildfire than others, and it nicely shows how wildfire risk is more intense in the rural areas just beyond the urbanized areas surrounding the greater Houston metropolitan area.

Figure 17-6. Properties at Risk from Wildfire



Source: (First Street 2024)

When examining the building stock across the planning area, it is clear that residential buildings retain the greatest threat to wildfire events, as detailed in Table 17-9. Table 17-10 summarizes the building stock in Harris County located within the low wildfire hazard threat area, sorted alphabetically by jurisdiction.

Table 17-9. Buildings in the Low Wildfire Threat Hazard Area by General Occupancy Class

Jurisdiction	Residential			Commercial			Industrial			Government, Religion, Agriculture, and Education		
	Low	Mod.	High	Low	Mod.	High	Low	Mod.	High	Low	Mod.	High
Baytown[b]	3,892	0	0	284	0	0	2	0	0	15	0	0
Bellaire	0	0	0	0	0	0	0	0	0	0	0	0
Bunker Hill Village	0	0	0	0	0	0	0	0	0	0	0	0
Deer Park	2,584	0	0	192	0	0	45	0	0	4	0	0
El Lago	0	0	0	0	0	0	0	0	0	0	0	0
Friendswood[a]	7,678	0	0	243	0	0	2	0	0	10	0	0
Galena Park	868	0	0	19	0	0	2	0	0	7	0	0
Hedwig Village	0	0	0	0	0	0	0	0	0	0	0	0
Hilshire Village	0	0	0	0	0	0	0	0	0	0	0	0
Houston[b]	55,416	232	33	3,605	2	2	246	0	0	196	0	0
Humble	2,875	0	0	399	0	0	35	0	0	18	0	0
Hunters Creek Village	0	0	0	0	0	0	0	0	0	0	0	0
Jacinto City	827	0	0	10	0	0	0	0	0	2	0	0
Jersey Village	0	0	0	0	0	0	0	0	0	0	0	0
Katy[a]	3,902	26	0	388	0	0	36	0	0	49	0	0
La Porte	3,135	0	0	338	0	0	27	0	0	10	0	0
League City[b]	369	0	0	7	0	0	0	0	0	1	0	0
Missouri City[a]	7,583	105	0	325	2	0	22	0	0	30	0	0
Morgan's Point	0	0	0	0	0	0	0	0	0	0	0	0
Nassau Bay	52	0	0	0	0	0	0	0	0	0	0	0
Pasadena	4,090	0	0	187	0	0	26	0	0	7	0	0
Pearland[b]	574	0	0	29	0	0	12	0	0	0	0	0
Piney Point Village	0	0	0	0	0	0	0	0	0	0	0	0
Seabrook	828	0	0	73	0	0	0	0	0	1	0	0

Jurisdiction	Residential			Commercial			Industrial			Government, Religion, Agriculture, and Education		
	Low	Mod.	High	Low	Mod.	High	Low	Mod.	High	Low	Mod.	High
Shoreacres	274	0	0	0	0	0	0	0	0	0	0	0
South Houston[b]	0	0	0	0	0	0	0	0	0	0	0	0
Southside Place	0	0	0	0	0	0	0	0	0	0	0	0
Spring Valley Village	0	0	0	0	0	0	0	0	0	0	0	0
Stafford[a]	249	0	0	103	0	0	5	0	0	4	0	0
Taylor Lake Village	352	0	0	1	0	0	0	0	0	0	0	0
Tomball	2,255	0	0	369	0	0	26	0	0	20	0	0
Waller[a]	632	0	0	147	0	0	14	0	0	8	0	0
Webster	662	0	0	153	0	0	2	0	0	7	0	0
West University Place	0	0	0	0	0	0	0	0	0	0	0	0
Unincorporated County	216,854	3,970	868	9,710	207	23	811	22	0	534	17	2
Harris County Total	315,951	4,333	901	16,582	211	25	1,313	22	0	923	17	2

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018; Texas A&M Forest Service

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Note: Census data was not available for non-municipal partners, and as such, this table does not capture the full scope of risks to buildings located in the Low Wildfire Threat Hazard Area.

Table 17-10. Wildfire Risk to Buildings in Harris County

Jurisdiction	Low Wildfire Threat Hazard Area				Moderate Wildfire Ignition Density				High Wildfire Ignition Density			
	# of Buildings	% of Total Buildings	RCV	% of Total Building RCV	# of Buildings	% of Total Buildings	RCV	% of Total Building RCV	# of Buildings	% of Total Buildings	RCV	% of Total Building RCV
Baytown[b]	4,193	20.3%	$3,157,412,145	25.3%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Bellaire	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Bunker Hill Village	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Deer Park	2,825	24.8%	$2,132,726,724	29.5%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
El Lago	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Friendswood[a]	7,933	59.3%	$4,731,276,647	63.2%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Galena Park	896	27.3%	$292,415,136	19.7%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Hedwig Village	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Hilshire Village	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Houston[b]	59,463	10.1%	$54,168,238,642	8.7%	234	<0.1%	$142,466,243	<0.1%	35	<0.1%	$134,163,410	<0.1%
Humble	3,327	68.9%	$3,316,500,365	60.5%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Hunters Creek Village	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Jacinto City	839	28.1%	$226,834,188	14.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Jersey Village	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Katy[a]	4,375	69.2%	$2,457,279,656	64.8%	26	0.4%	$8,785,383	0.2%	0	0.0%	$0	0.0%
La Porte	3,510	28.3%	$3,205,013,634	39.4%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
League City[b]	377	95.7%	$332,975,069	96.9%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Missouri City[a]	7,960	32.7%	$5,510,080,055	36.9%	107	0.4%	$48,522,716	0.3%	0	0.0%	$0	0.0%
Morgan's Point	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Nassau Bay	52	3.3%	$32,836,434	1.6%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Pasadena	4,310	10.6%	$4,688,551,047	16.8%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Pearland[b]	615	76.0%	$732,035,308	83.3%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%

Jurisdiction	Low Wildfire Threat Hazard Area				Moderate Wildfire Ignition Density				High Wildfire Ignition Density			
	# of Buildings	% of Total Buildings	RCV	% of Total Building RCV	# of Buildings	% of Total Buildings	RCV	% of Total Building RCV	# of Buildings	% of Total Buildings	RCV	% of Total Building RCV
Piney Point Village	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Seabrook	902	20.6%	$725,809,879	23.3%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Shoreacres	274	41.7%	$120,475,230	44.9%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
South Houston[b]	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Southside Place	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Spring Valley Village	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Stafford[a]	361	7.1%	$397,228,335	5.5%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Taylor Lake Village	353	25.4%	$222,987,006	26.1%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Tomball	2,670	73.5%	$3,286,371,358	76.6%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Waller[a]	801	73.0%	$248,763,657	91.8%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Webster	824	34.6%	$1,517,872,508	21.7%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
West Univ. Place	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%	0	0.0%	$0	0.0%
Unincorporated County	227,909	39.5%	$153,139,000,000	38.1%	4,216	0.7%	$4,290,771,481	1.1%	893	0.2%	$922,712,483	0.2%
Harris County Total	334,769	24.7%	$244,642,683,021	21.0%	4,583	0.3%	$4,490,545,823	0.4%	928	0.1%	$1,056,875,893	0.1%

Source: Harris County Appraisal District 2018, Harris County Flood Control District 2018, Fort Bend Central Appraisal District 2018, Fort Bend County 2016, Galveston Central Appraisal District 2018, RS Means 2024; Texas A&M Forest Service

a. City extends beyond Harris County limits and is a planning partner for this MJ-HMP. Values shown are for the entire city, including portions in other counties

b. City extends beyond Harris County limits and is not a planning partner for this MJ-HMP. Values shown are for the portion of city within Harris County only.

Note: Census data was not available for non-municipal partners, and as such, this table does not capture the full scope of risks to buildings located in the Wildfire Hazard Areas.

17.2.3 Community Lifelines and Other Critical Facilities

Community lifelines and other critical facilities in the low wildfire threat hazard area and the moderate wildfire ignition density hazard area are listed by jurisdiction in Table 17-11 and Table 17-12. Table 17-13 summarizes the countywide totals. In each category, energy and water system lifelines are more at-risk than other lifelines when it comes to wildfire risks.

17.2.4 Economy

Disruptions may impact transportation and logistics operations during evacuations or highway closures. They can also impact agricultural and ranching operations, particularly in western parts of Harris County. Other sectors of the economy which may be impacted by wildfire events are construction and development sites, where heavy equipment and open land increase ignition risks.

17.2.5 Natural, Historic and Cultural Resources

Several parks, cemeteries, and historic structures (e.g., San Jacinto Battleground, local heritage homes) are located near vegetated areas, and these resources are ones that will remain more at risk to wildfires simply due to their location.

17.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure the establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

17.3.1 Potential or Planned Development

As Harris County continues to grow, urban sprawl is pushing residential development into WUI zones. More homes and infrastructure in transitional areas (e.g., Cypress, Spring, Crosby) increase ignition sources and response complexity. Overgrown vegetation along rights-of-way, drainage channels, and flood control properties may elevate fire risk without routine maintenance. Redevelopment of vacant or foreclosed properties may leave unmanaged brush and debris that act as fire fuel. According to FirstStreet.org, there are 490,702 properties in Harris County that have wildfire risk, and over the next 30 years, this number is expected to increase to 614,953 properties (First Street 2024). Figure 17-7 provides a comparison of these changes in risk.

17.3.2 Projected Changes in Population

Urban sprawl is expected to continue across much of the planning area, with newer subdivisions and communities being built along the periphery of the greater metro area. Increasing population numbers in the planning area can mean an increase in wildfire risks – particularly in the new development around the periphery of existing development.

Table 17-11. Number of Facilities in Low Wildfire Threat Hazard Area, by Lifeline Category

Jurisdiction	Number of Facilities in Low Wildfire Threat Hazard Area, by Lifeline Category								Other Critical Facilities	Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems		Count	% of Jurisdiction Total
Baytown	42	201	46	28	6	21	37	6	39	426	38.9%
Bellaire	0	0	0	0	0	0	0	0	0	0	0.0%
Bunker Hill Village	0	0	0	0	0	0	0	0	0	0	0.0%
Deer Park	13	10	10	13	1	9	8	17	13	94	28.0%
El Lago	0	0	0	0	0	0	0	0	0	0	0.0%
Friendswood	3	9	5	2	1	6	6	11	6	49	81.7%
Galena Park	11	11	0	8	0	3	9	10	1	53	34.2%
Hedwig Village	0	0	0	0	0	0	0	0	0	0	0.0%
Hilshire Village	0	0	0	0	0	0	0	0	0	0	0.0%
Houston	453	1,340	272	290	36	111	393	190	191	3,276	17.7%
Humble	40	81	27	24	6	19	29	11	10	247	50.4%
Hunters Creek Village	0	0	0	0	0	0	0	0	0	0	0.0%
Jacinto City	0	0	0	1	0	0	2	2	2	7	10.3%
Jersey Village	0	0	0	0	0	0	0	0	0	0	0.0%
Katy	16	14	11	7	0	17	6	11	5	87	64.9%
La Porte	56	12	20	21	2	7	44	35	18	215	50.1%
League City	5	2	1	1	0	0	1	3	2	15	100.0%
Missouri City	1	0	0	0	1	1	5	4	3	15	31.9%
Morgan's Point	3	1	0	2	0	0	0	2	0	8	27.6%
Nassau Bay	1	0	0	1	0	0	0	1	0	3	8.1%
Pasadena	74	39	18	35	9	15	46	6	12	254	20.4%
Pearland	18	5	1	4	0	1	9	4	2	44	95.7%
Piney Point Village	0	0	0	0	0	0	0	0	0	0	0.0%

Jurisdiction	Number of Facilities in Low Wildfire Threat Hazard Area, by Lifeline Category								Other Critical Facilities	Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems		Count	% of Jurisdiction Total
Seabrook	0	8	0	2	2	0	2	0	0	14	16.3%
Shoreacres	3	0	0	0	0	1	3	4	3	14	43.8%
South Houston	0	0	0	0	0	0	0	0	0	0	0.0%
Southside Place	0	0	0	0	0	0	0	0	0	0	0.0%
Spring Valley Village	0	0	0	0	0	0	0	0	0	0	0.0%
Stafford	0	0	0	0	0	0	0	0	0	0	0.0%
Taylor Lake Village	0	0	0	0	0	0	0	0	0	0	0.0%
Tomball	16	143	30	13	13	17	28	36	15	311	78.7%
Waller	2	6	8	1	0	3	4	3	0	27	96.4%
Webster	17	11	8	21	4	4	17	8	10	100	44.6%
West University Place	0	0	0	0	0	0	0	0	0	0	0.0%
Unincorporated County	1,048	1,565	703	832	76	344	795	1,633	539	7,535	46.5%
Harris County Total	1,822	3,458	1,160	1,306	157	579	1,444	1,997	871	12,794	31.6%

Source: ATLAS 2022; BTS/USDOT 2023; CIMA 2024; City of La Porte 2025; City of Seabrook 2025; FleetCor 2024; Harris County 2020, 2024; Harris County Office of Emergency Management 2022-2025; Harris County Public Library, HCUS 2020, HCPCS 2024, HIFLD 2021-2024; METRO 2024; NPMS 2024; SETRAC 2024; TCEQ 2023-2024; Regional Catastrophic Preparedness Grant Program 2023; DSHS 2024; TEA 2023; THC 2023; USACE 2024; Texas A&M Forest Service

Note: The critical facilities analyzed are only within the boundaries of Harris County. Information pertaining to facilities owned/operated by the non-municipal planning partners was not available during the development of the risk assessment, and as such, the wildfire risks to these planning partners is not reflected in the table above.

Table 17-12. Number of Facilities in Moderate Wildfire Ignition Density Hazard Area, by Lifeline Category

Jurisdiction	Number of Facilities in Moderate Wildfire Ignition Density Hazard Area, by Lifeline Category								Other Critical Facilities	Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems		Count	% of Jurisdiction Total
Baytown	0	0	0	2	0	0	0	0	0	2	0.2%
Bellaire	0	0	0	0	0	0	0	0	0	0	0.0%
Bunker Hill Village	0	0	0	0	0	0	0	0	0	0	0.0%
Deer Park	0	0	0	0	0	0	0	0	0	0	0.0%
El Lago	0	0	0	0	0	0	0	0	0	0	0.0%
Friendswood	0	0	0	0	0	0	0	0	0	0	0.0%
Galena Park	0	0	0	0	0	0	0	0	0	0	0.0%
Hedwig Village	0	0	0	0	0	0	0	0	0	0	0.0%
Hilshire Village	0	0	0	0	0	0	0	0	0	0	0.0%
Houston	0	50	0	1	0	0	2	4	0	57	0.3%
Humble	1	7	0	0	0	0	0	0	0	8	1.6%
Hunters Creek Village	0	0	0	0	0	0	0	0	0	0	0.0%
Jacinto City	0	0	0	0	0	0	0	0	0	0	0.0%
Jersey Village	0	0	0	0	0	0	0	0	0	0	0.0%
Katy	0	0	0	0	0	0	0	0	0	0	0.0%
La Porte	0	0	0	0	0	0	0	0	0	0	0.0%
League City	0	0	0	0	0	0	0	0	0	0	0.0%
Missouri City	0	0	0	0	0	0	0	0	0	0	0.0%
Morgan's Point	0	0	0	0	0	0	0	0	0	0	0.0%
Nassau Bay	0	0	0	0	0	0	0	0	0	0	0.0%
Pasadena	0	0	0	0	0	0	0	0	0	0	0.0%
Pearland	0	0	0	0	0	0	0	0	0	0	0.0%
Piney Point Village	0	0	0	0	0	0	0	0	0	0	0.0%
Seabrook	0	0	0	0	0	0	0	0	0	0	0.0%

Jurisdiction	Number of Facilities in Moderate Wildfire Ignition Density Hazard Area, by Lifeline Category								Other Critical Facilities	Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems		Count	% of Jurisdiction Total
Shoreacres	0	0	0	0	0	0	0	0	0	0	0.0%
South Houston	0	0	0	0	0	0	0	0	0	0	0.0%
Southside Place	0	0	0	0	0	0	0	0	0	0	0.0%
Spring Valley Village	0	0	0	0	0	0	0	0	0	0	0.0%
Stafford	0	0	0	0	0	0	0	0	0	0	0.0%
Taylor Lake Village	0	0	0	0	0	0	0	0	0	0	0.0%
Tomball	0	4	1	0	0	0	1	0	0	6	1.5%
Waller	0	0	0	0	0	0	0	0	0	0	0.0%
Webster	0	0	0	0	0	0	0	0	0	0	0.0%
West University Place	0	0	0	0	0	0	0	0	0	0	0.0%
Unincorporated County	52	90	12	15	1	9	6	52	6	243	1.5%
Harris County Total	53	151	13	18	1	9	9	56	6	316	0.8%

Source: ATLAS 2022; BTS/USDOT 2023; CIMA 2024; City of La Porte 2025; City of Seabrook 2025; FleetCor 2024; Harris County 2020, 2024; Harris County Office of Emergency Management 2022-2025; Harris County Public Library, HCUS 2020, HCPCS 2024, HIFLD 2021-2024; METRO 2024; NPMS 2024; SETRAC 2024; TCEQ 2023-2024; Regional Catastrophic Preparedness Grant Program 2023; DSHS 2024; TEA 2023; THC 2023; USACE 2024; Texas A&M Forest Service

Note: The critical facilities analyzed are only within the boundaries of Harris County. Information pertaining to facilities owned/operated by the non-municipal planning partners was not available during the development of the risk assessment, and as such, the wildfire risks to these planning partners is not reflected in the table above.

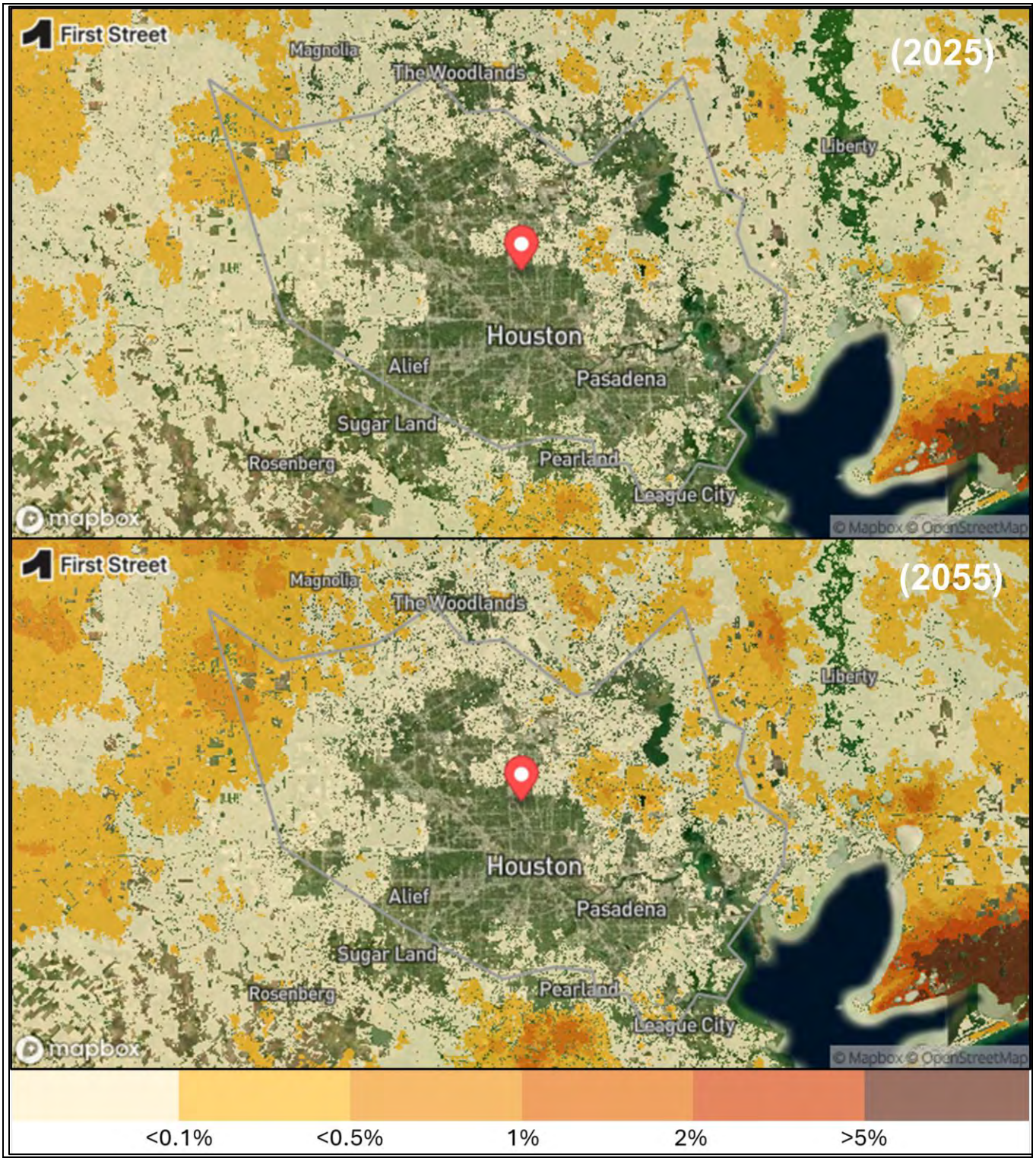
Table 17-13. Number of Facilities in Wildfire Ignition Density Area, by Lifeline Category

Fire Hazard Zone	Number of Facilities in Moderate Wildfire Ignition Density Hazard Area, by Lifeline Category									Total Facilities in Hazard Area	
	Communications	Energy	Food, Hydration, Shelter	Hazardous Materials	Health & Medical	Safety & Security	Transportation	Water Systems	Other Critical Facilities	Count	% of Total
Wildfire Threat Hazard Zones											
Low Wildfire Threat	1,822	3,458	1,160	1,306	157	579	1,444	1,997	871	12,794	31.6%
Wildfire Ignition Density Zones											
Very High	0	5	0	2	0	0	1	2	0	10	<0.1%
High	3	3	5	3	1	0	1	10	0	26	0.1%
Moderate	53	151	13	18	1	9	9	56	6	316	0.8%
Total For Ignition Density	56	159	18	23	2	9	11	68	6	352	0.9%

Source: ATLAS 2022; BTS/USDOT 2023; CIMA 2024; City of La Porte 2025; City of Seabrook 2025; FleetCor 2024; Harris County 2020, 2024; Harris County Office of Emergency Management 2022-2025; Harris County Public Library, HCUS 2020, HCPCS 2024, HIFLD 2021-2024; METRO 2024; NPMS 2024; SETRAC 2024; TCEQ 2023-2024; Regional Catastrophic Preparedness Grant Program 2023; DSHS 2024; TEA 2023; THC 2023; USACE 2024; Texas A&M Forest Service

Note: The critical facilities analyzed are only within the boundaries of Harris County. Information pertaining to facilities owned/operated by the non-municipal planning partners was not available during the development of the risk assessment, and as such, the wildfire risks to these planning partners is not reflected in the table above.

Figure 17-7. Changes in Percent Likelihood of Wildfires in Harris County (2025 to 2055)



Source: (First Street 2024)

18. ENERGY PIPELINE FAILURES

18.1 HAZARD PROFILE

18.1.1 Hazard Description

Energy Pipeline Failure refers to the disruption, damage, or failure of pipelines that transport natural gas, crude oil, refined petroleum products, and other energy resources. Failures may result from corrosion, mechanical damage, excavation accidents, natural hazards (e.g., flooding, hurricanes, or earthquakes), or intentional acts such as vandalism or cyberattacks. The consequences of pipeline failure can include explosions, fires, hazardous material spills, environmental contamination, service disruptions, and potential loss of life or property.

18.1.2 Location

Harris County is home to an extensive network of energy pipelines that transport oil, natural gas, and refined petroleum products. Key locations at risk include pipeline corridors, petrochemical & refining facilities, urban & suburban areas as well as at critical infrastructure sites.

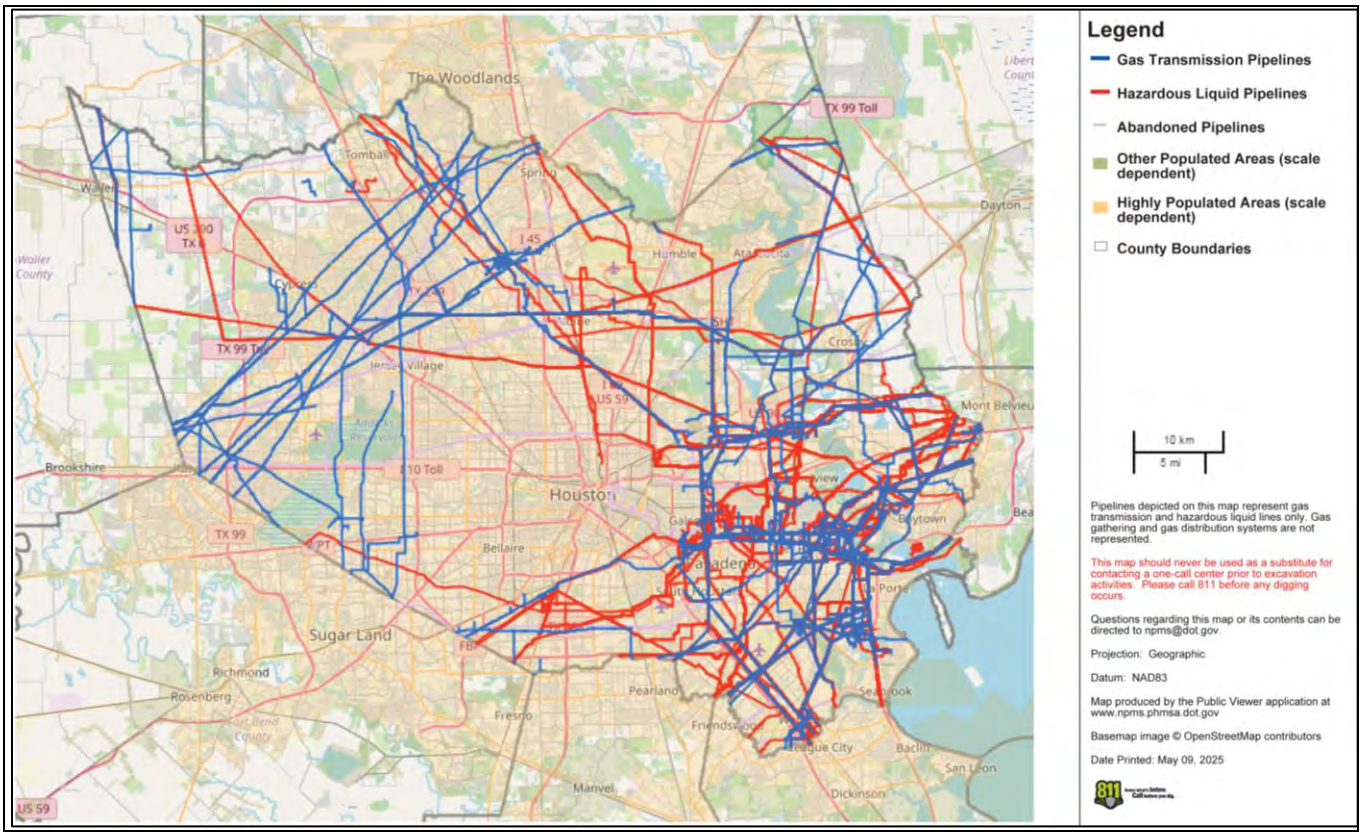
Major transmission and distribution pipelines traverse Harris County, particularly in industrial zones and along transportation corridors. The Houston Ship Channel area and surrounding industrial complexes house critical pipeline infrastructure. Pipelines run beneath residential and commercial developments, like in Pasadena, Deer Park, and La Porte. There are also additional pipelines that supply energy to power plants, hospitals, and emergency services (NPMS 2025). Figure 18-1 shows the locations of gas transmission pipelines, hazardous liquid pipelines, as well as abandoned pipelines within Harris County as depicted on the National Pipeline Mapping System (NPMS).

18.1.3 Extent

The severity of an Energy Pipeline Failure depends on several factors, including the type of material transported, the location of the failure, and the response time. While the extent of this hazard is not quantifiable, there are some key considerations that shape the overall extent of Energy Pipeline Failures. Those include the following:

- Size and pressure of the pipeline − Larger, high-pressure pipelines can cause more severe damage.
- Nature of the failure − Small leaks may go undetected, while ruptures can result in fires, explosions, or widespread environmental damage.
- Impact radius − The affected area may range from a few feet to several miles, depending on the type of failure.
- Service disruption duration − Outages may last hours to weeks, depending on the availability of repair resources and alternate supply routes.

Figure 18-1. Pipeline Locations in Harris County



Source: (NPMS 2025)

18.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and 2024, Harris County was not included in any major disaster (DR) or emergency (EM) declarations for energy pipeline failure-related events (FEMA 2025).

State Emergency Proclamations

Harris County has been included in zero Energy Pipeline Failure-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2020 and 2025, Harris County was included in zero USDA energy pipeline failure-related agricultural disaster declarations.

Previous Events

Known hazard events that impacted Harris County between January 2020 and May 2025 are discussed in Table 18-1. For events prior to 2020, refer to the 2020 HC MH-MAP.

Table 18-1. Energy Pipeline Failure Events in Harris County (2020 to 2025)

Event Date	Declaration or Proclamation Number	Location Impacted	Description
2022	N/A	Baytown and the surrounding areas	ExxonMobil Baytown Refinery Gas Line Leak
Sept. 16, 2024	N/A	Deer Park	An SUV struck Energy Transfer's pipeline in Deer Park and caused a massive explosion and fire.

Source: (Houston Landing 2025)

18.1.5 Probability of Future Occurrences

Information on previous energy pipeline failure occurrences in the County was used to calculate the probability of future occurrences of such events, as summarized in Table 18-2. Historical records for previous events were scattered across may online sources, including those used in the preceding section. Due to the limited historical data available, the probability of occurrence for energy pipeline failure in the County is considered "occasional."

Table 18-2. Probability of Future Energy Pipeline Failure Events in Harris County

Hazard Type	Number of Occurrences Between 2020 and 2025	Percent Chance of Occurring in Any Given Year
Energy pipeline failure	2	22.2%

Source: (Houston Landing 2025)

Note: Due to limitations in data, not all Energy Pipeline Failure events occurring between 1954 and 1996 are accounted for in the tally of occurrences. As a result, the number of hazard occurrences is calculated using the number of occurrences between 2020 and 2025

18.1.6 Cascading Impacts on Other Hazards

Energy pipeline failures can generate or exacerbate other hazards. Fire and Explosions can occur when there is immediate ignition of vapor clouds. These fires can cause massive damage to nearby buildings and critical infrastructure. Some energy products release volatile organic compounds (VOCs) or other hazardous air pollutants. There can also be waterway contamination when energy pipeline failures result in spills into bayous, creeks, or into the Houston Ship Channel. On occasion, events can lead to evacuations due to the events occurring in such close proximity to residential or commercial areas. When this occurs, it can trigger shelter-in-place orders or mass evacuations. Other cascading hazards include transportation interruption as well as utility disruptions.

18.2 VULNERABILITY AND IMPACT ASSESSMENT

To understand risk, a qualitative assessment was performed to evaluate what assets are exposed or vulnerable to energy pipeline failures. The entire County has been identified as exposed to hazard. Therefore, all assets in the County (population, structures, critical facilities, and lifelines), as described in Chapter 3 (County Profile), are exposed and potentially vulnerable.

18.2.1 Life, Health, and Safety

Population

Residents near major pipeline corridors face the greatest physical risk from explosions, toxic exposures, or evacuations. Urban pipelines near high-density neighborhoods (e.g., East Houston, Pasadena) increase casualty and displacement potential.

Low-income and minority communities, often located near industrial corridors or petrochemical facilities, may have less access to transportation during evacuations. They may also be more likely to suffer long-term health impacts or experience greater challenges in recovery. Other socially-vulnerable populations are those with limited English proficiency − this barrier may impede access to emergency alerts or shelter information. Residents with disabilities or chronic illnesses are also at higher risk during evacuations and shelter-in-place orders.

18.2.2 General Building Stock

Residential structures, schools, commercial facilities, and critical infrastructure near pipeline corridors may be directly damaged by blast impacts or fire. They can also be contaminated by airborne or ground pollutants. Energy Pipeline Failures can render some facilities unusable during cleanup or repair operations.

18.2.3 Economy

Harris County's economy is deeply tied to the energy and chemical sectors. Pipeline disruptions can halt refinery operations, increase fuel prices, and damage regional competitiveness. Large incidents may trigger supply chain interruptions, job losses, and public health expenditures.

18.2.4 Natural, Historic, and Cultural Resources

Aside from the environmental risks posed by energy pipeline failure, historical buildings and culturally significant sites located near pipeline corridors may also be at risk. This is especially the case in East End, Independence Heights, where impacts can come from fire damage or smoke contamination (Houston First Corporation 2025).

18.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure the establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

18.3.1 Potential or Planned Development

New residential and commercial growth in western and northern Harris County may expand into areas near transmission pipelines. Also, infilling around older industrial zones increases the proximity of people and buildings to high-risk areas. Mixed-use and high-density housing projects in Pasadena, Spring, and East Houston may be exposed if near older, pre-existing energy pipelines. Industrial expansion in the Ship Channel region increases the complexity of pipeline systems and associated failure risk

18.3.2 Climate Change

While there is no direct tie to climate change, energy pipeline failure hazards may potentially see future impacts from climate change as increased flooding or extreme rainfall events may lead to soil erosion and/or pipeline exposure. Heat stress and corrosion may also degrade pipeline materials faster, especially in aging infrastructure.

19. PUBLIC HEALTH EMERGENCY

19.1 HAZARD PROFILE

19.1.1 Hazard Description

A public health emergency is an event that can cause harm to a person's health or to the health of a community. These types of hazards can take many forms, each with different causes, transmission methods, and public health impacts (CDC 2024). These events can result in illness, fatalities, long-term disability, mental health crises, and severe economic and social disruptions. Public health emergencies can escalate rapidly and often require coordinated responses from public health agencies, emergency management agencies, health care systems, and local governments. Key functions during such emergencies include disease surveillance, public communication, vaccination or prophylaxis campaigns, quarantine or isolation measures, and surge medical care.

Public health emergencies may arise independently (e.g., flu pandemics, waterborne outbreaks) or as cascading consequences of other hazard events (e.g., floods, hurricanes, hazardous materials releases), increasing both their complexity and potential impact.

Types of Public Health Emergencies

Infectious Disease Outbreaks

Infectious disease outbreaks are the most common and widely recognized public health emergencies. Infectious disease outbreaks are transmitted through various routes, including airborne droplets, direct contact, or fecal-oral routes. Impacts are often observed as rapid person-to-person spread and can result in human illness and/or death. Severe outbreaks may put a strain on hospitals, close schools, and result in business disruptions (Tufts University 2025). Common types of infectious disease include the following:

- Viral diseases like influenza, COVID-19, norovirus, and measles
- Bacterial diseases like as tuberculosis or meningococcal disease
- Parasitic diseases like giardiasis, cryptosporidiosis
- Fungal Diseases

Vector-Borne Diseases

Spread by mosquitoes, ticks, or other insects, vector-borne diseases are a persistent risk in Harris County due to its warm, humid climate and frequent flooding. According to the Department of Public Health, Harris County vector surveillance/control centers primarily focus on mosquitoes and ticks (Harris Co. Department of Health, 2025). Mosquitos can carry diseases such as West Nile virus, Saint Louis encephalitis virus, eastern equine encephalitis virus, zika virus, chikungunya virus, and dengue virus (Baylor College of Medicine 2025). Ticks transmit rickettsia, anaplasmosis,

ehrlichiosis, and Lyme disease. These types of diseases can cause illnesses, birth defects, and disabilities and can generate long-term vector control costs (Harris County Public Health 2024).

Waterborne and Foodborne Illnesses

Outbreaks of diseases can be caused by contaminated food or water sources, poor sanitation, or aging water infrastructure. Primary diseases of concern include hepatitis A, cholera, Legionnaires' disease, listeria, and food poisoning from E coli or salmonella. Impacts can include localized outbreaks, restaurant or facility closures, boil-water notices, or public alerts (WHO 2023)

Environmental and Chemical Exposure

Public health emergencies can result from exposure to toxic substances through air, water, or soil as a result of accidents or natural disasters. Common agents can include benzene, dioxins, hydrogen sulfide, and volatile organic compounds (VOCs). The sources for these can be chemical plant incidents, refineries, Superfund sites, or post-disaster pollution. Impacts can include acute respiratory distress, long-term cancer risks, and mental health stress. Response to releases of such chemicals can result in the need for evacuations of affected areas (NIH 2025).

Heat-Related Illness and Air Quality Emergencies

NWS Houston/Galveston frequently issues air quality alerts or ozone action days. These occur when extreme heat combines with poor air quality to create large-scale public health risks, especially for people experiencing vulnerability. Hazards to residents include heat stroke, dehydration, ozone exposure, and asthma attacks. Some of the primary drivers for these types of public health emergencies are climate change, the urban heat island effect, and industrial emissions (Cleveland Clinic 2025). Impacts can be observed through high call volumes to emergency medical service (EMS) centers as well as increased mortality among older adults and/or outdoor workers.

Bioterrorism / Intentional Biological Release

While rare, intentional public health emergencies remain a concern due to Harris County's global profile. Agents of concern may include anthrax, smallpox, and ricin due to their negative health impacts they have on exposed populations. Impacts to the planning area may be the high mortality potential, widespread public panic, as well as the need for rapid detection and response.

Preparedness and Response

Preparedness and response activities related to public health emergencies include the following (Texas Health and Human Services 2025):

- Identifying and tracking outbreaks in the community.
- Providing immunizations or prophylactic medications to at risk communities. Such a response requires a great deal of coordination and communication between response agencies.
- Conducting inspections of grocery stores and restaurants to ensure those locations are abiding by health standards.

- Conducting inspections of shelters to ensure they are preparing food properly and are operating in a safe manner.
- Managing animal control activities to prevent the spread of zoonotic diseases such as rabies, hantavirus, and tularemia.
- Ensure EMS agencies have the staffing, equipment, and supplies they need to respond appropriately to calls for service.
- Conducting surveillance and targeted treatment for vector-borne diseases like West Nile virus, including larvicide application and mosquito spraying in affected areas.

19.1.2 Location

Public health emergencies are not location-bound. They can affect any or all parts of Harris County, depending on factors such as population density, access to health care, social vulnerability, proximity to sources of infection (airports, ports, major highways, etc.), and environmental conditions conducive to the spread of disease (e.g., standing water for mosquito breeding). The following locations may be more vulnerable:

- Dense population centers such as Houston, Pasadena, and Baytown, which pose a higher risk of rapid disease transmission
- Communities with limited access to health care or preventive care
- Homeless encampments or shelters, which often lack hygiene and sanitation infrastructure
- High-density facilities such as schools, long-term care facilities, and correctional institutions, where communicable diseases can spread quickly

19.1.3 Extent

The extent of a public health emergency varies significantly based on the type of health threat (e.g., viral, bacterial, chemical, vector-borne), the transmission characteristics (e.g., airborne, contact, waterborne), the duration of the outbreak or exposure, and the capacity of the health care system to respond.

A small-scale outbreak may affect a single community or facility (e.g., a foodborne illness in a school), while a large-scale emergency, such as the COVID-19 pandemic, may disrupt health care, education, commerce, and daily life across the entire County for months or years. Key indicators of severity include infection/fatality rates, economic losses, health care demand, and duration of disruption.

Harris County's large population, economic infrastructure, international connectivity, and environmental challenges make it highly susceptible to a wide range of public health emergencies. These events can spread rapidly, impact all aspects of life, and are often amplified by other hazard events, such as floods or chemical incidents.

19.1.4 Previous Occurrences

Significant Recent Events

Harris County has experienced a variety of public health emergencies in recent history: COVID-19 (2020-2022), H1N1 influenza pandemic (2009), West Nile VIRUS outbreaks, zika virus, and environmental health crises such as post-Hurricane Harvey conditions in 2017 or the ITC chemical fire in 2019. Table 19-1 lists significant hazard events that impacted Harris County between 2009 and 2025.

Table 19-1. Notable Public Health Emergency Events in Harris County (2009 to 2025)

Event Date	Declaration Number	Location Impacted	Description
2009	N/A	Harris County	H1N1 Influenza Pandemic: Affected tens of thousands in Harris County. Required extensive public health messaging, vaccination clinics, and coordination with school districts to manage outbreaks and absenteeism
2012, ongoing	N/A	Harris County	West Nile Virus: Harris County Public Health conducts annual mosquito surveillance and control to manage vector-borne disease threats. In 2012, dozens of human cases and several deaths occurred, prompting aerial spraying in high-risk areas.
2016, 2017	N/A	Harris County	Zika Virus: Imported cases confirmed in Harris County, raising concern due to the presence of Aedes mosquito vectors. Public education and vector control operations were launched to limit the spread
2017	N/A	Harris County	Following Hurricane Harvey in 2017, standing water, chemical spills, and sewage overflows created widespread health hazards, including mold exposure, injury risks, and potential waterborne diseases
2019	N/A	Harris County	The ITC chemical fire (2019) resulted in short-term air quality concerns and increased demand for public health advisories and monitoring
2019	N/A	Harris County	Measles outbreak
2020 – 2022	DR-4485 EM-3458	Harris County	The recent COVID-19 pandemic resulted in over 1 million reported cases and thousands of deaths across Harris County, and it led to widespread closures of schools and businesses, strained hospitals, and required mass vaccination campaigns. The pandemic prompted major public health interventions, including testing sites, personal protective equipment distribution, contact tracing, and emergency declarations at all levels of government
2022	N/A	Harris County	MPOX outbreak
2023	N/A	Harris County	The Shell plant fire (2023) resulted in short-term air quality concerns and increased demand for public health advisories and monitoring
2024	N/A	Harris County	Following Hurricane Beryl in 2024, standing water, chemical spills, and sewage overflows created widespread health hazards, including mold exposure, injury risks, and potential waterborne diseases
2025	N/A	Harris County	Measles outbreak

Source: (TSHA 2021) (UT Health Houston 2024) (NBC News 2016) (NIH 2023) (TCEQ 2020) (TCEQ 2023) (HCPH 2025)

FEMA Major Disaster and Emergency Declarations

Harris County has been included in two major disaster (DR) or emergency (EM) declarations for public health emergency-related events, as listed in Table 19-2 (FEMA 2025).

Table 19-2. FEMA Declarations for Public Health Emergency Events in Harris County (1954 to 2025)

Event Date	Declaration Date	Declaration Number	Description
January 20, 2020, to May 11, 2023	March 25, 2020	DR-4485	Biological disaster declaration for the COVID-19 pandemic
January 20, 2020, to May 11, 2023	March 13, 2020	EM-3458	Biological emergency declaration for the COVID-19 pandemic

Source: (FEMA 2025)

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Harris County has not been included in any public health emergency-related state emergency proclamations.

USDA Declarations

The U.S. Secretary of Agriculture is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2017 and 2025, Harris County was not included in any USDA public health emergency-related agricultural disaster declarations.

19.1.5 Probability of Future Occurrences

Probability Based on Previous Occurrences

Information on previous public health emergency occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 19-3. Based on historical records and input from the Steering Committee, the probability of occurrence for public health emergency hazards in the County is considered "occasional."

Table 19-3. Probability of Future Public Health Emergency Events in Harris County

Hazard Type	Number of Occurrences Between 2009 and 2025	Percent Chance of Occurring in Any Given Year
Public Health Emergency	11	65%

Source: (TSHA 2021) (UT Health Houston 2024) (NBC News 2016) (NIH 2023) (TCEQ 2020) (TCEQ 2023)
Note: The events listed in Table 19-1 were used to generate the probability statement described above.

Effect of Climate Change on Future Probability

The relationship between climate change and an increase in infectious diseases is difficult to predict with certainty, but there are scientific linkages between the two. Increased rainfall and heavy rainfalls increase the chances of standing water where mosquitos breed, leading to high probability of water-borne and vector-borne diseases. Moreover, a warming climate is likely to increase the length of the insect season, increasing the potential rates of transmission of insect-borne disease.

19.1.6 Cascading Impacts on Other Hazards

There are no known cascading impacts that public health emergencies can have on other hazards of concern for Harris County. However, other hazards can have impacts on the probability and impacts of public health emergencies:

- Flooding and Hurricanes—Floodwaters can contaminate drinking water and wastewater systems, spreading waterborne diseases. Standing water after storms creates breeding grounds for mosquitoes, increasing risk of insect-borne diseases. More information on flooding hazards can be found in Chapter 12; hurricanes and coastal storms are discussed in Chapter 13.
- Extreme Heat—Heat waves can trigger health emergencies, particularly for vulnerable groups such as older adults, children, or outdoor workers. More information on extreme heat hazards can be found in Chapter 11.
- Hazardous Materials Incidents—Chemical exposures may cause acute public health issues (e.g., respiratory distress), requiring mass medical attention and long-term monitoring. More information on toxic release/hazardous materials events can be found in Chapter 21.
- Utility Failures—Loss of water service can impede hygiene practices during infectious disease outbreaks. Power outages and loss of refrigeration may result in unsafe food or medical supply storage. Power outages can disrupt health care delivery, shut down ventilators or oxygen machines, and prevent access to digital health communications. More information on utility failure hazards can be found in Chapter 22.

In addition, any hazard that displaces people into shelters or crowded homes increases disease transmission potential.

19.2 VULNERABILITY AND IMPACT ASSESSMENT

Public health emergencies in Harris County pose significant risks due to the area's large and diverse population, dense urban development, and extensive international and interstate travel networks. The entire County is exposed to this hazard, so all assets in the County (population, structures, critical facilities, and lifelines), as described in Chapter 3, are potentially vulnerable.

19.2.1 Life, Health, and Safety

Vulnerability to public health emergencies is magnified by health disparities, aging infrastructure, and limited access to care in some areas, particularly among historically underserved communities. The following groups within Harris County are disproportionately vulnerable to public health emergencies:

- Low-income and minority populations—May lack access to health care, insurance, transportation, and paid leave, limiting their ability to seek treatment or isolate when sick.
- Elderly residents—At higher risk of serious illness and mortality during outbreaks of respiratory or vector-borne diseases.
- Children—Especially vulnerable to diseases transmitted in schools or daycare environments, such as measles or influenza.
- Individuals with chronic health conditions or disabilities—More likely to experience complications or require hospitalization during an outbreak.
- Unhoused populations—Often lack access to hygiene, nutrition, shelter, and routine medical care, increasing vulnerability to disease transmission and complications.
- Non-English-speaking residents—May experience delays in receiving accurate public health information and accessing care during emergencies.
- Pregnant and birthing people—More likely to experience a higher risk of infection and serious illness.

These groups are often concentrated in specific census tracts in northeast, southeast, and unincorporated parts of the County, and may face barriers to vaccination, quarantine, or accessing emergency alerts.

19.2.2 General Building Stock

No structures are anticipated to be directly affected by the public health emergency hazard.

19.2.3 Community Lifelines and Other Critical Facilities

Health care System

Harris County hosts a large and complex health care network, including the Texas Medical Center, the largest medical complex in the world. However, hospital surge capacity is limited during pandemics or mass exposure events. Long-term care facilities, clinics, and community health centers may struggle to manage rapid increases in patient loads, especially if staff are affected.

Public Services

Emergency services, public transportation, sanitation, and food distribution systems are vulnerable to workforce shortages during public health emergencies. Schools, jails, shelters, and other congregate facilities can be sites of rapid disease spread and require tailored response strategies.

Jurisdictional Vulnerabilities

Among the 44 participating jurisdictions in the MJ-HMP update, many smaller municipalities and special districts lack dedicated public health departments, relying on Harris County Public Health (HCPH) or the Houston Health Department for epidemiological support and disease response. School districts face particular vulnerabilities during infectious disease outbreaks, where closures and staff shortages can disrupt learning and operations. Schools also serve as mass dispensing or vaccination sites during some responses. Special districts such as municipal utility districts (MUDs) and emergency services districts may be affected if personnel fall ill or if supply chains are interrupted.

19.2.4 Economy

Costs associated with the activities and programs implemented to conduct surveillance and address pandemics have not been quantified in the available documentation. Public health emergencies can have economic impacts on supply chains and operations, which can then lead to potential price increases on supplies as well as daily needs. Small businesses and hourly wage workers may be disproportionately impacted by closures or illness-related absenteeism.

Smaller-scale public health emergencies can cause negative economic impacts, though the extent of the impact is variable. For example, an outbreak of mosquito or tick-borne diseases can impact Harris County's local economies associated with tourism and the use of parks and water bodies.

19.2.5 Natural, Historic, and Cultural Resources

Natural, historical, and cultural resources in Harris County are vulnerable to public health emergencies due to their physical location and the potential for damage during disasters. Table 19-4 identifies a few of the natural, historic, and/or cultural resources found within Harris County. Hazard vulnerabilities at many of these locations include public health emergencies due to dependence on tourism.

19.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure the establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

19.3.1 Potential or Planned Development

Harris County has been growing at a rapid rate over the last several years. That trend is expected to continue. As new homes and businesses are developed, infrastructure and public health services will have to keep pace to continue to track disease trends, to inspect restaurants, schools, and grocery stores for safety in food preparation and food storage, to provide immunizations, and to respond in the event of a public health emergency.

Table 19-4. Example Natural, Historic, and Cultural Resources in Harris County

Name	Location	Hazard Vulnerabilities
Sylvan Beach Pavilion	554 N Bayshore Dr. La Porte, TX	Coastal Erosion, Flooding, Hurricanes and Coastal Storms, Severe Weather, Tsunami
Walker House (Aka, The Treehouse; Isaiah P. Walker House)	3534 Miramar Dr. Shoreacres, TX 77571	Coastal Erosion, Flooding, Hurricanes and Coastal Storms, Severe Weather, Tsunami, Public Health Emergencies
Morgan's Point Historic District	89-835 Bayridge Rd and 300-322 Vinsonia, Houston, TX 77571	Coastal Erosion, Flooding, Hurricanes and Coastal Storms, Severe Weather, Tsunami, Public Health Emergencies
San Jacinto Battleground State Historic Site	3523 Independence Pkwy S, La Porte, TX	Coastal Erosion, Flooding, Hurricanes and Coastal Storms, Severe Weather, Tsunami, Public Health Emergencies
Washburn Tunnel	3100 Federal Rd, Houston/Pasadena, TX 77015	Flooding, Hurricanes and Coastal Storms, Toxic Release / Hazardous Materials, Utility Failure
Sammy Beckendorff Farm	6015 Katy Hockley Rd, Katy, TX 77493	Drought, Extreme Cold, Extreme Heat, Flooding, Severe Weather, Wildfire, Utility Failure, Public Health Emergencies
Wunderlich, Peter and Sophie, Farm	18202 Theiss Mail Rd, Klein, TX	Drought, Extreme Cold, Extreme Heat, Flooding, Severe Weather, Wildfire, Utility Failure, Public Health Emergencies
Houston National Cemetery	10410 Veterans Memorial Dr. Houston, TX 77038	Dam/Levee Failure, Drought, Flooding, Hurricanes and Coastal Storms, Mass Movement, Severe Weather, Wildfire
Sam Houston Park	1000 Bagby St, Houston, TX	Extreme Cold, Extreme Heat, Flooding, Hurricanes and Coastal Storms, Severe Weather, Tsunami, Utility Failure, Public Health Emergencies
Allen's Landing	1001 Commerce St, Houston, TX	Coastal Erosion, Flooding, Hurricanes and Coastal Storms, Mass Movement, Public Health Emergencies

Source: (TXDOT Esri 2025)

19.3.2 Projected Changes in Population

As the population of Harris County continues to grow, so do the risks of public health threats, particularly as populated areas become more compact. With population growth, public health services will need to also grow to offer new families essential public health services. If an outbreak should occur, public health resources will be needed to track the cause of the disease, track those that might have been infected, and connect them with needed medications and treatments. Public heath interventions and public information will be necessary to provide populations with the tools they need to stay healthy.

19.3.3 Climate Change

Climate change may exacerbate the frequency and impact of public health emergencies by expanding vector ranges, introducing new disease threats to Harris County (e.g., chikungunya, malaria), increasing heat-related illnesses and cardiovascular stress, or amplifying the severity and duration of storms, which can increase waterborne and mold-related illnesses. In addition to these

vector-related effects, rising temperatures can also worsen air quality, increase the frequency of temperature extremes, and contribute to a broader range of health risks, further straining public health systems and emergency response efforts.

19.3.4 Other Identified Conditions

All participating jurisdictions in Harris County are exposed to public health emergencies, but vulnerability varies depending on population density, persons exposed, and rates of exposure. Urban areas like Pasadena, Friendswood, and unincorporated Harris County have higher populations, increasing the potential exposure and elongating the public health emergency.

20. TECHNOLOGY FAILURE

20.1 HAZARD PROFILE

20.1.1 Hazard Description

Technology failure refers to the disruption, malfunction, or compromise of digital, mechanical, or communication systems essential to daily operations, public safety, and critical infrastructure. This category includes both hardware and software failures, cyberattacks, system overloads, information technology (IT) outages, and failures in digital communications networks (e.g., cellular, radio, internet). Technology failures may occur due to software bugs, outdated or misconfigured systems, human error, intentional sabotage or hacking, equipment failure, or cascading impacts from other hazards such as flooding, fire, or power outages (UNDRR 2021).

These failures can affect emergency response systems (e.g., 9-1-1 dispatch), local government operations, public communication platforms, critical infrastructure, and private sector services. In an increasingly digitized environment, dependence on technology makes communities more vulnerable to widespread disruption, data loss, and operational paralysis.

20.1.2 Location

Technology failures can occur throughout all jurisdictions in Harris County, given the widespread reliance on interconnected digital infrastructure and network-based services. Vulnerability is not limited to any single location; however, areas with concentrated critical infrastructure, government functions, medical centers, and financial or industrial operations may experience higher levels of potential impact. The following key assets and systems are vulnerable to technology failures:

- Government data centers and emergency operations platforms
- Hospitals and health care networks
- Water/wastewater treatment facility control systems
- Energy grid and pipeline control systems
- Transportation systems (e.g., traffic signals, rail, aviation)
- School district IT systems and remote learning platforms

20.1.3 Extent

Extent of technology failure hazards can be measured by the duration of the outage (minutes to days), systems affected (e.g., emergency communications, public services, infrastructure controls), geographic spread (localized, regional, or national systems), and the number of users or residents impacted. Technology failures can be localized, such as a city's 9-1-1 dispatch center going offline due to server failure. They can also be regional, like multi-county broadband internet outages or systemic failures such as a cyber-attack crippling IT systems across multiple jurisdictions. In

extreme cases, failures in public warning systems could result in cascading effects impacting public health, safety, and economic operations.

20.1.4 Previous Occurrences

Significant Recent Events

Although technology failure events are not always widely publicized, several incidents in recent years illustrate the hazard's relevance to Harris County. Table 20-1 lists significant hazard events that impacted Harris County between 2020 and 2025.

Table 20-1. Technology Failure Events in Harris County (2020 to 2025)

Event Date	Declaration Number	Harris County Included in Declaration?	Location Impacted	Description
May 2023	N/A	N/A	City of Houston	Cyberattack on Houston's IT Systems: A major cybersecurity breach affected multiple systems within the City of Houston, including police networks and email servers. While Harris County systems were not compromised, the event prompted increased investment in cybersecurity across regional jurisdictions
February 2021	DR-4586 EM-3554	Yes	Countywide	Winter Storm Uri: Technology systems controlling energy and water infrastructure were disrupted across Texas. In Harris County, widespread power outages disabled telecommunications and impacted systems at water and wastewater facilities, forcing operational shutdowns and manual overrides.
2020-2021	DR-4485 EM-3458	Yes	Multiple locations across Harris County	COVID-19 Pandemic: Remote work and education placed unprecedented strain on digital infrastructure, highlighting vulnerabilities in school district networks, public service platforms, and continuity of operations plans. School districts across Harris County faced disruptions due to bandwidth constraints, software failures, and cyber intrusions.
2020, 2022	N/A	N/A	Multiple locations in Harris County	911 System Disruptions have occurred multiple times since 2020. In 2020 and 2022, local jurisdictions in Harris County experienced temporary 9-1-1 outages due to fiber-optic line damage and software upgrades gone awry, delaying emergency response.

Source: (Government Technology 2024) (UH 2021) (NIH 2022) (ABC13 2024)

FEMA Major Disaster and Emergency Declarations

Harris County has not been included in any major disaster (DR) or emergency (EM) declarations for technology failure-related events (FEMA 2025).

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. There have been no technology failure-related state emergency proclamations for Harris County.

USDA Declarations

The U.S. Secretary of Agriculture is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2017 and 2025, Harris County was not included in any USDA technology failure-related agricultural disaster declarations.

20.1.5 Probability of Future Occurrences

Information on previous technology failure occurrences in the County was used to calculate the probability of future occurrences of such events, as summarized in Table 20-2. Based on historical records and input from the Steering Committee, the probability of occurrence for technology failure in the County is considered "occasional."

Table 20-2. Probability of Future Technology Failure Events in Harris County

Hazard Event Type	Number of Occurrences Between 2020 and 2025	Percent Chance of Occurring in Any Given Year
Technology Failure	4	67%

Source: (Government Technology 2024) (UH 2021) (NIH 2022) (ABC13 2024)
Note: Only the more significant events are listed, as there is no single source for historical technology failures in Harris County.

20.1.6 Cascading Impacts on Other Hazards

There are no known cascading impacts that technology failures can have on other hazards of concern for Harris County. However, other hazards can have impacts on the probability and impacts of technology failures:

Technology failures frequently result from or are worsened by natural hazard events. These cascading effects are especially important for mitigation planning:

- Flooding—Floodwaters can inundate data centers, emergency communications buildings, or utility control systems. Water damage to electrical and IT equipment often results in long-term outages.
- Hurricanes/Coastal Storms—Strong winds and storm surge can damage communication towers, fiber-optic cables, and electrical systems. Hurricane Harvey (2017) and Hurricane Nicholas (2021) led to telecom disruptions across the County.

- Winter Storms—During Winter Storm Uri (2021), extreme cold froze pipelines and disabled power grids, which in turn shut down internet, emergency services, and remote operations platforms.
- Extreme Heat—High temperatures can strain power grids, causing brownouts or blackouts that disable air-conditioned data centers or cloud storage systems without adequate backup power.
- Wildfires and Lightning—Fire and lightning can damage communications infrastructure, utility lines, and server farms, leading to localized or widespread outages.
- Tornadoes and Severe Weather—Direct strikes by tornadoes or debris can damage antennas, substations, or cable lines.

20.2 VULNERABILITY AND IMPACT ASSESSMENT

To understand risk, a qualitative assessment was performed to evaluate what assets are exposed or vulnerable to the technology failure hazard. The entire County is exposed to this hazard, so all assets in the County (population, structures, critical facilities, and lifelines), as described in Chapter 3, are potentially vulnerable.

Specific vulnerabilities to technology failures depend on factors such as the age and sophistication of IT infrastructure, existence of cybersecurity protocols, availability of redundant systems or cloud backups, and reliance on third-party service providers. Smaller jurisdictions and rural school districts may lack the resources for robust IT staffing or mitigation measures, increasing risk of prolonged outages or attacks.

20.2.1 Life, Health, and Safety

Technology failures are unlikely to significantly impact the health and safety of people in Harris County.

20.2.2 General Building Stock

Buildings and structures in general are usually not impacted by technology failures, though systems operated by electronics and computers are vulnerable.

20.2.3 Community Lifelines and Other Critical Facilities

Technology failures can compromise the operation of numerous critical systems, many of which are essential for public safety and continuity of operations. In Harris County, this includes the following:

- Emergency Communications—9-1-1 call centers, dispatch systems, and public safety radio networks rely on uninterrupted network and IT infrastructure.

- Hospitals and Medical Centers—Health care facilities depend on electronic health records, digital diagnostics, and telemedicine systems. Even brief IT outages can jeopardize patient care.
- Water/Wastewater Facilities—Control systems monitor and manage water quality, flow rates, and wastewater treatment. A technology failure could trigger overflows, contamination, or service interruptions.
- Transportation Systems—Intelligent traffic signal systems, rail operations, and traffic management centers depend on secure, functional communications infrastructure.
- Government Operations—Permit processing, emergency notifications, financial systems, and recordkeeping are all at risk from cyberattacks, outages, or hardware failures.
- Schools and Universities—Increasing reliance on digital instruction and remote learning platforms makes educational facilities particularly vulnerable to disruptions.

20.2.4 Economy

Ways in which technology failure events can affect the economy of Harris County include the following:

- Public Service Delays—Government operations such as licensing, permitting, utility billing, and emergency coordination may be halted or delayed.
- Business Disruptions—Private sector operations may be interrupted due to cyber-attack, internet outages, or data loss, especially in logistics, health care, finance, and manufacturing.
- Data Loss and Recovery Costs—Recovery from cyber-attacks or hardware failures can be expensive, especially if data is unrecoverable or requires forensic restoration.

20.2.5 Natural, Historic, and Cultural Resources

Technology failures typically will not impact natural, historic, or cultural resources. However, any that rely on technology for daily operation will remain vulnerable to a technology failure event without any redundant or back-up systems.

20.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure the establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability. Cyber attacks of the future are expected to increase in sophistication and complexity.

20.3.1 Potential or Planned Development

Generally, technology failure will not be affected by changes in land use and development.

20.3.2 Projected Changes in Population

An increase in the population will ultimately increase the number of persons who could be impacted by a technology failure event within the County. As people increase their reliance on technology, the likely impacts of a technology failure may increase.

20.3.3 Climate Change

Climate change could increase the frequency or intensity of severe weather events that can contribute to technology failures.

20.3.4 Other Identified Conditions

Cybersecurity threats are becoming increasingly frequent, sophisticated, and opportunistic, particularly those that target government and public-sector networks. There is also an increasing dependence on cloud and remote access capabilities; while providing flexibility, these services also introduce vulnerabilities through third-party platforms. There is also inherent risk from interdependence of various systems. For example, public safety, utilities, finance, and transportation all rely on IT platforms that can fail simultaneously if one system is compromised. Lastly, artificial intelligence and automation can produce their own vulnerabilities, because an increase in automation of services and infrastructure introduces new points of failure and complexity.

21. TOXIC RELEASE/HAZARDOUS MATERIALS

21.1 HAZARD PROFILE

21.1.1 Hazard Description

A toxic release or hazardous materials (HazMat) incident involves the accidental or intentional release of substances that pose a risk to public health, safety, property, or the environment. These substances may be chemical, biological, radiological, or otherwise dangerous, and can be released during production, storage, transport, or disposal. Hazardous materials can cause severe health effects, including respiratory distress, chemical burns, neurological damage, cancer, or death. Releases may occur as a result of equipment failure, transportation accidents, operator error, industrial accidents, natural disasters (e.g., flooding of chemical storage sites), or deliberate acts such as terrorism or sabotage.

Given its extensive industrial base, dense transportation infrastructure, and proximity to the Houston Ship Channel, Harris County is one of the most vulnerable areas in the United States to toxic releases and hazardous materials incidents.

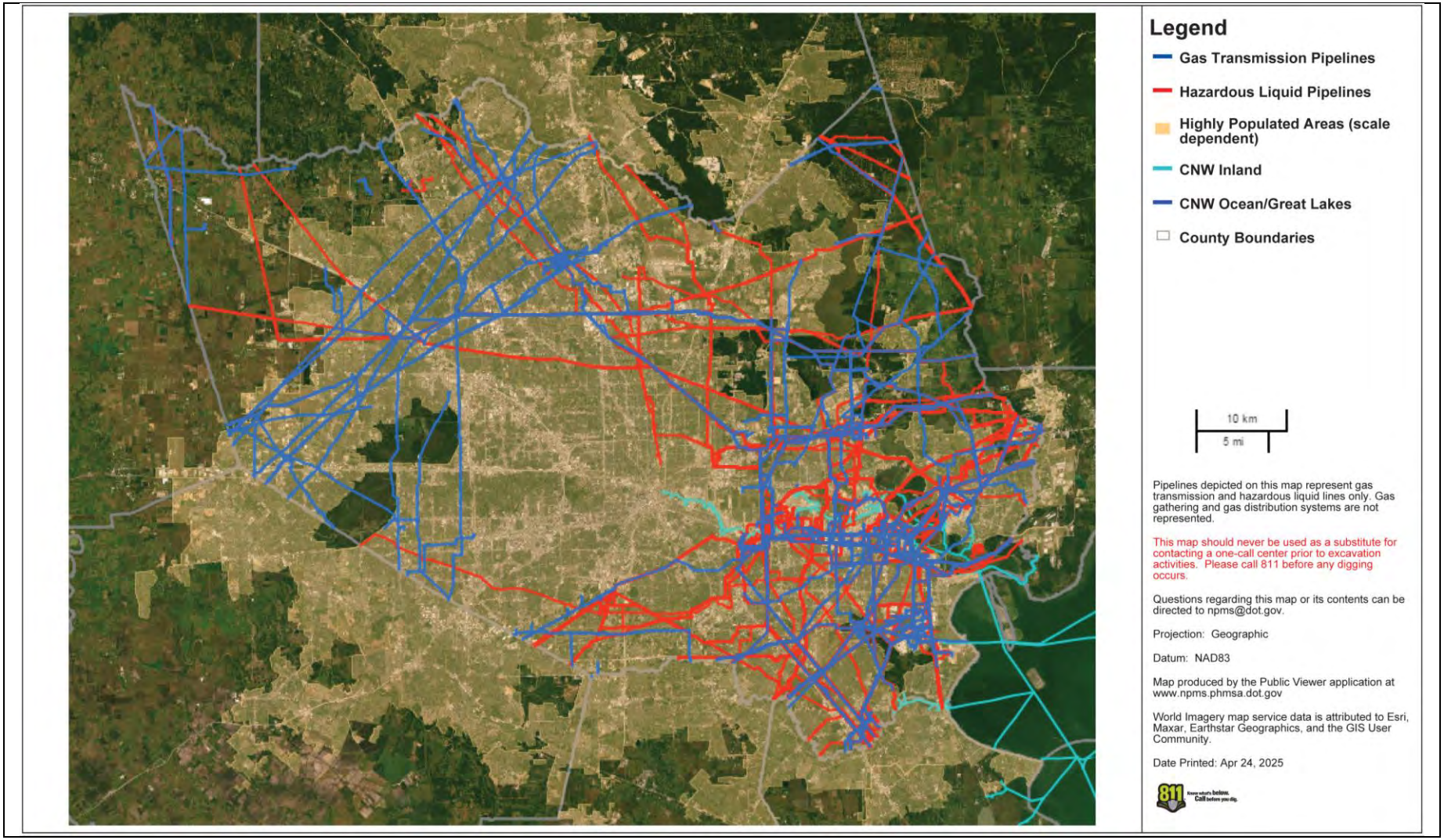
21.1.2 Location

Toxic release and hazardous materials risks are widespread across Harris County, but areas of particular concern include some of these areas:

- The Houston Ship Channel / Industrial Corridor: This region includes hundreds of chemical manufacturing, refining, and processing facilities regulated under the EPA's Risk Management Program
- Petrochemical complexes in Pasadena, Deer Park, and La Porte: These locations contain some of the highest concentrations of hazardous materials in the entire country
- Major highways and rail lines: Interstate 10, Interstate 45, State Highway 225, and major rail corridors carry large volumes of hazardous materials through urban and suburban areas on a daily basis.
- Pipeline networks: Harris County has a dense network of underground pipelines crisscrossing the planning area, with these lines carrying products such as crude oil, refined products, natural gas, as well as other chemicals.
- Schools and residential areas near the industrial zones: There are many vulnerable population groups in the County that live or go to school within close proximity to facilities or corridors where hazardous materials are stored and transported

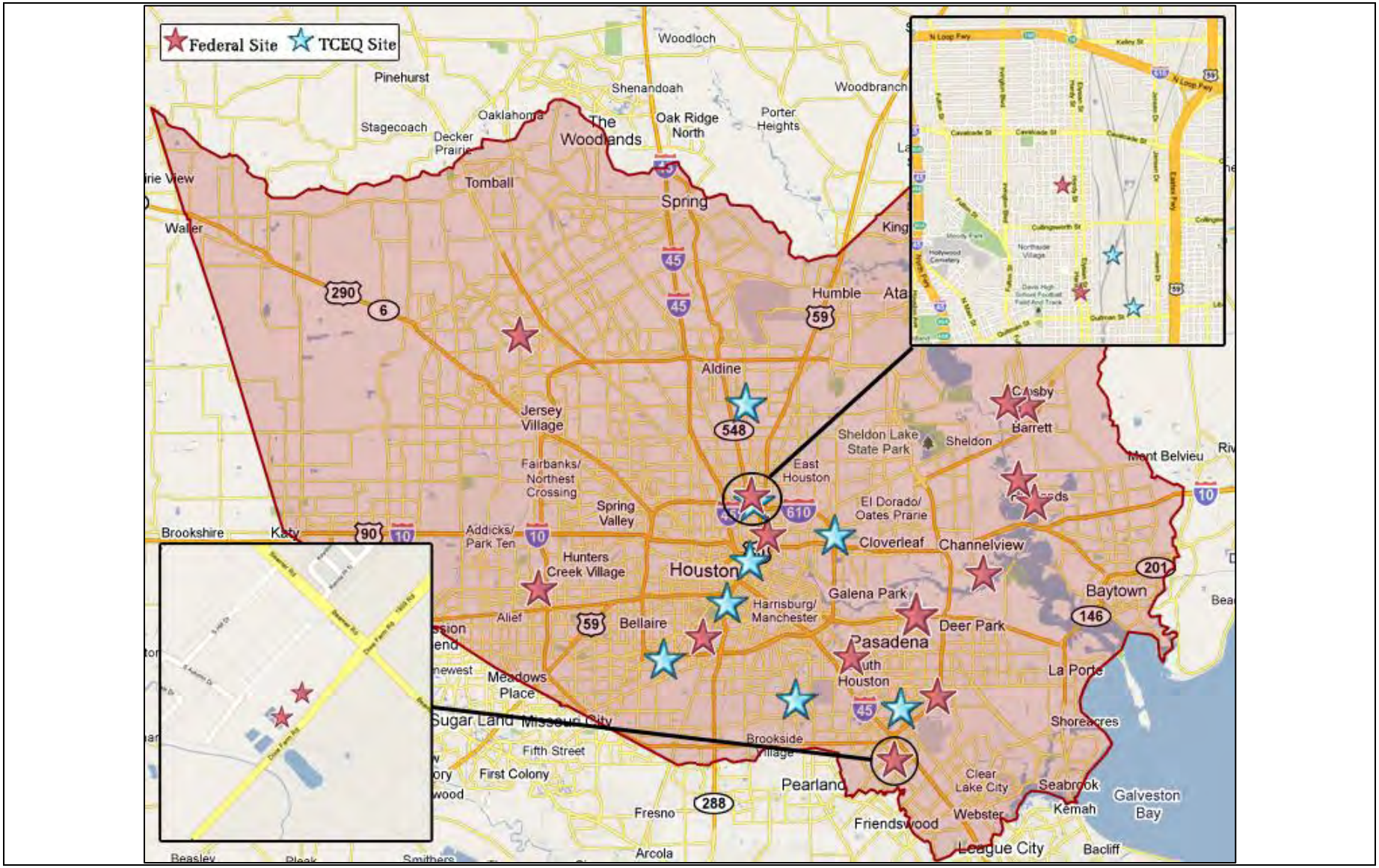
Toxic Release/Hazardous Materials events can originate from underground pipelines as well as known Superfund Sites located throughout the planning area. Figure 21-1 shows gas and hazardous liquid pipelines in Harris County. Figure 21-2 identifies known Superfund Sites. Details on each of the identified Superfunds can be found in Table 21-1.

Figure 21-1. Pipeline Mapping in Harris County



Source: (NPMS 2025)

Figure 21-2. Superfund Sites in Harris County



Source: (TCEQ 2025)

Table 21-1. Superfund Sites in Harris County

Site Name	Location	Type of Facility	Site Phase	Media Affected
Aluminum Finishing	13435 Pike Rd, Missouri City	Metal plating facility	Following cleanup, a certificate of completion was issued in 2011	Soil
Archem Thames/Chelsea	13103 Conklin Ln, Houston	Specialty chemical manufacturing	Cleanup complete	Soil
Brio Refining, Inc	2501 Dixie Farm Rd, Friendswood	Chemical reprocessing and refining facility	Removed from the National Priorities List in 2006	Chemical disposal practices contaminated groundwater, surface soils, and subsurface soils with hazardous chemicals.
Crystal Chemical Company	3502 Rogerdale Rd., Houston	Manufacturing arsenic-containing herbicides	Groundwater monitoring and cap inspections are ongoing. A review of data collected in 2020 indicates that the site is in compliance with the groundwater Remedial Action Objectives in the short term.	Soils and groundwater
Dixie Oil Processors	2505 Choate Rd, Friendswood	Copper recovery and hydrocarbon washing facility	EPA took the site off the Superfund program's National Priorities List in 2006. Operation and maintenance activities are ongoing	Site activities and waste disposal practices contaminated soil and groundwater with hazardous chemicals
Federated Metals	9200 Market St., Houston	Magnesium Dross/Sludge Disposal Inactive Landfill	Evaluation Under Way	Soil, Groundwater
French Limited	San Jacinto River Crossing, Crosby	Mining and operation of an industrial waste storage and disposal facility	Between 1966 and 1971, 90 million gallons of industrial waste from petrochemical companies were disposed of on the property. Following cleanup, operation and maintenance activities are ongoing	Site activities and waste disposal practices contaminated groundwater, surface water, soil, and sludge with hazardous chemicals and heavy metals.
Geneva Industries/ Fuhrmann Energy	9334 Canniff Rd, Houston	Historically used for petroleum exploration and petrochemical production. Currently status is inactive.	The site is drained by the Harris County Flood Control District Channel, which runs along the eastern boundary of the site.	
Gulf Metals Industries	Mykawa Rd and Almeda-Genoa Rd, Houston	Disposal of Hazardous Substances	No Further Superfund Environmental Response Actions Required/Referred to Volunteer Cleanup Program	Groundwater, Soil

Site Name	Location	Type of Facility	Site Phase	Media Affected
Harris (Farley Street)	1000 Genoa Red Bluff Rd, Houston	Abandoned landfill containing tars and sludges	EPA took the site off the Superfund program's National Priorities List in April 1988	A transporter leased the site in 1958 and disposed of over 550 tons of liquids and sludges from local chemical industries in two trenches. Waste disposal activities contaminated soil with hazardous substances
Highlands Acid Pit	611 Battlebell Clear Lake Rd, Highlands	Early in the 1950s, the area received an unknown quantity of industrial waste sludge, believed to be spent sulfuric acid, from oil and gas refining processes	Following cleanup, operation and maintenance activities and monitoring are ongoing	waste disposal activities contaminated soil and groundwater with hazardous chemicals
Houston Lead	300 Holmes Rd, Houston	Lead smelter and Lead-acid battery recycling	The site was remediated under the TCEQ Volunteer Cleanup Program	Soil
Houston Scrap	3799 Jensen Dr, Houston	Lead-acid battery recycling	Cleanup complete – operation and maintenance	Soil
Jensen Drive Scrap	3603 Jensen Dr, Houston	Scrap salvage	Cleanup complete – operation and maintenance	Shallow groundwater, soil
Jones Road Ground Water Plume	11600 Jones Rd, Houston	Former dry cleaning facility	Hazardous substances present at the Site include tetrachloroethylene, also known as perchloroethylene, and related breakdown products, trichloroethylene, 1,2-dichloroethylene, and vinyl chloride.	Groundwater, soils
La Pata Oil Company	1403 Ennis St, Houston	Recycling waste oil and industrial, and hazardous waste chemicals	Cleanup complete – no further Superfund environmental response actions are required	Soil
Many Diversified Interests, Inc	3617 Baer St, Houston	In 1926, Texas Electric Steel Casting Company (TESCO) began operating a metal casting foundry at the site		Soils due to air emissions containing particles of lead
North Cavalcade Street	Intersection of Cavalcade St & Maury St	Wood-preserving operations	Remedied with long-term groundwater cleanup ongoing	Groundwater

Site Name	Location	Type of Facility	Site Phase	Media Affected
Patrick Bayou	1 mile north of State Highway 225, Deer Park	Contaminants of concern at the Site include poly-chlorinated biphenyls (PCBs), mercury, total polyaromatic hydrocarbons (PAHs), dioxin, lead, and bis(2-ethylhexyl) phthalate.	Industrial wastewater discharged to the bayou is the primary source of contamination. Waters being monitored	Soil sediments and surface water
San Jacinto River Waste Pits	West Bank of San Jacinto River, North of Hwy 10, Channelview	Impoundments, or pits, built in the mid-1960s for disposing solid and liquid pulp and paper mill wastes that are contaminated with dioxins and furans	Ongoing community engagement and cleanup efforts	Surface water
Sikes Disposal Pits	2 miles southwest of Crosby, TX	From about 1961 until 1967, the Site was operated as an illegal open dump	Floods frequently, with floodwater from the San Jacinto River	Surface water and ground water
Sol Lynn— Industrial Transformers	1415 S Loop West, Houston	Electrical transformer salvage and recycling	Following cleanup, operation and maintenance activities and monitoring are ongoing.	Soil and groundwater contamination
South Cavalcade Street	Southeast of Cavalcade and Maury St, Houston	Former wood-treating plant	Following cleanup, operation and maintenance activities and monitoring are ongoing.	Soil and groundwater contaminated with volatile organic compounds (VOCs), polycyclic aromatic hydrocarbons (PAHs) and metal salts such as arsenic, chromium, copper, lead, and zinc.
U.S. Oil Recovery	200 N. Richey St, Pasadena	Oil processor and wastewater treatment facility;	Stabilization efforts continue	Variety of hazardous substances
Waste Oil Tank Service	2010 Hartwick Rd, Unincorporated Harris County	Waste Oil Collection and Transfer	Cleanup complete – no further environmental response required	Soils, surface waters, and groundwaters

Source: (EPA 2013) (TCEQ 2024)

21.1.3 Extent

The extent of a hazardous materials incident can vary on several factors, such as type and quantity of substance released or the mode of release (e.g., airborne, spilled into waterways or soil, fire/explosion). Extent can also be determined based on meteorological and environmental conditions, such as wind speed/direction, temperature, and proximity to bodies of water. It can

also be determined by the proximity of the event to population centers and/or critical infrastructure.

Impacts can range from a localized shelter-in-place order to multi-jurisdictional evacuations, environmental contamination, long-term public health concerns, or explosions causing fatalities and structural damage. The EPA and U.S. Department of Transportation (DOT) classify substances based on their toxicity, volatility, flammability, and environmental impact, guiding response measures.

21.1.4 Previous Occurrences

FEMA Major Disaster and Emergency Declarations

Between 1954 and April 2025, Harris County was included in no major disaster (DR) or emergency (EM) declarations for Toxic Release/Hazardous Materials-related events (FEMA 2023).

State Emergency Proclamations

Harris County has been included in no Toxic Release/Hazardous Materials-related state emergency proclamations.

USDA Declarations

The Secretary of Agriculture from the U.S. Department of Agriculture (USDA) is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Since 2013, Harris County was included in zero USDA Toxic Release/Hazardous Materials-related agricultural disaster declarations.

Previous Events

Harris County has experienced numerous high-profile hazardous materials incidents, some with major consequences. These events are summarized in Table 21-2. In addition to these significant Toxic Release/Hazardous Materials incidents, Harris County experiences frequent minor spills and leaks involving tankers, railcars, and trucks transporting hazardous materials. Many are contained quickly, but some require emergency response, lane closures, or brief evacuations.

According to the Harris County Fire Marshal's Office and the Local Emergency Planning Committee, hundreds of reportable hazmat incidents occur each year, ranging from small-scale spills to large facility fires.

Table 21-2. Toxic Release/Hazardous Materials Events in Harris County (2017 to 2025)

Event Date	Declaration or Proclamation Number	Location Impacted	Description
March 2019	N/A	Deer Park	A large chemical storage facility at ITC fire burned for several days, releasing dangerous air pollutants and prompting school closures and shelter-in-place orders. Dozens of tanks caught fire, and contaminated runoff entered the Houston Ship Channel.
April 19, 2019	N/A	Crosby	An explosion and subsequent fire at the KMCO Chemical Plant facility resulted in one fatality, multiple injuries, and a significant toxic plume. Nearby residents were advised to shelter in place.
August 29, 2017	N/A	Crosby	Arkema Chemical Plant Crisis, Crosby (August 2017, Hurricane Harvey): Flooding from Harvey disabled refrigeration systems at the plant, causing volatile chemicals to combust. The event highlighted how natural disasters can trigger secondary technological hazards
May 6-8, 2023	N/A	Deer Park	Shell Deer Park Fire (May 2023): A fire broke out in the olefins unit, releasing smoke and chemical vapors. Local air quality was affected, and the facility was temporarily shut down

Source: (CSB 2023) (CSB 2018) (C&EN 2023)

21.1.5 Probability of Future Occurrences

Information on previous Toxic Release/Hazardous Materials occurrences in the County was used to calculate the probability of future occurrence of such events, as summarized in Table 21-3. Based on historical records and input from the Steering Committee, the probability of occurrence for Toxic Release/Hazardous Materials in the County is considered "occasional."

Table 21-3. Probability of Future Toxic Release/Hazardous Materials Events in Harris County

Hazard Type	Number of Occurrences Between 2017 and 2025	Percent Chance of Occurring in Any Given Year
Toxic Release/Hazardous Materials	4	44%

Source: (CSB 2023) (CSB 2018) (C&EN 2023)

21.1.6 Cascading Impacts on Other Hazards

Toxic releases are frequently triggered or made worse by natural hazard events. Examples in Harris County include the following

- Flooding—Can submerge chemical storage tanks, leach hazardous waste from landfills, or disrupt secondary containment systems. During Hurricane Harvey (2017), dozens of facilities reported toxic spills, and floodwaters mixed with industrial runoff.

- Hurricanes and Coastal Storms—High winds, storm surge, and infrastructure damage can result in airborne chemical plumes, damaged pipelines, and emergency shutdowns that fail to secure hazardous inventories.
- Extreme Heat—May increase the volatility of flammable substances or overheat control systems. Aging infrastructure is particularly vulnerable to thermal stress.
- Lightning and Severe Weather—Direct strikes or power surges can ignite storage tanks or damage sensitive monitoring equipment.
- Earthquakes or Ground Movement—Although rare, subsidence or seismic activity can damage underground pipelines and tanks.

These cascading risks significantly increase the scope and complexity of planning and mitigation efforts

21.2 VULNERABILITY AND IMPACT ASSESSMENT

To understand risk, a qualitative assessment was performed to evaluate what assets are exposed or vulnerable to the Toxic Release or Hazardous Materials (HazMat) Incidents. The entire County has been identified as exposed to hazard. Therefore, all assets in the County (population, structures, critical facilities, and lifelines), as described in Chapter 3 (County Profile), are exposed and potentially vulnerable.

21.2.1 Life, Health, and Safety

Harris County is home to one of the largest petrochemical and hazardous materials corridors in the United States, including over a thousand regulated facilities and miles of pipeline, rail, and highway infrastructure. This dense industrial footprint increases vulnerability to both accidental and cascading toxic release incidents. Vulnerabilities are compounded by the proximity of schools, neighborhoods, and public infrastructure to these facilities and corridors. Moreover, natural hazard events like hurricanes, floods, and extreme heat can trigger or exacerbate hazardous materials incidents, adding complexity and urgency to emergency response and long-term recovery.

Population

Populations most vulnerable to toxic releases and hazardous materials incidents include the following:

- Residents in industrial-adjacent communities (e.g., Pasadena, La Porte, Deer Park): These communities may face acute exposure risks and may lack resources to relocate or shelter in place.
- Children and older adults: More sensitive to airborne toxicants and less able to evacuate quickly.
- People with pre-existing respiratory or cardiovascular conditions: At greater risk from chemical vapors and particulate matter.

- Non-English-speaking communities: May have limited access to emergency communications and protective guidance during an incident.
- Low-income and historically underserved populations: Often reside closer to hazardous sites and may face barriers to healthcare and post-disaster recovery.

Schools, nursing homes, and healthcare facilities near industrial zones are particularly vulnerable due to limited mobility and challenges in rapidly evacuating occupants.

21.2.2 Community Lifelines and Other Critical Facilities

Hazardous materials incidents can impact or be exacerbated by damage to critical infrastructure, including:

- Water supply systems: Contamination from spills or runoff can render drinking water unsafe.
- Wastewater treatment plants: Chemical spills can disrupt treatment processes or damage control systems.
- Transportation networks: Highways and rail lines affected by spills may be closed, affecting emergency response and economic activity.
- Public safety facilities: Police, fire, and EMS stations within impact zones may need to evacuate or reroute operations.
- Power plants and substations: Vulnerable to chemical fires, which could cause broader cascading power outages.

The loss or contamination of these assets may delay recovery and expand the impact footprint of an incident.

21.2.3 Economy

Major chemical releases may result in business closures, environmental fines, and loss of productivity in industrial sectors. Impacts to the Houston Ship Channel can disrupt national supply chains. Economic impacts may also arise due to environmental damage. Toxic substances released into the air, soil, or water can have long-term consequences for ecosystems. Wetlands and waterways near industrial areas, such as the San Jacinto River and Buffalo Bayou, are particularly at risk. Additionally, the cost of clean-up can be taxing. Hazardous materials incidents often involve complex and lengthy remediation processes that require coordination among local, state, and federal agencies.

21.2.4 Natural, Historical, and Cultural Resources

Toxic releases or hazardous materials incidents can have serious implications for natural, historical, and cultural resources located across the planning area. This is particularly the case due to the region's extensive industrial activities. These types of events can lead to environmental degradation, health risks, and the loss of heritage sites.

Natural resources, such as watersheds and waterways, can be polluted if toxic release events occur nearby, potentially resulting in contamination and/or disruption of the water supply. The numerous chemical plants and refineries increase the risk of toxic releases into the environment. Similarly, many industrial sites are located near waterways, making them susceptible to flooding, which can help spread contaminants. The Houston Ship Channel is a prime example of this risk. Others include the San Jacinto River Waste Pits Superfund site (EPA 2019). Other impacts to natural, historic, and cultural resources stem from the Creosote Contamination site in the Fifth Ward. After Union Pacific inherited the site in 1997, there have been repeated attempts to clean up contamination that is directly tied to historic groundwater and soil contamination. More than 11,000 tons or 22 million pounds of creosote-impacted soil was consolidated onsite, later capped to prevent further exposure (Houston Wood Preserving Works 2024). A complete list of the Superfund Sites and their current status can be found Table 21-1.

21.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure the establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

21.3.1 Potential or Planned Development

Toxic release and hazardous materials (hazmat) incidents significantly constrain future development in Harris County, especially near industrial corridors, transport routes, and petrochemical sites. Fixed-site risks (e.g., refineries, chemical plants) and mobile threats (e.g., rail and highway transport) heighten the chance of spills that threaten public health and the environment. These hazards can deter residential or commercial projects, increase permitting hurdles, or require costly mitigation like buffer zones and emergency systems. Without proactive, risk-informed planning, development may shift to less hazardous but more vulnerable areas, compounding climate and infrastructure risks.

21.3.2 Projected Changes in Population

Future increases in population will translate into an even denser population base across the planning area. As population numbers swell, so will the risk for Toxic Release or Hazardous Materials (HAZMAT) events to cause impacts. A larger population means more people living, working, and commuting near industrial zones and transportation corridors where hazmat incidents are more likely to occur. This amplifies the potential for mass exposure during an accidental release, complicates emergency response efforts, and increases the strain on public health and safety infrastructure. Additionally, higher population density can limit evacuation options and heighten the consequences of even minor incidents, making proactive risk management and land use planning more critical than ever.

21.3.3 Climate Change

As climate change projections become reality, so does the increased risk of toxic release and hazardous materials incidents by intensifying extreme weather events such as flooding, hurricanes, and extreme heat. These conditions can damage chemical facilities, disrupt containment systems, and increase the likelihood of accidental releases. As climate impacts grow, incorporating future risk into planning and infrastructure design will be critical to minimizing public health and environmental consequences.

22. UTILITY FAILURE

22.1 HAZARD PROFILE

22.1.1 Hazard Description

A utility failure occurs when essential services such as electricity, natural gas, water, wastewater, or telecommunications systems are interrupted or become unavailable, whether temporarily or for an extended period. These failures may result from various causes including severe weather events (e.g., hurricanes, ice storms), equipment or infrastructure failure, cyberattacks, human error, or incidents of other hazards (e.g., flood damage to substations or water mains).

Utility failures can significantly impact public health and safety, disrupt emergency response capabilities, hinder access to clean water and sanitation, disable life-sustaining medical devices, and shut down businesses and critical infrastructure.

In Harris County, utility systems are managed by a combination of public and private entities, including major providers such as CenterPoint Energy (electric and gas), municipal water systems, private water providers, and regional wastewater authorities. Because of the size and population density of the County, utility failures can have widespread and cascading effects.

22.1.2 Location

Utility failures have the potential to affect all jurisdictions within Harris County. However, impacts can vary depending on the type of utility, the geographic distribution of infrastructure, and redundancy within systems. The following general patterns apply:

- Electricity—Power outages can be widespread due to the interconnected grid. Both urban and rural areas can be affected, though urban areas tend to experience faster restoration due to higher population density and prioritization.
- Water and Wastewater—These systems are often more localized. Failures may be limited to specific municipalities or service areas, depending on the provider. Vulnerabilities exist in low-lying areas where infrastructure may be prone to flooding.
- Natural Gas—Gas line failures are generally localized but can cause widespread danger when leaks or explosions occur.
- Telecommunications—Failures can affect broad areas depending on the type and cause of disruption (e.g., fiber optic line damage, tower failures, cyberattacks).

Each participating jurisdiction is susceptible to at least one type of utility failure based on the infrastructure serving their community.

22.1.3 Extent

The severity of a utility failure is generally measured by three factors: the duration of the outage (ranging from a few minutes to several days or weeks), the geographic area affected, and by the number of customers impacted. Examples of potential events in Harris County include the following:

- Widespread blackouts can affect tens or hundreds of thousands of customers across the planning area, such as during Winter Storm Uri in 2021.
- Due to water system failures, boil-water notices can be issued to millions of residents during freeze events or after significant flood events when flood-related damage to water treatment plants knock them offline.
- Disruption to telecommunication systems can impact residents and businesses across the County. Extended cell service or internet outages frequently occur following hurricanes and strong wind events, as downed trees knock down towers and communication lines.

The extent of utility failure events is often exacerbated when multiple systems fail simultaneously—a situation that occurred during the 2021 winter storm when power, water, and natural gas systems were all significantly disrupted.

22.1.4 Previous Occurrences

Significant Recent Events

In recent years, there have been several notable utility failure events in Harris County. These events demonstrate the vulnerability of utility systems in Harris County and underscore the importance of mitigation efforts to enhance infrastructure resilience. Significant hazard events that impacted Harris County between 2017 and 2025 are described in Table 22-1.

FEMA Major Disaster and Emergency Declarations

Harris County has not been included in any major disaster (DR) or emergency (EM) declarations for utility failure-related events (FEMA 2023).

State Emergency Proclamations

The Governor of Texas is able to declare emergency proclamations to access powers authorized by the Emergency Services Act. Harris County has been included in zero utility failure-related state emergency proclamations.

Table 22-1. Utility Failure Events in Harris County (2017 to 2025)

Event Date	Declaration Number	Harris County Included in Declaration?	Location Impacted	Description
August 23 – September 15, 2017	DR-4332	Yes	Countywide	Flooding from Hurricane Harvey caused damage to key infrastructure assets across Harris County, leading to water service interruptions, wastewater treatment system failures, and some electrical outages due to submerged substations.
September 17-23, 2019	DR-4466	Yes	Countywide	Heavy rainfall from Tropical Storm Imelda led to localized power outages and overwhelmed some wastewater treatment facilities, resulting in discharges and system backups.
February 28, 2020	N/A	N/A	East Houston	Localized water main breaks occurred after a 96-inch water main break in East Houston resulted in a boil-water notice for over 2 million people. The event closed schools and disrupted businesses.
February 2021	DR-4586 EM-3554	Yes	Countywide	Winter Storm Uri led to a major power grid failure statewide and left over 4.5 million Texas customers without power at its peak. A large portion of these customers were residents in Harris County. Water systems failed as treatment plants lost power, and millions were under boil-water notices. Natural gas delivery was also disrupted due to frozen infrastructure.
September 2021	N/A	N/A	Countywide	Hurricane Nicholas caused power outages for more than 500,000 customers in Southeast Texas, including significant portions of Harris County
July 5-9, 2024	DR-4798	Yes	Countywide	Hurricane Beryl was the most direct hit on the city in decades, making landfall near Matagorda Bay before going north and knocking down thousands of trees while causing multiple deaths as well as widespread flooding and power outages. The eye of the storm passed over the west side of the Houston metro area, bringing the strongest winds over much of Harris County. Following the storm, power outages lingered for days while temperatures soared. Extreme heat coupled with widespread utility interruptions to produce dangerous post-storm conditions. Multiple cooling centers and shelters opened, allowing residents to cool off in air-conditioning and charge their electrical devices.

Source: (NOAA/NCEI 2025); (Houston PBS 2024) (Houston PBS 2024); (USA Today 2020)

USDA Declarations

The U.S. Secretary of Agriculture is authorized to designate counties as disaster areas to make emergency loans to producers suffering losses in those counties and in contiguous counties. Between 2017 and 2025, Harris County was not included in any USDA utility failure-related agricultural disaster declarations.

22.1.5 Probability of Future Occurrences

Information on previous utility failure occurrences in the County was used to calculate the probability of future occurrences of such events, as summarized in Table 22-2. Based on historical records and input from the Steering Committee, the probability of occurrence for utility failure events in the County is considered "occasional."

Table 22-2. Probability of Future Utility Failure Events in Harris County

Hazard Type	Number of Occurrences Between 2017 and 2025	Percent Chance of Occurring in Any Given Year
Utility Failure	6	67%

Source: (NOAA/NCEI 2025); (Houston PBS 2024) (Houston PBS 2024); (USA Today 2020)
Note: The number of hazard occurrences is calculated using the number of occurrences between 2017 and April 2025

22.1.6 Cascading Impacts on Other Hazards

One of the greatest vulnerabilities associated with this hazard lies in the interdependence between utility systems. For example, a power outage can disable water treatment plants, leading to boil-water notices. Natural gas delivery failures can disable backup power systems that rely on gas-fueled generators. Telecommunications failures can interrupt supervisory control and data acquisition (SCADA) systems used to operate utility infrastructure remotely. Flooding or high winds may simultaneously damage above-ground power lines and below-ground water infrastructure. These cascading impacts multiply the overall hazard and recovery complexity and should be addressed in both mitigation and emergency response planning.

22.2 VULNERABILITY AND IMPACT ASSESSMENT

Utility failures present a significant and increasing risk in Harris County due to aging infrastructure, rapid population growth, climate stressors, and interdependencies between critical systems. This qualitative assessment examines how utility failures affect people, infrastructure, services, and operations across jurisdictions, with particular focus on socially vulnerable populations and critical facilities.

22.2.1 Life, Health, and Safety

For vulnerable populations—such as older adults, medically dependent individuals, and low-income households—utility failures can be life-threatening and exacerbate existing inequities. Several demographic groups in Harris County are more vulnerable to the impacts of utility failures due to limited adaptive capacity, preexisting health conditions, and economic constraints:

- Medically Dependent Populations—Individuals who rely on electrically powered medical devices (e.g., ventilators, dialysis machines, oxygen concentrators) are at severe risk during extended power outages.

- Older Residents—Older adults are more susceptible to extreme temperatures if power failures disable heating or air conditioning. They may also be less able to access emergency resources.
- Low-Income Households—Families with fewer resources may lack backup systems (generators, bottled water, etc.) and face greater challenges recovering from loss of utilities.
- Non-English Speakers—These populations may experience communication barriers during advisories such as boil water notices, evacuation orders, or utility restoration updates.
- Rural and Underserved Areas—These communities may experience longer service restoration times due to lower infrastructure investment and service prioritization.

22.2.2 General Building Stock

All the building stock in the County is exposed to the utility failure hazard. Impacts sustained from utility failure are likely to be secondary impacts. Should water distribution be reduced or not available, then structures could be at increased risk for structural fire since current fire suppression is dependent on accessing water supply from hydrants. Further, households that rely on electricity to power in-home heating and cooling systems will be exposed to significantly colder or hotter indoor temperatures during a utility failure in the winter and summer, respectively. Households that use utility gas for home heating will be less vulnerable.

22.2.3 Community Lifelines and Other Critical Facilities

Utility failures directly impact the functionality and continuity of operations at critical facilities across Harris County. Key vulnerabilities include the following:

- Health Care Facilities—Hospitals and medical centers are reliant on electricity for life-saving equipment and clean water for sanitation. Backup generators may be limited in fuel capacity or fail to start.
- Water and Wastewater Treatment Plants—Power outages at these facilities can cause system-wide service disruptions, water contamination, or sanitary sewer overflows.
- Emergency Operations Centers, Police and Fire Stations—These community lifelines require uninterrupted power and communications to coordinate response efforts. Backup systems may not cover all essential functions.
- Schools and Shelters—Prolonged outages can interrupt school operations and force closures. Shelters may be unable to support evacuees if utilities fail.
- Transportation Infrastructure—Traffic signals and tunnel or bridge systems can fail during outages, increasing accident risks and hindering evacuations.

22.2.4 Economy

Utility failures pose serious risks to local economies and operational continuity for the public and private sectors. The following are examples of potential economic impacts:

- Business Interruptions—Small businesses and manufacturing sectors dependent on real-time operations, refrigeration, or digital systems are especially likely to experience interruptions.
- Loss of Public Services—City and County departments may face disruptions to IT systems, communications, and internal operations during outages.
- Disruption to Fuel Supply Chains—Fuel may become scarce, especially if gas stations cannot operate pumps or refueling stations are offline during grid failure.
- Telecommunication Outages—Any failure to telecommunication facilities would affect not only emergency response coordination but also remote work, education, and public alerting systems.

22.2.5 Natural, Historical, and Cultural Resources

Natural, historical, and cultural resources across Harris County are vulnerable to utility failure hazards in a similar way to county lifelines and critical facilities. This is particularly the case for utility failure events that involve prolonged or widespread electrical, water, wastewater, and communication system disruptions.

Water supply disruptions can affect natural ecosystems, including wetlands, bayous, and parks, that rely on utility-provided water systems. Other impacts on natural systems include habitat degradation. For example, any utility failure that leads to water contamination or disruption can harm habitats. These situations can also lead to a higher wildfire risk.

During hot, dry conditions, power outages that affect water pumps and communication systems can impede wildfire suppression efforts. Many historical buildings depend on HVAC systems to maintain proper humidity and temperature. They also help create an environment that is conducive for tourism/economic traffic, so utility failure events can mean these systems are knocked offline. Cultural resources, like libraries, cultural centers, and performance venues, rely on utilities for operations. Utility failure events can force closures, which disrupt community access and engagement (USACE 2020).

22.3 FUTURE CHANGES THAT MAY AFFECT RISK

Understanding future changes that affect vulnerability can assist in planning for future development and ensure the establishment of appropriate mitigation, planning, and preparedness measures. The following sections examine potential conditions that may affect hazard vulnerability.

22.3.1 Potential or Planned Development

Future development and plans for new utility assets will remain vulnerable unless plans include consideration of site-specific vulnerabilities.

22.3.2 Projected Changes in Population

Changes in the density of the population can impact the number of persons exposed to utility failure, as an increase in population would increase the demand or need for utilities.

22.3.3 Climate Change

Changing climate patterns are increasing the risk of utility failures in Harris County. More frequent and severe extreme weather (e.g., heat waves, freezes, and hurricanes) is placing greater stress on electric grids and water systems. Rising temperatures increase energy demand, heightening the risk of summer brownouts or blackouts. Flooding risks continue to threaten substations, water treatment facilities, and communication infrastructure. These conditions require forward-looking mitigation strategies to harden utilities and integrate climate resilience into infrastructure design.

22.3.4 Other Identified Conditions

Vulnerability to utility failure varies depending on factors such as infrastructure age and condition, backup power availability, operator capacity and funding, density of critical facilities, and localized hazard exposure (e.g., flood-prone substations or pipelines). Urban areas like Pasadena, Friendswood, and unincorporated Harris County have higher concentrations of infrastructure and population, increasing potential impacts. Rural or less-resourced jurisdictions may face longer recovery times due to fewer redundancies and emergency resources.

23. HAZARD RANKING

Hazard rankings have been used as one of the bases for identifying the jurisdictional hazard mitigation strategies included in Volume II. These rankings may vary among the jurisdictions. For example, a hazard may be ranked low in one municipality but, due to differences in vulnerability and impact, be ranked as high for the County or another municipality. Jurisdictional ranking results are presented in each jurisdictional annex in in Volume II.

23.1 HAZARD RANKING METHODOLOGY

Each jurisdiction participating in this HMP has differing levels of vulnerability to and potential impacts from each of the hazards assessed in this plan. Each jurisdiction needs to recognize the hazards that pose the greatest risk to its community and direct its attention and resources accordingly to manage risk and reduce losses. To achieve this, the hazards of concern were ranked using methodologies promoted by FEMA's hazard mitigation planning guidance and input from all participating jurisdictions. Relative ranking scores were generated by FEMA's Hazus risk assessment tool.

The total ranking score based on the categories described above is calculated using the following equation:

Risk Ranking Score Equation

Ranking Score= [(Consequence on Population x 3) + (Consequence on Property x 2) + (Consequence on Economy x 1) x 0.3] + [Adaptive Capacity x 0.3] + [Climate Change x 0.1] + [Probability of Occurrence x 0.3]

Using this equation, the highest possible ranking score is 6.9. The higher the number, the greater the relative risk. Based on the score for each hazard, a hazard ranking is assigned to each hazard of concern as follows:

- Low = Values less than 3.8
- Medium = Values between 3.9 and 4.9
- High = Values greater than 4.9.

All Planning Partners applied the same methodology to develop the hazard rankings to ensure consistency in the overall ranking of risk. However, each jurisdiction had the ability to alter rankings based on local knowledge and experience in handling each hazard.

23.2 HAZARD RANKING RESULTS

Using the methodology described above, the hazard ranking for the identified hazards of concern was determined for each planning partner. The hazard ranking for Harris County is detailed in the following tables that present the step-wise process for the ranking:

- Table 23-1 shows the unweighted numeric values assigned for the probability of occurrence for each hazard.
- Table 23-2 shows the numeric values assigned for each subcategory of consequence for each hazard. Results are shown for applying the subcategory factors, but not the category-wide weighting.
- Table 23-3 shows the unweighted numeric values assigned for adaptive capacity and climate change for each hazard.
- Table 23-4 shows the total weighted hazard ranking scores for each hazard of concern.

The countywide hazard ranking includes the entire planning area and may not reflect the highest risk for all Planning Partners. The overall ranking for each jurisdiction is included in Table 23-5 and in the annexes in Volume II.

Table 23-1. Probability of Occurrence for Hazards of Concern for Harris County

Hazard of Concern	Probability	Numeric Value
Extreme Heat	Frequent	3
Severe Weather	Frequent	3
Coastal Erosion	Occasional	2
Drought	Occasional	2
Extreme Cold	Occasional	2
Flooding	Occasional	2
Hurricane, Coastal Storm	Occasional	2
Wildfire	Occasional	2
Energy Pipeline Failure	Occasional	2
Public Health Emergency	Occasional	2
Technology Failure	Occasional	2
Toxic Release/Hazardous Materials	Occasional	2
Utility Failure	Occasional	2
Dam/Levee Failure	Rare	1
Mass Movement	Rare	1
Tsunami	Rare	1
Earthquake	Unlikely	0

Table 23-2. Consequence Rating for Hazards of Concern for Harris County

Hazard of Concern	Population			Property			Economy			Total Impact Rating (Population + Property + Economy)
	Consequence	Numeric Value	Multiplied by Factor (3)	Consequence	Numeric Value	Multiplied by Factor (2)	Consequence	Numeric Value	Multiplied by Factor (1)	
Coastal Erosion	Low	1	3	Low	2	4	Low	1	1	8
Dam/Levee Failure	Low	1	3	Low	2	4	Low	1	1	8
Drought	Low	1	3	Low	2	4	Medium	1	2	9
Earthquake	Low	1	3	Low	2	4	Low	1	1	8
Extreme Cold	High	3	9	Low	2	4	Low	1	1	14
Extreme Heat	High	3	9	Low	2	4	Low	1	1	14
Flooding	High	3	9	Low	2	4	Medium	2	1	14
Hurricane, Coastal Storm	High	3	9	High	2	4	Medium	3	1	14
Mass Movement	Low	2	6	Low	2	4	Low	1	1	11
Severe Weather	High	3	9	Medium	3	6	Medium	2	2	17
Tsunami	Low	1	3	Low	2	4	Low	1	1	8
Wildfire	Low	1	3	Low	2	4	Low	1	1	8
Energy Pipeline Failure	Low	1	3	Low	2	4	Low	1	1	8
Public Health Emergency	Medium	2	6	Low	2	4	Low	1	1	11
Technology Failure	Low	2	6	Low	2	4	Low	1	1	11
Toxic Release/ HazMat	Medium	2	6	Low	2	4	Low	1	1	11
Utility Failure	High	2	6	Low	2	4	Low	1	1	11

Table 23-3. Adaptive Capacity and Climate Change Ratings for Hazards of Concern for Harris County

Hazard of Concern	Adaptive Capacity		Climate Change	
	Level	Numeric Value	Level	Numeric Value
Coastal Erosion	Moderate	0	High	3
Dam/Levee Failure	Moderate	-1	Low	1
Drought	Moderate	0	High	3
Earthquake	Weak	0	High	3
Extreme Cold	Moderate	0	Low	1
Extreme Heat	Strong	1	High	3
Flooding	Strong	0	Moderate	2
Hurricane, Coastal Storm	Strong	-1	High	3
Mass Movement	Weak	-1	Moderate	2
Severe Weather	Strong	-1	High	3
Tsunami	Moderate	1	High	3
Wildfire	Moderate	0	Low	1
Energy Pipeline Failure	Moderate	0	Low	1
Public Health Emergency	Moderate	1	Low	1
Technology Failure	Moderate	0	Moderate	2
Toxic Release/ HazMat	Moderate	0	Moderate	2
Utility Failure	Moderate	0	Moderate	2

Table 23-4. Total Hazard Ranking Scores for the Hazards of Concern for Harris County

Hazard of Concern	Probability x 30%	Total Consequence x 30%	Adaptive Capacity x 30%	Changing Future Conditions x 10%	Total Hazard Ranking Score
Coastal Erosion	2	2.4	0	0.3	4.7
Dam/Levee Failure	1	2.4	-0.9	0.1	2.6
Drought	2	2,7	0	0.3	2.3
Earthquake	0	2.4	0	0.3	2.7
Extreme Cold	2	4.2	0	0.1	6.3
Extreme Heat	3	4.2	0.09	0.3	7.59
Flooding	2	4.2	0	0.2	6.4
Hurricane, Coastal Storm	2	4.2	-0.9	0.3	5.6
Mass Movement	1	3.3	-0.9	0.2	3.6
Severe Weather	3	5.1	-0.9	0.3	7.5
Tsunami	1	2.4	-0.9	0.3	2.8
Wildfire	2	2.4	0	0.1	4.5
Energy Pipeline Failure	2	2.4	0	0.1	4.5
Public Health Emergency	2	3.3	0.9	0.1	6.3
Technology Failure	2	3.3	0	0.2	5.5
Toxic Release/ HazMat	2	3.3	0	0.2	5.5
Utility Failure	2	3.3	0	0.2	5.5

Note: Low (yellow) = Values less than 3.9; Medium (orange) = Values between 3.9 and 4.9; High (red) = Values greater than 4.9

Table 23-5. Overall Ranking of Hazards by Jurisdiction

	Coastal Erosion	Dam/Levee Failure	Drought	Earthquake	Extreme Cold	Extreme Heat	Flooding	Hurricane, Coastal Storm	Mass Movement	Severe Weather	Tsunami	Wildfire	Energy Pipeline Failure	Public Health Emergency	Technology Failure	Toxic Release/ HazMat	Utility Failure
Harris County	Medium	Low	Low	Low	High	High	High	High	Low	High	Low	Medium	Medium	High	High	High	High
HCFCD[1]	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
Harris County OCA[2]	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of Bellaire	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of Bunker Hill Village	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of Deer Park	Low	Low	Low	Low	High	High	Medium	High	Low	High	Low	High	Low	Medium	High	Medium	High
City of El Lago	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	High	High	High	High
City of Friendswood	Low	Low	Medium	Low	Medium	Medium	High	High	Low	High	Low	Medium	Medium	Medium	Medium	Medium	Medium
City of Galena Park	Low	Low	Low	Low	High	High	Medium	High	Low	High	Low	Low	Medium	Medium	High	Medium	High
City of Hedwig Village	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	High	High	High	High
City of Hilshire Village	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Medium	Medium	High	High	High	High
City of Humble	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	Medium	High	Medium	High
City of Hunters Creek Village	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	High	High	Medium	High
City of Jacinto City	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	High	High	High	High	High
City of Jersey Village	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	Medium	High	Medium	High
City of Katy	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of La Porte	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	Medium	High	Medium	High
Cite of Missouri City	Low	Medium	Medium	Low	Medium	Medium	Medium	Medium	High	High	Low	Low	Medium	Low	Medium	Medium	Medium
City of Morgan's Point	Low	Low	Low	Low	High	High	Medium	High	Low	High	Low	Low	Low	Medium	Medium	Medium	Medium
City of Nassau Bay	Low	Low	Low	Low	High	High	High	High	Medium	High	High	Low	Medium	Medium	High	Medium	High
City of Pasadena	Low	Low	Medium	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of Piney Point Village	Low	Low	Low	Low	High	High	Medium	High	Low	High	Low	Low	Low	Medium	Medium	Medium	Medium
City of Seabrook	Medium	Low	Low	Low	Medium	Medium	High	High	Low	High	Low	Low	Medium	Medium	Medium	Medium	Medium

	Coastal Erosion	Dam/Levee Failure	Drought	Earthquake	Extreme Cold	Extreme Heat	Flooding	Hurricane, Coastal Storm	Mass Movement	Severe Weather	Tsunami	Wildfire	Energy Pipeline Failure	Public Health Emergency	Technology Failure	Toxic Release/ HazMat	Utility Failure
City of Shoreacres	Medium	Low	Medium	Low	Medium	High	High	High	Low	High	Low	Low	Medium	High	High	High	High
City of Southside Place	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of Spring Valley Village	Low	Low	Low	Low	High	High	Medium	High	Medium	High	Low	Low	Medium	Medium	High	Medium	High
City of Stafford	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Low	Medium	Medium	High	Medium	High
City of Taylor Lake Village	Medium	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	High	High	High	High
City of Tomball	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of Waller	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of Webster	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
City of West University Place	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	Medium	Medium	Medium	Medium
Cypress Fairbanks ISD	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Low	Medium	High	High	Medium	High
Goose Creek CISD	Medium	Low	Low	Low	Medium	High	High	High	Low	High	Low	Low	Medium	Medium	Medium	Medium	Medium
Spring ISD	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Medium	Medium	High	High	Medium	High
Varnett Public Schools	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
Houston Community College	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
Lee College	Low	Low	Low	Low	High	High	High	High	Low	High	Low	Medium	Medium	High	High	Medium	Medium
San Jacinto College	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
University of Houston	Medium	Low	Low	Low	High	High	High	High	Low	High	Low	Medium	Medium	High	High	High	High
Cypress Creek Drainage Improvement District	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
Harris Health	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High
Gulf Coast Authority	Low	Low	Low	Low	High	High	High	High	Medium	High	Low	Medium	Medium	Medium	High	Medium	High

Note: Low (yellow) = Values less than 3.9; Medium (orange) = Values between 3.9 and 4.9; High (red) = Values greater than 4.9

Part 3: Capability Assessment

24. CAPABILITY ASSESSMENT

A capability assessment is an inventory of a community's missions, programs, and policies and an analysis of its capacity to carry them out (FEMA 2003). This assessment identifies potential gaps in the jurisdiction's capabilities. This assessment is an integral part of the planning process. It enables identification, review, and analysis of current local and state programs, policies, regulations, funding, and practices that could either facilitate or hinder mitigation. Through assessing its capabilities, a jurisdiction learns how or whether it can implement certain mitigation actions by determining the following:

- Limitations that may exist on undertaking actions
- The range of local and/or state administrative, programmatic, regulatory, financial, and technical resources available to assist in implementing their mitigation actions
- Actions that are infeasible because they are outside the scope of current capabilities
- Types of mitigation actions that may be technically, legally, administratively, politically, or fiscally challenging or infeasible
- Opportunities to enhance local capabilities to support long-term mitigation and risk reduction

This chapter presents a summary and description of the existing plans, programs, and regulatory mechanisms at all levels of government (federal, state, county, local) that reduce hazard risks and support hazard mitigation within the planning area. These capabilities are presented in three categories:

- Planning and regulatory capabilities
- Administrative and technical capabilities
- Fiscal capabilities

Each Planning Partner's annex in Volume II also includes a capability assessment specific to those jurisdictions. While many communities currently operate within limited staffing or funding capacity, they remain committed to strengthening local mitigation and floodplain management programs. Through participation in this regional planning process and coordination with state and federal partners, jurisdictions have identified opportunities to improve enforcement of existing ordinances, update floodplain regulations, and pursue new programs or funding sources that enhance resilience. Each jurisdiction will continue to evaluate its administrative, technical, and financial capabilities and seek opportunities such as grant funding, interdepartmental coordination, and ordinance update to expand and improve their capacity to achieve long-term mitigation goals. In addition to the above categories, the annexes review capabilities in the more localized categories of adaptive capacity and education and outreach.

24.1 CAPABILITY ASSESSMENT PROCESS

Jurisdiction specific capabilities are assessed in each jurisdictional annex in Volume II. All participating jurisdictions were tasked with developing or updating their capability assessment for

this update, evaluating the effectiveness of their capabilities in supporting hazard mitigation and identifying opportunities to enhance local capabilities. Each jurisdiction identified how it has integrated hazard mitigation into its existing planning, regulatory, and operational/administrative framework and how it intends to promote ongoing integration.

The contracted consultant met with Harris County to review and update the information from the 2020 Plan. Virtual meetings were held with jurisdictions on an as-needed basis, and an in-person workshop was conducted at Houston TranStar to facilitate broader engagement and discussion. The consultant also reviewed plans, codes, and ordinances to enhance the information provided by the jurisdictions.

24.2 PLANNING AND REGULATORY CAPABILITIES

Planning and regulatory capabilities are based on ordinances, policies, local laws, state statutes, plans, and programs that relate to managing growth and development. Planning and regulatory capabilities refer not only to current plans and regulations, but also to the jurisdiction's ability to change and improve those plans and regulations as needed. This section summarizes planning and regulatory capabilities for Harris County. Further information is provided in the jurisdictional annexes in Volume II.

24.2.1 Federal

Americans with Disabilities Act

The Americans with Disabilities Act (ADA) seeks to prevent discrimination against people with disabilities in employment, transportation, public accommodation, communications, and government activities. Title II of the ADA deals with compliance with the Act in emergency management and disaster-related programs, services, and activities. It applies to state and local governments as well as third parties, including religious entities and private nonprofit organizations.

The ADA has implications for sheltering requirements and public notifications. During an emergency alert, officials must use a combination of warning methods to ensure that all residents have all necessary information. Those with hearing impairments may not hear radio, television, sirens, or other audible alerts, while those with visual impairments may not see flashing lights or other visual alerts. Two technical documents for shelter operators address physical accessibility needs of people with disabilities, as well as medical needs and service animals.

The ADA intersects with disaster preparedness programs in regard to transportation, social services, temporary housing, and rebuilding. Persons with disabilities may require additional assistance in evacuation and transit (e.g., vehicles with wheelchair lifts or paratransit buses). Evacuation and other response plans should address the unique needs of residents. Local governments may be interested in implementing a special-needs registry to identify the home addresses, contact information, and needs for residents who may require more assistance.

FEMA hazard mitigation project grant applications require full compliance with applicable federal acts. Any action identified in this plan that falls within the scope of this act will need to meet its requirements.

Bureau of Land Management

The U.S. Bureau of Land Management (BLM) funds and coordinates wildfire management programs and structural fire management and prevention on BLM lands. BLM works closely with the Forest Service and state and local governments to coordinate fire safety activities. The Interagency Fire Coordination Center in Boise, Idaho serves as the center for this effort.

Civil Rights Act of 1964

The Civil Rights Act of 1964 prohibits discrimination based on race, color, religion, sex or nation origin and requires equal access to public places and employment. The Act is relevant to emergency management and hazard mitigation in that it prohibits local governments from favoring the needs of one population group over another. Local government and emergency response must ensure the continued safety and well-being of all residents equally, to the extent possible. FEMA hazard mitigation project grant applications require full compliance with applicable federal acts. Any action identified in this plan that falls within the scope of this act will need to meet its requirements.

Clean Water Act

The federal Clean Water Act (CWA) employs regulatory and non-regulatory tools to reduce direct pollutant discharges into waterways, finance municipal wastewater treatment facilities, and manage polluted runoff. These tools are employed to achieve the broader goal of restoring and maintaining the chemical, physical, and biological integrity of the nation′s surface waters so that they can support "the protection and propagation of fish, shellfish, and wildlife and recreation in and on the water."

Evolution of CWA programs over the last decade has included a shift from a program-by-program, source-by-source, and pollutant-by-pollutant approach to more holistic watershed-based strategies. Under the watershed approach, equal emphasis is placed on protecting healthy waters and restoring impaired ones. Numerous issues are addressed, not just those subject to CWA regulatory authority. Involvement of stakeholder groups in the development and implementation of strategies for achieving and maintaining water quality and other environmental goals is a hallmark of this approach.

The CWA is important to hazard mitigation in several ways. There are often permitting requirements for any construction within 200 feet of water of the United States, which may have implications for mitigation projects identified by a local jurisdiction. Additionally, CWA requirements apply to wetlands, which serve important functions related to preserving and protecting the natural and beneficial functions of floodplains and are linked with a community′s floodplain management program. Finally, the National Pollutant Discharge Elimination System is part of the CWA and

addresses local stormwater management programs. Stormwater management plays a critical role in hazard mitigation by addressing urban drainage or localized flooding issues within jurisdictions.

FEMA hazard mitigation project grant applications require full compliance with applicable federal acts. Any action identified in this plan that falls within the scope of this act will need to meet its requirements.

Community Development Block Grant Disaster Resilience Program

In response to disasters, Congress may appropriate additional funding for the U.S. Department of Housing and Urban Development Community Development Block Grant programs to be distributed as Disaster Recovery grants (CDBG-DR). These grants can be used to rebuild affected areas and provide seed money to start the recovery process. CDBG-DR assistance may fund a broad range of recovery activities, helping communities and neighborhoods that otherwise might not recover due to limited resources. CDBG-DR grants often supplement disaster programs of FEMA, the Small Business Administration, and the U.S. Army Corps of Engineers. Housing and Urban Development generally awards noncompetitive, nonrecurring CDBG-DR grants by a formula that considers disaster recovery needs unmet by other federal disaster assistance programs. To be eligible for CDBG-DR funds, projects must meet the following criteria:

- Address a disaster-related impact (direct or indirect) in a presidentially declared county for the covered disaster
- Be a CDBG-eligible activity (according to regulations and waivers)
- Meet a national objective.

Incorporating preparedness and mitigation into these actions is encouraged, as the goal is to rebuild in ways that are safer and stronger. CDBG-DR funding is a potential alternative source of funding for actions identified in this plan.

Community Rating System

The CRS is a voluntary program within the NFIP that encourages floodplain management activities that exceed the minimum NFIP requirements. Flood insurance premiums are discounted to reflect the reduced flood risk resulting from community actions meeting the following three goals of the CRS:

- Reduce flood losses.
- Facilitate accurate insurance rating.
- Promote awareness of flood insurance.

For participating communities, flood insurance premium rates are discounted in increments of 5 percent. For example, a Class 1 community would receive a 45 percent premium discount, and a Class 9 community would receive a 5 percent discount. (Class 10 communities are those that do not participate in the CRS; they receive no discount.) The discount partially depends on location of the property. Properties outside the SFHA receive smaller discounts: a 10-percent discount if the

community is at Class 1 to 6 and a 5-percent discount if the community is at Class 7 to 9. The CRS classes for local communities are based on 18 creditable activities in the following categories:

- Public information
- Mapping and regulations
- Flood damage reduction
- Flood preparedness.

CRS activities can help to save lives and reduce property damage. Communities participating in the CRS represent a significant portion of the nation's flood risk; over 66 percent of the NFIP's policy base is located in these communities. Communities receiving premium discounts through the CRS range from small to large and represent a broad mixture of flood risks, including both coastal and riverine flood risks.

Disaster Mitigation Act

The DMA is the current federal legislation addressing hazard mitigation planning. It emphasizes planning for disasters before they occur. It specifically addresses planning at the local level, requiring plans to be in place before Hazard Mitigation Assistance grant funds are available to communities. This plan is designed to meet the requirements of DMA, improving eligibility for future hazard mitigation funds.

Emergency Relief for Federally Owned Roads Program

The U.S. Forest Service's Emergency Relief for Federally Owned Roads Program was established to assist federal agencies with repair or reconstruction of tribal transportation facilities, federal lands transportation facilities, and other federally owned roads that are open to public travel and have suffered serious damage by a natural disaster over a wide area or by a catastrophic failure. The program funds both emergency and permanent repairs (Office of Federal Lands Highway, 2016). Eligible activities under this program meet some of the goals and objectives for this plan and the program is a possible funding source for actions identified in this plan.

Emergency Watershed Program

The USDA Natural Resources Conservation Service (NRCS) administers the Emergency Watershed Protection (EWP) Program, which responds to emergencies created by natural disasters. Eligibility for assistance is not dependent on a national emergency declaration. The program is designed to help people and conserve natural resources by relieving imminent hazards to life and property caused by floods, fires, windstorms, and other natural occurrences. EWP is an emergency recovery program. Financial and technical assistance are available for the following activities (Natural Resources Conservation Service, 2016):

- Remove debris from stream channels, road culverts, and bridges
- Reshape and protect eroded banks
- Correct damaged drainage facilities
- Establish cover on critically eroding lands

- Repair levees and structures
- Repair conservation practices.

This federal program could be a possible funding source for actions identified in this plan.

Endangered Species Act

The federal Endangered Species Act (ESA) was enacted in 1973 to conserve species facing depletion or extinction and the ecosystems that support them. The act sets forth a process for determining which species are threatened and endangered and requires the conservation of the critical habitat in which those species live. The ESA provides broad protection for species of fish, wildlife and plants that are listed as threatened or endangered. Provisions are made for listing species, as well as for recovery plans and the designation of critical habitat for listed species. The ESA outlines procedures for federal agencies to follow when taking actions that may jeopardize listed species and contains exceptions and exemptions. It is the enabling legislation for the Convention on International Trade in Endangered Species of Wild Fauna and Flora. Criminal and civil penalties are provided for violations of the ESA and the Convention.

Federal agencies must seek to conserve endangered and threatened species and use their authorities in furtherance of the ESA's purposes. The ESA defines three fundamental terms:

- Endangered means that a species of fish, animal or plant is "in danger of extinction throughout all or a significant portion of its range." (For salmon and other vertebrate species, this may include subspecies and distinct population segments.)
- Threatened means that a species "is likely to become endangered within the foreseeable future." Regulations may be less restrictive for threatened species than for endangered species.
- Critical habitat means "specific geographical areas that are…essential for the conservation and management of a listed species, whether occupied by the species or not."

Five sections of the ESA are of critical importance to understanding it:

- Section 4: Listing of a Species—The National Oceanic and Atmospheric Administration Fisheries Service (NOAA Fisheries) is responsible for listing marine species; the U.S. Fish and Wildlife Service is responsible for listing terrestrial and freshwater aquatic species. The agencies may initiate reviews for listings, or citizens may petition for them. A listing must be made "solely on the basis of the best scientific and commercial data available." After a listing has been proposed, agencies receive comment and conduct further scientific reviews for 12 to 18 months, after which they must decide if the listing is warranted. Economic impacts cannot be considered in this decision, but it may include an evaluation of the adequacy of local and state protections. Critical habitat for the species may be designated at the time of listing.
- Section 7: Consultation—Federal agencies must ensure that any action they authorize, fund, or carry out is not likely to jeopardize the continued existence of a listed or proposed species or adversely modify its critical habitat. This includes private and public actions that require a federal permit. Once a final listing is made, non-federal actions are subject to the

same review, termed a "consultation." If the listing agency finds that an action will "take" a species, it must propose mitigations or "reasonable and prudent" alternatives to the action; if the proponent rejects these, the action cannot proceed.

- Section 9: Prohibition of Take—It is unlawful to "take" an endangered species, including killing or injuring it or modifying its habitat in a way that interferes with essential behavioral patterns, including breeding, feeding or sheltering.
- Section 10: Permitted Take—Through voluntary agreements with the federal government that provide protections to an endangered species, a non-federal applicant may commit a take that would otherwise be prohibited as long as it is incidental to an otherwise lawful activity (such as developing land or building a road). These agreements often take the form of a "Habitat Conservation Plan."
- Section 11: Citizen Lawsuits—Civil actions initiated by any citizen can require the listing agency to enforce the ESA's prohibition of taking or to meet the requirements of the consultation process.

FEMA hazard mitigation project grant applications require full compliance with applicable federal acts. Any action identified in this plan that falls within the scope of this act will need to meet its requirements.

Federal Energy Regulatory Commission Dam Safety Program

The Federal Energy Regulatory Commission (FERC) cooperates with a large number of federal and state agencies to ensure and promote dam safety. More than 3,000 dams are part of regulated hydroelectric projects in the FERC program. Two-thirds of these are more than 50 years old. As dams age, concern about their safety and integrity grows, so oversight and regular inspection are important. FERC inspects hydroelectric projects on an unscheduled basis to investigate the following:

- Potential dam safety problems
- Complaints about constructing and operating a project
- Safety concerns related to natural disasters
- Issues concerning compliance with the terms and conditions of a license.

Every five years, an independent engineer approved by the FERC must inspect and evaluate projects with dams higher than 32.8 feet (10 meters), or with a total storage capacity of more than 2,000 acre-feet.

FERC monitors seismic research and applies it in performing structural analyses of hydroelectric projects. FERC also evaluates the effects of potential and actual large floods on the safety of dams. During and following floods, FERC visits dams and licensed projects, determines the extent of damage, if any, and directs any necessary studies or remedial measures the licensee must undertake. The FERC publication Engineering Guidelines for the Evaluation of Hydropower Projects guides the FERC engineering staff and licensees in evaluating dam safety. The publication is frequently revised to reflect current information and methodologies.

FERC requires licensees to prepare emergency action plans and conducts training sessions on how to develop and test these plans. The plans outline an early warning system if there is an actual or potential sudden release of water from a dam due to failure. The plans include operational procedures that may be used, such as reducing reservoir levels and reducing downstream flows, as well as procedures for notifying affected residents and agencies responsible for emergency management. These plans are frequently updated and tested to ensure that everyone knows what to do in emergency situations.

National Environmental Policy Act

The National Environmental Policy Act requires federal agencies to consider the environmental impacts of proposed actions and reasonable alternatives to those actions, alongside technical and economic considerations. The National Environmental Policy Act established the Council on Environmental Quality, whose regulations (40 CFR Parts 1500-1508) set standards for compliance. Consideration and decision-making regarding environmental impacts must be documented in an environmental impact statement or environmental assessment. Environmental impact assessment requires the evaluation of reasonable alternatives to a proposed action, solicitation of input from organizations and individuals that could be affected, and an unbiased presentation of direct, indirect, and cumulative environmental impacts. FEMA hazard mitigation project grant applications require full compliance with applicable federal acts. Any action identified in this plan that falls within the scope of this act will need to meet its requirements.

Federal Wildfire Management Policy and Healthy Forests Restoration Act

The Federal Wildfire Management Policy and the Healthy Forests Restoration Act of 2003 collectively call for a unified and comprehensive federal fire policy to guide the fire management efforts of the Departments of the Interior and Agriculture. They mandate community-based collaboration to reduce risks from wildfire.

National Dam Safety Act

Potential for catastrophic flooding due to dam failures led to passage of the National Dam Inspection Act in 1972, creation of the National Dam Safety Program in 1996, and reauthorization of the program through the Dam Safety Act in 2006. National Dam Safety Program, administered by FEMA requires a periodic engineering analysis of the majority of dams in the country; exceptions include the following:

- Dams under jurisdiction of the Bureau of Reclamation, Tennessee Valley Authority, or International Boundary and Water Commission
- Dams constructed pursuant to licenses issued under the Federal Power Act
- Dams that the Secretary of the Army determines do not pose any threat to human life or property.

The goal of this FEMA-monitored effort is to identify and mitigate the risk of dam failure so as to protect lives and property of the public. The National Dam Safety Program is a partnership among the states, federal agencies, and other stakeholders that encourages individual and community responsibility for dam safety. Under FEMA's leadership, state assistance funds have allowed all

participating states to improve their programs through increased inspections, emergency action planning, and purchases of needed equipment. FEMA has also expanded existing and initiated new training programs. Grant assistance from FEMA provides support for improvement of dam safety programs that regulate most of the dams in the United States.

National Fire Plan (2001)

The 2001 National Fire Plan was developed based on the National Fire Policy. A major aspect of the National Fire Plan is joint risk reduction planning and implementation carried out by federal, state and local agencies and communities. The National Fire Plan presented a comprehensive strategy in five key initiatives:

- Firefighting—Be adequately prepared to fight fires each fire season.
- Rehabilitation and Restoration—Restore landscapes and rebuild communities damaged by wildfires.
- Hazardous Fuel Reduction—Invest in projects to reduce fire risk.
- Community Assistance—Work directly with communities to ensure adequate protection.
- Accountability—Be accountable and establish adequate oversight, coordination, program development, and monitoring for performance.

National Flood Insurance Program

The NFIP makes federally backed flood insurance available to homeowners, renters, and business owners in participating communities. For most participating communities, FEMA has prepared a detailed Flood Insurance Study. The study presents water surface elevations for floods of various magnitudes, including the 1-percent-annual-chance flood and the 0.2-percent-annual-chance flood. Base flood elevations and the boundaries of the flood hazard areas are shown on Flood Insurance Rate Maps, which are the principle tool for identifying the extent and location of the flood hazard. Flood Insurance Rate Maps are the most detailed and consistent data source available, and for many communities they represent the minimum area of oversight under the local floodplain management program. In recent years, Flood Insurance Rate Maps have been digitized as Digital Flood Insurance Rate Maps, which are more accessible to residents, local governments and stakeholders.

Participants in the NFIP must, at a minimum, regulate development in floodplain areas in accordance with NFIP criteria. Before issuing a permit to build in a floodplain, participating jurisdictions must ensure that three criteria are met:

- New buildings and those undergoing substantial improvements must, at a minimum, be elevated to protect against damage by the 1-percent-annual-chance flood.
- New floodplain development must not aggravate existing flood problems or increase damage to other properties.
- New floodplain development must exercise a reasonable and prudent effort to reduce its adverse impacts on threatened salmonid species.

National Incident Management System

The National Incident Management System (NIMS) is a systematic approach for government, nongovernmental organizations, and the private sector to work together to manage incidents involving hazards. The NIMS provides a flexible but standardized set of incident management practices. Incidents typically begin and end locally, and they are managed at the lowest possible geographical, organizational, and jurisdictional level. In some cases, success depends on the involvement of multiple jurisdictions, levels of government, functional agencies, and emergency responder disciplines. These cases necessitate coordination across a spectrum of organizations. Communities using NIMS follow a comprehensive national approach that improves the effectiveness of emergency management and response personnel across the full spectrum of potential hazards (including natural hazards, technological hazards, and human-caused hazards) regardless of size or complexity.

Although participation is voluntary, federal departments and agencies are required to make adoption of NIMS by local and state jurisdictions a condition to receive federal preparedness grants and awards. The content of this plan is considered to be a viable support tool for any phase of emergency management. The NIMS program is considered as a response function, and information in this hazard mitigation plan can support the implementation and update of all NIMS-compliant plans within the planning area.

Presidential Executive Order 11988, Floodplain Management

Executive Order 11988 requires federal agencies to avoid to the extent possible the long and short-term adverse impacts associated with the occupancy and modification of floodplains and to avoid direct and indirect support of floodplain development wherever there is a practicable alternative. It requires federal agencies to provide leadership and take action to reduce the risk of flood loss, minimize the impact of floods on human safety, health, and welfare, and restore and preserve the natural and beneficial values of floodplains. The requirements apply to the following activities (FEMA, 2015a):

- Acquiring, managing, and disposing of federal lands and facilities
- Providing federally undertaken, financed, or assisted construction and improvements
- Conducting federal activities and programs affecting land use, including but not limited to water and related land resources planning, regulation, and licensing.

Presidential Executive Order 11990, Protection of Wetlands

Executive Order 11990 requires federal agencies to provide leadership and take action to minimize the destruction, loss or degradation of wetlands, and to preserve and enhance the natural and beneficial values of wetlands. The requirements apply to the following activities (National Archives, 2016):

- Acquiring, managing, and disposing of federal lands and facilities
- Providing federally undertaken, financed, or assisted construction and improvements
- Conducting federal activities and programs affecting land use, including but not limited to water and related land resources planning, regulation, and licensing.

All actions identified in this plan will seek full compliance with all applicable presidential executive orders.

U.S. Army Corps of Engineers Dam Safety Program

The U.S. Army Corps of Engineers operates and maintains approximately 700 dams nationwide. It is also responsible for safety inspections of some federal and non-federal dams in the United States that meet the size and storage limitations specified in the National Dam Safety Act. The Corps has inventoried dams; surveyed each state and federal agency's capabilities, practices and regulations regarding design, construction, operation and maintenance of the dams; and developed guidelines for inspection and evaluation of dam safety. The Corps maintains the National Inventory of Dams, which contains information about a dam's location, size, purpose, type, last inspection and regulatory status (U.S. Army Corps of Engineers, 2017).

U.S. Army Corps of Engineers Flood Hazard Management

The U.S. Army Corps of Engineers has several civil works authorities and programs related to flood risk and flood hazard management:

- The Floodplain Management Services program offers 100-percent federally funded technical services such as development and interpretation of site-specific data related to the extent, duration and frequency of flooding. Special studies may be conducted to help a community understand and respond to flood risk. These may include flood hazard evaluation, flood warning and preparedness, or flood modeling.
- For more extensive studies, the Corps of Engineers offers a cost-shared program called Planning Assistance to States and Tribes. Studies under this program generally range from $25,000 to $100,000 with the local jurisdiction providing 50 percent of the cost.
- The Corps of Engineers has several cost-shared programs (typically 65 percent federal and 35 percent non-federal) aimed at developing, evaluating and implementing structural and non-structural capital projects to address flood risks at specific locations or within a specific watershed:
- The Continuing Authorities Program for smaller-scale projects includes Section 205 for Flood Control, with a $7 million federal limit and Section 14 for Emergency Streambank Protection with a $1.5 million federal limit. These can be implemented without specific authorization from Congress.
- Larger scale studies, referred to as General Investigations, and projects for flood risk management, for ecosystem restoration or to address other water resource issues, can be pursued through a specific authorization from Congress and are cost-shared, typically at 65 percent federal and 35 percent non-federal.
- Watershed management planning studies can be specifically authorized and are cost-shared at 50 percent federal and 50 percent non-federal.
- The Corps of Engineers provides emergency response assistance during and following natural disasters. Public Law 84-99 enables the Corps to assist state and local authorities in

flood fight activities and cost share in the repair of flood protective structures. Assistance is provided in the flowing categories:

- Preparedness—The Flood Control and Coastal Emergency Act establishes an emergency fund for preparedness for emergency response to natural disasters; for flood fighting and rescue operations; for rehabilitation of flood control and hurricane protection structures. Funding for Corps of Engineers emergency response under this authority is provided by Congress through the annual Energy and Water Development Appropriation Act. Disaster preparedness activities include coordination, planning, training and conduct of response exercises with local, state and federal agencies.
- Response Activities—Public Law 84-99 allows the Corps of Engineers to supplement state and local entities in flood fighting urban and other non-agricultural areas under certain conditions (Engineering Regulation 500-1-1 provides specific details). All flood fight efforts require a project cooperation agreement signed by the public sponsor and the sponsor must remove all flood fight material after the flood has receded. Public Law 84-99 also authorizes emergency water support and drought assistance in certain situations and allows for "advance measures" assistance to prevent or reduce flood damage conditions of imminent threat of unusual flooding.
- Rehabilitation—Under Public Law 84-99, an eligible flood protection system can be rehabilitated if damaged by a flood event. The flood system would be restored to its pre-disaster status at no cost to the federal system owner, and at 20-percent cost to the eligible non-federal system owner. All systems considered eligible for Public Law 84-99 rehabilitation assistance have to be in the Rehabilitation and Inspection Program prior to the flood event. Acceptable operation and maintenance by the public levee sponsor are verified by levee inspections conducted by the Corps on a regular basis. The Corps has the responsibility to coordinate levee repair issues with interested federal, state, and local agencies following natural disaster events where flood control works are damaged.

All of these authorities and programs are available to the planning partners to support any intersecting mitigation actions.

U.S. Fire Administration

There are federal agencies that provide technical support to fire agencies/organizations. For example, the U.S. Fire Administration, which is a part of FEMA, provides leadership, advocacy, coordination, and support for fire agencies and organizations.

U.S. Fish and Wildlife Service

The U.S. Fish and Wildlife Service fire management strategy employs prescribed fire to maintain early successional fire-adapted grasslands and other ecological communities throughout the National Wildlife Refuge System.

24.2.2 State

Texas Government Code Chapter 418; Texas Administrative Code Title 37, Part 1, Chapter 7

Under Texas Government Code Chapter 418 (Emergency Management), mayors and county judges serve as emergency management directors. In most jurisdictions, these officials appoint an emergency management coordinator to administer the program. The mayor and county judge are authorized by the Texas Disaster Act to declare a local disaster when conditions exist or when there is an immediate threat.

Texas government Code Chapter 418.014 grants the governor the authority to declare a state-level disaster. At the local level, an emergency management director serves as the governor's designated agent in the administration and supervision of duties under Chapter 418. An emergency management director may exercise the powers granted to the governor under this chapter on an appropriate local scale (Texas Government Code Chapter 418.015) and may declare a local state of disaster (Texas Government Code Chapter 418.108).

Jurisdictions must prepare emergency operations plans that follow TDEM's planning standards (Texas Government Code Chapter 418.102).

Texas Planning Law

The Texas Planning Law enables local governments to adopt comprehensive plans. A municipality may define the content and design of its comprehensive plan, but the state suggests including the following elements:

- Land use
- Transportation
- Public facilities.

The municipality may then, should it choose to do so, enact development regulations that are consistent with the plan. Consistency is defined by the local jurisdiction.

The law also authorizes the creation of a conservation easement that imposes limitations on land use for the purpose of retaining or protecting natural, scenic, agricultural, historical, or open space values of real property. If the conservation easement is limited in duration, as soon as it ends, an additional tax is imposed on the land equal to the tax break received for the five prior years, plus 7 percent annual interest.

Texas Property Code

Section 5.008 of the Texas Property Code establishes requirements for sellers to disclose whether a property is in the floodplain, has suffered flood damage, and has flood insurance coverage.

Texas Stormwater Management Program

The state of Texas and the Texas Commission on Environmental Quality were delegated authority by the U.S. EPA to administer the National Pollutant Discharge Elimination System permitting program for wastewater and stormwater discharges associated with construction and industrial activity, as well as Municipal Separate Storm Sewer System activity. Because the EPA and the Clean Water Act require a program for addressing the pollution caused by stormwater discharges, the Texas Commission on Environmental Quality implemented and managed the Texas Pollutant Discharge Elimination System permitting program to fulfill all Clean Water Act and federal mandates. The system provides a process for project owners and operators to attain permit coverage allowing the discharge of stormwater and wastewater into Texas surface waters

House Bill 108; Texas Water Code

Section 16.3145 of the Texas Water Code requires non-participating flood-prone communities to adopt a flood damage prevention ordinance and enroll in the National Flood Insurance Program.

HOME Program

The Texas Department of Housing and Community Affairs administers the HOME Program to expand the supply of decent, safe, affordable housing and strengthen public-private housing partnerships between units of local governments, public housing authorities, nonprofits, and for-profit entities. The Department of Housing and Community Affairs has set aside funding for disaster relief and persons with disabilities, among others.

Texas Water Development Board

The Texas Water Development Board manages state programs for flood hazard mapping under FEMA's CTP program. The Board also monitors NFIP program compliance on behalf of the State.

Texas Flash Flood Coalition

The Texas Flash Flood Coalition works to decrease the number of deaths caused by flash flooding in Texas. More than 30 representatives of higher education, media, private industry, local, state and federal governments participate in the coalition.

Texas Silver Jackets

This U.S. Army Corps of Engineers initiative provides a formal and consistent strategy for an interagency approach to planning and implementing measures in order to reduce the risks associated with flooding and other natural hazards.

24.2.3 County and Local Capability

All participating jurisdictions performed an assessment of existing authorities and capabilities. Table 24-1 summarizes the planning and regulatory capabilities available to Harris County at the local level. Other planning partners' assessments are included in the annexes in Volume II.

All core jurisdictional capabilities are considered to be adaptable to meet a jurisdiction's needs—every code can be amended, and every plan can be updated. If the capability assessment identified an opportunity to add a missing core capability or expand an existing one, then doing so has been selected as an action in the jurisdiction's action plan, which is included in the individual annexes presented in Volume II of this plan. Such adaptability is itself a capability.

Table 24-1. Harris County Planning and Regulatory Capabilities

Capability	Details	
Building Code, Certificates of Occupancy, and Zoning Ordinances – Verification Letter	Description:	Harris County does not issue certificates of occupancy, and the County has not adopted building codes for single-family residential developments.
	Responsible Agency:	N/A
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Harris County Regulations of Subdivision- Rules, Regulations, and Requirements relating to the approval and acceptance of Improvements in Subdivisions or Re-Subdivisions	Description:	Regulations of Harris County, Texas for the Approval and Acceptance of Infrastructure
	Responsible Agency:	Harris County Office of County Engineer
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Stormwater Management Plan	Description:	Policy, Criteria and Procedures Manual 2018—Harris County Flood Control District Stormwater Management Program
	Responsible Agency:	Harris County Flood Control District
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Stormwater Maintenance	Description:	Stormwater Maintenance: Regular clearing of drainage systems, ditches, and stormwater infrastructure to prevent flooding.
	Responsible Agency:	Harris County Office of Homeland Security & Emergency Management Harris County Flood Control District
	Provides Funding for Mitigation:	No
	Hazard:	Flood/Hurricane
Post-Disaster Recovery Plan	Description:	Annex J of the HC Basic Plan, 2025 2024 HCFCD Flood Operation & Hurricane Response Plan

Capability	Details	
	Responsible Agency:	Harris County Office of Homeland Security & Emergency Management Harris County Flood Control District
	Provides Funding for Mitigation:	Yes
	Hazard:	Flood/Hurricane
Environmental Protection Ordinance(s)	Description:	Regulations of Harris County, Texas for Stormwater Quality Management
	Responsible Agency:	N/A
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Flood Damage Prevention Ordinance	Description:	Regulations of Harris County, Texas for Floodplain Management
	Responsible Agency:	N/A
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Emergency Management Ordinance	Description:	Texas Gov Code, Title 4, Chapter 418, Emergency Management, 1975
	Responsible Agency:	Harris County Office of Homeland Security & Emergency Management
	Provides Funding for Mitigation:	No
	Hazard:	Drought, Flood, Wildfire, Earthquakes, Toxic Release/Hazmat, Severe Weather, Hurricane, Winter Storms, Mass Movement, Coastal Erosion
Other Code, Ordinance, or Regulation	Description:	2023 Rules Of Harris County, Including The Harris County Toll Road Authority, A Division Of Harris County, And The Harris County Flood Control District For The Construction Of Facilities Within Harris County And The Harris County Flood Control District Rights-Of-Way
	Responsible Agency:	Harris County Office of County Engineer, Harris County Toll Road Authority, Harris County Flood Control District.
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Comprehensive/Master Plan	Description:	Harris County Basic Plan, 2024
	Responsible Agency:	Harris County Office of Homeland Security & Emergency Management
	Provides Funding for Mitigation:	No
	Hazard:	All Hazards
Capital Improvement Plan	Description:	Interim Capital Improvements Program 2018 Bond Program, Community Development Block Grant Program
	Responsible Agency:	Harris County Office of Management & Budget Harris County Flood Control District

Capability	Details	
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Disaster Debris Management Plan	Description:	Annex K of the HC Basic Plan, 2024 2025 – HCFCD Disaster Debris Management Plan
	Responsible Agency:	Harris County Flood Control District Harris County Office of Homeland Security & Emergency Management
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Floodplain Management or Watershed Plan	Description:	Policy, Criteria and Procedures Manual Various studies within each of the District's 23 Watersheds. Identifies specific areas of flood risk and recommends appropriate mitigation strategies
	Responsible Agency:	Harris County Flood Control District
	Provides Funding for Mitigation:	No
	Hazard:	Flood
Stormwater Pollution Prevention Plan	Description:	2024 – HCFCD Stormwater Pollution Minimization Plans for Service Centers, Various Individual Project SWPPP for Construction Activities
	Responsible Agency:	Harris County Flood Control District
	Provides Funding for Mitigation:	N/A
	Hazard:	Flood
Community Wildfire Protection Plan	Description:	Harris County Community Wildfire Protection
	Responsible Agency:	Harris County Fire Marshal's Office
	Provides Funding for Mitigation:	No
	Hazard:	Wildfire
Transportation Plan	Description:	Harris County Transportation Plan
	Responsible Agency:	N/A
	Provides Funding for Mitigation:	No
	Hazard:	N/A
Climate Action/ Resiliency/Sustainability Plan	Description:	Harris County Climate Action Plan, 2023
	Responsible Agency:	Harris County Office of County Administration
	Provides Funding for Mitigation:	N/A
	Hazard:	All hazards
Emergency Operations Plan	Description:	Harris County Basic Plan, 2024
	Responsible Agency:	Harris County Office of Homeland Security & Emergency Management
	Provides Funding for Mitigation:	N/A
	Hazard:	All Hazards
Continuity of Operations Plan	Description:	Departmental Continuity of Operations Plans
	Responsible Agency:	Multiple County Departments

Capability	Details	
	Provides Funding for Mitigation:	N/A
	Hazard:	All Hazards
Substantial Damage Response Plan	Description:	Regulations of HC, TX for Floodplain Management
	Responsible Agency:	HC Office of County Engineer
	Provides Funding for Mitigation:	N/A
	Hazard:	Flood
Threat & Hazard Identification & Risk Assessment (THIRA)	Description:	Regional THIRA conducted annually for Urban Areas Security Initiative and H-GAC Southeast Texas Health and Medical Risk Assessment, December 2024
	Responsible Agency:	HCOHSEM SETRAC, DSHS Region 6/5 South, local health departments
	Provides Funding for Mitigation:	N/A
	Hazard:	N/A
Public Health Plan	Description:	Annex H of the Harris County Basic Plan, 2023
	Responsible Agency:	Harris County Public Health
	Provides Funding for Mitigation:	N/A
	Hazard:	Public Health Emergency

24.3 ADMINISTRATIVE AND TECHNICAL CAPABILITIES

Table 24-2 summarizes administrative and technical capabilities in Harris County. Further information is provided in the jurisdictional annexes in Volume II.

Table 24-2. Administrative and Technical Capability

Capability	Details
Planning Department	Harris County Engineering Department handles development planning functions. HCFCD Planning Division includes the following departments/functions: Property Acquisition, Planning, Risk Mitigation, Environmental Services, and Feasibility Study.
Harris County Flood Control District	Provide flood damage reduction projects that work, with appropriate regard for community and natural values. The Flood Control District accomplishes its mission by devising flood damage reduction plans, implementing the plans, and maintaining the infrastructure.
Harris County Office of Homeland Security & Emergency Management	HCOHSEM helps prepare, safeguard, and protect the residents and property of Harris County from the effects of disasters through effective planning, preparation, response, and recovery activities.
Harris County Fire Marshal's Office	Harris County Fire Marshal's Office is responsible for fire prevention, education, control and investigation in the unincorporated areas of Harris County. The Department also assists local municipalities when requested by their government representatives.
Harris County Office of County Engineer	Providing accessible services to all residents and partners while delivering innovative infrastructure solutions.
Harris County Office of County Administration	To provide community-centered services and infrastructure effectively, efficiently, and equitably, creating opportunities for all to be safe, healthy, and economically secure.

Capability	Details
Mitigation Planning Committee	Establish a Local Mitigation Planning Committee as part of this update which included representation from multiple County departments.
Public Works/Highway Department	N/A
Construction/Building/Code Enforcement Department	Limited to floodplain management and infrastructure regulations The Development Coordination and Inspection Department monitors and inspects construction activities within HCFCD Rights of Way.
Emergency Management/Public Safety Department	Harris County Office of Homeland Security & Emergency Management
Maintenance programs to reduce risk (stormwater maintenance, tree trimming, etc.)	The HCFCD Infrastructure and Operations staff are primarily focused on the safe and efficient maintenance of the District's flood risk reduction infrastructure. HCFCD employs preventative, repair & replace, service request, and forward looking maintenance functions to maintain its 2,500 miles of open water channels, over 280 storm water detention basins, 2 mitigation banks across the 23 watersheds in Harris County. Specific programs and departments that perform these functions include but are not limited to Citizen Service Center, Federal Concrete Lined Channel Program, Property Management, Facilities Maintenance Team, Infrastructure Repairs and Maintenance Engineering. Stormwater Maintenance: Regular clearing of drainage systems, ditches, and stormwater infrastructure to prevent flooding. Tree Trimming: Proactive vegetation management to reduce damage from storms and high winds. Infrastructure Maintenance: Inspection and upkeep of roads, bridges, and public assets to enhance resilience.
Mutual aid agreements	Numerous County depts.
Human Resources Manual	2024 – Harris County and Harris County Flood Control District Personnel Policies & Procedures Manual, 2022 – Harris County Flood Control District Personnel Guidelines. Numerous jobs at HCFCD relate directly to the identification, evaluation and implementation of flood risk reduction projects. These jobs include but are not limited to the Chief Resiliency Officer, Projects Managers, Risk Mitigation Manager, etc. Incorporated into the job description of HCOHSEM planner. Incorporated into job descriptions of Harris County Commissioner Precinct 2.
Housing and Community Development	Housing and Community Development works to strengthen communities by investing in housing, infrastructure, and services that promote resilience for residents of Harris County. They secure and manage grants from HUD and partner with other agencies to achieve these goals

24.4 FISCAL CAPABILITIES

This section summarizes fiscal capabilities in Harris County. Further information is provided in the jurisdictional annexes in Volume II.

Federal Hazard Mitigation Funding Opportunities

As noted on the FEMA hazard mitigation assistance website (FEMA n.d.), FEMA administers five programs that provide funding for eligible mitigation planning and projects that reduces disaster losses and protect life and property from future disaster damages. The programs are the Hazard Mitigation Grant Program (HMGP), the HMGP Post Fire Grant, the Flood Mitigation Assistance (FMA)

Program, and the Pre-Disaster Mitigation (PDM) Program. Table 24-3 provides an overview of program funding eligibility and cost share.

Table 24-3. FEMA HMA Grant Cost Share Requirements

Programs	Cost Share (Percent of Federal / Non-Federal Share)
HMGP	75 / 25
HMGP Post Fire	75 / 25
FMA (community flood mitigation, project scoping, individual mitigation of insured properties, and planning grants)	75 / 25
FMA—repetitive loss property(2)	90 / 10
FMA—severe repetitive loss property(2)	100 / 0
HHPD	65/35
PDM	75 / 25
PDM—small and impoverished community	Up to 90 / 10

Source: FEMA 2023; FEMA 2023

Subapplicants should consult their State Hazard Mitigation Officer for the amount of percentage of HMGP subrecipient management cost funding their State has determined to be passed through subrecipients.

To be eligible for an increased federal cost share, a FEMA-approved state or tribal (standard or enhanced) mitigation plan that addressed repetitive loss properties must be in effect at the time of award, and the property is being submitted for consideration must be a repetitive loss property.

HMGP assists in implementing long-term hazard mitigation planning and projects following a Presidential major disaster declaration. PDM provides funds for hazard mitigation planning and projects on an annual basis. FMA provides funds for planning and projects to reduce or eliminate risk of flood damage to buildings that are insured under the National Flood Insurance Program (NFIP) on an annual basis.

HMGP funding is generally 15 percent of the total amount of Federal assistance provided to a State, Territory, or federally recognized tribe following a major disaster declaration. PDM and FMA funding depends on the amount congress appropriates each year for those programs.

Individual homeowners and business owners may not apply directly to FEMA. Eligible local governments may apply on their behalf (FEMA 2023).

Federal mitigation grant funding is available to all communities with a current hazard mitigation plan (this plan); however, most of these grants require a "local share" in the range of 10-25 percent of the total grant amount. The FEMA mitigation grant programs are described below.

Hazard Mitigation Grant Program

The Hazard Mitigation Grant Program (HMGP) is a post-disaster mitigation program. It is made available to states by FEMA after each Federal disaster declaration. The HMGP can provide up to 75 percent funding for hazard mitigation measures. The HMGP can be used to fund cost-effective projects that will protect public or private property in an area covered by a federal disaster

declaration or that will reduce the likely damage from future disasters. Examples of projects include acquisition and demolition of structures in hazard-prone areas, flood-proofing or elevation to reduce future damage, minor structural improvements, and development of state or local standards. Projects must fit into an overall mitigation strategy for the area identified as part of a local planning effort. All applicants must have a FEMA-approved hazard mitigation plan (this plan).

Applicants who are eligible for the HMGP are state and local governments, certain nonprofit organizations or institutions that perform essential government services, and Indian tribes and authorized tribal organizations. Individuals or homeowners cannot apply directly for the HMGP; a local government must apply on their behalf. Applications are submitted to TDEM and placed in rank order for available funding and submitted to FEMA for final approval. Eligible projects not selected for funding are placed in an inactive status and may be considered as additional HMGP funding becomes available.

Flood Mitigation Assistance Program

The Flood Mitigation Assistance (FMA) program combines the previous Repetitive Flood Claims and Severe Repetitive Loss Grants into one grant program. The FMA provides funding to assist states and communities in implementing measures to reduce or eliminate the long-term risk of flood damage to buildings, manufactured homes, and other structures insurable under the NFIP. The FMA is funded annually; no federal disaster declaration is required. Only NFIP insured homes and businesses are eligible for mitigation in this program. Funding for FMA is very limited and, as with the HMGP, individuals cannot apply directly for the program. Applications must come from local governments or other eligible organizations. The federal cost share for an FMA project is at least 75 percent. At most, 25 percent of the total eligible costs must be provided by a non-federal source. Of this 25 percent, no more than half can be provided as in-kind contributions from third parties. At minimum, a FEMA-approved local flood mitigation plan is required before a project can be approved. The FMA funds are distributed from FEMA to the state.

Rehabilitation of High Hazard Potential Dams Program

The Rehabilitation of High Hazard Potential Dams (HHPD) grant program provides technical, planning, design, and construction assistance for eligible rehabilitation activities that reduce dam risk and increase community preparedness.

The HHPD Grant Program will provide assistance for technical, planning, design, and construction activities toward:

- Repair
- Removal
- Structural/nonstructural rehabilitation of eligible high hazard potential dams

Extraordinary Circumstances

For FMA project subawards, the FEMA Region may apply extraordinary circumstances when justification is provided and with concurrence from FEMA Headquarters (Risk Reduction and Risk Analysis Divisions) prior to granting an exception. If this exception is granted, a local mitigation

plan must be approved by FEMA within 12 months of the award of the project subaward to that community.

For HMGP and FMA, extraordinary circumstances exist when a determination is made by the Applicant and FEMA that the proposed project is consistent with the priorities and strategies identified in the State (Standard or Enhanced) Mitigation Plan and that the jurisdiction meets at least one of the criteria below. If the jurisdiction does not meet at least one of these criteria, the Region must coordinate with FEMA Headquarters (Risk Reduction and Risk Analysis Divisions) for HMGP; however, for FMA the Region must coordinate and seek concurrence prior to granting an exception:

- The jurisdiction meets the small, impoverished community criteria (see Part VIII, B.2).
- The jurisdiction has been determined to have had insufficient capacity due to lack of available funding, staffing, or other necessary expertise to satisfy the mitigation planning requirement prior to the current disaster or application deadline.
- The jurisdiction has been determined to have been at low risk from hazards because of low frequency of occurrence or minimal damage from previous occurrences as a result of sparse development.
- The jurisdiction experienced significant disruption from a declared disaster or another event that impacts its ability to complete the mitigation planning process prior to award or final approval of a project award.
- The jurisdiction does not have a mitigation plan for reasons beyond the control of the State, federally-recognized tribe, or local community, such as Disaster Relief Fund restrictions that delay FEMA from granting a subaward prior to the expiration of the local or Tribal Mitigation Plan.

For HMGP and FMA, the applicant must provide written justification that identifies the specific criteria or circumstance listed above, explains why there is no longer an impediment to satisfying the mitigation planning requirement and identifies the specific actions or circumstances that eliminated the deficiency.

When an HMGP project funding is awarded under extraordinary circumstances, the Recipient shall acknowledge in writing to the Regional Administrator that a plan will be completed within 12 months of the subaward. The Recipient must provide a work plan for completing the local or Tribal Mitigation Plan, including milestones and a timetable, to ensure that the jurisdiction will complete the plan in the required time. This requirement shall be incorporated into the award (both the planning and project subaward agreements if a planning subaward is also awarded).

Federal and State Disaster and Recovery Assistance Programs

Following a disaster, various types of assistance may be made available by local, state, and federal governments. The types and levels of disaster assistance depend on the severity of the damage and the declarations that result from the disaster event. Among the general types of assistance that may be provided should the President of the United States declare the event a major disaster includes the following:

Individual Assistance

Individual Assistance provides help for homeowners, renters, businesses, and some nonprofit entities after disasters occur. This program is largely funded by the U.S. Small Business Administration. For homeowners and renters, those who suffered uninsured or underinsured losses may be eligible for a Home Disaster Loan to repair or replace damaged real estate or personal property. Renters are eligible for loans to cover personal property losses. Individuals may borrow up to $200,000 to repair or replace real estate, $40,000 to cover losses to personal property, and an additional 20 percent for mitigation. For businesses, loans may be made to repair or replace disaster damages to property owned by the business, including real estate, machinery and equipment, inventory, and supplies. Businesses of any size are eligible. Nonprofit organizations such as charities, churches, private universities, etc. are also eligible. An Economic Injury Disaster Loan provides necessary working capital until normal operations resume after a physical disaster. These loans are restricted, by law, to small businesses only.

Public Assistance

Public Assistance (PA) provides cost reimbursement aid to local governments (state, county, local, municipal authorities, and school districts) and certain nonprofit agencies that were involved in disaster response and recovery programs or that suffered loss or damage to facilities or property used to deliver government-like services. This program is largely funded by FEMA with both local and state matching contributions required.

Small Business Administration Loans

The Small Business Administration (SBA) provides low-interest disaster loans to homeowners, renters, business of all sizes, and most private nonprofit organizations. SBA disaster loans can be used to repair or replace the following items damaged or destroyed in a declared disaster: real estate, personal property, machinery and equipment, and inventory and business assets.

Homeowners may apply for up to $200,000 to replace or repair their primary residence. Renters and homeowners may borrow up to $40,000 to replace or repair personal property (such as clothing, furniture, cars, and appliances) damaged or destroyed in a disaster. Physical disaster loans of up to $2 million are available to qualified businesses or most private nonprofit organizations.

Department of Homeland Security Grant Program

The Homeland Security Grant Program plays an important role in the implementation of the National Preparedness System by supporting the building, sustainment, and delivery of core capabilities essential to achieving the National Preparedness Goal of a secure and resilient nation. The program supports efforts to build and sustain core capabilities across the Prevention, Protection, Mitigation, Response, and Recovery mission areas. This includes two priorities: building and sustaining law enforcement terrorism prevention capabilities and maturation and enhancement of state and major urban area fusion centers. The program is composed of three interconnected grant programs including the State Homeland Security Program, Urban Areas Security Initiative, and the Operation Stonegarden. Together, these grant programs fund a range of preparedness

activities, including planning, organization, equipment purchase, training, exercises, and management and administration.

Community Development Block Grants

Community Development Block Grants (CDBG) are federal funds intended to provide low and moderate-income households with viable communities, including decent housing, as suitable living environment, and expanded economic opportunities. Eligible activities include community facilities and improvements, roads and infrastructure, housing rehabilitation and preservation, development activities, public services, economic development, planning, and administration. Public improvements may include flood and drainage improvements. In limited instances, and during the times of "urgent need" (e.g., post-disaster) as defined by the CDBG National Objectives, CDBG funding may be used to acquire a property located in a floodplain that was severely damaged by a recent flood, demolish a structure severely damaged by an earthquake, or repair a public facility severely damaged by a hazard event.

U.S. Economic Development Administration

The U.S. Economic Development Administration (USEDA) is an agency of the U.S. Department of Commerce that supports regional economic development in communities around the country. It provides funding to support comprehensive planning and makes strategic investments that foster employment creation and attract private investment in economically distressed areas of the United States. Through its Public Works Program, USEDA invests in key public infrastructure, such as in traditional public works projects, including water and sewer systems improvements, expansion of port and harbor facilities, brownfields, multitenant manufacturing and other facilities, business and industrial parks, business incubator facilities, redevelopment technology-based facilities, telecommunications, and development facilities. Through its Economic Adjustment Program, USEDA administers its Revolving Loan Fund Program, which supplies small businesses and entrepreneurs with the gap financing needed to start or expand their business, in areas that have experienced or are under threat of serious structural damage to the underlying economic base.

Federal Highway Administration Emergency Relief

Federal Highway Administration Emergency Relief (FHWA-ER) is a grant program that may be used for repair or reconstruction of Federal-aid highways and roads on Federal lands which have suffered serious damage as a result of a disaster.

Federal Transit Administration Emergency Relief

Federal Transit Administration Emergency Relief is a grant program that funds capital projects to protect, repair, reconstruct, or replace equipment and facilities of public transportation systems. Administered by the Federal Transit Authority at the U.S. Department of Transportation and directly allocated to metropolitan transit authorities and port authorities, this transportation-specific fund was created as an alternative to FEMA PA.

State Hazard Mitigation Funding Opportunities

Table 24-4 provides a list of programs, descriptions, and links for those jurisdictions seeking funding sources. This table is not intended to be a comprehensive list but rather a tool to help begin identifying potential sources of funding.

Table 24-4. List of State Funding Sources

Capability	Details	
Texas Water Development Board Flood Funding	Description:	The TWDB offers a variety of cost-effective loan and grant programs that provide for the planning, acquisition, design, and construction of water-related infrastructure and other water quality improvements.
	Responsible Agency:	Texas Water Development Board
	Provides Funding for Mitigation:	Yes
	Hazard:	Flooding
Texas A&M Forest Service Prescribed Burn Grants	Description:	Texas A&M Forest Service offers grants to landowners to complete prescribed fires on private land. Each grant targets landowners in different priority areas across the state.
	Responsible Agency:	Texas A&M Forest Service
	Provides Funding for Mitigation:	Yes
	Hazard:	Wildfire
Flood Control Dam Infrastructure Projects— Supplemental Funding	Description:	Projects to repair and rehabilitate flood control structures across Texas will now be funded due to a $150 million appropriations bill legislators passed this session.
	Responsible Agency:	Texas State Soil and Water Conservation Board
	Provides Funding for Mitigation:	Yes
	Hazard:	Flooding, Dam Failure
Flood Infrastructure Fund	Description:	Program provides financial assistance in the form of loans and grants for flood control, flood mitigation, and drainage projects.
	Responsible Agency:	Texas Water Development Board
	Provides Funding for Mitigation:	Yes
	Hazard:	Flooding
Texas Coastal Management Program Grant	Description:	Funding for projects that address environmental concerns and promote economic development within the Texas coastal zone.
	Responsible Agency:	Texas General Land Office
	Provides Funding for Mitigation:	Yes
	Hazard:	Erosion and Flooding
Texas Farm and Ranch Lands Conservation Program (TFRLCP)	Description:	Funding conserves natural resources by protecting working lands from fragmentation and development. TFRLCP maintains and enhances the ecological and agricultural productivity of these lands through Agricultural Conservation Easements.
	Responsible Agency:	Texas Parks and Wildlife
	Provides Funding for Mitigation:	Yes
	Hazard:	Flooding

24.4.1 County and Local

Harris County and individual jurisdictions are (legally, not necessarily practically) able to fund mitigation projects though existing local budgets, local appropriations (including referendums and bonding), and a variety of federal and state loan and grant programs.

24.4.2 Mitigation Funding Sources Summary

Table 24-5 provides a list of programs, descriptions, and links for those seeking funding to implement hazard mitigation strategies. This table is not a comprehensive list, but a starting point to help identify potential sources of funding for mitigation strategies.

Table 24-5. Mitigation Funding Sources

Program	Description	Lead Agency	Website
Hazard Mitigation Assistance (HMA)	Grants to provide funding for eligible mitigation activities that reduce disaster losses and protect life and property from future disaster damages—includes FMA and HMGP.	FEMA	https://www.fema.gov/hazard-mitigation-assistance
Flood Mitigation Assistance (FMA)	Program grants to States and communities for pre-disaster mitigation planning and projects to help reduce or eliminate the long-term risk of flood damage to structures insurable under the National Flood Insurance Program.	FEMA	https://www.fema.gov/flood-mitigation-assistance-grant-program
Hazard Mitigation Grant Program (HMGP)	Grants to States and communities for planning and projects providing long-term hazard mitigation measures following a major disaster declaration.	FEMA	https://www.fema.gov/hazard-mitigation-grant-program
Public Assistance: Hazard Mitigation Funding Under Section 404 and Section 406	Hazard mitigation discretionary funding available under Section 404 and 406 of the Robert T. Stafford Disaster Relief and Emergency Assistance Act following a Presidentially declared disaster.	FEMA	https://www.fema.gov/press-release/20250602/fema-hazard-mitigation-grants-404-and-406
Assistance to Firefighters Grant Program	The primary goal of the Assistance to Firefighters Grants is to enhance the safety of the public and firefighters with respect to fire-related hazards by providing direct financial assistance to eligible fire departments, nonaffiliated Emergency Medical Services organizations, and State Fire Training Academies. This funding is for critically needed resources to equip and train emergency personnel to recognized standards, enhance operations efficiencies, foster interoperability, and support community resilience.	FEMA	https://www.fema.gov/welcome-assistance-firefighters-grant-program
Disaster Housing Program	Emergency assistance for housing, including minor repair of home to establish livable conditions, mortgage, and rental assistance.	HUD	https://www.hud.gov/program_offices/public_indian_housing/publications/dhap
HUD Disaster Recovery Assistance	Grants to fund gaps in available recovery assistance after disasters (including mitigation).	HUD	https://www.hud.gov/info/disasterresources
Section 108 Loan Guarantee	Enables states and local governments participating in the Community Development Block Grant (CDBG) program to obtain federally guaranteed loans for disaster-distressed areas.	HUD	https://www.hudexchange.info/programs/section-108/
Smart-Growth Implementation Assistance program	The Smart-Growth Implementation Assistance program focuses on complex or cutting-edge issues, such as stormwater management, code revision, transit-oriented development, affordable housing, infill development, corridor planning, green building, and climate change. Applicants can submit proposals under 4 categories: community resilience to disasters, job creation, the role of manufactured homes in sustainable neighborhood design or medical and social service facilities siting.	EPA	https://www.epa.gov/smartgrowth

Program	Description	Lead Agency	Website
Partners for Fish and Wildlife	Financial and technical assistance to private landowners interested in pursuing restoration projects affecting wetlands and riparian habitats.	U.S. Fish and Wildlife Service	https://www.fws.gov/partners/
FHWA Emergency Relief Program	Fund for the repair or reconstruction of Federal-aid highways that have suffered serious damage as a result of (1) natural disasters or (2) catastrophic failures from an external cause.	U.S. Department of Transportation (DOT)	https://www.fhwa.dot.gov/programadmin/erelief.cfm
Better Utilizing Investments to Leverage Development (BUILD) Grant Program	Investing in critical road, rail, transit, and port projects across the nation	U.S. DOT	https://www.transportation.gov/BUILDgrants
Community Facilities Direct Loan & Grant Program	This program provides affordable funding to develop essential community facilities in rural areas. An essential community facility is defined as a facility that provides an essential service to the local community for the orderly development of the community in a primarily rural area, and does not include private, commercial, or business undertakings.	USDA	https://www.rd.usda.gov/programs-services/community-facilities-direct-loan-grant-program
Emergency Loan Program	USDA's Farm Service Agency provides emergency loans to help producers recover from production and physical losses due to drought, flooding, other natural disasters or quarantine.	USDA	https://www.fsa.usda.gov/programs-and-services/farm-loan-programs/emergency-farm-loans/index
Emergency Watershed Protection (EWP) Program	Provide assistance to relieve imminent hazards to life and property caused by floods, fires, drought, windstorms, and other natural occurrences.	NRCS	https://www.nrcs.usda.gov/programs-initiatives/ewp-emergency-watershed-protection
Financial Assistance	Financial assistance to help plan and implement conservation practices that address natural resource concerns or opportunities to help save energy, improve soil, water, plant, air, animal and related resources on agricultural lands and non-industrial private forest land.	NRCS	https://www.nrcs.usda.gov/wps/portal/nrcs/main/national/programs/financial/
Regional Conservation Partnership Program	The Regional Conservation Partnership Program promotes coordination of NRCS conservation activities with partners that offer value-added contributions to expand the collective ability to address on-farm, watershed, and regional natural resource concerns. Through this program, NRCS seeks to co-invest with partners to implement projects that demonstrate innovative solutions to conservation challenges and provide measurable improvements and outcomes tied to the resource concerns they seek to address.	NRCS	https://www.nrcs.usda.gov/wps/portal/nrcs/main/national/programs/financial/rcpp/
Emergency Management Performance Grants Program	Assist local, tribal, territorial, and state governments in enhancing and sustaining all-hazards emergency management capabilities.	U.S. DHS	https://www.fema.gov/emergency-management-performance-grant-program

Program	Description	Lead Agency	Website
Land & Water Conservation Fund	Matching grants to states and local governments for the acquisition and development of public outdoor recreation areas and facilities (as well as funding for shared federal land acquisition and conservation strategies).	National Park Service	https://www.nps.gov/subjects/lwcf/index.htm
Rehabilitation of High Hazard Potential Dams Grant Program	The main objective of the HHPD grant program is to provide technical, planning, design, and construction assistance in the form of grants to non-federal sponsors for rehabilitation of eligible high hazard potential dams.	FEMA	https://www.fema.gov/emergency-managers/risk-management/dam-safety/grants/resources

Part 4: Mitigation Strategy

25. MITIGATION STRATEGY

This chapter presents mitigation strategies for Harris County to reduce potential vulnerability and losses identified as concerns in the risk assessment portion of this plan. The Steering Committee reviewed the risk assessment and capability assessment to identify and develop these mitigation strategies.

Hazard mitigation reduces the potential impacts of, and costs associated with, emergency and disaster-related events. Mitigation actions address a range of impacts, including impacts on the population, property, the economy, and the environment.

Mitigation actions can include activities such as revisions to land-use planning, training and education, and structural and nonstructural safety measures.

25.1 PAST MITIGATION ACCOMPLISHMENTS

The County, through previous and ongoing hazard mitigation activities, has demonstrated that it is proactive in protecting its physical assets and the public against losses from natural hazards. Examples of previous and ongoing actions and projects include the following:

- The County facilitated the development of the original Harris County MJ-HMP. The current planning process represents the regulatory five-year plan update process, which includes the participation of 46 jurisdictions within the County, along with key County and regional stakeholders.
- The majority of municipalities participating in this MJ-HMP update participate in the National Flood Insurance Program (NFIP), which requires the adoption of FEMA floodplain mapping and certain minimum standards for building within the floodplain. Some participating jurisdictions may not yet meet all of the minimum NFIP standards or have fully adopted related local provisions; however, they have expressed their intent to do so in the future as part of ongoing efforts to expand local floodplain management capabilities and enhance program compliance.
- Reports, plans, and studies relating to or including information on natural hazards or natural hazard policies affecting Harris County have been reviewed and incorporated into this plan update as appropriate, as discussed in Chapter 2 and listed in the references.

25.2 REVIEW AND UPDATE OF MITIGATION GOALS AND OBJECTIVES

This section describes the process of updating hazard mitigation goals and objectives for reducing or avoiding long-term vulnerabilities to identified hazards. For the purposes of this plan, goals and objectives are defined as follows:

"The hazard mitigation strategy shall include a description of mitigation goals to reduce or avoid long-term vulnerabilities to the identified hazards."

44 CFR 201.6(c)(3)(i)

- Goals are general guidelines that explain what is to be achieved. They are usually broad, long-term, policy-type statements and represent global visions. Goals help define the benefits that the plan is trying to achieve. The success of the plan, once implemented, should be measured by the degree to which its goals have been met (that is, by the actual benefits in terms of hazard mitigation).
- Objectives are short-term aims that form a strategy or course of action to meet a goal. Unlike goals, objectives are stand-alone measurements of the effectiveness of a mitigation action. The objectives also are used to help establish priorities. Broadly defined mitigation objectives were eliminated from the updated strategy unless accompanied by discrete actions.

The Steering Committee reviewed the 2020 goals and objectives and made revisions for the 2025 update based on the following considerations:

- Hazard events and losses since the 2020 plan
- The updated hazard profiles and risk assessment
- The goals and objectives established in the 2023 Texas State Hazard Mitigation Plan
- The Planning Partnership's interests in integrating this plan with other planning mechanisms, including Harris County and local risk management plans
- Direct input from the Steering Committee, stakeholders, and the public on how the County and jurisdictions need to move forward to best manage their hazard risk
- Discussions and research on existing authorities, policies, programs, resources
- Support for mitigation through the protection of natural systems

As a result of this review process, the goals and objectives for the 2025 update were updated to the following:

- Goal 1—Improve and coordinate data collection efforts to fully maximize mitigation capabilities.
  - Objective 1a—Implement wildfire mitigation and watershed protection strategies through local, state, tribal, federal, and private partnerships
  - Objective 1b—Improve understanding of the locations, potential impacts, and linkages among threats, hazards, vulnerability, and measures needed to protect life safety, health, and property
- Goal 2—Enhance public outreach strategies to improve information sharing regarding hazards, including risk reduction strategies.
  - Objective 2a—Improve and expand systems that provide ongoing preparedness education, warning, and emergency communications to the whole community
- Goal 3—Reduce the impacts of hazards on people and property and protect natural and cultural resources.
  - Objective 3a—Reduce the impacts of hazards on individuals with disabilities and others with access and functional needs

  - Objective 3b—Coordinate state and local efforts to reduce greenhouse gas emissions and implement climate adaptation strategies through hazard mitigation plans and actions
  - Objective 3c—Control access and provide buffers to maximize resource protection where possible
  - Objective 3d—Encourage all state, regional, and local hazard mitigation projects and planning programs to protect the environment and promote the implementation of sustainable mitigation and climate actions
  - Objective 3e—Encourage hazard mitigation measures that promote and enhance natural processes and minimize adverse impacts on the ecosystem
  - Objective 3f—Encourage the building of redundant systems needed to support public health and quality of life
- Goal 4—Work to improve and coordinate existing local plans, policies, codes, and regulations to reduce hazard impacts
  - Objective 4a—Promote enforcement of relevant state regulations and development of local ordinances that significantly reduce life loss and injuries
  - Objective 4b—Encourage all cities, counties, special districts, Councils of Governments and tribal organizations to develop, adopt, and implement local hazard mitigation principles to be integrated with local general plan safety elements, local coastal plans, facilities master plans, and other local plan initiatives
  - Objective 4c—Implement wildfire mitigation and watershed protection strategies through local, state, tribal, federal and private partnerships
  - Objective 4d—Advance community resilience through preparation, adoption, and implementation of state, regional and local multi-hazard mitigation plans and projects
- Goal 5—Implement property protection measures to reduce or eliminate repetitive loss occurrences.
  - Objective 5a—Retrofit, elevate, purchase, or relocate structures in high-hazard areas, especially those known to be repetitively damaged
- Goal 6—Investigate and implement a range of structural projects that will reduce the effects of hazards on public and private property throughout the County
  - Objective 6a—Encourage projects that simultaneously reduce risk while increasing resilience and sustainability
  - Objective 6b—Support the protection of vital and essential records, and strengthening or replacement of buildings, infrastructure, and lifelines to minimize post-disaster disruption and facilitate short-term and long-term recovery
  - Objective 6c—Encourage life and property protection measures for all communities and structures located in the greater Harris County region
  - Objective 6d—Manage development in geologically hazardous areas and floodplains to protect life and property
- Goal 7—Investigate and implement a range of nature-based solutions that utilize, enhance, and protect natural resources and their ability to reduce hazard impacts.

  - Objective 7a—Support the protection of vital and essential records, and strengthening or replacement of buildings, infrastructure, and lifelines to minimize post-disaster disruption and facilitate short-term and long-term recovery
  - Objective 7b—Encourage life and property protection measures for all communities and structures located in the greater Harris County region
  - Objective 7c—Promote disaster resistant development
  - Objective 7d—Manage development in geologically hazardous areas and floodplains to protect life and property
- Goal 8—Advance hazard analysis and mitigation strategies to reflect interdependencies across infrastructure and community lifelines
  - Objective 8a—Encourage private and public sector agencies and organizations to work together in the development of plans, strategies, and projects to mitigate identified hazards
- Goal 9—Prioritize mitigation investments to inform and support effects to seek funding
  - Objective 9a—Identify funding sources and increase awareness of funding sources to obtain funding for mitigation projects from a variety of federal, state, regional, and local entities.

25.3 MITIGATION STRATEGY DEVELOPMENT AND UPDATE

25.3.1 Update of Local Jurisdiction Mitigation Strategies

Review of Previous Actions

FEMA defines *Mitigation Actions* as specific actions that help to achieve the mitigation goals and objectives.

To evaluate progress on local mitigation actions, each planning partner was provided with a Mitigation Action Plan Review Worksheet, pre-populated with the actions identified for their jurisdiction in the prior (2020) plan. The Planning Partners were asked to indicate the status of each action ("No Progress," "In Progress/Not Yet Complete," "Continuous," "Completed," "Discontinued"). They were requested to provide comments to quantify the extent of progress and provide reasons for the level of progress or why actions were discontinued. This information is included in the jurisdictional annexes.

Mitigation actions identified as "Complete" or "Discontinued" have been removed from the Planning Partners' updated mitigation strategies. Actions identified as "No Progress/Unknown" or "In Progress/Not Yet Complete" have been carried forward in their local updated mitigation strategies. Planning partners were asked to provide further details on these projects to help better define the projects, identify benefits and costs, and improve implementation.

Certain continuous or ongoing actions (Ongoing Capabilities) from the previous plan that represent programs that are now fully integrated into the normal operational and administrative framework of the community are identified in the capabilities assessment of each annex and removed from the updated mitigation strategy (marked as "Discontinued").

Identifying New Actions

Through Steering Committee Workshops and a series of email campaigns, all participating jurisdictions were further surveyed to identify completed mitigation actions, in-progress actions, or ongoing capabilities, as well as potential new actions. Communities also were made aware of potential new mitigation actions as such actions became evident during the plan update process (e.g., through the capability assessment, risk assessment, or the public and stakeholder outreach process). Furthermore, a newly-developed Mitigation Catalog for Harris County was distributed. This resource provided a variety of mitigation strategies to participants, all of which pertain to the 2025 Hazards of Concern.

Developing the Overall Strategy

Beginning in February 2025, members of the Steering Committee and contract consultants worked directly with each jurisdiction (by phone, email, or virtual meetings) to update their annex with mitigation strategies that focus on well-defined, implementable projects that meet the definition or characteristics of mitigation. Mitigation actions were selected with careful consideration of benefits (risk reduction, losses avoided), costs, and possible funding sources (including mitigation grant programs).

Addressing Known Vulnerabilities

To help support the selection of an appropriate risk-based mitigation strategy, each annex includes a summary of hazard vulnerabilities. These were identified during the plan update process by planning partner representatives, through review of available plans and reports, or through the hazard profiling and risk assessment process.

An online Survey123 form was created for all participating jurisdictions to support the development of focused problem statements based on the impacts of natural hazards in the County and their communities. The survey aimed at collecting these problem statements so there was a detailed description of a problem area, including its impacts on the jurisdiction, past damage, loss of service, etc. An effort was made to include the street address of the problem location, adjacent streets, water bodies, and well-known structures as well as a brief description of existing conditions (topography, terrain, hydrology) of the site. These problem statements form a bridge between the hazard risk assessment, which quantifies impacts on each community, and the development of actionable mitigation strategies.

Incorporating a Range of Action Types

Concerted efforts were made to ensure that Planning Partners develop updated mitigation strategies that cover the range of mitigation action types described in recent FEMA planning guidance (FEMA "Local Mitigation Planning Handbook" March 2013):

- Local Plans and Regulations—These actions include government authorities, policies or codes that influence the way land and buildings are developed and built.
- Structure and Infrastructure Project—These actions involve modifying existing structures and infrastructure to protect them from a hazard or remove them from a hazard

area. This could apply to public or private structures as well as community lifelines and other critical facilities. This type of action also involves projects to construct structures to reduce the impact of hazards.

- Natural Systems Protection—These are actions that minimize damage and losses to natural systems and preserve or restore their functions.
- Education and Awareness Programs—These are actions to inform and educate citizens, elected officials, and property owners about hazards and potential ways to mitigate them. These actions may also include participation in national programs, such as the National Flood Insurance Program, Community Rating System, StormReady (NOAA), and Firewise Communities.

Efforts were also made to develop mitigation strategies that cover the range of mitigation action types described in recent CRS guidance (FEMA 2018):

- Preventative Measures—Government, administrative or regulatory actions, or processes that influence the way land and buildings are developed and built. Examples include planning and zoning, floodplain local laws, capital improvement programs, open space preservation, and storm water management regulations.
- Property Protection—These actions include public activities to reduce hazard losses or actions that involve (1) modification of existing buildings or structures to protect them from a hazard or (2) removal of the structures from the hazard area. Examples include acquisition, elevation, relocation, structural retrofits, storm shutters, and shatter-resistant glass.
- Public Information—Actions to inform and educate citizens, elected officials, and property owners about hazards and potential ways to mitigate them. Such actions include outreach projects, real estate disclosure, hazard information centers, and educational programs for school-age children and adults.
- Natural Resource Protection—Actions that minimize hazard loss and also preserve or restore the functions of natural systems. These actions include sediment and erosion control, stream corridor restoration, watershed management, forest and vegetation management, and wetland restoration and preservation.
- Structural Flood Control Projects—Actions that involve the construction of structures to reduce the impact of a hazard. Such structures include dams, setback levees, floodwalls, retaining walls, and safe rooms.
- Emergency Services—Actions that protect people and property during and immediately following a disaster or hazard event. Services include warning systems, emergency response services, and the protection of essential facilities

Protecting Critical Facilities

Planning partner mitigation actions that address vulnerable critical facilities have been proposed in consideration of protection against future disasters. However, in the case of projects funded through federal mitigation programs, the level of protection may be influenced by cost-effectiveness as determined through a formal benefit-cost analysis. In the case of "self-funded" projects, local jurisdiction discretion must be recognized. Further, it must be recognized that the

County and jurisdictions have limited authority with regard to mitigation at any level of protection over privately owned critical facilities.

Accounting for Climate Change

As discussed in the hazard profiles in this MJ-HMP, the long-term effects of climate change are anticipated to exacerbate the impacts of weather-related hazards (e.g., flood, severe storm, severe winter storm, and wildfire). Communities are working to evaluate and recognize these long-term implications and to incorporate their mitigation strategies into planning and capital improvement updates.

25.3.2 Update of County Mitigation Strategy

The update of the County-level mitigation strategies included a review of progress on the actions/initiatives identified in the 2020 MJ-HMP using a process similar to that used to review local jurisdiction mitigation strategy progress. The County, through its various department representatives, was provided with a Mitigation Action Plan Review Worksheet identifying all County-level actions and initiatives from the 2020 plan. The County reviewed each action and provided progress. For each action, relevant County representatives were asked to indicate the status of each action (*No Progress/Unknown*, *In Progress/Not Yet Complete*, *Ongoing*, *Completed*, or *Discontinued*), and provide review comments on each.

Projects/initiatives identified as *Complete* or *Discontinued* have been removed from this plan update. Actions the County has identified as *No Progress/Unknown*, *In Progress/Not Yet Complete,* or *Ongoing* have been carried forward in the County's updated mitigation strategy. Actions considered ongoing capabilities were marked as *Discontinued* and included in the plan as ongoing capabilities.

Throughout the course of the plan update process, additional regional and County-level mitigation actions were identified by the following processes:

- Review of the results and findings of the updated risk assessment
- Review of available regional and County plans, reports, and studies
- Direct input from County departments and other regional agencies, including:
  - Harris County Office of Emergency Management and Homeland Security (HCOHSEM)
  - Harris County Judge's Office
  - Harris County Flood Control District (HCFCD)
  - Harris County Engineering Department (HCED)
  - Harris County Housing and Community Development (HCHCD)
  - Harris County Fire Marshal's Office (HCFMO)
  - Harris County Pollution Control (HCPC)
  - Harris County Public Health (HCPH)
  - Harris County Sheriff's Office (HCSO)
  - Harris County Human Resources & Risk Management (HRRM)

  - Harris County Office of County Administration (HCOCA)
  - Harris County Comm. Pct 1, Pct 2, Pct 3, and Pct 4
- Input received through the public and stakeholder outreach process

As discussed within the hazard profiles in this MJ-HMP, the long-term effects of climate change are anticipated to exacerbate the impacts of weather-related hazards including drought, flood, severe storm, and severe winter storm. The County has included mitigation actions, including continuing and long-term planning and emergency management support, to address these long-term implications and potential impacts.

Various County departments and agencies included mitigation actions to address vulnerable critical facilities. These actions were proposed in consideration of protection against 0.2-percent annual chance (500-year) events, or worst-case scenarios. It is recognized, however, that in the case of projects being funded through federal mitigation programs, the level of protection can be influenced by cost-effectiveness, as determined through a formal benefit-cost analysis. In the case of "self-funded" projects, local government authority can affect the ability to implement. Further, the County has limited authority over privately owned critical facility owners regarding mitigation at any level of protection.

25.3.3 Mitigation Best Practices

Catalogs of hazard mitigation best practices were developed that present a broad range of alternatives to be considered for use in the mitigation strategies, in compliance with 44 CFR Section 201.6(c)(3)(ii). The 2025 Harris County Mitigation Action Catalog included seventeen catalogs -- one for each hazard of concern evaluated in this plan. The catalogs present alternatives that are categorized in two ways:

- Base on who would have responsibility for implementation:
  - Individuals—personal scale
  - Businesses—corporate scale
  - Government—government scale
- Based on what the alternatives would do:
  - Manipulate the hazard
  - Reduce vulnerability to the hazard
  - Reduce impacts from the hazard
  - Build local capacity to respond to or be prepared for the hazard

The alternatives presented include actions that will mitigate current risk from hazards and actions that will help reduce risk from changes in the impacts of these hazards resulting from climate change. Hazard mitigation actions recommended in this plan were selected from among the alternatives presented in the catalogs. The catalogs provide a baseline of mitigation alternatives that are backed by a planning process, are consistent with the established goals and objectives, and are within the capabilities of the Planning Partners to implement. Some of these actions may not be feasible based on the selection criteria identified for this plan. The purpose of the catalogs

was to provide a list of what could be considered to reduce risk from natural hazards within the planning area. Actions in the catalog that are not included in the partnership's mitigation strategy were not selected for one or more of the following reasons:

- The action is not feasible
- The action is already being implemented
- There is an apparently more cost-effective alternative
- The action does not have public or political support.

The county's mitigation catalog can be found in Appendix G.

25.3.4 Mitigation Strategy Evaluation and Prioritization

Section 201.c.3.iii of 44 CFR establishes how mitigation strategies are to be prioritized, implemented, and administered by local jurisdictions. For this plan update, each mitigation strategy was prioritized using criteria suitable for evaluating hazard mitigation strategies. This method provided a systematic approach that considered the opportunities and constraints of implementing each mitigation action. The Steering Committee chose a set of 14 evaluation criteria for this process:

- Life Safety—How effective will the action be at protecting lives and preventing injuries? Will the proposed action adversely affect one segment of the population?
- Property Protection—How significant will the action be at eliminating or reducing damage to structures and infrastructure? For example: development in the floodplain or high-risk areas?
- Cost-Effectiveness—Are the costs to implement the action commensurate with the benefits achieved?
- Political—Is there overall public support for the action? Is there the political will to support it? Is the action at odds with development pressures?
- Legal—Does the jurisdiction have the authority to implement the action?
- Fiscal—Can the action be funded under existing program budgets (i.e., is this action currently budgeted for)? Or would it require a new budget authorization or funding from another source such as grants?
- Environmental—What are the potential environmental impacts of the action? Will it comply with environmental regulations? Are there co-benefits of this action?
- Social Vulnerability—Does the action benefit socially vulnerable populations and underserved communities? Additional considerations can include appropriate numerical measures of social vulnerability.
- Administrative—Does the jurisdiction have the personnel and administrative capabilities to implement the action and maintain it or will outside help be necessary? Does the scale and scope of the action align with the jurisdiction's capabilities?
- Hazards of Concern—Does the action address one or more of the jurisdiction's high-ranked hazards?

- Climate Change—Does the action incorporate climate change projections? Is the action designed to withstand/address long-term conditions?
- Timeline—Can the action be completed in less than five years?
- Community Lifelines—Does this action benefit community lifelines?
- Other Local Objectives—Does the action advance other local objectives, such as capital improvements, economic development, environmental quality, or open space preservation? Does it support the policies of other plans and programs?

Participating jurisdictions were asked to use these criteria to prioritize their identified mitigation actions. For each mitigation action, the jurisdictions assigned a numeric score for each of the 14 evaluation criteria:

- 1 = Highly effective or feasible
- 0 = Neutral
- -1 = Ineffective or not feasible

Jurisdictions were asked to provide a brief summary of the rationale behind the numeric rankings assigned. The numerical results were totaled and then used by each jurisdiction to help prioritize the action or strategy as *low*, *medium*, or *high*. Actions that had a numerical value between 0 and 6 were categorized as *low priority*; actions with numerical values between 7 and 10 were categorized as *medium priority*; and actions with numerical values between 11 and 14 were categorized as *high priority*. While this provided a consistent, systematic methodology to support the evaluation and prioritization of mitigation actions, jurisdictions may have additional considerations that could influence their overall prioritization of mitigation actions.

It is noted that jurisdictions may be carrying forward mitigation actions from prior mitigation strategies that were prioritized using a different, but not inherently contrary, approach. Mitigation actions in the prior (2020) Harris County MJ-HMP were "qualitatively evaluated against the mitigation goals and objectives and other evaluation criteria. They were then prioritized into three categories: high, medium, and low." At their discretion, jurisdictions carrying forward prior actions were encouraged to re-evaluate their priority, particularly if conditions that would affect the prioritization criteria had changed.

For the plan update there has been an effort to develop more clearly defined and action-oriented mitigation strategies. These local strategies include actions that are seen by the community as the most effective approaches to advance their local mitigation goals and objectives within their capabilities.

25.3.5 Benefit/Cost Review

Section 201.6.c.3iii of 44 CFR requires the prioritization of the mitigation strategy to emphasize the extent to which benefits are maximized according to a benefit/cost review of the proposed projects. For all actions identified in the local strategies, jurisdictions identified the associated costs and benefits as follows:

- Costs presented include the total project estimation. This can include administrative, construction (engineering, design, and permitting), and maintenance costs.
- Benefits are the savings from losses avoided attributed to project implementation. These can include life safety, structure and infrastructure damages, loss of service or function, and economic and environmental damage and losses.

When possible, jurisdictions were asked to identify the actual or estimated dollar costs and associated benefits. Where estimates of costs and benefits were available, the ratings were defined follows:

Low < = $10,000 Medium = $10,000 to $100,000 High > = $100,000

Often numerical costs and/or benefits were not identified and may be impossible to quantify. In this case, jurisdictions were asked to evaluate project cost-effectiveness using qualitative *high*, *medium*, and *low* ratings based on the following definitions

- Costs
  - High—Existing funding levels are not adequate to cover the costs of the proposed project, and implementation would require an increase in revenue through an alternative source (e.g., bonds, grants, and fee increases).
  - Medium—The project could be implemented with existing funding but would require a re-apportionment of the budget or a budget amendment, or the cost of the project would have to be spread over multiple years.
  - Low—The project could be funded under the existing budget. The project is part of or can be part of an existing, ongoing program.
- Benefits
  - High—Project will have an immediate impact on the reduction of risk exposure to life and property.
  - Medium—Project will have a long-term impact on the reduction of risk exposure to life and property or will provide an immediate reduction in the risk exposure to property.
  - Low—Long-term benefits of the project are difficult to quantify in the short-term.

Using this approach, projects with positive benefit versus cost ratios (such as high over high, high over medium, medium over low, etc.) are considered cost-effective.

For some of the Harris County actions identified, the Planning Partnership may seek financial assistance under FEMA's Hazard Mitigation Assistance (HMA) programs. These programs require detailed benefit/cost analysis as part of the application process. The benefit/cost review applied for the prioritization of actions in this update did not include the level of detail required by FEMA for project grant eligibility under HMA grant programs. These analyses will be performed when funding applications are prepared, using FEMA's Benefit-Cost Analysis model.

The Planning Partnership is committed to implementing mitigation strategies with benefits that exceed costs. For projects not seeking financial assistance from grant programs that require this sort of analysis, the Planning Partnership reserves the right to define benefits according to parameters that meet its needs and the goals and objectives of this plan.

25.4 RECOMMENDED AREA-WIDE ACTIONS

Seven area-wide actions were identified to improve local understanding of hazard risk and vulnerability, enhance mitigation capabilities, or reduce vulnerability of infrastructure. The action worksheets included at the end of this chapter describe the area-wide mitigation actions. These actions are incorporated as appropriate into each jurisdiction's mitigation strategy presented in the annexes in Volume II.

Table 25-1 lists the priority of each of the identified area-wide actions based on the prioritization process described in Section 25.3.4.

Table 25-1. Prioritization of Recommended Area-Wide Mitigation Actions

Project Number and Description	Scores for Evaluation Criteria															High / Medium / Low
	Life Safety	Property Protection	Cost-Effectiveness	Political	Legal	Fiscal	Environmental	Social Vulnerability	Administrative	Hazards of Concern	Climate Change	Timeline	Community Lifelines	Other Local Objectives	Total	
AW_1— Utilize the best available data and science as it becomes available, with relevance to the planning area to map the extent and location of all hazards.	1	1	1	1	1	1	1	1	1	1	1	1	1	0	13	High
AW_2—Continue to maintain and update the hazard mitigation page on HCOHSEM's ReadyHarris website with relevant information on the plan's implementation, such as annual progress reports, public comment opportunities, and plan integration opportunities. Each planning partner can support this initiative by creating a link to the County's hazard mitigation page on their jurisdictional websites	1	1	1	1	1	1	1	1	1	1	1	1	1	0	13	High
AW_3— Utilizing the existing public outreach capability of Harris County and Planning Partners; develop, deploy and disseminate targeted outreach projects promoting risk communication, mitigation and resilience to all the hazards of concern addressed by this plan.	1	1	1	1	1	1	1	1	1	1	1	1	1	0	13	High

Project Number and Description	Scores for Evaluation Criteria															High / Medium / Low
	Life Safety	Property Protection	Cost-Effectiveness	Political	Legal	Fiscal	Environmental	Social Vulnerability	Administrative	Hazards of Concern	Climate Change	Timeline	Community Lifelines	Other Local Objectives	Total	
AW_4—Strive to capture time-sensitive, perishable data—such as high-water marks, extent and location of hazard, and loss information—following hazard events to support future updates to the risk assessment as well as other plans and programs that utilize hazard extent and location data	1	1	1	1	1	1	1	1	1	1	1	1	1	0	13	High
AW_5—Continue to develop, improve and implement an enhanced mass public warning and alert system (iPAWS) within the Regional Joint Information Center to provide warning capability throughout Harris County to support the emergency management of all hazards of concern and disasters impacting critical infrastructure and residents	1	1	1	1	1	1	1	1	1	1	1	1	1	0	13	High
AW_6—Utilize viable and relevant information, data and tools (Hazus models) developed as part of the update to the risk assessment of this plan update to support training and exercise of the County's preparedness, response and recovery programs	1	1	1	1	1	1	1	1	1	1	1	1	1	0	13	High

Project Number and Description	Scores for Evaluation Criteria															High / Medium / Low
	Life Safety	Property Protection	Cost-Effectiveness	Political	Legal	Fiscal	Environmental	Social Vulnerability	Administrative	Hazards of Concern	Climate Change	Timeline	Community Lifelines	Other Local Objectives	Total	
AW_7—Utilize viable and relevant information, data and tools (Hazus models) developed as part of the update to the risk assessment of this plan update to support training and exercise of the County's preparedness, response and recovery programs	1	1	1	1	1	1	1	1	1	1	1	1	1	0	13	High

Area Wide Action #1 (AW_1)

Action Description	Utilize the best available data and science as it becomes available, with relevance to the planning area to map the extent and location of all hazards.
Lead Agency:	Harris County Office of Homeland Security & Emergency Management
Supporting Agencies:	Local, state, and federal agencies that capture and produce data on risk
Hazards of Concern:	Coastal Erosion, Dam/Levee Failure, Drought, Earthquake, Extreme Cold, Extreme Heat, Flooding, Hurricanes/Coastal Storms, Mass Movement, Severe Weather, Tsunami, Wildfire, Energy Pipeline Failures, Public Health Emergency, Technology Failure, Toxic Release/HazMat, Utilities Failure
Description of the Problem:	Harris County faces a variety of natural and man-made hazards that can threaten lives and property. Without accurate, up-to-date information on the location and extent of these hazards, the community may be more vulnerable to risks and have slower response and recovery times.
Description of the Solution:	By accurately mapping hazards and assessing risks across jurisdictions, the plan enables coordinated preparedness, targeted mitigation efforts, and informed decision-making to reduce vulnerabilities and enhance community resilience.
Estimated Cost:	Medium ($10,000 to $100,000)
Potential Funding Sources:	Local funding, grant funding
Implementation Timeline:	Long-term (>5 years)
Goals Met:	1, 2, 3, 4, 8
Benefits:	Medium ($10,000 to $100,000)
Impact on Socially Vulnerable Populations:	N/A
Impact on Future Development:	N/A
Impact on Critical Facilities/Lifelines:	N/A
Impact on Capabilities:	N/A
Climate Change Considerations:	N/A
Mitigation Category	Natural Systems Protection (NSP) Education and Awareness Programs (EAP) Local Plans and Regulations (LPR)
CRS Category	Public Information (PI)
Priority	High

Area Wide Action #2 (AW_2)

Action Description	Continue to maintain and update the hazard mitigation page on HCOHSEM's ReadyHarris website with relevant information on the plan's implementation, such as annual progress reports, public comment opportunities, and plan integration opportunities. Each planning partner can support this initiative by creating a link to the County's hazard mitigation page on their jurisdictional websites
Lead Agency	Harris County Office of Homeland Security & Emergency Management
Supporting Agencies:	All Planning Partners
Hazards of Concern:	Coastal Erosion, Dam/Levee Failure, Drought, Earthquake, Extreme Cold, Extreme Heat, Flooding, Hurricanes/Coastal Storms, Mass Movement, Severe Weather, Tsunami, Wildfire, Energy Pipeline Failures, Public Health Emergency, Technology Failure, Toxic Release/HazMat, Utilities Failure
Description of the Problem:	Without a well-maintained online resource and coordination among planning partners, important updates and opportunities for involvement risk being overlooked, limiting community awareness and collaboration in hazard mitigation efforts.
Description of the Solution:	This provides a centralized, continuously updated online resource that keeps the public and stakeholders informed about the progress and implementation of the hazard mitigation plan. By sharing annual reports and avenues for public involvement, it promotes transparency, engagement, and collaboration.
Estimated Cost:	Low (<$10,000)
Potential Funding Sources:	Local funding
Implementation Timeline:	Ongoing
Goals Met:	1, 2, 4, 8, and 9
Benefits:	Medium ($10,000 to $100,000)
Impact on Socially Vulnerable Populations:	N/A
Impact on Future Development:	N/A
Impact on Critical Facilities/Lifelines:	N/A
Impact on Capabilities:	N/A
Climate Change Considerations:	N/A
Mitigation Category	Local Plans and Regulations (LPR) Education and Awareness Programs (EAP)
CRS Category	Emergency Services (ES) Public Information (PI) Preventative Measures (PR)
Priority	High

Area Wide Action #3 (AW_3)

Action Description	Utilizing the existing public outreach capability of Harris County and Planning Partners; develop, deploy and disseminate targeted outreach projects promoting risk communication, mitigation and resilience to all the hazards of concern addressed by this plan.
Lead Agency	Harris County Office of Homeland Security & Emergency Management
Supporting Agencies:	N/A
Hazard(s) of Concern:	Coastal Erosion, Dam/Levee Failure, Drought, Earthquake, Extreme Cold, Extreme Heat, Flooding, Hurricanes/Coastal Storms, Mass Movement, Severe Weather, Tsunami, Wildfire, Energy Pipeline Failures, Public Health Emergency, Technology Failure, Toxic Release/HazMat, Utilities Failure
Description of the Problem:	N/A
Description of the Solution:	N/A
Estimated Cost:	Low (<$10,000)
Potential Funding Sources:	Local funding
Implementation Timeline:	Ongoing
Goals Met:	2 and 3
Benefits:	Medium ($10,000 to $100,000)
Impact on Socially Vulnerable Populations:	N/A
Impact on Future Development:	N/A
Impact on Critical Facilities/Lifelines:	N/A
Impact on Capabilities:	N/A
Climate Change Considerations:	N/A
Mitigation Category	Education and Awareness Programs (EAP)
CRS Category	Preventative Measures (PR) Public Information (PI)
Priority	High

Area Wide Action #4 (AW_4)

Action Description	Strive to capture time-sensitive, perishable data—such as high-water marks, extent and location of hazard, and loss information—following hazard events to support future updates to the risk assessment as well as other plans and programs that utilize hazard extent and location data
Lead Agency:	Harris County Office of Homeland Security & Emergency Management
Supporting Agencies:	N/A
Hazard(s) of Concern:	Coastal Erosion, Dam/Levee Failure, Drought, Earthquake, Extreme Cold, Extreme Heat, Flooding, Hurricanes/Coastal Storms, Mass Movement, Severe Weather, Tsunami, Wildfire, Energy Pipeline Failures, Public Health Emergency, Technology Failure, Toxic Release/HazMat, Utilities Failure
Description of the Problem:	After hazard events, critical data such as high-water marks, hazard extent, and loss information should be captured. Without this, future risk assessments and related planning efforts may lack accurate, detailed information, reducing their effectiveness in preparing for and mitigating future hazards.
Description of the Solution:	By capturing accurate, real-time information on hazard impacts, it strengthens future risk assessments and supports more precise updates to hazard mitigation plans and other related programs.
Estimated Cost:	Medium ($10,000 to $100,000)
Potential Funding Sources:	Local Funding, Grant Funding
Implementation Timeline:	Ongoing
Goals Met:	1 and 4
Benefits:	Medium ($10,000 to $100,000)
Impact on Socially Vulnerable Populations:	N/A
Impact on Future Development:	N/A
Impact on Critical Facilities/Lifelines:	N/A
Impact on Capabilities:	N/A
Climate Change Considerations:	N/A
Mitigation Category	Education and Awareness Programs (EAP) Local Plans and Regulations (LPR)
CRS Category	Emergency Services (ES) Preventative Measures (PR) Public Information (PI)
Priority	High

Area Wide Action 5 (AW_5)

Action Description	Continue to develop, improve and implement an enhanced mass public warning and alert system (iPAWS) within the Regional Joint Information Center to provide warning capability throughout Harris County to support the emergency management of all hazards of concern and disasters impacting critical infrastructure and residents
Lead Agency:	Harris County Office of Homeland Security & Emergency Management
Supporting Agencies:	All Planning Partners
Hazard(s) of Concern:	Coastal Erosion, Dam/Levee Failure, Drought, Earthquake, Extreme Cold, Extreme Heat, Flooding, Hurricanes/Coastal Storms, Mass Movement, Severe Weather, Tsunami, Wildfire, Energy Pipeline Failures, Public Health Emergency, Technology Failure, Toxic Release/HazMat, Utilities Failure
Description of the Problem:	Without a robust and reliable mass public warning system, timely communication during hazard events can be compromised, putting lives and property at greater risk.
Description of the Solution:	By expanding and strengthening this mass notification capability, the County can quickly disseminate critical warnings and information to residents, improving emergency response and public safety across all hazards of concern.
Estimated Cost:	Low (<$10,000)
Potential Funding Sources:	General Funds, EMPG, UASI
Implementation Timeline:	Ongoing
Goals Met:	2 and 3
Benefits:	High (>$100,000)
Impact on Socially Vulnerable Populations:	N/A
Impact on Future Development:	N/A
Impact on Critical Facilities/Lifelines:	N/A
Impact on Capabilities:	N/A
Climate Change Considerations:	N/A
Mitigation Category	Education and Awareness Programs (EAP) Structure and Infrastructure Projects (SIP)
CRS Category	Emergency Services (ES) Public Information (PI)
Priority	High

Area Wide Action #6 (AW_6)

Action Description	Utilize viable and relevant information, data and tools (Hazus models) developed as part of the update to the risk assessment of this plan update to support training and exercise of the County's preparedness, response and recovery programs
Lead Agency:	Harris County Office of Homeland Security & Emergency Management
Supporting Agencies:	N/A
Hazard(s) of Concern:	Coastal Erosion, Dam/Levee Failure, Drought, Earthquake, Extreme Cold, Extreme Heat, Flooding, Hurricanes/Coastal Storms, Mass Movement, Severe Weather, Tsunami, Wildfire, Energy Pipeline Failures, Public Health Emergency, Technology Failure, Toxic Release/HazMat, Utilities Failure
Description of the Problem:	Without access to up-to-date and relevant hazard risk data, training and exercise programs may lack realism, which could result in coordination gaps during emergencies.
Description of the Solution:	By integrating these resources, the County and Planning Partners can conduct scenario-based training that will improve the skills of emergency personnel.
Estimated Cost:	Low (<$10,000)
Potential Funding Sources:	Local funding / grant funding
Implementation Timeline:	Short-term (<5 years)
Goals Met:	1, 4 – 9
Benefits:	Medium ($10,000 to $100,000)
Impact on Socially Vulnerable Populations:	N/A
Impact on Future Development:	N/A
Impact on Critical Facilities/Lifelines:	N/A
Impact on Capabilities:	N/A
Climate Change Considerations:	N/A
Mitigation Category	Education and Awareness Programs (EAP) Local Plans and Regulations (LPR)
CRS Category	Emergency Services (ES) Preventative Measures (PR) Public Information (PI)
Priority	High

Area Wide Action #7 (AW_7)

Action Description	Foster the creation/updating of GIS datasets such as parcels/ownership, location of all structures, driveways with ingress/egress conditions, roads, ownership types, floodplains, utilities (power lines, gas lines and water lines), wetlands, water features, bridges and culverts.
Lead Agency:	Harris County Office of Homeland Security & Emergency Management
Supporting Agencies:	Local Units of Government, Local Fire Departments, County Commissioners, Harris County Department of Engineering, Harris County Department of Public Health, Utility companies
Hazard(s) of Concern:	Coastal Erosion, Dam/Levee Failure, Drought, Earthquake, Extreme Cold, Extreme Heat, Flooding, Hurricanes/Coastal Storms, Mass Movement, Severe Weather, Tsunami, Wildfire, Energy Pipeline Failures, Public Health Emergency, Technology Failure, Toxic Release/HazMat, Utilities Failure
Description of the Problem:	Existing hazard and infrastructure datasets are fragmented, outdated, or inconsistent across agencies. This limits the ability of Harris County to effectively assess risks, prioritize mitigation projects, and coordinate response during disasters.
Description of the Solution:	Develop and maintain a comprehensive, regularly updated GIS database that consolidates parcels, ownership, structures, utilities, floodplains, and critical infrastructure. Ensure multi-agency access and provide training for consistent use.
Estimated Cost:	Medium ($10,000 to $100,000)
Potential Funding Sources:	Local general funds, FEMA HMA, CDBG-MIT
Implementation Timeline:	Ongoing
Goals Met:	1, 2, 3, 6, 7
Benefits:	This action will improve hazard identification and risk assessments, enhance interagency coordination, and support faster disaster response and recovery.
Impact on Socially Vulnerable Populations:	Improved GIS datasets will help identify at-risk communities more accurately, allowing resources and mitigation projects to be directed where they are most needed. This reduces disproportionate impacts on socially vulnerable populations.
Impact on Future Development:	Comprehensive data will guide safer land use and infrastructure planning, helping to avoid development in high-risk areas and reduce long-term hazard exposure.
Impact on Critical Facilities/Lifelines:	Better mapping of utilities, roads, and critical infrastructure will improve protection of lifelines and ensure faster recovery after disasters.
Impact on Capabilities:	This action expands the County's technical and planning capacity by providing updated data tools that support stronger coordination across agencies.

Climate Change Considerations:	Enhanced GIS datasets will allow the County to incorporate future conditions modeling, including climate-driven flood and storm impacts, into planning.
Mitigation Category	Education and Awareness Programs (EAP) Local Plans and Regulations (LPR)
CRS Category	Emergency Services (ES) Preventative Measures (PR) Public Information (PI)
Priority	High

Part 5: Plan Maintenance

26. PLAN MAINTENANCE AND IMPLEMENTATION

This chapter details the process to ensure that the Hazard Mitigation Plan remains an active and relevant document and that the Planning Partnership maintains its eligibility for applicable funding sources. The plan maintenance process includes a schedule for monitoring and evaluating the plan annually and producing an updated plan every five years. In addition, this chapter describes how public participation will be integrated throughout the plan maintenance and implementation process. It explains how the mitigation strategies outlined in this plan update will be incorporated into existing planning mechanisms and programs, such as comprehensive land use planning processes, capital improvement planning, and building code enforcement and implementation.

26.1 PLAN ADOPTION

A hazard mitigation plan must document that it has been formally adopted by the governing bodies of the jurisdictions requesting federal approval of the plan (44 CFR Section 201.6(c)(5)). For multi-jurisdictional plans, each jurisdiction requesting approval must document that it has been formally adopted. Prior to adoption by the Harris County Commissioners' Court and the signatory planning partners, this plan was reviewed by the Texas Division of Emergency Management and FEMA Region 6 to ensure compliance with mitigation planning requirements. DMA compliance and its attendant benefits were dependent on completion of this adoption process. Copies of each planning partner's adoption resolution can be found in Appendix A of this volume.

26.2 MJ-HMP COORDINATOR

The Harris County Office of Homeland Security and Emergency Management (HCOHSEM) will serve as the lead agency responsible for overseeing the plan's progress. The MJ-HMP Coordinator is assigned to manage the maintenance and update of the plan during its approval period (the five-year period between FEMA's approval of the plan and its expiration), with the following responsibilities:

- Convene the Planning Partnership as needed.
- Be the primary point of contact for questions regarding the plan and its implementation.
- Coordinate the incorporation of additional information into the plan.
- Manage the monitoring, evaluation, and updating responsibilities identified in this section.

Currently, the Harris County MJ-HMP Coordinator is designated as:

Brittany Ray, Office of Emergency Management
Harris County Office of Homeland Security and Emergency Management
6922 Old Katy Rd
Houston, TX 77024
(713) 881-3100
Email: mitigation@oem.hctx.net

26.3 ACTION PLAN IMPLEMENTATION

The jurisdiction-specific action plans in Volume II present a range of action items for reducing loss from hazard events. The planning partners have prioritized actions and can begin to implement the highest-priority actions over the next five years. The effectiveness of the hazard mitigation plan depends on its effective implementation and incorporation of the outlined action items into all partners' existing plans, policies, and programs. Some action items do not need to be implemented through regulation but can be implemented through the creation of new educational programs, continued interagency coordination, or improved public participation.

Harris County will have the lead responsibility for overseeing the plan's implementation and maintenance strategy. Plan implementation will be a shared responsibility among all planning partnership members and agencies identified as lead agencies in the jurisdiction-specific action plans.

26.4 PLAN MAINTENANCE STRATEGY

A hazard mitigation plan must present a plan maintenance process that includes the following (44 CFR Section 201.6(C)(4)):

- A section describing the method and schedule of monitoring, evaluating, and updating the mitigation plan over a 5-year cycle
- A process by which local governments incorporate the requirements of the mitigation plan into other planning mechanisms, such as comprehensive or capital improvement plans, when appropriate
- A discussion on how the community will continue public participation in the plan maintenance process

This chapter describes the formal process by which planning partners will ensure that the MJ-HMP remains relevant as well as how public participation will be integrated throughout the plan maintenance and implementation process. It also explains how the mitigation strategies outlined in this plan will be incorporated into existing planning mechanisms and programs, such as comprehensive land-use planning processes, capital improvement planning, and building code enforcement and implementation.

The plan's format allows sections to be reviewed and updated when new data become available, resulting in a plan that will remain current and relevant. Pursuant to 44CFR 201.6(c)(4)(i), the plan maintenance matrix shown in Table 26-1 provides a synopsis of responsibilities for plan monitoring, integration, evaluation, and update, which are discussed in further detail in the sections below.

Table 26-1. Plan Maintenance Matrix

Task	Approach	Timeline	Lead Responsibility	Support Responsibility
Monitoring	Planning partners to recommend update of mitigation strategies, progress toward implementation of actions, identification of new actions, as part of submission for annual progress report.	Each summer, planning partners will provide status updates on existing projects, add new projects, and remove any if necessary.	Jurisdictional points of contact identified in Volume II	Jurisdictional implementation lead identified in Volume II
Integrating	Create a linkage between the hazard mitigation plan and individual jurisdictions' comprehensive plans or similar plans identified in the core capability assessments.	Ongoing during the performance period of this plan and as opportunities for integration become available, or according to timelines identified in the action plans for each planning partner	Jurisdictional points of contact identified in Volume II annexes	Jurisdictional implementation lead identified in Volume II annexes
Evaluating	Annual progress reports will be evaluated by an oversight steering committee annually.	Finalized progress report completed by June 1 of each year	HCOHSEM	Jurisdictional points of contact identified in Volume II annexes
Updating	Reconvene the planning partners, at a minimum, every 5 years to guide a full update to review and revise the plan.	Every 5 years or upon full update to the comprehensive plan or major disaster	Harris County	Jurisdictional points of contact identified in Volume II annexes
Grant Monitoring	Grant funding opportunities will be monitored via agency notifications, state associations and post-disaster response	Ongoing during the performance period of this plan	Harris County Jurisdictional points of contact identified in Volume II annexes	Jurisdictional points of contact identified in Volume II annexes
Public Involvement	Keep the website maintained and receive comments through it over the course of the plan. Maintain planning partner links to the website. Post county- wide progress report to the website.	Ongoing during the performance period of the plan. Progress reports are to be posted annually. Participating CRS communities are to provide the annual progress report to their governing bodies for their information	HCOHSEM will maintain the overall hazard mitigation website and post the Countywide progress report annually. Each planning partner will provide a link to the website and post their individual progress report at their discretion. Participating CRS communities will follow protocol identified under step 10 of the 10-step CRS planning requirements.	HCOHSEM and Jurisdictional points of contact identified in Volume II annexes

26.4.1 Monitoring

Each participating partner will track the status of the mitigation actions outlined in their respective action plans. Staff or departments with primary responsibility are identified in each jurisdictional annex (see Volume II). At a minimum, each planning partner will monitor the progress of their respective action plan during each 12-month performance period. This review may include the following:

- The internal process for monitoring and evaluating the plan's processes will be evaluated as part of the annual progress reporting process.
- Summary of any hazard events that occurred during the performance period and the impact these events had on the planning area.
- Review of mitigation success stories.
- Review of continuing public involvement.
- Brief discussion about why targeted strategies were not completed.
- Re-evaluation of the action plan to determine if the timeline for identified projects needs to be amended (such as changing a long-term project to a short-term one because of new funding).
- Recommendations for new projects.
- Changes in or potential for new funding options (grant opportunities).
- Impact of any other planning programs or actions that involve hazard mitigation.

The Core Planning Team has created a template that shows the minimum level of detail that will be sought for preparing a progress report. HCOHSEM will oversee progress reporting and will have the discretionary authority on how to capture this information at least annually over the performance period of the plan. This information may be captured by various means available to the planning partnership. This report should be:

- Posted on the Harris County website page dedicated to the hazard mitigation plan
- Presented to planning partner governing bodies to inform them of the progress of actions implemented during the reporting period
- Provided as part of the CRS annual re-certification package for planning partners that participate in the CRS. The CRS requires an annual recertification to be submitted by October 1 of every calendar year for which the community has not received a formal audit. To meet this recertification timeline, the planning team will strive to complete progress reports between June and September each year.

Uses of the progress report beyond those listed above will be at the discretion of each planning partner. Annual progress reporting is not a requirement specified under 44 CFR. However, it may enhance the planning partners' opportunities for funding. While failure to implement this component of the plan maintenance strategy will not jeopardize a planning partner's compliance under the DMA, it may jeopardize its opportunity to partner and leverage funding opportunities with the other partners. Each planning partner was informed of these protocols at the beginning of this planning process (in the "Planning Partner Expectations"

package provided at the start of the process), and each partner acknowledged these expectations with submittal of a letter of intent to participate in this process.

26.4.2 Incorporation into Other Planning Mechanisms

The information on hazard, risk, vulnerability, and mitigation contained in this plan is based on the best science and technology available at the time this plan was prepared. The comprehensive plans of the planning partners are considered to be integral parts of this plan. The planning partners, through adoption of comprehensive plans and zoning ordinances, have planned for the impact of natural hazards. The plan development process provided them with the opportunity to review and expand on policies contained within these planning mechanisms. The planning partners used their comprehensive plans and the hazard mitigation plan as complementary documents that work together to achieve the goal of reducing risk exposure to the citizens of the planning area. An update to a comprehensive plan may trigger an update to the hazard mitigation plan.

Municipal planning partners may create a linkage between the hazard mitigation plan and their individual comprehensive plans by identifying a mitigation action as such and giving that action a high priority. Other planning processes and programs to be coordinated with the recommendations of the hazard mitigation plan include the following:

- Emergency response plans
- Training and exercise of emergency response plans
- Debris management plans
- Recovery plans
- Capital improvement programs
- Municipal codes
- Community design guidelines
- Water-efficient landscape design guidelines
- Stormwater management programs
- Water system vulnerability assessments
- Community wildfire protection plans
- Comprehensive flood hazard management plans
- Resiliency plans
- Community Development Block Grant-Disaster Recovery action plans
- Public information/education plans

Some action items do not need to be implemented through regulation. Instead, these items can be implemented through the creation of new educational programs, continued interagency coordination, or improved public participation. As information becomes available from other planning mechanisms that can enhance this plan, that information may be incorporated via the update process.

26.4.3 Evaluating

HCOHSEM will oversee the evaluation process to ensure the effectiveness of the hazard mitigation plan in achieving its stated objectives and goals. HCOHSEM will coordinate with planning partners, including local government agencies, community organizations, and other stakeholders, on an annual basis. This annual review will allow planning partners to assess progress towards the plan's objectives and goals, identify any emerging needs, and make necessary adjustments. The evaluation process will involve compiling relevant data and progress updates to prepare the annual progress report summarizing the findings and recommendations, and implementing necessary adjustments to the plan based on feedback.

26.4.4 Plan Update

Federal regulations require that local hazard mitigation plans be reviewed, revised if appropriate, and resubmitted for approval to remain eligible for benefits awarded under the Disaster Mitigation Act (44 CFR Section 201.6.d(3) and Section 201.7(d)(3)). This plan's format allows the planning partnership to review and update sections when new data becomes available. New data can be easily incorporated, resulting in a plan that will remain current and relevant. The planning partnership intends to update the plan on a five-year cycle from the date of plan approval. This cycle may be accelerated to less than 5 years. The plan may be updated before the 5-year period if an event significantly alters the way a previously identified hazard affects the community.

It will not be the intent of the update process to develop a completely new hazard mitigation plan. Based on needs identified by the planning team, the update will, at a minimum, include the following elements:

- The update process will be convened through a new steering committee.
- The hazard risk assessment will be reviewed and, if necessary, updated using best available information and technologies.
- Action plans will be reviewed and revised to account for any actions completed, dropped, or changed and to account for changes in the risk assessment or planning partnership policies identified under other planning mechanisms (such as the comprehensive plan).
- The draft update will be sent to appropriate agencies and organizations for comment.
- The public will be given an opportunity to comment on the update prior to adoption.
- The plan will be submitted for approval to TDEM and FEMA.
- Partners' governing bodies will adopt their respective portions of the updated plan.

Because plan updates can require a year or more to complete, HCOHSEM will initiate efforts to update the plan before it expires. HCOHSEM will consider applying for funding to update the plan no later than 2 calendar years before plan expiration.

26.4.5 Grant Monitoring and Coordination

Harris County will send notification to participating entities on available funding opportunities as they come available. Once these opportunities are identified, planning partners interested in pursuing a grant opportunity should convene a local team to review the 2025 HC MJ-HMP and pursue a strategy to capture that grant funding. The grant coordination process will involve the following:

- HCOHSEM will continue to review the HC MJ-HMP annually to align with federal and state grant opportunities.
- Each planning partner should discuss any hazard events that occurred during the prior year and their impact on the planning area
- Impact of potential grant opportunities on the implementation of mitigation actions
- Re-evaluation of the action plans to determine if the timeline for identified actions need to be amended (such as changing a long-term action to a short-term action because of funding availability)
- Recommendations for new actions
- Impact of any other planning programs or initiatives that involve hazard mitigation.

If multiple planning partners decide to pursue the same grant funding opportunity, partnerships can be formed to utilize the hazard mitigation plan in the grant application.

26.4.6 Continued Public Involvement

The Planning Partners are committed to the continued involvement of the public in the hazard mitigation process. This MJ-HMP update will continue to be posted online at ReadyHarris. In addition, public outreach and dissemination of the MJ-HMP will include the following:

- Links to the plan on local websites of each jurisdiction with capability.
- Continued utilization of existing social media outlets (Facebook, Twitter) to inform the public of natural hazard events, such as floods and severe storms; the public can be educated via the jurisdictional websites on how these applications can be used in an emergency.
- Promotion of articles or workshops on hazards to educate the public and keep them aware of the dangers of hazards.

The MJ-HMP Coordinator will be responsible for receiving, tracking, and filing public comments regarding this MJ-HMP. The public will have an opportunity to comment on the plan via the hazard mitigation website at any time. The MJ-HMP Coordinator will ensure that:

- Public and stakeholder comments and input on the plan, and hazard mitigation in general, are collected, recorded, and addressed as appropriate.
- The Harris County MJ-HMP website is maintained and updated as appropriate.

ACRONYM DEFINITIONS

ACS—American Community Survey

ADA—Americans with Disabilities Act

BLM—U.S. Bureau of Land Management

BLS—U.S. Bureau of Labor Statistics

BTS—Bureau of Transportation Statistics

CCAHA—Conservation Center for Art and Historic Artifacts

CDBG—Community Development Block Grant

CDC—U.S. Centers for Disease Control and Prevention

CERPA—Coastal Erosion Planning and Response Act

CFR—Code of Federal Regulations

CIMA—Channel Industries Mutual Aid

CRS—Community Rating System

CSB—Chemical Safety and Hazard Investigation Board

CWA—Clean Water Act

DFIRM—Digital Flood Insurance Rate Map

DMA—Disaster Mitigation Act

DOT—U.S. Department of Transportation

DR—Presidential Disaster Declaration Number

DSHS—Texas Department of State Health Services

EF—Enhanced Fujita (Tornado Intensity Scale)

EMC—Emergency Management Coordinator

EM—Presidential Disaster Emergency Number

EMS—emergency medical service

EPA—U.S. Environmental Protection Agency

ESA—Endangered Species Act

EWP—Emergency Watershed Protection (Program)

FEMA—Federal Emergency Management Agency

FERC—Federal Energy Regulatory Commission

FHWA—Federal Highway Administration

FIRM—Flood Insurance Rate Map

FMA—Flood Mitigation Assistance (grant program)

GIS—geographic information system

GLO—Texas General Land Office

HCED—Harris County Engineering Department

HCFCD—Harris County Flood Control District

HCFMO—Harris County Fire Marshal's Office

HC—Harris County

HCHCD—Harris County Housing and Community Development

HCOCA—Harris County Office of County Administration

HCOHSEM—Harris County Office of Homeland Security and Emergency Management

HCPC—Harris County Pollution Control

HCPCS—Healthcare Common Procedure Coding System

HCPH—Harris County Public Health

HCSO—Harris County Sheriff's Office

HCUS—Harris County Universal Services

H-GAC—Houston-Galveston Area Council

HHPD—high hazard potential dams

HIFLD—Homeland Infrastructure Foundation-Level Data

HMA—Hazard Mitigation Assistance

HMAP—Hazard Mitigation Action Plan

HMGP—Hazard Mitigation Grant Program

HMP—Hazard Mitigation Plan

HRRM—Harris County Human Resources & Risk Management

HUD—U.S. Department of Housing and Urban Development

HVAC—heating, ventilation, and air conditioning

IPCC—Intergovernmental Panel on Climate Change

ISD—Independent School District

IT—information technology

KBDI—Keetch-Byram Drought Index

LAL—Lightning Activity Level

METRO—Metropolitan Transit Authority

MH-MAP —Harris County Multi-Hazard Mitigation Action Plan

MJ-HMP—Multi-Jurisdiction Hazard Mitigation Plan

MRP—mean return period

NASA—National Aeronautics and Space Administration

NCEI—National Centers for Environmental Information

NEHRP—National Earthquake Hazard Reduction Program

NFIP—National Flood Insurance Program

NHC—National Hurricane Center

NIH—National Institutes of Health

NIMS—National Incident Management System

NOAA—National Oceanic and Atmospheric Administration

NPMS—National Pipeline Mapping System

NRCS—U.S. Natural Resources Conservation Service

NRI—National Risk Index

NSSL—National Severe Storms Laboratory

NTHMP—National Tsunami Hazard Mitigation Program

NWCG—National Wildfire Coordinating Group

NWS—National Weather Service

OCA—Harris County Office of County Administration

PA—Public Assistance

PDM—Pre-Disaster Mitigation Program

PDSI—Palmer Drought Severity Index

PGA—peak ground acceleration

RCV—replacement cost value

RENS—Regional Emergency Notification System

RL—repetitive loss

SBA—U.S. Small Business Administration

SCADA—supervisory control and data acquisition

SETRAC—Southeast Texas Regional Advisory Committee

SFHA—special flood hazard area

SLOSH—Sea, Lake, and Overland Surges from Hurricanes

SPC—Storm Prediction Center

SRL—severe repetitive loss

SWPPP—Stormwater Pollution Prevention Plan

TAMU—Texas A&M University

TCEQ—Texas Commission on Environmental Quality

TDEM—Texas Department of Emergency Management

TEA—Texas Education Agency

TFRLCP—Texas Farm and Ranch Lands Conservation Program

THC—Texas Historical Commission

THIRA—Threat & Hazard Identification & Risk Assessment

TORRO—Tornado and Storm Research Organisation

TSHA—Texas State Historical Association

TWDB—Texas Water Development Board

UCAR—University Corporation for Atmospheric Research

UH—University of Houston

UNDRR—United Nations Office for Disaster Risk Reduction

UNESCO—United Nations Educational, Scientific and Cultural Organization

USACE—U.S. Army Corps of Engineers

USDA—U.S. Department of Agriculture

USDM—U.S. Drought Monitor

USDOT—U.S. Department of Transportation

USEDA—U.S. Economic Development Administration

USGS—U.S. Geological Survey

UT—University of Texas

WUI—wildland-urban interface

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APPENDIX A: ADOPTION RESOLUTIONS

U.S. Department of Homeland Security
FEMA Region 6
800 N. Loop 288
Denton, TX 76209

FEMA

January 28, 2026

Jennifer Charlton-Faia, Deputy State Hazard Mitigation Officer
Texas Division of Emergency Management
P.O. Box 285
Del Valle, Texas 78617-9998

RE: Approvable Pending Adoption of the Harris County, Texas Multi-Jurisdiction Hazard Mitigation Plan

Dear Ms. Charlton-Faia:

This office has concluded its review of the referenced plan, in conformance with the Final Rule on Mitigation Planning (44 CFR § 201.6). FEMA review does not include the review of content that exceeds the applicable FEMA mitigation planning requirements. Formal approval of this plan is contingent upon the adoption by the participants on Enclosure A, as well as the receipt of the final draft of the plan containing all plan components.

Adopting resolutions must be submitted to this agency for review and approval no later than one year from the date of this letter. Failure to submit these resolutions in a timely manner could lead to a required update of the plan prior to FEMA approval.

Once this final requirement has been met, a letter of official approval will be generated. The Local Hazard Mitigation Planning Tool, with the reviewer's comments, has been enclosed to further assist the jurisdictions in complying with planning requirements. If you have any questions, please contact David Reiff, HM Community Planner, at (940) 395-0731.

Sincerely,

Roberto E. Ramirez
Acting Chief, Risk Analysis Branch

Enclosures: Approved Participants
cc: Anne Lehnick

Participants

Attached is the list of participating local governments included in the January 28, 2026 review of the referenced Hazard Mitigation plan.

	Community Name
1)	Bellaire city
2)	Bunker Hill Village city
3)	Cypress Creek Drainage Improvement District
4)	Cypress-Fairbanks Independent School District
5)	Deer Park city
6)	El Lago city
7)	Friendswood city
8)	Galena Park city
9)	Goose Creek Independent School District
10)	Gulf Coast Authority
11)	Harris County
12)	Harris County Flood Control District
13)	Harris County Office of Administration
14)	Harris Health System
15)	Hedwig Village city
16)	Hilshire Village city
17)	Houston Community College
18)	Humble city
19)	Hunters Creek Village city
20)	Jacinto City city
21)	Jersey Village city
22)	Katy city
23)	La Porte city
24)	Lee College
25)	Missouri City city
26)	Morgan's Point city
27)	Nassau Bay city
28)	Pasadena city
29)	Piney Point Village city
30)	San Jacinto College
31)	Seabrook city

Community Name

32) Shoreacres city
33) Southside Place city
34) Spring Independent School District
35) Spring Valley Village city
36) Stafford city
37) Taylor Lake Village city
38) Tomball city
39) University of Houston
40) Varnett Public Schools
41) Waller city
42) Webster city
43) West University Place city

Adoption Submittal (Final)

Following the issuance this of Approvable Pending Adoption letter, all participants are provided one year to adopt the plan and submit it through the State to FEMA. For multi-jurisdictional plans, multiple adoptions should be submitted as a complete package as outlined below.

The State must submit the plan files via:

Risk Management Directorate (RMD) SharePoint:

https://rmd.msc.fema.gov/Regions/VI/Mitigation%20Planning/Forms/AllItems.aspx

Note: You will be requested to register if you have not already done so. All plans containing Protected Critical Infrastructure Information (PCII) must be submitted as an encrypted document with the password being sent separately in an email to ensure secure file submissions.

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1. Final draft of the plan in MS Word or pdf format containing:
   a. The final plan formatted as a single document.
   b. Documentation demonstrating adoption by the participating jurisdictions seeking approval. (i.e. copies of signed resolutions, official meeting minutes, etc.…) Note: Adoption resolutions can be separate files. Additional adoptions are not required to provide a copy of the plan.
   c. Remove strikethroughs, highlights and all Track Changes must be accepted in the final plan.

2. Send an email addressed to r6-mtd-planning@fema.dhs.gov as notification that the electronic file has been submitted. Please DO NOT send plans to the email inbox as it has very strict size limitations which will lock the inbox and not allow additional emails to be received. The email must include the following information:

   a. Include the follow when applicable: (Note: A submittal letter is no longer required.)
      i. Subject line [Approval Review for Name of Plan, State]
      ii. FEMA funding source, grant or disaster number, and project number (when applic
      iii. list of adopting jurisdictions
      iv. Plan File name (file name must include date submitted)

3. Submittals which do not conform to the above requirements will be returned to the State for resubmission

Adoption Resolutions to Be Inserted When Available

APPENDIX B: MEETING DOCUMENTATION

HCOHSEM Kickoff Meeting

April 12, 2024

Kickoff Meeting (Virtual) Registration -- April 12, 2024				
Full Name:	Position Title:	Agency:	Email:	Phone:
Brittany Ray	Resilience Planner	HCOHSEM	brittany.ray@oem.hctx.net	713-881-3100
Brian Murray	DEMC/PIO	HCOHSEM	brian.murray@oem.hctx.net	713-426-9543
Lindsey Drouet	Finance/Admin Supervisor	Harris County OHSEM	Lindsey.drouet@oem.hctx.net	713-426-9540
Keyona Stevenson	UASI Planner	HCOHSEM	Keyona.stevenson@oem.hctx.com	713-881-3100
Daniel Sanchez	GIS Specialist	HCOHSEM	Daniel.Sanchez1@oem.hctx.net	713-904-0011
James Anderson	UASI Planner	HCOHSEM	james.anderson1@oem.hctx.net	713-881-3100
Mel Bartis	Deputy EMC	HCOHSEM	Mel.bartis@oem.hctx.net	832-343-5988
Benzon John	Assist Deputy / PSC	HCOHSEM	benzon.john@oem.hctx.net	832-286-5620
Jennifer Suter	Training and Exercise Supervisor	HCOHSEM	Jennifer.suter@oem.hctx.net	713-426-9544
Charles Mistric	EMC	City of Waller	CMistric@WallerTexas.com	979-645-7557
Rees Jackson	Intern	HCOHSEM	rees.jackson@oem.hctx.net	832-370-5559
Kevin Rodgers	Director of Emergency Services	City of Seabrook	krodgers@seabrooktx.gov	281-291-5787
Brian Craig	Assistant public works director/ city engineer	City of Seabrook	Bcraig@seabrooktx.gov	281-291-5719
Allyson Lambert	Preparedness Coordinator	TDEM	Allyson.lambert@tdem.Texas.gov	830-500-1661
Cynthia Ward	Assistant City Administrator/ City Secretary	City of Waller	Cward@wallertexas.com	936-372-3880
Jacob Daily	Assistant Fire Chief / Assistant Fire Marshal / AEMC	Jersey Village Fire Department	Jdaily@jerseyvillagetx.com	346-679-1102
Kristina Clark	Planner	HCOHSEM	Kristina.clark@oem.hctx.net	713-881-3100
Eric Volkmann	Fire Marshal	City of Jersey Village Fire Dept	Evolkmann@jerseyvillagetx.com	713-466-2130
Mardie menke	Budget and Percurement	HCOHSEM	mardie.menke@oem.hctx.net	713-426-9539
David Potts	Mgr Emergency Management	Houston Community College	David.Potts@HCCS.edu	713-728-2195
Maritza AGUILAR	Hazard Mitigation Planner	City of Houston OEM	Maritza.aguilar@houstontx.gov	832-405-1276
Andrew Jaspers	Emergency Management Coordinator	Klein Independent School District	Ajaspers1@kleinisd.net	832-239-4230
Shain Carrizal	Executive Director	Human Resources & Risk Management	Shain.carrizal@harriscountytx.gov	713-274-5525
Jamie Galloway	Emergency Services Director	City of Deer Park	jgalloway@deerparktx.org	281-478-7298
Michael Mulligan	Chief of Department	Harris County ESD No. 46/Atascocita Fire Department	mmulligan@avfd.com	281-731-6896
Johnny Morales	Emergency Management Coordinator	City of La Porte	Moralesj@laportetx.gov	281-628-4249
James Lutz	Assistant Emergency Coordinator	Village Fire Department	Lutz@villagefire.org	281-415-2011
Johnathan Martin	ER Manager	PCS	johnathan.martin@pcs.hctx.net	713-274-6411
Greg Goedecker	Emergency Management Coordinator	City of Katy	ggoedecker@cityofkaty.com	832-418-1574
Steven Simmons	Depury Fire Marshal	City of Friendswood	Ssimmons@friendswood.com	281-996-3335
Stuart Mueller	Deputy Director- Technical	Harris County Pollution Control Services	Stuart.mueller@pcs.hctx.net	713-274-6414
Nicole Bradstreet	GIS Specialist	HCOHSEM	Nicole.Bradstreet@oem.hctx.net	713-881-3100
Robert Marquez	Director of Safe and Secure Schools	Goose Creek CISD	robert.marquez@gccisd.net	346-233-5158
Derrick Harris	Director	Lone Star College	derrick.d.harris@lonestar.edu	281-290-3667
Todd Ward	Risk Mitigation Department Manager	Harris County Flood Control District	Todd.ward@hcfcd.hctx.net	346-286-4858
Jason Cantu	Emergency Services Coordinator	Harris County District Clerks Office	Jason.Cantu@hcdistrictclerk.com	281-733-3556

Kickoff Meeting (Virtual) Registration -- April 12, 2024				
Full Name:	Position Title:	Agency:	Email:	Phone:
Merryl Holmes, Jr	Mitigation Coordinator	TDEM	Merryl.Holmes@tdem.texas.gov	281-794-4102
Trameka Jewett	Emc	Missouri city	Trameka.jewett@missouricitytx.gov	281-635-7158
Paul warner	Captain of police	Hedwig village PD	pwarner@hedwigtx.gov	832-309-9181
Casey Jimenez	Emergency management Specialist	City of Missouri City	Casey.jimenez@missouricitytx.gov	817-682-3349
Scott Hudson	Director of Emergency Management	Cypress-Fairbanks ISD Police Department	scott.hudson@cfisd.net	540-809-4167
Ingrid Calderon	Technical Specialist	Region 4	ingrid.calderon19@yahoo.com	832-844-9854
Andrew Aleman	DEMC	Pasadena OEM	aaleman@pasadenatx.gov	346-221-5940
Robert McCraken		Houston Community College	Robert.mccracken@hccs.edu	713-718-7520
Dena Mahan	EM Specialist	City of Webster	dmahan@cityofwebster.com	832-594-1751
Sam Bissett	Communication Planner	HCHSOEM	Sam.bissett@oem.hctx.net	281-748-1247
Azell Carter	EMC	Pasadena	acarter@pasadenatx.gov	713-475-4977
Danielle Calhoun	Operations Manager	Harris County Public Health	Danielle.calhoun@phs.hctx.net	361-944-0287
Katie Short	Deputy chief of staff	Harris County Commissioner Precinct One	katie.short@cp1.hctx.net	713-299-7566
Jamie Woodall	Executive director of school safety	Waller isd	jwoodall@wallerisd.net	979-229-2247

Steering Committee Meeting #1

July 24, 2024

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
1st Steering Committee Meeting
Wednesday, July 24, 2024
10:00 AM – 12:00 PM
Texas Department of Transportation (TxDOT)
7600 Washington Avenue Houston, TX 77007

Welcome and Introductions

FEMA's Local Mitigation Planning Policy Guide

The Steering Committee Role / Ground Rules

- Steering Committee Purpose, Ground Rules, and Expectations
- Meeting Schedule
- Steering Committee Charter

Texas State Hazard Mitigation Plan

Define the Planning Area

- Jurisdictions in Multiple Counties

Critical Facilities / Infrastructure Definition

- Review Proposed Definition
- Proposed Changes or Enhancements
- Data Sources

Hazards of Concern

- Review Hazards from the Previous Harris County HMP and State HMP
- Proposed Changes
- Hazard Scenarios for Risk Assessment

Public Involvement Strategy

- Public Engagement Meetings
- Planning Partner Coordination
- Additional Outreach Capabilities

Homework

- Review the 2020 Harris County Multi-Hazard Mitigation Action Plan
- Provide a list of Critical Facilities / Infrastructure
- Provide Local Public Engagement Strategy
- Begin Capability Assessment

Next Steps

- Mission Statement and Goal Setting
- Public Involvement Strategy

Adjourn

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
July 24th, 2024 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Brian Murray	HCOHSEM	DEMC	713-881-3100	brian.murray@hcohsem.harriscountytx.gov
Jamie Galloway	City of Deer Park	Dir Emergency Svc	281 478 7255	jgalloway@deerparktx.org
Michelle Dobbins	Houston Arts Alliance	Dir, Disister Services	713-581-6139	michelle@haatx.com
Brian Craig	City of Seabrook	Asst PW Dir/City Eng	281-291-5719	bcraig@seabrooktx.gov
Mary Anne Piacentini	Coastal Prairie Conservancy	President and CEO	281 851. 8762	maryanne@coastalprairieconservancy.org
Kevin Rodgers	City of Seabrook	Director Emergency Svc	281-291-5787	krodgers@seabrooktx.gov
Deann Tittle	City of Bellaire	Fire Chief / EOC	713-662-8201	dtittle@bellairetx.gov
Khenley Harvell	San Jacinto College	Emergency manage coord.	757 333 5117	khenley.harvell@sjcd.edu
Grant Burnett	Cypress-Fairbanks ISD PD	Emergency Mgmt. Coordinator	469-763-4156	grant.burnett@cfisd.net
James Meaux	SETRAC	Regional T&E coordinator	713- 876-7099	James.Meaux@setrac.org
~~M~~				
Dawn Blackmar, DVM	Houston SPCA	Director Emergency Mgt	281/772-1639	dblackmer@reagan.com
Amanda Ashcroft	H-GAC	Planner	713-993-4545	Amanda.Ashcroft@h-gac.com
Brittany Ray	HCOHSEM	Resilience Planner	713-881-3100	Brittany.Ray@oem.hctx.net

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
July 24th, 2024 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Jeremy Phillips	HC Flood Control	Asset Mgmt Div. Dir.	713 684 8780	jeremy.phillips@hcfcd.hctx.net
LEAH BARTON	Harris County office of County Admin	Deputy County Administrator	713 569 0697	leah.barton@harriscountytx.gov
Robert McCracken	Houston Community College	Associate Vice Chancellor, Risk Mgmt	713 718 7520	Robert.mccracken@hccs.edu
Adam Smith	West University FD	Fire Captain	713-662-5835	asmith@westutx.gov
CALVIN COBB	CYPRESS CREEK DRAINAGE DISTRICT	PRESIDENT	832-264-9141	CALVIN@CALVINCOBB.COM
Troy D. Harrison	City of Shoreacres	City Manager	713 410 5466	THarrison@cityofshoreacres.org
Steven Simmons	City of Friendswood	Deputy Fire Marshal	281-996-3335	SSimmons@friendswood.com
Colleen Gilbert	Greens Bayou Coalition	Exec Director	281-874-2143	colleen.gilbert@greensbayou.org
Marlisa Briggs	North Houston Association	President	281-851-5907	marlisa.briggs@north-houston.org
Tramaka Jewett	Missouri City	EMC	281-403-4378	tramaka.jewett@mctx.org
Mark Bitz	Jersey Village Fire	Fire Chief EMC	713 466 2143	mbitz@jerseyvillagetx.com

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
July 24th, 2024 / 10:00 AM – 12:00 PM

NON-STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Kristina Clark	HCOHSEM	FNSS Planner	(713)881-3100	Kristina.Clark@oem.hctx.net
Jackie Golz	HCOHSEM	Support Planner	2817532467	Jackie.golz@oem.hctx.net
Abbey Anderson	Spring Valley Village	Assistant to City Admin	832 585 9484	aanderson@springvalleytx.com
Lindsey Drouet	HCOHSEM	Finance/Admin Supervisor	(713) 426-9540	lindsey.drouet@oem.hctx.net
James White	TX Southern Univ	VP Gov't Relations	281 658 0540	JAMES.WHITE@TSU-EDU
Liz Cawhorn	HCOHSEM	APIO	713-426-9530	Liz.Cawhorn@oem.hctx.net

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Date / Time of Meeting:	Wednesday, July 24, 2024, 10:00 AM – 12:00 PM
Location:	TxDOT, 7600 Washington Avenue Houston, TX 77007
In Attendance:	30 total, of which 24 are Steering Committee Members
Summary Prepared By:	Brittany Ray
Quorum – YES / NO	Yes

Welcome and Introductions

- Brian Murray opened the meeting and facilitated group introductions.
- Distributed handouts included: Agenda, Draft - Steering Committee Charter, List of Participating Entities, Critical Facilities and Infrastructure Definition, and Hazards of Concern.

Background Information

- Brittany Ray provided an overview of the 2020 Harris County Multi-Hazard Mitigation Action Plan.
- Brittany Ray provided a summary of the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Kickoff Meeting held on April 12 and the list of participating entities.
- Brittany Ray provided the status of acquiring a contractor to assist in updating the Plan.

FEMA's Local Mitigation Planning Policy Guide

- Brittany Ray provided an overview of the updated policy guidance to include:
    - Stakeholder engagement needs to include representatives of businesses, academia, private organizations, and nonprofit organizations that work directly with and/or provide support to underserved communities and socially vulnerable populations.
    - Whole community approach, specifically for socially vulnerable and underserved communities.
    - Accounting for climate change in the risk assessment and climate adaptation strategies for mitigation action development.
    - Identified hazards, at a minimum, include any state or federal major disaster declarations.

Steering Committee Role / Ground Rules

- The purpose of the Steering Committee was discussed.
- The Chairperson (Brian Murray) and Vice Chairperson (Jeremy Phillips) were named.
- Quorum was established at 15 members plus at least one of the co-chairs.
- Alternates can be designated in the event a committee member cannot attend the meeting.
- Decision-making process will be a consensus, however, if a consensus cannot be reached, a decision will be confirmed by a majority vote.
- The spokesperson for the Steering Committee will be Brian Murray – all media requests should go through him at oemjic@oem.hctx.net.
- Meeting dates – will be the second Wednesday of every month. However, the August meeting may be adjusted due to the July meeting date being pushed back a few weeks.
- Public involvement – all meetings are open to the public. Each member of the public wishing to address the SC will be given a single allotment of 3 minutes to address the committee, with a total aggregated time not to exceed 15 minutes.

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Texas State Hazard Mitigation Plan

- Please review the Texas State Hazard Mitigation Plan, especially the hazards listed for our region.

Define the Planning Area

- TDEM has provided direction on how to address jurisdictions whose boundaries intersect more than 1 county. The risk assessment must extend into those counties outside of Harris.
- There are 5 cities in the Harris County planning partnership whose boundaries extend outside of Harris County - Friendswood, Katy, Missouri City, Seabrook, and Waller.

Critical Facilities / Infrastructure Definition

- The Steering Committee reviewed the critical facilities and infrastructure categories from the 2020 Harris County Multi-Hazard Mitigation Action Plan.
- The Steering Committee reviewed CISA CIKR Taxonomy to determine if changes to the critical facilities and infrastructure categories were needed.
- The Steering Committee agreed upon critical facilities and infrastructure categories for the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan.

Hazards of Concern

- Brittany Ray discussed the identified hazards in the 2020 Harris County Multi-Hazard Mitigation Action Plan.
- Brittany Ray discussed the identified hazards in the 2023 Texas State Hazard Mitigation Plan.
- Steering Committee approved:
    - Extreme Heat as its category rather than under Severe Weather.
    - Add: Extreme Cold, Technology Failure, and Utility Failure.

Public Involvement Strategy

- The Steering Committee was briefed on the requirements for public involvement throughout the plan update process.
- The Steering Committee discussed potential ways to have better public engagement.
- Brittany Ray briefed on the HCOHSEM mitigation page at https://www.readyharris.org/ which will showcase information related to Steering Committee meetings, public workshops, and other pertinent information.
- Participating entities will be asked to create a link to the above website on their websites.
- Participating entities will be asked to create their own local public engagement strategy.

Homework for Participating Entities:

- Review the 2020 Harris County Multi-Hazard Mitigation Action Plan.
- Provide a list of critical facilities/infrastructure within their jurisdiction.
- Provide a local public engagement strategy.
- Begin Capability Assessment.

Next Steps for the Steering Committee

- Development of the mission statement and goal setting.
- Further discussion of the public involvement strategy.

Steering Committee Meeting #2

September 11, 2024

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #2
Wednesday, Sept 11, 2024
10:00 AM – 12:00 PM
San Jacinto College – Central Campus
8060 Spencer Hwy Pasadena, TX 77505

Welcome and Introductions
- Group Introductions
- Review/Approve SC Meeting #1 Summary

Status of Participating Entities Deliverables from the July SC Meeting
- Critical Facilities and Infrastructure List
- Public Engagement Strategy
- Adding ReadyHarris - Mitigation Page as a Link on their Website

CDBG-MIT / Timeline Update

Public Engagement Strategy
- Public Survey
- Workshops
    - Educational Campaign (4)
    - Risk Assessment (4)
- Public Comment Review Period
- Additional Outreach Opportunities Discussion

Stakeholder / Partner Engagement Discussion
- Review Stakeholders from the 2020 HC MH-MAP
- Proposed Changes?

Mission Statement Development
- Review 2020 HC MH-MAP Mission Statement
- Changes or Enhancements?

Goal Setting:
- Review the goals from the 2020 HC MH-MAP
- Compare to State Hazard Mitigation Plan Goals
- Changes or Enhancements?

Homework:
- Review and Provide Feedback to Public Engagement Strategy
- Review and Provide Feedback to Stakeholder List
- Begin Capability Assessment

Next Steps:
- Approve HC MJ-HMP Mission Statement and Goals
- Capability Assessment Workshop

Adjourn

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
September 11th, 2024 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Brittany Ray	HCOHSEM	Resilience Planner	713-881-3100	Brittany.Ray@harriscountytx.gov
Jamie Galloway	Deer Park	Emer. Serv. Dir	281 478 7298	jgalloway@deerparktx.org
~~Kraig Winton~~	~~HCOHSEM~~	~~Planning Section~~	~~713 501 7825~~	~~Kraig.Winton@harriscountytx.gov~~
James Meaux	SETRAC	TEL Coordinator	713-876-7099	James.Meaux@setrac.org
Troy D. Harrison	Shoreacres	City Manager	713.410.5466	THarrison@shoreacrestx.com
Jeremy Phillips	HCFCD	Division Director, HMD	713 684 5780	jeremy.phillips@hcfcd.hctx.net
Calvin Cobbs	CCDID	President	832-244-6141	Calvin@Calvincobbs.com
Leslie Etheridge	HCDE	Director	917-642-6919	Letheridge@HCDE-Texas.org
Azell Carter	City of Pasadena	EMC	832-350-3282	acarter@pasadenatx.gov
Michelle Dobbins	Houston Arts Alliance	Director, Disaster Srv	713-	michelle@haatx.com
~~Sarai~~ Osorio	HGAC	Principal Planner	8326812528	sarai@osorio@h-gac.com
Colleen Gilbert	Greens Bayou Coalition	Exec Director	281 874 2142	colleen.gilbert@greensbayou.org
SCOTT HUDSON	CFISD POLICE DEPT.	DIRECTOR OF EMERG. MNGT.	540 809 4167	Scott.hudson@cfisd.net
~~Dana Markham~~	~~City of Webster~~	~~EM Spec~~	~~832 594-1751~~	~~dmarkham@cityofwebster.com~~
Grant Burnett	CFISD Police Dept	Coordinator of Emerg. Mgmt	469-763-4156	grant.Burnett@cfisd.net
~~Jesse Dickerman~~	~~Harris County~~	~~Managing Director~~	~~832-260-8050~~	~~jesse.dickerman@harriscountytx.gov~~

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
September 11th, 2024 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Khenley Harvell	San jacinto college	Emergency mg coordinator	737 333 5117	khenley.harvell@sjcd.edu
Brian Mweresa	HCPHSEM	Deputy EMC	713-581-3100	
Steven Simmons	City of Friendswood	Deputy Fire Marshal	281 996 3335	SSimmons@friendswood.com
Kimi Rodgers	City of Seabrook	Director Emergency Services	281 291 5787	krodgers@seabrooktx.gov
Mark Bitz	City of Jersey Village	Fire chief / EMC	713446-9993	Mbitz@jerseyvillagetx.com
BOB KOSAR	CITY OF EL LAGO + TAYLOR LAKE V.	Floodplain Admin	281 546 4705	cityfpa@ellago-tx.gov
Amanda Ashcraft	H-GAC	P		
LEAH BARTON	HCPH	EXEC DIR HCPH	713 569 0697	
JOHNNY MORALES	LPOEM	EMC		
ROBERT MCCRACKEN	HCC	AVC, RISK MGMT	713 718 7520	ROBERT.MCCRACKEN@HCCS.EDU
Marlisa Briggs	NHA	President	281 851 5909	on file
~~KARLA MARQUEZ~~	~~GOOSE CREEK CISD~~	~~DIRECTOR SAFETY SECURITY~~	~~346-232-5155~~	~~on file~~

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
September 11th, 2024 / 10:00 AM – 12:00 PM
NON-STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
ackie Golz	HCOHSEM	Support Planner	2817532467	Jackie.golz@harriscountytx.gov
Benzen John	HCOHSEM	Asst. Deputy & PSC	832-286-5620	benzen.john@harriscountytx.gov
Lizeth Hernandez	HCOHSEM	Communications Supervisor	713 881 3140	Lizeth.hernandez@hamscountytx.gov
Mel Bartis	HCOHSEM	Deputy EMC	8323435988	mel.bartis@hamscountytx.gov
Greg Goedecke	City of Katy	EMC	832-418-1574	ggoedecker@cityofkaty.com
Kiao Wynton	HCOHSEM	Planning Section	713 501 7225	Kiao.wynton@harriscountytx.gov
Brian Craig	Seabrook	Asst PW Dir/C.A. Eng	281 291 5719	bcraig@seabrooktx.gov
Kristina Clark	HCOHSEM			
Charles Mistric	Waller EMC	EMC	979-645-7557	CMistric@cityofwallertx.com
Archie Losey	Meshek + Assoc./Public	Planner	(713) 824-5219	alosey@meshekengr.com
Thomas Quinn	Lee College	ER + SAFety OPS manager	432-274-7494	tquinn@lee.edu
~~Sarah Osi~~				
Adam Nelson	HCED	Disaster Recovery	346-418-4494	adam.nelson@harriscountytx.gov
David Carbo	OCA	C.J Manager		David.Carbo@harriscountytx.gov
Dena Mahan	City of webster	EM spec.	832 594 1751	dmahan@cityofwebster.com
Jesse Dickerman	HC.OCA	Managing director	832-240-8050	jessedickerman@harris countytx.gov
Robert Marquez	goose creek CISD	Director, safe & secure schools	346 2335158	Robert.Marquez@gccisd.net

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Date / Time of Meeting:	Wednesday, Sept. 11, 2024, 10:00 AM – 12:00 PM
Location:	San Jacinto College (8060 Spencer Hwy, Pasadena, TX 77505)
In Attendance:	41 total, of which 22 are Steering Committee Members
Summary Prepared By:	Brittany Ray
Quorum – YES / NO	Yes

Welcome and Introductions

- Brian Murray opened the meeting and facilitated group introductions.
- Distributed handouts included the meeting agenda, July meeting summary, draft public survey questions, draft stakeholder list, draft mission statement, and draft goals of the 2025 HC MJ-HMP.
- Brian Murray called for a motion to approve the Steering Committee Meeting #1 Summary, which was approved unanimously.

Status of Participating Entities Deliverables from the July SC Meeting (Brittany Ray)

- Critical facilities and infrastructure list submissions:
  - Most of the entities have submitted their critical facilities and infrastructure list. A reminder was provided to those who have not completed this to submit as soon as possible, as HCOHSEM is compiling the information to turn over to the contractor.
- Local public engagement strategy submissions:
  - Many participating entities have submitted their strategy, and the extent of the outreach varied according to their capabilities.
  - At a minimum, each participating entity has a link to the ReadyHarris mitigation page on their website.

CDBG-MIT / Timeline Update (Brian Murray)

- Harris County is in the process of selecting a vendor, however, it is anticipated that the vendor may not be on board until November.
- Harris County is coordinating with the Texas Division of Emergency Management regarding the plan's expiration on July 22nd, 2025. If the plan is not completed until December 2025, we want to ensure that mitigation projects and CRS eligibility are not affected.

Public Engagement Strategy (Brittany Ray)

- Public survey:
  - The purpose is to gauge the public's knowledge of the hazards affecting them and use this data to inform the public engagement strategy and action items.
  - Harris County developed a draft public survey that utilized questions from the 2020 HC MH-MAP, some questions were removed or combined with others to shorten the survey.
  - The Steering Committee discussed possible revisions to the survey:
    - Question 1:
      - Change “coastal storms” to “tropical weather.”
      - Be more specific with terms by providing examples.
    - Question 2:
      - Remove the word “natural.”
      - Add an “Other” option with a fill-in box.
    - Question 3:
      - Considered redundant with the next three questions. Possibly remove or combine with questions 4 and 5.

        - Question 7:
            - Add options for local emergency management, word of mouth, community organizations, and radio.
            - Remove newspapers.
        - Question 8:
            - Add "word of mouth."
            - Change "EM" to "local government alert."
            - Clarify "audible" to "sirens" or "outdoor warning."
        - Question 10: Remove sub-question regarding the value of their home.
        - Question 13: Add a "none of the above" option.
        - Question 17: Remove "primary" language and change to a check all that apply.
        - Question 18:
            - Disconnect the email from the survey process.
            - Recommended having a link to the ReadyHarris website, where they can sign up to be notified about participatory opportunities for the Plan update.
    - The draft survey questions will be sent to the Steering Committee and participating entities for additional feedback.
    - The public survey will be launched on October 1st and closed on October 31st.
- Workshops:
    - Education Campaign:
        - Focus: The workshops will concentrate on educating the public about the hazard mitigation plan update, planning process, and successful projects in their communities.
        - Location: plan to hold one workshop in each Commissioner Precinct, preferably in a socially vulnerable area (total of 4).
        - Timeline: plan to complete before the contractor is onboarded, possibly in late October to early November.
        - The Steering Committee and participating entities will be asked to support these workshops in various ways such as having a booth, presenting, or disseminating the workshop information within their networks.
    - Risk Assessment:
        - Focus: Inform the public about their risks to hazards.
        - Location: plan to hold one workshop in each Commissioner Precinct, preferably in a socially vulnerable area (total of 4).
        - Timeline: plan to complete once the contractor is onboard to rely on their subject-matter expertise.
    - Discussed having a digital workshop to increase accessibility and participation.
- Public comment review period.
- Brian Murray made a motion to approve the public engagement strategy, contingent upon recommendations from the contractor, and the motion carried unanimously.

Stakeholder / Partner Engagement Discussion (Brittany Ray)

- The Steering Committee reviewed the 2020 HC MH-MAP stakeholder list to identify additional partners that should be notified about key milestones of the Plan update.
- Recommended adding Lone Star College, Harris County Long Term Recovery Committee, TGCRVOAD, Houston Food Bank, and Texas Forest Service.
- Recommended streamlining outreach to hospitals by coordinating with SETRAC.

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Mission Statement Development (Brian Murray)

- The Steering Committee reviewed and discussed the proposed Mission Statement for the 2025 HC MJ-HMP.
- The draft Mission Statement will be sent to participating entities for review.
- The goal is to approve the next Steering Committee meeting.

Goal Setting (Brittany Ray)

- The Steering Committee reviewed and discussed the proposed Goals for the 2025 HC MJ-HMP and compared them to the 2020 HC MH-MAP Goals.
- Recommended changes:
    - Goal 1: Remove "the intent of the efforts and to improve the."
    - Goal 3: Remove "Improve local government efforts to."
    - Goals 5&6: Possibly merge the goals for better clarity. HCFCD will revise these goals.
    - Goal 7: Add "protect."
    - Add a goal on prioritizing investments.
- The draft Goals will be sent to participating entities and the Steering Committee for further review.
- Aim to approve the HC MJ-HMP Goals next Steering Committee meeting.

Homework (Brittany Ray)

- Support the plan-wide Public Engagement Strategy
    - Workshops
    - Public survey
- Review and provide feedback to the stakeholder list.
- Review and provide feedback on the draft 2025 HC MJ-HMP mission statement and goals.
- Begin capability assessment:
    - Each participating entity should begin reviewing and updating their capability assessment from the last Plan.
    - Suggested beginning with an inventory of their current plans that integrate with hazard mitigation planning and identify opportunities for future integration.

Next Steps (Brittany Ray)

- Review and approve the 2025 HC MJ-HMP Mission Statement.
- Review and approve the 2025 HC MJ-HMP Goals.
- Launch the public survey.
- Continue planning for the public workshops.
- The October Steering Committee meeting may focus on planning for the public workshops or beginning the capability assessment, depending on when the contractor starts.

Adjourned

Steering Committee Meeting #3

October 2, 2024

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #3
Wednesday, October 2, 2024
10:00 AM – 11:00 AM
Virtual Meeting

Welcome and Introductions

Old Business

- Review/Approve SC Meeting #2 Summary
- Review/Approve HC MJ-HMP Mission Statement
- Review/Approve HC MJ-HMP Goals

CDBG-MIT / Timeline Update

Public Survey

- Launched: October 1st
- Closes: October 31st

Public Workshops – Education Campaign

- Workshop Locations and Dates:
    - Saturday, October 19th, 10:00 AM – 11:30 AM at San Jacinto College – Generation Park
    - Tuesday, October 22nd, 6:00 PM – 7:30 PM at Katherine Tyra Branch Library
    - Wednesday, October 30th, 6:00 PM – 7:30 PM at George HW Bush Community Center
    - Saturday, November 9th, 10:00 AM – 11:30 AM at San Jac. College – Central Campus
- Agenda Discussion

Homework:

- Socialize Public Survey
- Socialize Public Workshops

Open Discussion

Adjourn

October 2, 2024 Meeting Sign-in Sheet

2025 HC MJ-HMP Steering Committee Members						
ORGANIZATION	NAME	TITLE	Phone	Present?	Alternate	Present?
HCOHSEM	Brian Murray	Deputy EMC / PIO		yes		
HCOHSEM	Brittany Ray	Resilience Planner		yes		
HC OCA / HCPH	Leah Barton	Deputy County Administrator	713-569-0697	yes		
HC Flood Control District	Jeremy Phillips	Director, Asset Management Division	346-286-4092	yes		
City of Bellaire	Deacon Tittle	Fire Chief / Emergency Management Coordinator	281-932-0891	yes		
City of Deer Park	Jamie Galloway	Emergency Services Director	281-478-7298	yes		
City of Friendswood	Steven Simmons	Deputy Fire Marshal	281-996-3335	yes		
City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal		yes		
City of La Porte	Johnny Morales	Emergency Management Coordinator	281-628-4249	yes		
City of Missouri City	Trameka Jewett	Emergency Management Coordinator	281-635-7158	no		
City of Pasadena	Azell Carter	Emergency Management Coordinator	281-687-9193	yes		
City of Seabrook	Kevin Rodgers	Office of Emergency Management Director	346-500-1924	yes		
City of Shoreacres	Troy Harrison	City Manager	281-471-2244	no		
City of Taylor Lake Village & City of El Lago	Carl Nunn	Chief of Police / Emergency Management Coordinator	281-326-1098	no	Bob Kosar	yes
National Weather Service	Jeff Evans	Meteorologist in Charge	405-343-6693	no		
North Houston Association	Marlisa Briggs	President	281-851-5909	yes		
Houston Community College	Robert McCracken	Assoc. Vice Chancellor, Risk Mgmt & Resilience	713-718-7520	yes		
San Jacinto College	Ali Shah	Director, Emergency Management	281-914-3123	no	Khenley Harvell	yes
TranStar / Houston Public Works / COH OEM	Brian Mason	Executive Director / EMC	832-358-1996	no		
H-GAC	Amanda Ashcroft	Planner	713-993-4545	yes		
Cypress Creek Drainage Improvement District	Calvin Cobb	President	832-244-9141	yes		
Coastal Prairie Conservancy	Mary Anne Piacentini	President and CEO	281-851-8762	No	Elisa Donovan	yes
Greens Bayou Coalition	Colleen Gilbert	Executive Director	281-874-2142	yes		
Cy-Fair ISD	Scott Hudson	Director of Emergency Management	281-897-6448	yes		
SETRAC	James Meaux	Regional T&E Coordinator	713-876-7099	yes	Lisa Spivey	yes
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience	713-581-6139	yes		
Houston SPCA	Dawn Blackmar	Director, Emergency Management	713-869-7722	yes		
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Officer		No		
Harris County Dept. of Educ.	Leslie Etheridge	Director, Center for Safe and Secure Schools	917-642-6919	No		

Additional October 2nd SC Meeting Registrants			
ORGANIZATION	NAME	Phone	
HCOHSEM	Benzon John		
HCOHSEM	Jackie Golz		
HCOHSEM	Kial Whitton		
Memorial Villages	Howard Miller	713-468-7981	
Southwest Public Schools	JJ Bishop	936-828-6038	
Southwest Public Schools	Lionel Flores Jr.	781-686-4600	
City of Morgan's Point	Rick Ashe	346-222-7384 / 281-471-2171	
Lee College	Thomas Quinn	281-425-6409	
Public	Archie Losey		
Rice University	Lach Mullen		
HC CJO	Mikayla Panariello		
HC Comm Pct 1	Katie Short		
Houston Community College	David Potts	713-718-2195	
Cy-Fair ISD	Grant Burnett	469-763-4156	
Harris Health	Aaron Velasquez	832-654-7460	
HCFCD	Todd Ward	713-516-5039	
	Merryl Holmes Jr.		

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Date / Time of Meeting:	Wednesday, October 2, 2024, 10:00 AM – 11:00 AM
Location:	Virtual – MS Teams
In Attendance:	44 total, of which 24 are Steering Committee Members
Summary Prepared By:	Brittany Ray
Quorum – YES / NO	Yes

Old Business:

- The Steering Committee Meeting #2 Summary was approved unanimously.
- The 2025 HC MJ-HMP Mission Statement was approved unanimously.
- Goals of the 2025 HC MJ-HMP were approved unanimously.

CDBG-MIT / Timeline Update (Brian Murray)

- The County is on track to have the contractor start by late November.
- Contractor can complete the Plan update by the current plan's expiration date (July 22nd, 2025).

Public Survey (Brittany Ray)

- Went live on Tuesday, October 1st at 4:00 PM and have received 15 submissions thus far.
- The survey is available in English, Spanish, Chinese, Vietnamese, and Arabic. A vendor reviewed the additional languages to ensure proper translation.
- The survey can be found on the ReadyHarris Mitigation Page.
- A flyer has been developed to promote the public survey at community events, community centers, and other locations.
- A half-page document designed to promote the survey on social media posts has been created and is now available for the Steering Committee and participating entities.
- The Steering Committee reviewed the minor changes made to the survey since the September Steering Committee meeting, which included:
    - Adding an introductory statement explaining the purpose of the survey.
    - Added a question regarding climate change to align with the priorities in FEMA's Local Mitigation Planning Policy Guide.
    - Wind insurance was added as an answer to the insurance-related question.
    - Revised "Have you ever had problems securing homeowners or renters insurance due to risks from hazards in Harris County" to "Have you ever had problems securing or maintaining homeowners or renters insurance due to risks from hazards in Harris County." Answer choices were modified to:
        - Yes, I had trouble securing initial coverage.
        - Yes, my policy was dropped by my insurance provider.
        - Yes, my policy premiums significantly increased.
        - No, I have not experienced any issues.
    - Included a Nixel for the community to sign up for Plan updates once the survey is submitted. (Text HMP2025 to 888777).
- Due to one of the public workshops occurring in November, the Steering Committee discussed and approved extending the public survey deadline from October 31st to November 10th.

Public Workshops (Brittany Ray)

- Workshop Dates/Locations
    - October 19, 2024, 10:00 AM – 11:30 AM at San Jacinto College – Generation Park.
    - October 22, 2024, 6:00 PM – 7:30 PM at Katherine Tyra Branch Library.

  - October 30, 2024, 6:00 PM – 7:30 PM at George H.W. Bush Community Center.
  - November 9, 2024, 10:00 AM – 11:30 AM at San Jacinto College – Central Campus.
    - This workshop will have a virtual option to increase public participation.
- Workshop Agenda:
  - Attendees arrive, sign in, mingle/visit partner booths.
  - Partner Presentations
    - Harris County Office of Homeland Security & Emergency Management
      - Provide an overview of hazard mitigation planning and what it means to the community.
      - Status of the 2020 Harris County Multi-Hazard Mitigation Action Plan.
      - Ways for the community to get involved in the 2025 HC MJ-HMP update.
    - Harris County Flood Control District
      - Overview of the Flood Control District.
      - Highlight 1-2 mitigation projects that have been completed, are in progress, or want to be implemented within that community.
    - Harris County Engineering Department
      - Will provide an overview of mitigation projects they are working on in that area.
    - Harris County Office of County Administration
      - Will provide an overview of the Climate Justice Plan.
  - HCOHSEM – Q&A / mingle at partner booths.
  - End of Workshop.
- QR codes for the public survey will be available at the workshops.
- If any Steering Committee organizations or participating entities wish to have a booth at the workshops, please coordinate with HCOHSEM.

Homework:

- Actively Promote and Socialize the Public Survey.
- Actively Promote and Socialize the Upcoming Public Workshops.

Open Discussion:

- Adding Health-Related Hazard
  - A proposal was introduced to include a health-related hazard within the Plan.
  - Other large counties have incorporated a health-related hazard at various levels within their plan. However, it is not included as a hazard in the Texas State Hazard Mitigation Plan.
  - Adding it as a hazard, will allow for future funding opportunities.
  - This can be completed by the contractor with the assistance of Harris County Public Health.
  - The Steering Committee approved adding it to the other hazards of concern section but will not complete a full hazard profile.

Adjourned

Meeting Title: **Core Planning Team Kickoff**

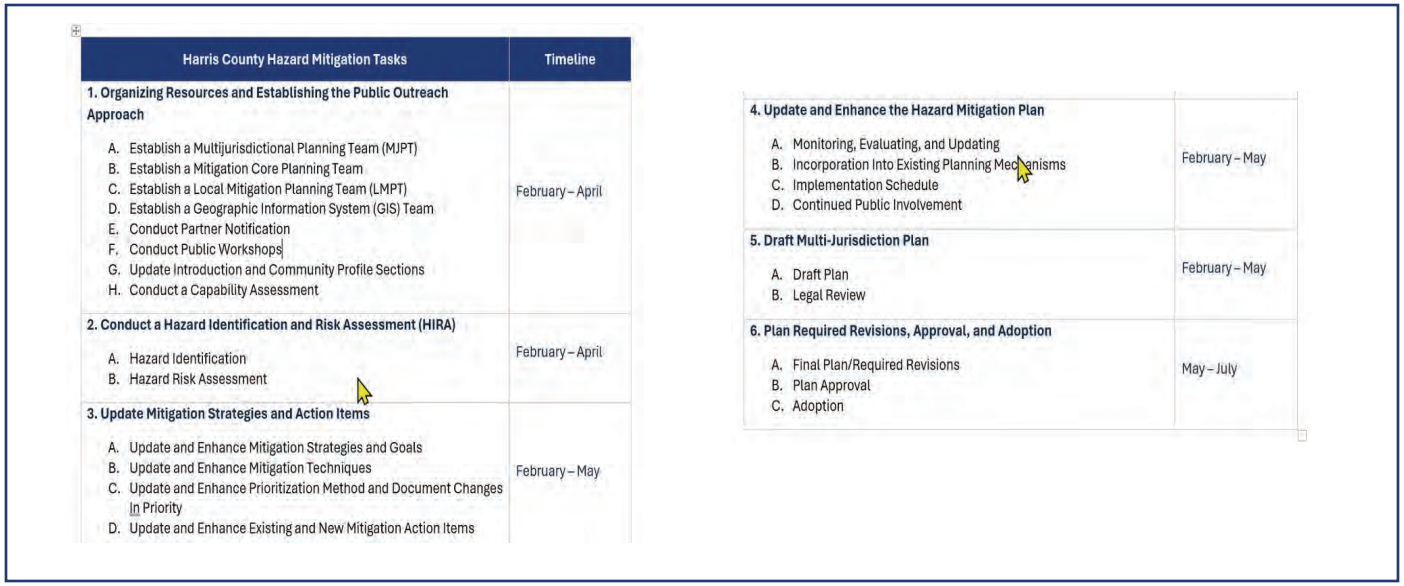
Date: **February 18, 2025**

Location: **Microsoft Teams (virtual)**

Attendees:

Name	Organization	Representative, Position
Brittany Ray	HCOHSEM	OEM
Brian Rutherford	Tetra Tech	Project Manager
Anita Machiavello	Tetra Tech	Technical Lead
Mark Stalcup	Tetra Tech	Planner

Summary: **Today's meeting centered around introducing the TT team to HCOHSEM, review a preliminary project timeline, and then answer questions about project material, transferring of data, schedules, and more. Tentative timeline discussed today is previewed below.**



Harris County Hazard Mitigation Tasks	Timeline
1. Organizing Resources and Establishing the Public Outreach Approach A. Establish a Multijurisdictional Planning Team (MJPT) B. Establish a Mitigation Core Planning Team C. Establish a Local Mitigation Planning Team (LMPT) D. Establish a Geographic Information System (GIS) Team E. Conduct Partner Notification F. Conduct Public Workshops G. Update Introduction and Community Profile Sections H. Conduct a Capability Assessment	February – April
2. Conduct a Hazard Identification and Risk Assessment (HIRA) A. Hazard Identification B. Hazard Risk Assessment	February – April
3. Update Mitigation Strategies and Action Items A. Update and Enhance Mitigation Strategies and Goals B. Update and Enhance Mitigation Techniques C. Update and Enhance Prioritization Method and Document Changes In Priority D. Update and Enhance Existing and New Mitigation Action Items	February – May
4. Update and Enhance the Hazard Mitigation Plan A. Monitoring, Evaluating, and Updating B. Incorporation Into Existing Planning Mechanisms C. Implementation Schedule D. Continued Public Involvement	February – May
5. Draft Multi-Jurisdiction Plan A. Draft Plan B. Legal Review	February – May
6. Plan Required Revisions, Approval, and Adoption A. Final Plan/Required Revisions B. Plan Approval C. Adoption	May – July

Meeting Notes:

No.	Description	Action item(s):
1	**Introductions** • Highlights and Discussion	
2	**Review of Initial Project Timeline** • Draft to be completed by mid-May • Steering Committee (SC) Workshops will occur in-person and on the second Wednesday of every month • Bi-Weekly Core Planning Team (CPT) Meetings are to happen every other week.	- Awaiting dates from HCOHSEM to schedule Bi-Weekly calls - Awaiting confirmation that Steering Committee Workshop will occur on March 12, 2025 at Transtar HQ
3	**Questions** • Two new participating entities: City of Tomball and the Gulf Coast Authority. Communication and information for their "next steps" has been given by HCOHSEM • Slight changes in the 2025 HOC: "Extreme Heat" is now no longer part of "Severe Weather" and is a standalone hazard for 2025. Also new are "Extreme Cold", "Utility Failure", and "Technology Failure". Natural hazards will have a full hazard profile. Those that are not natural hazards will have their own profile but the FEMA requirements won't be quite as rigorous. "Toxic Release", "Energy Pipeline Failure", "Utility Failure", and "Technology Failure" are those that will be briefer. • Public Survey closed in late November 2024. • HCOHSEM wishes to also add a "health-related hazard" • 2025 HMP document should be formatted similar to how it was done in 2020. No specific changes or objections were brought up • HCOHSEM has uploaded project information, files to the 2025 Update Material channel • SC members were last advised to review their individual annexes from 2020 HMP and prepare to discuss capabilities at the March workshop.	- Majority of HCOHSEM info uploaded to Teams. We'll upload social media work for various parts of the process. Also uploaded was a list of critical facilities received from participants - Tetra Tech to provide ideas for health-related hazards - HCOHSEM will provide survey results once those are available

No.	Description	Action item(s):
4	**Final Thoughts** • Introducing TT team to the SC via email.	- Tetra Tech will communicate with all SC members to introduce ourselves and prepare a project status update - email communications to SC should CC Brittany Ray as well as the OEM mitigation email, mitigation@oem.hctx.net
5	**Closing**	

GIS Kickoff Meeting

February 27, 2025

Meeting Title: GIS Kickoff Meeting

Date: February 27, 2025

Location: Microsoft Teams

Attendees:

Name	Organization	Role/Title
Benzon John	HCOHSEM	Deputy EM Coordinator
Brittany Ray	HCOHSEM	Assistant Planning Section Chief (Lead), Resilience Planner
Nicole Bradstreet	HCOHSEM	Geospatial Intelligence
Alyssa Wentz	Tetra Tech	GIS Lead
Anita Machiavello	Tetra Tech	Technical Lead Planner
Brian Rutherford	Tetra Tech	Project Manager
Mark Stalcup	Tetra Tech	Planner

Agenda Summary: Discussion between GIS teams to review dataset questions to identify any missing data needed before analysis begins.

Meeting Takeaways: Harris County

1. HCOHSEM to get Tetra Tech the Pipeline spatial data
2. HCOHSEM to get County Building spatial data to Tetra Tech by **March 7, 2025**[h]
    a. If date is missed, we will move forward with what was originally provided

Joint Actions

1. Reach out to Steering Committee to ask them about: Agenda items 3 – 6 in the table below
2. Meeting for Land Use and Land Cover Data: Nicole and TT to identify LU_Code in the dataset.

Meeting Notes:

No.	Description	Action item(s):

1	**GIS Wish List (Alyssa)**
	• Overview of what the Wish List means and its purpose
	HCOHSEM actions called out in bold orange font
	• HCOHSEM to get Tetra Tech the Pipeline spatial data
	• HCOHSEM to get County Building spatial data to Tetra Tech by the 7th
	  o If date is missed, we will move forward with what was originally provided
	• College/Universities: For Tetra Tech to do comparison of the 3 datasets
	• Public Health Department: Tetra Tech will incorporate the data from the last plan
	• Military Installations: Tetra Tech will used HIFLD source
	• Drug and Alcohol Treatment Centers: Tetra Tech will incorporate data from the last plan
	• Hospitals: There is data from SETRAC & DSHS. DSHS may include other Medical Facilities that are not hospitals.
	• Nursing Homes: Tetra Tech will use the dataset that is from DSHS
	• Food, Hydration, Shelter Lifelines: HCOHSEM to provide any of the data below
	  o Commercial Food Distribution
	  o Commercial Food Supply Chain
	  o Food Distribution Programs (e.g., food banks)
	  o Animals and Agriculture
	  o Temporary Hydration Missions (e.g., bottled water distribution)
	  o Commercial Water Supply Chain
	  o Housing (e.g., shelters homes,)
	  o Commercial Facilities (e.g., hotels, motels)
	• Water Systems
	  o Tetra Tech will include the following datasets from the last plan
	  ▪ Potable Water Supply Treatment Facilities
	  ▪ Potable Water Wells
	  o Nicole to provide an updated date for the Active Municipal Plants
	• Community Centers: Tetra Tech will use both dataset; one dataset are Community Centers from the
		• HCOEHSM to ask the Public Health Dept for an updated Harris Health layer.
		• Tetra Tech will use what was shown in the last plan and let HCOHSEM validate data.
		• Alyssa will provide an updated list of what datasets are missing or where additional questions remain
		• HCOHSEM to provide an updated list of buildings. Brittany is working to update the 2020 list. (county buildings = owned, operated, rented, leased, etc.)
		• HCOHSEM will aim for March 7th to have updated building data sent over.
		• Mark will work on adding Lat/Lon information to the critical facilities list compiled by the planning partners.

No.	Description	Action item(s):
	City of Houston (COH), and the other dataset is *Harris County (HC).* These datasets should be different. • Streams & Waterbodies: **HCOHSEM** to get information from the Flood Control District • Railways: Tetra Tech to use TXRRC • Land Use: Alyssa to connect with **HCOHSEM** about this data.	
2	**Dam Inundation Areas (Mark)** • Last plan only analyzed the Conroe Dam • Since the last plan, do you know if any other Dam Inundation Areas were created? Lake Houston? • 7 dams in HC, most of which is owned privately or by the Corps. Those inundation files are highly sensitive, and for 2025 plan, we'll analyze Conroe Dam like before. Lake Houston is covered in the COH HMP.	• Tetra Tech to proceed with analyzing dam failure of just Conroe Dam for now. Benzon might run this by Jeff Linder to see if the Addicks and Barker Reservoirs are also analyzed separately, but for now, Lake Conroe is it.
3	**Jurisdictions not participating in the Update: Baytown and COH** • Although they are not participating – do we still want to add the results or do we want to exclude their results • Likely want to include since there are unique opportunities from BRIC that allow Jurisdictions to cross boundaries to work together….lots of grant opportunities there. • If they are excluded, we can insert a disclaimer that they are excluded from the results	• HCOHSEM to ask Steering Committee but as they will be the ones who ultimately decide how to proceed • Brittany will look back at meeting notes to see if including the COH data was a topic discussed in earlier conversations
4	**Hazus Hurricane analysis** • last plan analyzed Hurricane Alicia 1983 • Available in Hazus (these are the storms that made landfall in Texas in the 2000s) ▪ Dolly (2008) - Category 2 ▪ Ike (2008) - Category 4 ▪ Harvey (2017) - Category 4 ▪ Hanna (2020) - Category 1	• HCOHSEM to ask SC which hurricane would be ideal for analyzing exposure. Alicia was a big flood event, but is a more recent storm (like Harvey) one that they'd like Tetra Tech to use for the Hurricane Hazard Profile
5	**Hazus & Exposure analysis for Hurricane Harvey flood scenario** • Is this hazard area something the client would like analyzed again	• HCOHSEM to ask SC whether or not the 2025 HMP includes an updated hazard analysis on the Harvey flood levels.
6	**Toxic Release/Hazardous Materials & Energy Pipeline Failure** • Last plan was a qualitative analysis (which included map figures) • Is it worth doing a quantitative analysis with the quick turn • we can buffer pipelines (La Porte fire was recent event) HC actively trying to identify what can be done in these areas where pipeline.	• Pipelines are a growing concern, Tetra Tech will proceed with analysis. • **Benzon** will ask (dept.) what a reasonable buffer zone would be best since it's such a

No.	Description	Action item(s):
		densely populated county. • **NICOLE** → Received data from NPMS and this can be shared with Tetra Tech for these hazard profiles
7	**Display of Results**	

Kickoff Meeting Notes – GIS
February 27, 2025

HCOHSEM actions called out in bold orange font

- **HCOHSEM** to get Tetra Tech the Pipeline spatial data
- **HCOHSEM** to get County Building spatial data to Tetra Tech by the 7th
    - If date is missed, we will move forward with what was originally provided
- College/Universities: For Tetra Tech to do comparison of the 3 datasets
- Public Health Department: Tetra Tech will incorporate the data from the last plan
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- Hospitals: There is data from SETRAC & DSHS. DSHS may include other Medical Facilities that are not hospitals.
- Nursing Homes: Tetra Tech will use the dataset that is from DSHS
- Food, Hydration, Shelter Lifelines: **HCOHSEM** to provide any of the data below
    - Commercial Food Distribution
    - Commercial Food Supply Chain
    - Food Distribution Programs (e.g., food banks)
    - Animals and Agriculture
    - Temporary Hydration Missions (e.g., bottled water distribution)
    - Commercial Water Supply Chain
    - Housing (e.g., shelters homes,)
    - Commercial Facilities (e.g., hotels, motels)
- Water Systems
    - Tetra Tech will include the following datasets from the last plan
        - Potable Water Supply Treatment Facilities
        - Potable Water Wells
    - **Nicole** to provide an updated date for the Active Municipal Plants
- Community Centers: Tetra Tech will use both dataset; one dataset are Community Centers from the *City of Houston (COH),* and the other dataset is *Harris County (HC)*. These datasets should be different.
- Streams & Waterbodies: **HCOHSEM** to get information from the Flood Control District
- Railways: Tetra Tech to use TXRRC
- Land Use: Alyssa to connect with **HCOHSEM** about this data.

Meeting Title: **Core Planning Team Meeting**

Date: **March 5, 2025**

Location: **Virtual/Microsoft Teams**

Attendees:

Name	Organization	Position, Title	Email
Brittany Ray	HCOHSEM	OEM	
Brian Rutherford	Tetra Tech	Project Manager	
Anita Machiavello	Tetra Tech	Technical Lead	
Mark Stalcup	Tetra Tech	Planner	

Meeting Notes:

No.	Description	Action item(s):
1	**SC Meeting Invitations Sent** •	•

2	**SC Meeting Agenda**
	• Project Status, Timeline Updates
	• Review 2025 Hazards of Concern
	• Steering Committee feedback on the 2025 Drafted Goals and Objectives (HCOHSEM)
	• Jurisdictional Annexes & Homework Worksheets
	  • Worksheet A. Hazard Event History
	  • Worksheet B. Capability Assessment
	  • Worksheet C. Mitigation Actions
	  • Worksheet D. NFIP Compliance
	  • Worksheet E. Building Permits
		•

3	**Capabilities Assessment Update** • Worksheets with pre-populated responses	•
4	**GIS Questions for SC Meeting** • Hazus analysis for Alicia vs more recent storm • Pipeline buffer zone • Including/Excluding Baytown, City of Houston in GIS analysis • Do SC members want Harvey flood extent analyzed against updated population stats?	•
5	**Meeting Logistics** • Printing out sign-in sheets, handouts, etc. • Arrival times next week at TranStar – meeting starts at 10am, when should we arrive?	•

Steering Committee Meeting #4

March 12, 2025

AGENDA

MARCH 12, 2025 @ 10:00 – 12:00 PM

1. **Welcome & Introductions**
   - Group introductions
   - Review & Approve Steering Committee #3 Meeting Summary
2. **Review and Approval of SC Meeting #3 Meeting Minutes**
3. **New Planning Partners**
4. **Project Status / Timeline Update**
   - Key agencies, partners, and stakeholders
5. **Review Public Survey Results**
6. **2025 Hazards of Concern**
7. **GIS Questions for the Steering Committee**
   - Identify storm for Hazus analysis
   - Re-analyzing Harvey with updated census
   - Planning area for GIS assessment
   - Pipeline buffer zone
8. **Overview & Presentation of Jurisdictional Annex Documents**
   - A – Hazard Event History
   - B – Capability Assessment
   - C – Mitigation Actions
   - D – NFIP Compliance
   - E – Building Permits
   - G – New Mitigation Actions
9. **Jurisdictional Annexes**
10. **Open Discussion**
11. **Adjourn**

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #4
March 12, 2025 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALT. NAME	SIGNATURE
Harris County OHSEM	Brian Murray	Deputy Emergency Management Coordinator / PIO		
Harris County OHSEM	Brittany Ray	Resilience Planner		
Harris Co. OCA / Harris Co. Pub. Health	Leah Barton	Deputy Co. Admin. /Executive. Director		
HCFCD	Jeremy Phillips	Director, Asset Management Division		
City of Bellaire	Deacon Tittle	Fire Chief / Emergency Management Coordinator		
City of Deer Park	Jamie Galloway	Emergency Services Director		
City of El Lago / Taylor Lake Village	Carl Nunn	Chief of Police / EMC		
City of Friendswood	Steven Simmons	Deputy Fire Marshal		
City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal		
City of La Porte	Johnny Morales	Emergency Management Coordinator		
City of Missouri City	Trameka Jewett	Emergency Management Coordinator		
City of Pasadena	Azell Carter	Emergency Management Coordinator		
City of Seabrook	Kevin Rodgers	Office of Emergency Management Director	Brian Cra	
City of Shoreacres	Troy Harrison	City Manager		
National Weather Service	Lance Wood	Science and Operations Officer		

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
March 12, 2025 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALT. NAME	SIGNATURE
North Houston Association	Marlisa Briggs	President		
Houston Community College	Robert McCracken	Assoc. Vice Chancellor, Risk Management & Resilience		
San Jacinto College	Ali Shah	Director, Emergency Management		
H-GAC	Amanda Ashcroft	Sr. Planner		Amanda A
Cypress Creek Drainage Improvement District	Calvin Cobb	President		
Coastal Prairie Conservancy	Mary Anne Piacentini	President and CEO		
Greens Bayou Coalition	Colleen Gilbert	Executive Director		
Cy-Fair ISD	Scott Hudson	Director of Emergency Management		
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Director		
SETRAC	James Meaux	Regional T&E Coordinator		
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience		
Houston SPCA	Dawn Blackmar	Director, Emergency Management		
Harris County Dept. of Education	Leslie Etheridge	Director, Center for Safe and Secure Schools		

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #4
March 12, 2025 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALT. NAME	SIGNATURE
Harris County OHSEM	Brian Murray	Deputy Emergency Management Coordinator / PIO		
Harris County OHSEM	Brittany Ray	Resilience Planner		
Harris Co. OCA / Harris Co. Pub. Health	Leah Barton	Deputy Co. Admin. /Executive. Director		
HCFCD	Jeremy Phillips	Director, Asset Management Division		
City of Bellaire	Deacon Tittle	Fire Chief / Emergency Management Coordinator		
City of Deer Park	Jamie Galloway	Emergency Services Director		
City of El Lago / Taylor Lake Village	Carl Nunn	Chief of Police / EMC		
City of Friendswood	Steven Simmons	Deputy Fire Marshal		
City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal		
City of La Porte	Johnny Morales	Emergency Management Coordinator	Amber Beltran	
City of Missouri City	Trameka Jewett	Emergency Management Coordinator		
City of Pasadena	Azell Carter	Emergency Management Coordinator		
City of Seabrook	Kevin Rodgers	Office of Emergency Management Director		
City of Shoreacres	Troy Harrison	City Manager		
National Weather Service	Lance Wood	Science and Operations Officer		

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
March 12, 2025 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALT. NAME	SIGNATURE
North Houston Association	Marlisa Briggs	President		
Houston Community College	Robert McCracken	Assoc. Vice Chancellor, Risk Management & Resilience		
San Jacinto College	Ali Shah	Director, Emergency Management	Kmenley Harvell	
H-GAC	Amanda Ashcroft	Planner		
Cypress Creek Drainage Improvement District	Calvin Cobb	President		
Coastal Prairie Conservancy	Mary Anne Piacentini	President and CEO		
Greens Bayou Coalition	Colleen Gilbert	Executive Director		
Cy-Fair ISD	Scott Hudson	Director of Emergency Management		
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Director		
SETRAC	James Meaux	Regional T&E Coordinator	Katherine Samuel	
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience		
Houston SPCA	Dawn Blackmar	Director, Emergency Management		
Harris County Dept. of Education	Leslie Etheridge	Director, Center for Safe and Secure Schools		
City of La Porte	[illegible]	Deputy EMC		

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #4
March 12, 2025 / 10:00 AM – 12:00 PM

PUBLIC SIGN-IN

NAME	CITY/JURISDICTION	ORGANIZATION	PHONE	EMAIL
ATRI SEN		RESILIRE	225. 281.4879	ATRI.SEN@RESILIREINFRASTRUCTURE.C
Nandini [illegible]		Urbanxt	504-473-2047	n-sen@outlook.com
Archie Losey	Unincorporated Harris County	Public	713-824-5219	archie.losey@wsbeng.com
Daniella Flanaga	Houston Health Dep	Public	832.923.8257	daniella.flanagan@houstonh.g

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #4
March 12, 2025 / 10:00 AM – 12:00 PM

NON-STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	PHONE	SIGNATURE
City of Waller	Charles Mistric	EMC	979-695-7557	
HC Engineering Dept.	Samuel Pena	DR	281.813.2746	
Cypress-Fairbanks ISD Police Dept.	Grant Burnett	EMC	469.763.4186	
★ Village FD/Mem Villages	Howard Milton	Fire Chief/EMC	713-468-7541	
★ Village FD	James Lutz	Asstiant EMC	713-468-7441	
City of Spring Valley Village	Kristina Watson	Director Publicworks	713-464-4951	
City of Tomball	Joe Sykora	FIRE CHIEF/EMC	2817261111	
City of Southside Place	Andrew Deavers	Assistant City Manage	8713668 2341	
City of Missouri City	Edward Nagorski	Assistant City Engineer	281-403-8546	Edward Nagorski
TDEM	Merryl Holmes	Mitigation Coor	281-794-4102	

Meeting Title: Harris County Steering Committee Meeting #4

Date: 3/12/2025

Location: Houston Transtar

Attendees:

Name	Organization	Representative, Position
Brittany Ray	HCOHSEM	OEM
Brian Rutherford	Tetra Tech	Project Manager
Anita Machiavello	Tetra Tech	Technical Lead
Mark Stalcup	Tetra Tech	Planner

Meeting Notes:

No.	Description	Action item(s):
1	**Meeting Topic** • Hazus analysis – are we omitting the communities in Hazus that are not participating in the HMP update process? • SC would like to include this capture because it is important to identifying the risk and hazards to buildings and assets in HC jurisdiction.	-
2	**Meeting Topic** • Timeline/Status update – no questions or follow-ups	-
3	**Meeting Topic** • Worksheets: Hazard profile – • Capabilities assessments – for jurisdictions to divide the worksheets between their mitigation teams. All jurisdictions are in	-requests for digital copies of all worksheets
4	**Meeting Topic** • there, all of their mitigation actions, FEMA has 45 days to review and provide feedback. They can still submit projects, and ongoing projects are not impacted by the plan review. .	-Prioritize each project deadline for each worksheet
5	**Meeting Topic** • NFIP only applies to NFIP participants – all jurisdictions will not have to fill this out	-
6	**Meeting Topic** • Building Permits – a way to gauge how development has changed or remained the same since the last plan. QR code has been created for participants to fill out this worksheet • New Mitigation Actions – a chance for participants to include new actions into the plan. HC recommends they use the BATool for this and the generative report feature to provide a status update. Any issues with can be directed to HC.	

No.	Description	Action item(s):
7	**Meeting Topic** • Next steps: a pulse check where we are, how many worksheets responses we have received, run a Hazus analysis on the Harvey extent. TT will be able to share some of the maps our GIS team was able to create. HC believes that Alicia is still the best option for Hurricane because Harvey was more of a rainstorm for them not a hurricane	
8	**Meeting Topic** • Capability Assessment walkthrough – it's okay to say no to several because it helps FEMA identify what they can help with. TT recommends sticking with FEMA guidance on sustainability and climate change	
9	**Meeting Topic** • Education and outreach mostly conducted by HC. All participating partners should have access to BATool	TT will send out contact information with presentation slides and other survey material

2025 Harris County *Multi-Jurisdictional Hazard Mitigation Plan Update*

March 12 - 2025

BULLETIN

March 12, 2025 Steering Committee Meeting Recap:

The Core Planning Team hosted the fourth Steering Committee Meeting for the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update with 31 persons in attendance. During the meeting the Steering Committee:

- Discussed information required to run HAZUS modeling
- Received an overview of the project timeline
- Reviewed jurisdictional annex worksheets

NEXT MEETING: WEDNESDAY, APRIL 9, 2025 2:00 PM - 4:00 PM
LOCATION: HOUSTON TRANSTAR - 6922 KATY RD. HOUSTON, TX 77024

Deliverables for Participating Entities:

- **Jurisdictional Annex Worksheets due by March 26, 2025**
  - Worksheet A: Hazard Event History
  - Worksheet B: Capability Assessment
  - Worksheet C: Previous Mitigation Actions
  - Worksheet D: NFIP Compliance
  - Worksheet E: Building Permits

- **Jurisdictional Annex Worksheets due by April 9, 2025**
  - Worksheet F: Hazard Risk Evaluation
  - Worksheet G: New Mitigation Actions

The most up-to-date information can be found on our website at: **www.readyharris.org**
If you have any questions, please don't hesitate to contact the team at: **mitigation@oem.hctx.net**

MARCH 12 - 2025

BULLETIN

Participating Entities (46 total):

Harris County, Harris County Flood Control District, Harris County Office of County Administration, Bellaire, Bunker Hill Village, Deer Park, El Lago, Friendswood, Galena Park, Hedwig Village, Hilshire Village, Humble, Hunters Creek Village, Jacinto City, Jersey Village, Katy, La Porte, Missouri City, Morgan's Point, Nassau Bay, Pasadena, Piney Point Village, Seabrook, Shoreacres, Southside Place, Spring Valley Village, Stafford, Taylor Lake Village, Tomball, Waller, Webster, West University Place, A+ Unlimited Potential Charter Schools, Cypress Fairbanks ISD, Goose Creek CISD, Pasadena ISD, Southwest Public Schools, Spring ISD, Varnett Public Schools, Houston Community College, Lee College, San Jacinto College, University of Houston, Cypress Creek Drainage District, Harris Health, and Gulf Coast Authority.

Community Rating System (CRS) Jurisdictions:

Bellaire, Deer Park, El Lago, Friendswood, Harris County, Jersey Village, La Porte, Missouri City, Nassau Bay, Pasadena, Seabrook, Shoreacres, Taylor Lake Village, and West University Place.

CORE PLANNING TEAM

Chairperson & Spokesperson
Brian Murray
Brian.Murray@harriscountytx.gov

Vice Chairperson
Jeremy Phillips
Jeremy.Phillips@hcfcd.net

Lead Planner
Brittany Ray
Brittany.Ray@harriscountytx.gov

Project Manager
Brian Rutherford
Brian.Rutherford@tetratech.com

Technical Lead Planner
Anita Machiavello
Anita.Machiavello@tetratech.com

Planner
Mark Stalcup
Mark.Stalcup@tetratech.com

Harris County is committed to creating and sustaining communities that are more resilient to disasters. To fulfill this pledge, *Harris County Office of Homeland Security and Emergency Management* (HCOHSEM) has contracted with Tetra Tech to assist in updating the Harris County Multi-Jurisdictional Hazard Mitigation Plan in partnership with cities and special purpose districts throughout the County. Federal regulations require this plan be updated every five years.

Office of Homeland Security and Emergency Management

www.readyharris.org

Capabilities Workshop

March 21, 2025

Capabilities Workshop -- March 21, 2025

	Jurisdiction/Agency	Name	Title/Position
1	Harris County		
2	Harris County Flood Control District		
3	Harris County OCA		
4	City of Bellaire		
5	City of Bunker Hill Village	G. BARRETA	CITY ADMINISTRATOR
6	City of Deer Park		
7	City of El Lago		
8	City of Friendswood		
9	City of Galena Park		
10	City of Hedwig Village		
11	City of Hilshire Village		
12	City of Humble		
13	City of Hunters Creek Village		
14	City of Jacinto City		
15	City of Jersey Village		
16	City of Katy		
17	City of La Porte		
18	City of Missouri City		
19	City of Morgan's Point	Kevin Lagana	Police Chief
20	City of Nassau Bay		
21	City of Pasadena		
22	City of Piney Point Village		
23	City of Seabrook		
24	City of Shoreacres		
25	City of Southside Place		
26	City of Spring Valley Village	Jose Gomez	Dev. Services Manager
27	City of Stafford		
28	City of Taylor Lake Village		
29	City of Tomball		
30	City of Waller		
31	City of Webster		
32	City of West University Place		
33	Cypress Creek Drainage District		
34	Gulf Coast Authority		
35	Harris Health		
36	A+ Unlimited Potential Charter Schools		
37	Cypress Fairbanks ISD		
38	Goose Creek ISD		
39	Houston Community College	David Potts	Mgr EM
40	Lee College		
41	Pasadena ISD		
42	San Jacinto College		
43	Southwest Public Schools		
44	Spring ISD		
45	University of Houston		
46	Varnett Public Schools		

Steering Committee Meeting #5

April 9, 2025

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
April 9, 2025 / 2:00 PM – 4:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALTERNATE NAME (OPTIONAL)	SIGNATURE
Harris County Office of Homeland Security & Emergency Management	Brian Murray	Deputy EMC / PIO		
Harris County Office of Homeland Security & Emergency Management	Brittany Ray	Resilience Planner		
Harris County Office of County Administration / Harris County Public Health	Leah Barton	Deputy County Administrator / Executive Director Harris County Public Health		
Harris County Flood Control District	Jeremy Phillips	Director, Asset Management Division		
City of Bellaire	Deacon Tittle	Fire Chief / Emergency Management Coordinator		
City of Deer Park	Jamie Galloway	Emergency Services Director		
City of El Lago / Taylor Lake Village	Carl Nunn	Chief of Police / EMC		
City of Friendswood	Steven Simmons	Deputy Fire Marshal		
City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal		
City of La Porte	Johnny Morales	Emergency Management Coordinator	Amber Cosby Bethea Deputy EMC	
City of Missouri City	Trameka Jewett	Emergency Management Coordinator	EM Spec.	
City of Pasadena	Azell Carter	Emergency Management Coordinator		

ORGANIZATION	NAME	TITLE	ALTERNATE NAME (OPTIONAL)	SIGNATURE
City of Seabrook	Kevin Rodgers	Office of Emergency Management Director	Kevin Rodgers / Brian Bray	
City of Shoreacres	Troy Harrison	City Manager		
National Weather Service	Lance Wood	Science and Operations Officer		
North Houston Association	Marlisa Briggs	President		
Houston Community College	Robert McCracken	Associate Vice Chancellor, Risk Mgmt & Resilience		
San Jacinto College	Ali Shah	Director, Emergency Management	Kimberley Harvell	
H-GAC	Amanda Ashcroft	Planner		
Cypress Creek Drainage Improvement District	Calvin Cobb	President		
Coastal Prairie Conservancy	Mary Anne Piacentini	President and CEO		
Greens Bayou Coalition	Colleen Gilbert	Executive Director		
Cy-Fair ISD	Scott Hudson	Director of Emergency Management		
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Director		
SETRAC	James Meaux	Regional T&E Coordinator		
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience		
Houston SPCA	Dawn Blackmar	Director, Emergency Management		
Harris County Dept. of Education	Leslie Etheridge	Director, Center for Safe and Secure Schools		

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
April 9, 2025 / 2:00 PM – 4:00 PM

NON-STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	PHONE	SIGNATURE
City of Waller	Charles Mistrik	EMC	979-645-7557	
City of Katy	Greg Goodeckei	EMC	832-418-1574	
City of Tomball	Joe Sykora	Fire Chief / EMC	281 290 1063	
Grant Cy-Fair ISD	Grant Brandt	EMC	469-768-4156	
Village FD	Howard Miller	Fire Chief / EMC	713-468-7741	
west university FD	James Novak	Fire Chief / EMC	713-662-5836	
GREATER SE MGMT DISTRICT	DR. ABDUL HALEEM Muhammad	Vice Chair	8322369956	
west University FD	Aaron Rosenfield	Asst. Chief	713-662-5837	
Southside Place	Andrew Deavers	Assistant City Mgr	7136682341	
Bunzon John	Bunzon John	Deputy EMC	832-286-5620	
~~Kristina Clark~~ HCOHSEM	Kristina Clark	Planning Supervisor	713.881.3100	

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
April 9, 2025 / 2:00 PM – 4:00 PM

PUBLIC SIGN-IN

NAME	CITY/JURISDICTION	ORGANIZATION	PHONE	EMAIL
ATRI SEN	RESALIRE →		220.281.4879	
Jennifer Leighton	Resalire →		832 646 7390	

AGENDA

APRIL 16, 2025 @ 11:00 – 12:00 PM

1. **Worksheets A-E Status Report**

2. **Review of 2025 Goals and Objectives**

3. **Summary of Outreach Material**
   - SCM #5 Bulletin
   - Worksheet F: Hazard Evaluation (since 2020) *Due: April 30, 2025*
   - Worksheet G: New Mitigation Actions *Due: April 30, 2025*
   - Worksheet G QR Code
   - Handout 1: Prioritization of Actions
     - Strategy for outreach
   - Handout 2: Analysis of Actions by Hazard and by Category
   - Mitigation Catalog
   - New Development QR Code *Due: April 30, 2025*
   - Problem Area QR Code *Due: April 30, 2025*

4. **Hazard Risk Ranking**
   - Overview of R.R. spreadsheet
   - GIS results

5. **New Action Prioritization**
   - To be discussed further at next week's CPT call

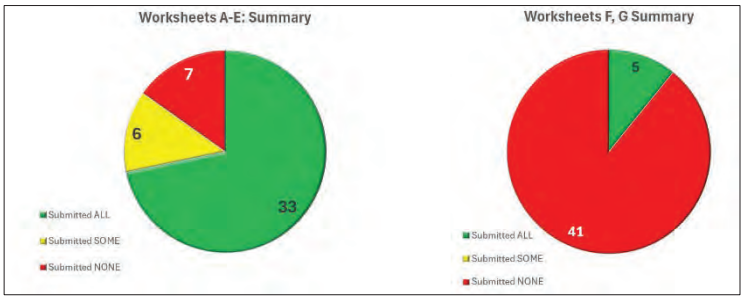
6. **Need for Planning Partner Workshops**
   - Virtual? In-person?
   - When?

7. **Closing Discussion / Adjourn**
   - Steering Committee Meeting #6 Adoption Resolution Example

AGENDA

APRIL 23, 2025 @ 11:15 – 12:15 PM

1. **Project Timeline and Status Update**
   - Worksheet Status Report –
     i. A+UPC School
     ii. Piney Point Village
     iii. Taylor Lake Village
     iv. Harris Health
     v. Southwest Pub. Schools
     vi. Spring ISD
     vii. Varnett Public Schools
   - Set SC Meeting #6 date/time/location
     i. Presentation of the HMP DRAFT
     ii. Adoption Resolution Letters
     iii. Plan Maintenance and Integration
   - Public Review and Comment Period – announcements online?



2. **Poll Results from Planning Partner Poll**

Name2	Email Address	Jurisdiction/Agency	Would it be beneficial for there to be a Planning Partner Workshop organized in your area over the coming weeks?	If yes, would you prefer the Workshop to be held in-person in a location near you? Or virtually through Microsoft Teams?	What kinds of assistance would be most helpful during a Planning Partner Workshop? Please choose any/all that apply.	Generally speaking, if a Planning Partner Workshop were to be organized, what day(s) of the week would work best for you? (Select all that apply)	What time of day would be most helpful for the Planning Partner Workshop to be held?	Additional comments or requests
Thomas Quinn	tquinn@lee.edu	Lee College	Yes	Virtually	Worksheet G: New Mitigation Actions;	Monday;Tuesday;	Any	N/A
Tricia Times	tricia.times@gccisd.net	Goose Creek CISD	Yes	Virtually	Worksheet A B;	Monday;	9 am	None
Jamie Galloway	jgalloway@deerparktx.org	City of Deer Park	No	In-person	Worksheet G: New Mitigation Actions;	Thursday;	Afternoon	None
Charles Mistric	CMistric@WallerTexas.Gov	Waller / Harris County	Yes	In-person	Worksheet G: New Mitigation Actions;	Thursday;Friday;	1400 hours.	No.
SHERRI DITRICH	SHERRI.DITRICH@NASSAUBAY.COM	CITY OF NASSAU BAY	No	Virtually	Worksheet F: Hazard Risk Evaluation; Worksheet G: New Mitigation Actions;	Monday;Wednesday;	MORNING-MID-AFTERNOON	NONE
Tom Ehlers	tehlers@cityofgalenapark-tx.gov	City of Galena Park	Yes	Either	Worksheet F: Hazard Risk Evaluation; Worksheet G: New Mitigation Actions;	Tuesday;Wednesday;Thursday;	9-3	N/A
Calvin Cobb	calvin@calvincobb.com	Cypress Creek Drainage Improvement District	No	Not Applicable	Not Applicable;	Not Applicable;	Not Applicable	Not Applicable
Todd Ward	Todd.ward@hcfcd.hctx.net	Harris County Flood Control District	Yes	Virtually	Worksheet G: New Mitigation Actions;	Wednesday;Friday;	10-5	N/A

3. **Harris County Department Webinar**
   - TT Availability: Monday (April 28th) or Tuesday (April 29th)
   - Calendar Invites for Webinar – HCOHSEM or TT?
   - Presentation will be made available on Shared Teams Channel for approval

4. **HMP Graphics and Imagery**

5. **Closing Discussion / Adjourn**

Meeting Title: Harris County CPT Weekly Check-in

Date: April 23, 2025, 11:15am to 12:15pm

Location: Teams

Attendees:

Name	Organization	Representative, Position
Brittany Ray	HCOHSEM	Resilience Planner
Benzon John	HCOHSEM	Assistant Deputy EMC
Brian Rutherford	Tetra Tech	Project Manager
JaLeesa Tate	Tetra Tech	Director
Anita Machiavello	Tetra Tech	Technical Lead
Mark Stalcup	Tetra Tech	Planner

Meeting Notes:

No.	Description	Action item(s):
1	**Project Timeline and Status Update** • We have not receive any worksheets from seven participating partners: o A+UPC School o Piney Point Village stated they did submit their worksheets and Brittany has someone working to retrieve them. o Taylor Lake Village (is a CRS community) o Harris Health has had a VP change and were not sure if they could participate but later received approval. We need to touch base with them. o Southwest Pub. Schools: no response so far o Spring ISD: Brittany spoke with them Monday and they are working on it this week. o Varnett Public Schools o Pasadena ISD: two of the three have not been received. • SC Meeting #6 date/time/location o Last week of May is the TDEM conference o June 4th ▪ Announcements and meeting materials can wait until 1st of May to assess where we are with worksheets. o SC meeting review – 10 days, o Public Comment review – 2 weeks ▪ Public engagement subcontractor to be used during these two weeks ▪ TT and HCOSHEM to discuss outreach strategy	▪ Brittany to reach out to Taylor Lake Village POC and Brian R. offered to assist the Village as needed to complete the worksheets. ▪ Benzon wants TT to send an email today letting all jurisdictions with missing worksheets know that they must submit by Friday. ▪ For those that do not make the deadline Brittany will draft a letter on Monday, April 28th letting them know that we have to move forward without their participation. ▪ TT to update visual timeline based on feedback and send to HC.
2	**Poll Results from Planning Partners** • Mark proposed one on one consultation with partners that responded, and Benzon and Brittany agreed.	▪ TT to spearhead direct assistance via phone
3	**Harris County Department Webinar** • Calendar Invites will be sent by HCOHSEM • Date: Tuesday April 29th, 10-11:30 or 1-2:30p • Presentation materials made available to everyone prior to webinar	▪ HCOHSEM to send out invites and communication

No.	Description	Action item(s):
4	**HMP Graphics and Imagery** • TT had a meeting with BES and they are on track to complete static maps by the first week of May. • HCOHSEM to send TT any graphics or imagery that we want incorporated into the plan.	▪ HCOHSEM to send graphics and imagery to include in HMP
5	**Requirements for CRS jurisdictions** • Requirements are focused on the process to develop the plan and overall stakeholder engagement. TT can send HCOHSEM these requirements for their review.	
6	**Plan Maintenance Strategy**	▪ Brittany working to incorporate changes in their strategy from the previous plan and will send that over to TT soon. ▪ TT to send an additional list of questions needed for Planning Process narrative

Tetra Tech Webinar for County Departments

April 29, 2025

Hazard Mitigation Planning
in
Harris County

Maximizing the Value of the 2025 Multi-Jurisdictional Hazard Mitigation Plan (MJ-HMP)

Agenda

- Welcome
- What is Hazard Mitigation?
- Hazard Mitigation Planning Process
- Purpose of the 2025 MJ-HMP Update
- Benefits of an Updated MJ-HMP
- Eligible Mitigation Actions by County Department
- Effectively Utilizing the MJ-HMP
- What's new in 2025?
- Next Steps for County Departments
- Open Discussion

What is Hazard Mitigation?

HARRIS COUNTY
Office of Homeland Security & Emergency Management

- Actions taken to reduce or eliminate long-term risk to people and property from hazards
- Focuses on breaking the cycle of disaster damage, reconstruction, and repeated damage

» Hazard

- Something that has the potential to cause damage, injuries, and losses.

» Exposure

- People, property, infrastructure, and other resources located within areas that could be affected by hazards.

» Vulnerability

- The probability that people, property, infrastructure, and other resources will be harmed or damaged.

» Risk

- Probability and severity of damages to people, property, and infrastructure.

Hazard Mitigation Planning Process

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Upcoming Milestones	Expected Completion
~~Risk Assessment and Vulnerability Assessment~~	~~April 25 , 2025~~
Capabilities Assessment	April 30, 2025
Steering Committee Review and Comment Period	Early June 2025
Public Review & Comment Period	Mid-June 2025
HMP Submitted to TDEM for formal review	Early July 2025
HMP Submitted to FEMA for formal review	Early to Mid July 2025
Approved-Pending-Adoption (APA)	August 2025

Purpose of the 2025 MJ-HMP Update

HARRIS COUNTY
Office of Homeland Security & Emergency Management

» Purpose of the Mitigation Strategy

»The **Mitigation Strategy** is the County's roadmap to reduce the risk of hazards identified in the HMP. The strategy is based on hazard impacts, asset vulnerability, and the County's capabilities.

»**Mitigation Actions** are specific activities, such as policies, projects, and studies, that stakeholders identify to reduce risk.

»Need to think about the following…

- »**Goals** - What outcomes do you want to achieve?
- »**Actions** - What actions will be taken to reduce risk?
- »**Action Plan** - How will the action be prioritized and implemented?

» How do you reduce risk for a hazard?

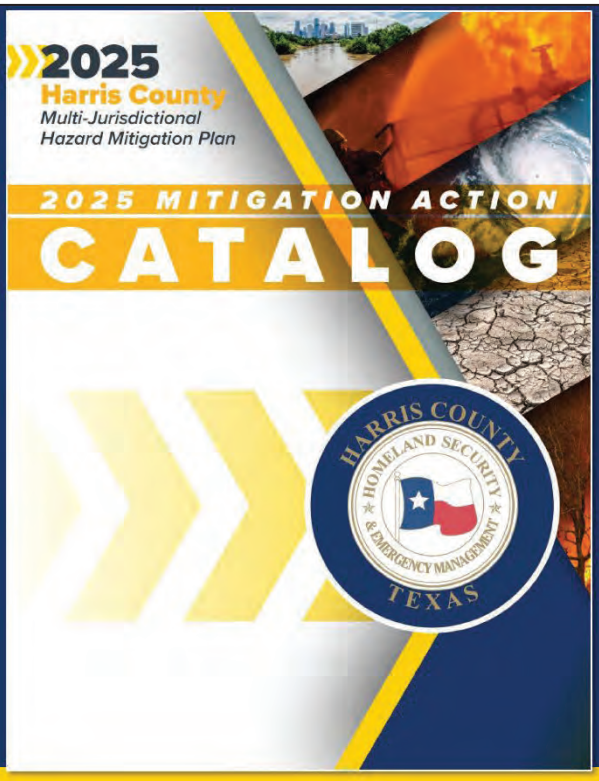
»**Manipulate the Hazard**: Structural flood control (e.g., flood walls, levees)

»**Reduce/Eliminate Exposure**: Acquire flood-prone properties

»**Reduce Vulnerability**: Retrofit (e.g., floodproofing structures)

»**Increase Capability**: $, preparation, technical assistance, planning enforcement

2025 HC MJ-HMP Mitigation Action Catalog



Benefits of an Updated HMP

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Using Your Mitigation Strategy to Reduce Risk?

What is a Mitigation Strategy?

A group of activities, such as projects or actions, to reduce the impacts of the hazards of concern on your community

- Plans and Regulations
- Structure and Infrastructure Studies and Projects
- Natural Systems Protection Studies and Projects
- Education and Awareness Programs

Terms to describe the Mitigation Strategy include:

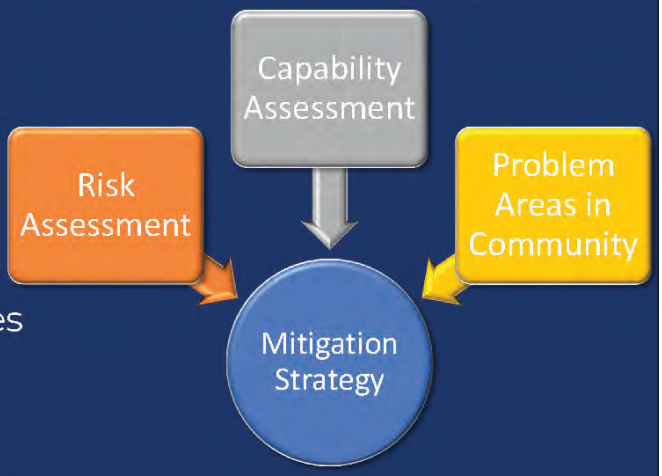
- Mitigation Action Plan or Action Plan
- Mitigation Projects or Initiatives or Actions

Connecting to the Mitigation Strategy

Need a clear connection between vulnerabilities identified in the risk assessment and proposed mitigation actions.

The capability assessment can provide insight into challenges and opportunities for the mitigation strategy.

All actions proposed in the mitigation strategy must have a factual basis tied to hazards (this shouldn't be a wish list!)



Eligible Mitigation Actions by County Department

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Hazard Mitigation Action Examples

Planning and Regulations	Structure and Infrastructure Projects	Education and Awareness Programs	Natural Systems Protection
• Shoreline setbacks • Subdivision regulations • Building codes and enforcement • Capital improvement programs • Open space preservation • Stormwater management regulations and master plans	• Acquisitions and elevations of flood prone structures • Utility undergrounding • Structural retrofits • Floodwalls and retaining walls • Detention and retention structures • Bridges and culverts • Safe rooms	• Radio or television spots • Websites with maps and information • Real estate disclosure • Training • NFIP outreach • StormReady or Firewise Communities	• Sediment and Erosion control • Stream Corridor Restoration • Forest Management • Conservation Easements • Wetland Restoration and Preservation • Greenways and Blueways • Living Shorelines

» Public Infrastructure / Engineering Dept.

- Drainage Improvements
- Road elevations
- Stormwater detention

» Public Health / Emergency Services

- Generators
- Shelter retrofits
- Cooling or Warming Centers
- Outreach and Education

» Flood Control District

- Channel improvements
- Floodplain restoration

» IT / Utilities

- Backup power
- Comms hardening
- Cybersecurity

» Planning / Housing

- Property acquisition
- Resilient zoning
- Wildfire mitigation

» Other Mitigation Actions to Consider:

Local Disaster Debris Management Plans

Substantial Damage Management Plan

Actions to address high-hazard or significant-hazard dams

Public education and outreach programs

Generators at critical facilities and community lifelines

Floodproofing critical facilities and community lifelines

Addressing repetitive and severe repetitive loss properties

Utilizing the 2025 MJ-HMP *Effectively*

HARRIS COUNTY
Office of Homeland Security & Emergency Management

- Strategic Planning: Guide for capital improvements and resilience
- Grant Writing: Use risk data to strengthen applications
- Policy alignment: integrate with other plans

- Department Coordination: Align mitigation work across agencies
- Community Outreach – share findings to engage the public

What's New in 2025?

HARRIS COUNTY
Office of Homeland Security & Emergency Management

- **Updated hazard data:**
  - Coastal Erosion
  - Extreme Heat
  - Utility Failure
  - Public Health Emergency
- **Climate-informed risk assessments**
- **Jurisdiction-specific mitigation actions**

Next Steps for County Departments

HARRIS COUNTY
Office of Homeland Security & Emergency Management

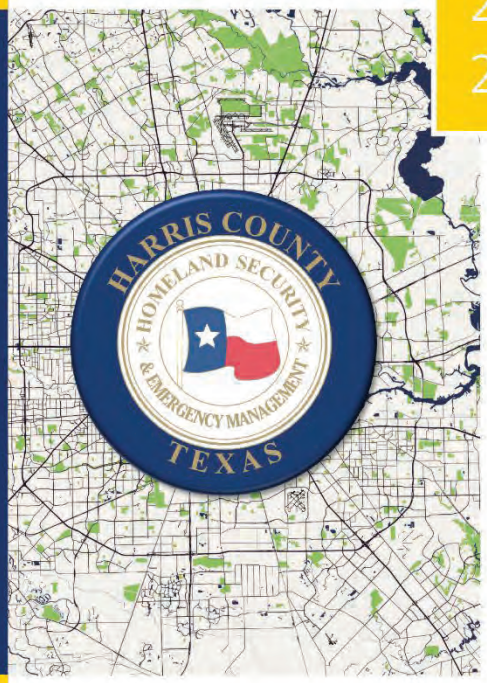
Open Discussion and Questions

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Contact Us

mitigation@oem.hctx.net

CorePlanningTeam@tetratech.com



2025

Meeting Title: **Core Planning Team Meeting**

Date: **Wednesday, April 30, 2025**

Location: **Microsoft Teams**

Attendees:

Name	Organization	Representative, Position
Brittany Ray	HCOHSEM	Assistant Planning Chief
Brian Murray	HCOHSEM	Deputy EMC/PIO
Benzon John	HCOHSEM	Deputy EMC
JaLeesa Tate	Tetra Tech	Director
Brian Rutherford	Tetra Tech	Project Manager
Anita Machiavello	Tetra Tech	Technical Lead
Mark Stalcup	Tetra Tech	Planner

Meeting Notes:

No.	Description	Action item(s):
1	**Worksheets Update** • 11 jurisdictions have submitted at least one of the two worksheets (F and G) that are due today	• Brittany plans to send out communication to the jurisdictions about the deadline extension to Friday, May 2, with the Worksheet . • Benzon would like to see the timeline changes that are discussed and be able to see how it will impact our overall timeline.
2	**SC meeting #6 June 4th** • CPT discussed whether the last meeting should be at TranStar or somewhere else.	• HCOHSEM to decide where the location will be tomorrow. • TT will send the invite to SC #6 meeting
3	**Gaps in information for Worksheets Submissions** • Highlights and Discussion	• TT plans to provide a summary of the missing information we need from each jurisdiction after we've reviewed those that have been submitted.
4	**Public Review Period** • Public Review and Comment period on schedule to be from June 18 to July 2 • Four Public Meetings will need to take place, one in each commissioner precinct. At these meetings, a digital PDF version of the MJ-HMP will be made available for the public to review while the digitized, interactive tool will be completed in conjunction with Bent Ear Solutions at a later date • Collaboration between TT and HCOHSEM will facilitate the creation of flyers which will announce each of the four meetings/times/locations, and that flyer will be shared on the County's website as well as on the County's social media platforms • Follow-up conversations will be had at the next CPT meeting to further discuss these details	• TT will put together a proposal for what we recommend for the public meeting will consist of. • HCOHSEM will begin identifying potential locations and venues for each of the four meetings to take place
5	**Hazus – Earthquake results** • Highlights and Discussion It was returning zero results for the County, our recommendation is to go to the next higher magnitude event so that we can show impacts with the plan.	• Benzon is okay with this as long as we mention it within the plan.

No.	Description	Action item(s):
6	**Other Notes** - TT recommend sending a memo that shares the overall public comments that were received and our plan to incorporate them into the plan. - TT provided an update on BES where they are prioritizing static maps first and on track to have them complete by May 5th. - TT confirmed that there will be a digital version of the plan for public comment.	•

2025 Harris County MJ-HMP Core Planning Team Meeting

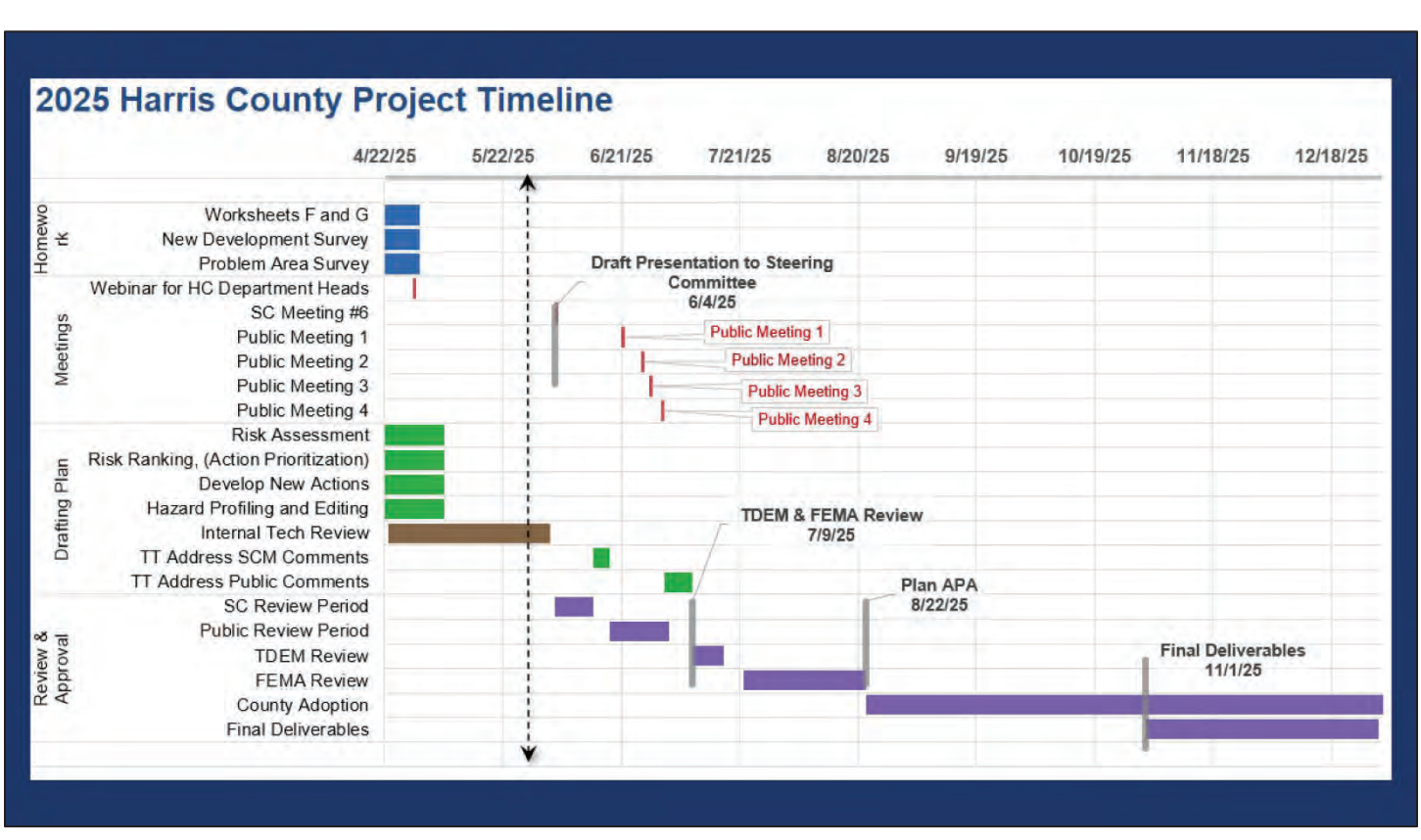
May 28, 2025

Agenda

- Remaining Timeline
- SCM 6 Logistics
- SCM 6 Agenda
- Hazard Risk Ranking for the County
- Prioritization of Action Iteam
- MJ-HMP Graphics/Figures
- Adoption Resolution Examples
- Community Workshops

Project Timeline & *Remaining Tasks*

HARRIS COUNTY
Office of Homeland Security & Emergency Management



SC Meeting #6 – *Meeting Logistics*

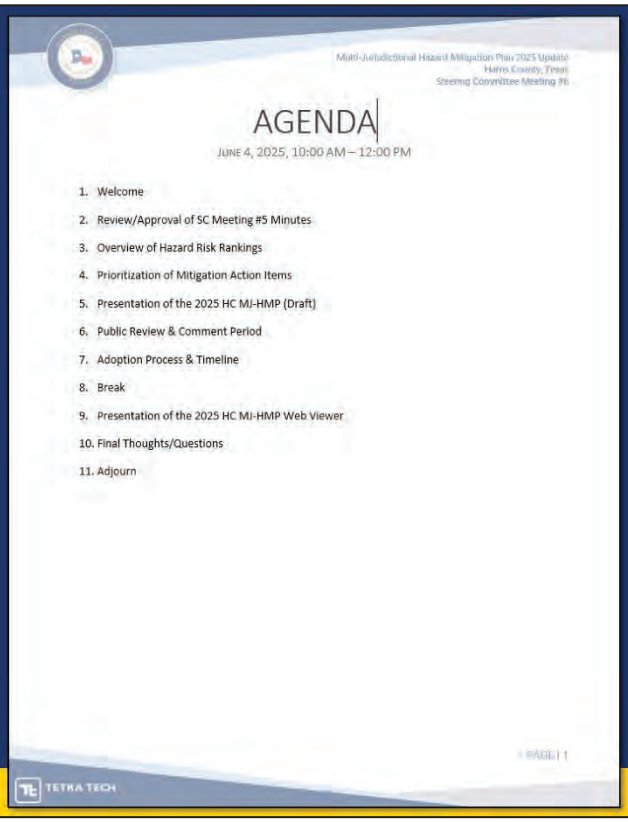
HARRIS COUNTY
Office of Homeland Security & Emergency Management

- **Arrival time at Pasadena City Hall**
- **Meeting materials provided by TT:**
  - Sign-in Sheets, Agendas
  - Printed copy of the MJ-HMP Draft, poster prints
  - Adoption Resolution Example/Template
  - Projector? Cords?
- **Meeting materials provided by HCOHSEM:**
  - Printer?
  - Internal Laptop for residents to explore the Interactive viewer

SC Meeting #6 – *Agenda Preview*

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Preliminary Agenda



Multi-Jurisdictional Hazard Mitigation Plan 2025 Update
Harris County, Texas
Steering Committee Meeting #6

AGENDA

JUNE 4, 2025, 10:00 AM – 12:00 PM

1. Welcome
2. Review/Approval of SC Meeting #5 Minutes
3. Overview of Hazard Risk Rankings
4. Prioritization of Mitigation Action Items
5. Presentation of the 2025 HC MJ-HMP (Draft)
6. Public Review & Comment Period
7. Adoption Process & Timeline
8. Break
9. Presentation of the 2025 HC MJ-HMP Web Viewer
10. Final Thoughts/Questions
11. Adjourn

PAGE | 1

TETRA TECH

County Hazard Risk Ranking

HARRIS COUNTY
Office of Homeland Security & Emergency Management

» Hazard Rankings: *2020 vs 2025*

Coastal Erosion	Low
Dam/Levee Failure	Low
Drought	High
Earthquake	Low
Flood	High
Hurricane /Coastal Storms	High
Mass Movements	Low
Severe Weather	High
Tsunami	Low
Wildfire	High

Flooding	HIGH
Hurricane, Coastal Storm	HIGH
Extreme Heat	HIGH
Severe Weather	HIGH
Drought	HIGH
Extreme Cold	MODERATE
Dam/Levee Failure	MODERATE
Toxic Release / Hazardous Materials	MODERATE
Technology Failure	MODERATE
Energy Pipeline Failure	MODERATE
Coastal Erosion	LOW
Earthquakes	LOW
Mass Movement	LOW
Tsunami	LOW
Wildfire	LOW
Public Health Emergency	LOW
Utility Failure	LOW

Prioritizing Action Items

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Meeting Graphics and MJ-HMP Figures

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Maps to include in the SCM #6 Presentation

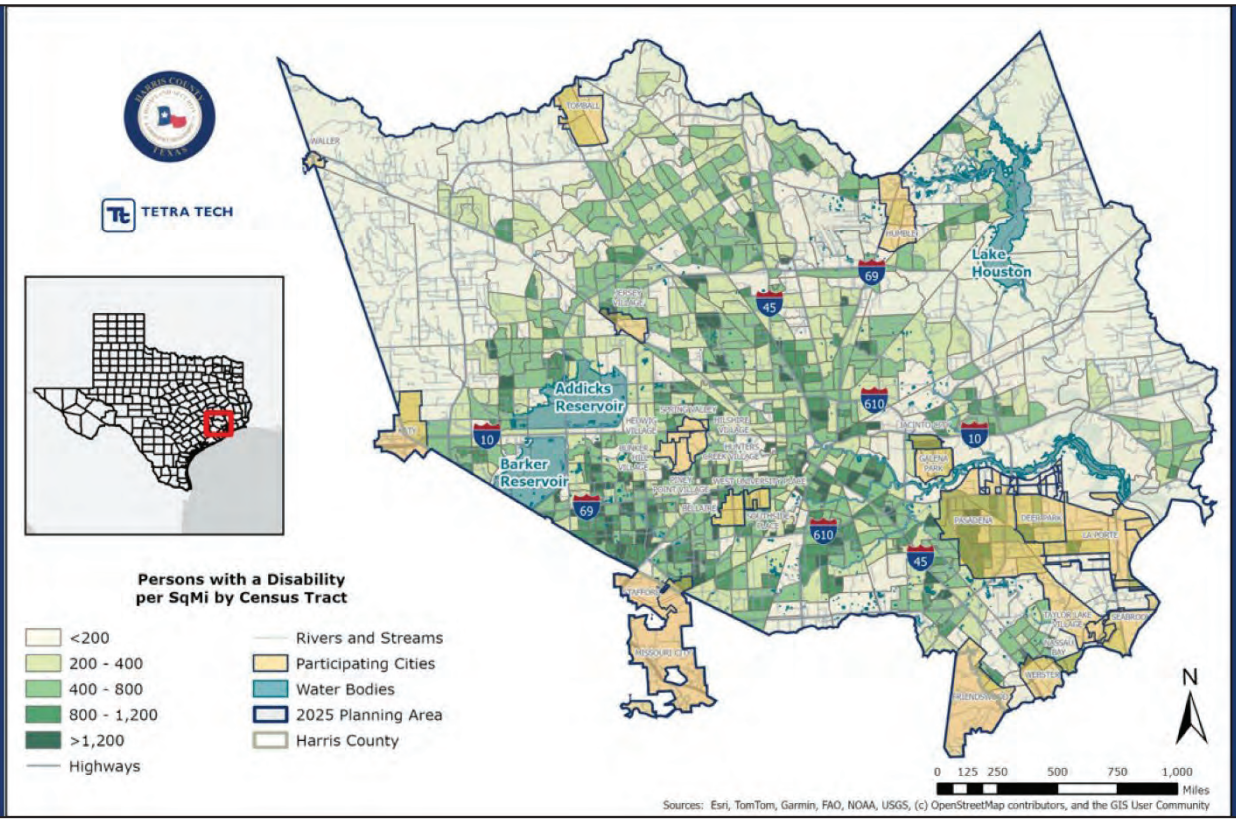
1. Vulnerable Population Maps

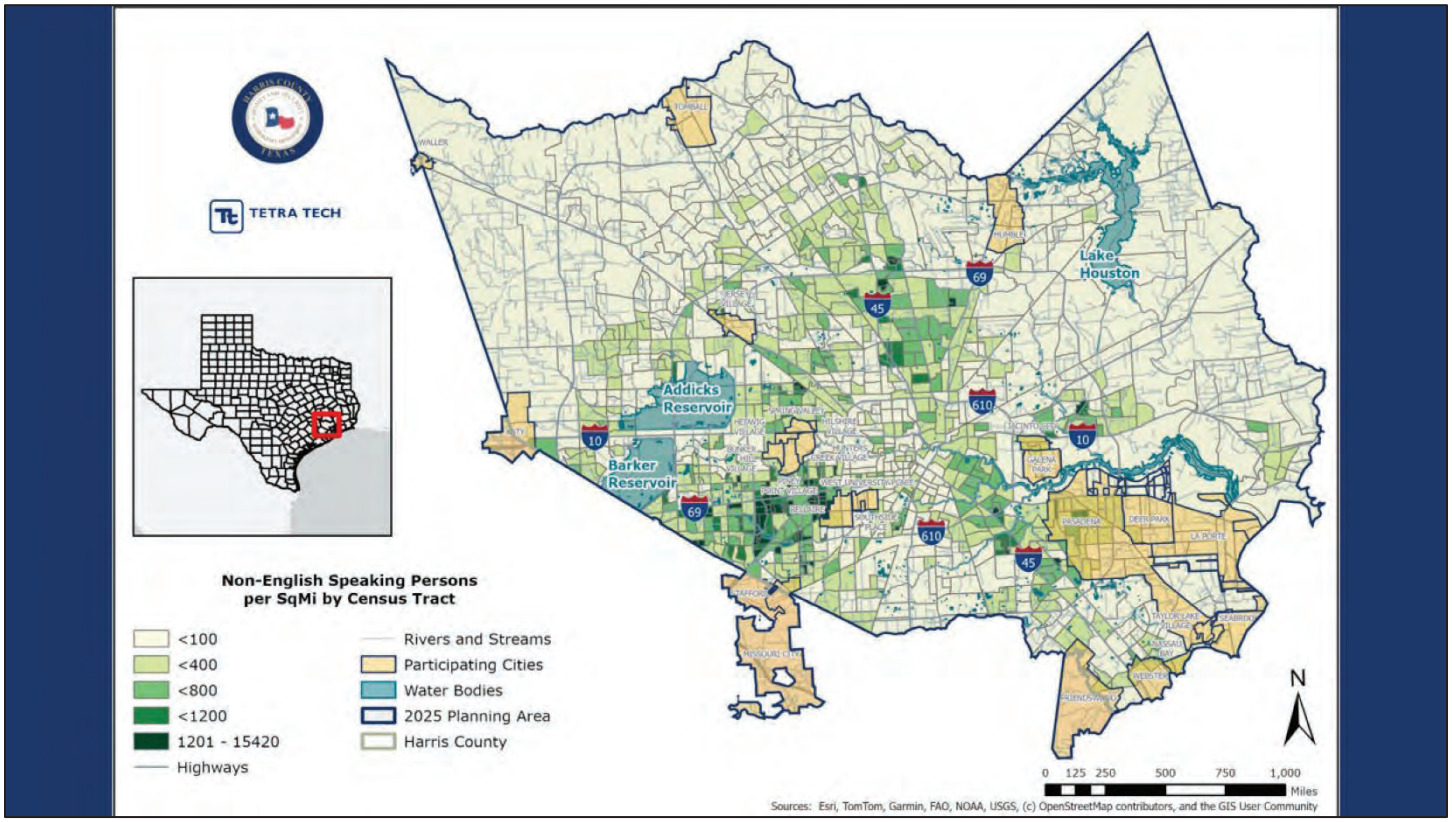
Disabilities, Non-English, Young, Elderly, Poverty, Total Population

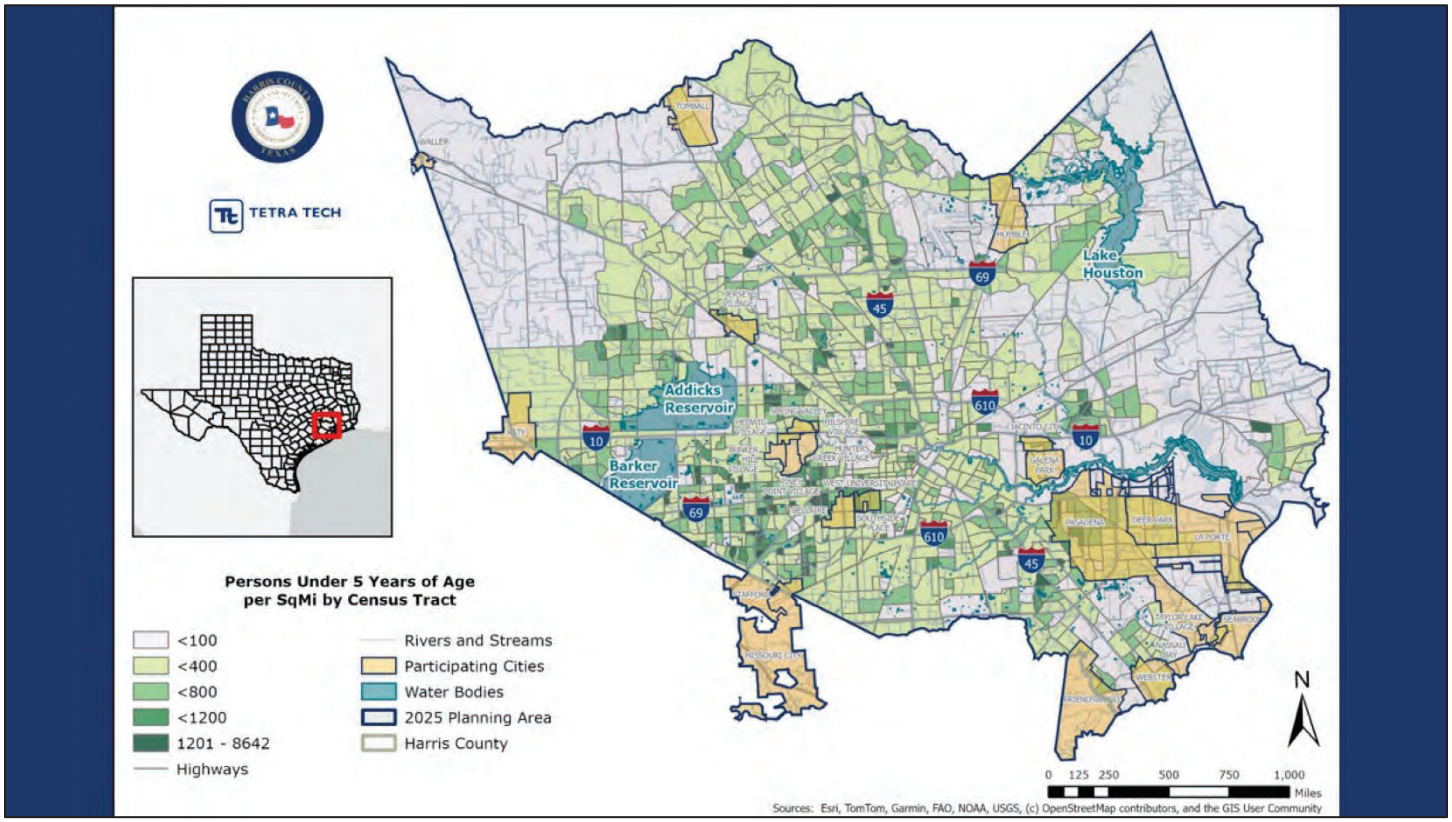
2. Community Lifeline maps?

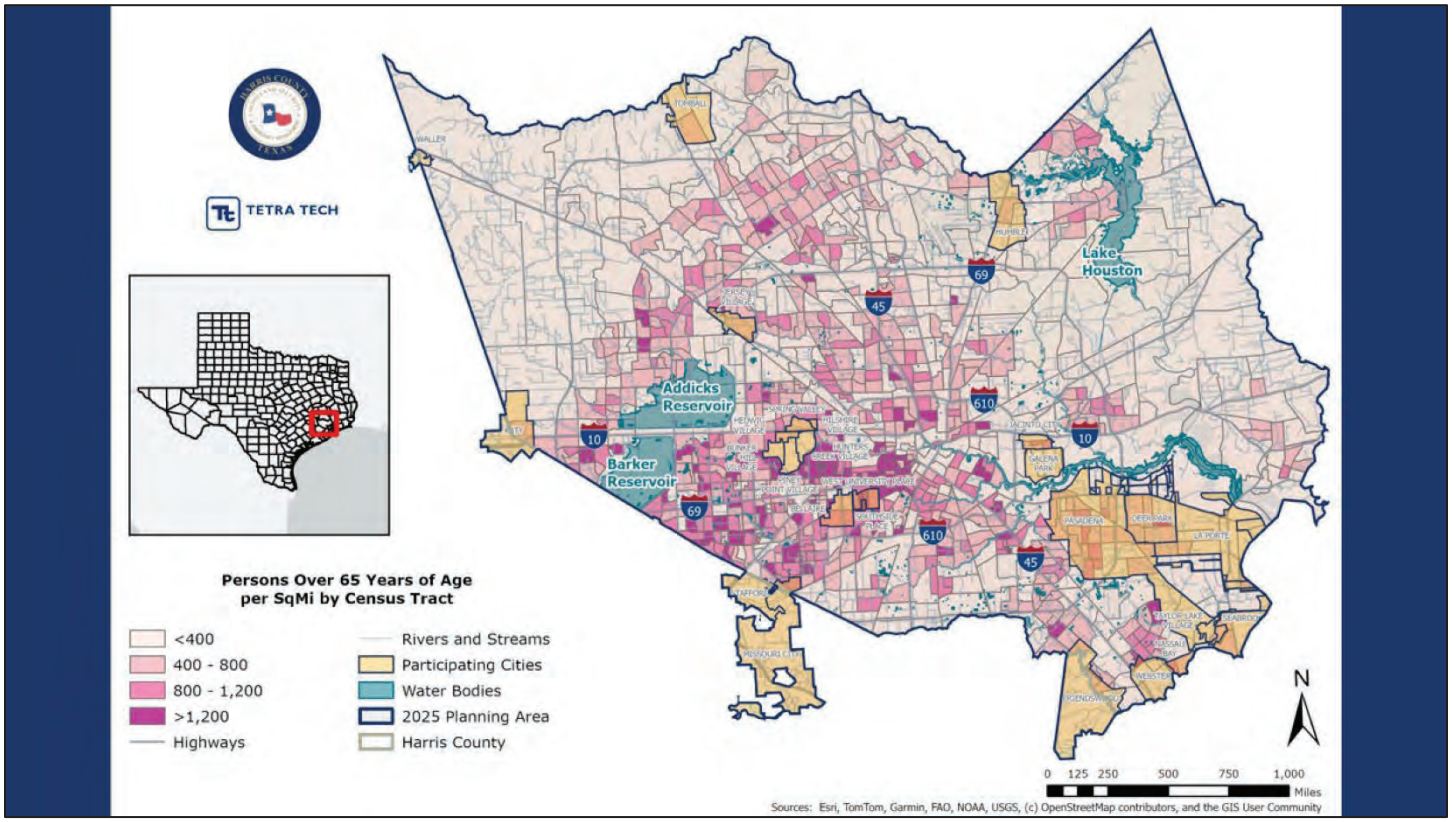
Communications, Energy, Food, Health, HazMat, Safety, Transportation, Water System

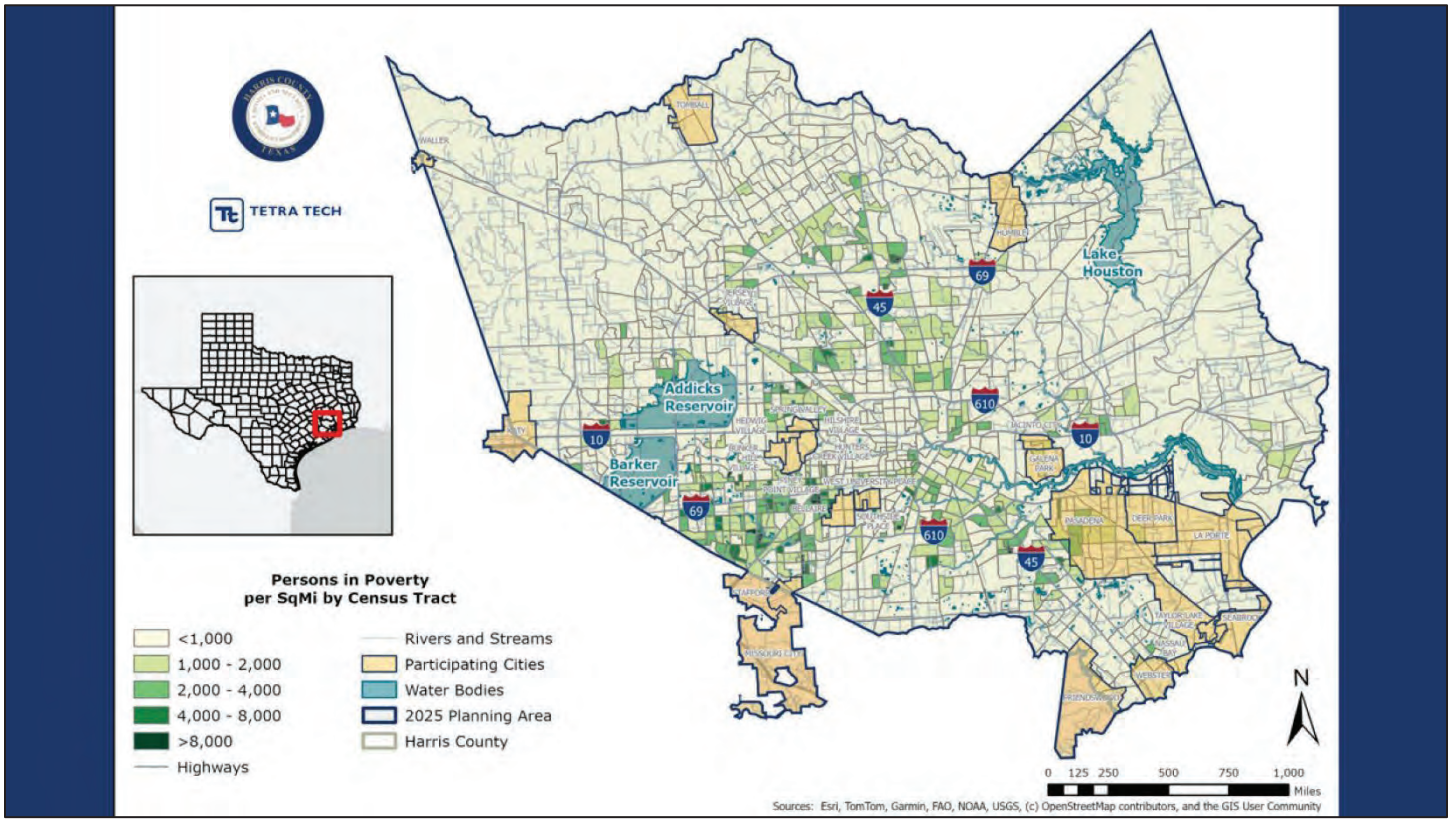
3. What else?

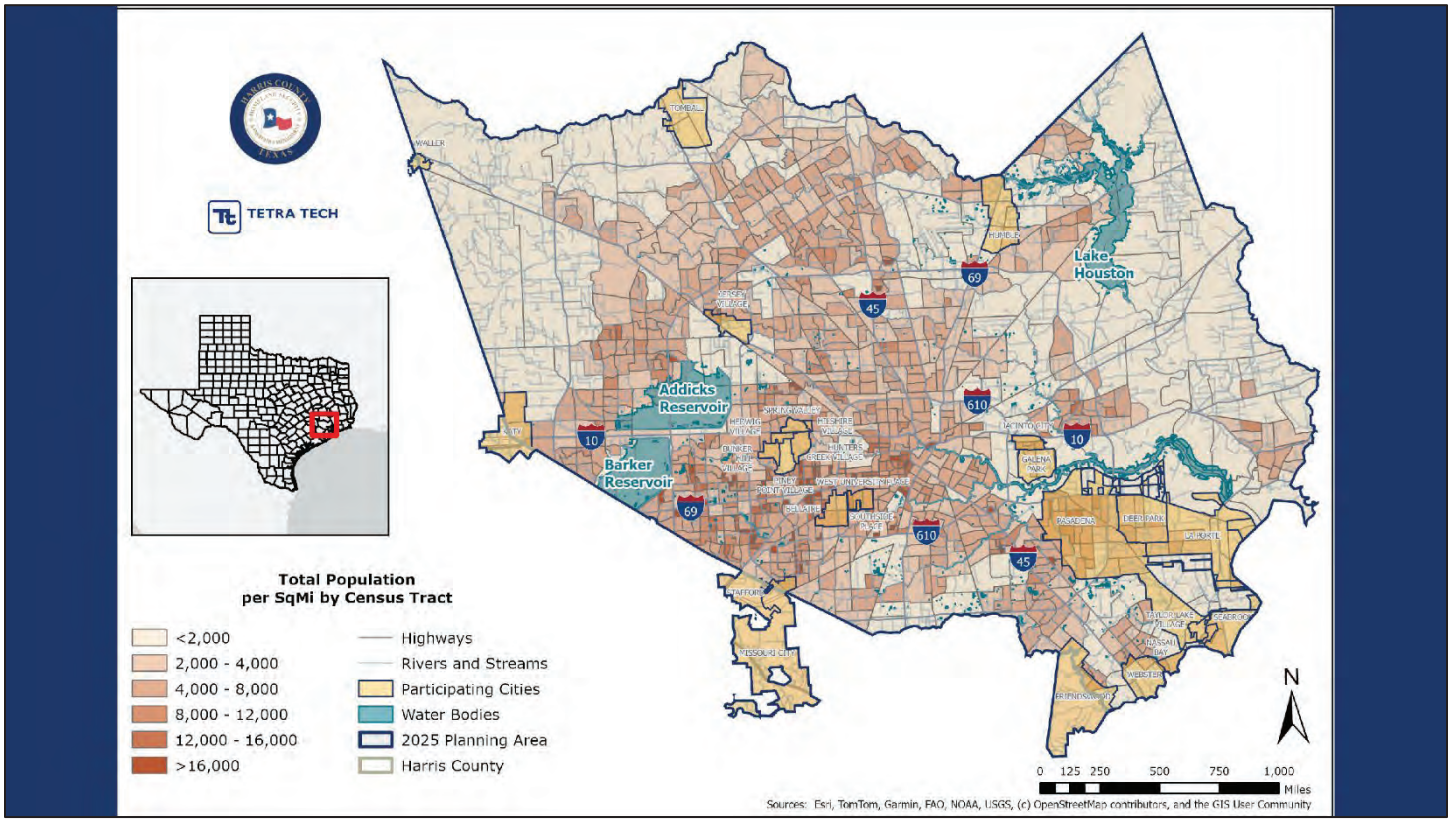


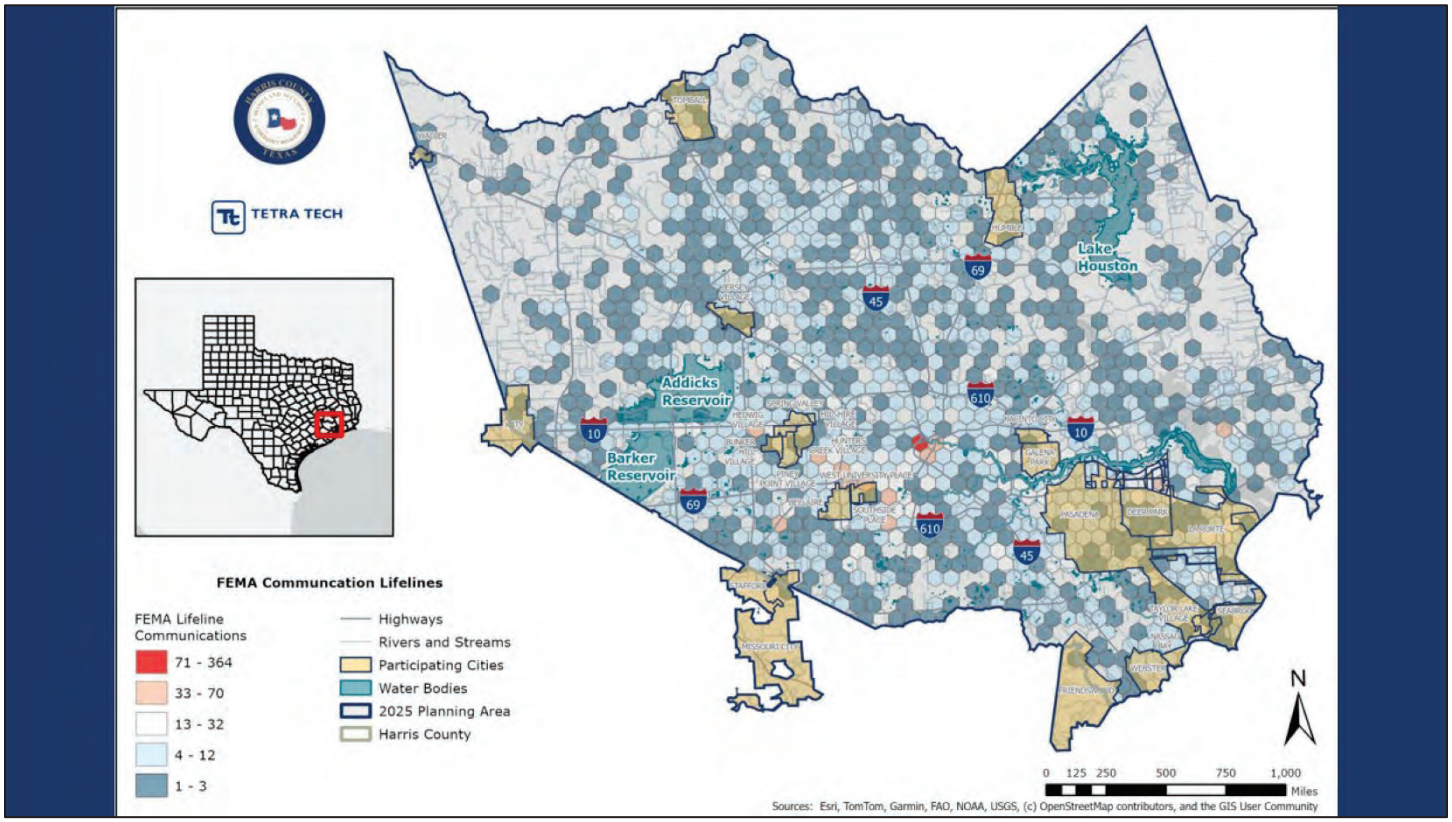

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Non-English Speaking Persons per SqMi by Census Tract
<100
<400
<800
<1200
1201 - 15420
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

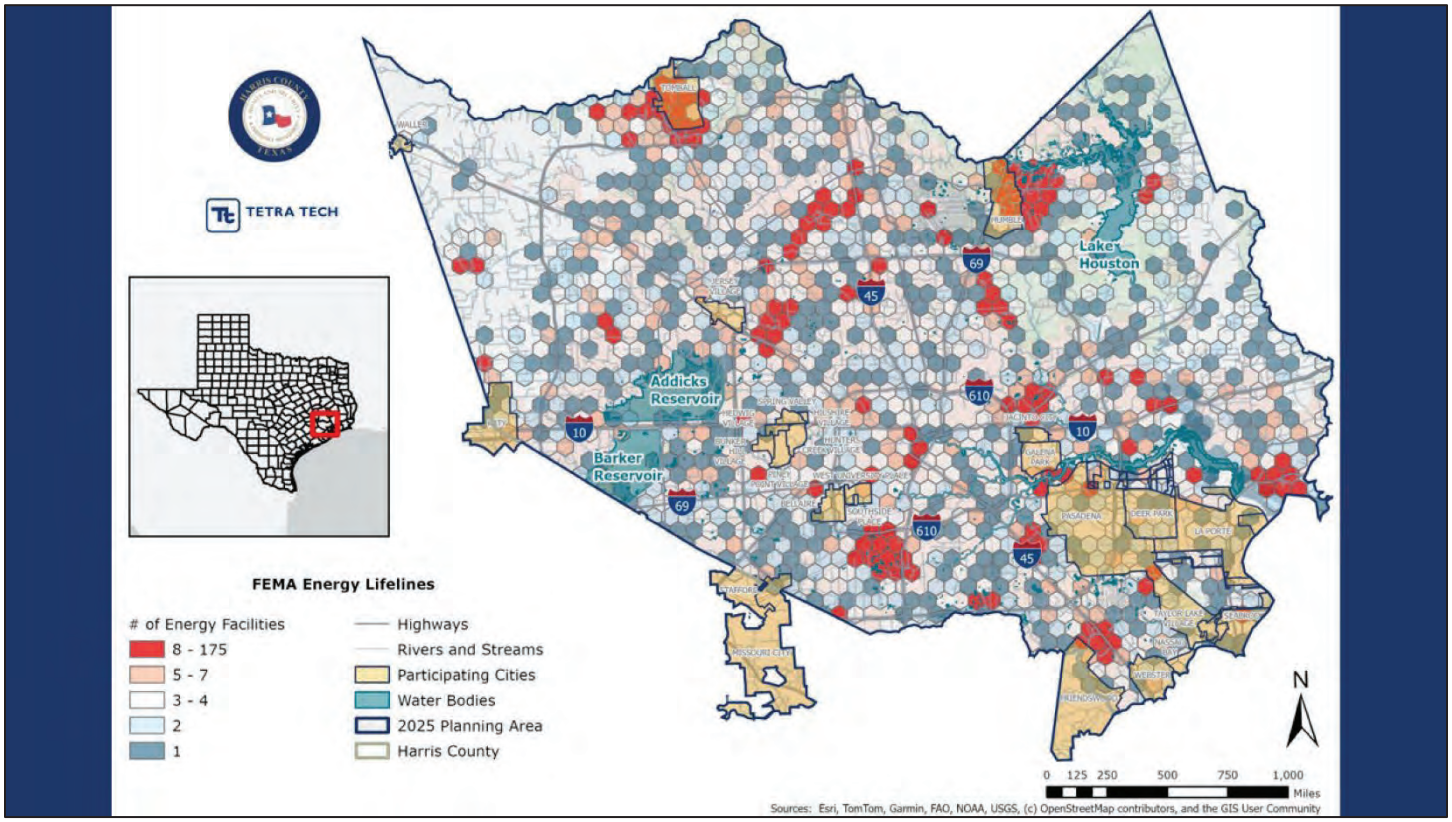

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Persons Under 5 Years of Age per SqMi by Census Tract
<100
<400
<800
<1200
1201 - 8642
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

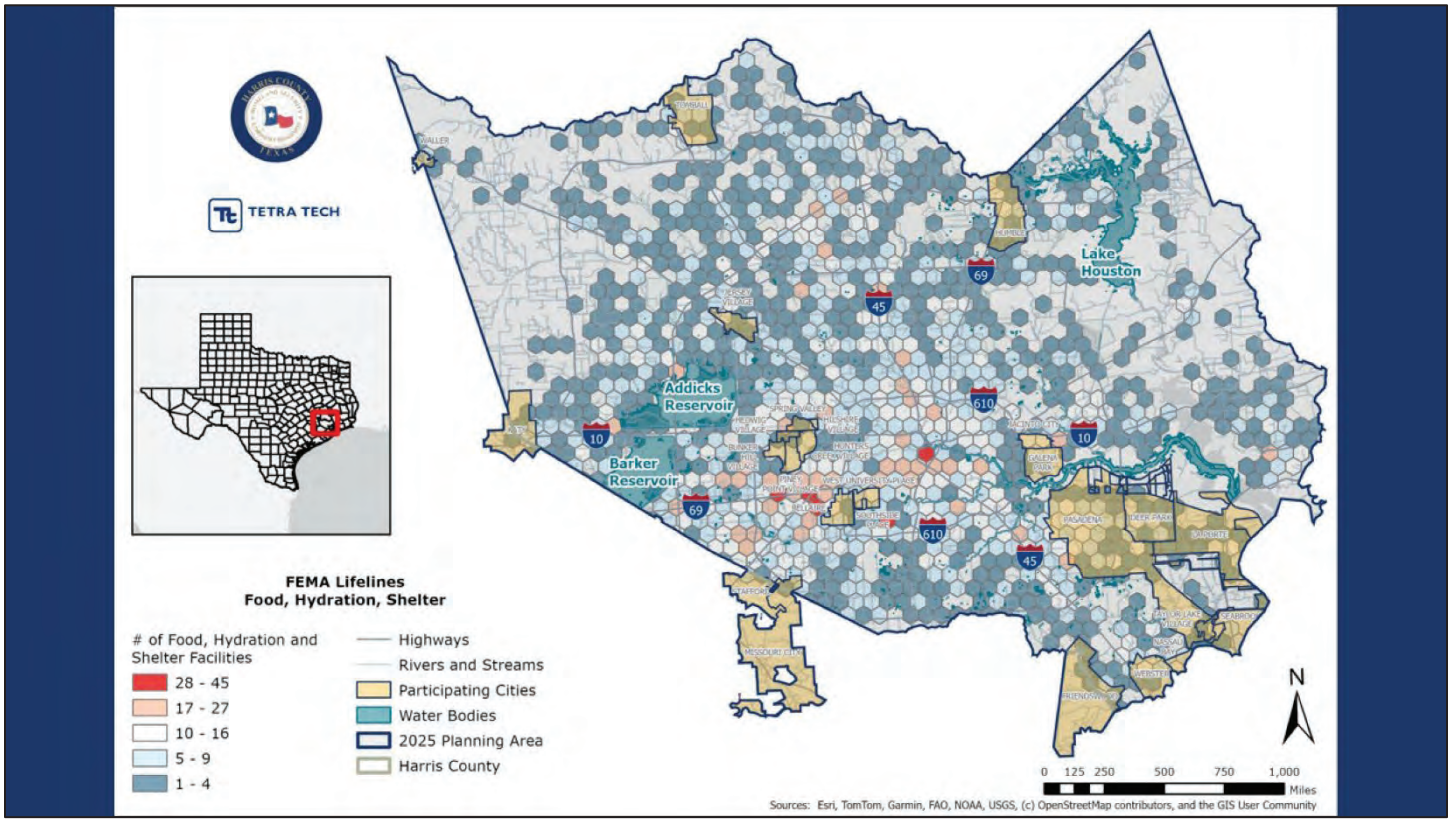

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Persons Over 65 Years of Age per SqMi by Census Tract
<400
400 - 800
800 - 1,200
>1,200
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

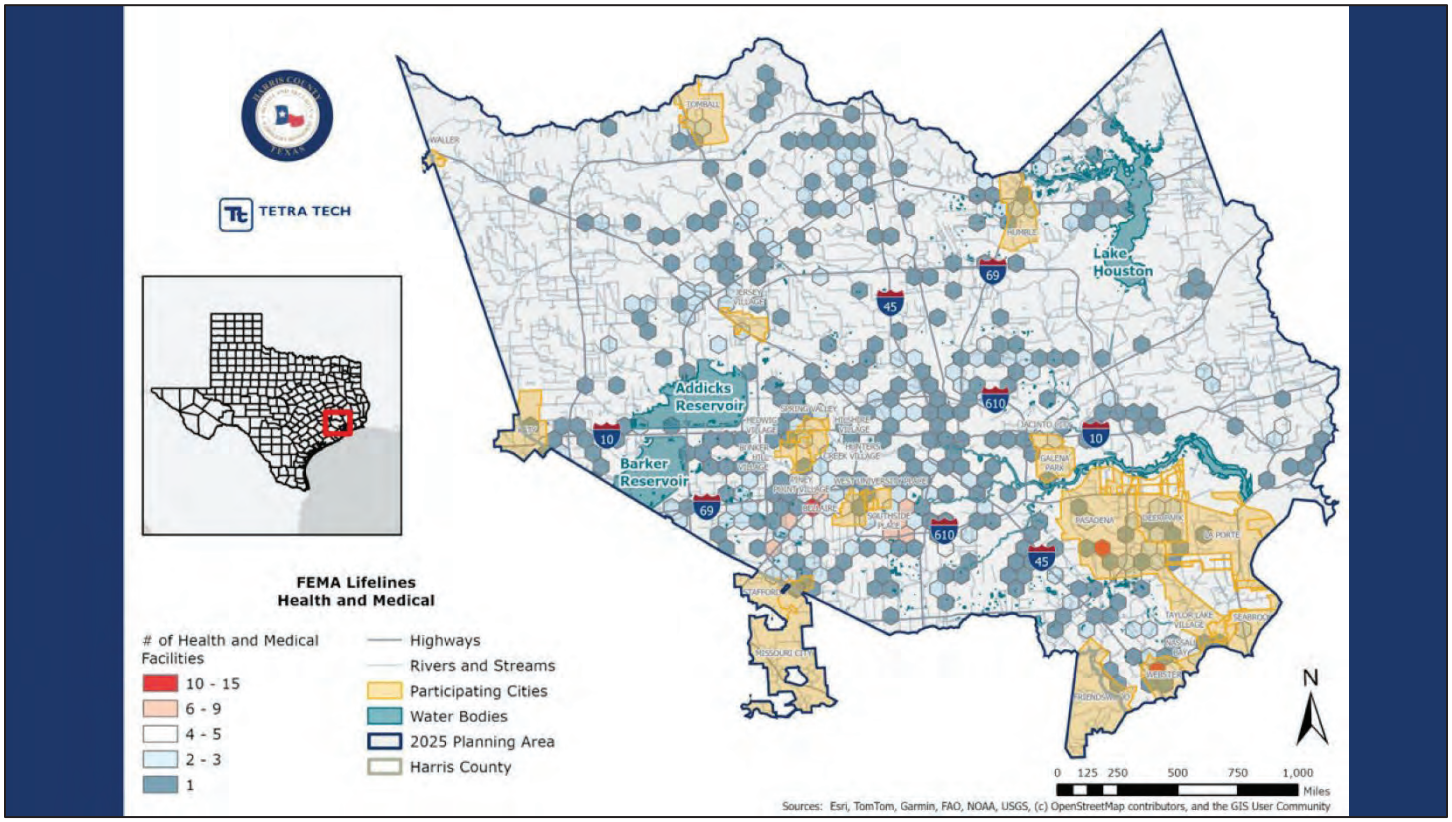

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Persons in Poverty per SqMi by Census Tract
<1,000
1,000 - 2,000
2,000 - 4,000
4,000 - 8,000
>8,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

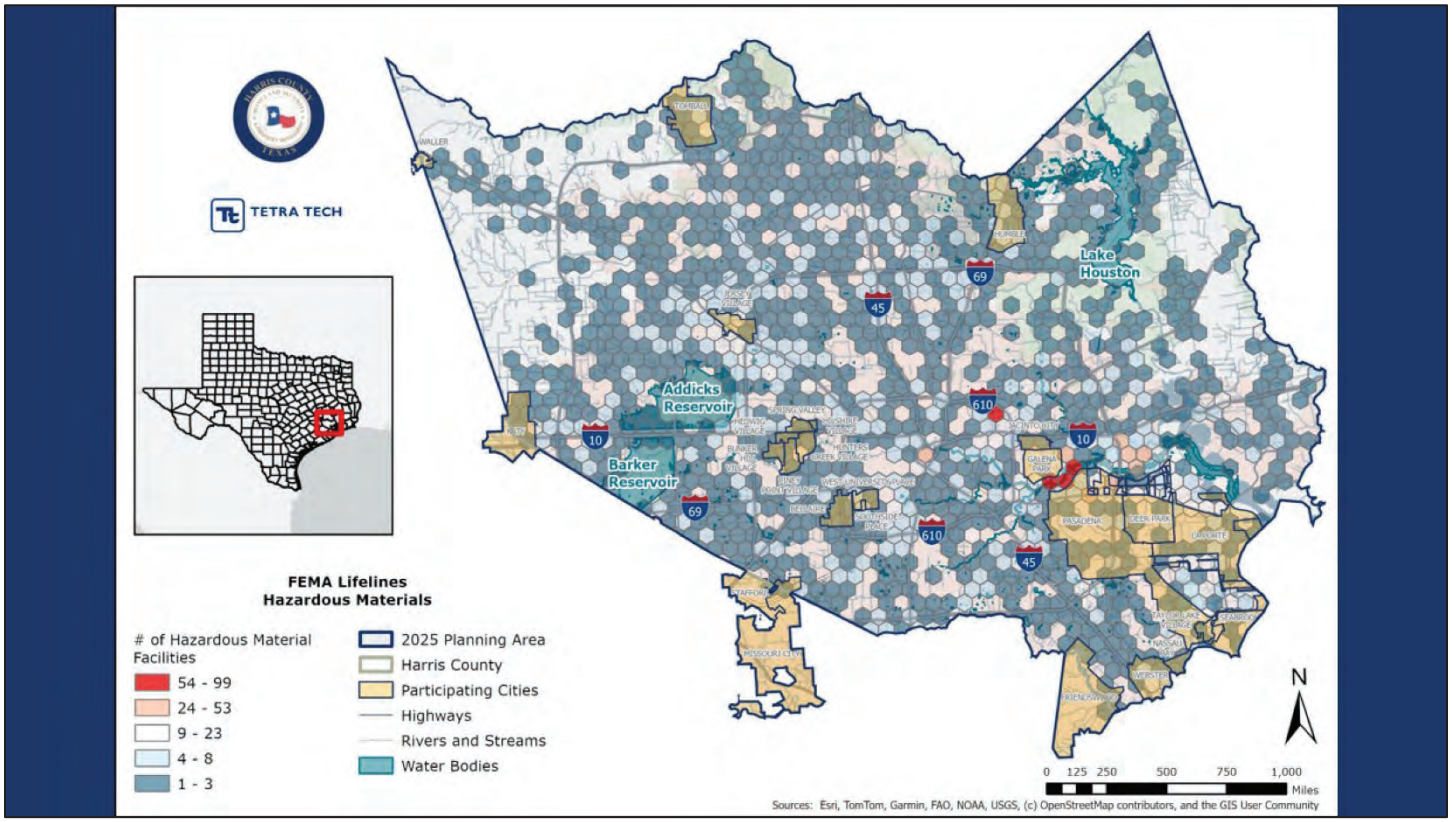

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Total Population
per SqMi by Census Tract
<2,000
2,000 - 4,000
4,000 - 8,000
8,000 - 12,000
12,000 - 16,000
>16,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

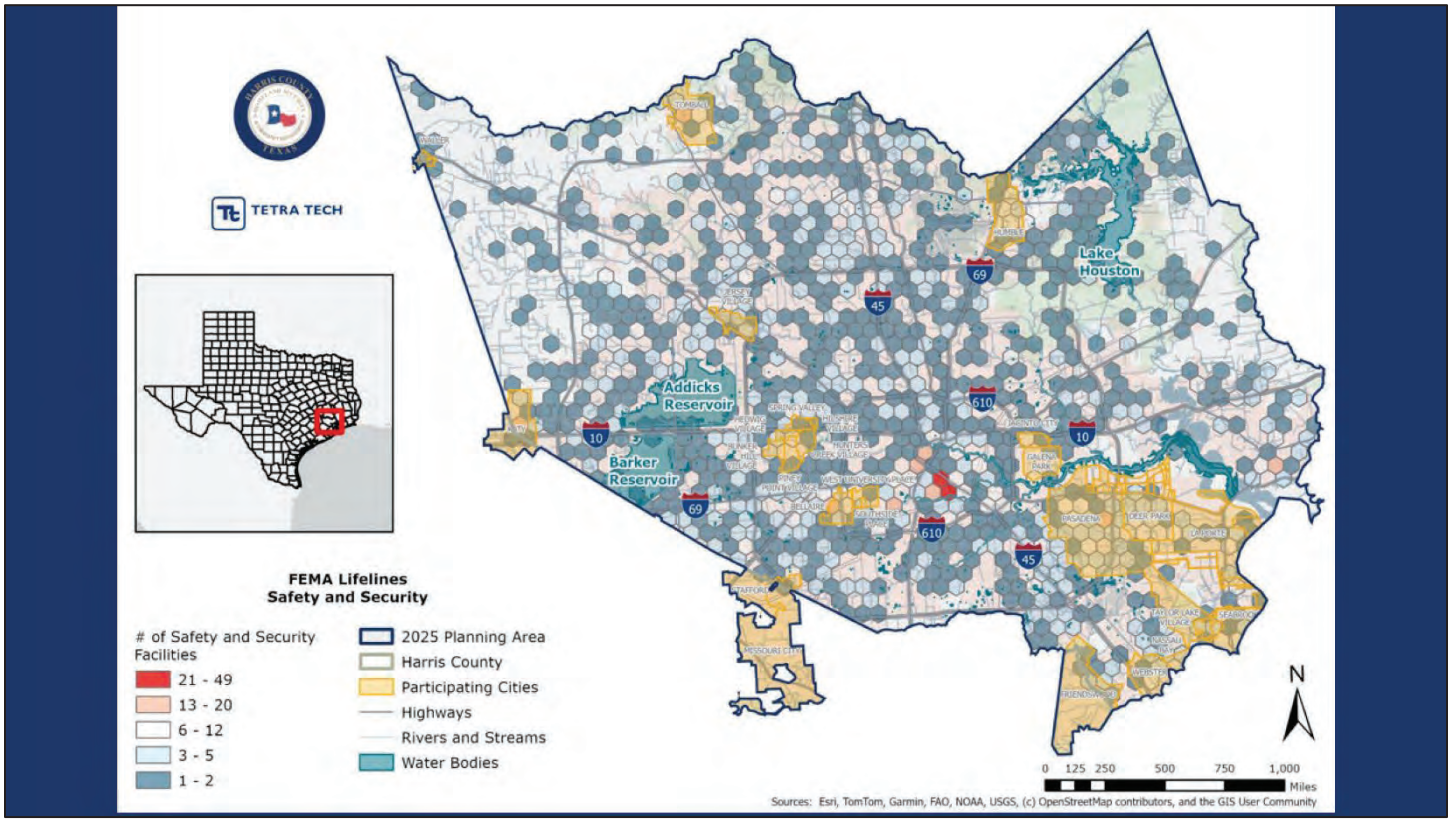

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Communcation Lifelines
FEMA Lifeline
Communications
71 - 364
33 - 70
13 - 32
4 - 12
1 - 3
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

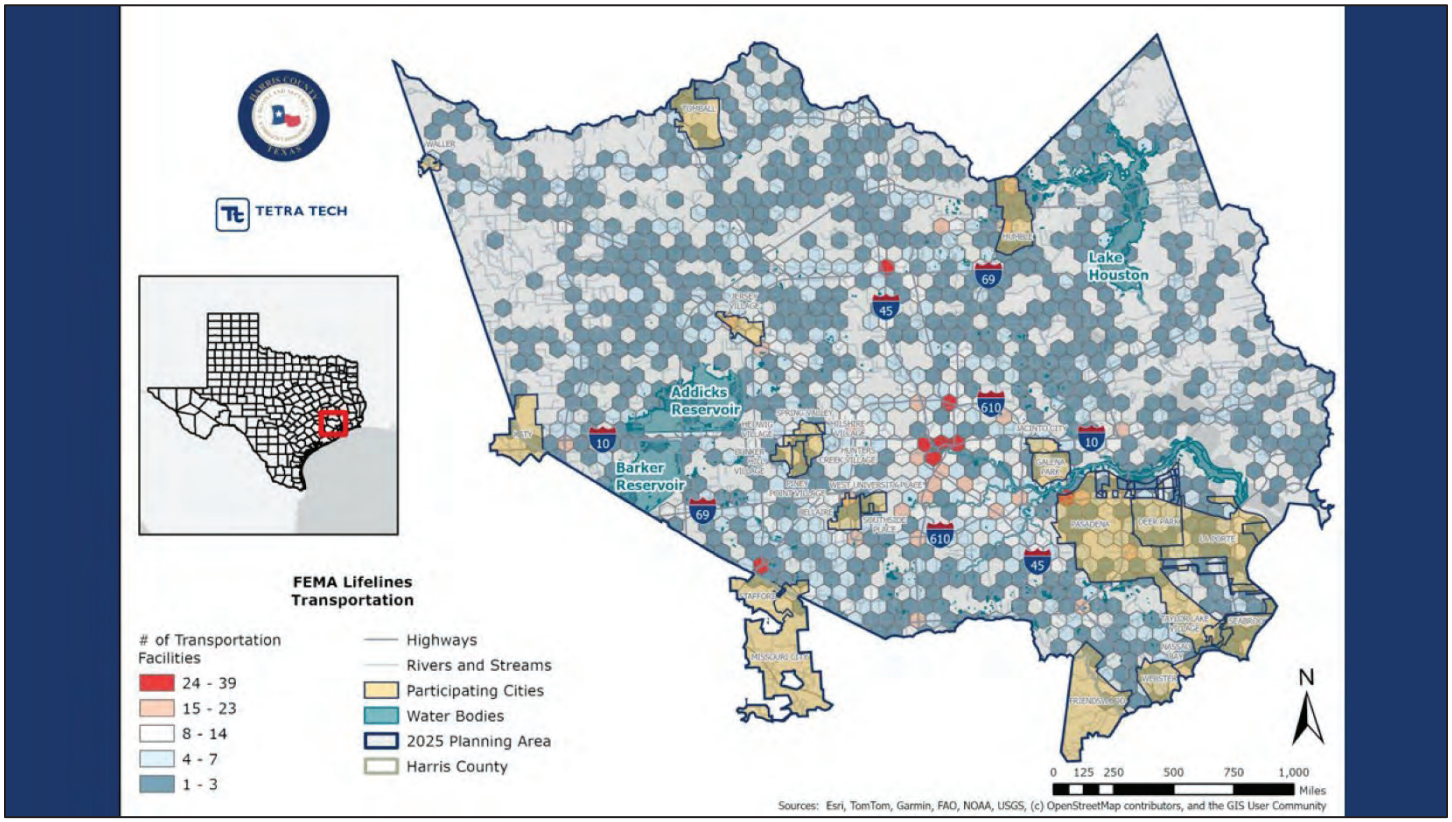

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Energy Lifelines
# of Energy Facilities
8 - 175
5 - 7
3 - 4
2
1
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

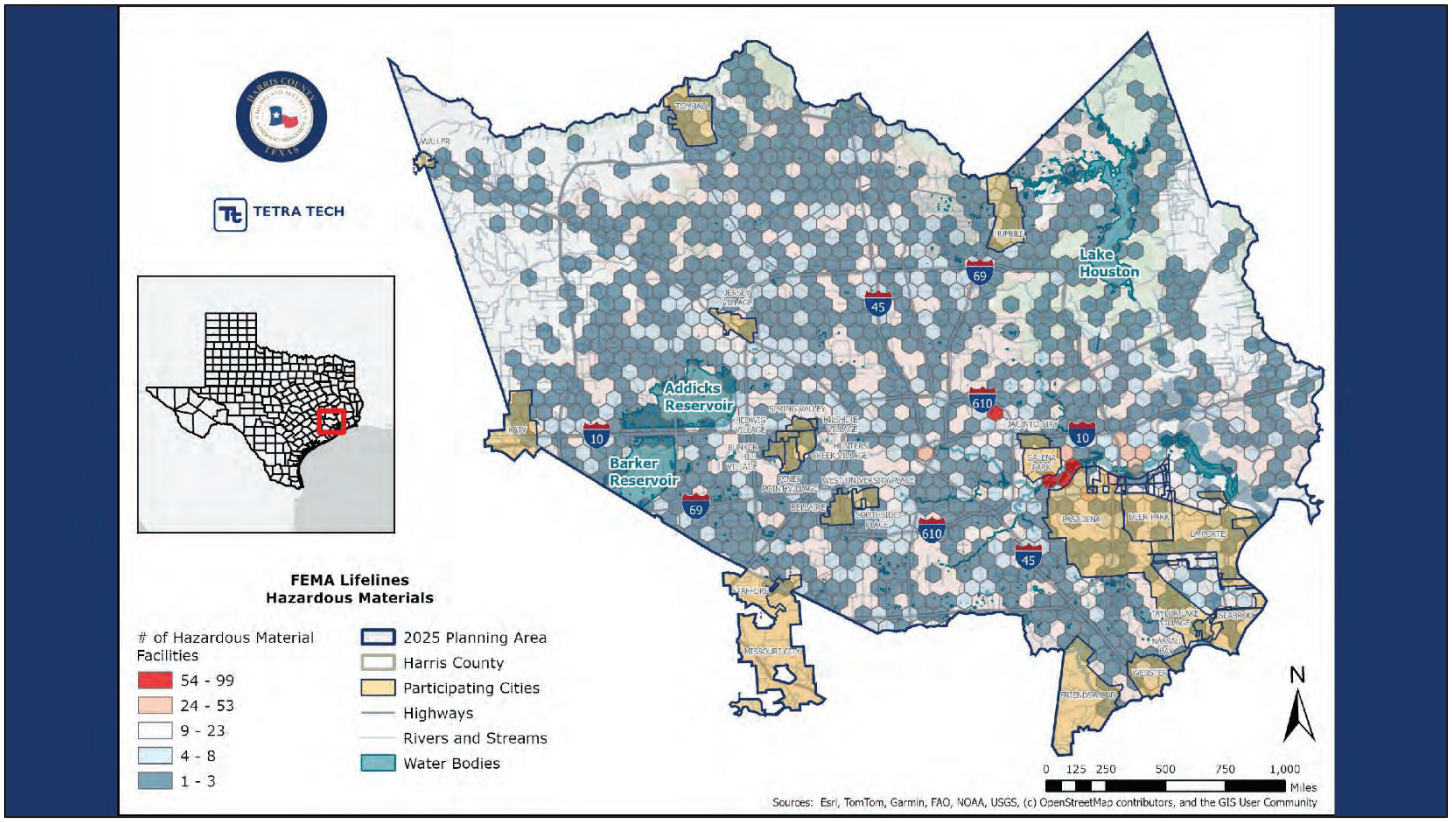

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Food, Hydration, Shelter
# of Food, Hydration and Shelter Facilities
28 - 45
17 - 27
10 - 16
5 - 9
1 - 4
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

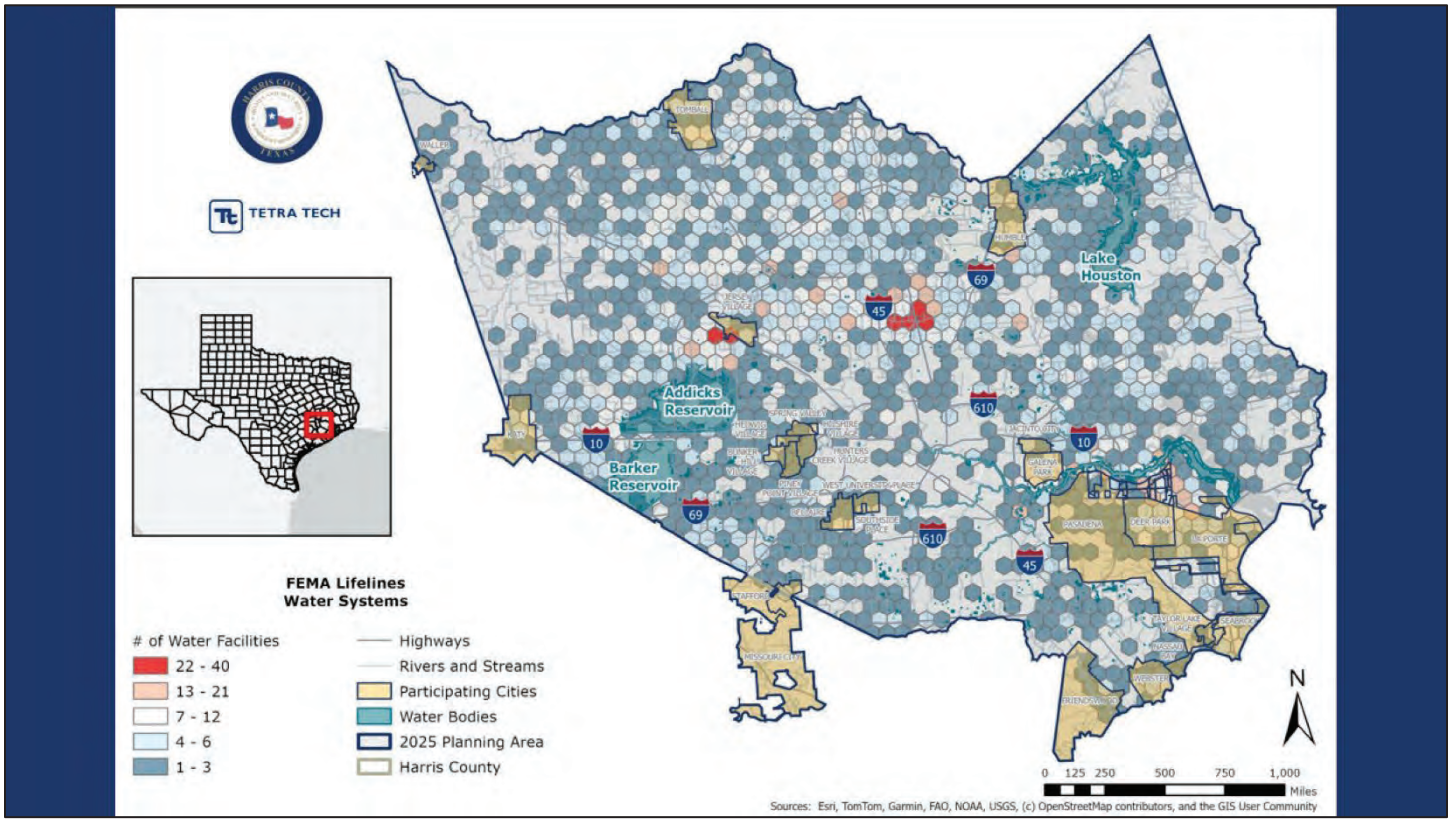

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Health and Medical
# of Health and Medical Facilities
10 - 15
6 - 9
4 - 5
2 - 3
1
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Hazardous Materials
# of Hazardous Material Facilities
54 - 99
24 - 53
9 - 23
4 - 8
1 - 3
2025 Planning Area
Harris County
Participating Cities
Highways
Rivers and Streams
Water Bodies
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Safety and Security
# of Safety and Security Facilities
21 - 49
13 - 20
6 - 12
3 - 5
1 - 2
2025 Planning Area
Harris County
Participating Cities
Highways
Rivers and Streams
Water Bodies
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

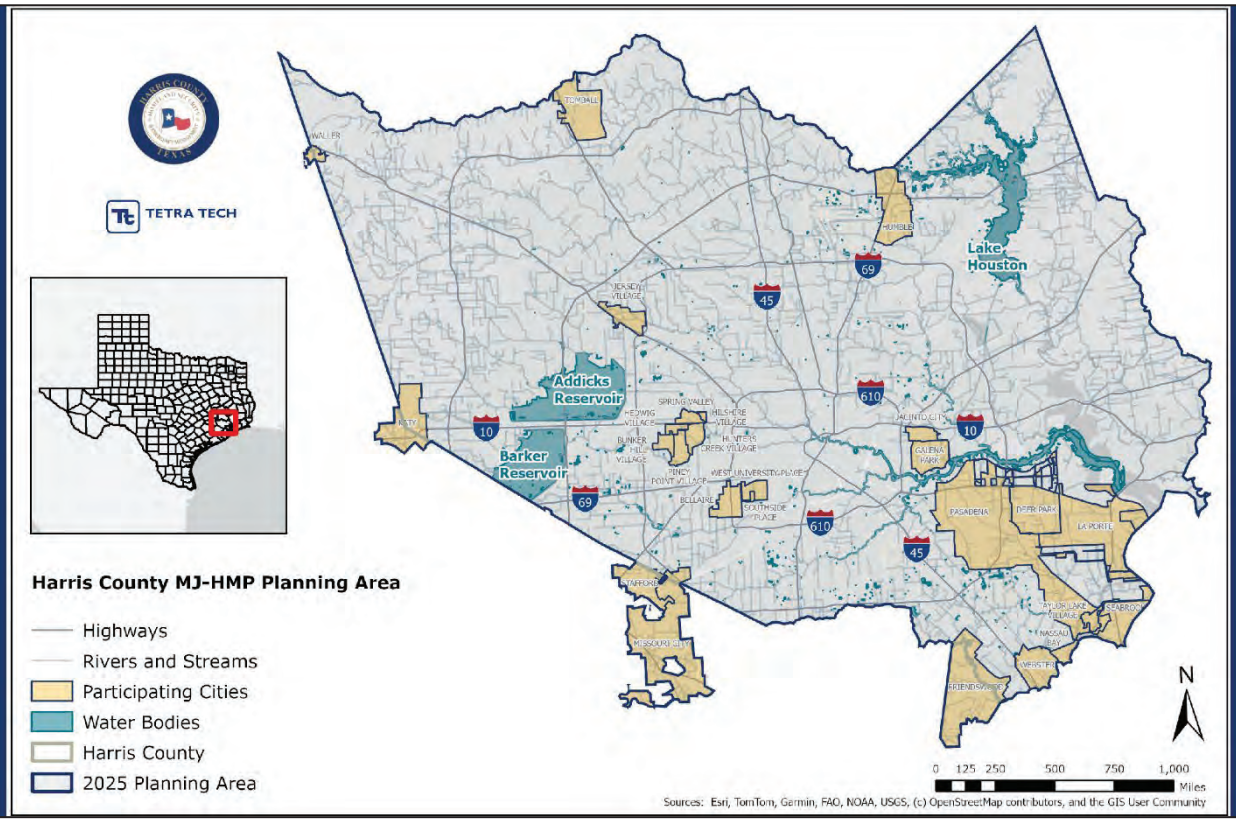

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Transportation
# of Transportation Facilities
24 - 39
15 - 23
8 - 14
4 - 7
1 - 3
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

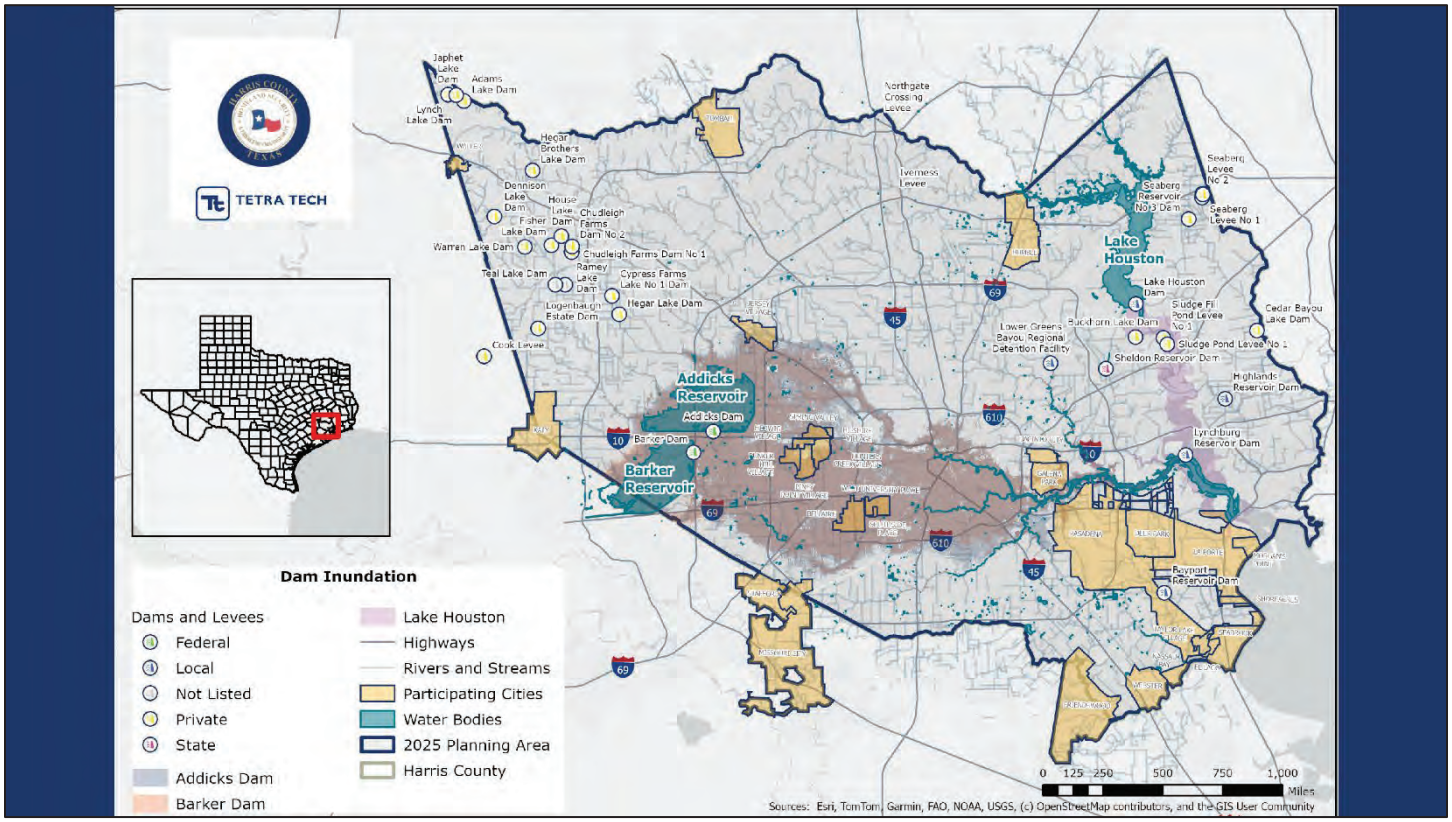

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Hazardous Materials
# of Hazardous Material Facilities
54 - 99
24 - 53
9 - 23
4 - 8
1 - 3
2025 Planning Area
Harris County
Participating Cities
Highways
Rivers and Streams
Water Bodies
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

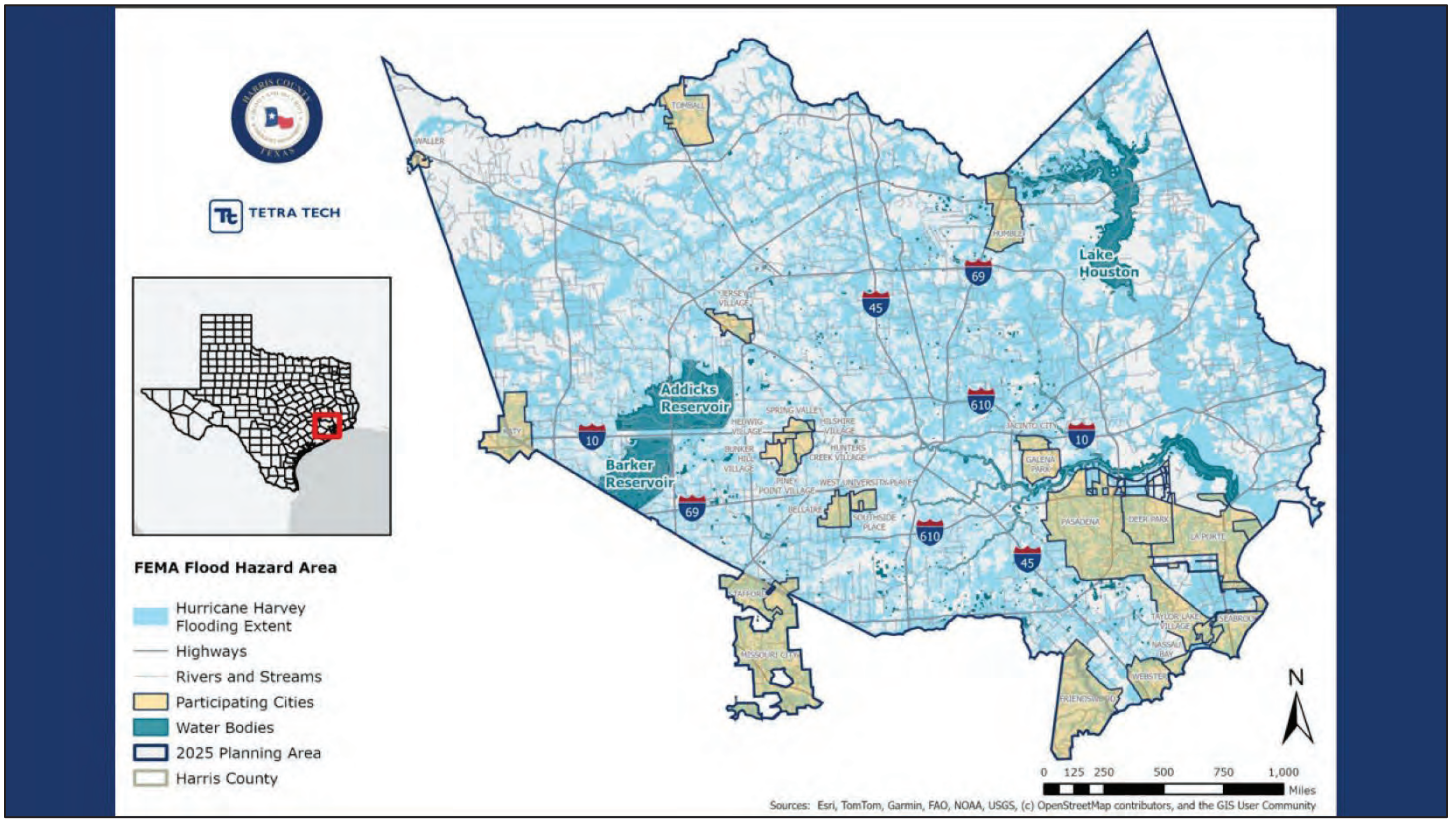

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Water Systems
# of Water Facilities
22 - 40
13 - 21
7 - 12
4 - 6
1 - 3
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

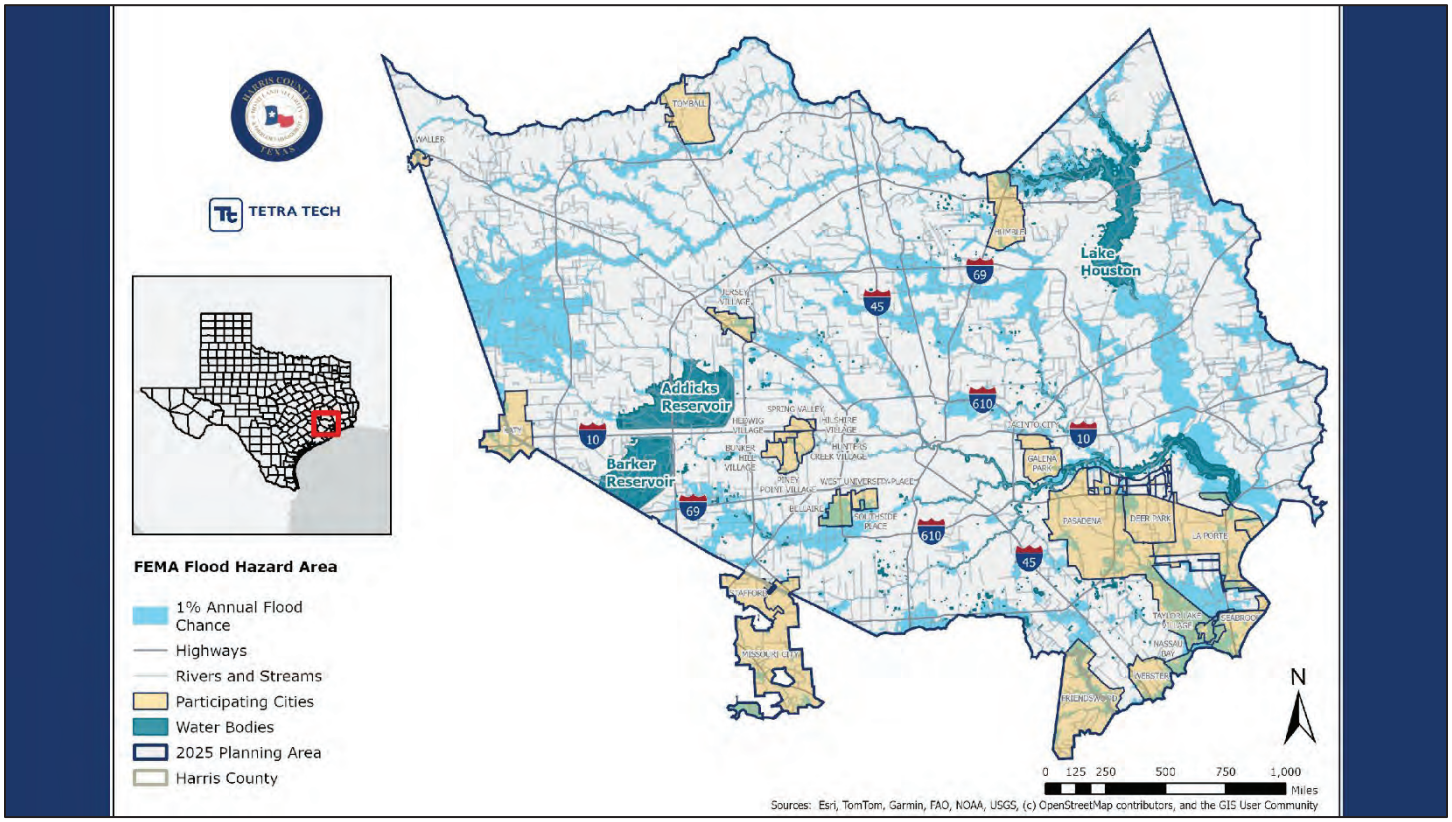
Maps to print for SC Meeting #6:

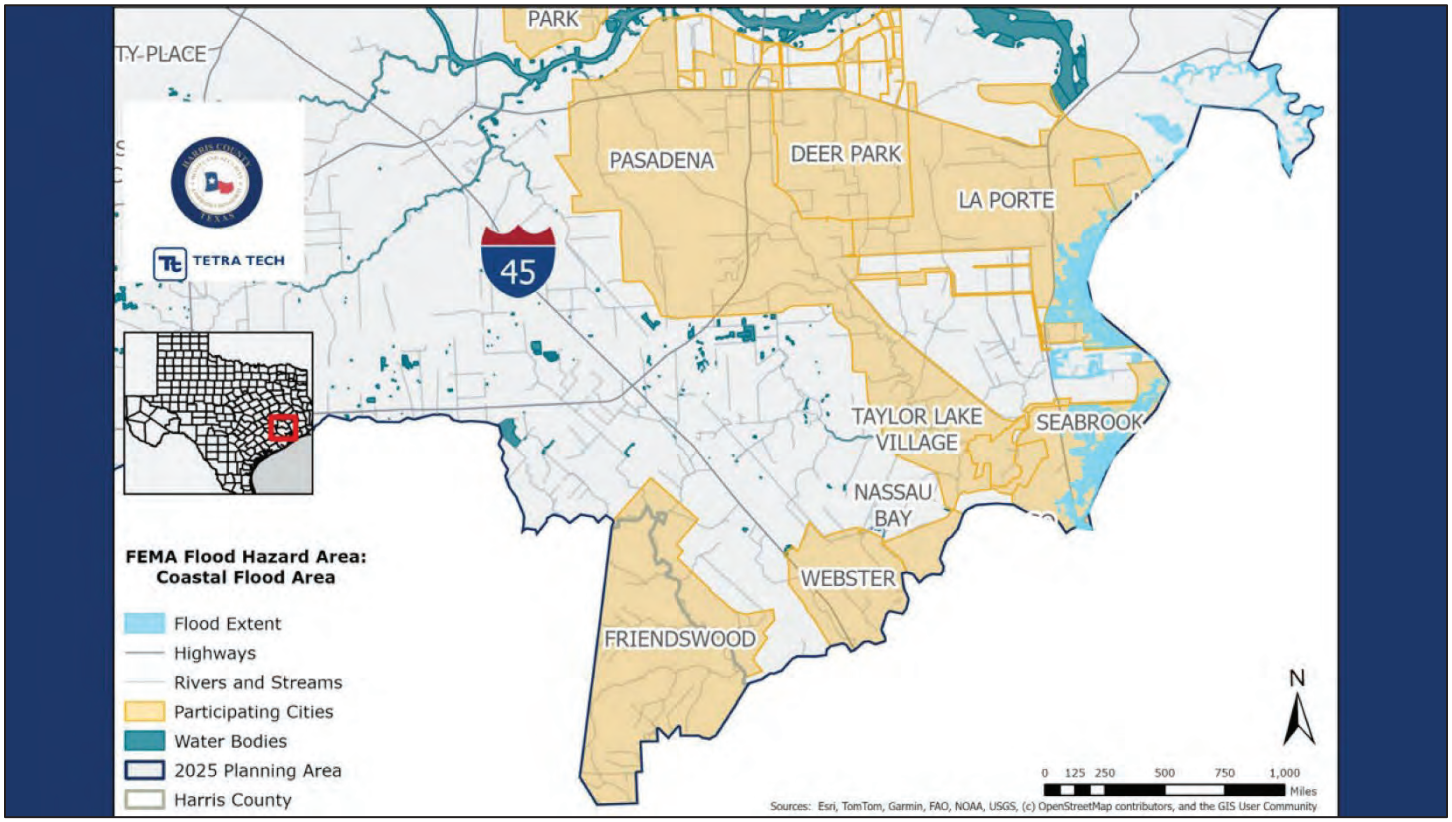
1. **Planning area**
2. **Dam Locations**
3. **Flooding**
   - **(1%, Harvey, Coastal)**
4. **Wildfire**
   - **Ignition Density**
   - **Wildfire Threat**
5. **What else?**

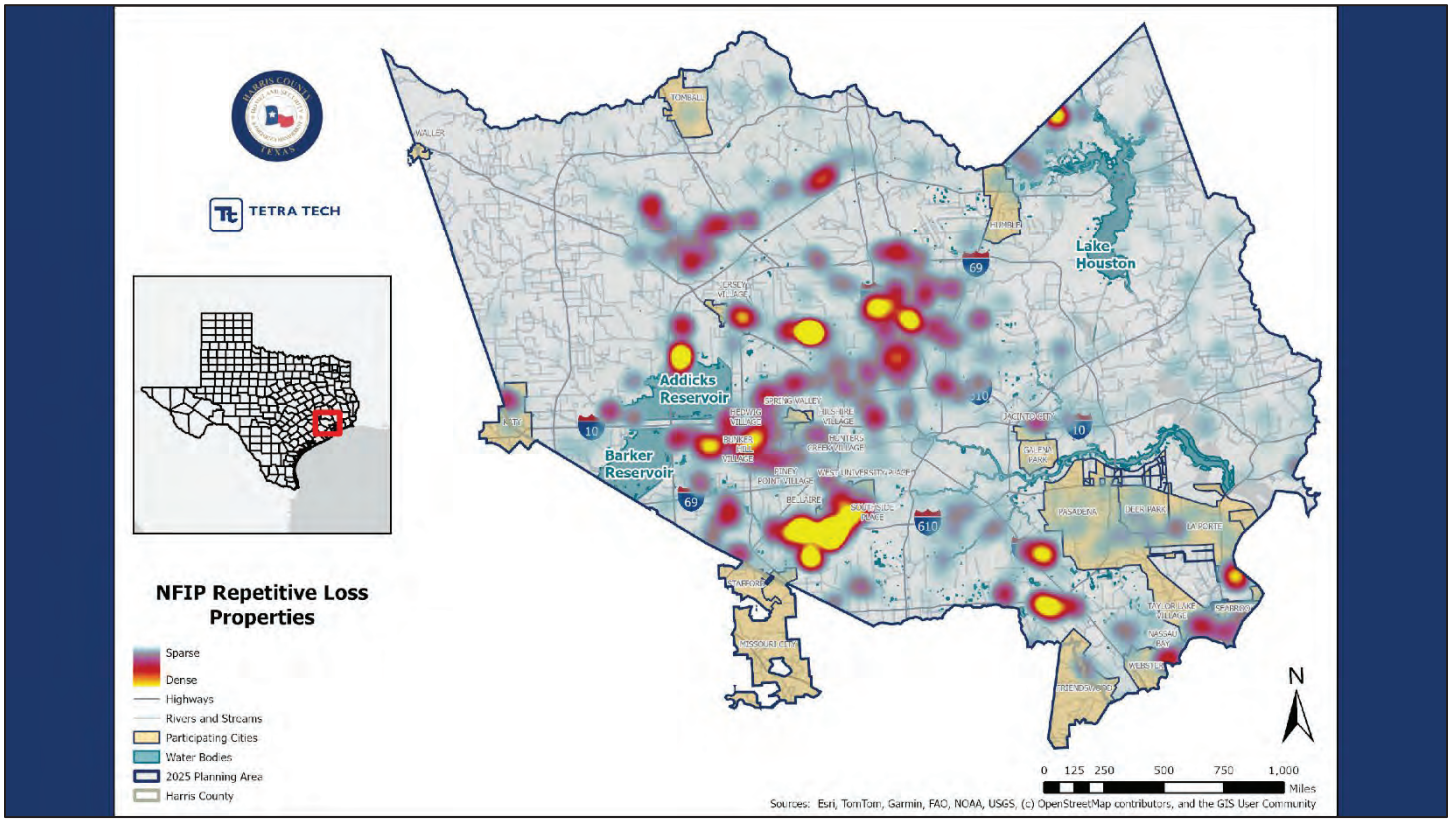


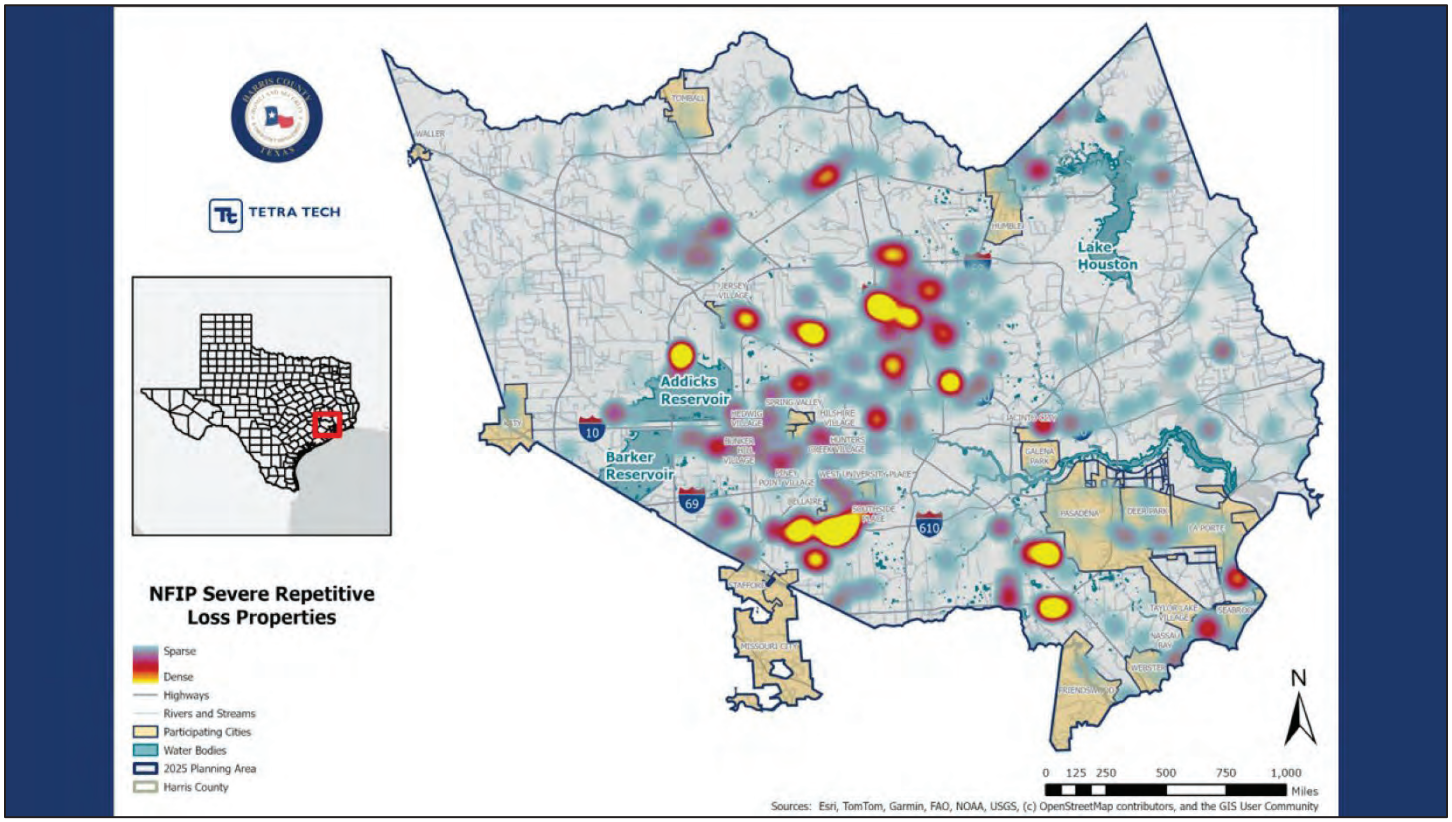

TETRA TECH
Japhet Lake Dam
Adams Lake Dam
Lynch Lake Dam
Hegar Brothers Lake Dam
Dennison Lake Dam
House Lake Dam
Fisher Lake Dam
Chudleigh Farms Dam No 2
Warren Lake Dam
Chudleigh Farms Dam No 1
Ramey Lake Dam
Teal Lake Dam
Cypress Farms Lake No 1 Dam
Logenbaugh Estate Dam
Hegar Lake Dam
Cook Levee
Northgate Crossing Levee
Iverness Levee
Seaberg Levee No 2
Seaberg Reservoir No 3 Dam
Seaberg Levee No 1
Lake Houston
Lake Houston Dam
Sludge Fl Pond Levee No 1
Cedar Bayou Lake Dam
Lower Greens Bayou Regional Detention Facility
Buckhorn Lake Dam
Sludge Pond Levee No 1
Sheldon Reservoir Dam
Highlands Reservoir Dam
Addicks Reservoir
Addicks Dam
Barker Dam
Barker Reservoir
Lynchburg Reservoir Dam
Bayport Reservoir Dam
Dam Inundation
Dams and Levees
Federal
Local
Not Listed
Private
State
Addicks Dam
Barker Dam
Lake Houston
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

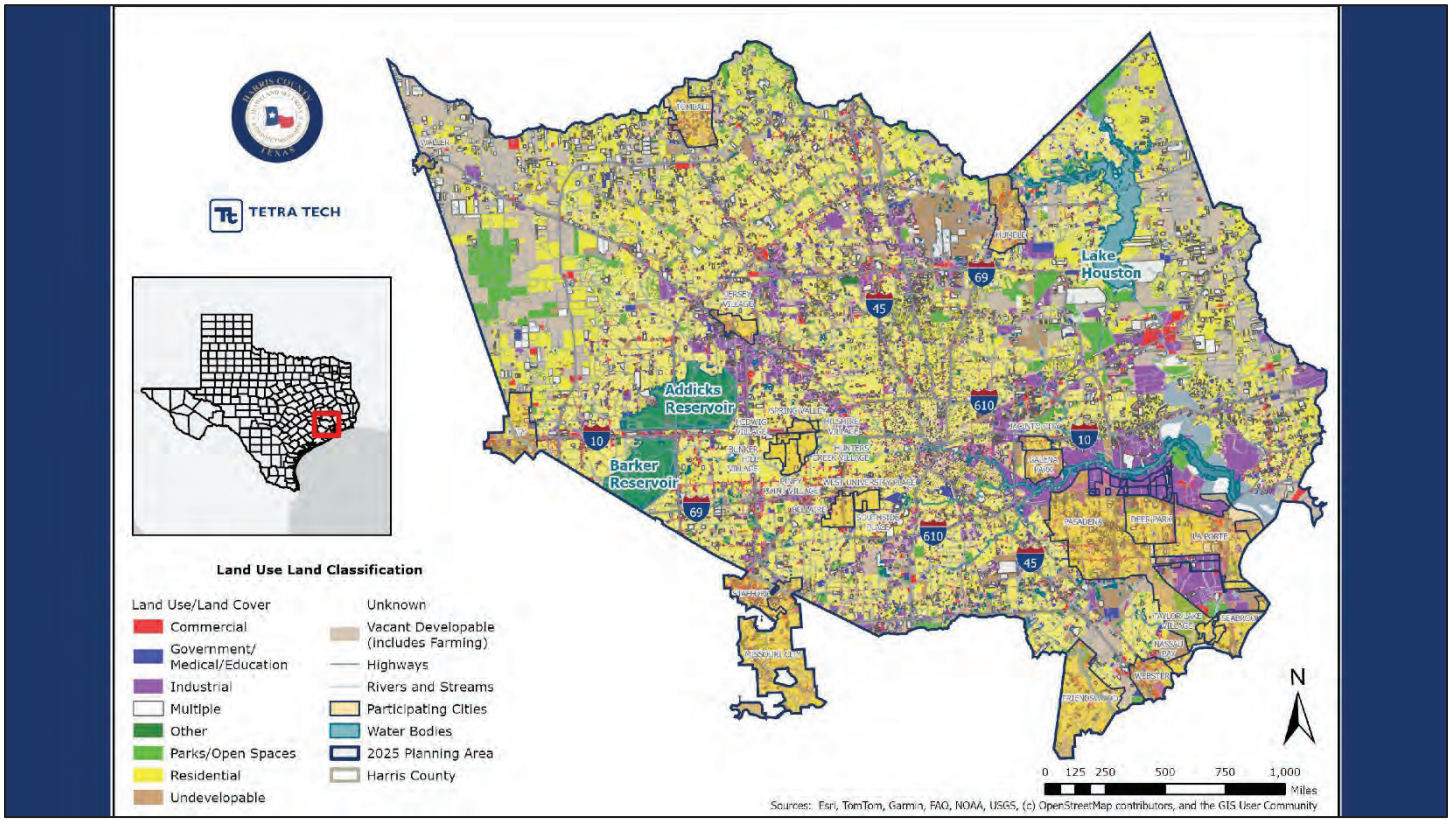

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Flood Hazard Area
Hurricane Harvey Flooding Extent
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

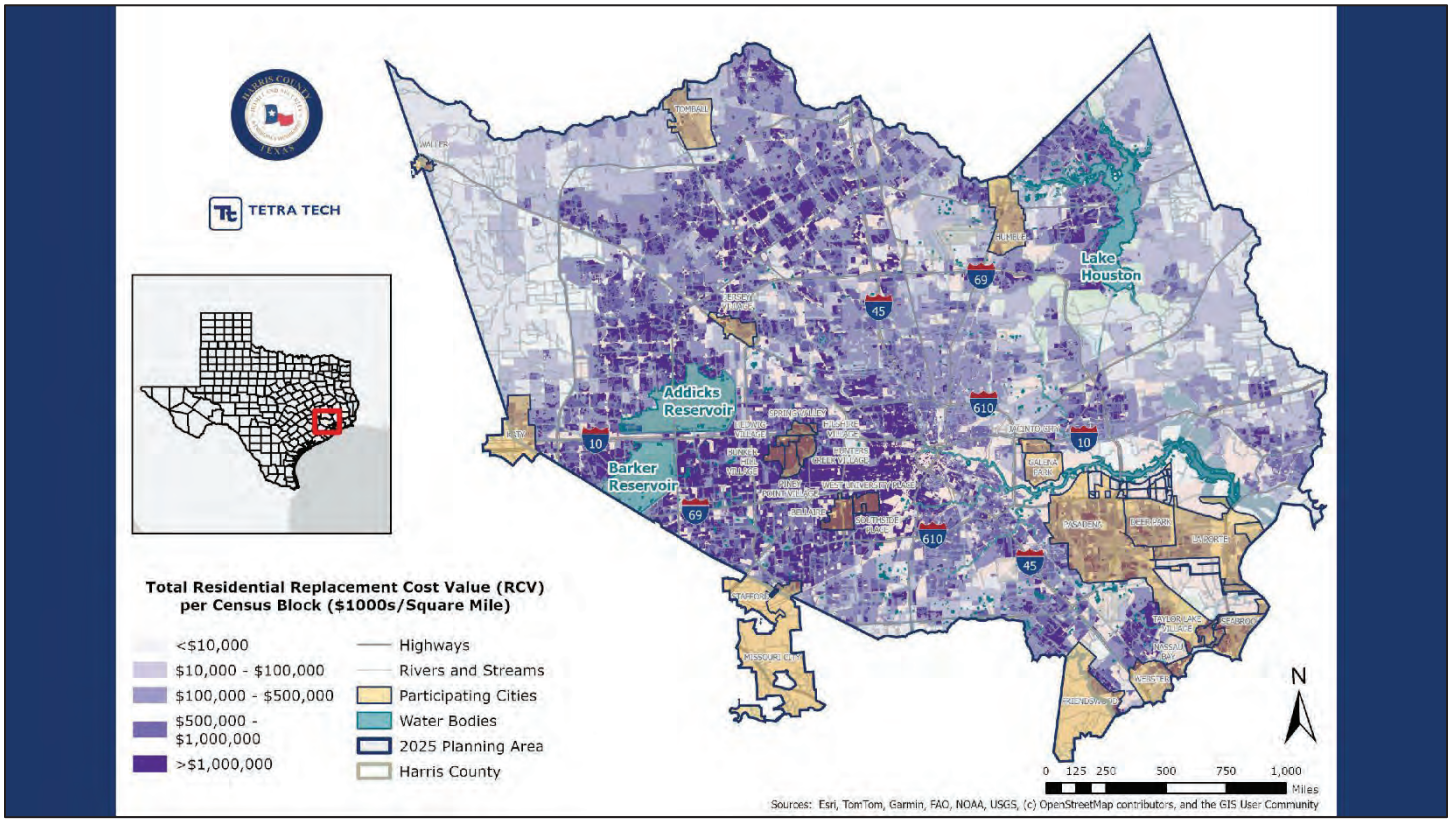

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Flood Hazard Area
1% Annual Flood Chance
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

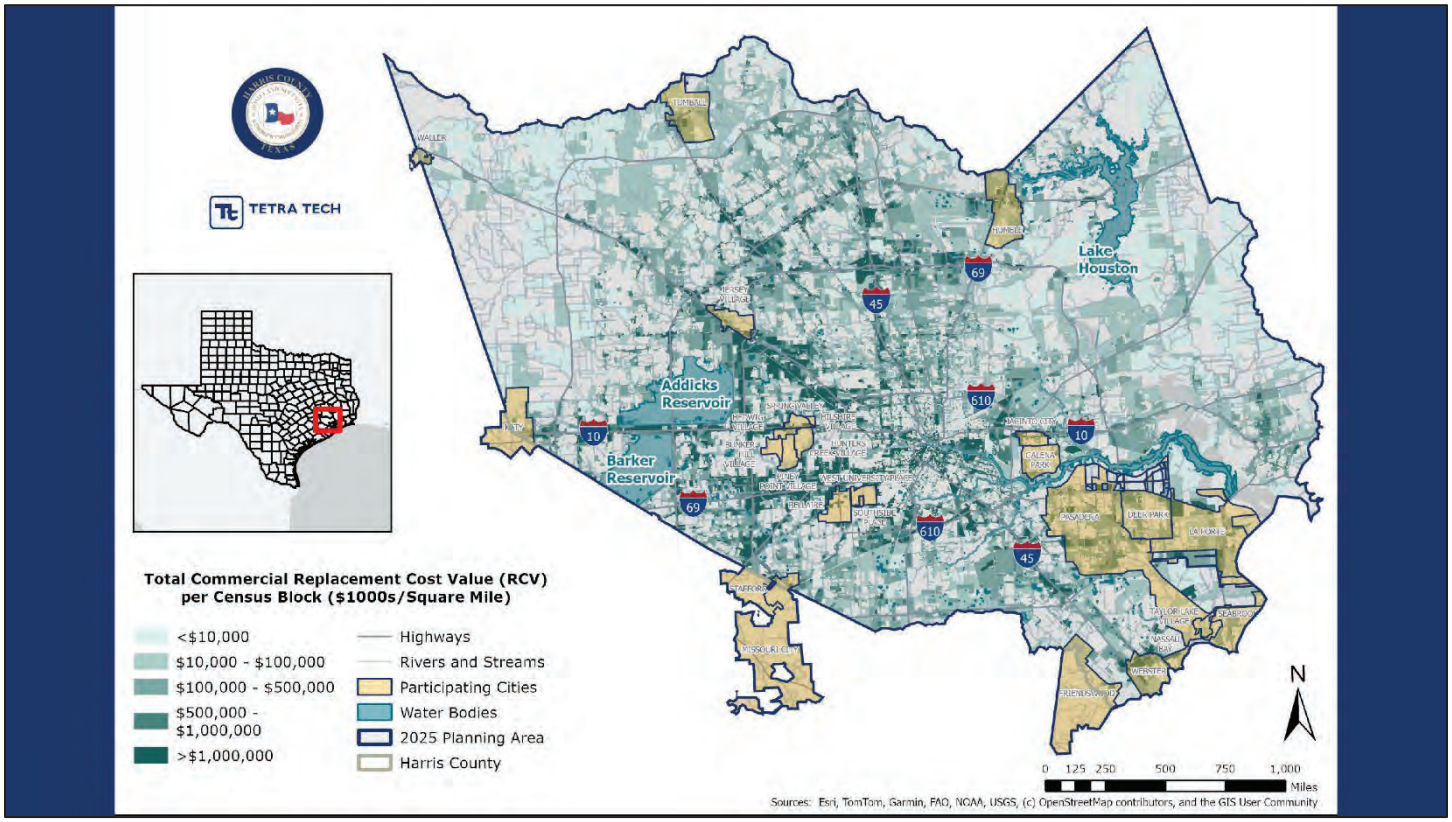

TETRA TECH
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
45
FEMA Flood Hazard Area:
Coastal Flood Area
Flood Extent
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

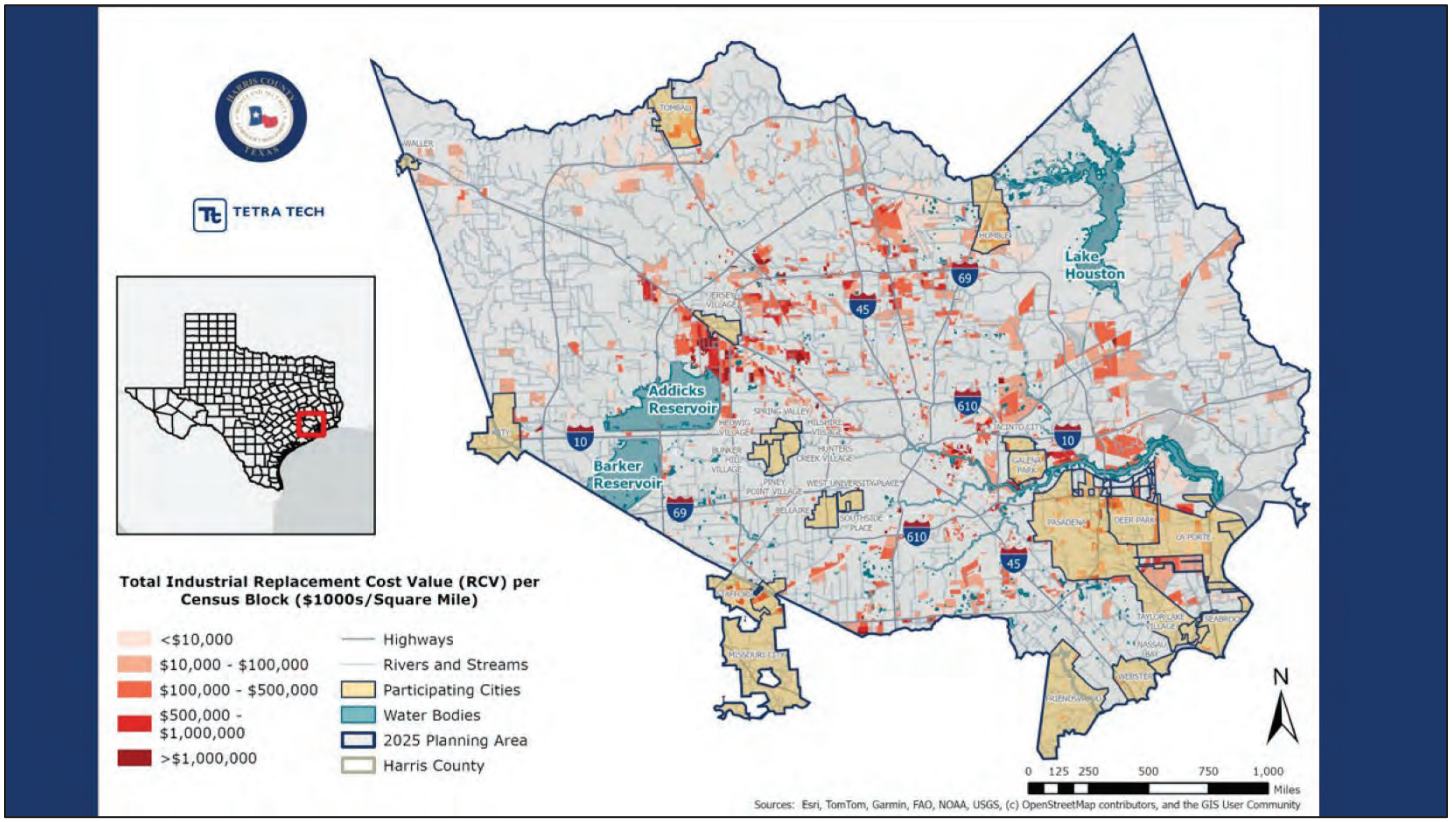

TETRA TECH
NFIP Repetitive Loss Properties
Sparse
Dense
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

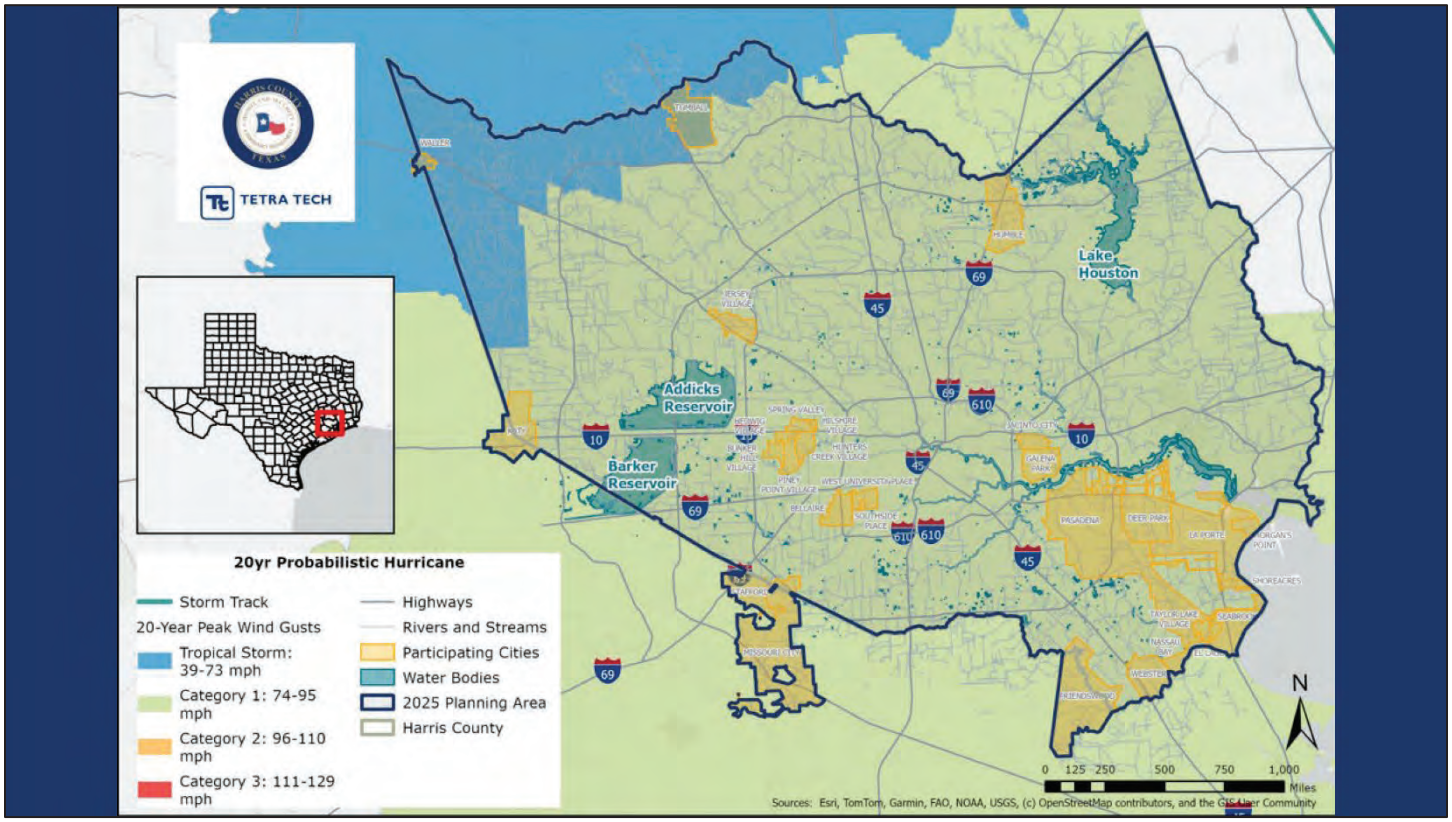

TETRA TECH
NFIP Severe Repetitive Loss Properties
Sparse
Dense
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

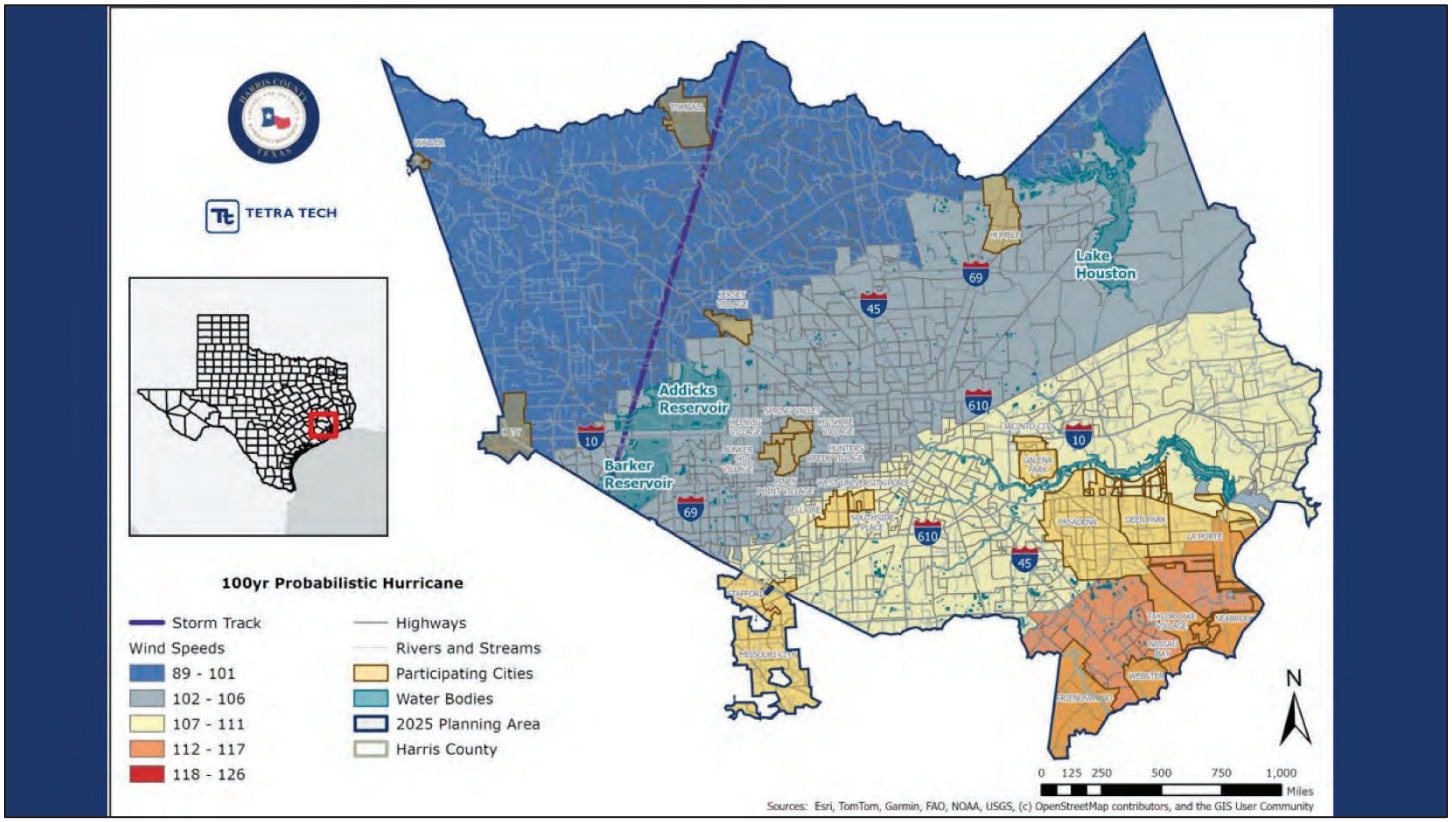

TETRA TECH
Addicks Reservoir
Barker Reservoir
Lake Houston
Land Use Land Classification
Land Use/Land Cover
Commercial
Government/ Medical/Education
Industrial
Multiple
Other
Parks/Open Spaces
Residential
Undevelopable
Unknown
Vacant Developable (Includes Farming)
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

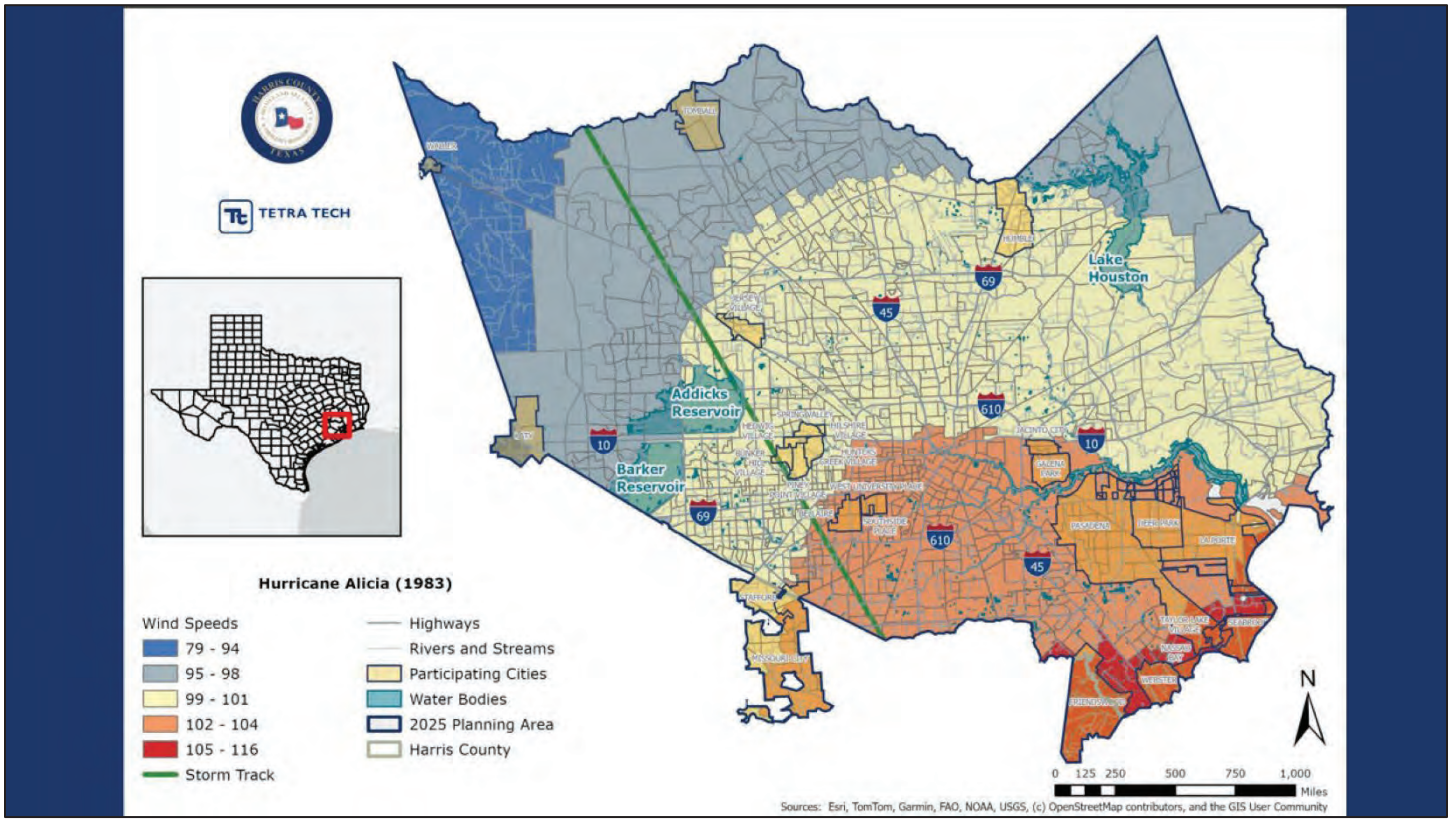

TETRA TECH
Addicks Reservoir
Barker Reservoir
Lake Houston
Total Residential Replacement Cost Value (RCV) per Census Block ($1000s/Square Mile)
<$10,000
$10,000 - $100,000
$100,000 - $500,000
$500,000 - $1,000,000
>$1,000,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

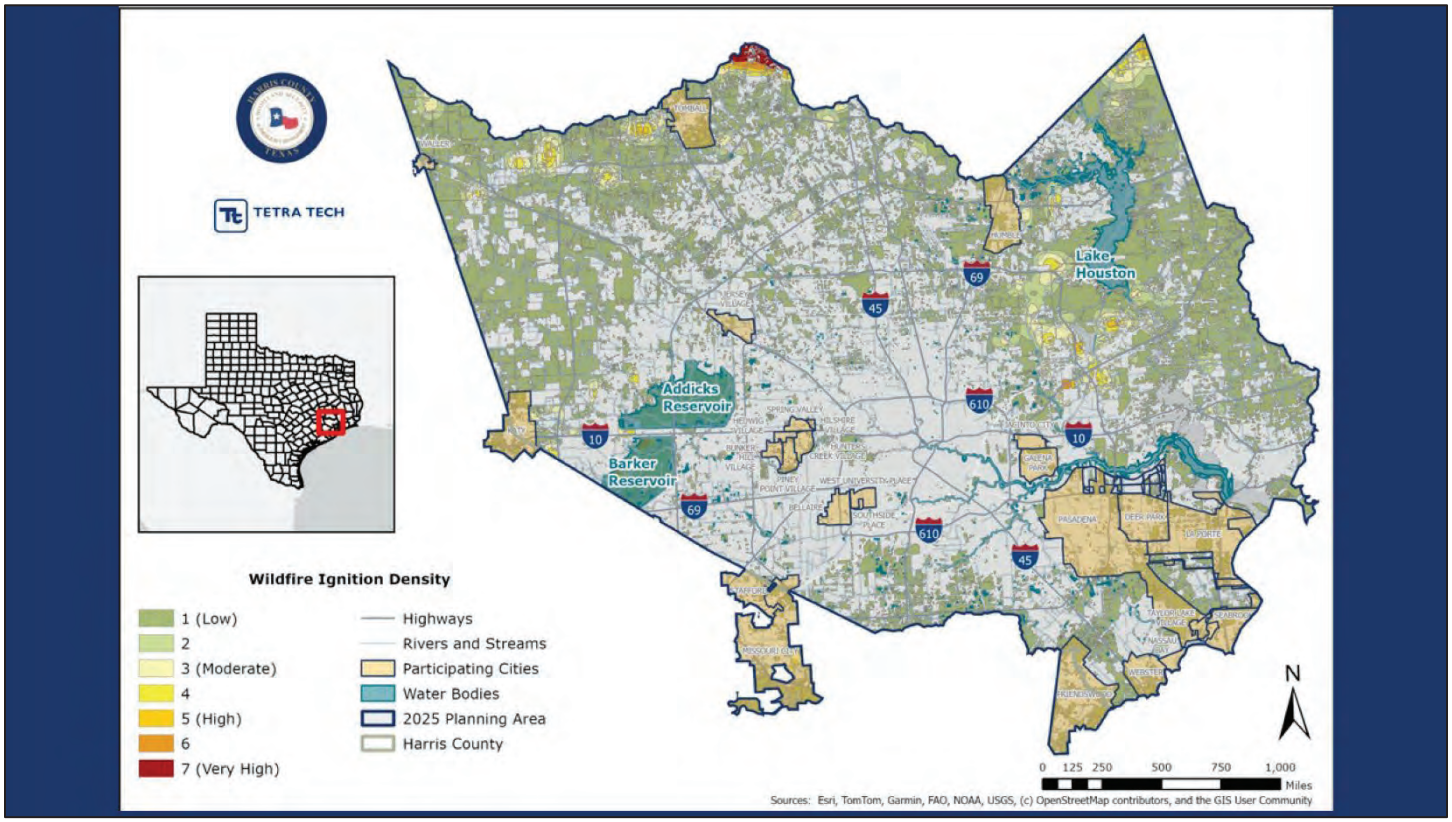

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Total Commercial Replacement Cost Value (RCV) per Census Block ($1000s/Square Mile)
<$10,000
$10,000 - $100,000
$100,000 - $500,000
$500,000 - $1,000,000
>$1,000,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

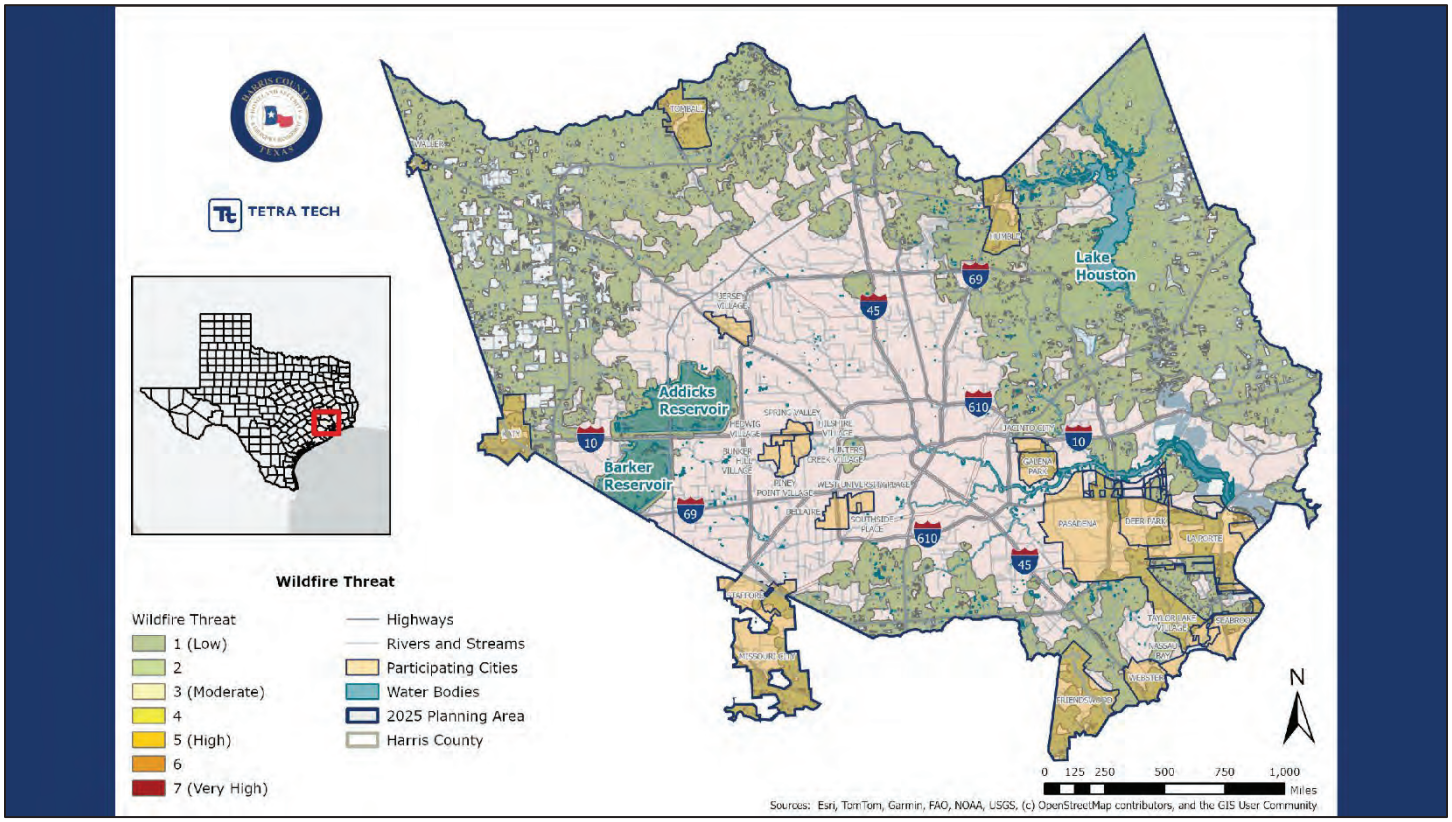

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Total Industrial Replacement Cost Value (RCV) per Census Block ($1000s/Square Mile)
<$10,000
$10,000 - $100,000
$100,000 - $500,000
$500,000 - $1,000,000
>$1,000,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
20yr Probabilistic Hurricane
Storm Track
20-Year Peak Wind Gusts
Tropical Storm: 39-73 mph
Category 1: 74-95 mph
Category 2: 96-110 mph
Category 3: 111-129 mph
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
100yr Probabilistic Hurricane
Storm Track
Wind Speeds
89 - 101
102 - 106
107 - 111
112 - 117
118 - 126
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Hurricane Alicia (1983)
Wind Speeds
79 - 94
95 - 98
99 - 101
102 - 104
105 - 116
Storm Track
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Wildfire Ignition Density
1 (Low)
2
3 (Moderate)
4
5 (High)
6
7 (Very High)
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

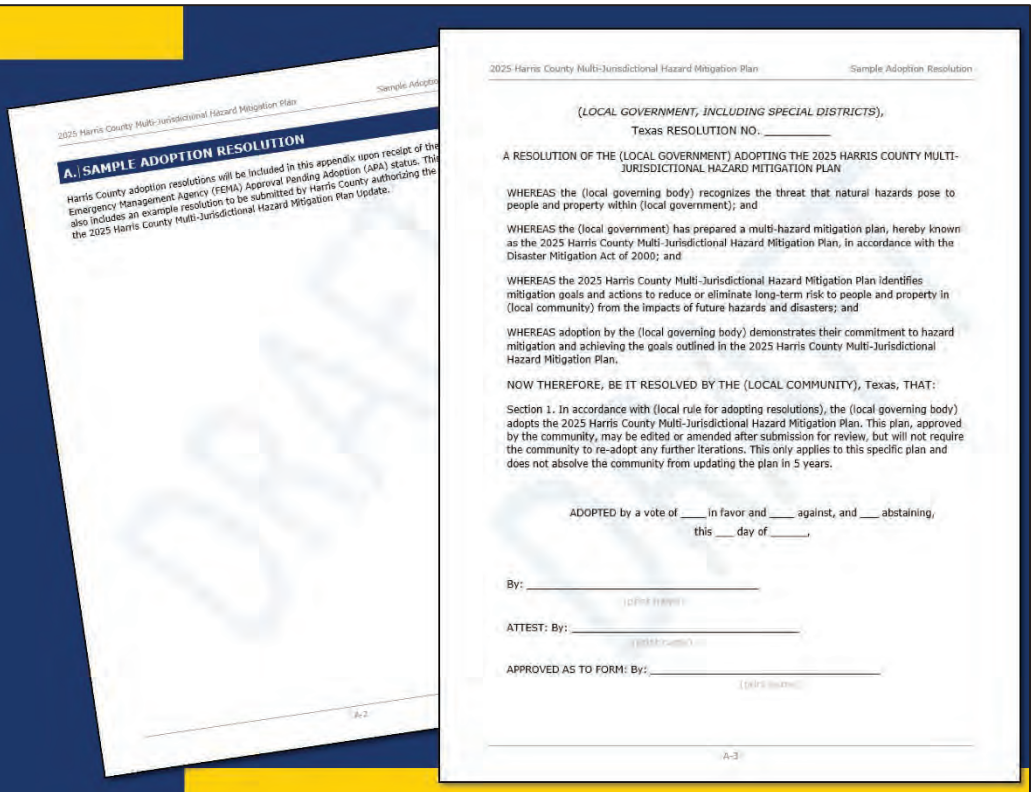


Adoption Resolution Example

HARRIS COUNTY

Office of Homeland Security & Emergency Management

Example Adoption Resolution



2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan

A. SAMPLE ADOPTION RESOLUTION

Harris County adoption resolutions will be included in this appendix upon receipt of the Emergency Management Agency (FEMA) Approval Pending Adoption (APA) status. This also includes an example resolution to be submitted by Harris County authorizing the the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update.

A-2

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan — Sample Adoption Resolution

(LOCAL GOVERNMENT, INCLUDING SPECIAL DISTRICTS),

Texas RESOLUTION NO. _____

A RESOLUTION OF THE (LOCAL GOVERNMENT) ADOPTING THE 2025 HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN

WHEREAS the (local governing body) recognizes the threat that natural hazards pose to people and property within (local government); and

WHEREAS the (local government) has prepared a multi-hazard mitigation plan, hereby known as the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan, in accordance with the Disaster Mitigation Act of 2000; and

WHEREAS the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan identifies mitigation goals and actions to reduce or eliminate long-term risk to people and property in (local community) from the impacts of future hazards and disasters; and

WHEREAS adoption by the (local governing body) demonstrates their commitment to hazard mitigation and achieving the goals outlined in the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan.

NOW THEREFORE, BE IT RESOLVED BY THE (LOCAL COMMUNITY), Texas, THAT:

Section 1. In accordance with (local rule for adopting resolutions), the (local governing body) adopts the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan. This plan, approved by the community, may be edited or amended after submission for review, but will not require the community to re-adopt any further iterations. This only applies to this specific plan and does not absolve the community from updating the plan in 5 years.

ADOPTED by a vote of ____ in favor and ____ against, and ___ abstaining, this ___ day of _____,

By: _____

ATTEST: By: _____

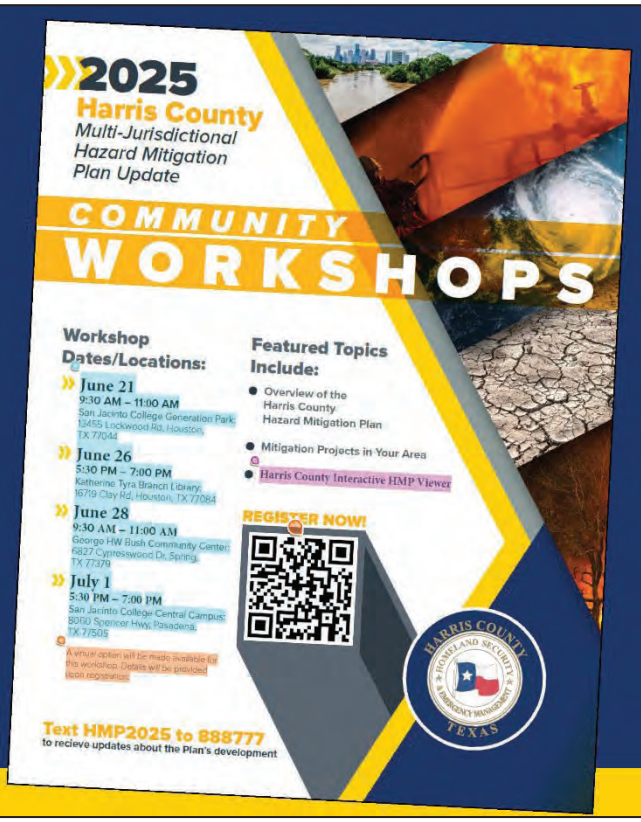
APPROVED AS TO FORM: By: _____

A-3

Public Meetings 1-4 & Public Review/Comment Period

HARRIS COUNTY
Office of Homeland Security & Emergency Management

https://forms.office.com/r/hEx4hwBbp0



Registration Survey Question

2025 Harris County MJ-HMP Community Workshop Registration

Harris County invites members of the public as well as local residents who are interested in previewing the newly-drafted 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan (MJ-HMP) update. Before the plan is formally reviewed by FEMA, members of the community are invited to attend any of the four workshops being held to welcome public comment & feedback. To register to attend, please answer the following questions.

* Required

1. First Name *

Enter your answer

2. Last Name *

Enter your answer

3. Jurisdiction or municipality of residence *

Enter your answer

4. Phone *

Enter your answer

5. Email *

Enter your answer

6. Please select which community workshop you are attending. *

- (Location 1) on June 21 from 9:30 AM to 11:00 AM
- (Location 2) on June 26 (5:30 PM - 7:00 PM)
- (Location 3) on June 28 (9:30 AM - 11:00 AM)
- (Location 4) on July 1 (5:30 PM - 7:00 PM)

You can print a copy of your answer after you submit

Submit

Additional Public Workshops

HARRIS COUNTY
Office of Homeland Security & Emergency Management

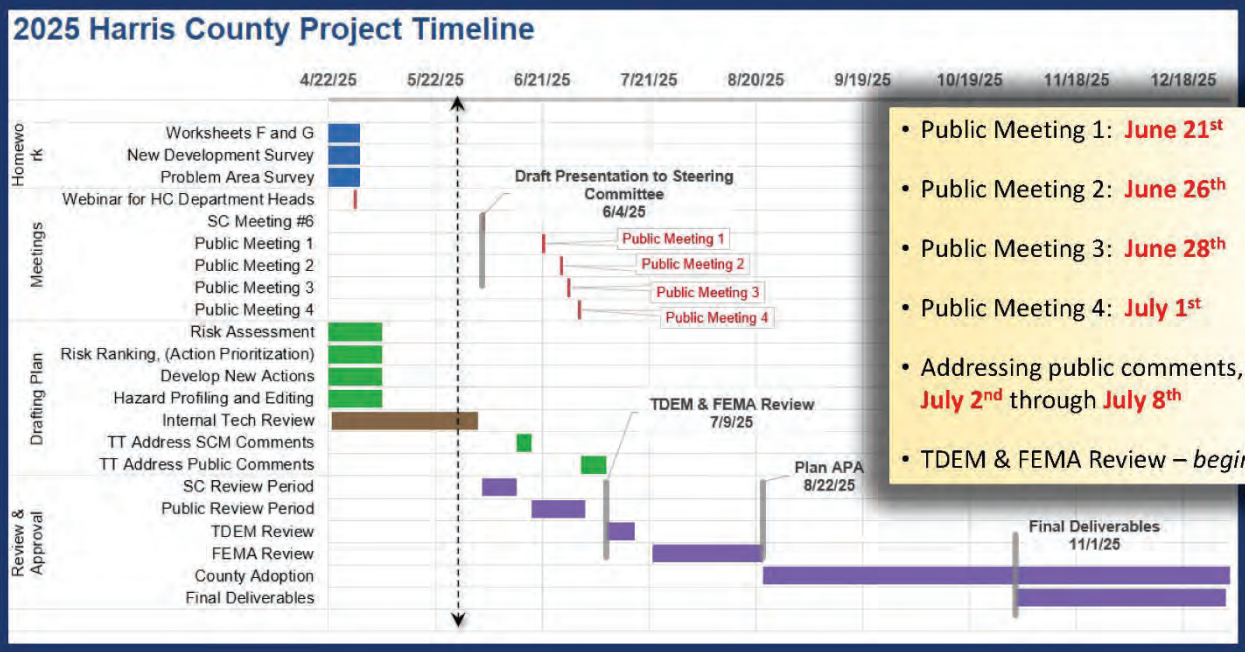
Municipal/Jurisdictional Workshops (optional)

If participants opt to host additional public workshops for review/comment, they will need to supply the following:

- Screenshots of workshop meeting invitations
- Workshop details (time, location, etc.)
- Workshop sign-in sheets and any related agendas
- List of public feedback/comments collected during the workshop(s)

With limited time remaining, how can TT provide assistance to these efforts?

Limited Time Remaining



- Public Meeting 1: **June 21st**
- Public Meeting 2: **June 26th**
- Public Meeting 3: **June 28th**
- Public Meeting 4: **July 1st**
- Addressing public comments, feedback: **July 2nd** through **July 8th**
- TDEM & FEMA Review – *begins* **July 9th**

Next Steps

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Next Steps

- ✓ Drafting/technical review of HMP Draft (ongoing)
- ✓ Hazard Risk Ranking (in progress)
- ✓ Prioritization of Action Items (in progress)
- ✓ Working with BES to finish last round of edits to figures
- ✓ Prepare SC Meeting #6 (meeting materials, poster print-offs, hard copy of the MJ-HMP).

HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

THANK YOU!

2025 Harris County MJ-HMP Core Planning Team Meeting

May 28, 2025

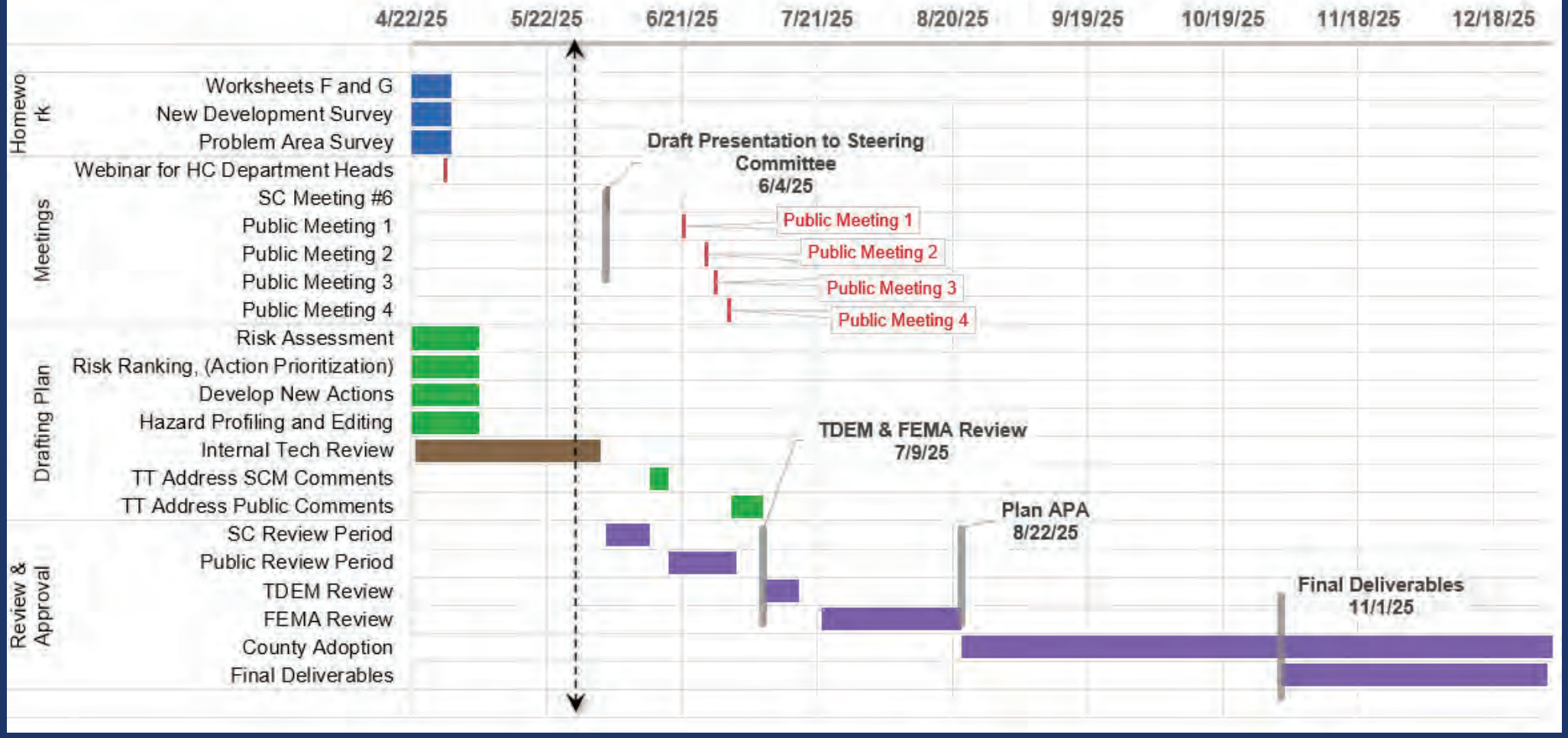
Agenda

- Remaining Timeline
- SCM 6 Logistics
- SCM 6 Agenda
- Hazard Risk Ranking for the County
- Prioritization of Action Iteam
- MJ-HMP Graphics/Figures
- Adoption Resolution Examples
- Community Workshops

Project Timeline & *Remaining Tasks*

HARRIS COUNTY
Office of Homeland Security & Emergency Management

2025 Harris County Project Timeline



SC Meeting #6 – *Meeting Logistics*

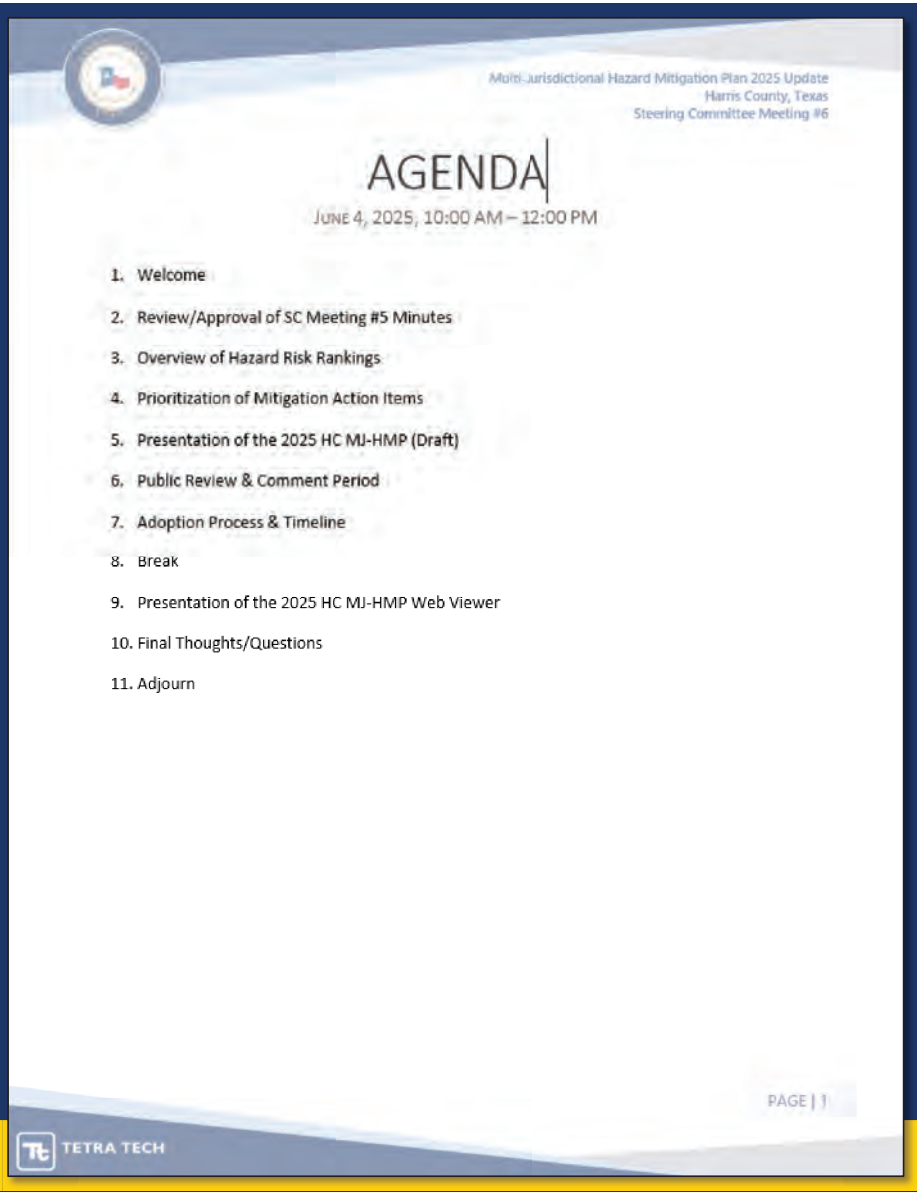
HARRIS COUNTY
Office of Homeland Security & Emergency Management

- **Arrival time at Pasadena City Hall**
- **Meeting materials provided by TT:**
  - Sign-in Sheets, Agendas
  - Printed copy of the MJ-HMP Draft, poster prints
  - Adoption Resolution Example/Template
  - Projector? Cords?
- **Meeting materials provided by HCOHSEM:**
  - Printer?
  - Internal Laptop for residents to explore the Interactive viewer

SC Meeting #6 – *Agenda Preview*

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Preliminary Agenda



Multi-Jurisdictional Hazard Mitigation Plan 2025 Update
Harris County, Texas
Steering Committee Meeting #6

AGENDA

JUNE 4, 2025, 10:00 AM – 12:00 PM

1. Welcome
2. Review/Approval of SC Meeting #5 Minutes
3. Overview of Hazard Risk Rankings
4. Prioritization of Mitigation Action Items
5. Presentation of the 2025 HC MJ-HMP (Draft)
6. Public Review & Comment Period
7. Adoption Process & Timeline
8. Break
9. Presentation of the 2025 HC MJ-HMP Web Viewer
10. Final Thoughts/Questions
11. Adjourn

TETRA TECH

County Hazard Risk Ranking

HARRIS COUNTY

Office of Homeland Security & Emergency Management

Hazard Rankings:
2020 vs 2025

Coastal Erosion	Low
Dam/Levee Failure	Low
Drought	High
Earthquake	Low
Flood	High
Hurricane /Coastal Storms	High
Mass Movements	Low
Severe Weather	High
Tsunami	Low
Wildfire	High

Flooding	HIGH
Hurricane, Coastal Storm	HIGH
Extreme Heat	HIGH
Severe Weather	HIGH
Drought	HIGH
Extreme Cold	MODERATE
Dam/Levee Failure	MODERATE
Toxic Release / Hazardous Materials	MODERATE
Technology Failure	MODERATE
Energy Pipeline Failure	MODERATE
Coastal Erosion	LOW
Earthquakes	LOW
Mass Movement	LOW
Tsunami	LOW
Wildfire	LOW
Public Health Emergency	LOW
Utility Failure	LOW

Prioritizing Action Items

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Meeting Graphics and MJ-HMP Figures

HARRIS COUNTY
Office of Homeland Security & Emergency Management

Maps to include in the SCM #6 Presentation

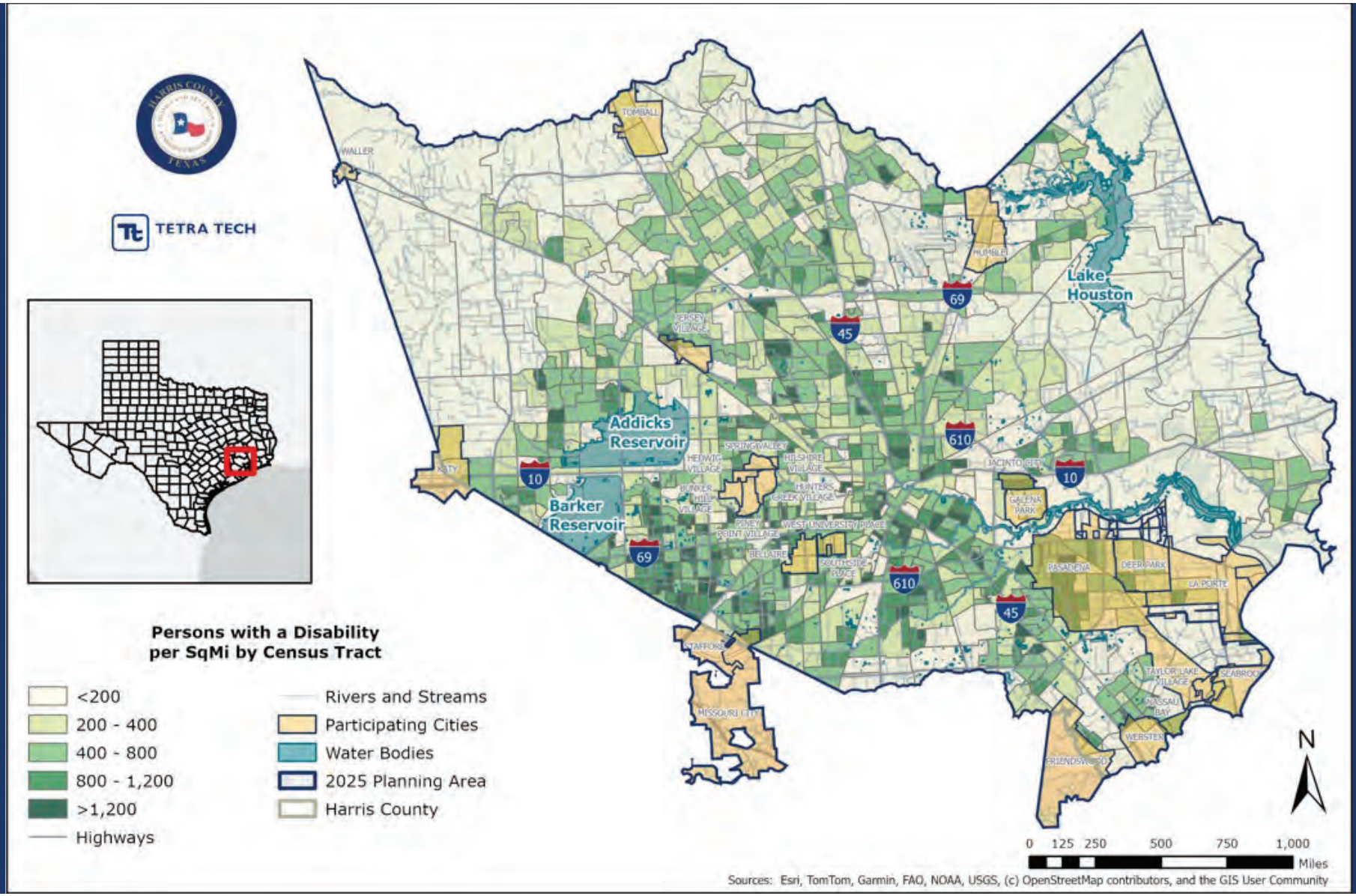
1. Vulnerable Population Maps

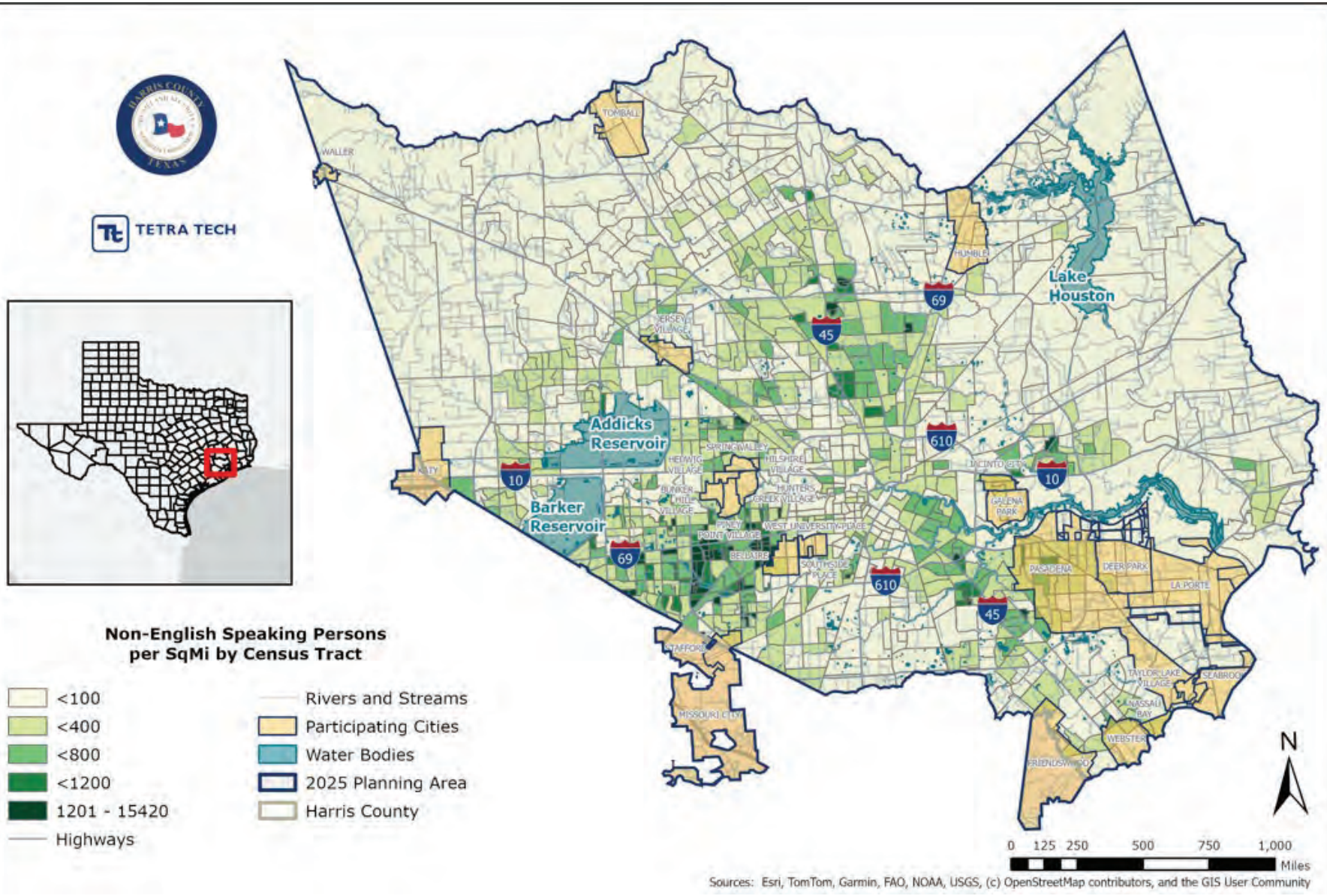
Disabilities, Non-English, Young, Elderly, Poverty, Total Population

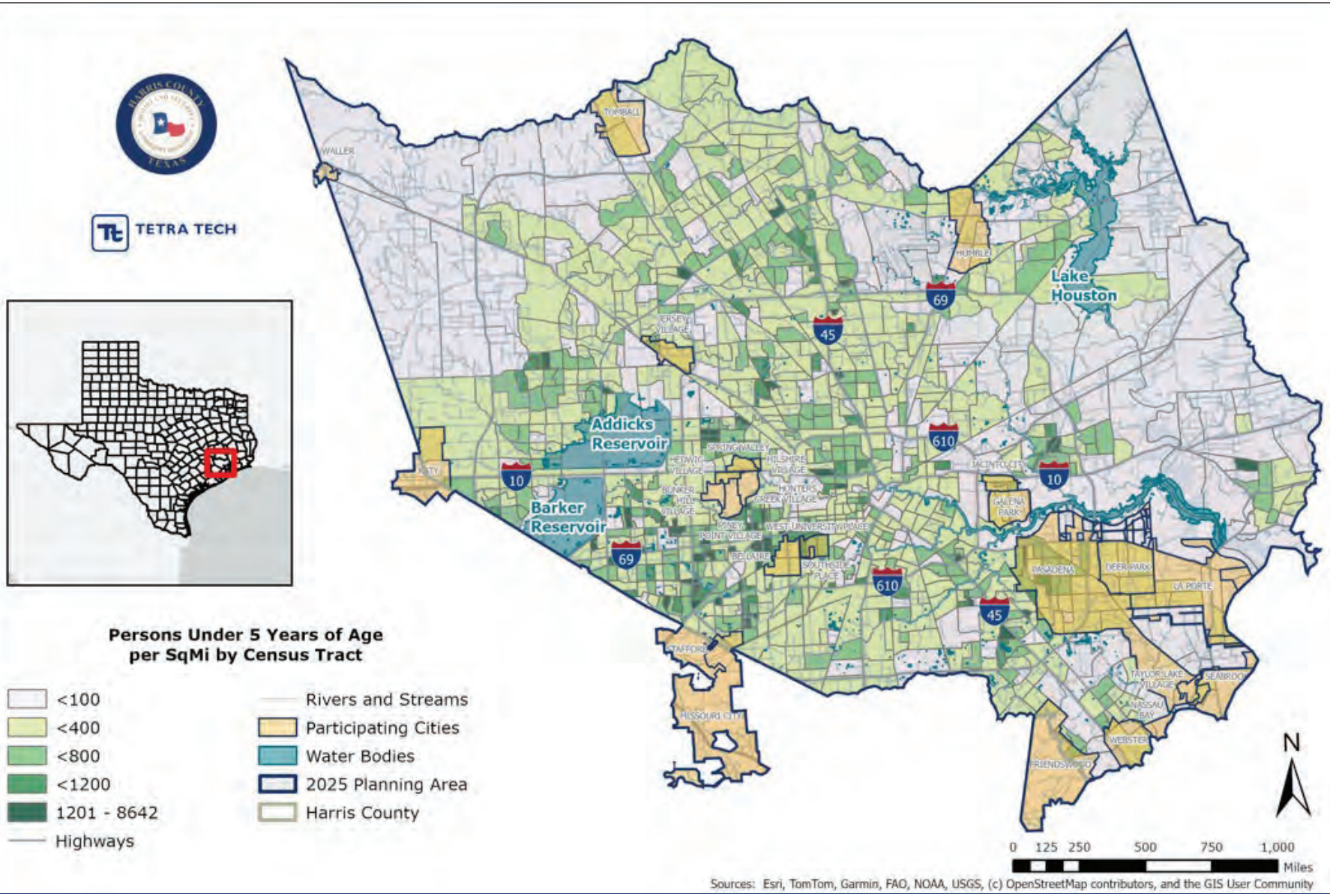
2. Community Lifeline maps?

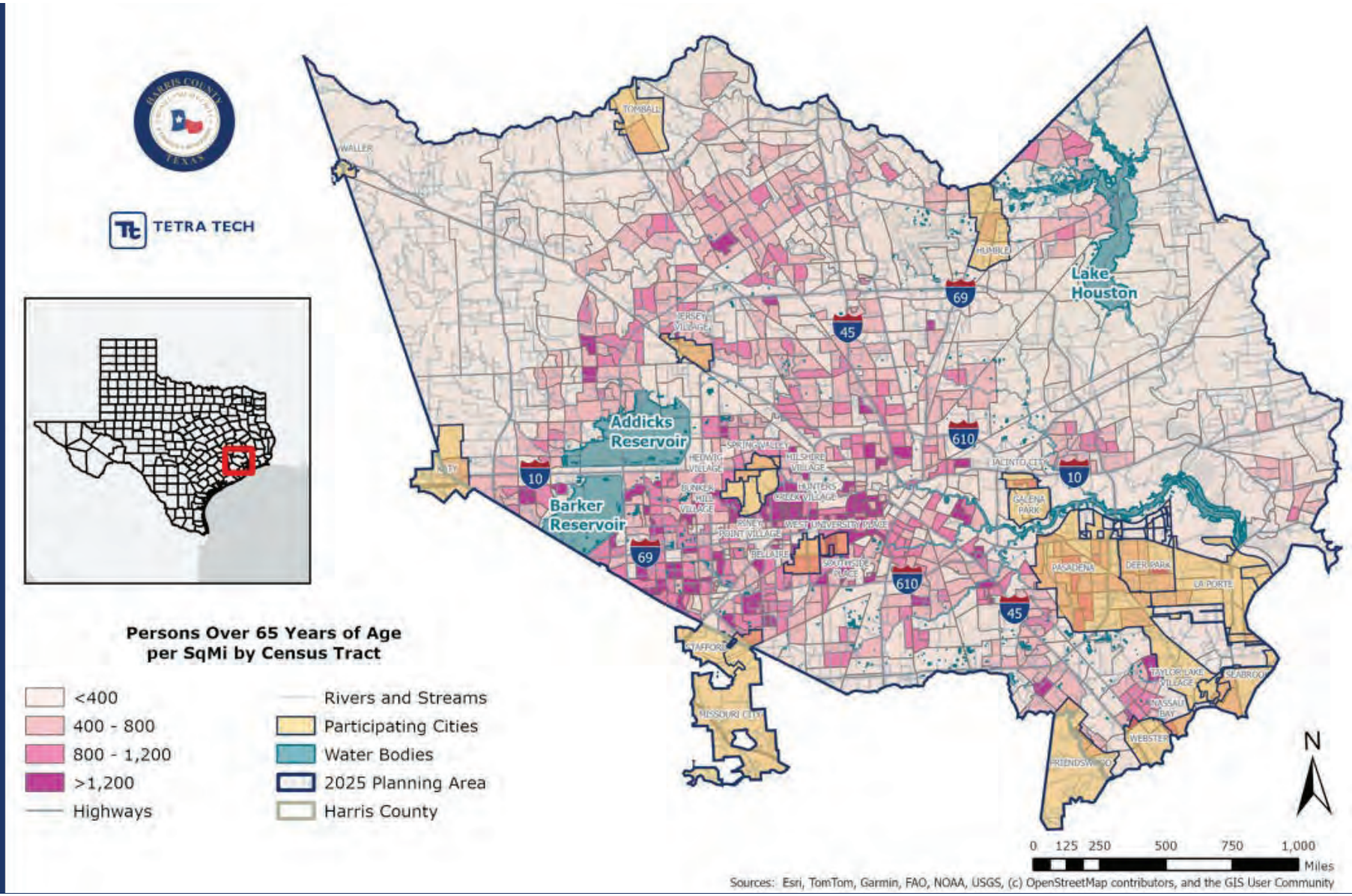
Communications, Energy, Food, Health, HazMat, Safety, Transportation, Water System

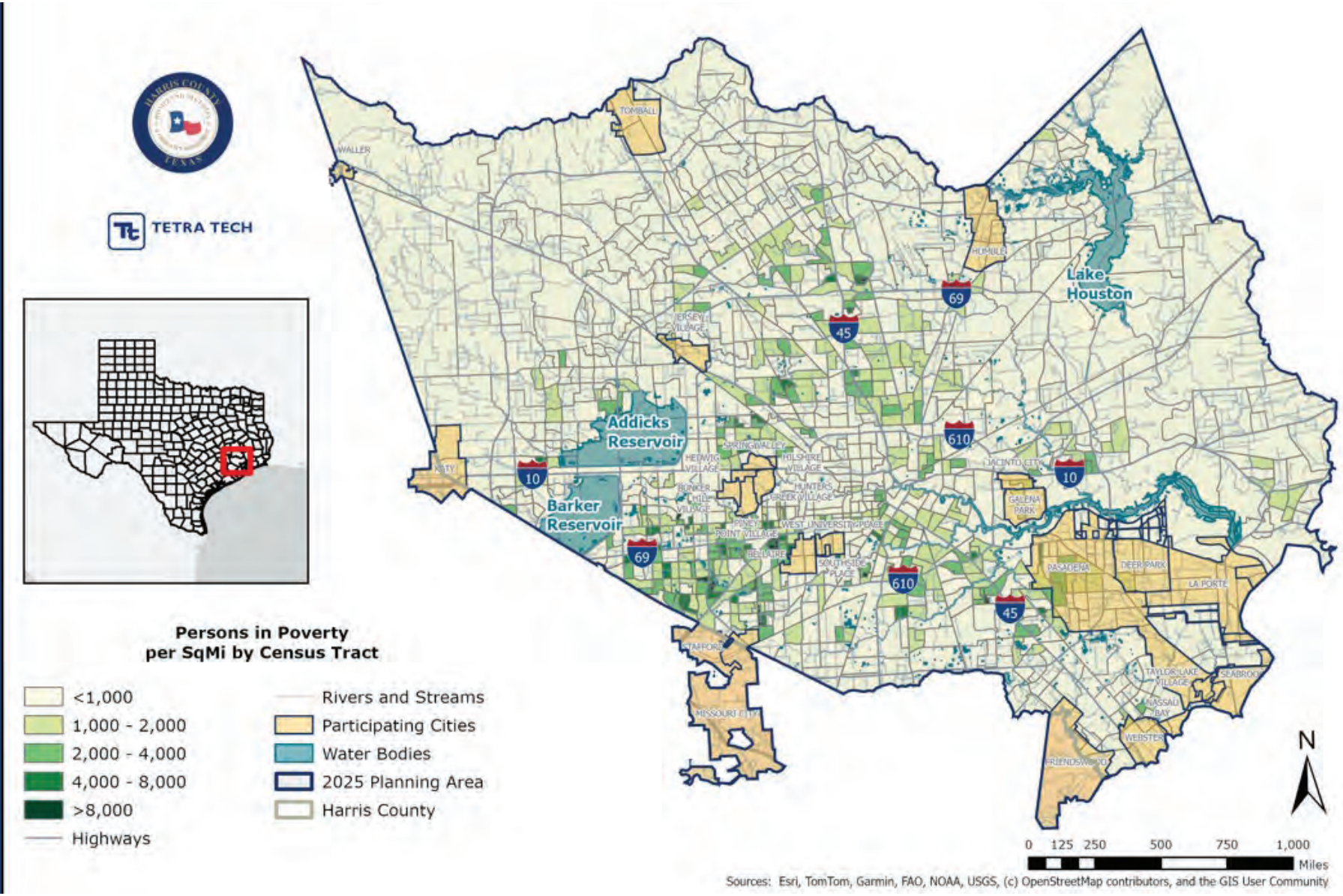
3. What else?

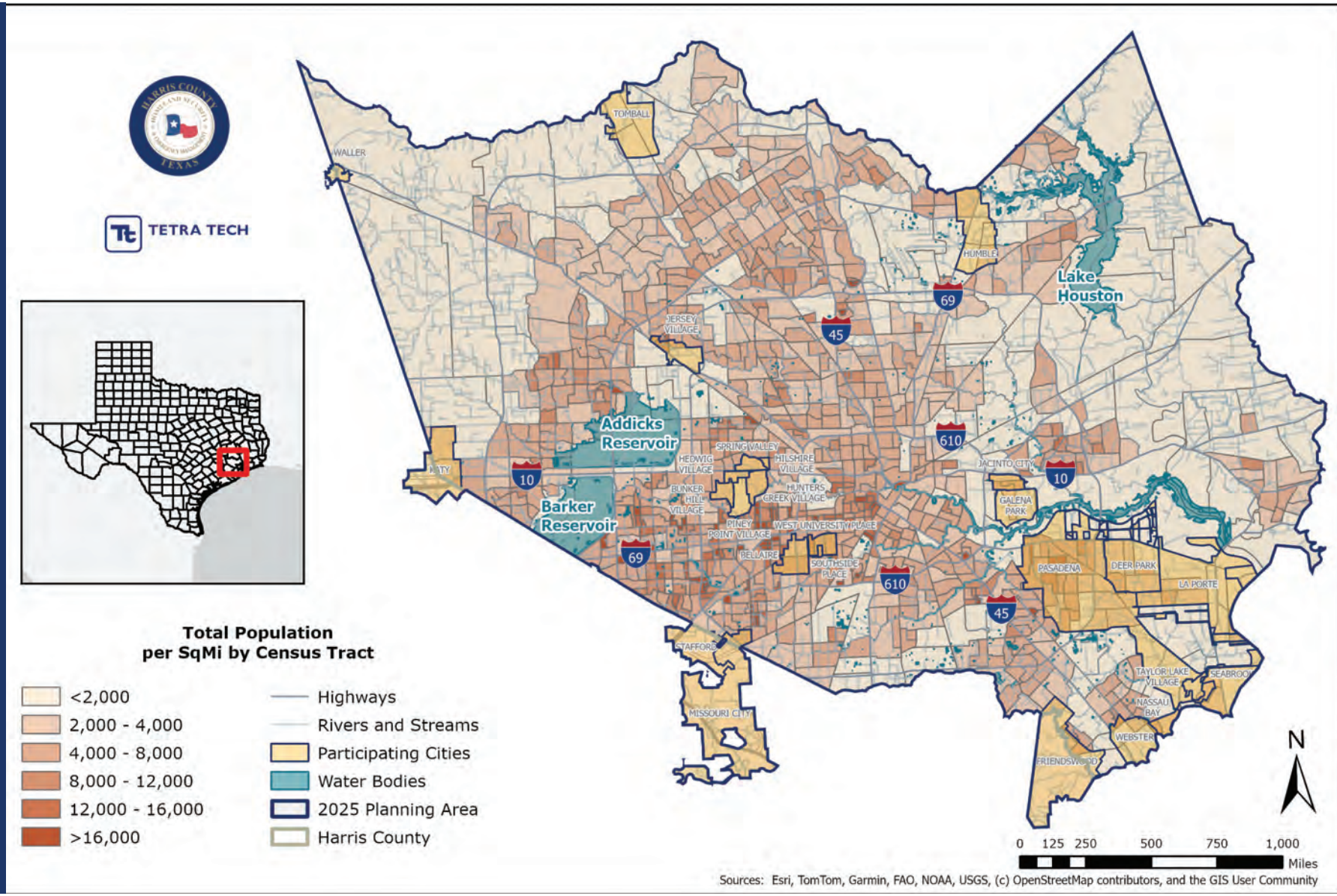

TETRA TECH
Persons with a Disability
per SqMi by Census Tract
<200
200 - 400
400 - 800
800 - 1,200
>1,200
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
SEABROOK
TAYLOR LAKE VILLAGE
NASSAU BAY
WEBSTER
FRIENDSWOOD
MISSOURI CITY
STAFFORD
SOUTH HOUSTON
BELLAIRE
WEST UNIVERSITY PLACE
PINEY POINT VILLAGE
HUNTERS CREEK VILLAGE
HILSHIRE VILLAGE
HEDWIG VILLAGE
SPRING VALLEY
JERSEY VILLAGE
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

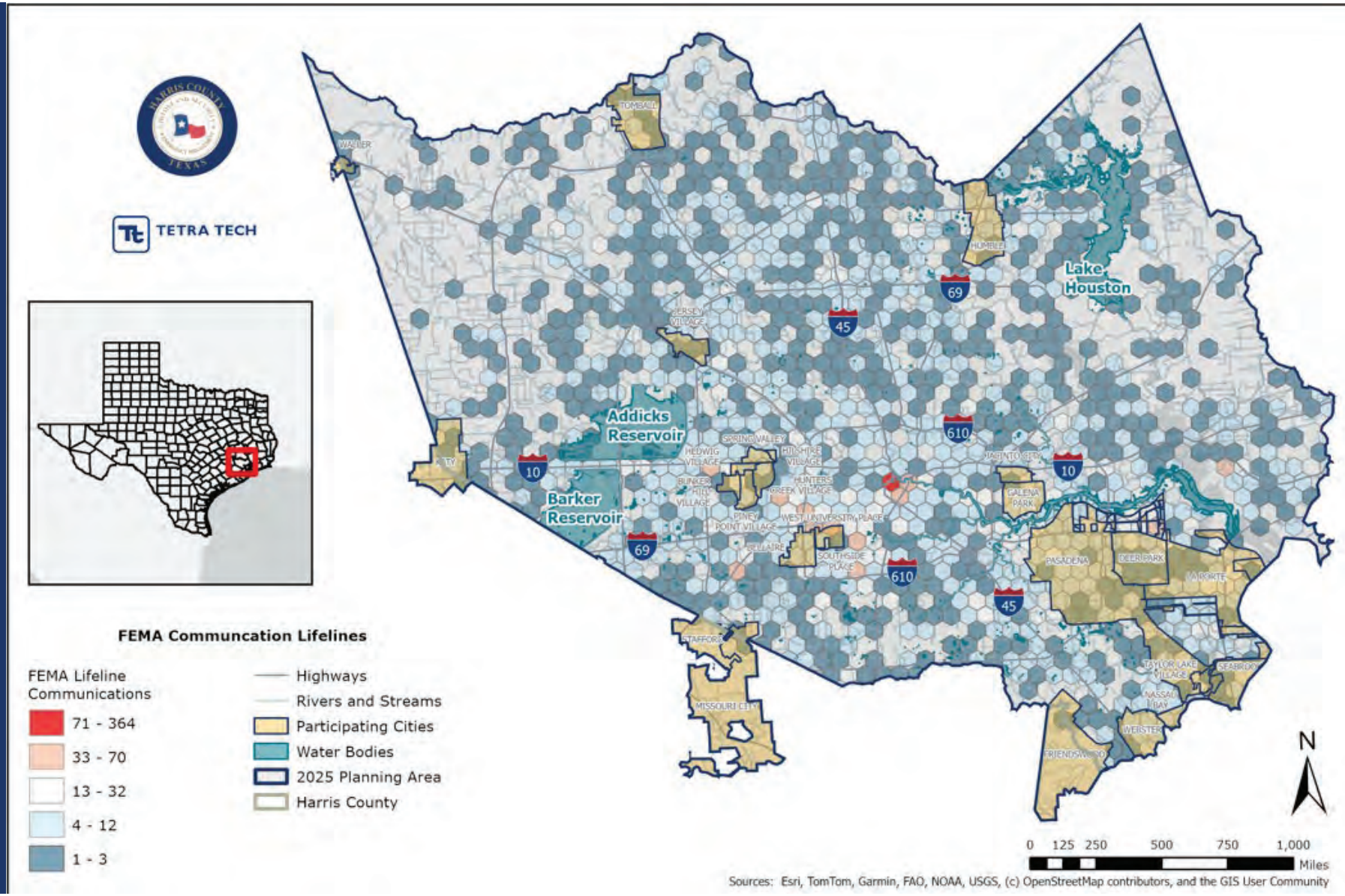

TETRA TECH
Non-English Speaking Persons
per SqMi by Census Tract
<100
<400
<800
<1200
1201 - 15420
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
HEDWIG VILLAGE
SPRING VALLEY
HILSHIRE VILLAGE
HUNTERS CREEK VILLAGE
BUNKER HILL VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
MISSOURI CITY
STAFFORD
KATY
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

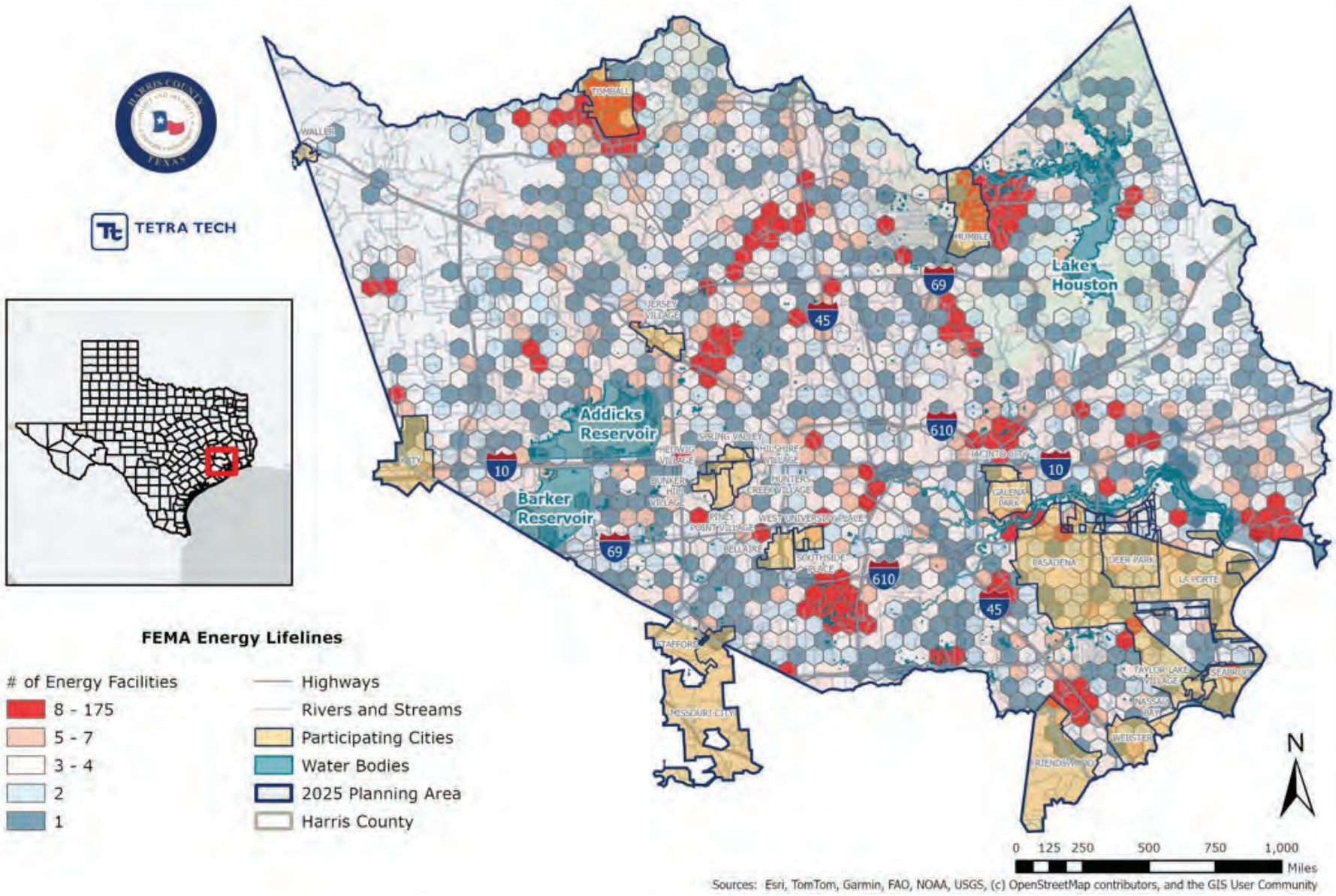

TETRA TECH
Persons Under 5 Years of Age
per SqMi by Census Tract
<100
<400
<800
<1200
1201 - 8642
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
WALLER
TOMBALL
HUMBLE
KATY
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
MISSOURI CITY
SUGAR VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
HUNTERS CREEK VILLAGE
BUNKER HILL VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
TAFFORD
10
45
69
610
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

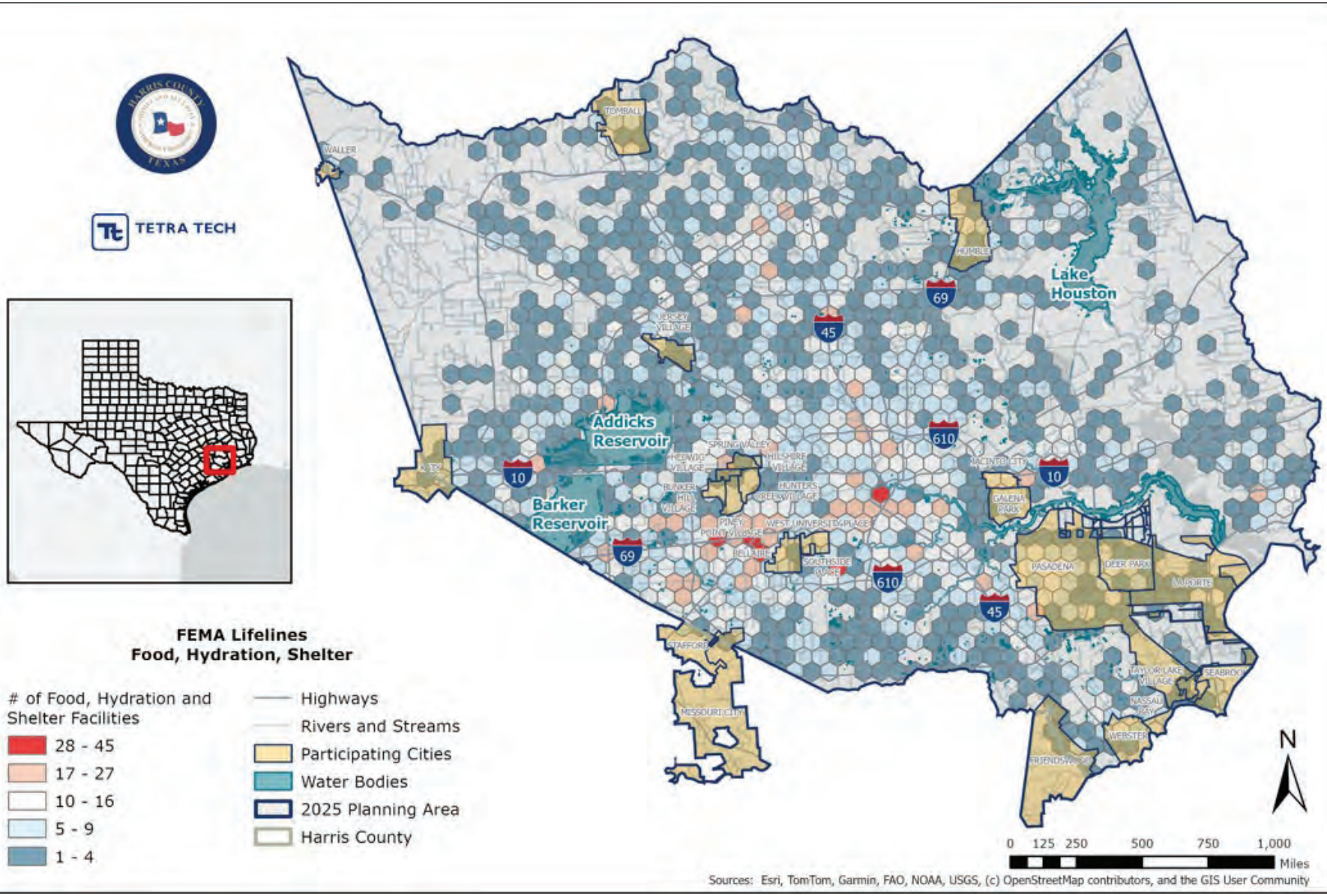

TETRA TECH
Addicks Reservoir
Barker Reservoir
Lake Houston
Persons Over 65 Years of Age per SqMi by Census Tract
<400
400 - 800
800 - 1,200
>1,200
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

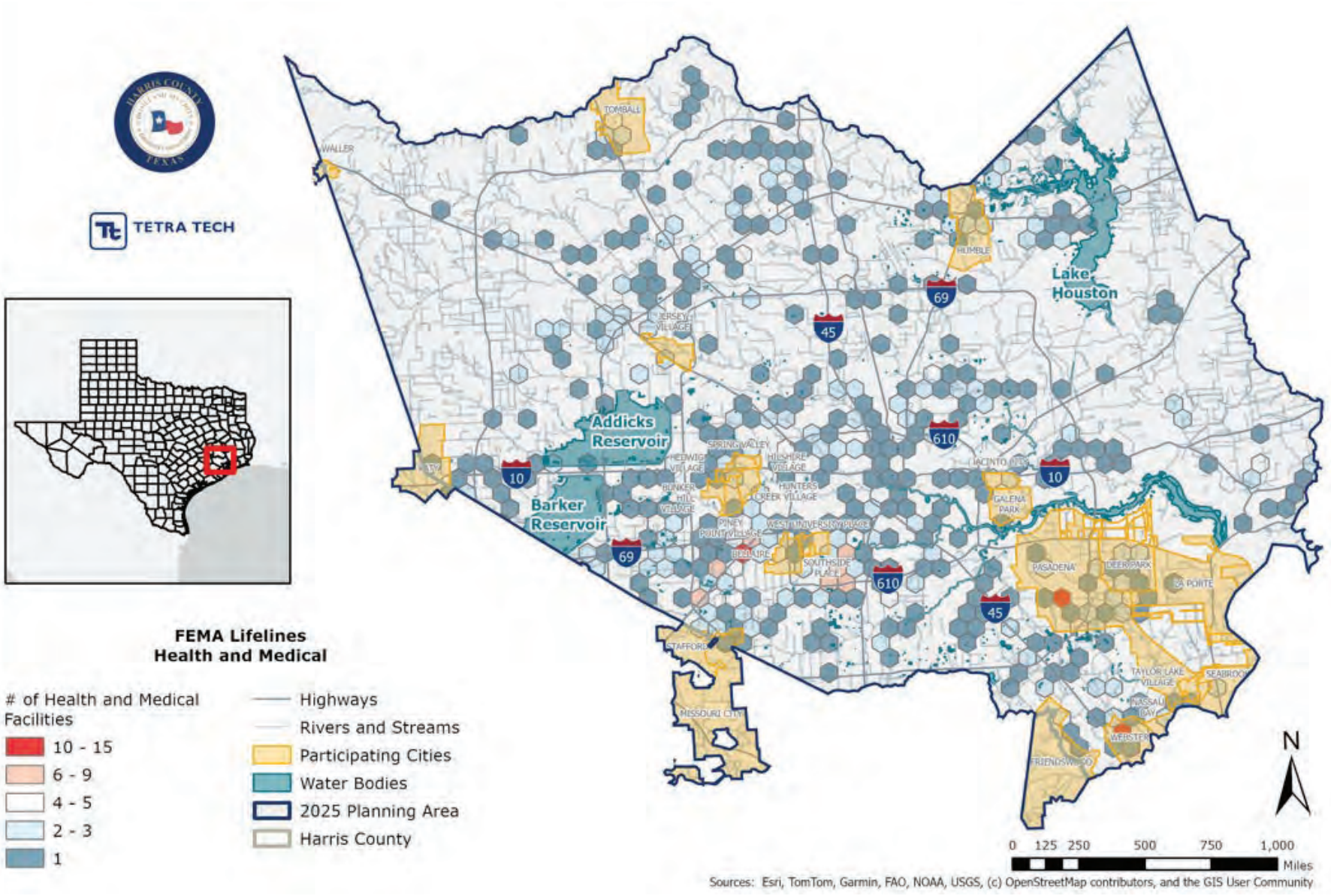

TETRA TECH
Persons in Poverty
per SqMi by Census Tract
<1,000
1,000 - 2,000
2,000 - 4,000
4,000 - 8,000
>8,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
KATY
PASADENA
DEER PARK
LA PORTE
SEABROOK
TAYLOR LAKE VILLAGE
NASSAU BAY
WEBSTER
FRIENDSWOOD
MISSOURI CITY
SOUTHSIDE PLACE
BELLAIRE
GALENA PARK
JACINTO CITY
10
45
69
610
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

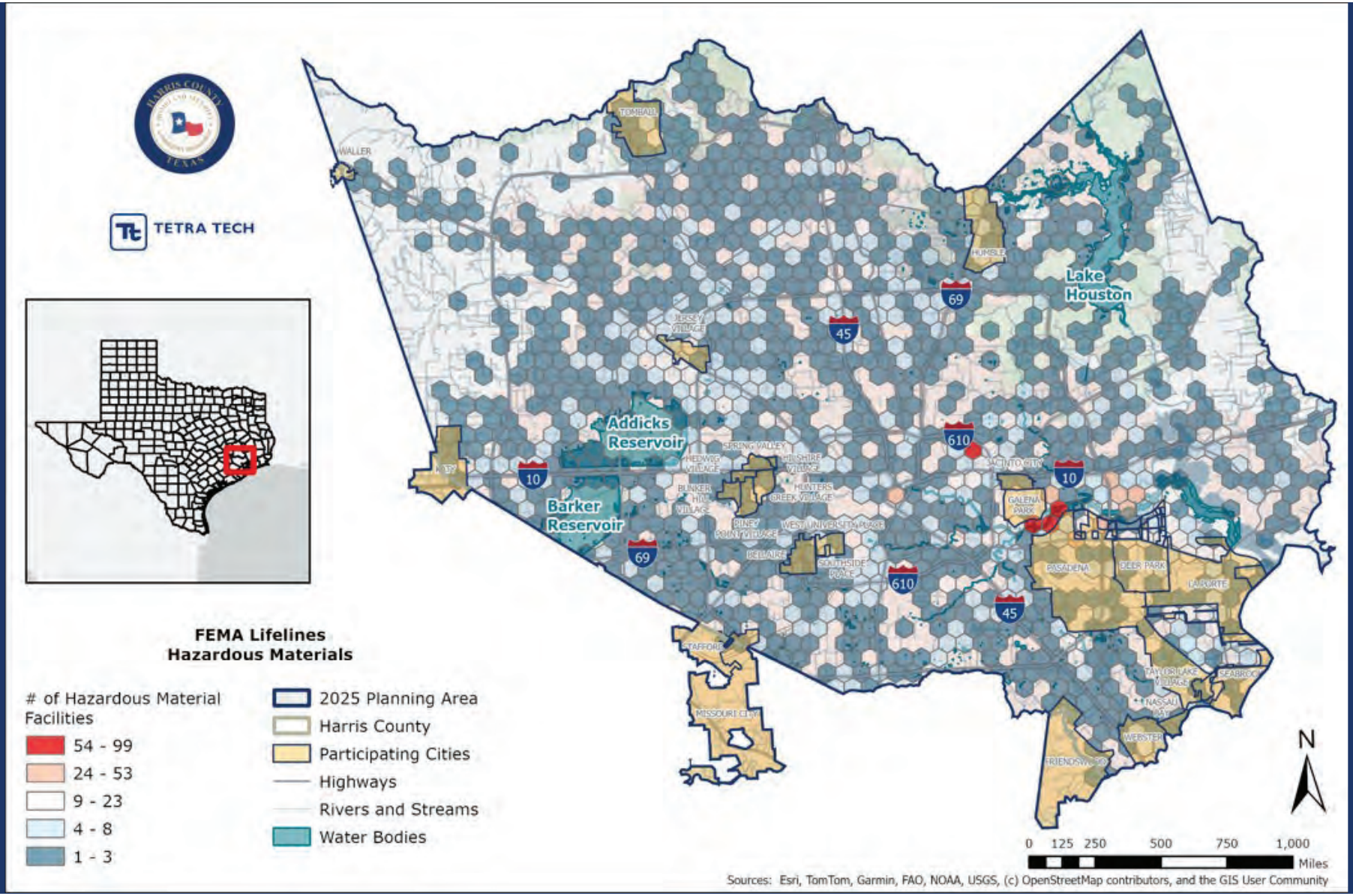

TETRA TECH
Total Population
per SqMi by Census Tract
<2,000
2,000 - 4,000
4,000 - 8,000
8,000 - 12,000
12,000 - 16,000
>16,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
BUNKER HILL VILLAGE
HUNTERS CREEK VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
STAFFORD
MISSOURI CITY
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

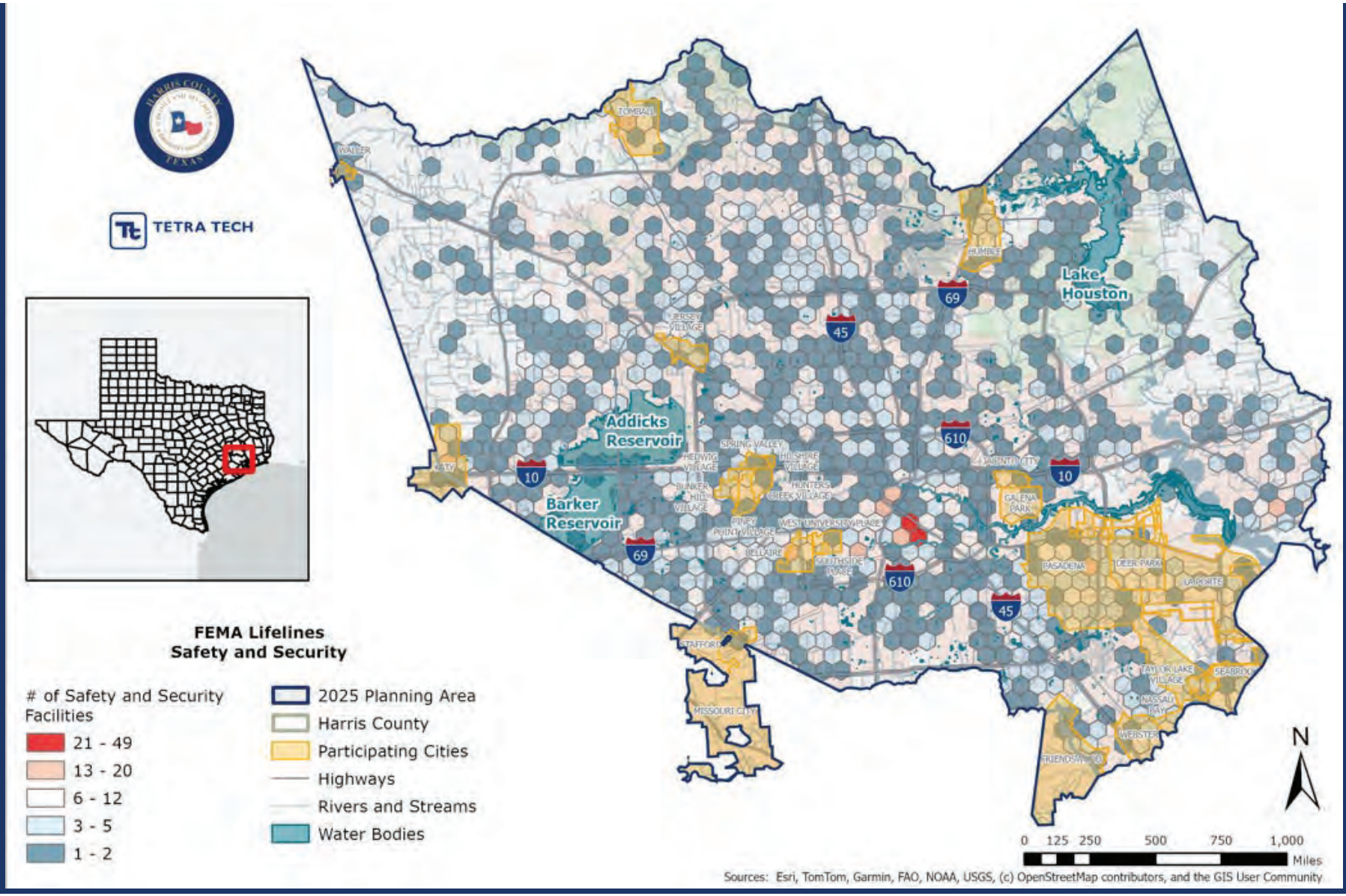

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Communcation Lifelines
FEMA Lifeline Communications
71 - 364
33 - 70
13 - 32
4 - 12
1 - 3
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

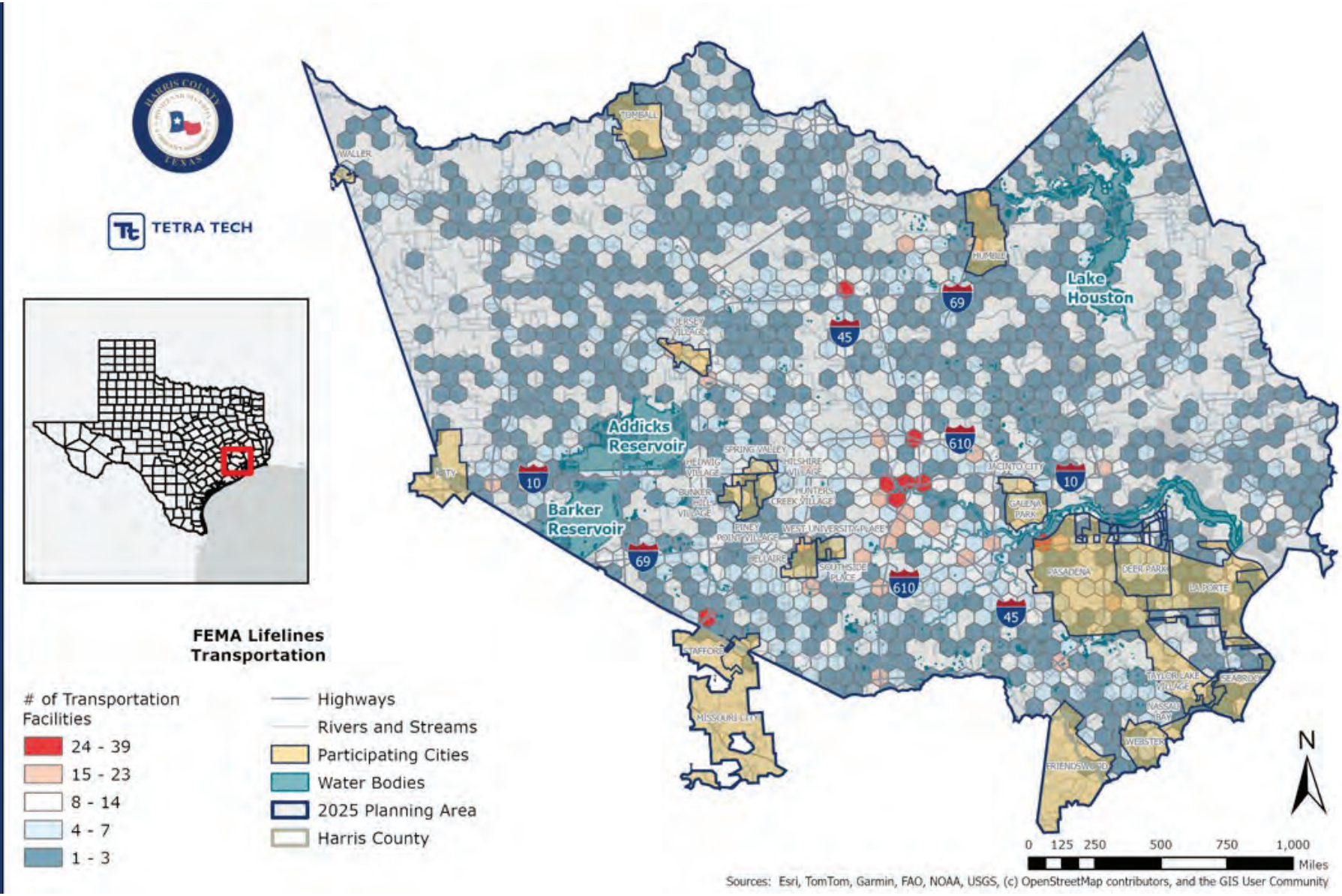

TETRA TECH
FEMA Energy Lifelines
# of Energy Facilities
8 - 175
5 - 7
3 - 4
2
1
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
BUNKER HILL VILLAGE
HUNTERS CREEK VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
HOUSTON
IACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
STAFFORD
MISSOURI CITY
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

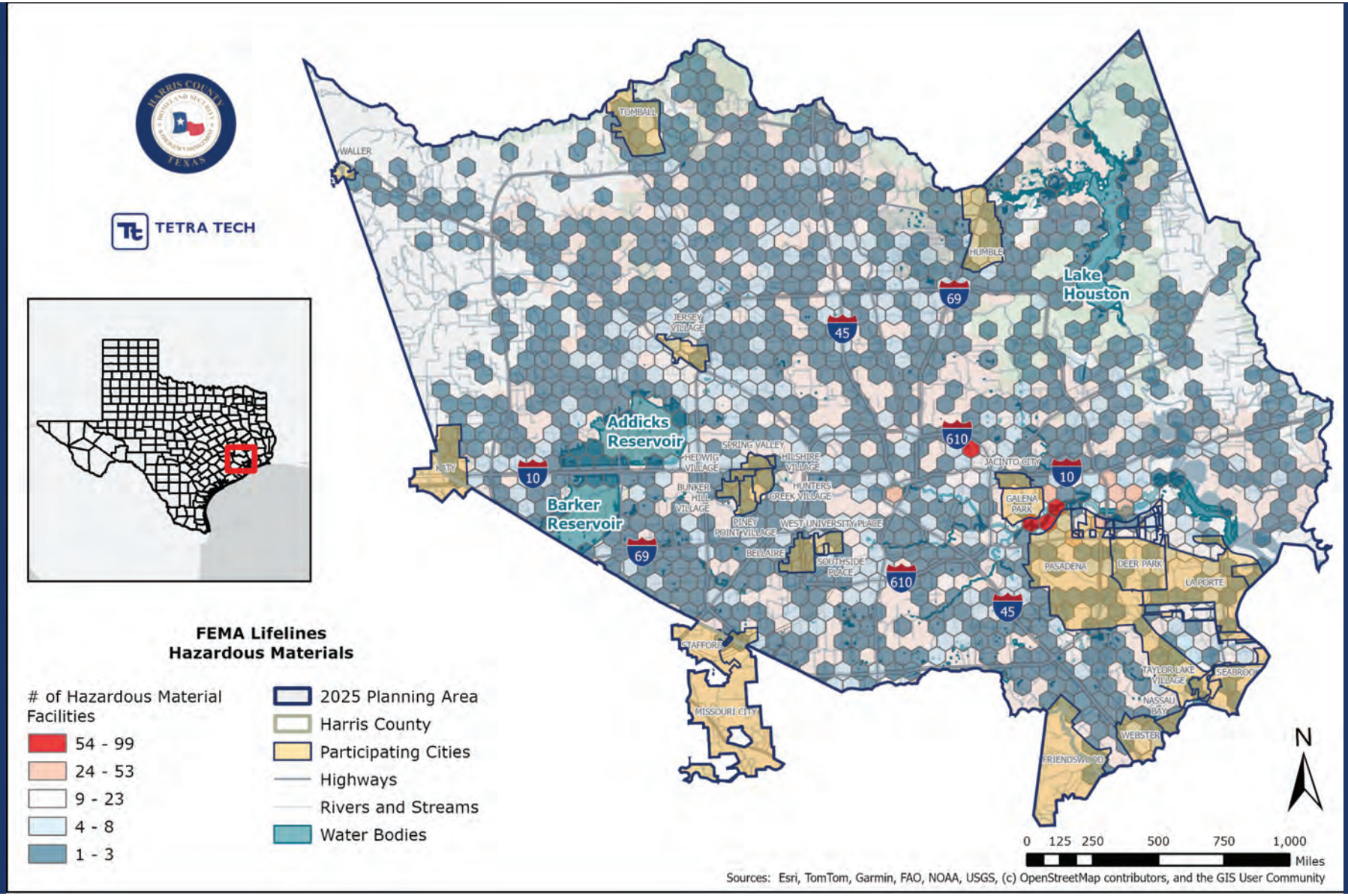

TETRA TECH
Addicks Reservoir
Barker Reservoir
Lake Houston
FEMA Lifelines
Food, Hydration, Shelter
# of Food, Hydration and Shelter Facilities
28 - 45
17 - 27
10 - 16
5 - 9
1 - 4
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

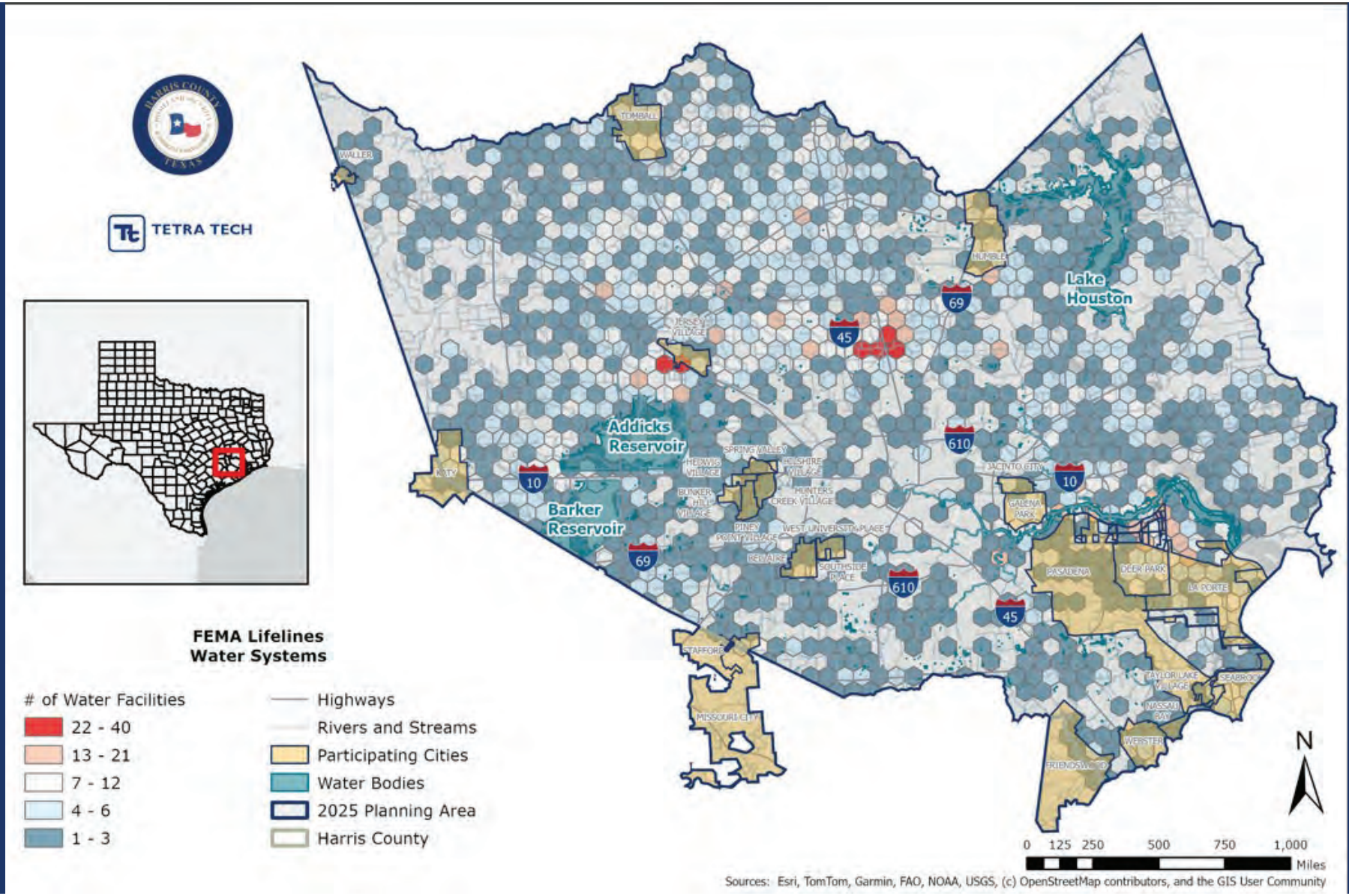

TETRA TECH
FEMA Lifelines
Health and Medical
# of Health and Medical Facilities
10 - 15
6 - 9
4 - 5
2 - 3
1
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
HUNTERS CREEK VILLAGE
BUNKER HILL VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAFFORD
MISSOURI CITY
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Addicks Reservoir
Barker Reservoir
Lake Houston
FEMA Lifelines
Hazardous Materials
# of Hazardous Material Facilities
54 - 99
24 - 53
9 - 23
4 - 8
1 - 3
2025 Planning Area
Harris County
Participating Cities
Highways
Rivers and Streams
Water Bodies
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Addicks Reservoir
Barker Reservoir
Lake Houston
FEMA Lifelines
Safety and Security
# of Safety and Security Facilities
21 - 49
13 - 20
6 - 12
3 - 5
1 - 2
2025 Planning Area
Harris County
Participating Cities
Highways
Rivers and Streams
Water Bodies
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

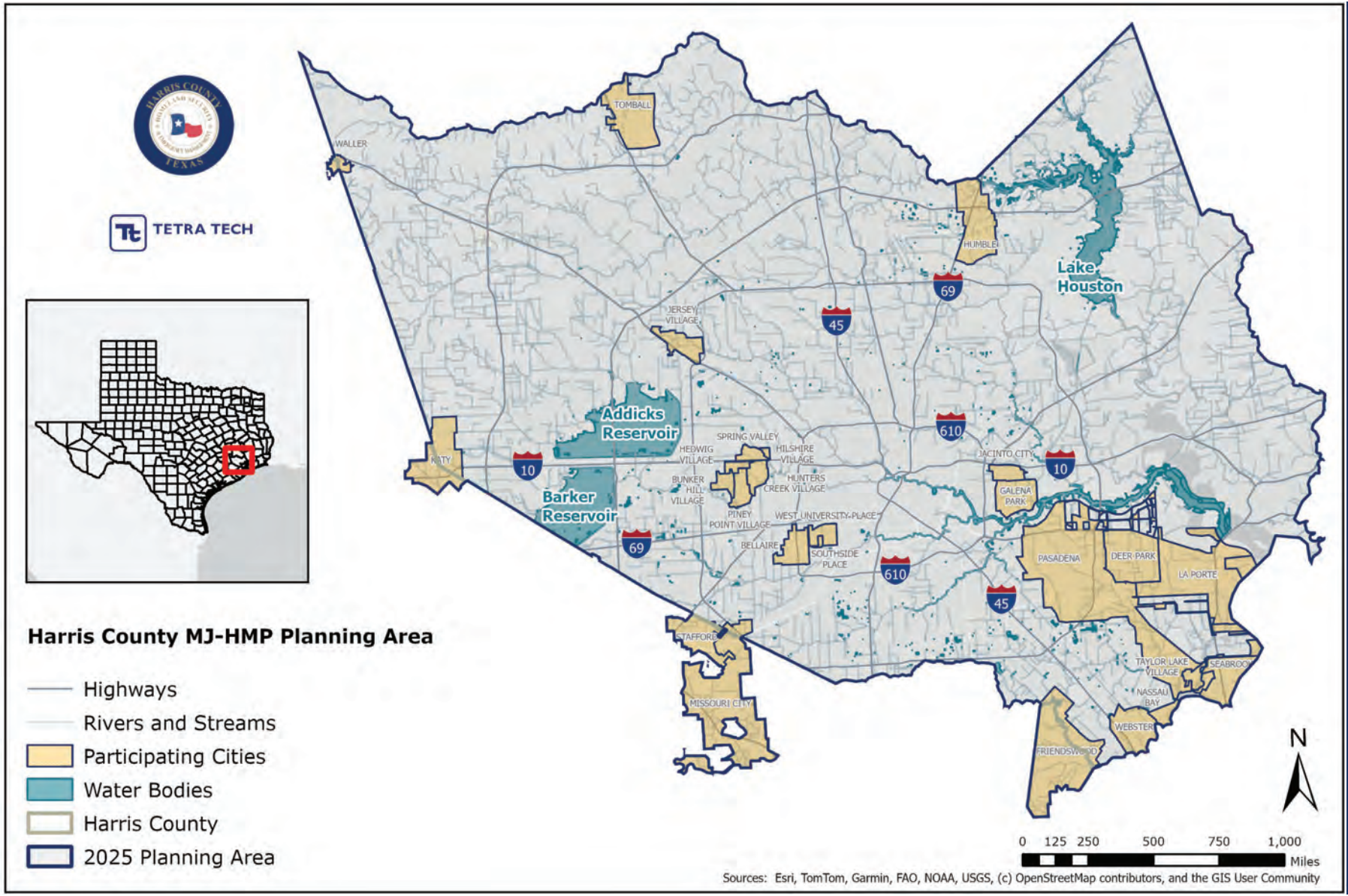

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Transportation
# of Transportation Facilities
24 - 39
15 - 23
8 - 14
4 - 7
1 - 3
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

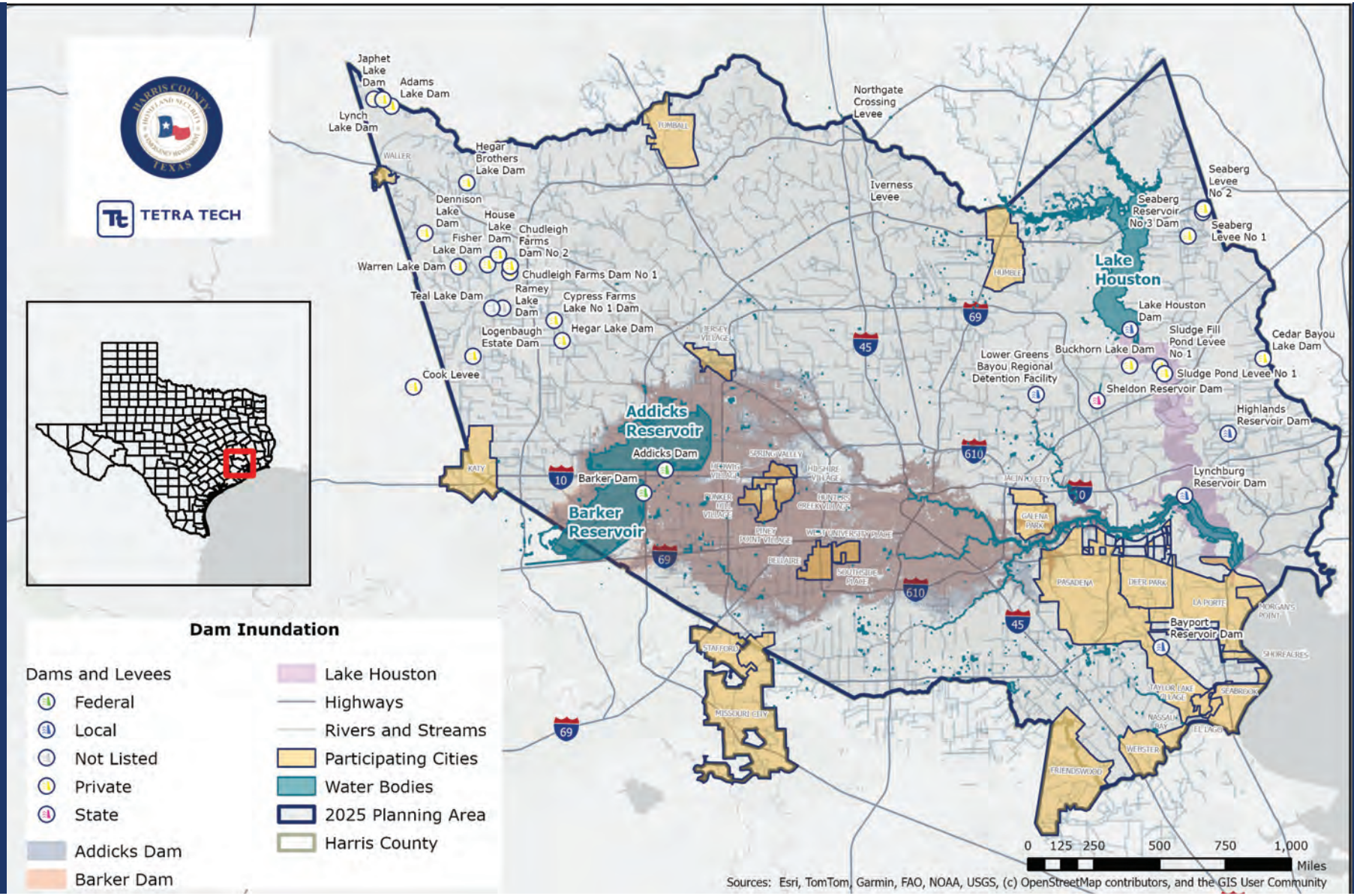

TETRA TECH
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
HEDWIG VILLAGE
SPRING VALLEY
HILSHIRE VILLAGE
HUNTERS CREEK VILLAGE
BUNKER HILL VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
STAFFORD
MISSOURI CITY
KATY
69
45
610
10
FEMA Lifelines
Hazardous Materials
# of Hazardous Material Facilities
54 - 99
24 - 53
9 - 23
4 - 8
1 - 3
2025 Planning Area
Harris County
Participating Cities
Highways
Rivers and Streams
Water Bodies
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

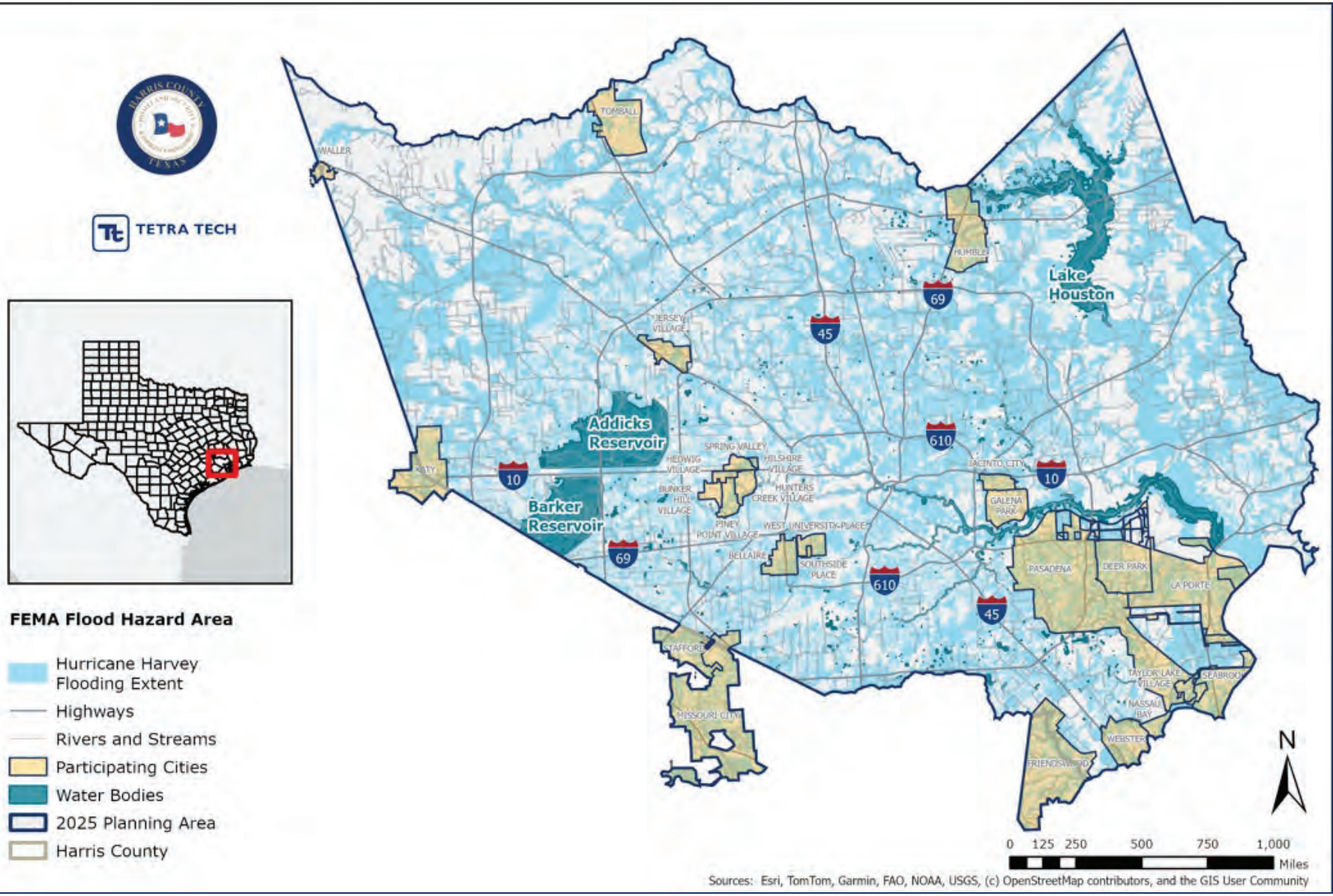

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
FEMA Lifelines
Water Systems
# of Water Facilities
22 - 40
13 - 21
7 - 12
4 - 6
1 - 3
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

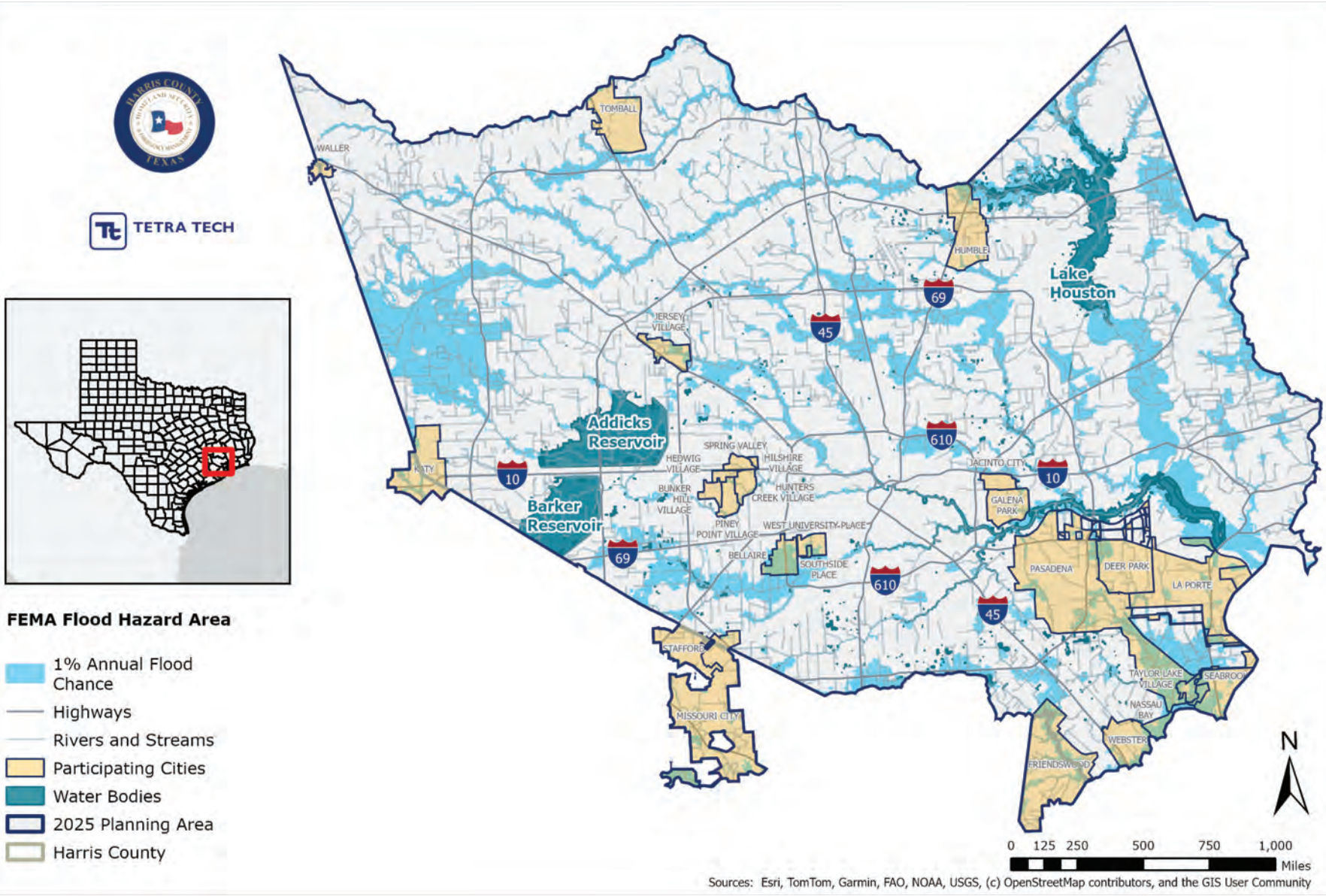
Maps to print for SC Meeting #6:

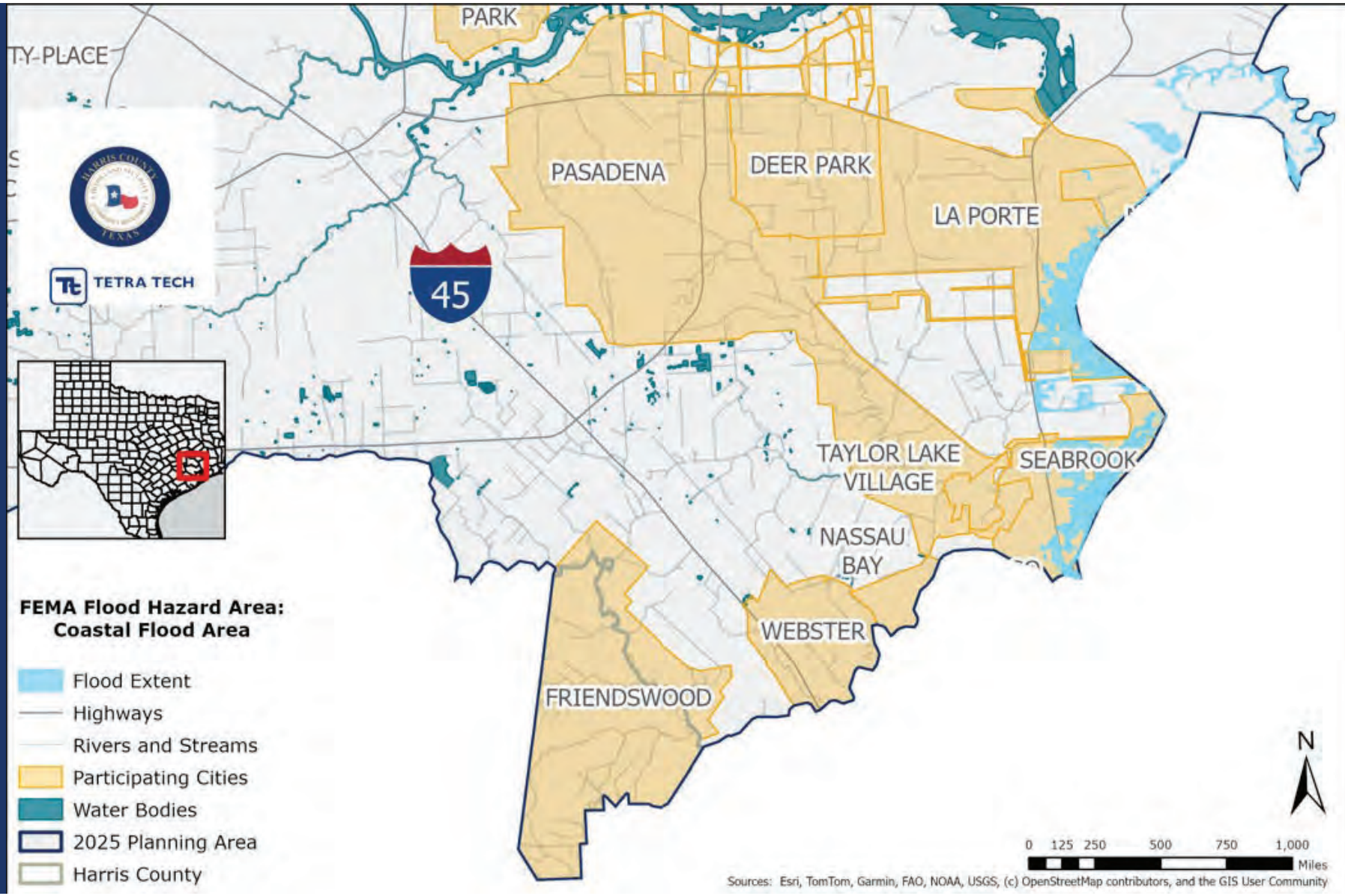
1. Planning area
2. Dam Locations
3. Flooding
   (1%, Harvey, Coastal)
4. Wildfire
   Ignition Density
   Wildfire Threat
5. What else?

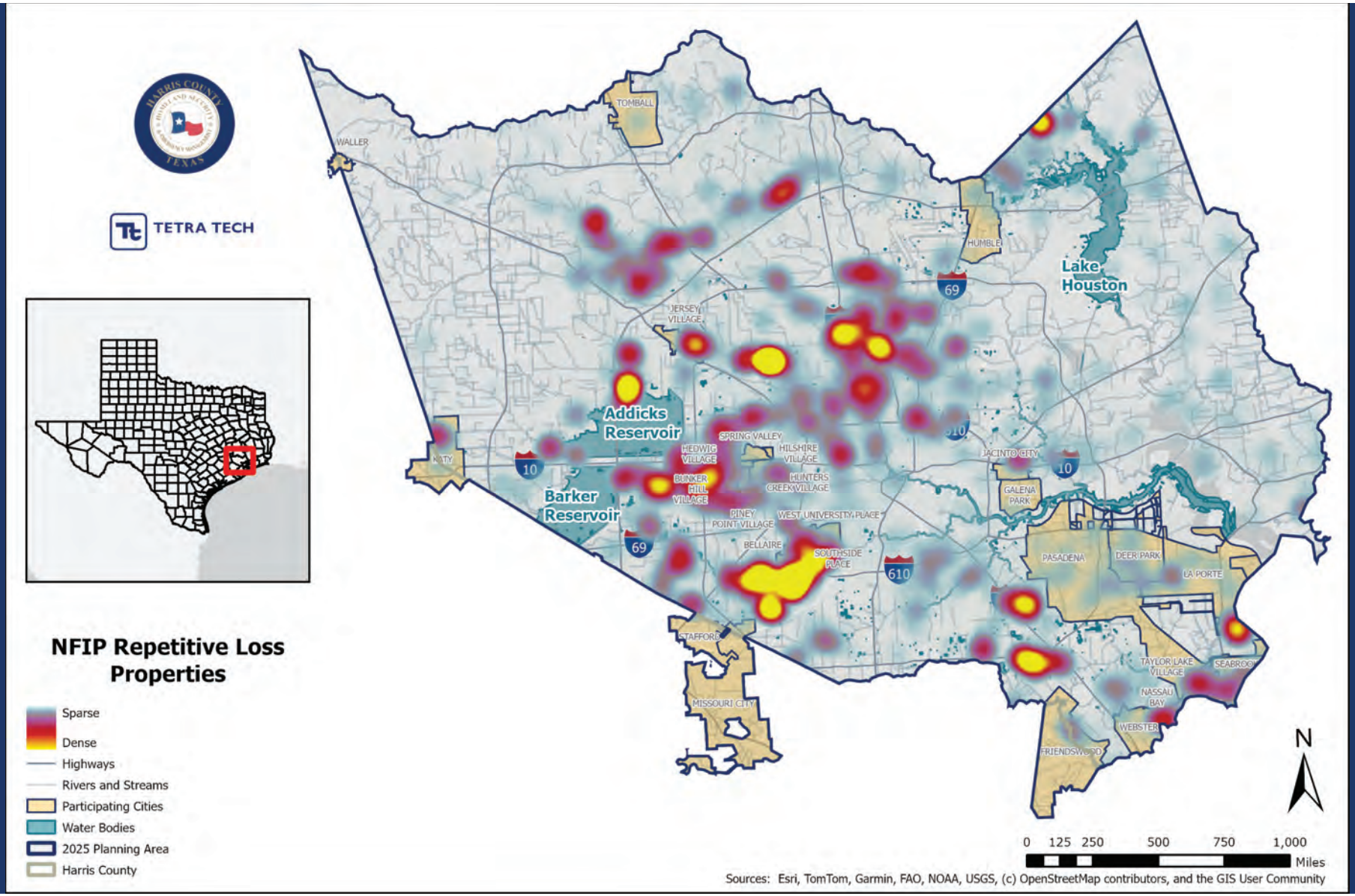

TETRA TECH
Harris County MJ-HMP Planning Area
Highways
Rivers and Streams
Participating Cities
Water Bodies
Harris County
2025 Planning Area
TOMBALL
WALLER
HUMBLE
Lake Houston
JERSEY VILLAGE
Addicks Reservoir
Barker Reservoir
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
BUNKER HILL VILLAGE
HUNTERS CREEK VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
KATY
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAFFORD
MISSOURI CITY
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
45
69
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

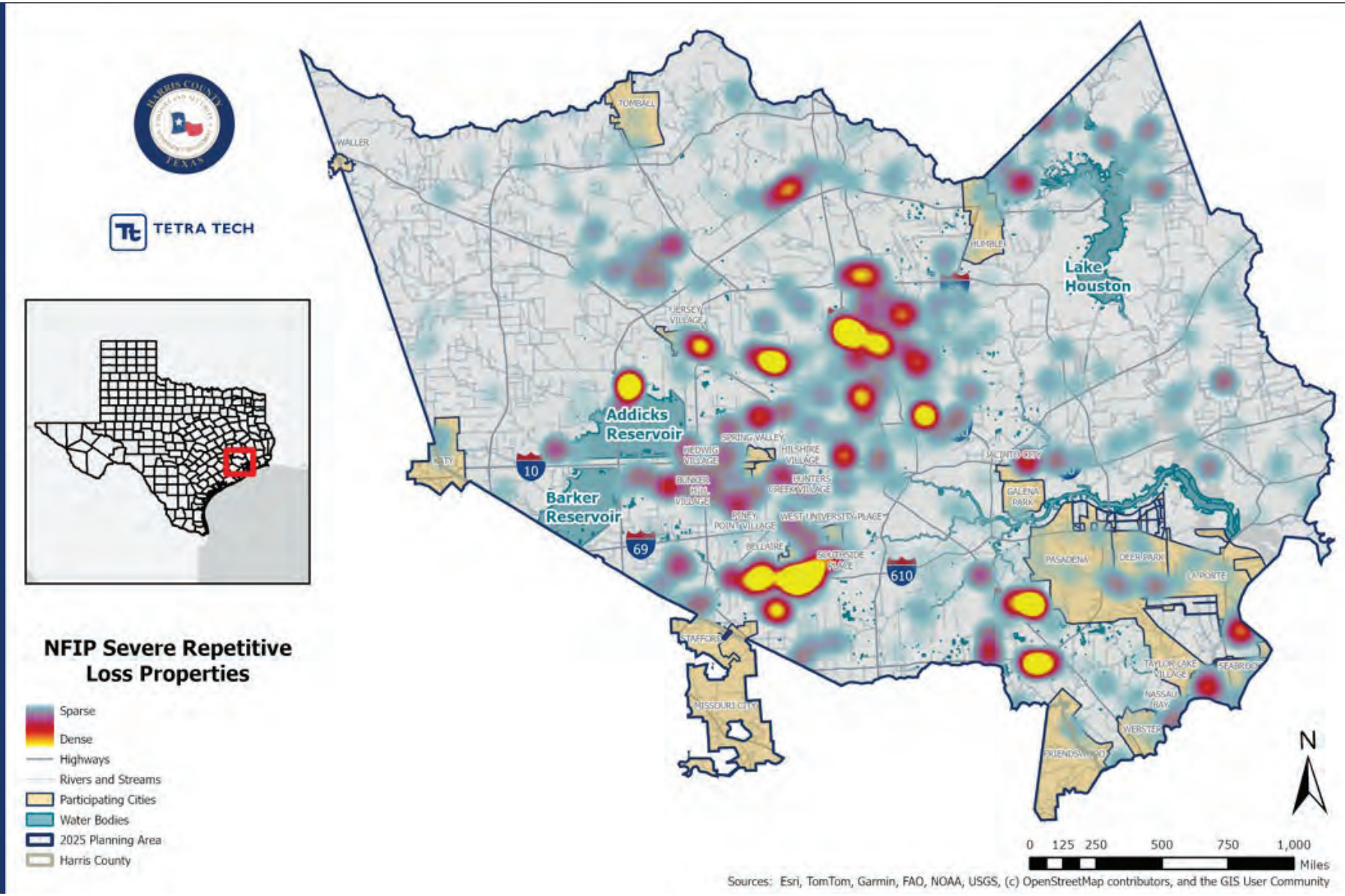

TETRA TECH
Japhet Lake Dam
Adams Lake Dam
Lynch Lake Dam
Hegar Brothers Lake Dam
Dennison Lake Dam
House Lake Dam
Chudleigh Farms Dam No 2
Fisher Lake Dam
Warren Lake Dam
Chudleigh Farms Dam No 1
Teal Lake Dam
Ramey Lake Dam
Cypress Farms Lake No 1 Dam
Logenbaugh Estate Dam
Hegar Lake Dam
Cook Levee
Northgate Crossing Levee
Iverness Levee
Seaberg Levee No 2
Seaberg Reservoir No 3 Dam
Seaberg Levee No 1
Lake Houston
Lake Houston Dam
Sludge Fill Pond Levee No 1
Cedar Bayou Lake Dam
Buckhorn Lake Dam
Sludge Pond Levee No 1
Lower Greens Bayou Regional Detention Facility
Sheldon Reservoir Dam
Highlands Reservoir Dam
Addicks Reservoir
Addicks Dam
Barker Dam
Barker Reservoir
Lynchburg Reservoir Dam
Bayport Reservoir Dam
Dam Inundation
Dams and Levees
Federal
Local
Not Listed
Private
State
Addicks Dam
Barker Dam
Lake Houston
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

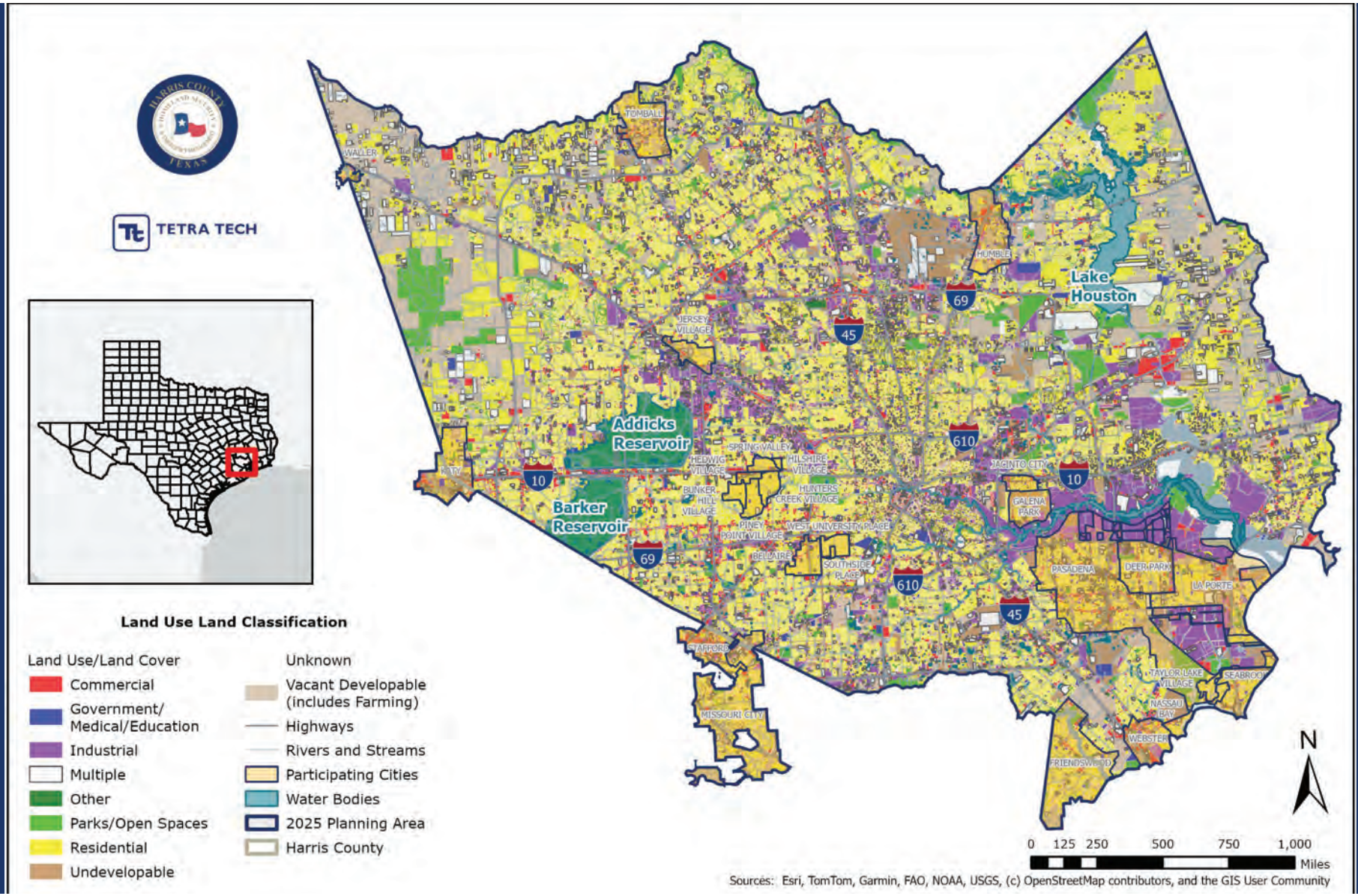

TETRA TECH
FEMA Flood Hazard Area
Hurricane Harvey Flooding Extent
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
HUNTERS CREEK VILLAGE
BUNKER HILL VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
BACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAFFORD
MISSOURI CITY
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

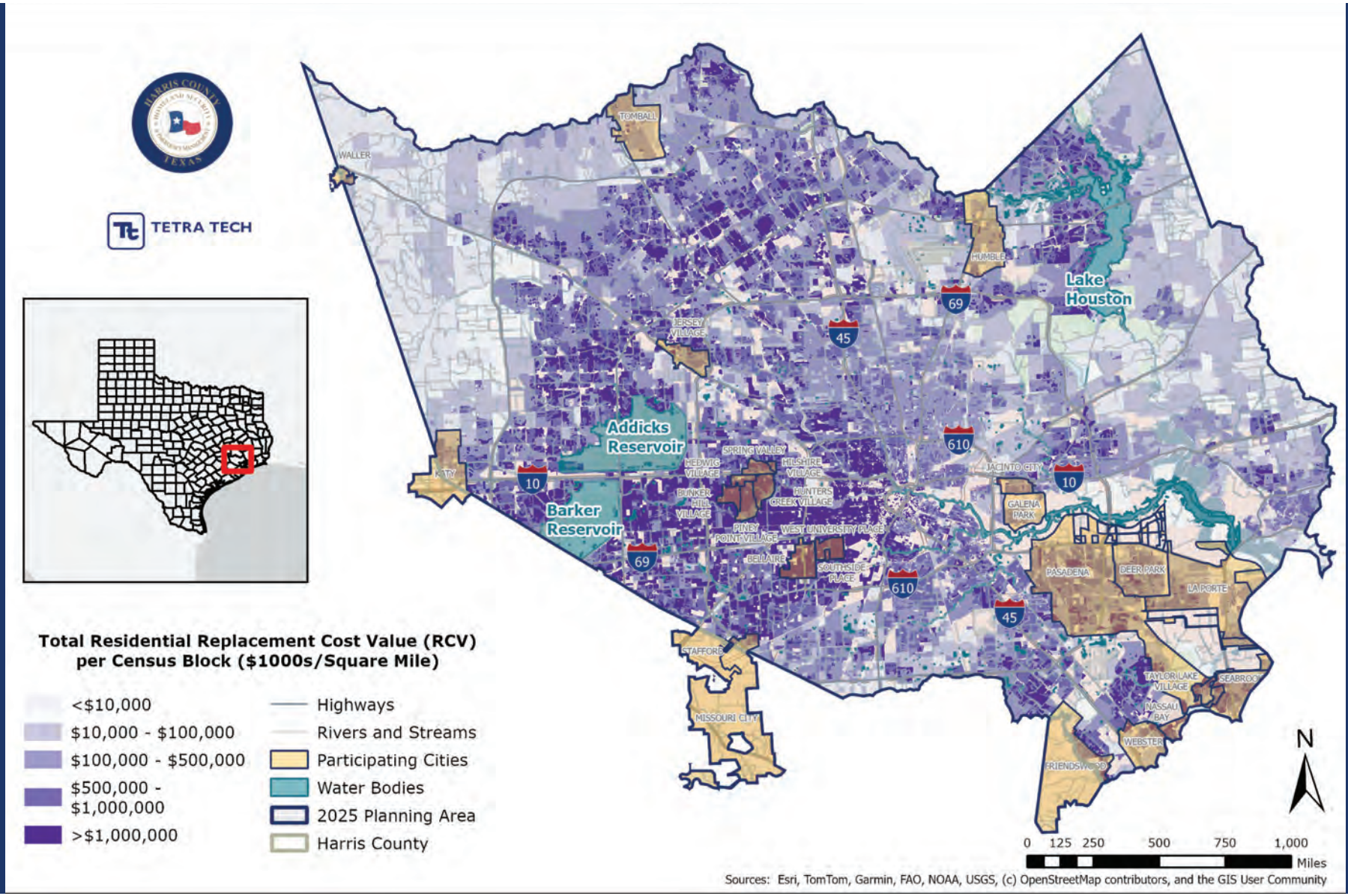

TETRA TECH
FEMA Flood Hazard Area
1% Annual Flood Chance
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
TOMBALL
WALLER
HUMBLE
Lake Houston
JERSEY VILLAGE
Addicks Reservoir
Barker Reservoir
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
BUNKER HILL VILLAGE
HUNTERS CREEK VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
KATY
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
STAFFORD
MISSOURI CITY
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


PARK
TY-PLACE
PASADENA
DEER PARK
LA PORTE
45
TETRA TECH
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
FEMA Flood Hazard Area:
Coastal Flood Area
Flood Extent
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
NFIP Repetitive Loss Properties
Sparse
Dense
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
HUNTERS CREEK VILLAGE
BUNKER HILL VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
STAFFORD
MISSOURI CITY
KATY
69
10
610
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
NFIP Severe Repetitive Loss Properties
Sparse
Dense
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
TOMBALL
WALLER
HUMBLE
Lake Houston
JERSEY VILLAGE
Addicks Reservoir
Barker Reservoir
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
HUNTERS CREEK VILLAGE
BUNKER HILL VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
MISSOURI CITY
STAFFORD
KATY
10
69
610
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Land Use Land Classification
Land Use/Land Cover
Commercial
Government/ Medical/Education
Industrial
Multiple
Other
Parks/Open Spaces
Residential
Undevelopable
Unknown
Vacant Developable (includes Farming)
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
HEDWIG VILLAGE
SPRING VALLEY
HILSHIRE VILLAGE
HUNTERS CREEK VILLAGE
BUNKER HILL VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
MISSOURI CITY
STAFFORD
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Total Residential Replacement Cost Value (RCV) per Census Block ($1000s/Square Mile)
<$10,000
$10,000 - $100,000
$100,000 - $500,000
$500,000 - $1,000,000
>$1,000,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
KATY
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
BUNKER HILL VILLAGE
HUNTERS CREEK VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
STAFFORD
MISSOURI CITY
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
10
45
69
610
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

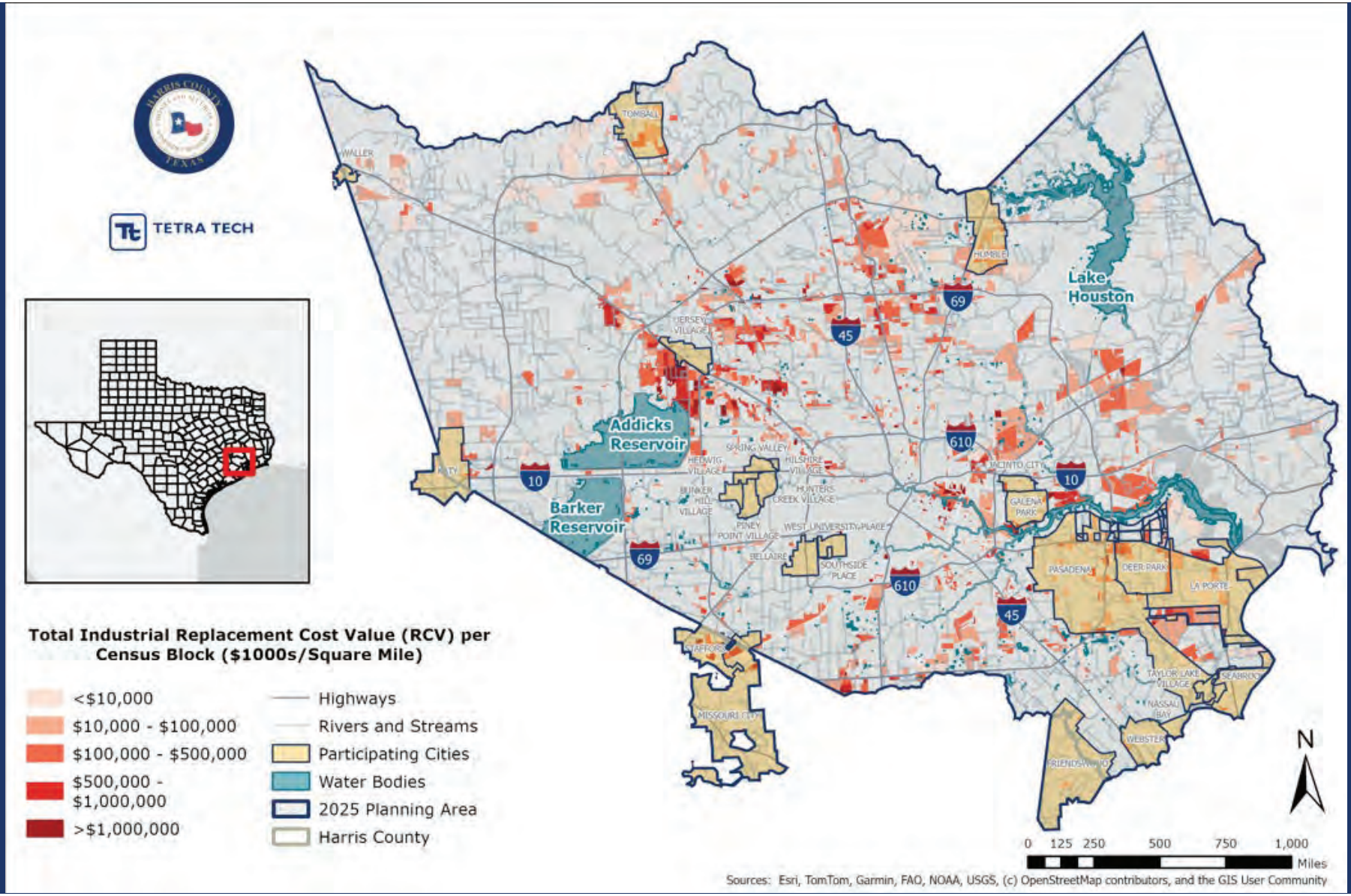
Total Commercial Replacement Cost Value (RCV) per Census Block ($1000s/Square Mile)

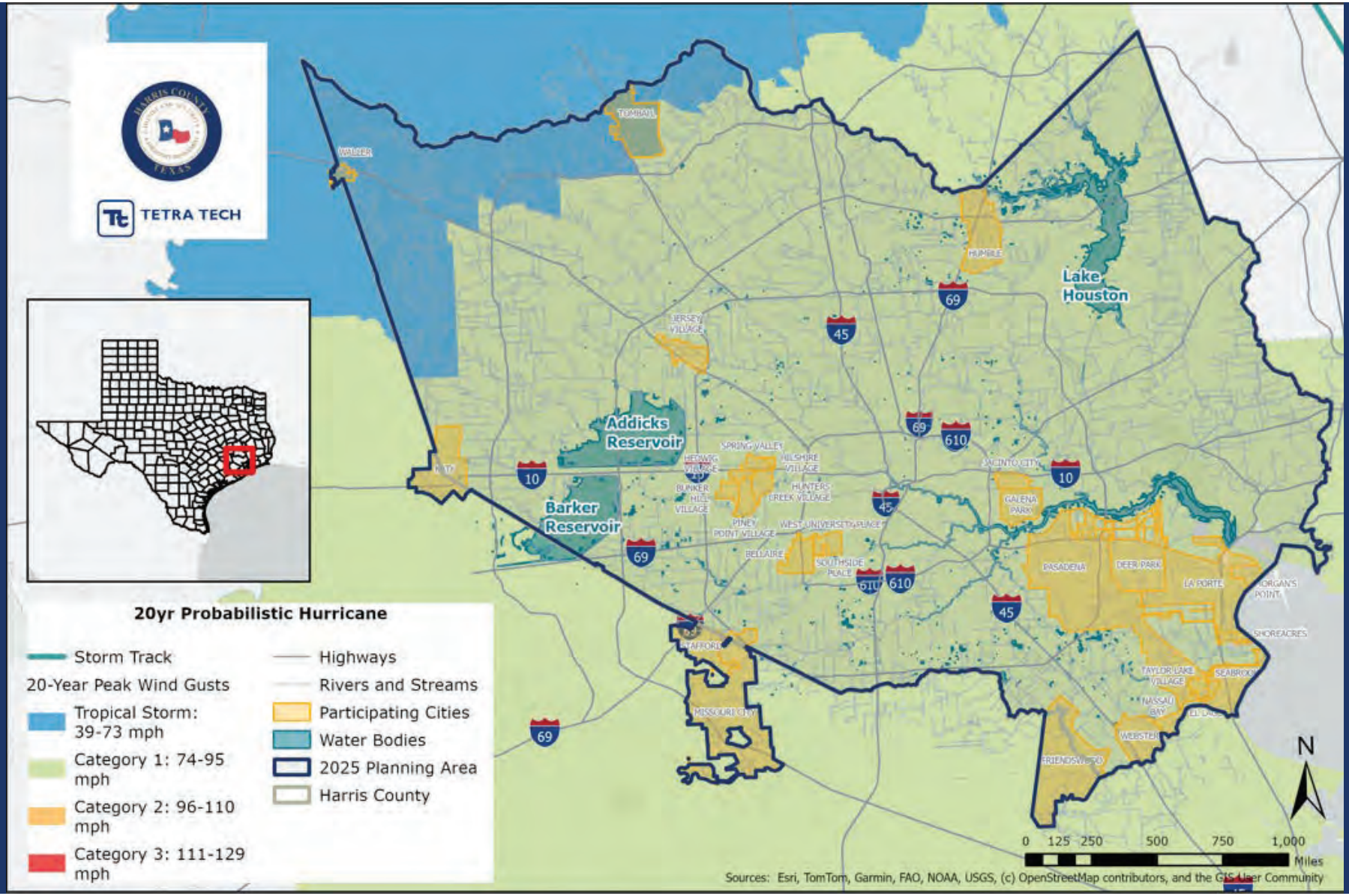
- <$10,000
- $10,000 - $100,000
- $100,000 - $500,000
- $500,000 - $1,000,000
- >$1,000,000

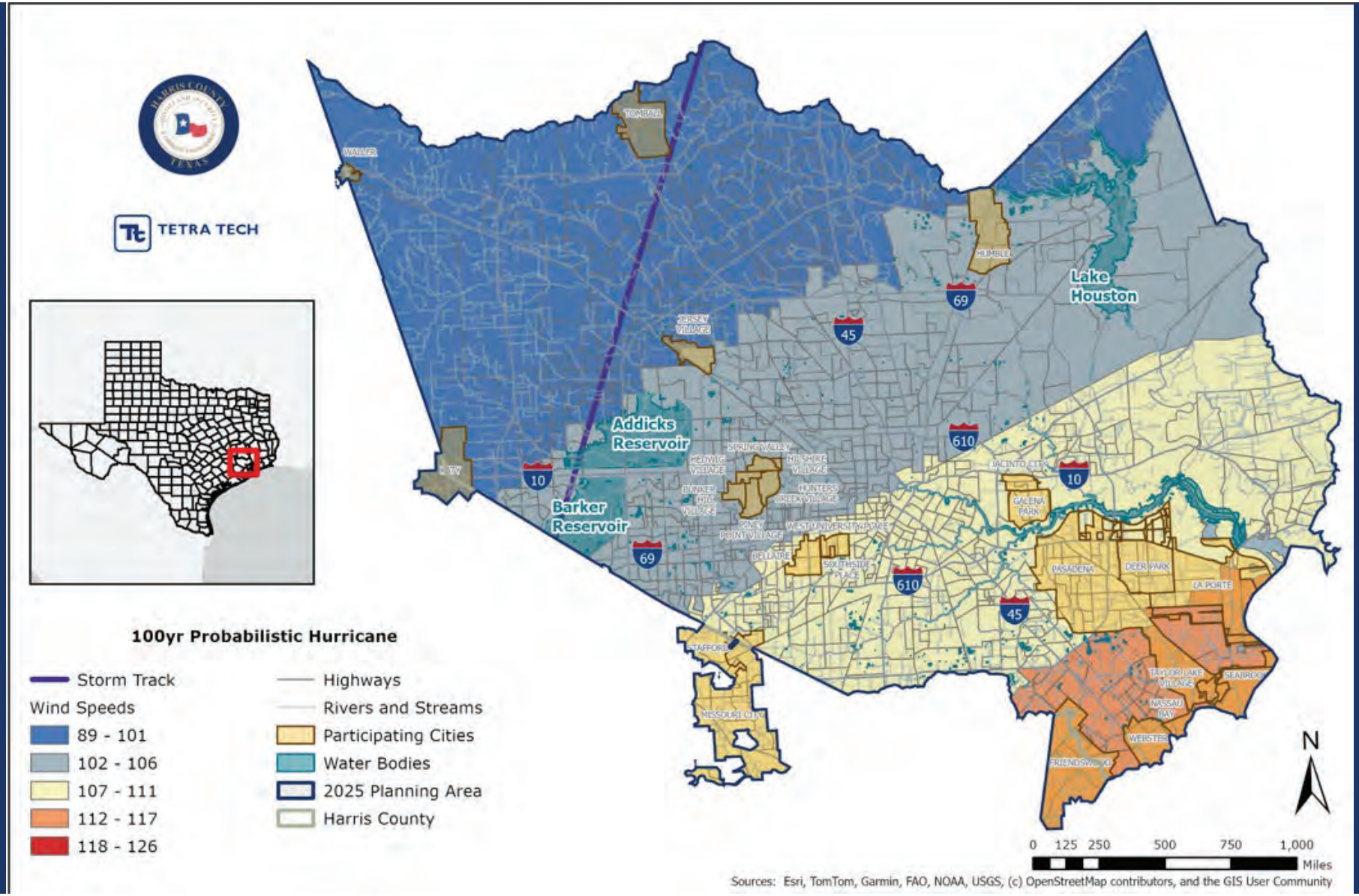
- Highways
- Rivers and Streams
- Participating Cities
- Water Bodies
- 2025 Planning Area
- Harris County

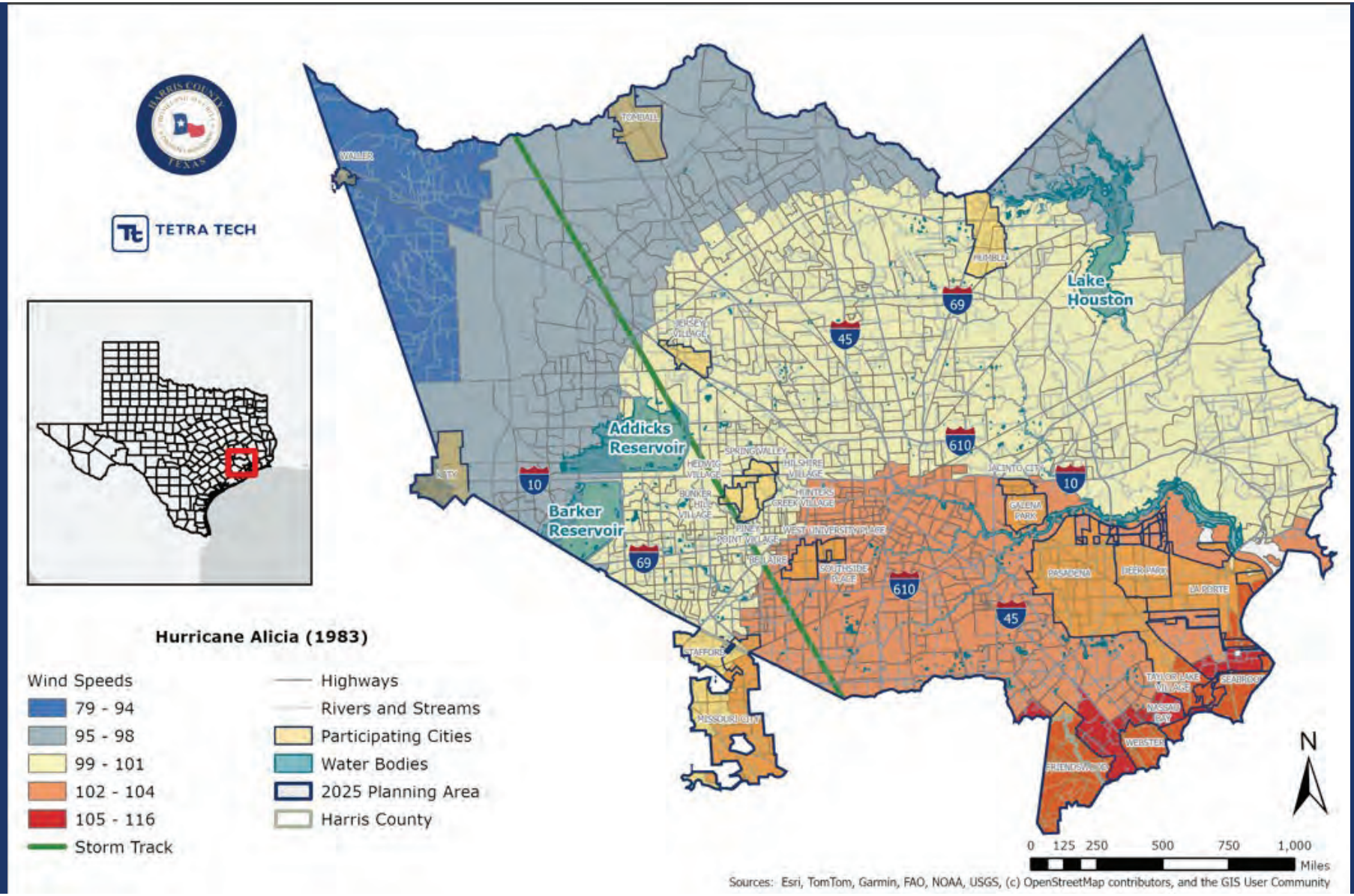
0 125 250 500 750 1,000 Miles

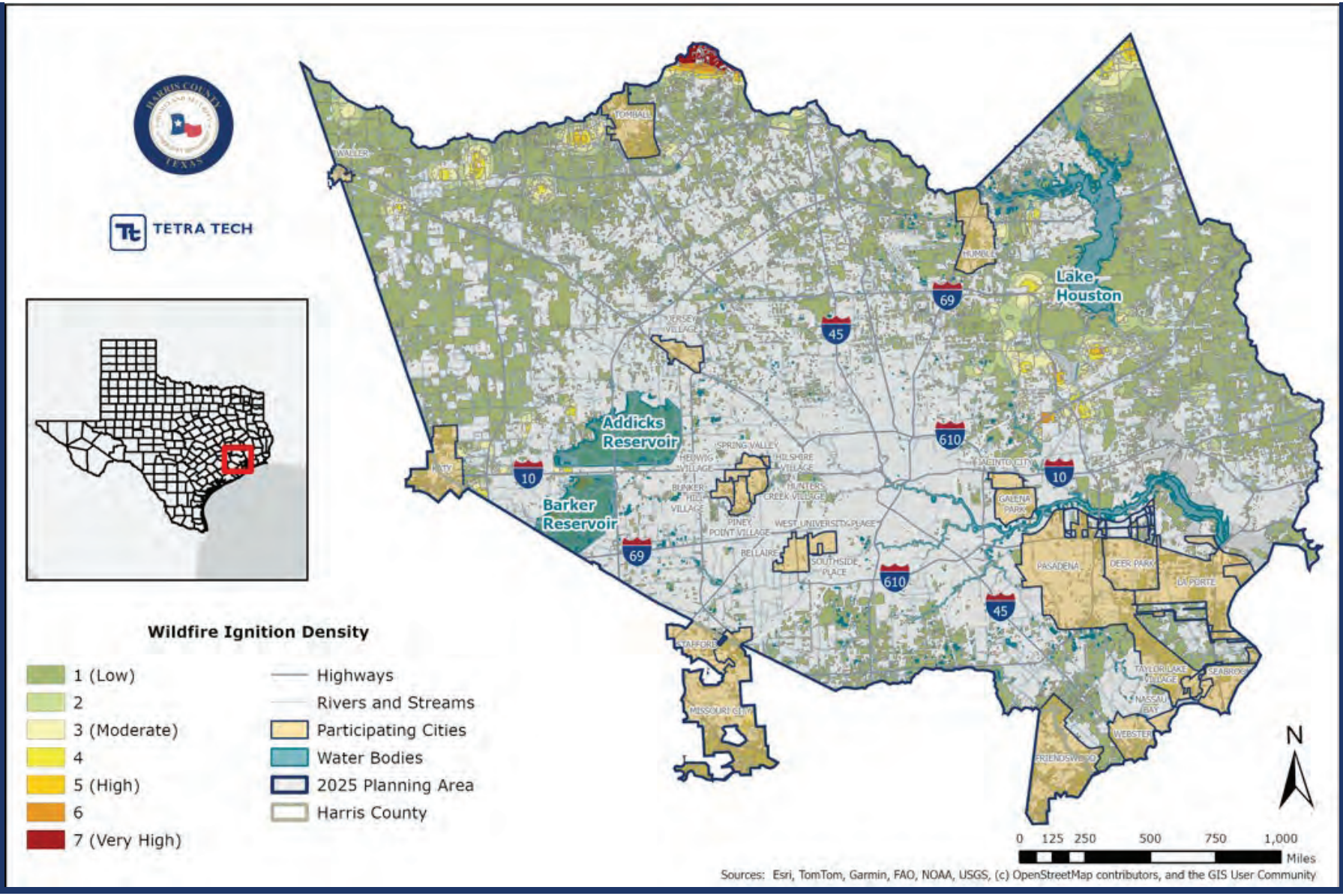
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

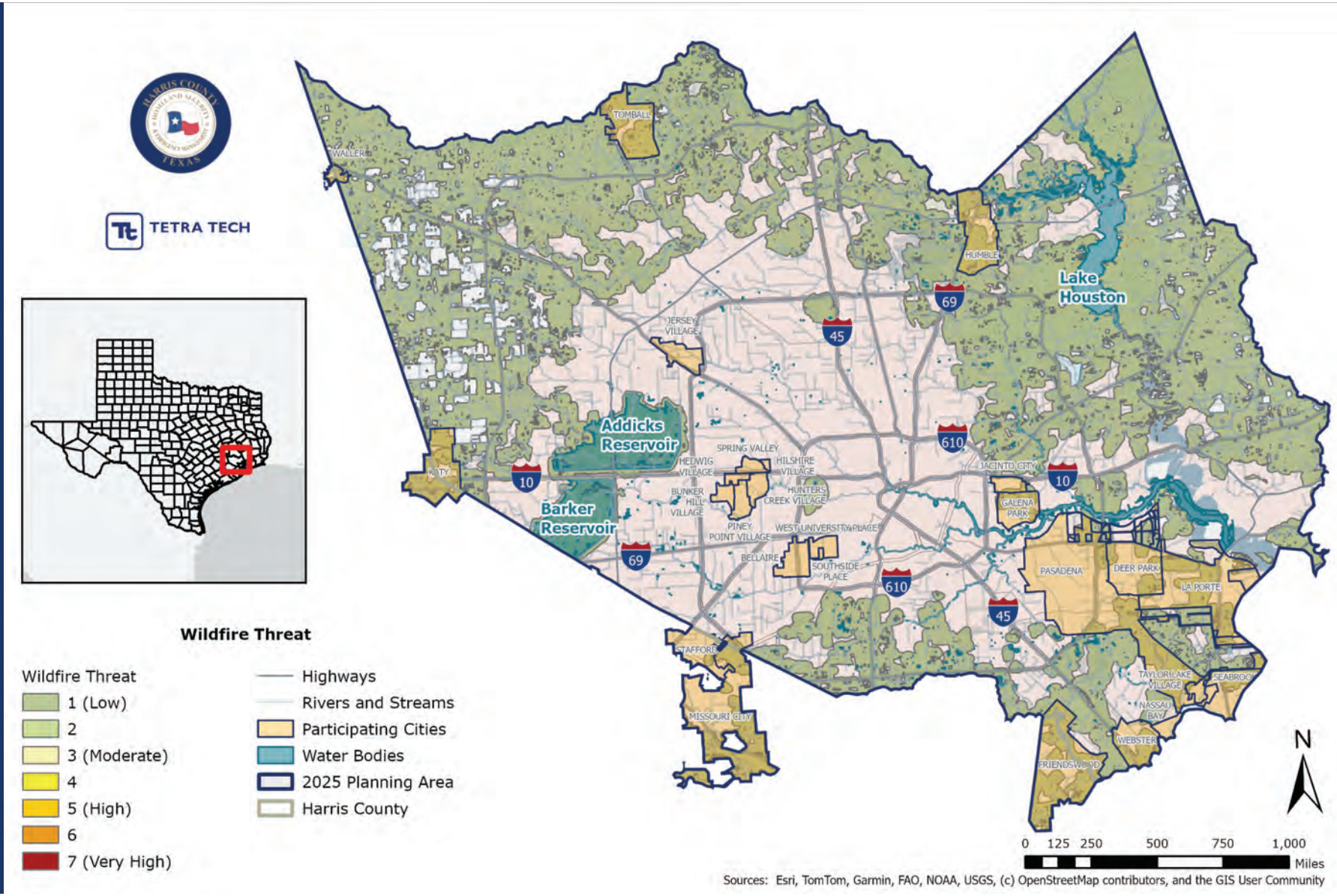

TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Total Industrial Replacement Cost Value (RCV) per Census Block ($1000s/Square Mile)
<$10,000
$10,000 - $100,000
$100,000 - $500,000
$500,000 - $1,000,000
>$1,000,000
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
20yr Probabilistic Hurricane
Storm Track
20-Year Peak Wind Gusts
Tropical Storm: 39-73 mph
Category 1: 74-95 mph
Category 2: 96-110 mph
Category 3: 111-129 mph
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
SPRING VALLEY
HEDWIG VILLAGE
WILSHIRE VILLAGE
BUNKER HILL VILLAGE
HUNTERS CREEK VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
MORGAN'S POINT
SHOREACRES
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
MISSOURI CITY
STAFFORD
KATY
10
45
69
610
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Addicks Reservoir
Barker Reservoir
Lake Houston
100yr Probabilistic Hurricane
Storm Track
Wind Speeds
89 - 101
102 - 106
107 - 111
112 - 117
118 - 126
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Hurricane Alicia (1983)
Wind Speeds
79 - 94
95 - 98
99 - 101
102 - 104
105 - 116
Storm Track
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
PASADENA
BAYTOWN
LA PORTE
SEABROOK
STAFFORD
MISSOURI CITY
0 125 250 500 750 1,000
Miles
N
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community


TETRA TECH
Lake Houston
Addicks Reservoir
Barker Reservoir
Wildfire Ignition Density
1 (Low)
2
3 (Moderate)
4
5 (High)
6
7 (Very High)
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

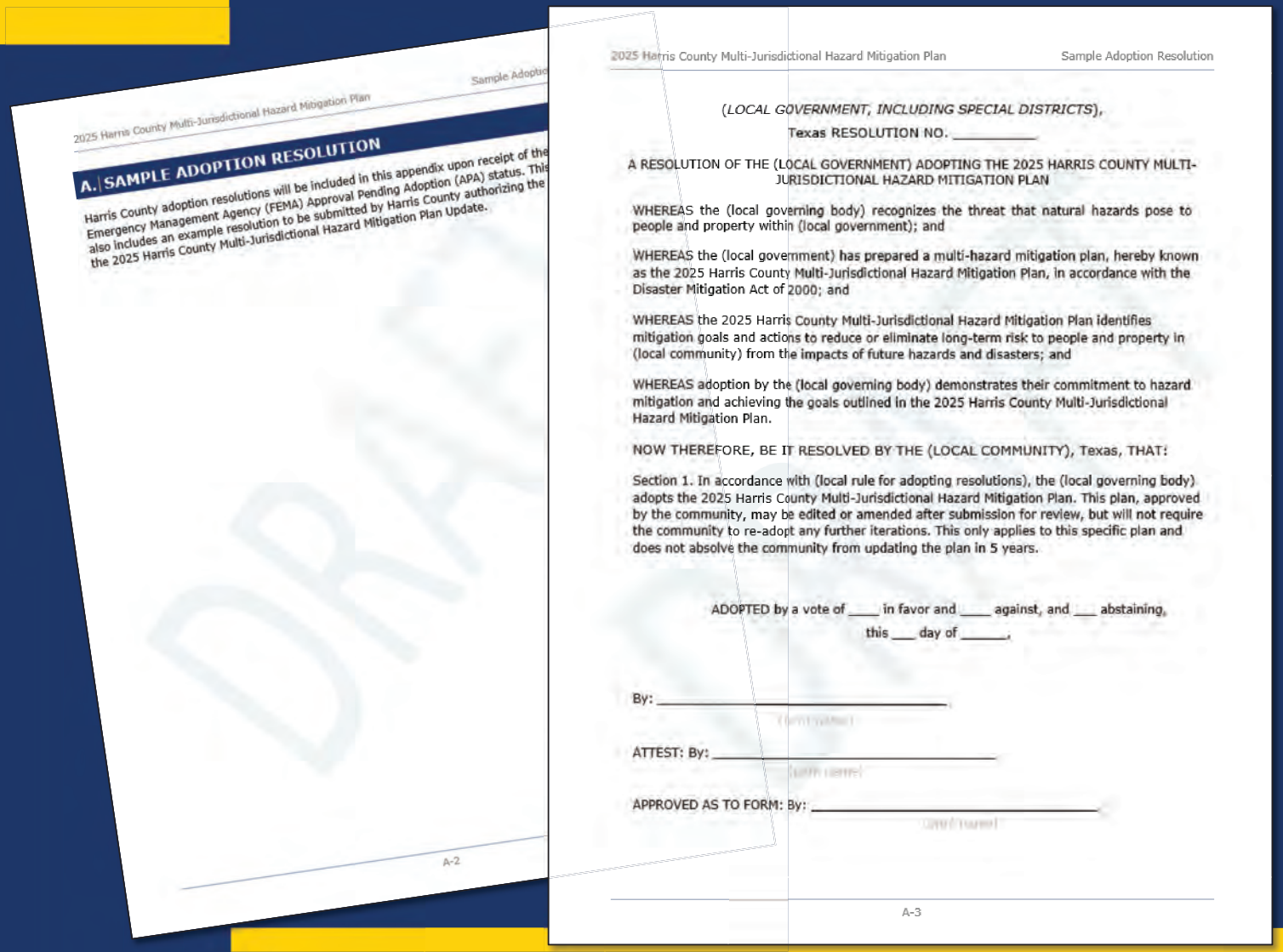

TETRA TECH
Wildfire Threat
Wildfire Threat
1 (Low)
2
3 (Moderate)
4
5 (High)
6
7 (Very High)
Highways
Rivers and Streams
Participating Cities
Water Bodies
2025 Planning Area
Harris County
Addicks Reservoir
Barker Reservoir
Lake Houston
TOMBALL
WALLER
HUMBLE
JERSEY VILLAGE
SPRING VALLEY
HEDWIG VILLAGE
HILSHIRE VILLAGE
BUNKER HILL VILLAGE
HUNTERS CREEK VILLAGE
PINEY POINT VILLAGE
WEST UNIVERSITY PLACE
BELLAIRE
SOUTHSIDE PLACE
JACINTO CITY
GALENA PARK
PASADENA
DEER PARK
LA PORTE
TAYLOR LAKE VILLAGE
SEABROOK
NASSAU BAY
WEBSTER
FRIENDSWOOD
STAFFORD
MISSOURI CITY
KATY
69
45
610
10
N
0 125 250 500 750 1,000
Miles
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

Adoption Resolution Example

HARRIS COUNTY

Office of Homeland Security & Emergency Management

Example Adoption Resolution



2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan

A. SAMPLE ADOPTION RESOLUTION

Harris County adoption resolutions will be included in this appendix upon receipt of the Emergency Management Agency (FEMA) Approval Pending Adoption (APA) status. This also includes an example resolution to be submitted by Harris County authorizing the the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update.

A-2

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Sample Adoption Resolution

(LOCAL GOVERNMENT, INCLUDING SPECIAL DISTRICTS),

Texas RESOLUTION NO. _____

A RESOLUTION OF THE (LOCAL GOVERNMENT) ADOPTING THE 2025 HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN

WHEREAS the (local governing body) recognizes the threat that natural hazards pose to people and property within (local government); and

WHEREAS the (local government) has prepared a multi-hazard mitigation plan, hereby known as the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan, in accordance with the Disaster Mitigation Act of 2000; and

WHEREAS the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan identifies mitigation goals and actions to reduce or eliminate long-term risk to people and property in (local community) from the impacts of future hazards and disasters; and

WHEREAS adoption by the (local governing body) demonstrates their commitment to hazard mitigation and achieving the goals outlined in the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan.

NOW THEREFORE, BE IT RESOLVED BY THE (LOCAL COMMUNITY), Texas, THAT:

Section 1. In accordance with (local rule for adopting resolutions), the (local governing body) adopts the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan. This plan, approved by the community, may be edited or amended after submission for review, but will not require the community to re-adopt any further iterations. This only applies to this specific plan and does not absolve the community from updating the plan in 5 years.

ADOPTED by a vote of ____ in favor and ____ against, and ___ abstaining,
this ___ day of _____,

By: _____

ATTEST: By: _____

APPROVED AS TO FORM: By: _____

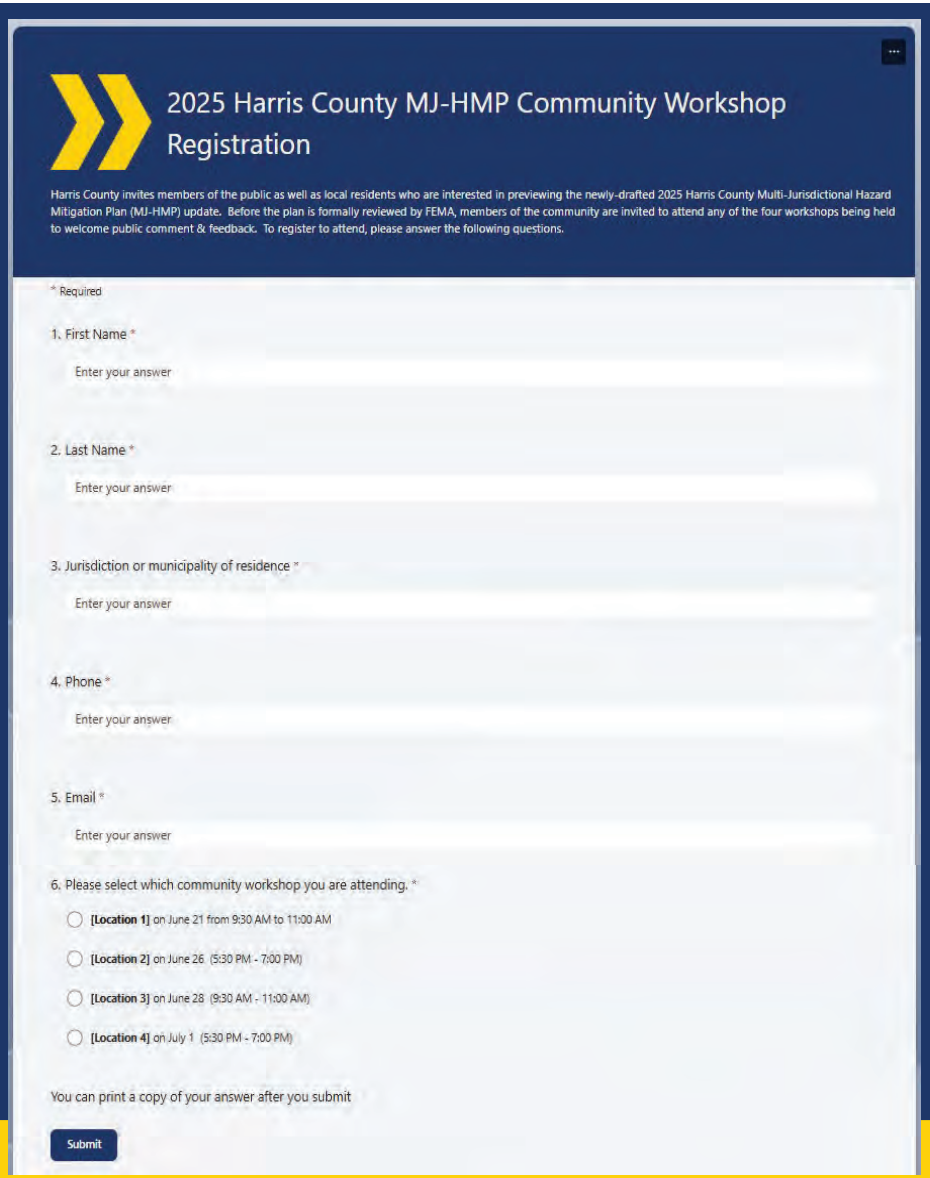
A-3

Public Meetings 1-4 & Public Review/Comment Period

HARRIS COUNTY
Office of Homeland Security & Emergency Management

https://forms.office.com/r/hEx4hwBbp0

Registration Survey Question



2025 Harris County MJ-HMP Community Workshop Registration

Harris County invites members of the public as well as local residents who are interested in previewing the newly-drafted 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan (MJ-HMP) update. Before the plan is formally reviewed by FEMA, members of the community are invited to attend any of the four workshops being held to welcome public comment & feedback. To register to attend, please answer the following questions.

* Required

1. First Name *

Enter your answer

2. Last Name *

Enter your answer

3. Jurisdiction or municipality of residence *

Enter your answer

4. Phone *

Enter your answer

5. Email *

Enter your answer

6. Please select which community workshop you are attending. *

- [Location 1] on June 21 from 9:30 AM to 11:00 AM
- [Location 2] on June 26 (5:30 PM - 7:00 PM)
- [Location 3] on June 28 (9:30 AM - 11:00 AM)
- [Location 4] on July 1 (5:30 PM - 7:00 PM)

You can print a copy of your answer after you submit

Submit

Additional Public Workshops

HARRIS COUNTY
Office of Homeland Security & Emergency Management

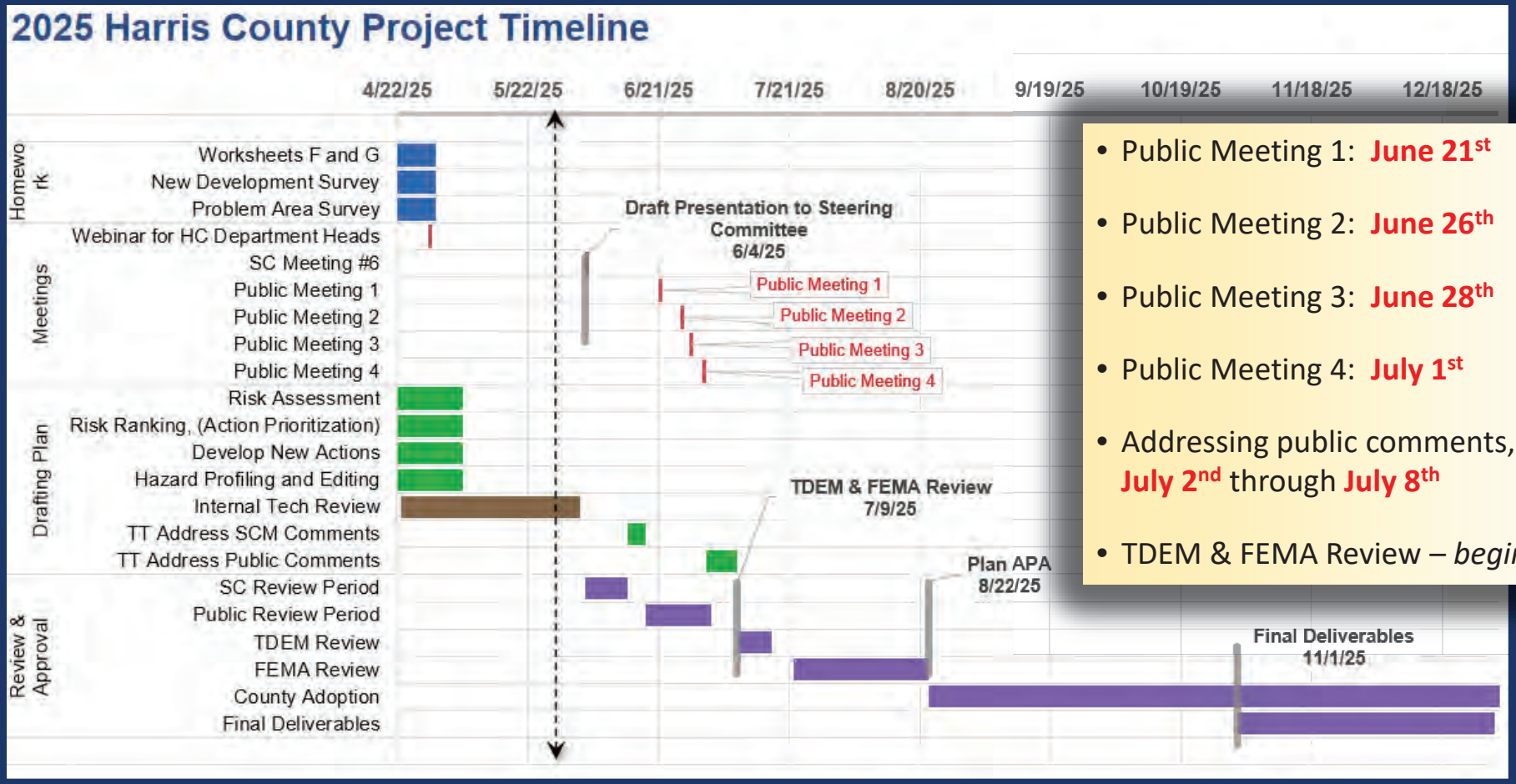
Municipal/Jurisdictional Workshops (optional)

If participants opt to host additional public workshops for review/comment, they will need to supply the following:

- Screenshots of workshop meeting invitations
- Workshop details (time, location, etc.)
- Workshop sign-in sheets and any related agendas
- List of public feedback/comments collected during the workshop(s)

With limited time remaining, how can TT provide assistance to these efforts?

Limited Time Remaining



- Public Meeting 1: **June 21st**
- Public Meeting 2: **June 26th**
- Public Meeting 3: **June 28th**
- Public Meeting 4: **July 1st**
- Addressing public comments, feedback: **July 2nd** through **July 8th**
- TDEM & FEMA Review – *begins* **July 9th**

Next Steps

HARRIS COUNTY

Office of Homeland Security & Emergency Management

Next Steps

- ✓ Drafting/technical review of HMP Draft (ongoing)
- ✓ Hazard Risk Ranking (in progress)
- ✓ Prioritization of Action Items (in progress)
- ✓ Working with BES to finish last round of edits to figures
- ✓ Prepare SC Meeting #6 (meeting materials, poster print-offs, hard copy of the MJ-HMP).

HARRIS COUNTY
HOMELAND SECURITY
& EMERGENCY MANAGEMENT
TEXAS

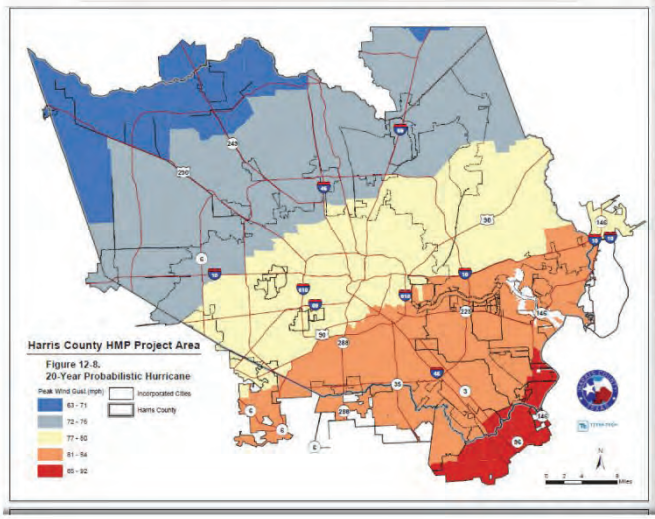
THANK YOU!

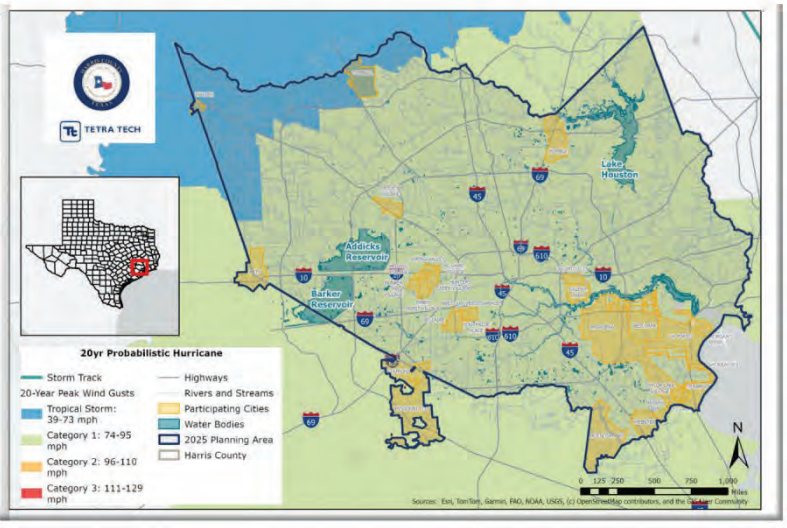
Coordination Meeting
with
Bent Ear Solutions and HCOHSEM

12:00 PM – 1:00 PM CDT

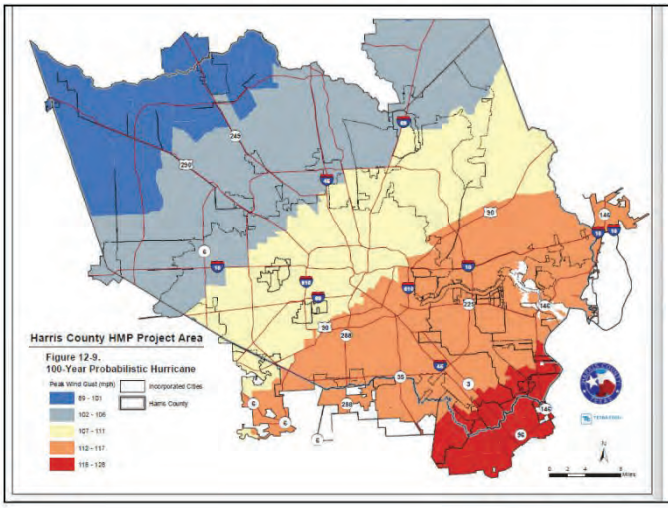
FRIDAY, MAY 30, 2025

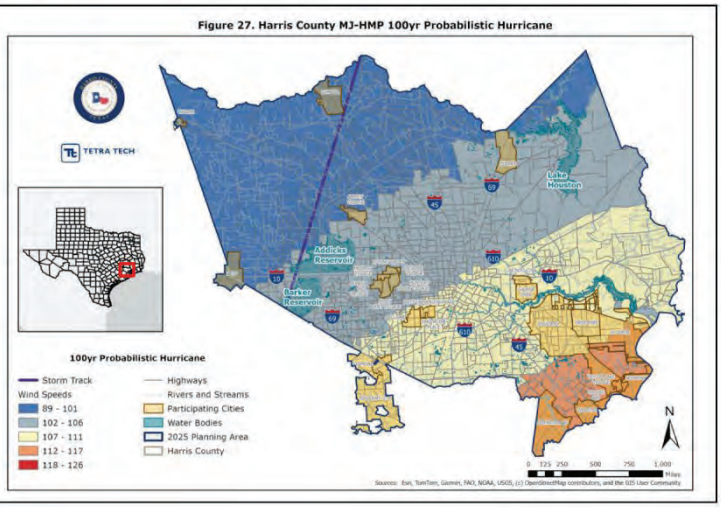
20yr Probabilistic Hurricane



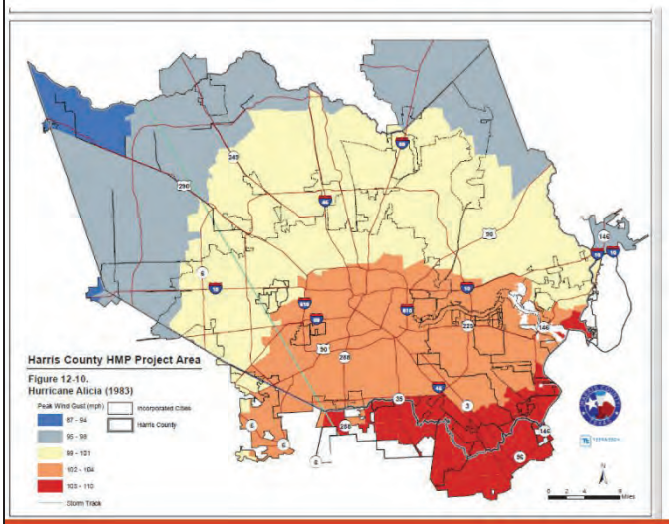


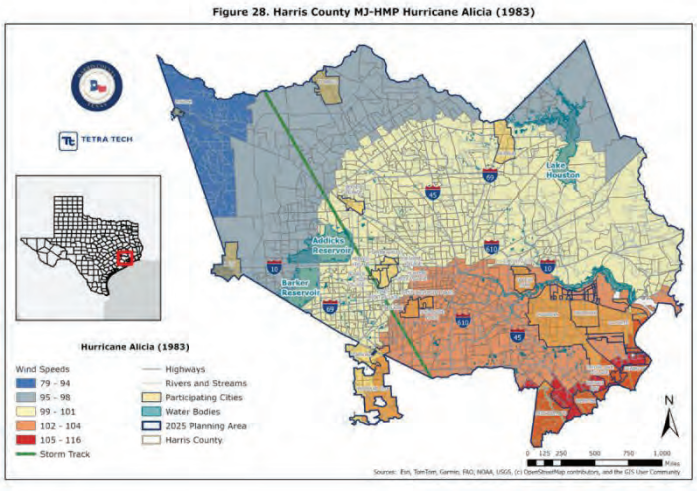
100yr Probabilistic Hurricane



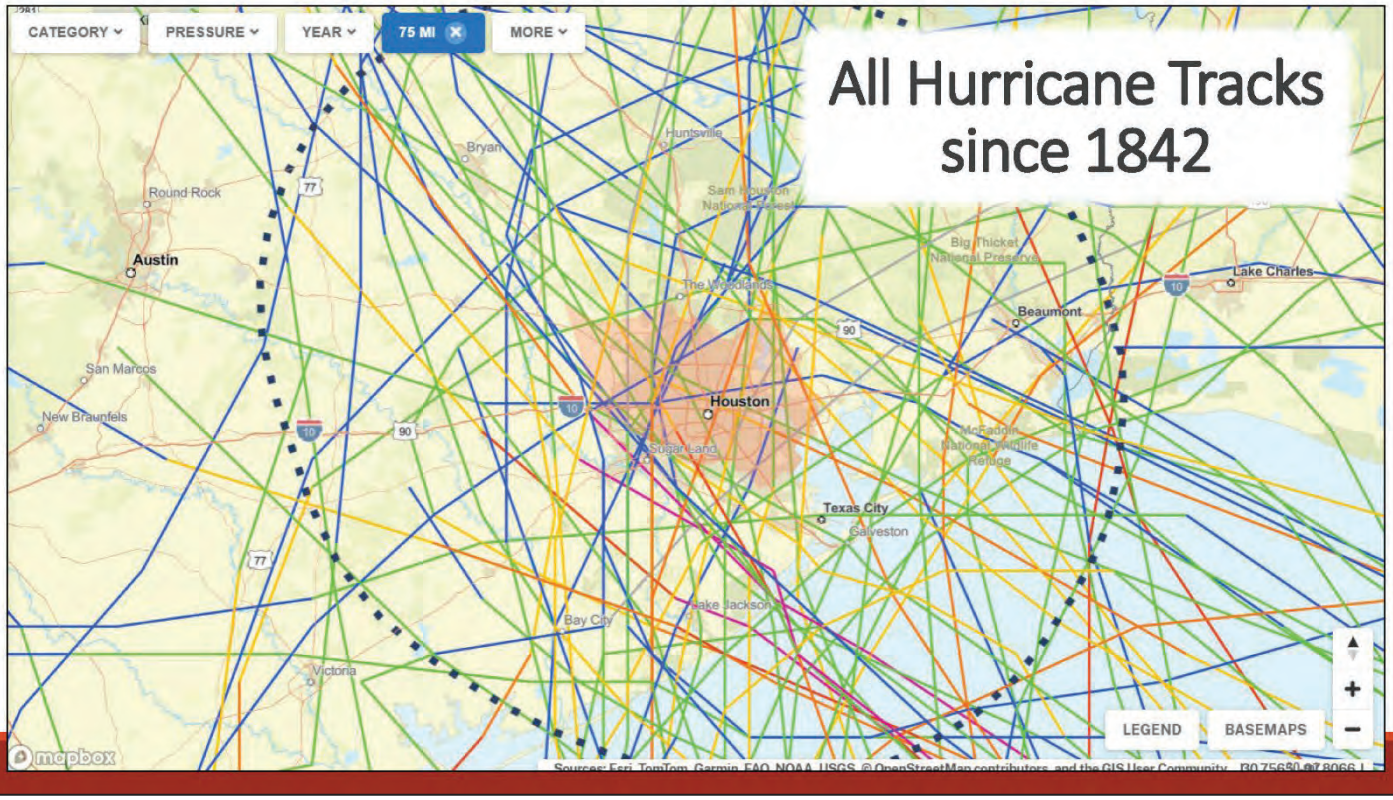


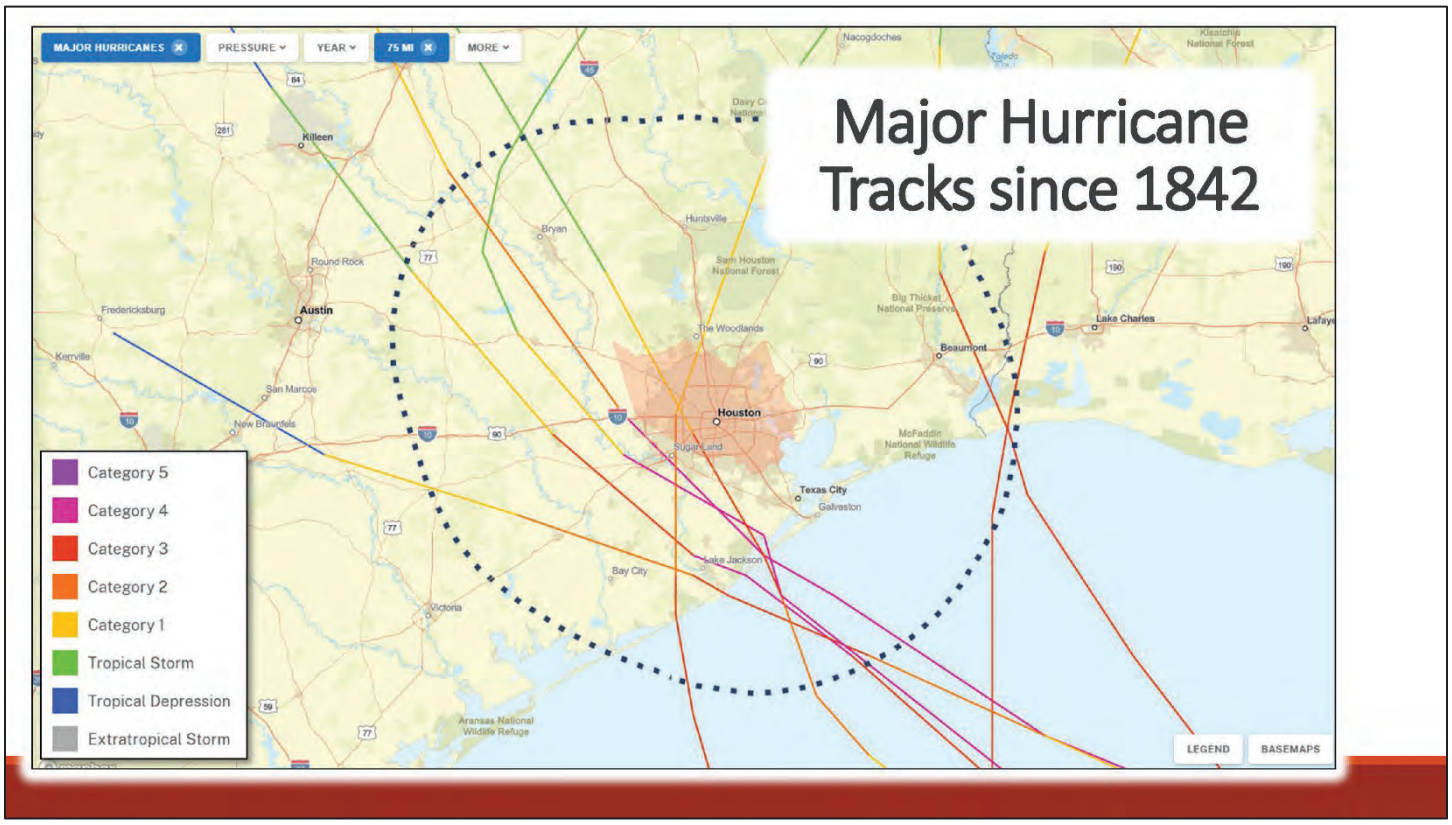
Hurricane Alicia (1983)

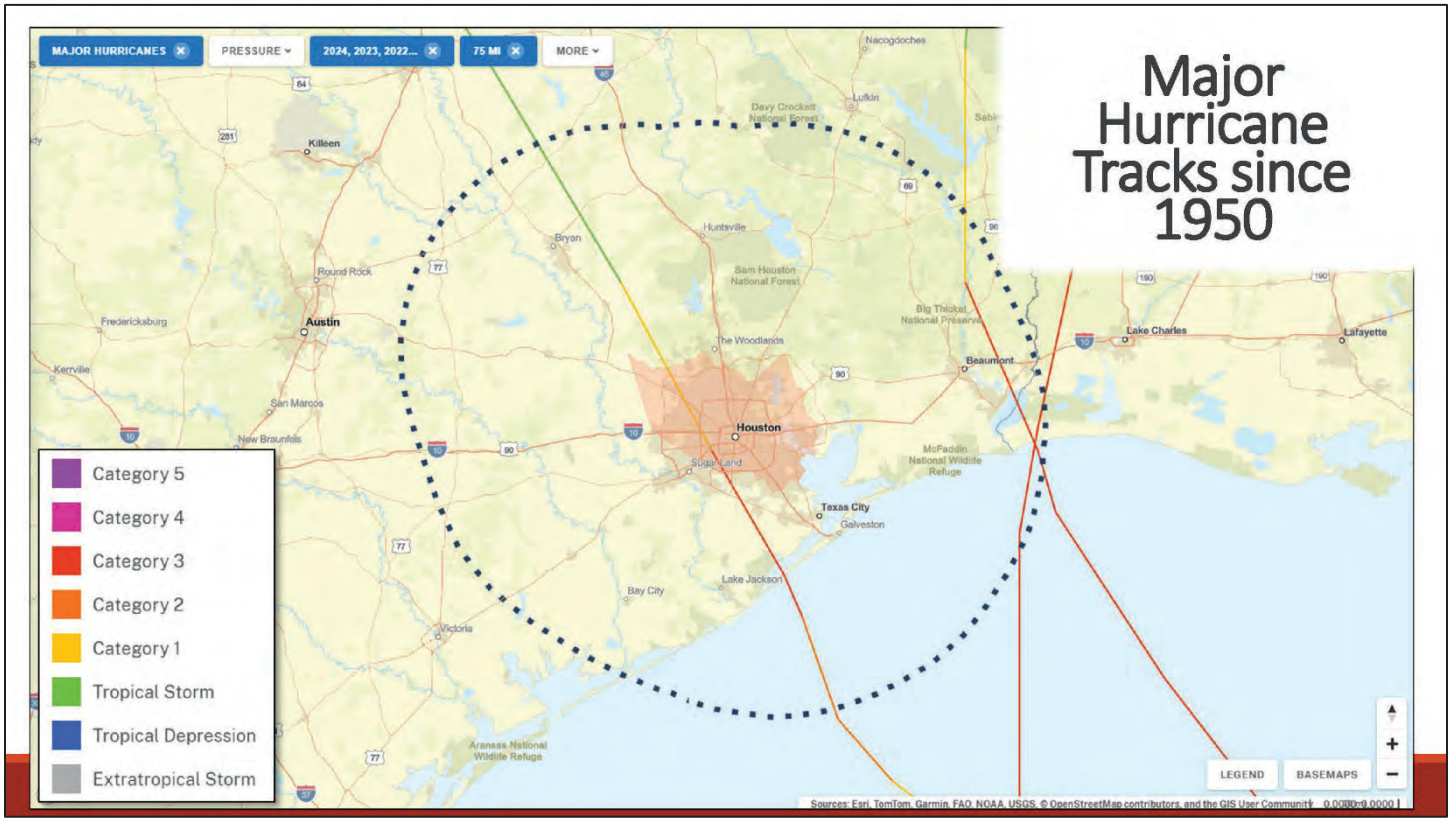


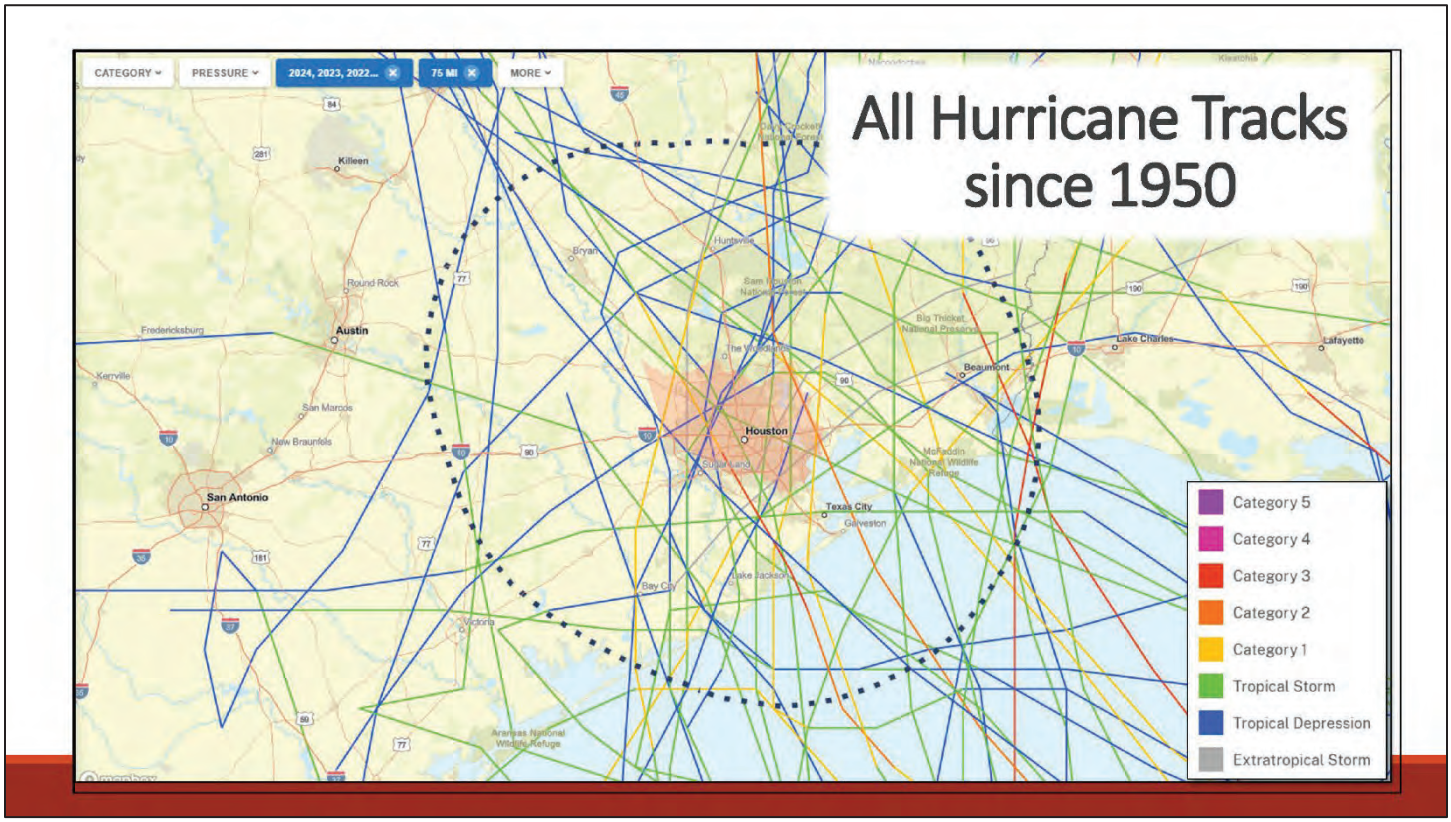


A "20-year probabilistic hurricane" refers to the probability of a hurricane of a certain intensity occurring within a specific timeframe, in this case, 20 years.




MAJOR HURRICANES
PRESSURE
YEAR
75 MI
MORE
Major Hurricane Tracks since 1842
Category 5
Category 4
Category 3
Category 2
Category 1
Tropical Storm
Tropical Depression
Extratropical Storm
LEGEND
BASEMAPS
Killeen
Austin
Round Rock
Bryan
Huntsville
Houston
The Woodlands
Sugar Land
Beaumont
Lake Charles
Texas City
Galveston
Victoria
Bay City
Lake Jackson
San Marcos
New Braunfels
Fredericksburg
Kerrville


MAJOR HURRICANES
PRESSURE
2024, 2023, 2022...
75 MI
MORE
Major Hurricane Tracks since 1950
Category 5
Category 4
Category 3
Category 2
Category 1
Tropical Storm
Tropical Depression
Extratropical Storm
LEGEND
BASEMAPS
Killeen
Austin
Round Rock
Bryan
Huntsville
Houston
The Woodlands
Sugar Land
Beaumont
Lake Charles
Lafayette
Texas City
Galveston
Victoria
Bay City
Lake Jackson
San Marcos
New Braunfels
Fredericksburg
Kerrville


CATEGORY
PRESSURE
2024, 2023, 2022...
75 MI
MORE
All Hurricane Tracks since 1950
Killeen
Austin
Round Rock
Fredericksburg
Kerrville
San Marcos
New Braunfels
San Antonio
Bryan
Houston
Sugar Land
Texas City
Galveston
Bay City
Victoria
Lake Jackson
Beaumont
Lake Charles
Lafayette
Category 5
Category 4
Category 3
Category 2
Category 1
Tropical Storm
Tropical Depression
Extratropical Storm

Steering Committee Meeting #6

June 4, 2025

AGENDA

JUNE 4, 2025, 10:00 AM – 12:00 PM

1. Welcome

2. Review/Approve Steering Committee Meeting #5 Minutes

3. Timeline & Status Update

4. Presentation of the 2025 Harris County MJ-HMP (Draft)
   - Hazard Risk Ranking
   - Prioritizing Action Items

5. Partner & Public Review Period

6. Adoption Process & Timeline
   - Adoption Resolution Letters
   - Local Adoption Requirements
   - Approval Timeline

7. Demonstration of the 2025 MJ-HMP Web Viewer

8. Questions / Next Steps

9. Adjourn

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #6
June 4, 2025 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALTERNATE NAME (OPTIONAL)	SIGNATURE
HCOHSEM	Brian Murray	Deputy EMC / PIO		
HCOHSEM	Brittany Ray	Assistant Planning Section Chief		
Harris County OCA / Public Health	Leah Barton	Deputy County Administrator / Executive Director Harris County Public Health		
HCFCD	Jeremy Phillips	Director, Asset Management Division		
City of Bellaire	Deacon Tittle	Fire Chief / Emergency Management Coordinator		
City of Deer Park	Jamie Galloway	Emergency Services Director		
City of El Lago / Taylor Lake Village	Carl Nunn	Chief of Police / EMC		
City of Friendswood	Steven Simmons	Deputy Fire Marshal	Brian Mansfield	
City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal		
City of La Porte	Johnny Morales	Emergency Management Coordinator	Amber Castro Deputy EMC	
City of Missouri City	Trameka Jewett	Emergency Management Coordinator	Casey Jimenez	
City of Pasadena	Azell Carter	Emergency Management Coordinator		
City of Seabrook	Kevin Rodgers	Office of Emergency Management Director		
City of Shoreacres	Troy Harrison	City Manager	James Tidwell (jtidwell@shoreacres.tx.us)	
National Weather Service	Lance Wood	Science and Operations Officer		
North Houston Association	Marlisa Briggs	President		
Houston Community College	Robert McCracken	Associate Vice Chancellor, Risk Mgmt & Resilience		
San Jacinto College	Ali Shah	Director, Emergency Management		

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #6
June 4, 2025 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALTERNATE NAME (OPTIONAL)	SIGNATURE
H-GAC	Amanda Ashcroft	Planner		
Cypress Creek Drainage Improvement District	Calvin Cobb	President		Calvin B. Cobb
Coastal Prairie Conservancy	Mary Anne Piacentini	President and CEO		
Greens Bayou Coalition	Colleen Gilbert	Executive Director		
Cy-Fair ISD	Scott Hudson	Director of Emergency Management		
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Director		
SETRAC	James Meaux	Regional T&E Coordinator		
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience		
Houston SPCA	Dawn Blackmar	Director, Emergency Management		
Harris County Dept. of Education	Leslie Etheridge	Director, Center for Safe and Secure Schools		

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #6

June 4, 2025 / 10:00 AM – 12:00 PM

NON-STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	PHONE	SIGNATURE
TDEM	Merryl Holmes JR	Mitigation Coordinator	281-794-4102	
Village Fire Dept	James Culty	EMC	281-415-2011	
County Judge's Office	Zac Nguyen	EM Liaison	713 303 9123	
City of La Porte	Jeremy Borja	EM Planner		
Cy-Fair ISD OEM	Grant Burnett	EM Coordinator	281-897-6450	
Seabrook	Brian Craig	PW Dir/City E	281-291-5719	

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #6

June 4, 2025 / 10:00 AM – 12:00 PM

PUBLIC SIGN-IN

NAME	CITY/JURISDICTION	ORGANIZATION	PHONE	EMAIL
Charles Mistrot	City of Waller - Waller-Harris	City of Waller	979-645-7557	CMistrot@wallerTexas.gov
Joseph Leonard	Resalire	Resalire	281-992-7568	Joseph.Leonard@resalireinfrastructure.com
Jennifer Leighton	Resalire		832 646 7390	Jennifer.Leighton@resalireinfrastructur.com
Kevin Lagana	Morgan's Point P.D	Police	346-222-7384	KLAGANA@MORGANSPOINT-TX.COM
Kiael Whitton	Harris County OEM	HCOHSEM	713 501 7825	Kiael.Whitton@harriscountytx.gov
Atri Sen	RESALIRE	RESALIRE	225. 281 4879	
Maritza Aguilar V.	City of Houston/OEM	city of Houston	832 405 12 76	maritza.aguilar@houstontx.gov.
David Potts	HCC	HCC	281 9604375	David.Potts@HCCS.edu
Archie Losey	Unincorporated Harris Co.	Resident	(713) 824-5219	archie.losey@wsbeng.com
Cassie Stephens	Hilshire Village	Hilshire Village	713-973-1701	Cassie.stephens@hilshirevillagetexas.com
Drew Castillo	GCA		281-667-2446	acastillo@GCATX.org
Dena Mahan	Webster	Webster OEM	832-594-1751	dmahan@cityofwebster.com

Public Review Workshops

June 26 – July 1, 2025

2025 Harris County *Multi-Jurisdictional Hazard Mitigation Plan Update*

COMMUNITY WORKSHOPS

Workshop Dates/Locations:

October 19
10:00 AM – 11:30 AM
San Jacinto College Generation Park:
13455 Lockwood Rd, Houston,
TX 77044

October 22
6:00 PM – 7:30 PM
Katherine Tyra Branch Library:
16719 Clay Rd, Houston, TX 77084

October 30
6:00 PM – 7:30 PM
George HW Bush Community Center:
6827 Cypresswood Dr, Spring,
TX 77379

November 9
10:00 AM – 11:30 AM
San Jacinto College Central Campus:
8060 Spencer Hwy, Pasadena,
TX 77505

A virtual option will be made available for this workshop. Details will be provided upon registration.

Featured Topics Include:

- Overview of the Harris County Hazard Mitigation Plan
- Mitigation Projects in Your Area
- Harris County Climate Justice Plan

REGISTER NOW!

HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

Text HMP2025 to 888777
to recieve updates about the Plan's development

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

Public Review Workshop #1

June 26, 2025 / 5:30 PM – 7:00 PM

PUBLIC SIGN-IN

ORGANIZATION	NAME	TITLE	PHONE	SIGNATURE
HCOHSEM	Brittany Ray	Assistant Planning Section Chief	713-881-3100	
HCOHSEM	Kelsie Suter	Planning Specialist	713-881-3100	
HCOHSEM	Jennifer Suter	Training & Exercise Supervisor	713-881-3100	
HCOHSEM	Brian Murray	Deputy EMC	713-881-3100	
HCFCD	Jeremy Phillips	HCFCD Asset Management Div. Dir.	346-[illegible]-274-4092	
HCFCD	JESSICA LAZO	HCFCD PCT3 COORDINATOR	C: 713 226 9294	
County Judge's Office	Zac Nguyen	Policy Advisor	713 303 9123	
Tetra Tech	Anita Machiavello	technical lead planner	281-448-6189	
Tetra Tech	Mark Stalcup	planner	713-391-7324	

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Public Review Workshop #2
June 28, 2025 / 9:30 AM – 11:00 AM

PUBLIC SIGN-IN

Name	Zip Code	E-mail	Telephone
Jeff Syptak	77429	jrsyptak@gmail.com	281-455-5604
Shaun Williams	77065	Stwilliams021@gmail.com	832.287.9675
LaTeef Williams	77016	912wwllc@gmail.com	346-837-6238
Brittany Ray		Brittany.Ray@harriscountytx.gov	713-881-3100
Sam Bissett	77080	Sam.bissett@harriscountytx.gov	

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Public Review Workshop #3
July 1, 2025 / 5:30 PM – 7:00 PM

PUBLIC SIGN-IN

Name	Zip Code	E-mail	Telephone
Brittany Ray	77024	Brittany.Ray@harriscountytx.gov	713-881-3100
Lindsey Drouet	77024	lindsey.drouet@harriscountytx.gov	713-881-3100
Ivonne Schneider	77062	ivonne.schneider@harriscountytx.gov	360-359-1428
MARIO SALINAS	77092	MARIO.SALINAS@PCT2.HCTX.NET	713-687-0209
José Lozano	77059	Joseph.Lozano@operationinfrastructure.com	281-552-7520
Benzon Koshy John	77584	BENZON.JOHN@HARRISCOUNTYTX.GOV	713-881-3100
Elicia Winkler	77007	Elicia.Winkler@phs.hctx.net	713-274-4974
Madison Stazo	77036	Madison.Stazo@PHS.hctx.net	
Hilda De la Rosa	77028	esteesmitay1000@gmail.com	832-434-6996
Pilar Fuentes	77007	pilar.fuentes@harriscountytx.gov	832-414-5728
Andy Escobar	77065	andy@ceerhouston.org	832 881 0486
ATTI SEN	77494		225-281-4879

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

Public Review Workshop #4

July 2, 2025 / 5:00 PM – 6:30 PM

PUBLIC SIGN-IN

Name	Zip Code	E-mail	Telephone
ALBERT LATSON	77078	MRJACK2010@LIVE.COM	281-799-1635
BENZON KOSHY - JOHN	77584	BENZON.JOHN@HARRISCOUNTYTX.GOV	713-436-3100
ATRI SEN	77494		281.281.4879
Jennifer Leighton	77433	jennifer.leighton@resalireinfrastructure.com	832 646 7590
Isaac Eguia	77094	isaac.eguia@harriscountytx.gov	713-854-8789
LISA LIN	77009	lisa.lin@harriscountytx.gov	713.274.1140
Dennis Sias	77016	dennissias@yahoo.com	8325170871
Leah Barton	77008	leah.barton@harriscountytx.gov	7135690697
Nicole Branfeul	47020	msqb31@yahoo.com	7859-1610
Myrtala Tristan	77078	mTristan03@gmail.com	713 7051579
VINCE DUNCAN	77029	VINCEFORUS3@GMAIL.COM	713.252.2436

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Public Review Workshop #4
July 2, 2025 / 5:00 PM – 6:30 PM

PUBLIC SIGN-IN

Name	Zip Code	E-mail	Telephone
Lowell Williams	77020	LP.2[illegible]/cop/@Yahoo.com	2815488732
Gregg Gutierrez	77379	gregg.gutierrez@cpl.hctx.net	281 6050 75
Lora Clack	77054	lclack@unitedwayhouston.org	[illegible]
Laura de la Roche	77204	ldelaroc@uh.edu	
Virginia Lamott	77016	v.lamott@sbcglobal.net	832 491 4754
Dee Manning	77047	deeamanning@yahoo.com	
carolina manning	77047	carolinacmanning@yahoo.com	
Debra LaDay Boggs	77026	debradboggs2@gmail.com	346-420-0515
Stephanie Davidson	77017	Stephanieeneedavidson@gmail	2814613853
Kortney Carmouche	77013	Kortney.carmouche@hccs.edu	281-216-4300
CHARLOTTE WILLIAMS	77020		
Wanda Sanders	77028	Grandsanders@gmail.com	832 2028673

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Public Review Workshop #4
July 2, 2025 / 5:00 PM – 6:30 PM

PUBLIC SIGN-IN

Name	Zip Code	E-mail	Telephone
Lizeth Hernandez	77024	Lizeth.Hernandez@harriscountytx.gov	713-881-3100
Lindsey Drouet	77024	lindsey.drouet@harriscountytx.gov	713-881-3100
Dylan McNally	77008	dylan.mcnally@sba.gov	571-369-8895
Todd Ward	77064	Todd.Ward@hcfcd.hctx.net	713-296-4858
Lyanne Ortiz	77055	lortiz@sbpusa.org	787-420-4901
Kevin	77356	KEVIN.MCCOY@REDCROSS.ORG	713-896-0484
María Elena	77530	mzuniga@connectivetx.org	504-261-6098
Mikaela Hamilton	77373	mikaela.hamilton@fmo.hctx.net	713-449-1660
Betsabe Campos	77017	betsabe.campos@fmo.hctx.net	713-274-1735
Brandi Dumas	77041	brandi.dumas@fmo.hctx.net	832-509-7400
Jasmin Andrade	77047	jasmin.andrade@phs.hctx.net	281-753-2551
Alex Edwards	77098	AEdwards@unitedwayhouston.org	211

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Public Review Workshop #4
July 2, 2025 / 5:00 PM – 6:30 PM

PUBLIC SIGN-IN

Name	Zip Code	E-mail	Telephone
Zoe Washington	77047	ZWashington@unitedwayhouston.org	211

APPENDIX C: PUBLIC AND STAKEHOLDER OUTREACH

2025
Harris County
Multi-Jurisdictional
Hazard Mitigation
Plan Update
STEERING COMMITTEE
MEETING
Date: July 10, 2024
Time: 10:00 AM - 12:00 PM
Location: Houston TranStar
6922 Katy Rd | Houston, TX 77024
Purpose of Hazard Mitigation Planning
Identify impacts from hazards.
Identify actions and activities to reduce losses from those hazards.
Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
Requirement for federal disaster grant assistance.
Meeting Attendees may Include:
Steering Committee Members
Participating entities (municipalities, schools, etc.)
General Public
REGISTER NOW!
HARRIS COUNTY
HOMELAND SECURITY & EMERGENCY MANAGEMENT
TEXAS
Questions? Email mitigation@oem.hctx.net

ReadyHarris
Published by Daisy Campos
· 17m ·

Our office has started working on the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update.

The first Steering Committee meeting will be on July 24, 2024; the details are listed below. This collaborative process includes multiple jurisdictions and public participation, and all Steering Committee meetings will be open to the public.

For more information: https://bit.ly/2025HazardMitigation

See insights and ads

Boost post

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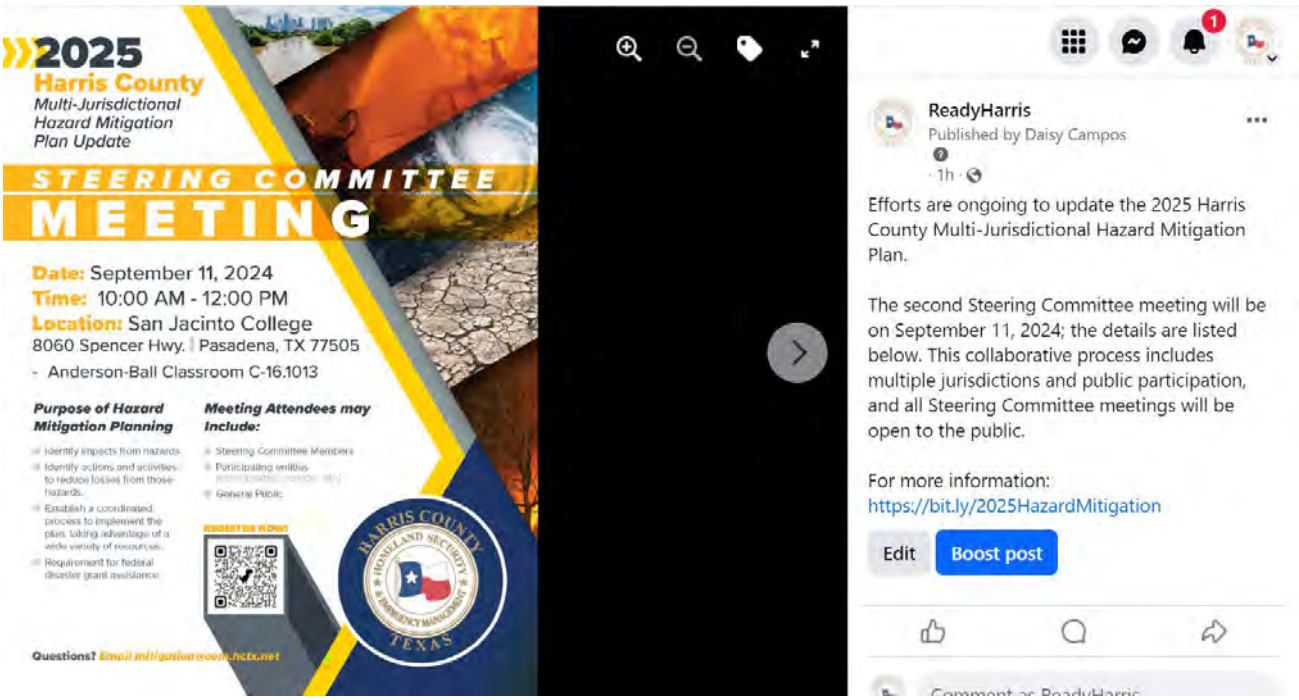
2025
Harris County
Multi-Jurisdictional
Hazard Mitigation
Plan Update
STEERING COMMITTEE
MEETING
Date: September 11, 2024
Time: 10:00 AM - 12:00 PM
Location: San Jacinto College
8060 Spencer Hwy. | Pasadena, TX 77505
- Anderson-Ball Classroom C-16.1013
Purpose of Hazard Mitigation Planning
Identify impacts from hazards.
Identify actions and activities to reduce losses from those hazards.
Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
Requirement for federal disaster grant assistance.
Meeting Attendees may Include:
Steering Committee Members
Participating entities
General Public
REGISTER NOW!
Questions? Email mitigation@oem.hctx.net
HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS
ReadyHarris
Published by Daisy Campos
1h
Efforts are ongoing to update the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan.
The second Steering Committee meeting will be on September 11, 2024; the details are listed below. This collaborative process includes multiple jurisdictions and public participation, and all Steering Committee meetings will be open to the public.
For more information:
https://bit.ly/2025HazardMitigation
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2025
Harris County
Multi-Jurisdictional
Hazard Mitigation
Plan Update
STEERING COMMITTEE
MEETING
Date: September 11, 2024
Time: 10:00 AM - 12:00 PM
Location: San Jacinto College
8060 Spencer Hwy. | Pasadena, TX 77505
- Anderson-Ball Classroom C-16.1013
Purpose of Hazard Mitigation Planning
Identify impacts from hazards.
Identify actions and activities to reduce losses from those hazards.
Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
Requirement for federal disaster grant assistance.
Meeting Attendees may Include:
Steering Committee Members
Participating entities
General Public
REGISTER NOW!
Questions? Email mitigation@oem.hctx.net
HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS
readyharris
readyharris Efforts are ongoing to update the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan.
The second Steering Committee meeting will be on September 11, 2024; the details are listed below. This collaborative process includes multiple jurisdictions and public participation, and all Steering Committee meetings will be open to the public.
For more information: https://bit.ly/2025HazardMitigation
1h
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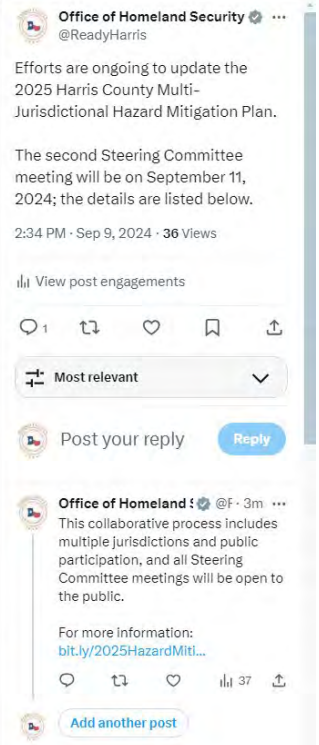
2025
Harris County
Multi-Jurisdictional
Hazard Mitigation
Plan Update
STEERING COMMITTEE
MEETING
Date: September 11, 2024
Time: 10:00 AM - 12:00 PM
Location: San Jacinto College
8060 Spencer Hwy. | Pasadena, TX 77505
- Anderson-Ball Classroom C-16.1013
Purpose of Hazard Mitigation Planning
Identify impacts from hazards.
Identify actions and activities to reduce losses from those hazards.
Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
Requirement for federal disaster grant assistance.
Meeting Attendees may Include:
Steering Committee Members
Participating entities
(municipalities, schools, etc.)
General Public
REGISTER NOW!
Questions? Email mitigation@oem.hctx.net
HARRIS COUNTY
HOMELAND SECURITY & EMERGENCY MANAGEMENT
TEXAS
1
36


Office of Homeland Security
@ReadyHarris
Efforts are ongoing to update the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan.
The second Steering Committee meeting will be on September 11, 2024; the details are listed below.
2:34 PM · Sep 9, 2024 · 36 Views
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Reply
Office of Homeland S @F · 3m
This collaborative process includes multiple jurisdictions and public participation, and all Steering Committee meetings will be open to the public.
For more information:
bit.ly/2025HazardMiti...
37
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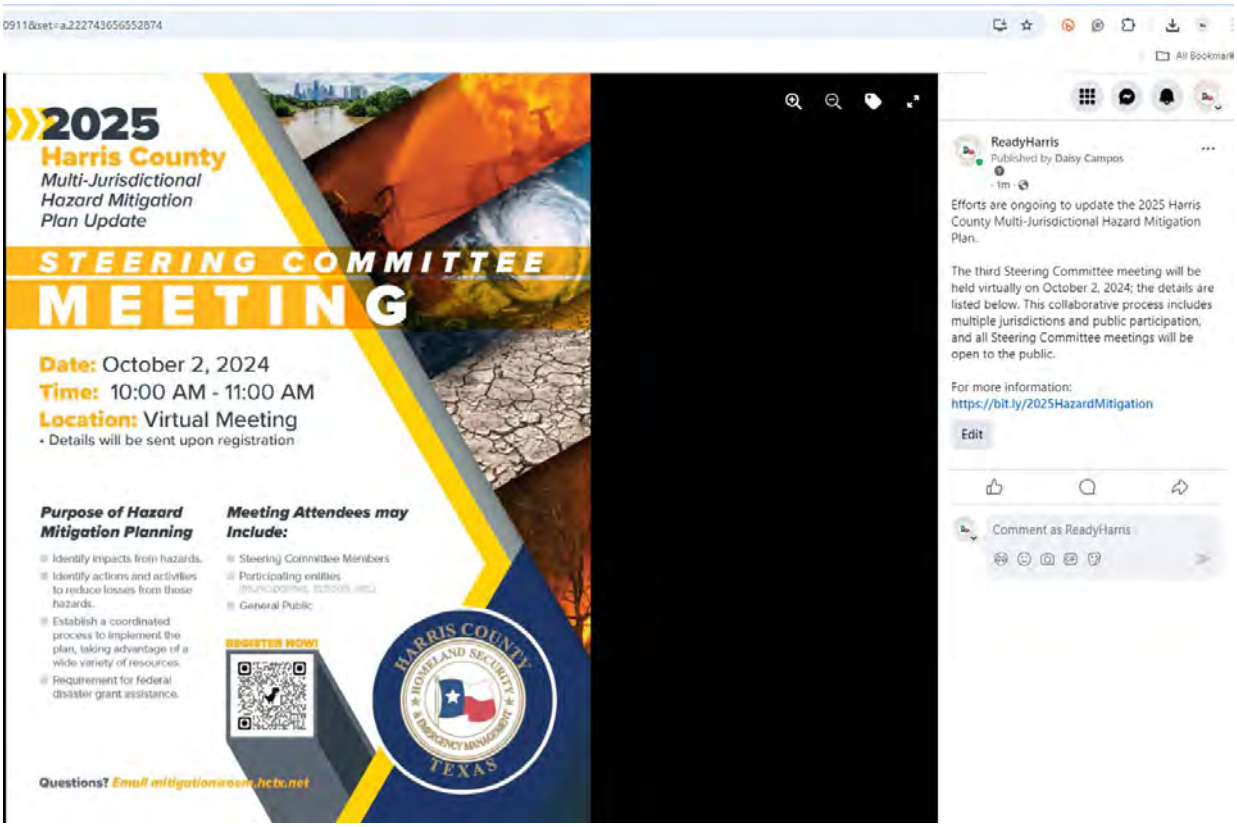
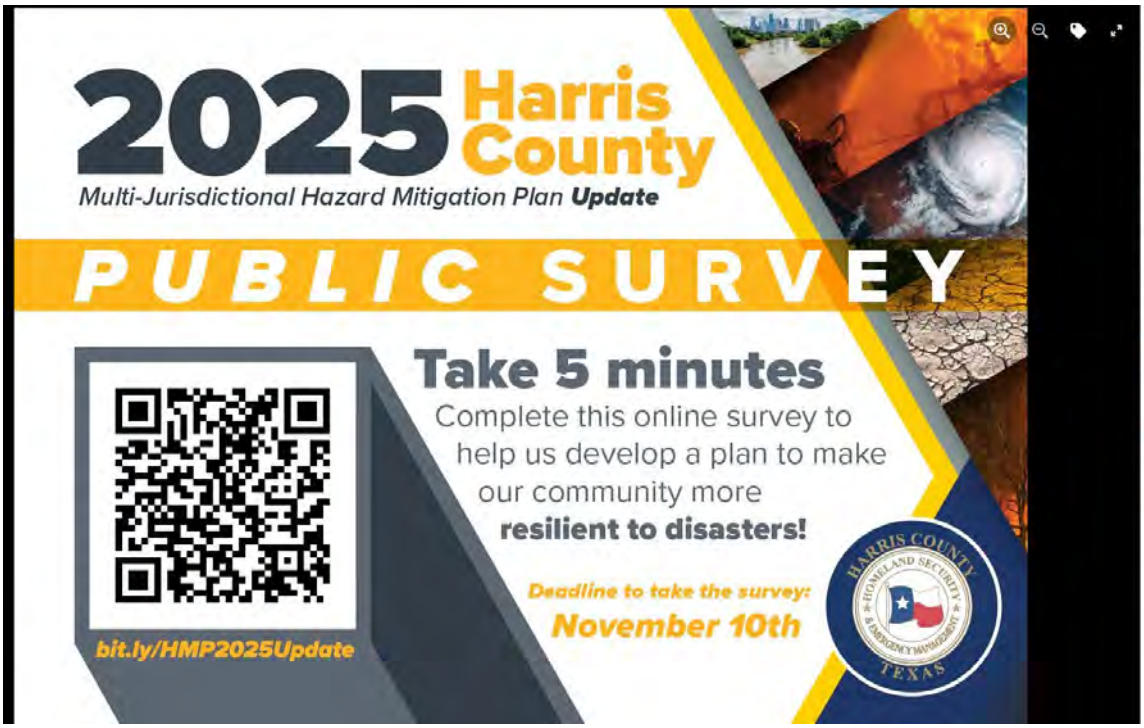

0911&set=a.222743656552874
2025
Harris County
Multi-Jurisdictional
Hazard Mitigation
Plan Update
STEERING COMMITTEE
MEETING
Date: October 2, 2024
Time: 10:00 AM - 11:00 AM
Location: Virtual Meeting
• Details will be sent upon registration
Purpose of Hazard Mitigation Planning
Identify impacts from hazards.
Identify actions and activities to reduce losses from those hazards.
Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
Requirement for federal disaster grant assistance.
Meeting Attendees may Include:
Steering Committee Members
Participating entities
General Public
REGISTER NOW!
Questions? Email mitigationwork.hctx.net
HARRIS COUNTY
HOMELAND SECURITY & EMERGENCY MANAGEMENT
TEXAS
ReadyHarris
Published by Daisy Campos
· 1m ·
Efforts are ongoing to update the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan.
The third Steering Committee meeting will be held virtually on October 2, 2024; the details are listed below. This collaborative process includes multiple jurisdictions and public participation, and all Steering Committee meetings will be open to the public.
For more information:
https://bit.ly/2025HazardMitigation
Edit
Comment as ReadyHarris


2025 Harris County
Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC SURVEY
Take 5 minutes
Complete this online survey to help us develop a plan to make our community more resilient to disasters!
Deadline to take the survey:
November 10th
bit.ly/HMP2025Update
HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

ReadyHarris
Published by Daisy Campos
October 2
#GetInvolved | The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan is actively being updated and we need YOUR input! Complete this short survey (https://bit.ly/HMP2025Update) so the county and our partners can have a better idea of:
- Residents' level of awareness regarding disasters.
- Areas vulnerable to various types of disasters.
For more information on the Multi-Jurisdictional Hazard Mitigation Plan, visit: https://bit.ly/2025HazardMitigation
#2025H... See more
Edit
Boost post
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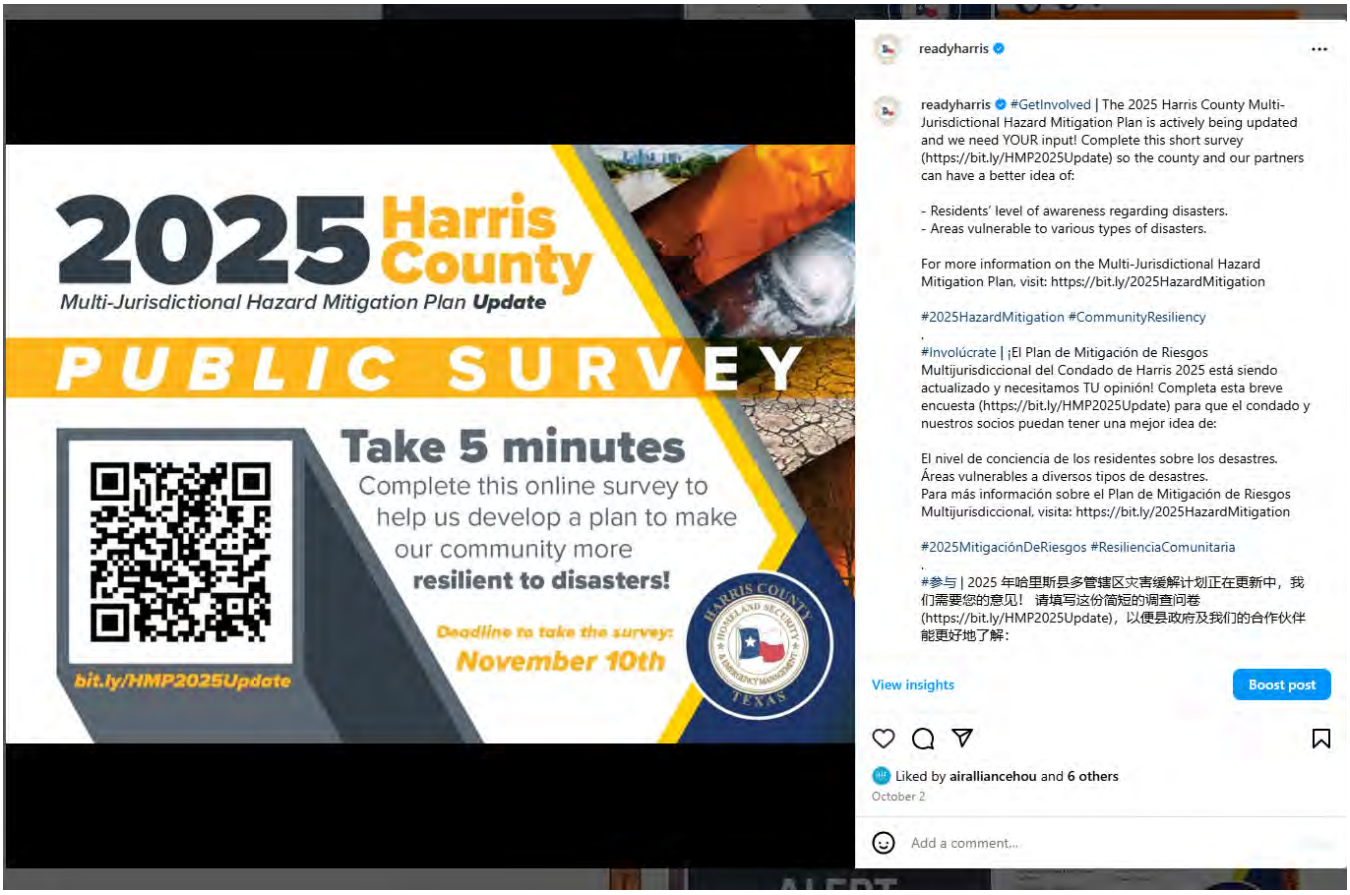
2025 Harris County
Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC SURVEY
Take 5 minutes
Complete this online survey to help us develop a plan to make our community more resilient to disasters!
Deadline to take the survey:
November 10th
bit.ly/HMP2025Update
HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

readyharris
readyharris #GetInvolved | The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan is actively being updated and we need YOUR input! Complete this short survey (https://bit.ly/HMP2025Update) so the county and our partners can have a better idea of:
- Residents' level of awareness regarding disasters.
- Areas vulnerable to various types of disasters.
For more information on the Multi-Jurisdictional Hazard Mitigation Plan, visit: https://bit.ly/2025HazardMitigation
#2025HazardMitigation #CommunityResiliency
#Involúcrate | ¡El Plan de Mitigación de Riesgos Multijurisdiccional del Condado de Harris 2025 está siendo actualizado y necesitamos TU opinión! Completa esta breve encuesta (https://bit.ly/HMP2025Update) para que el condado y nuestros socios puedan tener una mejor idea de:
El nivel de conciencia de los residentes sobre los desastres.
Áreas vulnerables a diversos tipos de desastres.
Para más información sobre el Plan de Mitigación de Riesgos Multijurisdiccional, visita: https://bit.ly/2025HazardMitigation
#2025MitigaciónDeRiesgos #ResilienciaComunitaria
#参与 | 2025 年哈里斯县多管辖区灾害缓解计划正在更新中，我们需要您的意见！ 请填写这份简短的调查问卷 (https://bit.ly/HMP2025Update)，以便县政府及我们的合作伙伴能更好地了解：
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Liked by airalliancehou and 6 others
October 2
Add a comment...

ReadyHarris
Published by Daisy Campos
· October 7 at 3:28 PM ·

The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update continues to the next phase of the planning process. A series of Community Workshops will be hosted throughout the county, encouraging community members to learn of featured topics, including:

- An overview of the Harris County Mitigation Plan
- Mitigation Projects in your area
- Harris County Climate Justice Plan

To register, complete the survey:
https://bit.ly/HMPWorkshopRegistration
... See more

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Hope it includes tornado sirens in Cypress
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readyharris The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update continues to the next phase of the planning process. A series of Community Workshops will be hosted throughout the county, encouraging community members to learn of featured topics, including:
- An overview of the Harris County Mitigation Plan
- Mitigation Projects in your area
- Harris County Climate Justice Plan
To register, complete the survey:
https://bit.ly/HMPWorkshopRegistration
Text HMP2025 to 888777 to receive updates about the plan's development.
4w

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October 7

Add a comment...

Office of Homeland Security &
@ReadyHarris

The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update continues to the next phase of the planning process. A series of Community Workshops will be hosted throughout the county to engage residents in the planning process.

To register: bit.ly/HMPWorkshopReg...

3:34 PM · Oct 7, 2024 · **1,101** Views

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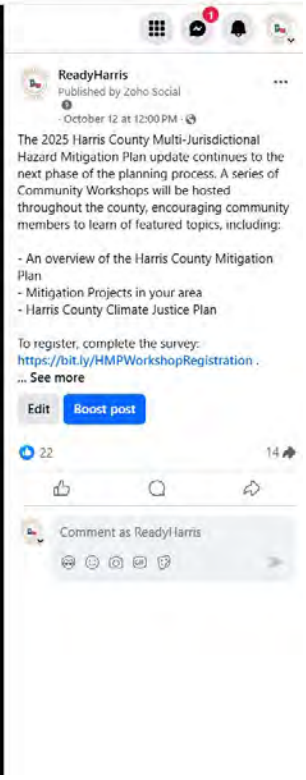

ReadyHarris
Published by Zoho Social
· October 12 at 12:00 PM ·

The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update continues to the next phase of the planning process. A series of Community Workshops will be hosted throughout the county, encouraging community members to learn of featured topics, including:

- An overview of the Harris County Mitigation Plan
- Mitigation Projects in your area
- Harris County Climate Justice Plan

To register, complete the survey:
https://bit.ly/HMPWorkshopRegistration .
... See more

Edit Boost post

22 14

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ReadyHarris
Published by Zoho Social
· October 14 at 10:00 AM ·

#GetInvolved | The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan is actively being updated and we need YOUR input! Help our community be more resilient and let us know your awareness of the disasters that impact you.

Link to Survey: https://bit.ly/HMP2025Update
For more info on the 2025HMP:
https://bit.ly/2025HazardMitigation

#2025HazardMitigation #CommunityResiliency

.

#Involúcrate | ¡El Plan de Mitigación de Riesgos Multijurisdiccional del Condado de ... **See more**

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1 6

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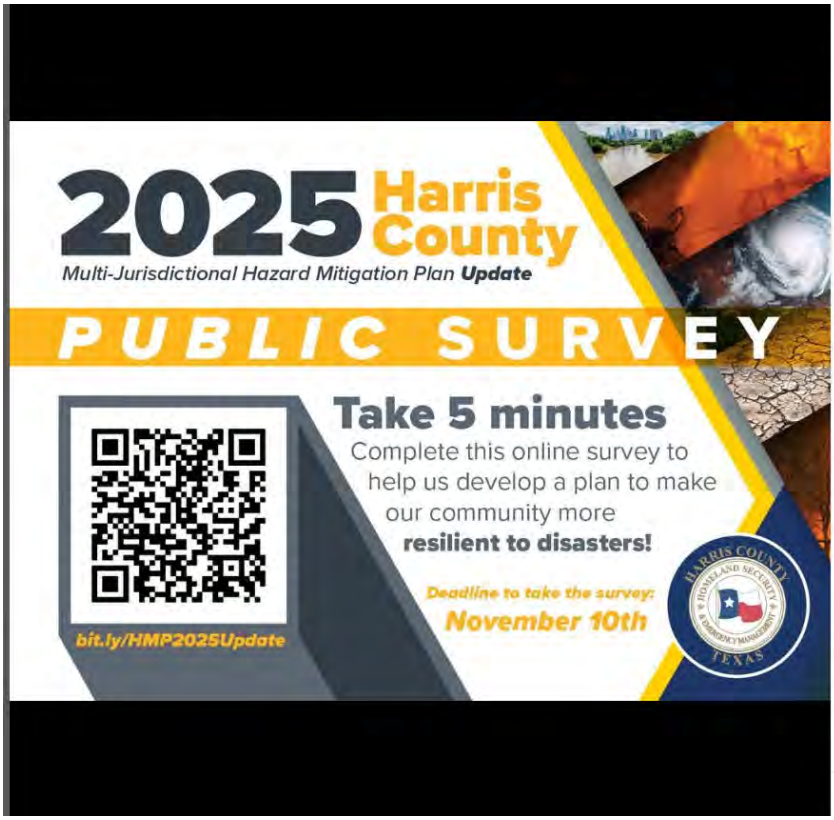

readyharris

readyharris #GetInvolved | The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan is actively being updated and we need YOUR input! Help our community be more resilient and let us know your awareness of the disasters that impact you.

Link to Survey: https://bit.ly/HMP2025Update
For more info on the 2025HMP:
https://bit.ly/2025HazardMitigation

#2025HazardMitigation #CommunityResiliency

.

#Involúcrate | ¡El Plan de Mitigación de Riesgos Multijurisdiccional del Condado de Harris 2025 está siendo actualizado activamente y necesitamos TU opinión! Ayuda a nuestra comunidad a ser más resiliente y cuéntanos tu conocimiento sobre los desastres que te afectan.

Enlace a la encuesta: https://bit.ly/HMP2025Update
Para más información sobre el 2025HMP:
https://bit.ly/2025HazardMitigation

#2025MitigaciónDeRiesgos #ResilienciaComunitaria

.

#参与其中 | 2025年哈里斯县多缇区危害缓解计划正在积极更新中，我们需要您的意见！帮助我们的社区更具韧性，并告诉我们您对影响您的灾害的了解。

调查链接: https://bit.ly/HMP2025Update
关于2025HMP的更多信息: https://bit.ly/2025HazardMitigation

#2025灾害缓解 #社区韧性

.

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October 14

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Office of Homeland Security &
@ReadyHarris

The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

👇 Upcoming Community Workshops 👇

Register at bit.ly/HMPWorkshopReg...

Text HMP2025 to 888777 to receive updates about the plan's development.

#Houston #HarrisCounty #EmergencyManagement #Community

3:01 PM · Oct 15, 2024 · **1,251** Views

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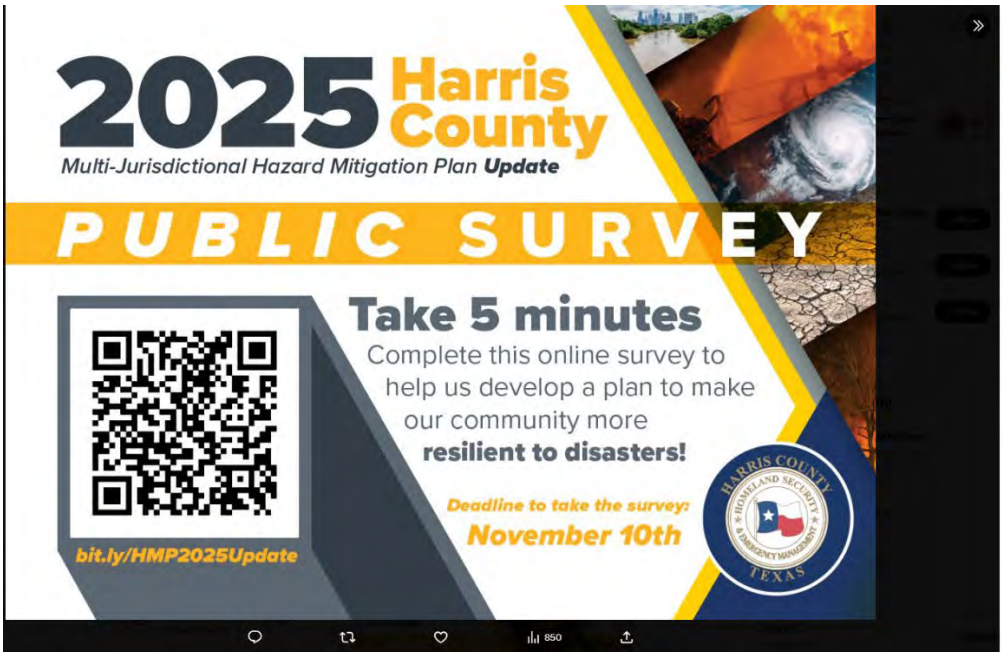

Office of Homeland Security &
@ReadyHarris

#GetInvolved | Take 5 minutes and complete this online survey to help our community become more resilient to disasters. Your participation will highlight strengths and areas for improvement, shaping future safety and risk reduction initiatives.

Survey: bit.ly/HMP2025Update

3:13 PM · Oct 15, 2024 · **850** Views

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2025 Harris County
Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC SURVEY
Take 5 minutes
Complete this online survey to help us develop a plan to make our community more resilient to disasters!
Deadline to take the survey:
November 10th
bit.ly/HMP2025Update
HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

ReadyHarris
Published by Daisy Campos
October 16 at 1:55 PM
#GetInvolved | The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan is actively being updated and we need YOUR input! Help our community be more resilient and let us know your awareness of the disasters that impact you.
Link to Survey: https://bit.ly/HMP2025Update
For more info on the 2025HMP: https://bit.ly/2025HazardMitigation
#2025HazardMitigation #CommunityResiliency
#Involúcrate | ¡El Plan de Mitigación de Riesgos Multijurisdiccional del Condado de ... See more
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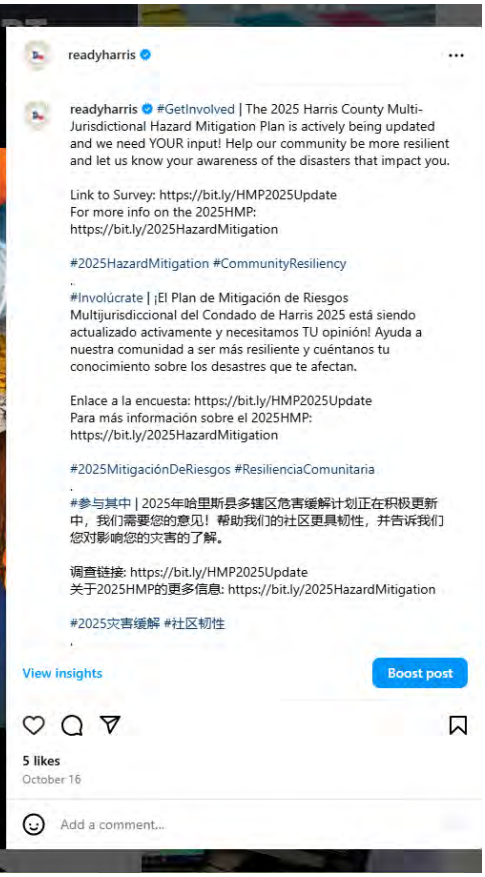
3w Like Reply Hide
Hugo Felix
We should focus on fixing slope and pipe failure. I fear that broken and clogged (vegetation or debris) pipes that storm drains are supposed to lead to —increase the risk of
Comment as ReadyHarris

2025 Harris County
Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC SURVEY
Take 5 minutes
Complete this online survey to help us develop a plan to make our community more resilient to disasters!
Deadline to take the survey:
November 10th
bit.ly/HMP2025Update
HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS


readyharris
readyharris #GetInvolved | The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan is actively being updated and we need YOUR input! Help our community be more resilient and let us know your awareness of the disasters that impact you.
Link to Survey: https://bit.ly/HMP2025Update
For more info on the 2025HMP: https://bit.ly/2025HazardMitigation
#2025HazardMitigation #CommunityResiliency
#Involúcrate | ¡El Plan de Mitigación de Riesgos Multijurisdiccional del Condado de Harris 2025 está siendo actualizado activamente y necesitamos TU opinión! Ayuda a nuestra comunidad a ser más resiliente y cuéntanos tu conocimiento sobre los desastres que te afectan.
Enlace a la encuesta: https://bit.ly/HMP2025Update
Para más información sobre el 2025HMP: https://bit.ly/2025HazardMitigation
#2025MitigaciónDeRiesgos #ResilienciaComunitaria
#参与其中 | 2025年哈里斯县多辖区危害缓解计划正在积极更新中，我们需要您的意见！帮助我们的社区更具韧性，并告诉我们您对影响您的灾害的了解。
调查链接: https://bit.ly/HMP2025Update
关于2025HMP的更多信息: https://bit.ly/2025HazardMitigation
#2025灾害缓解 #社区韧性
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October 16
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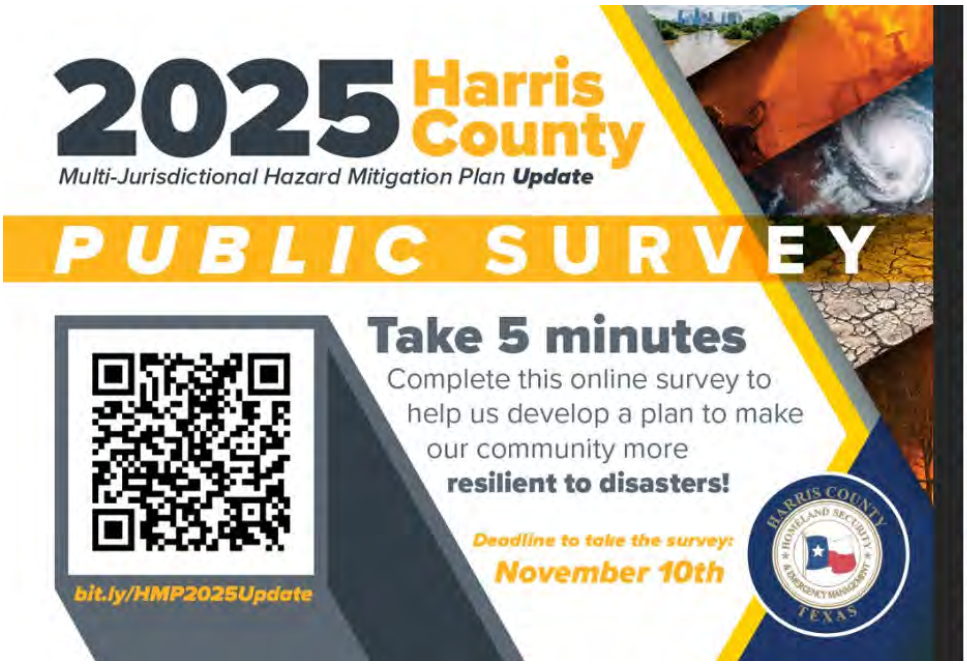

ReadyHarris
Harris County Office of Homeland Security a... · 16 Oct

#GetInvolved | The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan is actively being updated and we need YOUR input! Help our community be more resilient and let us know your awareness of the disasters that impact you.

Link to Survey: https://bit.ly/HMP2025Update
For more info on the 2025HMP:
https://bit.ly/2025HazardMitigation

#2025HazardMitigation #CommunityResiliency

.

#Involúcrate | ¡El Plan de Mitigación de Riesgos Multijurisdiccional del Condado de Harris 2025 está siendo actualizado activamente y necesitamos TU opinión! Ayuda a nuestra comunidad a ser más resiliente y cuéntanos tu conocimiento sobre los desastres que te afectan.

Enlace a la encuesta: https://bit.ly/HMP2025Update
Para más información sobre el 2025HMP:
https://bit.ly/2025HazardMitigation

#2025MitigaciónDeRiesgos #ResilienciaComunitaria

.

#参与其中 | 2025年哈里斯县多辖区危害缓解计划正在积极更新中，我们需要您的意见！帮助我们的社区更具韧性，并告诉我们您对影响您的灾害的了解。

调查链接: https://bit.ly/HMP2025Update
关于2025HMP的更多信息:
https://bit.ly/2025HazardMitigation

#2025灾害缓解 #社区韧性

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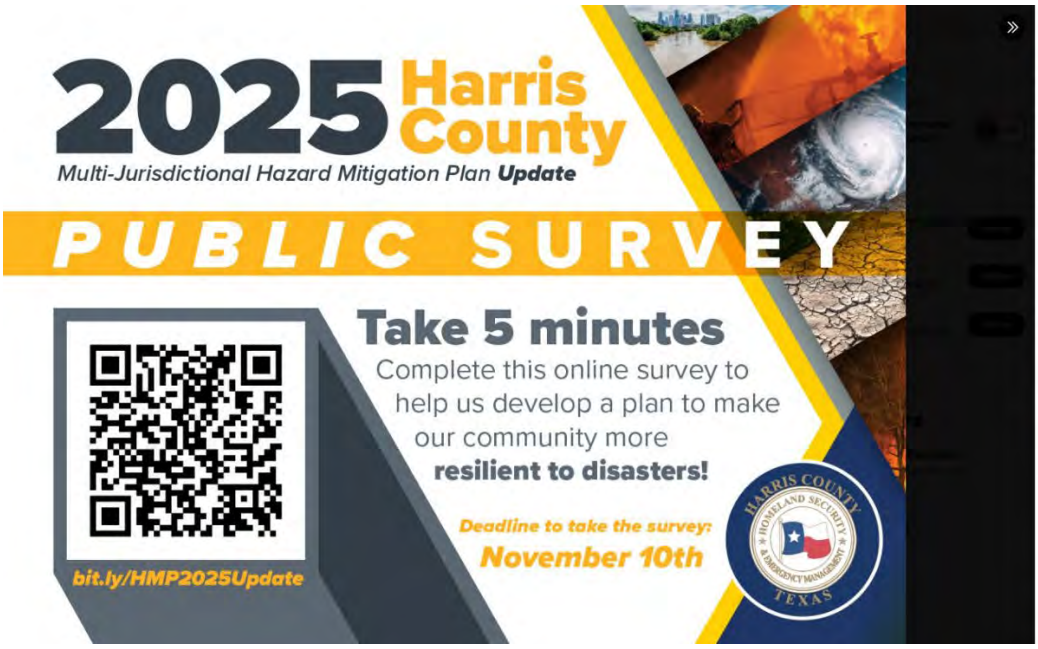
#ThamGia | Kế hoạch Giảm thiểu Rủi ro Đa khu vực



Office of Homeland Security
@ReadyHarris

#GetInvolved | Take 5 minutes and complete this online survey to help our community become more resilient to disasters. Your participation will highlight strengths and areas for improvement, shaping future safety and risk reduction initiatives.

Survey: bit.ly/HMP2025Update

1:58 PM · Oct 16, 2024 · 776 Views

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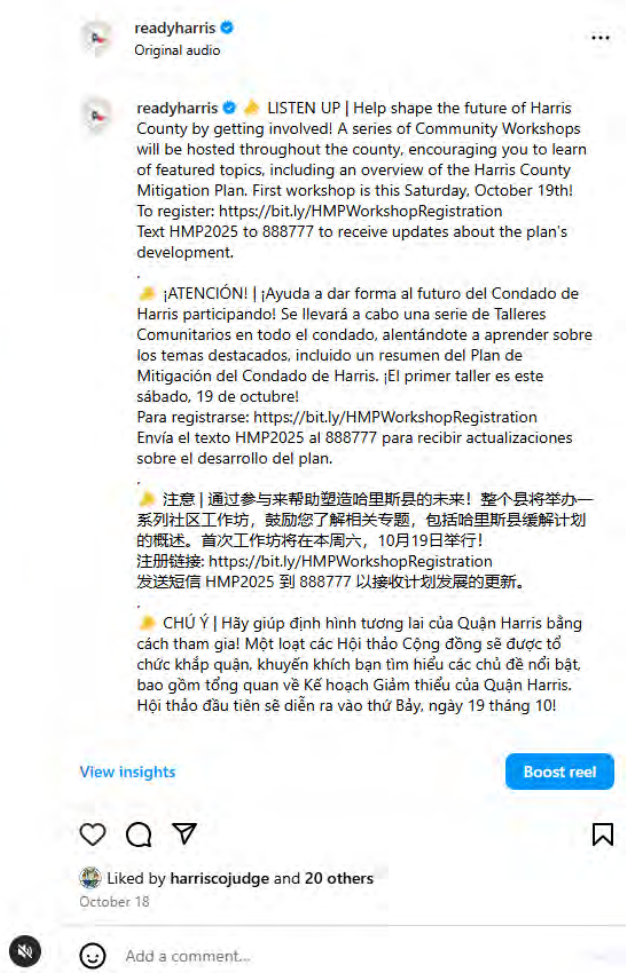
Office of Homeland · Oct 16

#Involúcrate | Tómate 5 minutos y completa esta encuesta en línea para ayudar a que nuestra comunidad sea más resiliente frente a los desastres. Tu participación destacará fortalezas y áreas de mejora.

Encuesta:

survey123.arcgis.co
Harris County Hazard Mitigation ...



readyharris
Original audio

readyharris LISTEN UP | Help shape the future of Harris County by getting involved! A series of Community Workshops will be hosted throughout the county, encouraging you to learn of featured topics, including an overview of the Harris County Mitigation Plan. First workshop is this Saturday, October 19th!
To register: https://bit.ly/HMPWorkshopRegistration
Text HMP2025 to 888777 to receive updates about the plan's development.
.
¡ATENCIÓN! | ¡Ayuda a dar forma al futuro del Condado de Harris participando! Se llevará a cabo una serie de Talleres Comunitarios en todo el condado, alentándote a aprender sobre los temas destacados, incluido un resumen del Plan de Mitigación del Condado de Harris. ¡El primer taller es este sábado, 19 de octubre!
Para registrarse: https://bit.ly/HMPWorkshopRegistration
Envía el texto HMP2025 al 888777 para recibir actualizaciones sobre el desarrollo del plan.
.
注意 | 通过参与来帮助塑造哈里斯县的未来！整个县将举办一系列社区工作坊，鼓励您了解相关专题，包括哈里斯县缓解计划的概述。首次工作坊将在本周六，10月19日举行！
注册链接: https://bit.ly/HMPWorkshopRegistration
发送短信 HMP2025 到 888777 以接收计划发展的更新。
.
CHÚ Ý | Hãy giúp định hình tương lai của Quận Harris bằng cách tham gia! Một loạt các Hội thảo Cộng đồng sẽ được tổ chức khắp quận, khuyến khích bạn tìm hiểu các chủ đề nổi bật, bao gồm tổng quan về Kế hoạch Giảm thiểu của Quận Harris. Hội thảo đầu tiên sẽ diễn ra vào thứ Bảy, ngày 19 tháng 10!

View insights

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Liked by **harriscojudge** and **20 others**
October 18

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📣 LISTEN UP | A series of Community Workshops will be hosted throughout the county, encouraging you to learn of featured topics, including an overview of the Harris County Mitigation Plan. First workshop is this Saturday, October 19th!

To register: bit.ly/HMPWorkshopReg...

5:26 PM · Oct 18, 2024 · **975** Views

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readyharris 👍 LISTEN UP | We are continuing our series of Community Workshops across the county, and we want YOUR input. Learn about the 2025 Hazard Mitigation Plan and speak to our partners about how they minimize risk in your community.

Register to attend tonight's meeting, October 22 at 6 pm (Katherine Tyra Branch Library): https://bit.ly/HMPWorkshopRegistration

#CommunityEngagement #GetInvolved

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🔔 ¡ESCUCHA! | Continuamos nuestra serie de Talleres Comunitarios en todo el condado, y queremos TU opinión. Aprende sobre el Plan de Mitigación de Riesgos 2025 y habla con nuestros socios sobre cómo minimizan los riesgos en tu comunidad.

Regístrate para asistir a la reunión de esta noche, 22 de octubre a las 6 p.m. (Biblioteca Sucursal Katherine Tyra): https://bit.ly/HMPWorkshopRegistration

#ParticipaciónComunitaria #Involúcrate

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🔔 注意 | 我们正在全县继续开展社区工作坊系列活动，我们需要您的意见。了解2025年风险减缓计划，并与我们的合作伙伴讨论他们如何在您的社区降低风险。

注册参加今晚的会议，10月22日下午6点（Katherine Tyra分馆图书馆）：https://bit.ly/HMPWorkshopRegistration

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Register: bit.ly/HMPWorkshopReg...

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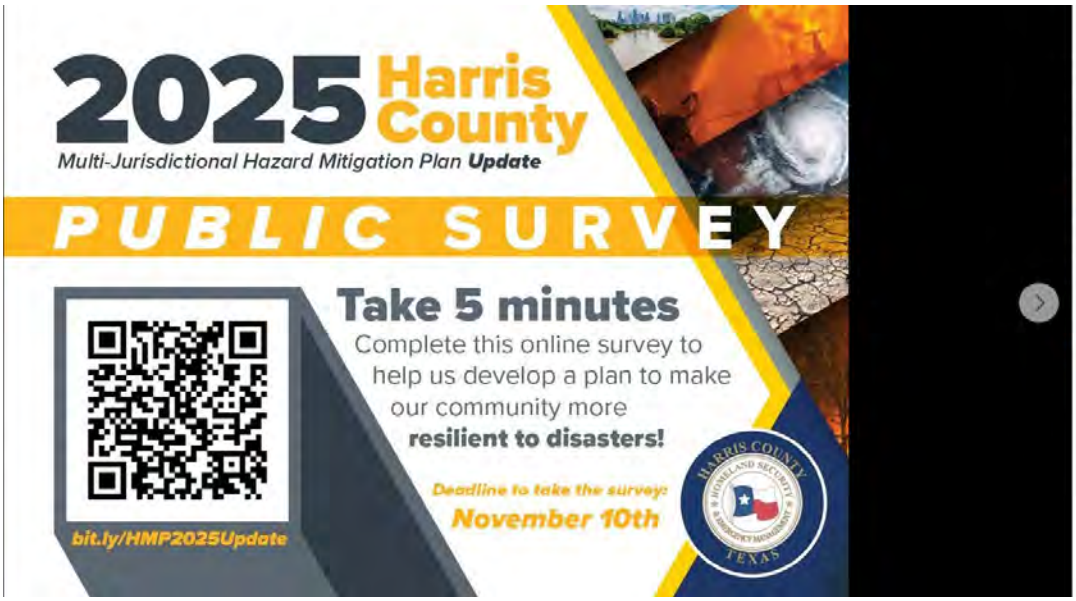

Published by Zoho Social
· October 25 at 9:30 AM ·

#GetInvolved | The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan is actively being updated and we need YOUR input! Help our community be more resilient and let us know your awareness of the disasters that impact you.

Link to Survey: https://bit.ly/HMP2025Update
For more info on the 2025HMP:
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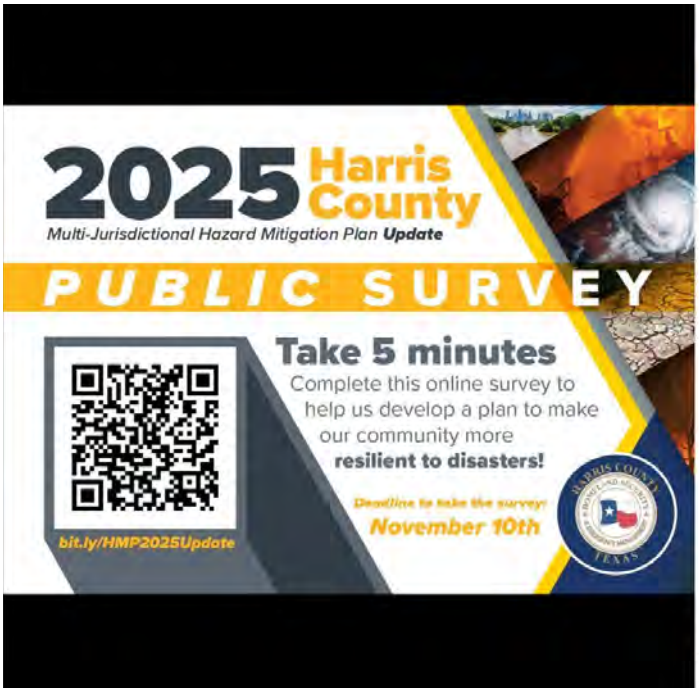

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#2025HazardMitigation #CommunityResiliency

#Involúcrate | ¡El Plan de Mitigación de Riesgos Multijurisdiccional del Condado de Harris 2025 está siendo actualizado activamente y necesitamos TU opinión! Ayuda a nuestra comunidad a ser más resiliente y cuéntanos tu conocimiento sobre los desastres que te afectan.

Enlace a la encuesta: https://bit.ly/HMP2025Update
Para más información sobre el 2025HMP:
https://bit.ly/2025HazardMitigation

#2025MitigaciónDeRiesgos #ResilienciaComunitaria

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调查链接: https://bit.ly/HMP2025Update
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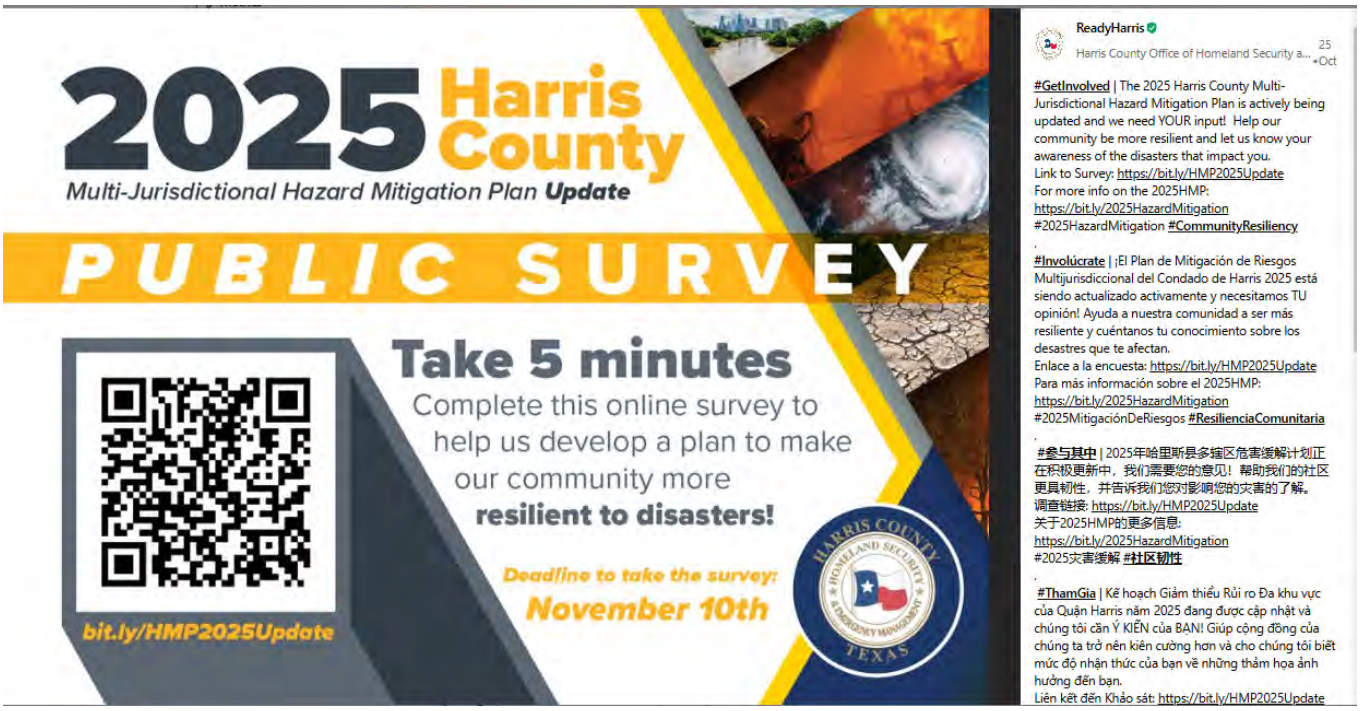

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关于2025HMP的更多信息:
https://bit.ly/2025HazardMitigation
#2025灾害缓解 #社区韧性

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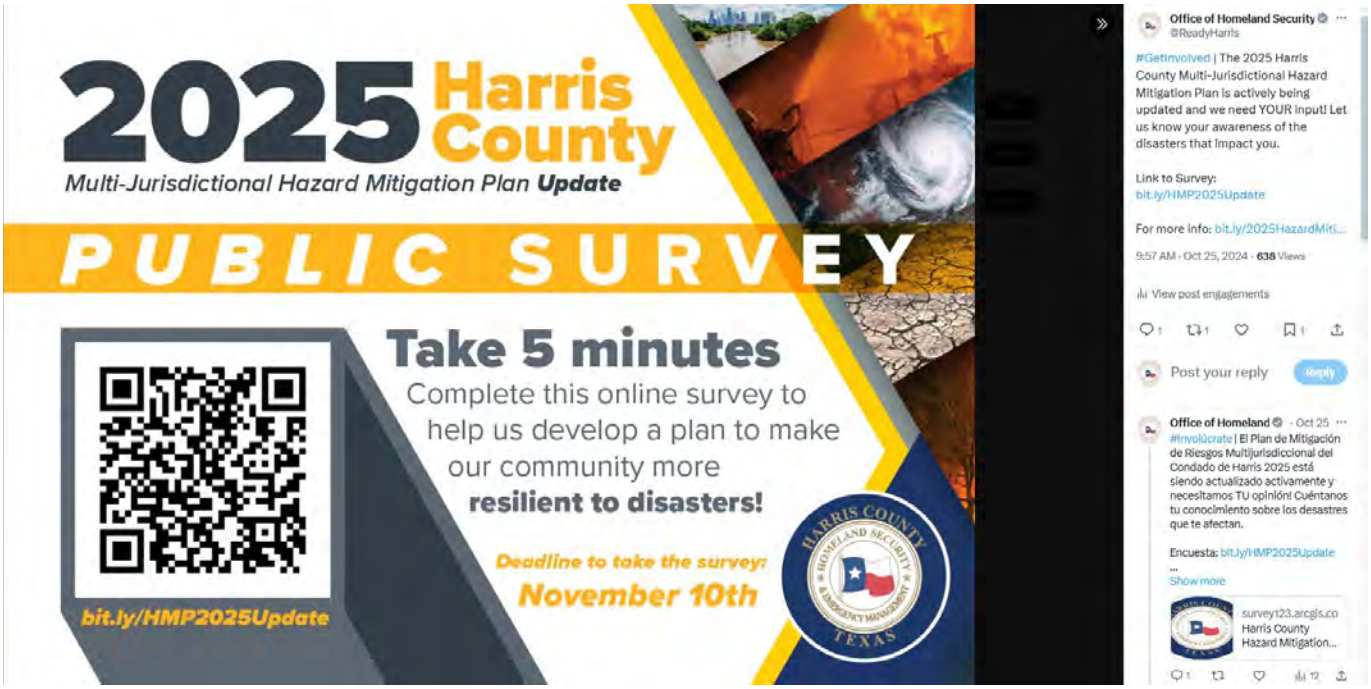
#ThamGia | Kế hoạch Giảm thiểu Rủi ro Đa khu vực của Quận Harris năm 2025 đang được cập nhật và chúng tôi cần Ý KIẾN của BẠN! Giúp cộng đồng của chúng ta trở nên kiên cường hơn và cho chúng tôi biết mức độ nhận thức của bạn về những thảm họa ảnh hưởng đến bạn.
Liên kết đến Khảo sát: https://bit.ly/HMP2025Update



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ReadyHarris
Published by Daisy Campos
October 28 at 3:40 PM

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https://bit.ly/HMPWorkshopRegistration

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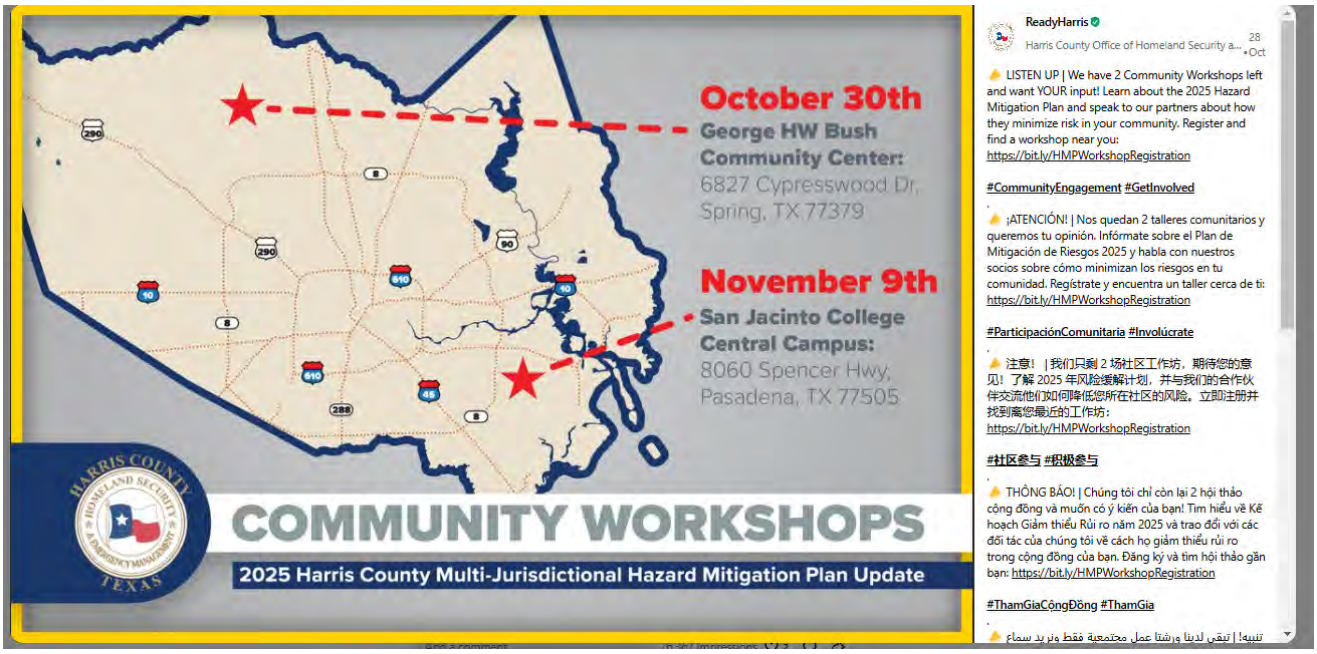

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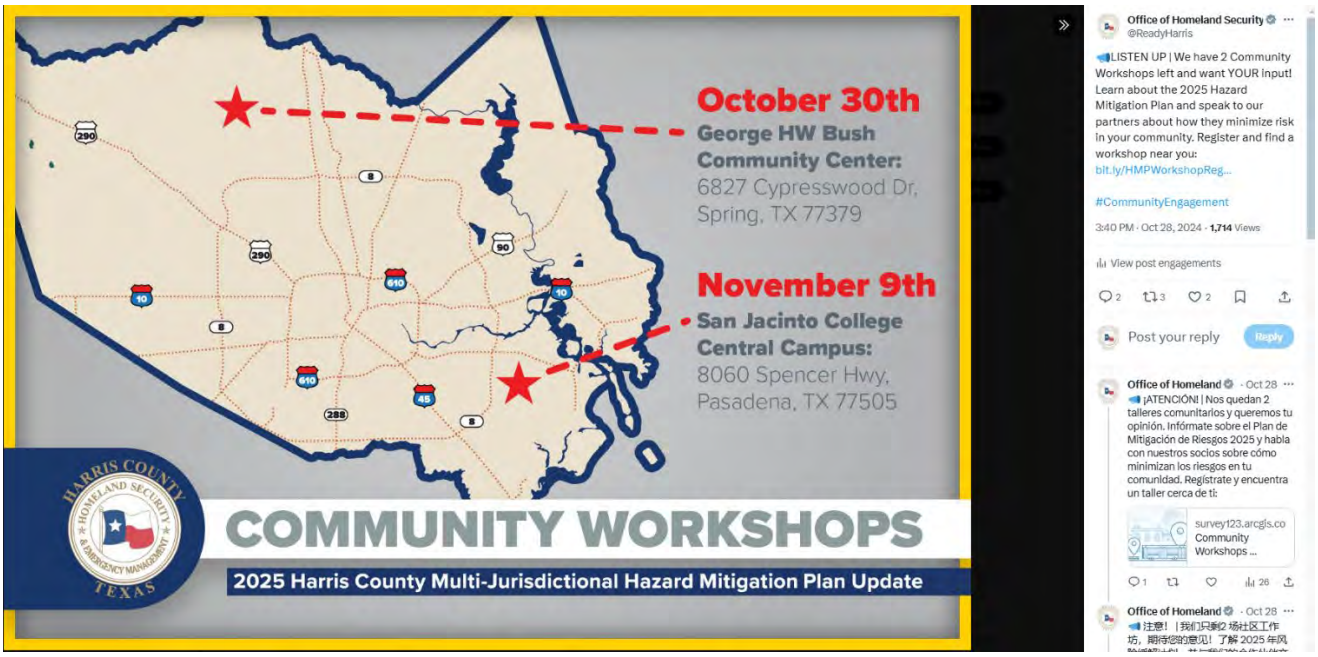
👍 تنبيه! | تبقى لدينا ورشتا عمل مجتمعية فقط ونريد سماع



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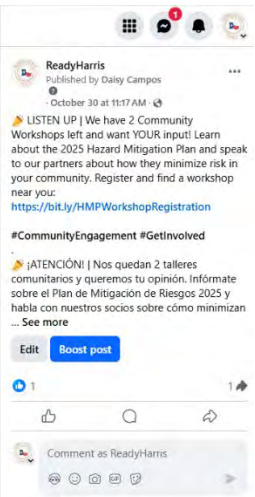
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Published by Daisy Campos
October 30 at 11:17 AM
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Oct 30, 6:00 PM - 7:00 PM

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan - Community Workshop

6827 Cypresswood Drive, Spring, TX

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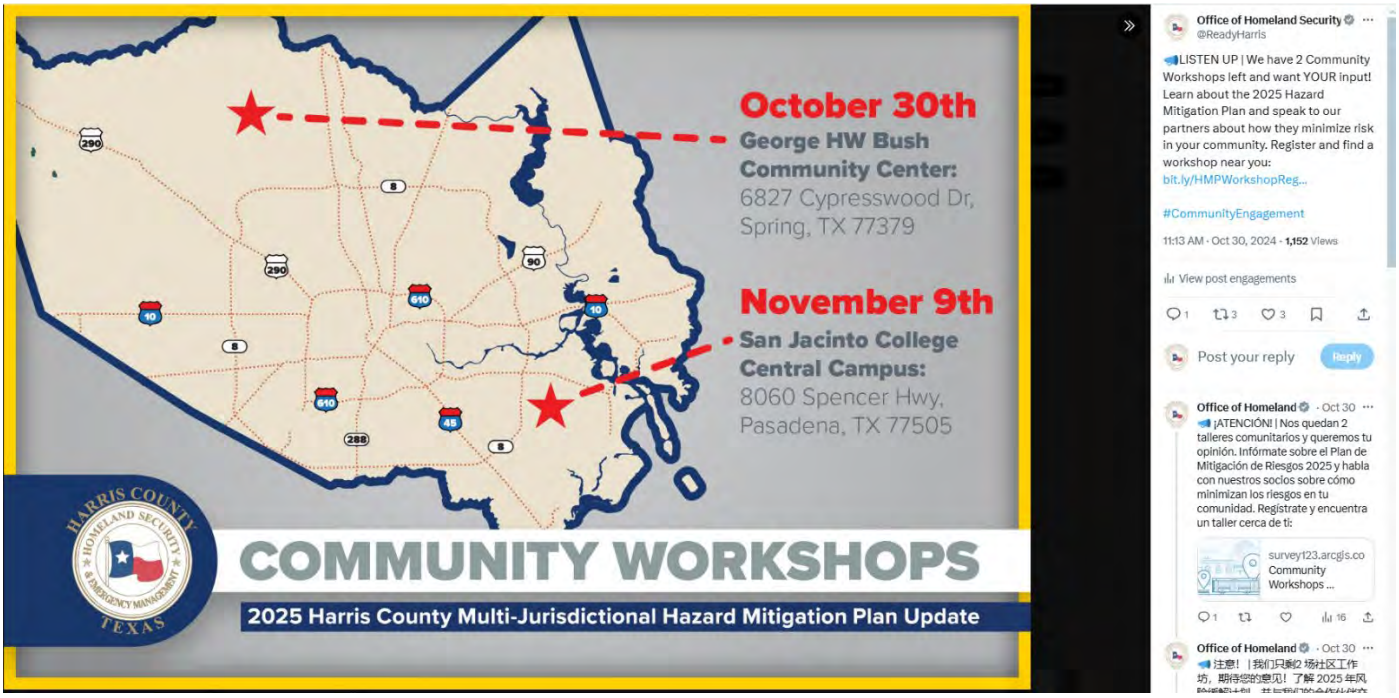
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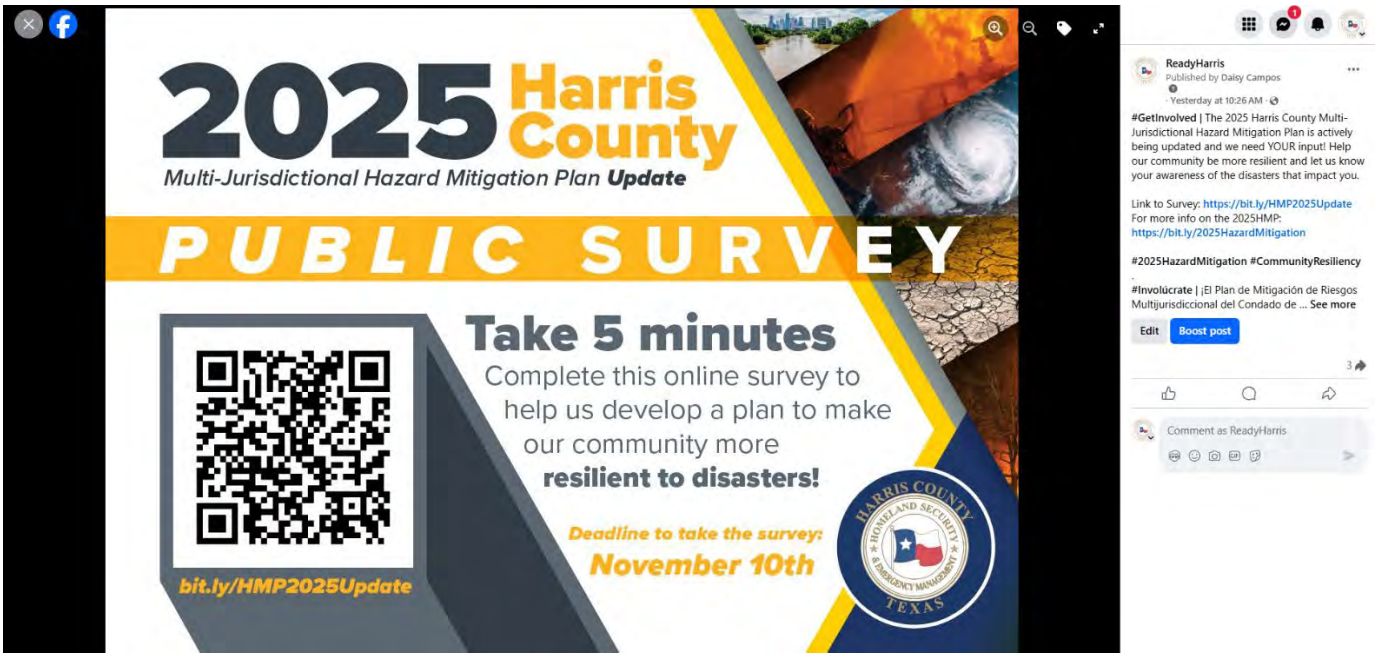
2025 Harris County
Multi-Jurisdictional Hazard Mitigation Plan Update
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Take 5 minutes
Complete this online survey to help us develop a plan to make our community more resilient to disasters!
Deadline to take the survey:
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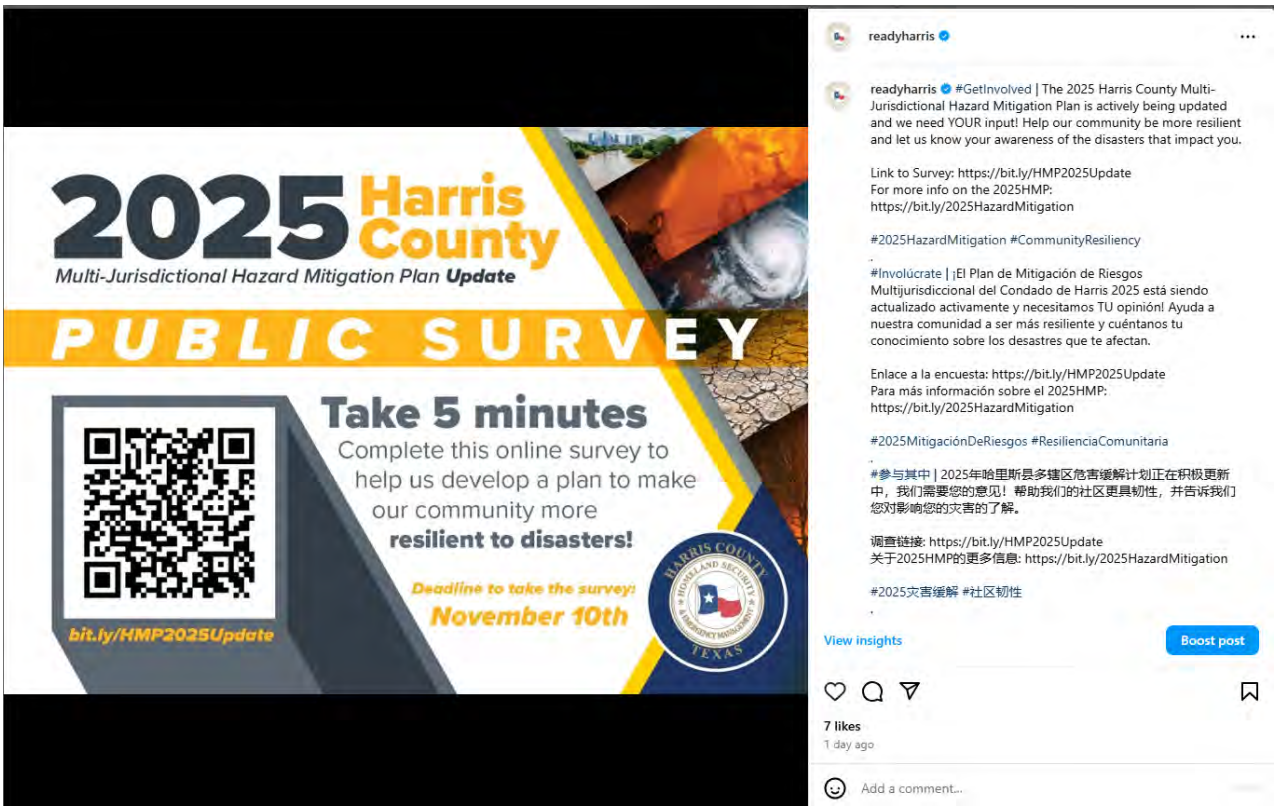

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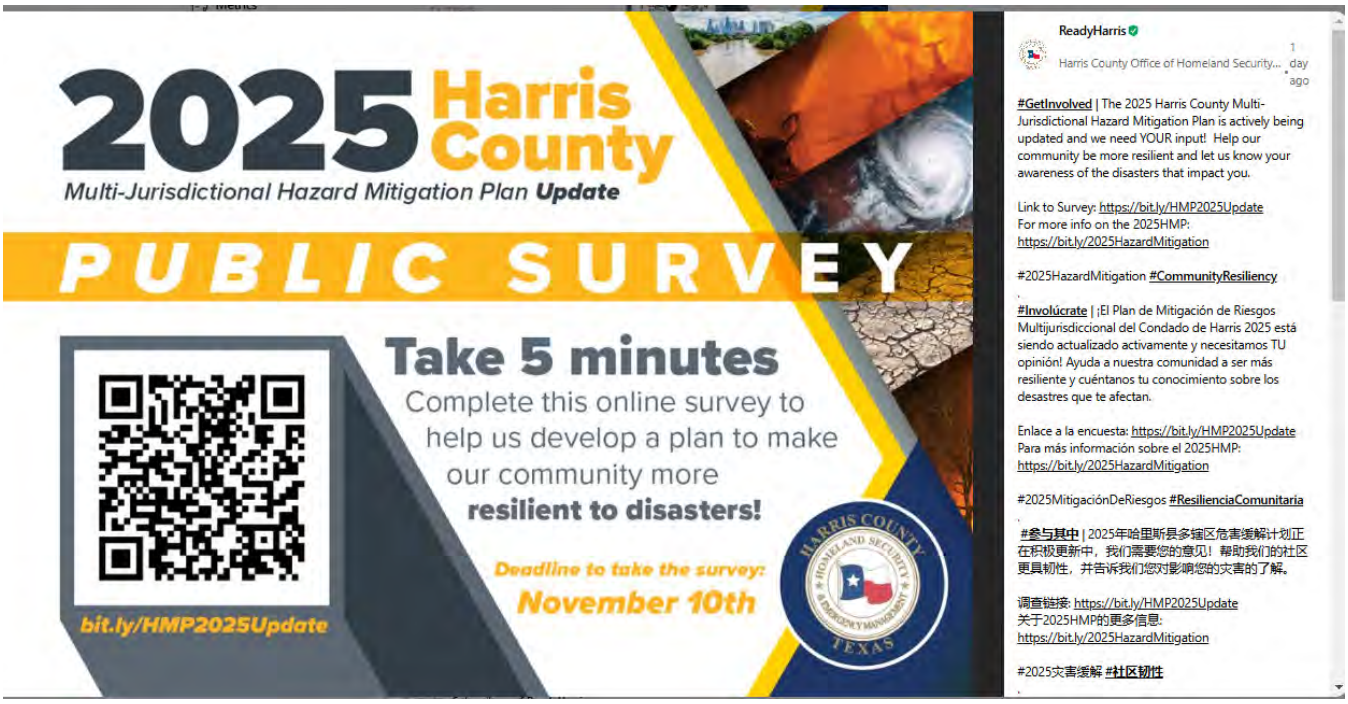

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#2025HazardMitigation #CommunityResiliency

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#Involúcrate | ¡El Plan de Mitigación de Riesgos Multijurisdiccional del Condado de Harris 2025 está siendo actualizado activamente y necesitamos TU opinión! Ayuda a nuestra comunidad a ser más resiliente y cuéntanos tu conocimiento sobre los desastres que te afectan.

Enlace a la encuesta: https://bit.ly/HMP2025Update
Para más información sobre el 2025HMP:
https://bit.ly/2025HazardMitigation

#2025MitigaciónDeRiesgos #ResilienciaComunitaria

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#参与其中 | 2025年哈里斯县多辖区危害缓解计划正在积极更新中，我们需要您的意见！帮助我们的社区更具韧性，并告诉我们您对影响您的灾害的了解。

调查链接: https://bit.ly/HMP2025Update
关于2025HMP的更多信息:
https://bit.ly/2025HazardMitigation

#2025灾害缓解 #社区韧性

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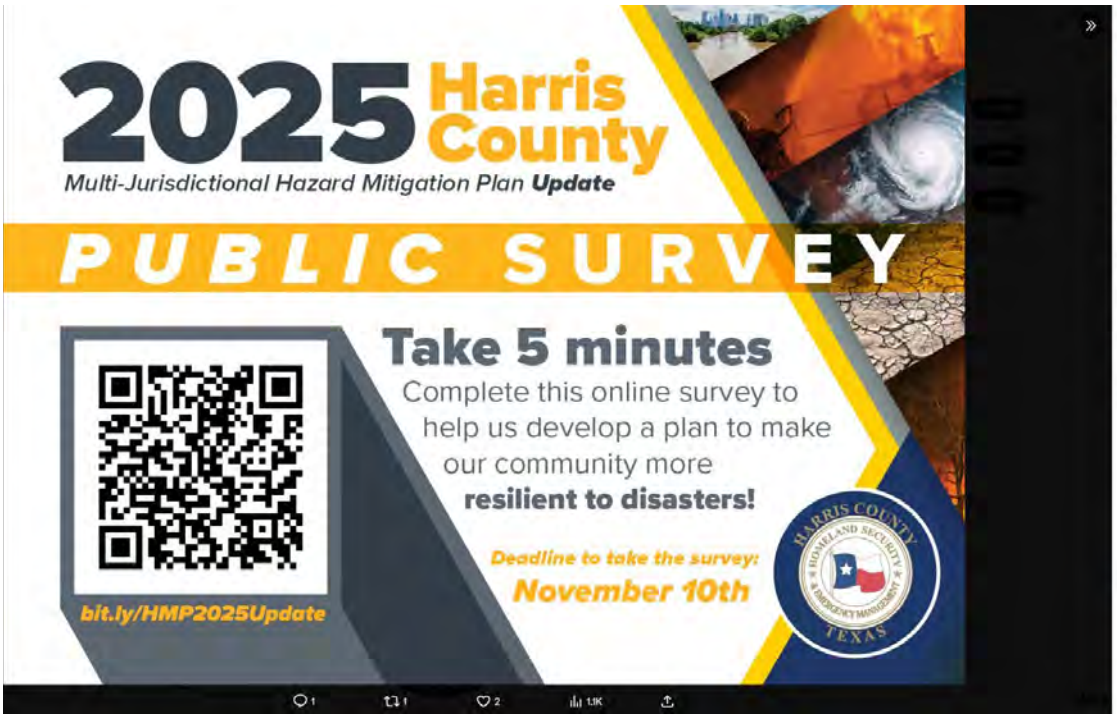
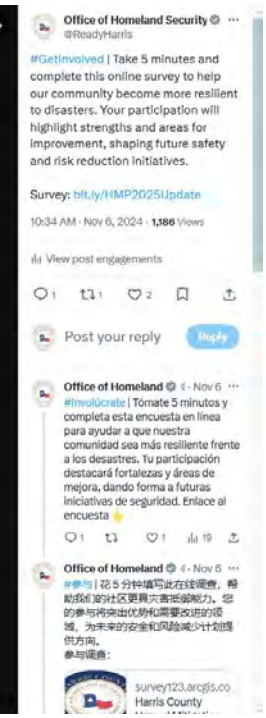

2025 Harris County
Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC SURVEY
Take 5 minutes
Complete this online survey to help us develop a plan to make our community more resilient to disasters!
Deadline to take the survey:
November 10th
bit.ly/HMP2025Update
HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS


Office of Homeland Security
@ReadyHarris
#GetInvolved | Take 5 minutes and complete this online survey to help our community become more resilient to disasters. Your participation will highlight strengths and areas for improvement, shaping future safety and risk reduction initiatives.
Survey: bit.ly/HMP2025Update
10:34 AM · Nov 6, 2024 · 1,186 Views
View post engagements
Post your reply
Reply
Office of Homeland · Nov 6
#InvolúcrateTómate 5 minutos y completa esta encuesta en línea para ayudar a que nuestra comunidad sea más resiliente frente a los desastres. Tu participación destacará fortalezas y áreas de mejora, dando forma a futuras iniciativas de seguridad. Enlace al encuesta
Office of Homeland · Nov 6
#参与 | 花 5 分钟填写此在线调查，帮助我们的社区更具灾害抵御能力。您的参与将突出优势和需要改进的领域，为未来的安全和风险减少计划提供方向。
参与调查：
survey123.arcgis.co
Harris County

2025 Harris County
Multi-Jurisdictional Hazard Mitigation Plan Update
COMMUNITY WORKSHOPS
November 9
10:00 AM – 11:30 AM
San Jacinto College Central Campus:
8060 Spencer Hwy, Pasadena, TX 77505
A virtual option will be made available for this workshop. Details will be provided upon registration.
REGISTER NOW!
Text HMP2025 to 888777
to recieve updates about the Plan's development
HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

ReadyHarris
Published by Daisy Campos
· 29m ·
Final Call! | Don't miss the last Community Workshop this Saturday (11/9)! We want YOUR input! Learn about the 2025 Hazard Mitigation Plan and speak to our partners about how they minimize risk in your community.
Register to attend in person or virtually: https://bit.ly/HMPWorkshopRegistration.
#CommunityEngagement #GetInvolved
¡Última oportunidad! | No te pierdas el último Taller Comunitario este sábado (11/9). ¡Queremos TU opinión! Aprende sobre el Plan de Mitig... See more
Edit
Boost post

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

REGISTER NOW

Date: April 12th, 2024
Time: 9:30 AM - 11:00 AM
Location: Houston TranStar
6922 Katy Rd | Houston, TX 77024

Purpose of Hazard Mitigation Planning

- Identify impacts from hazards.
- Identify actions and activities to reduce losses from those hazards.
- Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
- Requirement for federal disaster grant assistance.

Eligible Participating Jurisdictions

- Municipalities
- Special Purpose Districts *(Schools, WCIDs, MUDs, etc.)*

SCAN ME!

HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

Questions? *Email oemmitigation@oem.hctx.net*

2025 Harris County *Multi-Jurisdictional Hazard Mitigation Plan Update*

JULY 24 · 2024

BULLETIN

July 24, 2024 Steering Committee Meeting Recap:

On July 24, 2024, the Core Planning Team hosted the first Steering Committee Meeting for the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan with 30 persons in attendance. The Steering Committee established the following:

- Reviewed and approved the Steering Committee Charter.
- Described the overall mitigation planning process and new requirements per FEMA's Local Mitigation Planning Policy Guide effective April 2023.
- Steering Committee meetings will be held on the second Wednesday of every month.
- Critical facilities and infrastructure categories presented and discussed.
- Identified hazards of concern.
- Provided a brief overview of the public involvement strategy, which will be discussed further at the next Steering Committee meeting.

NEXT MEETING DATE IS · WEDNESDAY, SEPT. 11TH 10:00 AM - 12:00 PM
LOCATION · TBD

Upcoming Deliverables for Participating Entities:

- Review the 2020 Harris County Multi-Hazard Mitigation Action Plan.
  - Volume 1 – Area Wide Elements.
  - Volume 2 – Jurisdiction Specific Elements.
- Provide a list of critical facilities/infrastructure within your jurisdiction to **mitigation@oem.hctx.net** by August 14, 2024.
- Provide your local public engagement strategy to **mitigation@oem.hctx.net** by August 14, 2024.
- Add a link to your website for the mitigation plan update as part of the public outreach strategy.
- Begin Capability Assessment.

The most up-to-date information can be found on our website at: **www.readyharris.org**
If you have any questions, please don't hesitate to contact the team at: **mitigation@oem.hctx.net**

JULY 24 · 2024

BULLETIN

Participating Entities (44 Total):

Harris County, Harris County Flood Control District, Harris County Office of County Administration, Bellaire, Bunker Hill Village, Deer Park, El Lago, Friendswood, Galena Park, Hedwig Village, Hilshire Village, Humble, Hunters Creek Village, Jacinto City, Jersey Village, Katy, La Porte, Missouri City, Morgan's Point, Nassau Bay, Pasadena, Piney Point Village, Seabrook, Shoreacres, Spring Valley Village, Stafford, Taylor Lake Village, Waller, Webster, West University Place, A+ Unlimited Potential Charter Schools, Cypress Fairbanks ISD, Goose Creek ISD, Klein ISD, Pasadena ISD, Southwest Public Schools, Spring ISD, Varnett Public Schools, Houston Community College, Lee College, San Jacinto College, University of Houston, Cypress Creek Drainage District, and Harris Health.

Community Rating System (CRS) Jurisdictions:

Bellaire, Deer Park, El Lago, Friendswood, Harris County, Jersey Village, La Porte, Missouri City, Nassau Bay, Pasadena, Seabrook, Shoreacres, Taylor Lake Village, and West University Place.

CORE PLANNING TEAM

Chairperson & Spokesperson
Brian Murray
Brian.Murray@harriscountytx.gov

Vice Chairperson
Jeremy Phillips
Jeremy.Phillips@hcfcd.hctx.net

Lead Planner
Brittany Ray
Brittany.Ray@oem.hctx.net

HCOHSEM Mitigation
mitigation@oem.hctx.net

Media Inquiries
oemjic@oem.hctx.net

Harris County is committed to creating and sustaining communities that are more resilient to disasters. To fulfill this pledge, *Harris County Office of Homeland Security and Emergency Management* (HCOHSEM) is in the process of updating the Harris County Multi-Jurisdictional Hazard Mitigation Plan in partnership with cities and special purpose districts throughout the County. Federal rules require this plan be updated every five years.

Office of Homeland Security and Emergency Management

www.readyharris.org

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

STEERING COMMITTEE MEETING

Date: July 24, 2024
Time: 10:00 AM - 12:00 PM
Location: TxDOT
7600 Washington Ave. | Houston, TX 77007

Purpose of Hazard Mitigation Planning

- Identify impacts from hazards.
- Identify actions and activities to reduce losses from those hazards.
- Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
- Requirement for federal disaster grant assistance.

Meeting Attendees may Include:

- Steering Committee Members
- Participating entities *(municipalities, schools, etc.)*
- General Public

REGISTER NOW!

HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

Questions? *Email mitigation@oem.hctx.net*

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

SEPTEMBER 11 · 2024

BULLETIN

September 11, 2024 Steering Committee Meeting Recap:

The Core Planning Team hosted the second Steering Committee Meeting for the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update with 41 persons in attendance. During the meeting, the Steering Committee:

- Drafted the 2025 HC MJ-HMP Mission Statement.
- Developed draft goals of the 2025 HC MJ-HMP.
- Reviewed and approved the public engagement strategy.

NEXT MEETING DATE IS - WEDNESDAY, OCT. 2ND 10:00 AM - 12:00 PM
LOCATION - HOUSTON TRANSTAR - 6922 KATY RD. HOUSTON, TX 77024

Upcoming Deliverables for Participating Entities:

- **Submit Critical Facilities and Infrastructure List:** If you haven't done so already, please provide your list as soon as possible.
- **Provide Feedback on Mission Statement and Goals:** Review and comment on the draft 2025 HC MJ-HMP Mission Statement and Goals by COB Sept. 27.
- **Review Draft Public Survey:** Send your feedback on the draft public survey by COB September 20th.
- **Actively Promote and Support the Public Engagement Strategy:**
  - The public survey will launch on October 1st.
  - 4 public workshops may be held in late October.
- **Begin Capability Assessment.**

The most up-to-date information can be found on our website at: **www.readyharris.org**
If you have any questions, please don't hesitate to contact the team at: **mitigation@oem.hctx.net**

SEPTEMBER 11 - 2024

BULLETIN

Participating Entities (44 Total):

Harris County, Harris County Flood Control District, Harris County Office of County Administration, Bellaire, Bunker Hill Village, Deer Park, El Lago, Friendswood, Galena Park, Hedwig Village, Hilshire Village, Humble, Hunters Creek Village, Jacinto City, Jersey Village, Katy, La Porte, Missouri City, Morgan's Point, Nassau Bay, Pasadena, Piney Point Village, Seabrook, Shoreacres, Spring Valley Village, Stafford, Taylor Lake Village, Waller, Webster, West University Place, A+ Unlimited Potential Charter Schools, Cypress Fairbanks ISD, Goose Creek CISD, Klein ISD, Pasadena ISD, Southwest Public Schools, Spring ISD, Varnett Public Schools, Houston Community College, Lee College, San Jacinto College, University of Houston, Cypress Creek Drainage District, and Harris Health.

Community Rating System (CRS) Jurisdictions:

Bellaire, Deer Park, El Lago, Friendswood, Harris County, Jersey Village, La Porte, Missouri City, Nassau Bay, Pasadena, Seabrook, Shoreacres, Taylor Lake Village, and West University Place.

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Brian Murray
Brian.Murray@harriscountytx.gov

Vice Chairperson
Jeremy Phillips
Jeremy.Phillips@hcfcd.hctx.net

Lead Planner
Brittany Ray
Brittany.Ray@harriscountytx.gov

HCOHSEM Mitigation
mitigation@oem.hctx.net

Media Inquiries
oemjic@oem.hctx.net

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Office of Homeland Security and Emergency Management

www.readyharris.org

2025
Harris County
Multi-Jurisdictional
Hazard Mitigation
Plan Update
STEERING COMMITTEE
MEETING
Date: September 11, 2024
Time: 10:00 AM - 12:00 PM
Location: San Jacinto College
8060 Spencer Hwy. | Pasadena, TX 77505
- Anderson-Ball Classroom C-16.1013
Purpose of Hazard Mitigation Planning
Identify impacts from hazards.
Identify actions and activities to reduce losses from those hazards.
Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
Requirement for federal disaster grant assistance.
Meeting Attendees may Include:
Steering Committee Members
Participating entities (municipalities, schools, etc.)
General Public
REGISTER NOW!
HARRIS COUNTY
HOMELAND SECURITY & EMERGENCY MANAGEMENT
TEXAS
Questions? Email mitigation@oem.hctx.net

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

OCTOBER 2 · 2024

BULLETIN

October 2, 2024 Steering Committee Meeting Recap:

The Core Planning Team hosted the third Steering Committee Meeting for the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update with 44 persons in attendance. During the meeting the Steering Committee:

- Approved the 2025 HC MJ-HMP Mission Statement.
- Approved the goals of the 2025 HC MJ-HMP.
- Discussed the 4 upcoming public workshops.

NEXT MEETING: WEDNESDAY, NOVEMBER 13, 2024 10:00 AM – 12:00 PM
LOCATION: HOUSTON TRANSTAR - 6922 KATY RD. HOUSTON, TX 77024

Deliverables for Participating Entities:

- **Actively Promote and Socialize the Public Survey**
  - Launched on October 1st and will remain open until November 10th
    - Link to survey: bit.ly/HMP2025Update
- **Actively Promote and Socialize the Upcoming Public Workshops**
  - October 19, 2024, 10:00 AM – 11:30 AM at San Jacinto College – Generation Park
  - October 22, 2024, 6:00 PM – 7:30 PM at Katherine Tyra Branch Library
  - October 30, 2024, 6:00 PM – 7:30 PM at George H.W. Bush Community Center
  - November 9, 2024, 10:00 AM – 11:30 AM at San Jacinto College – Central Campus

The most up-to-date information can be found on our website at: **www.readyharris.org**
If you have any questions, please don't hesitate to contact the team at: **mitigation@oem.hctx.net**

OCTOBER 2 - 2024

BULLETIN

Participating Entities (44 Total):

Harris County, Harris County Flood Control District, Harris County Office of County Administration, Bellaire, Bunker Hill Village, Deer Park, El Lago, Friendswood, Galena Park, Hedwig Village, Hilshire Village, Humble, Hunters Creek Village, Jacinto City, Jersey Village, Katy, La Porte, Missouri City, Morgan's Point, Nassau Bay, Pasadena, Piney Point Village, Seabrook, Shoreacres, Spring Valley Village, Stafford, Taylor Lake Village, Waller, Webster, West University Place, A+ Unlimited Potential Charter Schools, Cypress Fairbanks ISD, Goose Creek CISD, Pasadena ISD, Southwest Public Schools, Spring ISD, Varnett Public Schools, Houston Community College, Lee College, San Jacinto College, University of Houston, Cypress Creek Drainage District, and Harris Health.

Community Rating System (CRS) Jurisdictions:

Bellaire, Deer Park, El Lago, Friendswood, Harris County, Jersey Village, La Porte, Missouri City, Nassau Bay, Pasadena, Seabrook, Shoreacres, Taylor Lake Village, and West University Place.

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Office of Homeland Security and Emergency Management

www.readyharris.org

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

STEERING COMMITTEE MEETING

Date: October 2, 2024
Time: 10:00 AM - 11:00 AM
Location: Virtual Meeting
- Details will be sent upon registration

Purpose of Hazard Mitigation Planning

- Identify impacts from hazards.
- Identify actions and activities to reduce losses from those hazards.
- Establish a coordinated process to implement the plan, taking advantage of a wide variety of resources.
- Requirement for federal disaster grant assistance.

Meeting Attendees may Include:

- Steering Committee Members
- Participating entities *(municipalities, schools, etc.)*
- General Public

REGISTER NOW!

HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

Questions? *Email mitigation@oem.hctx.net*

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

STEERING COMMITTEE MEETING

Date: March 12, 2025
Time: 10:00 AM - 12:00 PM
Location: Houston TranStar
6922 Katy Rd. Houston, TX 77024
Conference Room 120

Purpose of Hazard Mitigation Planning

- Identify impacts from hazards.
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REGISTER NOW!

HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

Questions? *Email mitigation@oem.hctx.net*

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

APRIL 9 - 2025

BULLETIN

April 9, 2025 Steering Committee Meeting Recap:

The Core Planning Team hosted the fifth Steering Committee Meeting for the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update with a total of 38 persons in attendance. During the meeting the Steering Committee:

- Approved Minutes from Steering Committee Meeting #4
- Received an overview of the project timeline
- Received an overview of Hazard Risk Ranking and S.W.O.O. (Strengths, Weaknesses, Obstacles, and Opportunities)
- Reviewed and updated 2025 HC MJ-HMP Mitigation Objectives
- Previewed Worksheets F & G for jurisdictional annex
- Reviewed and Approved 2025 Areawide action

NEXT MEETING: TBD
LOCATION: TBD

Deliverables for Participating Entities:

- **Jurisdictional Annex Worksheets due by April 17, 2025**
  - Worksheet A: Hazard Event History
  - Worksheet B: Capability Assessment
  - Worksheet C: Previous Mitigation Actions
  - Worksheet D: NFIP Compliance
  - Worksheet E: Building Permits

- **Jurisdictional Annex Worksheets due by April 25, 2025**
  - Worksheet F: Hazard Risk Evaluation
  - Worksheet G: New Mitigation Actions

- **Two new surveys (New Development & Problem Areas) due by April 25, 2025**

The most up-to-date information can be found on our website at: **www.readyharris.org**
If you have any questions, please don't hesitate to contact the team at: **mitigation@oem.hctx.net**

APRIL 9 - 2025

BULLETIN

Participating Entities (46 total):

Harris County, Harris County Flood Control District, Harris County Office of County Administration, Bellaire, Bunker Hill Village, Deer Park, El Lago, Friendswood, Galena Park, Hedwig Village, Hilshire Village, Humble, Hunters Creek Village, Jacinto City, Jersey Village, Katy, La Porte, Missouri City, Morgan's Point, Nassau Bay, Pasadena, Piney Point Village, Seabrook, Shoreacres, Southside Place, Spring Valley Village, Stafford, Taylor Lake Village, Tomball, Waller, Webster, West University Place, A+ Unlimited Potential Charter Schools, Cypress Fairbanks ISD, Goose Creek CISD, Pasadena ISD, Southwest Public Schools, Spring ISD, Varnett Public Schools, Houston Community College, Lee College, San Jacinto College, University of Houston, Cypress Creek Drainage District, Harris Health, and Gulf Coast Authority.

Community Rating System (CRS) Jurisdictions:

Bellaire, Deer Park, El Lago, Friendswood, Harris County, Jersey Village, La Porte, Missouri City, Nassau Bay, Pasadena, Seabrook, Shoreacres, Taylor Lake Village, and West University Place.

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Lead Planner
Brittany Ray
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Project Manager
Brian Rutherford
Brian.Rutherford@tetratech.com

Technical Lead Planner
Anita Machiavello
Anita.Machiavello@tetratech.com

Planner
Mark Stalcup
Mark.Stalcup@tetratech.com

Harris County is committed to creating and sustaining communities that are more resilient to disasters. To fulfill this pledge, *Harris County Office of Homeland Security and Emergency Management* (HCOHSEM) has contracted with Tetra Tech to assist in updating the Harris County Multi-Jurisdictional Hazard Mitigation Plan in partnership with cities and special purpose districts throughout the County. Federal regulations require this plan be updated every five years.

Office of Homeland Security and Emergency Management

www.readyharris.org

2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Update

STEERING COMMITTEE MEETING

Date: April 9, 2025
Time: 2:00 PM - 4:00 PM
Location: Houston TranStar
6922 Katy Rd. Houston, TX 77024
Conference Room 120

Purpose of Hazard Mitigation Planning

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Meeting Attendees may Include:

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- Participating entities *(municipalities, schools, etc.)*
- General Public

REGISTER NOW!

HARRIS COUNTY HOMELAND SECURITY & EMERGENCY MANAGEMENT TEXAS

Questions? *Email mitigation@oem.hctx.net*

Local HC MJ-HMP Public Engagement Strategies:

City of Friendswood:

- Social media: general updates and items requiring public feedback will be posted on the City of Friendswood's Facebook page with appropriate links.
- Email newsletters: general updates requiring public feedback will be distributed via mass email to the City of Friendswood's newsletter distribution lists.
- May host public meetings at the Friendswood Public Library.
- Open meetings: general updates may be made to the Mayor and Council at City Council meetings by the City Manager.

City of Galena Park:

- Fact sheets.
- Publish data on the city website.
- Public meetings/workshops.
- Convene focus groups.
- Develop polls.
- Create citizen advisory boards.

City of Jacinto City:

- Social media – Facebook, X, Instagram
- City Signage – 5 programmable 5x9 LED signs (meeting invites, QR codes to surveys/forms)
- Civic and Service Groups:
    - JC & GP CIP Partnership
    - Chamber of Commerce
    - Heritage Hall Forum
    - JC / NC LEPC
- Public hearing – available on Zoom
- Local media – news story and public meeting notice.

City of Jersey Village:

- Update the City website with mitigation planning updates and meeting minutes.
- Host public comment periods.
- Meetings will be posted in the neighborhood newsletter, social media, and City marquees.

City of Katy:

- Survey:
    - Aimed at collecting input on local mitigation priorities and concerns.
    - KTALERT Notification to send the survey link to residents through email and the app.
    - Post the survey on the City's official website and dedicated OEM webpage.
    - Share the survey across the City's social media platforms, including Facebook, Twitter, and Instagram.
- Public Service Announcements
- Roundtables/Forums:
    - Roundtable 1: Kick-off Meeting: provide an overview of the risks addressed in the Plan, proposed actions, and benefits of the plan. Gather public thoughts, suggestions, and concerns to inform the planning process.

    - Roundtable 2: Discuss the draft HMP and collect public input on the proposed mitigation actions and priorities.
- Public Review of the Final Draft Plan
    - Available for public review on the City's website and social media platforms.
    - Provide a designated period for public comments.
- Provide regular progress updates on social media and OEM webpage.

City of La Porte:

- Host a town hall and targeted workshops (e.g., on flooding, and hurricanes).
- Link on City of La Porte website: Hazard Mitigation | La Porte, TX - Official Website.
- Run a social media campaign and update the City website (X, Facebook, Nextdoor).
- Collaborate with local organizations for outreach.
- Issue a press release and create shareable information for local outlets.
- Share information and data provided by HCHOHSEM to all platforms and echo messages.

City of Morgan's Point:

- Give updates to Council on progress which will also be emailed to the community.

City of Nassau Bay:

- Utilizing the City website or related social media to announce any maintenance and/or periodic review activities taking place.
- Providing copies of the plan for reference at City Hall.

City of Pasadena

- Create a QR code for residents to use to take them to the ReadyHarris Mitigation Page.
- Send out information via our mass messaging notification system to residents informing them of steering meetings that are open to the public.
- Disseminate mitigation activity to our local community advisory councils (BAYCAP, PCAC, and SEACAP).

City of Shoreacres:

- Social media, primarily Next Door

City of Webster:

- Stakeholder Identification
    - Target Groups: Residents, local businesses, community organizations, schools, emergency responders, and vulnerable populations.
    - Partnerships: Collaborate with local schools, churches, business associations, and non-profits to reach diverse community members
- Launched a dedicated webpage on City of Webster's official site providing information on the mitigation planning process, including timelines, goals, and opportunities for input.
- Utilize social media platforms to share updates, publicize meetings, and gather feedback through polls and surveys.
- Target Groups: Residents, local businesses, community organizations, schools, emergency responders, and vulnerable populations.
- Partnerships: Collaborate with local schools, churches, business associations, and non-profits to reach diverse community members.

- Online Survey: Develop and distribute a community survey to assess residents' concerns, past experiences with local hazards, and ideas for mitigation.
- Feedback Forms: Provide feedback forms at all public meetings and online to ensure ongoing input.
- Engagement Events: Participate in existing community events, when possible, to increase awareness and gather input.

Lee College:

- Establish a mitigation work group comprising Lee College Safety Committee members and key partners. The mitigation work group will meet monthly to discuss mitigation actions, annual progress, and grant opportunities.
- The public will be invited to learn about the hazard mitigation plan, respond to polling questions, and be able to comment on the draft plan.
- News release to the Baytown Sun upon completing the draft plan to solicit comments.
- Periodic website updates
- A Community Survey will be developed to solicit input on types of mitigation partnerships that Lee College could grow.
- Draft mitigation plan presentation will be posted on the website and social media to solicit input to help finalize the plan.

San Jacinto College

- Plan updates shared on their website.

Southwest Public Schools:

- We utilize social media, news outlets, newsletters, website postings, and two-way communication engagement tools through Apptegy Rooms and Thrillshare to engage our public and school community.
- We provide surveys for feedback from teachers, families, and community members.
- We have parent fairs quarterly hosted by the district and individual campuses such as the Back to School Bash, Houston Food Bank distribution, and Community Closet distributions, and we use these events to solicit feedback from our parents and provide information to them as needed.
- We house parent meetings monthly in English and Spanish on topics such as Safety.
- Safety information is distributed to our community from our Superintendent quarterly.
- We have an established Safety community which includes parents, employees, and community partners to guide our work and solicit feedback.

Cypress Creek Drainage Improvement District:

- 3 town hall meetings throughout the watershed in 2023.
- 2 workshops in the first half of 2024.
- Will hold an additional 3 workshops in the second half of 2024.
- Update their website – links, topics, background information, copies of presentations, and videos of public meetings.

Public Workshops Sign In Sheets

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC WORKSHOPS SIGN-IN SHEET
October 19, 2024, 10:00 AM – 11:30 AM
San Jacinto College – Generation Park

FIRST NAME	LAST NAME	PHONE (OPTIONAL)	EMAIL (OPTIONAL)
Marvelou	Lee-Gatherson	713 557-7841	Marveloulee@outlook.com
Linda	Woods	281 7703088	l_woods98@yahoo.com
Holly Streety		713 542 9342	hhstreet1@yahoo.com

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC WORKSHOPS SIGN-IN SHEET

October 22, 2024, 6:00 PM – 7:30 PM

Katherine Tyra Branch Library

FIRST NAME	LAST NAME	PHONE (OPTIONAL)	EMAIL (OPTIONAL)
Melissa	Knox	915 425 3688	melissakknox@gmail.com
Rev.	Kim		
Mark	Stalcup	713-391-7324	mstalcup4@gmail.com mark.stalcup@tetratech.com
Cassie	Stephens	832-273-5081	Cassie.Stephens@HilshireVillageTexas.com
Michele	Fortenberry	253 606 5553	Jnchele@gmail.com

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC WORKSHOPS SIGN-IN SHEET

October 30, 2024, 6:00 PM – 7:30 PM

George H.W. Bush Community Center

FIRST NAME	LAST NAME	PHONE (OPTIONAL)	EMAIL (OPTIONAL)
Joe	Perusic	281-250-1505	jperusic3@hotmail.com
Doyadira	Arellano	281-323-0968	Tejascomms@gmail.com

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
PUBLIC WORKSHOPS SIGN-IN SHEET
November 9, 2024, 10:00 AM – 11:30 AM

San Jacinto College – Central Campus

FIRST NAME	LAST NAME	PHONE (OPTIONAL)	EMAIL (OPTIONAL)
Emily	Woodell		emily.woodell@hcfcd.org
ROSALIE	BROCKMAN		ROSALIE.BROCKMAN@HARRISCOUNTYTX.GOV
BENZON	JOHN		Benzon.j.dm@aem.hctx.net
Archie	Losey		alosey@meshekengr.com
Heaven	Dominguez		heaven.dominguez@yahoo.com
Sandra	Pursley		sandrapursley@yahoo.com
Sheila	Matteson		smatt554@aol.com
Emily	McGee		Emilyrosegarcia1352005@gmail.com
Nalleli ~~Hidalgo~~	Hidalgo	—	nalleli@tejasbarnos.org
Fatimah	El-Ibrahim		fatimah.El-Ibrahim@mail.house.gov
Samantha	Haritos		Sharitos@gcadd.dst.tx.us
Debbie	Vasick		Debbie.vasick@yahoo.com
Unknown			

Steering Committee Meeting Materials

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Date / Time of Meeting:	Wednesday, July 24, 2024, 10:00 AM – 12:00 PM
Location:	TxDOT, 7600 Washington Avenue Houston, TX 77007
In Attendance:	30 total, of which 24 are Steering Committee Members
Summary Prepared By:	Brittany Ray
Quorum – YES / NO	Yes

Welcome and Introductions

- Brian Murray opened the meeting and facilitated group introductions.
- Distributed handouts included: Agenda, Draft - Steering Committee Charter, List of Participating Entities, Critical Facilities and Infrastructure Definition, and Hazards of Concern.

Background Information

- Brittany Ray provided an overview of the 2020 Harris County Multi-Hazard Mitigation Action Plan.
- Brittany Ray provided a summary of the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Kickoff Meeting held on April 12 and the list of participating entities.
- Brittany Ray provided the status of acquiring a contractor to assist in updating the Plan.

FEMA's Local Mitigation Planning Policy Guide

- Brittany Ray provided an overview of the updated policy guidance to include:
    - Stakeholder engagement needs to include representatives of businesses, academia, private organizations, and nonprofit organizations that work directly with and/or provide support to underserved communities and socially vulnerable populations.
    - Whole community approach, specifically for socially vulnerable and underserved communities.
    - Accounting for climate change in the risk assessment and climate adaptation strategies for mitigation action development.
    - Identified hazards, at a minimum, include any state or federal major disaster declarations.

Steering Committee Role / Ground Rules

- The purpose of the Steering Committee was discussed.
- The Chairperson (Brian Murray) and Vice Chairperson (Jeremy Phillips) were named.
- Quorum was established at 15 members plus at least one of the co-chairs.
- Alternates can be designated in the event a committee member cannot attend the meeting.
- Decision-making process will be a consensus, however, if a consensus cannot be reached, a decision will be confirmed by a majority vote.
- The spokesperson for the Steering Committee will be Brian Murray – all media requests should go through him at oemjic@oem.hctx.net.
- Meeting dates – will be the second Wednesday of every month. However, the August meeting may be adjusted due to the July meeting date being pushed back a few weeks.
- Public involvement – all meetings are open to the public. Each member of the public wishing to address the SC will be given a single allotment of 3 minutes to address the committee, with a total aggregated time not to exceed 15 minutes.

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Texas State Hazard Mitigation Plan

- Please review the Texas State Hazard Mitigation Plan, especially the hazards listed for our region.

Define the Planning Area

- TDEM has provided direction on how to address jurisdictions whose boundaries intersect more than 1 county. The risk assessment must extend into those counties outside of Harris.
- There are 5 cities in the Harris County planning partnership whose boundaries extend outside of Harris County - Friendswood, Katy, Missouri City, Seabrook, and Waller.

Critical Facilities / Infrastructure Definition

- The Steering Committee reviewed the critical facilities and infrastructure categories from the 2020 Harris County Multi-Hazard Mitigation Action Plan.
- The Steering Committee reviewed CISA CIKR Taxonomy to determine if changes to the critical facilities and infrastructure categories were needed.
- The Steering Committee agreed upon critical facilities and infrastructure categories for the 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan.

Hazards of Concern

- Brittany Ray discussed the identified hazards in the 2020 Harris County Multi-Hazard Mitigation Action Plan.
- Brittany Ray discussed the identified hazards in the 2023 Texas State Hazard Mitigation Plan.
- Steering Committee approved:
    - Extreme Heat as its category rather than under Severe Weather.
    - Add: Extreme Cold, Technology Failure, and Utility Failure.

Public Involvement Strategy

- The Steering Committee was briefed on the requirements for public involvement throughout the plan update process.
- The Steering Committee discussed potential ways to have better public engagement.
- Brittany Ray briefed on the HCOHSEM mitigation page at https://www.readyharris.org/ which will showcase information related to Steering Committee meetings, public workshops, and other pertinent information.
- Participating entities will be asked to create a link to the above website on their websites.
- Participating entities will be asked to create their own local public engagement strategy.

Homework for Participating Entities:

- Review the 2020 Harris County Multi-Hazard Mitigation Action Plan.
- Provide a list of critical facilities/infrastructure within their jurisdiction.
- Provide a local public engagement strategy.
- Begin Capability Assessment.

Next Steps for the Steering Committee

- Development of the mission statement and goal setting.
- Further discussion of the public involvement strategy.

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting

July 24th, 2024 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Brian Murray	HCOHSEM	DEMC	713-881-3100	brian.murray@harriscountytx.gov
Jamie Galloway	City of Deer Park	Dir Emergency Svc	2814787255	jgalloway@deerparktx.org
Michelle Dobbins	Houston Arts Alliance	Dir, Disaster Services	713-581-6139	michelle@haatx.com
Brian Craig	City of Seabrook	Asst PW Dir/City Eng	281-291-5719	bcraig@seabrooktx.gov
Mary Anne Piacentini	Coastal Prairie Conservancy	President and CEO	281 851. 8762	maryanne@coastalprairieconservancy.org
Kevin Rodgers	City of Seabrook	Director Emergency Svc	281-291-5787	Krodgers@seabrooktx.gov
Deacon Tittle	City of Bellaire	Fire Chief / EOC	713-662-8201	dtittle@bellairetx.gov
Khenley Harvell	San Jacinto College	Emergency manage coord.	757 333 5117	khenley.harvell@sjcd.edu
Grant Burnett	Cypress-Fairbanks ISD PD	Emergency Mgmt. Coordinator	469-763-4156	grant.burnett@cfisd.net
James Meaux	SETRAC	Regional T&C coordinator	713-876-7099	James.Meaux@setrac.org
~~M~~				
Dawn Blackmar, DVM	Houston SPCA	Director Emergency Mgt	281/772-1639	dblackmar@reagan.com
Amanda Ashcroft	H-GAC	Planner	713-993-4545	Amanda.Ashcroft@h-gac.com
Brittany Ray	HCOHSEM	Resilience Planner	713-881-3100	Brittany.Ray@oem.hctx.net

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting

July 24th, 2024 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Jeremy Phillips	HC Flood Control	Asset Mangemt Div. Dir.	FD 614 8780	jeremy.phillips@hcfcd.hctx.net
LEAH BARTON	Harris Cnty office of Cnty Admin	Deputy County Administrator	713 569 0697	leah.barton@harriscountytx.gov
Robert mmccracken	Houston Community College	Associate Vice Chancellor, Risk Mgmt	713 718 7520	Robert.mccracken@hccs.edu
Adam Smith	West University FD	Fire Captain	713-662-5835	a-smith@westutx.gov
CALVIN COBB	CYPRESS CREEK DRAINAGE DISTRICT	PRESIDENT	832-264-9141	CALVIN@CALVINCOBB.COM
Troy D. Harrison	CITY of Shoreacres	CITY MANAGER	713 410 5466	THarrison@cityofshoreacres.org
Steven Simmons	city of Friendswood	Deputy Fire Marshal	281.996-3335	SSimmons@friendswood.com
Colleen Gilbert	Greens Bayou Coalition	Exec. Director	281-874-2143	colleen.gilbert@greensbayou.org
Marlisa Briggs	North Houston Association	President	281-851-5907	marlisa.briggs@north-houston.org
Tramaka Jewett	Missouri City	EMC	281-403-4370	tramaka.jewett@mctx.gov
Mark Bitz	Jersey Village Fire	Fire Chief EMC	713 466 2143	mbitz@jerseyvillagetx.com

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting

July 24th, 2024 / 10:00 AM – 12:00 PM

NON-STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Kristina Clark	HCOHSEM	FNSS Planner	(713)881-3100	Kristina.Clark@oem.hctx.net
Jackie Golz	HCOHSEM	Support Planner	2817532467	Jackie.golz@oem.hctx.net
Abbey Anderson	Spring Valley Village	Assistant to City Admin	832 585 9484	aanderson@springvalleytx.com
Lindsey Drouet	HCOHSEM	Finance/Admin Supervisor	(713) 426-9540	lindsey.drouet@oem.hctx.net
James White	TX Southern Univ	VP Gov't Relations	281 658 0540	JAMES.WHITE@TSU.EDU
Liz Cawhorn	HCOHSEM	APIO	713-426-9530	Liz.Cawhorn@oem.hctx.net

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Date / Time of Meeting:	Wednesday, Sept. 11, 2024, 10:00 AM – 12:00 PM
Location:	San Jacinto College (8060 Spencer Hwy, Pasadena, TX 77505)
In Attendance:	41 total, of which 22 are Steering Committee Members
Summary Prepared By:	Brittany Ray
Quorum – YES / NO	Yes

Welcome and Introductions

- Brian Murray opened the meeting and facilitated group introductions.
- Distributed handouts included the meeting agenda, July meeting summary, draft public survey questions, draft stakeholder list, draft mission statement, and draft goals of the 2025 HC MJ-HMP.
- Brian Murray called for a motion to approve the Steering Committee Meeting #1 Summary, which was approved unanimously.

Status of Participating Entities Deliverables from the July SC Meeting (Brittany Ray)

- Critical facilities and infrastructure list submissions:
    - Most of the entities have submitted their critical facilities and infrastructure list. A reminder was provided to those who have not completed this to submit as soon as possible, as HCOHSEM is compiling the information to turn over to the contractor.
- Local public engagement strategy submissions:
    - Many participating entities have submitted their strategy, and the extent of the outreach varied according to their capabilities.
    - At a minimum, each participating entity has a link to the ReadyHarris mitigation page on their website.

CDBG-MIT / Timeline Update (Brian Murray)

- Harris County is in the process of selecting a vendor, however, it is anticipated that the vendor may not be on board until November.
- Harris County is coordinating with the Texas Division of Emergency Management regarding the plan's expiration on July 22nd, 2025. If the plan is not completed until December 2025, we want to ensure that mitigation projects and CRS eligibility are not affected.

Public Engagement Strategy (Brittany Ray)

- Public survey:
    - The purpose is to gauge the public's knowledge of the hazards affecting them and use this data to inform the public engagement strategy and action items.
    - Harris County developed a draft public survey that utilized questions from the 2020 HC MH-MAP, some questions were removed or combined with others to shorten the survey.
    - The Steering Committee discussed possible revisions to the survey:
        - Question 1:
            - Change “coastal storms” to “tropical weather.”
            - Be more specific with terms by providing examples.
        - Question 2:
            - Remove the word “natural.”
            - Add an “Other” option with a fill-in box.
        - Question 3:
            - Considered redundant with the next three questions. Possibly remove or combine with questions 4 and 5.

        - Question 7:
            - Add options for local emergency management, word of mouth, community organizations, and radio.
            - Remove newspapers.
        - Question 8:
            - Add "word of mouth."
            - Change "EM" to "local government alert."
            - Clarify "audible" to "sirens" or "outdoor warning."
        - Question 10: Remove sub-question regarding the value of their home.
        - Question 13: Add a "none of the above" option.
        - Question 17: Remove "primary" language and change to a check all that apply.
        - Question 18:
            - Disconnect the email from the survey process.
            - Recommended having a link to the ReadyHarris website, where they can sign up to be notified about participatory opportunities for the Plan update.
    - The draft survey questions will be sent to the Steering Committee and participating entities for additional feedback.
    - The public survey will be launched on October 1st and closed on October 31st.
- Workshops:
    - Education Campaign:
        - Focus: The workshops will concentrate on educating the public about the hazard mitigation plan update, planning process, and successful projects in their communities.
        - Location: plan to hold one workshop in each Commissioner Precinct, preferably in a socially vulnerable area (total of 4).
        - Timeline: plan to complete before the contractor is onboarded, possibly in late October to early November.
        - The Steering Committee and participating entities will be asked to support these workshops in various ways such as having a booth, presenting, or disseminating the workshop information within their networks.
    - Risk Assessment:
        - Focus: Inform the public about their risks to hazards.
        - Location: plan to hold one workshop in each Commissioner Precinct, preferably in a socially vulnerable area (total of 4).
        - Timeline: plan to complete once the contractor is onboard to rely on their subject-matter expertise.
    - Discussed having a digital workshop to increase accessibility and participation.
- Public comment review period.
- Brian Murray made a motion to approve the public engagement strategy, contingent upon recommendations from the contractor, and the motion carried unanimously.

Stakeholder / Partner Engagement Discussion (Brittany Ray)

- The Steering Committee reviewed the 2020 HC MH-MAP stakeholder list to identify additional partners that should be notified about key milestones of the Plan update.
- Recommended adding Lone Star College, Harris County Long Term Recovery Committee, TGCRVOAD, Houston Food Bank, and Texas Forest Service.
- Recommended streamlining outreach to hospitals by coordinating with SETRAC.

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Mission Statement Development (Brian Murray)

- The Steering Committee reviewed and discussed the proposed Mission Statement for the 2025 HC MJ-HMP.
- The draft Mission Statement will be sent to participating entities for review.
- The goal is to approve the next Steering Committee meeting.

Goal Setting (Brittany Ray)

- The Steering Committee reviewed and discussed the proposed Goals for the 2025 HC MJ-HMP and compared them to the 2020 HC MH-MAP Goals.
- Recommended changes:
    - Goal 1: Remove "the intent of the efforts and to improve the."
    - Goal 3: Remove "Improve local government efforts to."
    - Goals 5&6: Possibly merge the goals for better clarity. HCFCD will revise these goals.
    - Goal 7: Add "protect."
    - Add a goal on prioritizing investments.
- The draft Goals will be sent to participating entities and the Steering Committee for further review.
- Aim to approve the HC MJ-HMP Goals next Steering Committee meeting.

Homework (Brittany Ray)

- Support the plan-wide Public Engagement Strategy
    - Workshops
    - Public survey
- Review and provide feedback to the stakeholder list.
- Review and provide feedback on the draft 2025 HC MJ-HMP mission statement and goals.
- Begin capability assessment:
    - Each participating entity should begin reviewing and updating their capability assessment from the last Plan.
    - Suggested beginning with an inventory of their current plans that integrate with hazard mitigation planning and identify opportunities for future integration.

Next Steps (Brittany Ray)

- Review and approve the 2025 HC MJ-HMP Mission Statement.
- Review and approve the 2025 HC MJ-HMP Goals.
- Launch the public survey.
- Continue planning for the public workshops.
- The October Steering Committee meeting may focus on planning for the public workshops or beginning the capability assessment, depending on when the contractor starts.

Adjourned

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting

September 11th, 2024 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Brittany Ray	HCOHSEM	Resilience Planner	713-881-3100	Brittany.Ray@harriscountytx.gov
Jamie Galloway	Deer Park	Emer Serv. Dir	281 478 7298	jgalloway@deerparktx.org
~~Kael Watton~~	~~HCOHSEM~~	~~Planning Lead Team~~	~~713 501 7825~~	~~Kael.Watton@harriscountytx.gov~~
James Meaux	SETRAC	TEL Coordinator	713-876-7099	James.Meaux@setrac.org
Troy D. Harrison	Shoreacres	CITY MANAGER	713.410.5466	THarrison@shoreacrestx.com
Jeremy Phillips	HCFCD	~~Division Director, HMP~~	713 684 4780	jeremy.phillips@hcfcd.hctx.net
CALVIN Corbs	CCDID	PRESIDENT	832-744-6141	CALVIN@CALVINCORBS.COM
Leslie Etheridge	HCDE	Director	917-647-6919.	LEtheridge@HCDE-TEXAS.ORG
Azell Carter	City of Pasadena	EMC	832-350-3282	acarter@pasadenatx.gov
Michelle Dobbins	Houston Arts Alliance	Director, Disaster Srv	713-	michelle@haatx.com
~~D~~arai Osorio	HGAC	Principal Planner	8326812528	saraiosorio@h-gac.com
Colleen Gilbert	Greens Bayou Coalition	Exec Director	281 874 2142	colleen.gilbert@greensbayou.org
SCOTT HUDSON	CFISD POLICE DEPT.	DIRECTOR OF EMERG. MNGT.	540 809 4167	Scott.hudson@cfisd.net
~~Dena Mahan~~	~~City of Webster~~	~~EM Spec~~	~~832 594-1751~~	~~dmahan@cityofwebster.com~~
Grant Burkett	CFISD Police Dept.	Coordinator of Emerg. Mgmt	469-763-4156	grant.Burkett@cfisd.net
~~Jesse Dickerman~~	~~Harris County~~	~~Managing Director~~	~~832-260-8058~~	~~jesse.dickerman@harriscountytx.gov~~

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting

September 11th, 2024 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Khenley Harvell	San Jacinto College	Emergency Mg Coordinator	737 333 5117	Khenley.harvell@sjcd.edu
Brian Murray	HCOHSEM	Deputy EMC	713-581-3100	
Steven Simmons	City of Friendswood	Deputy Fire Marshal	281-996-3335	SSimmons@Friendswood.com
Kim Rodgers	City of Seabrook	Director Emergency Services	281 291 5787	krodgers@seabrooktx.gov
Mark Bitz	City of Jersey Village	Fire chief / EMC	713 466-9993	Mbitz@jerseyvillagetx.com
Bob Kosar	City of El Lago + Taylor Lake V.	Floodplain Admin	281 546 4705	cityfpa@ellago-tx.gov
Amanda Ashcraft	H-GAC	?		
Leah Barton	HCPH	Exec Dir HCPH	713 569 0697	
Johnny Morales	LPOEM	EMC		
Robert McCrackle	HCC	AVC, Risk Mgmt	713 718 7520	Robert.mccrackle@hccs.edu
Madison Briggs	NHA	President	281 851 5909	on file
~~Ronda Marquez~~	~~Goose Creek CISD~~	~~Director Safety/Security~~	~~346-232-5155~~	~~on file~~

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting

September 11th, 2024 / 10:00 AM – 12:00 PM

NON-STEERING COMMITTEE MEMBERS SIGN-IN

NAME	ORGANIZATION	TITLE	PHONE	EMAIL
Jackie Golz	HCOHSEM	Support Planner	2817532467	Jackie.golz@harriscountytx.gov
Benzow John	HCOHSEM	Asst. Deputy & PSC	832-286-5620	benzeon.john@harriscountytx.gov
Licett Hernandez	HCOHSEM	Communications Supervisor	713 881 3100	Licett.hernandez@harriscountytx.gov.
Mel Bartis	HCOHSEM	Deputy EMC	8323435988	mel.bartis@harriscountytx.gov
Greg Goedecker	City of Katy	EMC	832-418-1574	ggoedecker@cityofkaty.com
Kiao Whitton	HCOHSEM	Planning division	713 501 7825	Kiao.whitton@harriscountytx.gov
Brian Craig	Seabrook	Asst PW Dir/City Eng	281 291 5719	bcraig@seabrooktx.gov
Kristina Clark	HCOHSEM			
Charles Mistric	Waller EMC	EMC	979-645-7557	CMistric@cityofwaller.com
Archie Losey	Meshek + Assoc/Public	Planner	(713) 824-5219	alosey@meshekengr.com
Thomas Quinn	Lee College	ER + Safety Ops manager	439-234-7494	tquinn@lee.edu
~~Sarah Oshi~~				
Adam Nelson	HCED	Disaster Recovery	501-418-4496	adam.nelson@harriscountytx.gov
David Carbie	OCA	CJ Manager		David.Carbie@harriscountytx.gov
Dena Mahan	City of Webster	EM Spec.	832 594 1751	dmahan@cityofwebster.com
Jesse Dickerman	HC OCA	Managing director	832-768-8050	jessedickerman@harris countytx.gov
Robert Marquez	goose creek CISD	Director, Safe & secure schools	346 2335158	Robert.Marquez@gccisd.net

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Date / Time of Meeting:	Wednesday, October 2, 2024, 10:00 AM – 11:00 AM
Location:	Virtual – MS Teams
In Attendance:	44 total, of which 24 are Steering Committee Members
Summary Prepared By:	Brittany Ray
Quorum – YES / NO	Yes

Old Business:

- The Steering Committee Meeting #2 Summary was approved unanimously.
- The 2025 HC MJ-HMP Mission Statement was approved unanimously.
- Goals of the 2025 HC MJ-HMP were approved unanimously.

CDBG-MIT / Timeline Update (Brian Murray)

- The County is on track to have the contractor start by late November.
- Contractor can complete the Plan update by the current plan's expiration date (July 22nd, 2025).

Public Survey (Brittany Ray)

- Went live on Tuesday, October 1st at 4:00 PM and have received 15 submissions thus far.
- The survey is available in English, Spanish, Chinese, Vietnamese, and Arabic. A vendor reviewed the additional languages to ensure proper translation.
- The survey can be found on the ReadyHarris Mitigation Page.
- A flyer has been developed to promote the public survey at community events, community centers, and other locations.
- A half-page document designed to promote the survey on social media posts has been created and is now available for the Steering Committee and participating entities.
- The Steering Committee reviewed the minor changes made to the survey since the September Steering Committee meeting, which included:
    - Adding an introductory statement explaining the purpose of the survey.
    - Added a question regarding climate change to align with the priorities in FEMA's Local Mitigation Planning Policy Guide.
    - Wind insurance was added as an answer to the insurance-related question.
    - Revised "Have you ever had problems securing homeowners or renters insurance due to risks from hazards in Harris County" to "Have you ever had problems securing or maintaining homeowners or renters insurance due to risks from hazards in Harris County." Answer choices were modified to:
        - Yes, I had trouble securing initial coverage.
        - Yes, my policy was dropped by my insurance provider.
        - Yes, my policy premiums significantly increased.
        - No, I have not experienced any issues.
    - Included a Nixel for the community to sign up for Plan updates once the survey is submitted. (Text HMP2025 to 888777).
- Due to one of the public workshops occurring in November, the Steering Committee discussed and approved extending the public survey deadline from October 31st to November 10th.

Public Workshops (Brittany Ray)

- Workshop Dates/Locations
    - October 19, 2024, 10:00 AM – 11:30 AM at San Jacinto College – Generation Park.
    - October 22, 2024, 6:00 PM – 7:30 PM at Katherine Tyra Branch Library.

- October 30, 2024, 6:00 PM – 7:30 PM at George H.W. Bush Community Center.
- November 9, 2024, 10:00 AM – 11:30 AM at San Jacinto College – Central Campus.
  - This workshop will have a virtual option to increase public participation.

- Workshop Agenda:
  - Attendees arrive, sign in, mingle/visit partner booths.
  - Partner Presentations
    - Harris County Office of Homeland Security & Emergency Management
      - Provide an overview of hazard mitigation planning and what it means to the community.
      - Status of the 2020 Harris County Multi-Hazard Mitigation Action Plan.
      - Ways for the community to get involved in the 2025 HC MJ-HMP update.
    - Harris County Flood Control District
      - Overview of the Flood Control District.
      - Highlight 1-2 mitigation projects that have been completed, are in progress, or want to be implemented within that community.
    - Harris County Engineering Department
      - Will provide an overview of mitigation projects they are working on in that area.
    - Harris County Office of County Administration
      - Will provide an overview of the Climate Justice Plan.
  - HCOHSEM – Q&A / mingle at partner booths.
  - End of Workshop.
- QR codes for the public survey will be available at the workshops.
- If any Steering Committee organizations or participating entities wish to have a booth at the workshops, please coordinate with HCOHSEM.

Homework:

- Actively Promote and Socialize the Public Survey.
- Actively Promote and Socialize the Upcoming Public Workshops.

Open Discussion:

- Adding Health-Related Hazard
  - A proposal was introduced to include a health-related hazard within the Plan.
  - Other large counties have incorporated a health-related hazard at various levels within their plan. However, it is not included as a hazard in the Texas State Hazard Mitigation Plan.
  - Adding it as a hazard, will allow for future funding opportunities.
  - This can be completed by the contractor with the assistance of Harris County Public Health.
  - The Steering Committee approved adding it to the other hazards of concern section but will not complete a full hazard profile.

Adjourned

HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Date / Time of Meeting:	Wednesday, June 4, 2025, 10:00 AM – 12:00 PM
Location:	Pasadena City Hall, 1149 Elsworth Drive, Pasadena, Texas 77506
In Attendance:	32 total, of which 18 are Steering Committee Members
Summary Prepared By:	Anita Machiavello
Quorum – YES / NO	Yes

Welcome *Brian Murray*

-

Review and Approval of SC Meeting #5 Minutes *Brian Murray*

-

Prioritizing Action Items *Tetra Tech*

-

Presentation of the 2025 Harris County MJ-HMP (Draft) *Tetra Tech*

-
-
-
-
-

Public Review & Comment *Tetra Tech*

-

Adoption Process & Timeline *Tetra Tech*

-

Presentation of 2025 HC MJ-HMP Interactive Web Viewer *Tetra Tech*

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HARRIS COUNTY MULTI-JURISDICTIONAL HAZARD MITIGATION PLAN
STEERING COMMITTEE MEETING SUMMARY

Date / Time of Meeting:	Wednesday, April 9, 2025, 2:00 PM – 4:00 PM
Location:	Houston TranStar, 6922 Katy Rd, Houston, TX 77024
In Attendance:	38 total, of which 18 are Steering Committee Members
Summary Prepared By:	Anita Machiavello
Quorum – YES / NO	Yes

Old Business:

- The Steering Committee Meeting #4 Summary was approved unanimously.

Welcome (Brian Murray)

- Brian welcomed everyone and explained an overview of the material that will be discussed today.

2025 Harris County MJ-HMP Presentation (Tetra Tech)

- Project Timeline – Mark and Anita (TT) provided an update on progress made to date, noting that two subcontractors, Resalire and Bent Ear Solutions, have started working. Resalire is currently assisting with developing Volume II of the HC MJ-HMP, while Bent Ear Solutions will focus on developing a web-based interactive tool for the public to explore data collected for the HC MJ-HMP.
    - TT mentioned that the Draft HC MJ-HMP is scheduled to be ready in mid-May for review, and both Volume I and Volume II development are underway. Additionally, the TT GIS team is on track to complete the GIS assessments by April 25th.
    - Chair Brian Murray noted to the committee that Worksheets A-E are due on April 17th.
    - TT mentioned that 21 complete submissions have been received, 10 partial, and 15 remaining for Worksheets A–E.
- Planning Partners will receive a New Development and Problem Areas Questionnaire. These responses will be incorporated into the plan and added to the maps that will be produced.
- HAZUS Risk Ranking – TT explained the process of identifying and ranking hazard risk based on three key factors. An overview was provided on the types of data used to calculate risk and TT plans to provide a visual of the hazard risk ranking results to the next SC meeting.
- 2025 HC MJ-HMP Objectives
    - The Steering Committee reviewed the 2025 mitigation goals to confirm that the new hazards of concern were identified and addressed. Then they read through each goal objective to determine if any revisions were necessary.
    - TT plans to capture the revisions the SC decided on and provide them for the next SC meeting in May, where they can be voted on for approval.
- Mitigation Strategy – TT provided an overview of how a community could reduce risk for hazards of concern and a high-level walk-through of the Mitigation Action Catalog that will be distributed after today's meeting. TT explained how the catalog can help Planning Partners brainstorm action items.
- SWOO – TT provided a walkthrough of SWOO and how jurisdictions can focus their mitigation strategies.
- Additional Worksheets Overview and Activity I
    - TT provided an overview of Worksheets F and G that the Planning Partners will need to complete. TT also shared that a new BATool will be released soon, so it is not a priority for new participants to have a login at this time.

  - The Steering Committee reviewed and approved the Area-Wide Actions for the 2025 HC MJ-HMP.

- Homework:
  - Participating entities will need to complete the Jurisdictional Annex Worksheets.
- Next Meeting: TBD

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
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STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALTERNATE NAME (OPTIONAL)	SIGNATURE
Harris County Office of Homeland Security & Emergency Management	Brian Murray	Deputy EMC / PIO		
Harris County Office of Homeland Security & Emergency Management	Brittany Ray	Resilience Planner		
Harris County Office of County Administration / Harris County Public Health	Leah Barton	Deputy County Administrator / Executive Director Harris County Public Health		
Harris County Flood Control District	Jeremy Phillips	Director, Asset Management Division		
City of Bellaire	Deacon Tittle	Fire Chief / Emergency Management Coordinator		
City of Deer Park	Jamie Galloway	Emergency Services Director		
City of El Lago / Taylor Lake Village	Carl Nunn	Chief of Police / EMC		
City of Friendswood	Steven Simmons	Deputy Fire Marshal		
City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal		
City of La Porte	Johnny Morales	Emergency Management Coordinator	Amber Cook Bethea Deputy EMC	
City of Missouri City	Trameka Jewett	Emergency Management Coordinator	EM Spec.	
City of Pasadena	Azell Carter	Emergency Management Coordinator		

ORGANIZATION	NAME	TITLE	ALTERNATE NAME (OPTIONAL)	SIGNATURE
City of Seabrook	Kevin Rodgers	Office of Emergency Management Director	Kevin Rodgers | Brian Bray	
City of Shoreacres	Troy Harrison	City Manager		
National Weather Service	Lance Wood	Science and Operations Officer		
North Houston Association	Marlisa Briggs	President		
Houston Community College	Robert McCracken	Associate Vice Chancellor, Risk Mgmt & Resilience		
San Jacinto College	Ali Shah	Director, Emergency Management	Khenley Harvell	
H-GAC	Amanda Ashcroft	Planner		
Cypress Creek Drainage Improvement District	Calvin Cobb	President		
Coastal Prairie Conservancy	Mary Anne Piacentini	President and CEO		
Greens Bayou Coalition	Colleen Gilbert	Executive Director		
Cy-Fair ISD	Scott Hudson	Director of Emergency Management		
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Director		
SETRAC	James Meaux	Regional T&E Coordinator		
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience		
Houston SPCA	Dawn Blackmar	Director, Emergency Management		
Harris County Dept. of Education	Leslie Etheridge	Director, Center for Safe and Secure Schools		

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Steering Committee Meeting
April 9, 2025 / 2:00 PM – 4:00 PM

NON-STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	PHONE	SIGNATURE
City of Waller	Charles Mistrik	EMC	979-645-7557	
City of Katy	Greg Goodeckert	EMC	832-418-1574	
City of Tomball	Joe Sykora	Fire Chief / EMC	281 290 1063	
Cy-Fair ISD	Grant Brandt	EMC	469-768-4156	
Village FD	Howard Miller	Fire Chief / EMC	713-468-7541	
west university PD	James Novak	Fire Chief / EMC	713-662-5836	
GREATER SE MGMT DISTRICT	Dr. ABDUL HALEEM Muhammad	Vice Chair	8322369956	
west University FD	Aaron Rosenfield	Asst. Chief	713-662-5837	
SouthSide Place	Andrew Deavers	Assistant City Mgr	7136682341	
Benzon John	Benzon John	Deputy EMC	832-286-5620	
~~Kristina Clark~~ HCOHSEM	Kristina Clark	Planning Supervisor	713.881.3100	

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting
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PUBLIC SIGN-IN

NAME	CITY/JURISDICTION	ORGANIZATION	PHONE	EMAIL
ATM SEN	RESALIRE		225.281.4879	
Jennifer Leighton	Resalire		832 646 7390	

Harris County Multi-Jurisdictional Hazard Mitigation Plan Update
Steering Committee Meeting #4
March 12, 2025 / 10:00 AM – 12:00 PM

STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	ALT. NAME	SIGNATURE
Harris County OHSEM	Brian Murray	Deputy Emergency Management Coordinator / PIO		
Harris County OHSEM	Brittany Ray	Resilience Planner		
Harris Co. OCA / Harris Co. Pub. Health	Leah Barton	Deputy Co. Admin. /Executive. Director		
HCFCD	Jeremy Phillips	Director, Asset Management Division		
City of Bellaire	Deacon Tittle	Fire Chief / Emergency Management Coordinator		
City of Deer Park	Jamie Galloway	Emergency Services Director		
City of El Lago / Taylor Lake Village	Carl Nunn	Chief of Police / EMC		
City of Friendswood	Steven Simmons	Deputy Fire Marshal		
City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal		
City of La Porte	Johnny Morales	Emergency Management Coordinator		
City of Missouri City	Trameka Jewett	Emergency Management Coordinator		
City of Pasadena	Azell Carter	Emergency Management Coordinator		
City of Seabrook	Kevin Rodgers	Office of Emergency Management Director	Brian Cralj	
City of Shoreacres	Troy Harrison	City Manager		
National Weather Service	Lance Wood	Science and Operations Officer		

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North Houston Association	Marlisa Briggs	President		
Houston Community College	Robert McCracken	Assoc. Vice Chancellor, Risk Management & Resilience		
San Jacinto College	Ali Shah	Director, Emergency Management		
H-GAC	Amanda Ashcroft	Sr. Planner		Amanda A
Cypress Creek Drainage Improvement District	Calvin Cobb	President		
Coastal Prairie Conservancy	Mary Anne Piacentini	President and CEO		
Greens Bayou Coalition	Colleen Gilbert	Executive Director		
Cy-Fair ISD	Scott Hudson	Director of Emergency Management		
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Director		
SETRAC	James Meaux	Regional T&E Coordinator		
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience		
Houston SPCA	Dawn Blackmar	Director, Emergency Management		
Harris County Dept. of Education	Leslie Etheridge	Director, Center for Safe and Secure Schools		

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City of Jersey Village	Mark Bitz	Fire Chief / Fire Marshal		
City of La Porte	Johnny Morales	Emergency Management Coordinator	Amber Beltran	
City of Missouri City	Trameka Jewett	Emergency Management Coordinator		
City of Pasadena	Azell Carter	Emergency Management Coordinator		
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Cy-Fair ISD	Scott Hudson	Director of Emergency Management		
Partnership Lake Houston	Stephanie Wiggins	Chief Economic Development Director		
SETRAC	James Meaux	Regional T&E Coordinator	Katherine Samuel	
Houston Arts Alliance	Michelle Dobbins	Director, Disaster Recovery & Resilience		
Houston SPCA	Dawn Blackmar	Director, Emergency Management		
Harris County Dept. of Education	Leslie Etheridge	Director, Center for Safe and Secure Schools		
City of La Porte	[illegible]	Deputy EMC		

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PUBLIC SIGN-IN

NAME	CITY/JURISDICTION	ORGANIZATION	PHONE	EMAIL
ATRI SEN		RESILIRE	225. 281.4879	ATRI.SEN@ RESILIRE INFRASTRUCTURE.C
[illegible]		Urbanxt	504-473-2047	n-sen@outlook.com
Archie Losey	Unincorporated Harris County	Public	713-824-5219	archie.losey@wsbeng.com
Daniella Flanaga	Houston Health Dep-	Public	832.923.8257	daniella.flanagan@houstonh.g

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NON-STEERING COMMITTEE MEMBERS SIGN-IN

ORGANIZATION	NAME	TITLE	PHONE	SIGNATURE
City of Waller	Charles Mistric	EMC	979-695-7557	
HC Engineering Dept.	Samuel Pena	DR	281.813.2746	
Cypress-Fairbanks ISD Police Dept.	Grant Burnett	EMC	469.763.4186	
★ Village FD/Mem Villages	Howard Miller	Fire Chief/EMC	713-468-7541	
★ Village FD	James Lutz	Asstant EMC	713-468-7441	
City of Spring Valley Village	Kristina Watson	Director Publicworks	713-454-4951	
City of Tomball	Joe Sykora	FIRE CHIEF/EMC	2817261111	
City of Southside Place	Andrew Deavers	Assistant City Manage	713668 2341	
City of Missouri City	Edward Nagorski	Assistant City Engineer	281-403-8546	Edward Nagorski
TDEM	Merryl Holmes	Mitigation Coor	281-794-4102	

Posting at the Friendswood City Hall -- June 27, 2025

Posting at the Friendswood City Facebook -- June 27, 2025

ReadyHarris

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The 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan update continues to the next phase of the planning process. A series of Community Workshops will be hosted throughout the county to review the updated plan.
To register for a workshop visit: https://linktr.ee/HarrisCountyHMP

1 1

Like Comment

City of Friendswood - Government

14h ·

Make your voice heard!
Harris County is vulnerable to many natural hazards including flooding, hurricanes, tropical storms, wildfires, and tornadoes. These hazards threaten the safety of residents and have the potential to damage property, disrupt the local economy, and impact the quality of life of Harris County residents. While we cannot eliminate natural hazards, hazard mitigation planning identifies ways to reduce the risk of these hazards.The 2025 Harris County Multi-J... **See more**

Posting at the Friendswood City Facebook -- June 27, 2025

Posting at the Greens Bayou Collection Facebook Page -- June 26, 2025

High Hazard Potential Dam (HHPD) Rehabilitation Program

Summary of Benefits and Eligibility

Program Purpose

The High Hazard Potential Dam (HHPD) Rehabilitation Program provides federal assistance to repair, rehabilitate, or remove high hazard potential dams whose failure could cause loss of life. Participation requires the development of a Floodplain Management Plan (FPMP), which focuses on the unique impacts of the project dam and ensures long-term community safety.

Benefits of the HHPD Program

Safety and Risk Reduction: Supports structural improvements and operational measures that reduce the risk of catastrophic dam failure, protecting downstream communities from flooding and loss of life.

Flood Resilience: Encourages actions that mitigate impacts of a wide range of flood events (including the base 1% annual chance flood), both upstream & downstream of the project area

Community Preparedness: Strengthens public safety by enhancing emergency planning, warning, and evacuation procedures tied to dam safety

Environmental and Economic Protection: Promotes sensible floodplain management practices that limit property damage, reduce public expenditures, and preserve natural floodplain values

Technical and Financial Support: FEMA provides technical assistance, and grant funds may be applied to planning, design, construction, and post-project assessments.

Eligibility Through the HHPD Process

Floodplain Management Plan Requirement: To qualify for HHPD funding, dam owners (or their sponsoring agencies) must prepare a dam-specific FPMP that identifies hazard boundaries, potential impacts, and long-term management measures.

Timeline: The plan must be developed within two years of the project agreement and implemented within two years of project completion

Regulatory Compliance: The FPMP ensures the project meets federal floodplain regulations (EO 11988) as well as applicable state and local requirements for floodplain development and dam safety

Integration and Adoption: Plans must be reviewed, adopted, and integrated into local/state operations, with roles and responsibilities clearly defined among dam owners, local governments, and stakeholders

Why Participate?

By engaging in the HHPD planning process, dam owners gain access to federal rehabilitation grants, meet compliance obligations under national floodplain policy, and establish a safer, more resilient future for their communities.

Source: FEMA – *High Hazard Potential Dam Floodplain Management Job Aid (2022)*

APPENDIX D: PARTICIPATION MATRIX

Meeting Participation

First Name	Last Name	Title/ Position	Jurisdiction/ Agency	Kickoff	SC Workshop #1 07-24-2024	SC Workshop #2 09-11-2024	SC Workshop #3 10-02-2024	SC Workshop #4 03-12-2025	SC Workshop #5 04-09-2025	SC Workshop #6 06-04-2025	Community Workshop #1 10-19-2024	Community Workshop #2 10-22-2024	Community Workshop #3 10-30-2024	Community Workshop #4 11-09-2024	Public Meeting #1 06-26-2025	Public Meeting #2 06-28-2025	Public Meeting #3 07-01-2025	Public Meeting #4 07-02-2025
Aaron	Rosenfield	Assistant Chief, Fire Department	City of West University Place	Y	--	--	--	--	Y	--	--	--	--	--	--	--	--	--
Aaron	Velasquez	Emergency Management Specialist	Harris Health System	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Abbey	Anderson	Assistant to the City Administrator	City of Spring Valley Village	--	Y	--	--	--	--	--	--	--	--	--	--	--	--	--
Abdul	Haleem Muhammad, M.D.	Vice Chair	Greater SE Management District	--	--	--	--	--	Y	--	--	--	--	--	--	--	--	--
Adam	Smith	Fire Captain	City of West University Place Police Department	Y	Y	--	--	--	--	--	--	--	--	--	--	--	--	--
Adam	Nelson	Disaster Recovery	Harris County Department of Education	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Ali	Shah	Director, Emergency Management	San Jacinto College	--	--	--	--	Y	--	--	--	--	--	--	--	--	--	--
Alison	Hare	Director, PH Preparedness and Response Division	Harris County Public Health	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Allyson	Lambert	Preparedness Coordinator	Texas Division of Emergency Management (TDEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Amanda	Ashcroft	Senior Planner	Houston-Galveston Area Council (H-GAC)	--	Y	Y	--	Y	Y	--	--	--	--	--	--	--	--	--
Amber	Castro-Beltran	Deputy Emergency Management Coordinator	City of La Porte	--	--	--	--	Y	Y	Y	--	--	--	--	--	--	--	--
Andrew	Deavers	Assistant City Manager	City of Southside Place	--	--	--	--	Y	Y	--	--	--	--	--	--	--	--	--
Andrew	Aleman	Deputy Emergency Management Coordinator	City of Pasadena Office of Emergency Management	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Andrew	Jaspers	Emergency Management Coordinator	Klein Independent School District	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Anna	Brown	Facilities Manager	Gulf Coast Authority	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Anna	Brewster	Commissioner Precinct 1	Harris County		--			--		--	--	--	--	--	--	--	--	--
Anthony	Pineda	Director Safety, Operations & Emergency Management	Varnett Public Schools	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Archie	Losey	Planner at Meshek & Associates, LLC	Public	--	--	Y	--	Y	--	Y	--	--	--	Y	--	--	--	--
Ashley	Lopez	Permit Clerk	City of Jersey Village	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Meeting Participation

First Name	Last Name	Title/ Position	Jurisdiction/ Agency	Kickoff	SC Workshop #1 07-24-2024	SC Workshop #2 09-11-2024	SC Workshop #3 10-02-2024	SC Workshop #4 03-12-2025	SC Workshop #5 04-09-2025	SC Workshop #6 06-04-2025	Community Workshop #1 10-19-2024	Community Workshop #2 10-22-2024	Community Workshop #3 10-30-2024	Community Workshop #4 11-09-2024	Public Meeting #1 06-26-2025	Public Meeting #2 06-28-2025	Public Meeting #3 07-01-2025	Public Meeting #4 07-02-2025
Atri	Sen	Planner	Resalire Infrastructure		--			Y	Y	Y	--	--	--	--	--	--	--	--
Aubrey	Harbin	Director of Community Development	City of Friendswood	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Austin	Bleess	City Manager	City of Jersey Village	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Azell	Carter	Emergency Management Coordinator	City of Pasadena	Y	--	Y		--	Y	Y	--	--	--	--	--	--	--	--
Benzon	John	Deputy Emergency Management Coordinator	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	Y		--	Y	--	--	--	--	Y	--	--	--	--
Betsy	DeAnda		City of Deer Park	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Bob	Kosar	Floodplain Administrator	City of El Lago & Taylor Lakeview	--	--	Y	--	--	--	--	--	--	--	--	--	--	--	--
Bobby	Pennington	City Administrator	City of Piney Point Village	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Brian	Mansfield		City of Friendswood	--	--	--	--	--	--	Y	--	--	--	--	--	--	--	--
Brian	Rouane	City Floodplain Administrator	City of Friendswood	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Brian	Schneider	City Administrator	City of Morgan's Point	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Brian	Craig	Assistant Public Works Director / City Engineer	City of Seabrook	Y	Y	Y		Y	Y	Y	--	--	--	--	--	--	--	--
Brian	Murray	Deputy Emergency Management Coordinator / Public Information Officer	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	Y	Y		Y	Y	Y	--	--	--	--	--	--	--	--
Brian	Mason	Executive Director / Emergency Management Coordinator	TranStar / Houston Public Works / City of Houston Office of Emergency Management		--			--		--	--	--	--	--	--	--	--	--
Brittany	Ray	Assistant Planning Section Chief	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	Y	Y		Y	Y	Y	--	--	--	--	--	--	--	--
Calvin	Cobb	President	Cypress Creek Drainage Improvement District		Y	Y		Y	Y	Y	--	--	--	--	--	--	--	--
cannot read			Urban "xt" (Transportation??)		--			Y		--	--	--	--	--	--	--	--	--

Meeting Participation

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Carl	Nunn	Chief of Police / Emergency Management Coordinator	City of Taylor Lake Village & City of El Lago		--			Y		--	--	--	--	--	--	--	--	--
Casey	Jimenez	Emergency Management Speicialist	City of Missouri City	Y	--	--	--	--	Y	Y	--	--	--	--	--	--	--	--
Cassie	Stephens	City Administrator, Secretary, Treasurer	City of Hilshire Village	--	--	--	--	--	--	Y	--	Y	--	--	--	--	--	--
Charles	Mistric	Emergency Management Coordinator	City of Waller	Y	--	Y	--	Y	Y	Y	--	--	--	--	--	--	--	--
Christopher	Turrentine	Fire Marshal Office	Harris County		--			--		--	--	--	--	--	--	--	--	--
Colleen	Gilbert	Executive Director / Emergency Management Coordinator	Greens Bayou Coalition		Y	Y		Y	Y	Y	--	--	--	--	--	--	--	--
Cynthia	Ward	Assistant City Administrator / City Secretary	City of Waller	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Daniel	Sanchez	GIS Specialist	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Daniella	Flanagan	Community Engagement Coordinator	City of Houston Health Department	--	--	--	--	Y	--	--	--	--	--	--	--	--	--	--
Danielle	Calhoun	Operations Manager	Harris County Public Health	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Daniiel	Rios-Lazo	Combination Inspector	City of Deer Park	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
David	Moss	City Manager	City of Southside Place	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
David	Car…	CJ Manager	Harris County Office of County Administration	--	--	Y	--	--	--	--	--	--	--	--	--	--	--	--
David	Potts	Manager of Emergency Management	Houston Community College	Y	--	--	--	--	--	Y	--	--	--	--	--	--	--	--
Dawn	Blackmar	Director, Emergency Management	Houston SPCA		Y			--		--	--	--	--	--	--	--	--	--
Deacon	Tittel	Fire Chief / Emergency Management Coordinator	City of Bellaire	--	Y			--		Y	--	--	--	--	--	--	--	--
Debbie	Vascik	Local Resident	Public	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Dena	Mahan	Emergency Management Specialist	City of Webster	Y	--	Y	--	--	--	Y	--	--	--	--	--	--	--	--
Derrick	Harris	Director	Lone Star College	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

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Deyadira	Arellano	Local Resident	Public	--	--	--	--	--	--	--	--	--	Y	--	--	--	--	--
Don	McCall	Chief of Police	City of Southside Place	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Drew	Castillo	Operations Supervisor	Gulf Coast Authority	--	--	--	--	--	--	Y	--	--	--	--	--	--	--	--
Edward	Nagurski	Assistant City Engineer	City of Missouri City	--	--	--	--	Y	--	--	--	--	--	--	--	--	--	--
Elvin	Hernandez	Director of Public Works & Community Development	Harris County Commissioner Precinct 3	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Emily	Woodell	Chief External Affairs Officer	Harris County Flood Control District (HCFCD)	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Emily	McGee	Local Resident	Public	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Eric	Volkmann	Fire Marshall	City of Jersey Village Fire Department	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Evan	Duvall	Building Official	City of Hedwig Village	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Fatimah	El-Ibrahim	Senior Constituent Advocate with the U.S. House of Representatives	Public	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Gerardo	Barrera	City Administrator	City of Bunker Hill Village	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Ginger	Walker	Director, Emergency Management	University of Houston	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Grant	Burnett	Emergency Management Coordinator	Cypress-Fairbanks Independent School District Police Department	--	Y	Y	--	Y	Y	Y	--	--	--	--	--	--	--	--
Greg	Goedecker	Emergency Management Coordinator	City of Katy	Y	--	Y	--	--	Y	--	--	--	--	--	--	--	--	--
Heaven	Dominguez	Local Resident	Public	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Holly	Streety	Local Resident	Public	--	--	--	--	--	--	--	Y	--	--	--	--	--	--	--
Howard	Miller	Fire Chief/Emergency Management Coordinator	Village Fire Department	--	--	--	--	--	Y	--	--	--	--	--	--	--	--	--
Ingrid	Calderon	Technical Specialist	Region 4 Education Service Center	Y	--			--		--	--	--	--	--	--	--	--	--
Jackie	Golz	Support Planner	Harris County Office of Homeland Security and	--	Y	Y	--	--	--	--	--	--	--	--	--	--	--	--

Meeting Participation

First Name	Last Name	Title/ Position	Jurisdiction/ Agency	Kickoff	SC Workshop #1 07-24-2024	SC Workshop #2 09-11-2024	SC Workshop #3 10-02-2024	SC Workshop #4 03-12-2025	SC Workshop #5 04-09-2025	SC Workshop #6 06-04-2025	Community Workshop #1 10-19-2024	Community Workshop #2 10-22-2024	Community Workshop #3 10-30-2024	Community Workshop #4 11-09-2024	Public Meeting #1 06-26-2025	Public Meeting #2 06-28-2025	Public Meeting #3 07-01-2025	Public Meeting #4 07-02-2025
			Emergency Management (HCOHSEM)															
Jacob	Daily	Assistant Fire Chief / Assistant Fire Marshall / Assistant Emergency Management Coordinator	City of Jersey Village Fire Department	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
James	Tidwell	Police Sergeant	City of Shoreacres	--	--	--	--	--	--	Y	--	--	--	--	--	--	--	--
James	Novak	Fire Chief/Emergency Management Coordinator	City of West University Place	Y	--	--	--	--	Y	--	--	--	--	--	--	--	--	--
James	Lutz	Emergency Management Coordinator	City of Jersey Village Fire Department	Y	--	--	--	--	--	Y	--	--	--	--	--	--	--	--
James	Nykaza	Emergency Management Coordinator	City of Humble	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
James	Anderson	Urban Area Security Initiative (UASI) Planner	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
James	Meaux	Regional T&E Coordinator	SouthEast Texas Regional Advisory Council (SETRAC)		Y	Y		--	Y	Y	--	--	--	--	--	--	--	--
James	White	VP of Government Relations	Texas Southern University	--	Y	--	--	--	--	--	--	--	--	--	--	--	--	--
Jamie	Galloway	Emergency Services Director	City of Deer Park	Y	Y	Y		Y	Y	Y	--	--	--	--	--	--	--	--
Jamie	Woodall	Executive Director of School Safety	Waller ISD	Y	--			--		--	--	--	--	--	--	--	--	--
Jason	Cantu	Emergency Services Coordinator	Harris County Flood Control District (HCFCD)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Jeff	Evans	Meteorologist in Charge	National Weather Service		--			--		--	--	--	--	--	--	--	--	--
Jennifer	Gomez	AICP, Director of Development Services Department	City of Missouri City	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Jennifer	Suter	Training and Exercise Supervisor	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Jennifer	Leighton	Planner	Resalire Infrastructure	--	--	--	--	--	Y	Y	--	--	--	--	--	--	--	--
Jeremy	Barja	Emergency Management Planner	City of La Porte	--	--	--	--	--	--	Y	--	--	--	--	--	--	--	--

Meeting Participation

First Name	Last Name	Title/ Position	Jurisdiction/ Agency	Kickoff	SC Workshop #1 07-24-2024	SC Workshop #2 09-11-2024	SC Workshop #3 10-02-2024	SC Workshop #4 03-12-2025	SC Workshop #5 04-09-2025	SC Workshop #6 06-04-2025	Community Workshop #1 10-19-2024	Community Workshop #2 10-22-2024	Community Workshop #3 10-30-2024	Community Workshop #4 11-09-2024	Public Meeting #1 06-26-2025	Public Meeting #2 06-28-2025	Public Meeting #3 07-01-2025	Public Meeting #4 07-02-2025
Jeremy	Phillips	Director, Asset Management Division	Harris County Flood Control District (HCFCD)		Y	Y		Y	Y	Y	--	--	--	--	--	--	--	--
Jesse	Dickerman	Interim County Administrator	Harris County Office of County Administration		--	Y		--		--	--	--	--	--	--	--	--	--
Joe	Sykoza	Fire Chief / Emergency Management Coordinator	City of Tomball	--	--	--	--	Y	Y	--	--	--	--	--	--	--	--	--
Joe	Perusil	Local Resident	Public	--	--	--	--	--	--	--	--	--	Y	--	--	--	--	--
Johnathan	Martin	County Engineer	Harris County Pollution Control Services	Y	--			--		--	--	--	--	--	--	--	--	--
Johnny	Morales	Emergency Management Coordinator	City of La Porte	Y	--	Y		Y	Y	Y	--	--	--	--	--	--	--	--
Joseph	Leonard	Planner	Resalire	--	--	--	--	--	--	Y	--	--	--	--	--	--	--	--
Katherine	Samuel	Emergency Management Operations – West Corridor	SouthEast Texas Regional Advisory Council (SETRAC)	--	--	--	--	Y	--	--	--	--	--	--	--	--	--	--
Kathy	Holcomb		City of Deer Park	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Katie	Short	Deputy Chief of Staff	Harris County Commissioner Precinct 1	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Kevin	Laguna	Police Chief	City of Morgan's Point	--	--	--	--	--	--	Y	--	--	--	--	--	--	--	--
Kevin	Garza	Health and Safety Manager	Harris County	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Kevin	Rodgers	Office of Emergency Management Director	City of Seabrook	Y	Y	Y		Y		Y	--	--	--	--	--	--	--	--
Keyona	Stevenson	Urban Area Security Initiative (UASI) Planner	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Khenley	Harvell	Emergency Management Coordinator	San Jacinto College	--	Y	Y	--	Y	Y	--	--	--	--	--	--	--	--	--
Kial	Whitton	Planning Intern	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	--	--	Y	--	--	--	Y	--	--	--	--	--	--	--	--
Kris	Jones	Director of Operations	Varnett Public Schools	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Kristina	Watson	Director of Public Works	City of Spring Valley Village	--	--	--	--	Y	--	--	--	--	--	--	--	--	--	--

Meeting Participation

First Name	Last Name	Title/ Position	Jurisdiction/ Agency	Kickoff	SC Workshop #1 07-24-2024	SC Workshop #2 09-11-2024	SC Workshop #3 10-02-2024	SC Workshop #4 03-12-2025	SC Workshop #5 04-09-2025	SC Workshop #6 06-04-2025	Community Workshop #1 10-19-2024	Community Workshop #2 10-22-2024	Community Workshop #3 10-30-2024	Community Workshop #4 11-09-2024	Public Meeting #1 06-26-2025	Public Meeting #2 06-28-2025	Public Meeting #3 07-01-2025	Public Meeting #4 07-02-2025
Kristina	Clark	Planner	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	Y	Y	--	--	Y	--	--	--	--	--	--	--	--	--
L?????	Hernandez	Communications Supervisor	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	--	--	Y	--	--	--	--	--	--	--	--	--	--	--	--
Latrice	Babin	Executive Director	Harris County Pollution Control Services	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Leah	Barton	Deputy County Administrator	Harris County Office of County Administration / Public Health		Y	Y		--	Y	Y	--	--	--	--	--	--	--	--
Leslie	Etheridge	Director, Center for Safe and Secure Schools	Harris County Department of Education		--	Y		--		--	--	--	--	--	--	--	--	--
Linda	Woods	Local Resident	Public	--	--	--	--	--	--	--	Y	--	--	--	--	--	--	--
Lindsey	Drouet	Finance/Admin Supervisor	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	Y	--	--	--	--	--	--	--	--	--	--	--	--	--
Lisa	Spivey	Director of Preparedness	SouthEast Texas Regional Advisory Council (SETRAC)		--			--		--	--	--	--	--	--	--	--	--
Liz	Lawhorn	Assistant Public Information Officer	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	--	Y	--	--	--	--	--	--	--	--	--	--	--	--	--
Lon	Squyres	City Manager/Emergency Management Coordinator	City of Jacinto City	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Malik	Michel	Engineering Manager	City of La Porte	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Marcus	Snell	City Engineer	City of Missouri City	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Mardie	Menke	Budget and Procurement	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maritza	Aguilar	Hazard Mitigation Planner	City of Houston Office of Emergency Management	Y	--	--	--	--	--	Y	--	--	--	--	--	--	--	--
Mark	Bitz	Fire Chief / Fire Marshal	City of Jersey Village		Y	Y		--	Y	--	--	--	--	--	--	--	--	--
Mark	Stalcup	Local Resident	Public	--	--	--	--	--	--	--	--	Y	--	--	--	--	--	--

Meeting Participation

First Name	Last Name	Title/ Position	Jurisdiction/ Agency	Kickoff	SC Workshop #1 07-24-2024	SC Workshop #2 09-11-2024	SC Workshop #3 10-02-2024	SC Workshop #4 03-12-2025	SC Workshop #5 04-09-2025	SC Workshop #6 06-04-2025	Community Workshop #1 10-19-2024	Community Workshop #2 10-22-2024	Community Workshop #3 10-30-2024	Community Workshop #4 11-09-2024	Public Meeting #1 06-26-2025	Public Meeting #2 06-28-2025	Public Meeting #3 07-01-2025	Public Meeting #4 07-02-2025
Marlisa	Briggs	President	North Houston Association		Y	Y		--	Y	--	--	--	--	--	--	--	--	--
Marvelou	Lee-Gatterson	Local Resident	Public	--	--	--	--	--	--	--	Y	--	--	--	--	--	--	--
Mary Anne	Piacentini	President and CEO	Coastal Prairie Conservancy		Y			--	Y	Y	--	--	--	--	--	--	--	--
Megan	Mayes	City Secretary	City of Morgan's Point		--			--	--	--	--	--	--	--	--	--	--	--
Melanie	Bartis	Assistant Director OEM	City of Houston	Y	--	Y	--	--	--	--	--	--	--	--	--	--	--	--
Melissa	Knox	Local Resident	Public	--	--	--	--	--	--	--	--	Y	--	--	--	--	--	--
Meryl	Holmes, Jr.	Mitigation Coordinator	Texas Division of Emergency Management (TDEM)	Y	--	--	--	Y	--	Y	--	--	--	--	--	--	--	--
Michael	Mulligan	Chief of Department	Harris County Emergency Service District (ESD) No. 46 / Atascocita Fire Department	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Michele	Fontenberry	Local Resident	Public	--	--	--	--	--	--	--	--	Y	--	--	--	--	--	--
Michelle	Dobbins	Director, Disaster Recover & Resilience	Houston Arts Alliance (HAA)		Y	Y		Y		Y	--	--	--	--	--	--	--	--
Miesha	Johnson	Community Development Coordinator	City of Jersey Village	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Miesha	Thorne	Administrative Assistant Coordinator	Varnett Public Schools	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Nalleli	Hidalgo	Community Outreach and Education Liaison	Texas Environmental Justice Advocacy Services (TEJAS)	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Nandini	Seth	Planner	Resalire Infrastructure	--	--	--	--	Y	--	--	--	--	--	--	--	--	--	--
Nathan	Alston	GIS Technician	Harris County Flood Control District (HCFCD)	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Nicole	Bradstreet	GIS Specialist	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Nita (Dr.)	White	Superintendent	Varnett Public Schools	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Paul	Warner	Captain of Police	City of Hedwig Village Police Department	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Meeting Participation

First Name	Last Name	Title/ Position	Jurisdiction/ Agency	Kickoff	SC Workshop #1 07-24-2024	SC Workshop #2 09-11-2024	SC Workshop #3 10-02-2024	SC Workshop #4 03-12-2025	SC Workshop #5 04-09-2025	SC Workshop #6 06-04-2025	Community Workshop #1 10-19-2024	Community Workshop #2 10-22-2024	Community Workshop #3 10-30-2024	Community Workshop #4 11-09-2024	Public Meeting #1 06-26-2025	Public Meeting #2 06-28-2025	Public Meeting #3 07-01-2025	Public Meeting #4 07-02-2025
Rachel	Lazo	Certified Floodplain Manager/City Planner	City of Katy	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Rees	Jackson	Intern	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Rev.	Kim	Local Resident	Public	--	--	--	--	--	--	--	--	Y	--	--	--	--	--	--
Robert	McCracken	Associate Vice Chancellor, Risk Management & Resilience	Houston Community College	Y	Y	Y		--		--	--	--	--	--	--	--	--	--
Robert	Marquez	Director of Safe and Secure Schools	Goose Creek Consolidated Independent School District		--	Y		--		--	--	--	--	--	--	--	--	--
Rosalie	Brockman	Senior Director, Building Resiliency & Sustainability Operations	Harris County ?	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Ryan	Hvitlok	Planning & Development Director	City of La Porte	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Sam	Bissett	Communications Planner	Harris County Office of Homeland Security and Emergency Management (HCOHSEM)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Samantha	Haritos	COO Financial Procurement at Galveston Co. Consolidated Drainage District	Public	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Samuel	Peña	Safety Consultant	Harris County Engineering Department	--	--	--	--	Y	--	--	--	--	--	--	--	--	--	--
Sandra	Pursley	Local Resident	Public	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Sarai	Osorio	Principal Planner	Houston-Galveston Area Council (H-GAC)		--	Y		--		--	--	--	--	--	--	--	--	--
Scott	Hudson	Director of Emergency Management	Cypress-Fairbanks Independent School District Police Department	Y	--	Y	--	--	Y	Y	--	--	--	--	--	--	--	--
Shain	Carrizal	Executive Director	Harris County HR & Risk Management	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Sheila	Matteson	Local Resident	Public	--	--	--	--	--	--	--	--	--	--	Y	--	--	--	--
Sherri	Ditrich	EM & Communications Specialist	City of Nassau Bay	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Steven	Simmons	Deputy Fire Marshal	City of Friendswood	Y	Y	Y	--	Y	Y	Y	--	--	--	--	--	--	--	--

Meeting Participation

First Name	Last Name	Title/ Position	Jurisdiction/ Agency	Kickoff	SC Workshop #1 07-24-2024	SC Workshop #2 09-11-2024	SC Workshop #3 10-02-2024	SC Workshop #4 03-12-2025	SC Workshop #5 04-09-2025	SC Workshop #6 06-04-2025	Community Workshop #1 10-19-2024	Community Workshop #2 10-22-2024	Community Workshop #3 10-30-2024	Community Workshop #4 11-09-2024	Public Meeting #1 06-26-2025	Public Meeting #2 06-28-2025	Public Meeting #3 07-01-2025	Public Meeting #4 07-02-2025
Stuart	Mueller	Deputy Director - Technical	Harris County Pollution Control Services	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Thomas	Quinn	ER & Safety Operations Manager	Lee College	--	--	Y	--	--	--	--	--	--	--	--	--	--	--	--
Thuy	La	Deputy Emergency Management Coordinator	Houston Public Works		--			--		--	--	--	--	--	--	--	--	--
Tiffany	Stasney		City of Deer Park	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Todd	Ward	Risk Mitigation Department Manager	Harris County Flood Control District (HCFCD)	Y	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Tom	Ehlers	Fire Chief/Emergency Management Coordinator	City of Galena Park	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Tom	Fullen	City Administrator	City of Hunters Creek Village	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Trameka	Jewett	Emergency Management Coordinator	City of Missouri City	Y	Y			--	Y	Y	--	--	--	--	--	--	--	--
Travis	Tanner	Director of Development Services	City of Bellaire	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Troy	Harrison	City Manager	City of Shoreacres		Y	Y		Y		Y	--	--	--	--	--	--	--	--
Wendy	Baimbridge	City Administrator	City of Hedwig Village	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Zac	Nguyen	Emergency Management Liasison, Judge's Office	Harris County		--			--		Y	--	--	--	--	--	--	--	--

APPENDIX E: ACTION WORKSHEET TEMPLATE AND INSTRUCTIONS

Please send all electronic Word versions to the emails below by Wednesday, April 30th, to:

Tetra Tech | CorePlanningTeam@tetratech.com **HCOHSEM | mitigation@oem.hctx.net**

Jurisdiction: Choose an item.

Name/Title of Individual Completing Worksheet:

Instructions:

The purpose of this worksheet is to help the planning team assemble a list of the new potential action items that will be incorporated in the 2025 Harris County Multi-Jurisdictional (MJ) Hazard Mitigation Plan (HMP) Update. The table below will help planners collect and catalog new mitigation actions, and later, once all action items are drafted, actions will be evaluated and prioritized in Volume I of the Harris County MJ-HMP. For each new mitigation action, please fill out the table below.

Jurisdiction/ Planning Partner:	Choose an item.		
Action Description:			
Lead Agency			
Supporting Agency:			
Hazard(s) Addressed:	☐ Coastal Erosion	☐ Flooding	☐ Energy Pipeline Failure
	☐ Dam/Levee Failure	☐ Hurricane/Coastal Storm	☐ Public Health Emergency
	☐ Drought	☐ Mass Movement	☐ Technology Failure
	☐ Earthquake	☐ Severe Weather	☐ Toxic Release/Hazardous Materials
	☐ Extreme Cold	☐ Tsunami	☐ Utility Failure
	☐ Extreme Heat	☐ Wildfire	
Asset Type	Choose an item.		
Goal(s) Met	☐ **GOAL 1:** *Improve and coordinate data collection efforts to fully maximize mitigation capabilities*		
	☐ **GOAL2:** *Enhance public outreach strategies to improve information sharing regarding hazards, including risk reduction strategies*		
	☐ **GOAL 3:** *Reduce the impacts of hazards on people and property and protect natural and cultural resources*		
	☐ **GOAL 4:** *Work to improve and coordinate existing local plans, policies, codes, and regulations to reduce hazard impacts*		
	☐ **GOAL 5:** *Implement property protection measures to reduce or eliminate repetitive loss occurrences.*		
	☐ **GOAL 6:** *Investigate and implement a range of structural projects that will reduce the effects of hazards on public and private property throughout the County*		
	☐ **GOAL 7:** *Investigate and implement a range of nature-based solutions that utilize, enhance, and protect natural resources and their ability to reduce hazard impacts.*		
	☐ **GOAL 8:** *Advance hazard analysis and mitigation strategies to reflect interdependencies across infrastructure and community lifelines*		
	☐ **Goal 9:** *Prioritize mitigation investments to inform and support efforts to seek funding*		

Description of the Problem:	
Description of the Solution:	
Benefits:	Choose an item.
Benefits Comments (Optional)	
Cost:	Choose an item.
Cost Comments (Optional)	
Potential Funding Sources:	
Implementation Timeline:	Choose an item.
Timeline Comments (Optional)	
Climate Change Considerations:	
FEMA Mitigation Category:	☐ Education and Awareness (EAP)
	☐ Local Plans and Regulations (LPR)
	☐ Natural Systems Protection (NSP)
	☐ Structure and Infrastructure Projects (SIP)
CRS Mitigation Category:	☐ Emergency Services (ES)
	☐ Preventative Measures (PR)
	☐ Property Protection (PP)
	☐ Public Information (PI)
	☐ Natural Resource Protection (NR)
	☐ Structural Flood Control Projects (SP)
Priority:	Choose an item.

APPENDIX F: PLAN REVIEW TOOLS

Local Mitigation Planning Policy Guide

FP-206-21-0002

Effective April 11, 2025

OMB Collection #1660-0062

FEMA

Appendix A: Local Mitigation Plan Review Tool

Cover Page

The Local Mitigation Plan Review Tool (PRT) demonstrates how the local mitigation plan meets the regulation in 44 CFR § 201.6 and offers states and FEMA Mitigation Planners an opportunity to provide feedback to the local governments, including special districts.

1. The Multi-Jurisdictional Summary Sheet is a worksheet that is used to document how each jurisdiction met the requirements of the plan elements (Planning Process; Risk Assessment; Mitigation Strategy; Plan Maintenance; Plan Update; and Plan Adoption).
2. The Plan Review Checklist summarizes FEMA's evaluation of whether the plan has addressed all requirements.

For greater clarification of the elements in the Plan Review Checklist, please see Section 4 of this guide. Definitions of the terms and phrases used in the PRT can be found in Appendix E of this guide.

Plan Information	
Jurisdiction(s)	Click or tap here to enter text.
Title of Plan	Click or tap here to enter text.
New Plan or Update	Click or tap here to enter text.
Single- or Multi-Jurisdiction	Choose an item.
Date of Plan	Click or tap to enter a date.
Local Point of Contact	
Title	Click or tap here to enter text.
Agency	Click or tap here to enter text.
Address	Click or tap here to enter text.
Phone Number	Click or tap here to enter text.
Email	Click or tap here to enter text.

Additional Point of Contact	
Title	Click or tap here to enter text.
Agency	Click or tap here to enter text.
Address	Click or tap here to enter text.
Phone Number	Click or tap here to enter text.
Email	Click or tap here to enter text.

Review Information	
State Review	
State Reviewer(s) and Title	Click or tap here to enter text.
State Review Date	Click or tap to enter a date.
FEMA Review	
FEMA Reviewer(s) and Title	Click or tap here to enter text.
Date Received in FEMA Region	Click or tap to enter a date.
Plan Not Approved	Click or tap to enter a date.
Plan Approvable Pending Adoption	Click or tap to enter a date.
Plan Approved	Click or tap to enter a date.

Multi-Jurisdictional Summary Sheet

#	Jurisdiction Name	A. Planning Process	B. Risk Assessment	C. Mitigation Strategy	D. Plan Maintenance	E. Plan Update	F. Plan Adoption	G. HHPD Requirements	H. State Requirements
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									

Plan Review Checklist

The Plan Review Checklist is completed by FEMA. States and local governments are encouraged, but not required, to use the PRT as a checklist to ensure all requirements have been met prior to submitting the plan for review and approval. The purpose of the checklist is to identify the location of relevant or applicable content in the plan by element/sub-element and to determine if each requirement has been "met" or "not met." FEMA completes the "required revisions" summary at the bottom of each element to clearly explain the revisions that are required for plan approval. Required revisions must be explained for each plan sub-element that is "not met." Sub-elements in each summary should be referenced using the appropriate numbers (A1, B3, etc.), where applicable. Requirements for each element and sub-element are described in detail in Section 4: Local Plan Requirements of this guide.

Plan updates must include information from the current planning process.

If some elements of the plan do not require an update, due to minimal or no changes between updates, the plan must document the reasons for that.

Multi-jurisdictional elements must cover information unique to all participating jurisdictions.

Element A: Planning Process

Element A Requirements	Location in Plan (section and/or page number)	Met / Not Met
A1. Does the plan document the planning process, including how it was prepared and who was involved in the process for each jurisdiction? (Requirement 44 CFR § 201.6(c)(1))		
A1-a. Does the plan document how the plan was prepared, including the schedule or time frame and activities that made up the plan's development, as well as who was involved?	Click or tap here to enter text.	Choose an item.
A1-b. Does the plan list the jurisdiction(s) participating in the plan that seek approval, and describe how they participated in the planning process?	Click or tap here to enter text.	Choose an item.

Element A Requirements	Location in Plan (section and/or page number)	Met / Not Met
A2. Does the plan document an opportunity for neighboring communities, local and regional agencies involved in hazard mitigation activities, and agencies that have the authority to regulate development as well as businesses, academia, and other private and non-profit interests to be involved in the planning process? (Requirement 44 CFR § 201.6(b)(2))		
A2-a. Does the plan identify all stakeholders involved or given an opportunity to be involved in the planning process, and how each stakeholder was presented with this opportunity?	Click or tap here to enter text.	Choose an item.
A3. Does the plan document how the public was involved in the planning process during the drafting stage and prior to plan approval? (Requirement 44 CFR § 201.6(b)(1))		
A3-a. Does the plan document how the public was given the opportunity to be involved in the planning process and how their feedback was included in the plan?	Click or tap here to enter text.	Choose an item.
A4. Does the plan describe the review and incorporation of existing plans, studies, reports, and technical information? (Requirement 44 CFR § 201.6(b)(3))		
A4-a. Does the plan document what existing plans, studies, reports and technical information were reviewed for the development of the plan, as well as how they were incorporated into the document?	Click or tap here to enter text.	Choose an item.

ELEMENT A REQUIRED REVISIONS
Required Revision**:** Click or tap here to enter text.

Element B: Risk Assessment

Element B Requirements	Location in Plan (section and/or page number)	Met / Not Met
B1. Does the plan include a description of the type, location, and extent of all natural hazards that can affect the jurisdiction? Does the plan also include information on previous occurrences of hazard events and on the probability of future hazard events? (Requirement 44 CFR § 201.6(c)(2)(i))		
B1-a. Does the plan describe all natural hazards that can affect the jurisdiction(s) in the planning area, and does it provide the rationale if omitting any natural hazards that are commonly recognized to affect the jurisdiction(s) in the planning area?	Click or tap here to enter text.	Choose an item.
B1-b. Does the plan include information on the location of each identified hazard?	Click or tap here to enter text.	Choose an item.
B1-c. Does the plan describe the extent for each identified hazard?	Click or tap here to enter text.	Choose an item.
B1-d. Does the plan include the history of previous hazard events for each identified hazard?	Click or tap here to enter text.	Choose an item.
B1-e. Does the plan include the probability of future events for each identified hazard, including the type, location and range of anticipated intensities?	Click or tap here to enter text.	Choose an item.
B1-f. For participating jurisdictions in a multi-jurisdictional plan, does the plan describe any hazards that are unique to and/or vary from those affecting the overall planning area?	Click or tap here to enter text.	Choose an item.
B2. Does the plan include a summary of the jurisdiction's vulnerability and the impacts on the community from the identified hazards? Does this summary also address NFIP-insured structures that have been repetitively damaged by floods? (Requirement 44 CFR § 201.6(c)(2)(ii))		
B2-a. Does the plan provide an overall summary of each jurisdiction's vulnerability to the identified hazards?	Click or tap here to enter text.	Choose an item.
B2-b. For each participating jurisdiction, does the plan describe the potential impacts of each of the identified hazards on each participating jurisdiction?	Click or tap here to enter text.	Choose an item.
B2-c. Does the plan address NFIP-insured structures within each jurisdiction that have been repetitively damaged by floods?	Click or tap here to enter text.	Choose an item.

ELEMENT B REQUIRED REVISIONS
Required Revision: Click or tap here to enter text.

Element C: Mitigation Strategy

Element C Requirements	Location in Plan (section and/or page number)	Met / Not Met
C1. Does the plan document each participant's existing authorities, policies, programs and resources and its ability to expand on and improve these existing policies and programs? (Requirement 44 CFR § 201.6(c)(3))		
C1-a. Does the plan describe how the existing capabilities of each participant are available to support the mitigation strategy? Does this include a discussion of the existing building codes and land use and development ordinances or regulations?	Click or tap here to enter text.	Choose an item.
C1-b. Does the plan describe each participant's ability to expand and improve the identified capabilities to achieve mitigation?	Click or tap here to enter text.	Choose an item.
C2. Does the plan address each jurisdiction's participation in the NFIP and continued compliance with NFIP requirements, as appropriate? (Requirement 44 CFR § 201.6(c)(3)(ii))		
C2-a. Does the plan contain a narrative description or a table/list of their participation activities?	Click or tap here to enter text.	Choose an item.
C3. Does the plan include goals to reduce/avoid long-term vulnerabilities to the identified hazards? (Requirement 44 CFR § 201.6(c)(3)(i))		
C3-a. Does the plan include goals to reduce the risk from the hazards identified in the plan?	Click or tap here to enter text.	Choose an item.

Element C Requirements	Location in Plan (section and/or page number)	Met / Not Met
C4. Does the plan identify and analyze a comprehensive range of specific mitigation actions and projects for each jurisdiction being considered to reduce the effects of hazards, with emphasis on new and existing buildings and infrastructure? (Requirement 44 CFR § 201.6(c)(3)(ii))		
C4-a. Does the plan include an analysis of a comprehensive range of actions/projects that each jurisdiction considered to reduce the impacts of hazards identified in the risk assessment?	Click or tap here to enter text.	Choose an item.
C4-b. Does the plan include one or more action(s) per jurisdiction for each of the hazards as identified within the plan's risk assessment?	Click or tap here to enter text.	Choose an item.
C5. Does the plan contain an action plan that describes how the actions identified will be prioritized (including a cost-benefit review), implemented, and administered by each jurisdiction? (Requirement 44 CFR § 201.6(c)(3)(iv)); (Requirement §201.6(c)(3)(iii))		
C5-a. Does the plan describe the criteria used for prioritizing actions?	Click or tap here to enter text.	Choose an item.
C5-b. Does the plan provide the position, office, department or agency responsible for implementing/administrating the identified mitigation actions, as well as potential funding sources and expected time frame?	Click or tap here to enter text.	Choose an item.

ELEMENT C REQUIRED REVISIONS
Required Revision**:** Click or tap here to enter text.

Element D: Plan Maintenance

Element D Requirements	Location in Plan (section and/or page number)	Met / Not Met
D1. Is there discussion of how each community will continue public participation in the plan maintenance process? (Requirement 44 CFR § 201.6(c)(4)(iii))		
D1-a. Does the plan describe how communities will continue to seek future public participation after the plan has been approved?	Click or tap here to enter text.	Choose an item.
D2. Is there a description of the method and schedule for keeping the plan current (monitoring, evaluating and updating the mitigation plan within a five-year cycle)? (Requirement 44 CFR § 201.6(c)(4)(i))		
D2-a. Does the plan describe the process that will be followed to track the progress/status of the mitigation actions identified within the Mitigation Strategy, along with when this process will occur and who will be responsible for the process?	Click or tap here to enter text.	Choose an item.
D2-b. Does the plan describe the process that will be followed to evaluate the plan for effectiveness? This process must identify the criteria that will be used to evaluate the information in the plan, along with when this process will occur and who will be responsible.	Click or tap here to enter text.	Choose an item.
D2-c. Does the plan describe the process that will be followed to update the plan, along with when this process will occur and who will be responsible for the process?	Click or tap here to enter text.	Choose an item.
D3. Does the plan describe a process by which each community will integrate the requirements of the mitigation plan into other planning mechanisms, such as comprehensive or capital improvement plans, when appropriate? (Requirement 44 CFR § 201.6(c)(4)(ii))		
D3-a. Does the plan describe the process the community will follow to integrate the ideas, information and strategy of the mitigation plan into other planning mechanisms?	Click or tap here to enter text.	Choose an item.
D3-b. Does the plan identify the planning mechanisms for each plan participant into which the ideas, information and strategy from the mitigation plan may be integrated?	Click or tap here to enter text.	Choose an item.

Element D Requirements	Location in Plan (section and/or page number)	Met / Not Met
D3-c. For multi-jurisdictional plans, does the plan describe each participant's individual process for integrating information from the mitigation strategy into their identified planning mechanisms?	Click or tap here to enter text.	Choose an item.
ELEMENT D REQUIRED REVISIONS		
Required Revision: Click or tap here to enter text.		

Element E: Plan Update

Element E Requirements	Location in Plan (section and/or page number)	Met / Not Met
E1. Was the plan revised to reflect changes in development? (Requirement 44 CFR § 201.6(d)(3))		
E1-a. Does the plan describe the changes in development that have occurred in hazard-prone areas that have increased or decreased each community's vulnerability since the previous plan was approved?	Click or tap here to enter text.	Choose an item.
E2. Was the plan revised to reflect changes in priorities and progress in local mitigation efforts? (Requirement 44 CFR § 201.6(d)(3))		
E2-a. Does the plan describe how it was revised due to changes in community priorities?	Click or tap here to enter text.	Choose an item.
E2-b. Does the plan include a status update for all mitigation actions identified in the previous mitigation plan?	Click or tap here to enter text.	Choose an item.
E2-c. Does the plan describe how jurisdictions integrated the mitigation plan, when appropriate, into other planning mechanisms?	Click or tap here to enter text.	Choose an item.
ELEMENT E REQUIRED REVISIONS		
Required Revision: Click or tap here to enter text.		

Element F: Plan Adoption

Element F Requirements	Location in Plan (section and/or page number)	Met / Not Met
F1. For single-jurisdictional plans, has the governing body of the jurisdiction formally adopted the plan to be eligible for certain FEMA assistance? (Requirement 44 CFR § 201.6(c)(5))		
F1-a. Does the participant include documentation of adoption?	Click or tap here to enter text.	Choose an item.
F2. For multi-jurisdictional plans, has the governing body of each jurisdiction officially adopted the plan to be eligible for certain FEMA assistance? (Requirement 44 CFR § 201.6(c)(5))		
F2-a. Did each participant adopt the plan and provide documentation of that adoption?	Click or tap here to enter text.	Choose an item.
ELEMENT F REQUIRED REVISIONS		
Required Revision**:** Click or tap here to enter text.		

Element G: High Hazard Potential Dams (Optional)

HHPD Requirements	Location in Plan (section and/or page number)	Met / Not Met
HHPD1. Did the plan describe the incorporation of existing plans, studies, reports and technical information for HHPDs?		
HHPD1-a. Does the plan describe how the local government worked with local dam owners and/or the state dam safety agency?	Click or tap here to enter text.	Choose an item.
HHPD1-b. Does the plan incorporate information shared by the state and/or local dam owners?	Click or tap here to enter text.	Choose an item.
HHPD2. Did the plan address HHPDs in the risk assessment?		
HHPD2-a. Does the plan describe the risks and vulnerabilities to and from HHPDs?	Click or tap here to enter text.	Choose an item.
HHPD2-b. Does the plan document the limitations and describe how to address deficiencies?	Click or tap here to enter text.	Choose an item.

HHPD Requirements	Location in Plan (section and/or page number)	Met / Not Met
HHPD3. Did the plan include mitigation goals to reduce long-term vulnerabilities from HHPDs?		
HHPD3-a. Does the plan address how to reduce vulnerabilities to and from HHPDs as part of its own goals or with other long-term strategies?	Click or tap here to enter text.	Choose an item.
HHPD3-b. Does the plan link proposed actions to reducing long-term vulnerabilities that are consistent with its goals?	Click or tap here to enter text.	Choose an item.
HHPD4-a. Did the plan include actions that address HHPDs and prioritize mitigation actions to reduce vulnerabilities from HHPDs?		
HHPD4-a. Does the plan describe specific actions to address HHPDs?	Click or tap here to enter text.	Choose an item.
HHPD4-b. Does the plan describe the criteria used to prioritize actions related to HHPDs?	Click or tap here to enter text.	Choose an item.
HHPD4-c. Does the plan identify the position, office, department or agency responsible for implementing and administering the action to mitigate hazards to or from HHPDs?	Click or tap here to enter text.	Choose an item.
HHPD Required Revisions		
Required Revision**:** Click or tap here to enter text.		

Element H: Additional State Requirements (Optional)

Element H Requirements	Location in Plan (section and/or page number)	Met / Not Met
This space is for the State to include additional requirements.		
Click or tap here to enter text.	Click or tap here to enter text.	Choose an item.

Plan Assessment

These comments can be used to help guide your annual/regularly scheduled updates and the next plan update.

Element A. Planning Process

Strengths

- [insert comments]

Opportunities for Improvement

- [insert comments]

Element B. Risk Assessment

Strengths

- [insert comments]

Opportunities for Improvement

- [insert comments]

Element C. Mitigation Strategy

Strengths

- [insert comments]

Opportunities for Improvement

- [insert comments]

Element D. Plan Maintenance

Strengths

- [insert comments]

Opportunities for Improvement

- [insert comments]

Element E. Plan Update

Strengths

- [insert comments]

Opportunities for Improvement

- [insert comments]

Element G. HHPD Requirements (Optional)

Strengths

- [insert comments]

Opportunities for Improvement

- [insert comments]

Element H. Additional State Requirements (Optional)

Strengths

- [insert comments]

Opportunities for Improvement

- [insert comments]

APPENDIX G: RISK ASSESSMENT SUPPLEMENTARY DATA

Table G-1. Hazard Locations for Non-Municipal Planning Partners

Planning Partner	Facility Name	Wildfire Threat Zone[a]	Wildfire Ignition Density Zone[b]	1% Annual Chance Flood Zone	Hurricane Harvey Flood Extent	Coastal Flood Zone	100-Year Probabilistic Hurricane Wind Zone[c]	20-Year Probabilistic Hurricane Wind Zone[c]	Hurricane Alicia Wind Zone[c]	Addicks Dam Inundation Zone	Barker Dam Inundation Zone	Lake Houston Dam Inundation Zone
Unincorporated Harris County	—	0 to 1 (Low)	0 to 7 (Very High)	Yes	Yes	Yes	89-117	55-83	79-116	Yes	Yes	Yes
Bellaire (C)	—	0	0 to 1 (Low)	Yes	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Bunker Hill Village (C)	—	0	0 to 1 (Low)	No	Yes	No	102-106	66-70	99-101	Yes	Yes	No
Deer Park (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	107-111	71-74	102-104	Yes	Yes	Yes
El Lago (C)	—	0	0 to 1 (Low)	Yes	Yes	Yes	112-117	71-83	105-116	No	No	No
Friendswood (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	112-117	71-83	102-116	No	No	No
Galena Park (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	107-111	66-70	102-104	Yes	Yes	Yes
Hedwig Village (C)	—	0	0 to 1 (Low)	No	Yes	No	102-106	66-70	99-101	Yes	Yes	No
Hilshire Village (C)	—	0	0 to 1 (Low)	Yes	Yes	No	102-106	66-70	99-101	Yes	Yes	No
Humble (C)	—	0 to 1 (Low)	0 to 3 (Moderate)	Yes	Yes	No	102-106	62-70	99-101	No	No	No
Hunters Creek Village (C)	—	0	0 to 1 (Low)	Yes	Yes	No	102-106	66-70	99-101	Yes	Yes	No
Jacinto City (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	107-111	66-70	102-104	Yes	Yes	Yes
Jersey Village (C)	—	0	0 to 1 (Low)	Yes	Yes	No	89-106	62-66	99-101	Yes	Yes	No
Katy (C)	—	0 to 1 (Low)	0 to 4 (Moderate-High)	Yes	Yes	No	89-101	62-66	79-98	No	Yes	No
La Porte (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	Yes	107-117	71-83	102-116	No	No	Yes

Planning Partner	Facility Name	Wildfire Threat Zone[a]	Wildfire Ignition Density Zone[b]	1% Annual Chance Flood Zone	Hurricane Harvey Flood Extent	Coastal Flood Zone	100-Year Probabilistic Hurricane Wind Zone[c]	20-Year Probabilistic Hurricane Wind Zone[c]	Hurricane Alicia Wind Zone[c]	Addicks Dam Inundation Zone	Barker Dam Inundation Zone	Lake Houston Dam Inundation Zone
Missouri City (C)	—	0 to 1 (Low)	0 to 4 (Moderate-High)	Yes	Yes	No	107-111	66-74	99-104	Yes	Yes	No
Morgan's Point (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	Yes	107-117	71-74	102-116	No	No	No
Nassau Bay (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	112-117	71-74	105-116	No	No	No
Pasadena (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	107-117	71-74	102-116	Yes	Yes	Yes
Piney Point Village (C)	—	0	0 to 1 (Low)	Yes	Yes	No	102-106	66-70	99-101	Yes	Yes	No
Seabrook (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	112-117	71-83	105-116	No	No	No
Shoreacres (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	Yes	112-117	71-83	102-116	No	No	No
Southside Place (C)	—	0	0	Yes	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Spring Valley Village (C)	—	0	0 to 1 (Low)	Yes	Yes	No	102-106	66-70	99-101	Yes	Yes	No
Stafford (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	107-111	66-70	99-104	No	Yes	No
Taylor Lake Village (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	112-117	71-74	105-116	No	No	No
Tomball (C)	—	0 to 1 (Low)	0 to 3 (Moderate)	Yes	Yes	No	89-101	55-66	95-98	No	No	No
Waller (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	89-101	55-62	79-94	No	No	No
Webster (C)	—	0 to 1 (Low)	0 to 1 (Low)	Yes	Yes	No	112-117	71-74	105-116	No	No	No
West University Place (C)	—	0	0 to 1 (Low)	Yes	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Cy-Fair ISD	Adam Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Alternative Learning Ctr - West	0	0	No	Yes	No	89-101	62-66	95-98	No	No	No

Planning Partner	Facility Name	Wildfire Threat Zone[a]	Wildfire Ignition Density Zone[b]	1% Annual Chance Flood Zone	Hurricane Harvey Flood Extent	Coastal Flood Zone	100-Year Probabilistic Hurricane Wind Zone[c]	20-Year Probabilistic Hurricane Wind Zone[c]	Hurricane Alicia Wind Zone[c]	Addicks Dam Inundation Zone	Barker Dam Inundation Zone	Lake Houston Dam Inundation Zone
Cy-Fair ISD	Alternative Learning Ctr-East	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Andre Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Anthony Middle	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Aragon Middle	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Arnold Middle	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Ault Elementary	0	0	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Bane Elementary	0	0	No	No	No	102-106	66-70	99-101	Yes	Yes	No
Cy-Fair ISD	Bang Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Birkes Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Black Elementary	1 (Low)	1 (Low)	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Bleyl Middle	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Bridgeland High School	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Brosnahan Elementary	1 (Low)	3 (Moderate)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Campbell Middle	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Cook Middle	0	0	No	Yes	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Copeland Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Cy-Fair High School	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Cypress Creek High School	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Cypress Falls High School	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Cypress Lakes High School	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Cypress Park High School	0	1 (Low)	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Cypress Ranch High School	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Cypress Ridge High School	0	0	No	No	No	89-101	62-66	99-101	No	No	No

Planning Partner	Facility Name	Wildfire Threat Zone[a]	Wildfire Ignition Density Zone[b]	1% Annual Chance Flood Zone	Hurricane Harvey Flood Extent	Coastal Flood Zone	100-Year Probabilistic Hurricane Wind Zone[c]	20-Year Probabilistic Hurricane Wind Zone[c]	Hurricane Alicia Wind Zone[c]	Addicks Dam Inundation Zone	Barker Dam Inundation Zone	Lake Houston Dam Inundation Zone
Cy-Fair ISD	Cypress Springs High School	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Cypress Woods High School	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Cypress-Fairbanks J J A E P	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Danish Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Dean Middle	0	0	No	No	No	102-106	66-70	99-101	Yes	Yes	No
Cy-Fair ISD	Duryea Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Emery Elementary	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Emmott Elementary	0	0	No	Yes	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Farney Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Fiest Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Francone Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Frazier Elementary	0	0	No	Yes	No	102-106	62-66	99-101	No	No	No
Cy-Fair ISD	Gleason Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Goodson Middle	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Hairgrove Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Hamilton Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Hamilton Middle	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Hancock Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Hemmenway Elementary	1 (Low)	1 (Low)	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Holbrook Elementary	0	0	No	No	No	102-106	66-70	99-101	Yes	Yes	No
Cy-Fair ISD	Holmsley Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Hoover Elementary	0	0	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Hopper Middle	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Horne Elementary	0	0	No	No	No	89-101	62-66	99-101	Yes	No	No
Cy-Fair ISD	Jersey Village High School	0	0	No	No	No	102-106	62-66	99-101	Yes	No	No
Cy-Fair ISD	Jowell Elementary	0	0	No	Yes	No	89-101	62-66	95-98	No	No	No

Planning Partner	Facility Name	Wildfire Threat Zone[a]	Wildfire Ignition Density Zone[b]	1% Annual Chance Flood Zone	Hurricane Harvey Flood Extent	Coastal Flood Zone	100-Year Probabilistic Hurricane Wind Zone[c]	20-Year Probabilistic Hurricane Wind Zone[c]	Hurricane Alicia Wind Zone[c]	Addicks Dam Inundation Zone	Barker Dam Inundation Zone	Lake Houston Dam Inundation Zone
Cy-Fair ISD	Kahla Middle	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Keith Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Kirk Elementary	0	0	No	No	No	102-106	62-66	99-101	No	Yes	No
Cy-Fair ISD	Labay Middle	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Lamkin Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Langham Creek High School	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Lee Elementary	0	0	No	No	No	102-106	62-66	99-101	Yes	Yes	No
Cy-Fair ISD	Lieder Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Lowery Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Matzke Elementary	0	1 (Low)	No	Yes	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	McFee Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	McGown Elementary	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Metcalf Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Millsap Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Moore Elementary	1 (Low)	0	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	New Elementary	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Owens Elementary	0	0	No	Yes	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Pope Elementary	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Post Elementary	0	0	Yes	No	No	102-106	62-66	99-101	Yes	No	No
Cy-Fair ISD	Postma Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Reed Elementary	0	0	No	No	No	102-106	62-66	99-101	No	No	No
Cy-Fair ISD	Rennell Elementary	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Robinson Elementary	1 (Low)	3 (Moderate)	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Robison Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Rowe Middle	0	1 (Low)	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Salyards Middle	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Sampson Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No

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Cy-Fair ISD	Sheridan Elementary	1 (Low)	0	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Smith Middle	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Spillane Middle	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Sprague Middle	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Swenke Elementary	1 (Low)	1 (Low)	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Thornton Middle	1 (Low)	0	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Tipps Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Truitt Middle	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Walker Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Warner Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Watkins Middle	0	0	No	No	No	89-101	62-66	95-98	Yes	No	No
Cy-Fair ISD	Wells Elementary	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Willbern Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Cy-Fair ISD	Wilson Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Woodard Elementary	0	1 (Low)	No	No	No	89-101	62-66	95-98	No	No	No
Cy-Fair ISD	Yeager Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Goose Creek CISD	Alamo Elementary	1 (Low)	1 (Low)	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Ashbel Smith Elementary	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Banuelos Elementary	1 (Low)	1 (Low)	No	Yes	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Baytown Junior High	1 (Low)	0	No	Yes	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Carver Elementary	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Cedar Bayou Junior High	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Clark Elementary	1 (Low)	1 (Low)	No	N/A	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Crockett Elementary	1 (Low)	0	No	No	No	107-111	71-74	99-101	No	No	No

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Goose Creek CISD	De Zavala Elementary	1 (Low)	0	No	No	No	107-111	71-74	102-104	No	No	No
Goose Creek CISD	E F Green Junior School	1 (Low)	0	No	Yes	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Excel Academy (Murworth)	0	0	No	Yes	No	107-111	71-74	102-104	Yes	Yes	No
Goose Creek CISD	GCCISD Virtual Academy	1 (Low)	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Gentry Junior High	1 (Low)	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Goose Creek Memorial	1 (Low)	1 (Low)	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Harlem Elementary	1 (Low)	0	No	No	No	107-111	66-70	99-101	No	No	No
Goose Creek CISD	High Point School	1 (Low)	0	No	Yes	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Highlands Elementary	0	0	No	No	No	107-111	66-70	99-101	No	No	No
Goose Creek CISD	Highlands Junior High	0	0	No	No	No	107-111	66-70	99-101	No	No	No
Goose Creek CISD	Hopper Pri	0	0	No	No	No	107-111	66-70	99-101	No	No	No
Goose Creek CISD	Horace Mann Junior High	0	0	No	No	No	107-111	71-74	102-104	No	No	No
Goose Creek CISD	Impact Early College High School	0	0	Yes	Yes	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	James Bowie Elementary	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Jessie Lee Pumphrey Elementary	1 (Low)	1 (Low)	No	Yes	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Lamar Elementary	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Lee High School	0	0	No	No	No	107-111	71-74	99-101	No	No	No

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Goose Creek CISD	Peter E Hyland Center	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Point Alternative Center	1 (Low)	1 (Low)	No	Yes	No	107-111	66-70	99-101	No	No	No
Goose Creek CISD	San Jacinto Elementary	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Sheila Liles Early Learning Academy	1 (Low)	1 (Low)	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Stephen F Austin Elementary	1 (Low)	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Sterling High School	1 (Low)	0	No	Yes	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Stuart Career Tech High School	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Travis Elementary	1 (Low)	0	No	No	No	107-111	71-74	99-101	No	No	No
Goose Creek CISD	Victoria Walker Elementary	1 (Low)	0	No	Yes	No	107-111	71-74	99-101	No	No	No
Spring ISD	Anderson Elementary	1 (Low)	0	No	Yes	No	89-101	62-66	95-98	No	No	No
Spring ISD	Andy Dekaney High School	1 (Low)	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Bammel Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Bammel Middle	1 (Low)	0	Yes	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Beneke Elementary	0	0	No	Yes	No	89-101	62-66	99-101	No	No	No
Spring ISD	Carl Wunsche Sr High School	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Carolee Booker Elementary	1 (Low)	1 (Low)	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Chet Burchett Elementary	1 (Low)	2 (Low-Moderate)	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Clark Elementary	0	1 (Low)	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Deloras E Thompson Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No

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Spring ISD	Donna Lewis Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Dueitt Middle	0	2 (Low-Moderate)	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Edward Roberson Middle	0	1 (Low)	No	Yes	No	89-101	62-66	99-101	No	No	No
Spring ISD	Edwin M Wells Middle	1 (Low)	3 (Moderate)	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Ginger McNabb Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Gloria Marshall Elementary	0	1 (Low)	No	Yes	No	89-101	62-66	95-98	No	No	No
Spring ISD	Helen Major Elementary	1 (Low)	1 (Low)	No	Yes	No	89-101	62-66	99-101	No	No	No
Spring ISD	Heritage Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Hoyland Elementary	0	1 (Low)	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Joan Link Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	John Winship Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Meyer Elementary	1 (Low)	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Mildred Jenkins Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Milton Cooper Elementary	0	1 (Low)	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Momentum High School	1 (Low)	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Northgate Elementary	1 (Low)	2 (Low-Moderate)	No	Yes	No	89-101	62-66	95-98	No	No	No
Spring ISD	Pat Reynolds Elementary	1 (Low)	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Pearl M Hirsch Elementary	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Ponderosa Elementary	0	0	No	No	No	89-101	62-66	99-101	No	No	No

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Spring ISD	Ralph Eickenroht Elementary	0	1 (Low)	No	Yes	No	89-101	62-66	99-101	No	No	No
Spring ISD	Richey Academy	0	1 (Low)	No	Yes	No	89-101	62-66	99-101	No	No	No
Spring ISD	Rickey C Bailey Middle	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Salyers Elementary	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	School For Accelerated Learning	0	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Spring ISD	Smith Elementary	1 (Low)	0	No	Yes	No	89-101	62-66	95-98	No	No	No
Spring ISD	Spring Early College Academy	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	Spring High School	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Spring Leadership Academy	0	0	No	Yes	No	89-101	62-66	99-101	No	No	No
Spring ISD	Springwoods Village Middle	1 (Low)	1 (Low)	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Stelle Claughton Middle	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Spring ISD	The School for International Studies at Bammel	0	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Twin Creeks Middle	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Spring ISD	Westfield High School	1 (Low)	0	No	No	No	89-101	62-66	99-101	No	No	No
Varnett Public School	The Varnett Public Schools	1 (Low)	0	No	Yes	No	107-111	66-70	102-104	Yes	No	No
Varnett Public School	Northeast	0	0	No	Yes	No	102-106	66-70	99-101	No	No	No
Varnett Public School	Southwest Early Childhood Center	1 (Low)	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Varnett Public School	Southeast	0	0	No	No	No	107-111	71-74	102-104	No	No	No
Varnett Public School	East	1 (Low)	0	No	Yes	No	107-111	66-70	99-101	No	No	No
Varnett Public School	Southwest	0	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No

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Varnett Public School	Administration	1 (Low)	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Houston Community College	Katy Campus	1 (Low)	3 (Moderate)	No	Yes	No	89-101	62-66	95-98	No	No	No
Houston Community College	Spring Branch Campus	0	0	No	Yes	No	102-106	66-70	99-101	Yes	Yes	No
Houston Community College	Alief Hayes Campus	0	0	No	No	No	102-106	66-70	99-101	Yes	Yes	No
Houston Community College	Alief Bissonnet Campus	0	0	No	Yes	No	102-106	66-70	99-101	No	No	No
Houston Community College	Brays Oaks Campus	0	1 (Low)	No	Yes	No	107-111	66-70	99-101	Yes	No	No
Houston Community College	Stafford Campus	1 (Low)	1 (Low)	No	Yes	No	107-111	66-70	99-101	No	No	No
Houston Community College	Missouri City Campus	0	1 (Low)	No	Yes	No	107-111	66-70	102-104	No	No	No
Houston Community College	South Campus	1 (Low)	1 (Low)	No	Yes	No	107-111	71-74	102-104	No	No	No
Houston Community College	Coleman Campus	0	0	No	No	No	107-111	66-70	102-104	Yes	Yes	No
Houston Community College	West Loop Campus	0	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Houston Community College	Central Campus	0	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Houston Community College	District Administration Office	0	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No

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Houston Community College	Eastside Campus	0	0	No	No	No	107-111	66-70	102-104	Yes	Yes	No
Houston Community College	Felix Fraga Academic Campus	0	0	No	No	No	107-111	71-74	102-104	Yes	Yes	No
Houston Community College	Northeast Campus	1 (Low)	1 (Low)	Yes	Yes	No	107-111	66-70	99-101	No	No	No
Houston Community College	Northline Campus	0	0	No	Yes	No	102-106	66-70	99-101	No	No	No
Houston Community College	Automotive Technology Training Center	0	0	No	Yes	No	102-106	66-70	99-101	No	No	No
Houston Community College	North Forest Campus	0	0	No	Yes	No	102-106	66-70	99-101	No	No	No
Houston Community College	Acres Homes Campus	0	0	No	Yes	No	102-106	66-70	99-101	No	No	No
Lee College	Lee College	0	0	No	No	No	107-111	71-74	99-101	No	No	No
Lee College	Lee College McNair Center	1 (Low)	1 (Low)	Yes	Yes	No	107-111	66-70	99-101	No	No	No
Lee College	Lee College	0	0	No	Yes	No	107-111	71-74	99-101	No	No	No
San Jacinto Community College	South	0	0	No	Yes	No	112-117	71-74	102-104	No	No	No
San Jacinto Community College	Central Campus	1 (Low)	1 (Low)	No	No	No	112-117	71-74	102-104	No	No	No
San Jacinto Community College	District Office	0	0	No	No	No	107-111	71-74	102-104	No	No	No
San Jacinto Community College	North	1 (Low)	0	No	Yes	No	107-111	66-70	99-101	No	No	No

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San Jacinto Community College	Maritime Campus	1 (Low)	1 (Low)	Yes	Yes	Yes	112-117	75-83	105-116	No	No	No
San Jacinto Community College	Generation Park	1 (Low)	2 (Low-Moderate)	No	Yes	No	102-106	66-70	99-101	No	No	No
University of Houston	UH Downtown	0	0	No	No	No	107-111	66-70	102-104	Yes	Yes	No
University of Houston	UH Main Campus	0	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No
CCDID	N/A	1 (Low)	1 (Low) to 5 (High)	Yes	Yes	No	89-106	55-66	79-101	No	No	No
Harris Health	Acres Home Health Center	0	0	No	No	No	102-106	66-70	99-101	No	No	No
Harris Health	Aldine Health Center	0	0	No	No	No	102-106	66-70	99-101	No	No	No
Harris Health	Ben Taub Hospital	0	0	No	No	No	107-111	66-70	102-104	Yes	Yes	No
Harris Health	Casa de Amigos Health Center	0	0	No	No	No	107-111	66-70	102-104	Yes	Yes	No
Harris Health	Cypress Health Center	0	0	No	No	No	89-101	62-66	99-101	No	No	No
Harris Health	Danny Jackson Health Center	1 (Low)	0	No	No	No	89-101	62-66	95-98	No	No	No
Harris Health	El Franco Lee Health Center	0	0	No	Yes	No	102-106	66-70	99-101	Yes	Yes	No
Harris Health	Gulfgate Health Center	0	0	No	No	No	107-111	71-74	102-104	Yes	Yes	No
Harris Health	Humble "Squatty Lyons" Health Center	0	0	No	No	No	102-106	62-66	99-101	No	No	No
Harris Health	Lyndon B. Johnson (LBJ) Hospital	0	0	Yes	No	No	102-106	66-70	99-101	No	No	No
Harris Health	Martin Luther King Health Center	1 (Low)	1 (Low)	No	No	No	107-111	71-74	102-104	No	No	No
Harris Health	Monroe Clinic	0	0	No	Yes	No	107-111	71-74	102-104	No	No	No
Harris Health	Northwest Health Center	0	0	No	No	No	102-106	66-70	99-101	No	No	No

Planning Partner	Facility Name	Wildfire Threat Zone[a]	Wildfire Ignition Density Zone[b]	1% Annual Chance Flood Zone	Hurricane Harvey Flood Extent	Coastal Flood Zone	100-Year Probabilistic Hurricane Wind Zone[c]	20-Year Probabilistic Hurricane Wind Zone[c]	Hurricane Alicia Wind Zone[c]	Addicks Dam Inundation Zone	Barker Dam Inundation Zone	Lake Houston Dam Inundation Zone
Harris Health	Outpatient Center (adjacent to LBJ Hospital)	0	0	Yes	No	No	102-106	66-70	99-101	No	No	No
Harris Health	Pasadena Pediatric & Adolescent Health Center	0	0	No	No	No	107-111	71-74	102-104	No	No	No
Harris Health	Settegast Health Center	0	0	No	No	No	102-106	66-70	99-101	No	No	No
Harris Health	Smith Clinic	0	0	No	Yes	No	107-111	66-70	102-104	Yes	Yes	No
Harris Health	Strawberry Health Center	0	0	No	Yes	No	107-111	71-74	102-104	Yes	No	No
Harris Health	Thomas Street at Quentin Mease Health Center	0	0	No	No	No	107-111	66-70	102-104	Yes	Yes	No
Harris Health	Valbona Health Center	0	0	Yes	No	No	107-111	66-70	99-101	Yes	Yes	No
Gulf Coast Authority	Bayport Industrial Wastewater Treatment Facility	1 (Low)	0	Yes	Yes	No	112-117	71-74	102-104	No	No	No
Gulf Coast Authority	Washburn Tunnel Wastewater Treatment Facility	0	0	Yes	No	No	107-111	71-74	102-104	Yes	Yes	No
Gulf Coast Authority	Vince Bayou Receiving Station	0	0	Yes	No	No	107-111	71-74	102-104	Yes	Yes	No
Gulf Coast Authority	Blackhawk Regional Wastewater Treatment Plant	1 (Low)	1 (Low)	No	No	No	112-117	71-74	105-116	No	No	No
Gulf Coast Authority	GCA Central Office (administrative)	0	0	No	Yes	No	112-117	71-74	105-116	No	No	No

***Source**: 2025 Harris County Multi-Jurisdictional Hazard Mitigation Plan Web Viewer*

***Notes**: A designation of "yes" means part or all of the identified facility/jurisdiction is identified has being at risk. Conversely, "no" or "0" denotes areas with no hazard threat as described in Part 2: Risk Assessment; (C)=City boundary; CCISD=Cypress Creek Drainage Improvement District; GCA=Gulf Coast Authority;*

a=see Figure 17-2 for legend description

b=see Figure 17-1 for legend description

c=wind speeds are given in miles per hour (mph)

APPENDIX H: LINKAGE PROCEDURES

Local Mitigation Planning Policy Guide

FP-206-21-0002

Effective April 11, 2025

OMB Collection #1660-0062

FEMA

Appendix D: Amendment and Joining Procedures

A mitigation plan may need to be amended after it is approved by FEMA (or a state that was delegated approval authority under a PAS Agreement) and adopted by the local government(s). Amending an approved and adopted plan does not necessarily result in the need to reevaluate the entire plan against all requirements. FEMA expects local governments to conduct regularly scheduled reviews and amendments to their mitigation plan. This may result in modifications to the risk assessment or adding/removing mitigation actions, especially in preparation for submitting applications to FEMA for assistance and ensuring the project conforms with the mitigation plan. Participants are encouraged to keep the state and FEMA informed, but these amendments do not need to be reviewed and approved by the state and FEMA. If these changes identify new mitigation actions that might be eligible for FEMA assistance programs, then advise FEMA and the state. FEMA will acknowledge and note the receipt of the added action(s), where appropriate, but does not need to formally review or approve the action(s).

In addition, after a multi-jurisdictional mitigation plan has been adopted and approved by FEMA, other jurisdictions may wish to "join" the mitigation plan. Jurisdictions may be added to an existing, approved mitigation plan only if the conditions below are met.

- The jurisdiction asking to be included is within the boundaries of, or adjacent to, the area covered by the multi-jurisdictional mitigation plan.
- The organization responsible for preparing and submitting the multi-jurisdictional mitigation plan to the state and FEMA agrees with adding the requesting jurisdiction(s) to the mitigation plan.
- When the multi-jurisdictional mitigation plan was developed, the risk assessment included an analysis of the natural hazards that have the potential to affect the additional jurisdiction(s).

If all three conditions are not met, the jurisdiction may develop its own mitigation plan. If these conditions can be met, three options exist to add jurisdictions to an approved mitigation plan. For any of these options, each jurisdiction joining a multi-hazard planning process and seeking to receive approval from FEMA for a mitigation plan must satisfy all of the local mitigation plan requirements in 44 CFR § 201.6 and this guide at time of approval.

Option 1 - Participating jurisdiction that did not adopt the plan within one year of the Approved Pending Adoption (APA) date

This option is best suited to local jurisdictions that participated throughout the plan's development, but failed to adopt the plan within one year after the APA date. Participating jurisdictions adopting the plan more than a year after the APA date must either:

- Validate that the information in the plan remains current with respect to both the risk assessment (no recent hazard events, no changes in development) and mitigation strategy (no changes necessary); or
- Make the necessary updates before submitting the adoption resolution to state and FEMA.

However, this late adoption does not affect the plan expiration date. The plan will still expire five years from the date the first adoption was received.

Option 2 – Adding a jurisdiction that did not participate in the original planning process

This option is best suited to a multi-jurisdictional mitigation plan that has been recently approved by FEMA (and thus most of the plan's five-year approval period remains). In this case, the jurisdictions that participated in the multi-jurisdictional planning process are not required to take any action. Plan content specific to any new jurisdiction is included in a new annex or appendix to the existing mitigation plan, and no other changes are made to the previously approved mitigation plan.

The following actions must be taken to add new jurisdictions to an existing multi-jurisdictional mitigation plan and enable them to receive approval as part of the mitigation plan:

1. The requesting jurisdiction(s) must review the multi-jurisdictional hazard analysis and determine if any additional hazards that have not been addressed threaten the jurisdiction(s). If none exist, the jurisdiction(s) must document their review process and state that no additional hazards exist. If the review reveals additional hazards, the jurisdiction(s) must analyze the risks they face associated with those hazards and include this analysis in their written appendix to the multi-jurisdictional mitigation plan. The existing risk assessment cannot be resubmitted without this additional documentation.
2. The requesting jurisdiction(s) must document their agreement with the stated mitigation goals of the multi-jurisdictional mitigation plan. Additional goals specific to the requesting jurisdiction may be added. Each additional jurisdiction must also develop a list of proposed mitigation actions appropriate for that jurisdiction. These can include the common actions outlined in the multi-jurisdictional mitigation plan but must include specific mitigation actions for each profiled hazard for the jurisdiction itself.
3. The requesting jurisdiction(s) must document the involvement of both the general public and the local government in the planning process, in accordance with 44 CFR § 201.6. The level of participation by the additional jurisdiction(s) must be consistent with those in the multi-jurisdictional mitigation plan.
4. The annex or appendix, along with the multi-jurisdictional mitigation plan and correspondence of concurrence from the agency or organization responsible for the mitigation plan, must be submitted to the state for formal review. When the state finds the mitigation plan approvable, it will forward it to FEMA. When FEMA's review finds the mitigation plan “approvable pending adoption,” the new jurisdiction can officially adopt the full mitigation plan and its jurisdiction-

specific annex or appendix and submit the mitigation plan in final form through the state to FEMA for approval.

The mitigation plan expiration date for the added jurisdictions will be the date on which the originally approved multi-jurisdictional mitigation plan expires. This means that the added jurisdictions will have less than the entire five-year plan approval period before they will need to engage in the required update of the full multi-jurisdictional mitigation plan.

Option 3 - Revise and Update Full Plan

This option is best suited to situations in which new jurisdictions are added to the multi-jurisdictional mitigation plan more than one year into the approval period of the mitigation plan. In this case, the organization responsible for the mitigation plan determines that it is an appropriate time to begin the plan update process, and the new jurisdiction(s) can participate in that update with the original jurisdictions.

APPENDIX I: PUBLIC SURVEY RESULTS

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Start Date:	End Date:	Total # of Responses:
10/1/24	11/23/24	1,194

Question 1: Which of the following hazards have you experienced within the past 20 years in Harris County? (select all that apply)

	Very Concerned	Concerned	Somewhat Concerned	Not Concerned
Hurricane and Tropical Weather	60%	29%	10%	1%
Flooding (street, riverine, coastal, etc.)	42%	32%	22%	4%
Severe Weather (hail, thunderstorms, etc.)	31%	38%	26%	5%
Extreme Cold	24%	30%	33%	13%
Extreme Heat	40%	27%	24%	9%
Drought	26%	30%	30%	14%
Earthquake	2%	1%	7%	90%
Dam/Levee Failure	4%	4%	17%	75%
Coastal Erosion	8%	16%	33%	43%
Mass Movement (subsidence, landslide)	4%	7%	18%	71%
Tsunami	2%	2%	9%	87%
Toxic Release/HazMat	32%	25%	28%	14%
Energy Pipeline Failure	26%	27%	31%	16%
Utility Failure (prolonged disruption in electricity, water, or gas services)	59%	27%	12%	2%
Technology Failure (cellular, internet)	39%	35%	21%	6%

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 3: How concerned are you about the impacts of climate change on the frequency and intensity of hazards in Harris County?

Survey Response	County	Percentage
Concerned	811	71%
Not Concerned	334	29%
Total Responses	**1,145**	

Question 4: How many days of food and water does your household have on hand in the event of a disaster?

Survey Response	County	Percentage
0 – 2 days (not at all prepared)	***94***	***8%***
3 – 4 days (somewhat prepared)	***478***	***41%***
5 – 7 days (adequately prepared)	***397***	***34%***
8+ days (well prepared)	***209***	***18%***

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 5: How many days of vital supplies, such as medicine, does your household have on hand in the event of a disaster?

Survey Response	County	Percentage
0 – 2 days (not at all prepared)	83	*7%*
3 – 4 days (somewhat prepared)	226	*19%*
5 – 7 days (adequately prepared)	366	*31%*
8+ days (well prepared)	497	*42%*

Question 6: How prepared is your household to get along without electricity?

Survey Response	County	Percentage
0 – 2 days (not at all prepared)	366	*31%*
3 – 4 days (somewhat prepared)	376	*32%*
5 – 7 days (adequately prepared)	201	*17%*
8+ days (well prepared)	231	*20%*

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 7: Where would you expect to find useful information to help you get prepared? (Please rank the options below from 1 being your first choice to 11 being your last choice.)

Many free responses referenced problems they had when trying to make selections on Question 7.

Question 7a: Please specify any expected information sources not listed above.

AM Radio	Cell phone	Employer	HOA	Sirens
Amateur radio	Churches	Family	Prior Experience	Library
Books, Journals, newspapers	Emergency Text Messages	FEMA	Jeff Linder	Weather Channel

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 8: How would you expect to be notified in the case of an immediate threat caused by a hazard?

Many free responses referenced problems they had when trying to make selections on Question 8.

Question 8a: Please specify any expected notification sources not listed above.

Alert on cell phone	Cell Phone	Local news	Nextdoor
Cannot rearrange or pick one	Churches	Family Members	San Jacinto Police Department
AM Radio	Employer	HOA	Severe Weather Texts
Can't change priorities	Internet	neighbors	Telephone call from the City

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

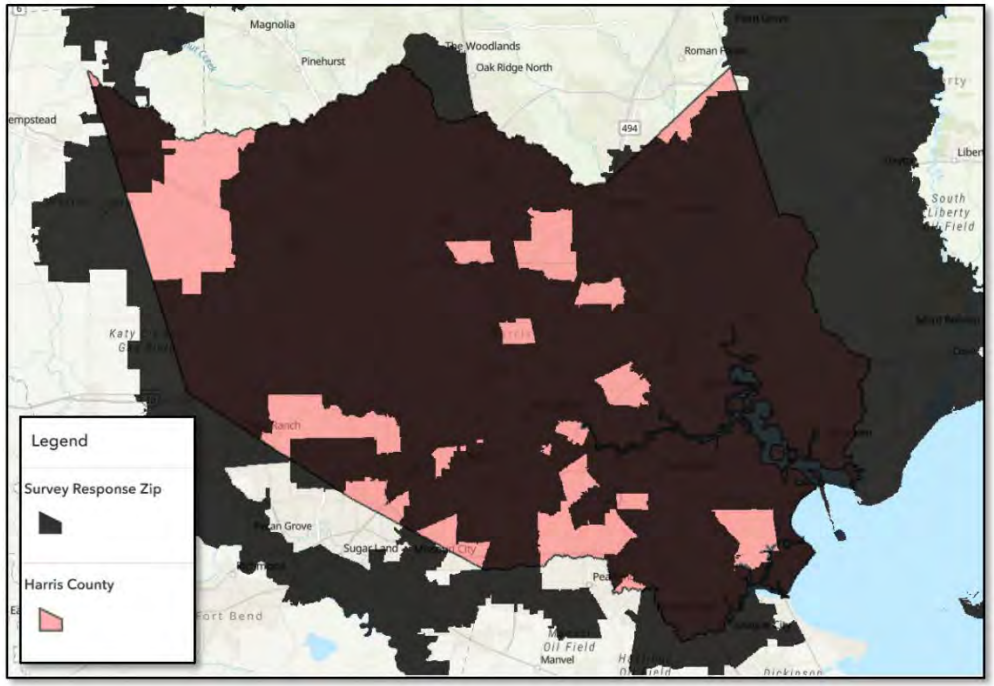
Worksheet H – Public Survey Results

Question 9: In what zip code do you live?

- 158 different zip code responses
- Top five zip codes shown below

Zip Code	Count	Percentage
77536	44	4%
77586	46	4%
77040	62	5%
77546	99	8%
77571	276	23%

- The map shows which zip codes responded in the survey. Areas NOT colored in black are zip codes that had no feedback.



Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

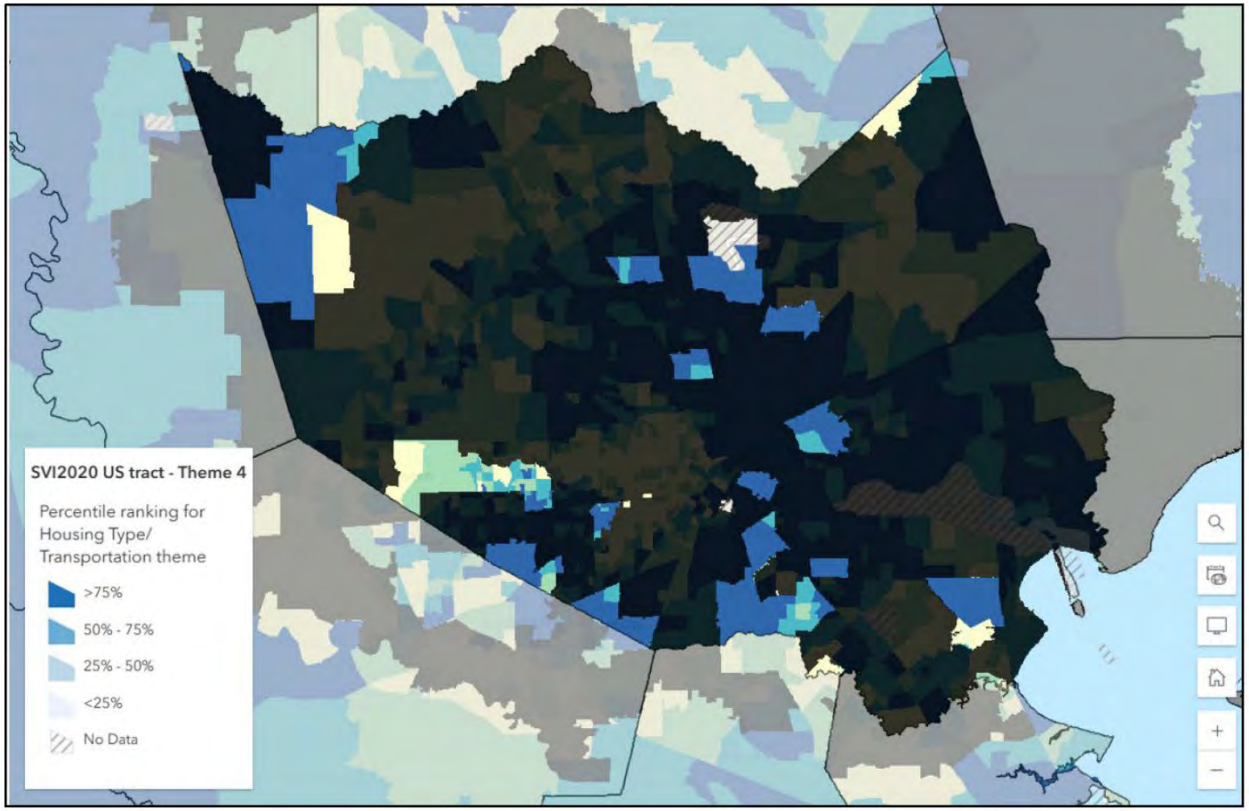
Harris County, Texas

Worksheet H – Public Survey Results

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

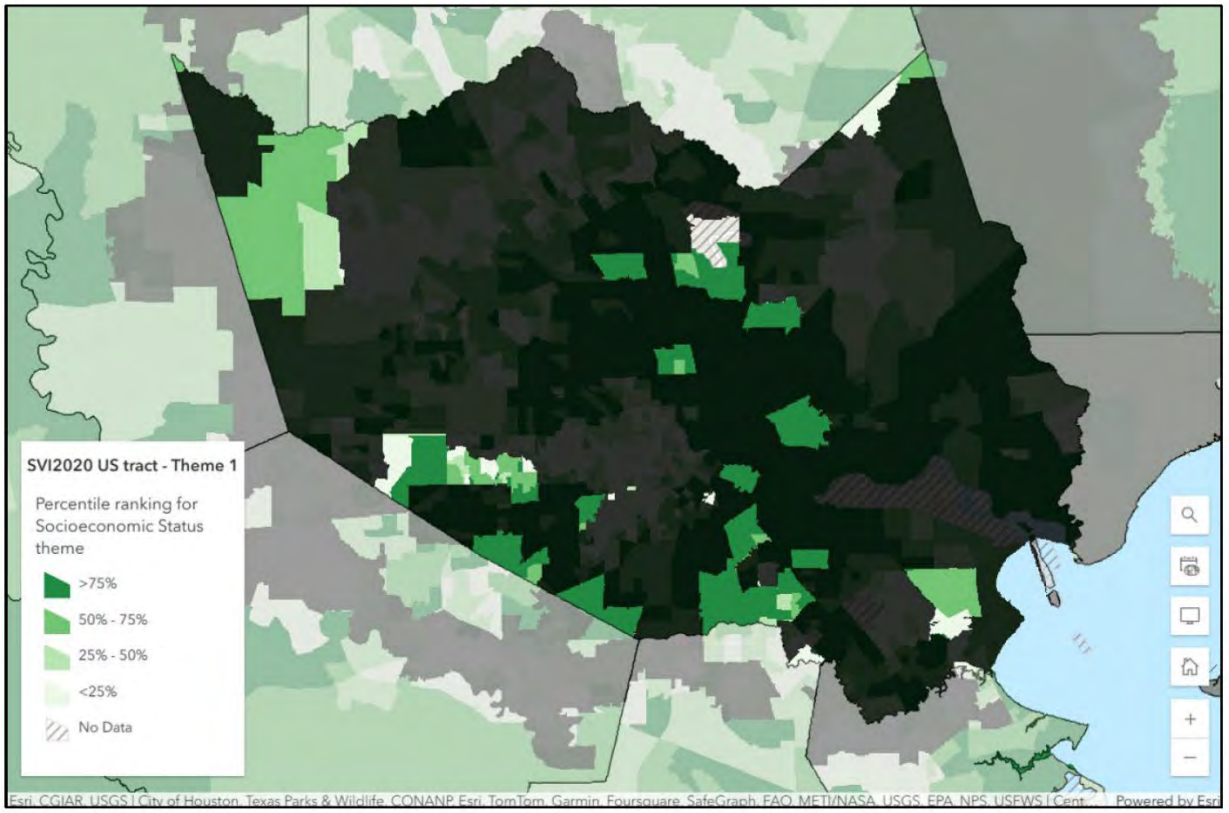
Worksheet H – Public Survey Results



Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

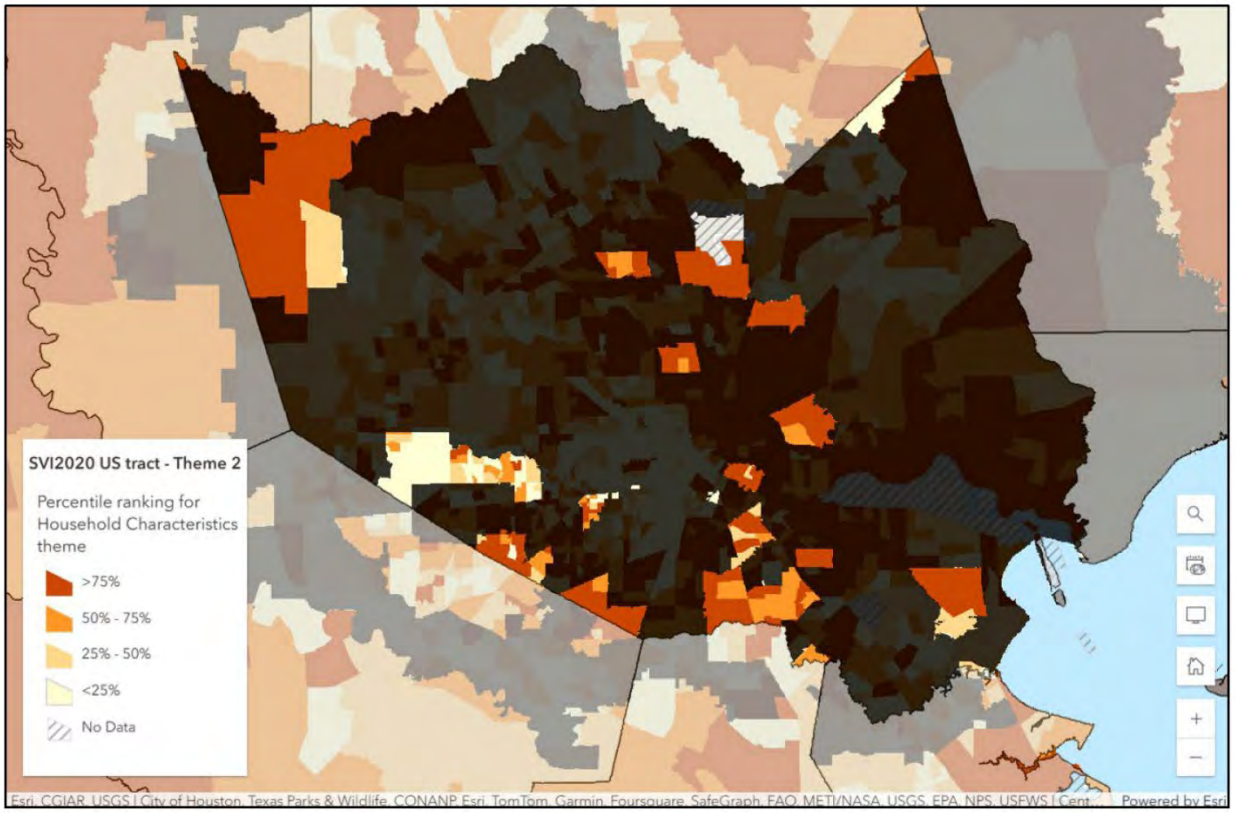
Worksheet H – Public Survey Results



Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

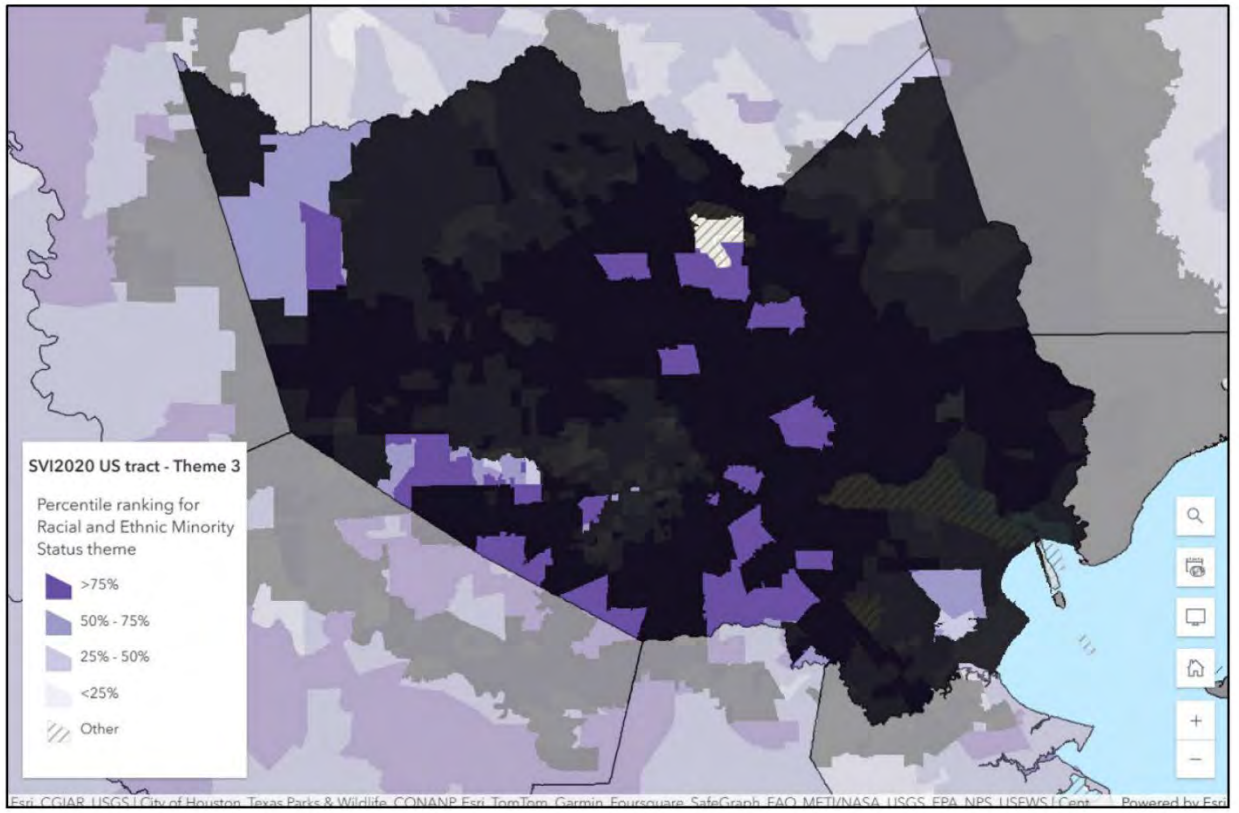
Worksheet H – Public Survey Results



Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

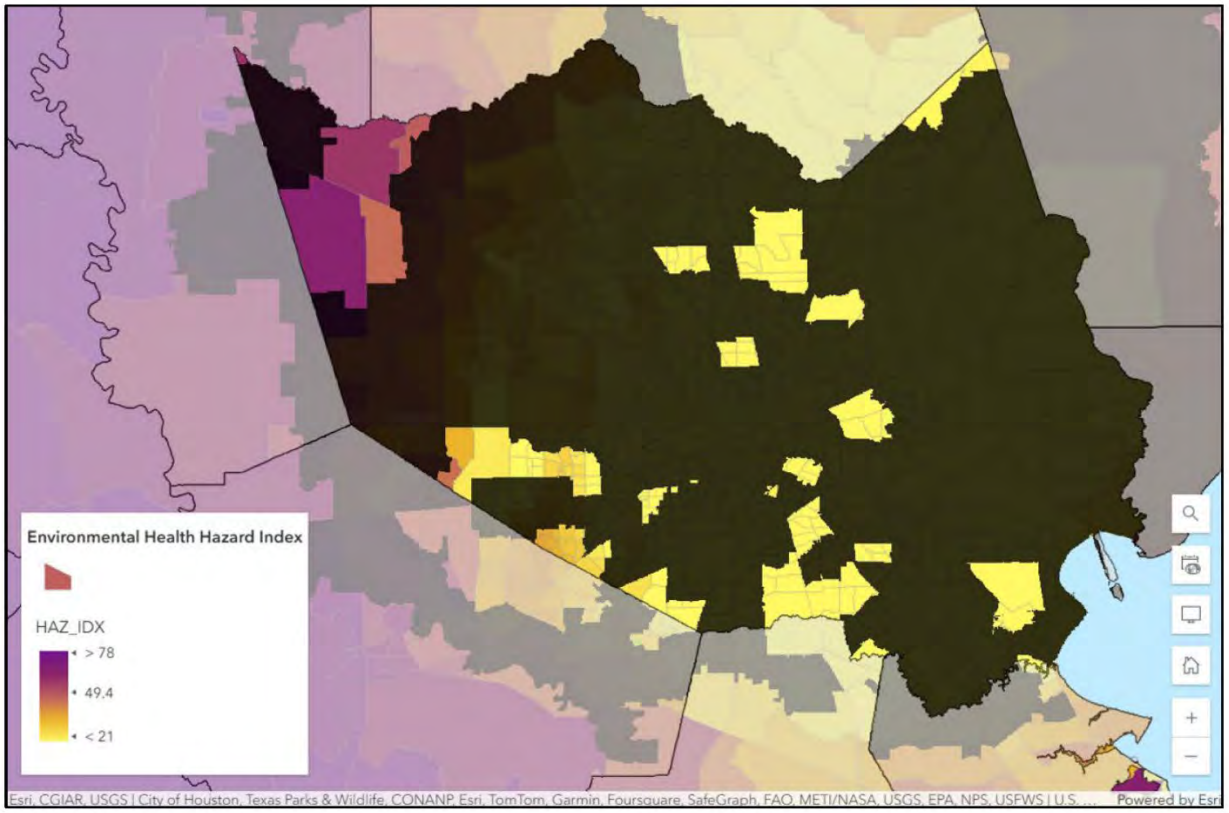
Worksheet H – Public Survey Results



Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results



Environmental Health Hazard Exposure Index summarizes potential exposure to harmful toxins at a neighborhood level

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 10: Do you own or rent your home?

- 1,045 responses (89%) were "owners
- 134 responses (11%) were "renters"

Question 11: When renting or buying your home, did you consider the impact a disaster could have on your home?

- 342 responses (29%) were "no"
- 830 responses (71%) were "yes"

Question 12: Was the risk of a natural hazard disclosed to you by a real estate agent, seller, or landlord before you purchased or moved?

- 791 responses (69%) were "no"
- 350 responses (31%) were "yes"

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 13: Is your residence? (select all that apply)

Survey Response	Count
at risk for wildfires	7
at_risk_for_wildfires,at_risk_of_subsidence	2
at_risk_for_wildfires,Unsure_if_located_in_any_of_the	1
at_risk_of_subsidence	27
at_risk_of_subsidence,Unsure_if_located_in_any_of_the	3
dam_failure_zone	2
FEMA_floodplain	368
FEMA_floodplain,at_risk_for_wildfires	5
FEMA_floodplain,at_risk_for_wildfires,at_risk_of_subsidence	3
FEMA_floodplain,at_risk_of_subsidence	60
FEMA_floodplain,at_risk_of_subsidence,Unsure_if_located_in_any_of_the	1
FEMA_floodplain,dam_failure_zone	2
FEMA_floodplain,dam_failure_zone,at_risk_for_wildfires	2
FEMA_floodplain,dam_failure_zone,at_risk_for_wildfires,at_risk_of_subsidence	2
FEMA_floodplain,dam_failure_zone,at_risk_of_subsidence	1
FEMA_floodplain,None	3
FEMA_floodplain,Unsure_if_located_in_any_of_the	25
None	331
Unsure_if_located_in_any_of_the	328
Unsure_if_located_in_any_of_the,None	3
TOTAL RESPONSES	**1,176**

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 14: Do you have? (select all that apply)

Survey Response	Count
Flood_insurance	3
Homeowner's_insurance	267
Homeowner's_insurance,Flood_insurance	323
Homeowner's_insurance,Flood_insurance,Wind_insurance	323
Homeowner's_insurance,Flood_insurance,Wind_insurance,None	1
Homeowner's_insurance,Renter's_insurance	4
Homeowner's_insurance,Renter's_insurance,Flood_insurance	6
Homeowner's_insurance,Renter's_insurance,Flood_insurance,Wind_insurance	1
Homeowner's_insurance,Wind_insurance	80
None	83
Renter's_insurance	84
Renter's_insurance,Flood_insurance	3
Renter's_insurance,Flood_insurance,Wind_insurance	1
Renter's_insurance,None	1
TOTAL RESPONSES	**1,180**

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 15: Have you ever experienced difficulties with securing or maintaining homeowner's or renter's insurance due to risks from hazards in Harris County? (select all that apply)

Survey Response	Count	Percentage of Total
No_no_issues	547	47%
Yes_policy_was_dropped	27	2%
Yes_policy_was_dropped,Yes_premiums_increased	77	7%
Yes_premiums_increased	350	30%
Yes_premiums_increased,No_no_issues	7	1%
Yes_trouble_securing_coverage	55	5%
Yes_trouble_securing_coverage,No_no_issues	2	0%
Yes_trouble_securing_coverage,Yes_policy_was_dropped	2	0%
Yes_trouble_securing_coverage,Yes_policy_was_dropped,Yes_premiums_increased	53	5%
Yes_trouble_securing_coverage,Yes_premiums_increased	36	3%
TOTAL RESPONSES	**1,156**	

Multi-Jurisdictional Hazard Mitigation Plan 2025 Update

Harris County, Texas

Worksheet H – Public Survey Results

Question 16: Please indicate the languages spoken in your household. (select all that apply)

Survey Response	Count	Percentage of Total
English	1006	84%
English,Arabic	2	<1%
English,Chinese	3	<1%
English,Chinese,other	1	<1%
English,other	26	2%
English,Spanish	136	11%
English,Spanish,Arabic,other	1	<1%
English,Spanish,Chinese	3	<1%
English,Spanish,Chinese,Arabic,Vietnamese	1	<1%
English,Spanish,other	7	1%
English,Spanish,Vietnamese	2	<1%
English,Vietnamese	4	<1%
other	1	<1%
Spanish	1	<1%
TOTAL	1,194	

APPENDIX J: NFIP GUIDANCE SURVEY RESULTS

Survey Question	Survey Response
Name	Brian Schneider
Which participating jurisdiction do you represent?	City of Morgan's Point
Who is your Floodplain Administrator or Manager?	Cobb Fendley, Carl Ahrendt, CAhrendt@cobbfendley.com
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	Access it via FEMA's website.
Has your community adopted the most recent FIRM?	No
When was the adoption?	—
If yes, where is that information stored?	—
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Yes
Does your municipality provide assistance with local floodplain determinations?	No, rely on registered surveyors to provide the determination.
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	No
If yes, what office or position is responsible?	—
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	No
Does the municipality conduct substantial damage assessments in the SFHA?	No
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Yes
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	No
Which provisions do you intend to expand?	—
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	—

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Jamie Galloway
Which participating jurisdiction do you represent?	City of Deer Park
Who is your Floodplain Administrator or Manager?	Greg Melching Deputy Building Official 281-478-7235 710 E San Augustine Deer Park, Tx 77536
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	Planning & Development Offices
Has your community adopted the most recent FIRM?	Yes
When was the adoption?	August 1, 2022
If yes, where is that information stored?	Planning & Development Offcies
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Yes
Does your municipality provide assistance with local floodplain determinations?	Yes -- Planning & Development Department reviews all building plans and, if approved, will issue the building permit. Exceeds, due to our free-board requirements. We require the lowest floor be elevated to the 500-hundred-year water surface elevation or two feet above the center line of the street or one-foot above the BFE, WHICHEVER IS HIGHER
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	Yes
If yes, what office or position is responsible?	Community Development & Planning
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Yes
Does the municipality conduct substantial damage assessments in the SFHA?	Yes
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Unsure
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	Yes
Which provisions do you intend to expand?	Permitting
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	—

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Dena Mahan
Which participating jurisdiction do you represent?	City of Webster
Who is your Floodplain Administrator or Manager?	James A Stewart Chief Building Official, City of Webster 101 Pennsylvania, Webster, TX 77598 jstewart@webstertx.gov 281-316-4131
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	Building Officials Office and Central Filing System
Has your community adopted the most recent FIRM?	Yes
When was the adoption?	Adopted by ordinance reference when map became available.
If yes, where is that information stored?	Maps are stored in the Building Officials Office and in our GIS System.
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Yes
Does your municipality provide assistance with local floodplain determinations?	Yes, either by providing a determination or by referring property owners to the tools available from FEMA.
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	Yes
If yes, what office or position is responsible?	James A Stewart Chief Building Official, City of Webster 101 Pennsylvania, Webster, TX 77598 jstewart@webstertx.gov 281-316-4131
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Yes
Does the municipality conduct substantial damage assessments in the SFHA?	Yes
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Yes
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	Yes
Which provisions do you intend to expand?	—
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	—

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Cassie Stephens
Which participating jurisdiction do you represent?	City of Hilshire Village
Who is your Floodplain Administrator or Manager?	Efrain Him HDR Engineering efrain.him@hdrinc.com
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	Online
Has your community adopted the most recent FIRM?	No
When was the adoption?	—
If yes, where is that information stored?	—
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Unsure
Does your municipality provide assistance with local floodplain determinations?	Yes, we have not needed to provide assistance in the past but can.
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	Yes
If yes, what office or position is responsible?	Permit Coordinator / Building Official/ City Engineers
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	No
Does the municipality conduct substantial damage assessments in the SFHA?	Unsure
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Unsure
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	Yes
Which provisions do you intend to expand?	Unsure, but it has been a while since we have assessed our ordinances on this topic.
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	—

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Charles Mistric
Which participating jurisdiction do you represent?	City of Waller
Who is your Floodplain Administrator or Manager?	City of Waller / Mayor Danny Marburger
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	No.
Has your community adopted the most recent FIRM?	No
When was the adoption?	—
If yes, where is that information stored?	—
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	No
Does your municipality provide assistance with local floodplain determinations?	Yes, the city of Waller Engineering Firm IDS reviews site plans and advises during review prior to building.
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	Yes
If yes, what office or position is responsible?	The city of Waller Engineering Firm IDS.
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Yes
Does the municipality conduct substantial damage assessments in the SFHA?	Unsure
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Unsure
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	No
Which provisions do you intend to expand?	—
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	Unknown at this time.

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Jose Magana
Which participating jurisdiction do you represent?	City of Stafford
Who is your Floodplain Administrator or Manager?	Christopher Riggs Director of Public Works 281-261-3940
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	N/A
Has your community adopted the most recent FIRM?	No
When was the adoption?	—
If yes, where is that information stored?	—
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Unsure
Does your municipality provide assistance with local floodplain determinations?	No, there is a lack of information to answer the question accurately, and no one knows the answer.
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	No
If yes, what office or position is responsible?	—
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Unsure
Does the municipality conduct substantial damage assessments in the SFHA?	Unsure
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Unsure
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	No
Which provisions do you intend to expand?	—
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	Over the last four years, we have had three different public works directors; the current one just started a couple of months ago. I accepted the position of Fire Chief on September 17, 2025. So, this was handed to me to fill out. The information necessary to answer the questions is unavailable. We look forward to any assistance you can provide moving forward.

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Kristina Watson
Which participating jurisdiction do you represent?	City of Spring Valley Village
Who is your Floodplain Administrator or Manager?	Safe Built/Scott Williams 281-740-7865
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	Harris County
Has your community adopted the most recent FIRM?	Yes
When was the adoption?	May 1, 2007
If yes, where is that information stored?	Harris County
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Yes
Does your municipality provide assistance with local floodplain determinations?	Yes, in partnership with Harris County
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	No
If yes, what office or position is responsible?	—
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Yes
Does the municipality conduct substantial damage assessments in the SFHA?	No
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	No
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	Yes
Which provisions do you intend to expand?	We would make changes to be current with the newest passed regulations
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	—

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Troy D. Harrison
Which participating jurisdiction do you represent?	City of Shoreacres
Who is your Floodplain Administrator or Manager?	Troy D. Harrison City Manager
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	City Hall
Has your community adopted the most recent FIRM?	Yes
When was the adoption?	November, 2024
If yes, where is that information stored?	City Hall
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Yes
Does your municipality provide assistance with local floodplain determinations?	Yes, when requested
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	Yes
If yes, what office or position is responsible?	City Manager
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Yes
Does the municipality conduct substantial damage assessments in the SFHA?	Yes
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Yes
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	No
Which provisions do you intend to expand?	—
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	—

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Lon Squyres
Which participating jurisdiction do you represent?	City of Jacinto City
Who is your Floodplain Administrator or Manager?	Lon Squyres
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	City Hall
Has your community adopted the most recent FIRM?	No
When was the adoption?	—
If yes, where is that information stored?	—
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Yes
Does your municipality provide assistance with local floodplain determinations?	No, we have never disagreed with the determinations.
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	Yes
If yes, what office or position is responsible?	Building Official
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Yes
Does the municipality conduct substantial damage assessments in the SFHA?	Yes
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Yes
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	Yes
Which provisions do you intend to expand?	Permitting
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	Limited staffing restricts our ability to meet the needs following a major flood event.

Note: "—" is used where there was no survey response

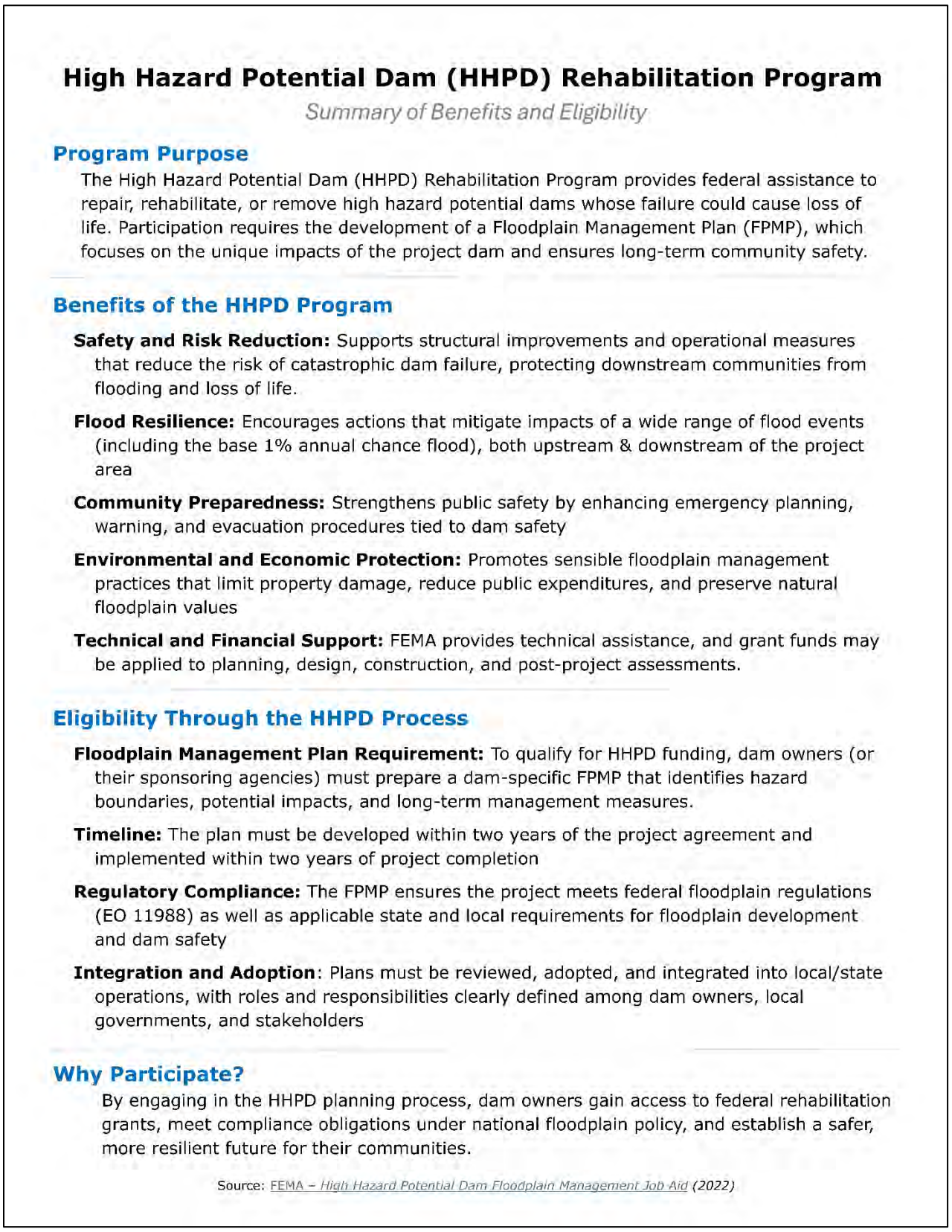
Survey Question	Survey Response
Name	Kevin Rodgers
Which participating jurisdiction do you represent?	City of Seabrook
Who is your Floodplain Administrator or Manager?	Nick Kondejewski
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	On the City of Seabrook internal network.
Has your community adopted the most recent FIRM?	Yes
When was the adoption?	November 1, 2016
If yes, where is that information stored?	On the City of Seabrook internal network
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Yes
Does your municipality provide assistance with local floodplain determinations?	Yes, our administrator provides information to potential developers during predevelopment meetings, and he will provide information upon request.
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	Yes
If yes, what office or position is responsible?	The City of Seabrook permit office along with the flood plain administrator.
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Yes
Does the municipality conduct substantial damage assessments in the SFHA?	Yes
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Yes
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	No
Which provisions do you intend to expand?	—
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	Additional federal funding.

Note: "—" is used where there was no survey response

Survey Question	Survey Response
Name	Tom Fullen
Which participating jurisdiction do you represent?	City of Hunters Creek Village
Who is your Floodplain Administrator or Manager?	Cobb Fendley - City Engineer, Steve Byington 713-485-8073
Where does your jurisdiction keep your Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report?	Online
Has your community adopted the most recent FIRM?	Yes
When was the adoption?	2014
If yes, where is that information stored?	Online
Has your community updated the floodplain ordinance language to include the current FIRM and FIS?	Yes
Does your municipality provide assistance with local floodplain determinations?	Yes, we help them look at the map to determine BFE
Does your municipality issue permits for all proposed development in the Special Flood Hazard Area (SFHA)?	Yes
If yes, what office or position is responsible?	CFM
Does your community have a coordinated process to determine substantial damage and to permit repair and improvement?	Unsure
Does the municipality conduct substantial damage assessments in the SFHA?	Unsure
Does your community have a plan for who will conduct substantial damage assessments and a procedure for assessment?	Yes
Does your municipality intend to revise future flood ordinances to include additional provisions and expanded capabilities (permitting, damage determinations, etc.)? If so, which provisions to you in	Yes
Which provisions do you intend to expand?	Adopt new FIRM
Would you like to include any challenges or any unmet needs that would assist in expanding the capabilities of your municipality? If so, please explain. (OPTIONAL)	—

Note: "—" is used where there was no survey response

Figure J-1. HHPD Rehabilitation Program Summary Handout



High Hazard Potential Dam (HHPD) Rehabilitation Program

Summary of Benefits and Eligibility

Program Purpose

The High Hazard Potential Dam (HHPD) Rehabilitation Program provides federal assistance to repair, rehabilitate, or remove high hazard potential dams whose failure could cause loss of life. Participation requires the development of a Floodplain Management Plan (FPMP), which focuses on the unique impacts of the project dam and ensures long-term community safety.

Benefits of the HHPD Program

Safety and Risk Reduction: Supports structural improvements and operational measures that reduce the risk of catastrophic dam failure, protecting downstream communities from flooding and loss of life.

Flood Resilience: Encourages actions that mitigate impacts of a wide range of flood events (including the base 1% annual chance flood), both upstream & downstream of the project area

Community Preparedness: Strengthens public safety by enhancing emergency planning, warning, and evacuation procedures tied to dam safety

Environmental and Economic Protection: Promotes sensible floodplain management practices that limit property damage, reduce public expenditures, and preserve natural floodplain values

Technical and Financial Support: FEMA provides technical assistance, and grant funds may be applied to planning, design, construction, and post-project assessments.

Eligibility Through the HHPD Process

Floodplain Management Plan Requirement: To qualify for HHPD funding, dam owners (or their sponsoring agencies) must prepare a dam-specific FPMP that identifies hazard boundaries, potential impacts, and long-term management measures.

Timeline: The plan must be developed within two years of the project agreement and implemented within two years of project completion

Regulatory Compliance: The FPMP ensures the project meets federal floodplain regulations (EO 11988) as well as applicable state and local requirements for floodplain development and dam safety

Integration and Adoption: Plans must be reviewed, adopted, and integrated into local/state operations, with roles and responsibilities clearly defined among dam owners, local governments, and stakeholders

Why Participate?

By engaging in the HHPD planning process, dam owners gain access to federal rehabilitation grants, meet compliance obligations under national floodplain policy, and establish a safer, more resilient future for their communities.

Source: FEMA – *High Hazard Potential Dam Floodplain Management Job Aid (2022)*